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(54) **HETEROCYCLIC COMPOUND AND ORGANIC LIGHT EMITTING DEVICE COMPRISING SAME**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

2016/0141515 A1 5/2016 Hayama et al.

2017/0133601 A1 5/2017 Sim et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 112194670 A 1/2021

KR 10-2016-0002408 A 1/2016

(Continued)

OTHER PUBLICATIONS

International Search Report for PCT/KR2020/009424 mailed on Nov. 3, 2020.

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(57) **ABSTRACT**

The present specification relates to a heterocyclic compound represented by Chemical Formula 1, and an organic light emitting device including the same.

11 Claims, 2 Drawing Sheets



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(56)

References Cited

U.S. PATENT DOCUMENTS

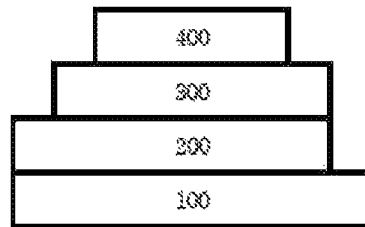
2017/0133602 A1 5/2017 Lee et al.
2019/0103560 A1 4/2019 Jung et al.
2020/0066996 A1 2/2020 La et al.

FOREIGN PATENT DOCUMENTS

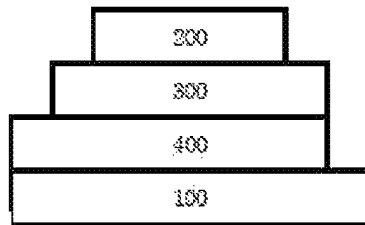
KR 10-2016-0018458 A 2/2016
KR 10-2016-0046703 A 4/2016
KR 10-2017-0053759 A 5/2017
KR 10-2017-0086277 A 7/2017
KR 10-2017-0090139 A 8/2017
KR 10-2018-0075398 A 7/2018
KR 10-2018-0076358 A * 7/2018

* cited by examiner

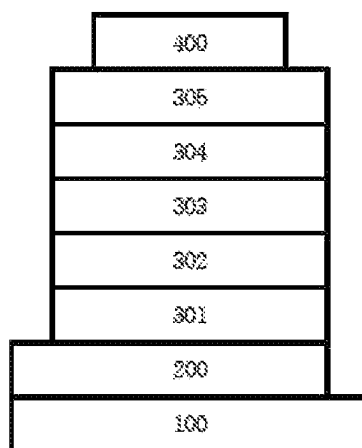
【FIG. 1】



【FIG. 2】



【FIG. 3】



【FIG. 4】

CATHODE
ELECTRON INJECTION LAYER
SECOND ELECTRON TRANSFER LAYER
SECOND HOLE BLOCKING LAYER
SECOND STACK LIGHT EMITTING LAYER
SECONE ELECTRON BLOCKING LAYER
SECOND HOLE TRANSFER LAYER
P-TYPE CHARGE GENERATION LAYER
N-TYPE CHARGE GENERATION LAYER
FIRST ELECTRON TRANSFER LAYER
FIRST HOLE BLOCKING LAYER
FIRST STACK LIGHT EMITTING LAYER
FIRST ELECTRON BLOCKING LAYER
FIRST HOLE TRANSFER LAYER
FIRST HOLE INJECTION LAYER
ANODE
SUBSTRATE

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HETEROCYCLIC COMPOUND AND ORGANIC LIGHT EMITTING DEVICE COMPRISING SAME

TECHNICAL FIELD

The present specification relates to a heterocyclic compound, and an organic light emitting device including the same.

The present specification claims priority to and the benefits of Korean Patent Application No. 10-2019-0088238, filed with the Korean Intellectual Property Office on Jul. 22, 2019, the entire contents of which are incorporated herein by reference.

BACKGROUND ART

An electroluminescent device is one type of self-emissive display devices, and has an advantage of having a wide viewing angle, and a high response speed as well as having an excellent contrast.

An organic light emitting device has a structure disposing an organic thin film between two electrodes. When a voltage is applied to an organic light emitting device having such a structure, electrons and holes injected from the two electrodes bind and pair in the organic thin film, and light emits as these annihilate. The organic thin film may be formed in a single layer or a multilayer as necessary.

A material of the organic thin film may have a light emitting function as necessary. For example, as a material of the organic thin film, compounds capable of forming a light emitting layer themselves alone may be used, or compounds capable of performing a role of a host or a dopant of a host-dopant-based light emitting layer may also be used. In addition thereto, compounds capable of performing roles of hole injection, hole transfer, electron blocking, hole blocking, electron transfer, electron injection and the like may also be used as a material of the organic thin film.

Development of an organic thin film material has been continuously required for enhancing performance, lifetime or efficiency of an organic light emitting device.

DISCLOSURE

Technical Problem

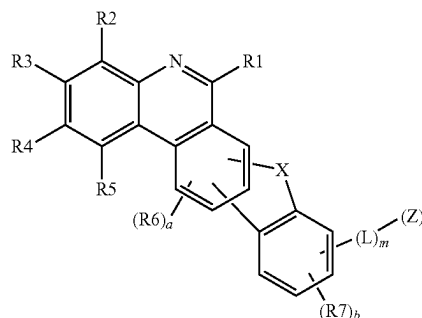
The present specification is directed to providing a heterocyclic compound, and an organic light emitting device including the same.

Technical Solution

One embodiment of the present specification provides a heterocyclic compound represented by the following Chemical Formula 1.

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[Chemical Formula 1]



In Chemical Formula 1,

X is O or S,

L is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group,

Z and R1 are each independently a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2 to C60 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group,

R2 to R7 are each independently hydrogen; deuterium; a halogen group; a cyano group; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group,

a is 1 or 2,

b is an integer of 1 to 3,

m and n are each an integer of 1 to 5, and

when a, b, m and n are each 2 or greater, substituents in the parentheses are the same as or different from each other.

Another embodiment of the present application provides an organic light emitting device including a first electrode; a second electrode provided opposite to the first electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes the heterocyclic compound represented by Chemical Formula 1.

Another embodiment of the present application provides an organic light emitting device including a first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and a second electrode provided on the second stack, wherein the charge generation layer includes the heterocyclic compound represented by Chemical Formula 1.

Advantageous Effects

A compound described in the present specification can be used as a material of an organic material layer of an organic

light emitting device. In the organic light emitting device, the compound is capable of performing a role of a hole injection material, a hole transfer material, a light emitting material, an electron transfer material, an electron injection material or the like. Particularly, the compound can be used as an electron transfer layer material or a charge generation layer material of an organic light emitting device.

Particularly, Chemical Formula 1 has a structure in which dibenzofuran or dibenzothiophene is fused to quinoline, and HOMO and LUMO levels can be adjusted by having various substituents in the benzene ring which is not fused to the quinoline in the dibenzofuran or dibenzothiophene. Accordingly, electrical properties of the structure can be enhanced by adjusting an energy band gap. In addition, electron transfer capability of a material can be enhanced through substituents having strong electron migration properties.

Specifically, when using the compound represented by Chemical Formula 1 in an organic material layer, a driving voltage of the device can be lowered, light efficiency can be enhanced, and lifetime properties of the device can be enhanced.

DESCRIPTION OF DRAWINGS

FIG. 1 to FIG. 4 each illustrate a lamination structure of an organic light emitting device according to one embodiment of the present specification.

- 100: Substrate
- 200: Anode
- 300: Organic Material Layer
- 301: Hole Injection Layer
- 302: Hole Transfer Layer
- 303: Light Emitting Layer
- 304: Electron Transfer Layer
- 305: Electron Injection Layer
- 400: Cathode

MODE FOR DISCLOSURE

Hereinafter, the present specification will be described in more detail.

In the present specification, a certain part “including” certain constituents means capable of further including other constituents, and does not exclude other constituents unless particularly stated on the contrary.

In the present specification, the term “substitution” means a hydrogen atom bonding to a carbon atom of a compound being changed to another substituent, and the position of substitution is not limited as long as it is a position at which a substituent can substitute, and when two or more substituents substitute, the two or more substituents may be the same as or different from each other.

In the present specification, a “T1 value” means an energy level value in a triplet state.

In the present specification, “substituted or unsubstituted” means being substituted with one or more substituents selected from the group consisting of a halogen group; a cyano group; a C1 to C60 linear or branched alkyl group; a C2 to C60 linear or branched alkenyl group; a C2 to C60 linear or branched alkynyl group; a C3 to C60 monocyclic or polycyclic cycloalkyl group; a C2 to C60 monocyclic or polycyclic heterocycloalkyl group; a C6 to C60 monocyclic or polycyclic aryl group; a C2 to C60 monocyclic or polycyclic heteroaryl group; a silyl group; a phosphine oxide group; and an amine group, or being unsubstituted, or being

substituted with a substituent linking two or more substituents selected from among the substituents illustrated above, or being unsubstituted.

In the present specification, the halogen may be fluorine, chlorine, bromine or iodine.

In the present specification, the alkyl group includes linear or branched having 1 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkyl group may be from 1 to 60, specifically from 1 to 40 and more specifically from 1 to 20. Specific examples thereof may include a methyl group, an ethyl group, a propyl group, an n-propyl group, an isopropyl group, a butyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a sec-butyl group, a 1-methyl-butyl group, a 1-ethylbutyl group, a pentyl group, an n-pentyl group, an isopentyl group, a neopentyl group, a tert-pentyl group, a hexyl group, an n-hexyl group, a 1-methylpentyl group, a 2-methylpentyl group, a 4-methyl-2-pentyl group, a 3,3-dimethylbutyl group, a 2-ethylbutyl group, a heptyl group, an n-heptyl group, a 1-methylhexyl group, a cyclopentylmethyl group, a cyclohexylmethyl group, an octyl group, an n-octyl group, a tert-octyl group, a 1-methylheptyl group, a 2-ethylhexyl group, a 2-propylpentyl group, an n-nonyl group, a 2,2-dimethylheptyl group, a 1-ethyl-propyl group, a 1,1-dimethyl-propyl group, an isohexyl group, a 2-methylpentyl group, a 4-methylhexyl group, a 5-methylhexyl group and the like, but are not limited thereto.

In the present specification, the alkenyl group includes linear or branched having 2 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkenyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 2 to 20. Specific examples thereof may include a vinyl group, a 1-propenyl group, an isopropenyl group, a 1-butenyl group, a 2-butenyl group, a 3-butenyl group, a 1-pentenyl group, a 2-pentenyl group, a 3-pentenyl group, a 3-methyl-1-butenyl group, a 1,3-butadienyl group, an allyl group, a 1-phenylvinyl-1-yl group, a 2-phenylvinyl-1-yl group, a 2,2-diphenylvinyl-1-yl group, a 2-phenyl-2-(naphthyl-1-yl)vinyl-1-yl group, a 2,2-bis(diphenyl-1-yl)vinyl-1-yl group, a stilbenyl group, a styrenyl group and the like, but are not limited thereto.

In the present specification, the alkynyl group includes linear or branched having 2 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkynyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 2 to 20.

In the present specification, the cycloalkyl group includes monocyclic or polycyclic having 3 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the cycloalkyl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a cycloalkyl group, but may also be different types of cyclic groups such as a heterocycloalkyl group, an aryl group and a heteroaryl group. The number of carbon groups of the cycloalkyl group may be from 3 to 60, specifically from 3 to 40 and more specifically from 5 to 20. Specific examples thereof may include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a 3-methylcyclopentyl group, a 2,3-dimethylcyclopentyl group, a cyclohexyl group, a 3-methylcyclohexyl group, a 4-methylcyclohexyl group, a 2,3-dimethylcyclohexyl group, a 3,4,5-trimethylcyclohexyl group, a 4-tert-butylcyclohexyl group, a cycloheptyl group, a cyclooctyl group and the like, but are not limited thereto.

In the present specification, the heterocycloalkyl group includes O, S, Se, N or Si as a heteroatom, includes

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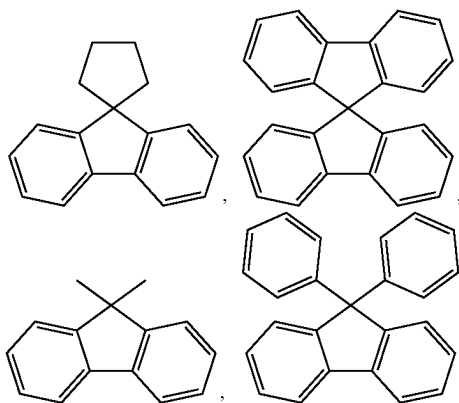
monocyclic or polycyclic having 2 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the heterocycloalkyl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a heterocycloalkyl group, but may also be different types of cyclic groups such as a cycloalkyl group, an aryl group and a heteroaryl group. The number of carbon atoms of the heterocycloalkyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 3 to 20.

In the present specification, the aryl group includes monocyclic or polycyclic having 6 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the aryl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be an aryl group, but may also be different types of cyclic groups such as a cycloalkyl group, a heterocycloalkyl group and a heteroaryl group. The aryl group includes a spiro group. The number of carbon atoms of the aryl group may be from 6 to 60, specifically from 6 to 40 and more specifically from 6 to 25. Specific examples of the aryl group may include a phenyl group, a biphenyl group, a triphenyl group, a naphthyl group, an anthryl group, a chrysenyl group, a phenanthrenyl group, a perylenyl group, a fluoranthenyl group, a triphenylenyl group, a phenalenyl group, a pyrenyl group, a tetracenyl group, a pentacenyl group, a fluorenyl group, an indenyl group, an acenaphth-ylenyl group, a benzofluorenyl group, a spirobifluorenyl group, a 2,3-dihydro-1H-indenyl group, a fused ring thereof, and the like, but are not limited thereto.

In the present specification, the silyl group is a substituent including Si, having the Si atom directly linked as a radical, and is represented by $\text{—SiR}_{104}\text{R}_{105}\text{R}_{106}$. R_{104} to R_{106} are the same as or different from each other, and may be each independently a substituent formed with at least one of hydrogen; deuterium; a halogen group; an alkyl group; an alkenyl group; a cycloalkyl group; an aryl group; and a heterocyclic group. Specific examples of the silyl group may include a trimethylsilyl group, a triethylsilyl group, a t-butyl dimethylsilyl group, a vinyl dimethylsilyl group, a propyl dimethylsilyl group, a triphenylsilyl group, a diphenylsilyl group, a phenylsilyl group and the like, but are not limited thereto.

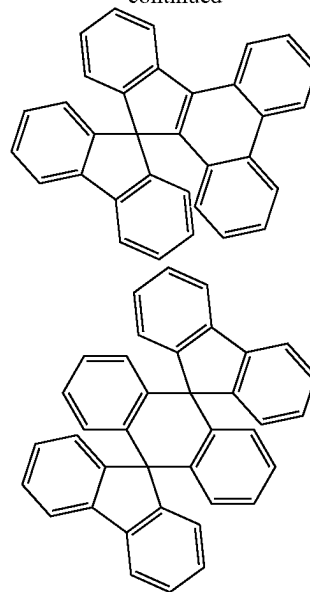
In the present specification, the fluorenyl group may be substituted, and adjacent substituents may bond to each other to form a ring.

When the fluorenyl group is substituted,



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-continued



and the like may be included, however, the structure is not limited thereto.

In the present specification, the heteroaryl group includes O, S, Se, N or Si as a heteroatom, includes monocyclic or polycyclic having 2 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the heteroaryl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a heteroaryl group, but may also be different types of cyclic groups such as a cycloalkyl group, a heterocycloalkyl group and an aryl group. The number of carbon atoms of the heteroaryl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 3 to 25. Specific examples of the heteroaryl group may include a pyridyl group, a pyrrolyl group, a pyrimidyl group, a pyridazinyl group, a furanyl group, a thiophene group, an imidazolyl group, a pyrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, a triazolyl group, a furazanyl group, an oxadiazolyl group, a thiadiazolyl group, a dithiazolyl group, a tetrazolyl group, a pyranyl group, a thiopyranyl group, a diazanyl group, an oxazinyl group, a thiazinyl group, a dioxynyl group, a triazinyl group, a tetrazinyl group, a quinolyl group, an isoquinolyl group, a quinazolinyl group, an isoquinazolinyl group, a quinoxalinyl group, a naphthyridyl group, an acridinyl group, a phenanthridinyl group, an imidazopyridinyl group, a diazanaphthalenyl group, a triazaindene group, an indolyl group, an indolizinyl group, a benzothiazolyl group, a benzoxazolyl group, a benzimidazolyl group, a benzothiophene group, a benzofuran group, a dibenzothiophene group, a dibenzofuran group, a carbazolyl group, a benzo-carbazolyl group, a dibenzocarbazolyl group, a phenazinyl group, a dibenzosilole group, spirobi(dibenzosilole), a dihydrophenazinyl group, a phenoxazinyl group, a phenanthridyl group, an imidazopyridinyl group, a thienyl group, an indolo [2,3-a]carbazolyl group, an indolo[2,3-b]carbazolyl group, an indolinyl group, a 10,11-dihydro-dibenzo[b,f]azepine group, a 9,10-dihydroacridinyl group, a phenanthrazinyl group, a phenothiazinyl group, a phthalazinyl group, a naphthylidiny group, a phenanthrolinyl group, a benzo[c][1,2,5]thiadiazolyl group, a 5,10-dihydrobenzo[b,e][1,4]azasilinyl group, a pyrazolo[1,5-c]quinazolinyl group, a

pyrido[1,2-b]indazolyl group, a pyrido[1,2-a]imidazo[1,2-e]indolyl group, a 5,11-dihydroindeno[1,2-b]carbazolyl group and the like, but are not limited thereto.

In the present specification, the phosphine oxide group may be specifically substituted with an aryl group, and examples described above may be applied to the aryl group. Examples of the phosphine oxide group may include a diphenylphosphine oxide group, a dinaphthylphosphine oxide group and the like, but are not limited thereto.

In the present specification, the amine group may be selected from the group consisting of a monoalkylamine group; a monoarylamine group; a monoheteroarylamine group; —NH₂; a dialkylamine group; a diarylamine group; a diheteroarylamine group; an alkylarylamine group; an alkylheteroarylamine group; and an arylheteroarylamine group, and although not particularly limited thereto, the number of carbon atoms is preferably from 1 to 30. Specific examples of the amine group may include a methylamine group, a dimethylamine group, an ethylamine group, a diethylamine group, a phenylamine group, a naphthylamine group, a biphenylamine group, a dibiphenylamine group, an anthracenylamine group, a 9-methyl-anthracenylamine group, a diphenylamine group, a phenylnaphthylamine group, a ditolylamine group, a phenyltolylamine group, a triphenylamine group, a biphenylnaphthylamine group, a phenylbiphenylamine group, a biphenylfluorenylamine group, a phenyltriphenylenylamine group, a biphenyltriphenylenylamine group and the like, but are not limited thereto.

In the present specification, the structures illustrated as the aryl group and the heteroaryl group described above may be used as the arylene group and the heteroarylene group except that they are not a monovalent group.

In the present specification, an “adjacent” group may mean a substituent substituting an atom directly linked to an atom substituted by the corresponding substituent, a substituent sterically most closely positioned to the corresponding substituent, or another substituent substituting an atom substituted by the corresponding substituent. For example, two substituents substituting ortho positions in a benzene ring, and two substituents substituting the same carbon in an aliphatic ring may be interpreted as groups “adjacent” to each other.

As an aliphatic or aromatic hydrocarbon ring or heteroring that adjacent groups may form, the structures illustrated as the cycloalkyl group, the heterocycloalkyl group, the aryl group and the heteroaryl group described above may be applied except for those that are not a monovalent group.

One embodiment of the present specification provides a heterocyclic compound represented by Chemical Formula 1.

Particularly, Chemical Formula 1 has a structure in which dibenzofuran or dibenzothiophene is fused to quinoline, and HOMO and LUMO levels can be adjusted by having various substituents in the benzene ring which is not fused to the quinoline in the dibenzofuran or dibenzothiophene. Accordingly, electrical properties of the structure may be enhanced by adjusting an energy band gap. In addition, electron transfer capability of a material may be enhanced through substituents having strong electron migration properties.

In one embodiment of the present specification, X is O or S.

In one embodiment of the present specification, X is O.

In another embodiment, X is S.

In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group.

By the Z substituent and the core structure being separated by L in the compound, hole migration properties of the core structure and electron migration properties of the substituent are distinguished in the electron cloud distribution of the molecule. This is an excellent bipolar structure facilitating electron migration, and therefore, efficiency, lifetime and driving voltage may be significantly improved.

In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted C6 to C30 arylene group; or a substituted or unsubstituted C2 to C30 heteroarylene group.

In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted phenylene group; a substituted or unsubstituted biphenylene group; a substituted or unsubstituted terphenylene group; a substituted or unsubstituted naphthylene group; a substituted or unsubstituted anthracenylene group; a substituted or unsubstituted triphenylenylene group; a substituted or unsubstituted phenanthrenylene group; a substituted or unsubstituted divalent pyridine group; a substituted or unsubstituted divalent pyrimidine group; or a substituted or unsubstituted divalent triazine group.

In one embodiment of the present specification, L is a direct bond; a phenylene group unsubstituted or substituted with an aryl group or a heteroaryl group; a biphenylene group; a terphenylene group; a naphthylene group; an anthracenylene group; a triphenylenylene group; a phenanthrenylene group; a divalent pyridine group unsubstituted or substituted with an aryl group or a heteroaryl group; a divalent pyrimidine group unsubstituted or substituted with an aryl group; or a divalent triazine group unsubstituted or substituted with an aryl group or a heteroaryl group.

In one embodiment of the present specification, L is a direct bond; a phenylene group unsubstituted or substituted with a phenyl group or a carbazole group; a biphenylene group; a terphenylene group; a naphthylene group; an anthracenylene group; a triphenylenylene group; a phenanthrenylene group; a divalent pyridine group unsubstituted or substituted with a phenyl group or a pyridine group; a divalent pyrimidine group unsubstituted or substituted with a phenyl group or a biphenyl group; or a divalent triazine group unsubstituted or substituted with a phenyl group unsubstituted or substituted with a cyano group, a biphenyl group, a naphthyl group, or a carbazole group unsubstituted or substituted with a phenyl group.

In one embodiment of the present specification, Z is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2 to C60 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

In one embodiment of the present specification, Z is a substituted or unsubstituted C1 to C30 alkyl group; a substituted or unsubstituted C2 to C30 alkenyl group; a substituted or unsubstituted C2 to C30 alkynyl group; a substituted or unsubstituted C3 to C30 cycloalkyl group; a substituted or unsubstituted C2 to C30 heterocycloalkyl group; a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group; a substituted or unsubstituted C1 to C10 alkylamine group; a substituted or

unsubstituted C6 to C30 arylamine group; a substituted or unsubstituted C2 to C30 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

In one embodiment of the present specification, Z is a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group; or a substituted or unsubstituted phosphine oxide group.

In one embodiment of the present specification, Z is a substituted or unsubstituted phenyl group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a substituted or unsubstituted pyridine group; a substituted or unsubstituted pyrimidine group; a substituted or unsubstituted triazine group; a substituted or unsubstituted carbazole group; a substituted or unsubstituted benzocarbazole group; a substituted or unsubstituted phenanthrolinyl group; a substituted or unsubstituted dibenzofuran group; a substituted or unsubstituted dibenzothiophene group; a substituted or unsubstituted dihydroacridine group; or a substituted or unsubstituted phosphine oxide group.

In one embodiment of the present specification, Z is a phenyl group unsubstituted or substituted with a cyano group, a phenyl group or a carbazole group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a pyridine group unsubstituted or substituted with a phenyl group or a pyridine group; a pyrimidine group unsubstituted or substituted with a phenyl group or a biphenyl group; a triazine group unsubstituted or substituted with a phenyl group unsubstituted or substituted with a naphthyl group or a cyano group, a biphenyl group, a naphthyl group, a dimethylfluorenyl group, a dibenzofuran group, a dibenzothiophene group, or a carbazole group unsubstituted or substituted with a phenyl group; a carbazole group unsubstituted or substituted with a phenyl group; a benzocarbazole group; a phenanthrolinyl group unsubstituted or substituted with a phenyl group; a dibenzofuran group unsubstituted or substituted with a phenyl group; a dibenzothiophene group unsubstituted or substituted with a phenyl group; a dihydroacridine group unsubstituted or substituted with a methyl group or a phenyl group; or a phosphine oxide group unsubstituted or substituted with a phenyl group.

In one embodiment of the present specification, R1 is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2 to C60 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

In one embodiment of the present specification, R1 is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl

group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

In one embodiment of the present specification, R1 is a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

In one embodiment of the present specification, R1 is a substituted or unsubstituted C6 to C30 aryl group; or a substituted or unsubstituted C2 to C30 heteroaryl group.

In one embodiment of the present specification, R1 is a substituted or unsubstituted C6 to C30 aryl group.

In one embodiment of the present specification, R1 is a substituted or unsubstituted phenyl group; a substituted or unsubstituted naphthyl group; a substituted or unsubstituted anthracenyl group; a substituted or unsubstituted phenanthrenyl group; a substituted or unsubstituted pyrenyl group; or a substituted or unsubstituted perylene group.

In one embodiment of the present specification, R1 is a phenyl group; a naphthyl group; an anthracenyl group; a phenanthrenyl group; a pyrenyl group; or a perylene group.

In one embodiment of the present specification, R2 to R7 are each independently hydrogen; deuterium; a halogen group; a cyano group; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

In one embodiment of the present specification, R2 to R7 are each independently hydrogen; deuterium; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

In one embodiment of the present specification, R2 to R7 are each independently hydrogen; or deuterium.

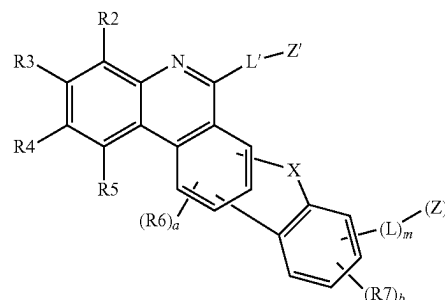
In one embodiment of the present specification, R2 to R7 are hydrogen.

In one embodiment of the present specification, m may be 1 or 2.

In one embodiment of the present specification, n may be an integer of 1 to 5.

In one embodiment of the present specification, Chemical Formula 1 is represented by the following Chemical Formula 1-1.

[Chemical Formula 1-1]



In Chemical Formula 1-1,

X, L, Z, R2 to R7, a, b, m and n have the same definitions as in Chemical Formula 1,

L' is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group,

Z' is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl

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group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

In one embodiment of the present specification, L' is a direct bond.

In one embodiment of the present specification, Z' is a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

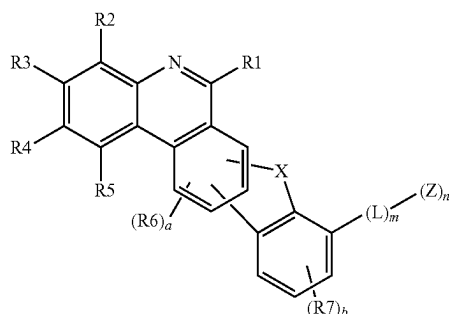
In one embodiment of the present specification, Z' is a substituted or unsubstituted C6 to C60 aryl group.

In one embodiment of the present specification, Z' is a substituted or unsubstituted C6 to C30 aryl group.

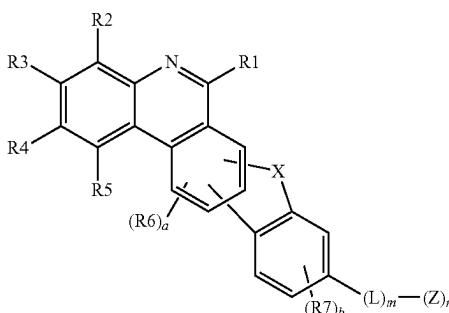
In one embodiment of the present specification, Z' is a phenyl group; a naphthyl group; an anthracenyl group; a phenanthrenyl group; a pyrenyl group; or a perylene group.

In the heterocyclic compound provided in one embodiment of the present specification, Chemical Formula 1 is represented by any one of the following Chemical Formulae 1-2 to 1-5.

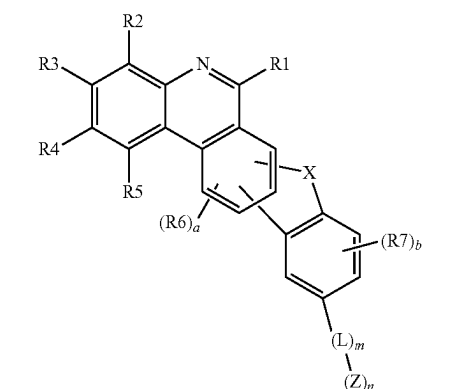
[Chemical Formula 1-2]



[Chemical Formula 1-3]



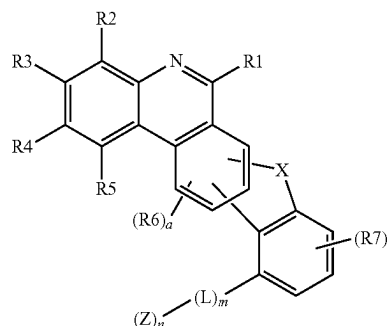
[Chemical Formula 1-4]



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[Chemical Formula 1-5]

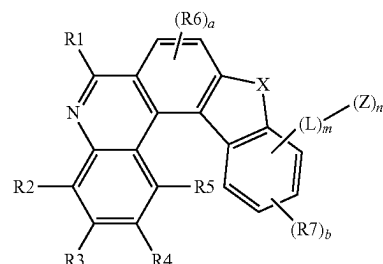


In Chemical Formulae 1-2 to 1-5,

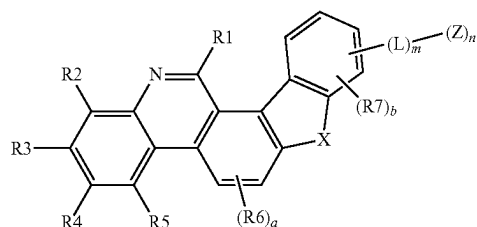
X, L, Z, R1 to R7, a, b, m and n have the same definitions as in Chemical Formula 1.

In the heterocyclic compound provided in one embodiment of the present specification, Chemical Formula 1 is represented by any one of the following Chemical Formulae 2 to 4.

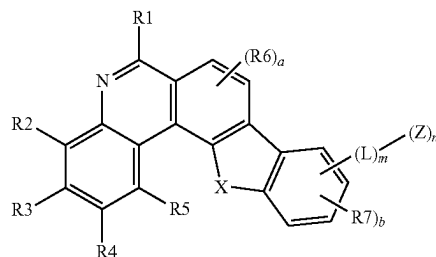
[Chemical Formula 2]



[Chemical Formula 3]



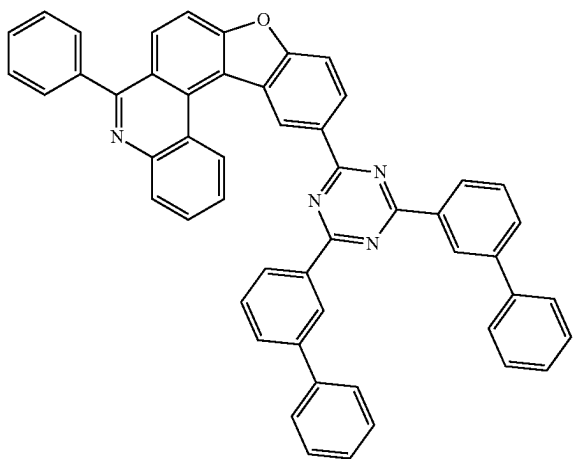
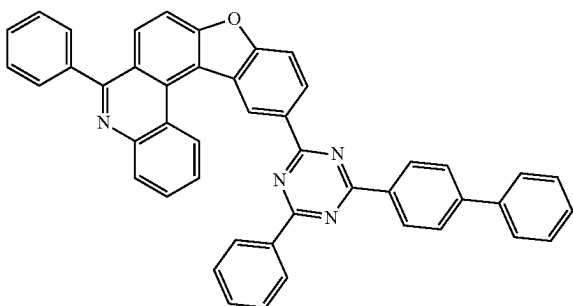
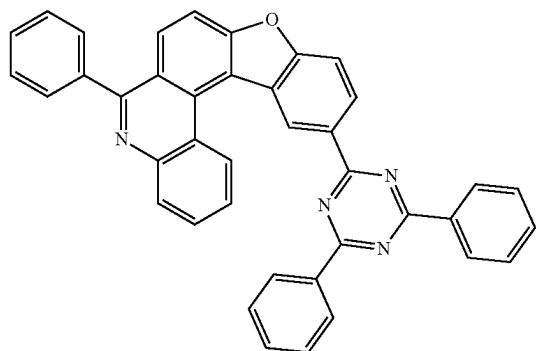
[Chemical Formula 4]



In Chemical Formulae 2 to 4,

X, L, Z, R1 to R7, a, b, m and n have the same definitions as in Chemical Formula 1.

In one embodiment of the present specification, Chemical Formula 1 may be represented by any one of the following compounds, but is not limited thereto.

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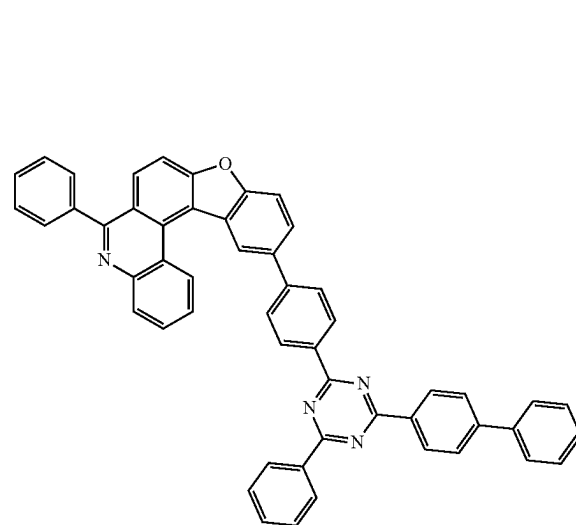
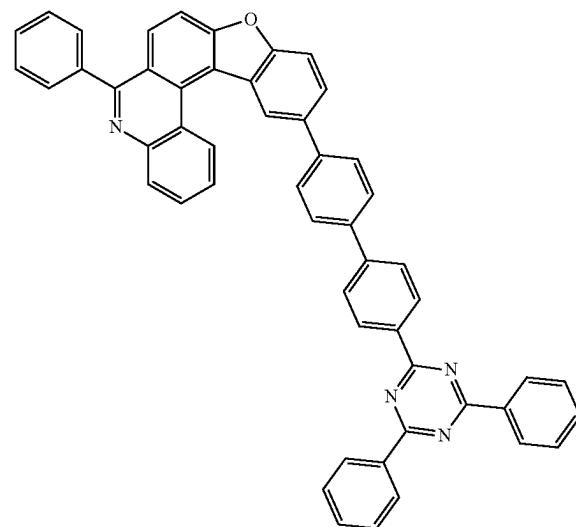
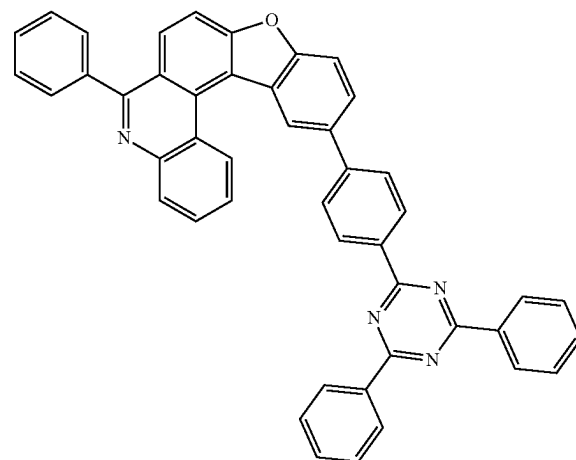
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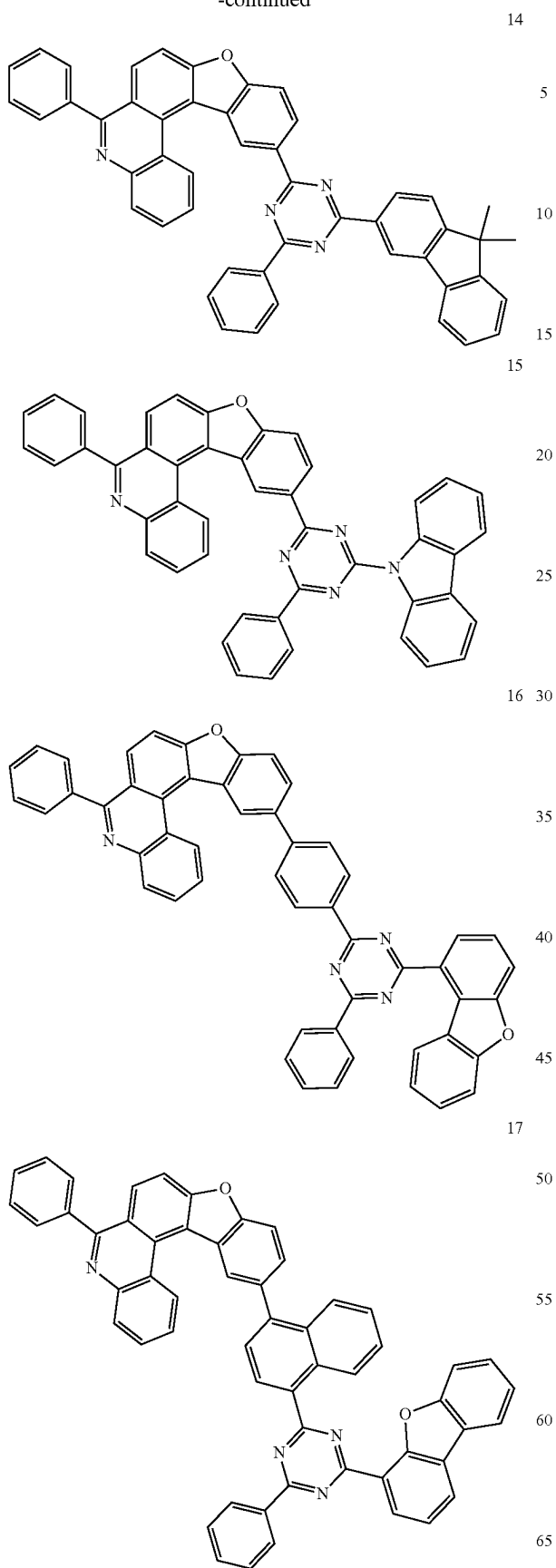
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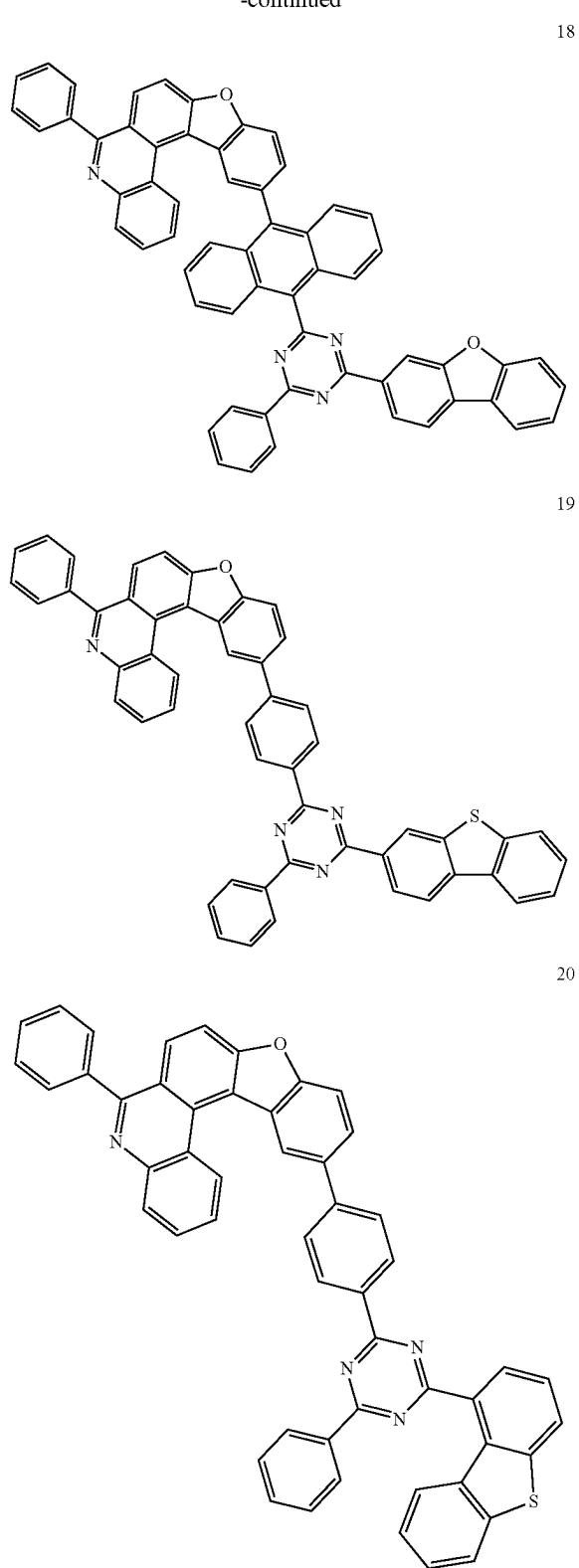


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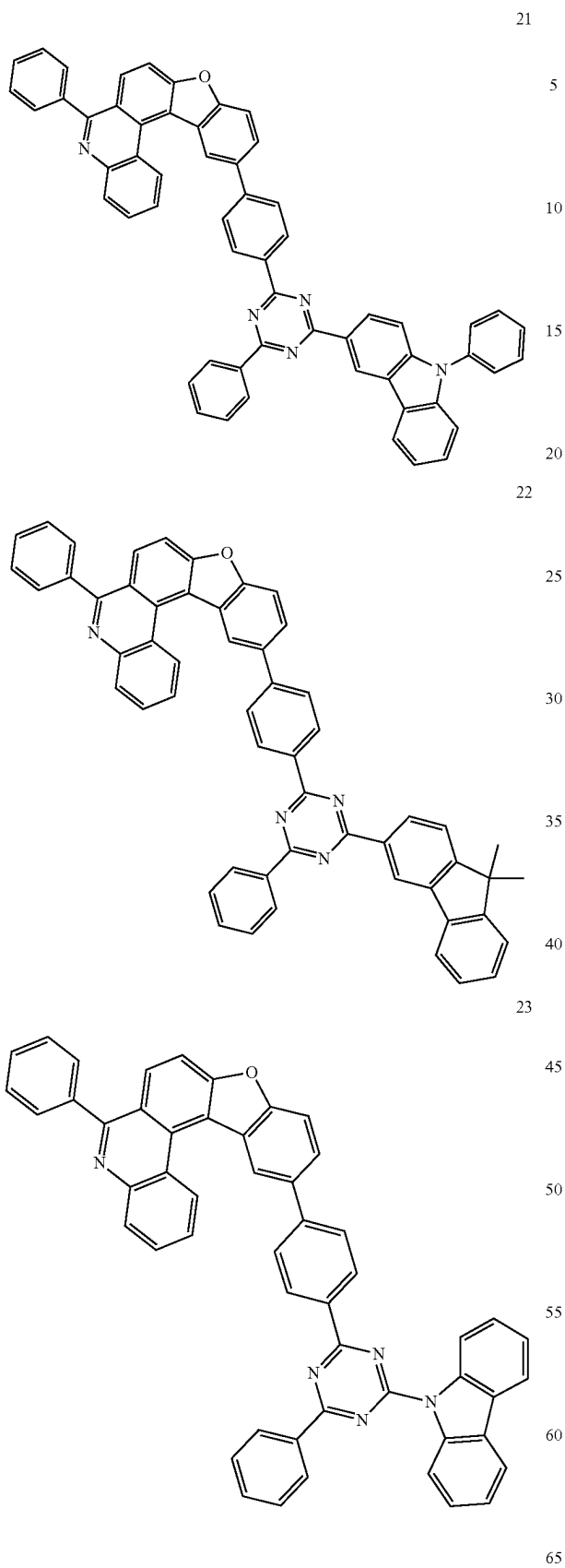
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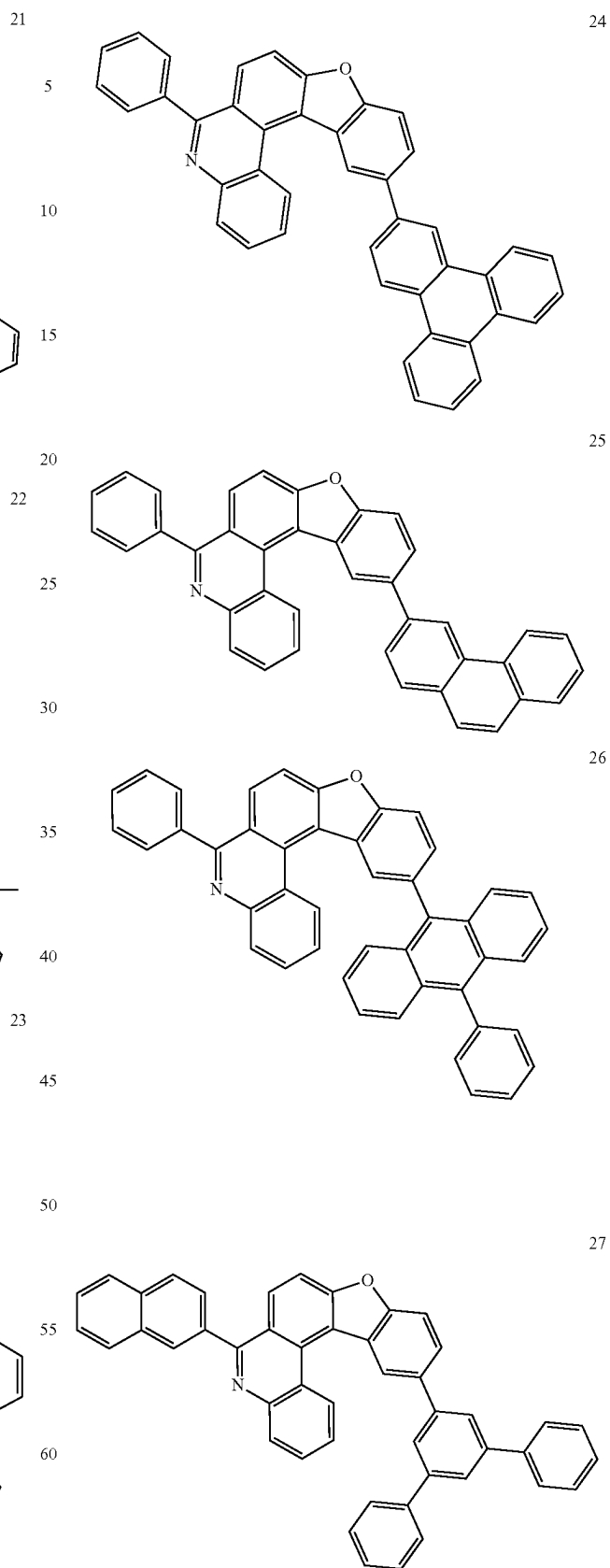


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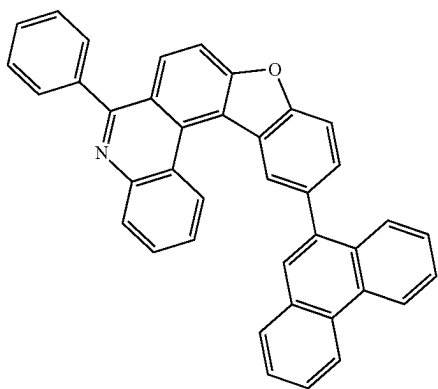
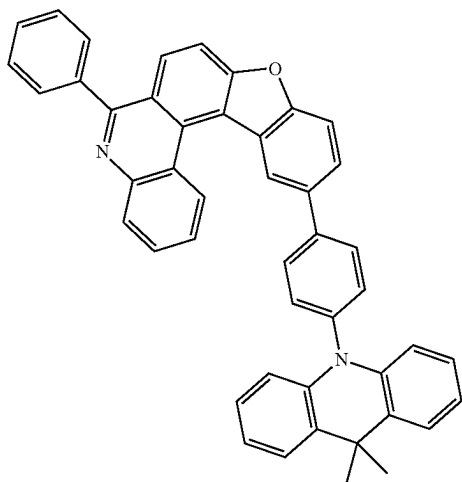
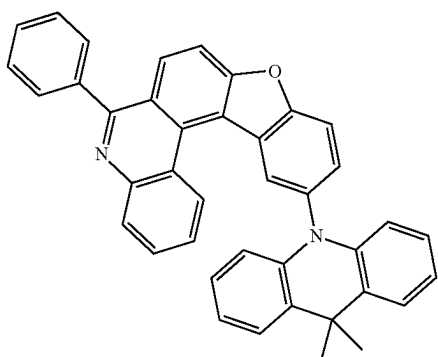
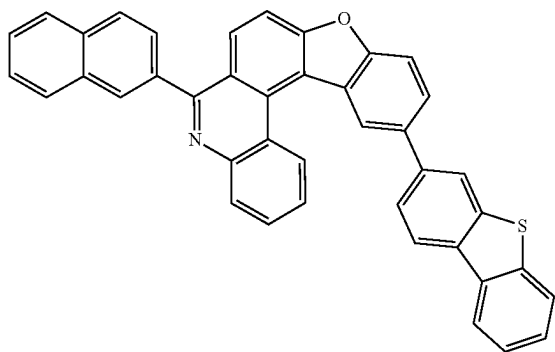
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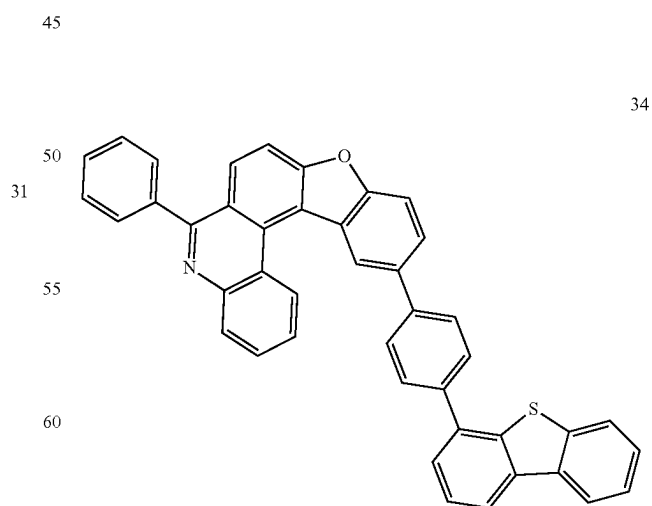
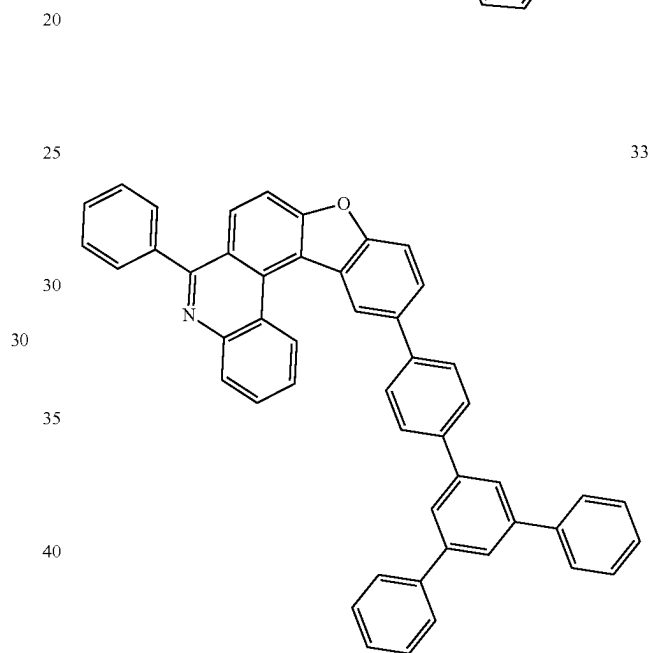
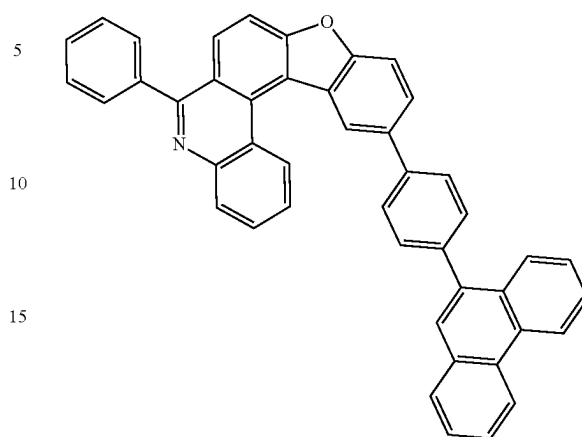


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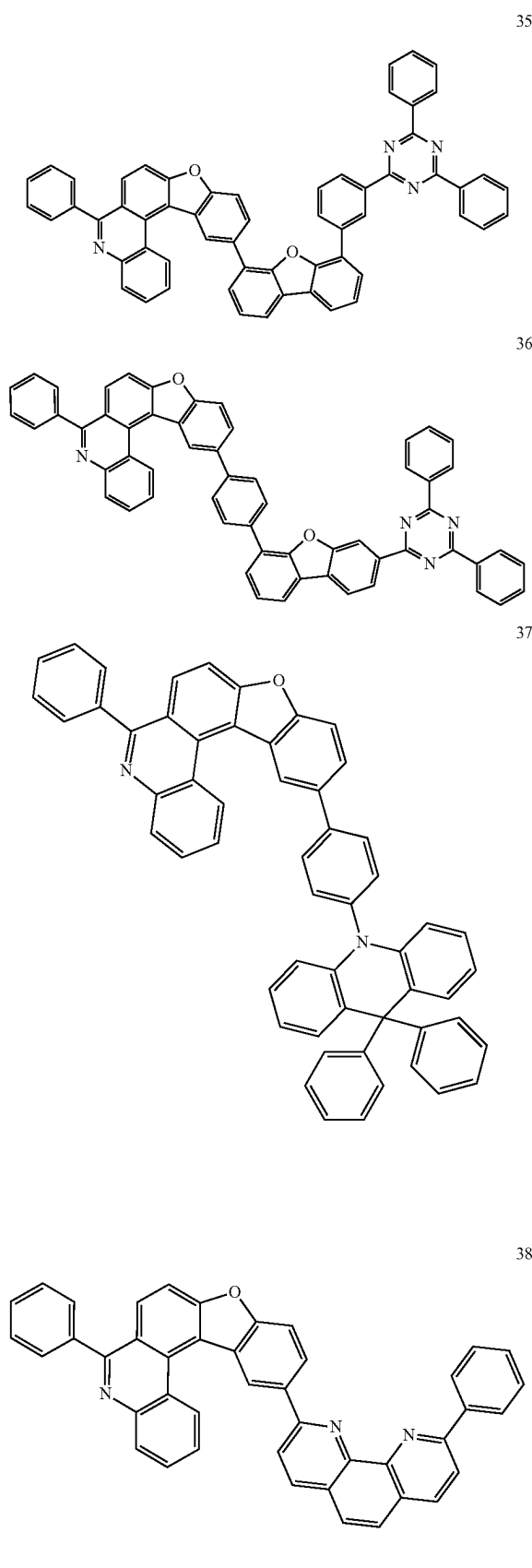
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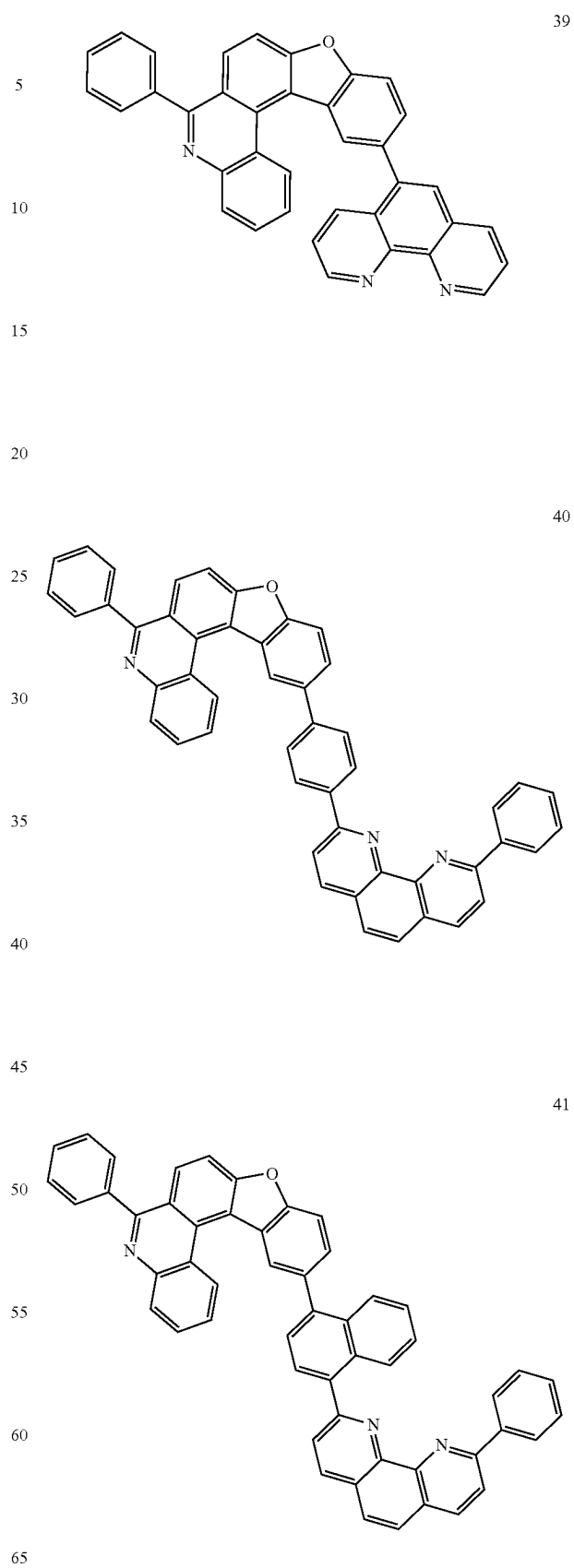
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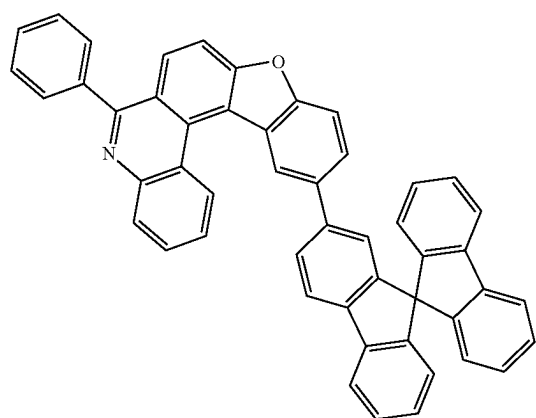
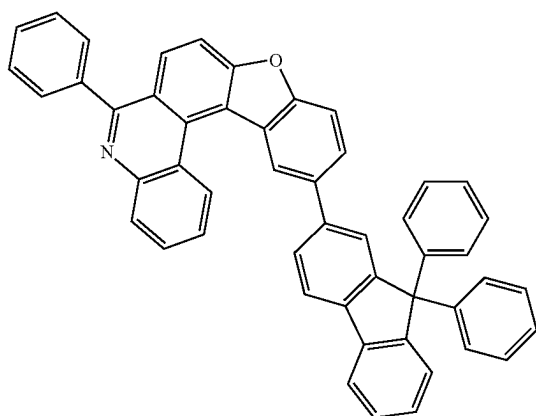
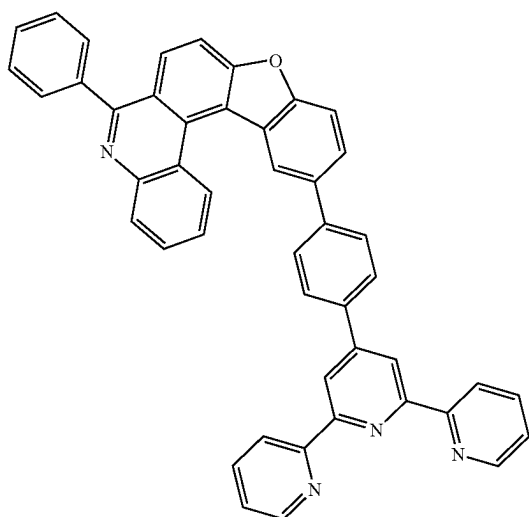
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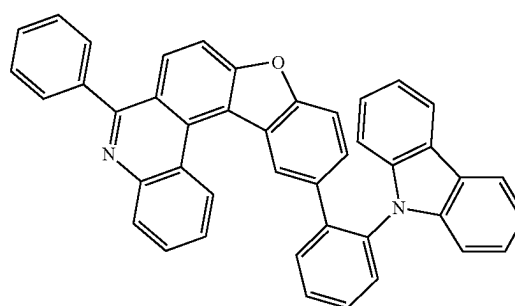
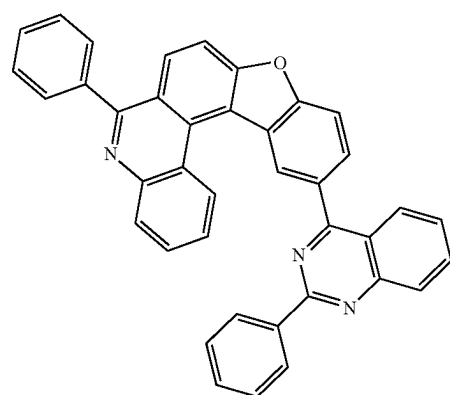
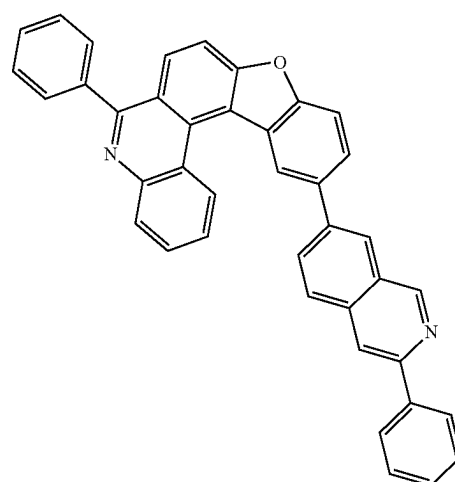
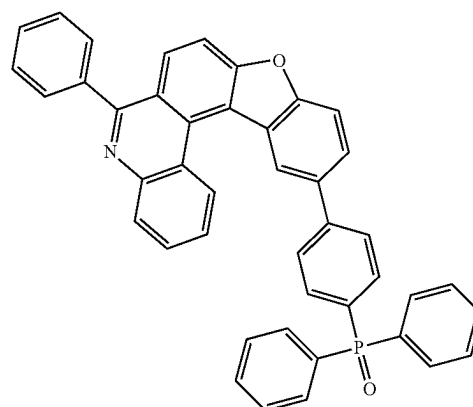


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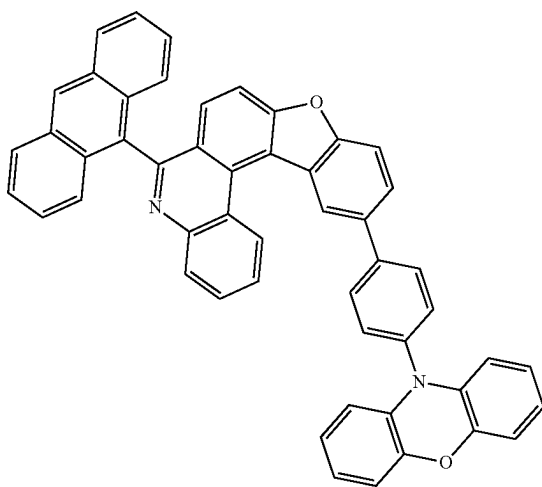
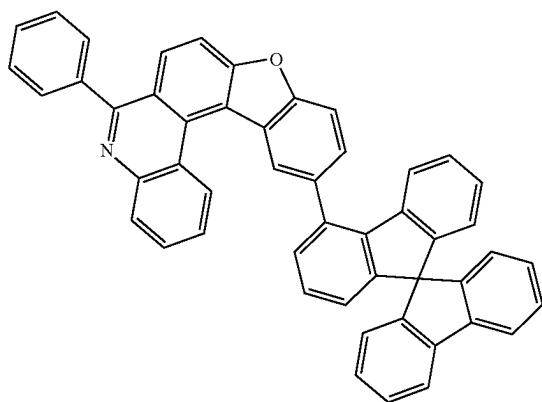
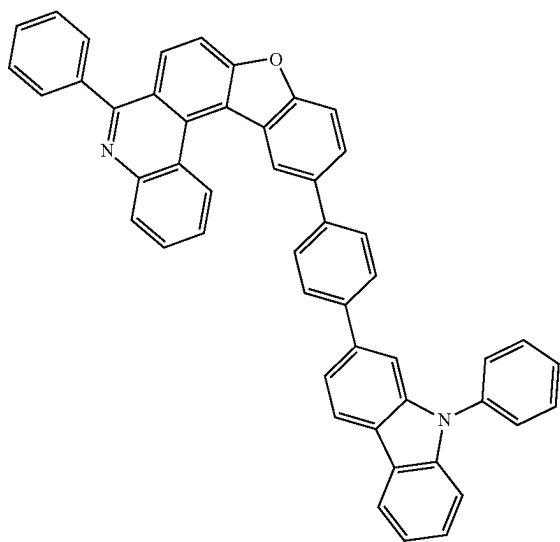
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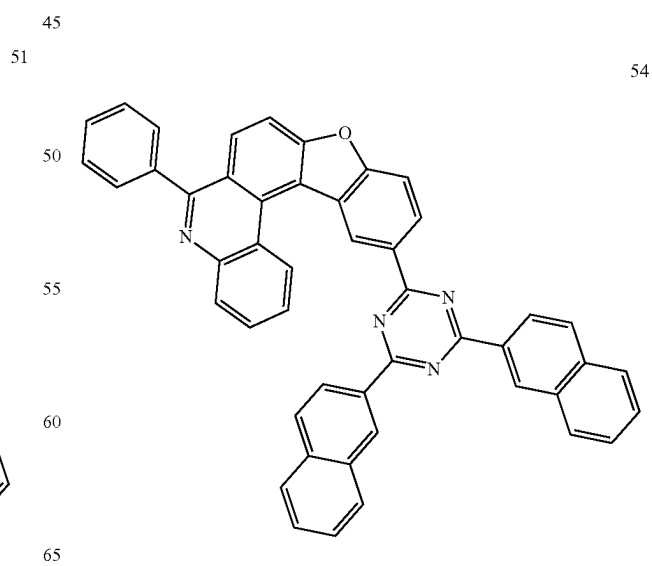
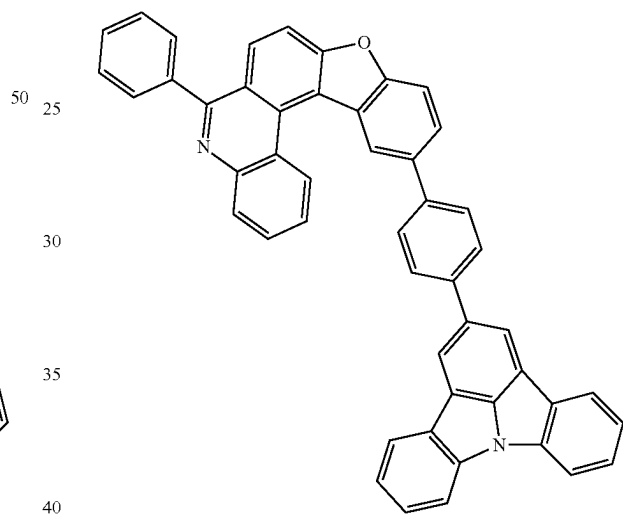
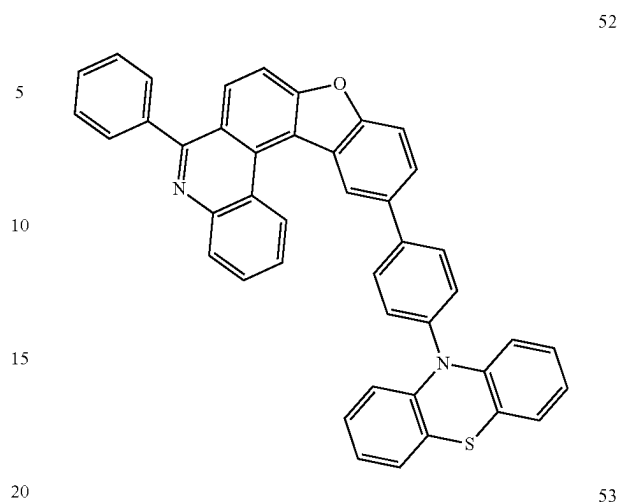


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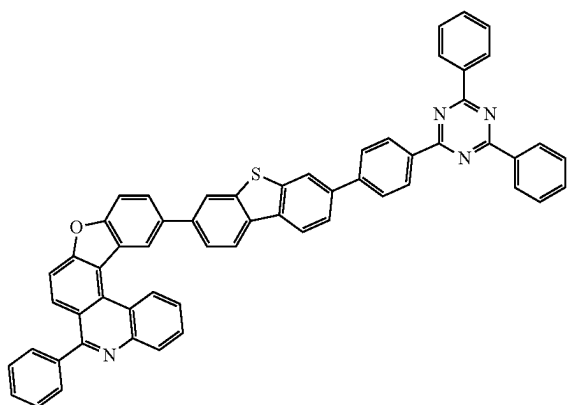
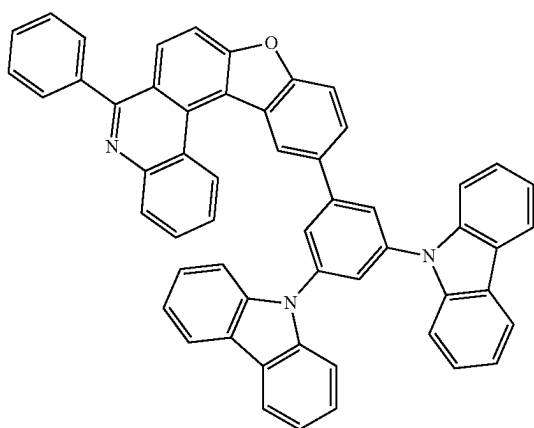
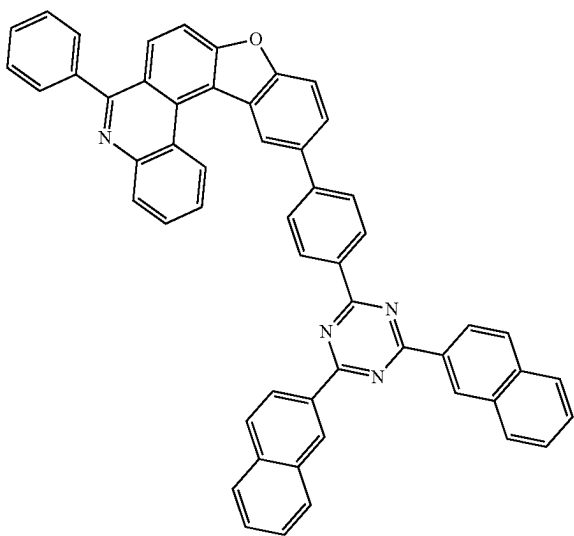
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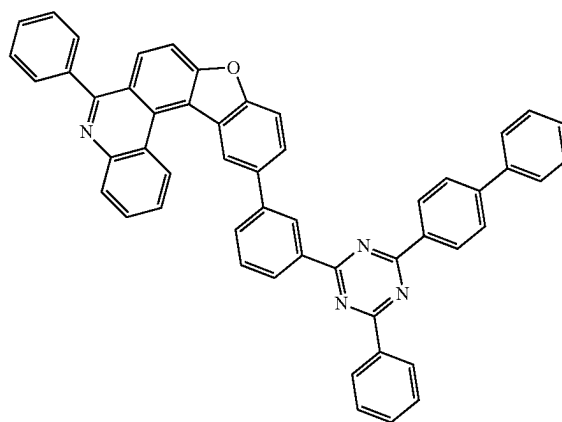
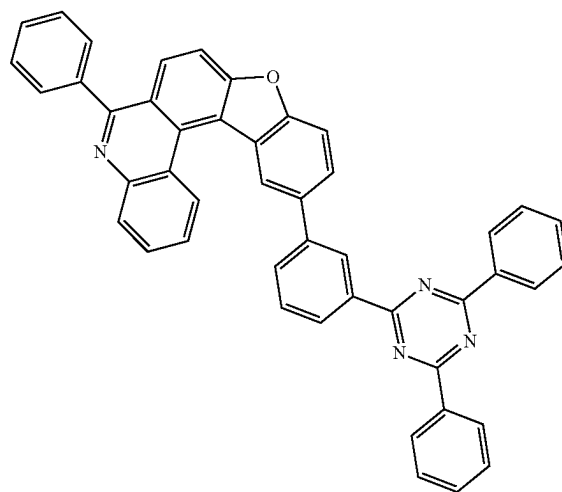
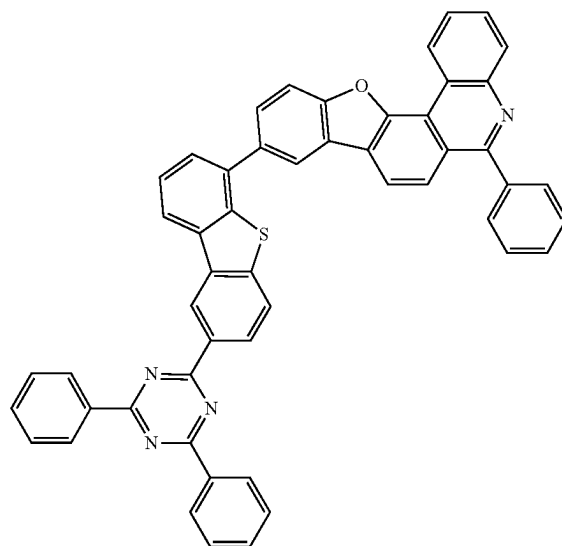


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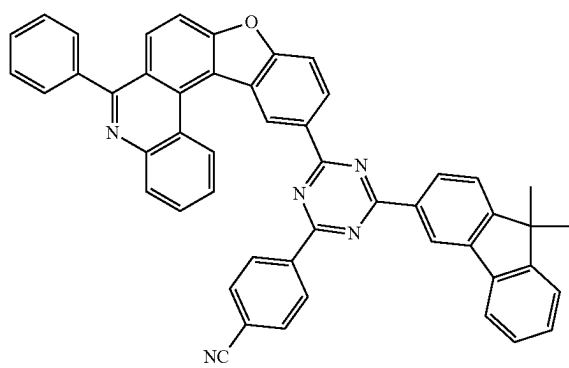
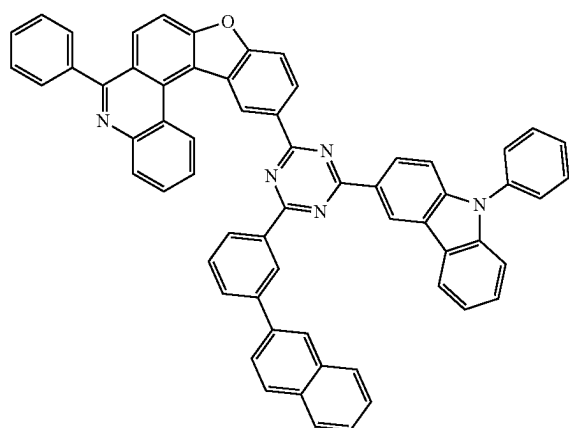
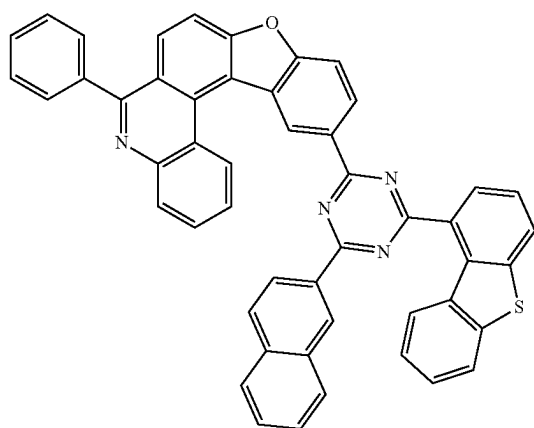
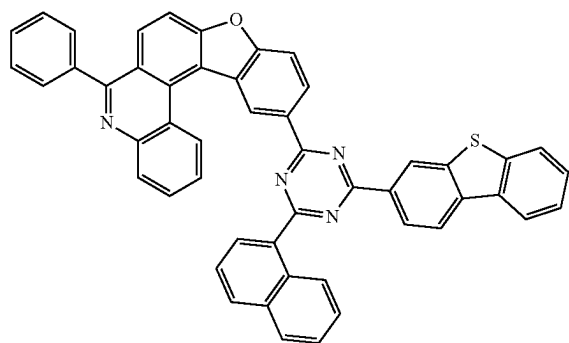
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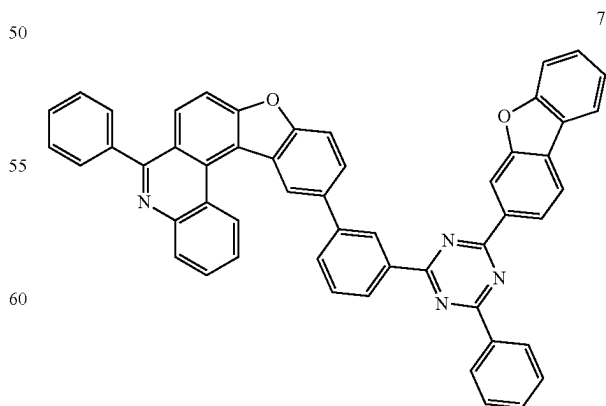
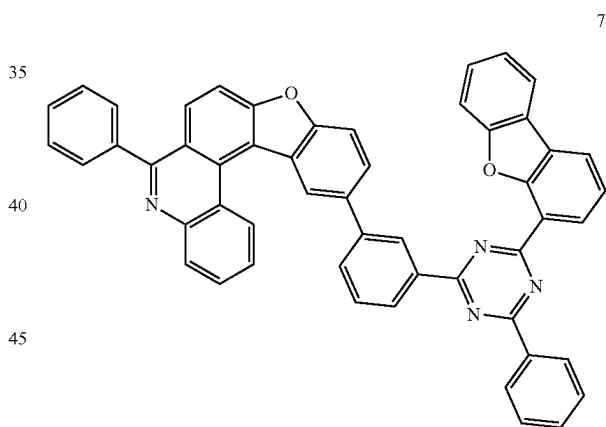
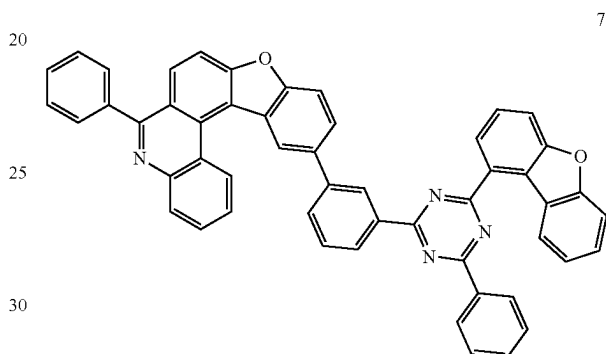
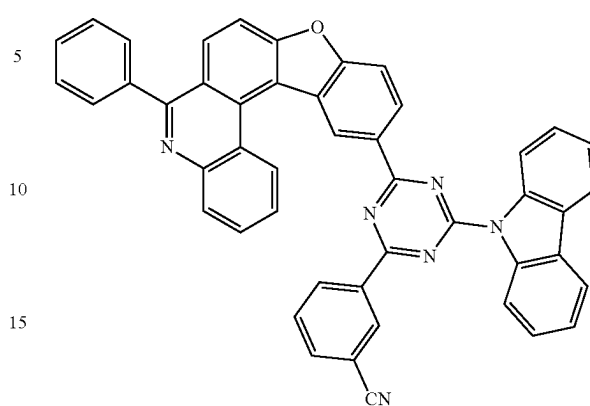


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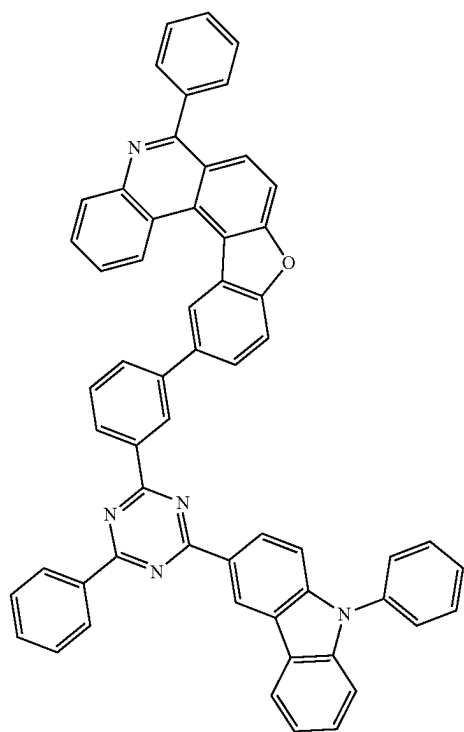
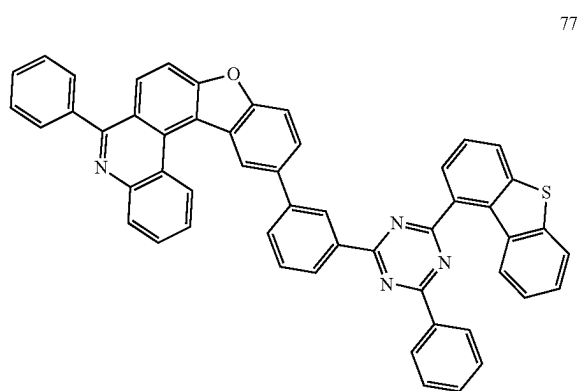
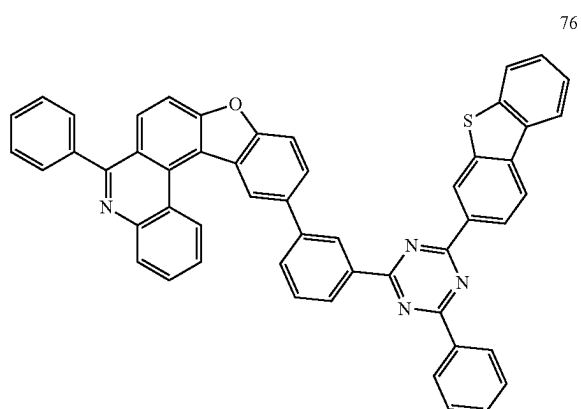
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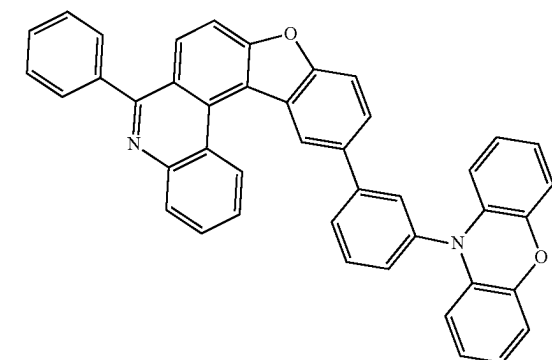
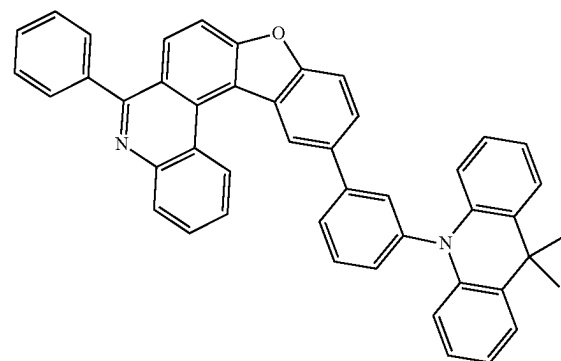
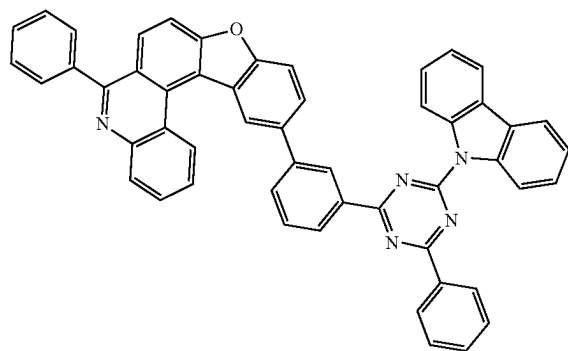
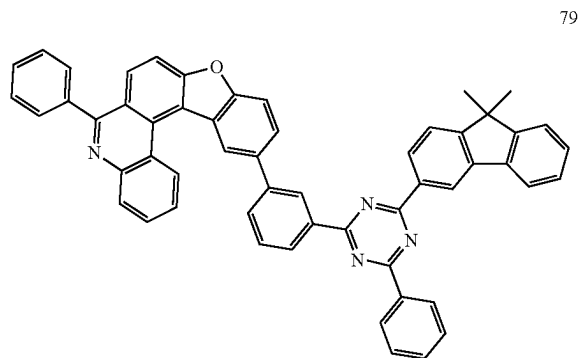


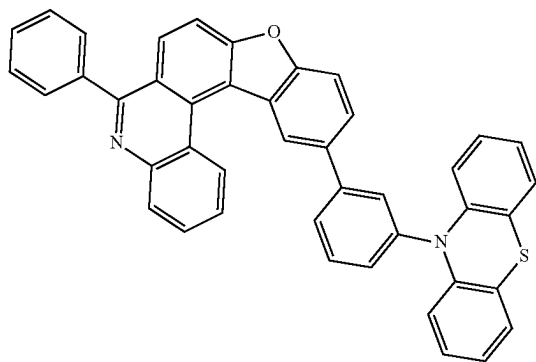
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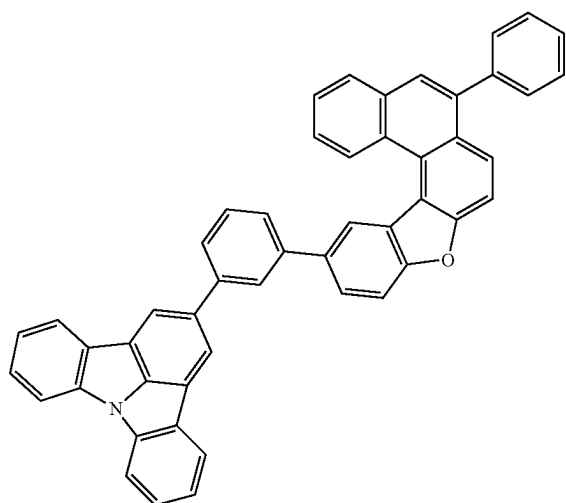
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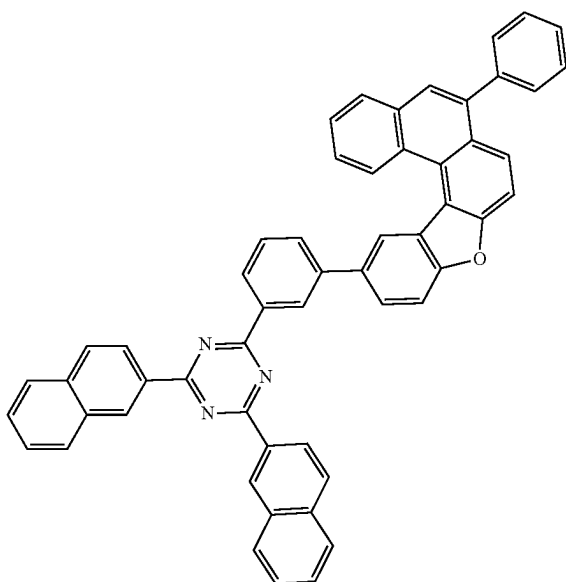




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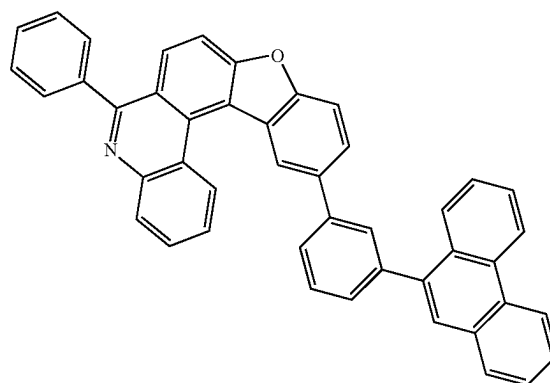
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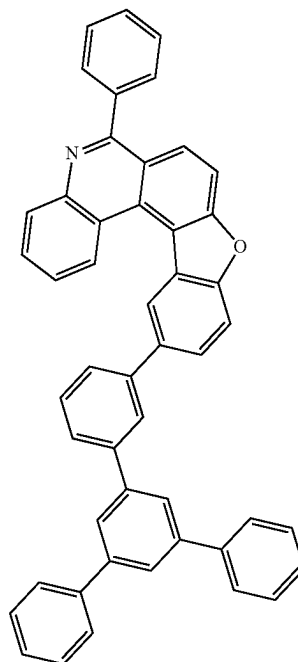
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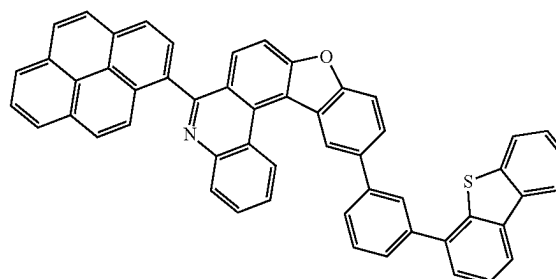
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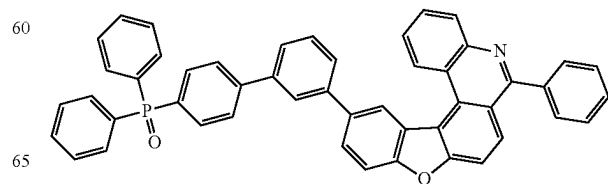
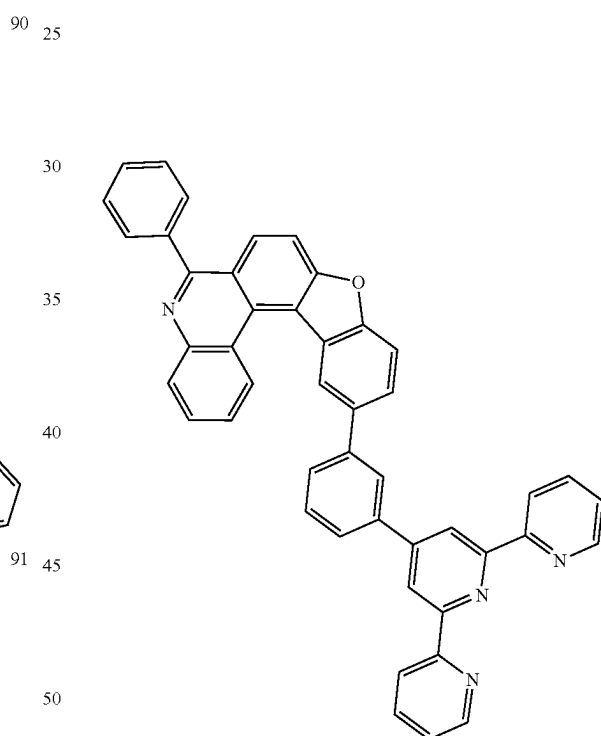
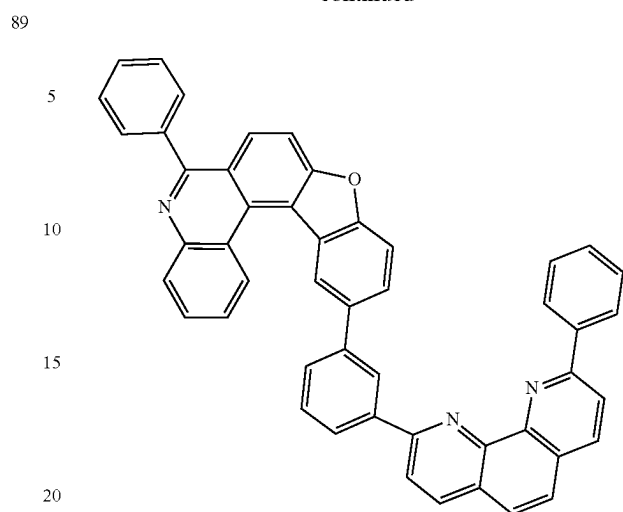
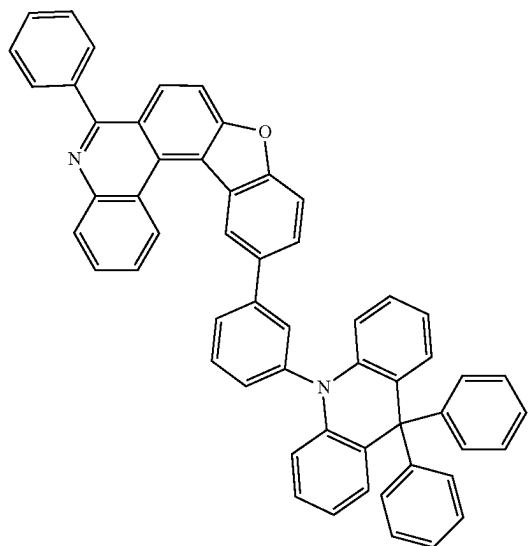
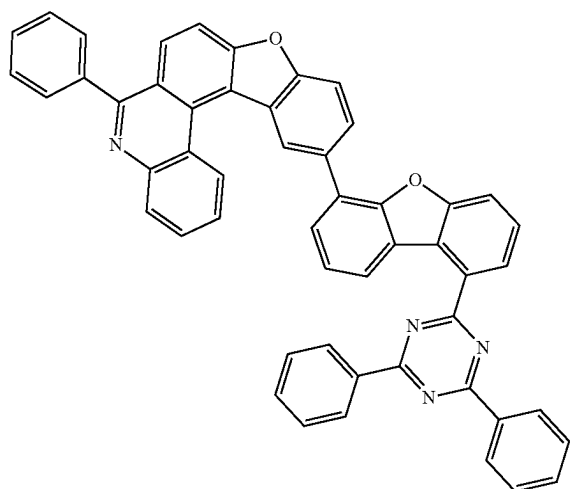
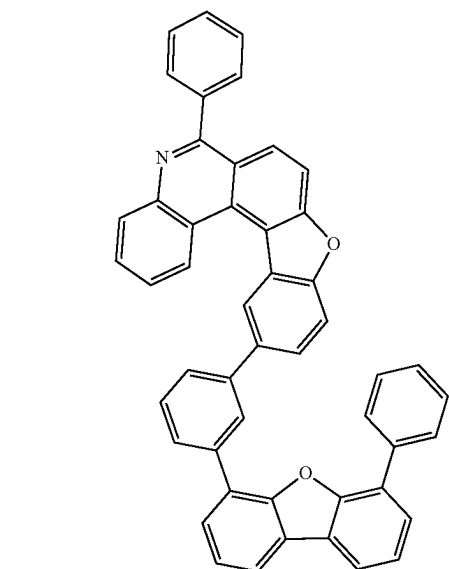
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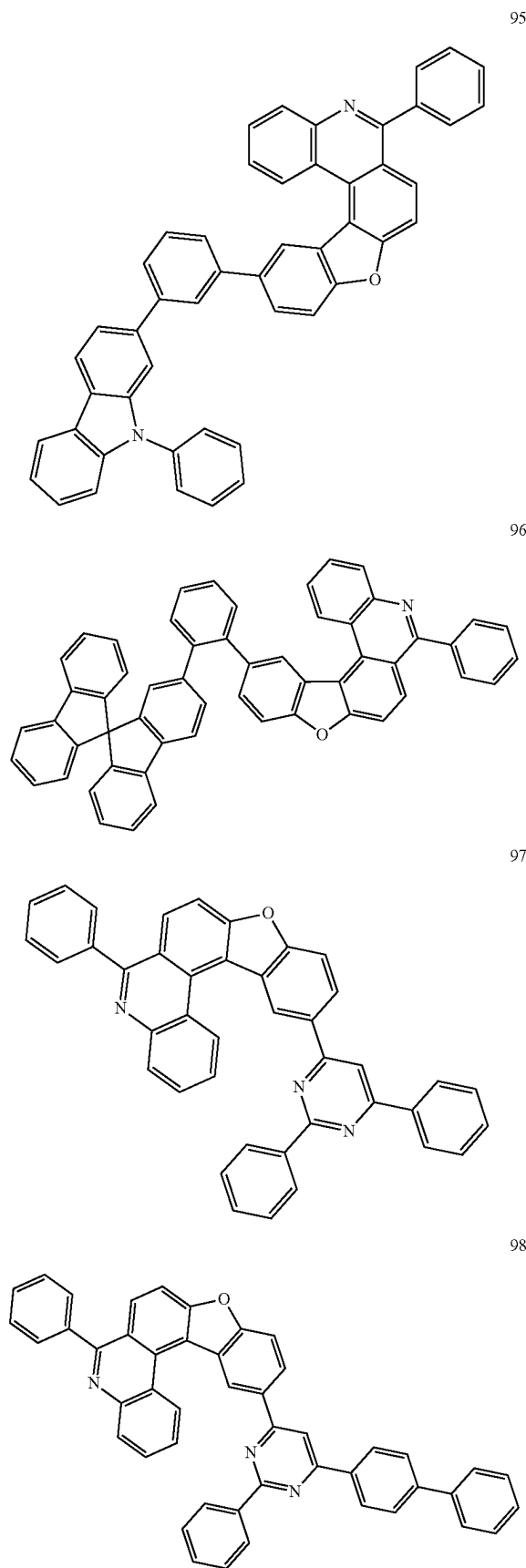


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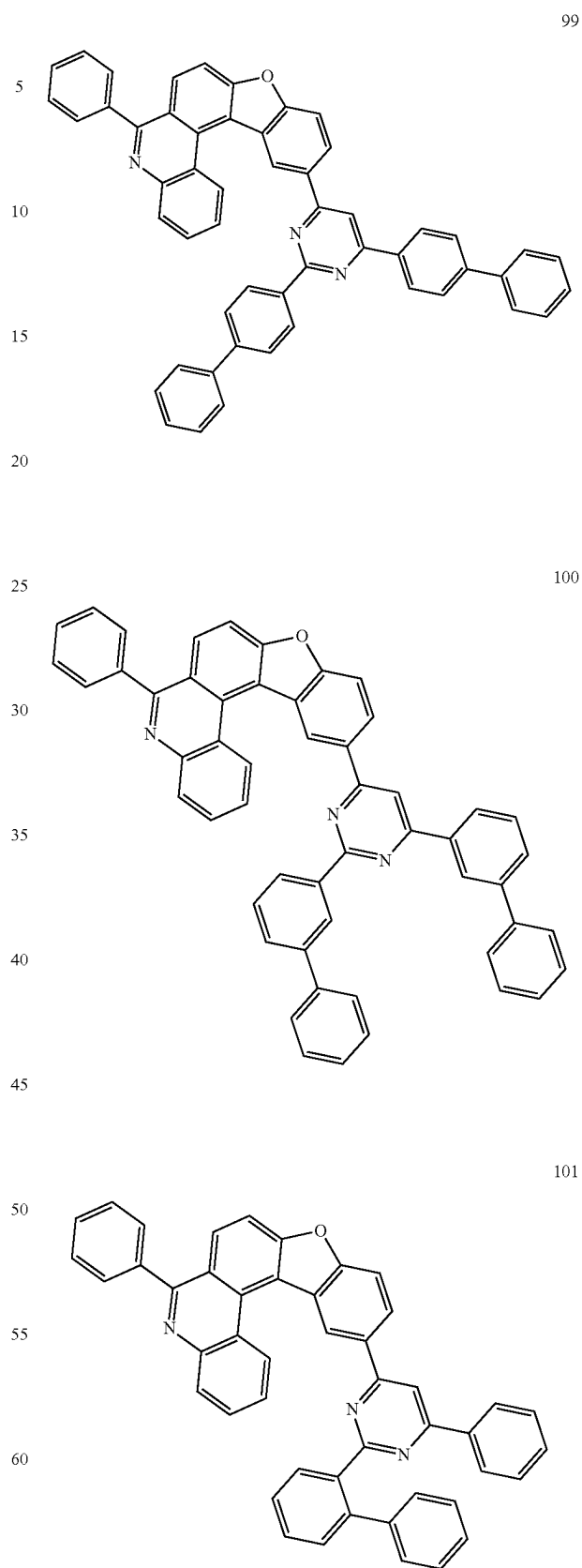


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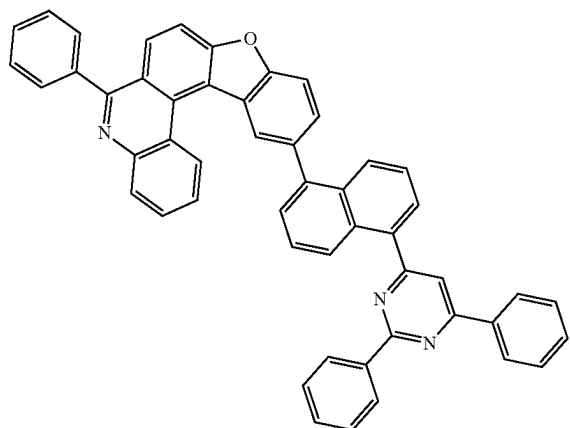
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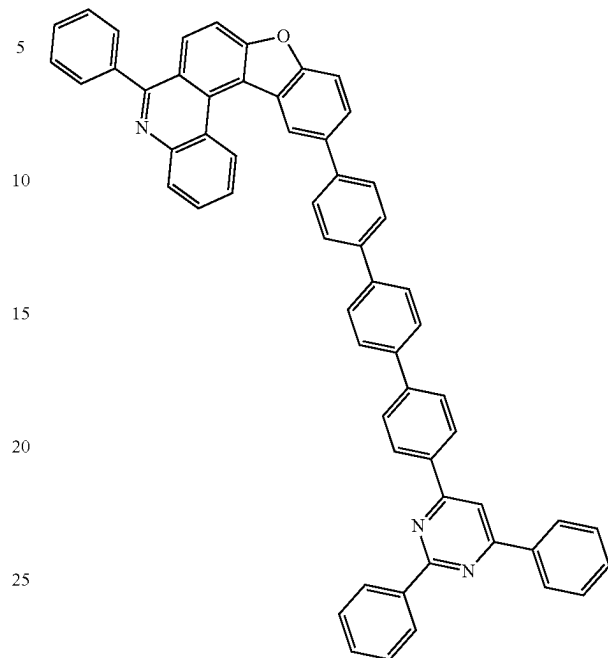
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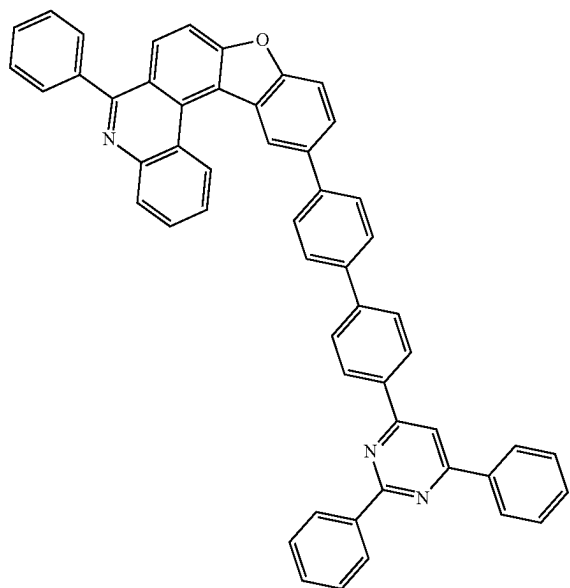


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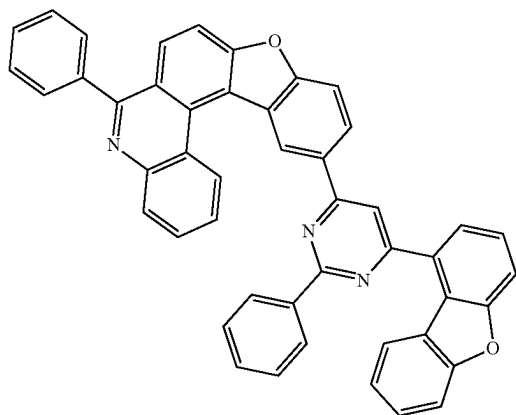
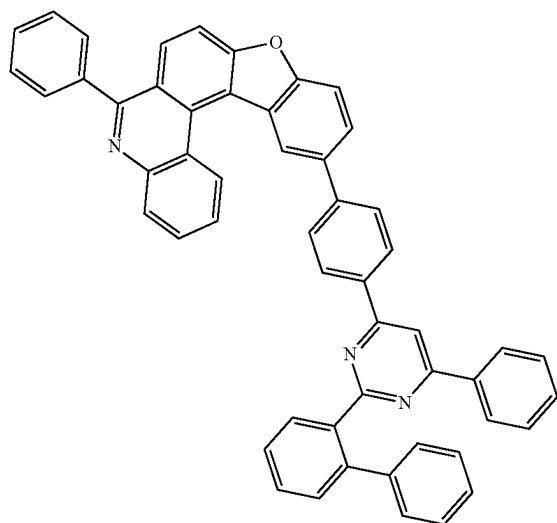
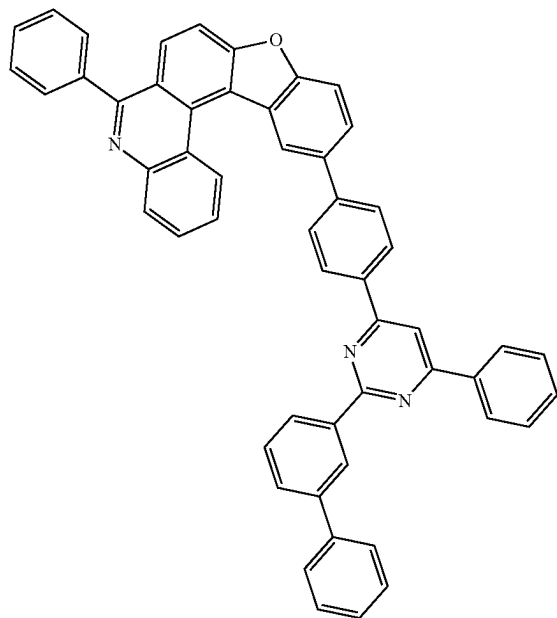
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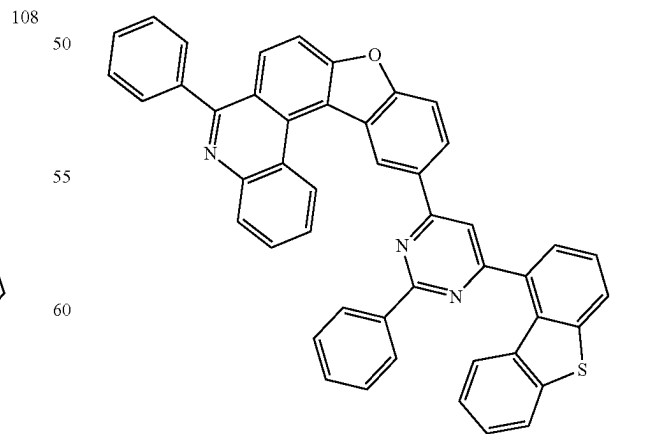
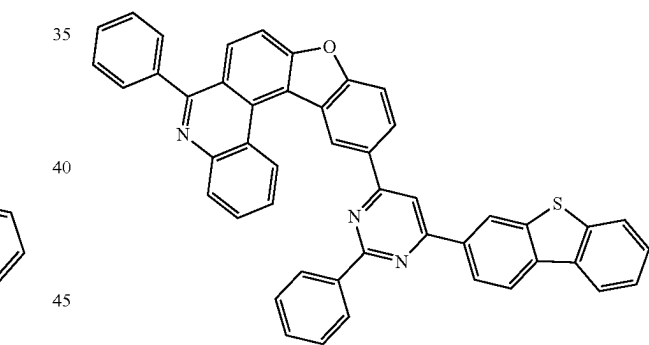
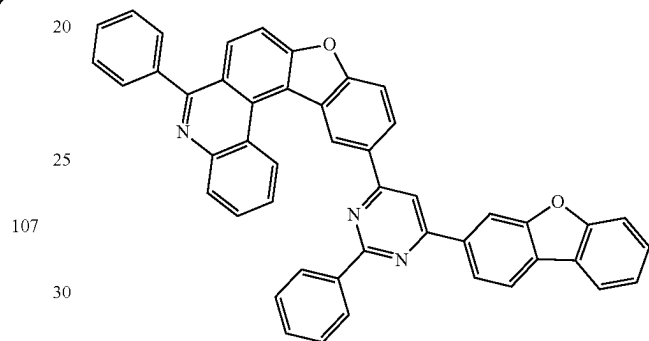
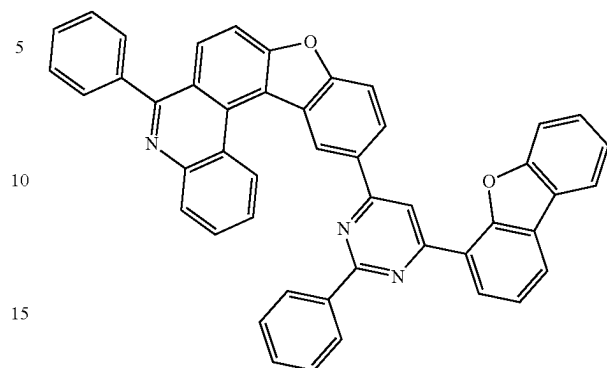
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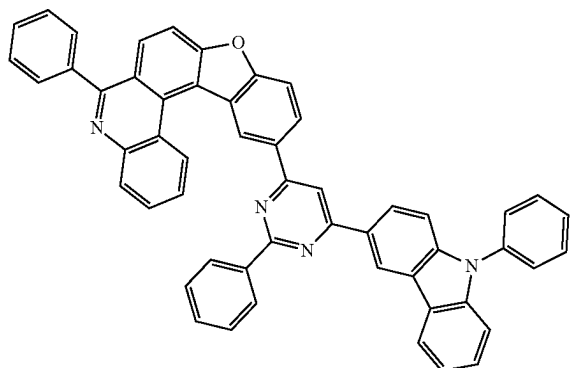
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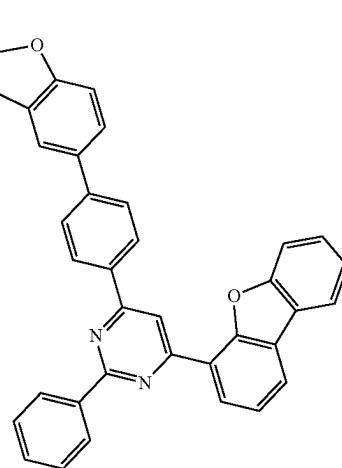
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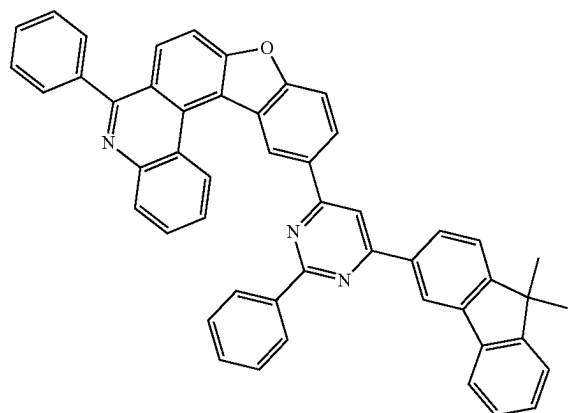
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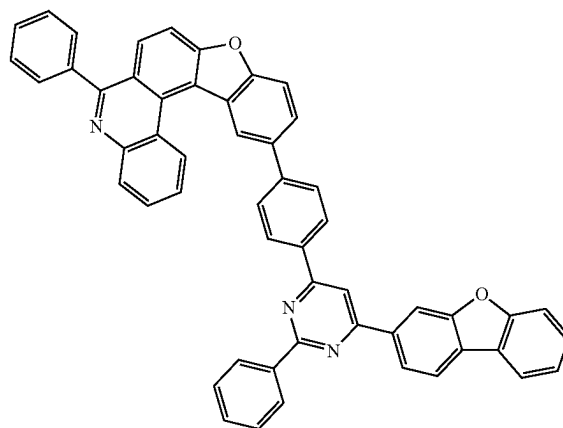


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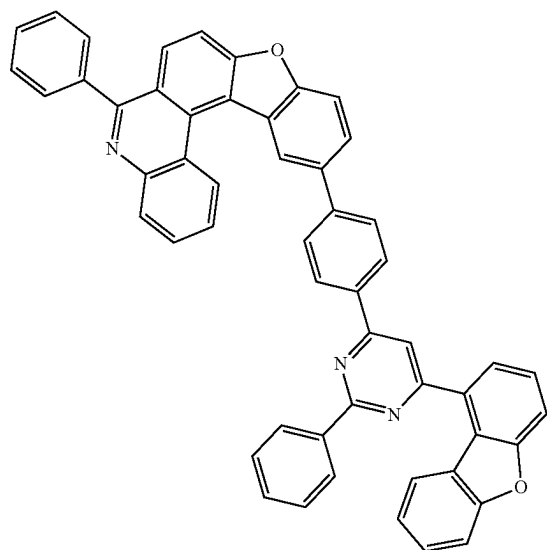
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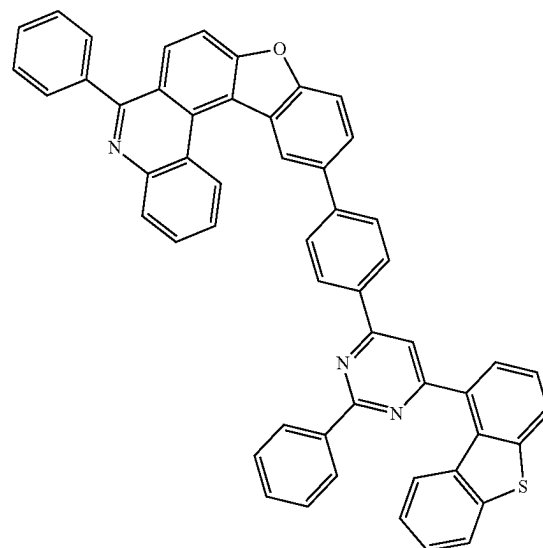
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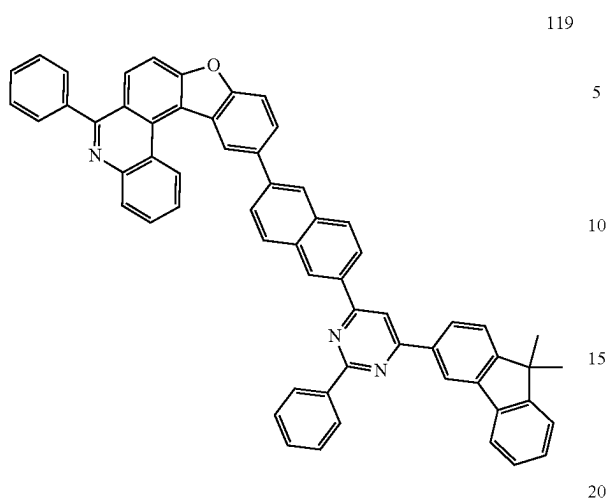
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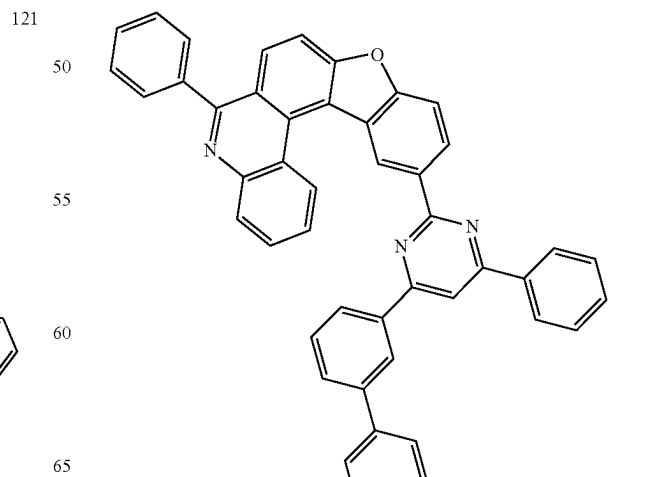
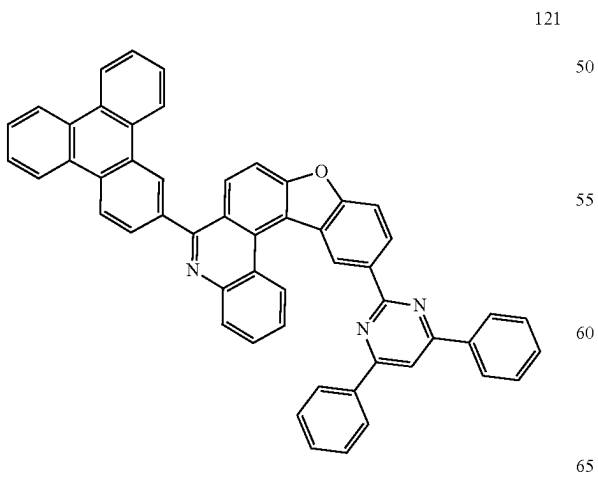
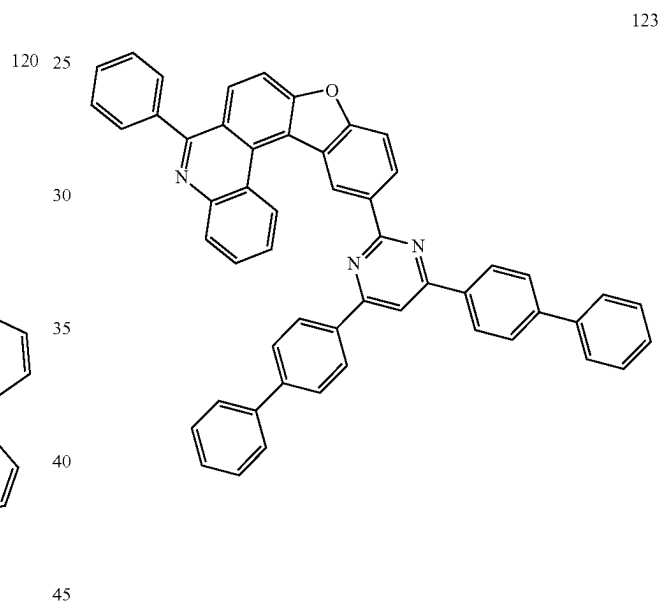
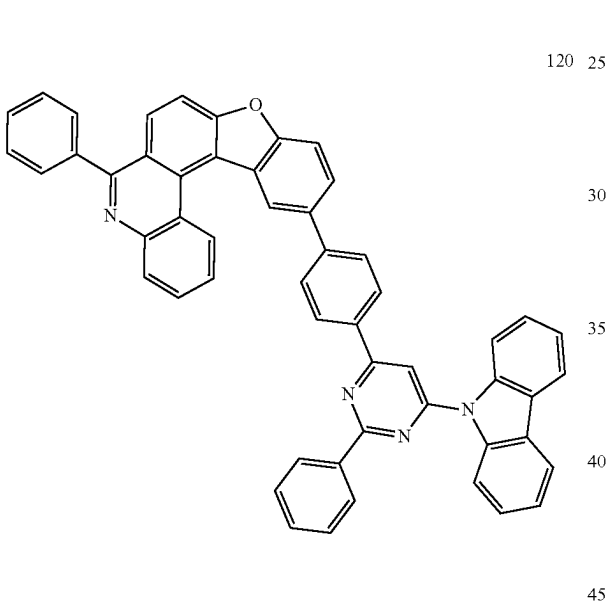
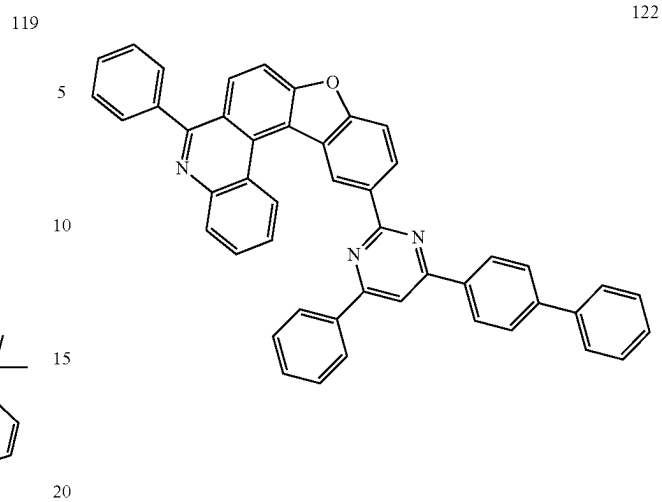
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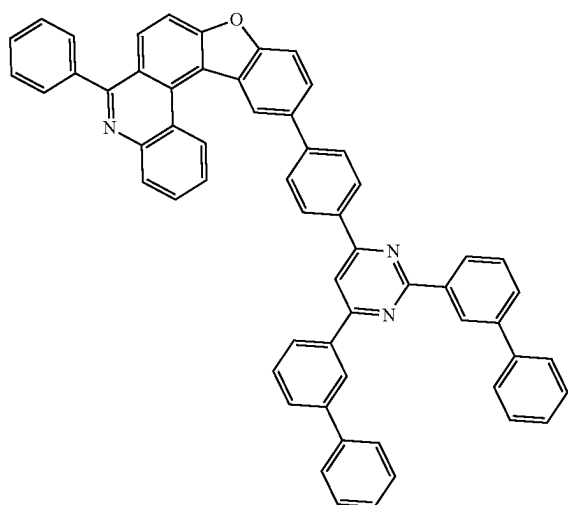
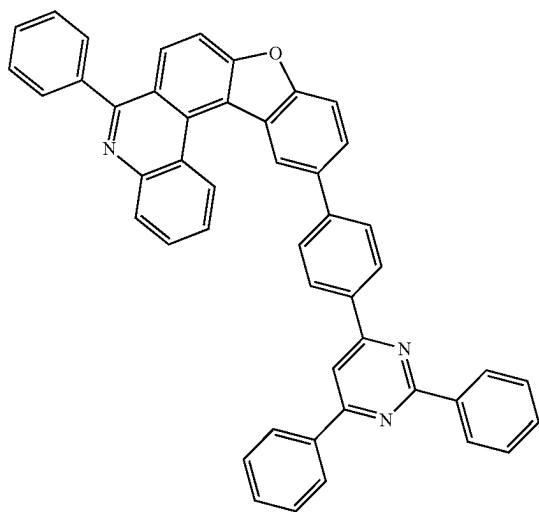
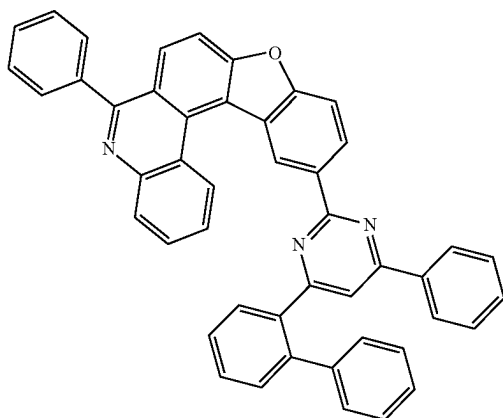
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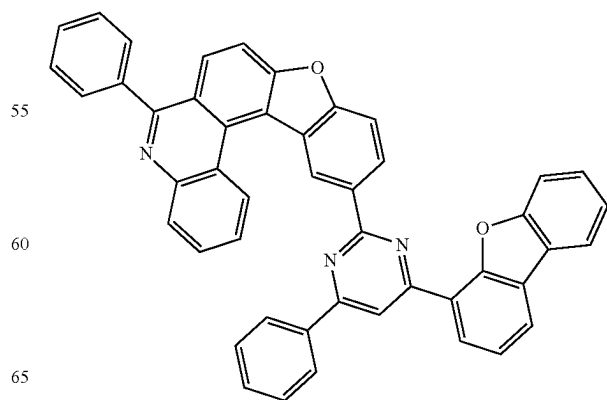
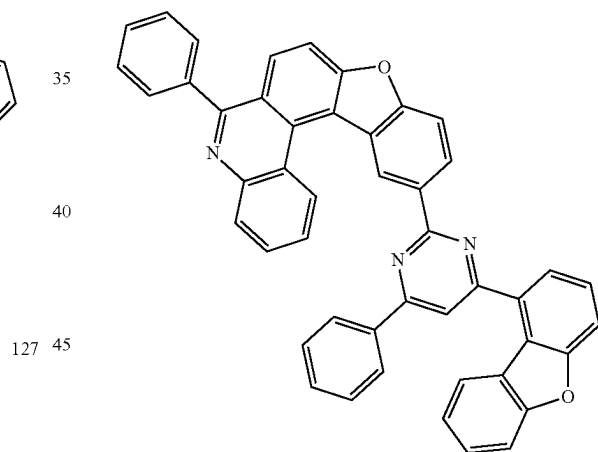
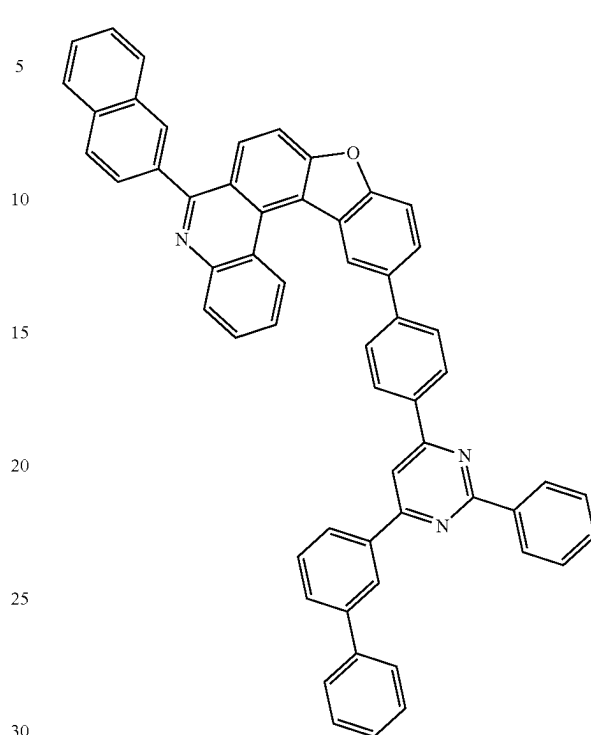


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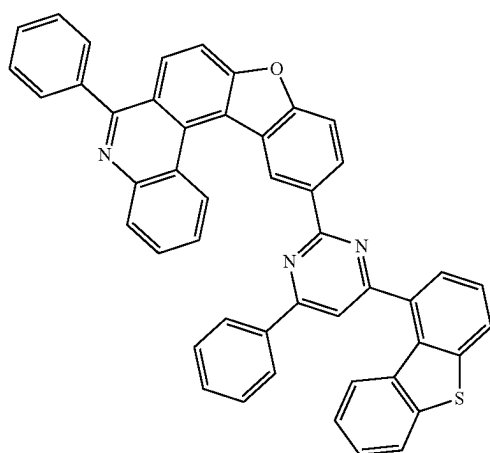
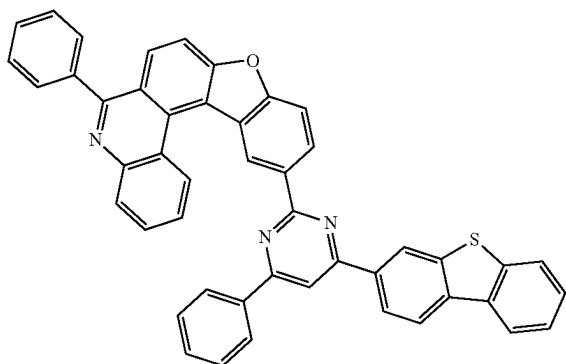
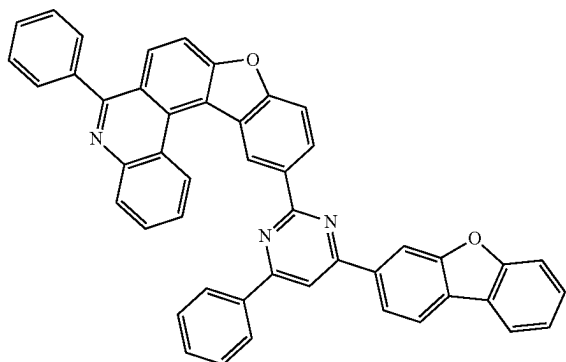
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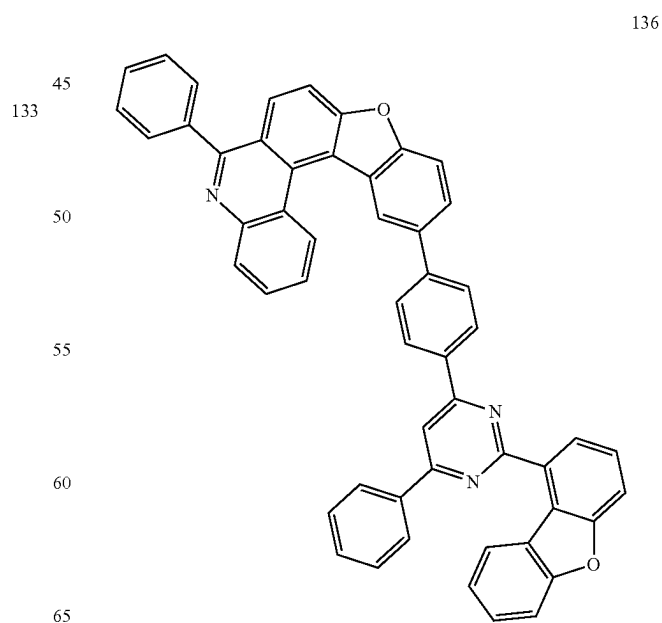
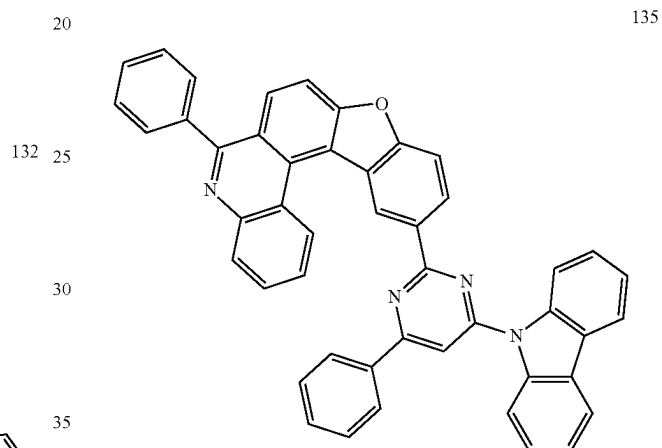
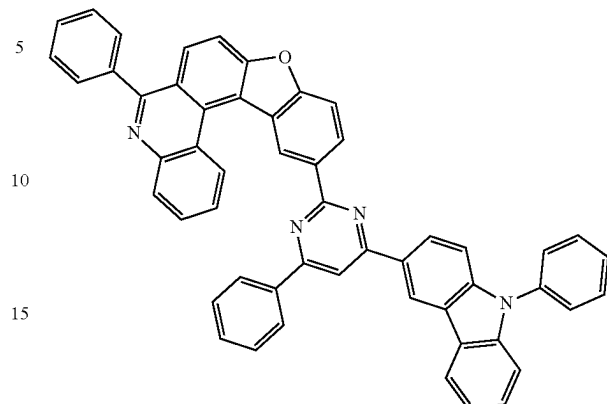


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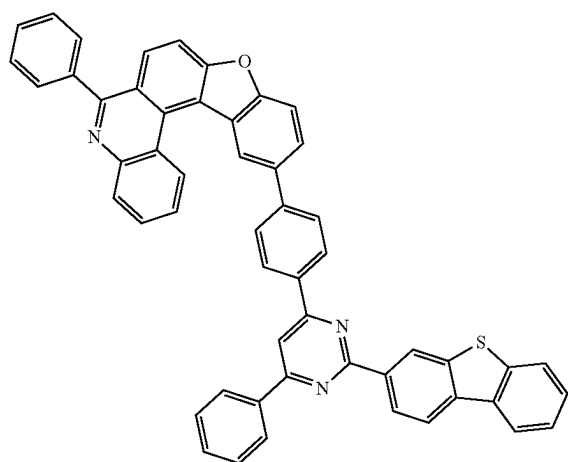
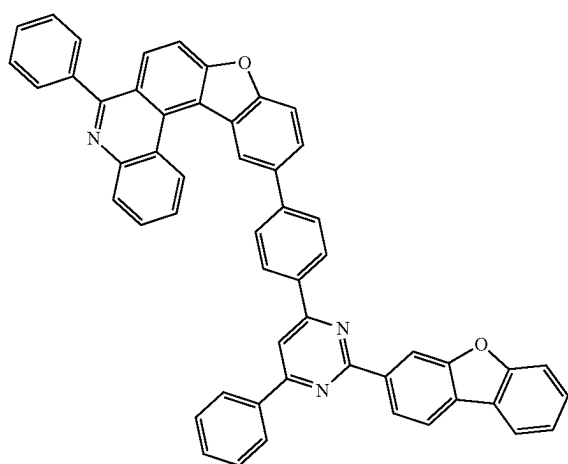
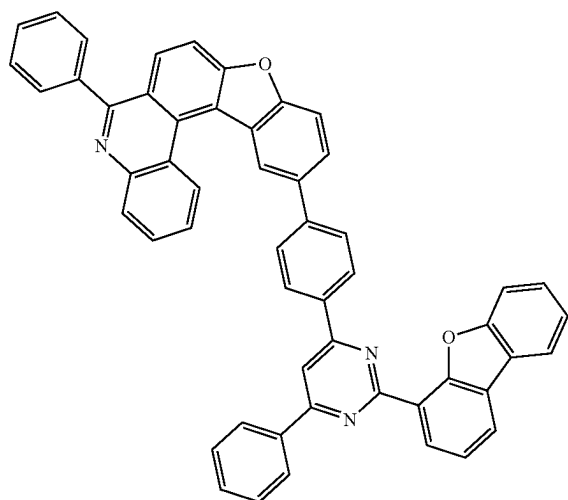
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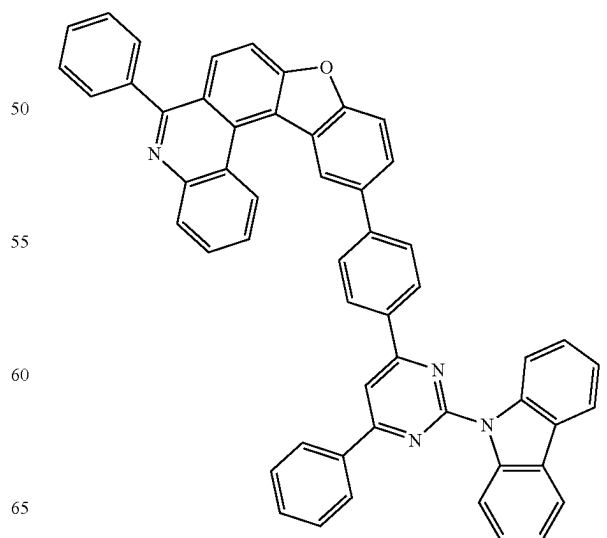
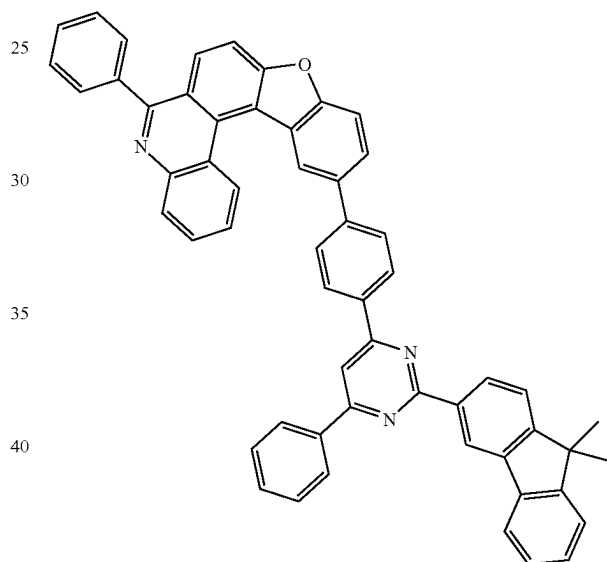
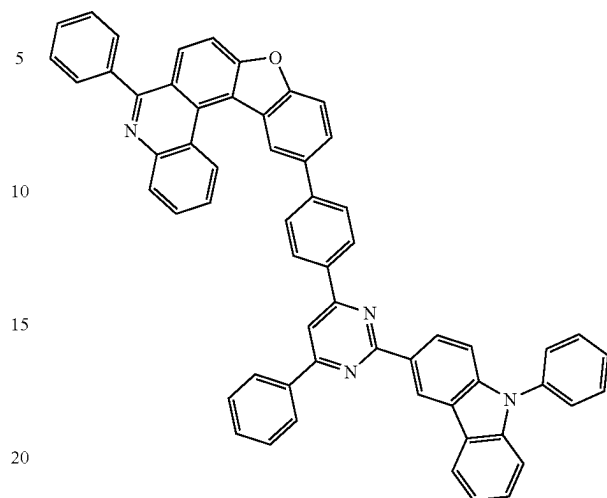


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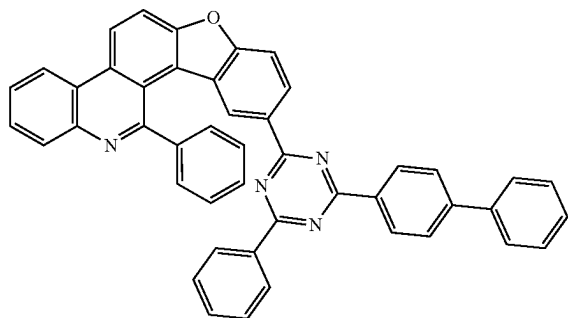
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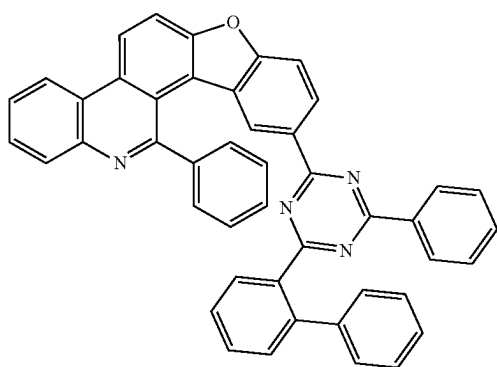
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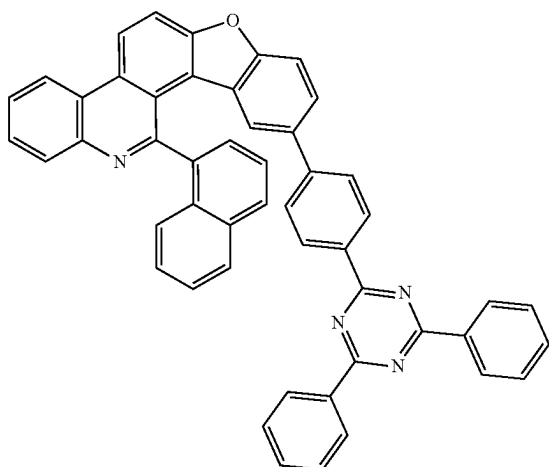


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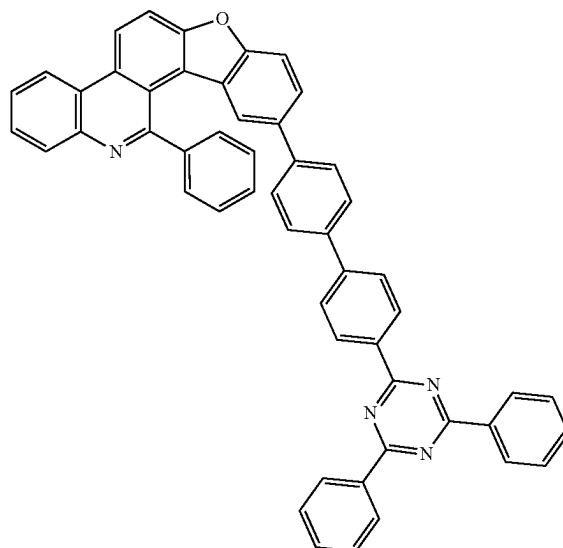
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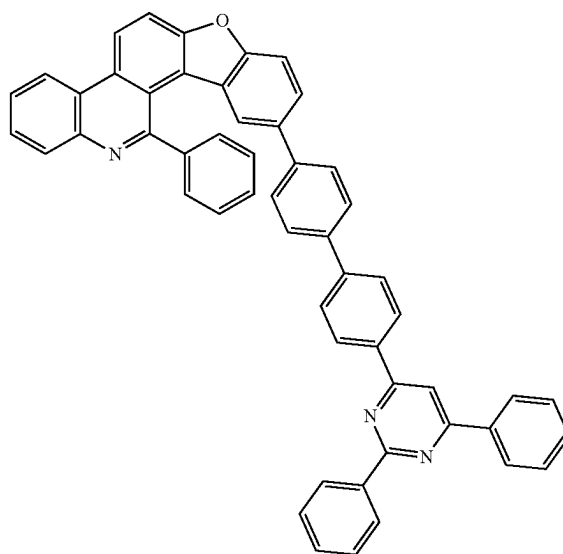
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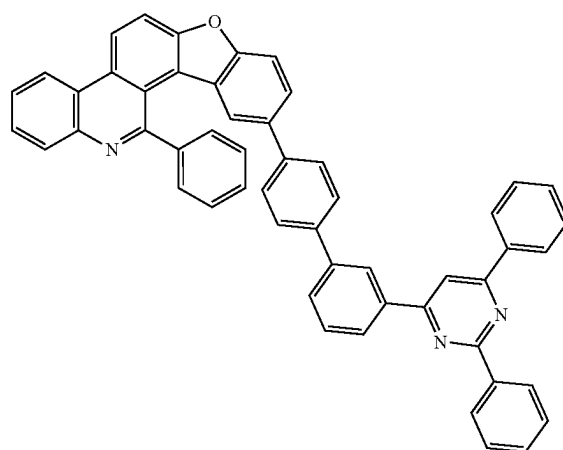


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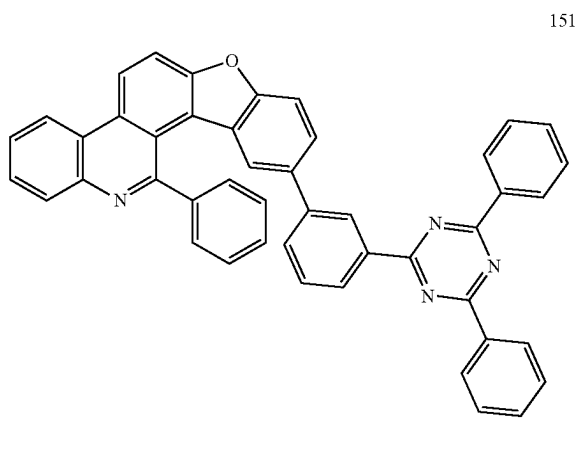
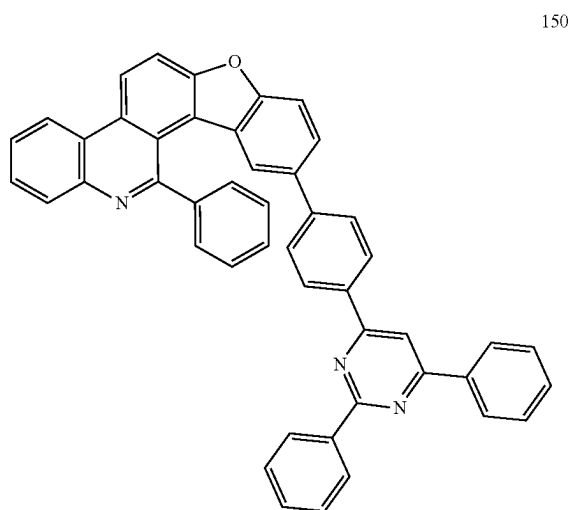
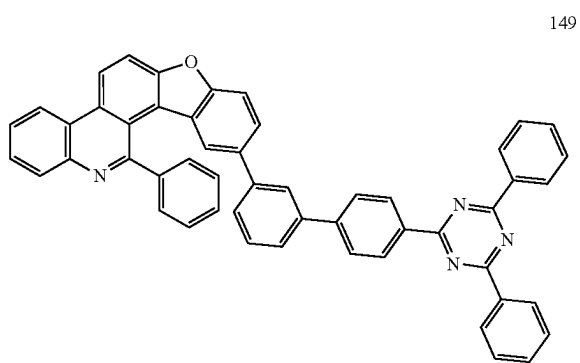


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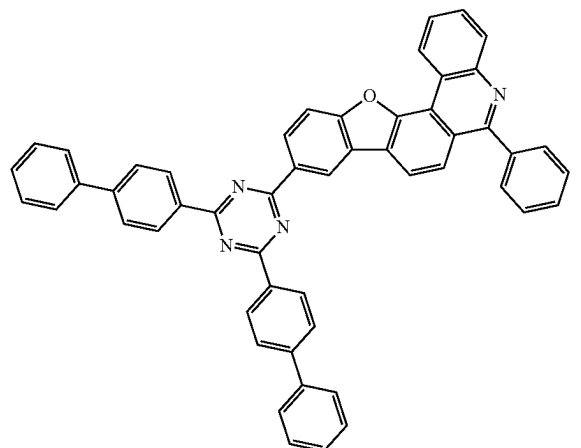
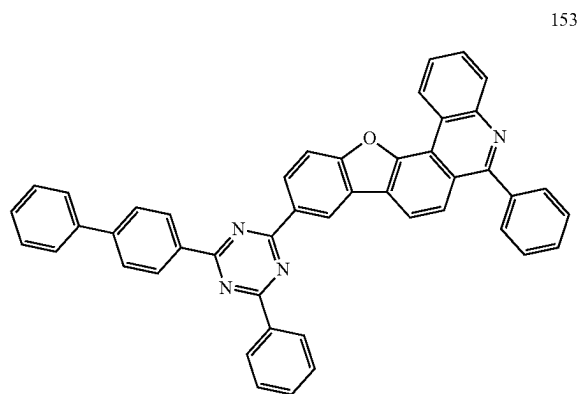
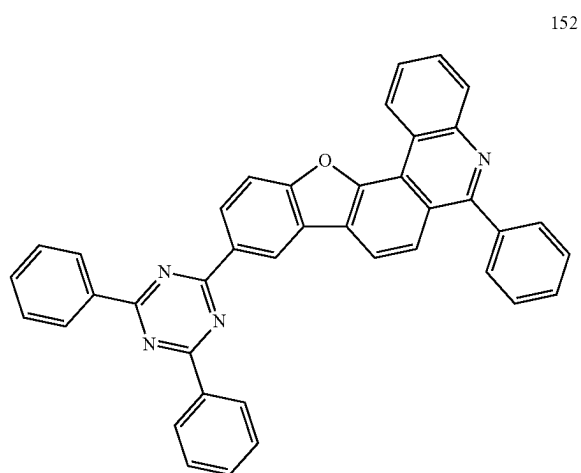


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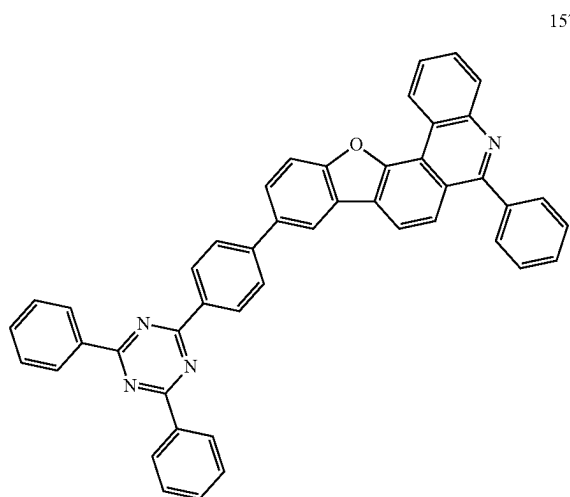
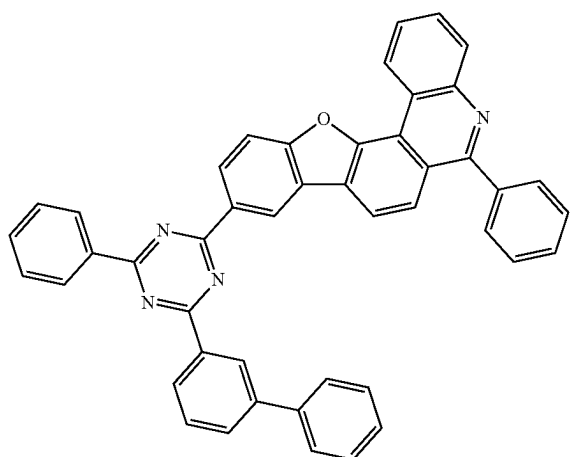
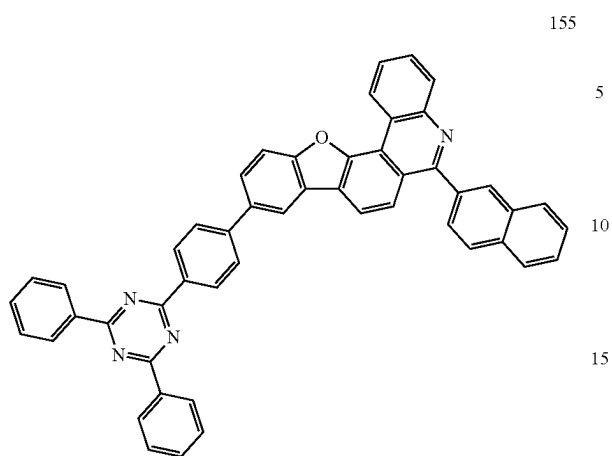
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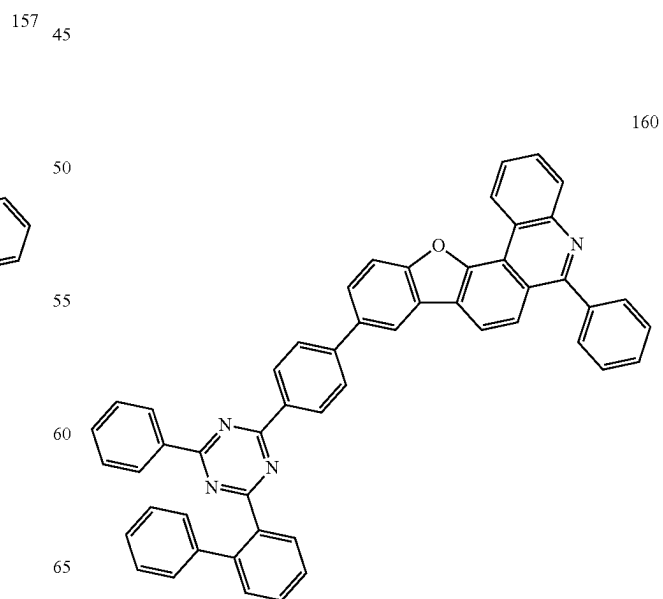
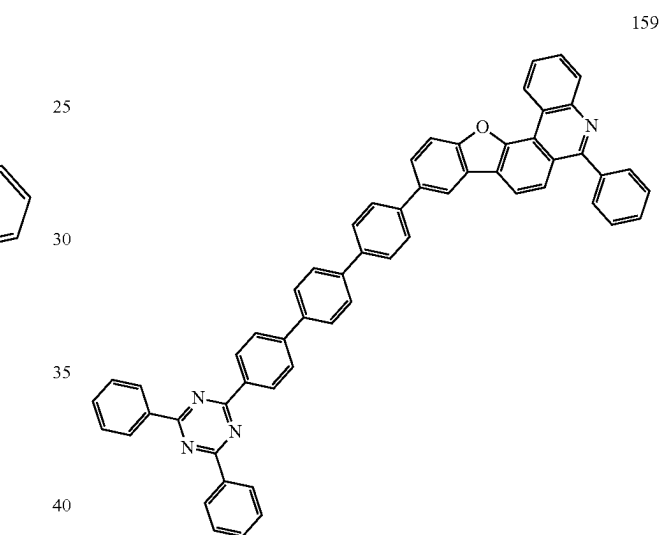
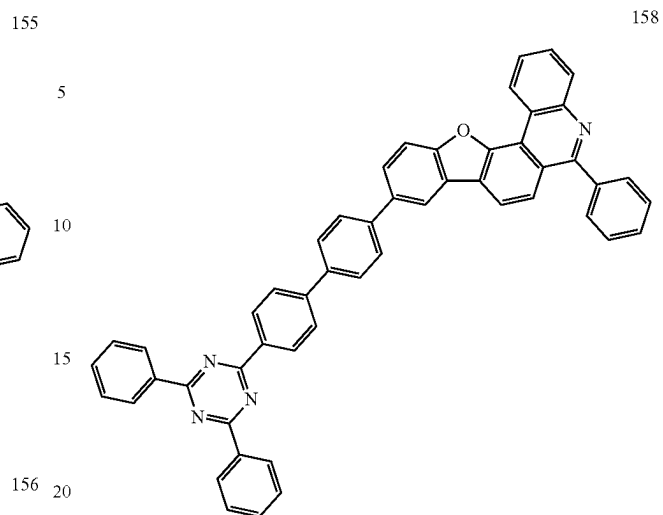


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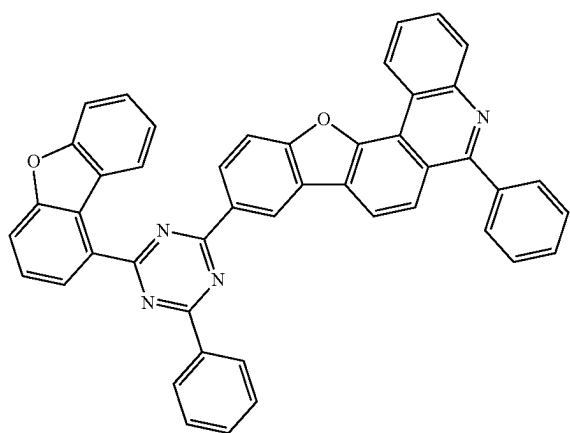
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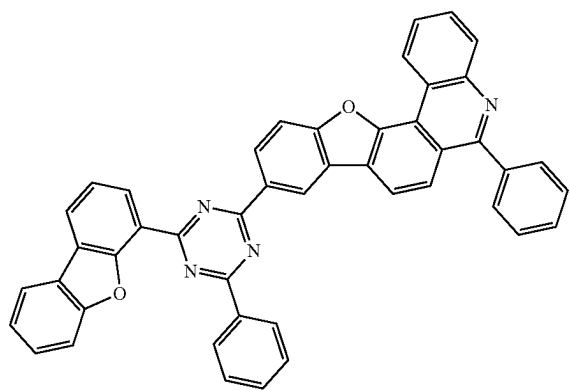


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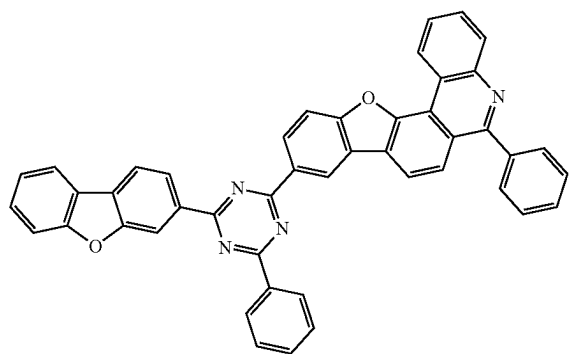


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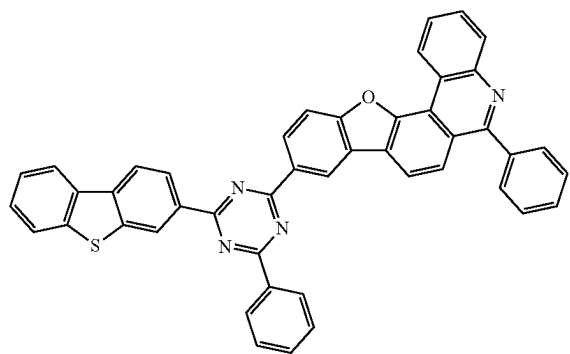


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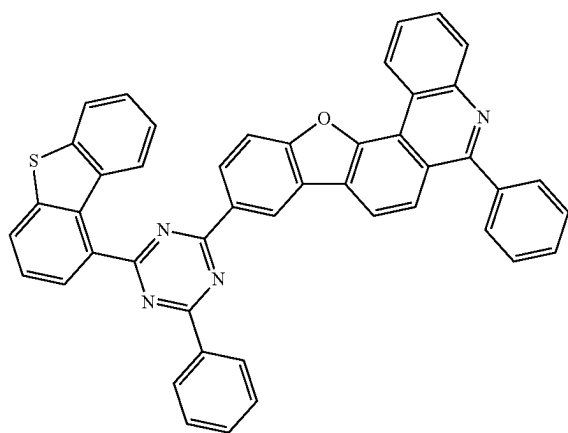
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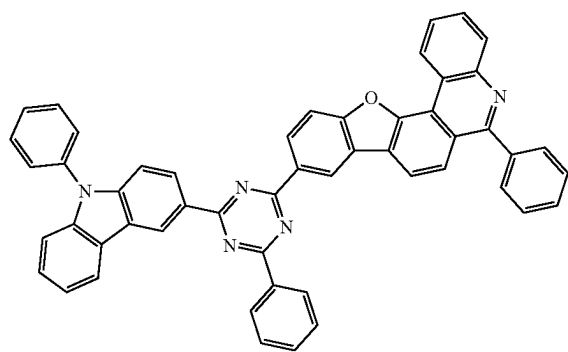
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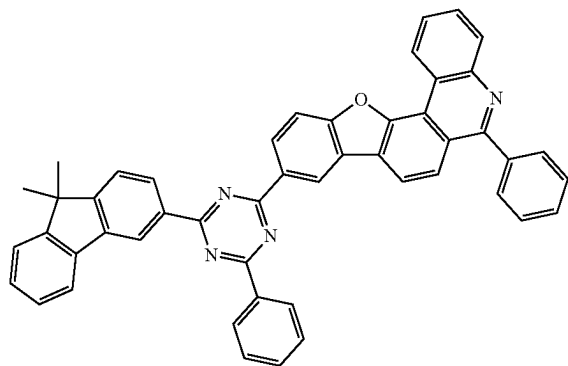


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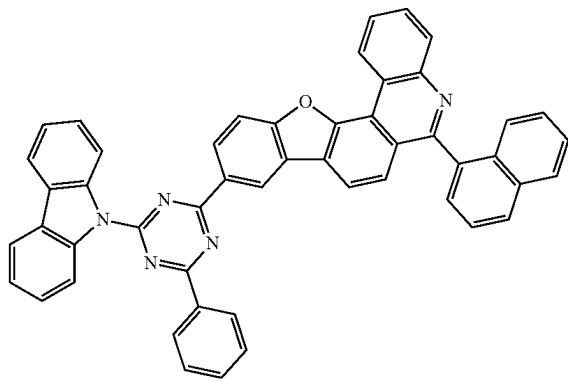
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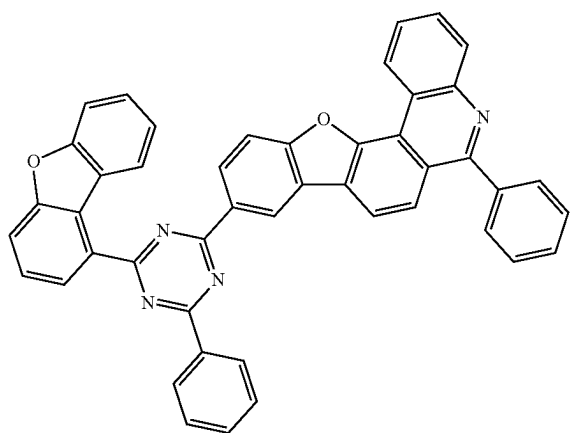


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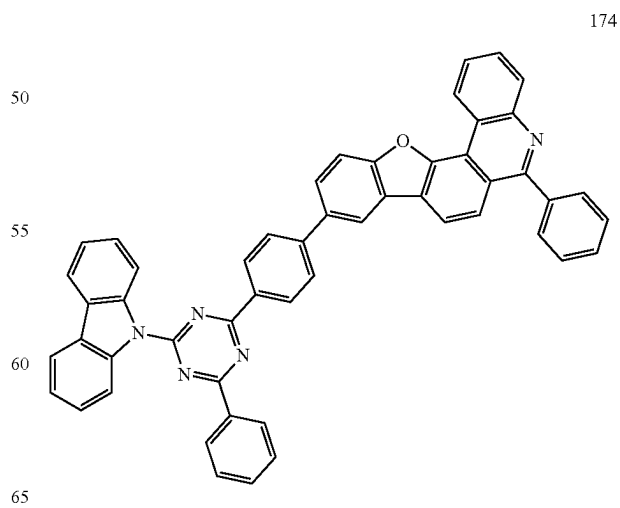
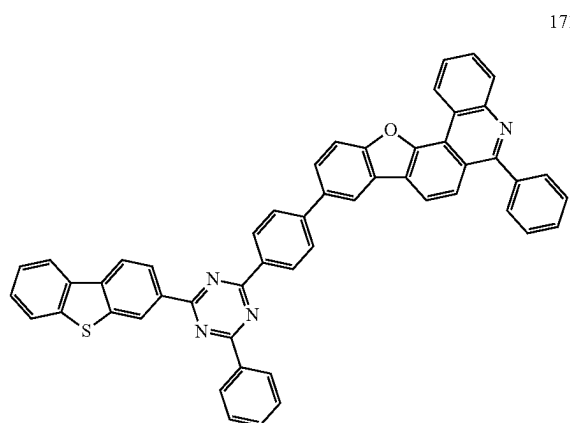
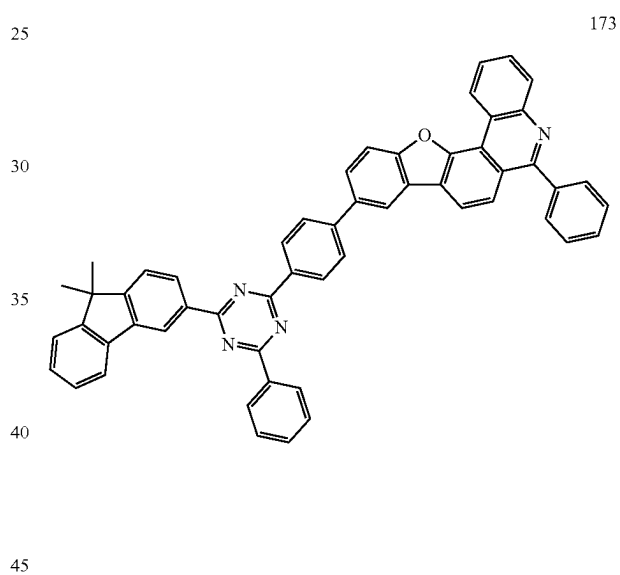
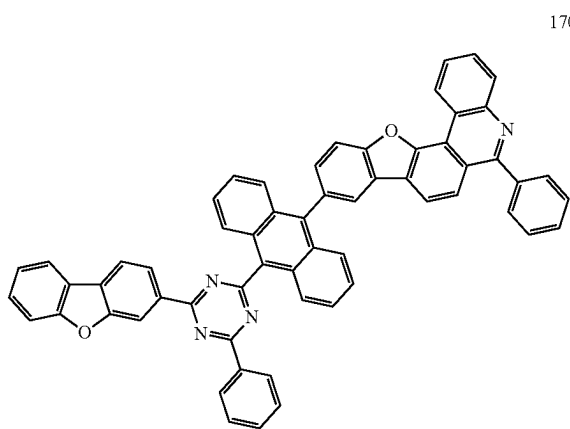
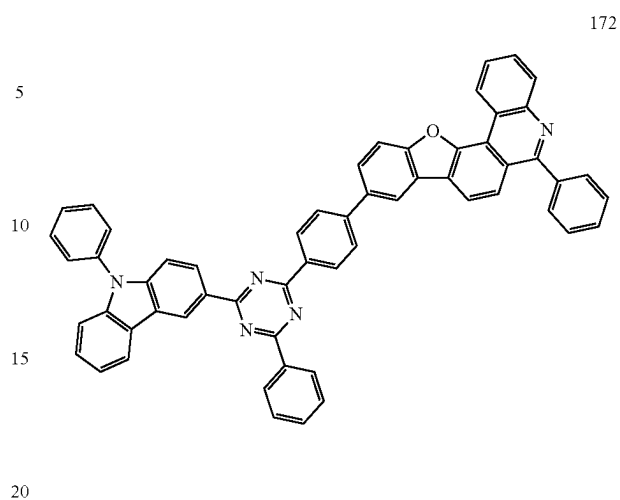


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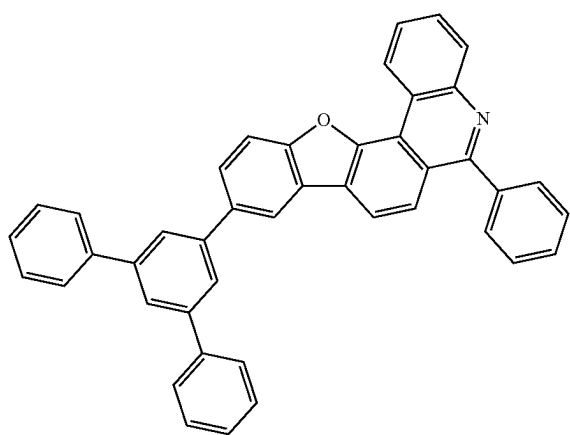
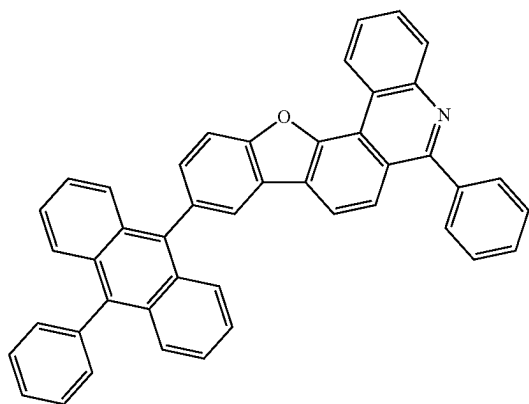
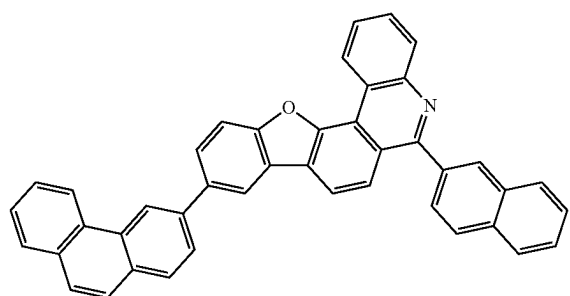
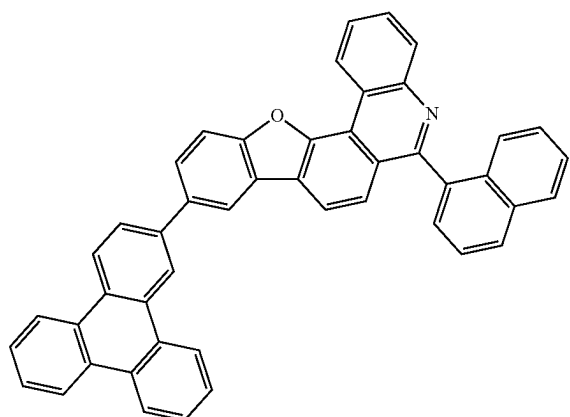
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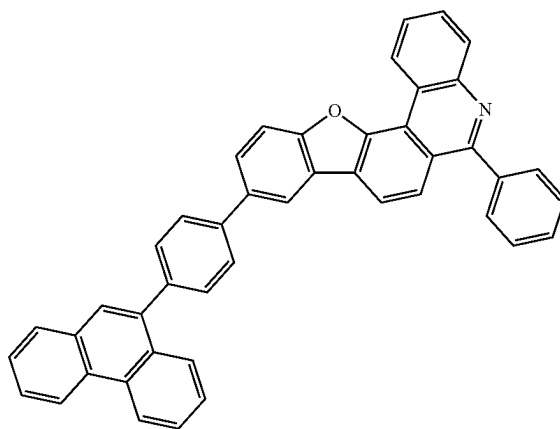
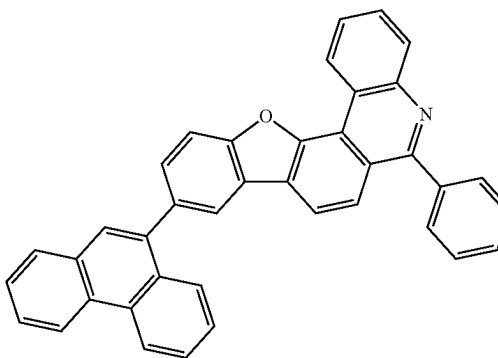
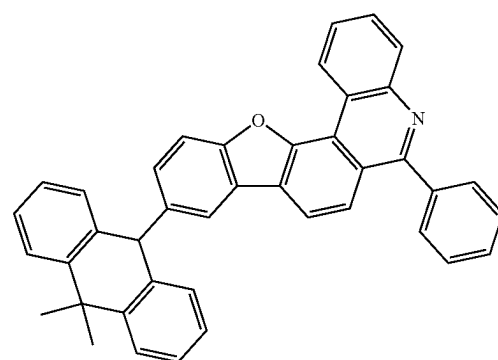
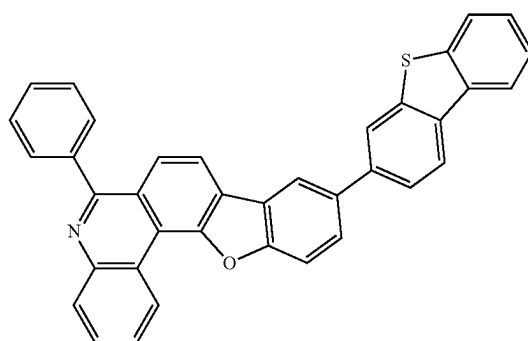


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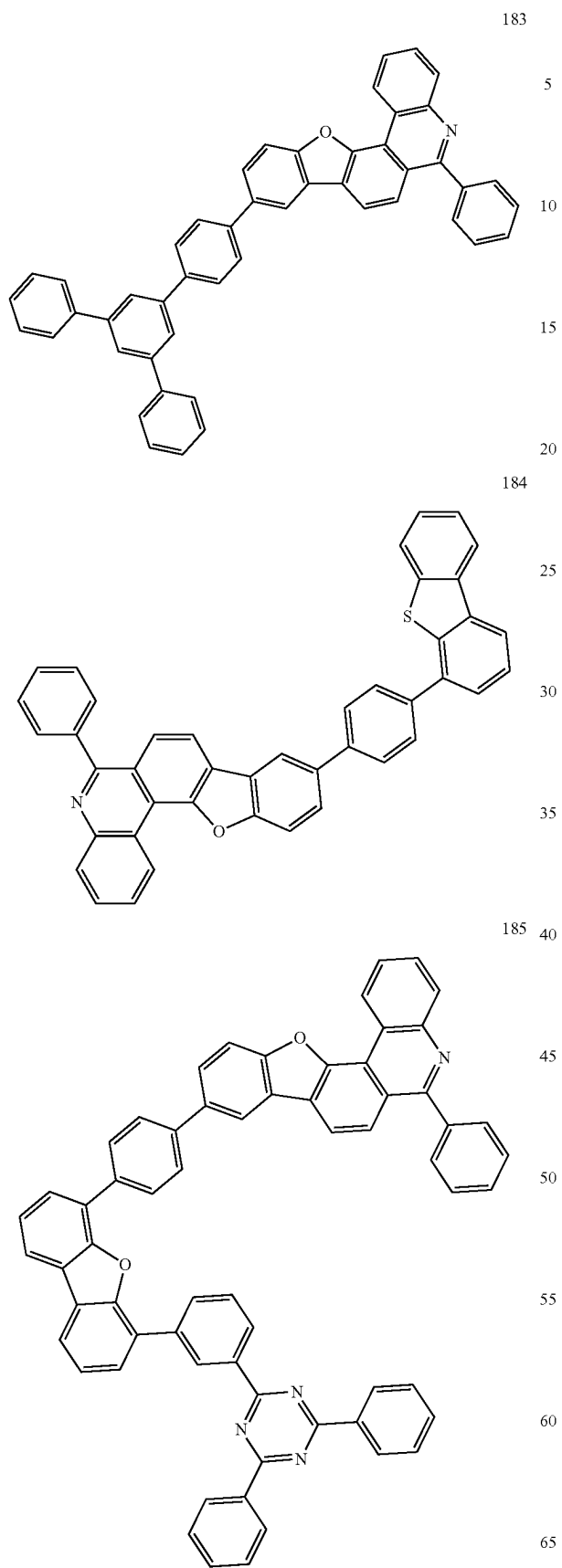
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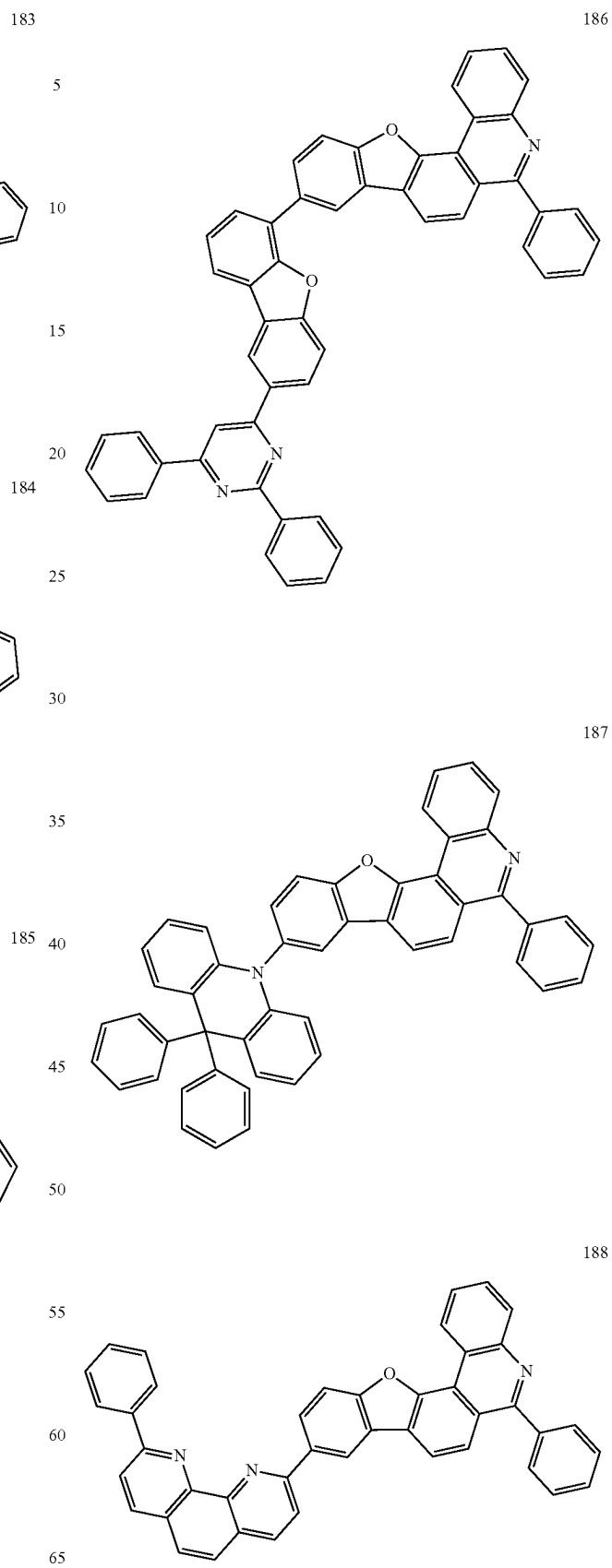


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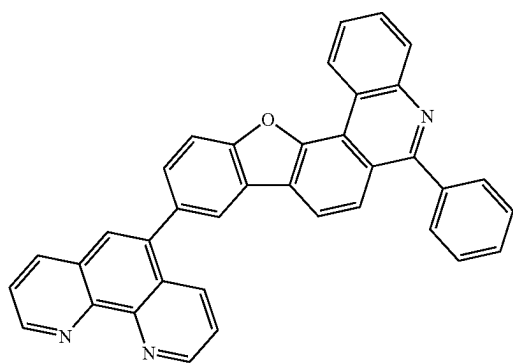
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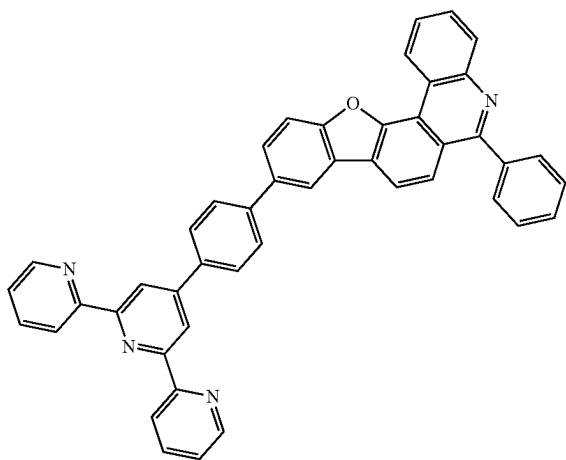
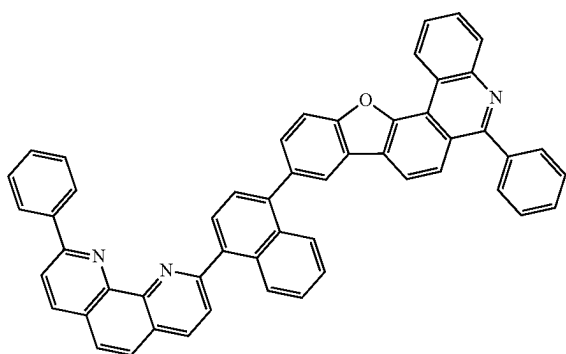
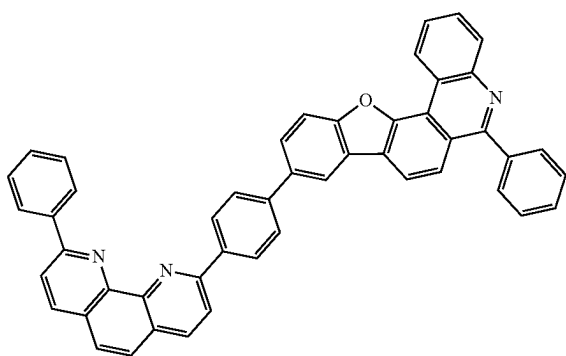
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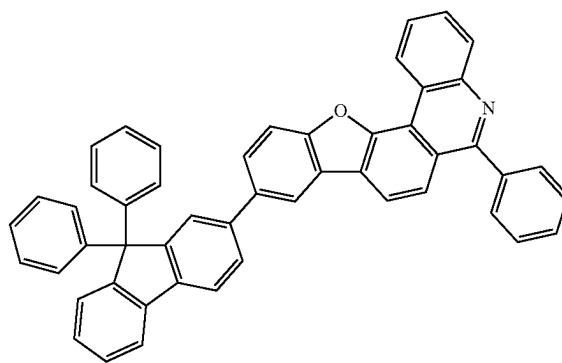
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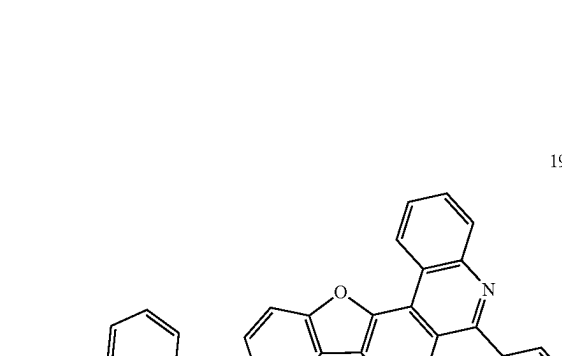
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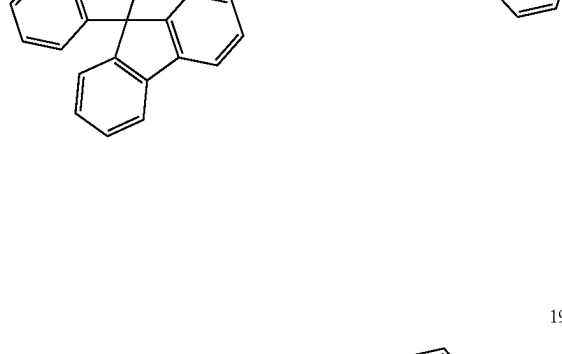
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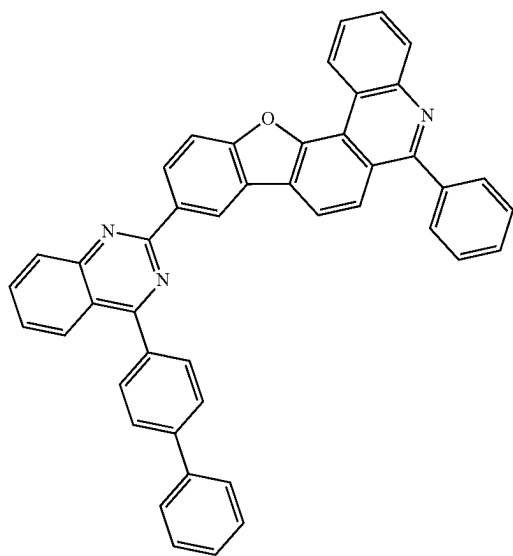
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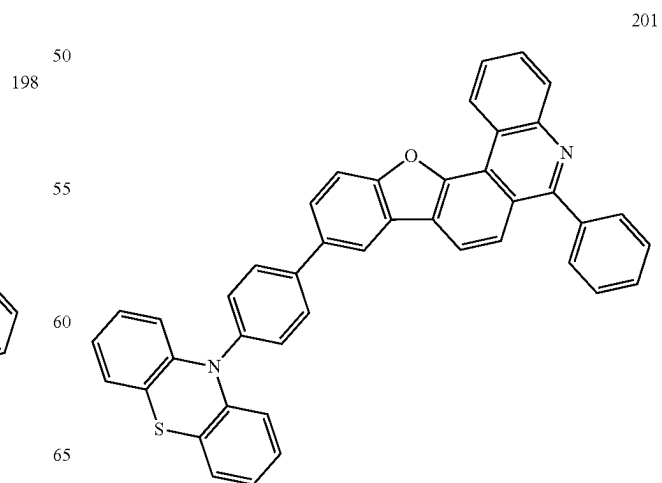
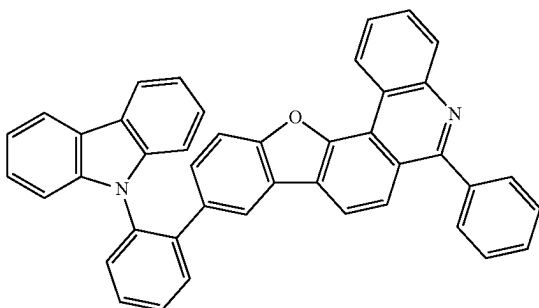
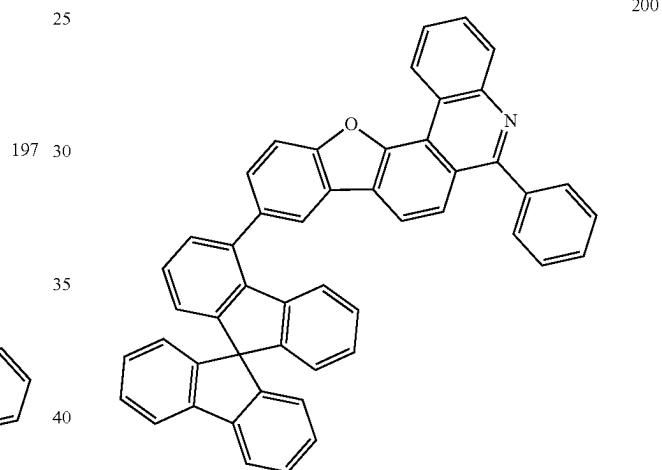
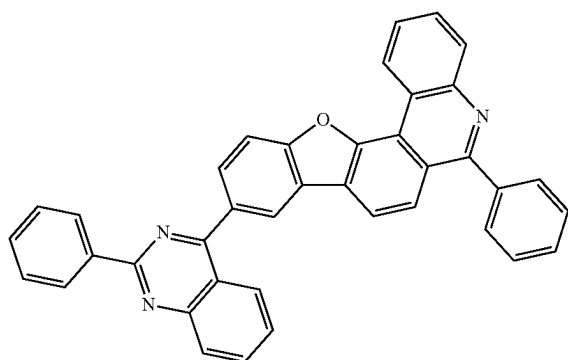
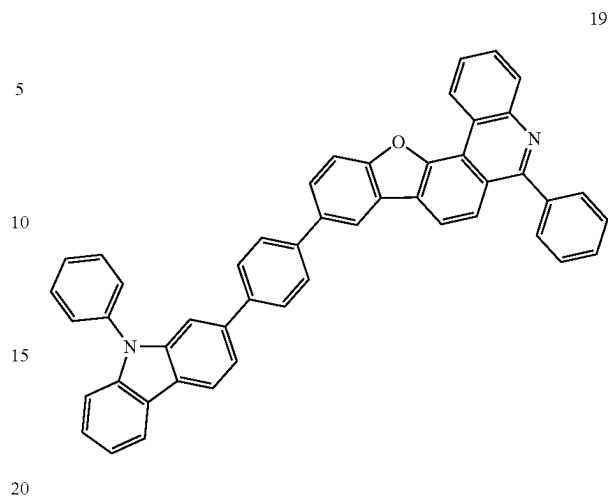
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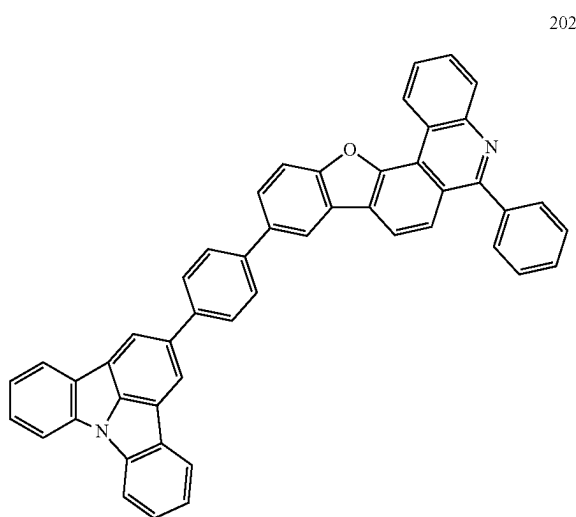
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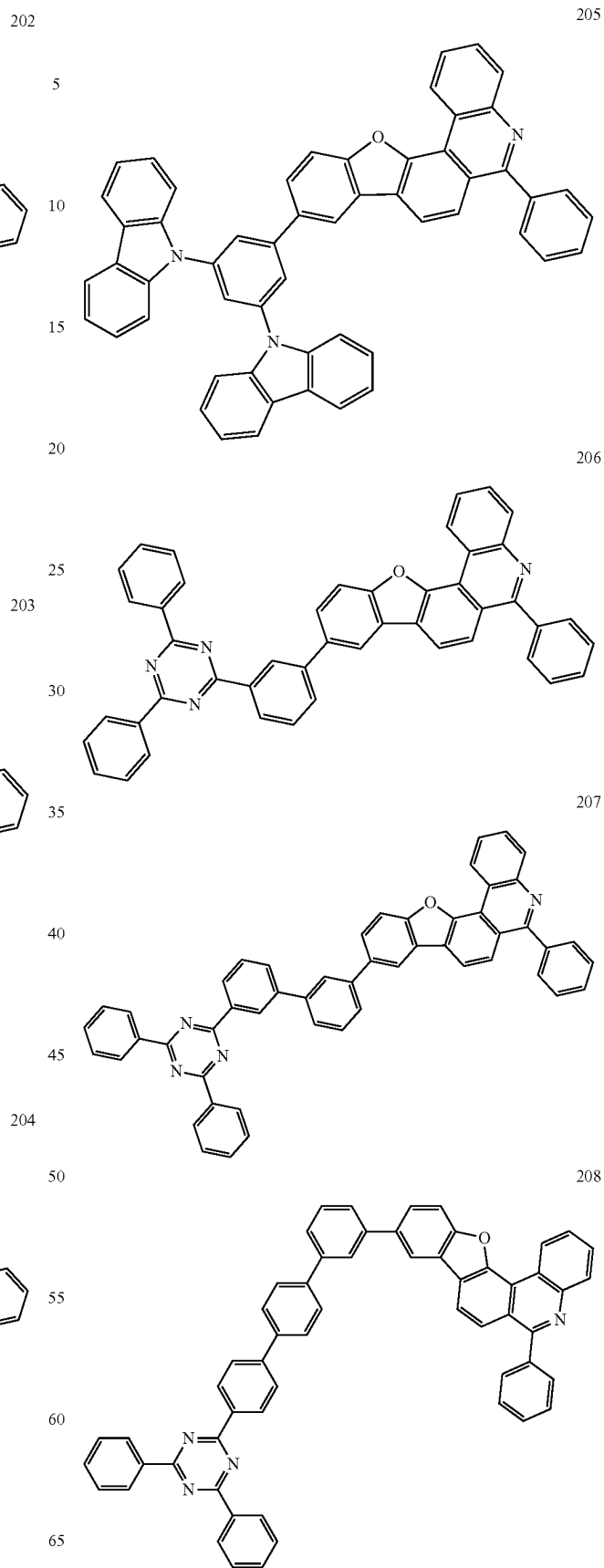


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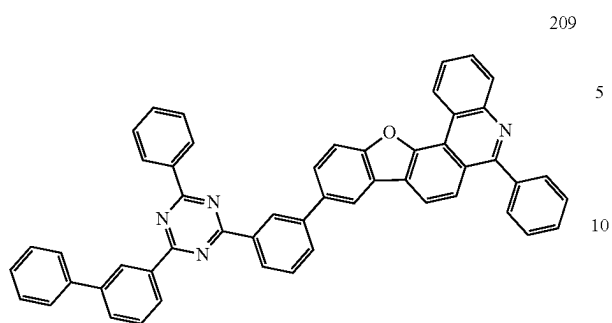
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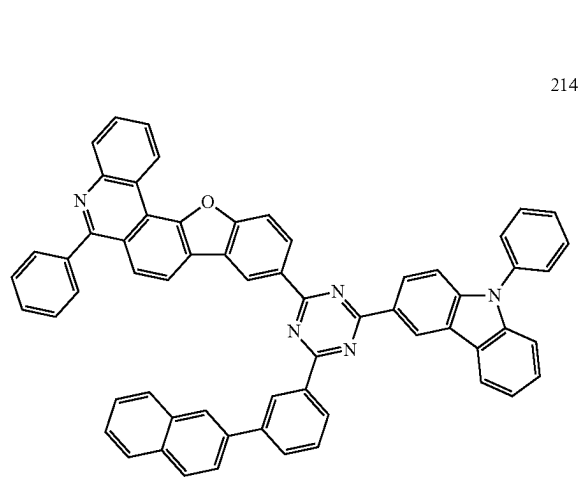
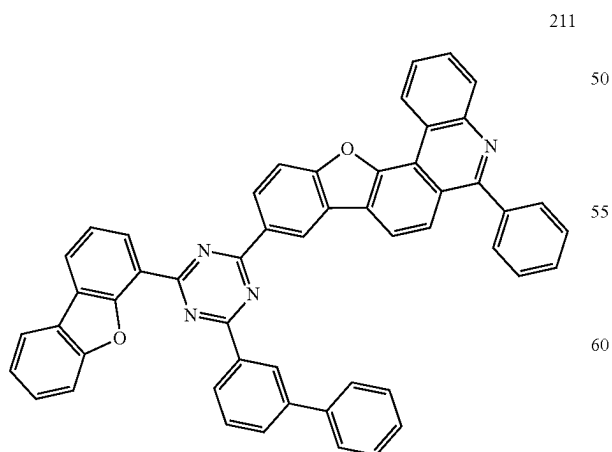
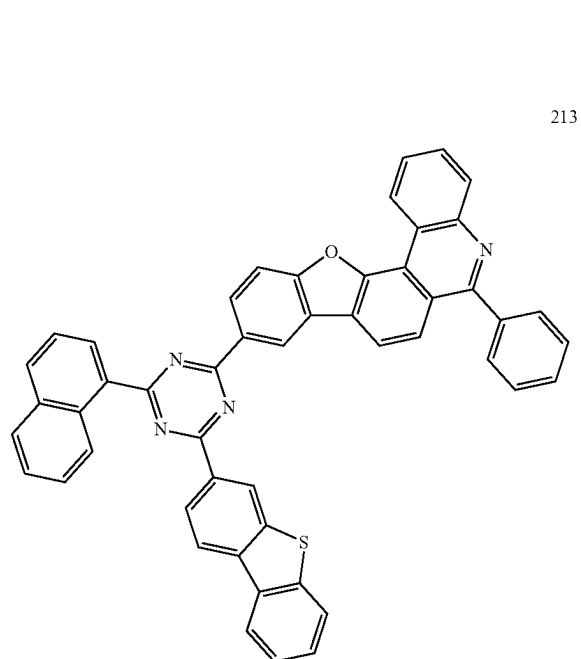
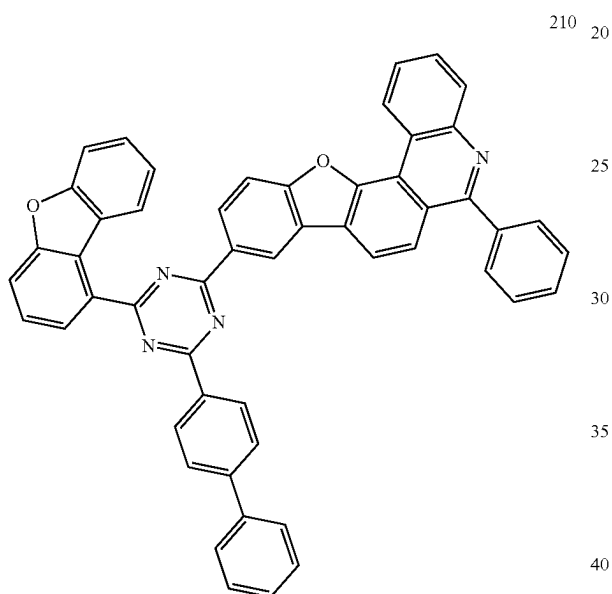
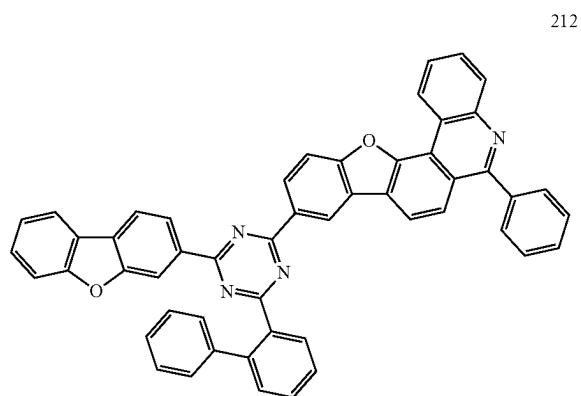


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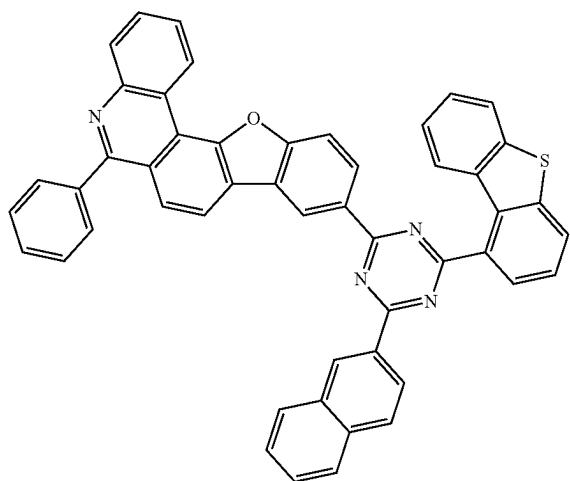
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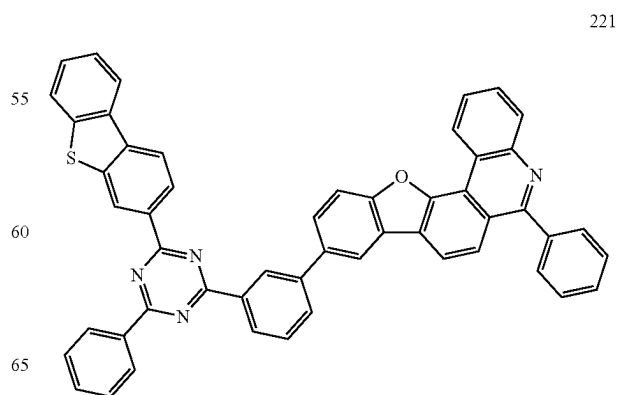
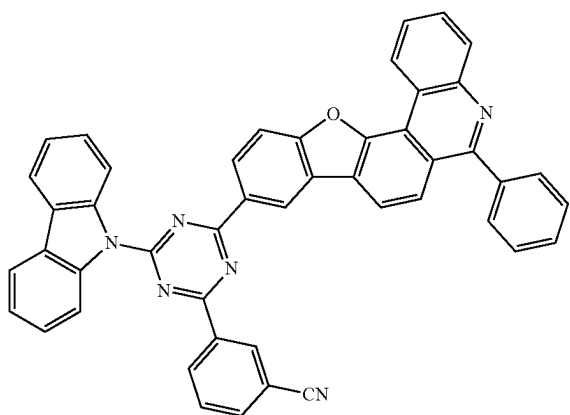
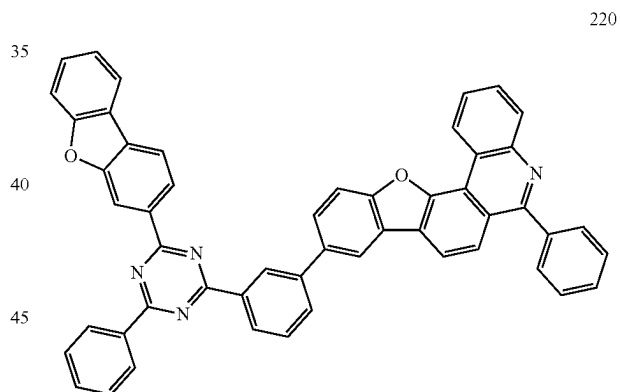
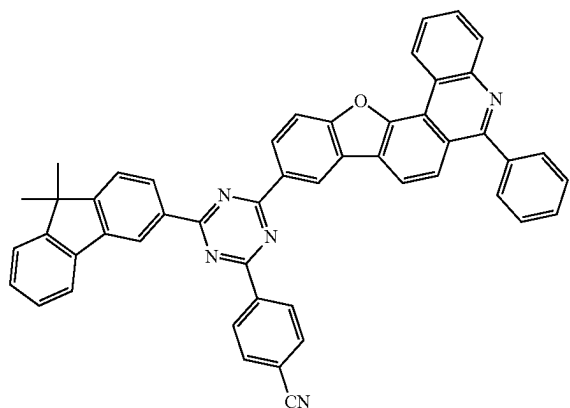
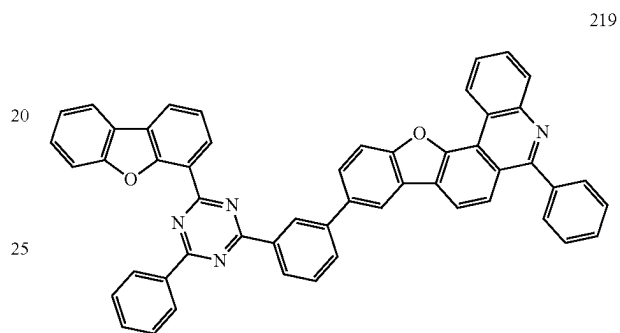
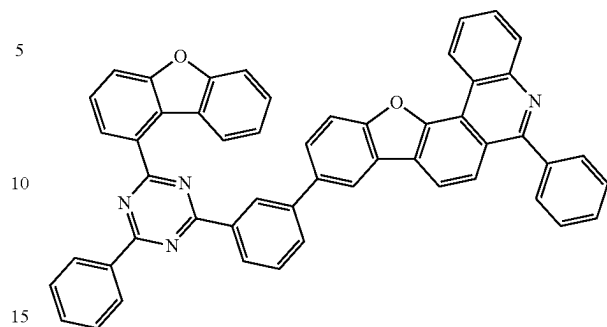


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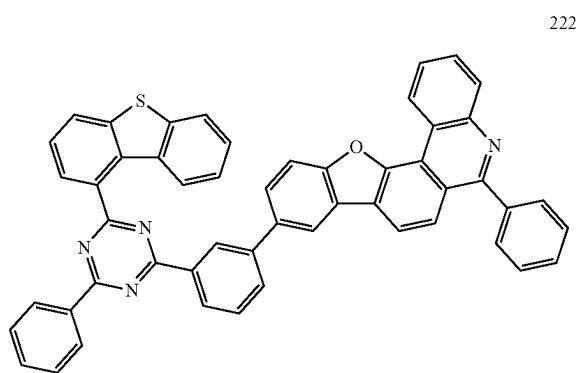
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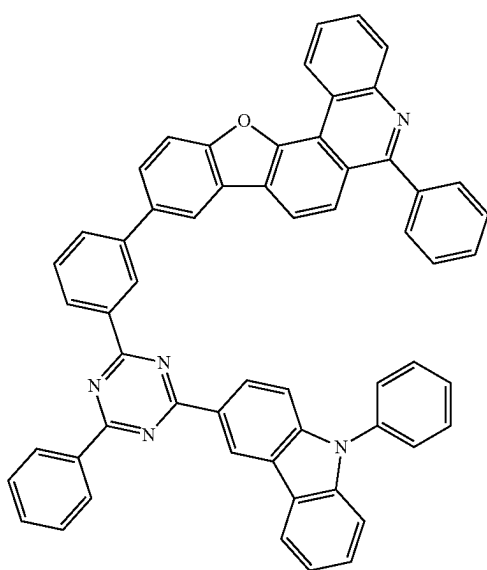
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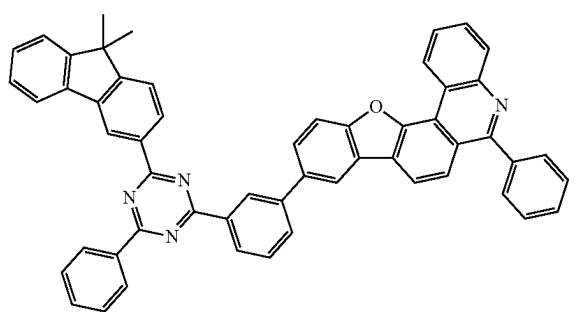
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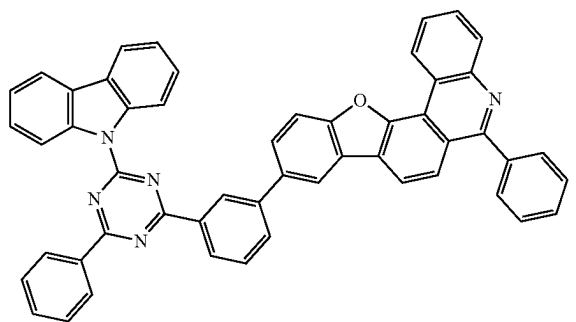


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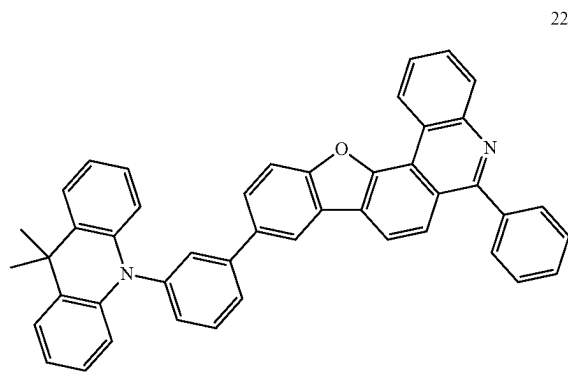


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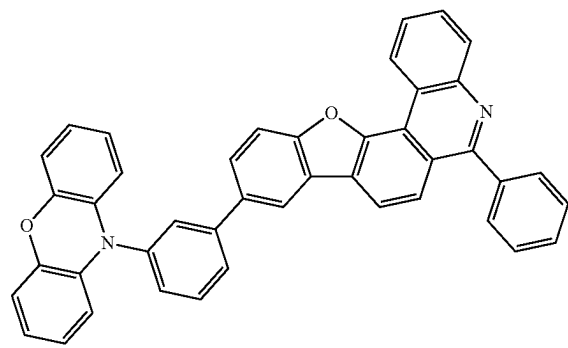
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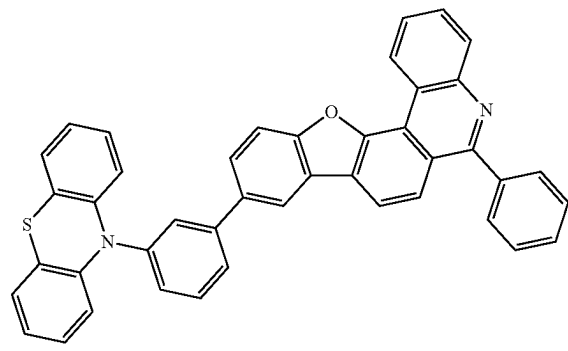
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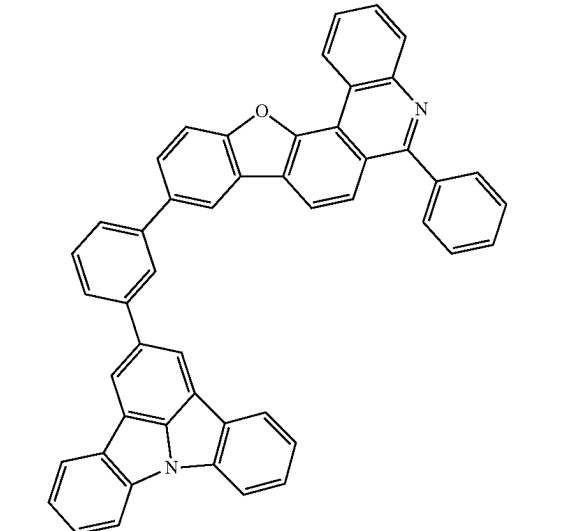
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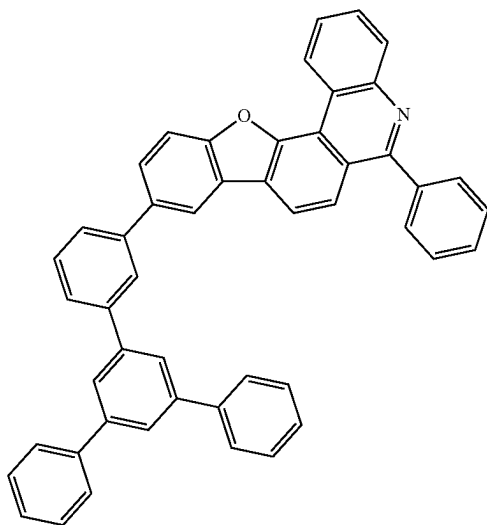
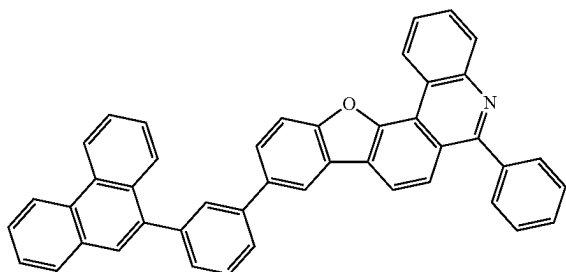
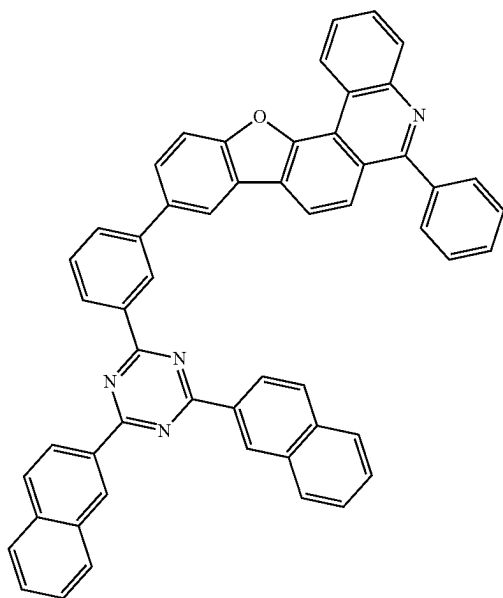
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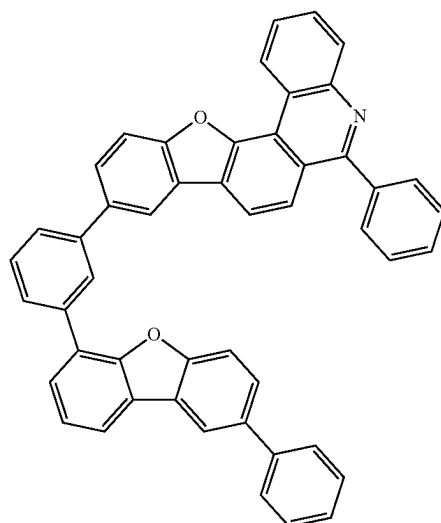
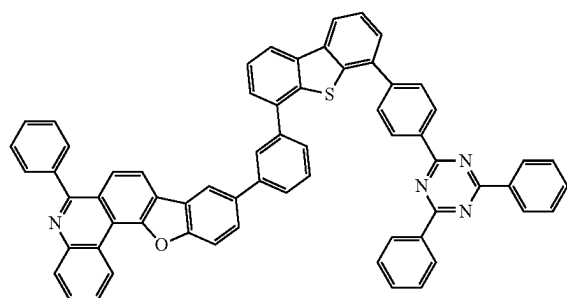
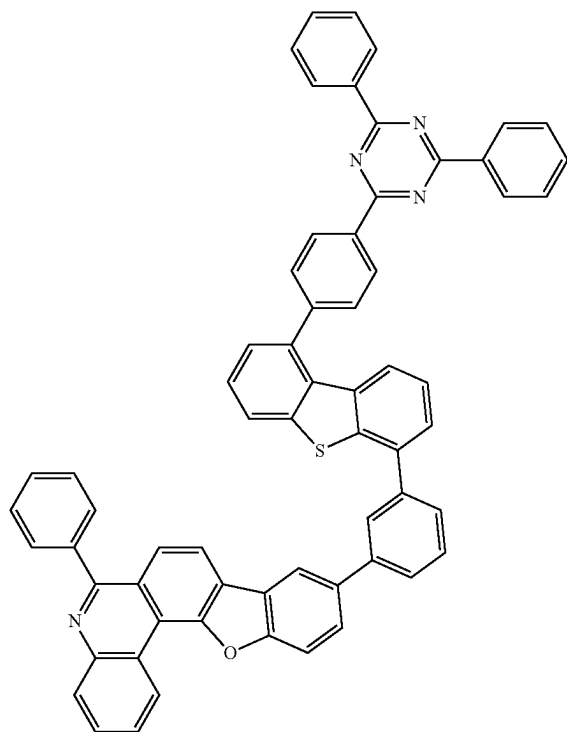
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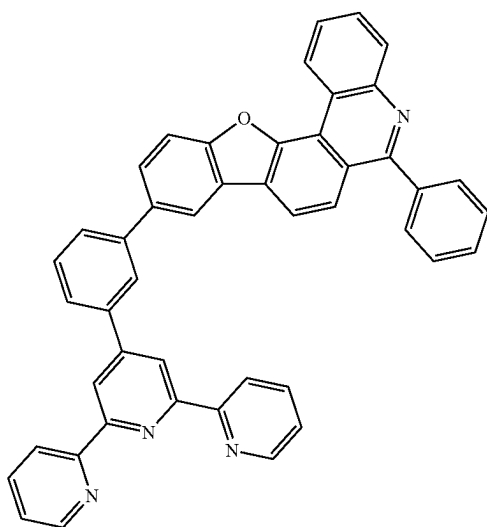
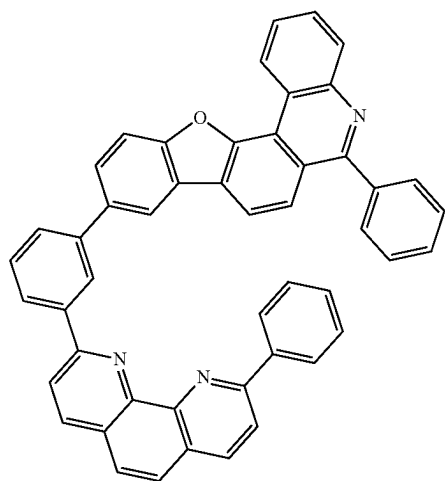
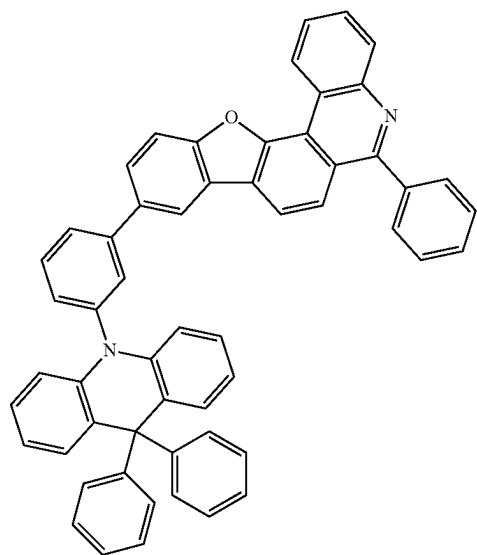
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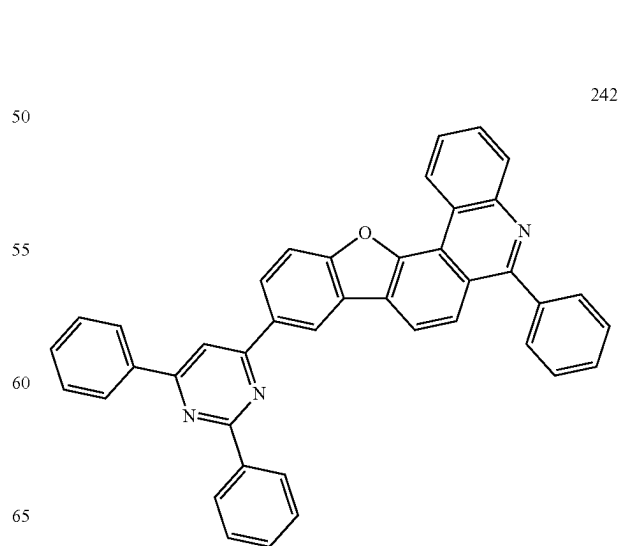
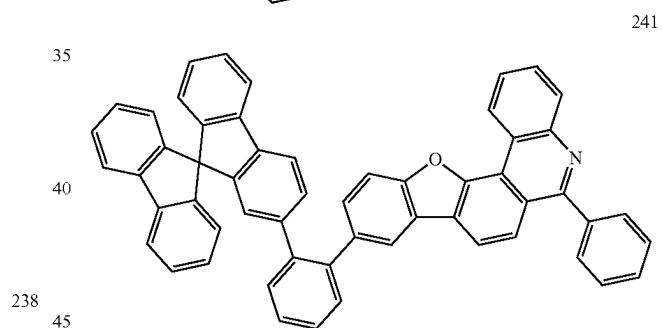
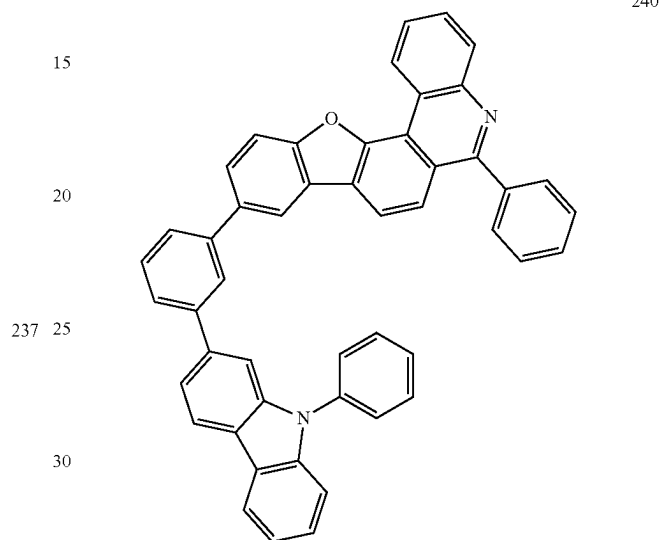
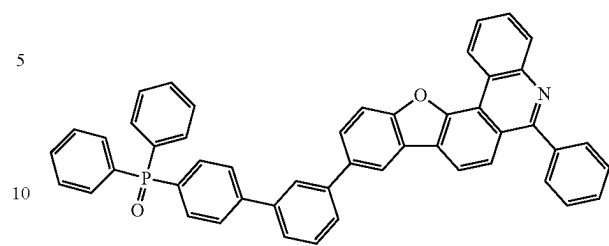


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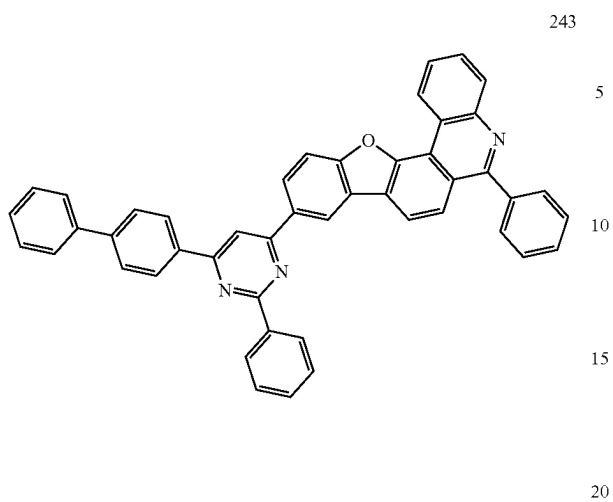
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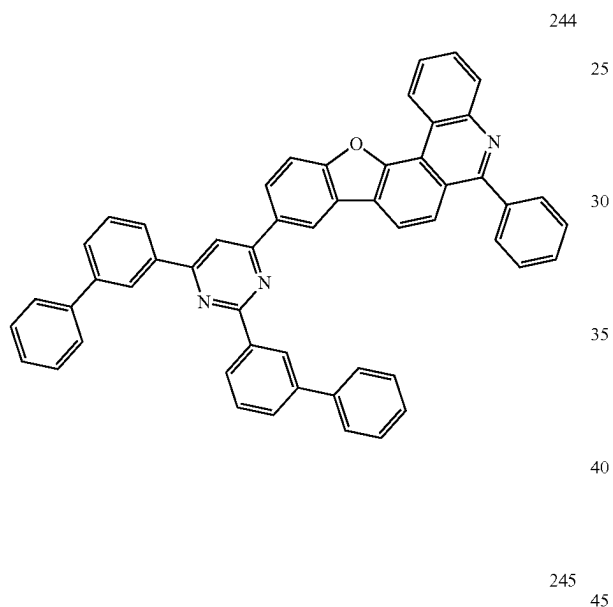
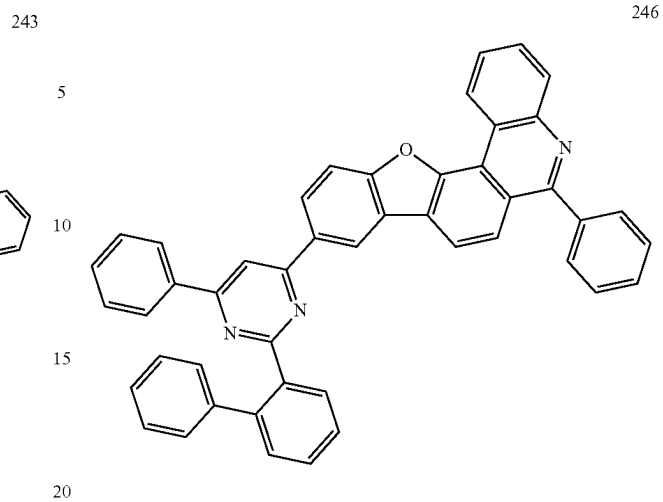


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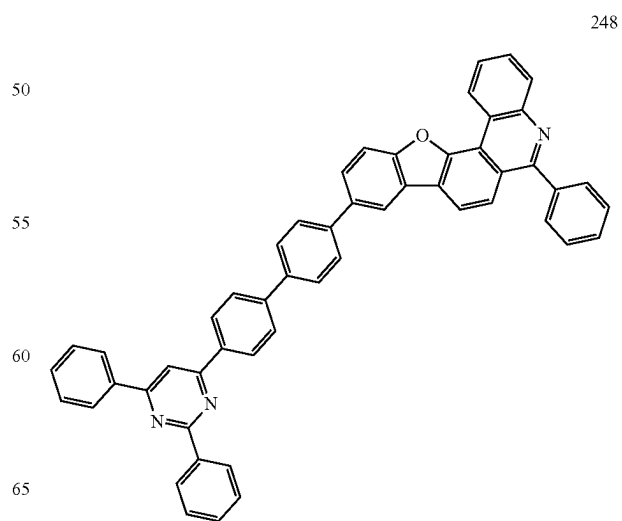


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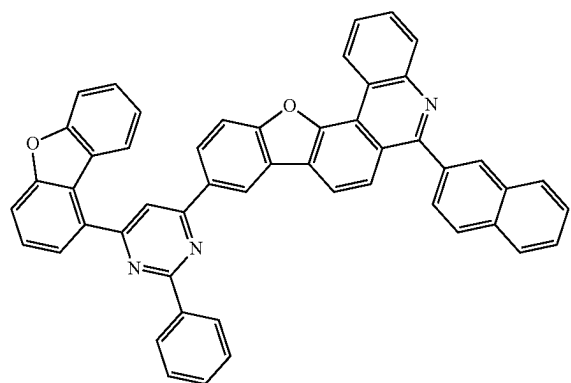
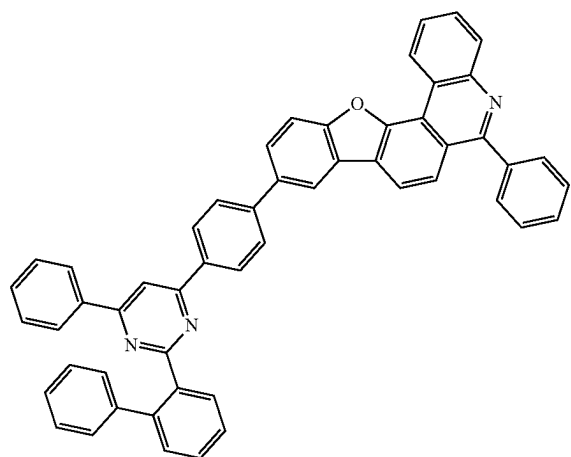
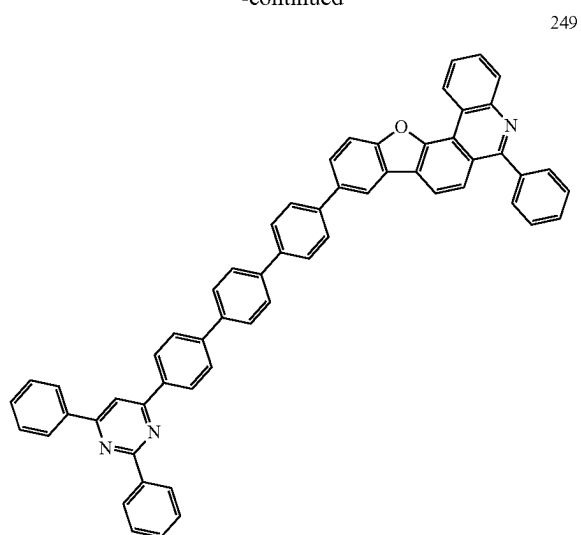
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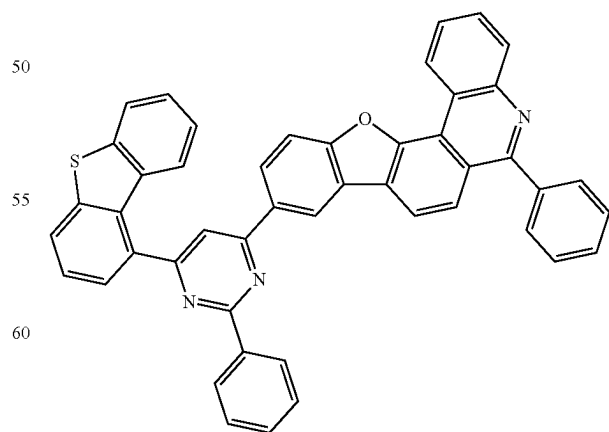
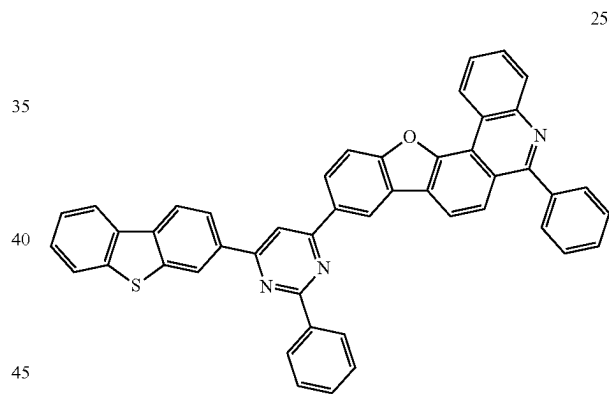
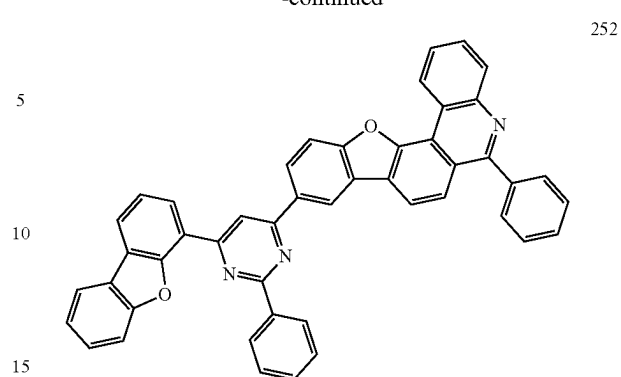
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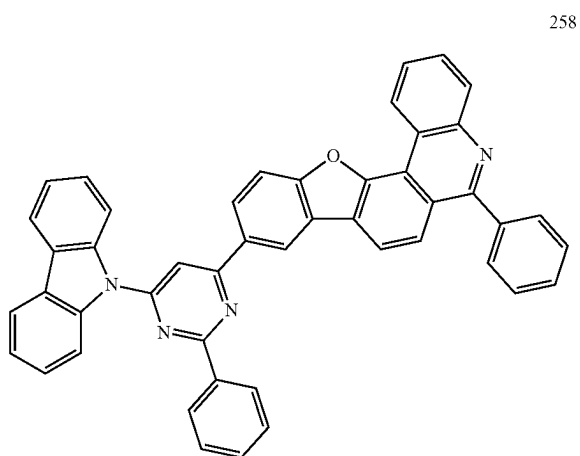
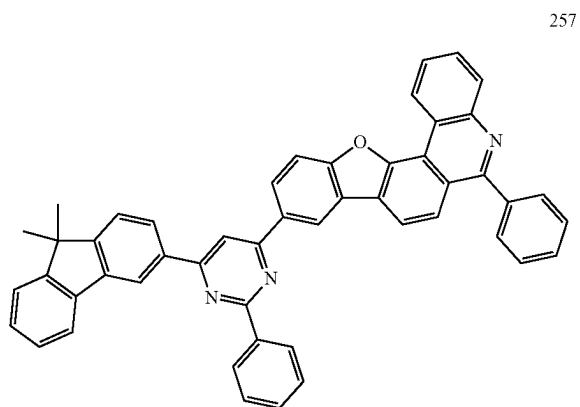
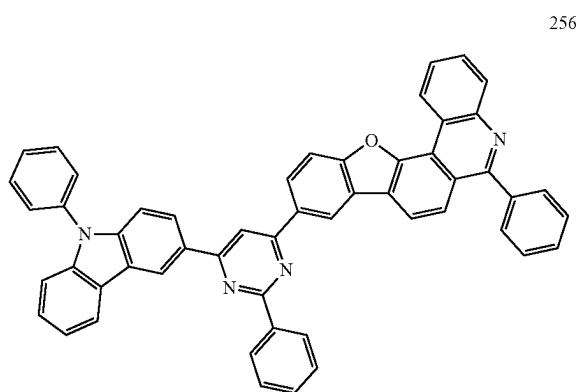
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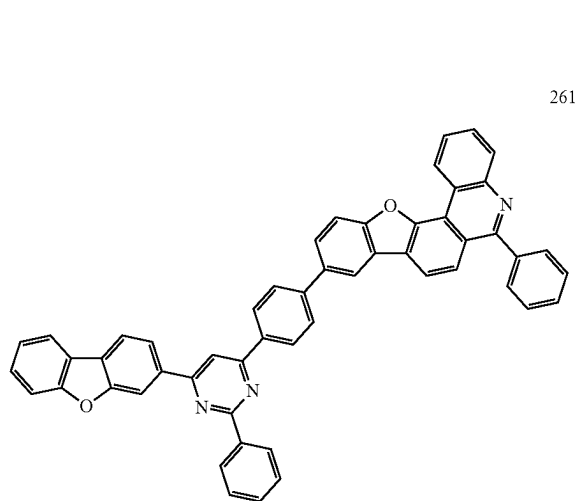
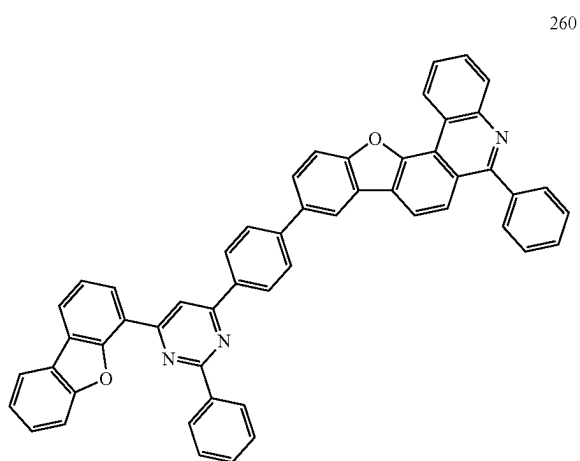
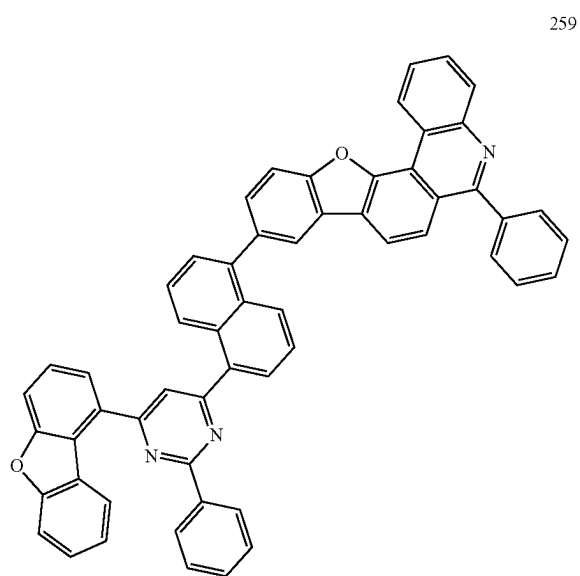


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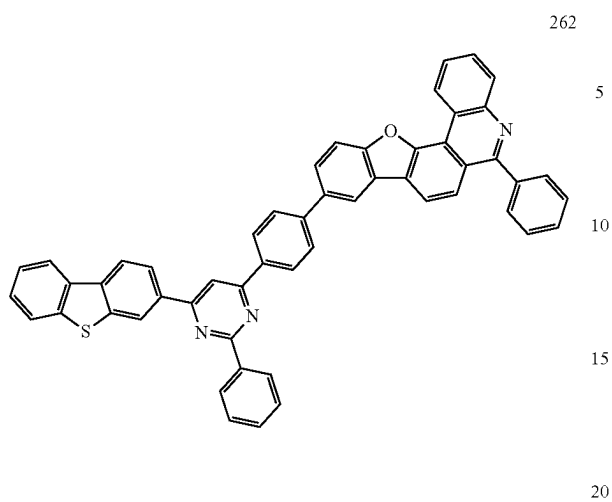
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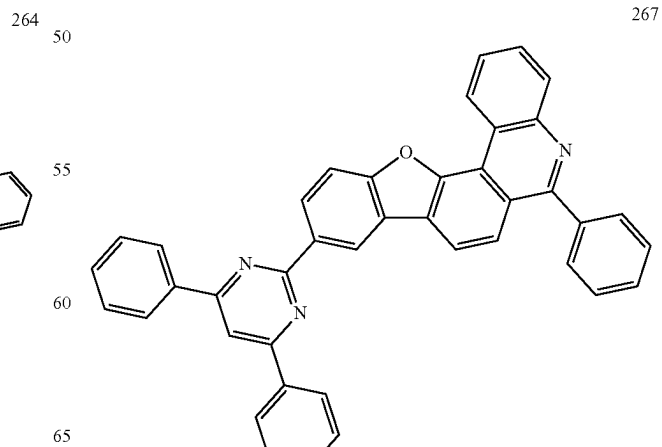
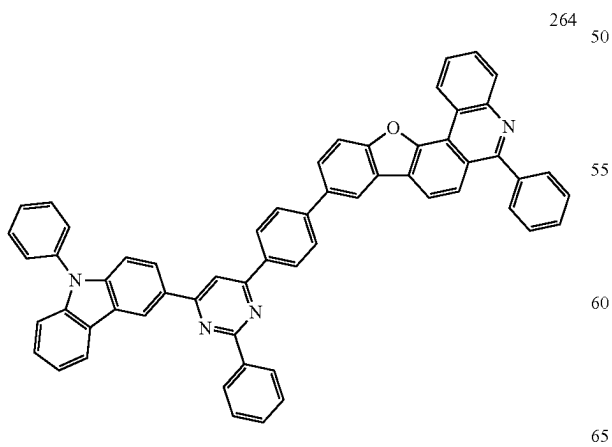
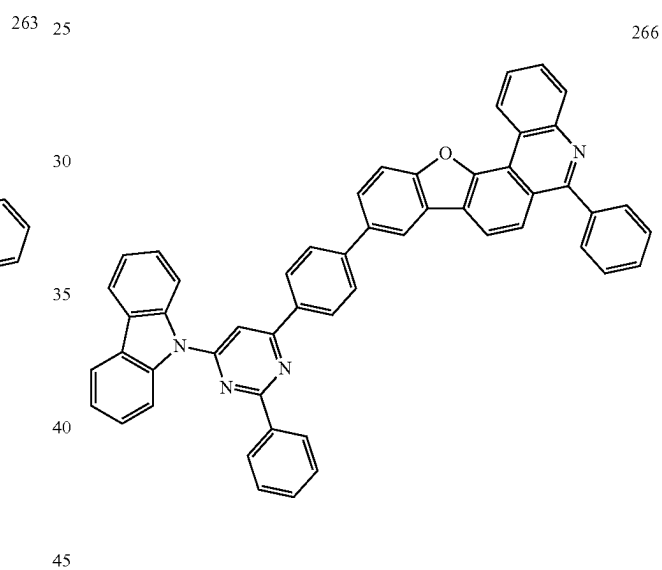
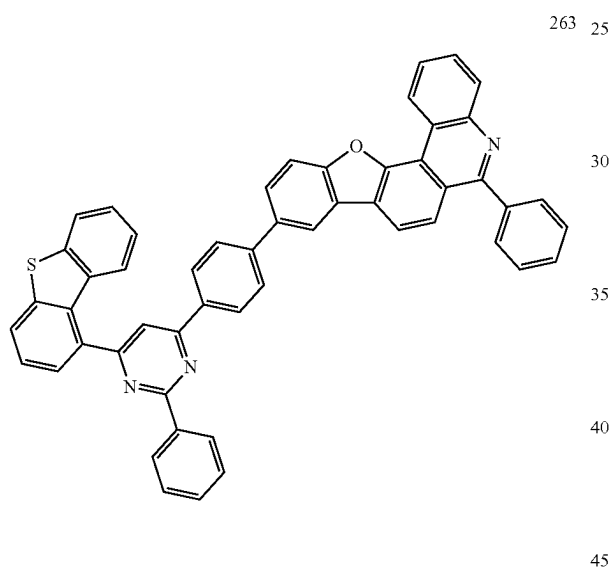
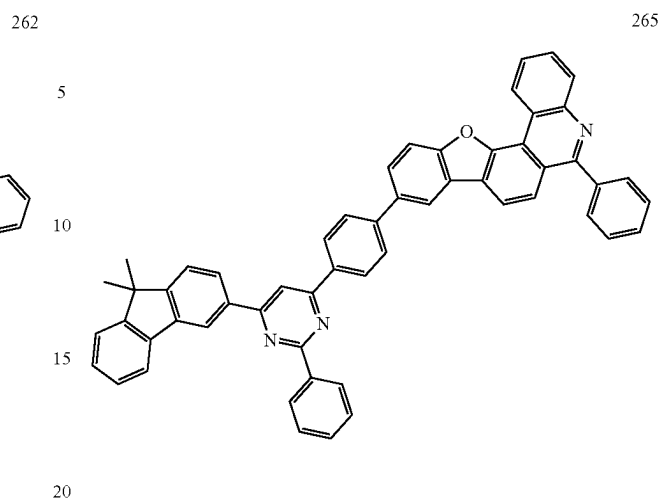


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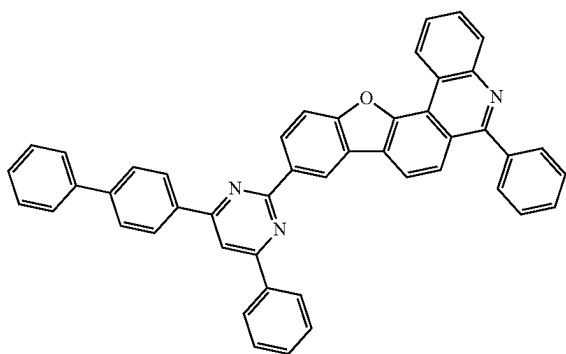
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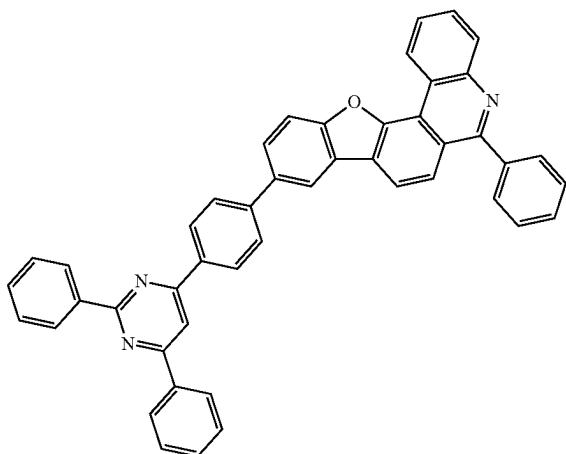
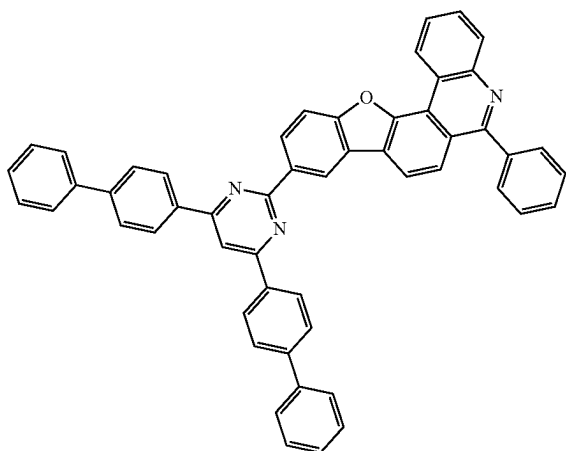
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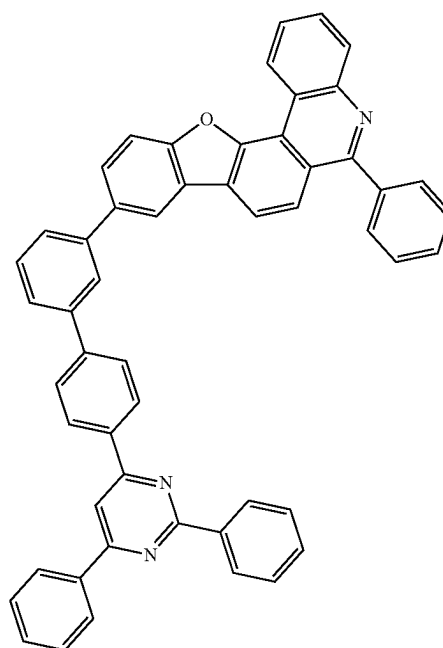
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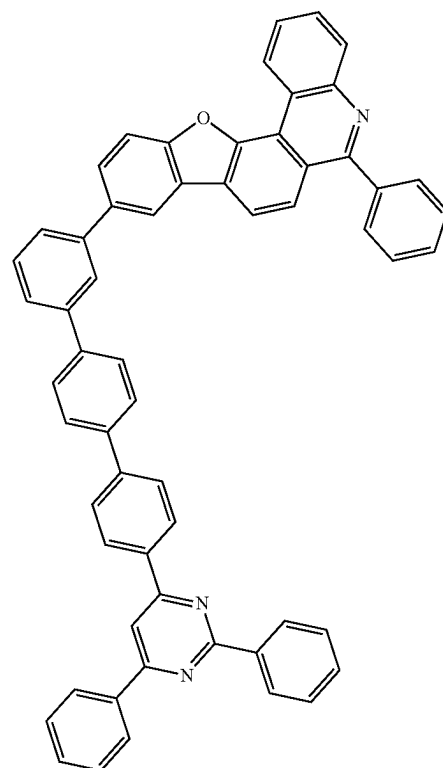
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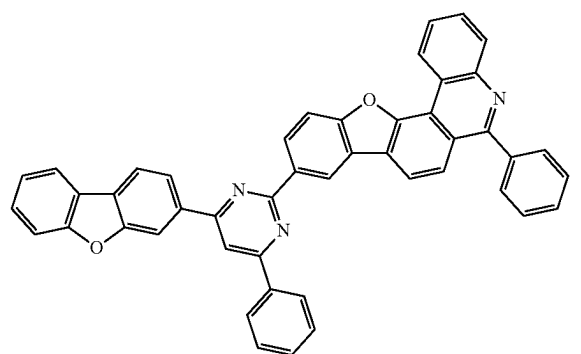
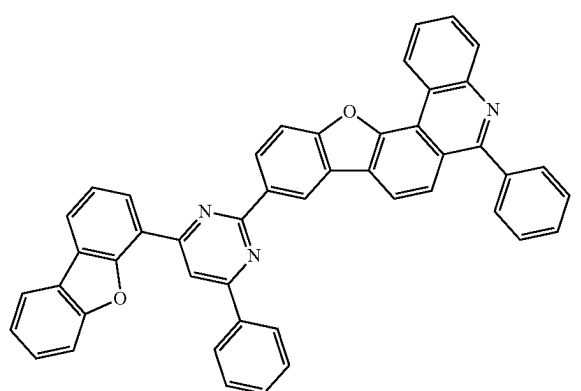
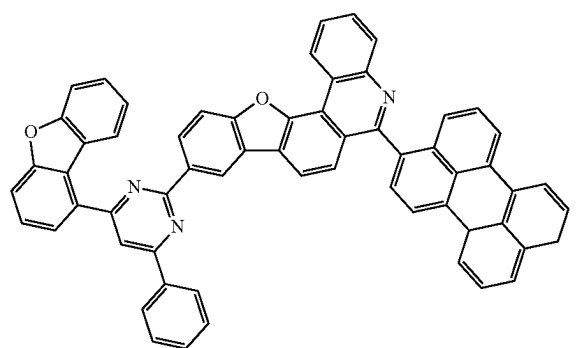
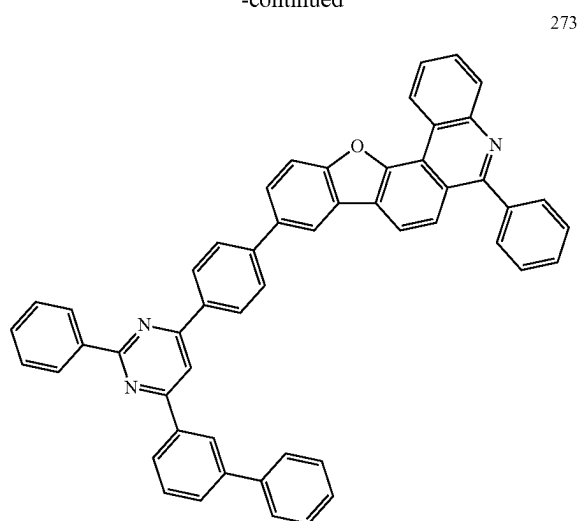


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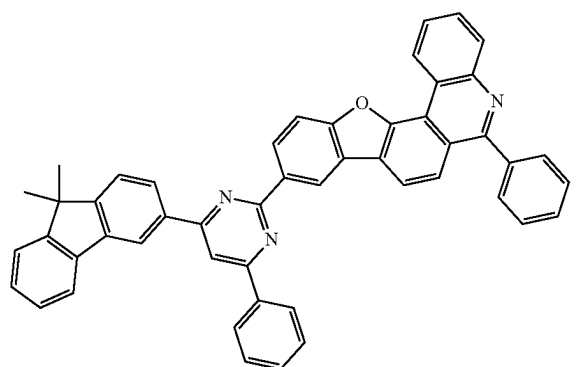
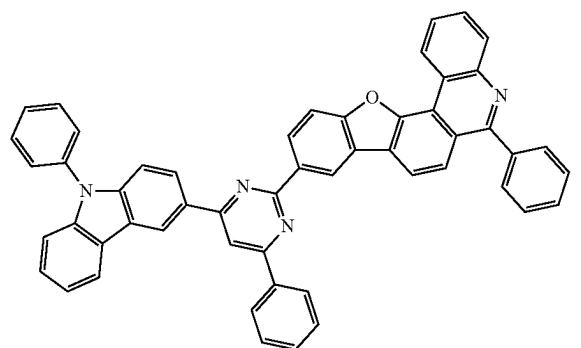
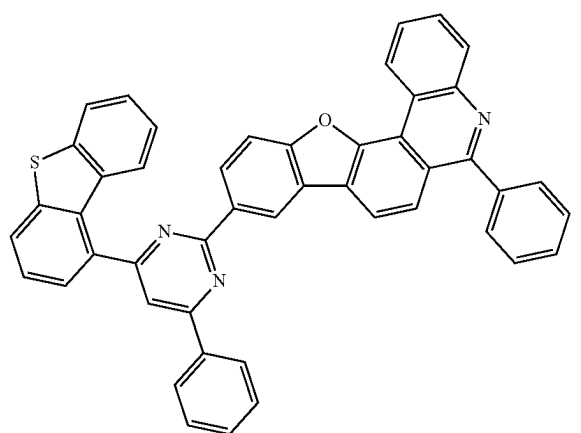
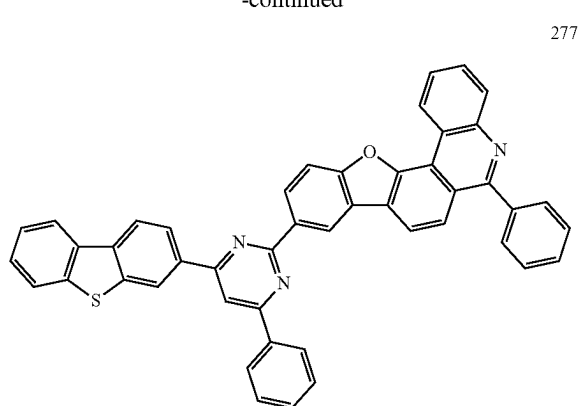


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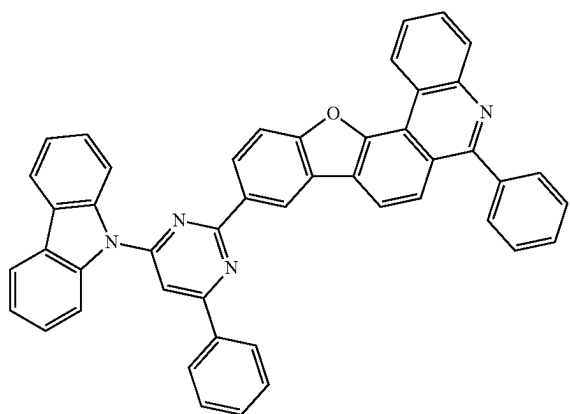
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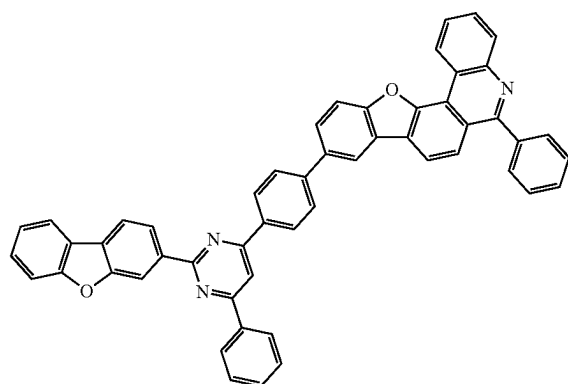
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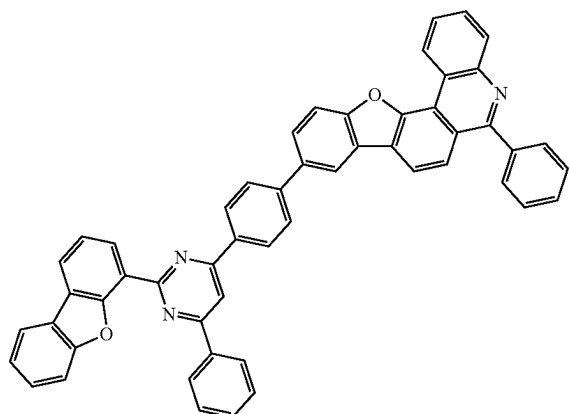
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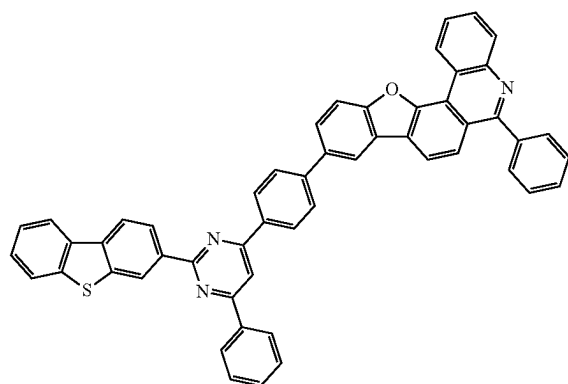
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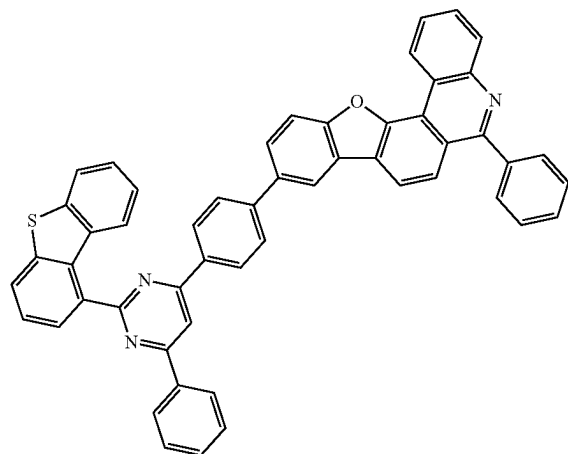
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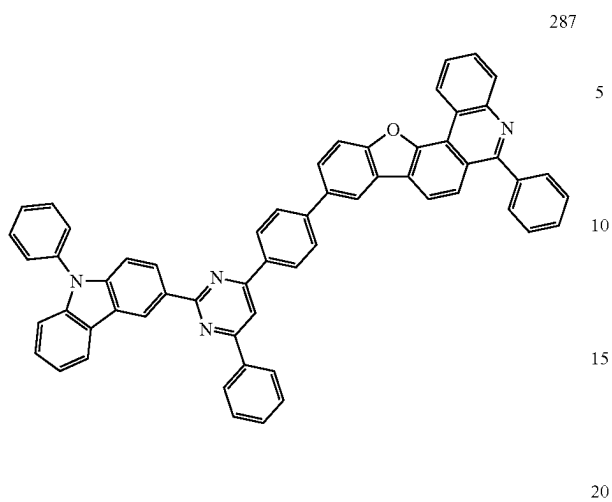


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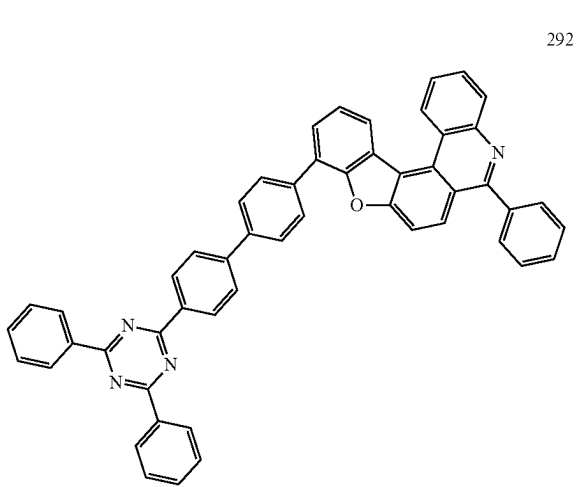
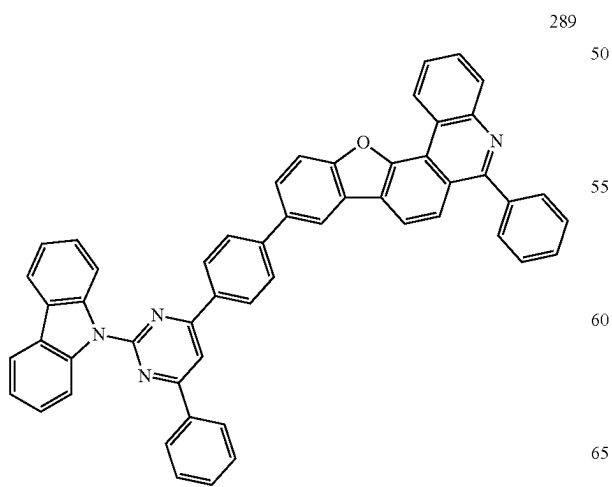
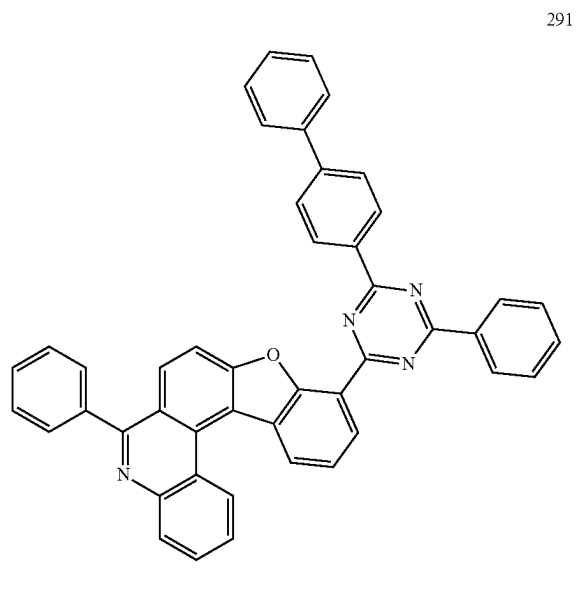
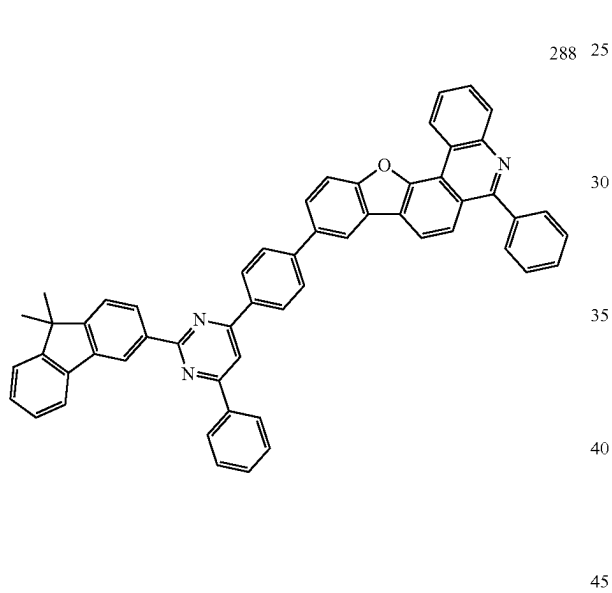
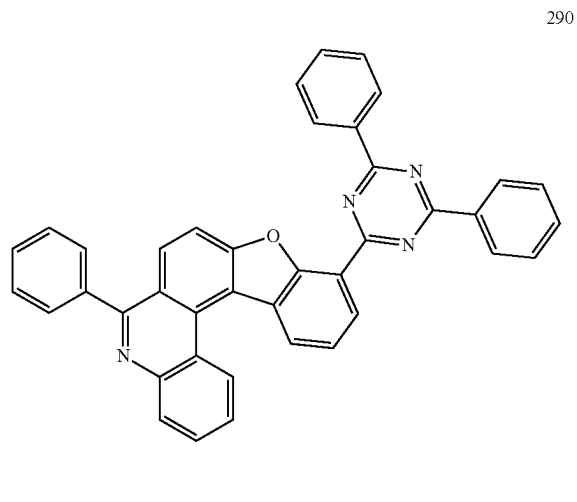


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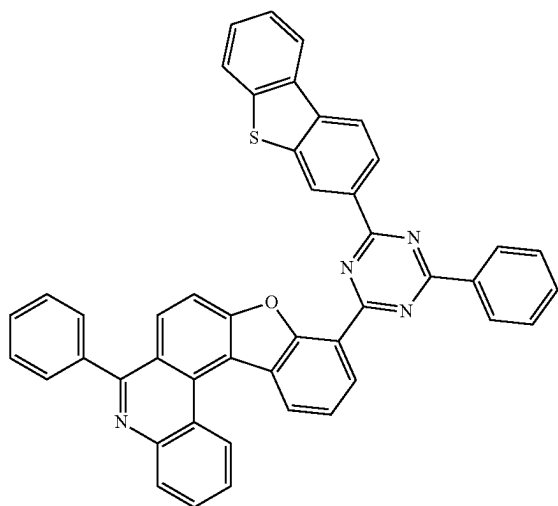
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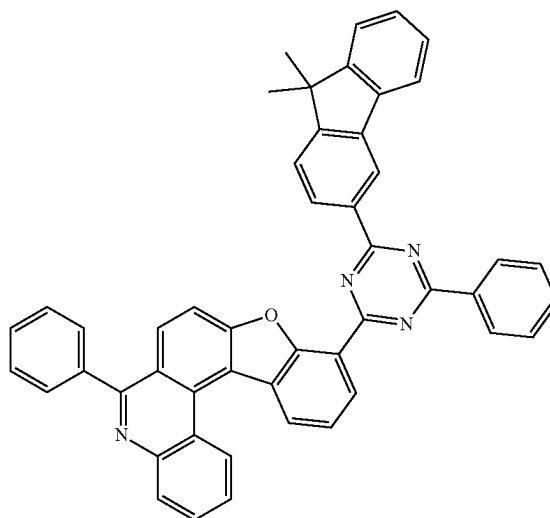
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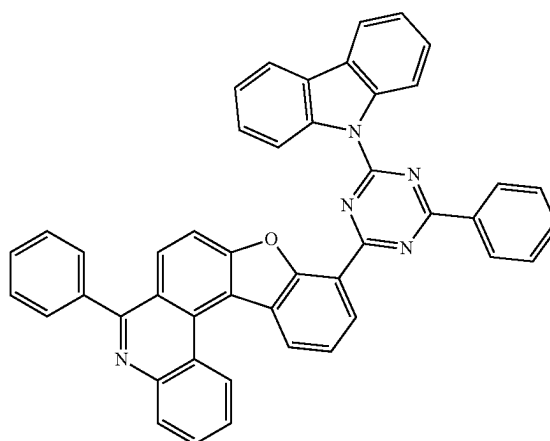
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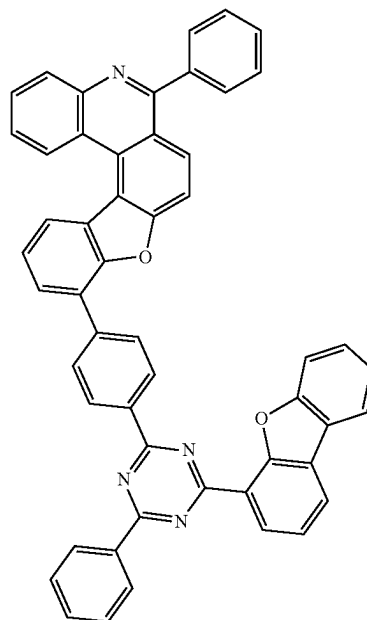
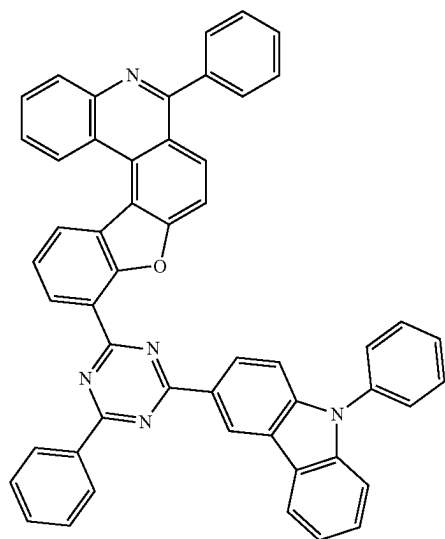
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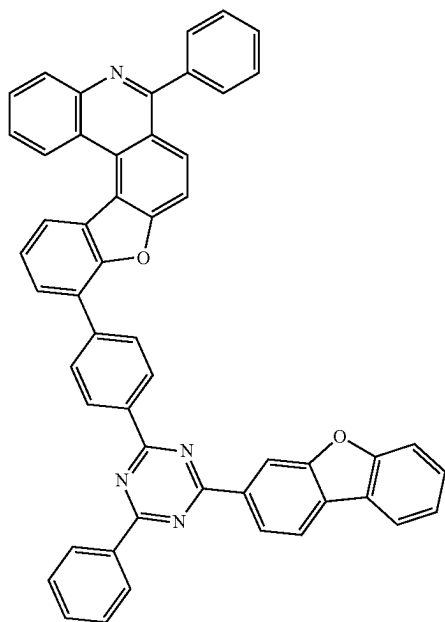
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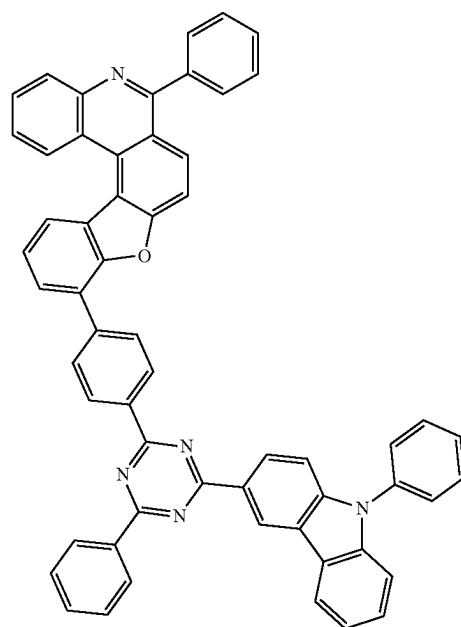
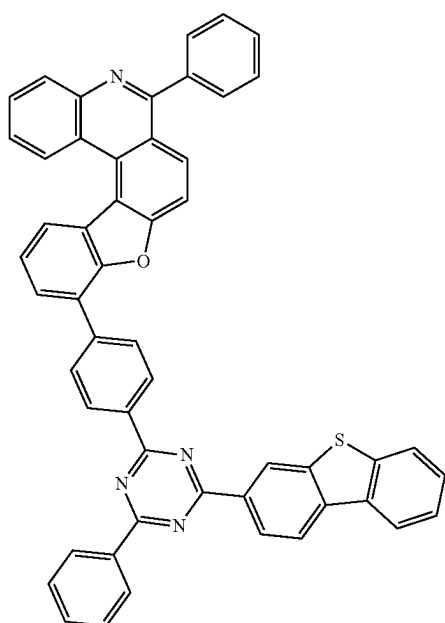
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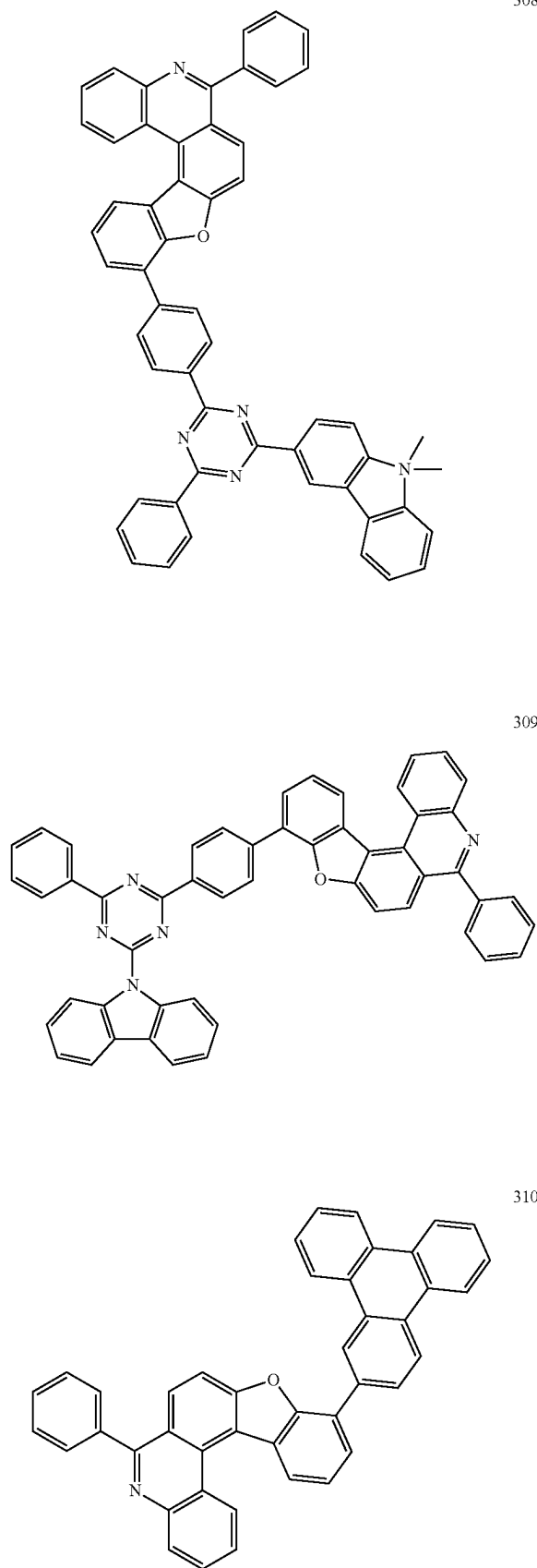
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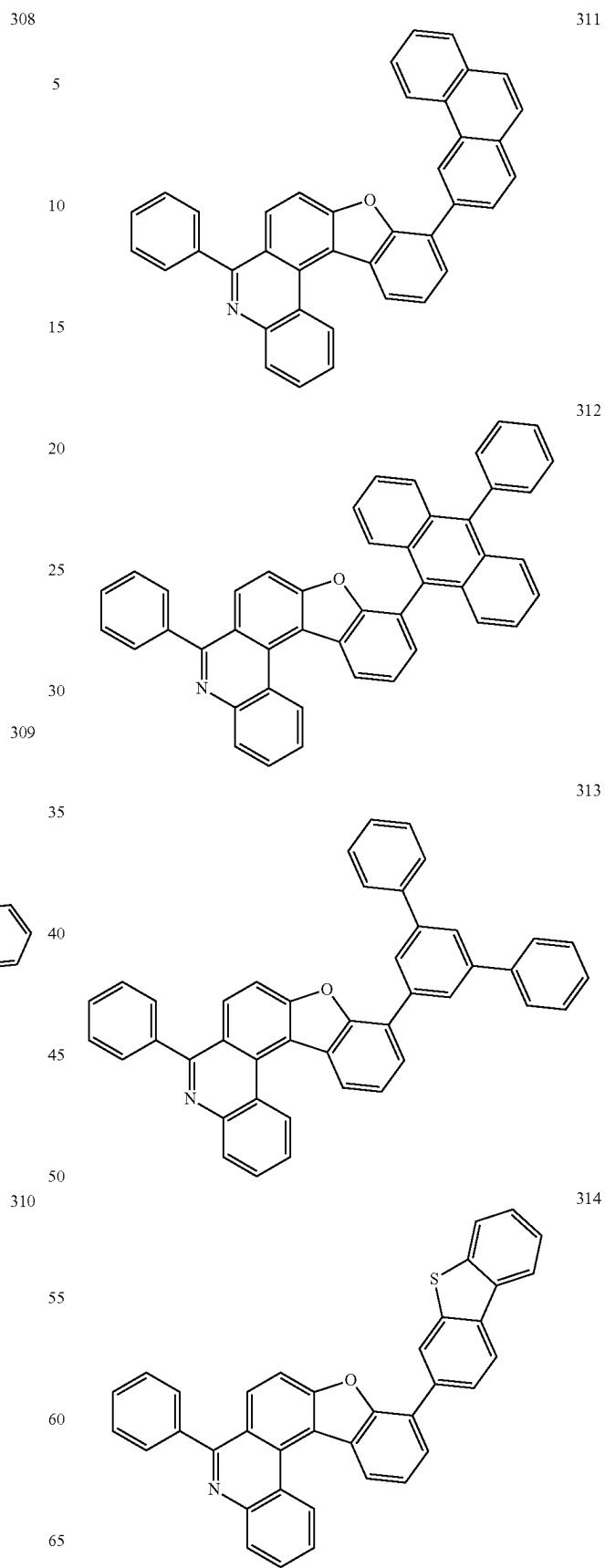


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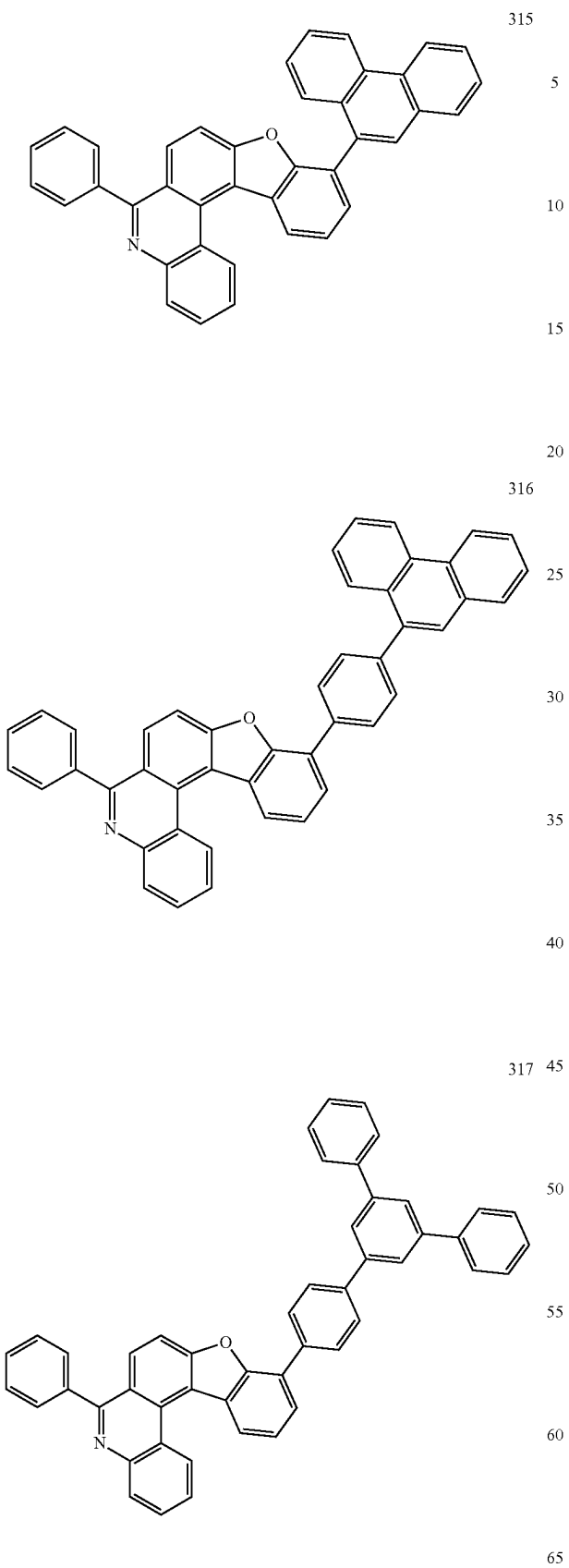
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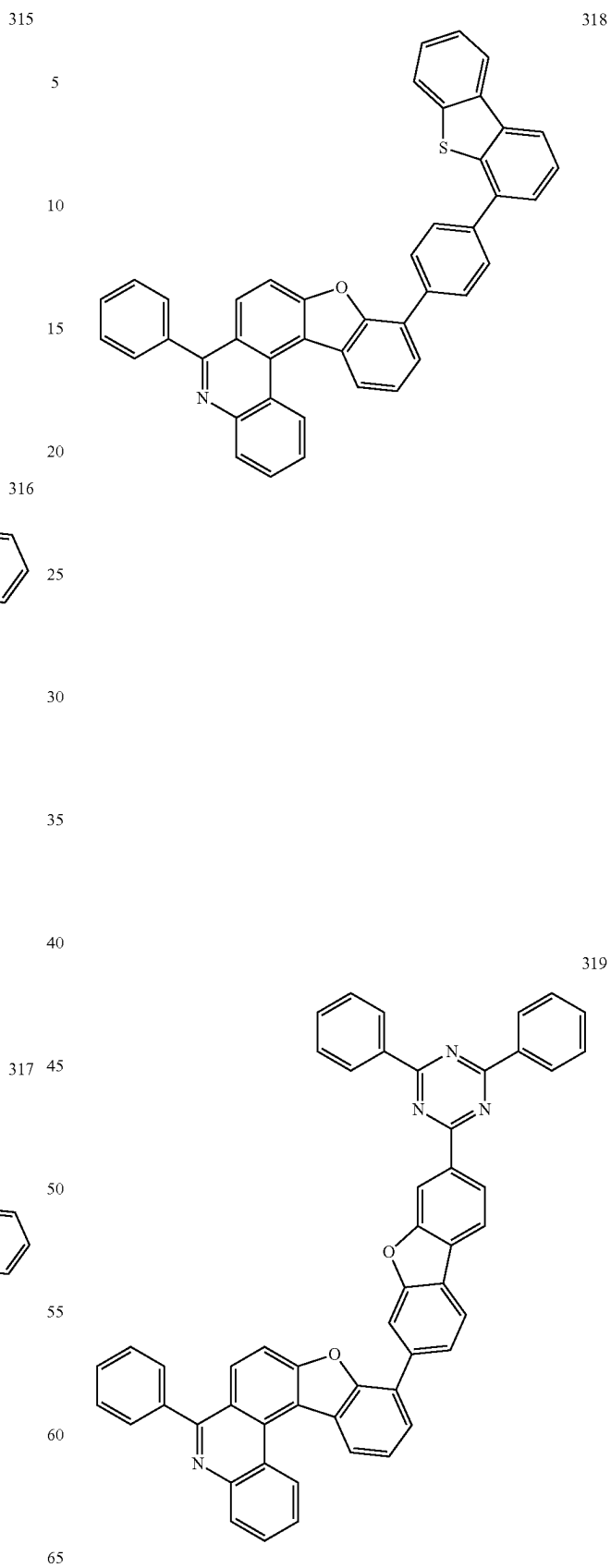


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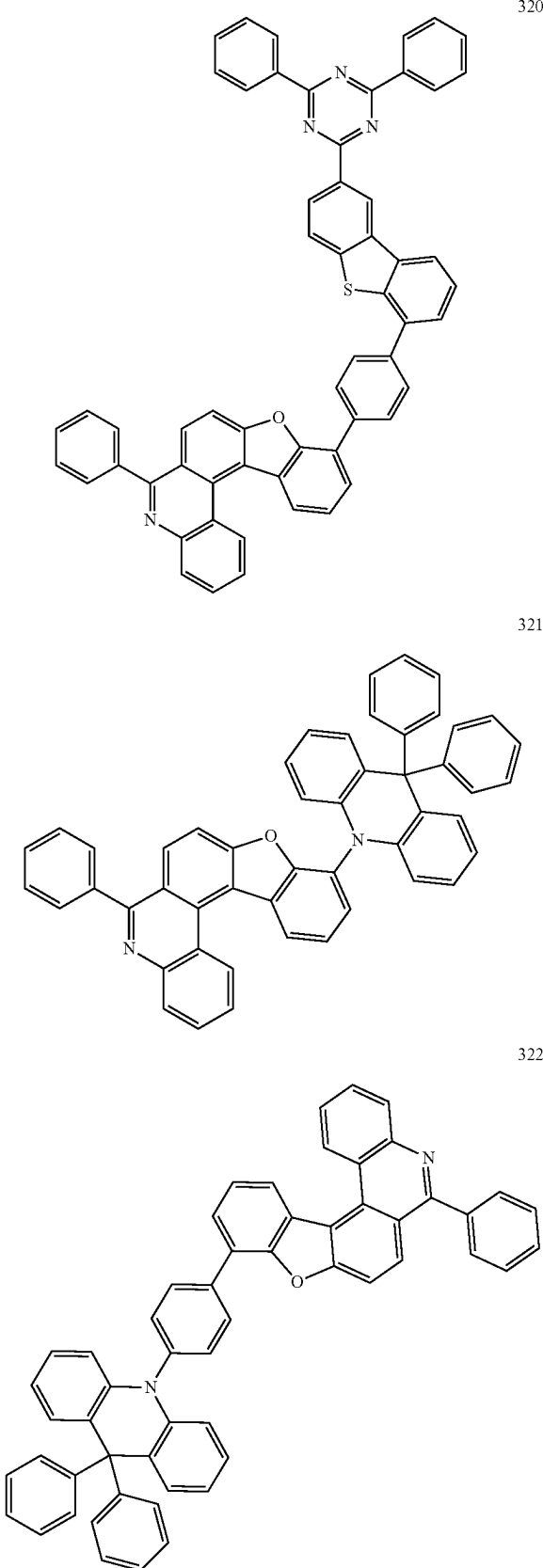
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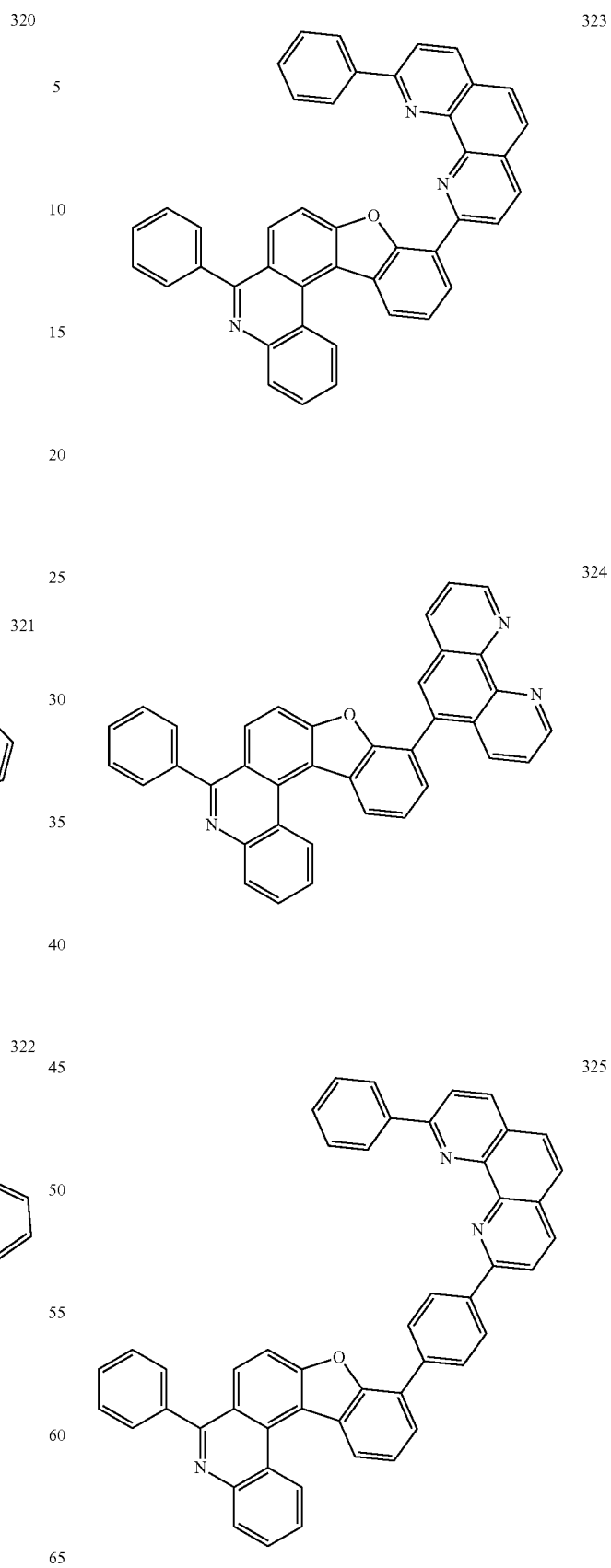


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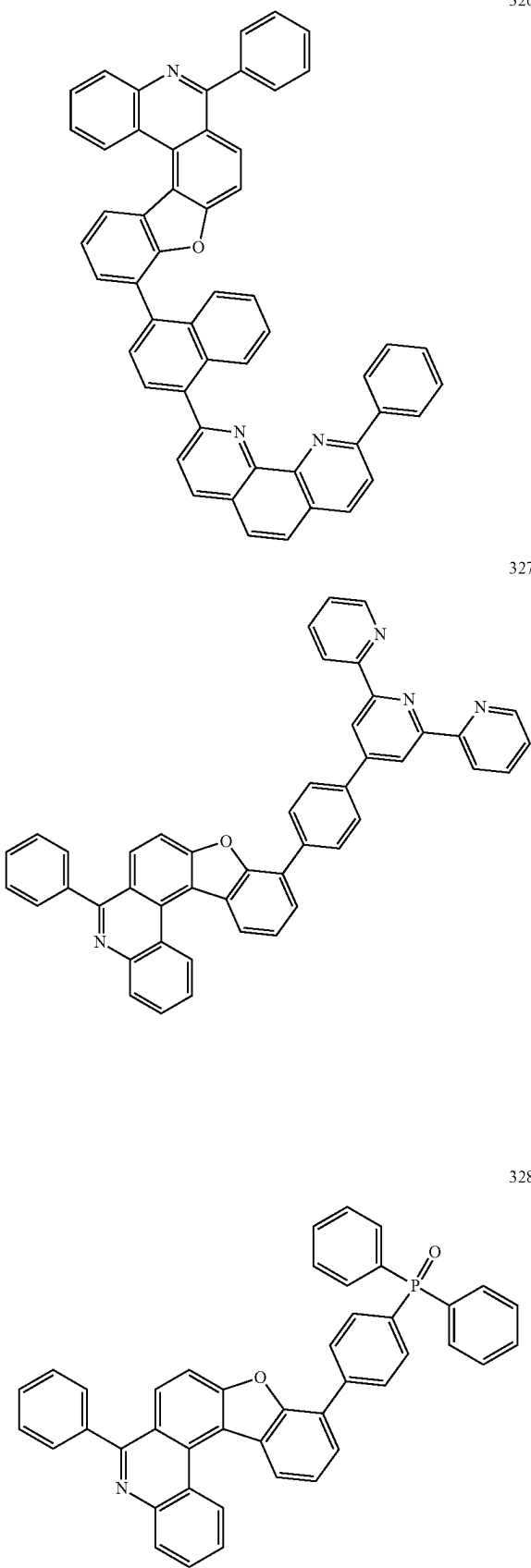
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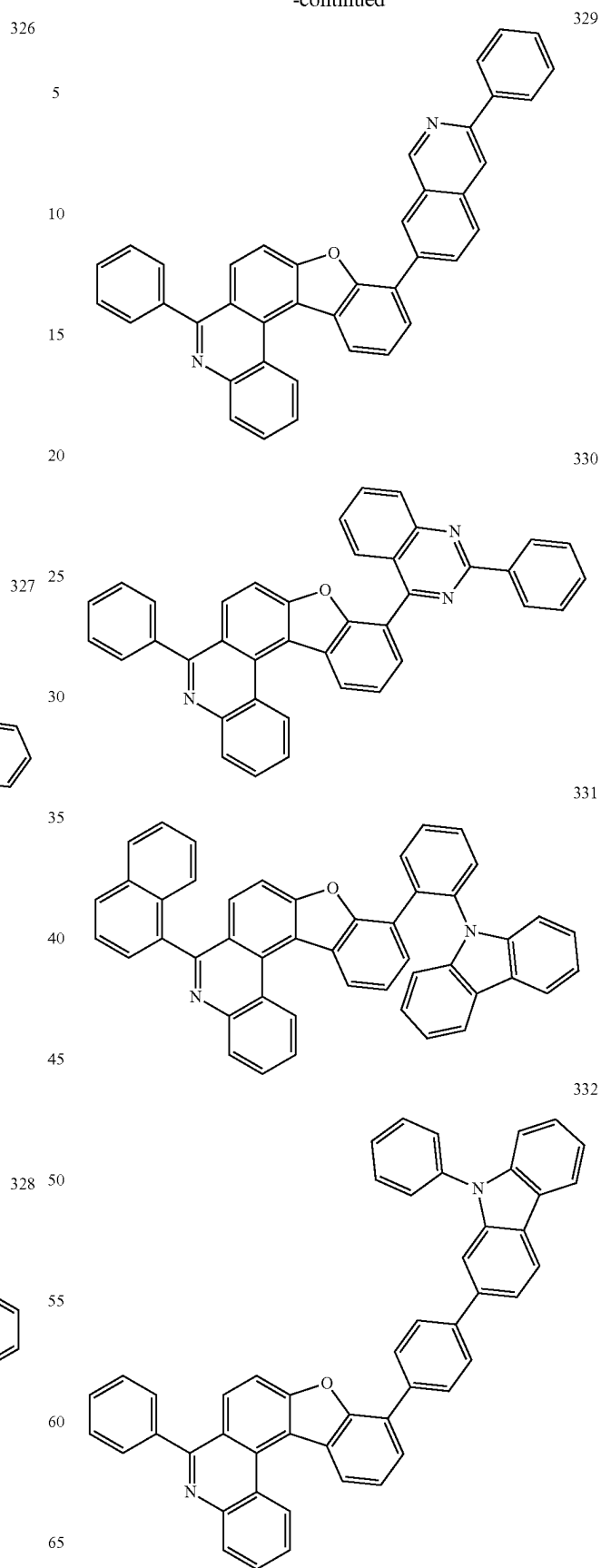


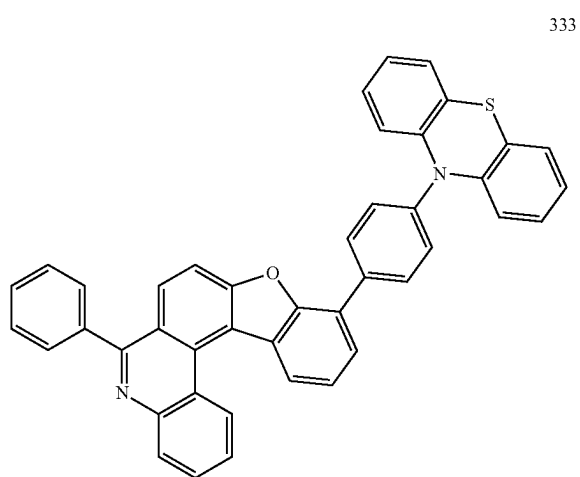
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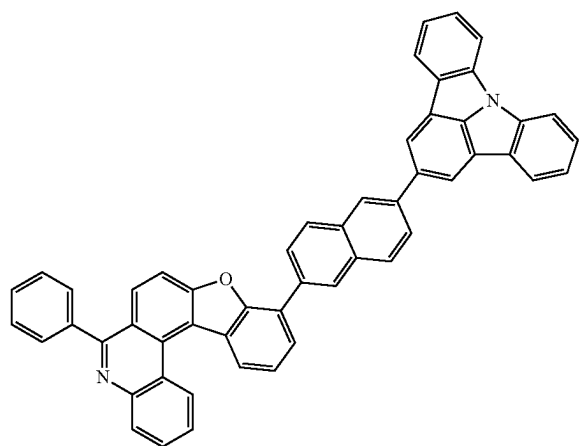
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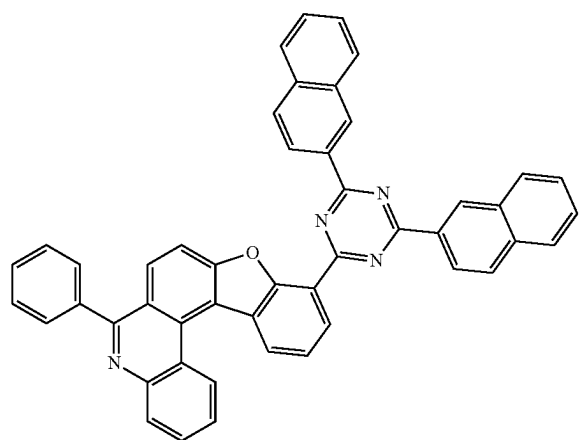
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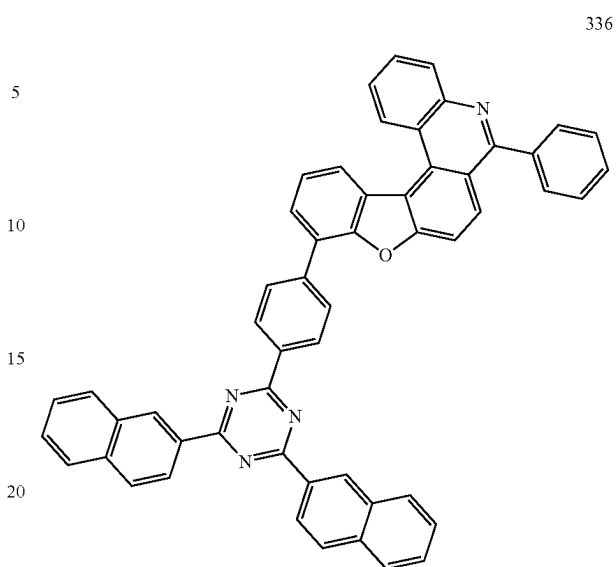
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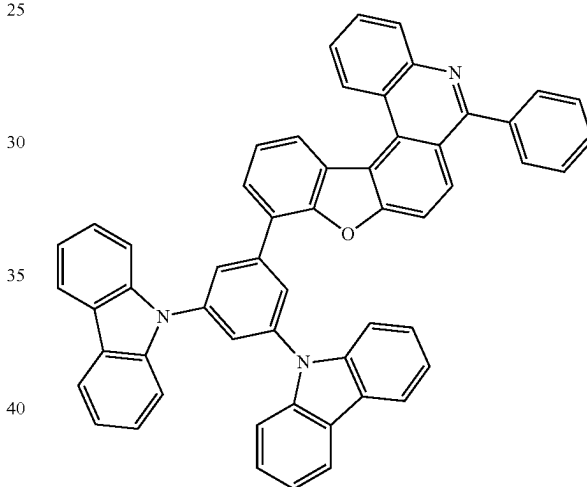
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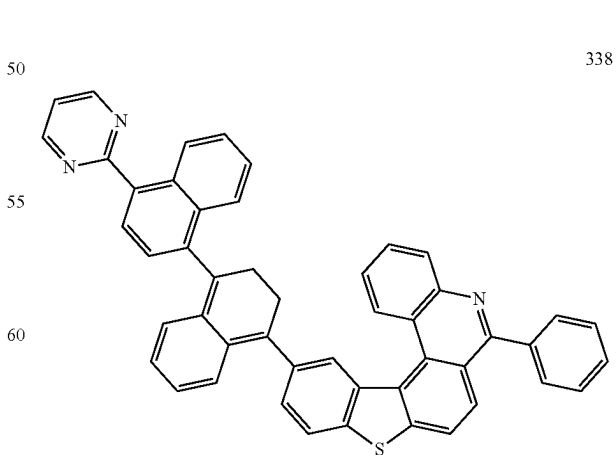
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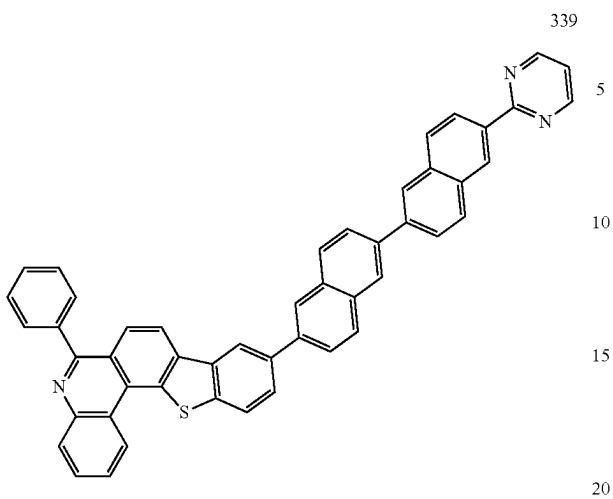
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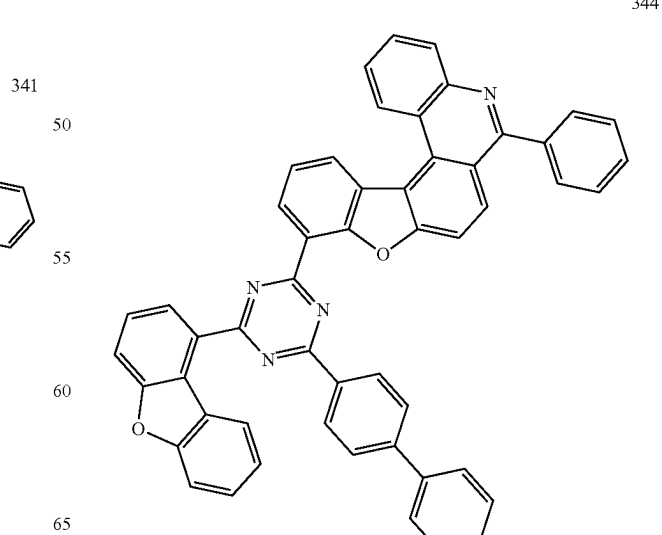
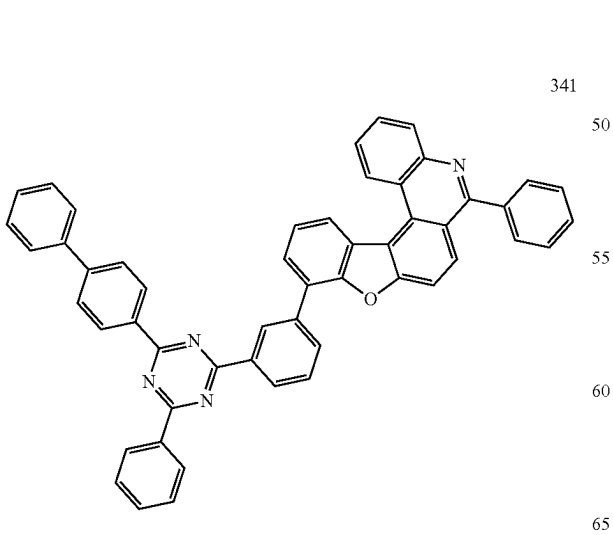
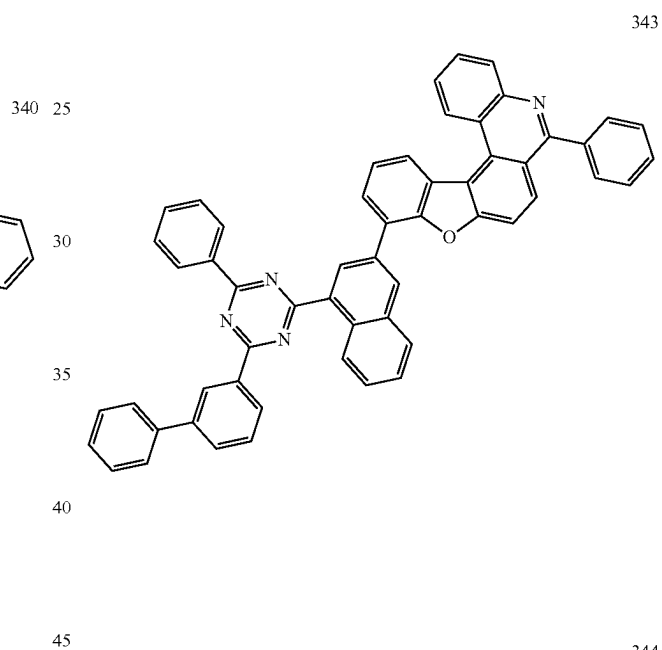
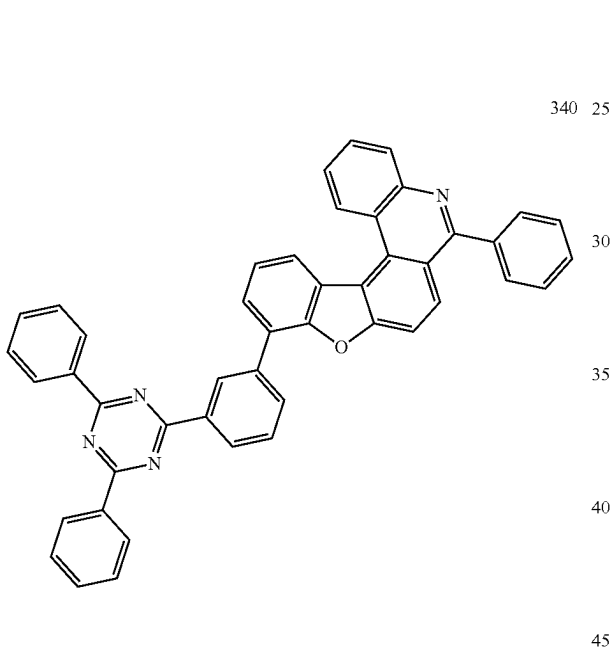
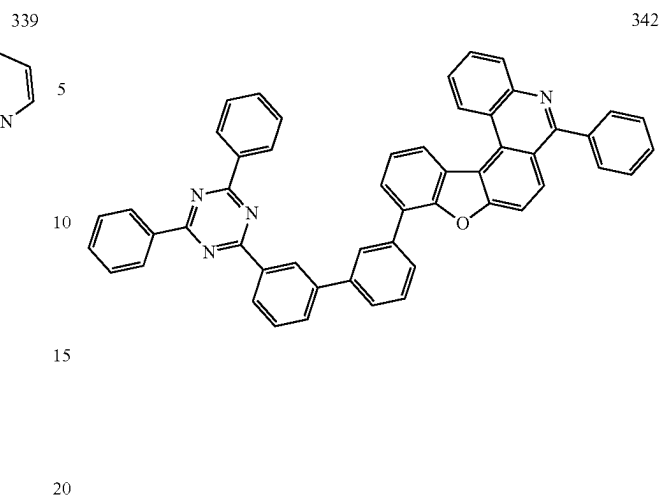
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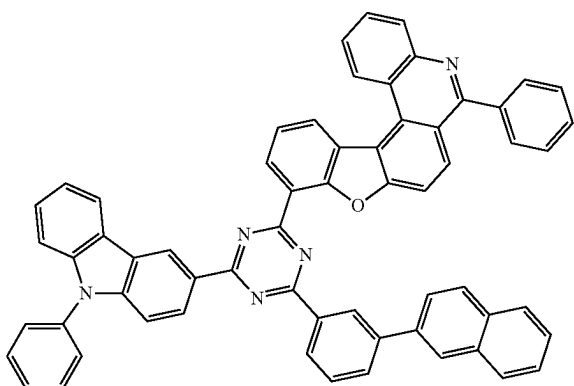
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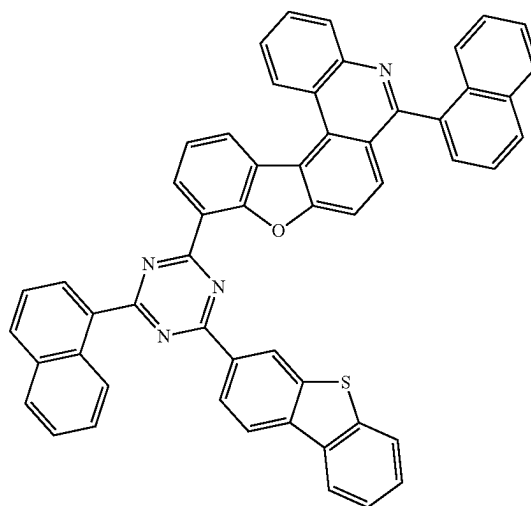
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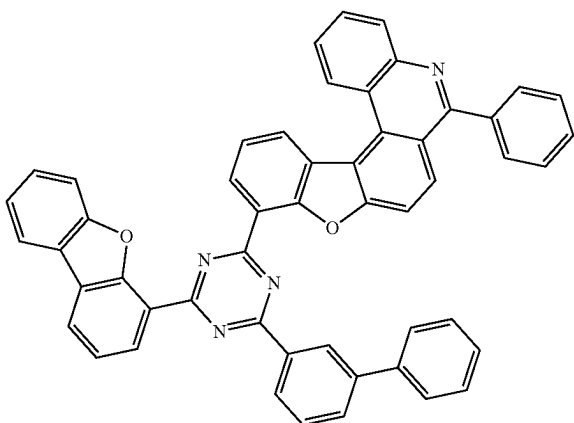
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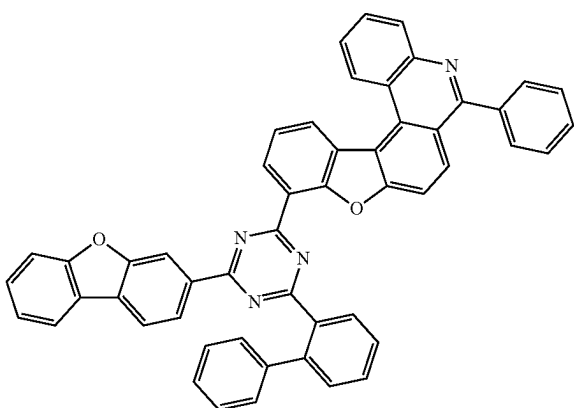
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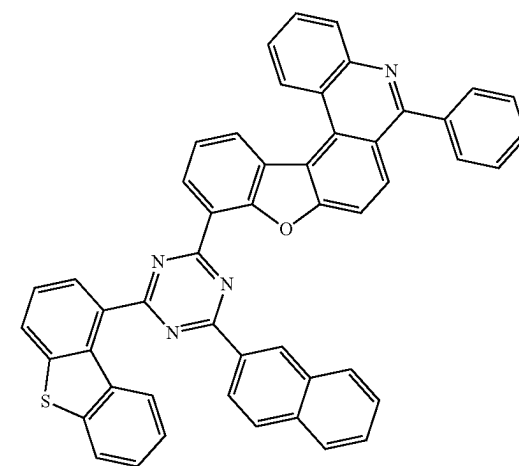
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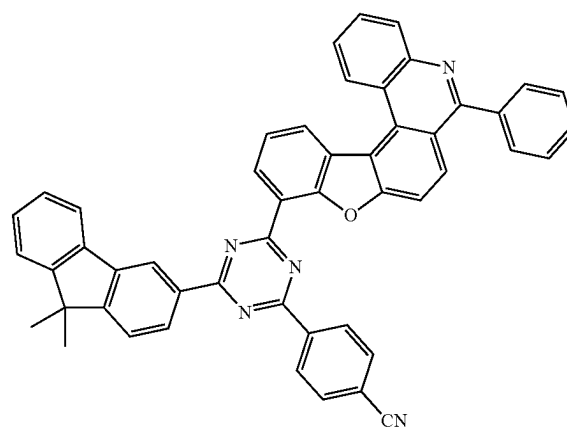
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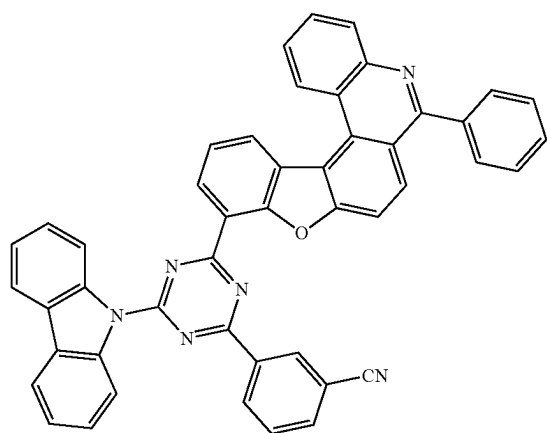


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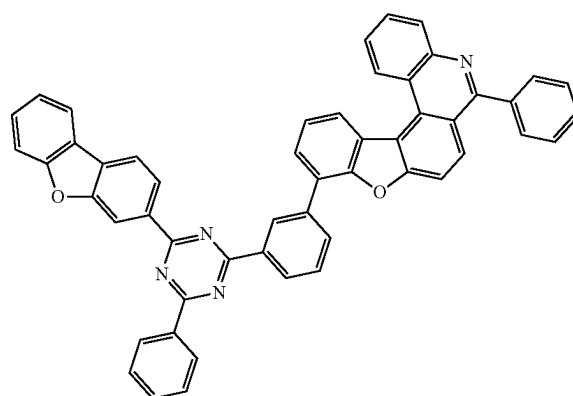
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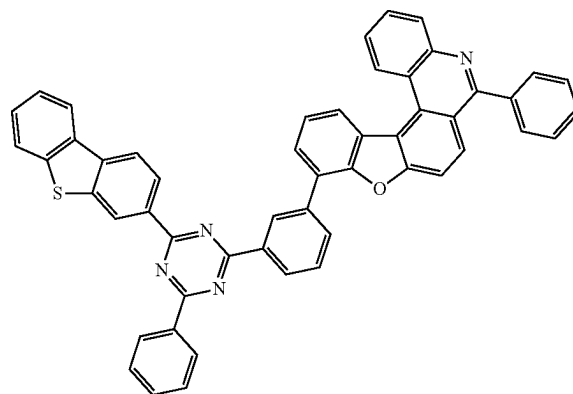
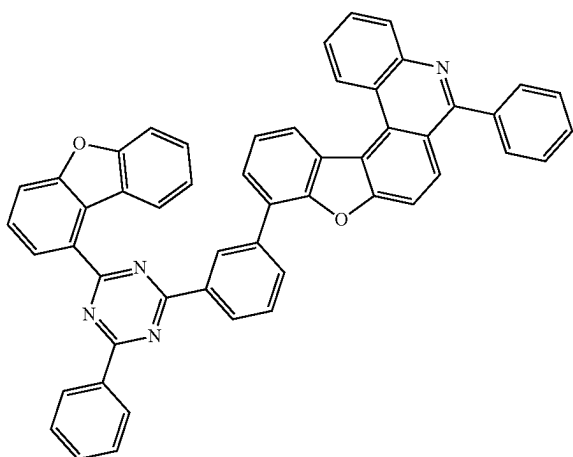
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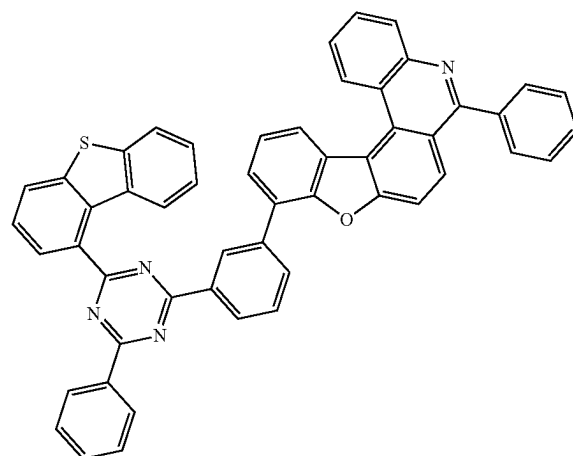
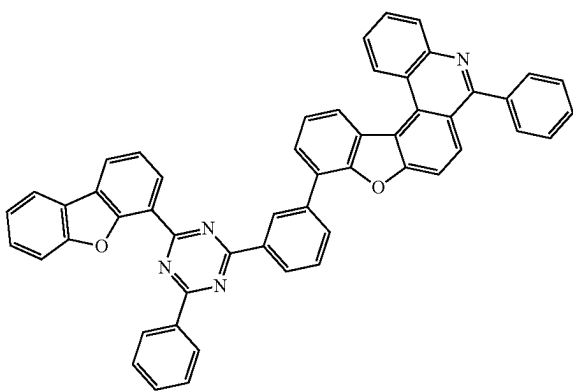
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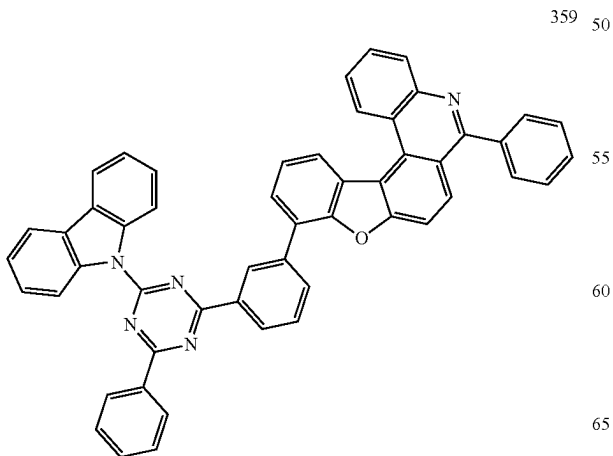
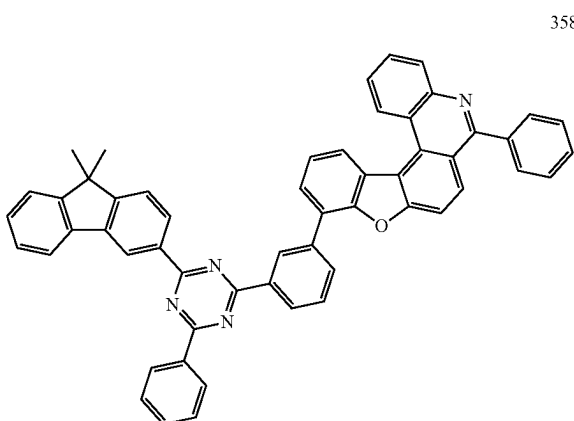
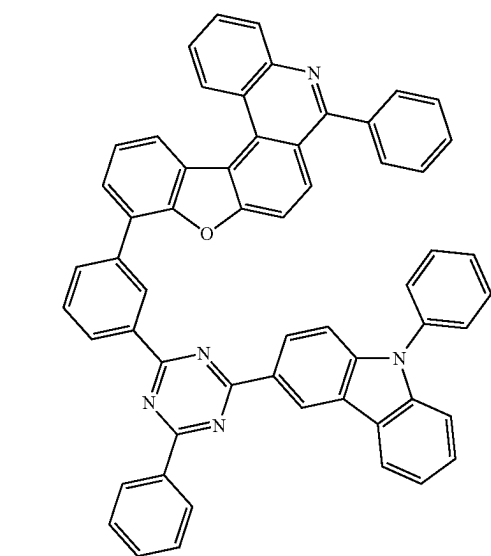
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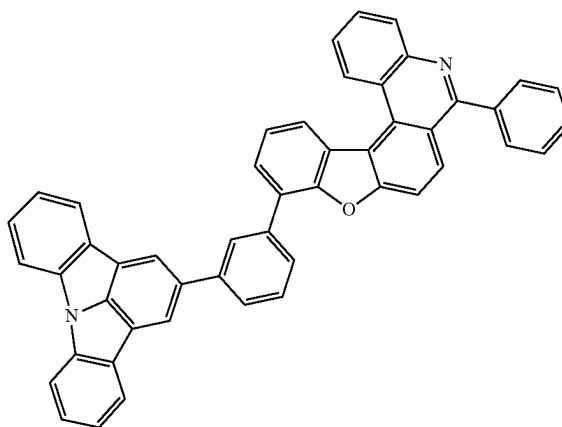
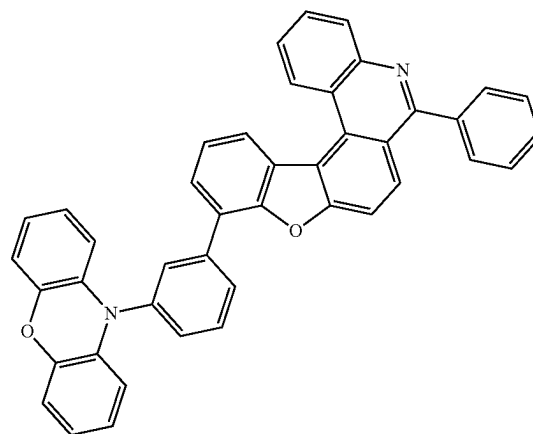
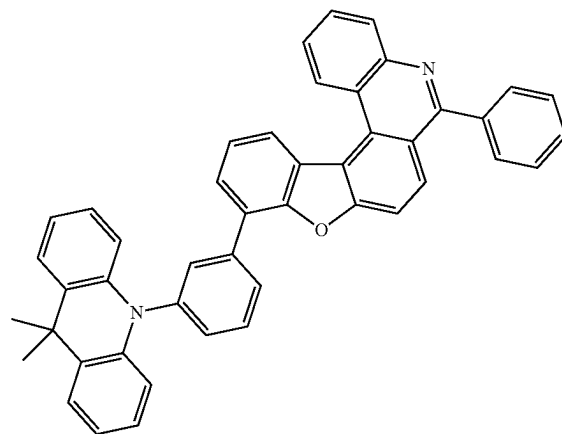
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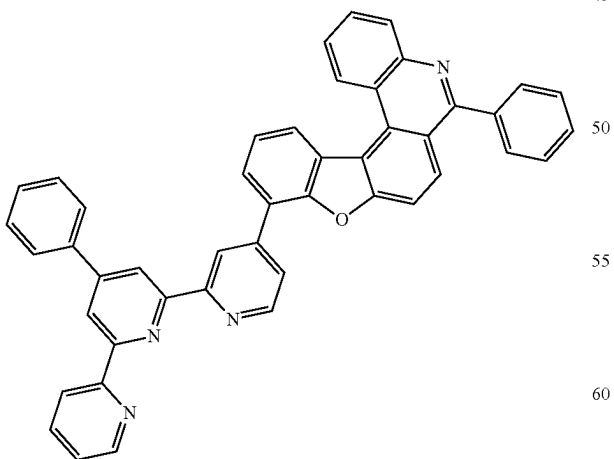
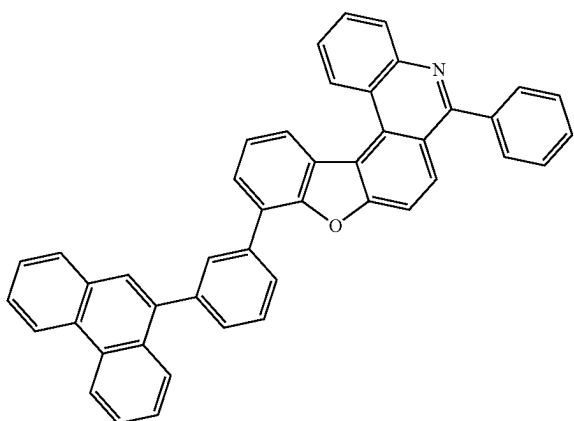
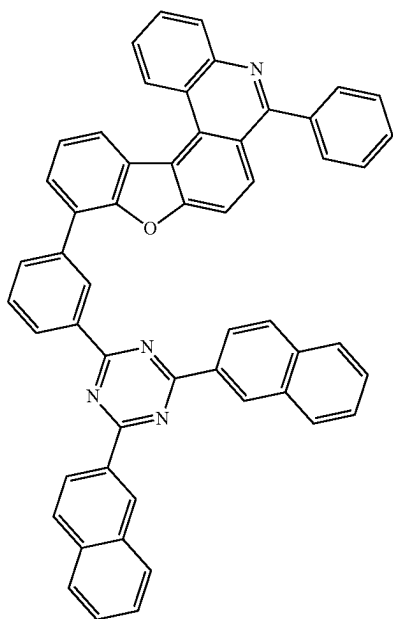
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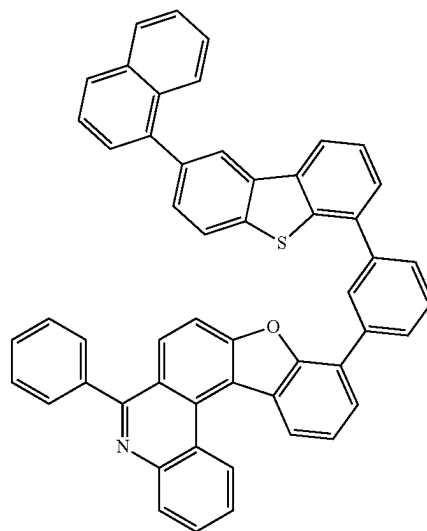
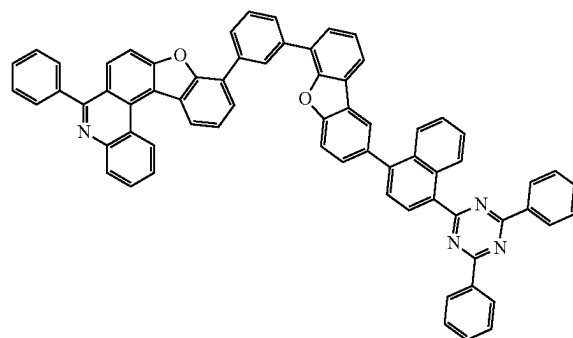
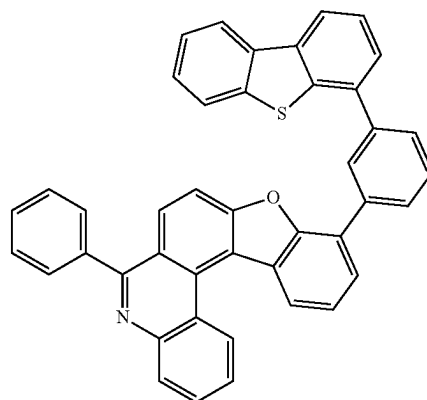


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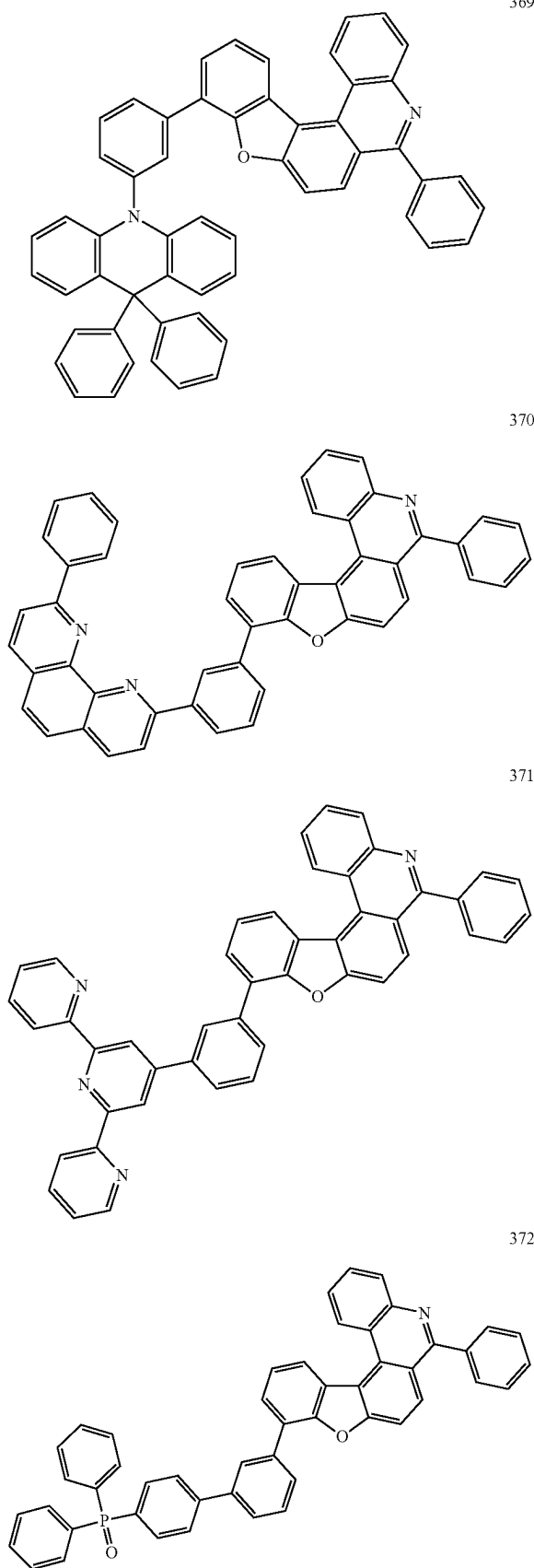
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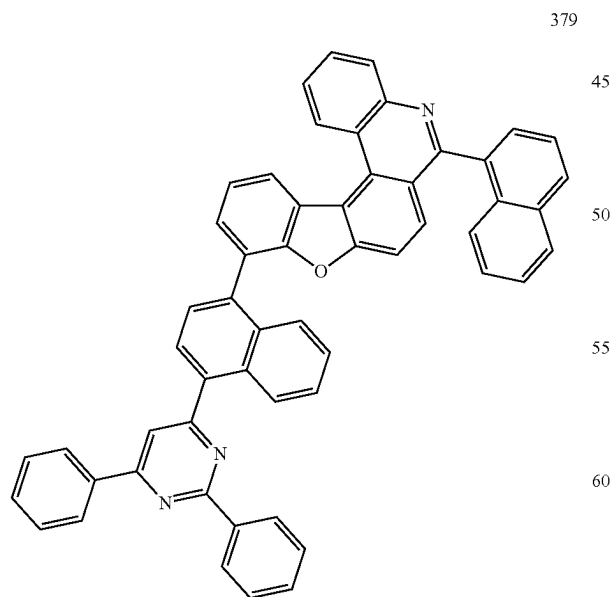
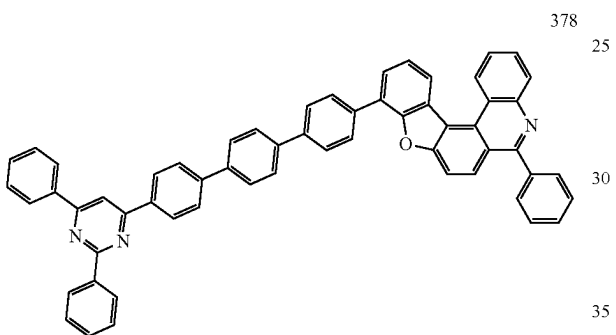
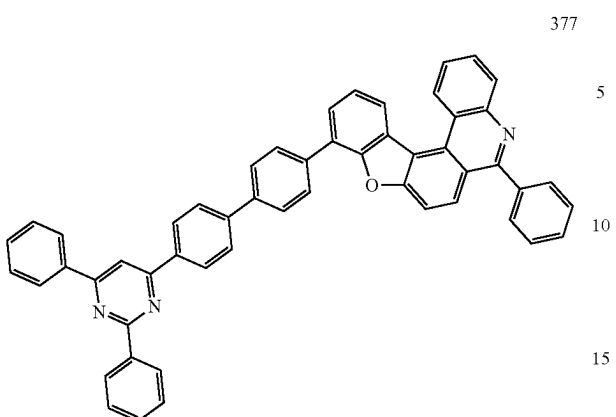
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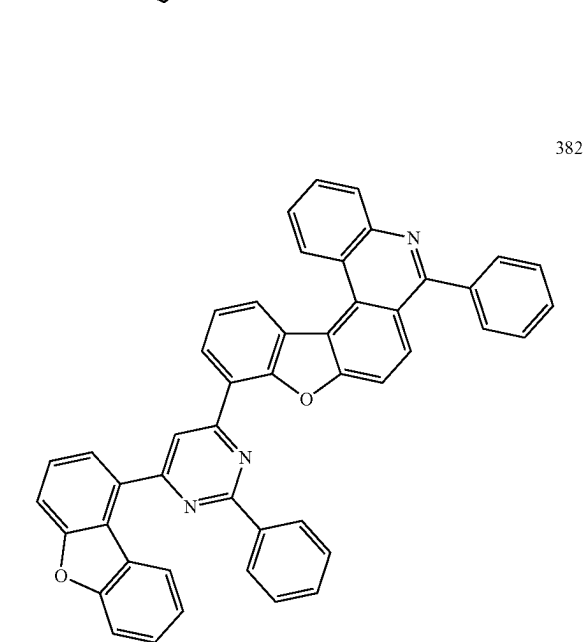
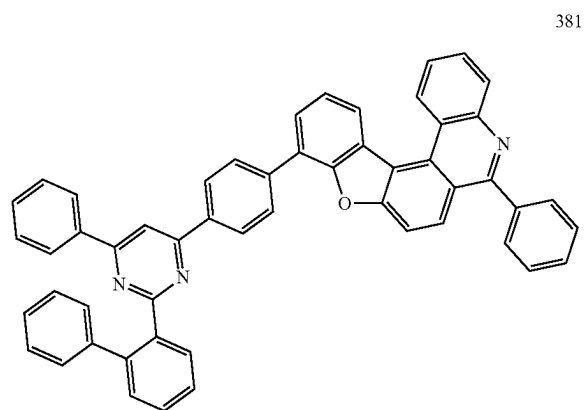
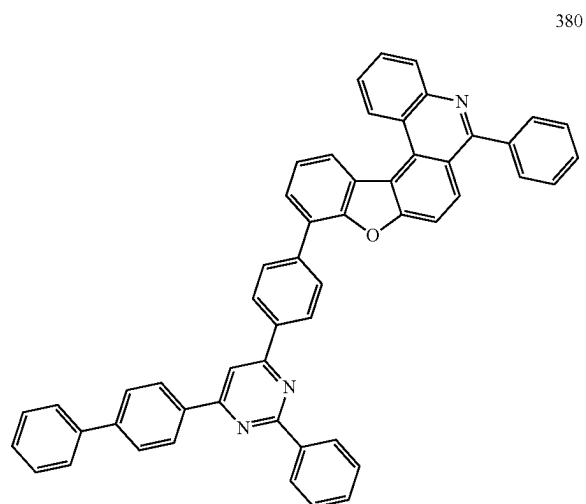


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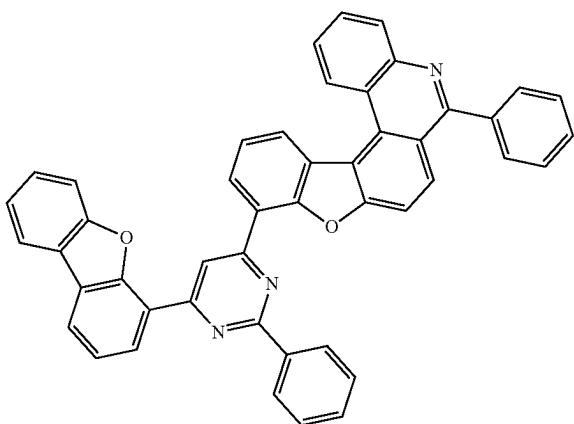
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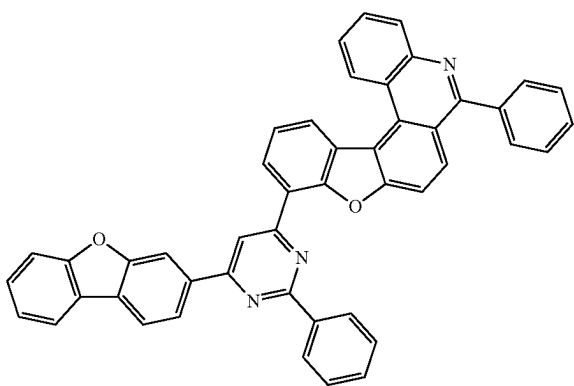
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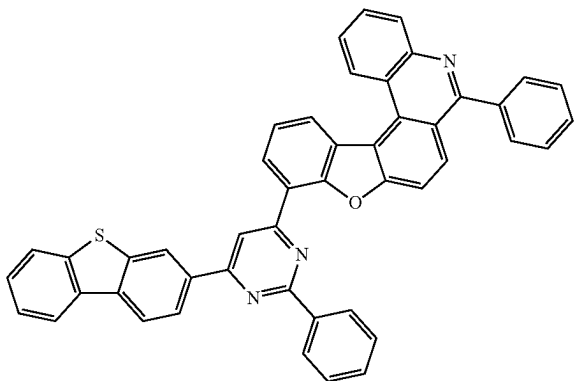
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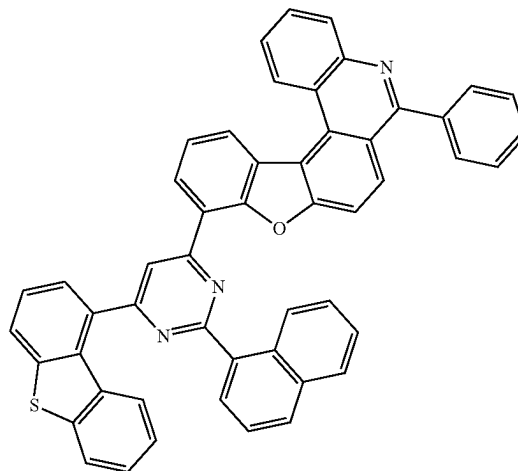
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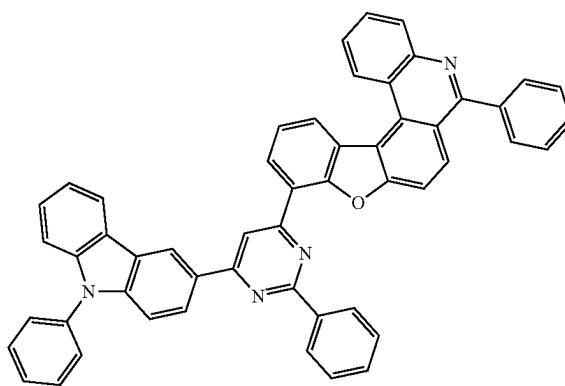
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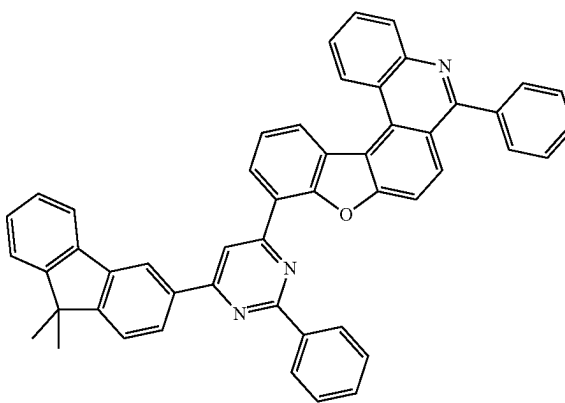


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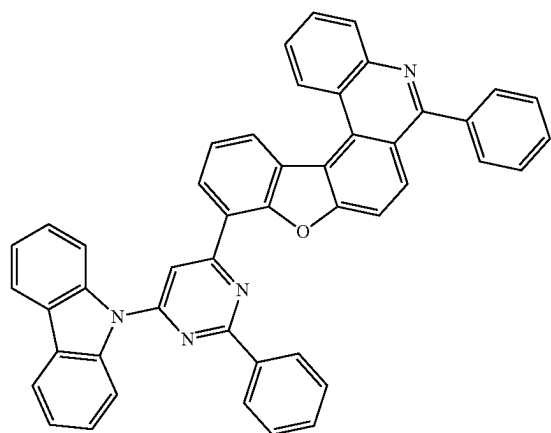
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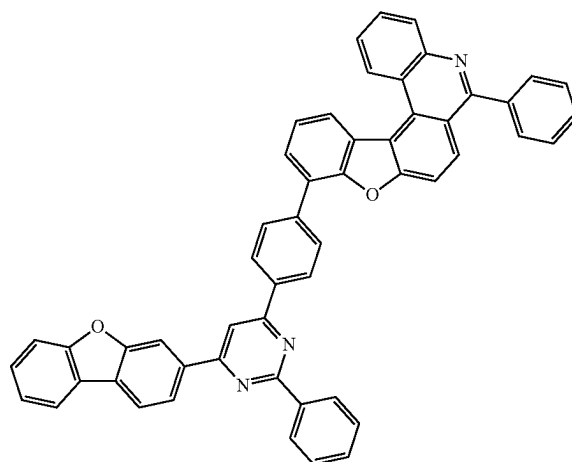
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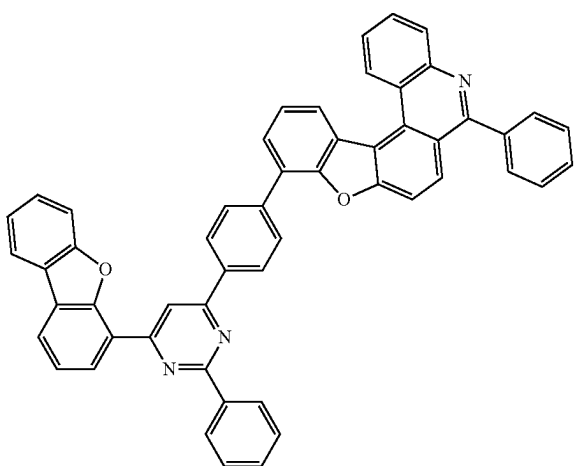
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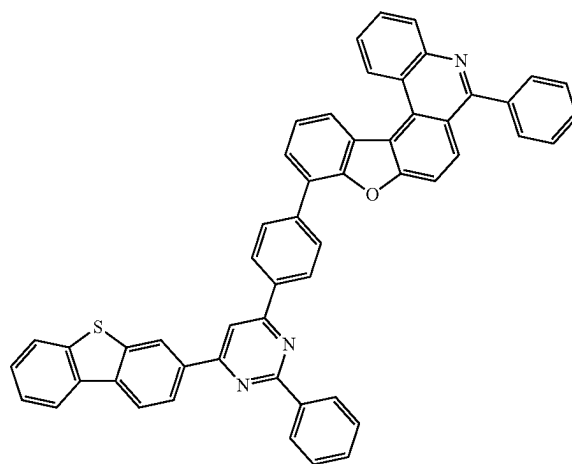
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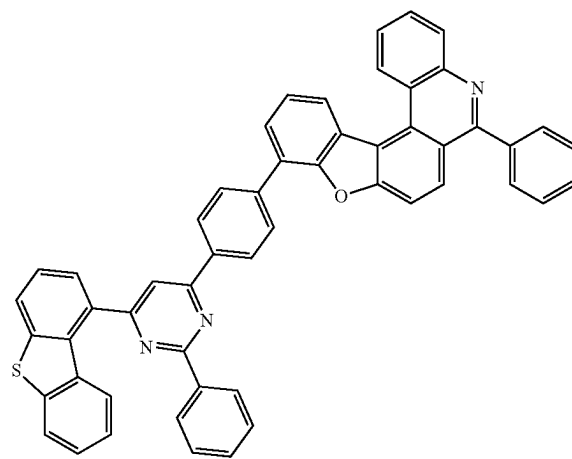
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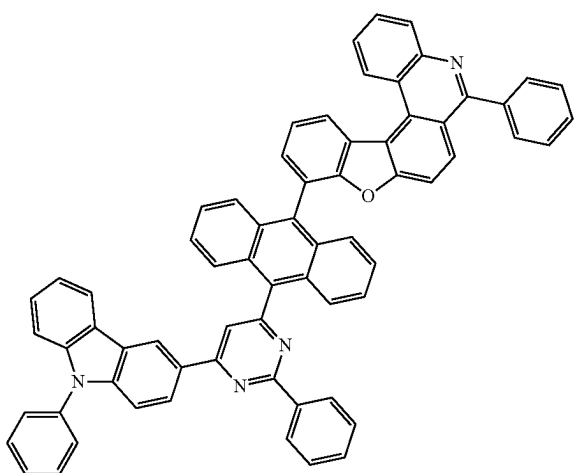


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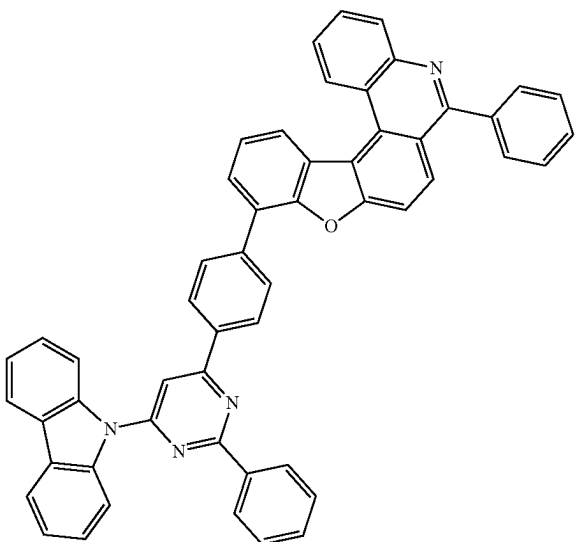
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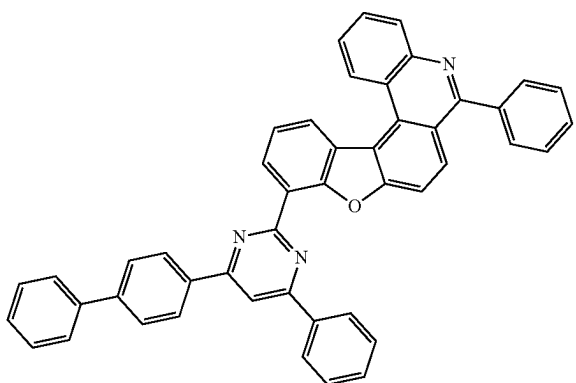
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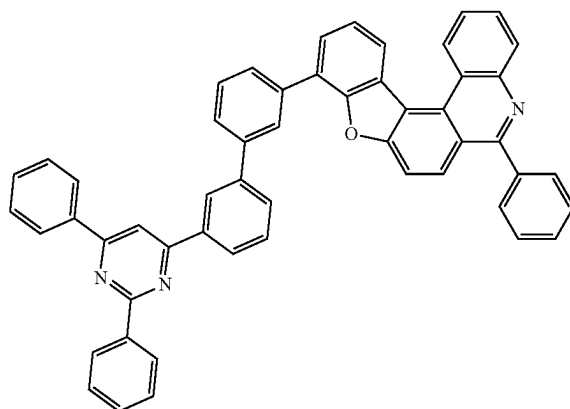


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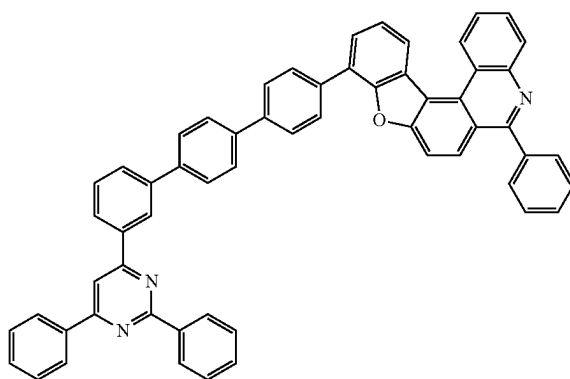
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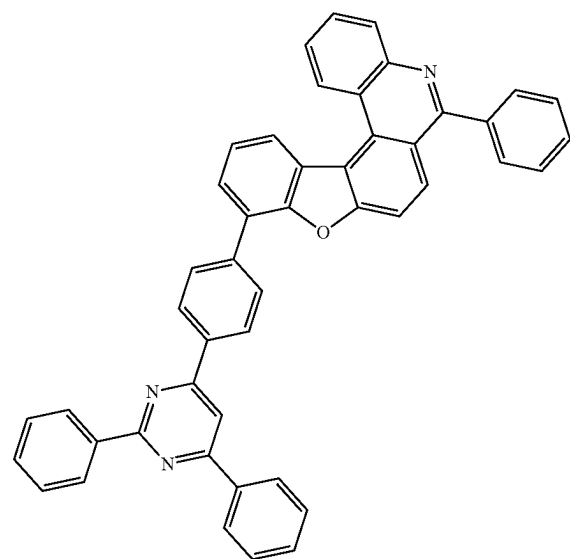
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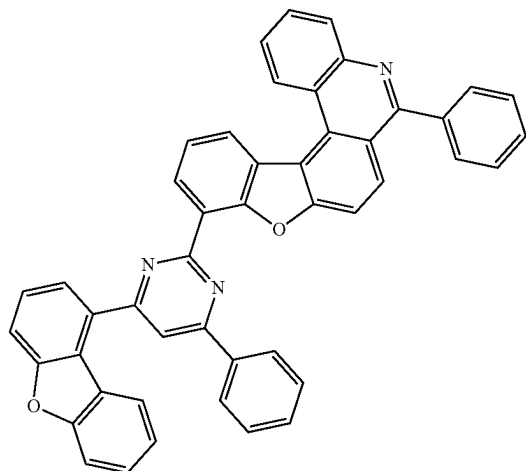
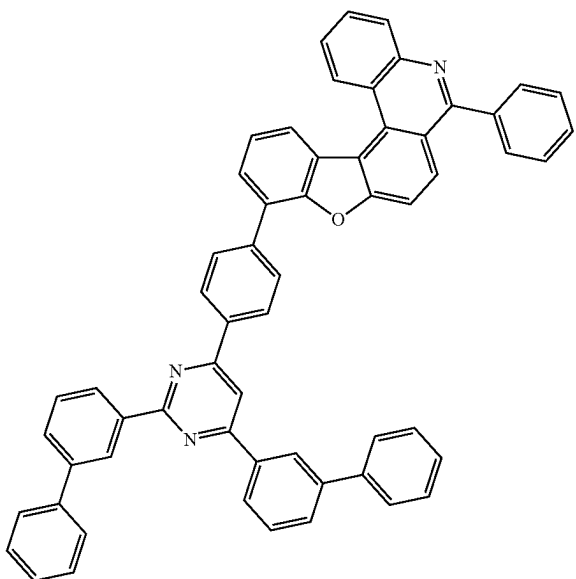
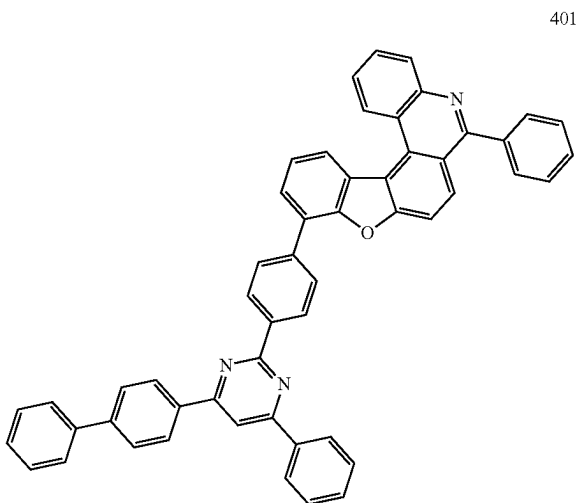
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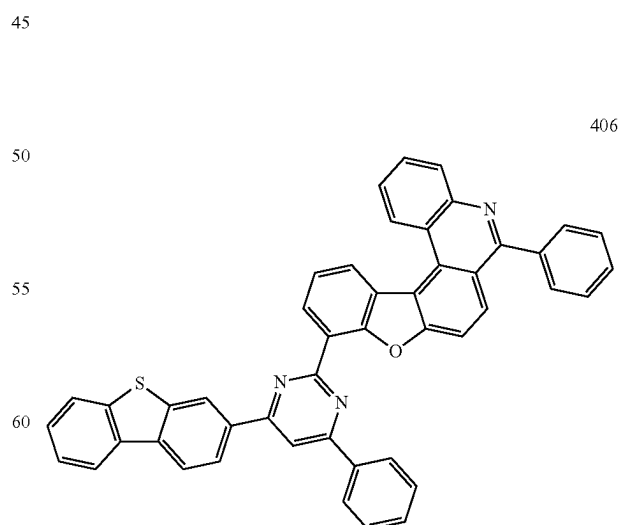
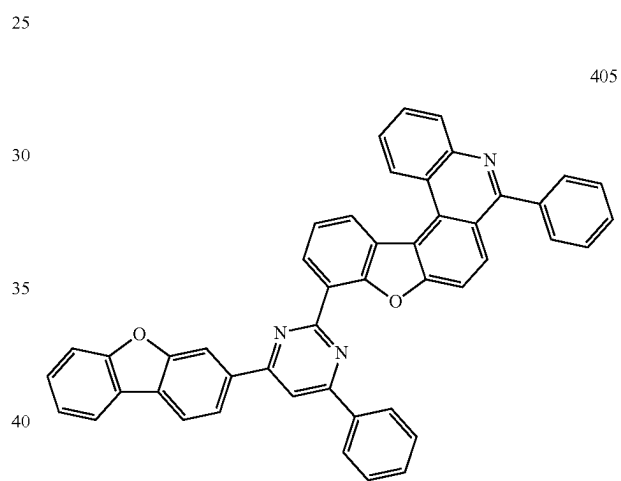
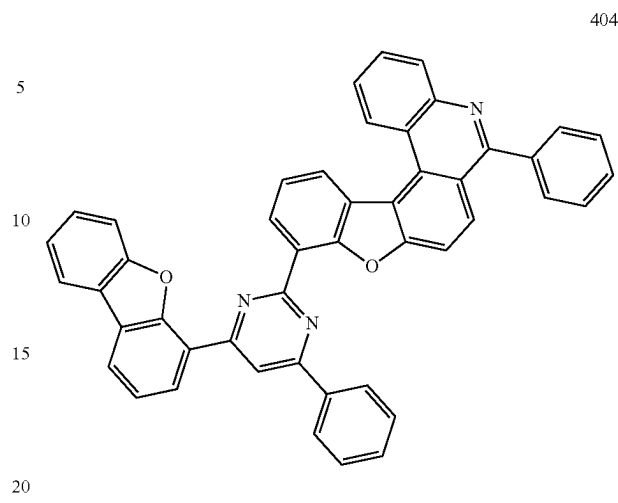
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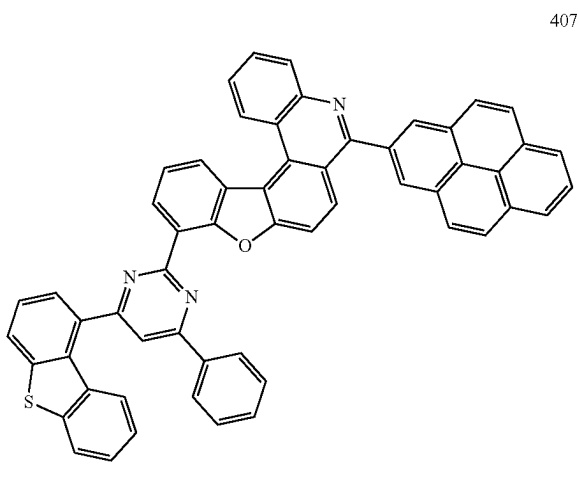
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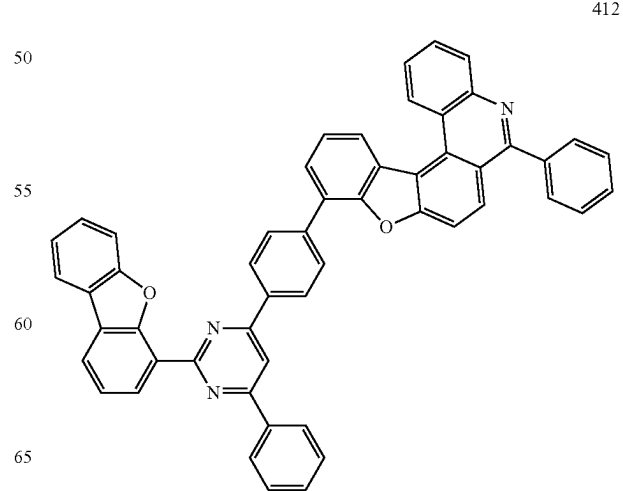
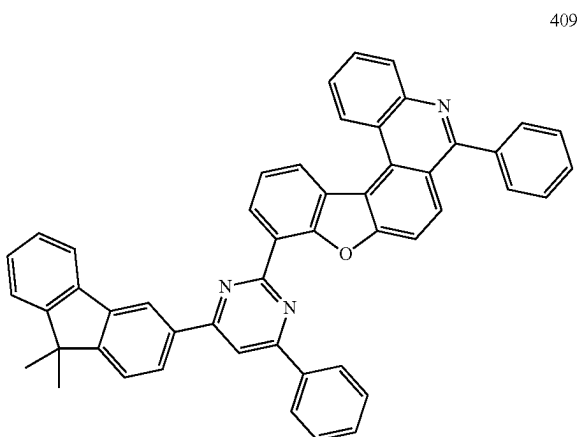
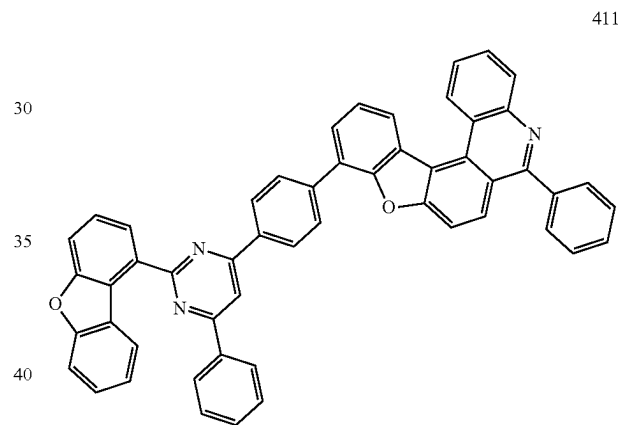
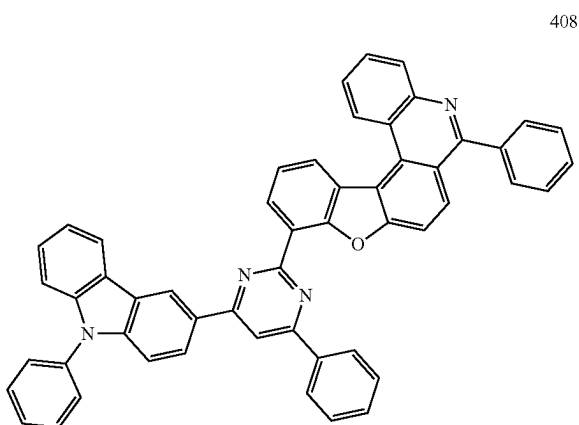
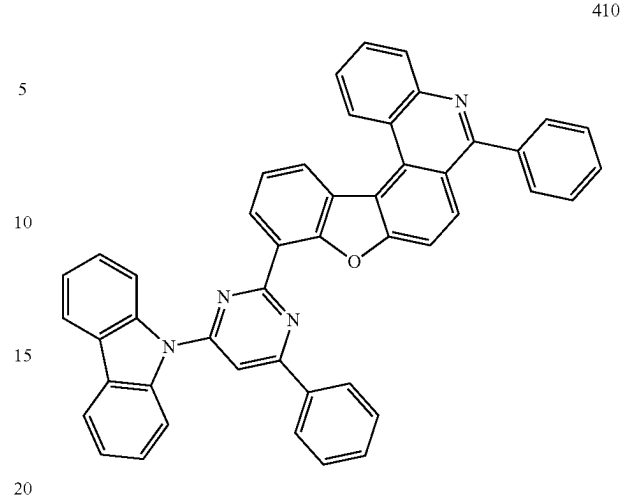
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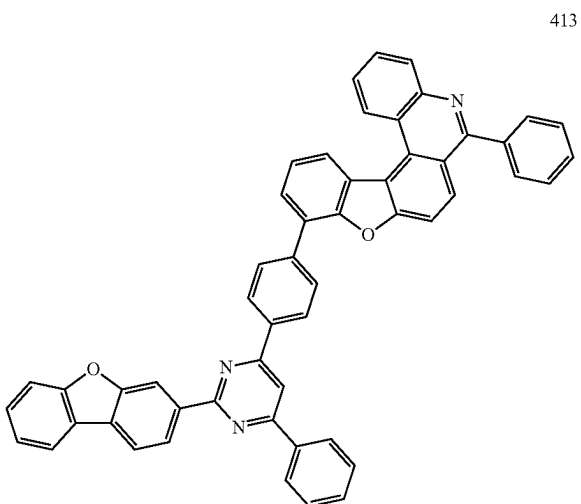
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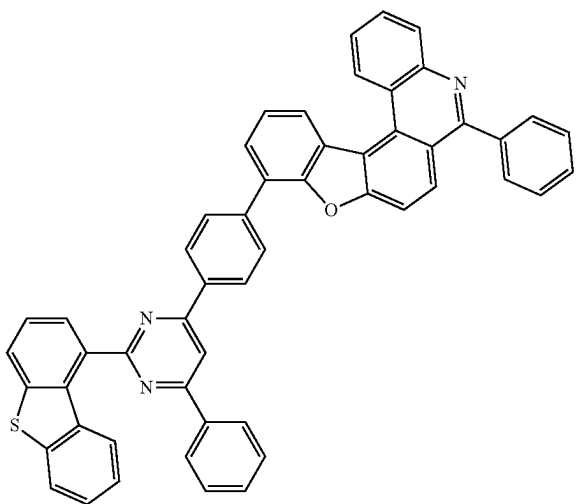
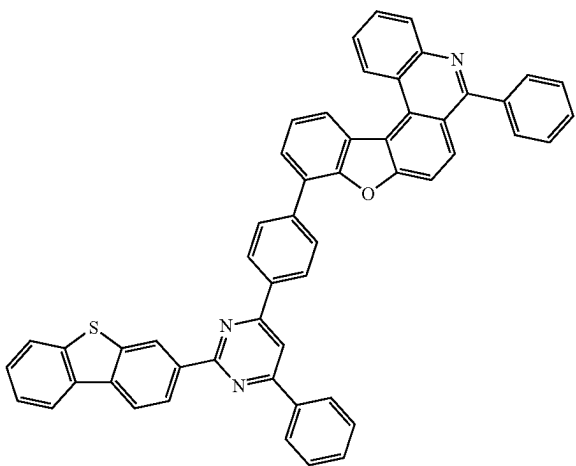


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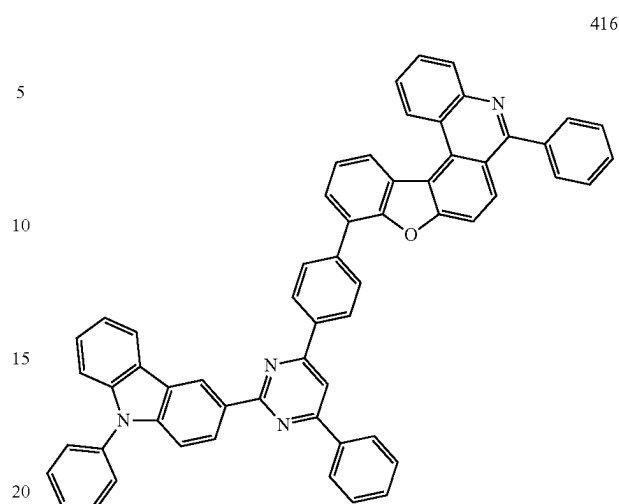
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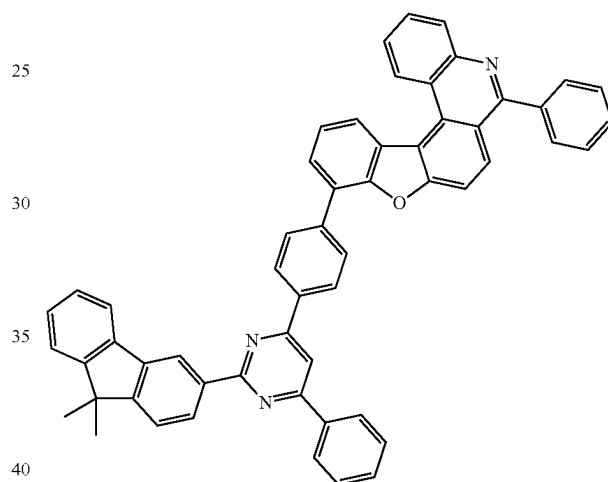
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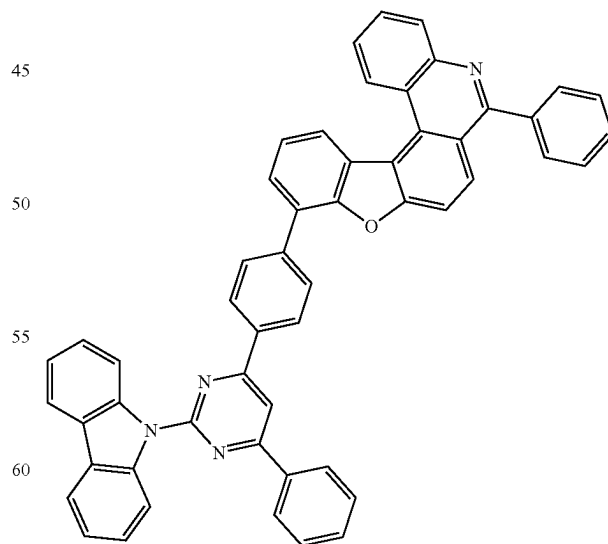
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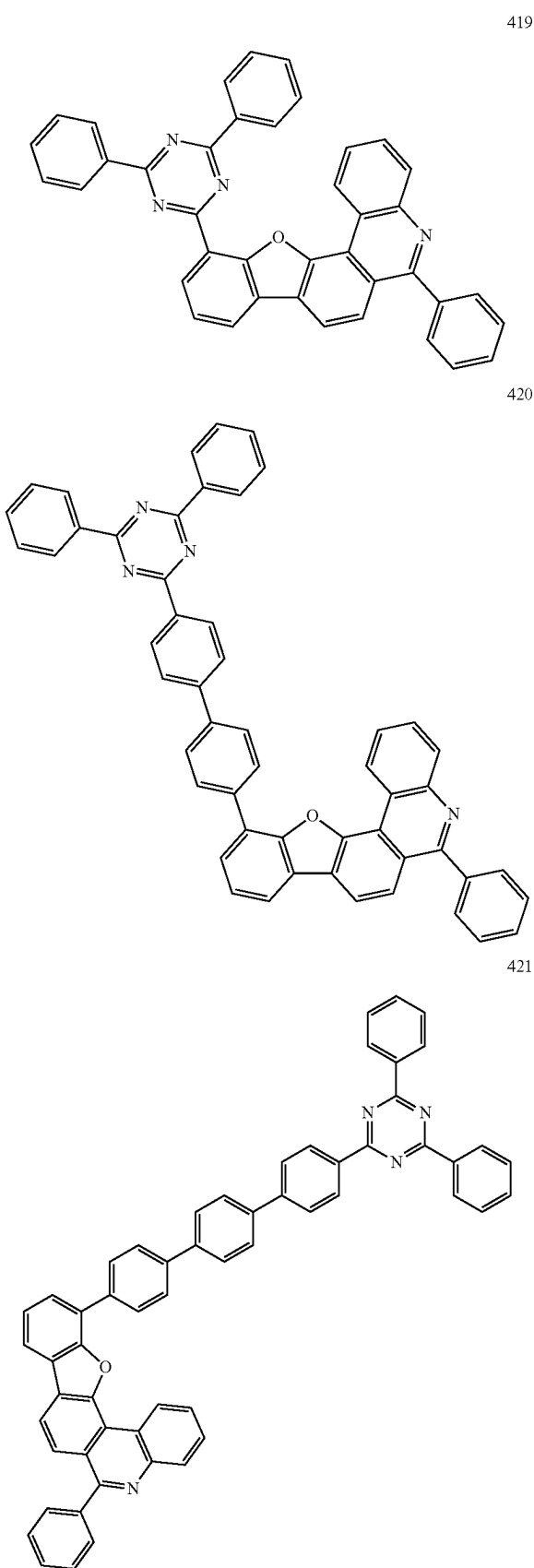


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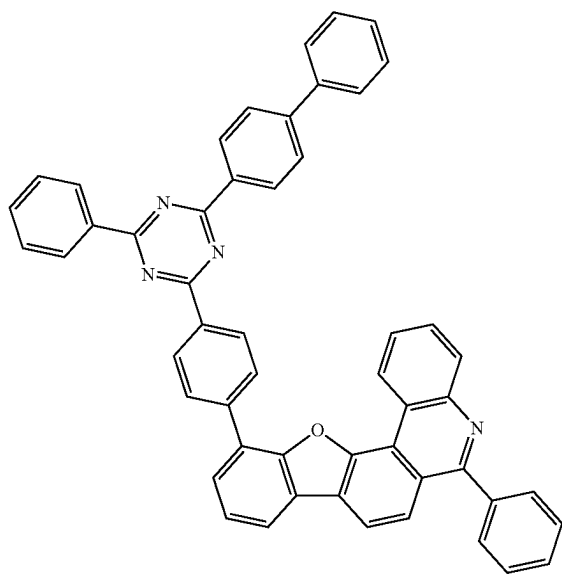
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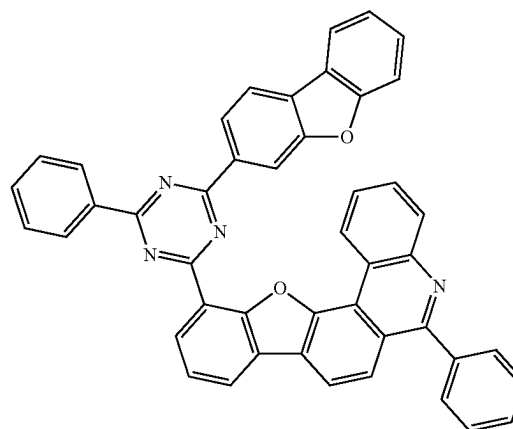
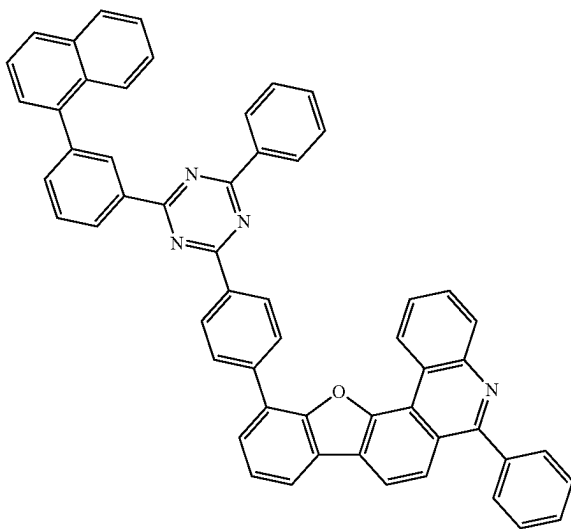
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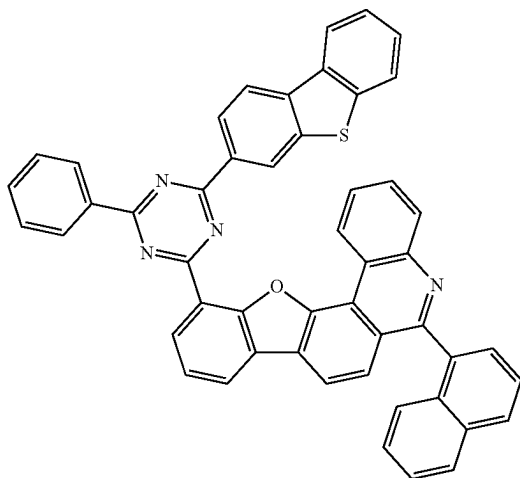
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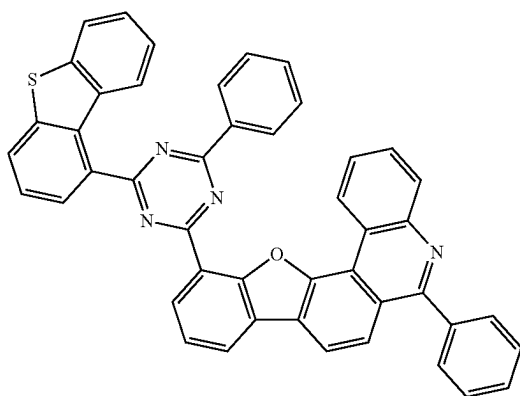
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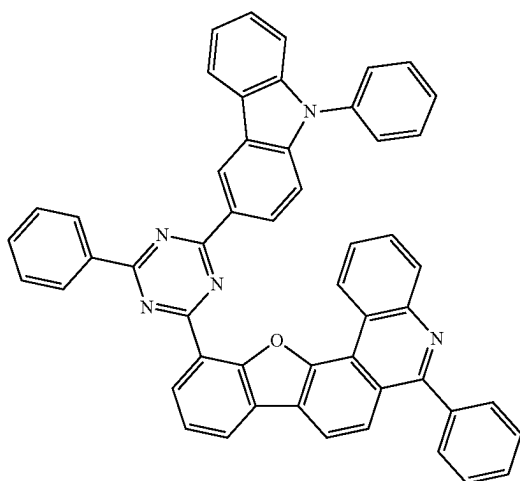
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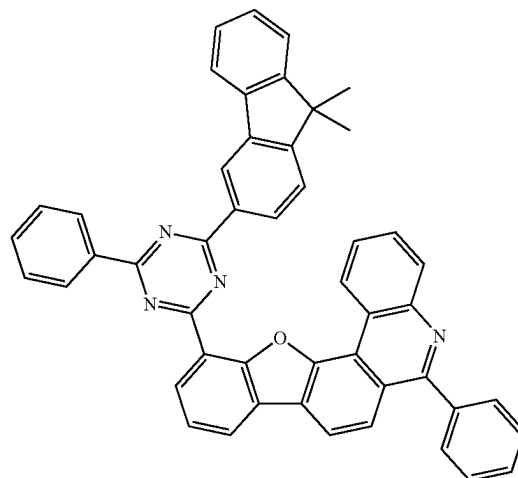
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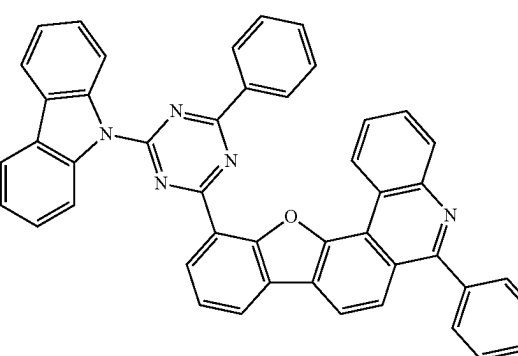
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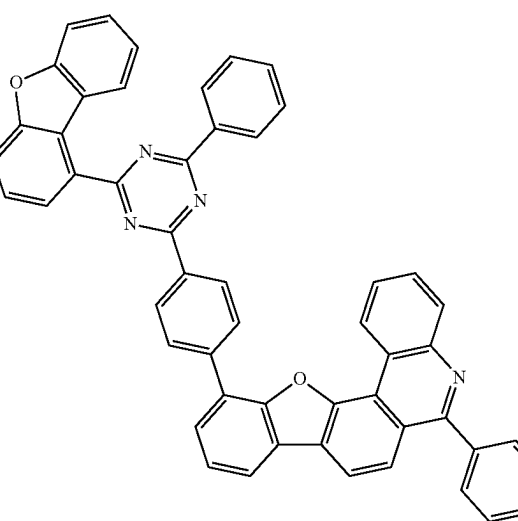
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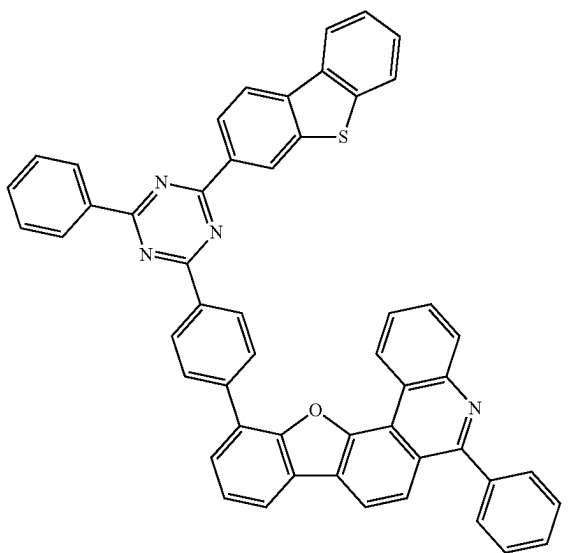
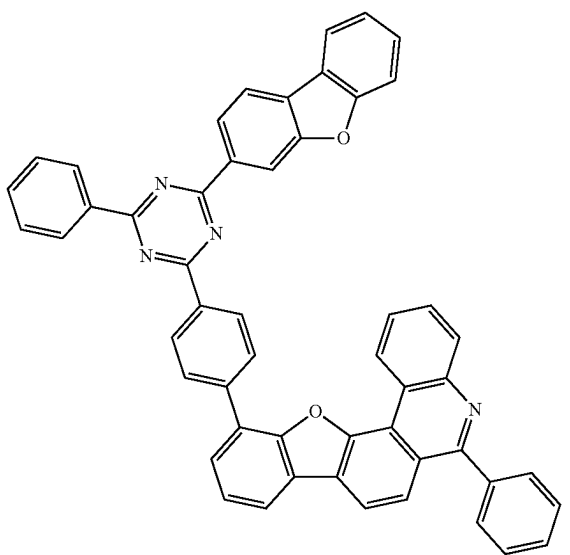
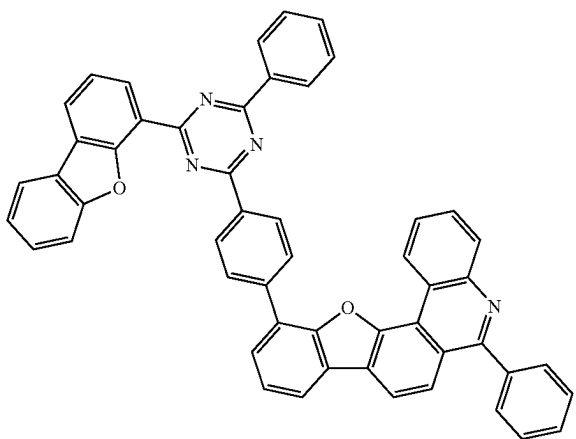
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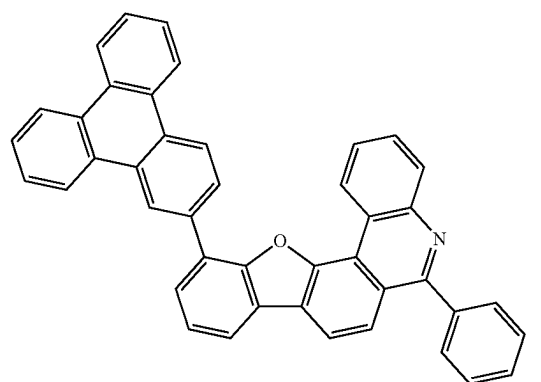
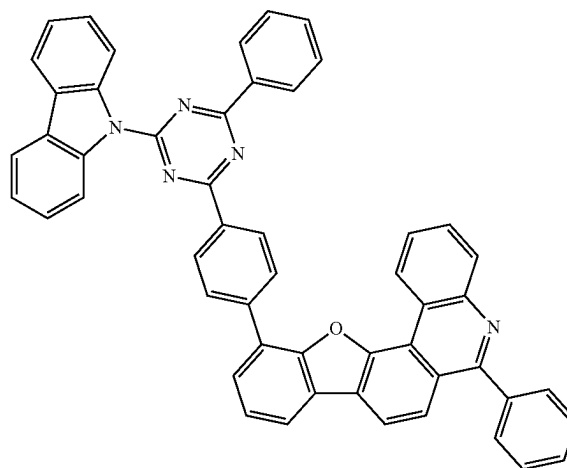
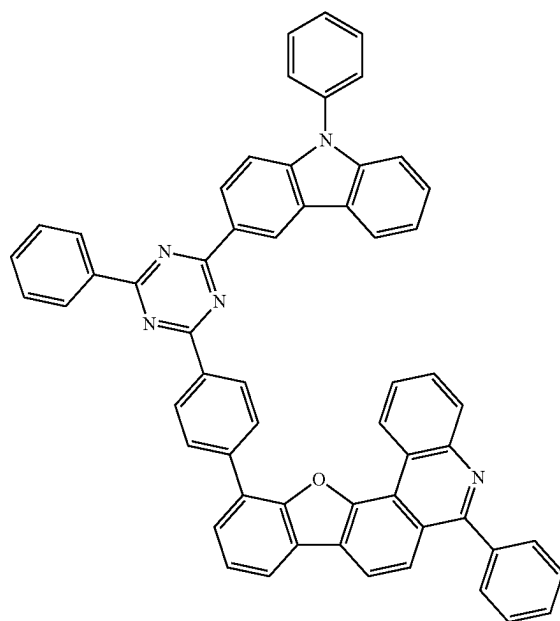
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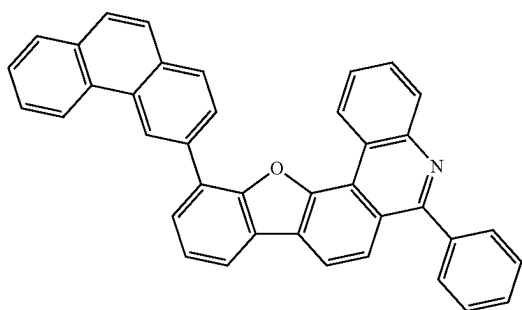
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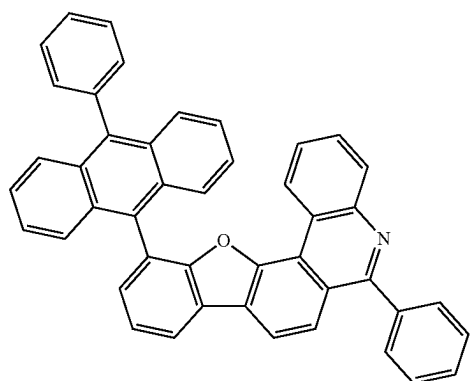


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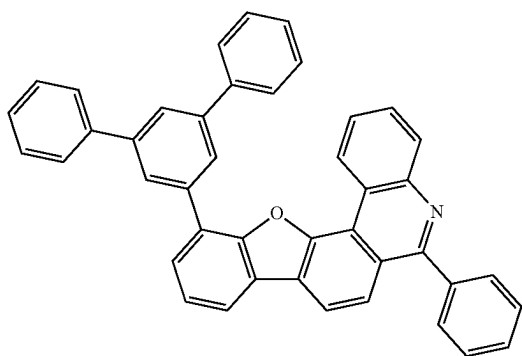
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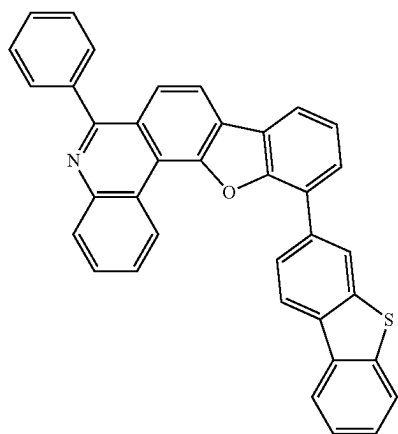
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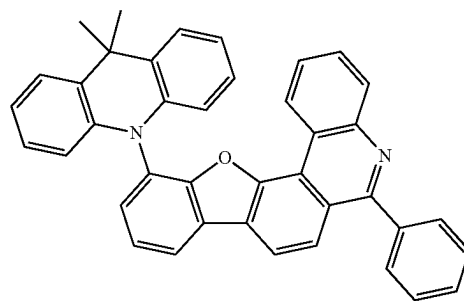
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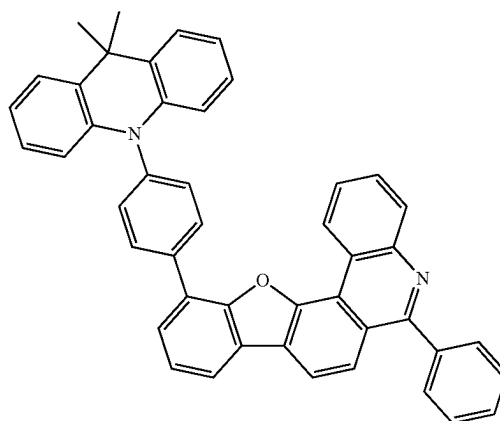
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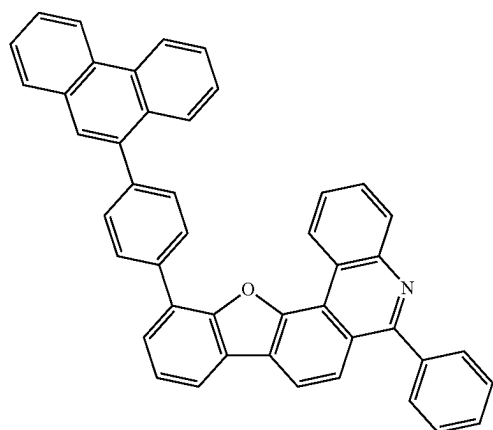
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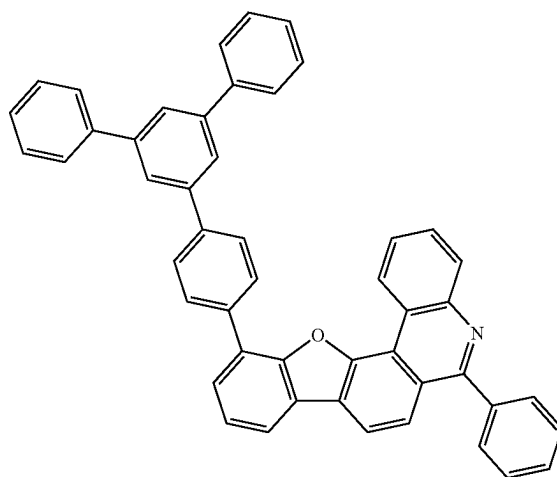
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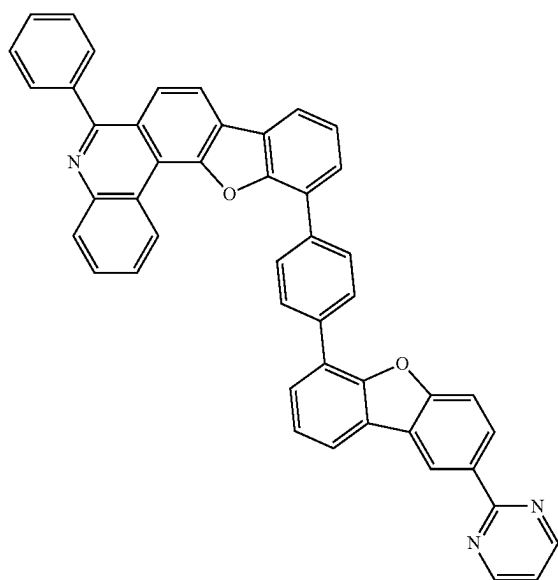
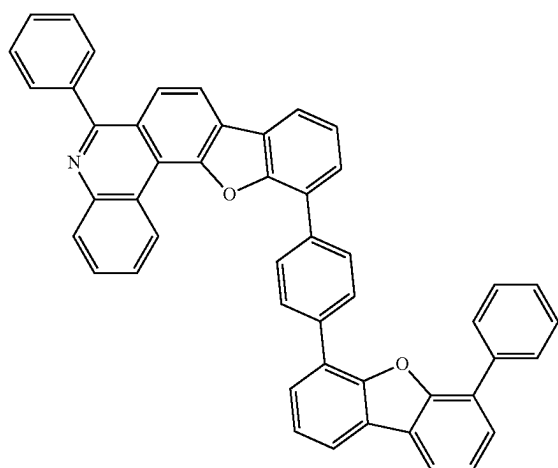
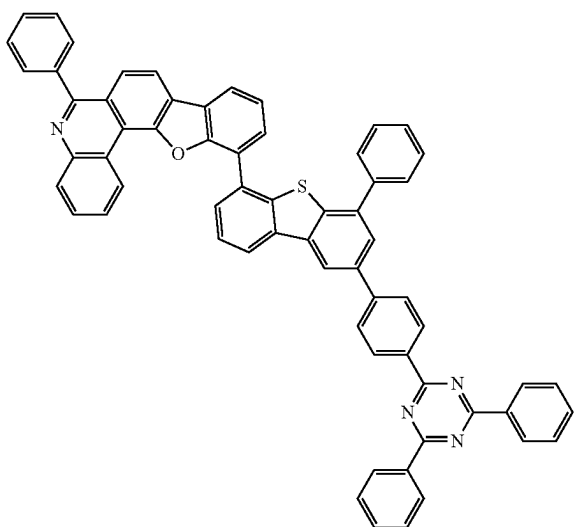


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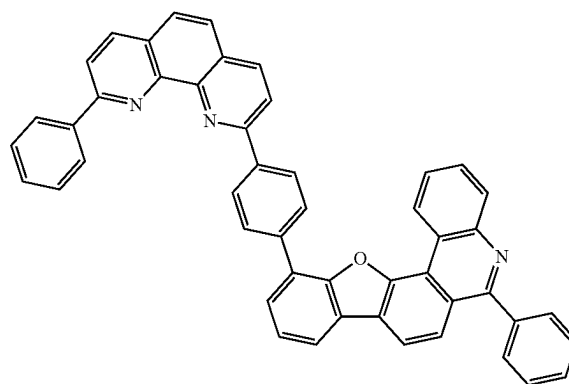
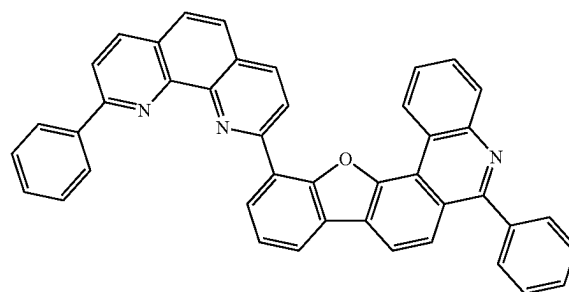
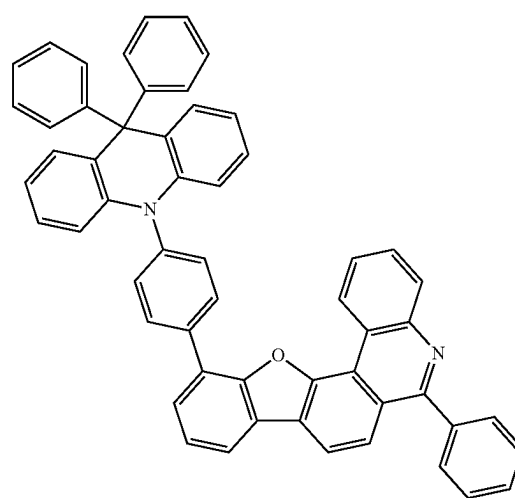
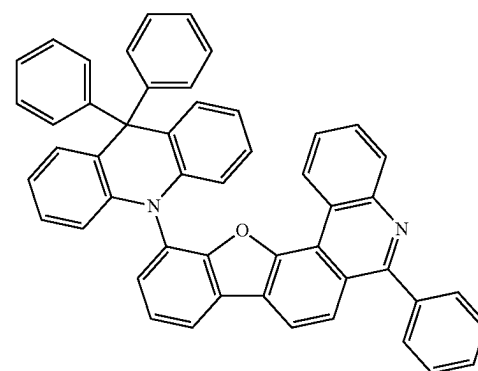
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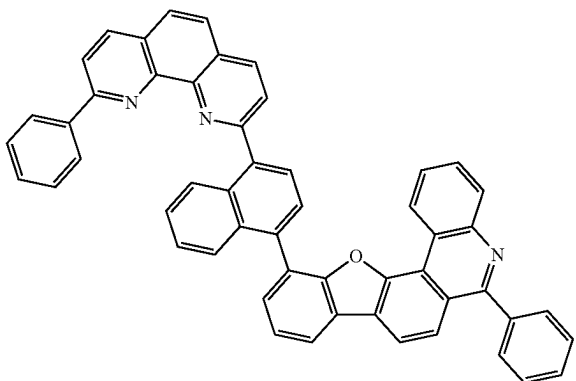
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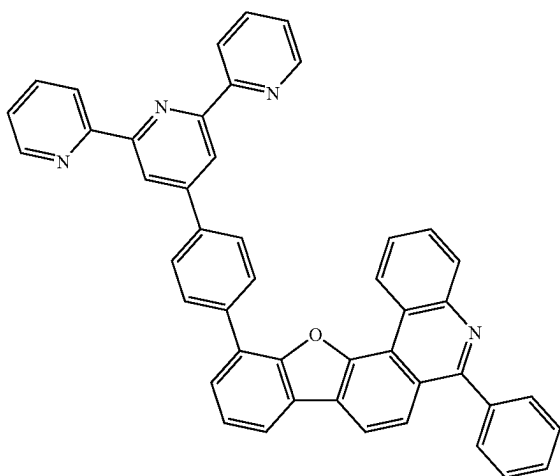
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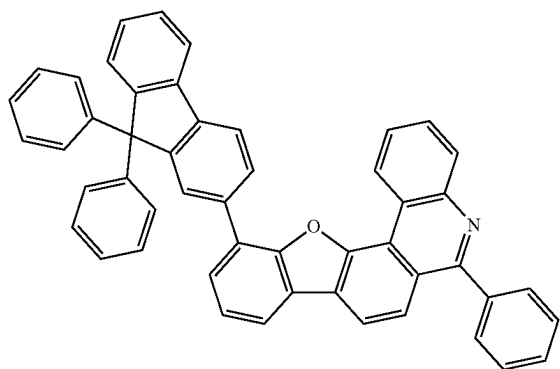
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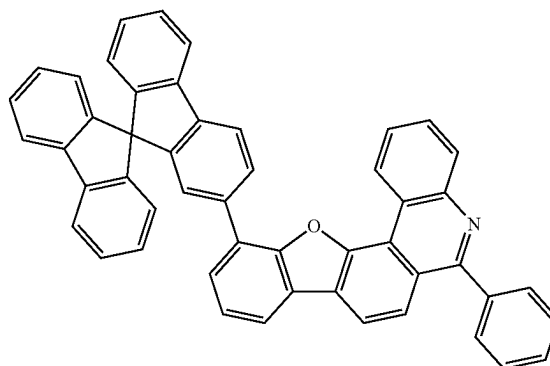
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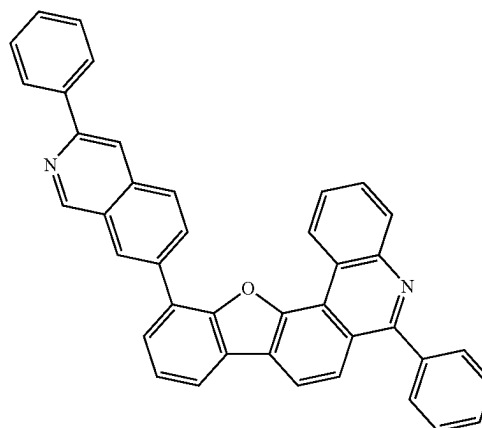
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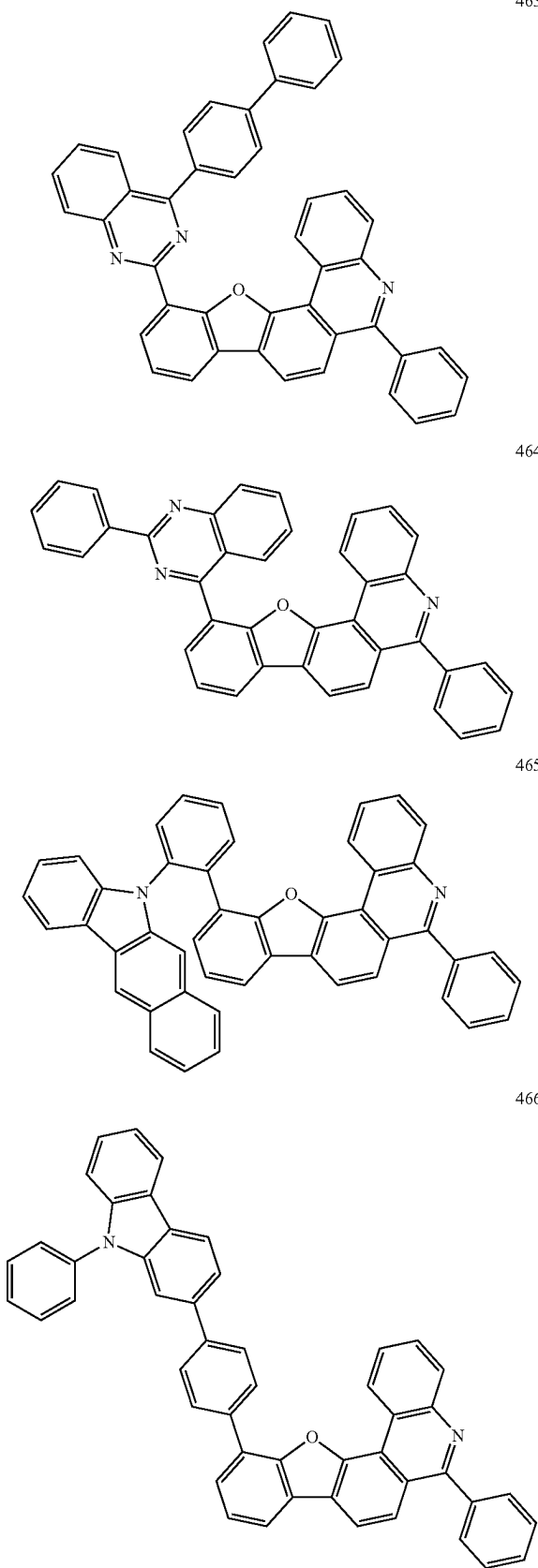
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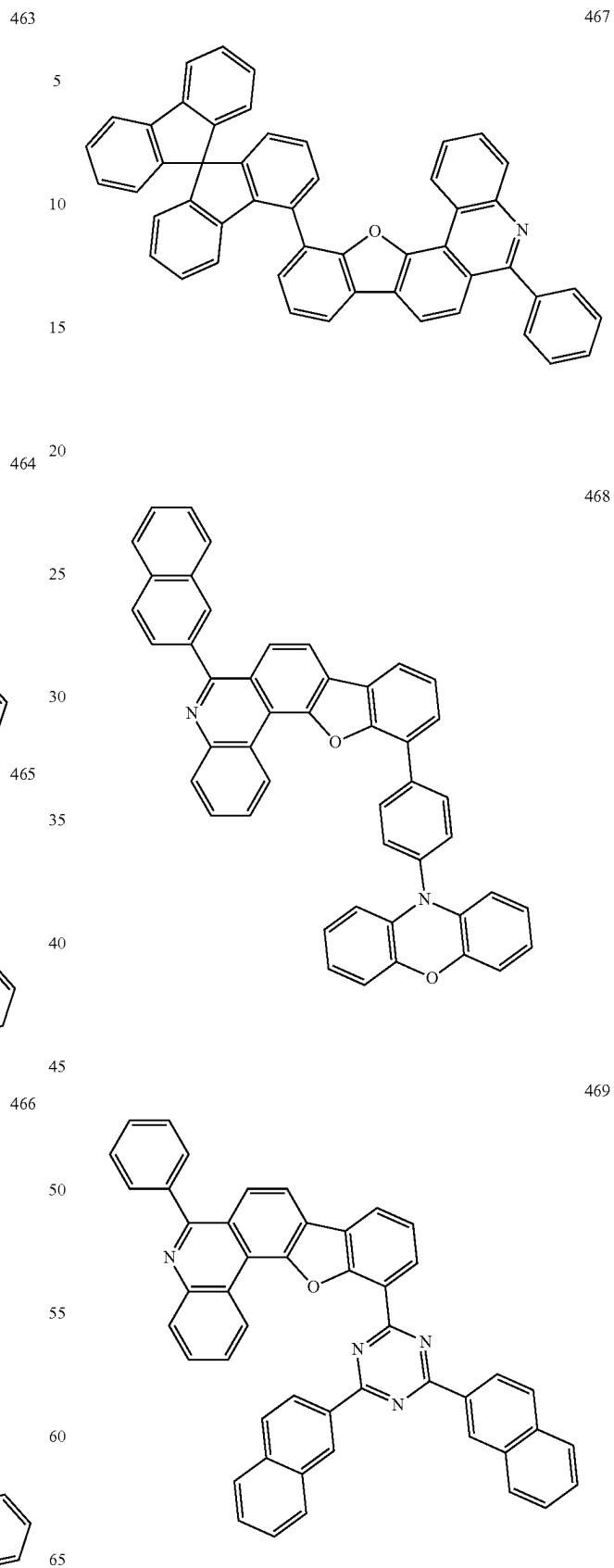


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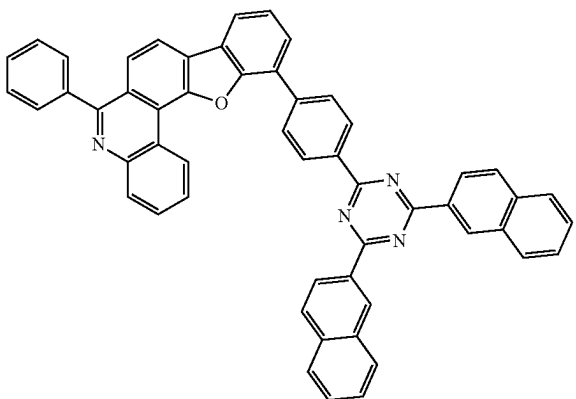
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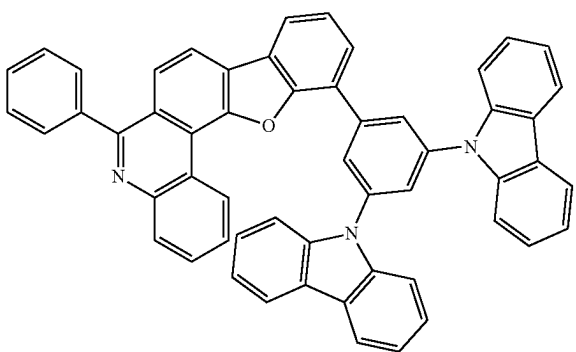
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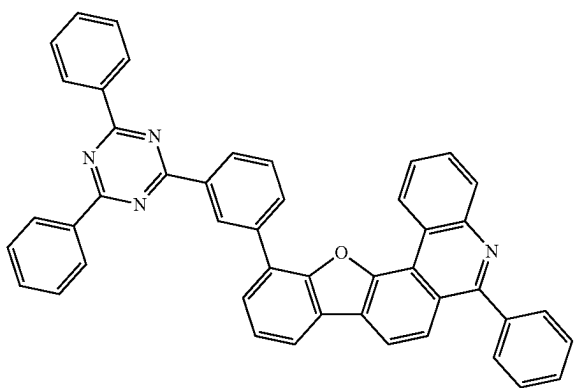
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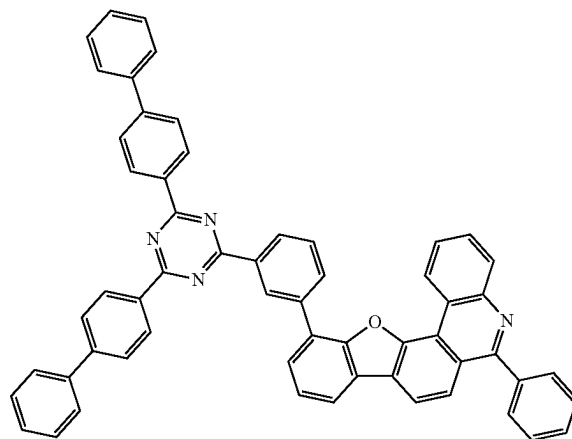
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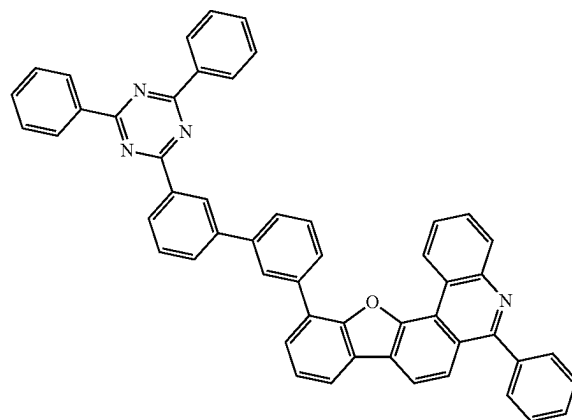
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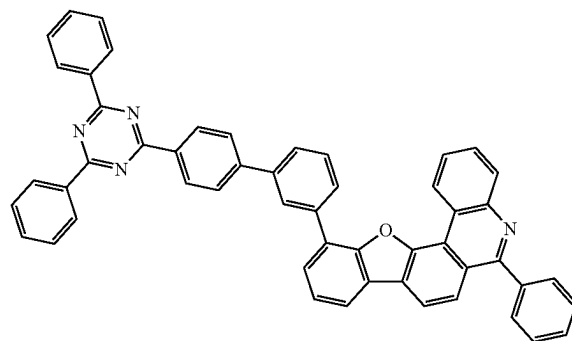
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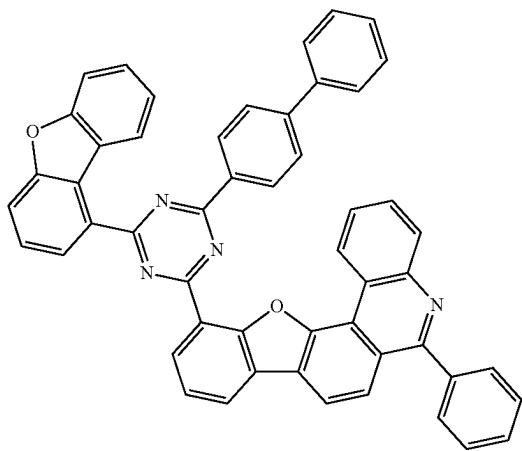


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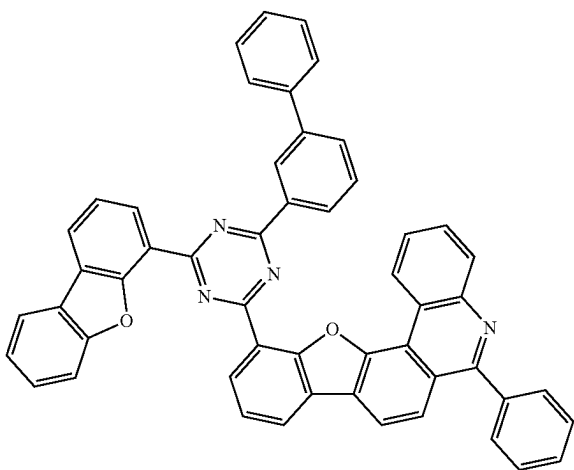


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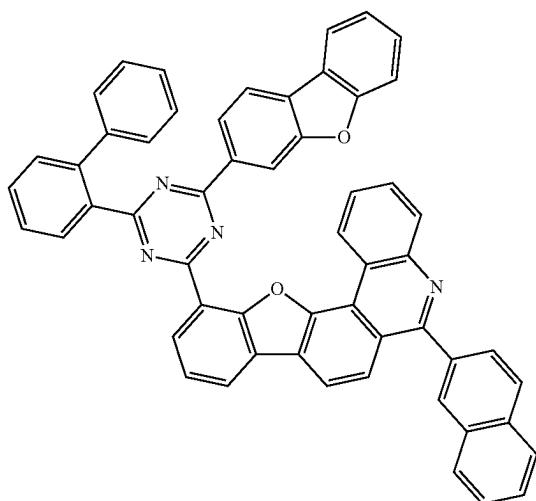
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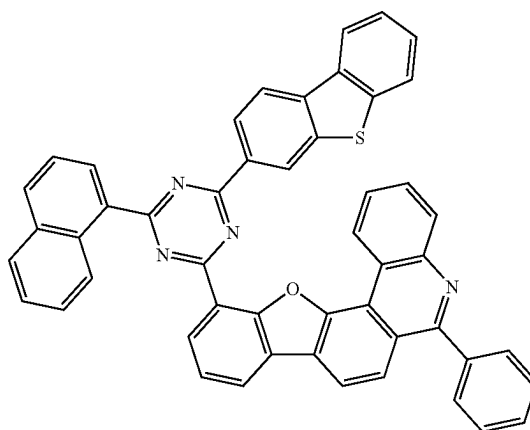
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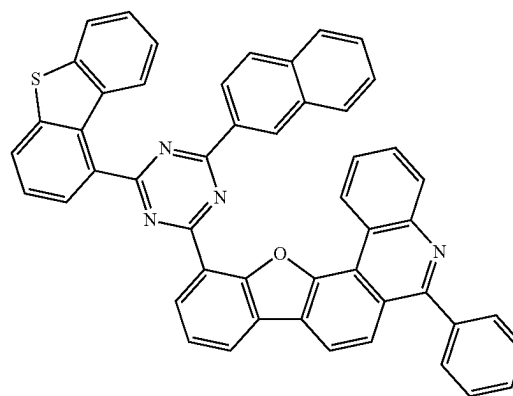
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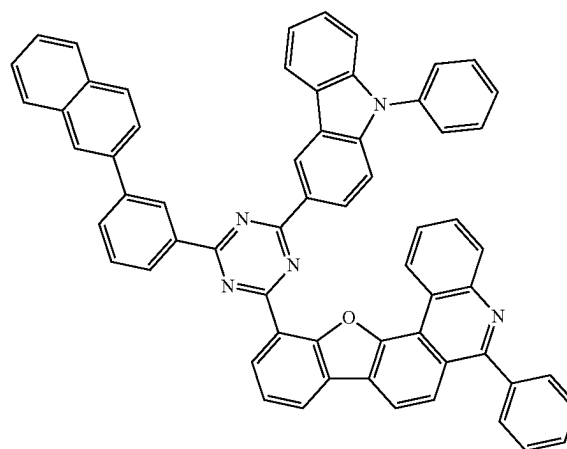
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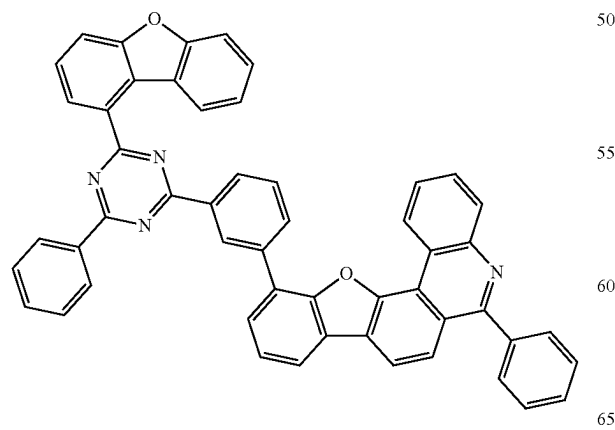
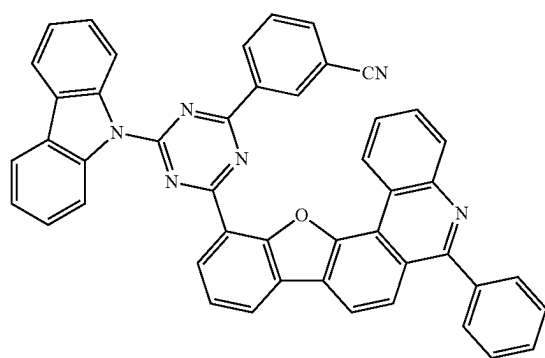
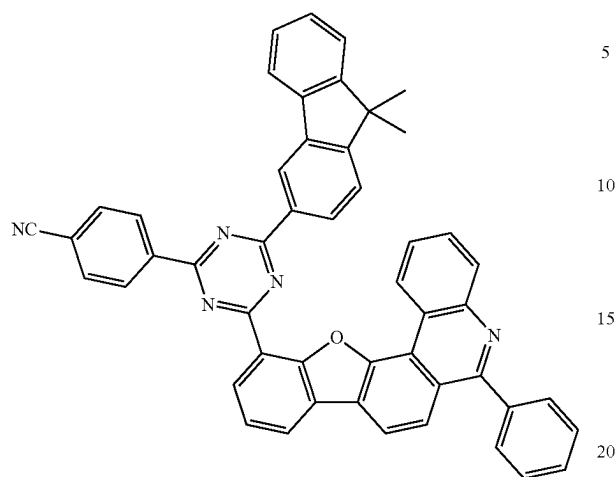


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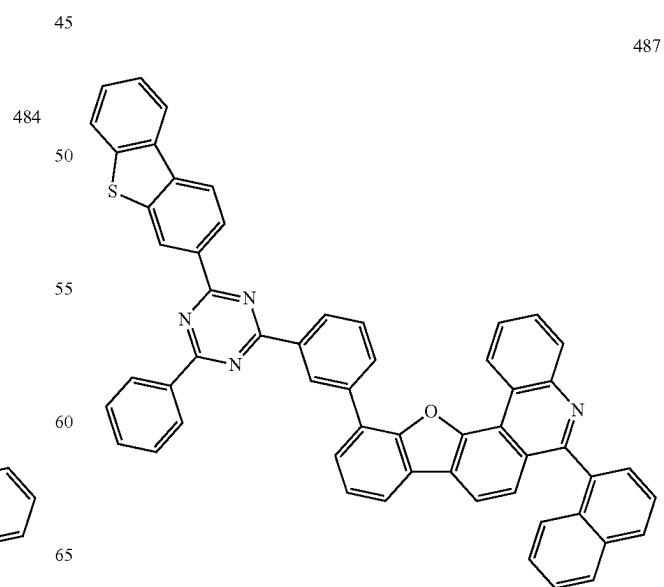
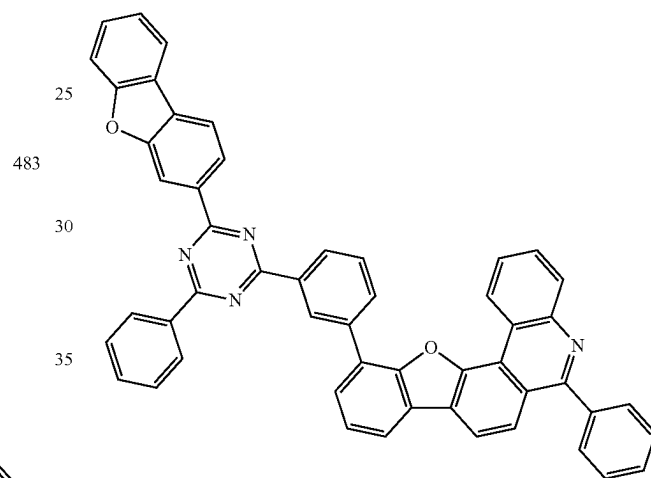
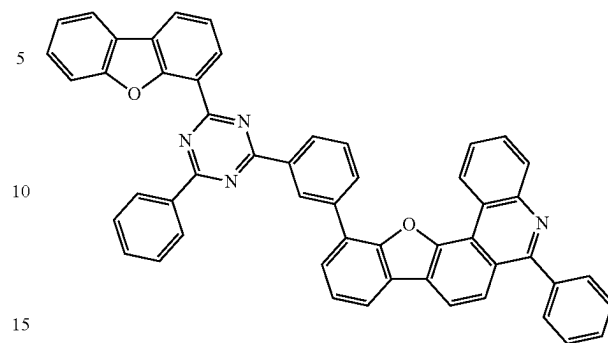


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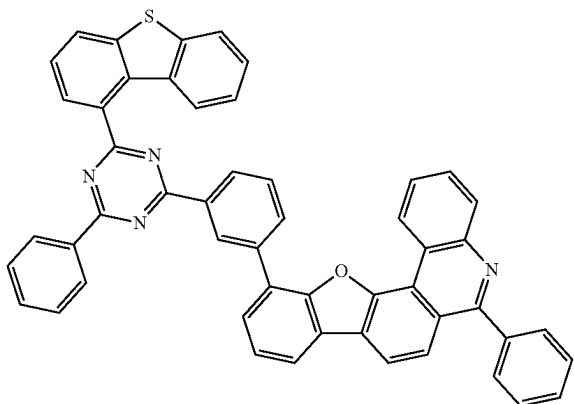
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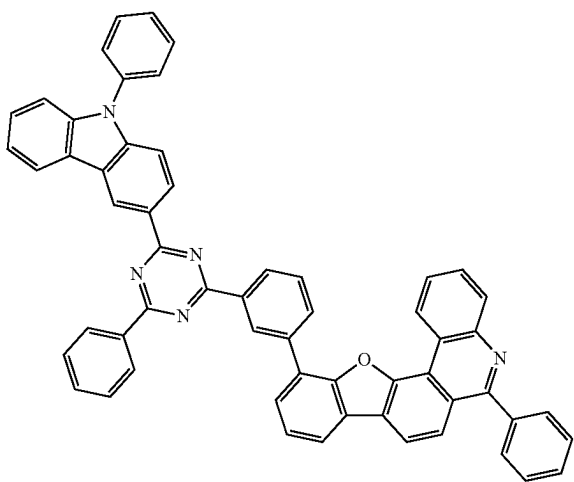


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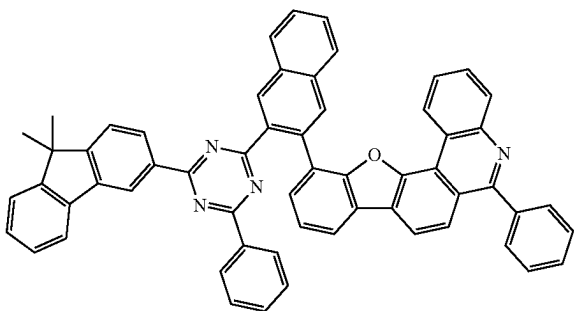
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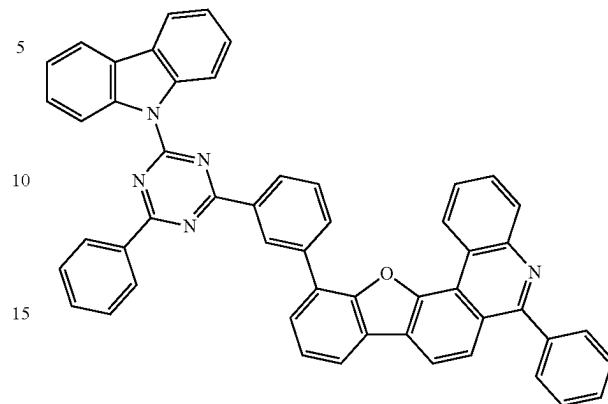
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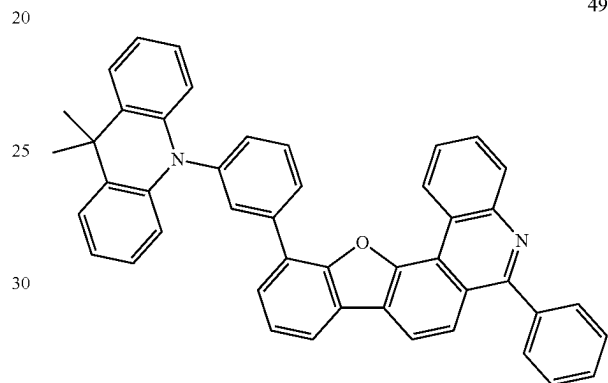
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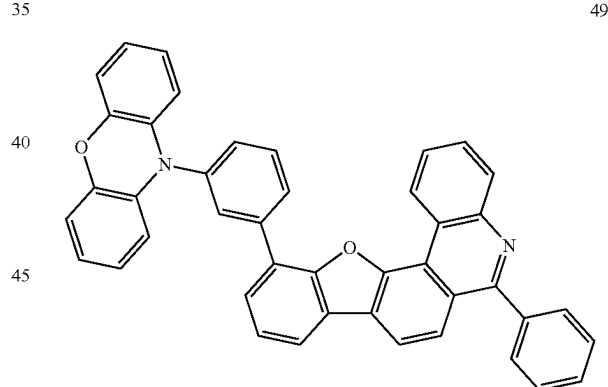
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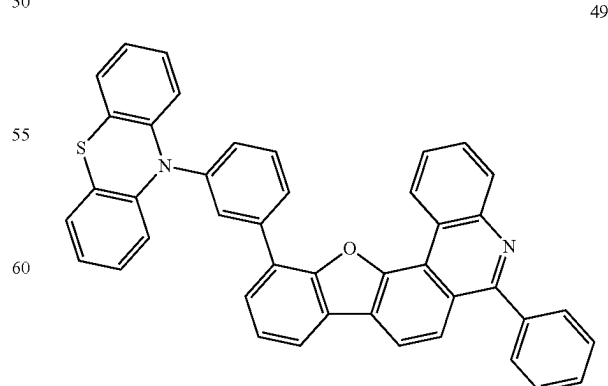
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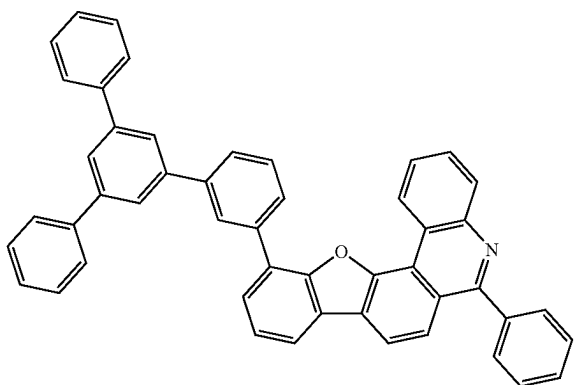
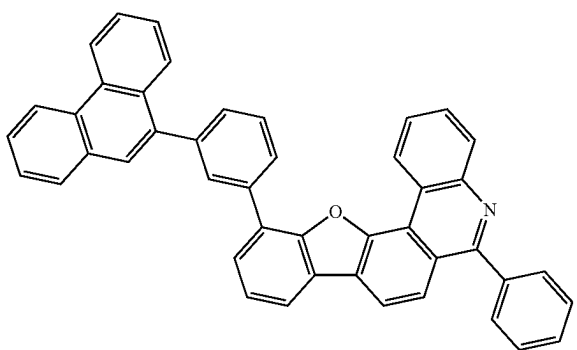
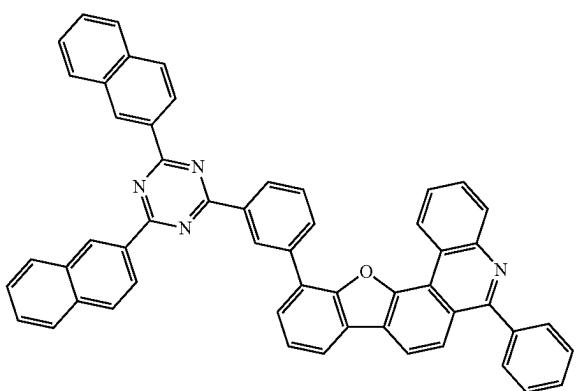
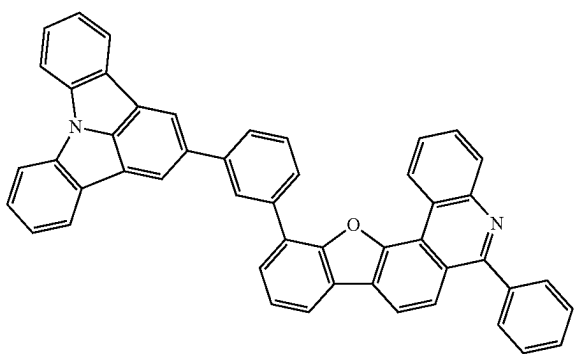


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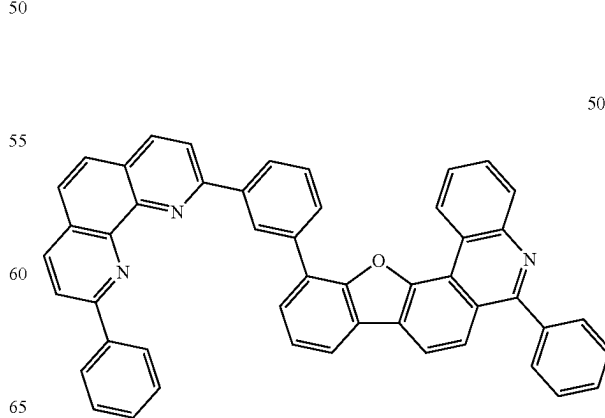
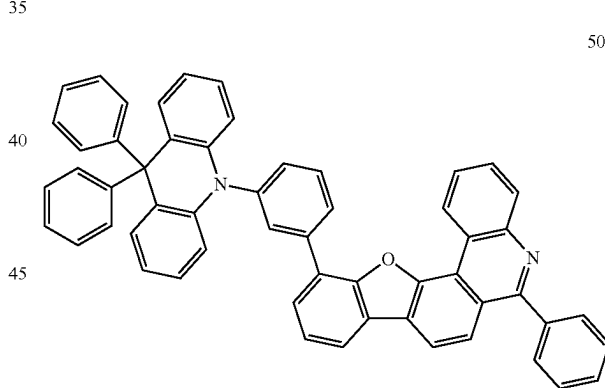
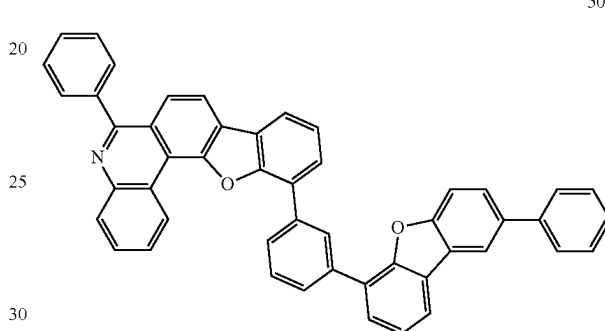
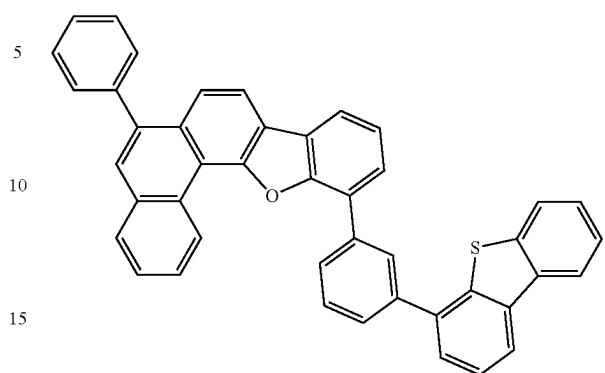
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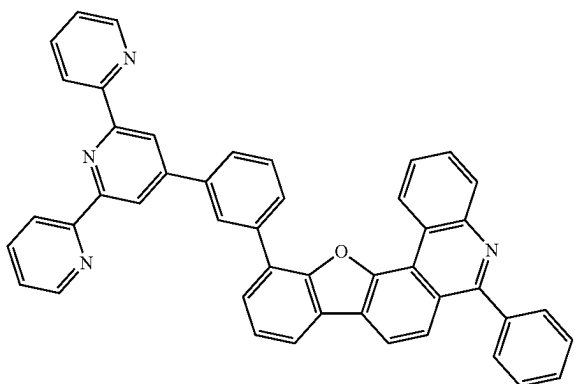
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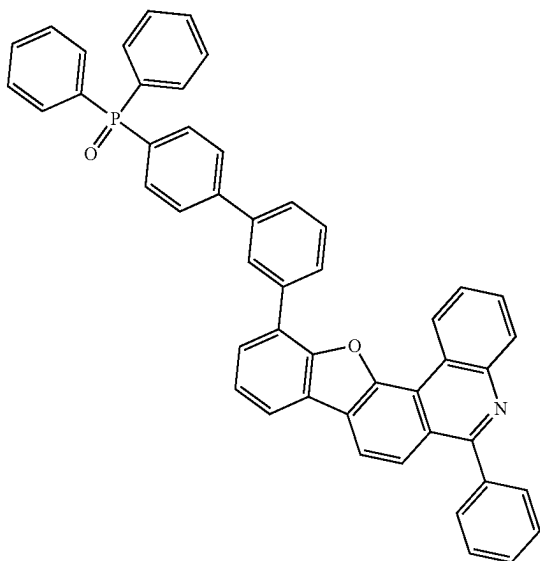
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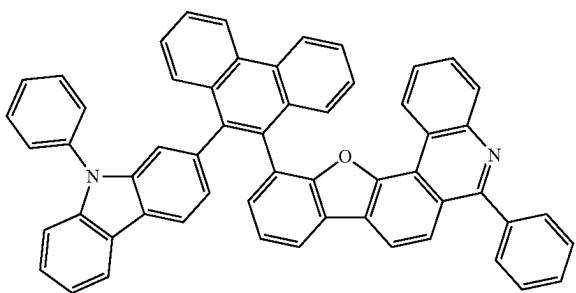
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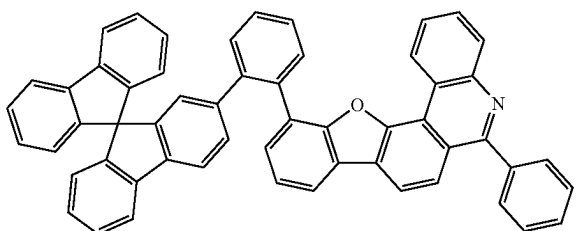
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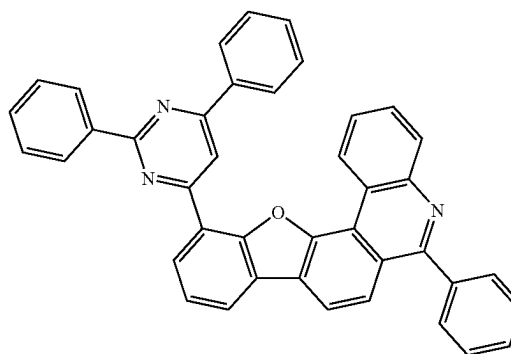


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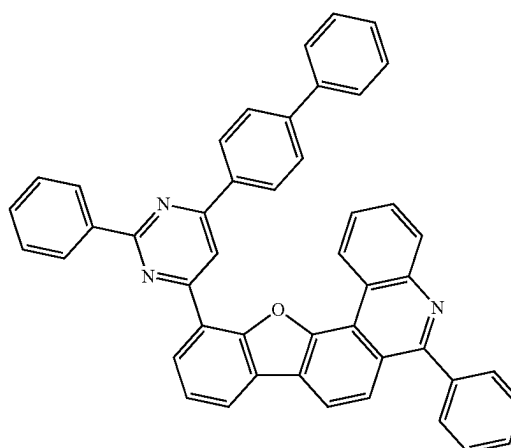
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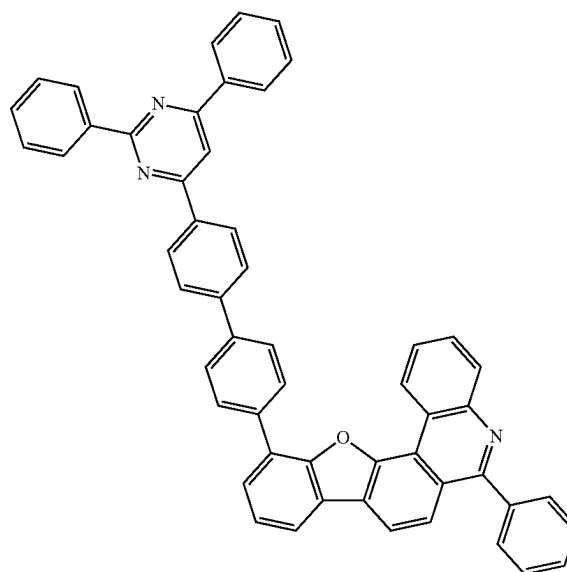
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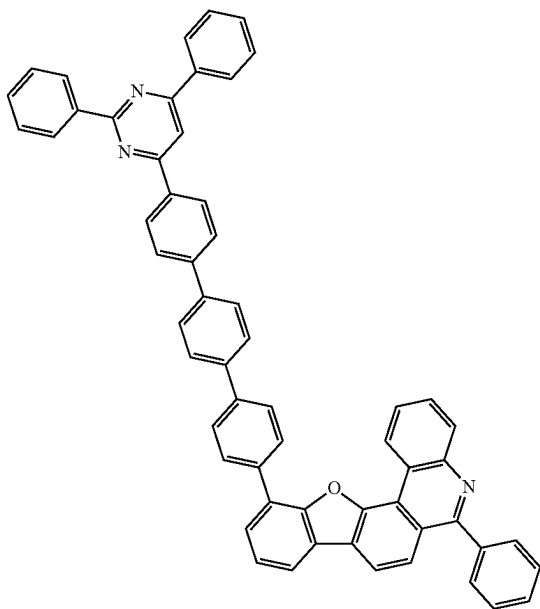
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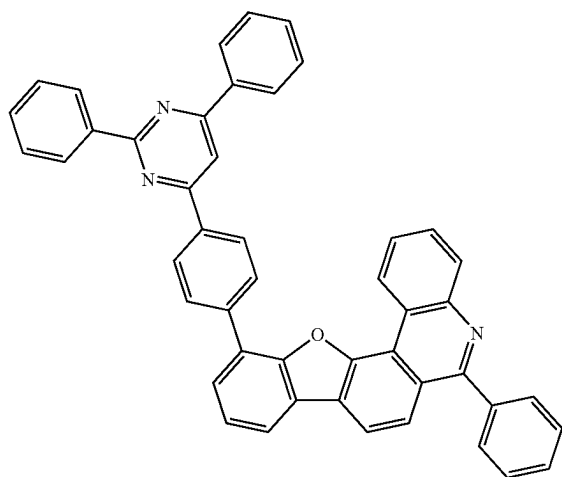


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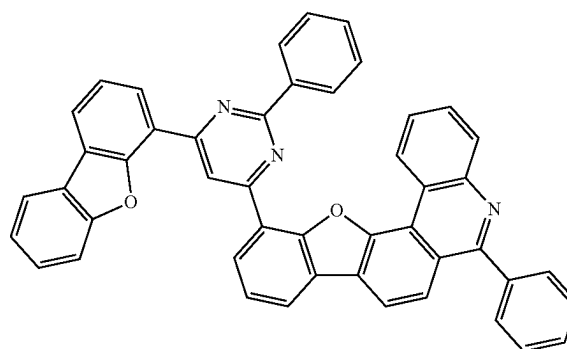
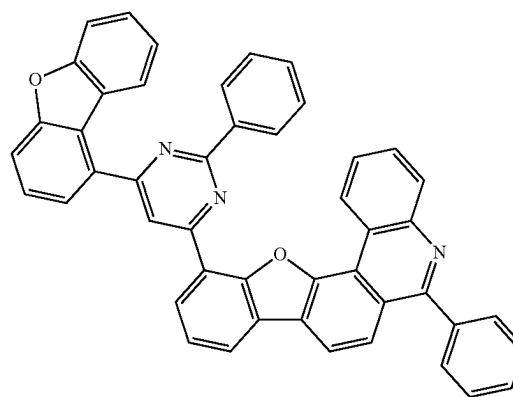
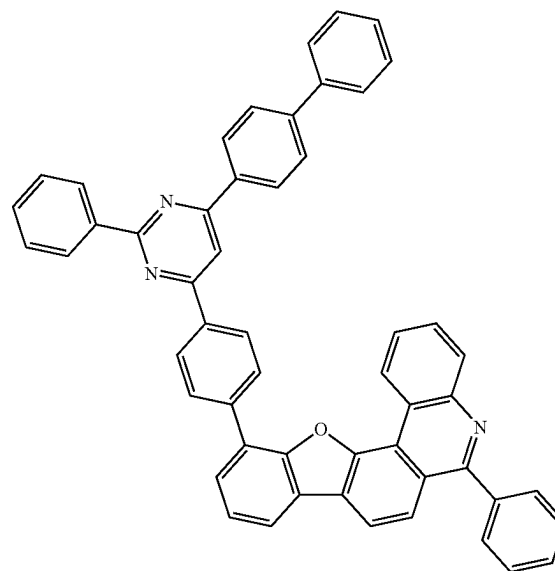
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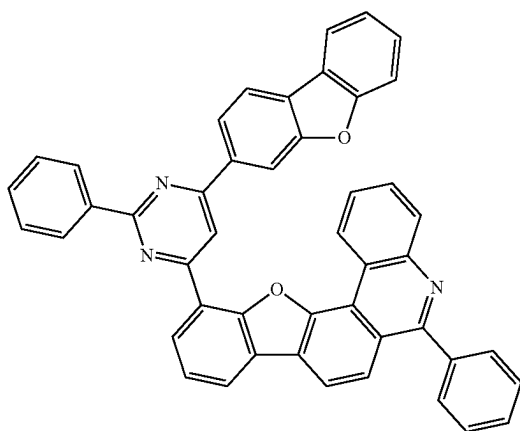
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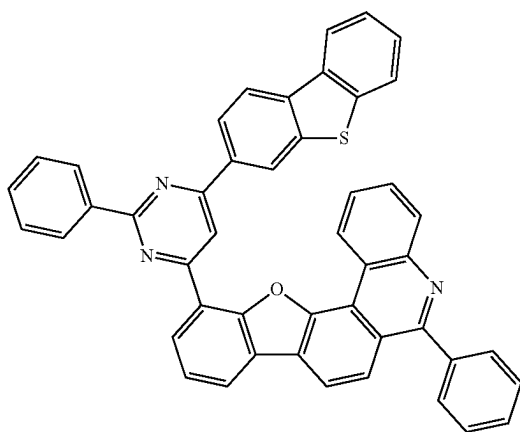
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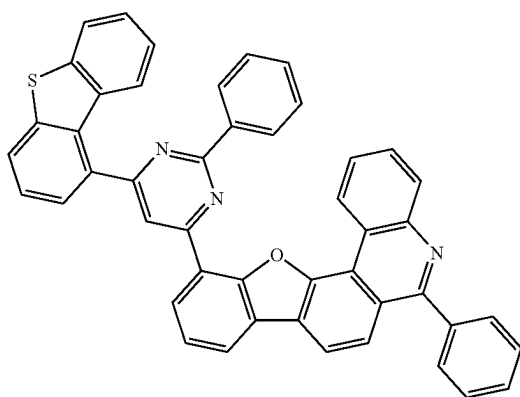
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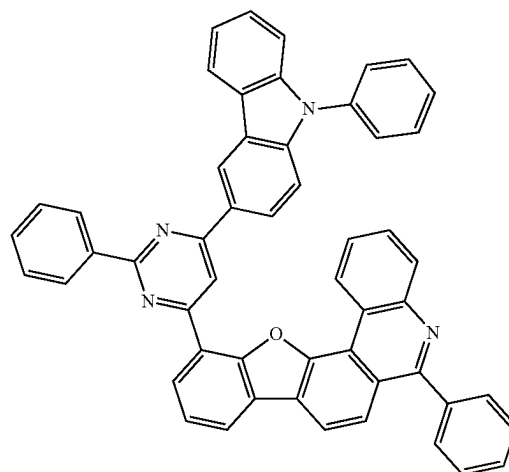
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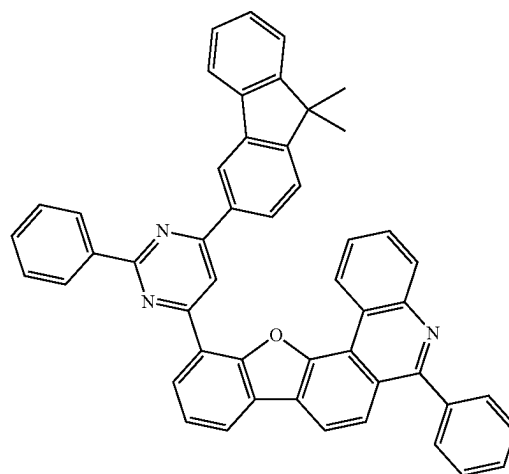
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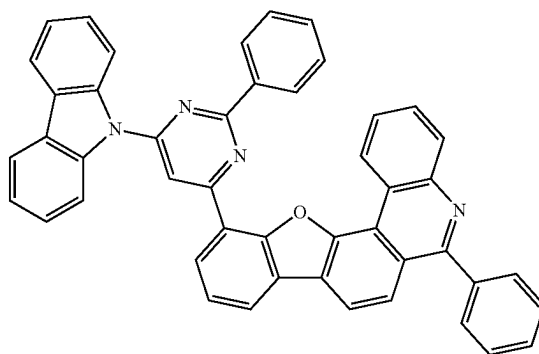
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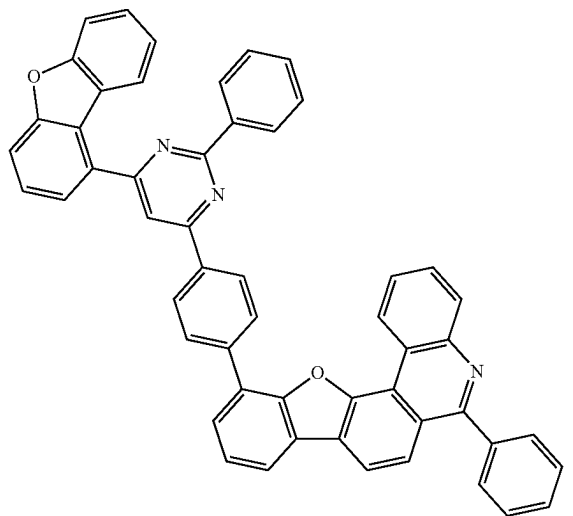
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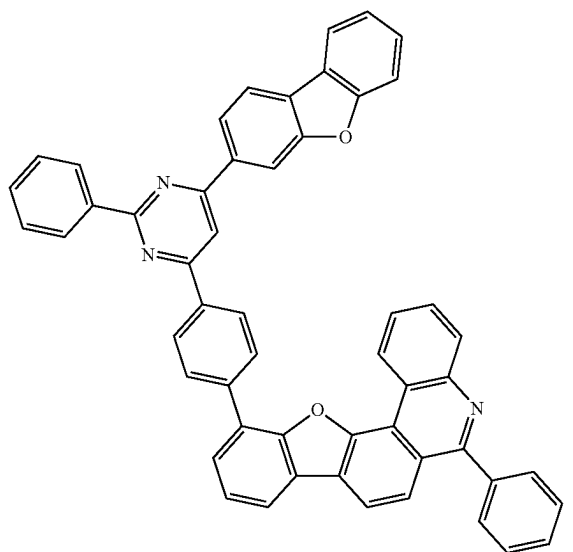
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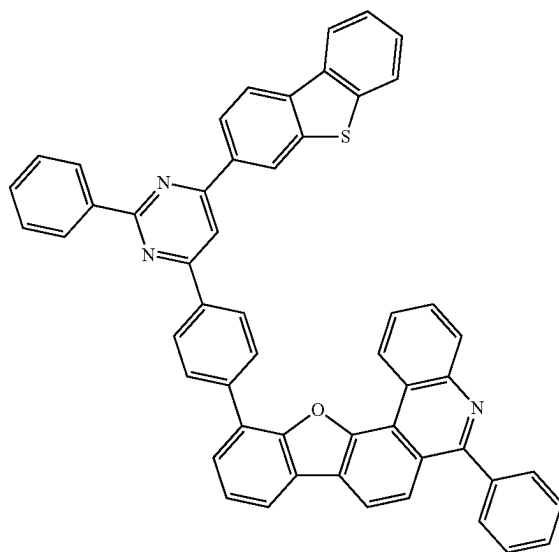
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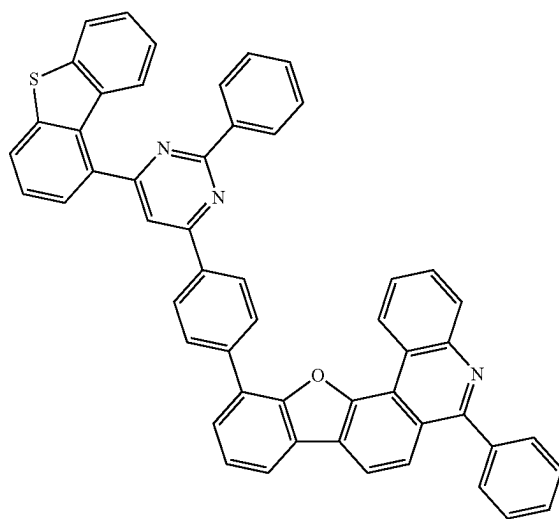
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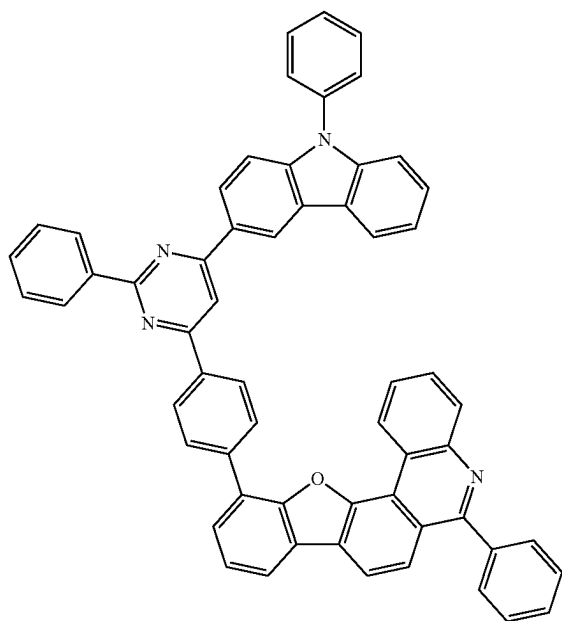
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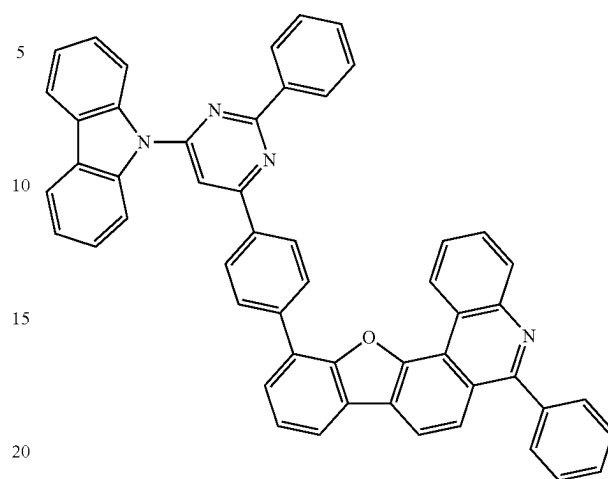
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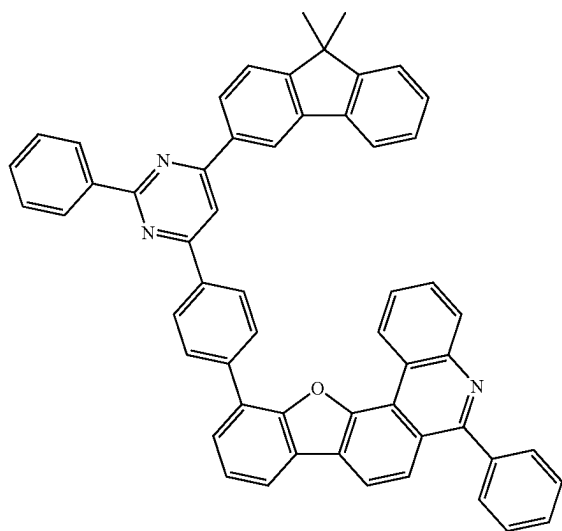


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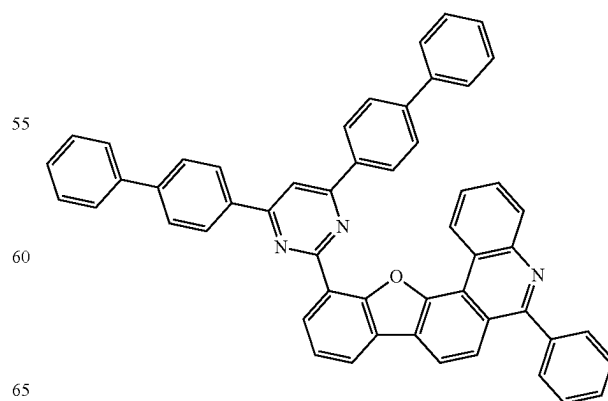


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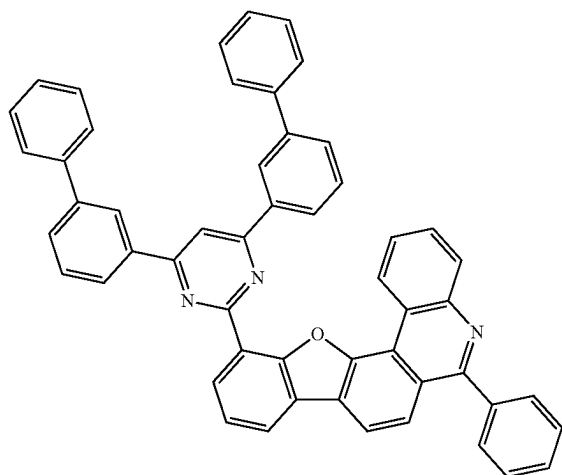
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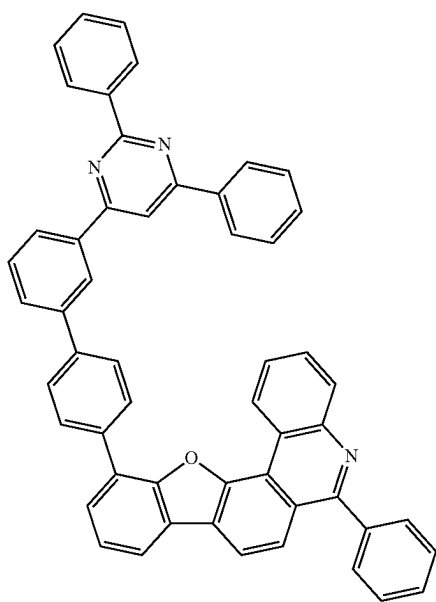
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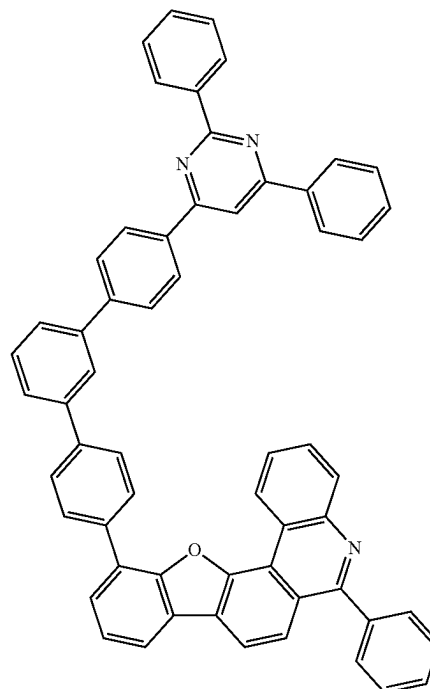
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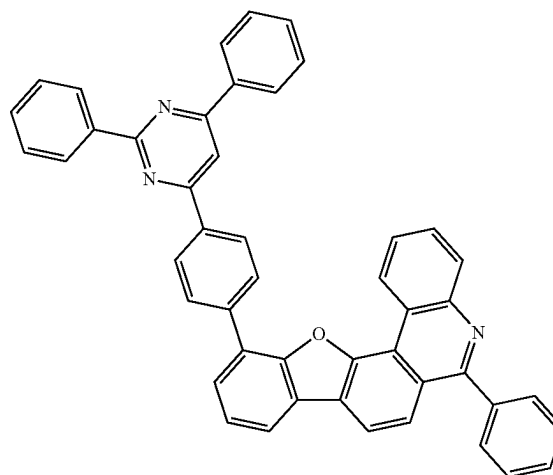
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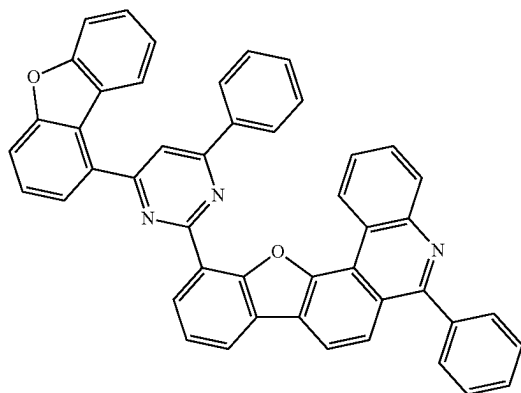
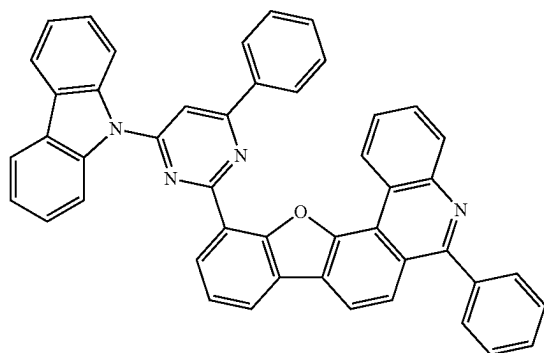
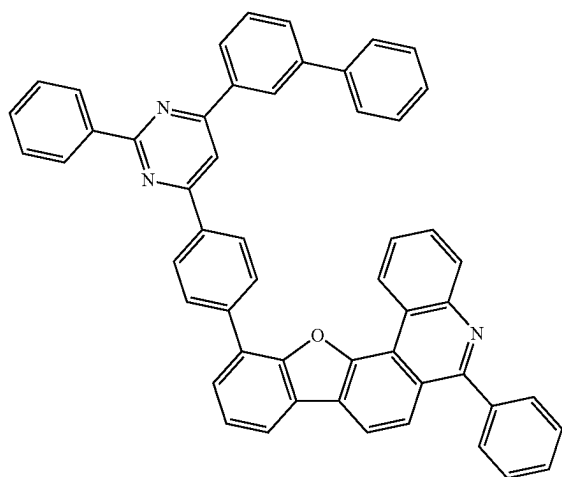
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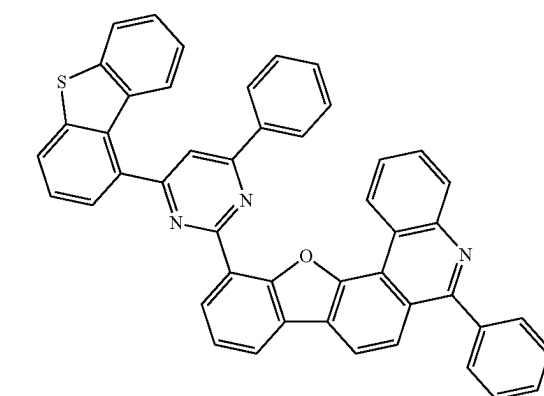
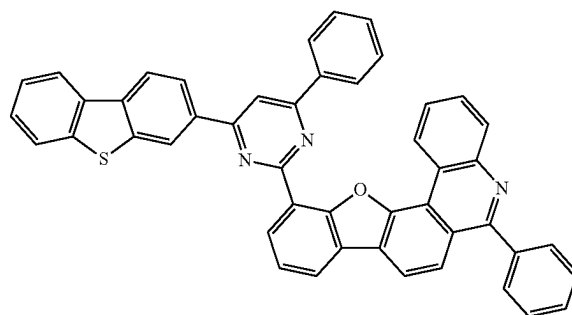
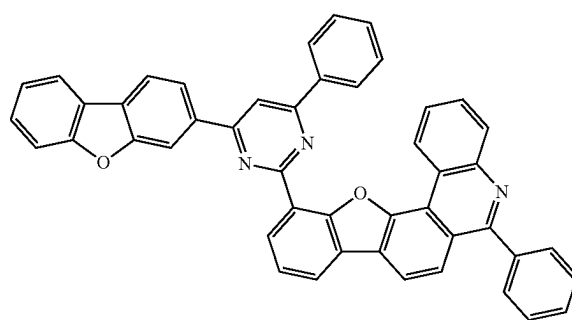
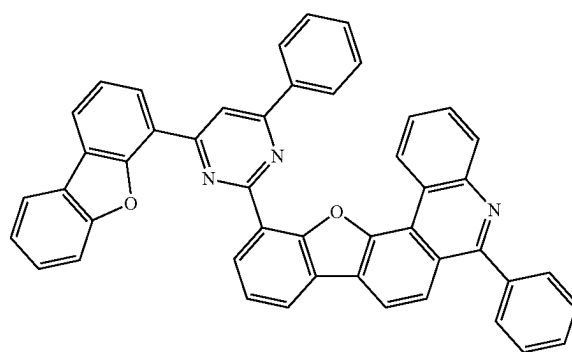
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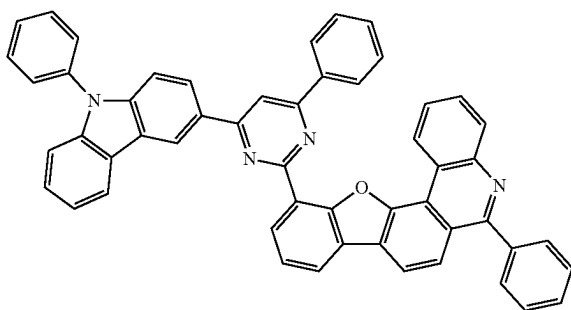
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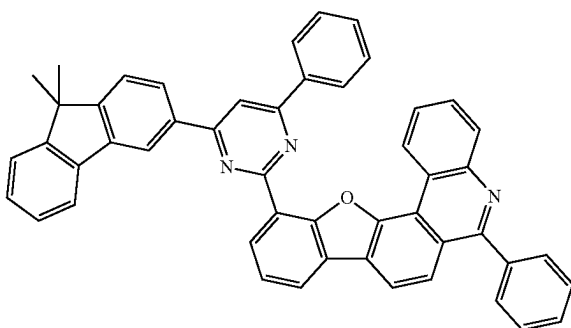
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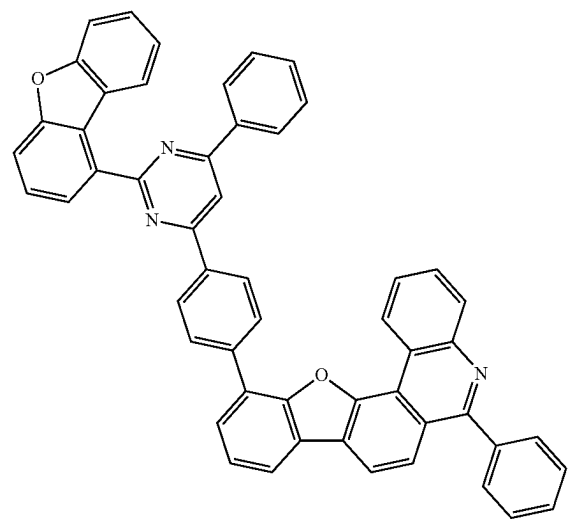
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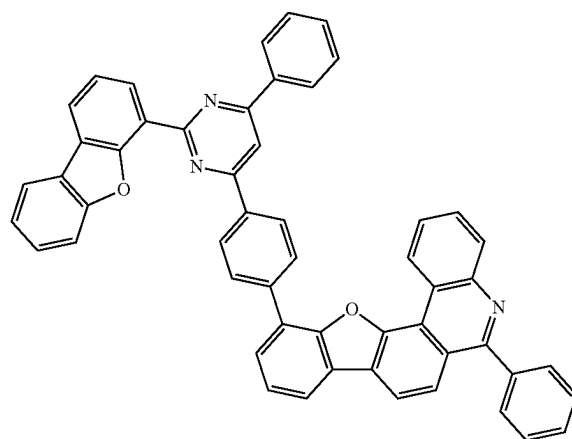
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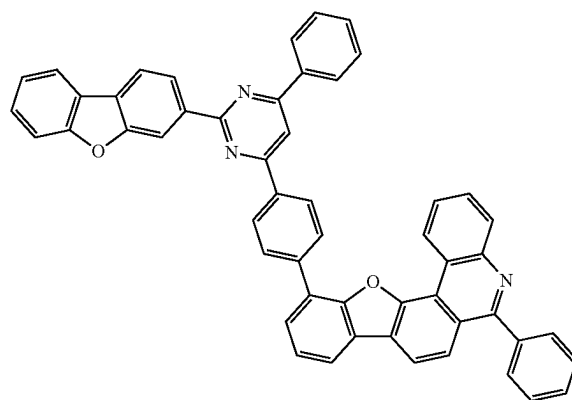
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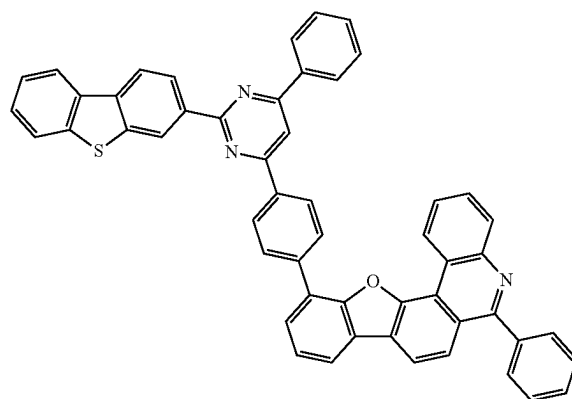
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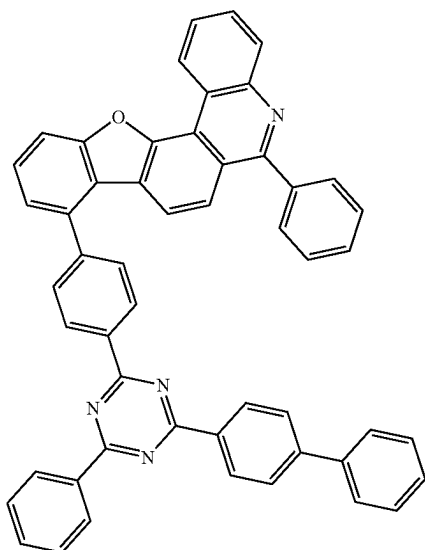
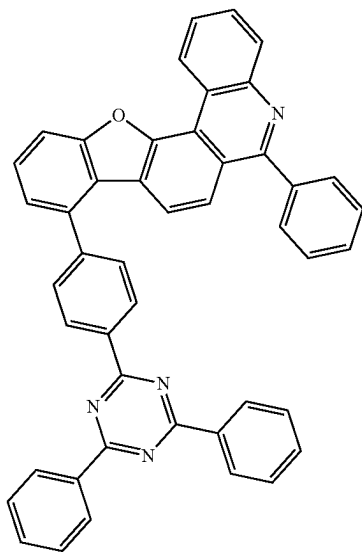
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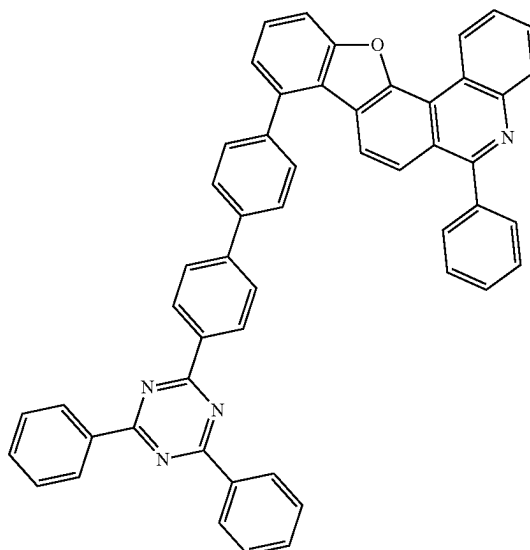
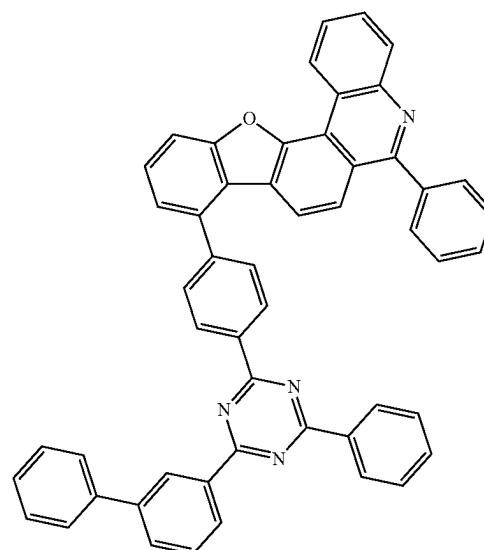
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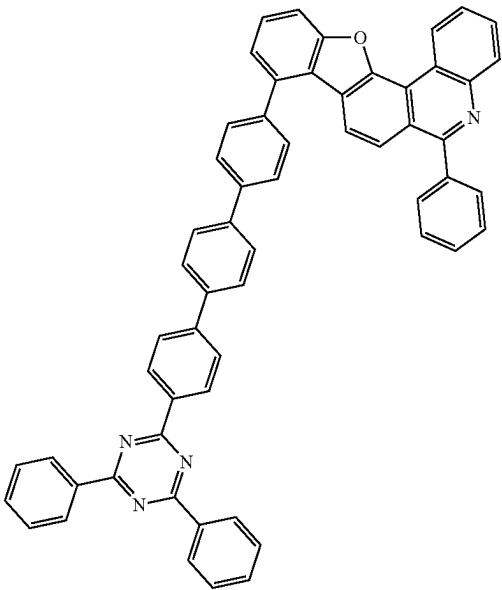
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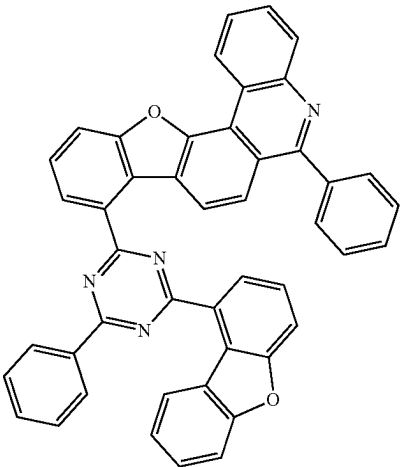
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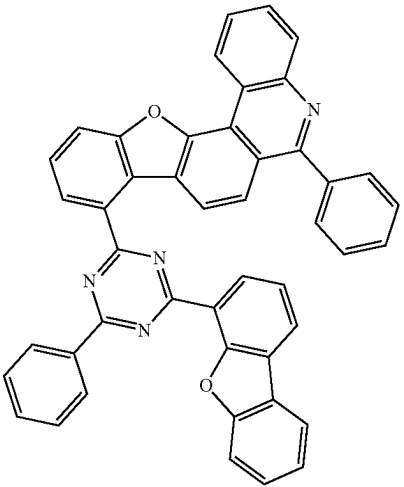
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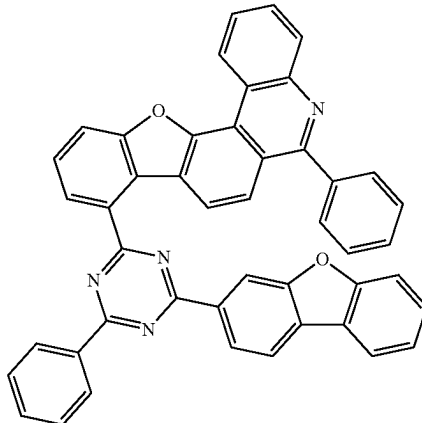
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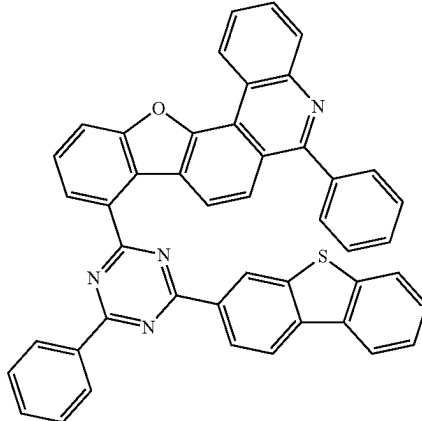
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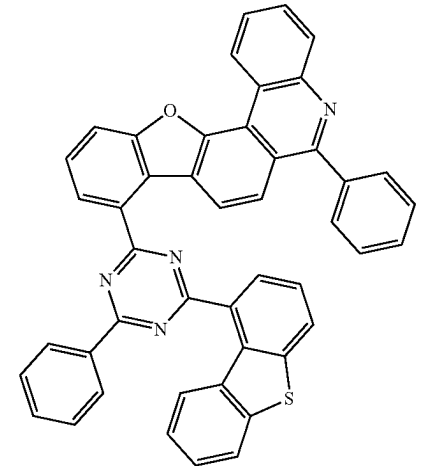


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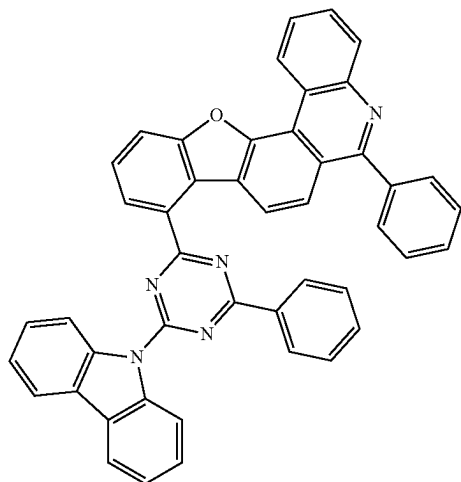
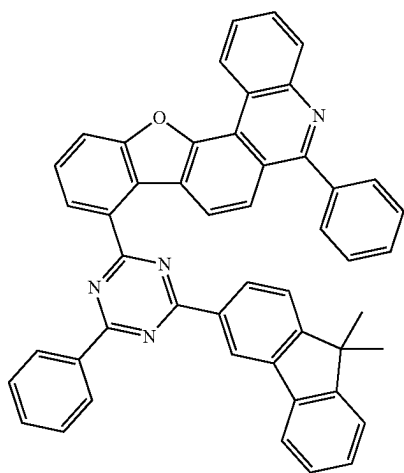
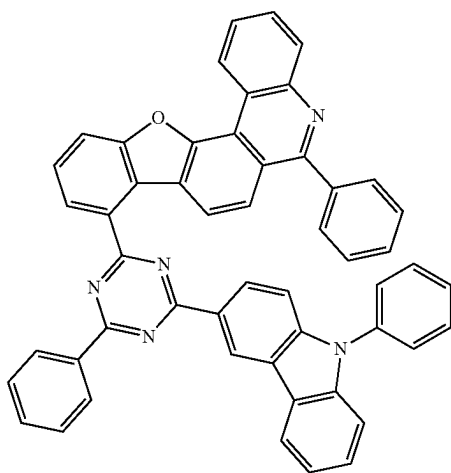


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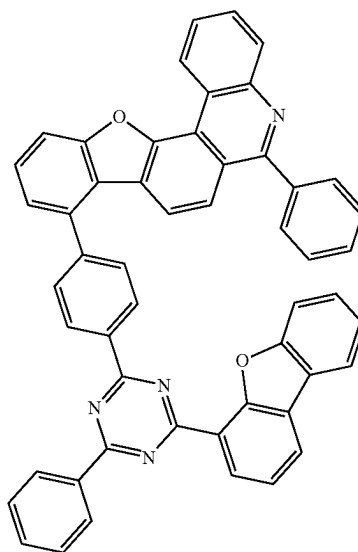
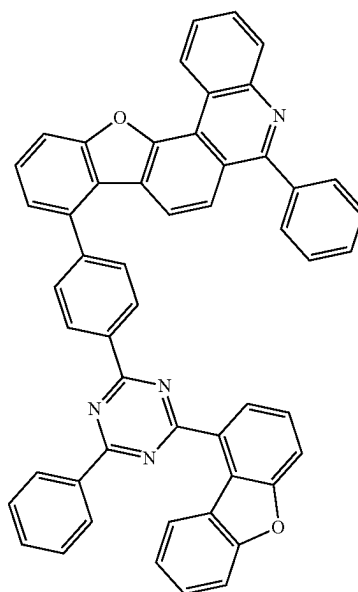


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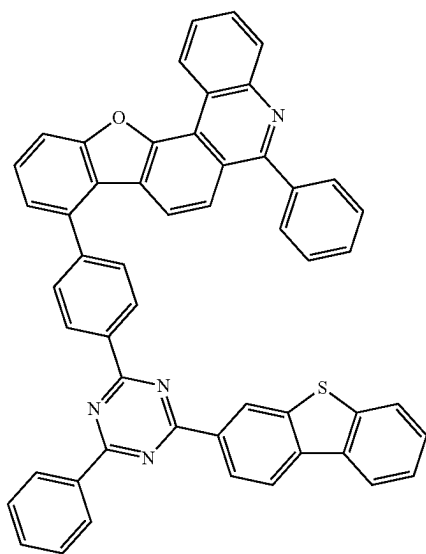
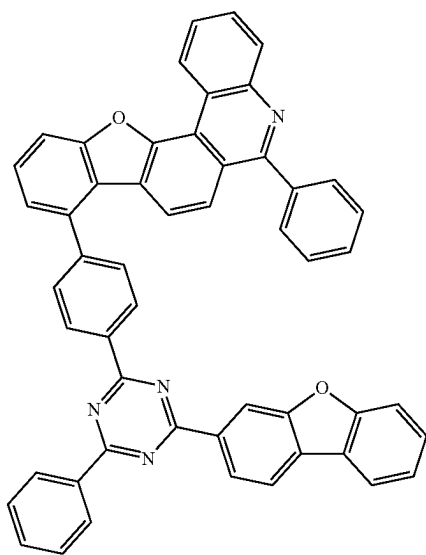
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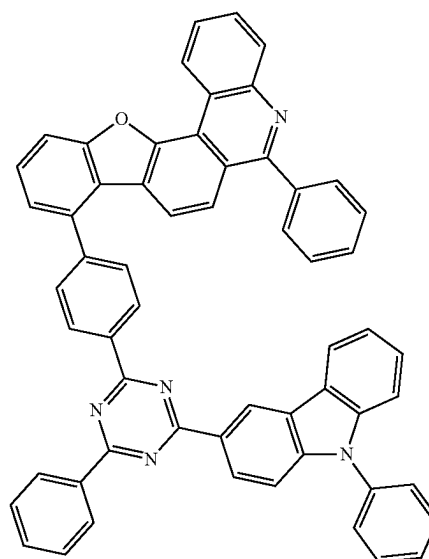
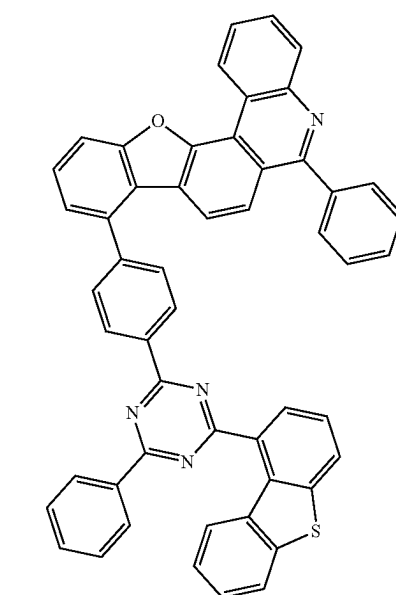
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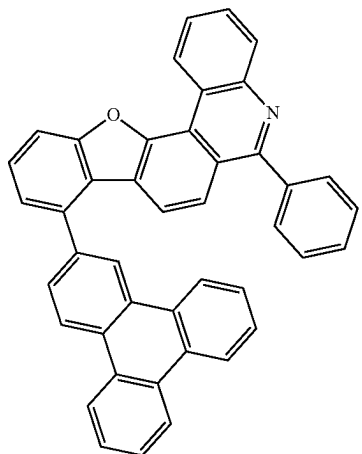
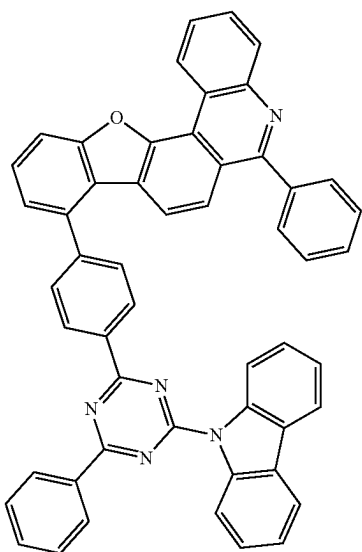
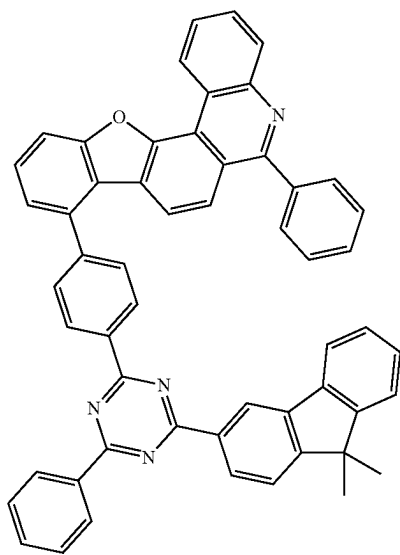
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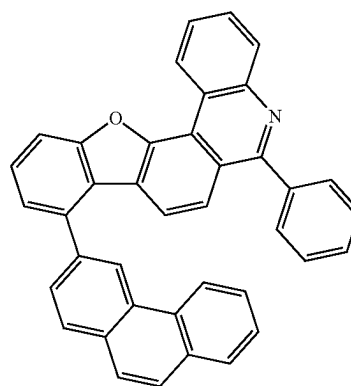
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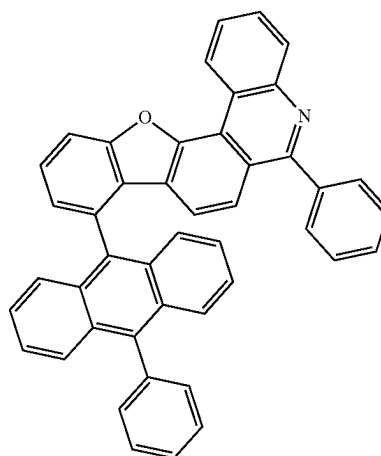
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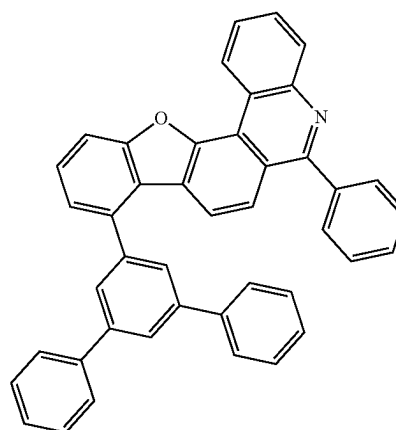
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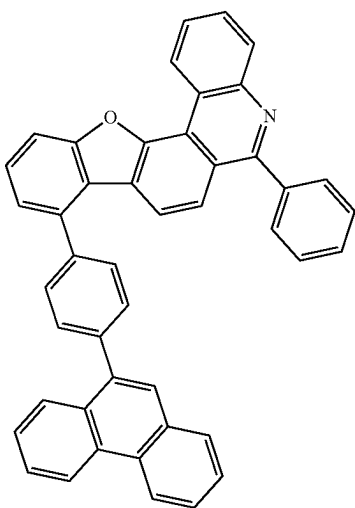
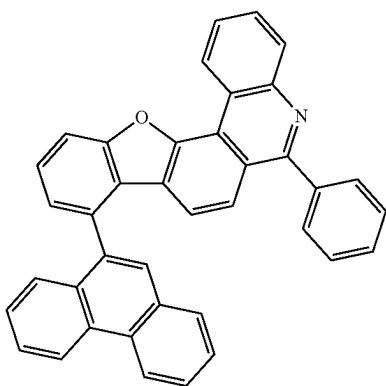
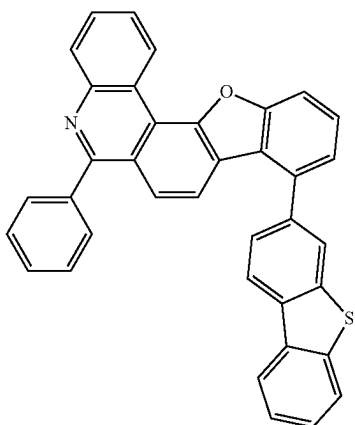


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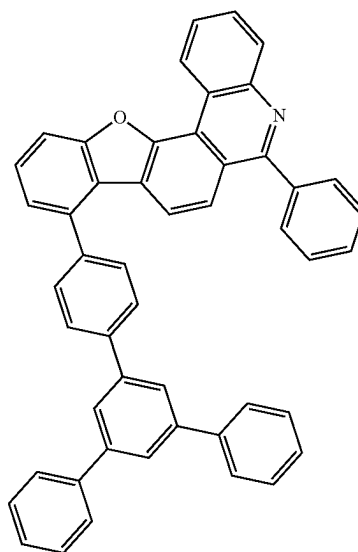
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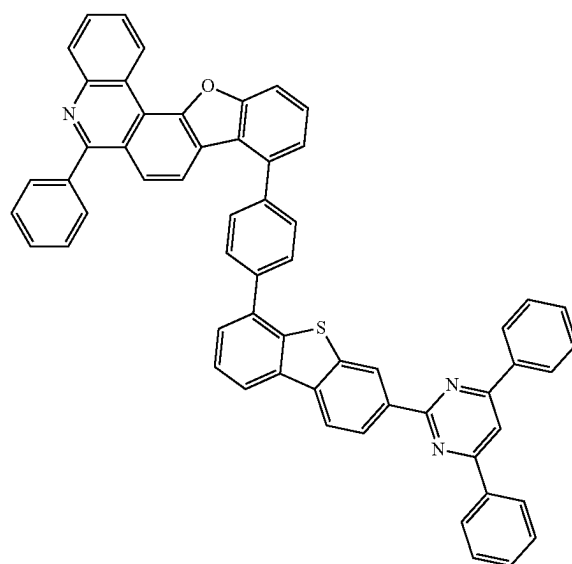
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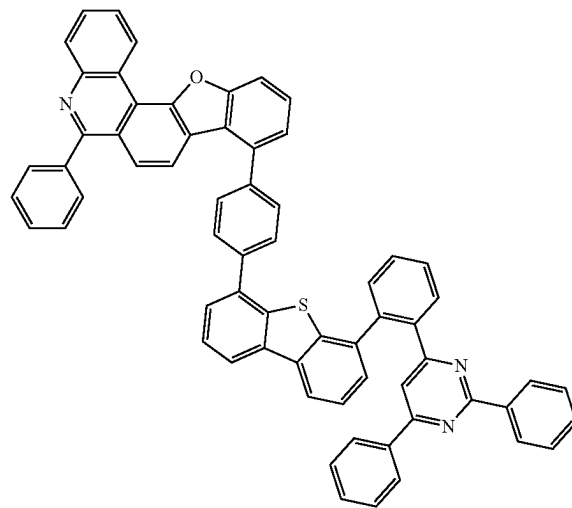
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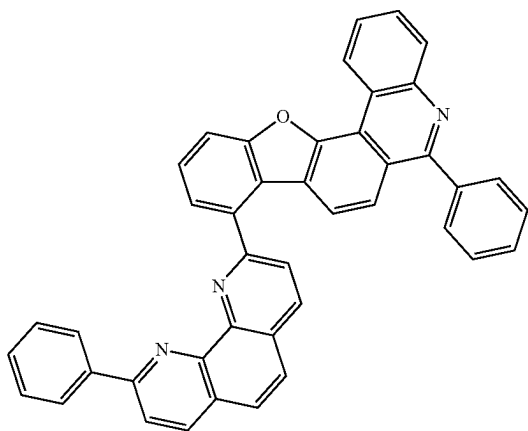
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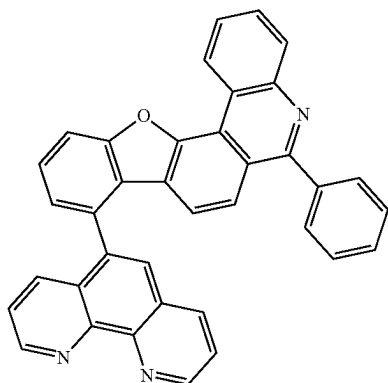


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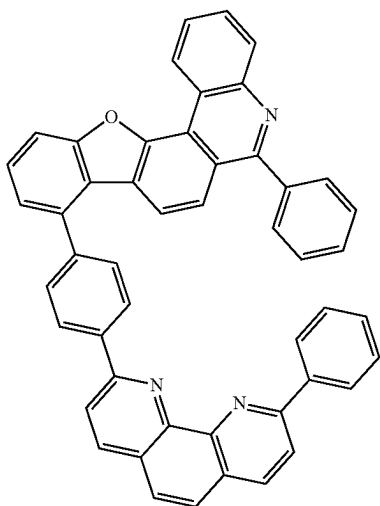
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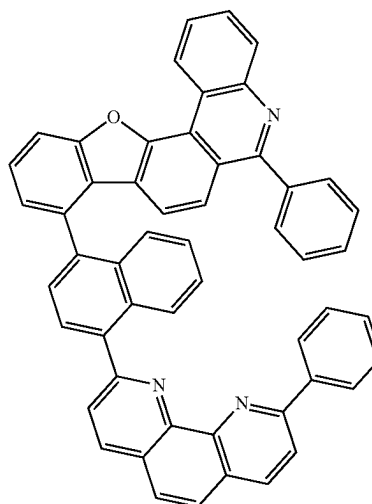
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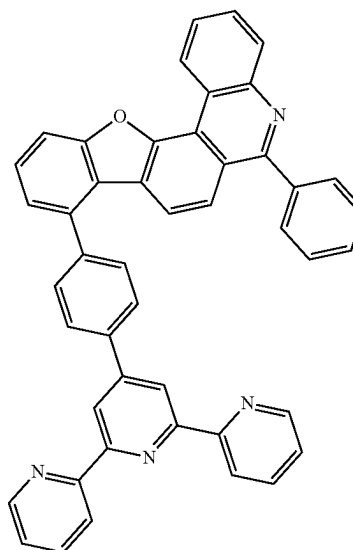
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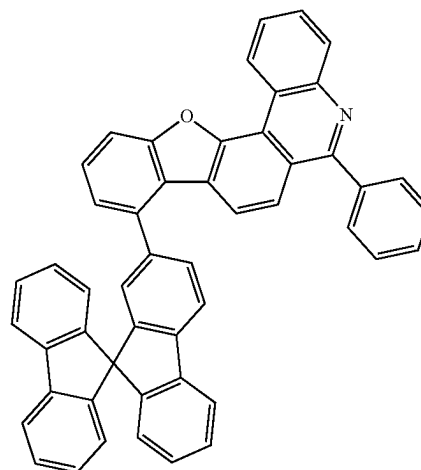
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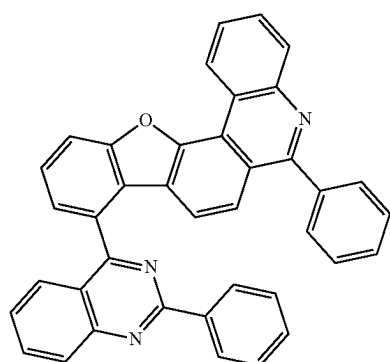
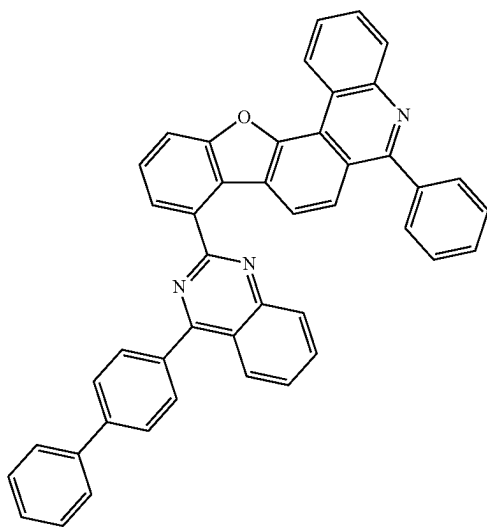
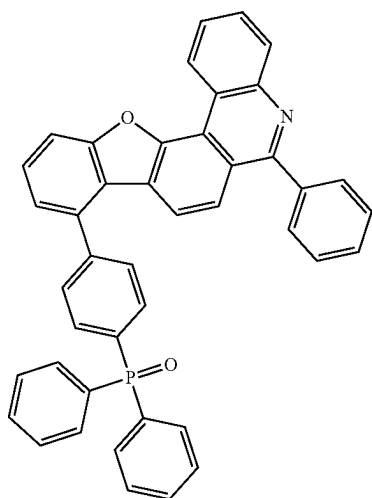
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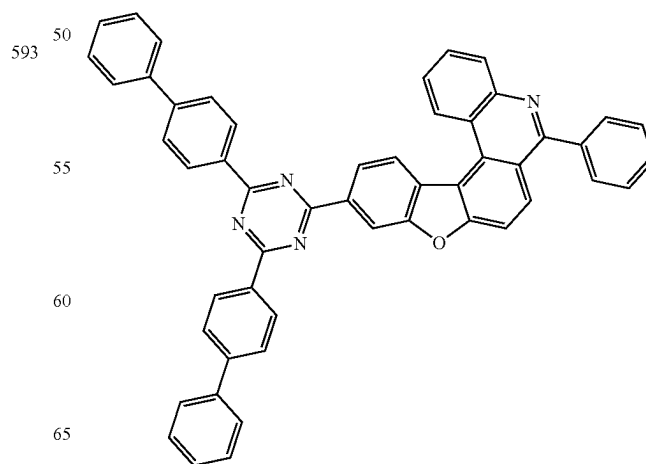
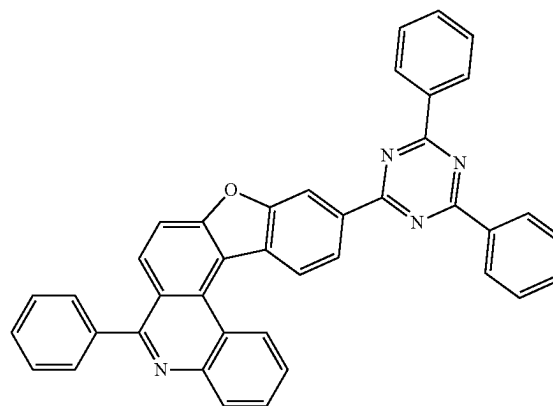
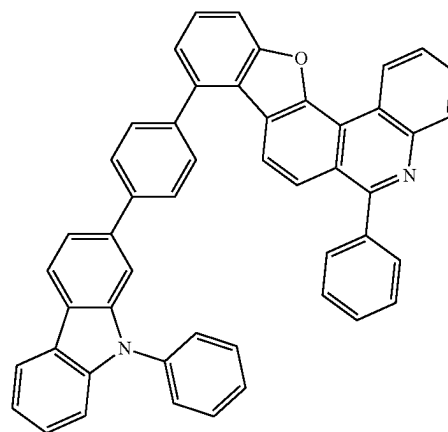
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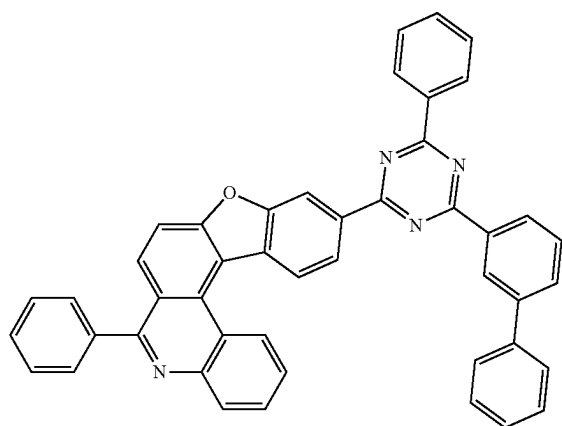
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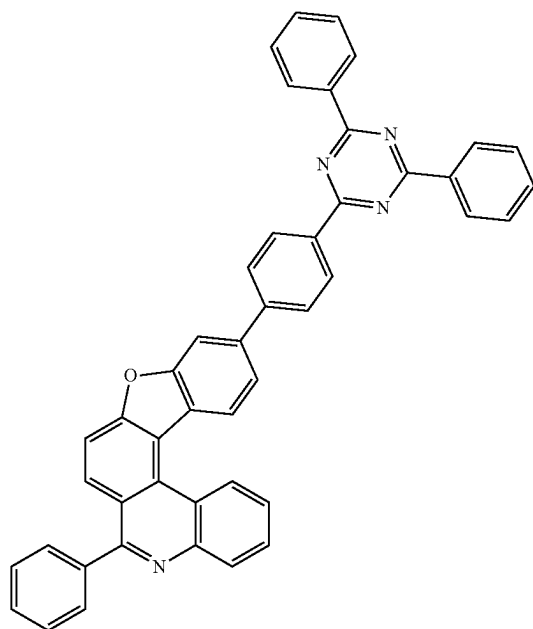
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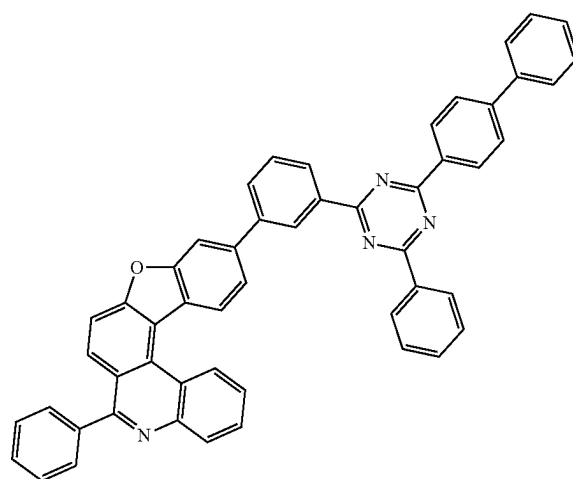
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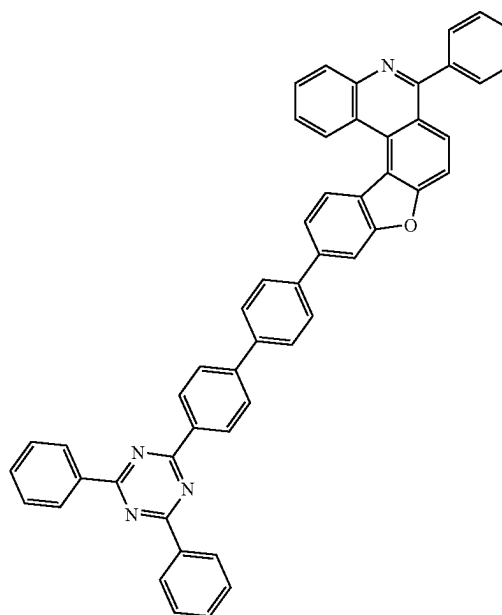
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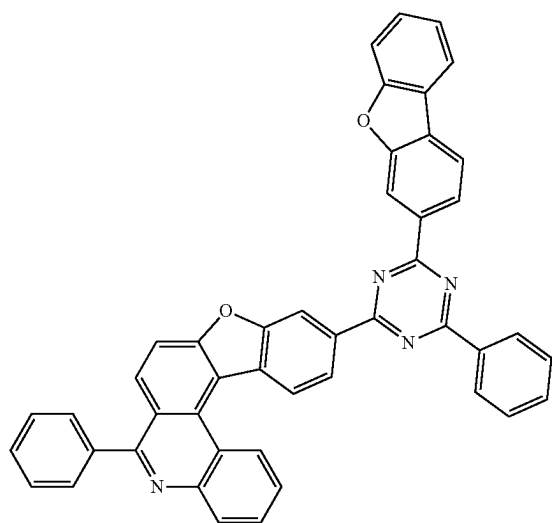
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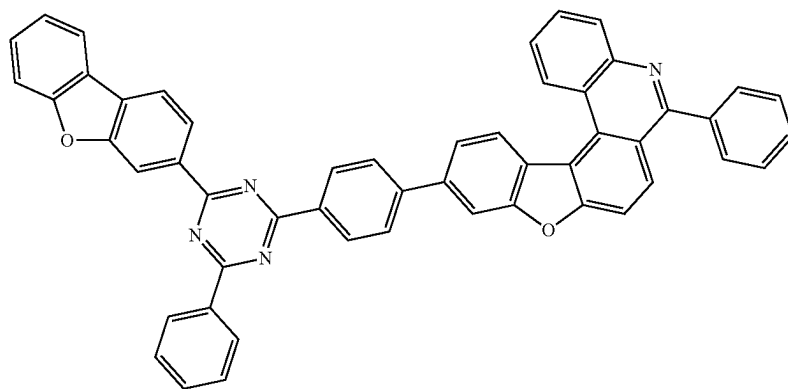
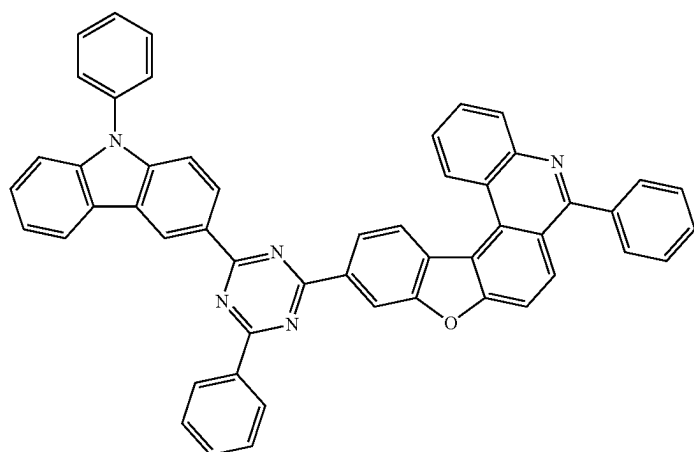
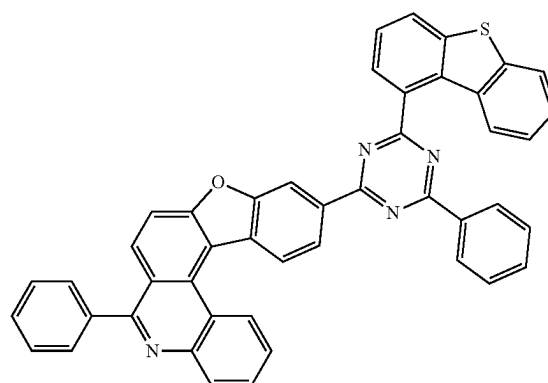
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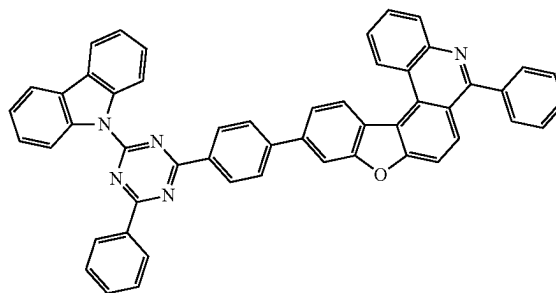
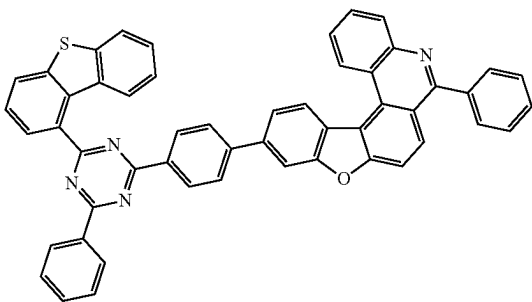
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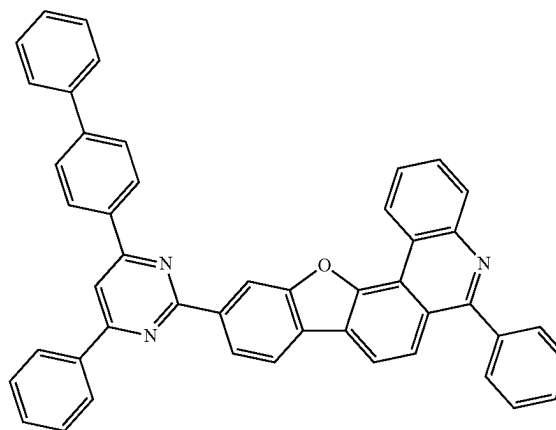
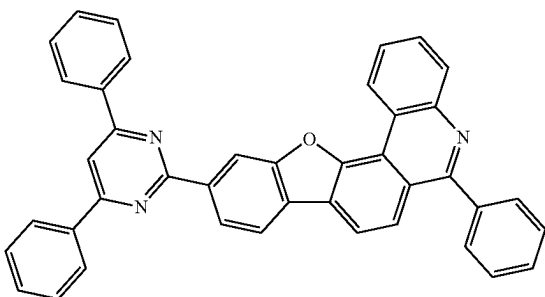
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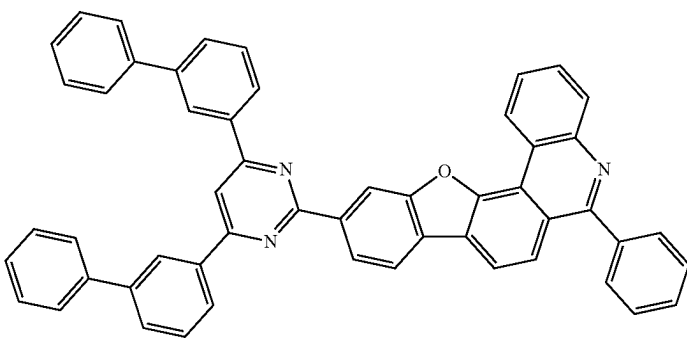


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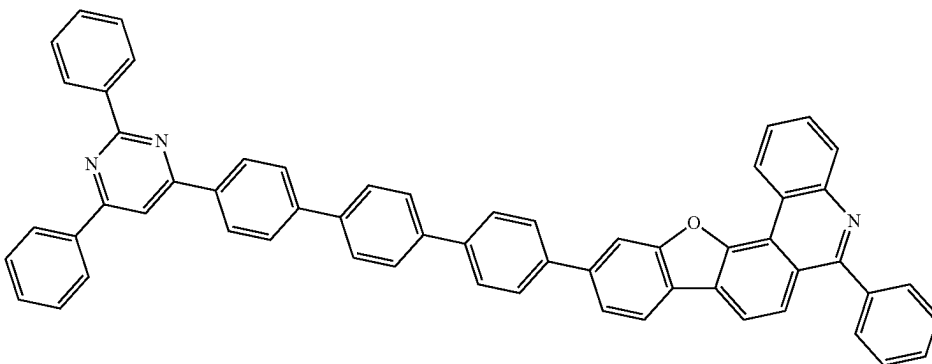
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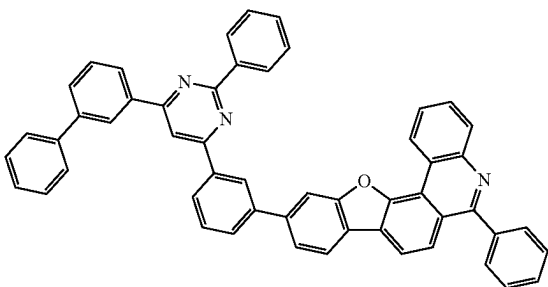
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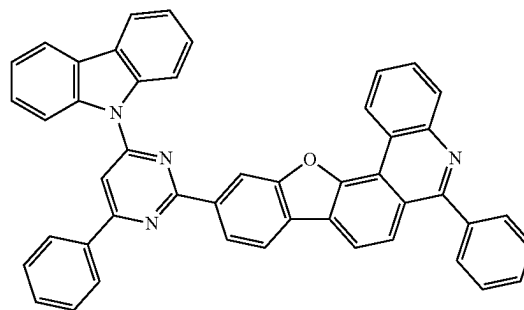
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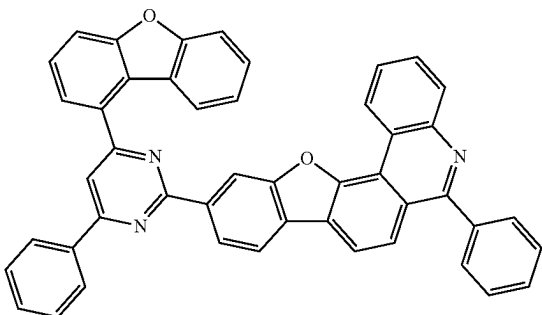
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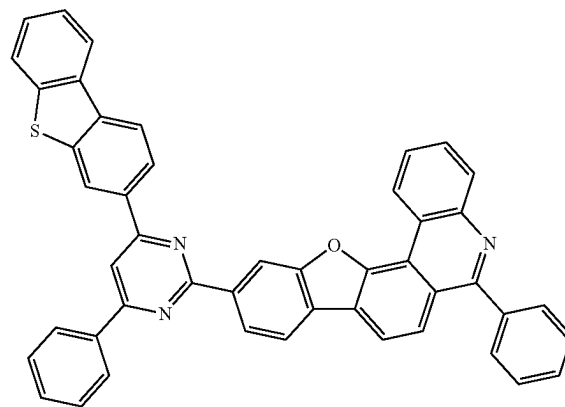
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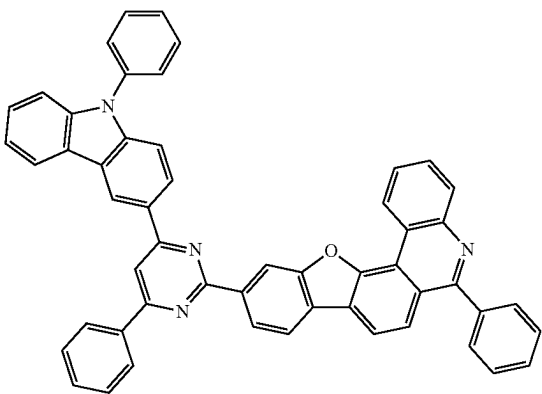
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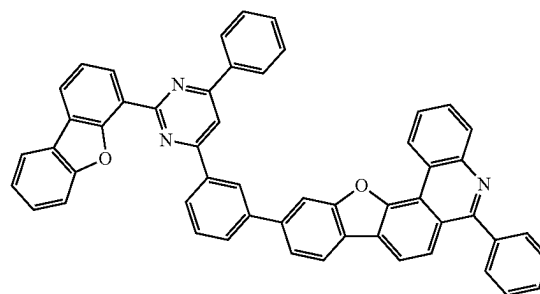
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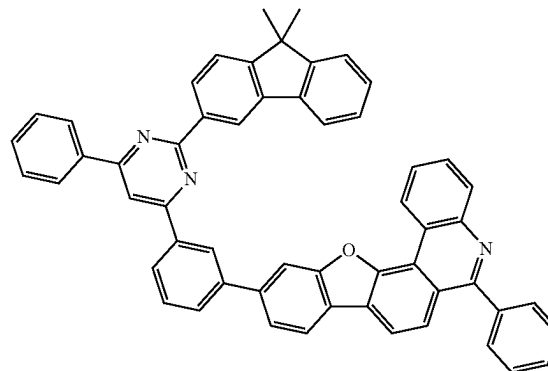
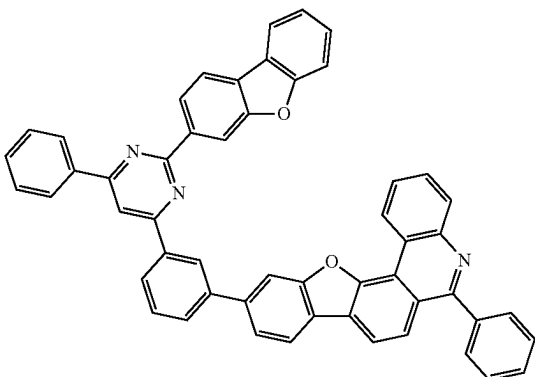
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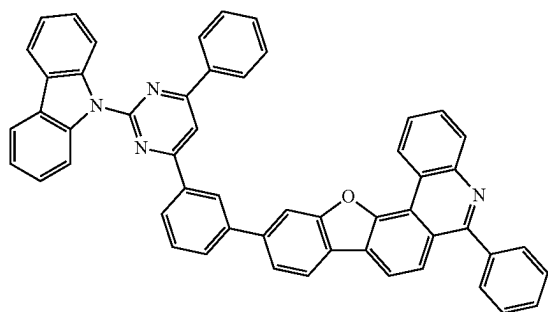
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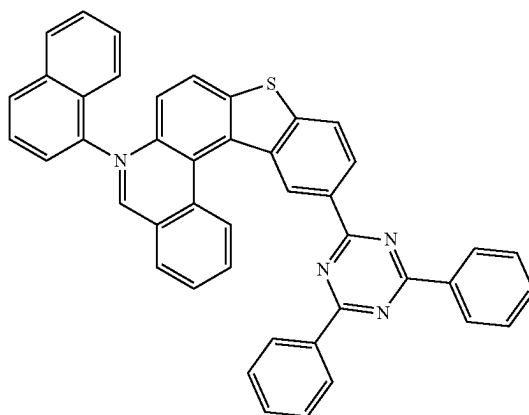
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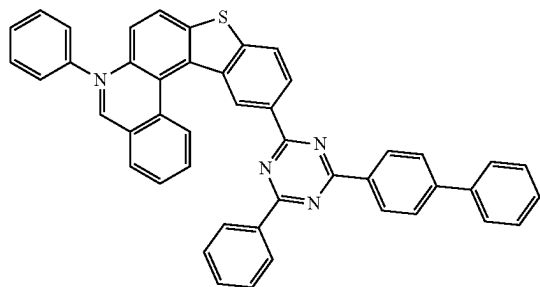
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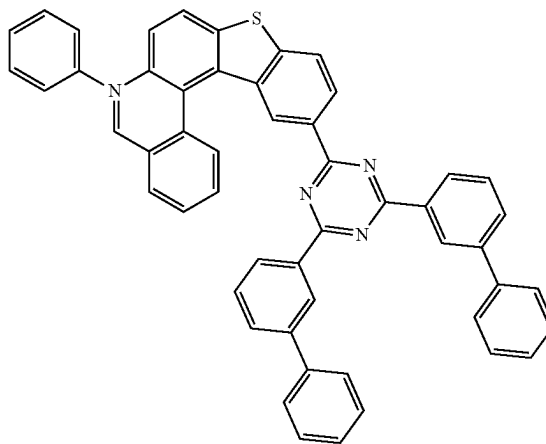


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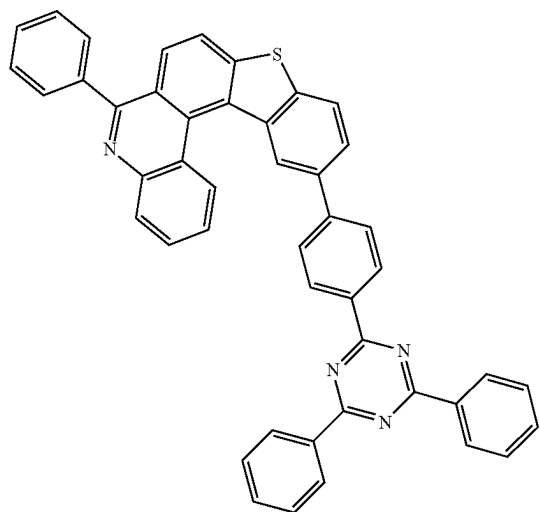
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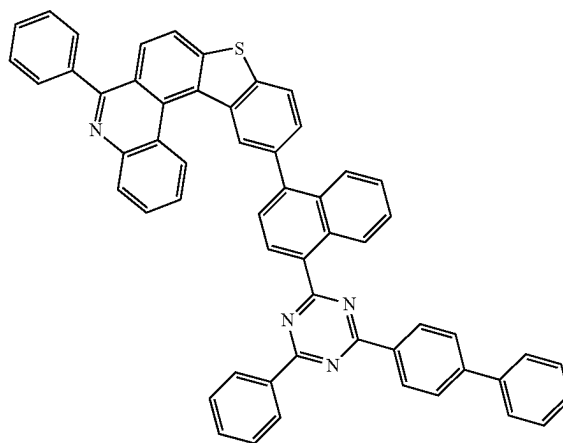
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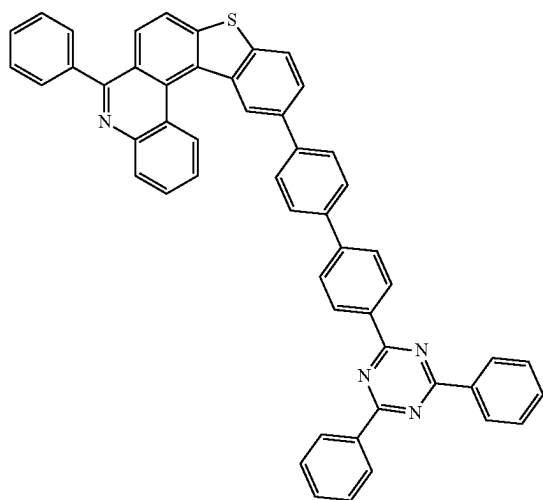


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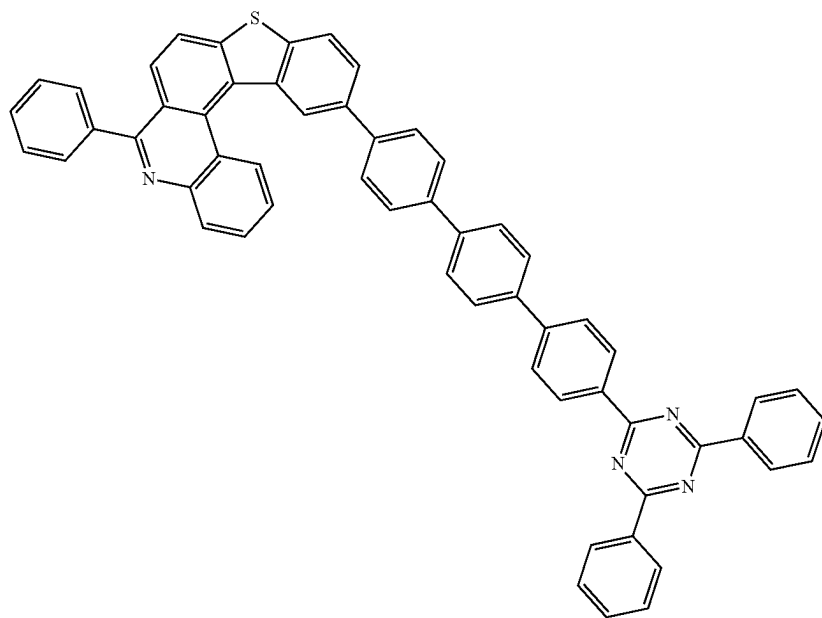
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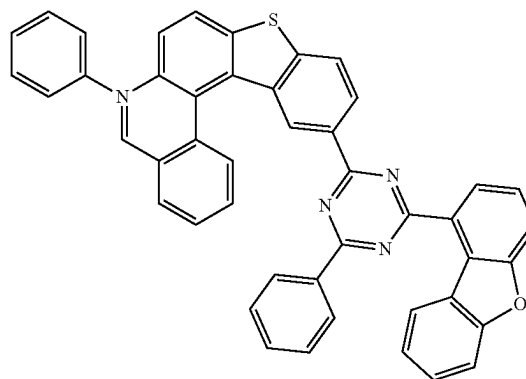
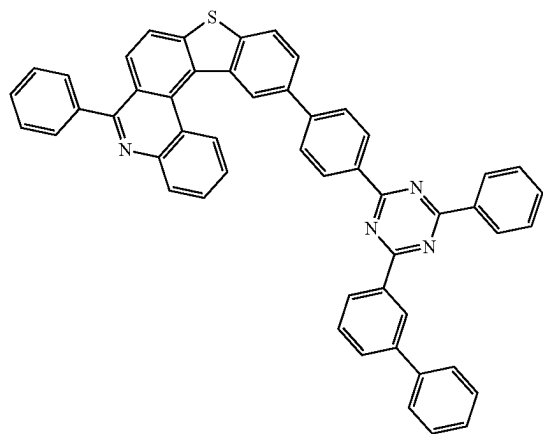


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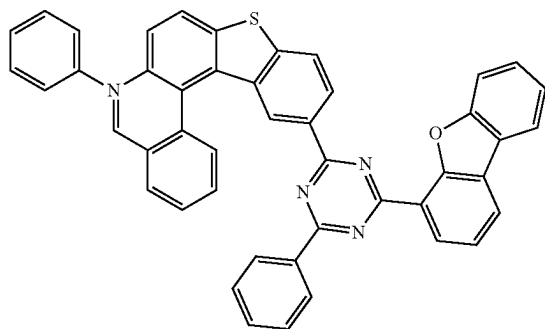


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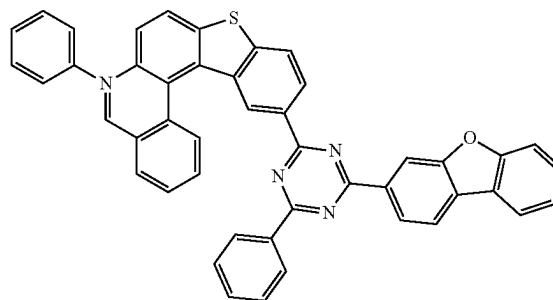
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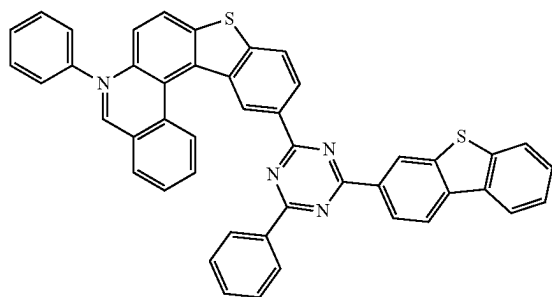
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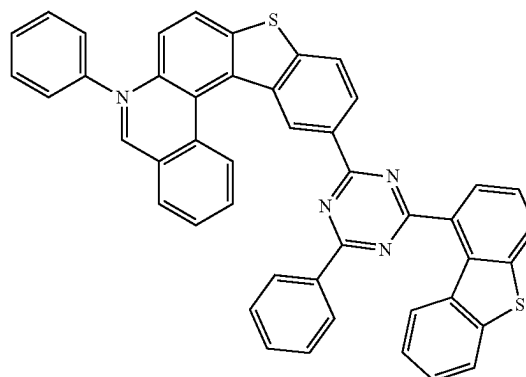


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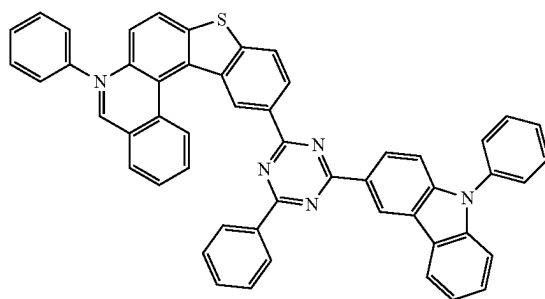
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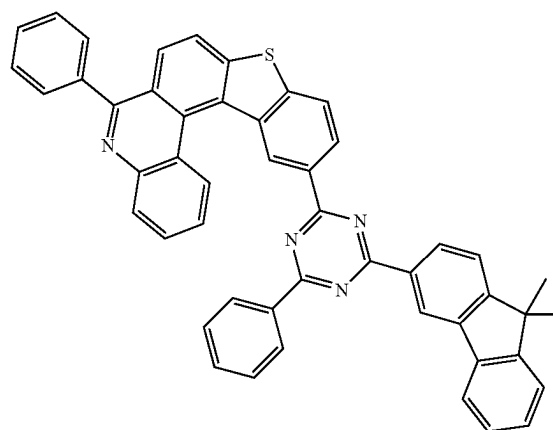
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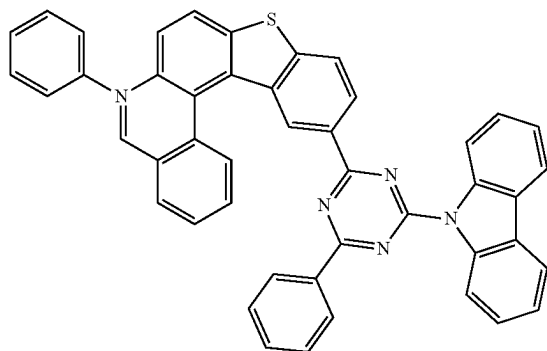
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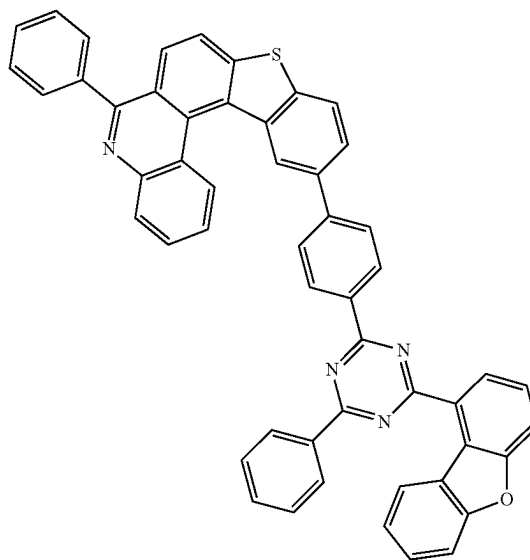
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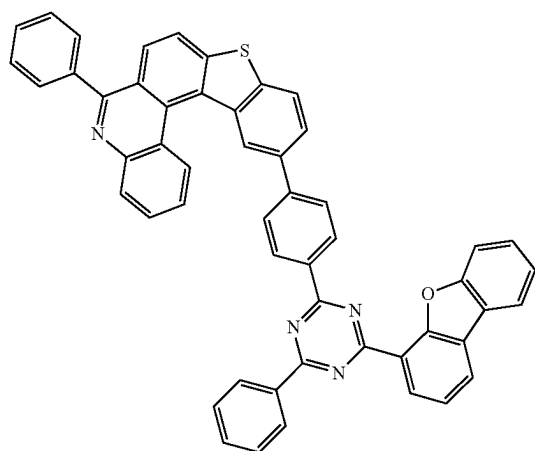
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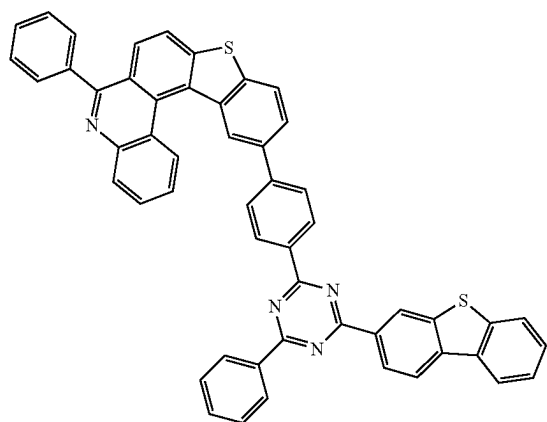
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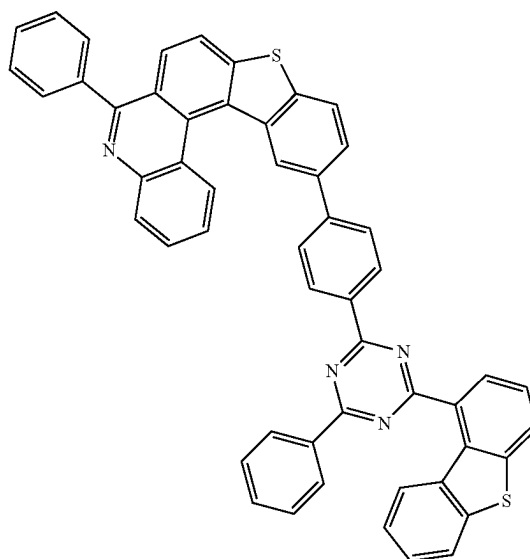


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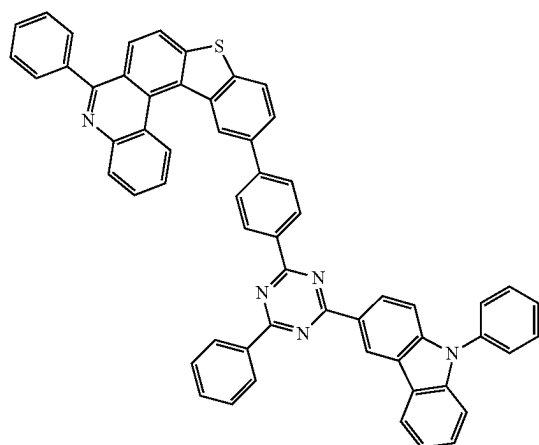
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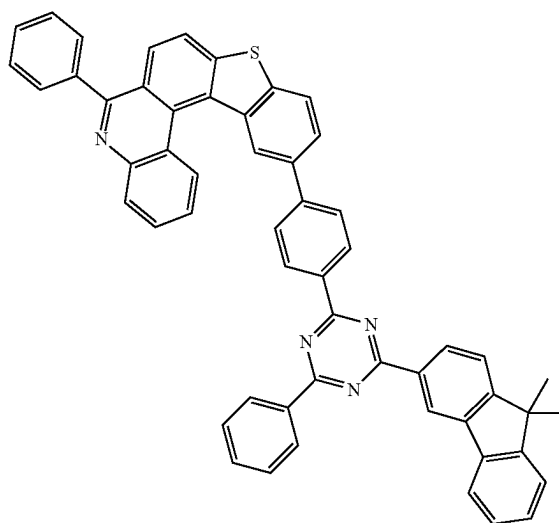


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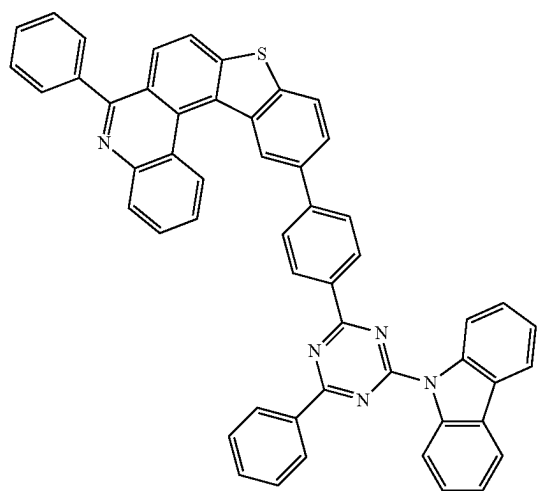


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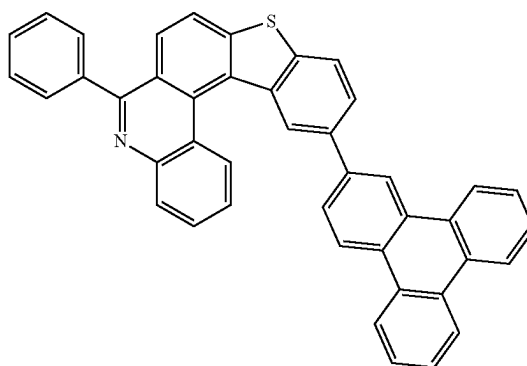
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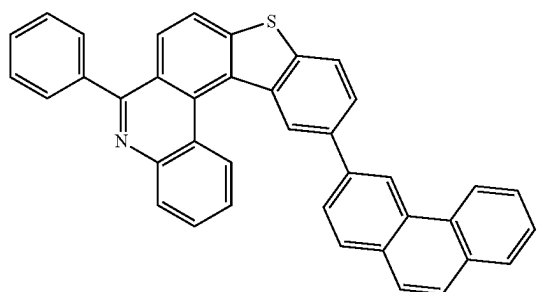
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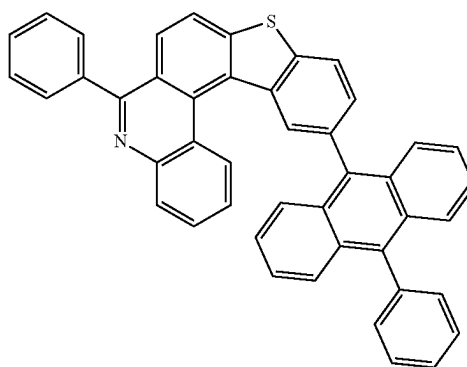
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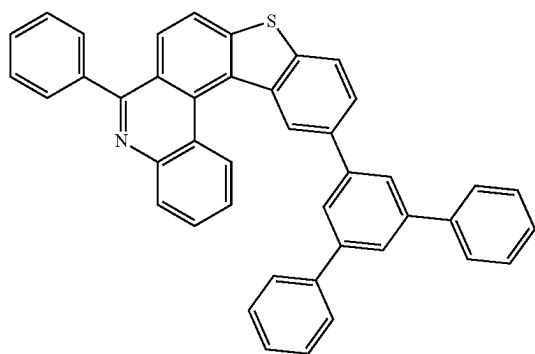
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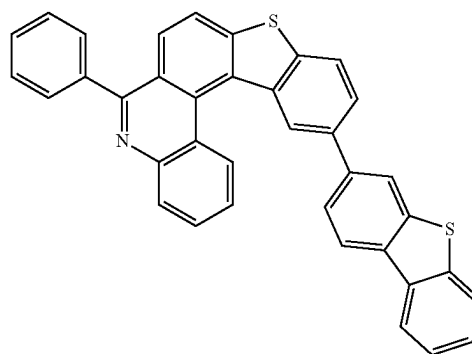


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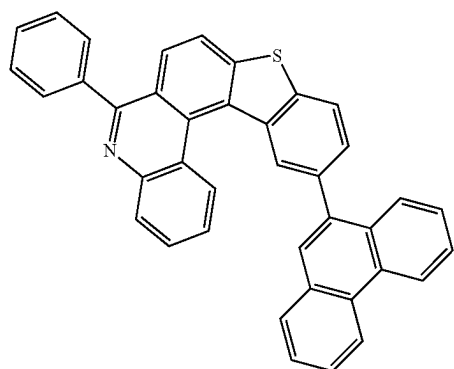
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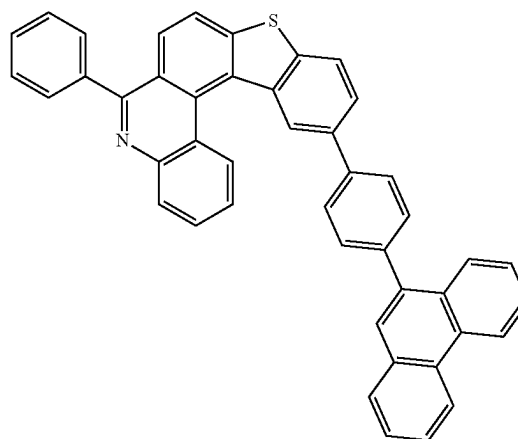
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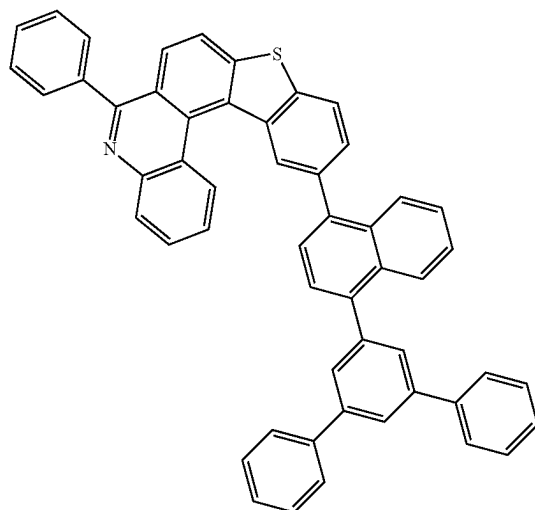
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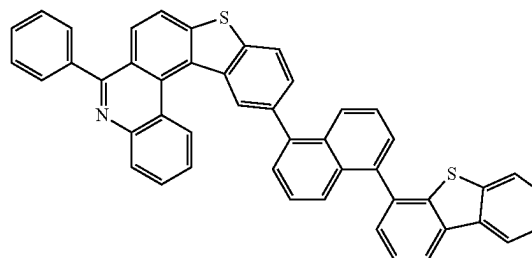
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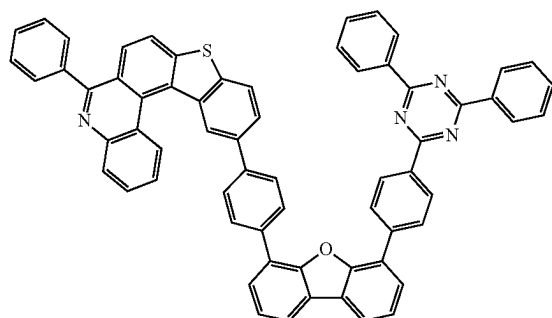


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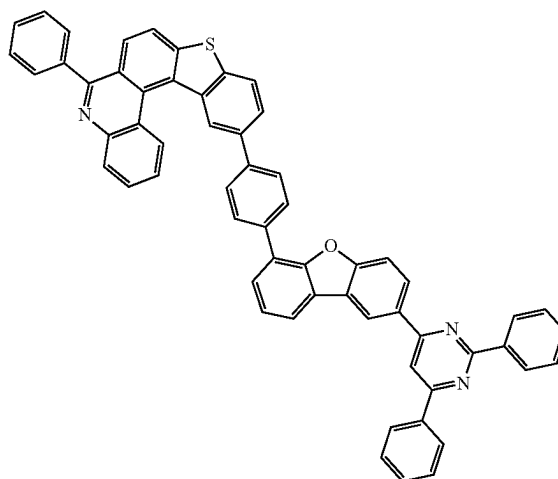
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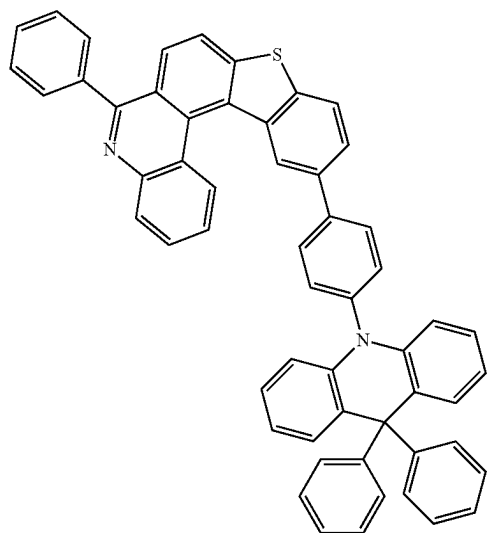


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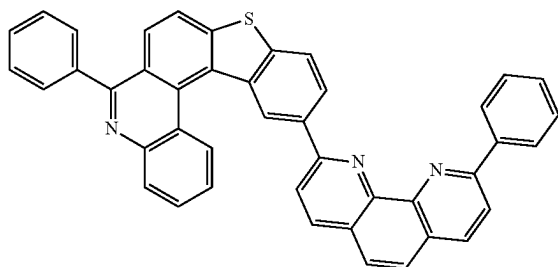
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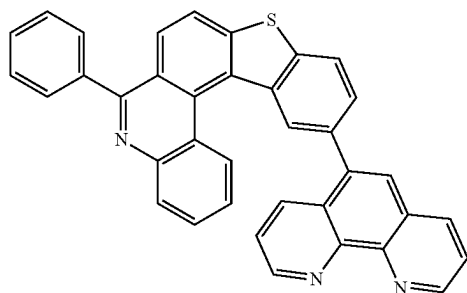
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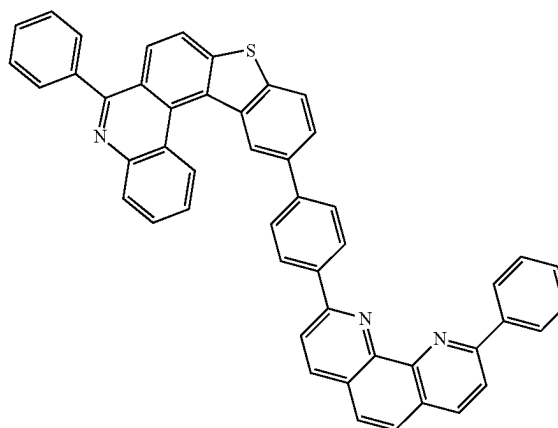
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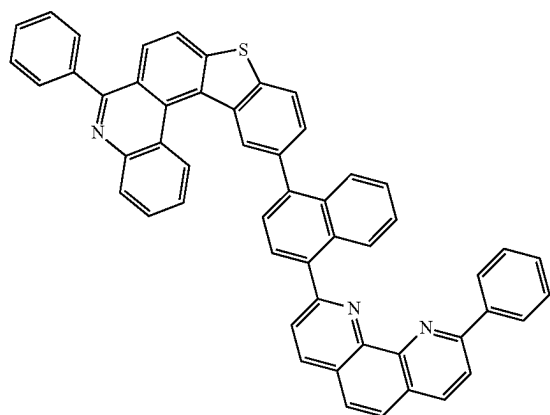


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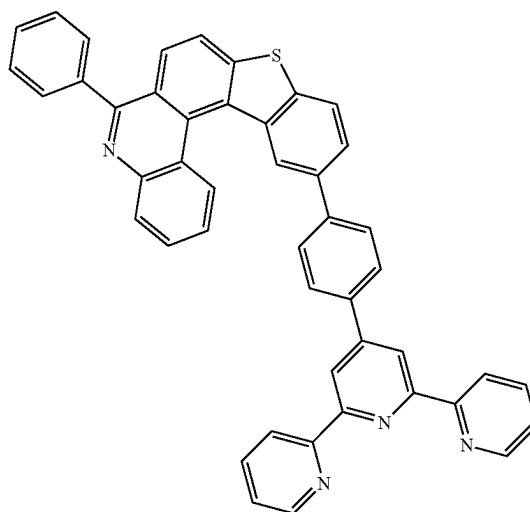


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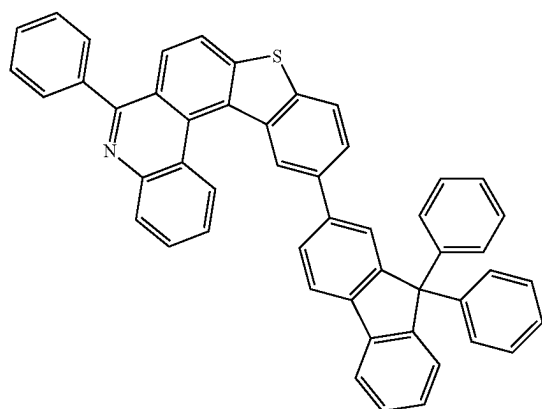
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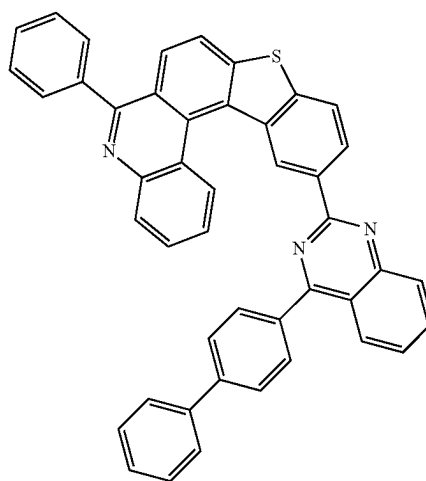
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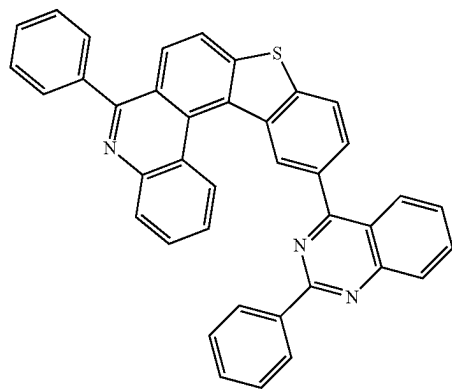
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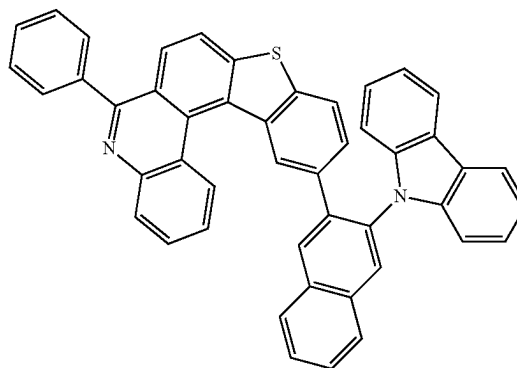
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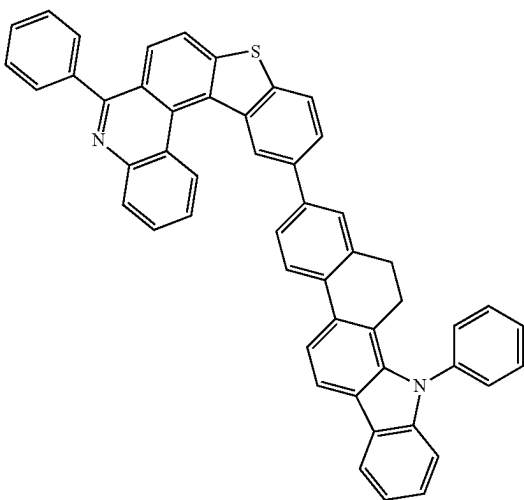


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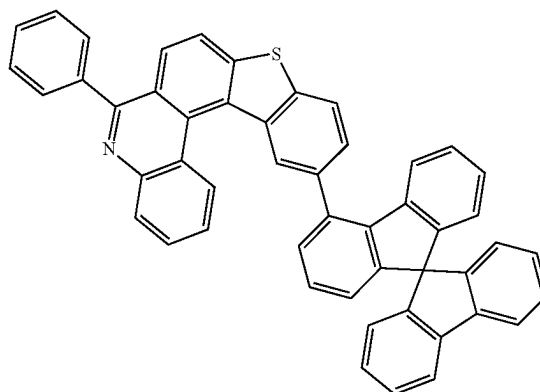


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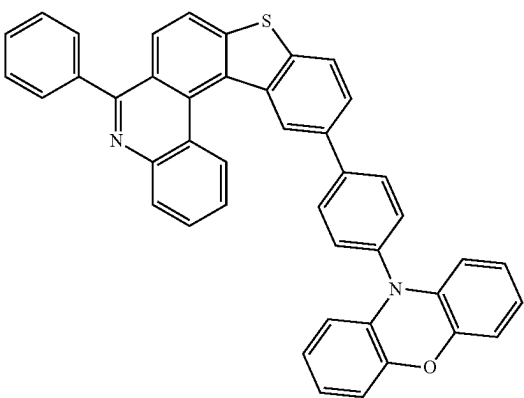
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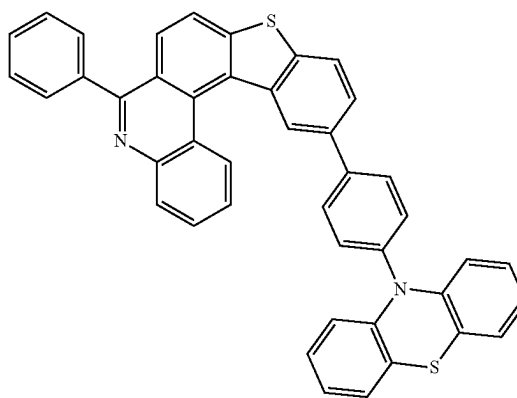
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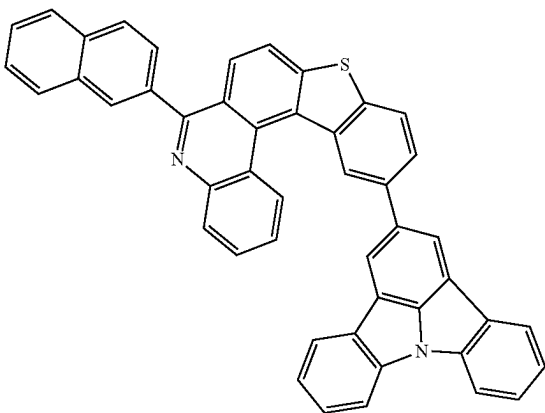
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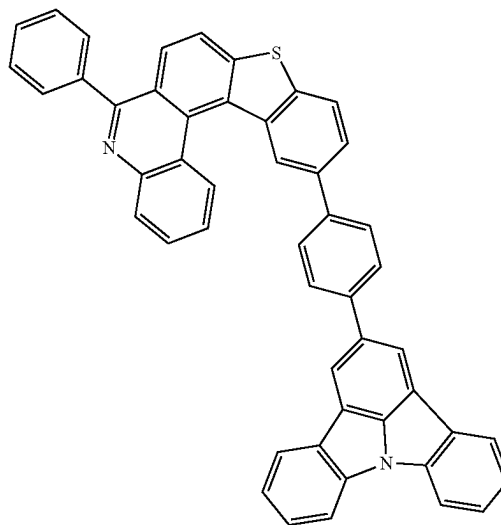
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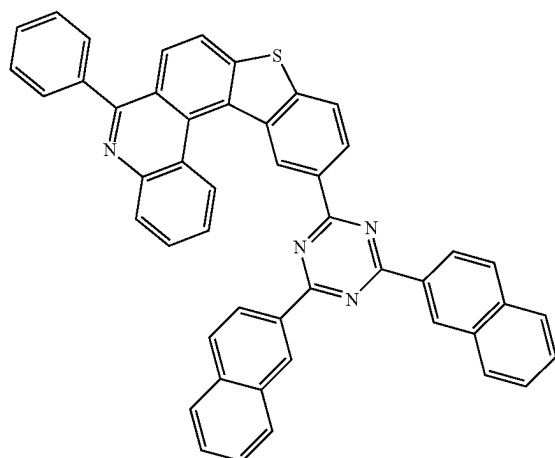


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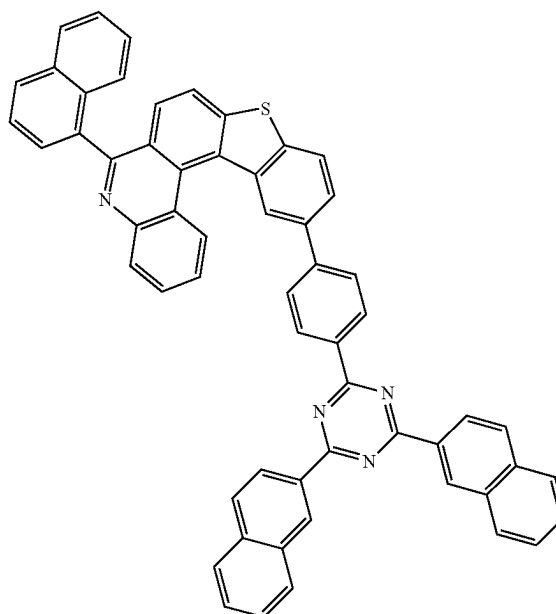
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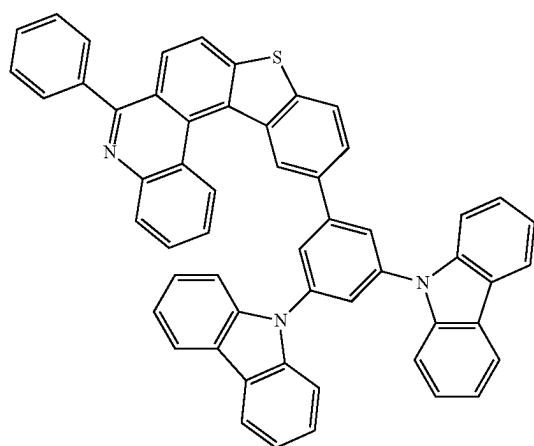


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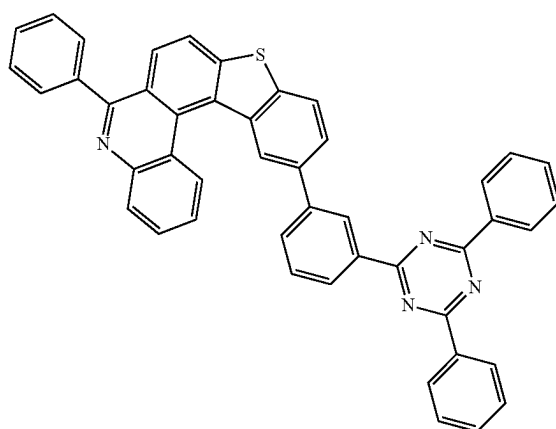
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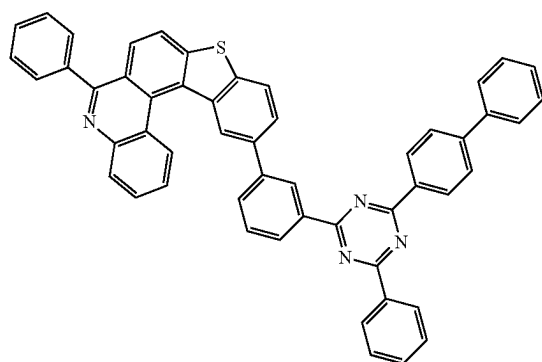
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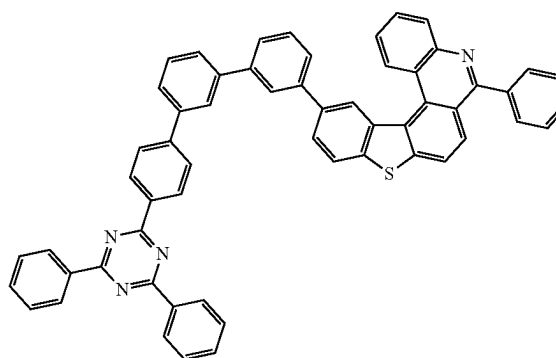
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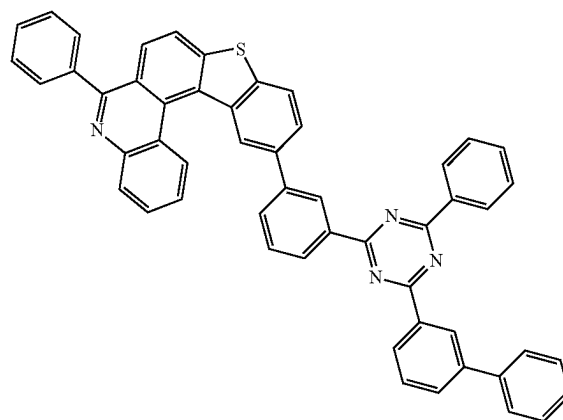
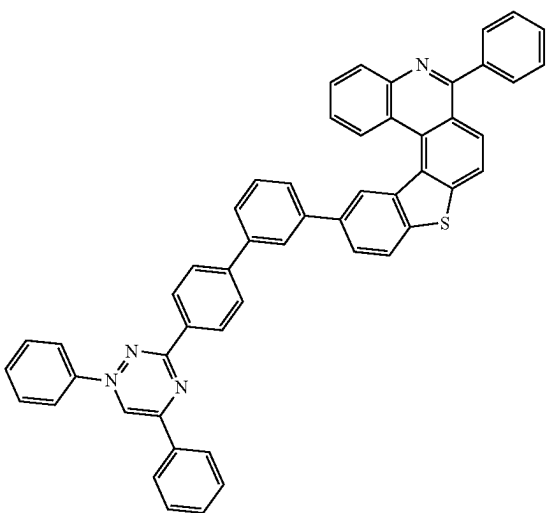
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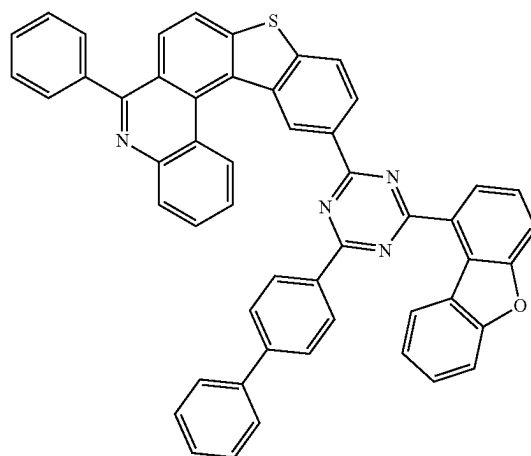
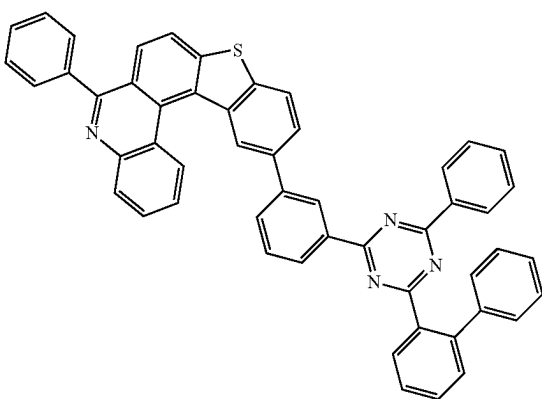
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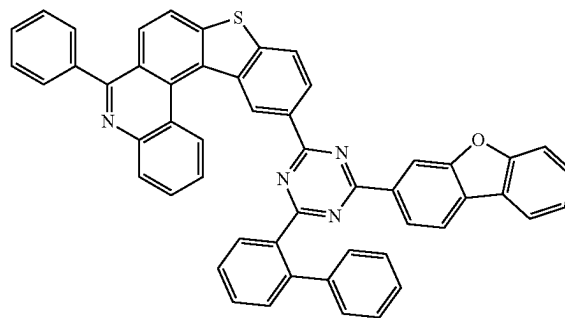
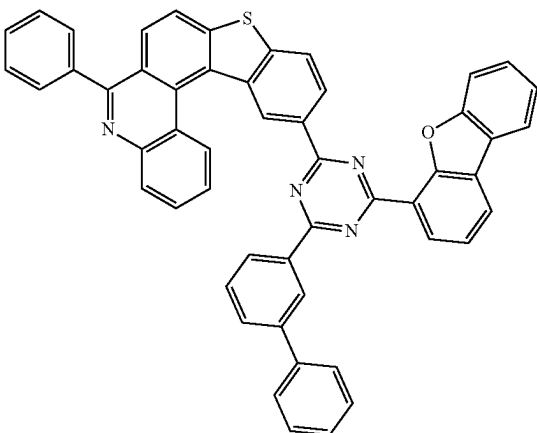
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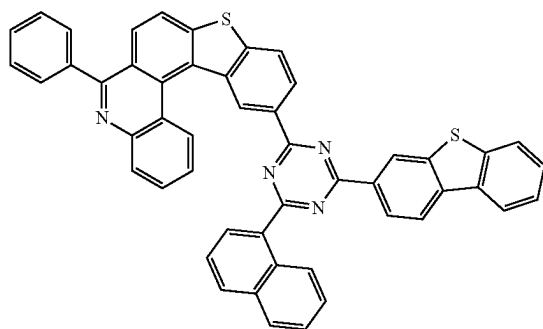


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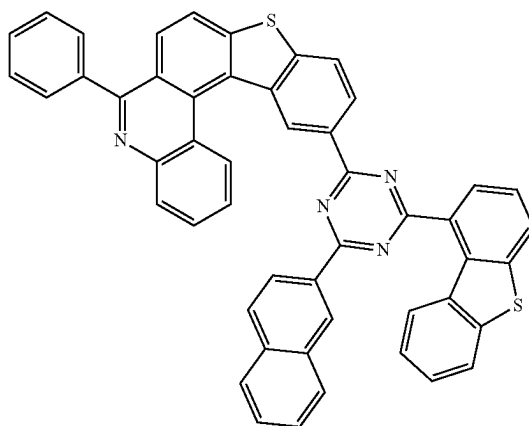
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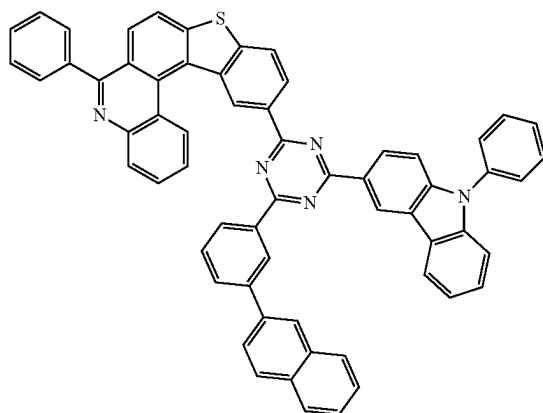
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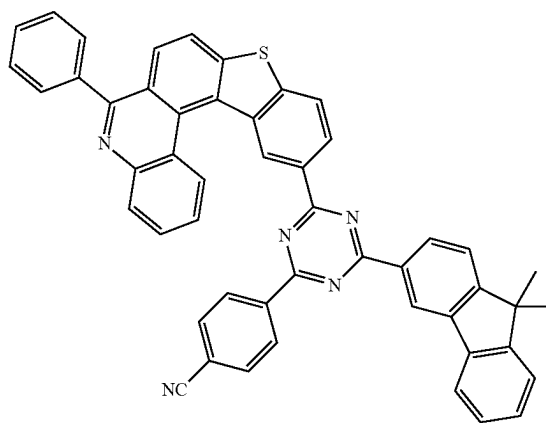
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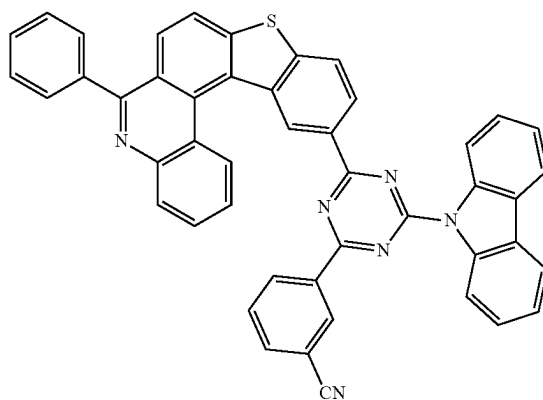
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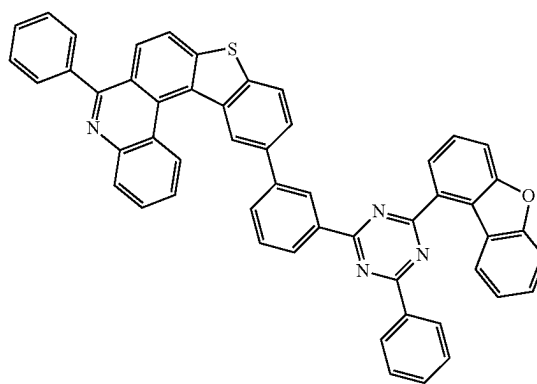
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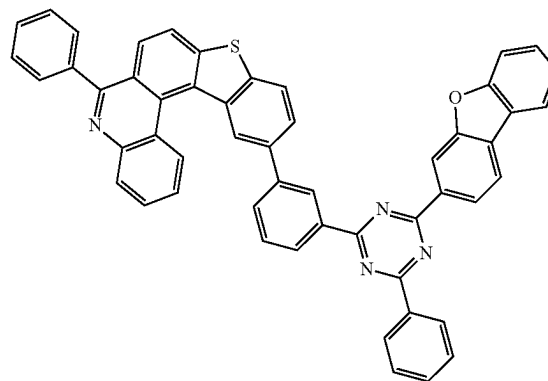
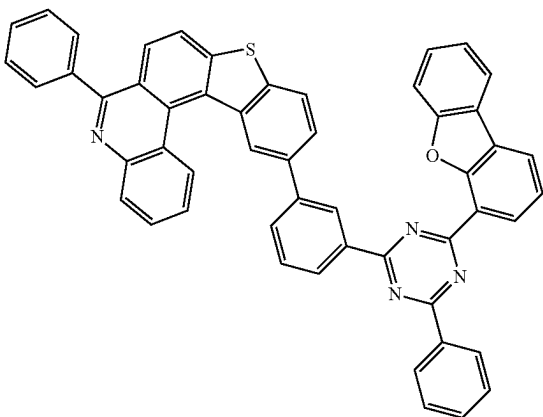
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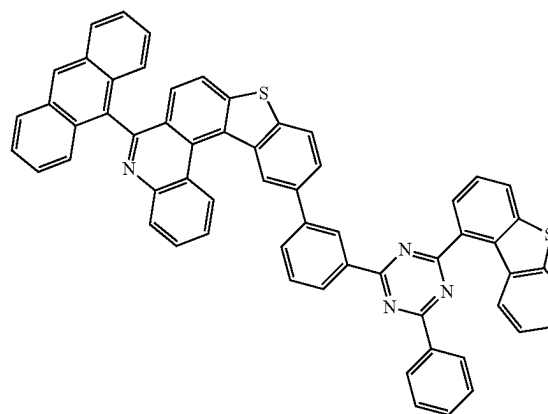
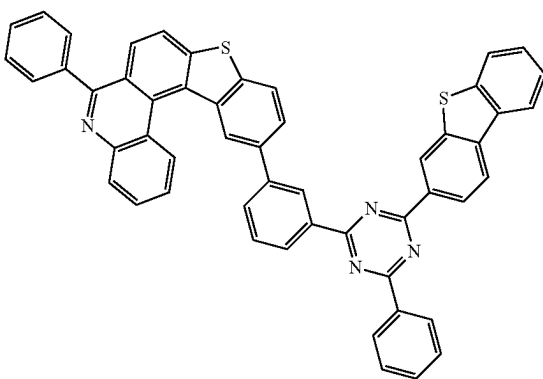
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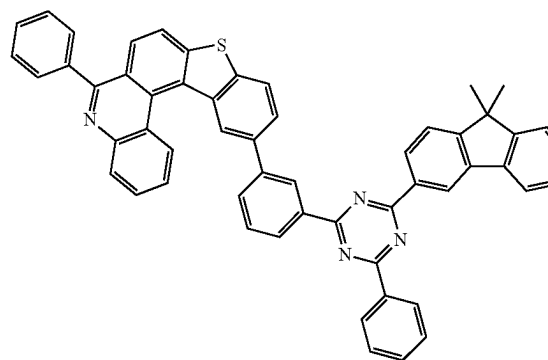
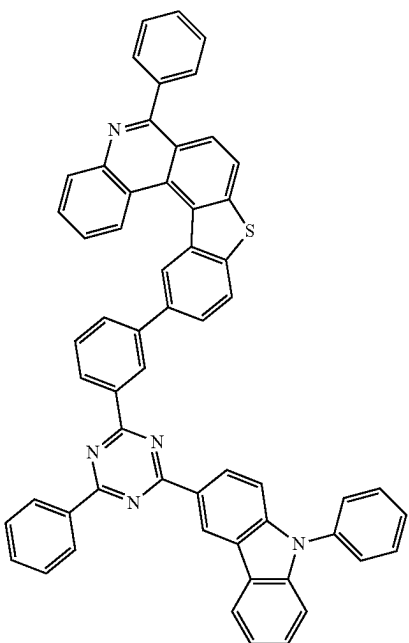
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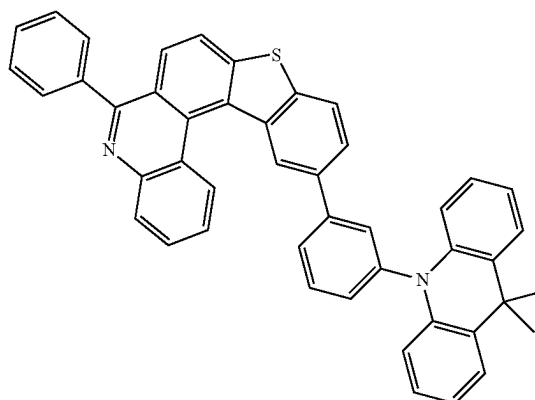
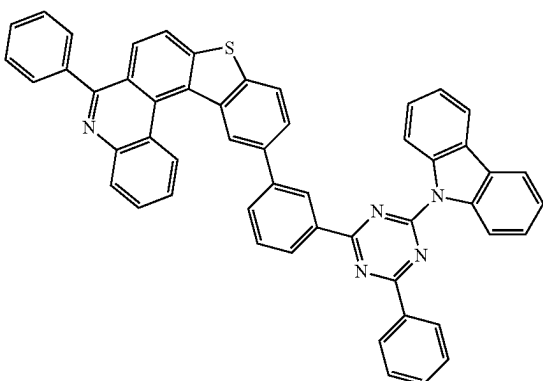
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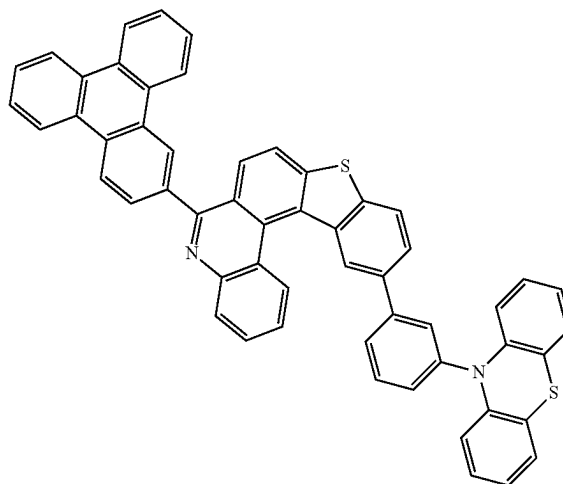
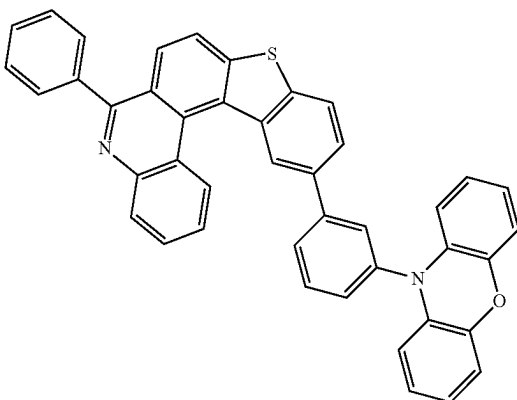
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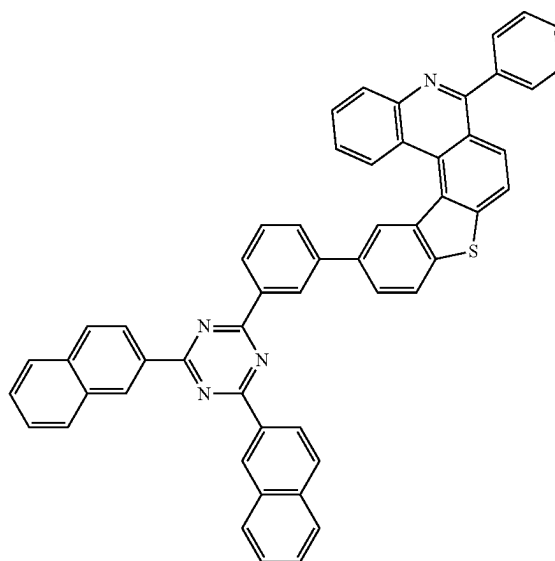
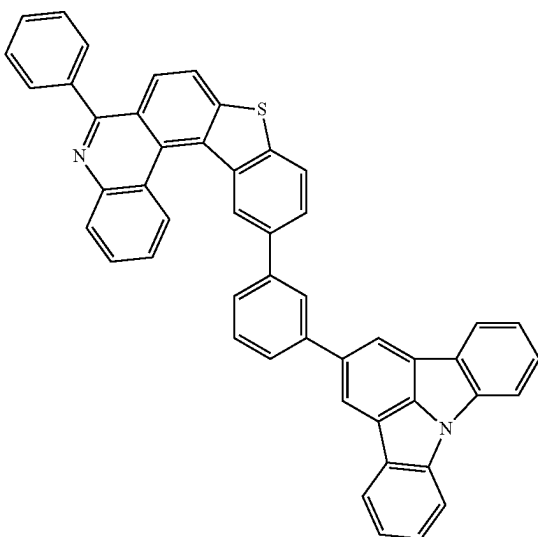
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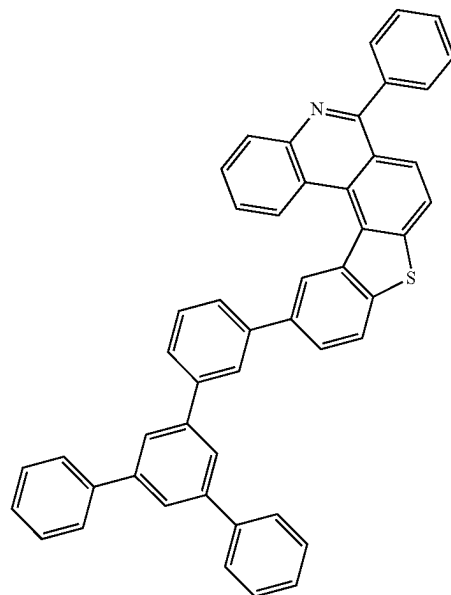
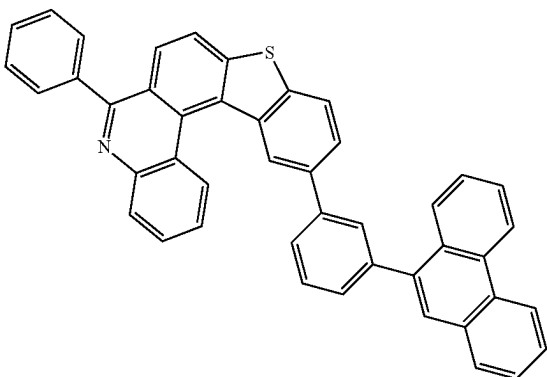


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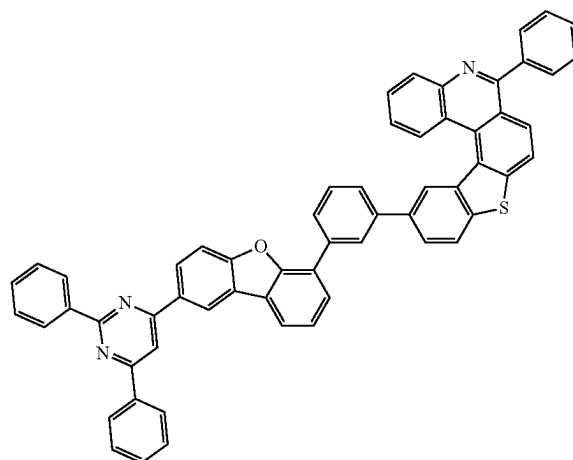
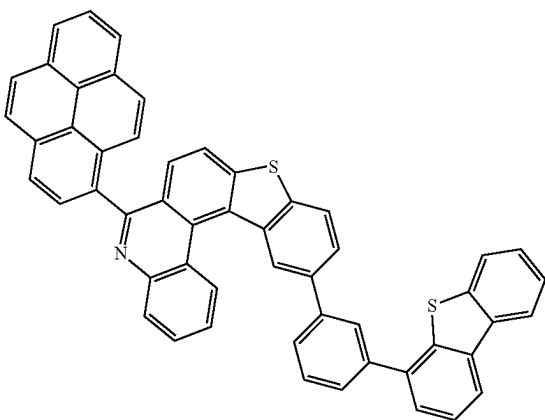
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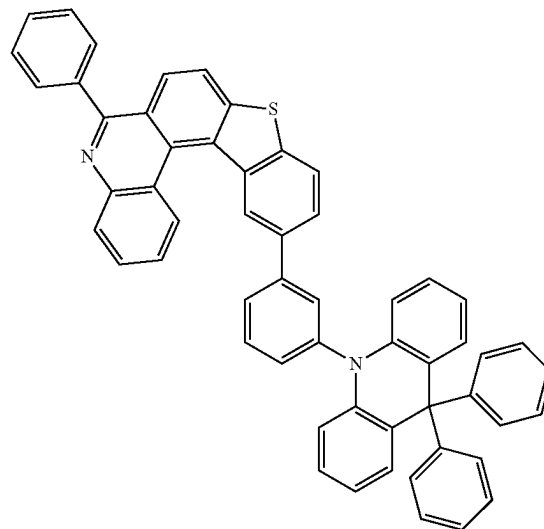
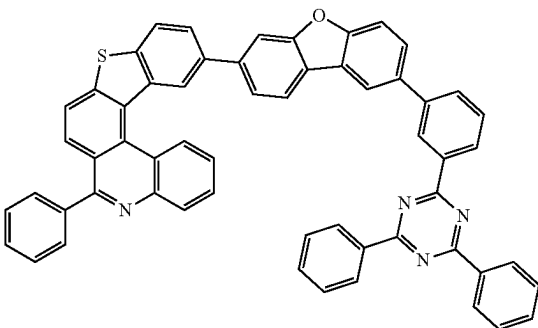
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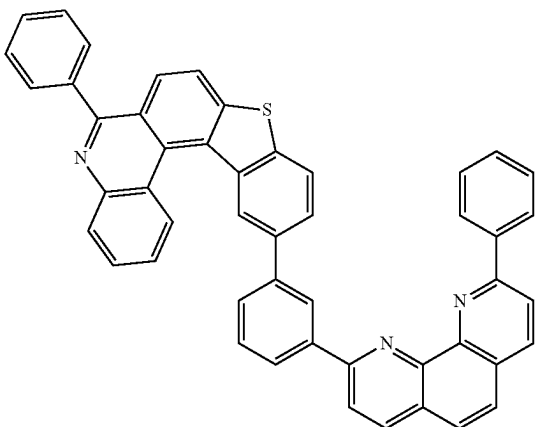


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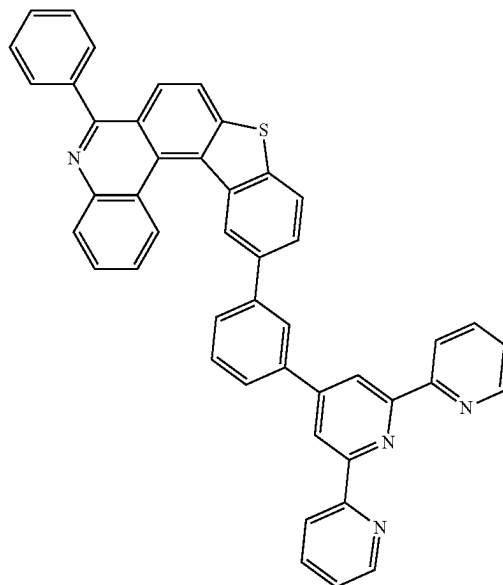
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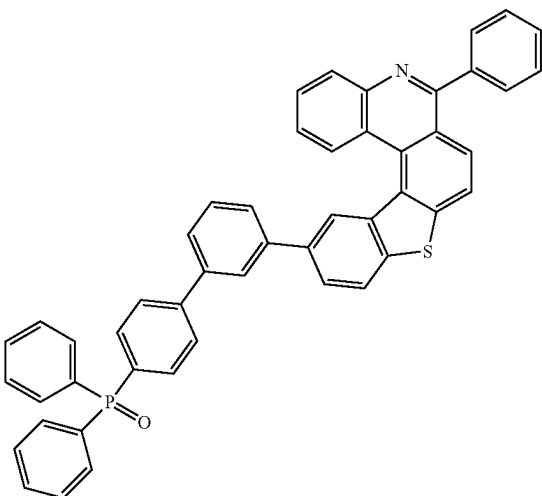
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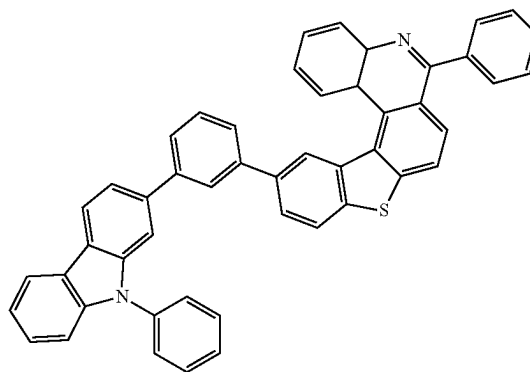


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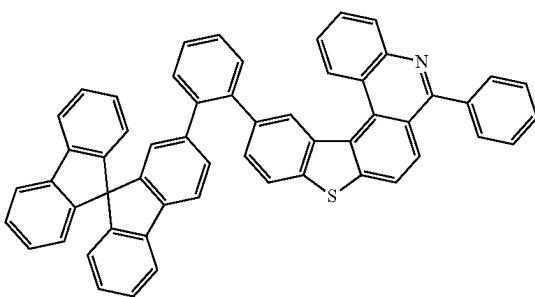
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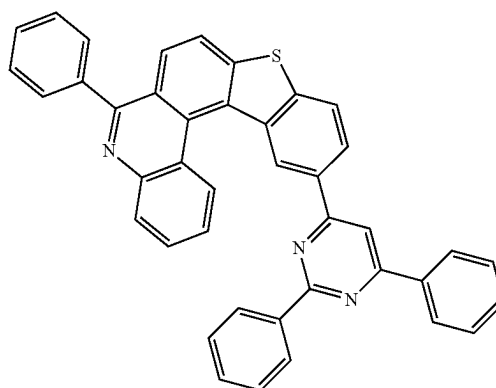
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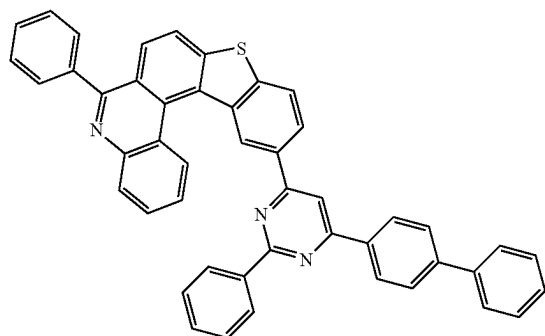
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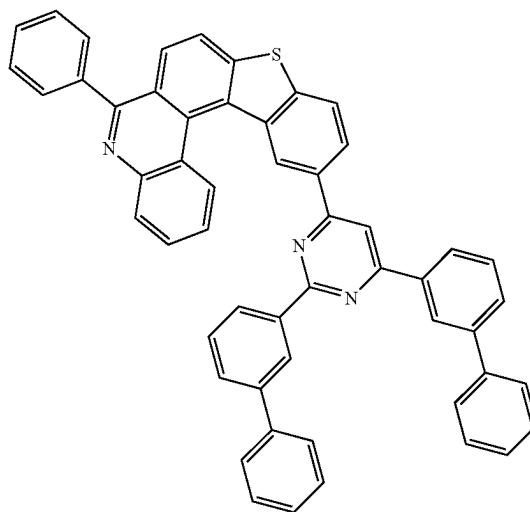
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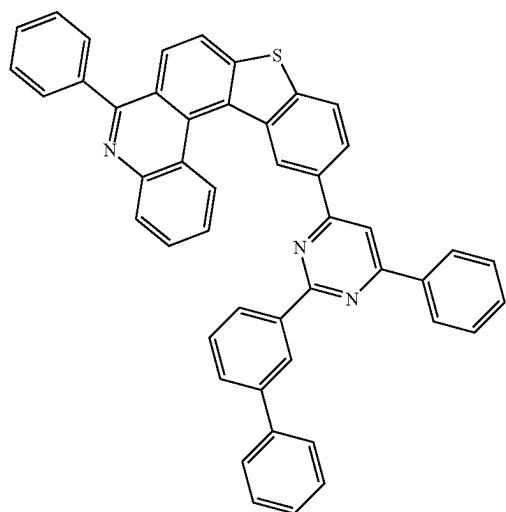
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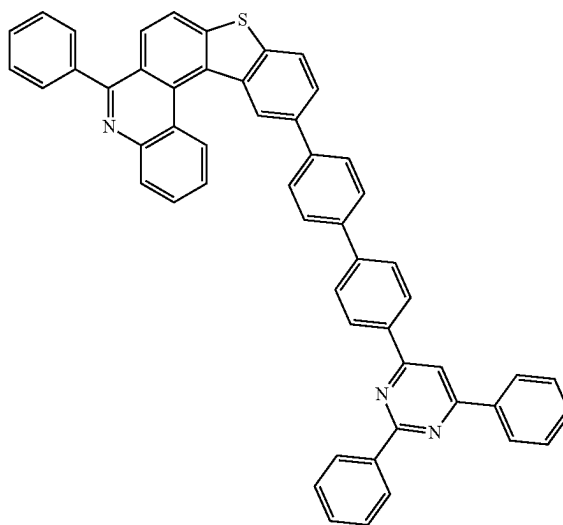


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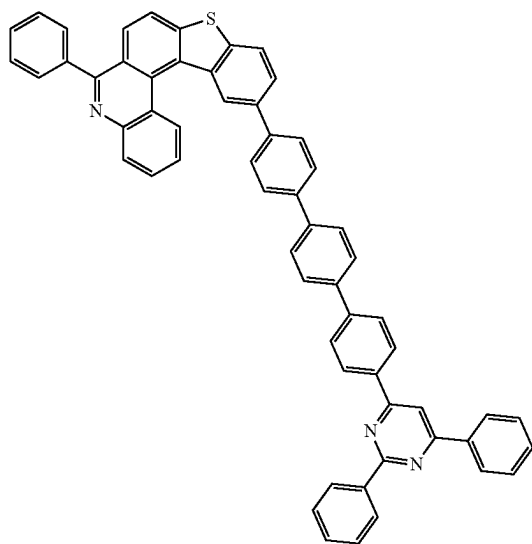
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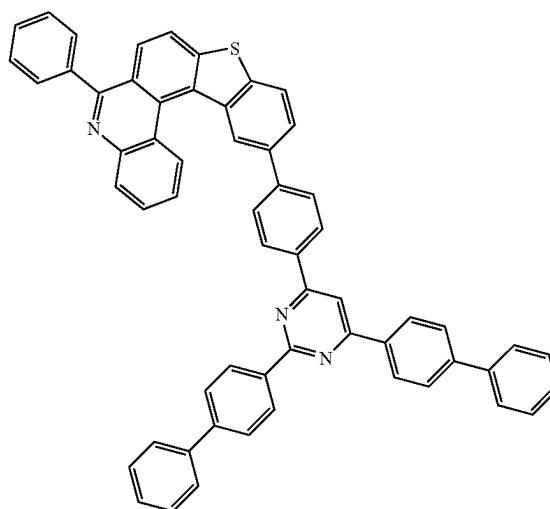
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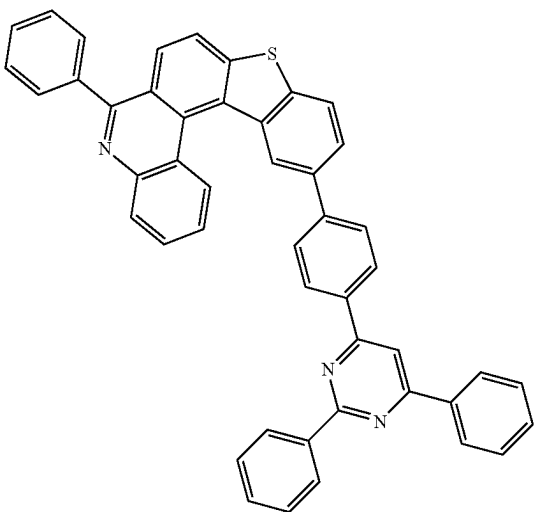
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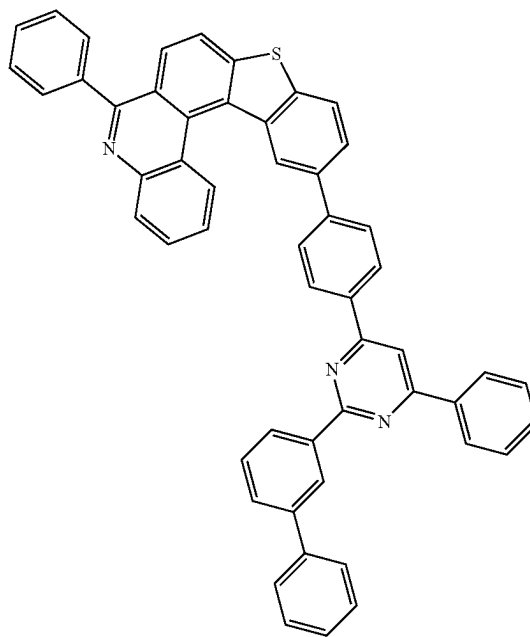
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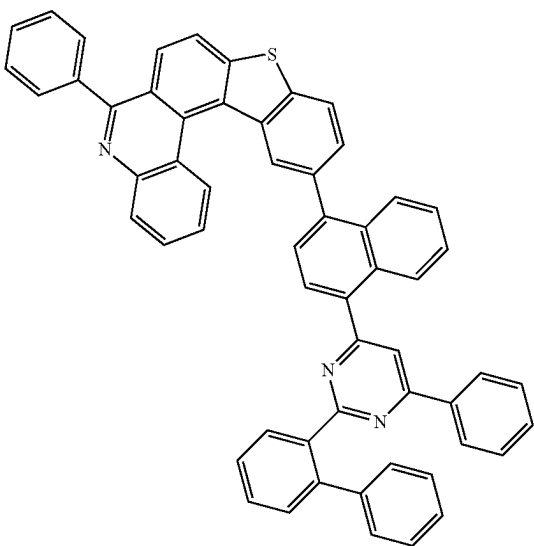
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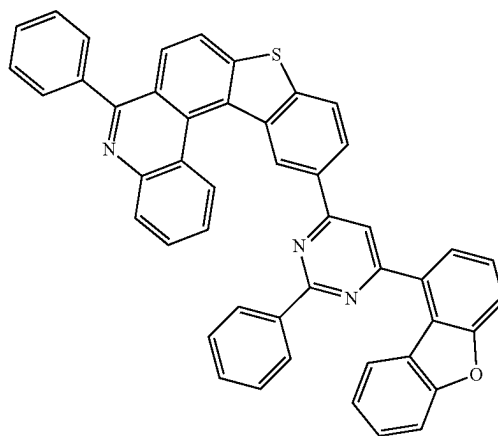
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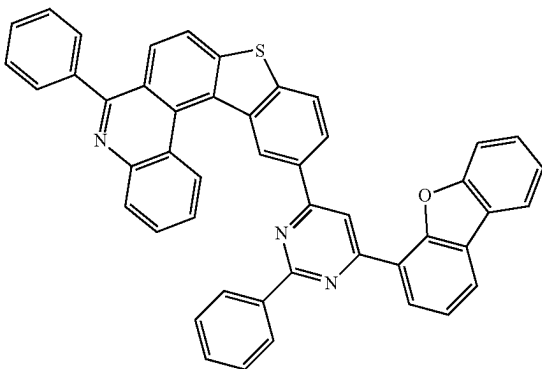
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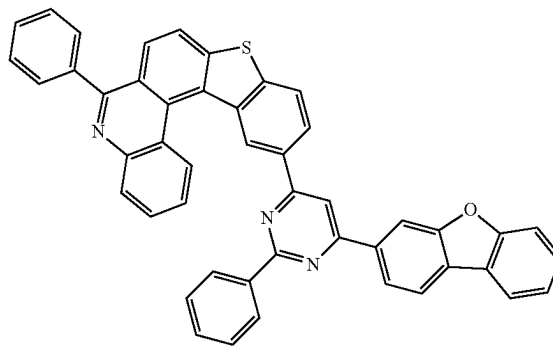
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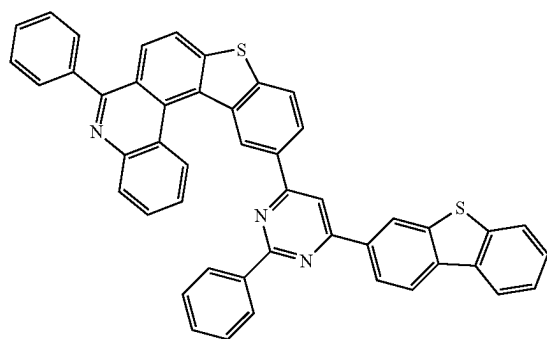


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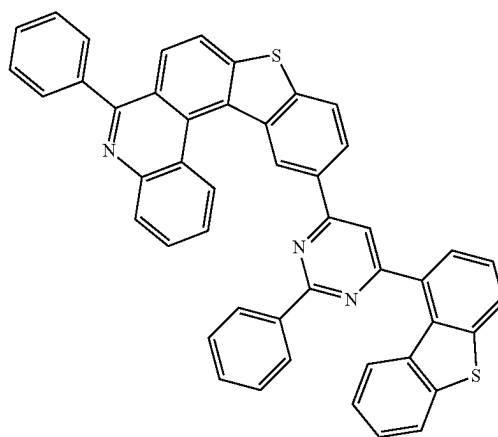
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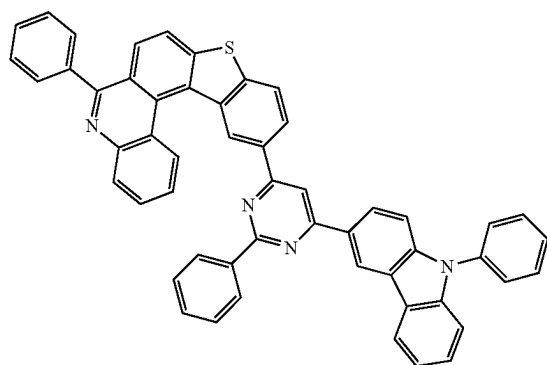
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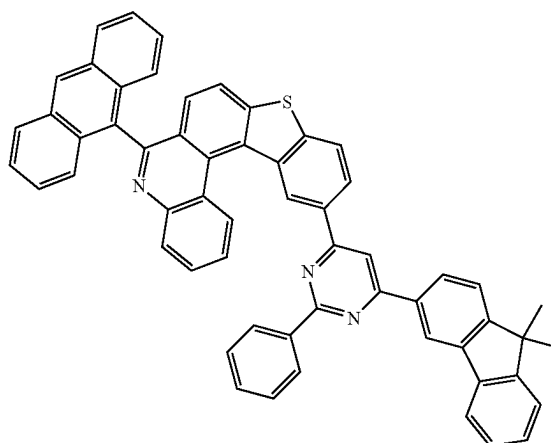


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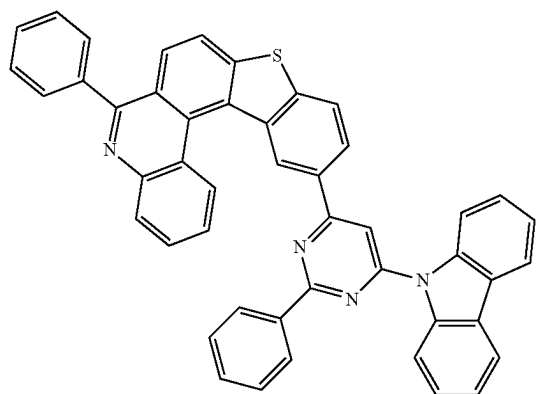
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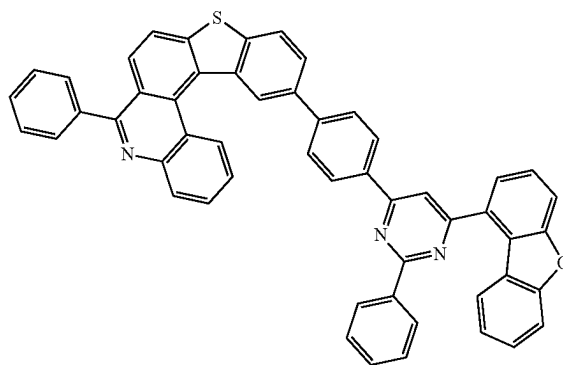
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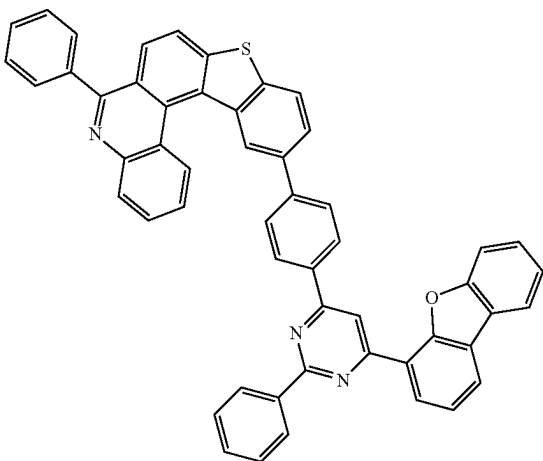
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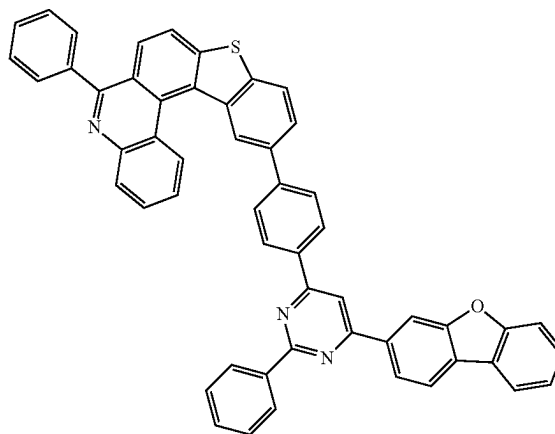
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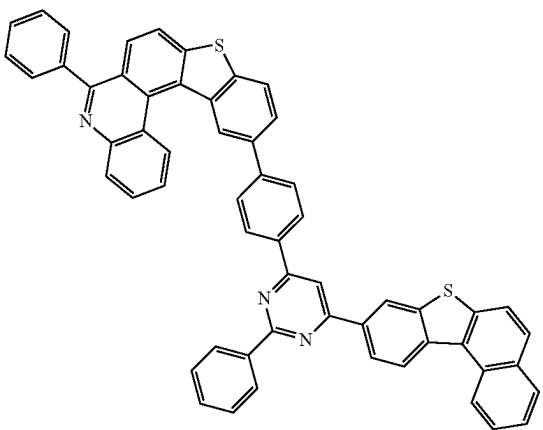
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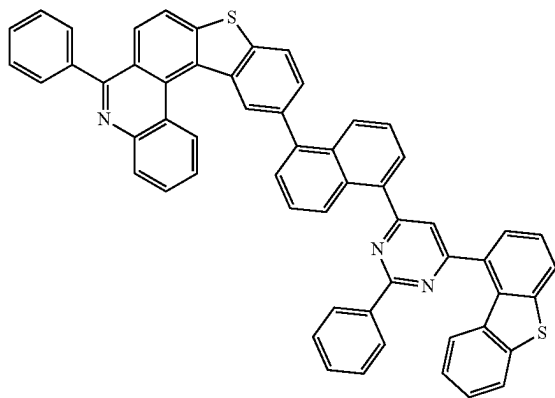


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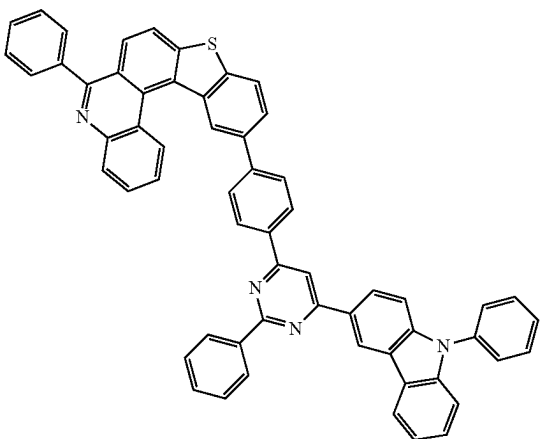
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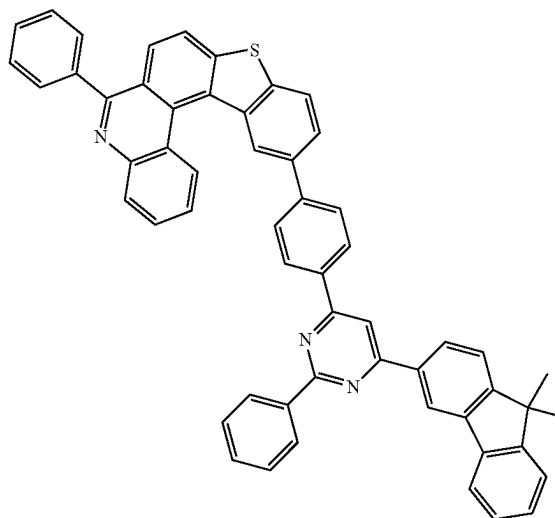
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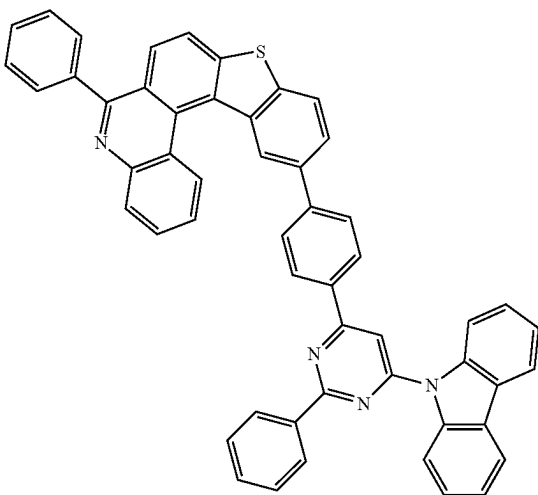
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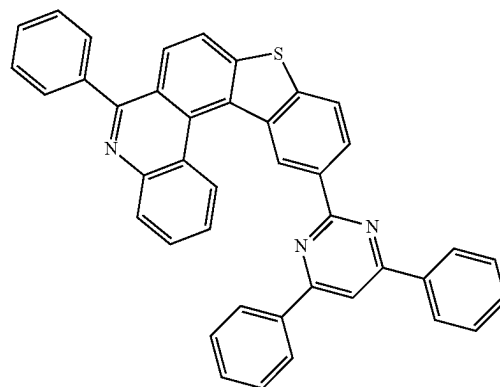
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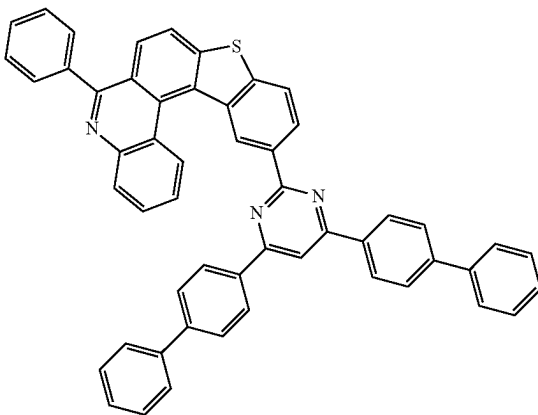
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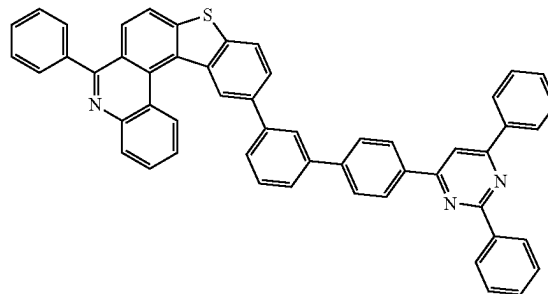


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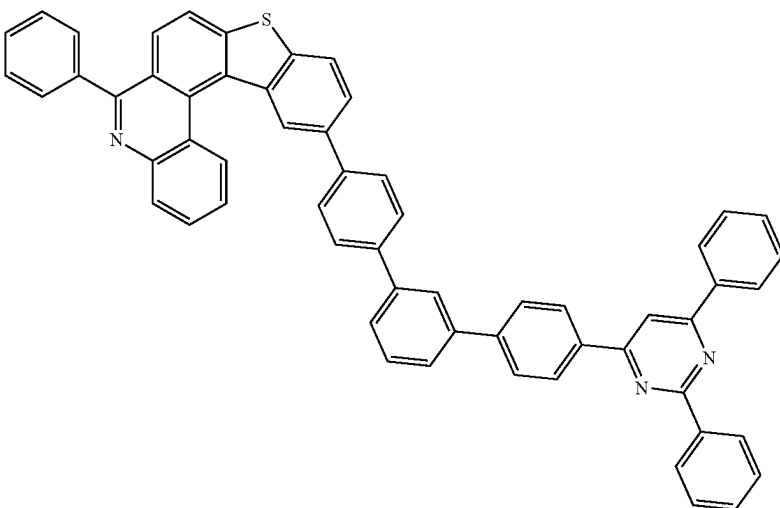
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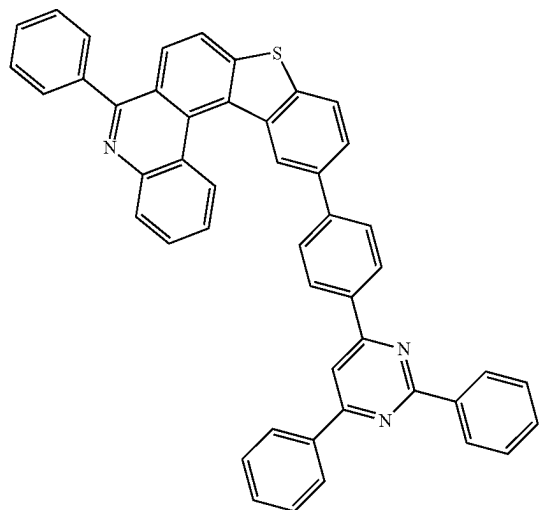
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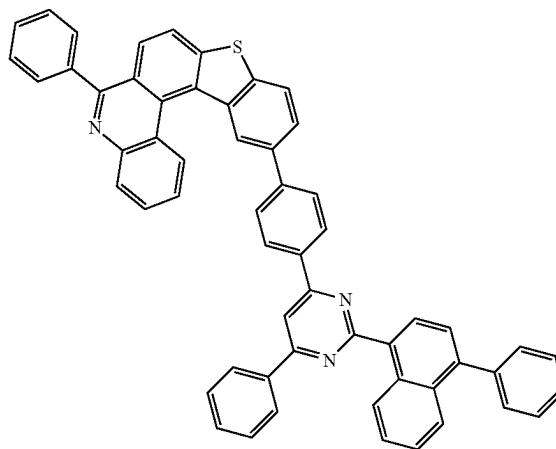
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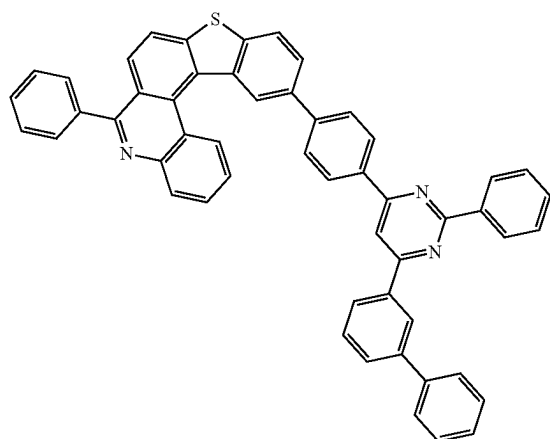
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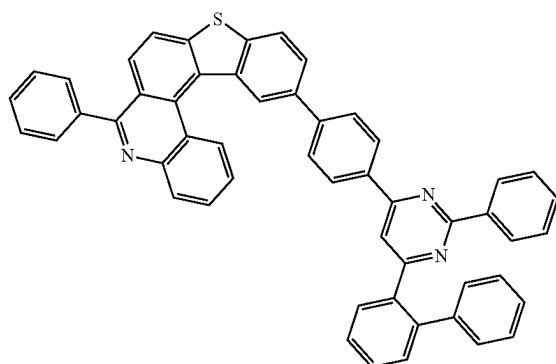
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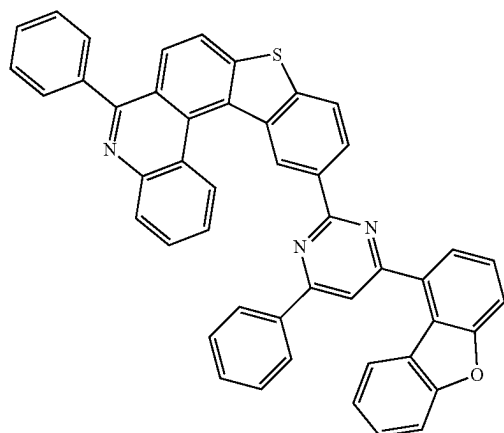
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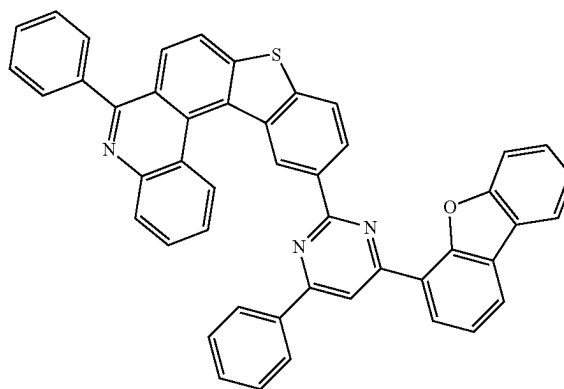
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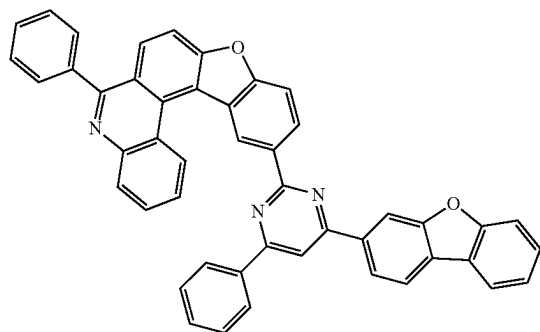


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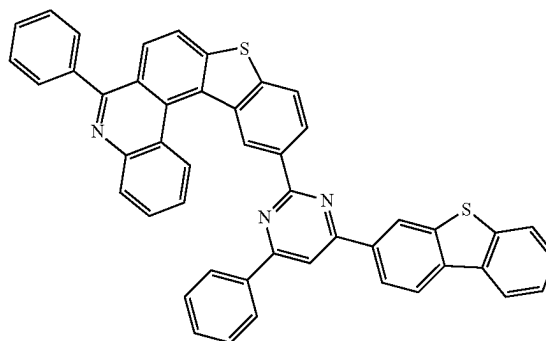
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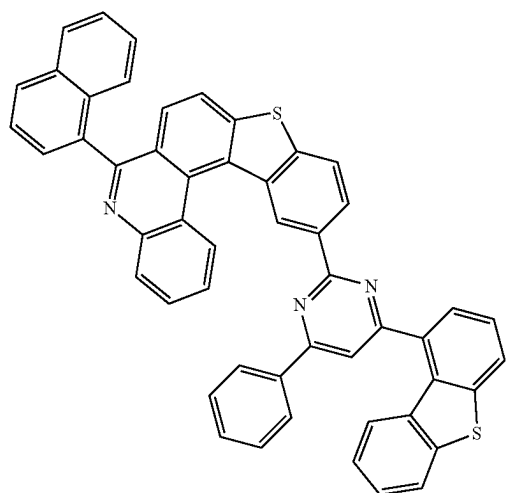
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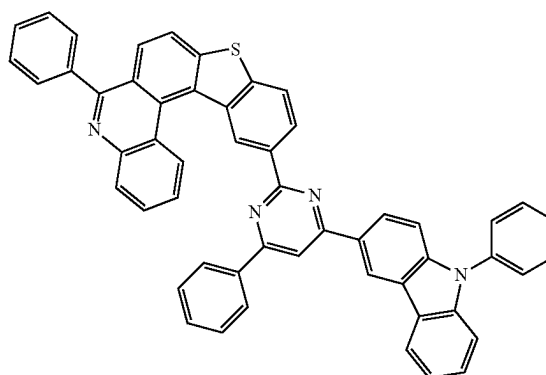
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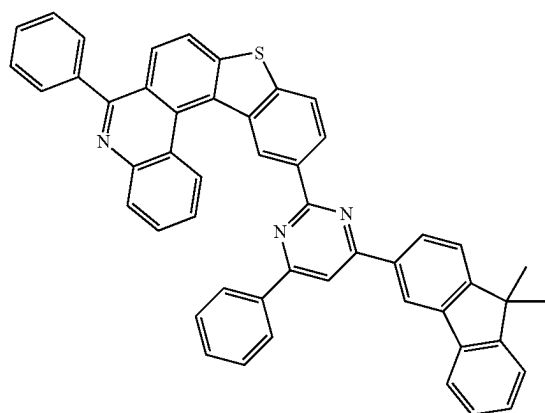
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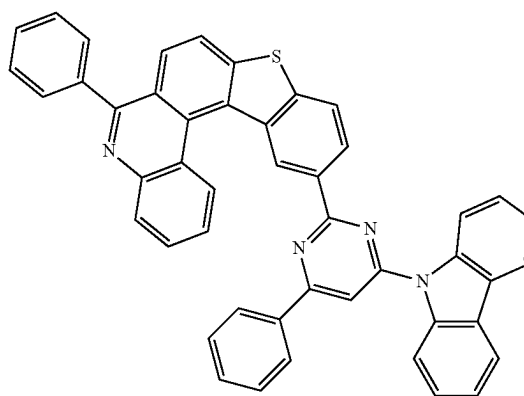
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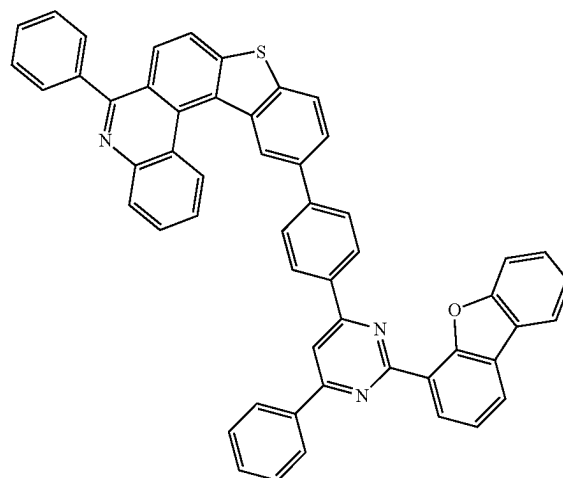
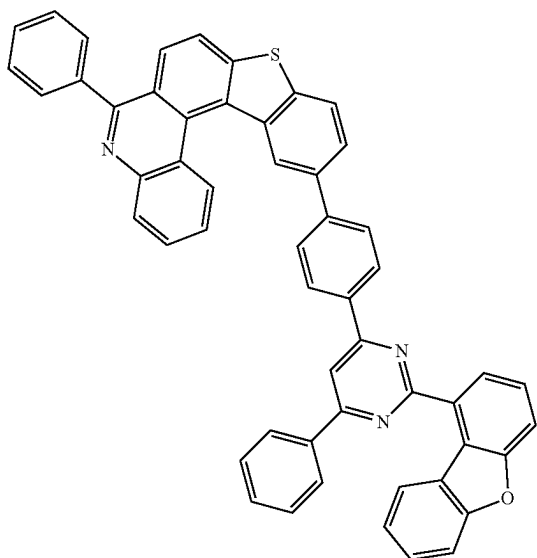
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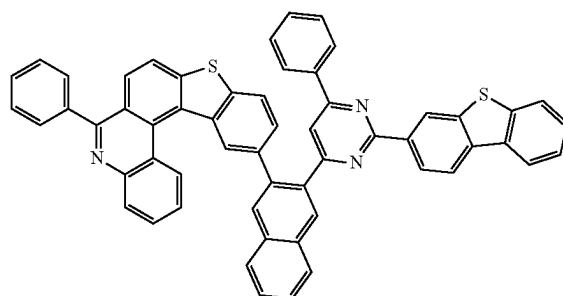
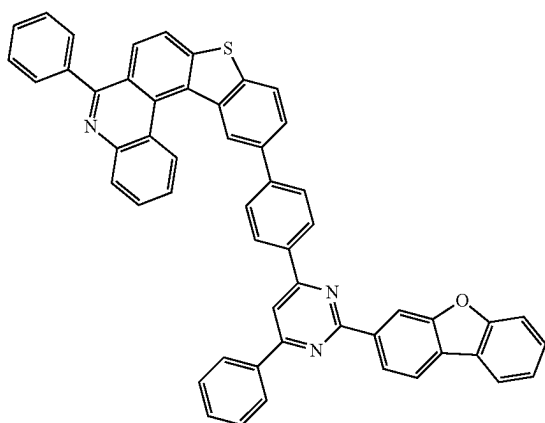
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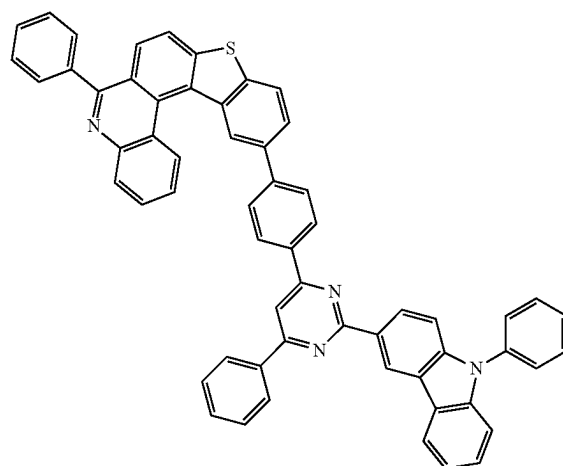
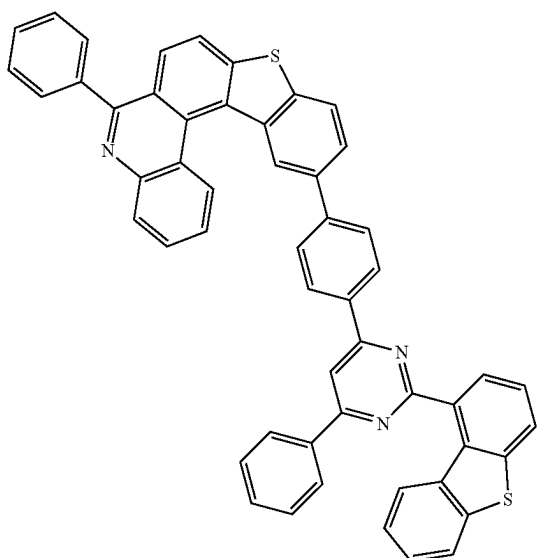
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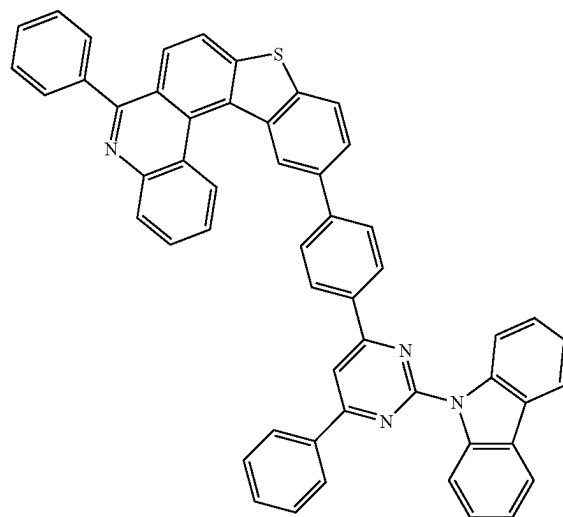
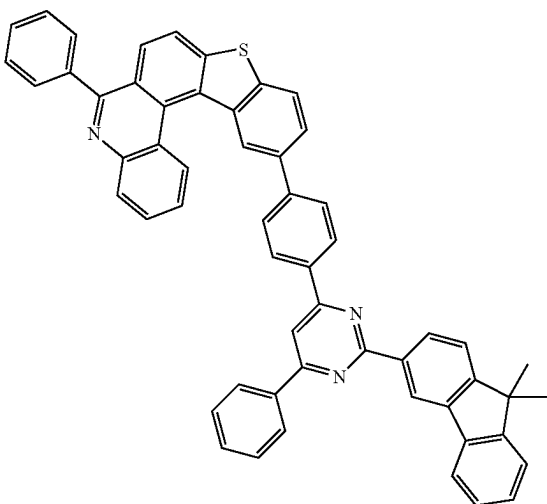
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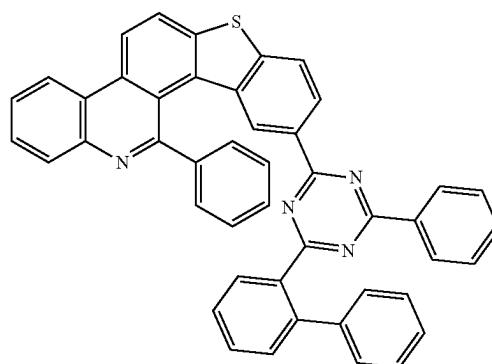
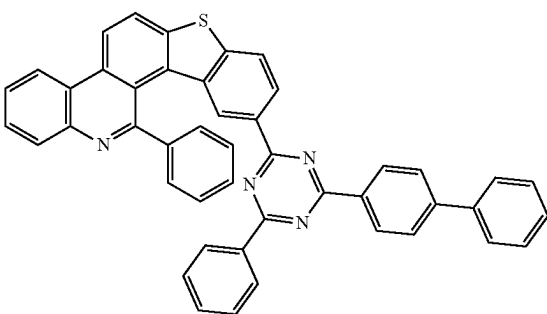
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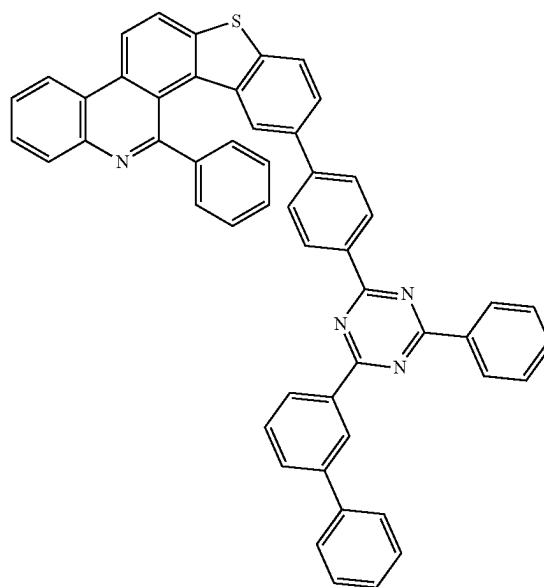
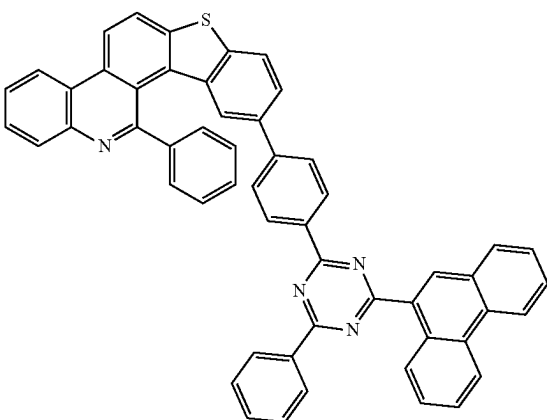
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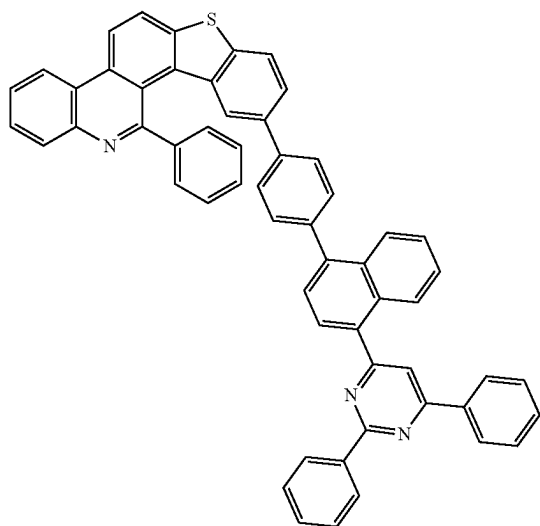


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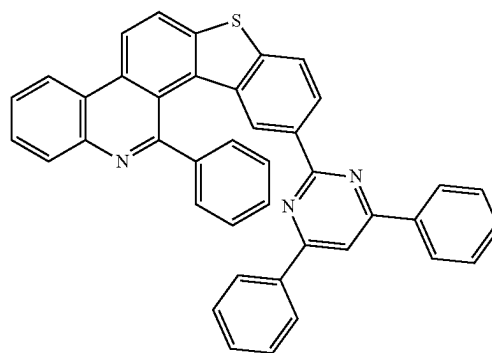
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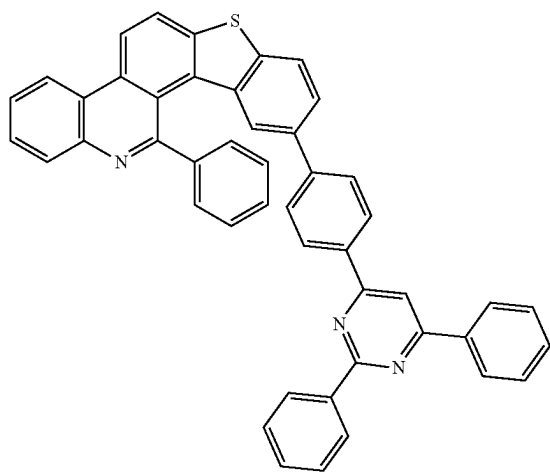
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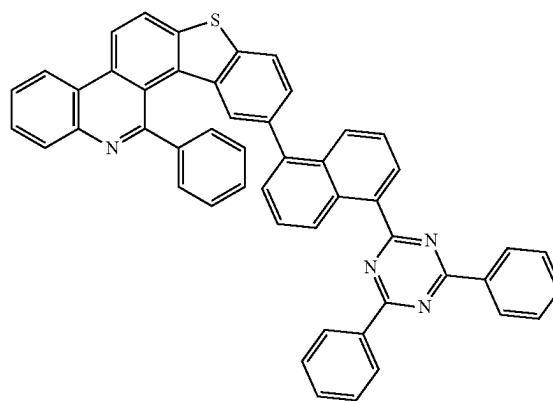


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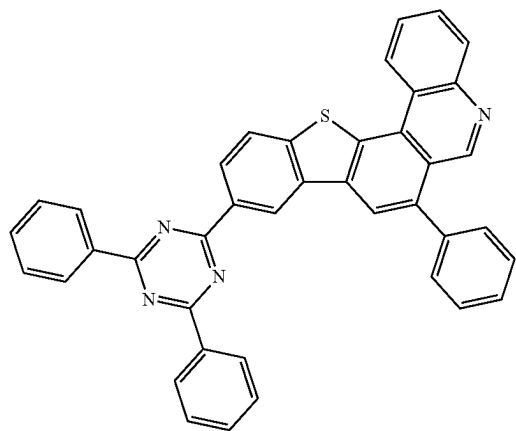
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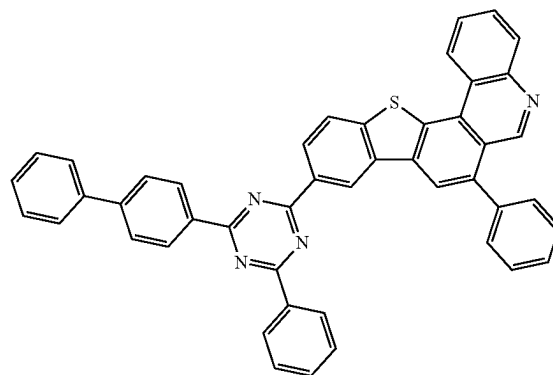
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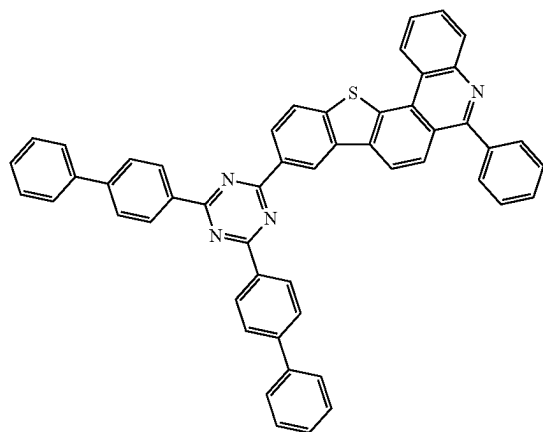
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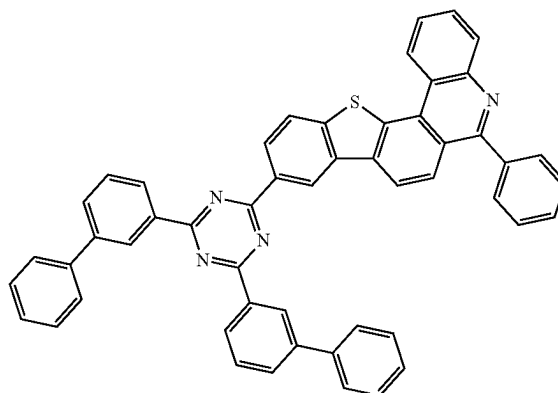
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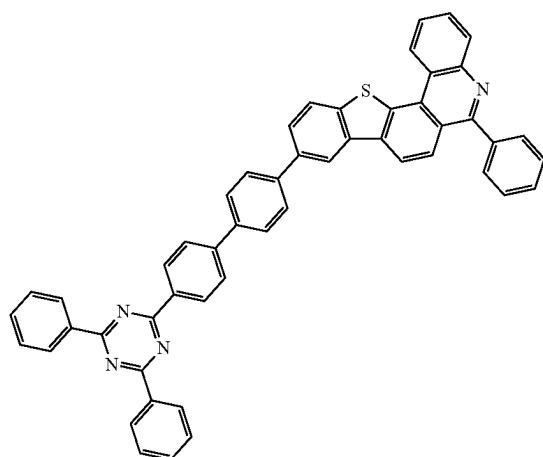


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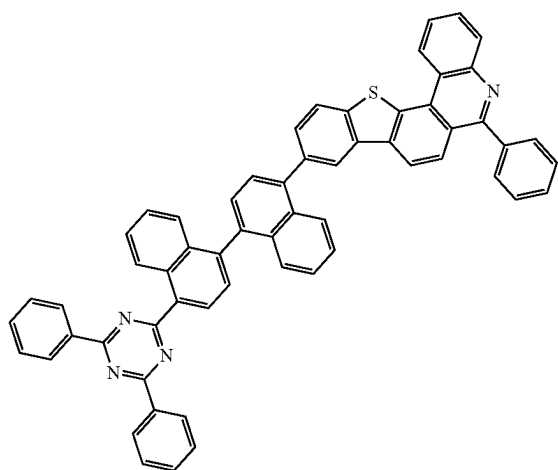
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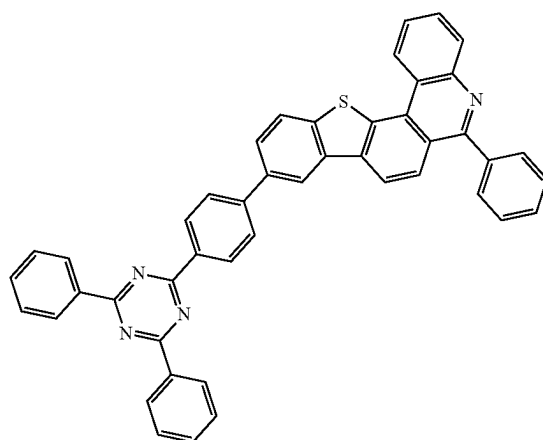
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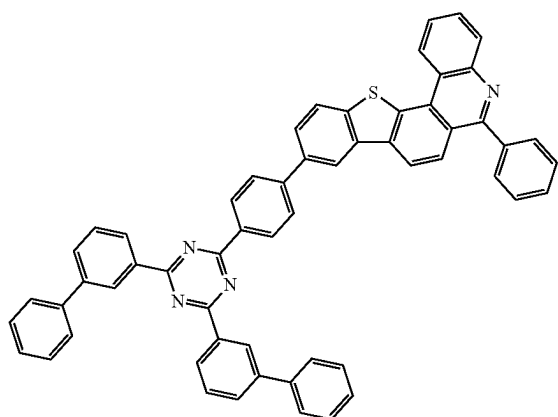
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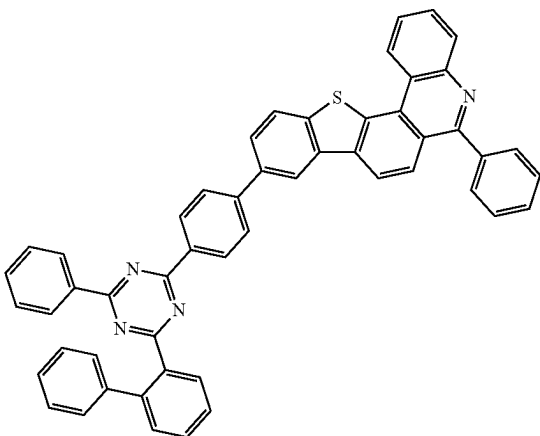
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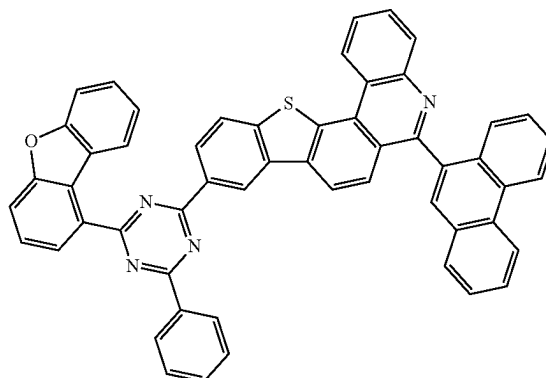
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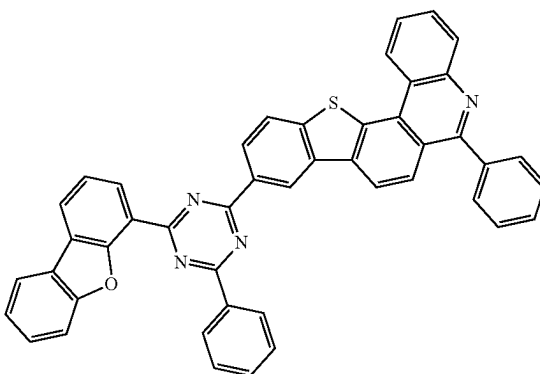


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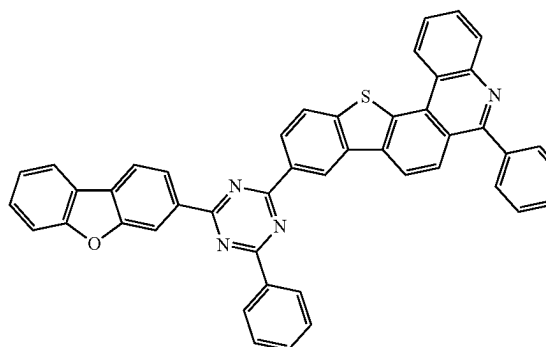
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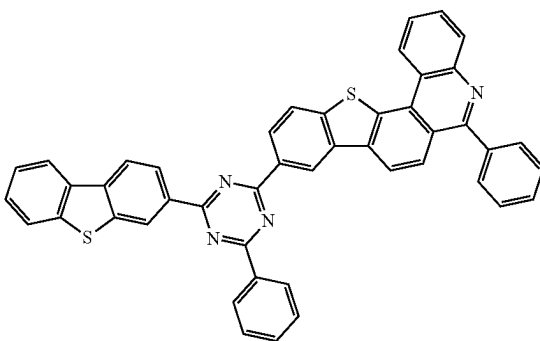
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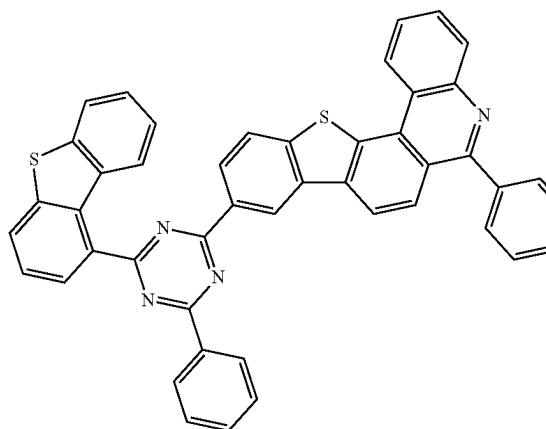
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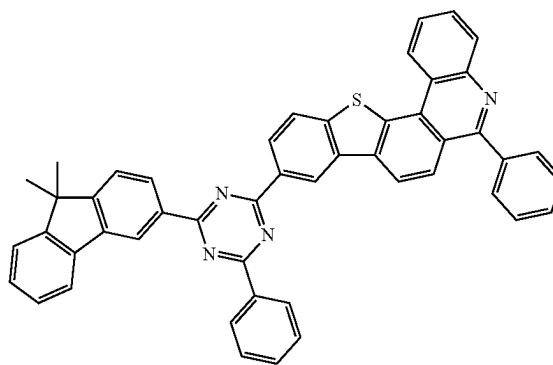
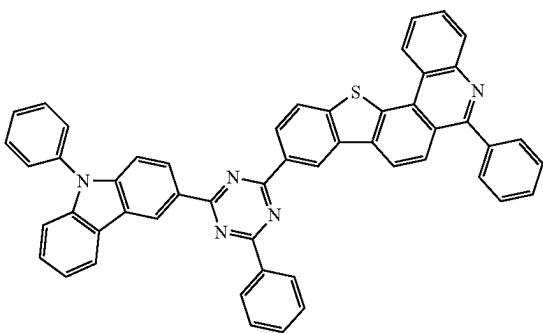
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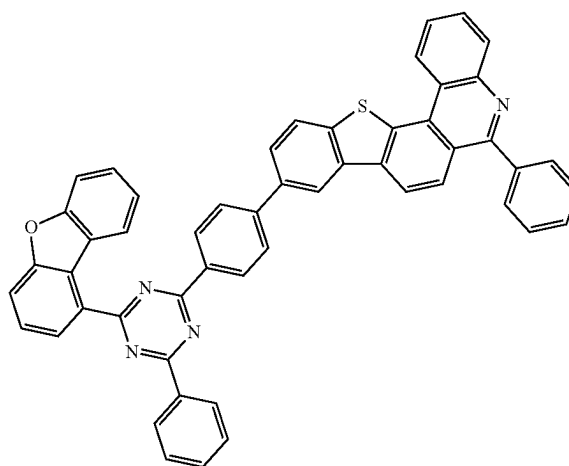
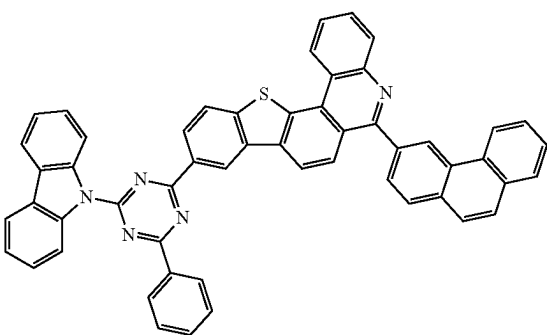
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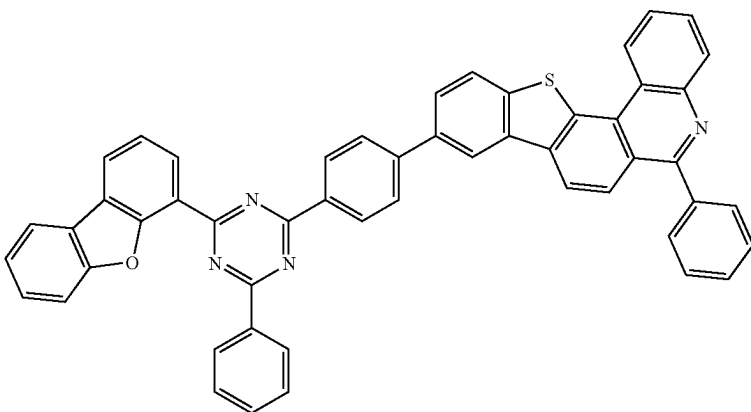


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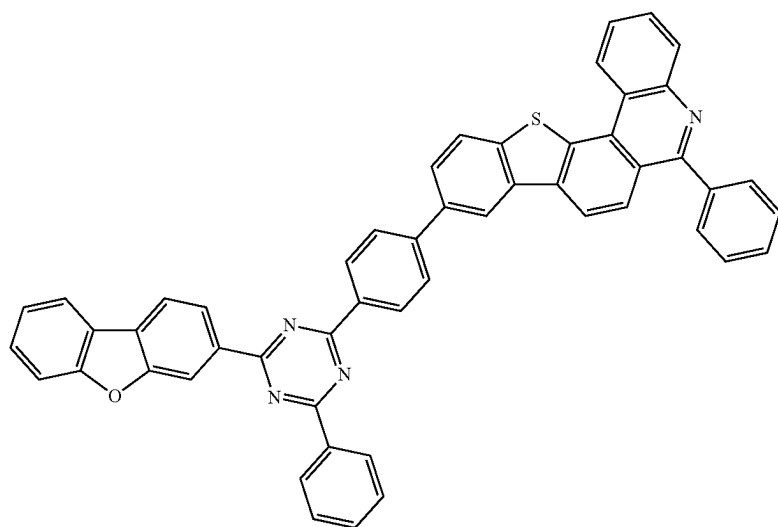


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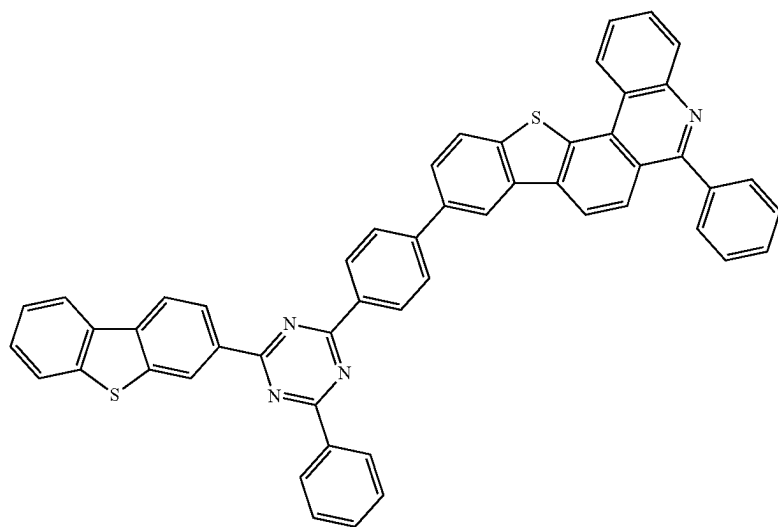
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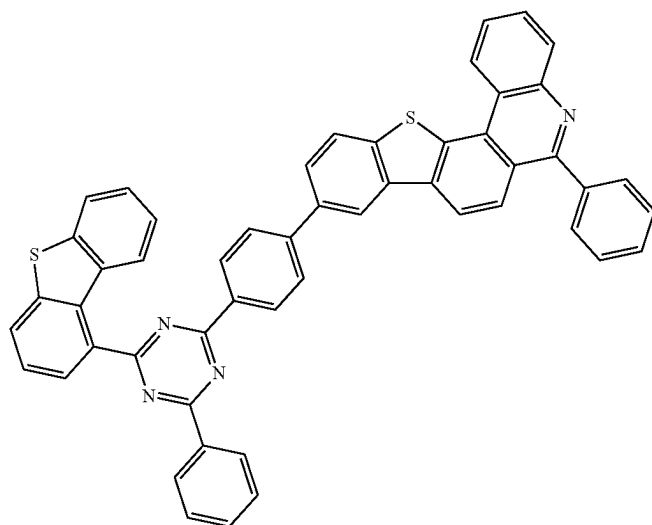
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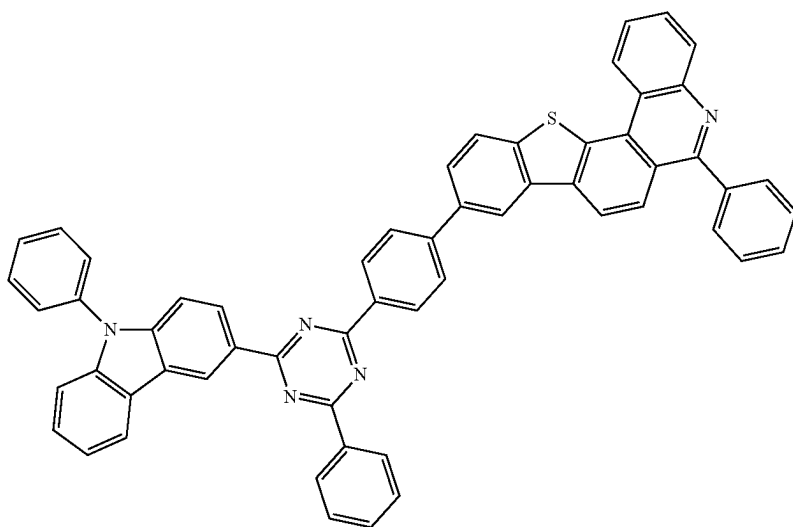


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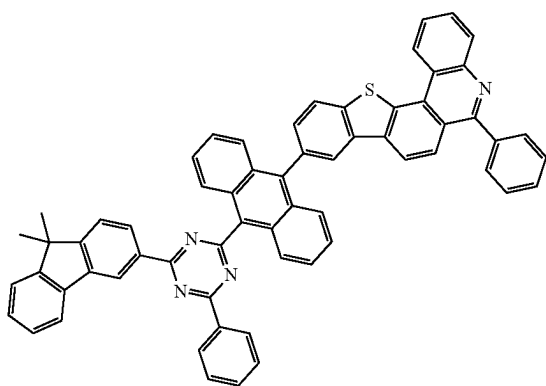
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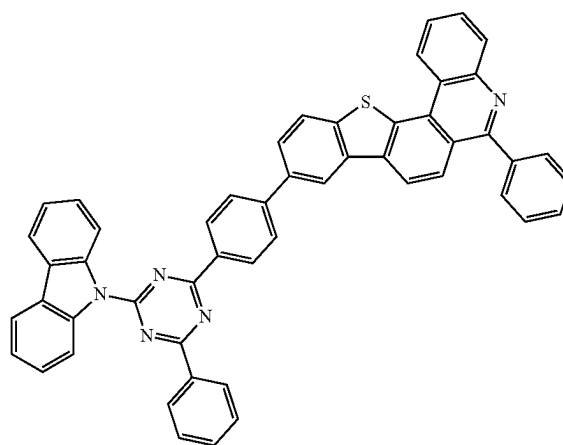
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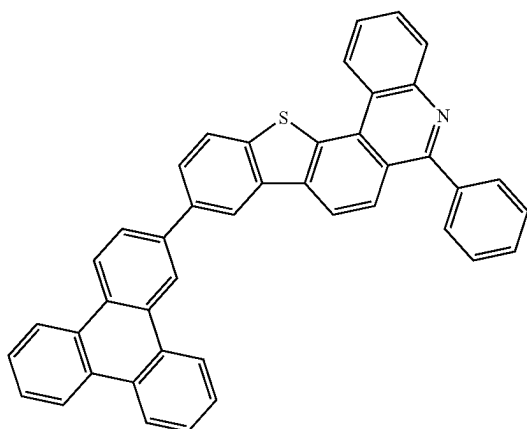
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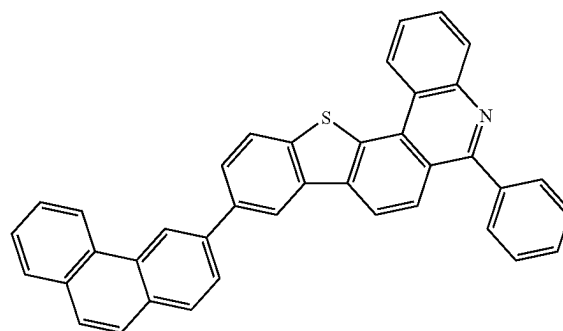
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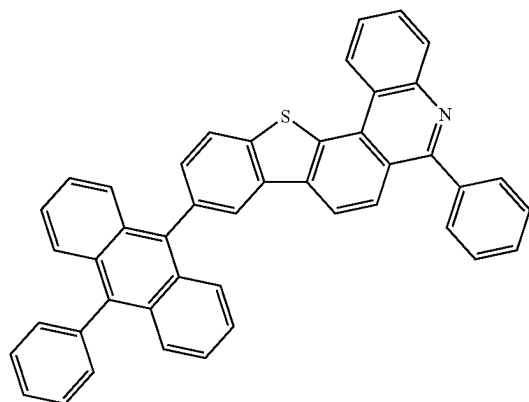
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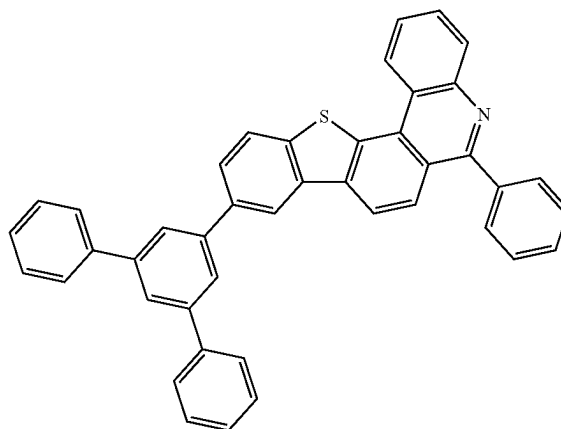


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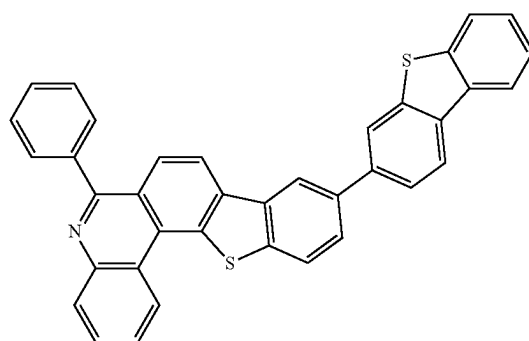
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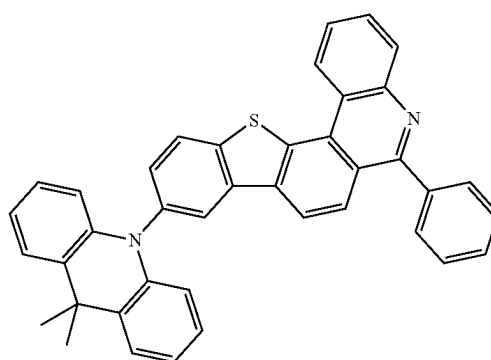
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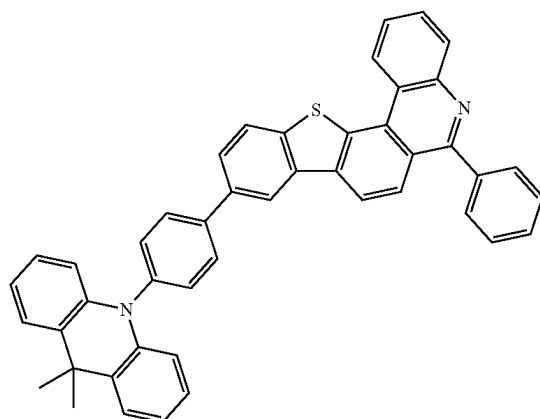
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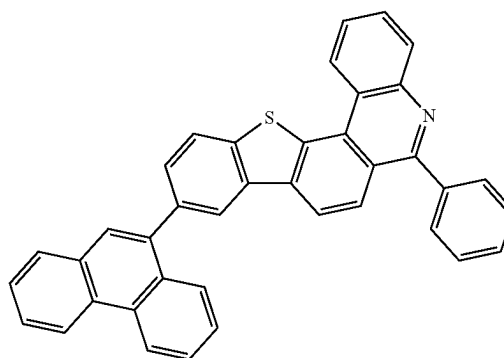
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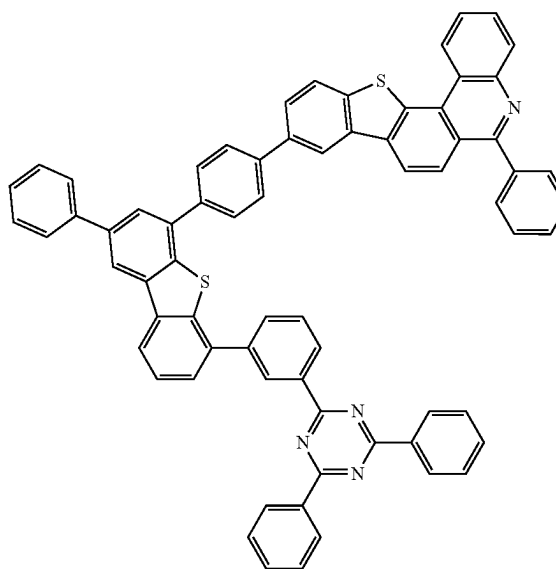
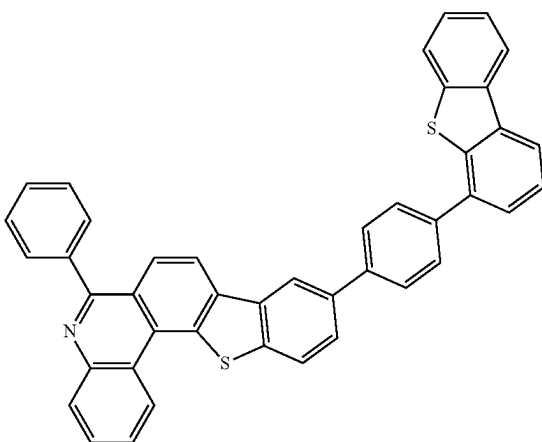
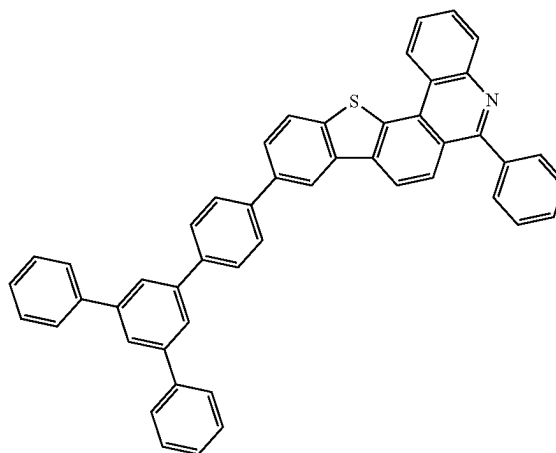
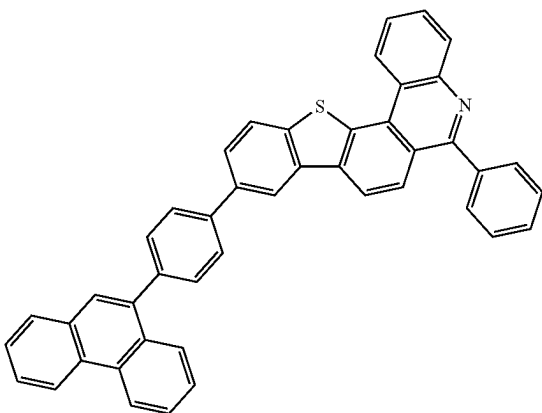
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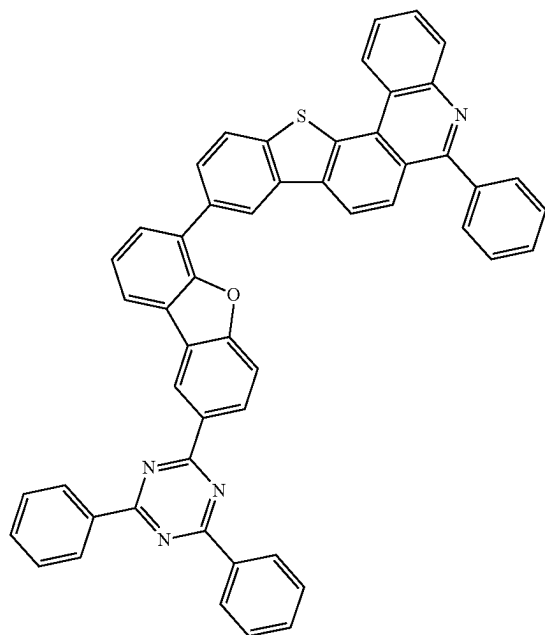
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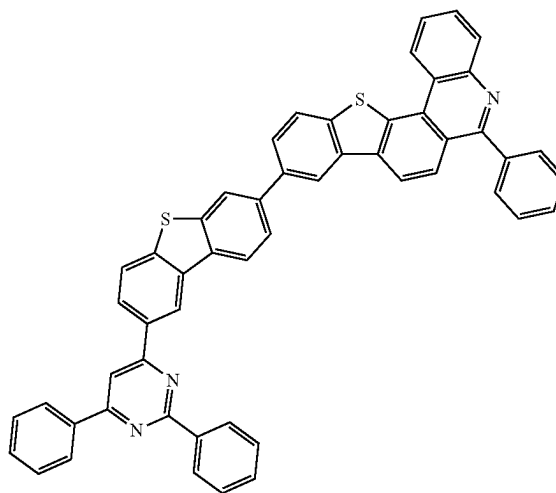
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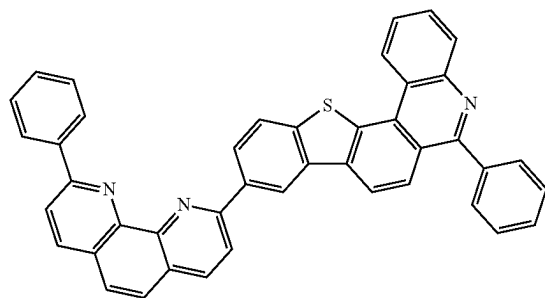


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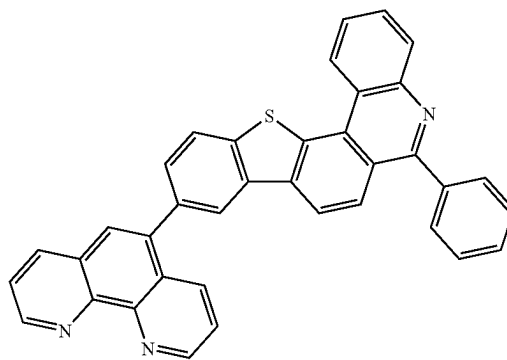
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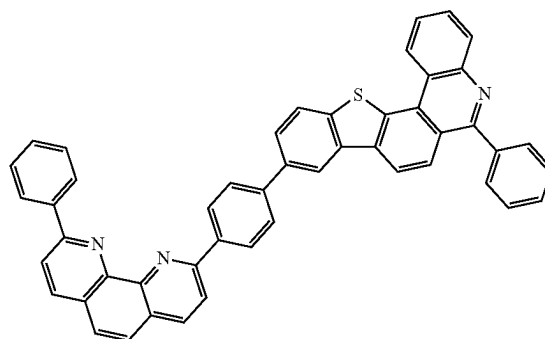
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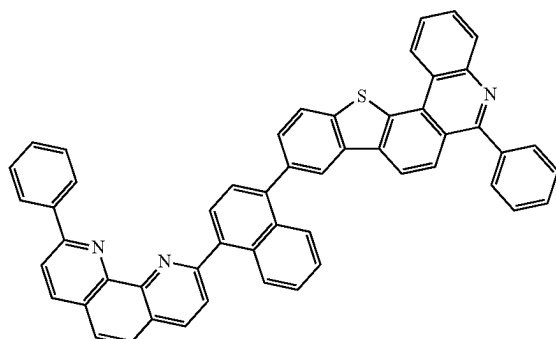
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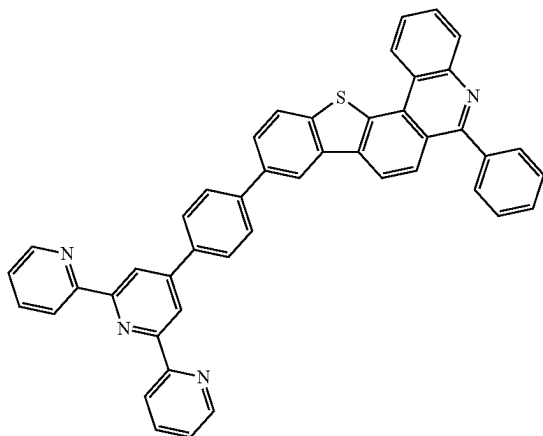
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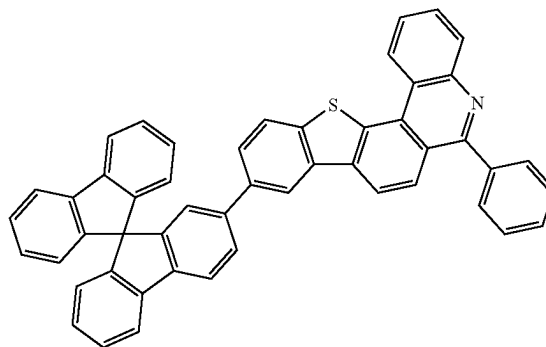


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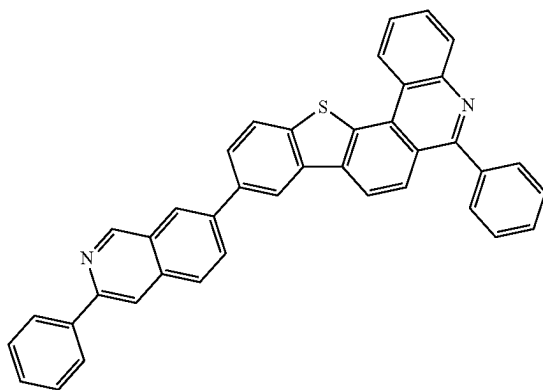
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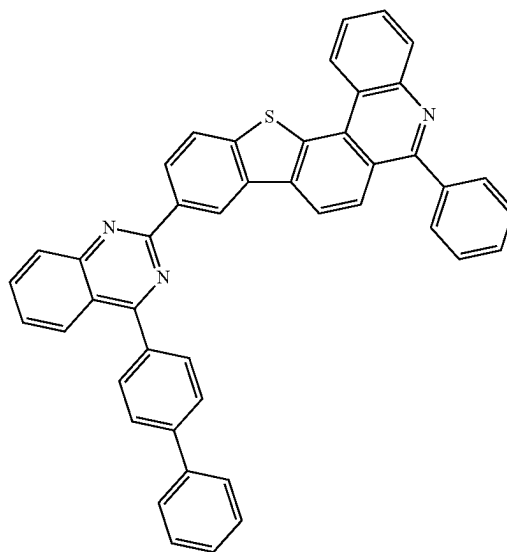
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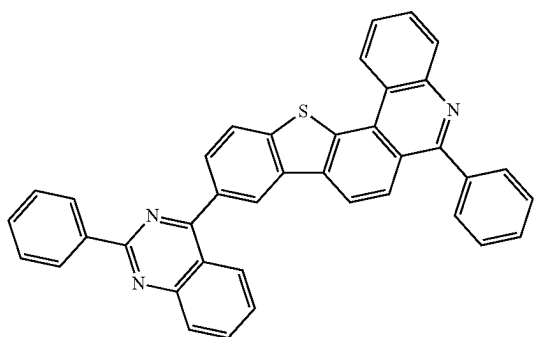
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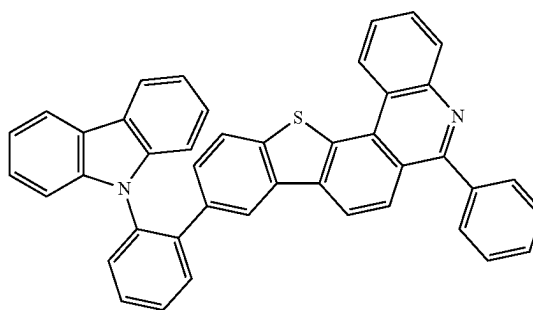
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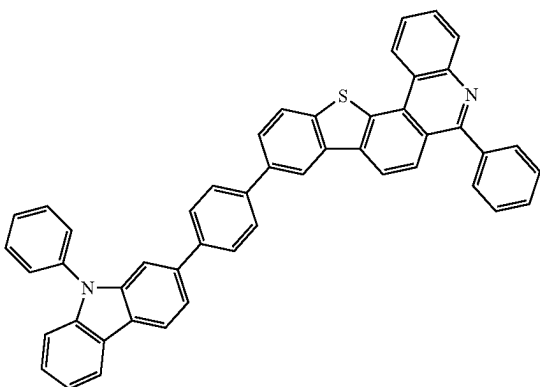
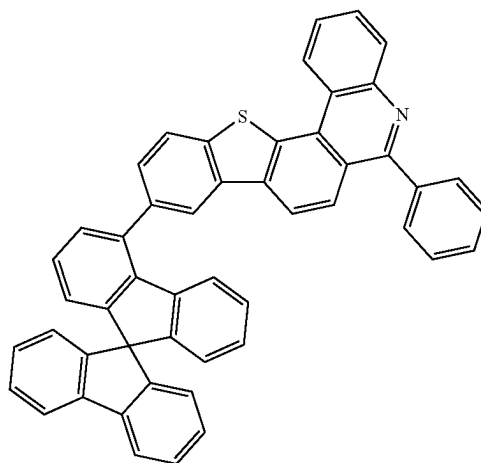


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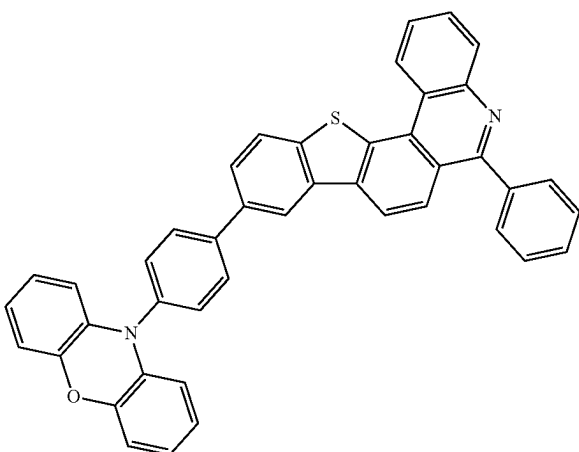


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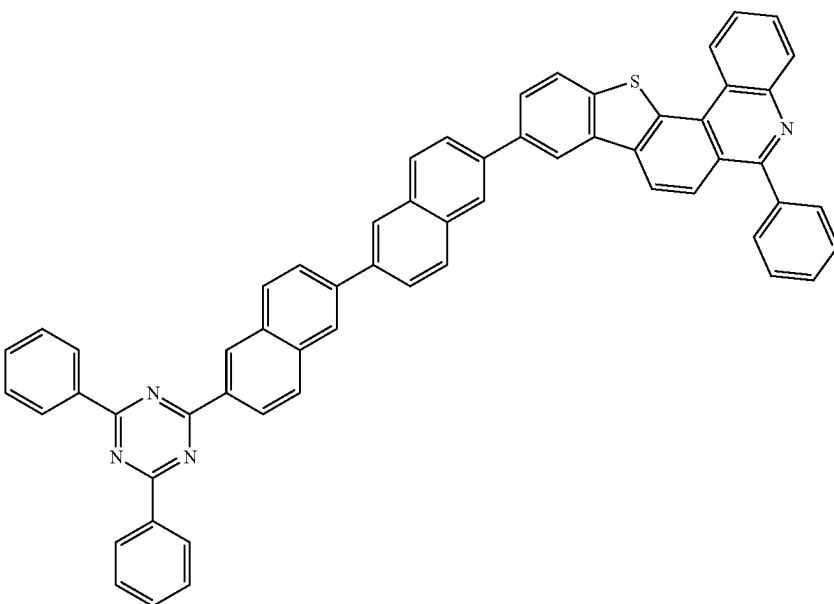


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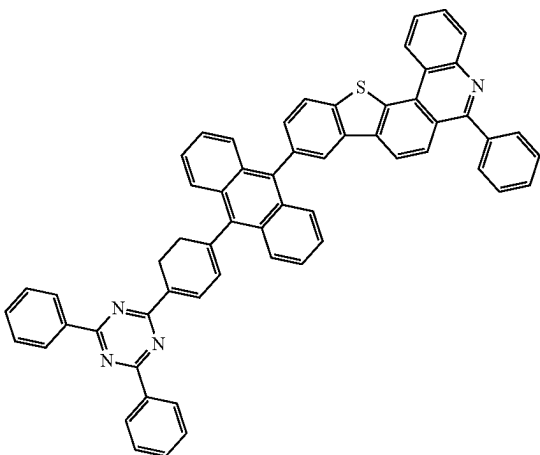


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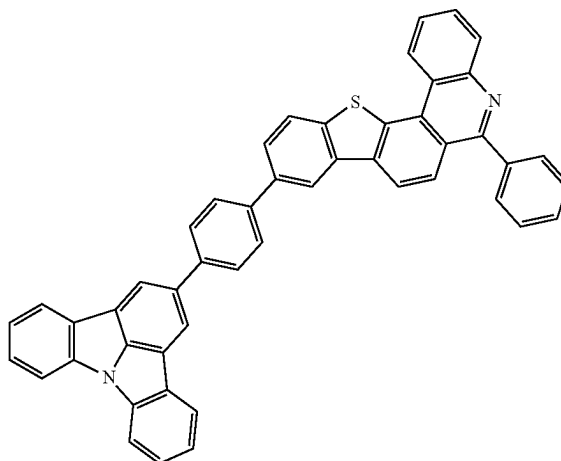
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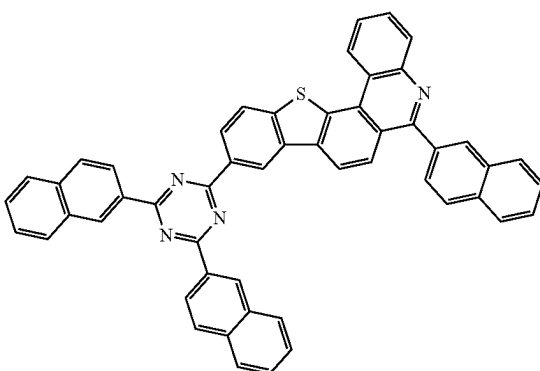
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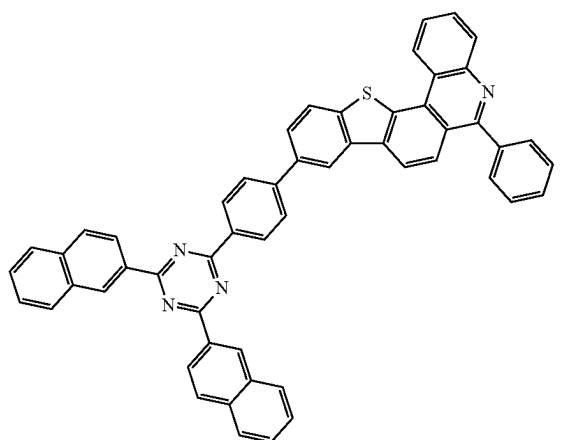
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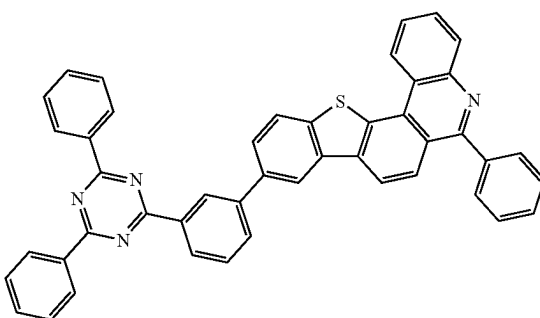
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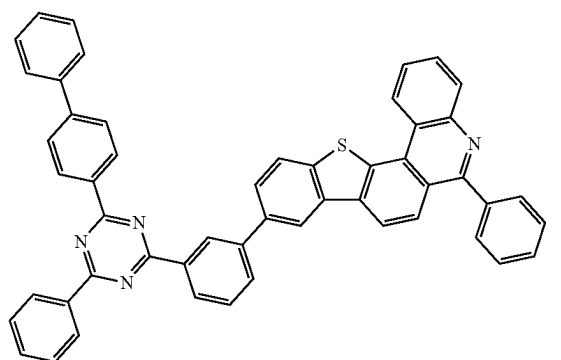
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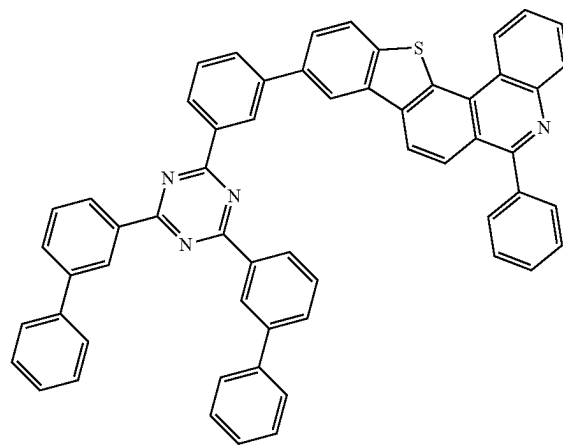
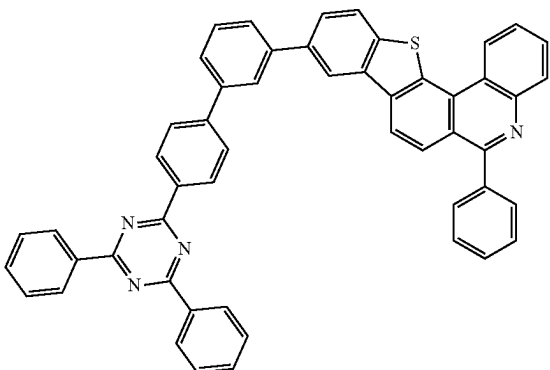
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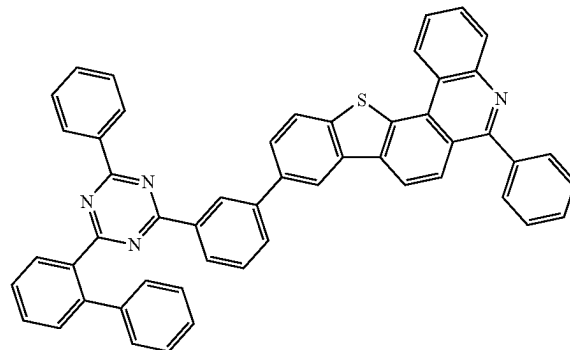
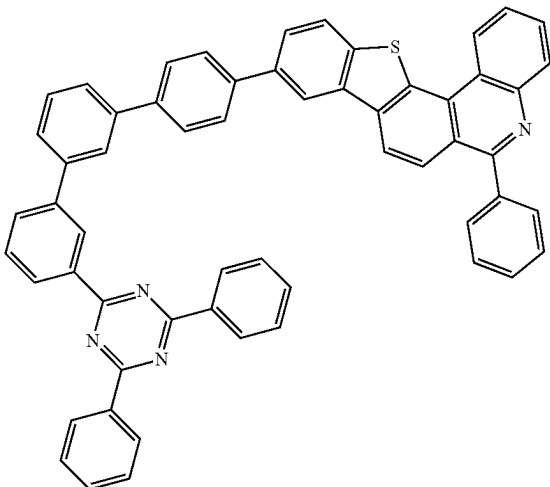
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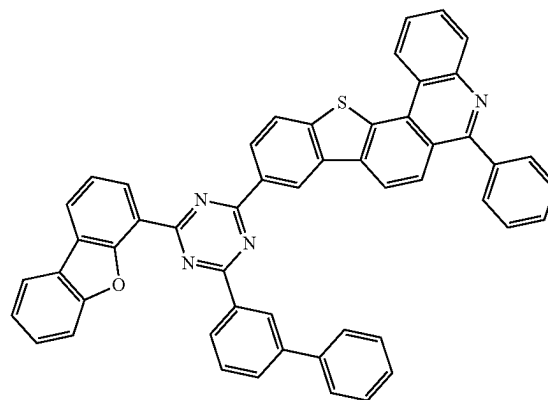
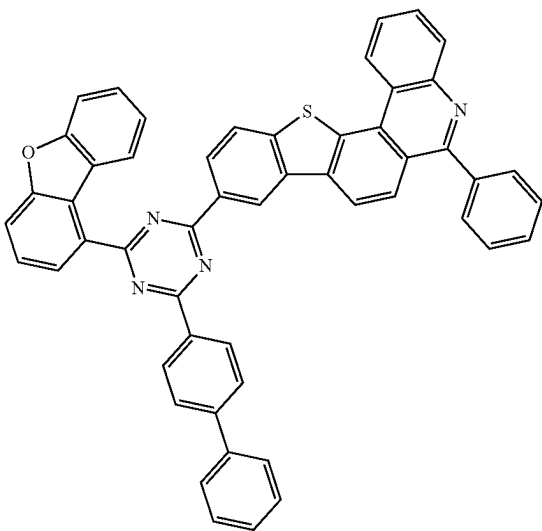
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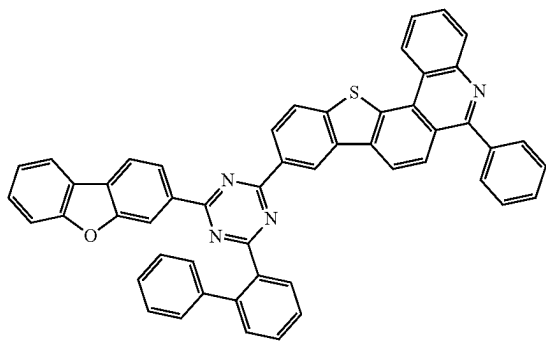


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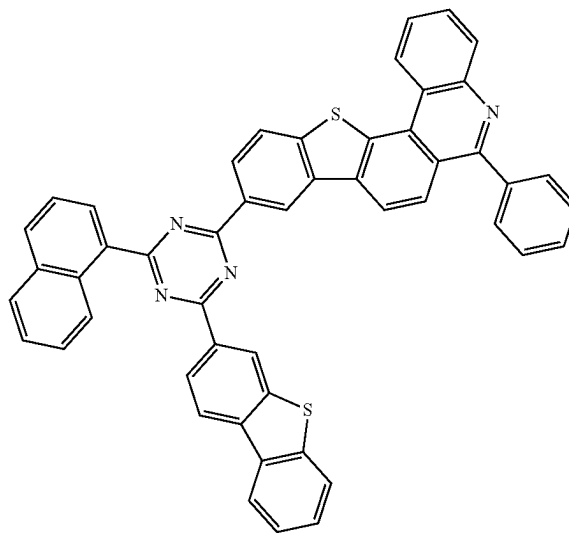


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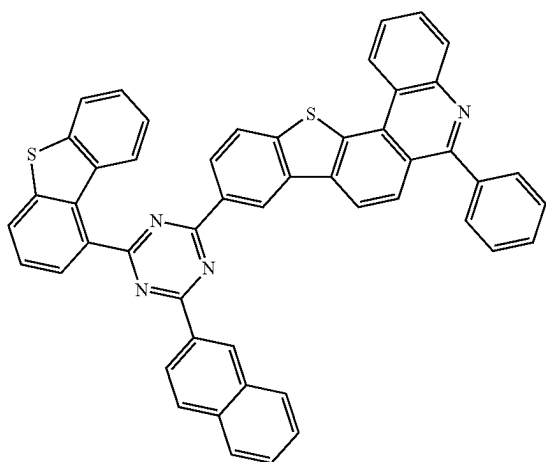
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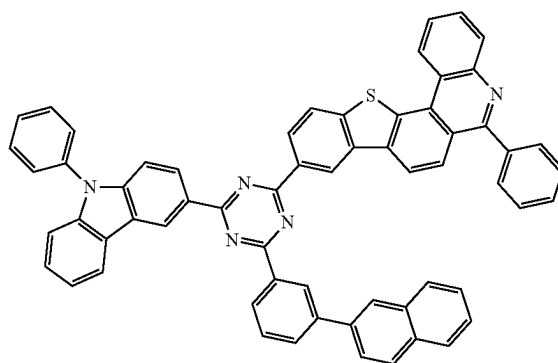
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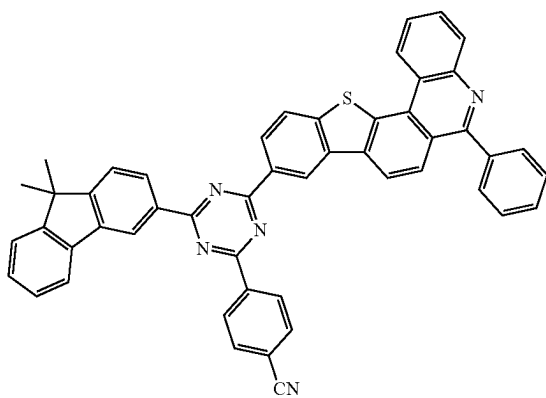
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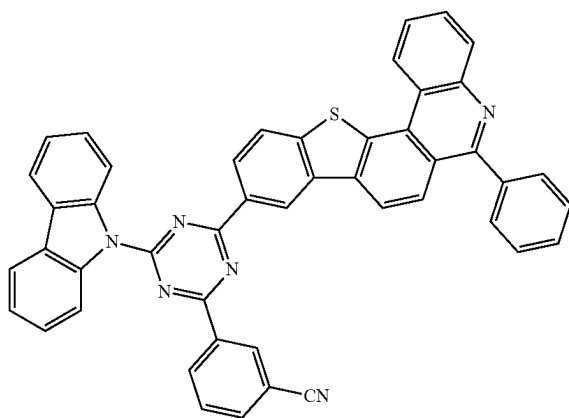
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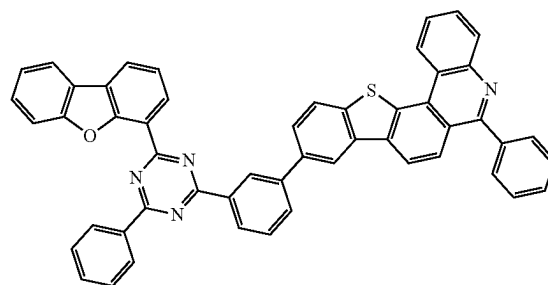
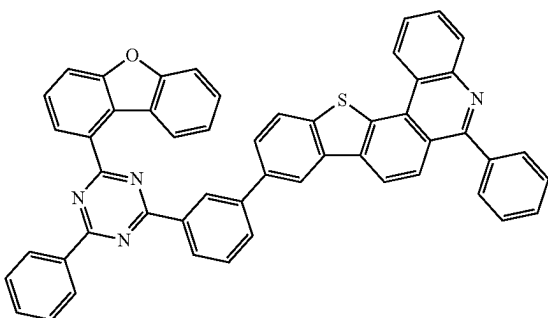


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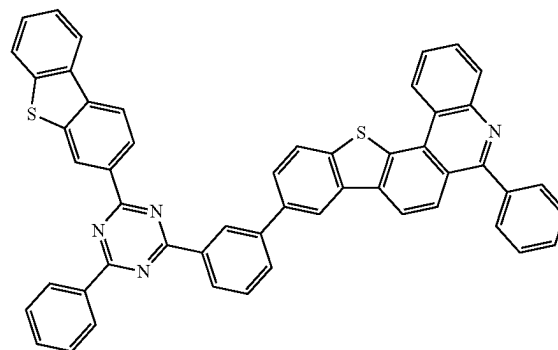
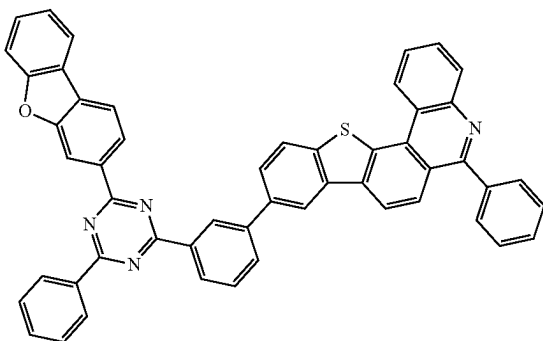
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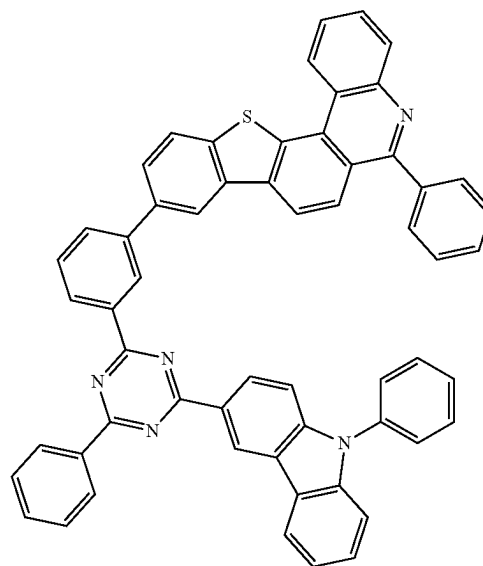
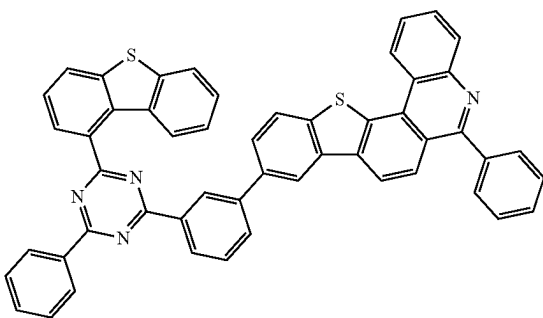
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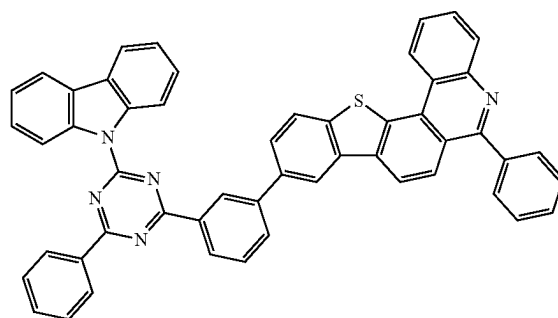
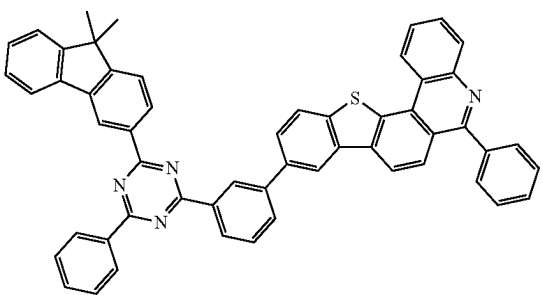
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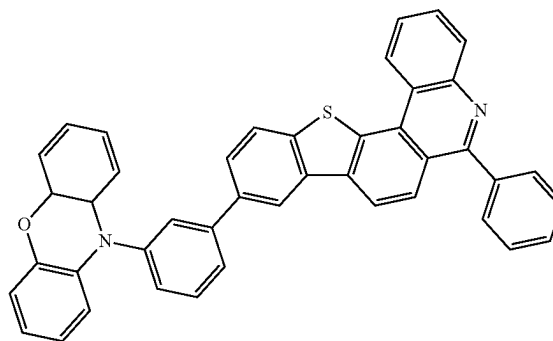
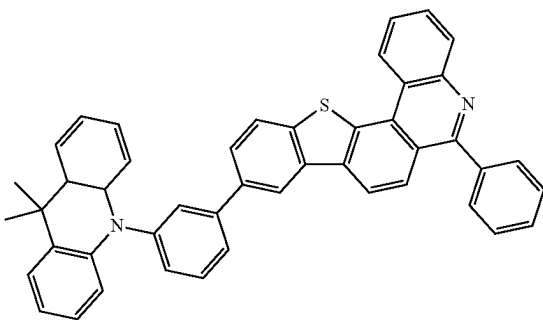
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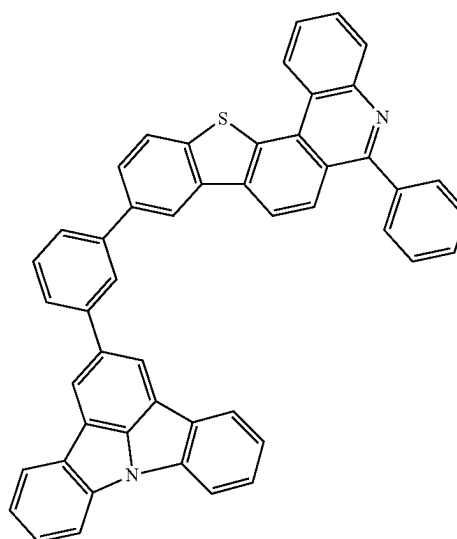
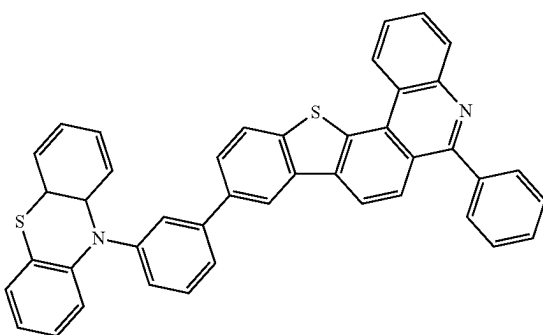
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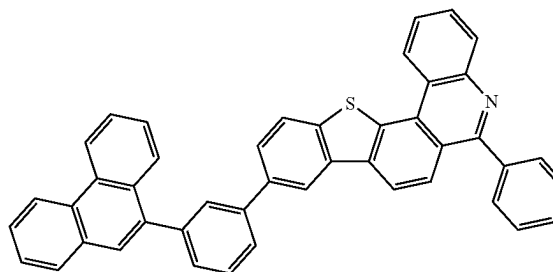
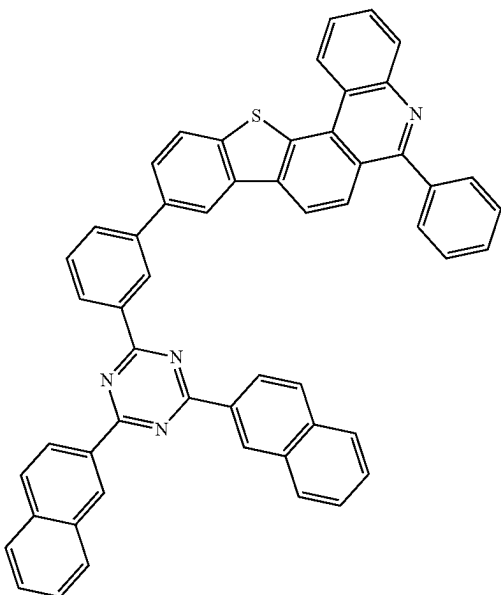
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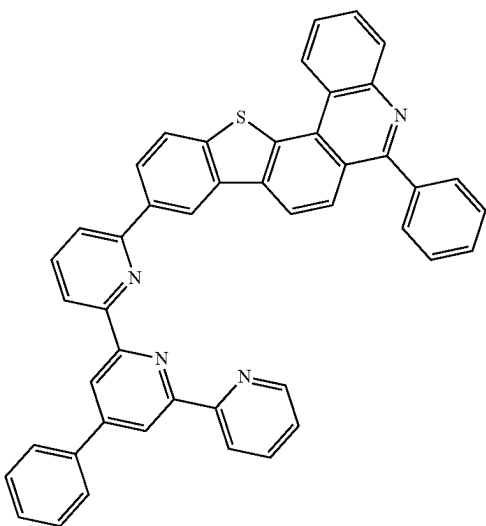


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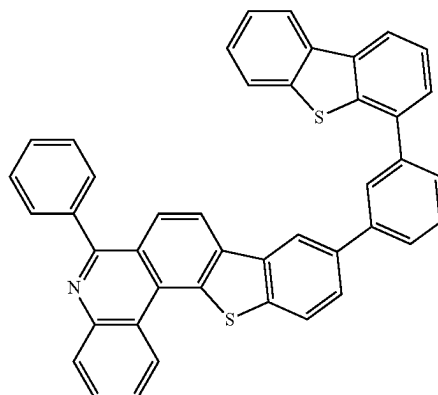


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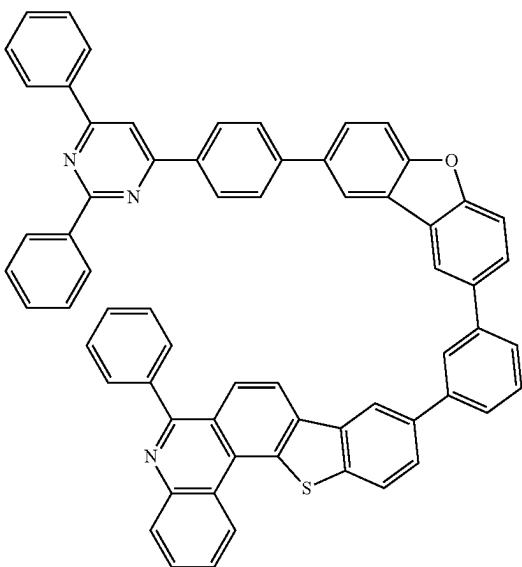


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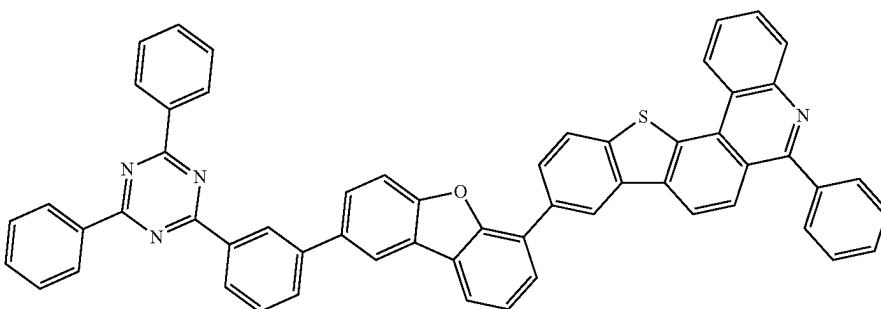
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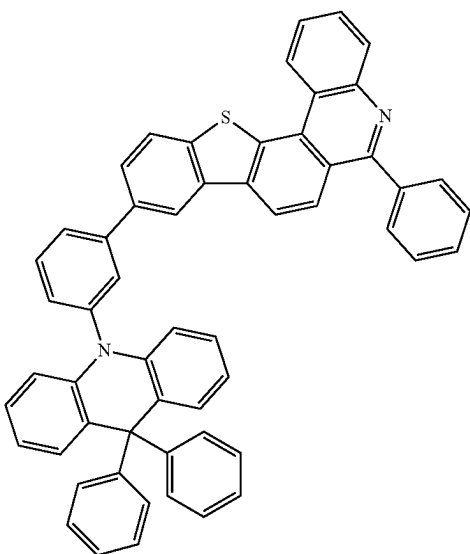


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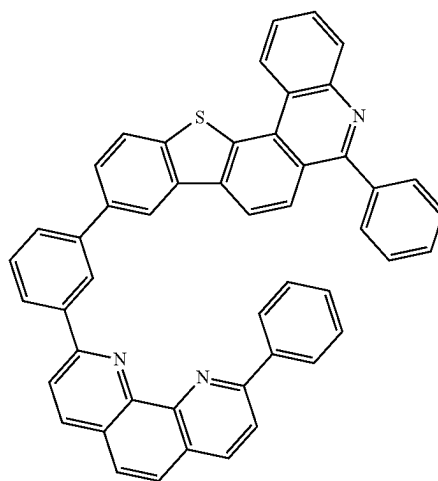
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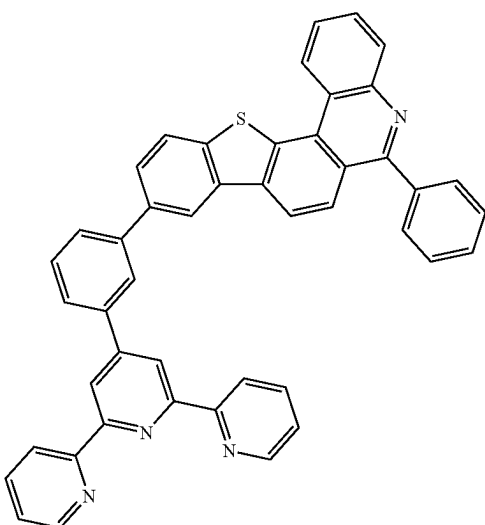
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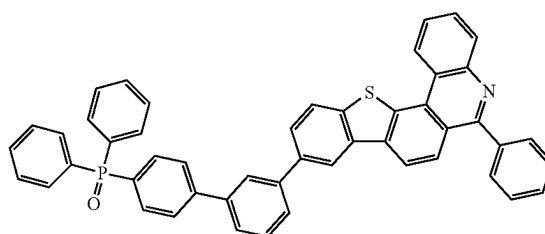


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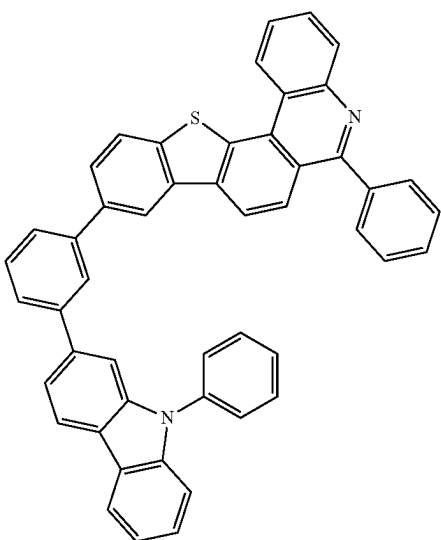
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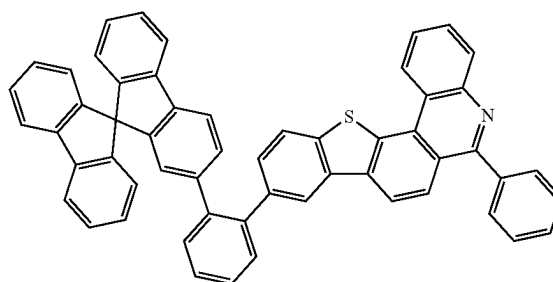
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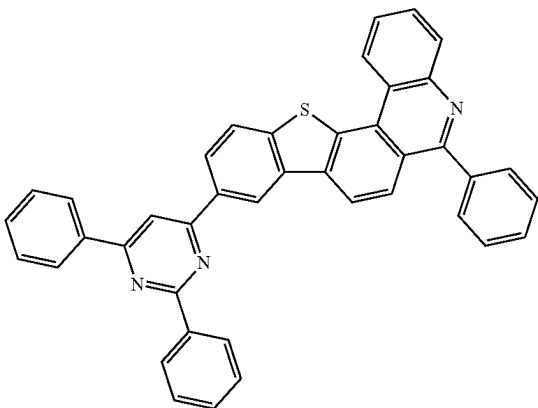
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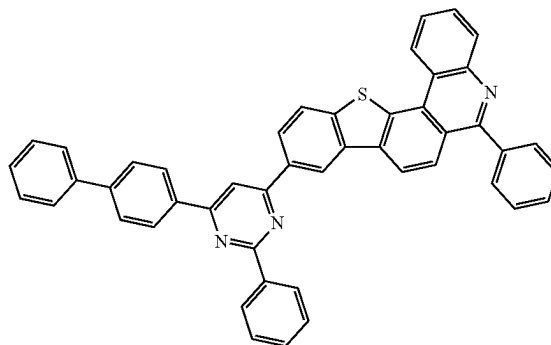


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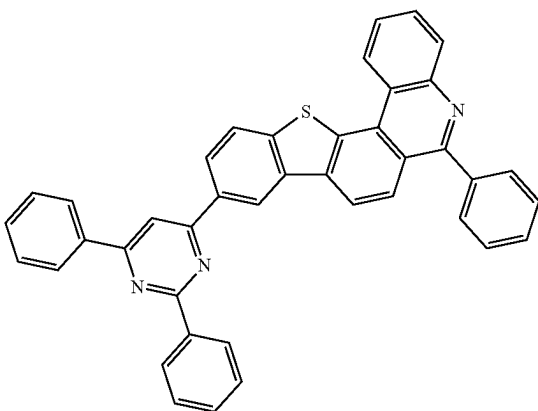
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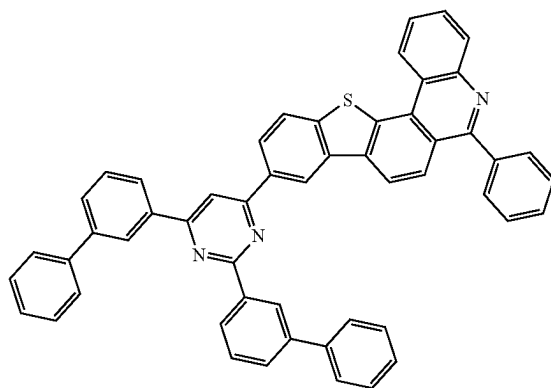
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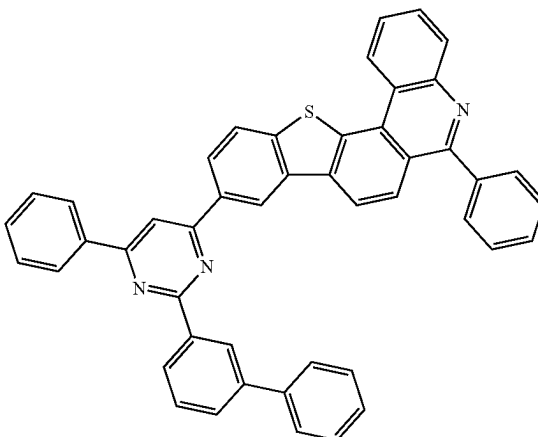
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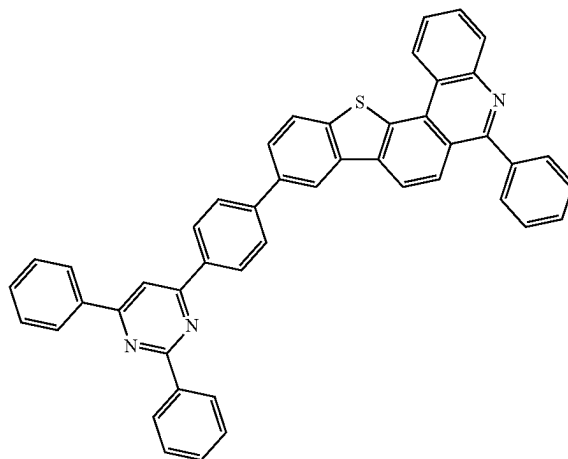
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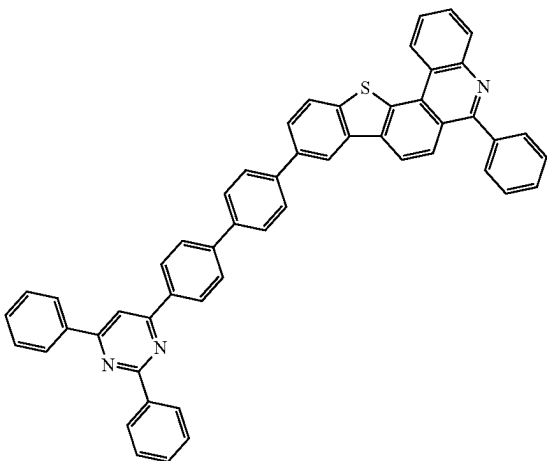
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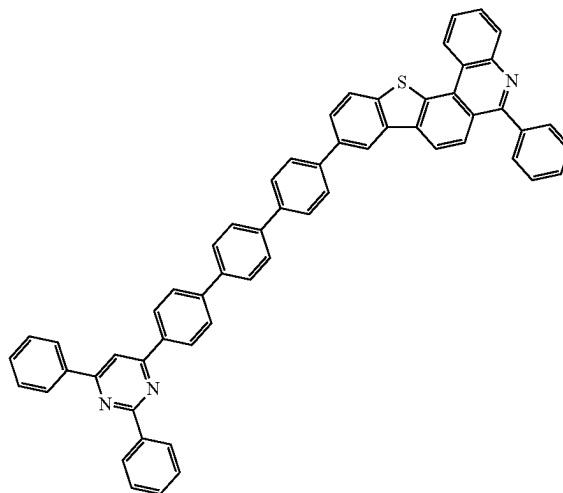
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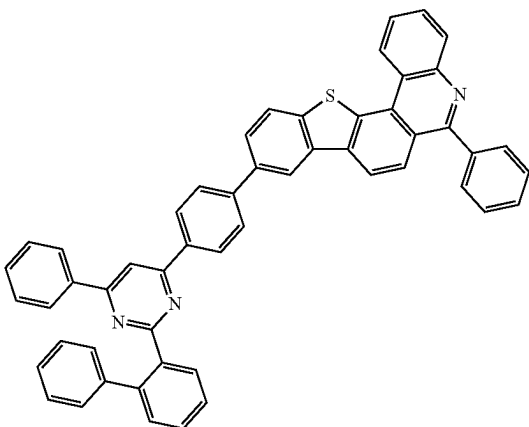


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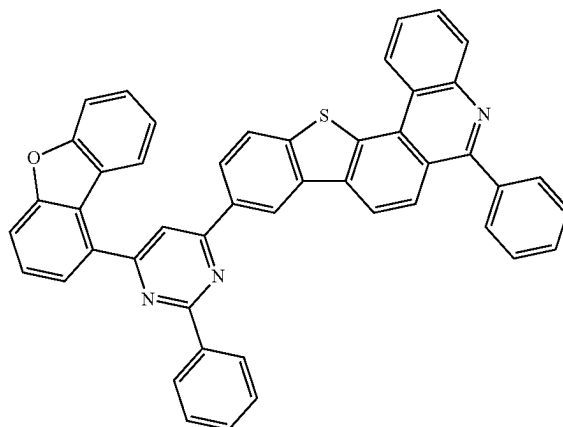
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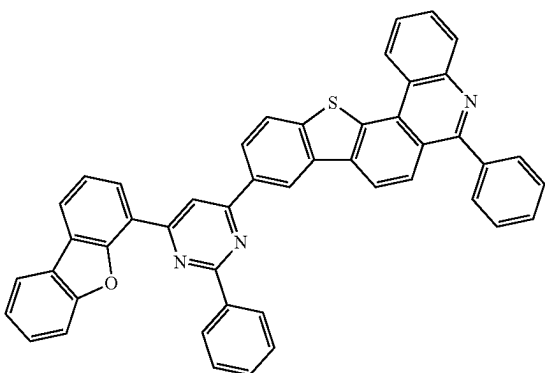
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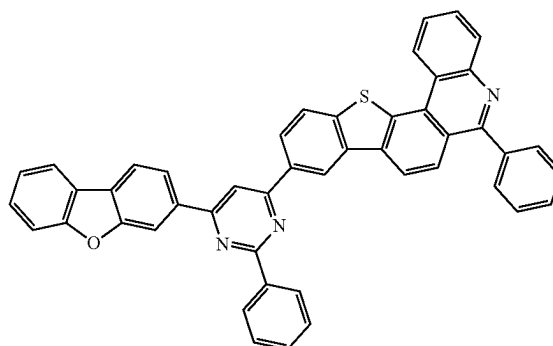
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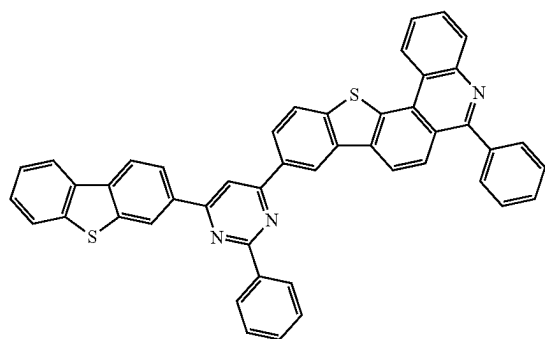
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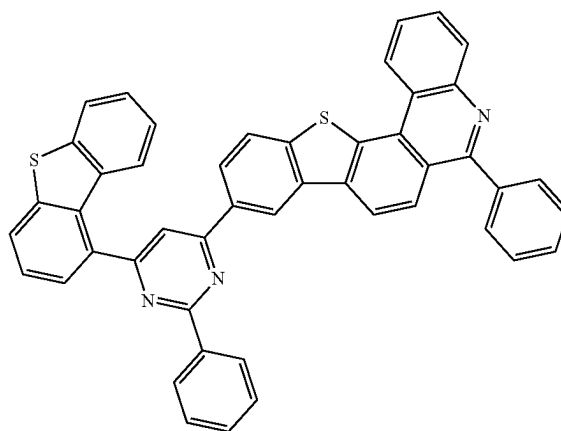


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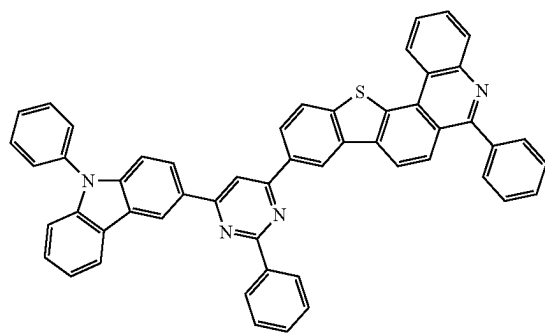
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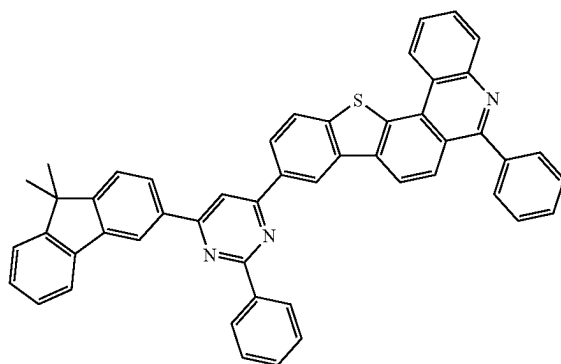
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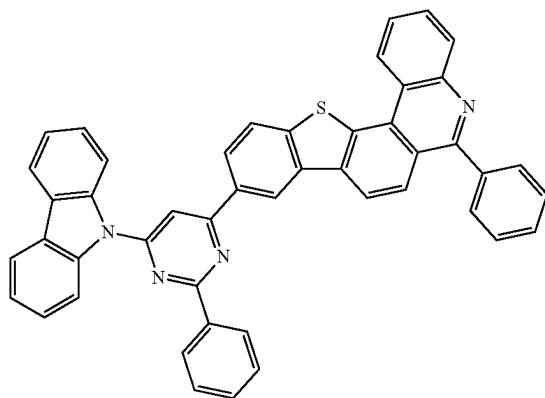
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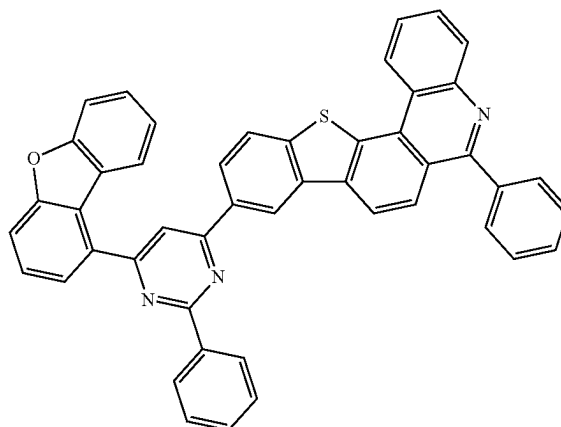
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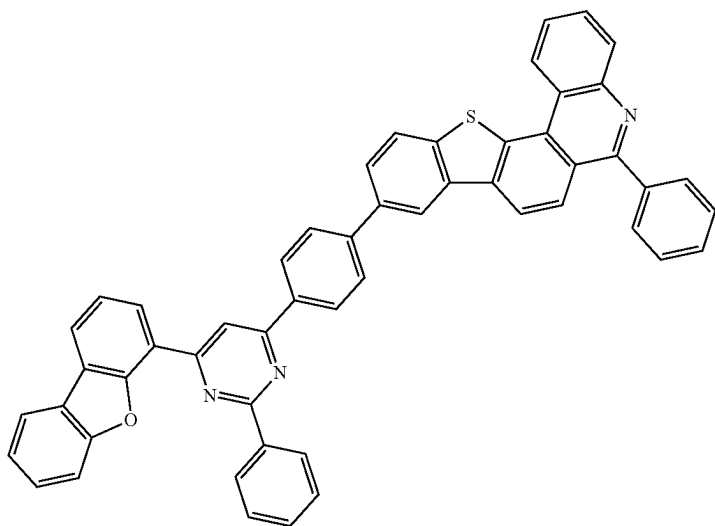


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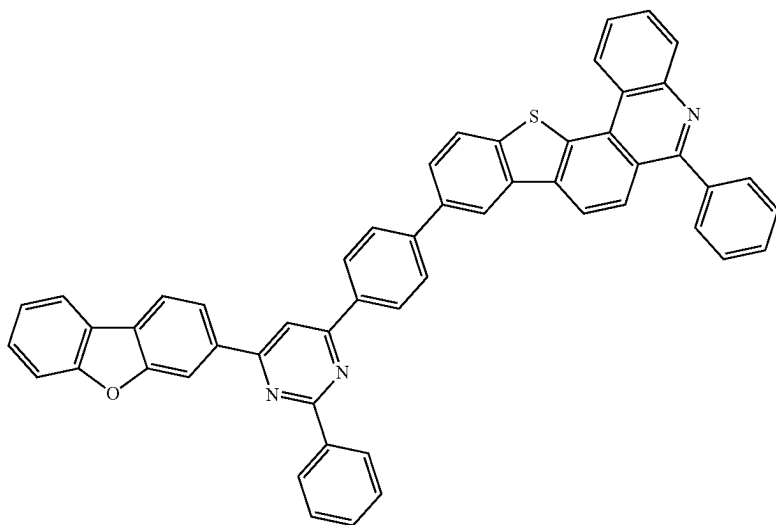
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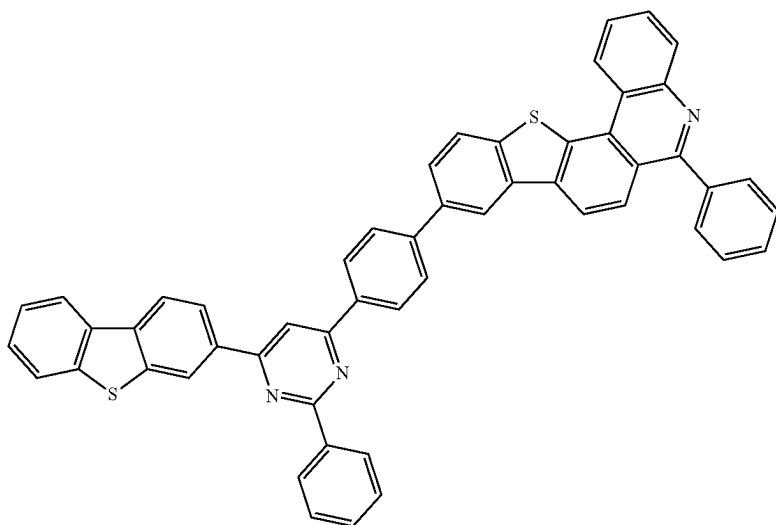
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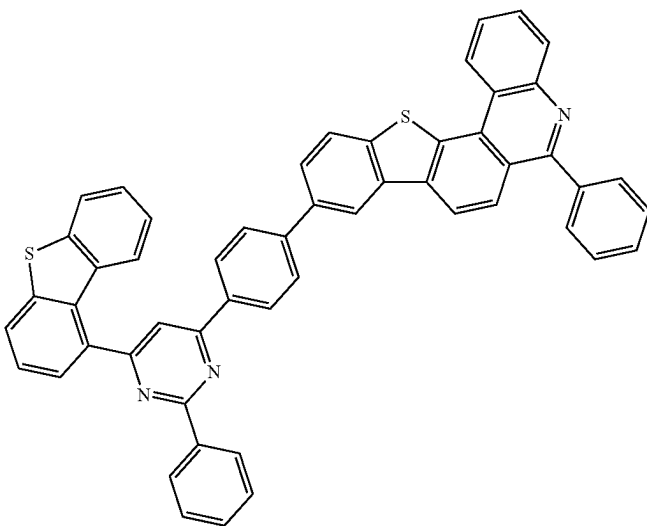


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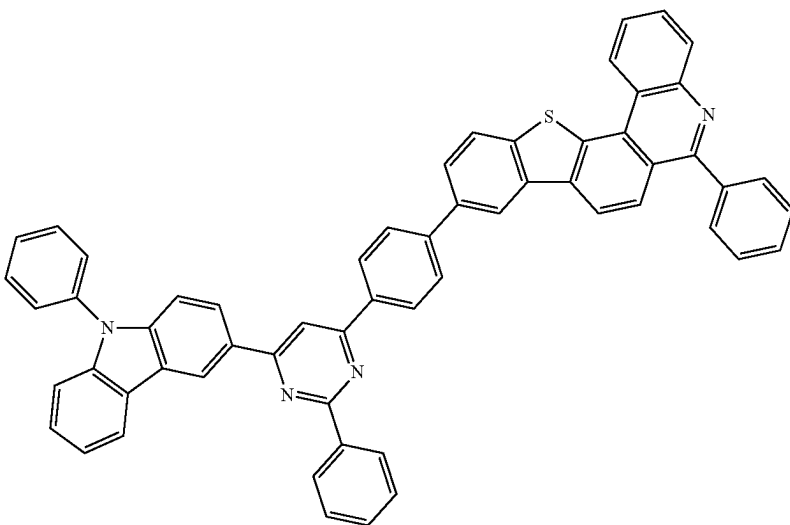
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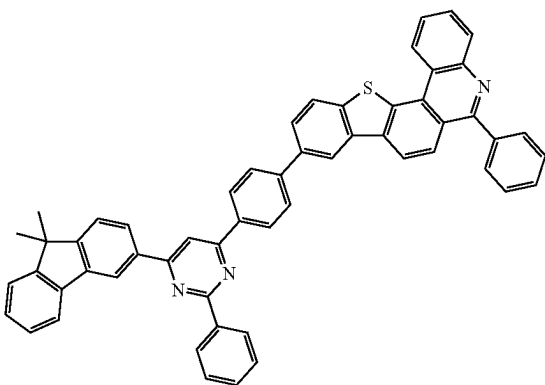
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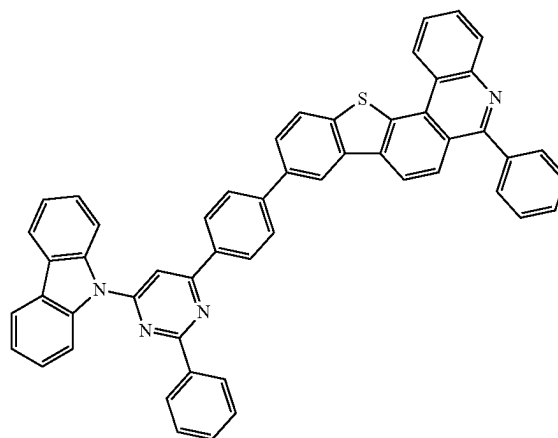
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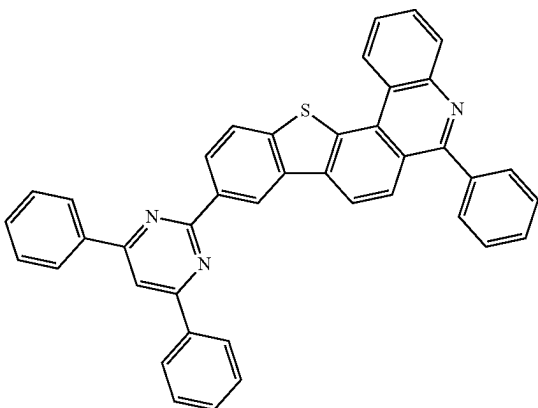
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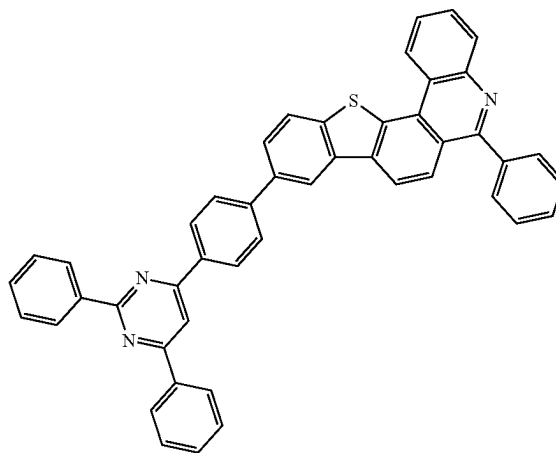


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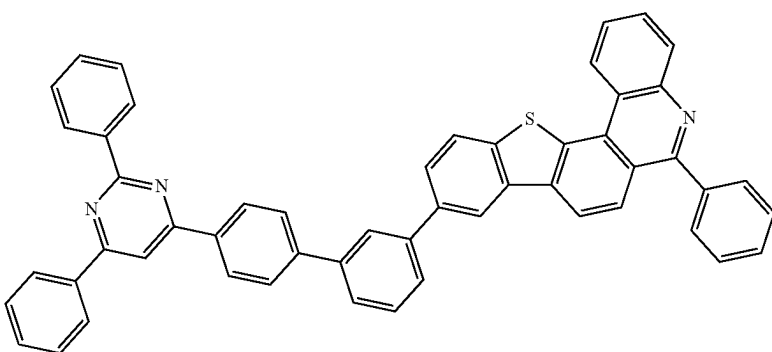
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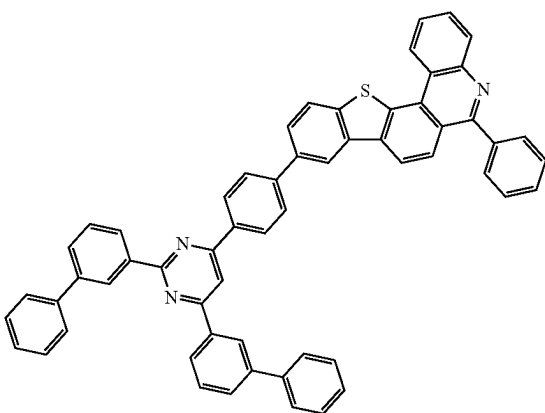
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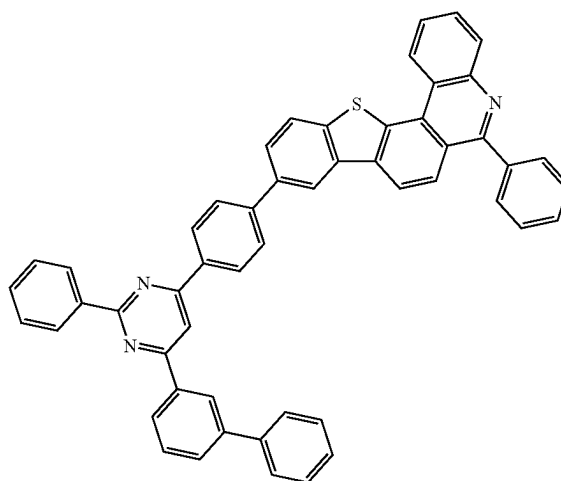
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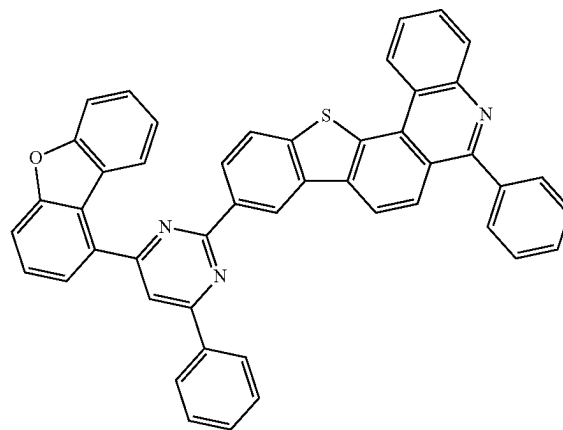
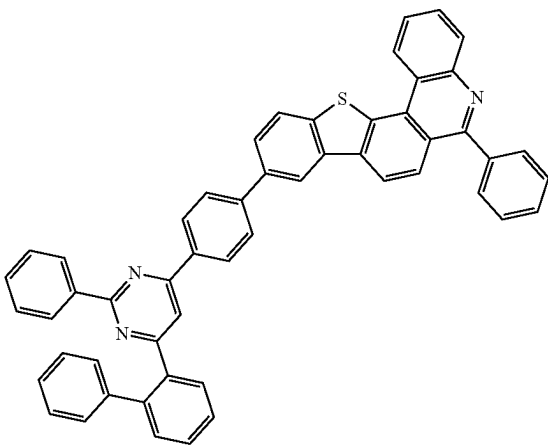
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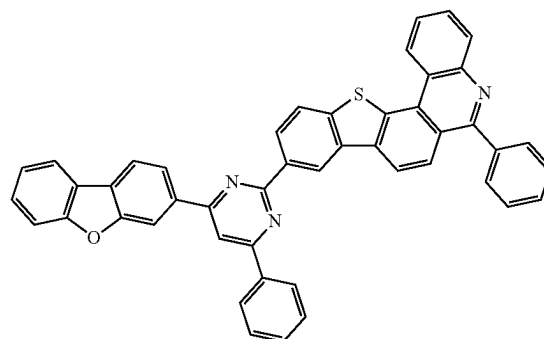
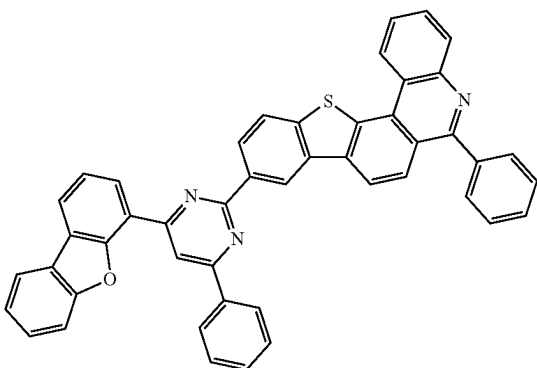
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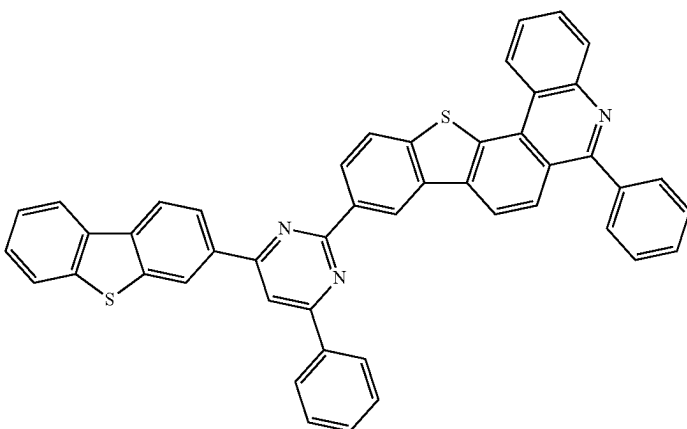


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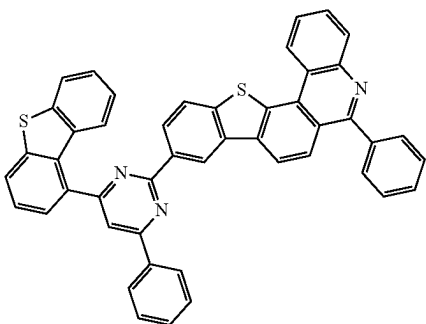
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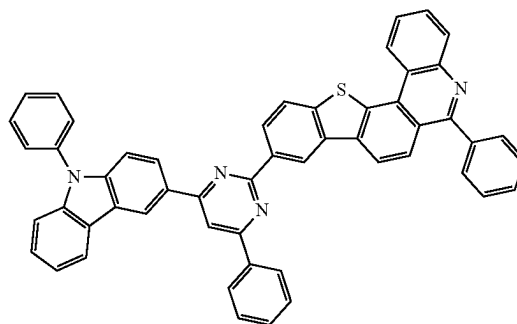


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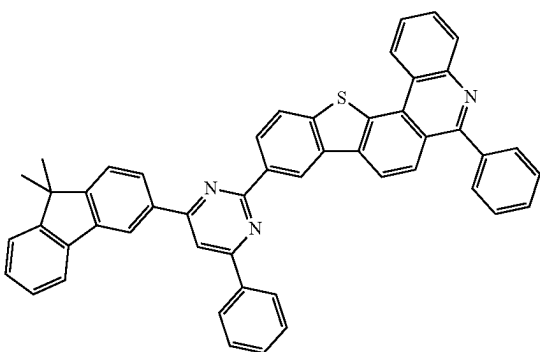


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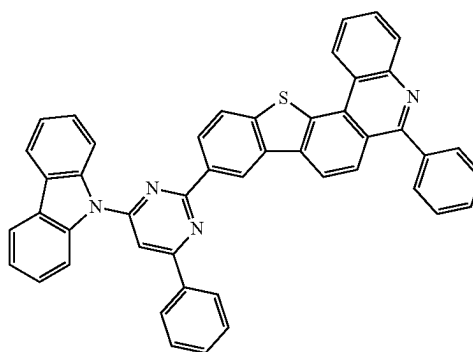
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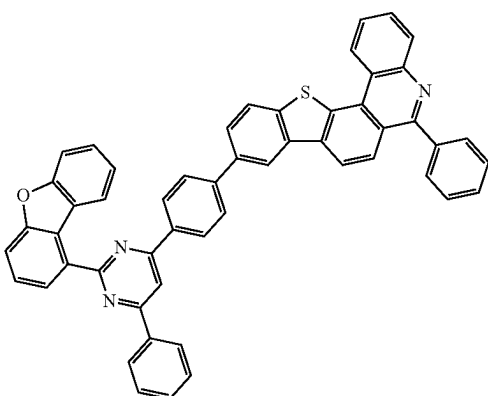
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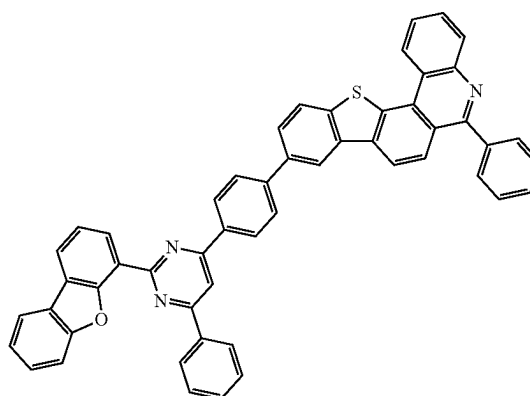
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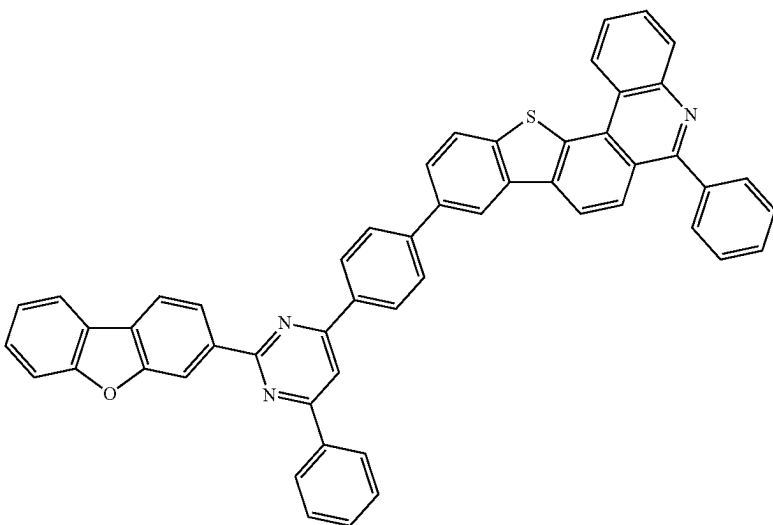
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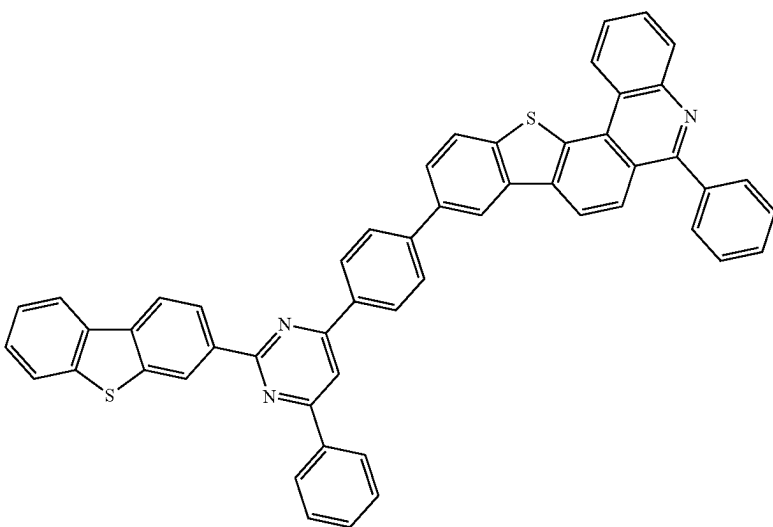
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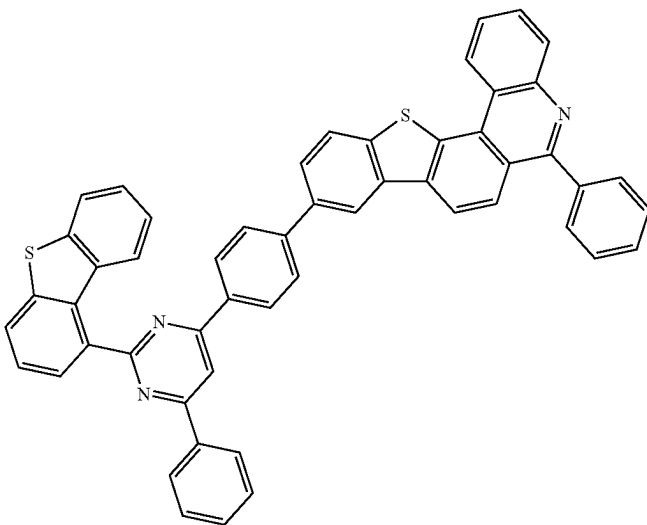
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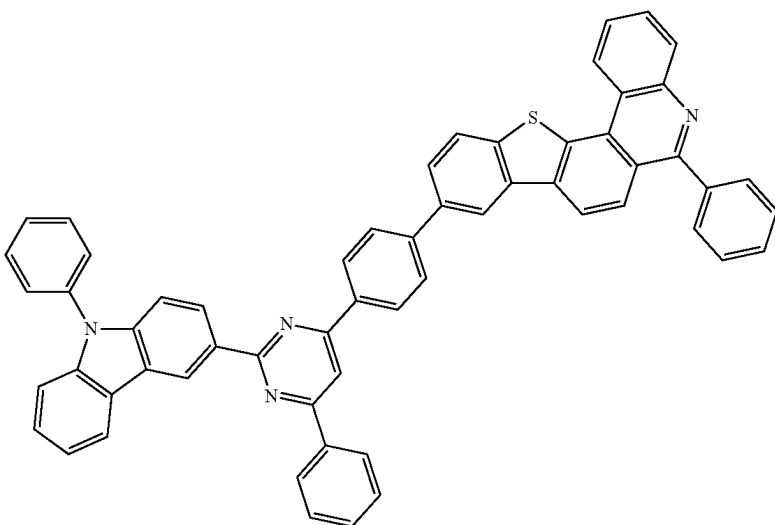
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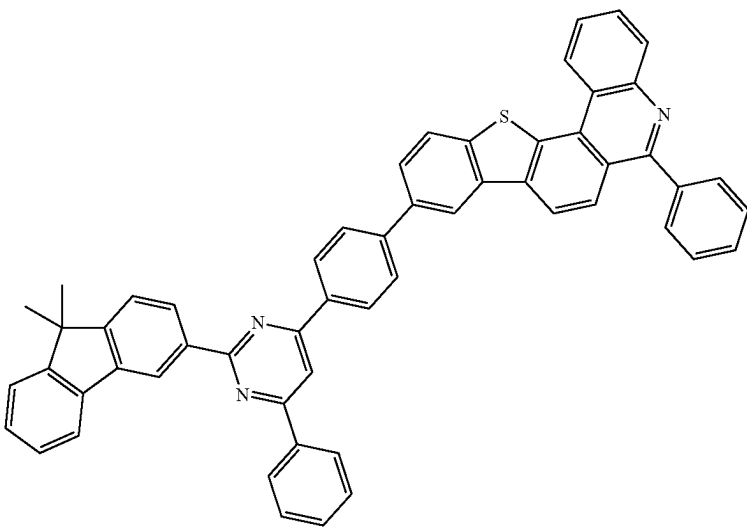


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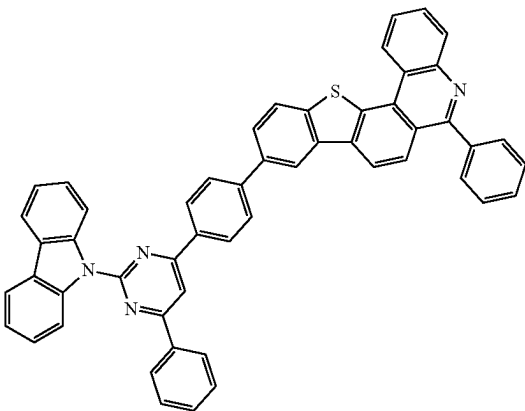
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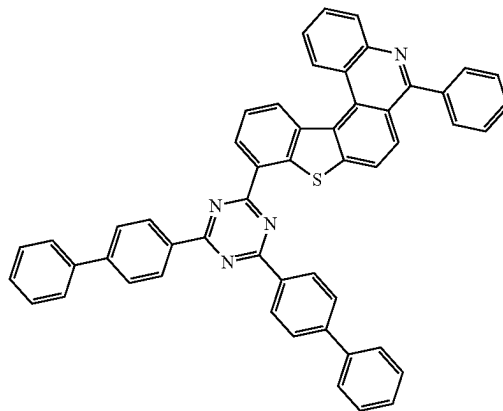
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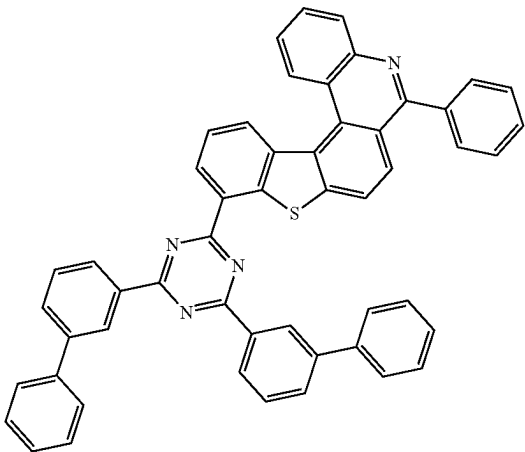
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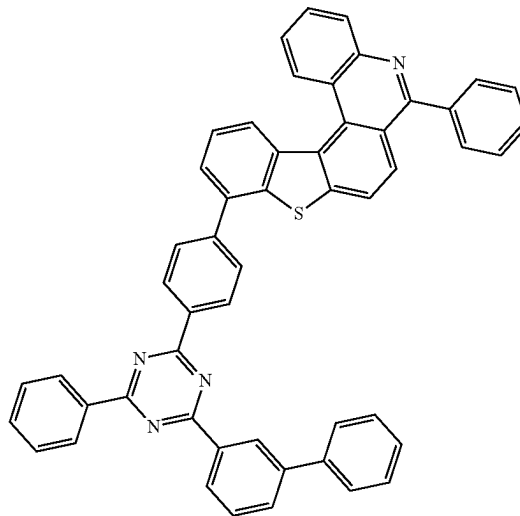
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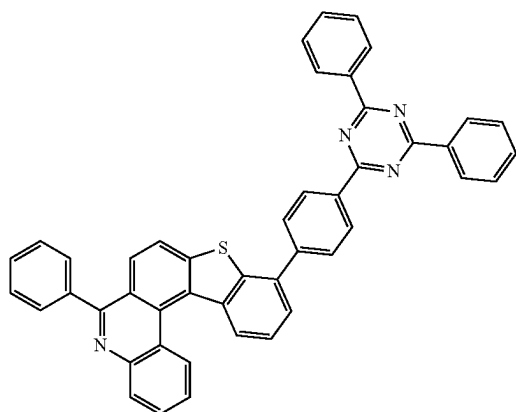
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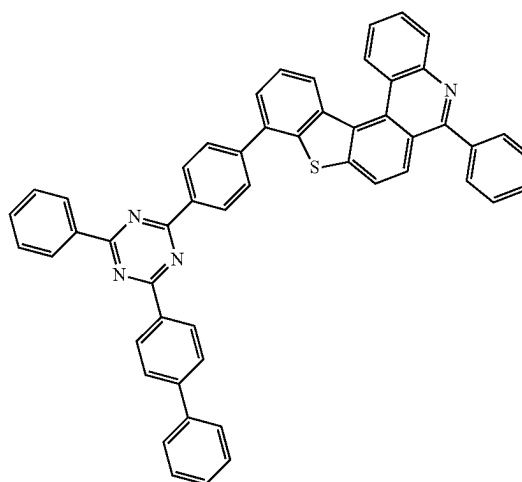


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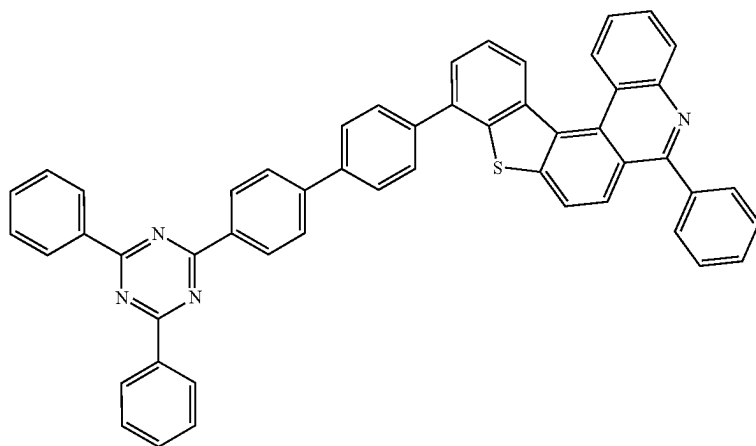
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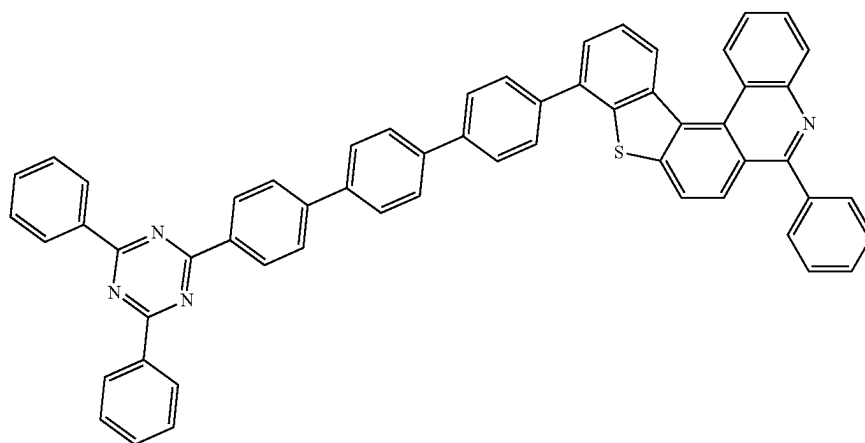
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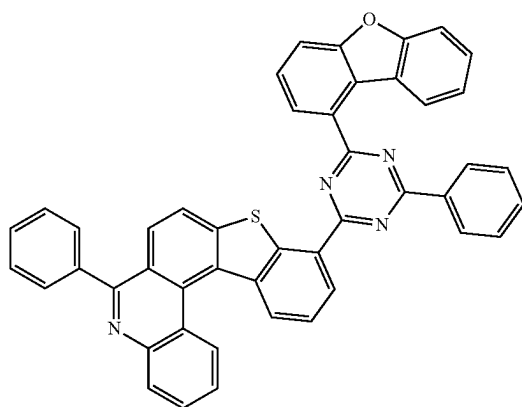
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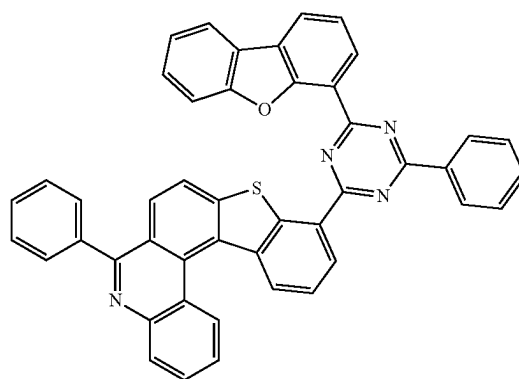
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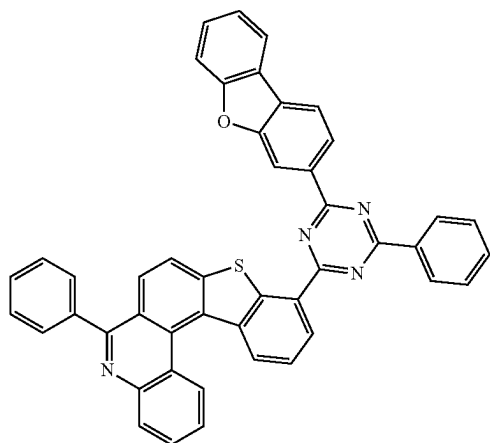
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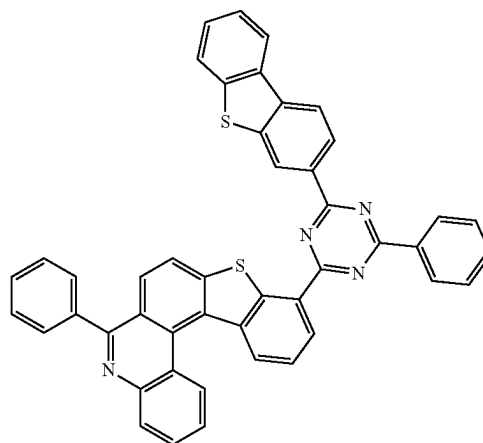


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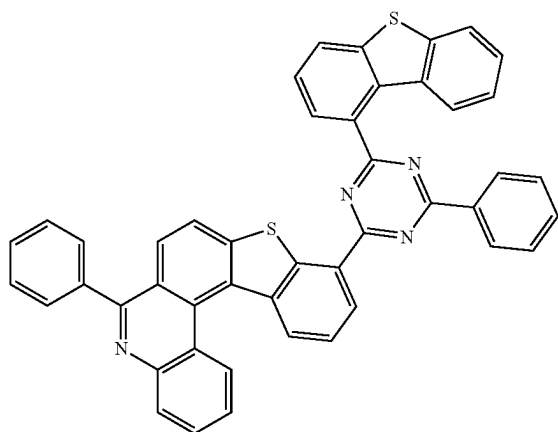
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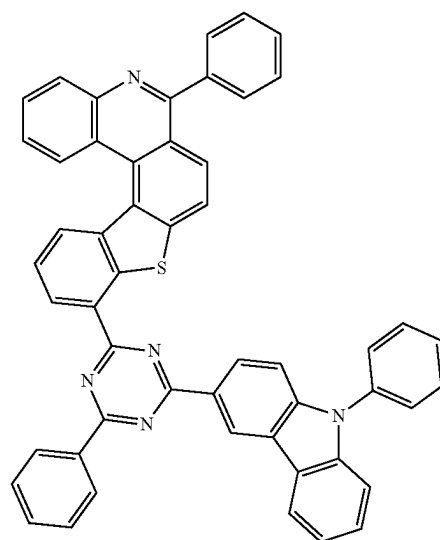
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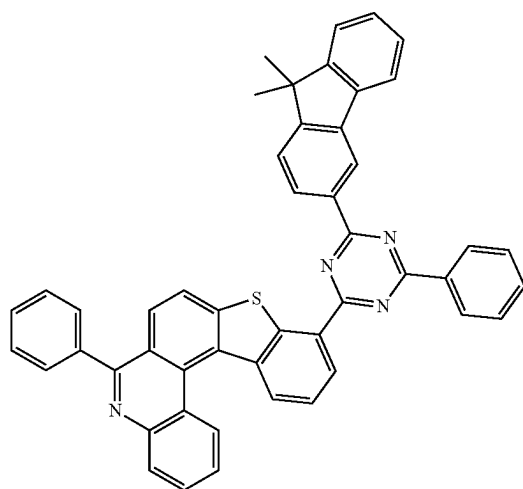
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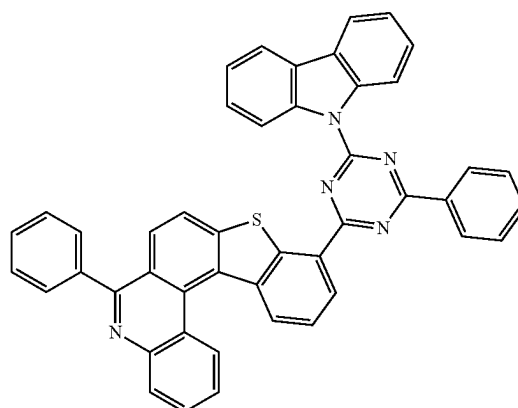
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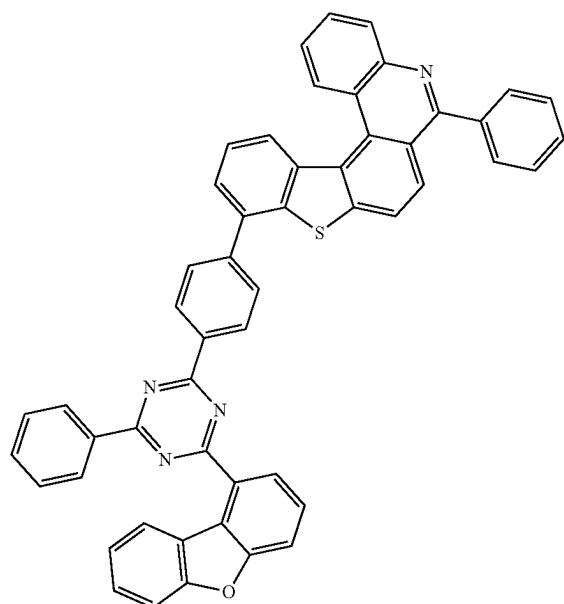
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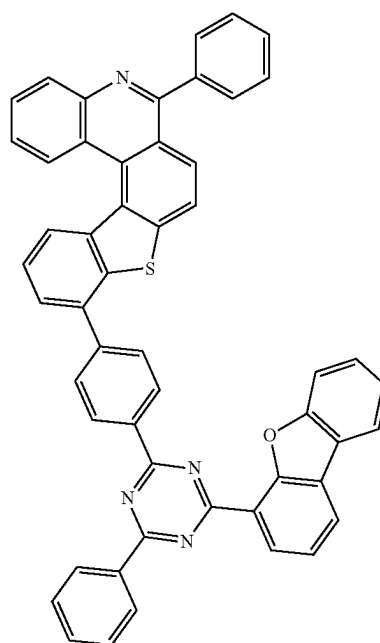


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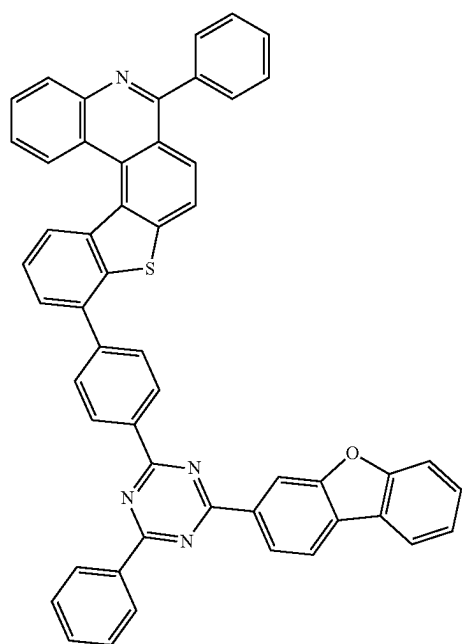
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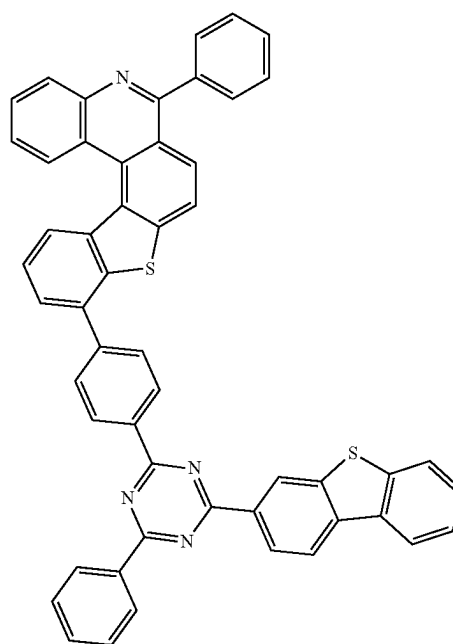


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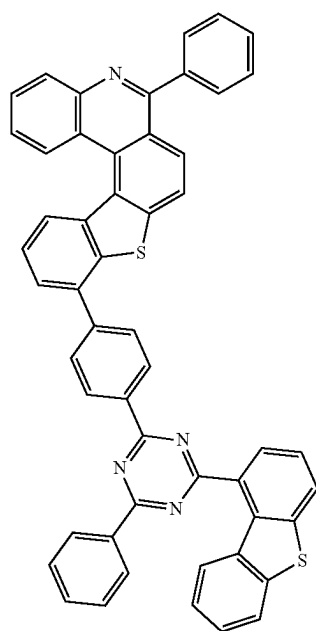
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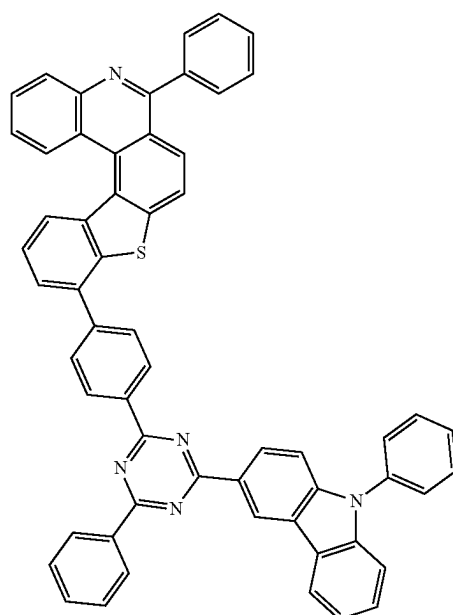


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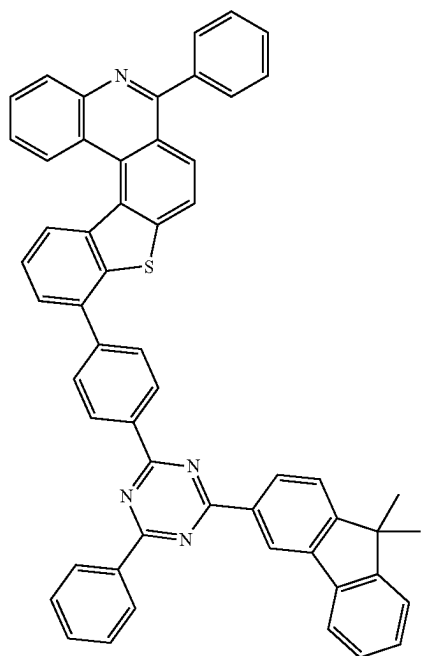
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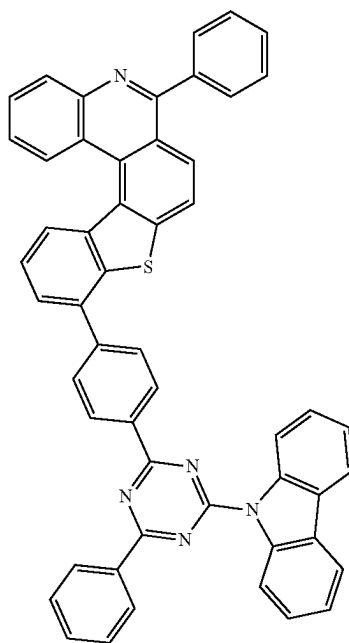
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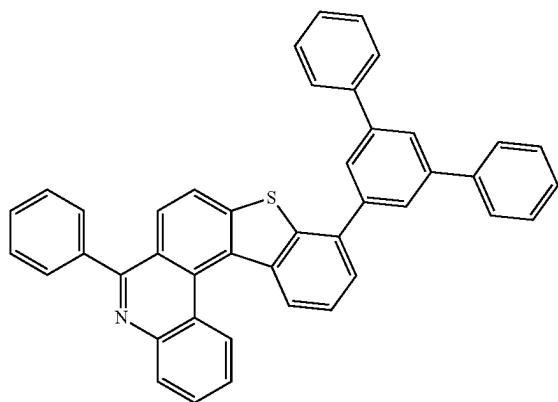
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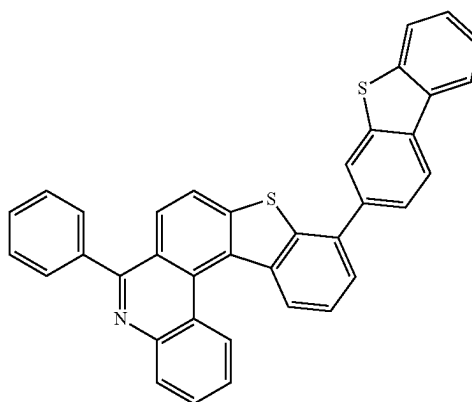


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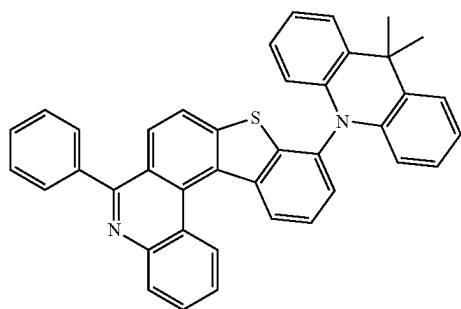
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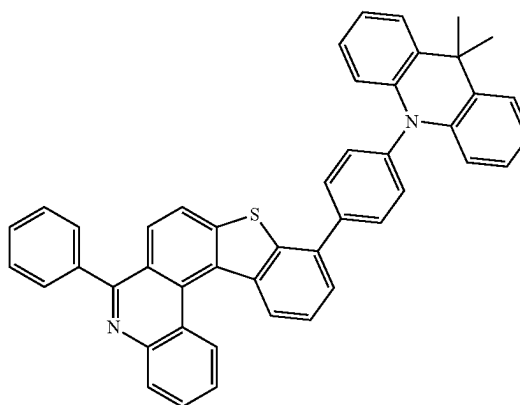
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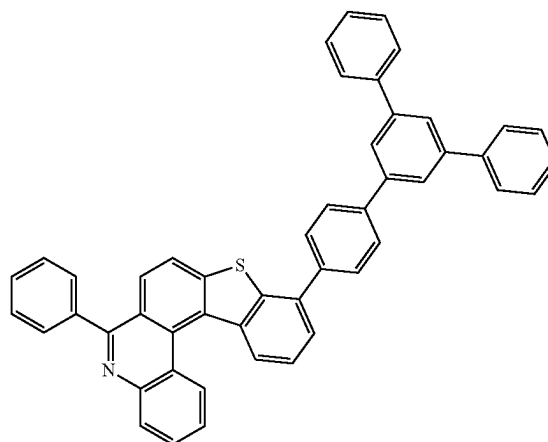
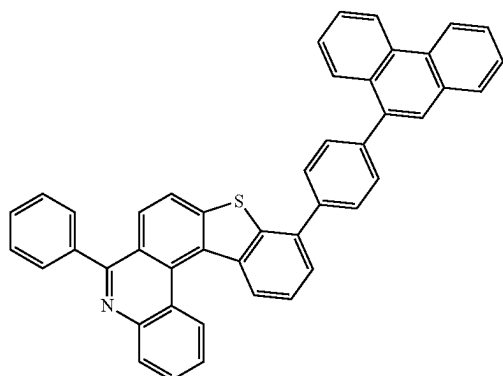
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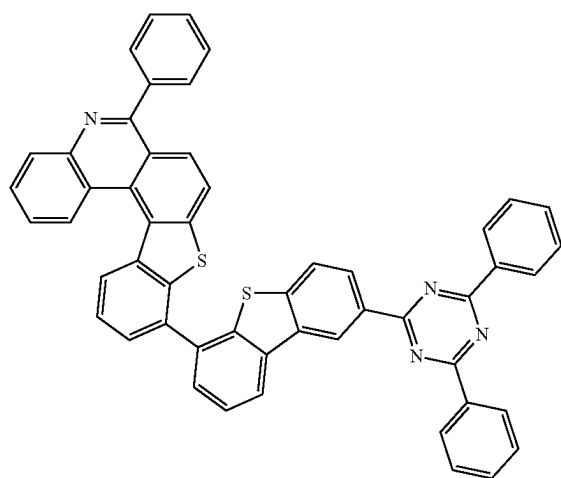
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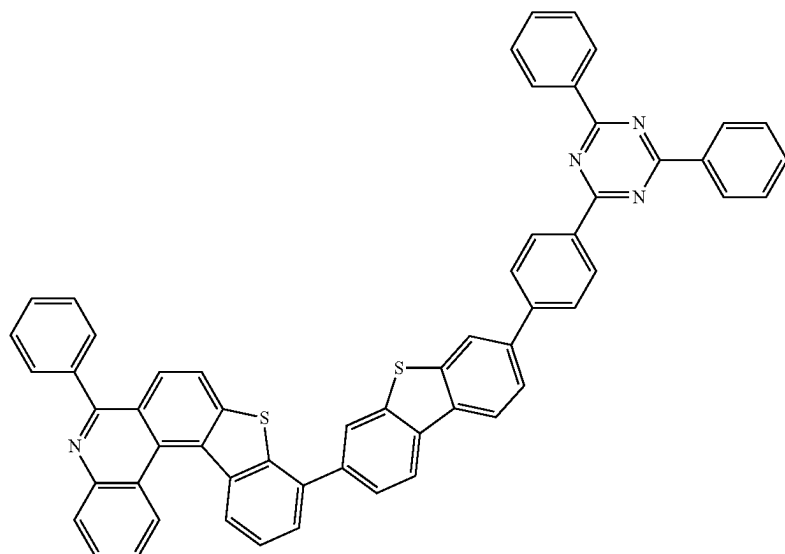
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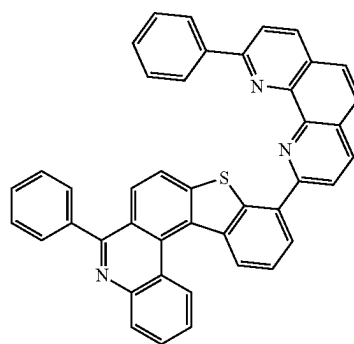
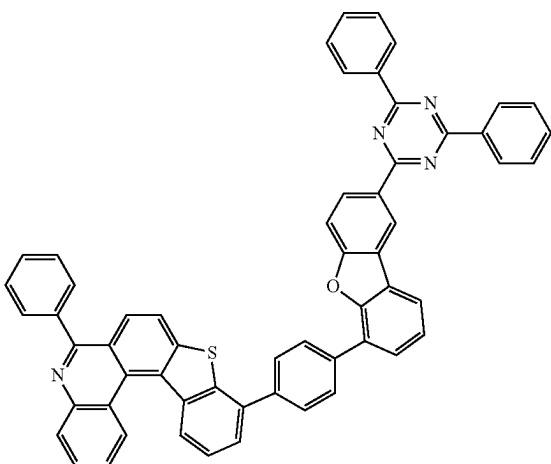
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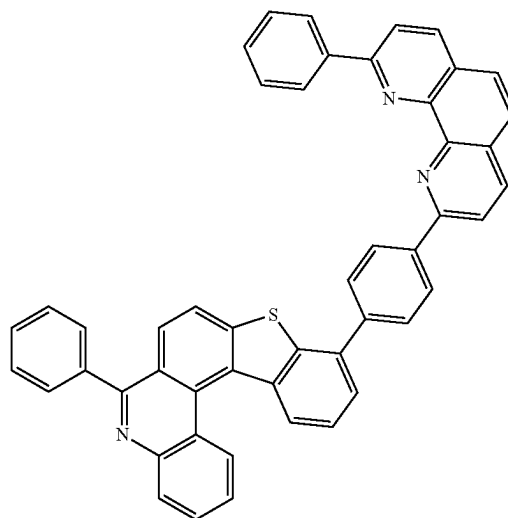
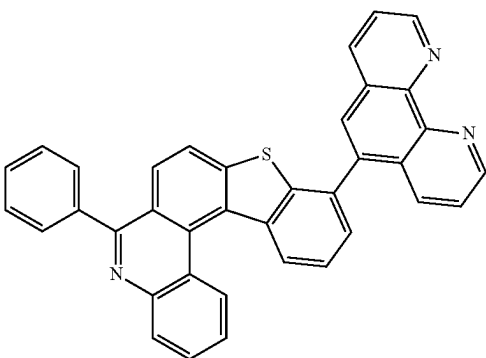
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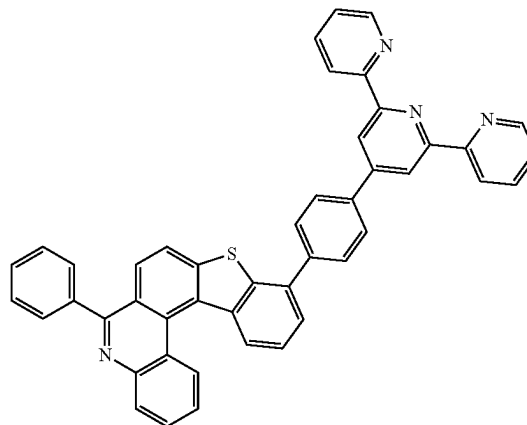
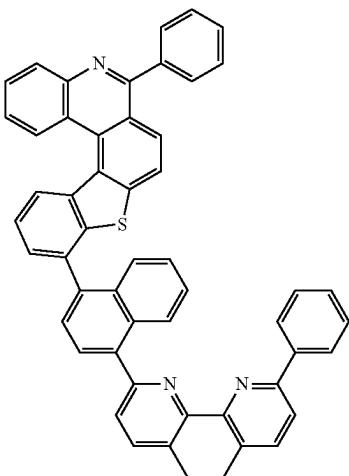
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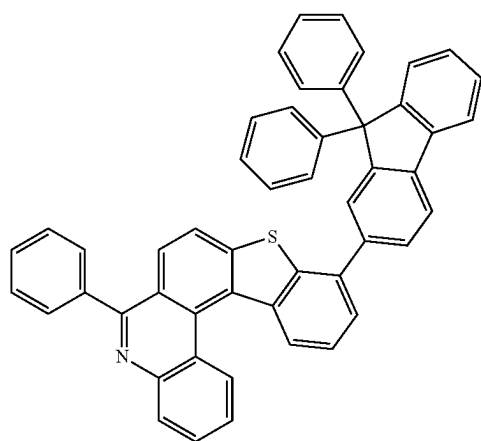


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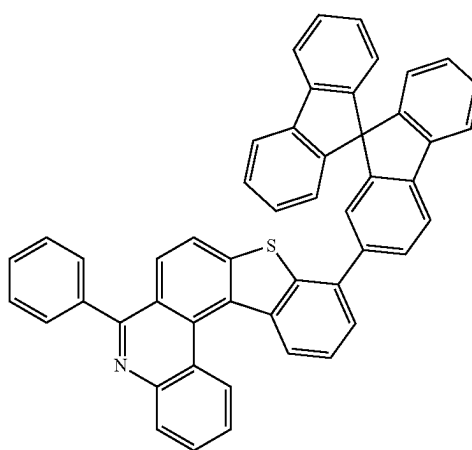


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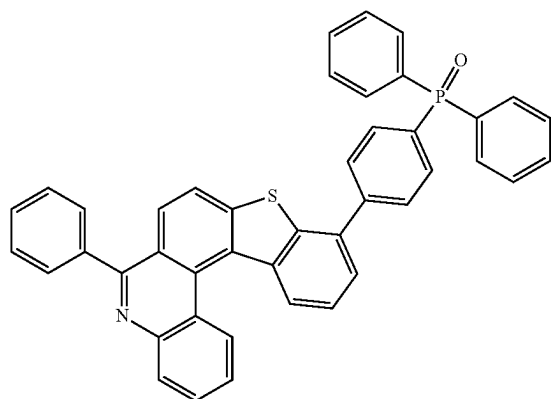
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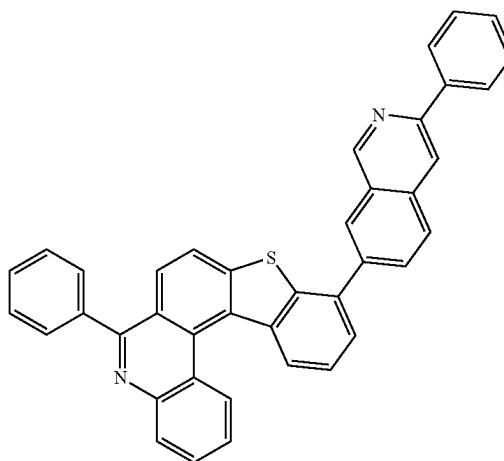


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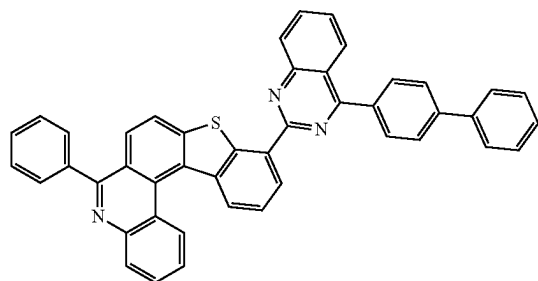
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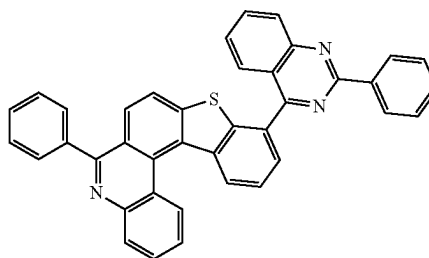
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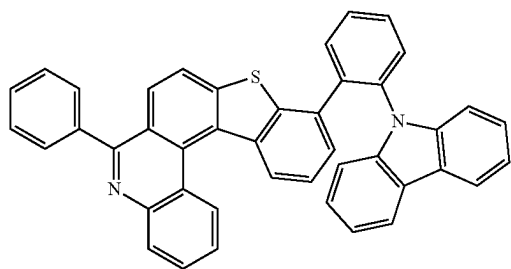
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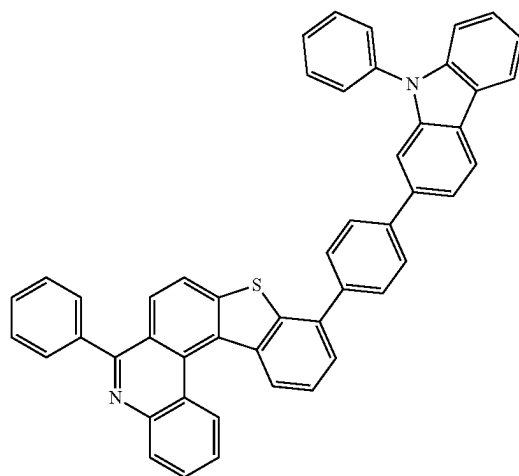


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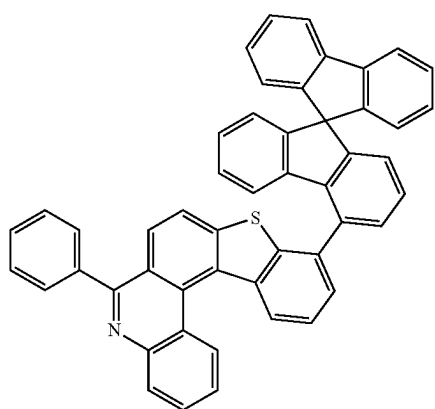


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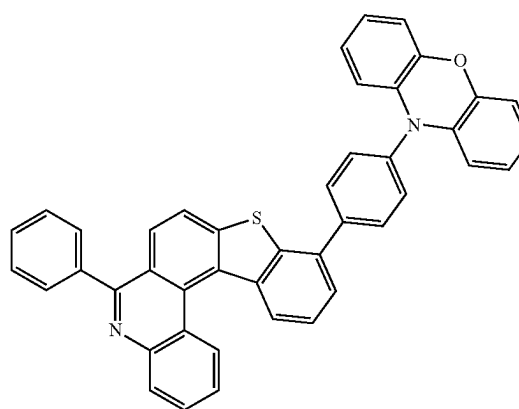
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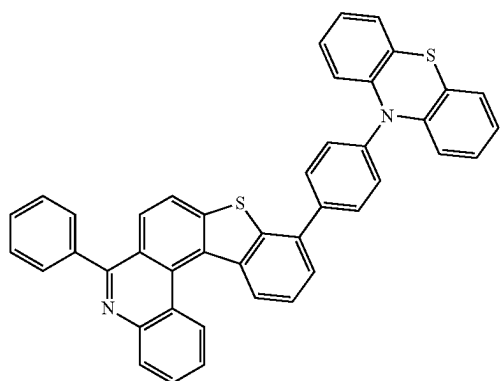
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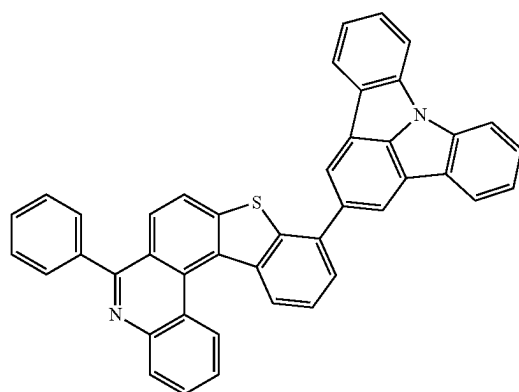
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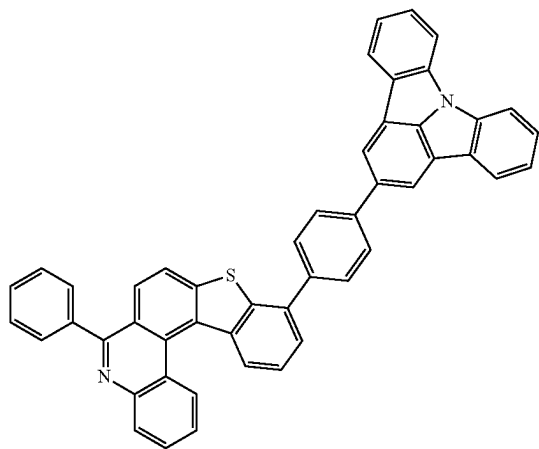


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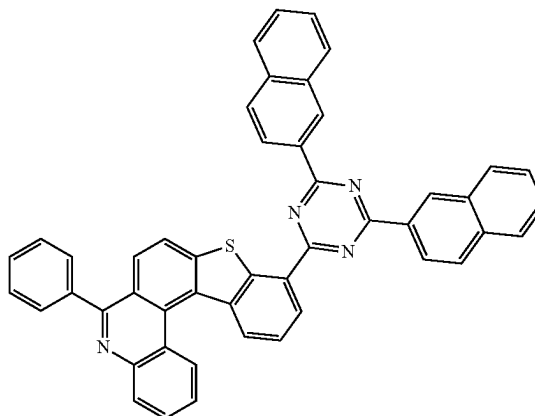
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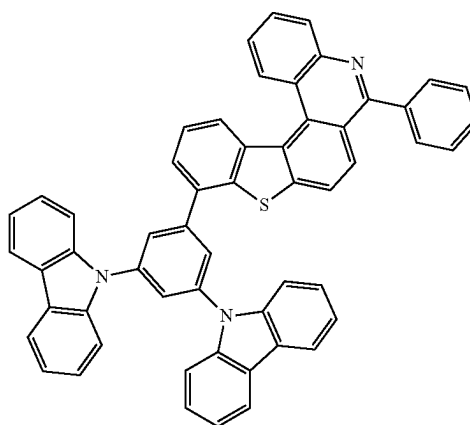
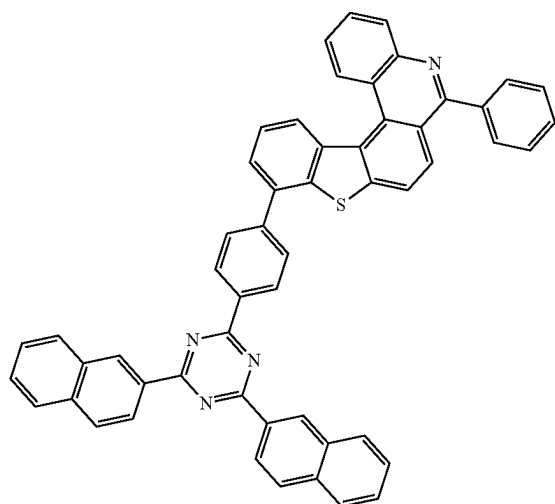
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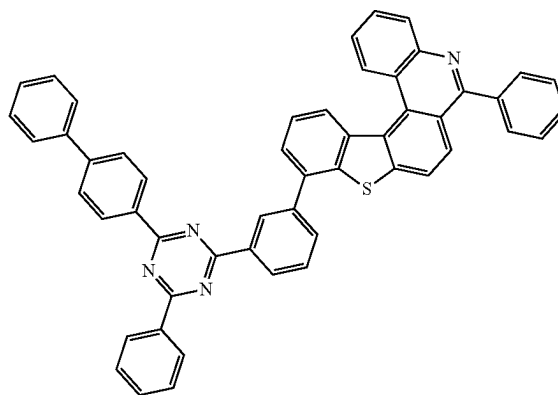
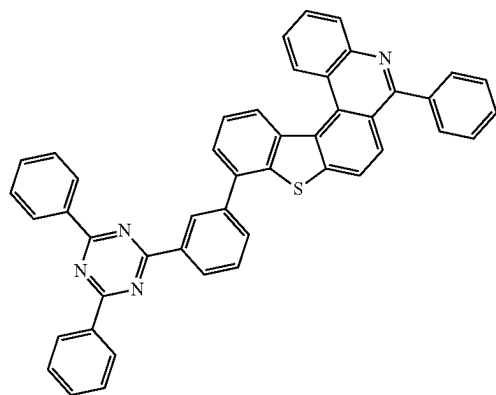
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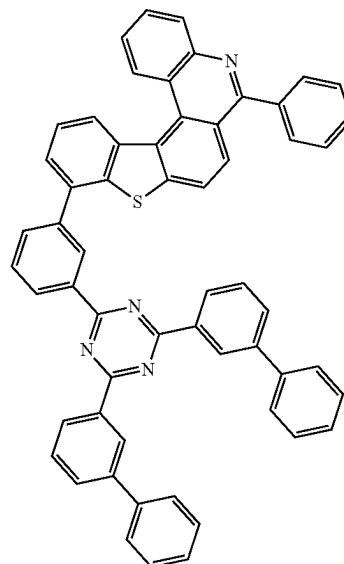
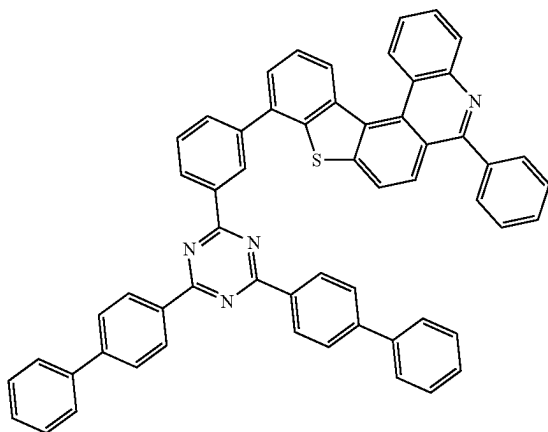
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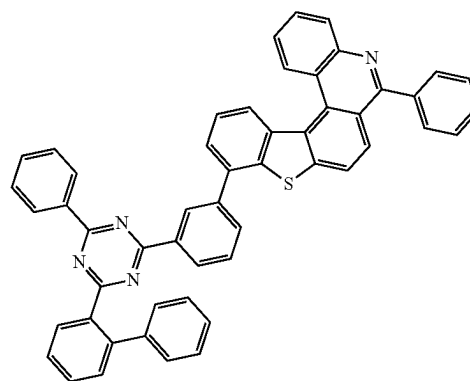
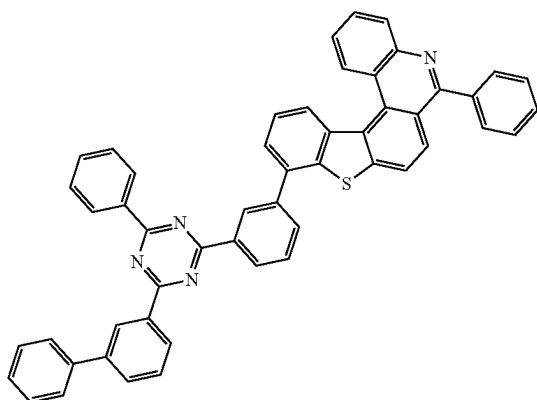
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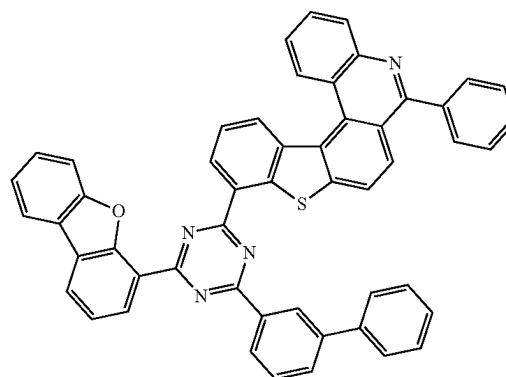
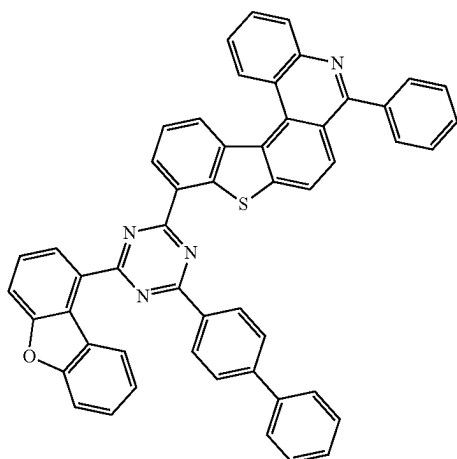
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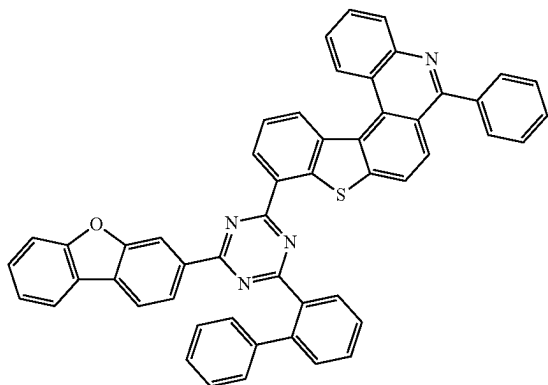


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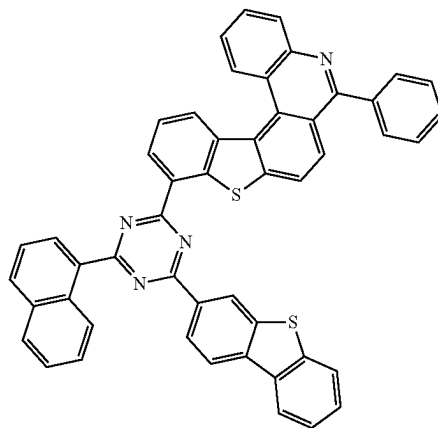


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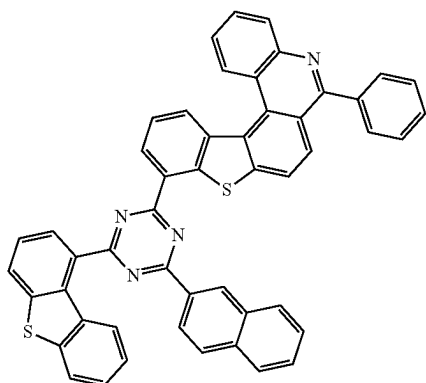
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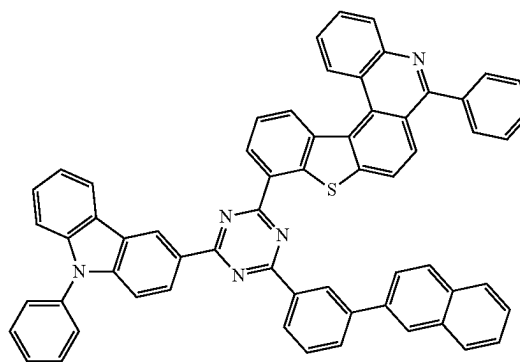
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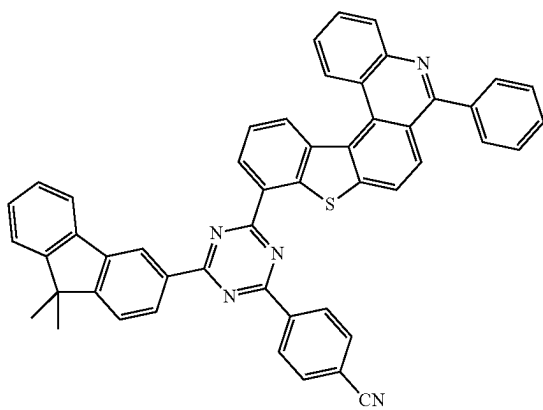
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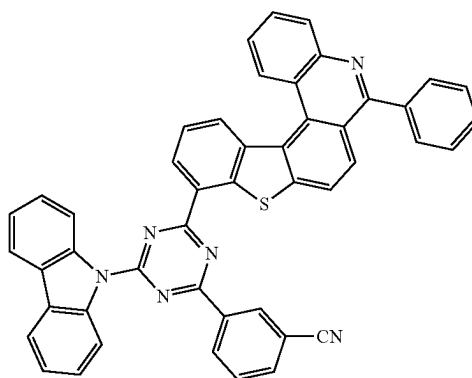
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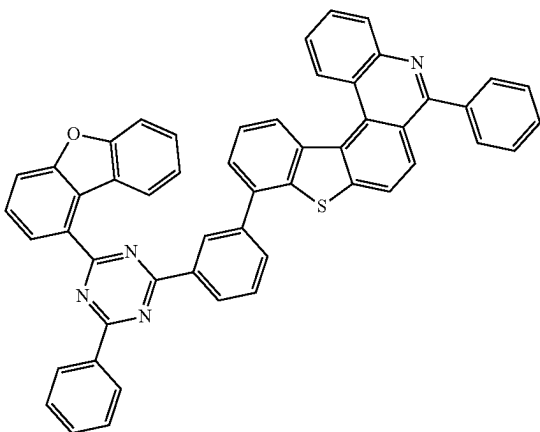


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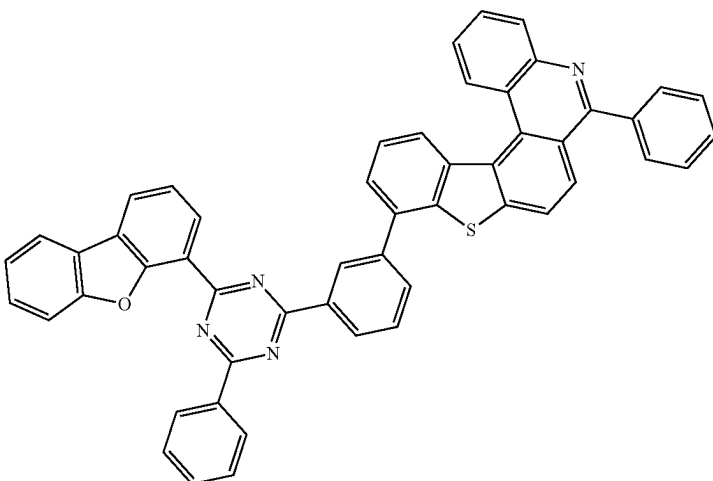
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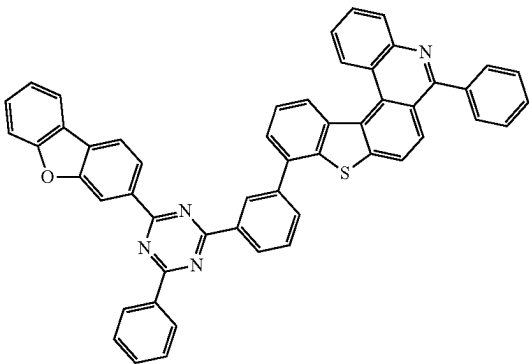
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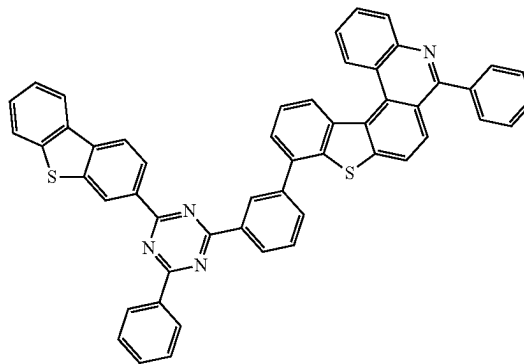
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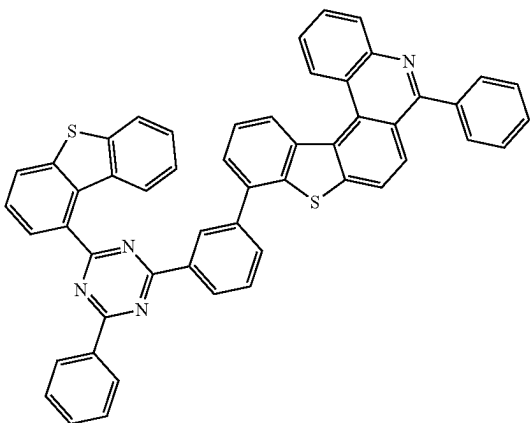
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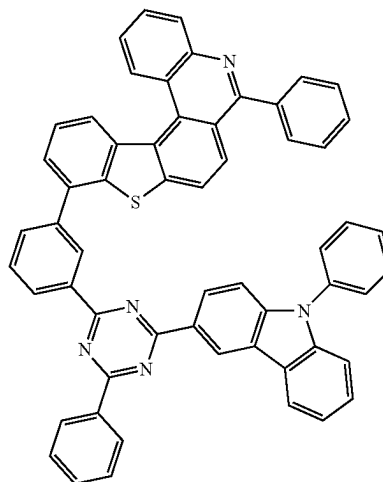
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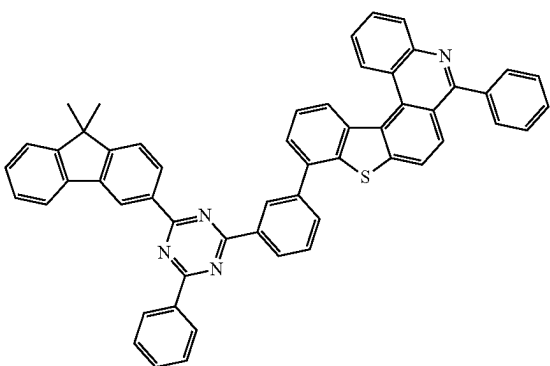
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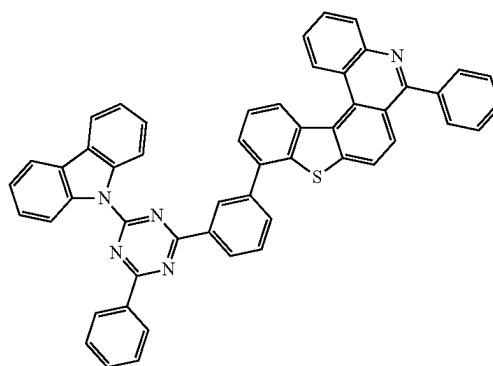


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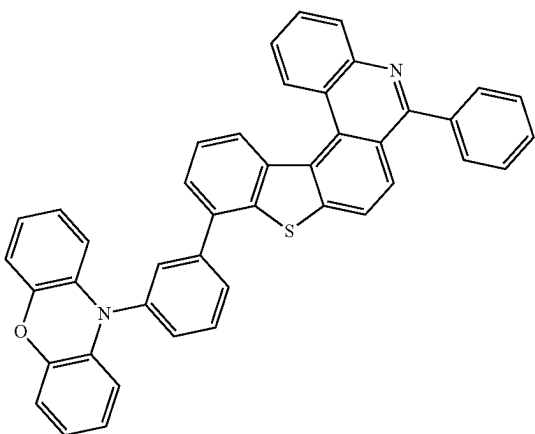
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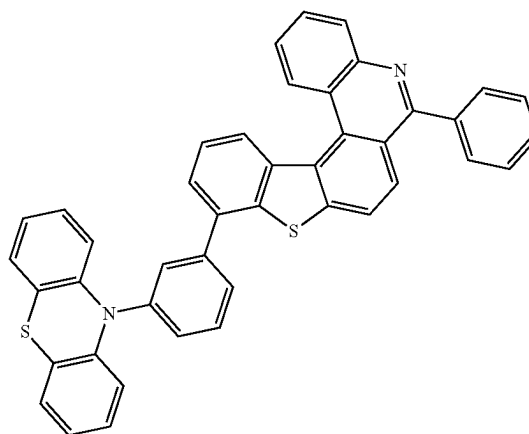
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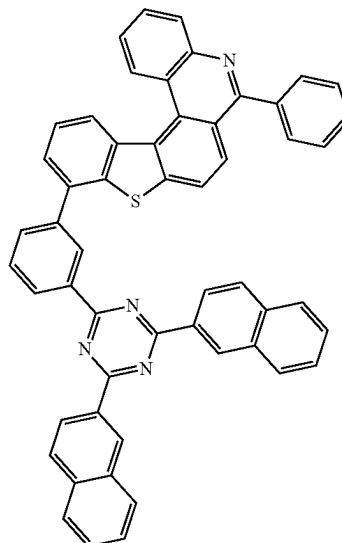
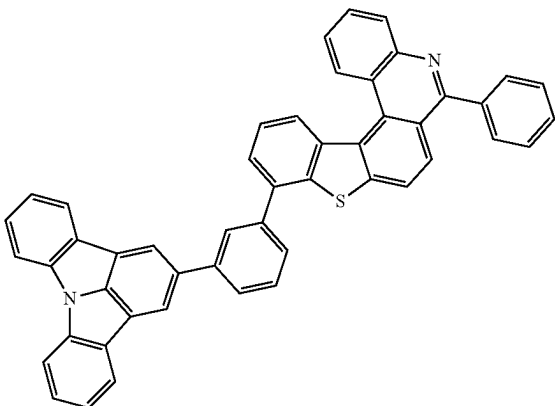
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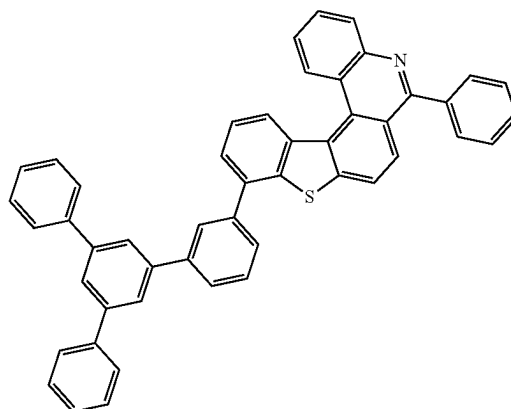
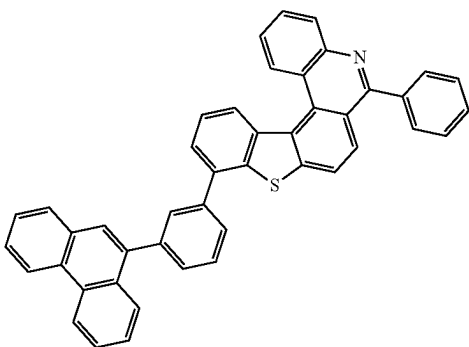
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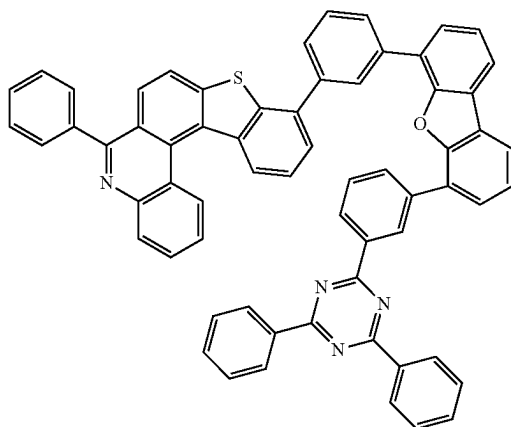
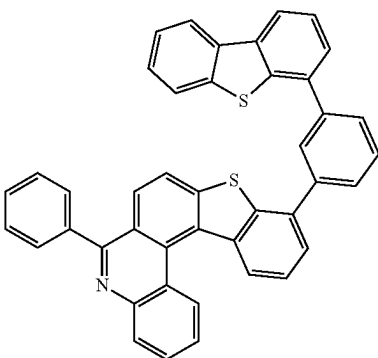
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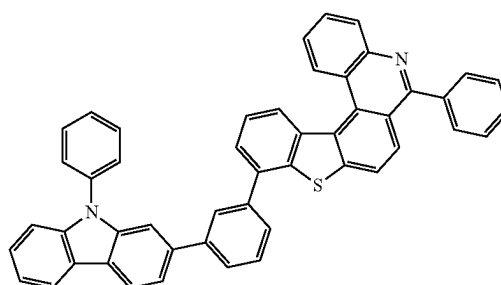
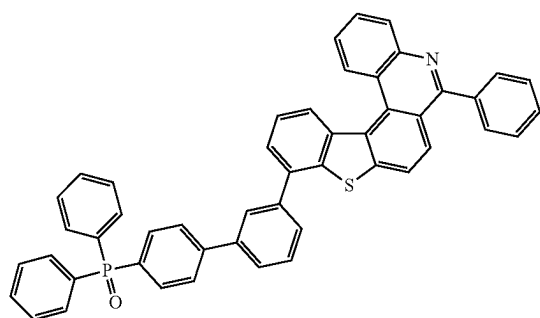
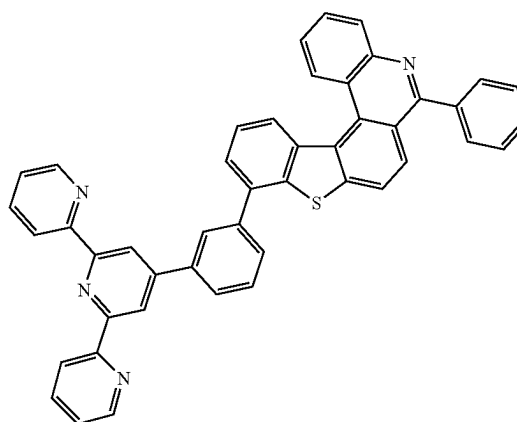
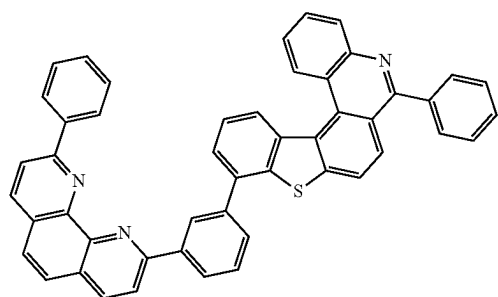
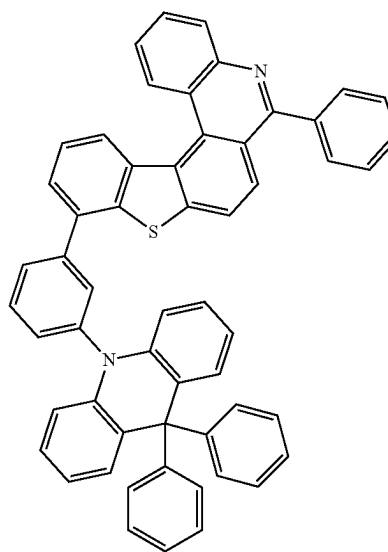
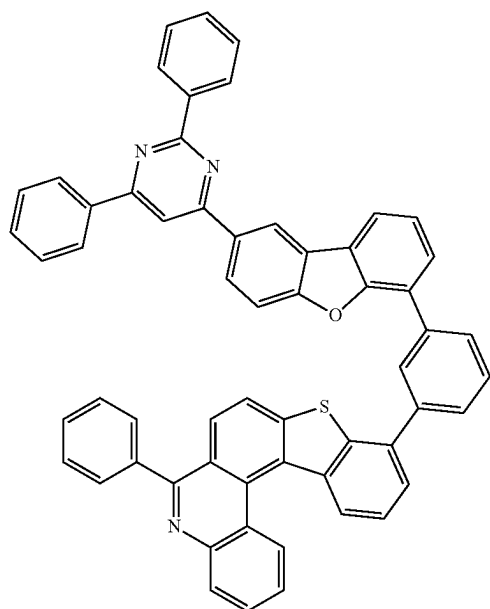


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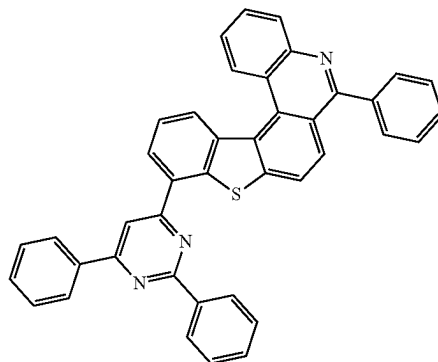
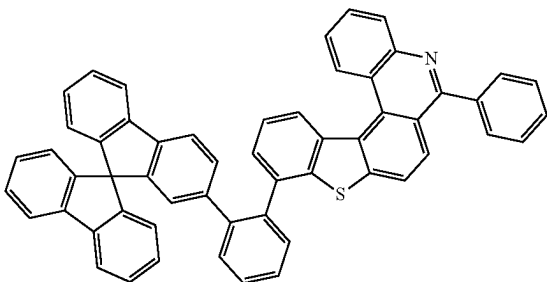
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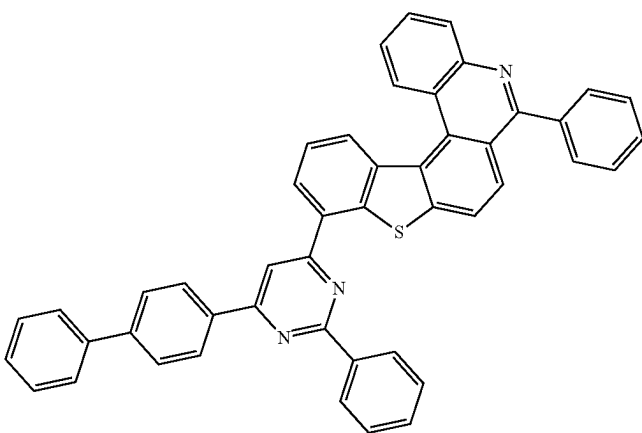
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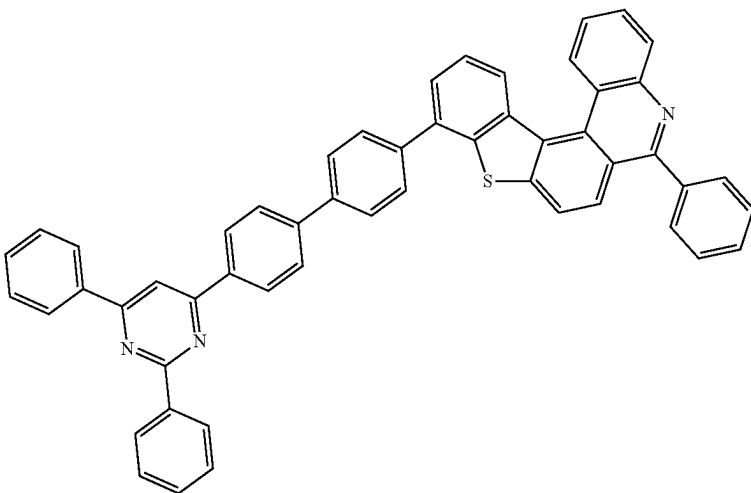
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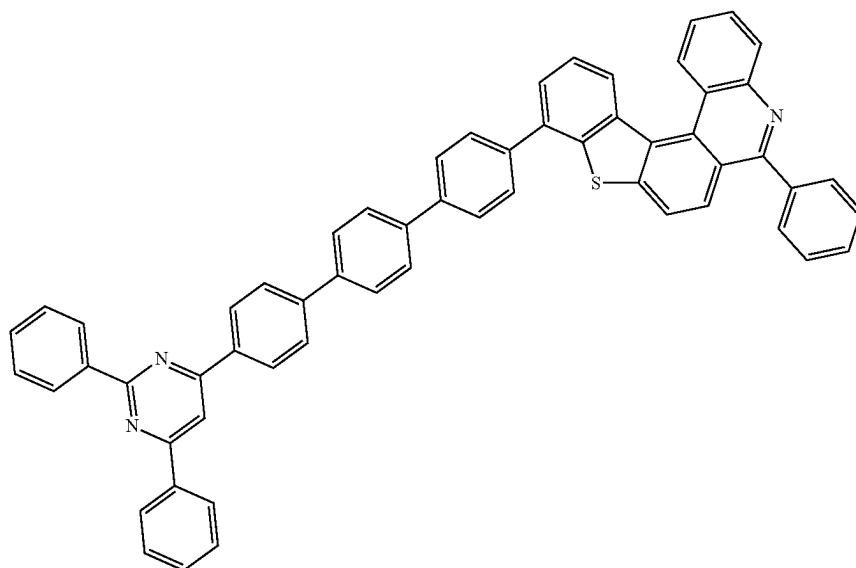


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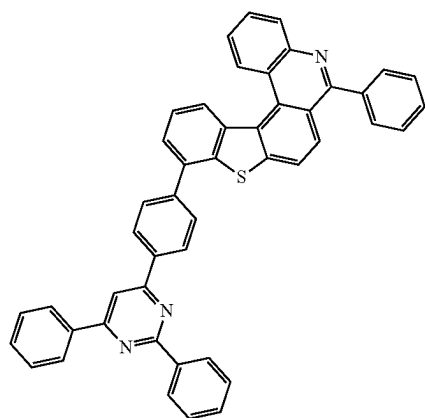
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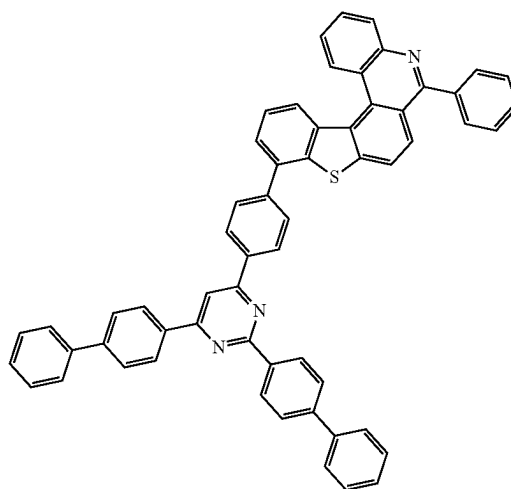
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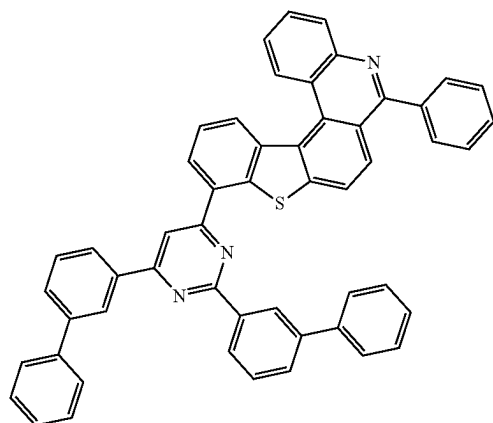
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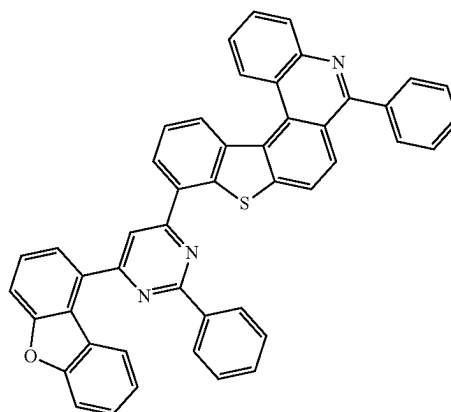
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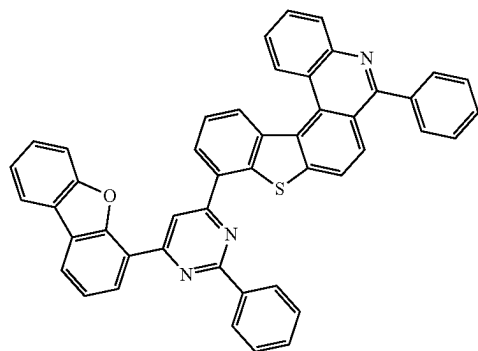


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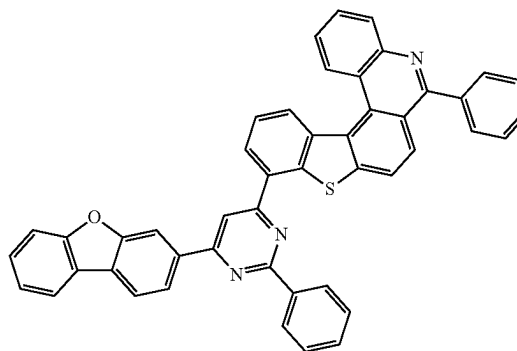
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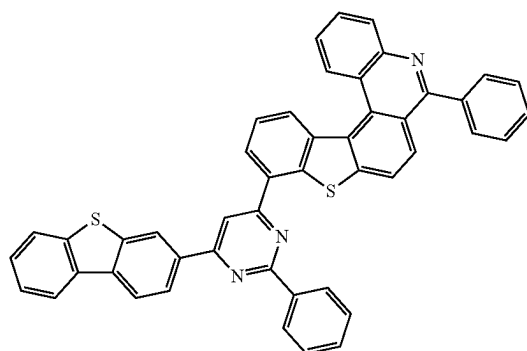
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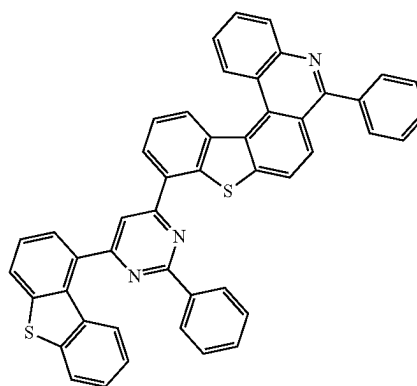
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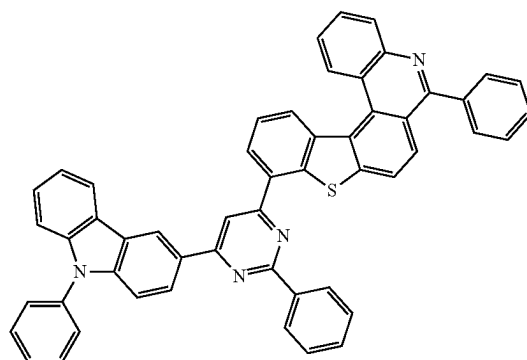
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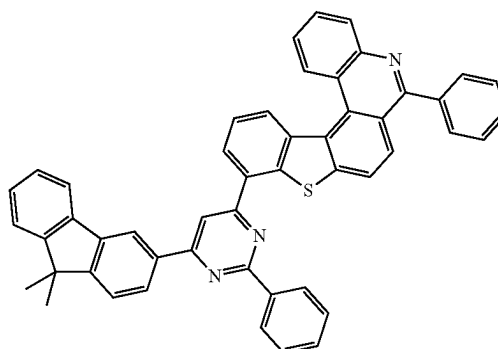
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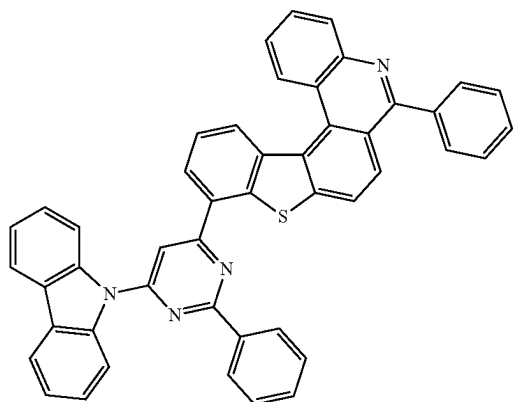
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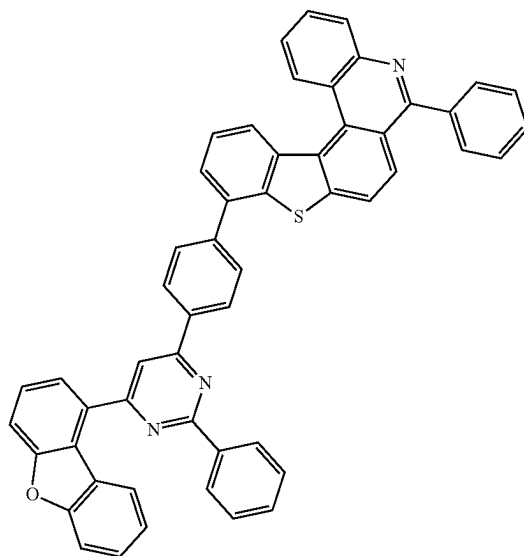


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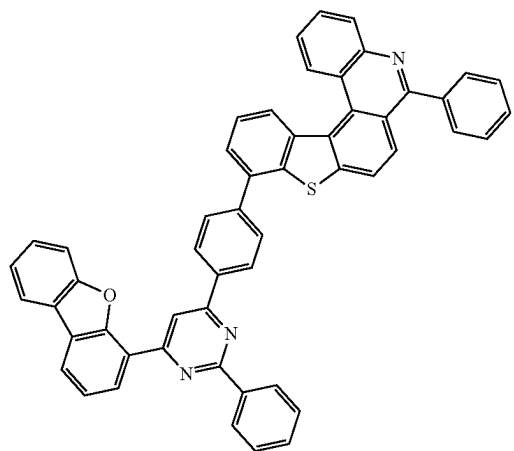
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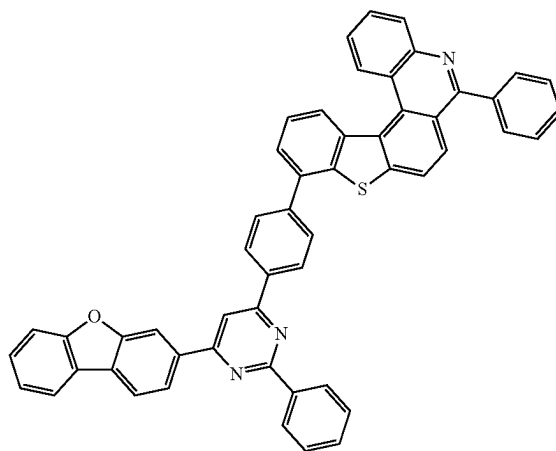


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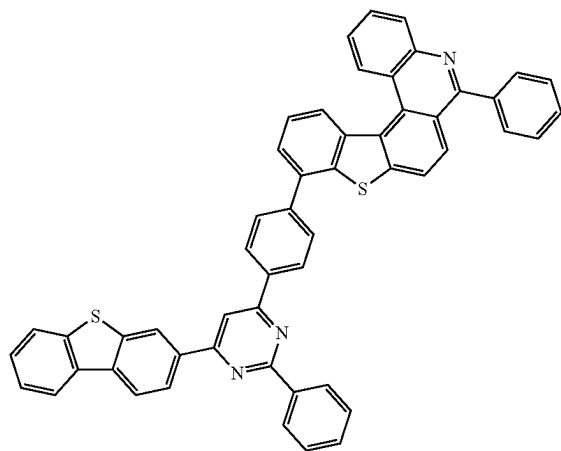
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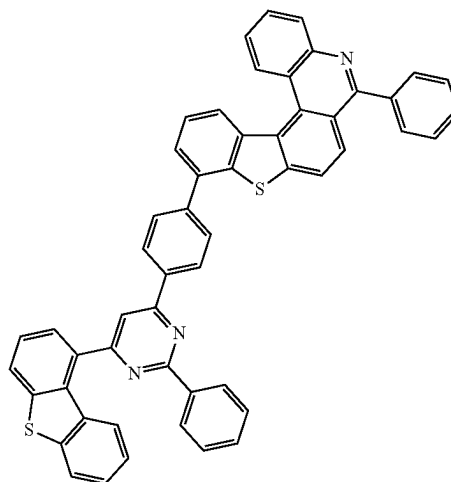
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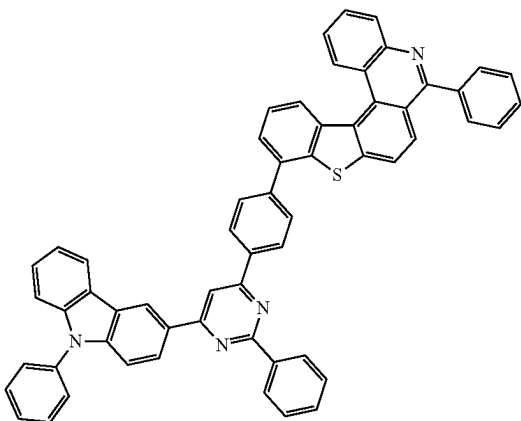
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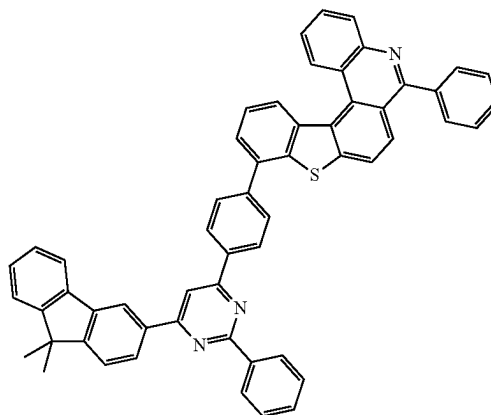


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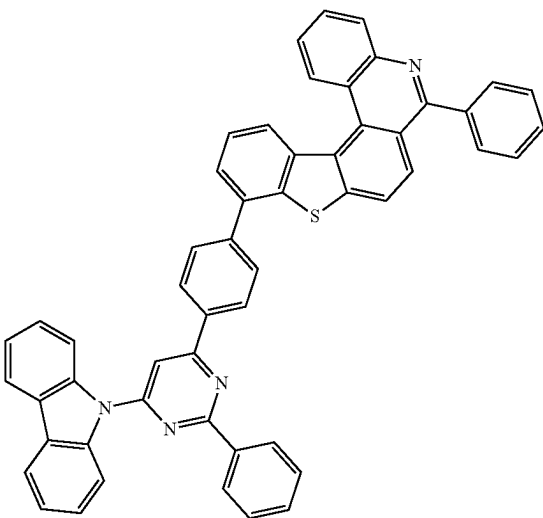
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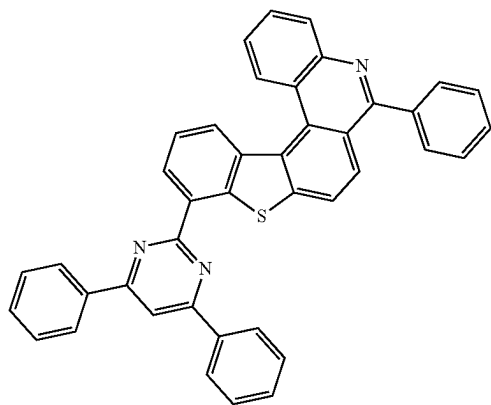
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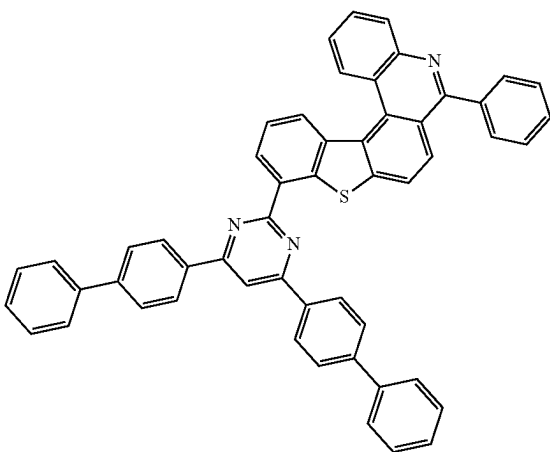
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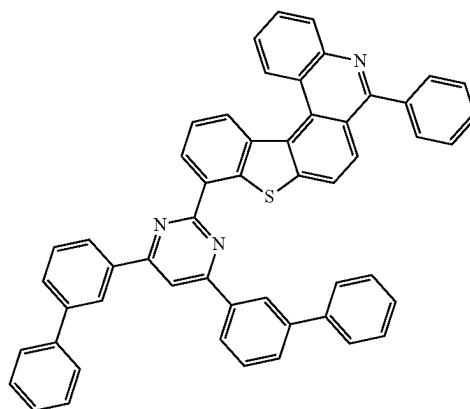
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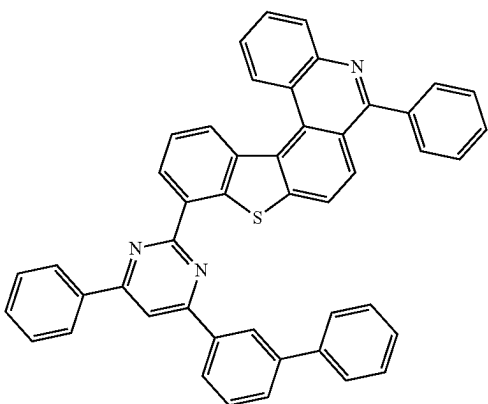
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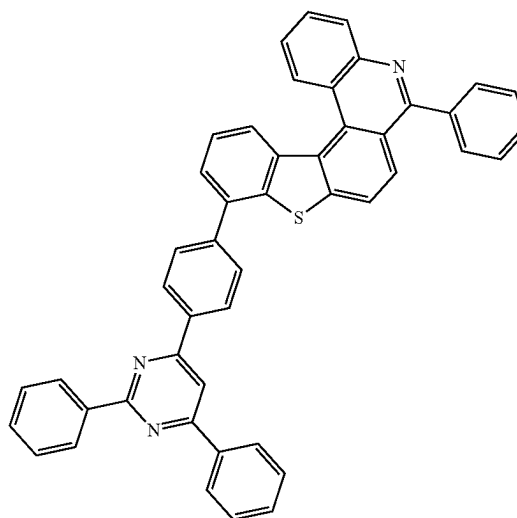
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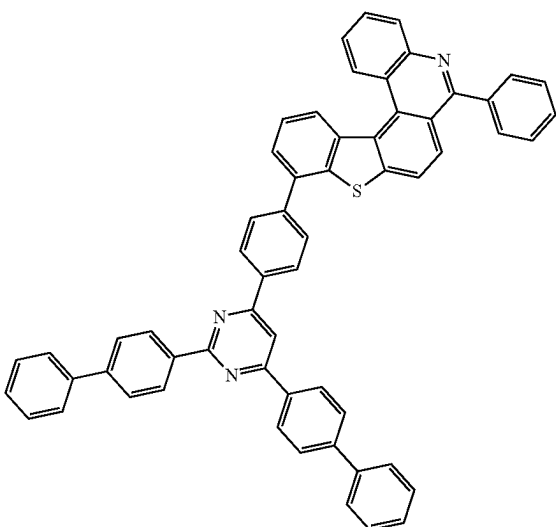
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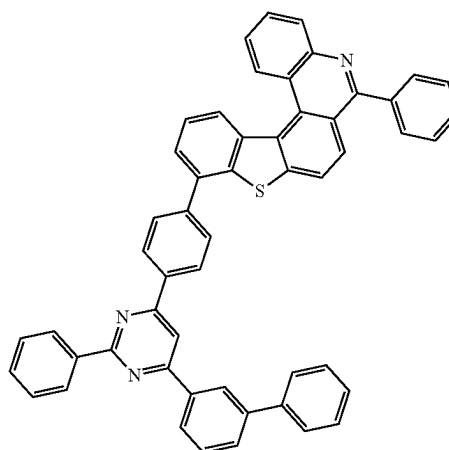


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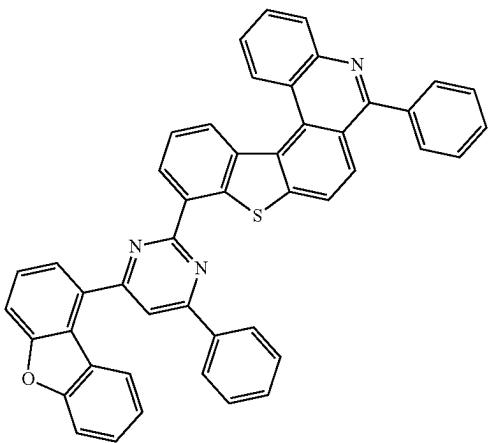
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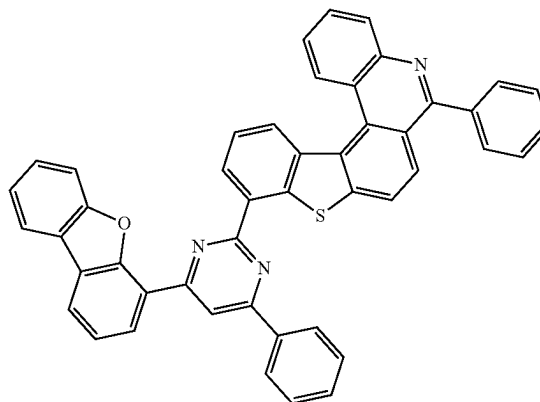
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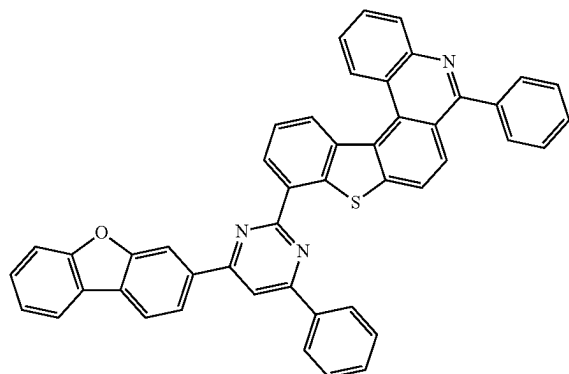
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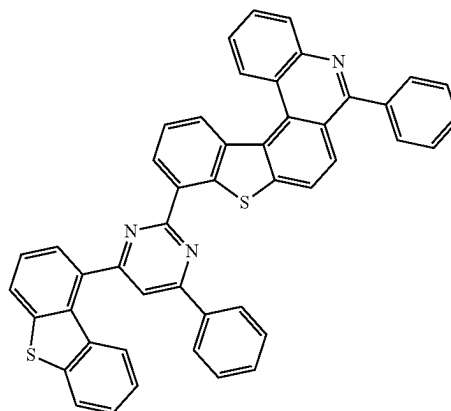
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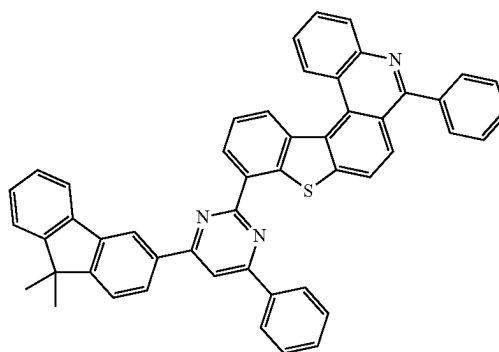
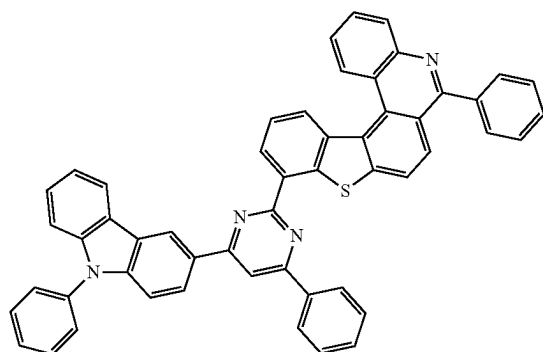
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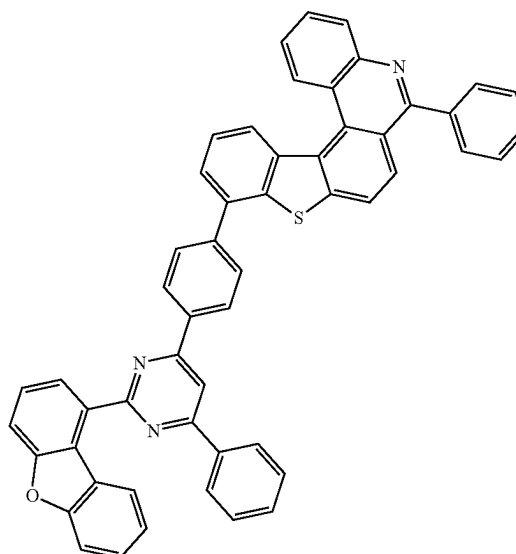
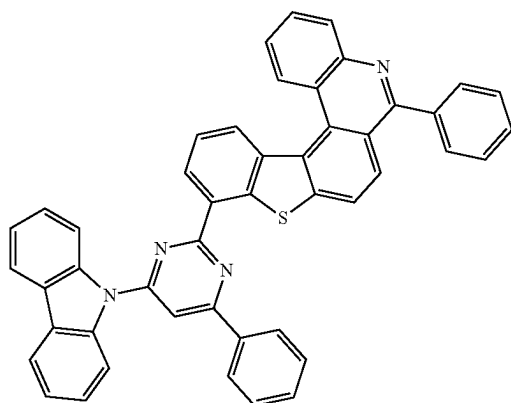
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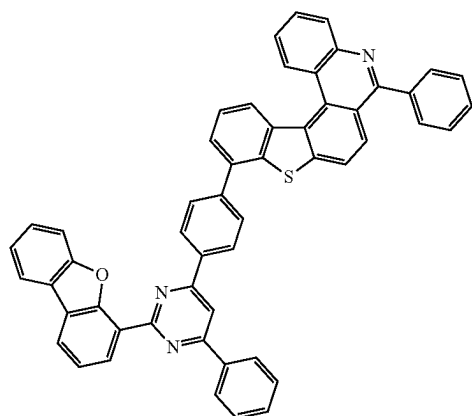


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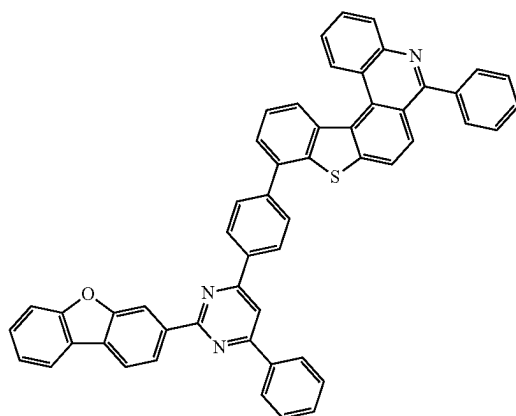
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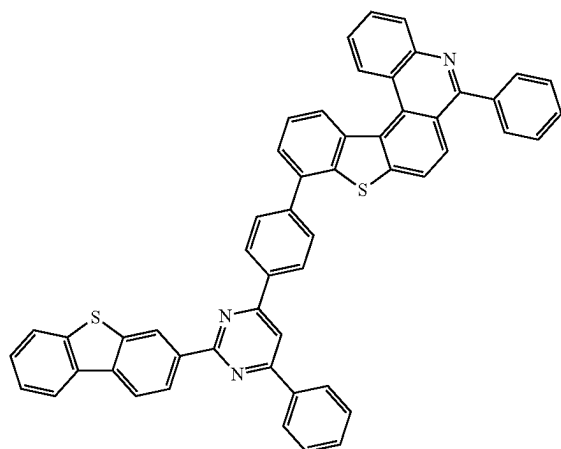
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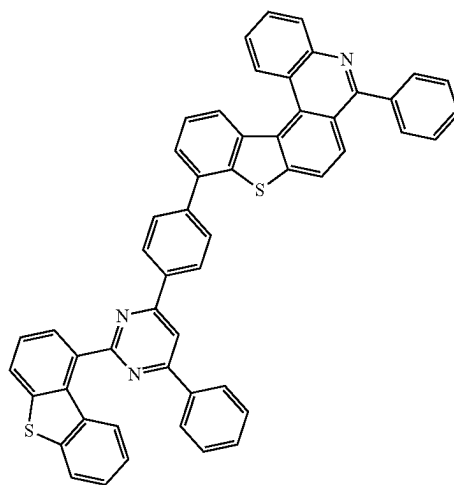


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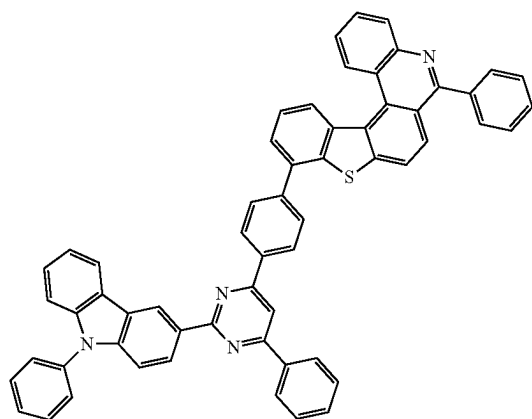
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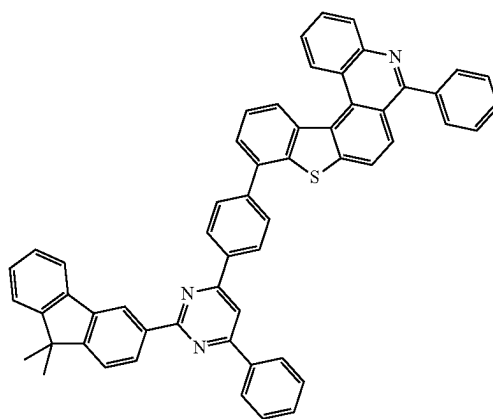
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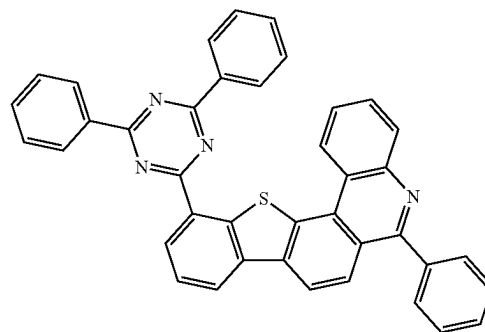
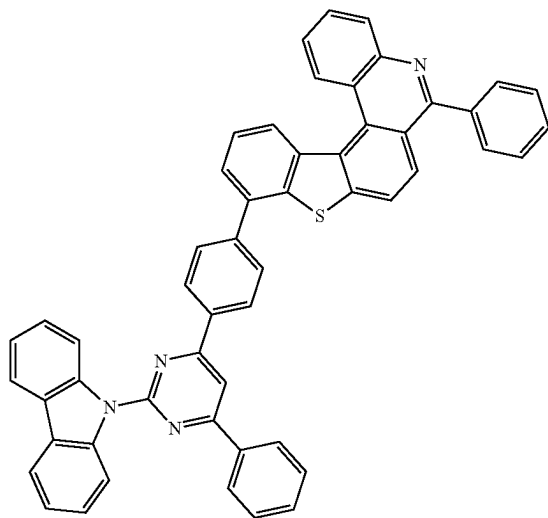
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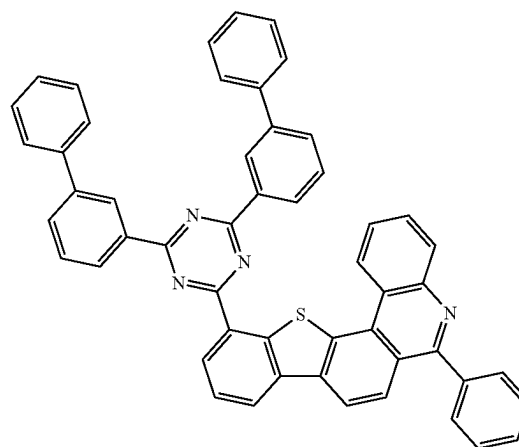
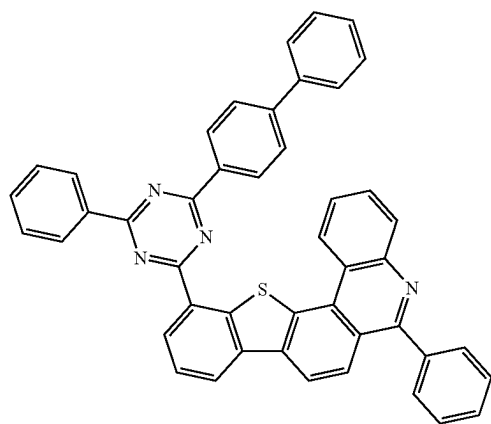
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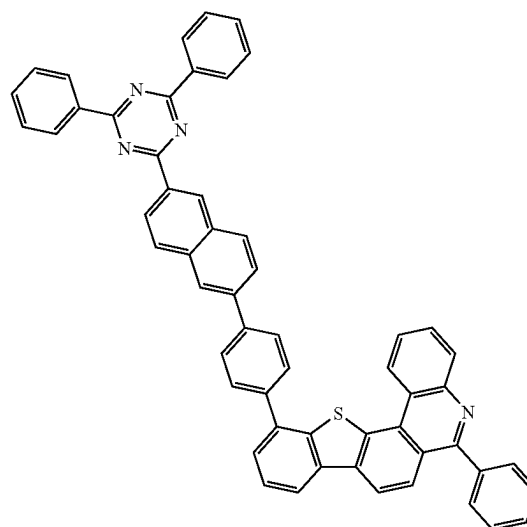
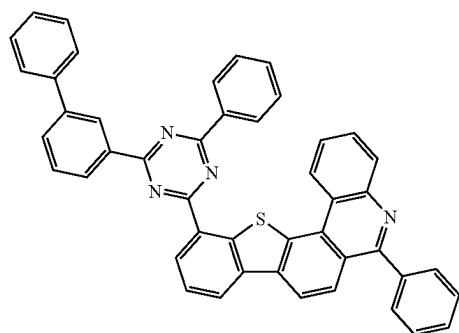
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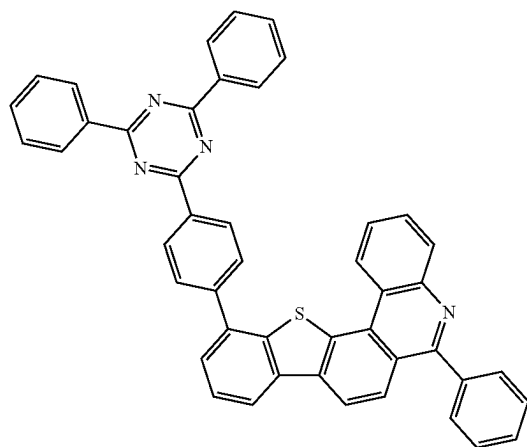


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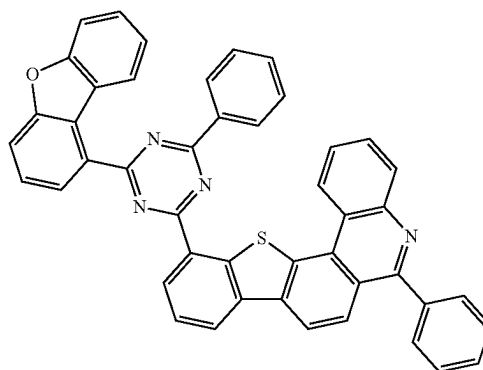
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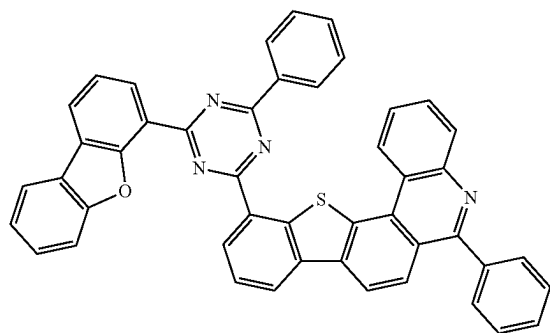
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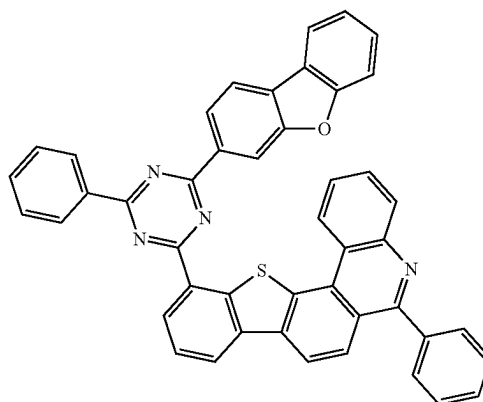


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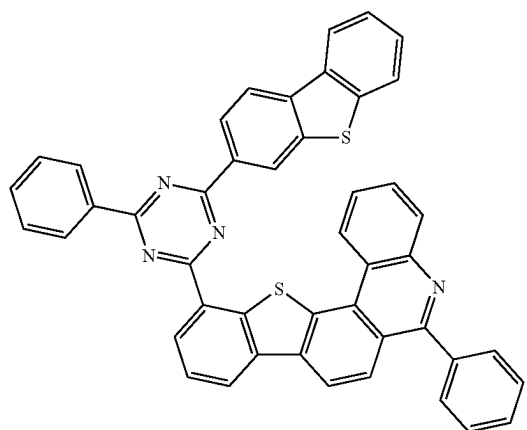
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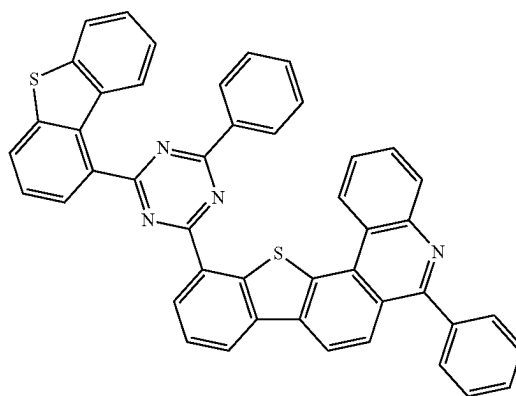
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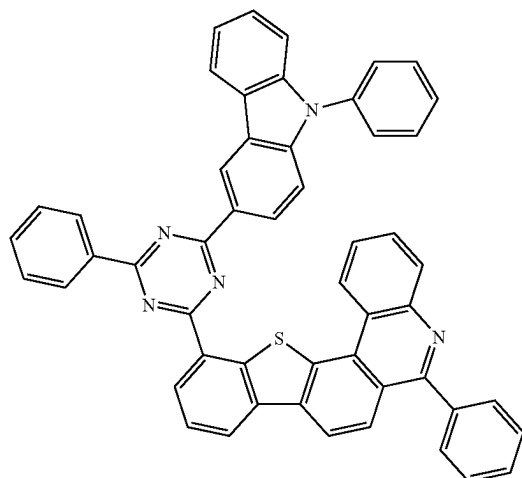
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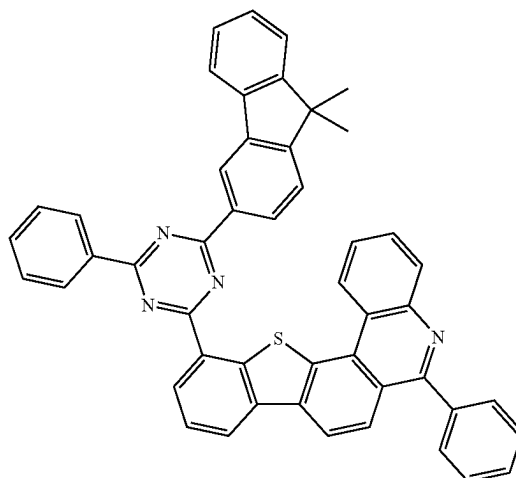


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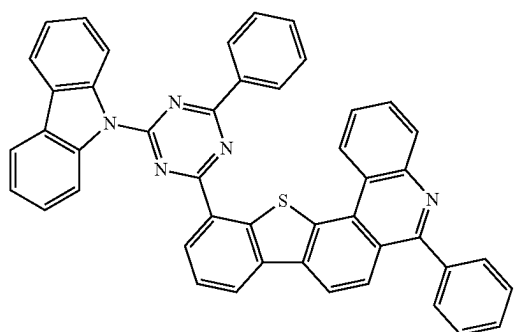
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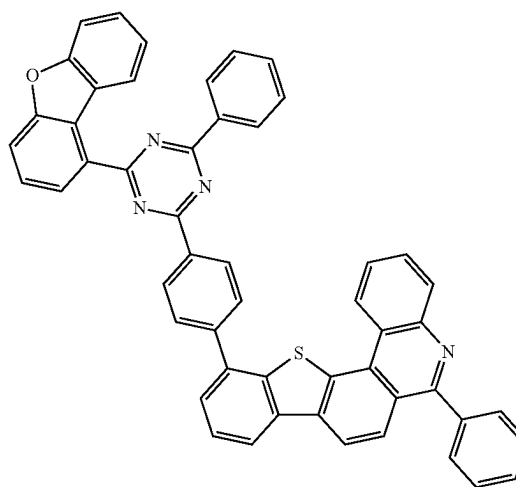
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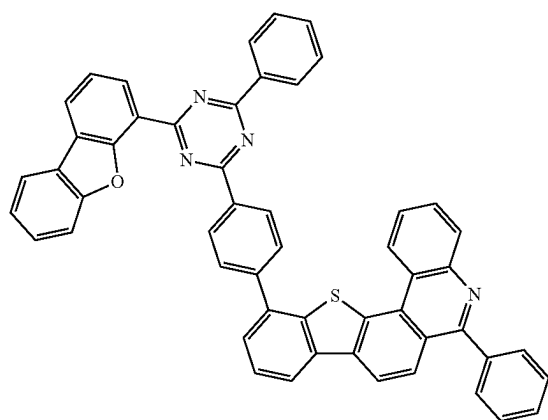
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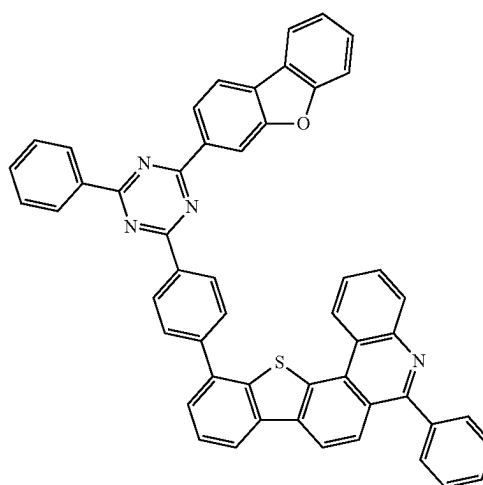
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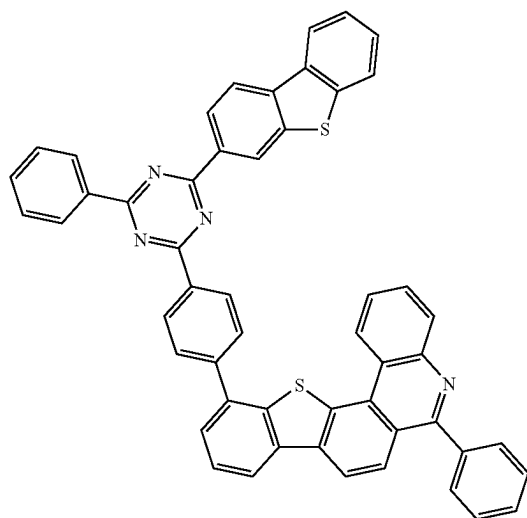
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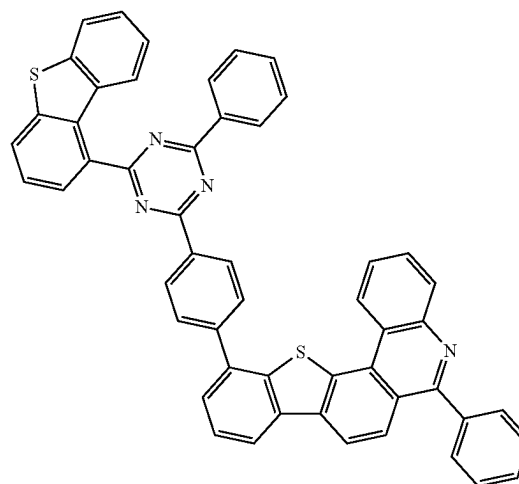
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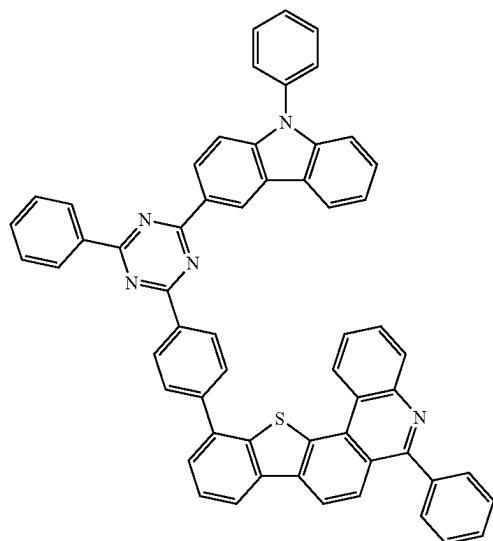
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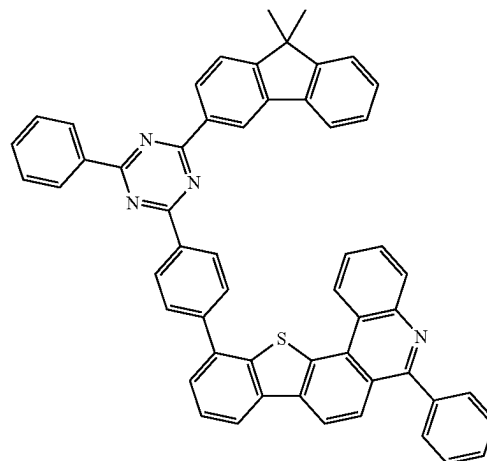


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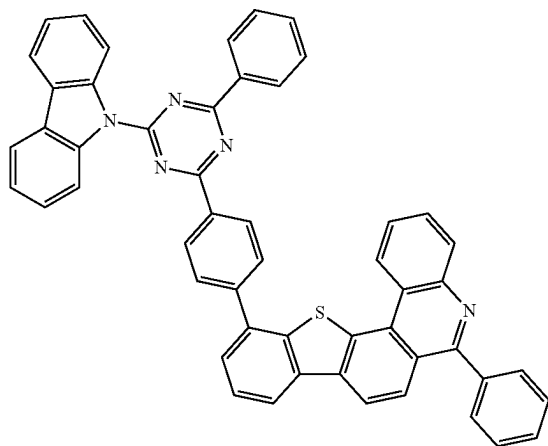
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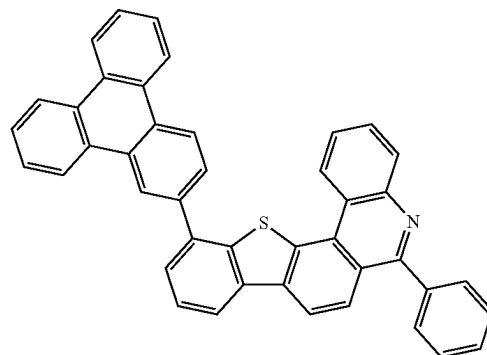
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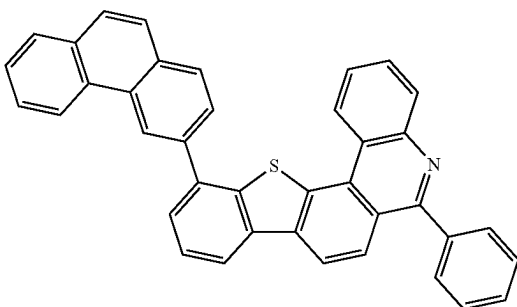
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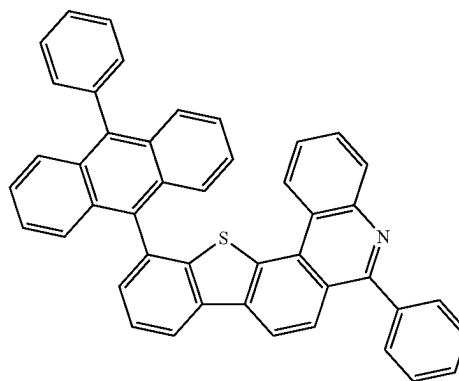


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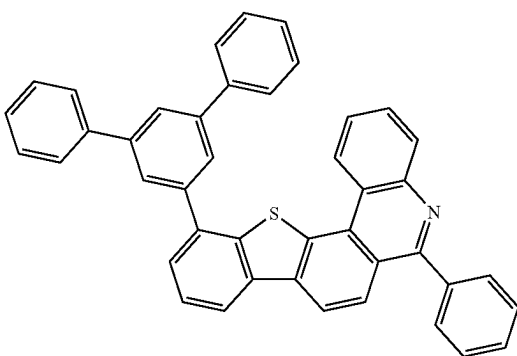


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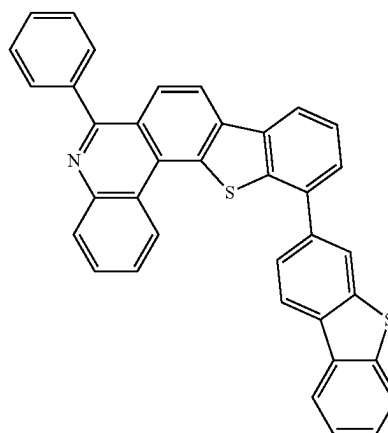
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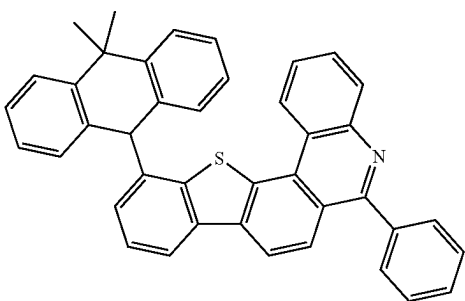
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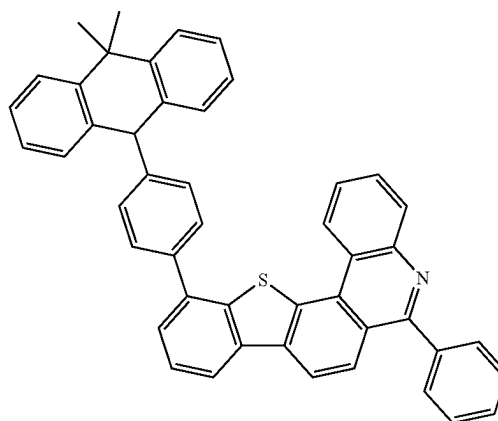
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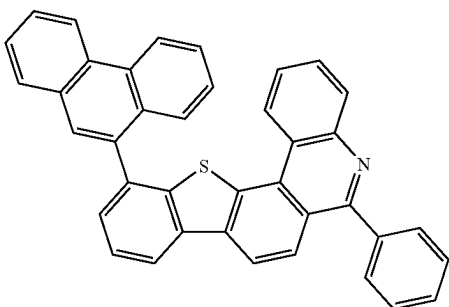


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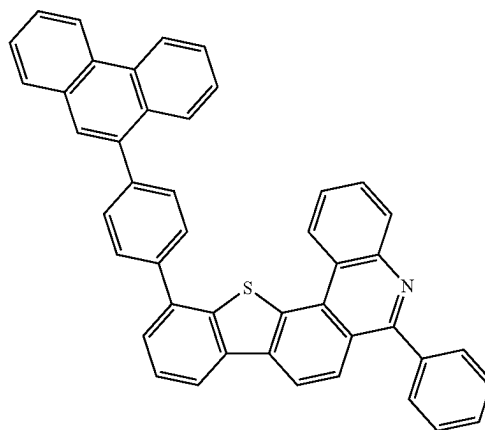


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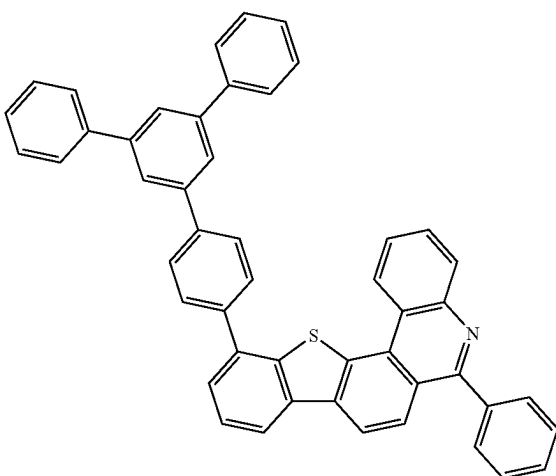
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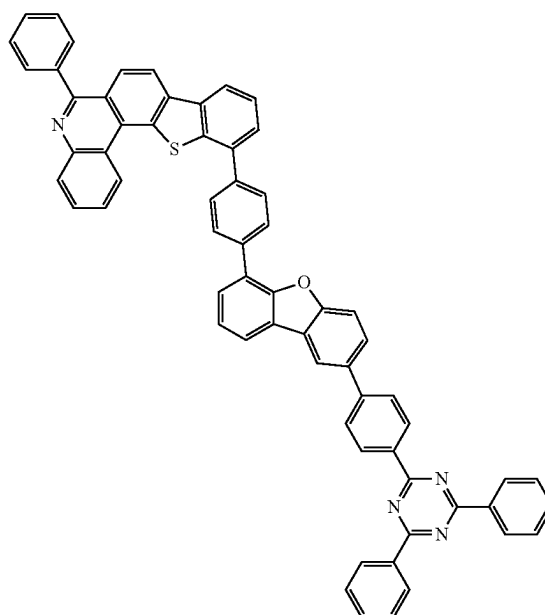
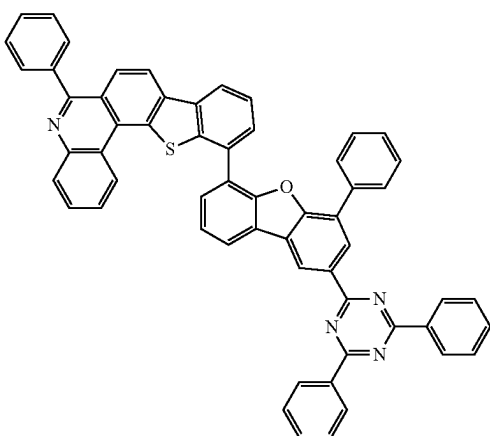
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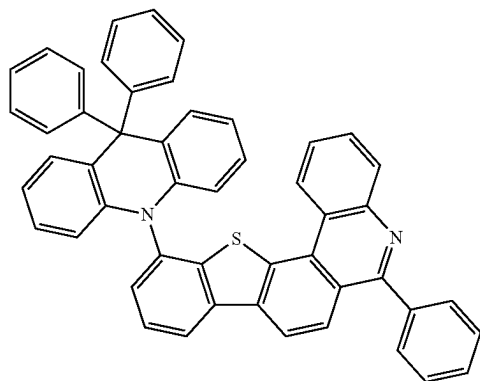


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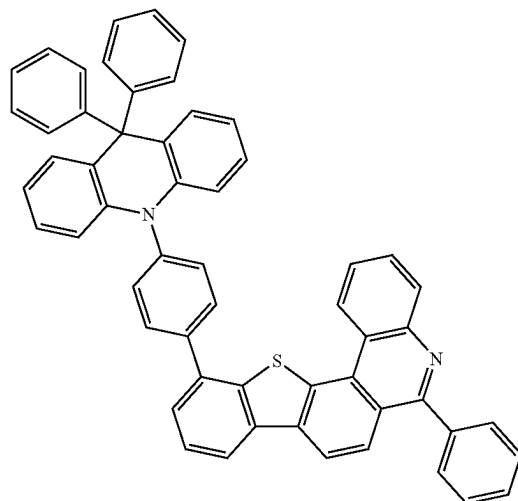


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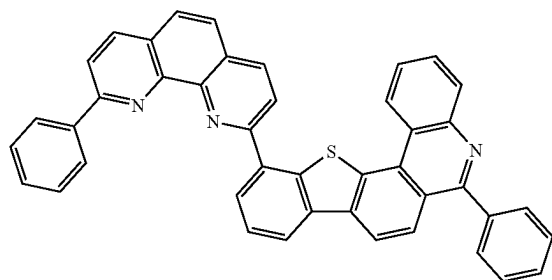
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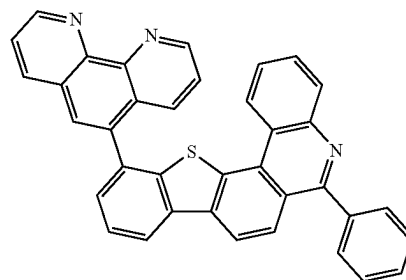


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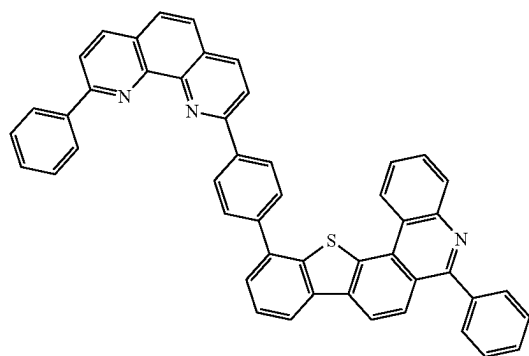
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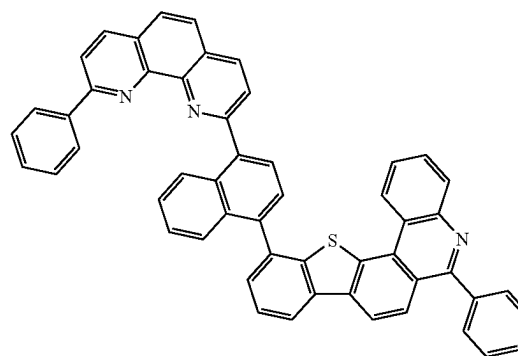
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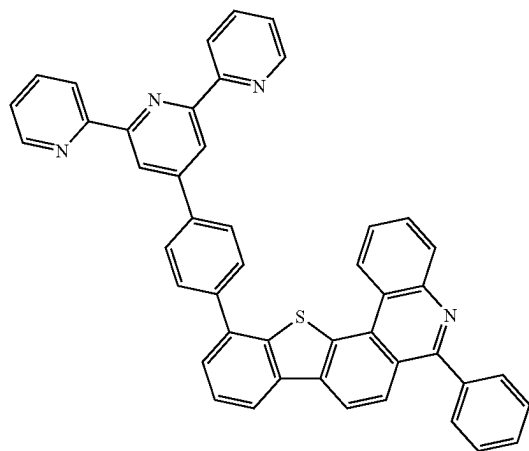
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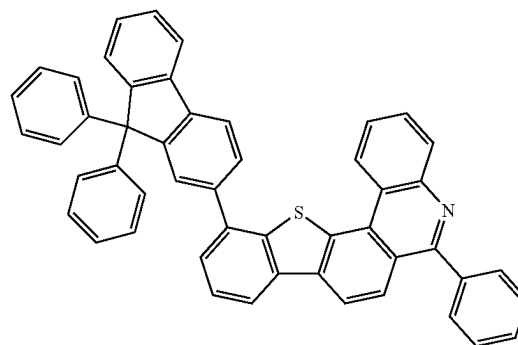
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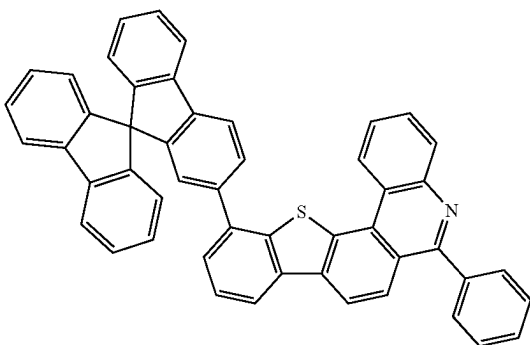
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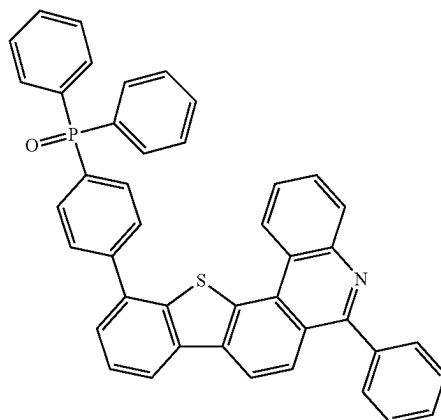
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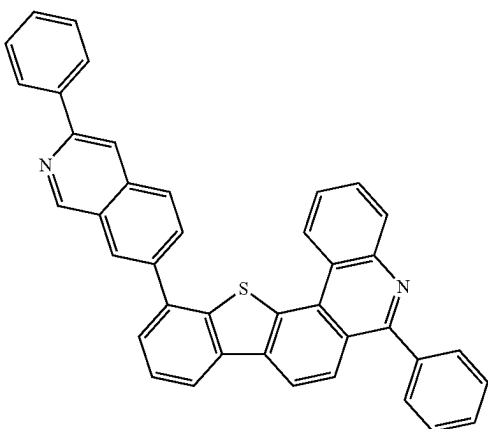
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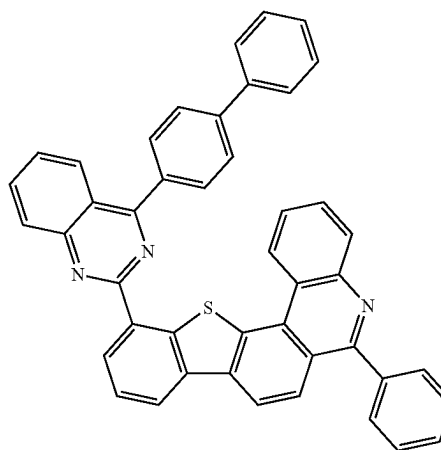


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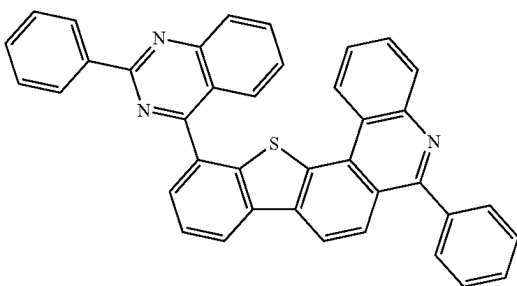
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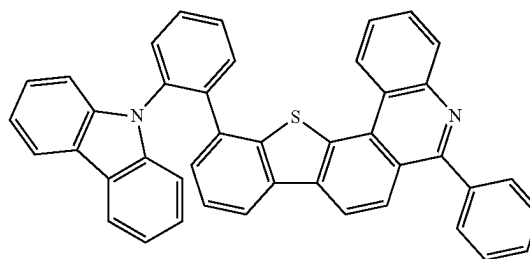
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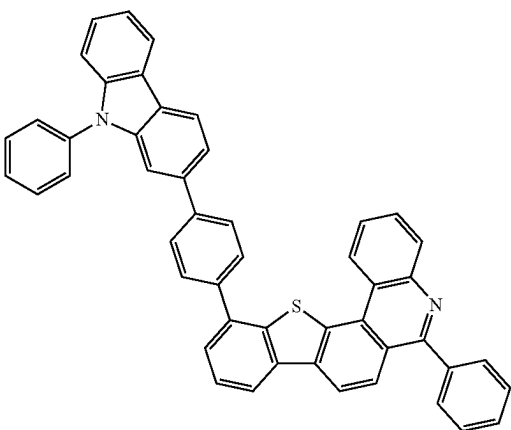
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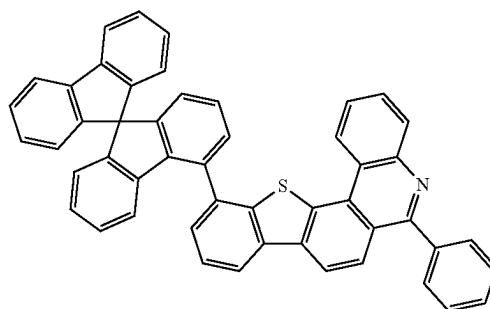
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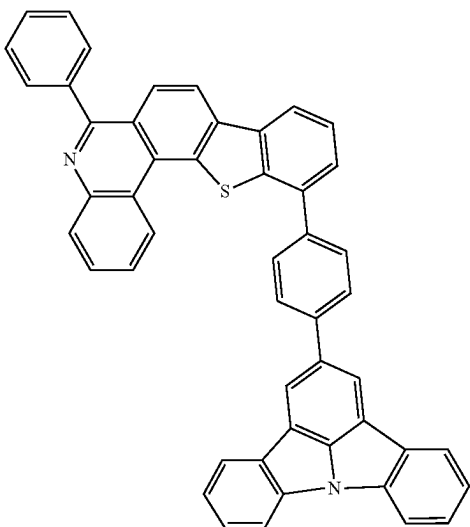
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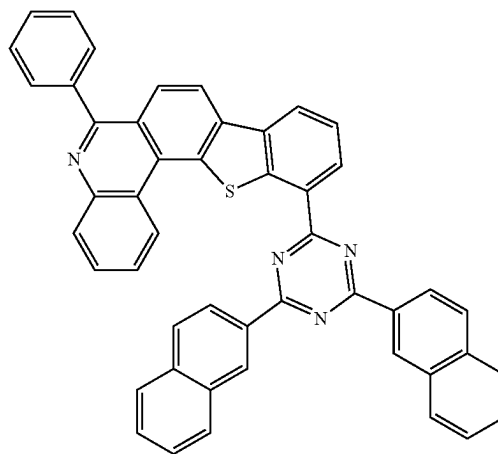
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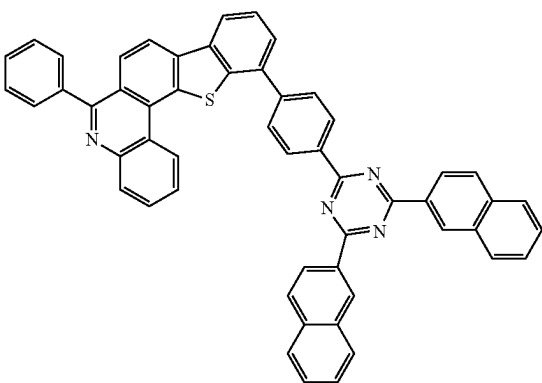
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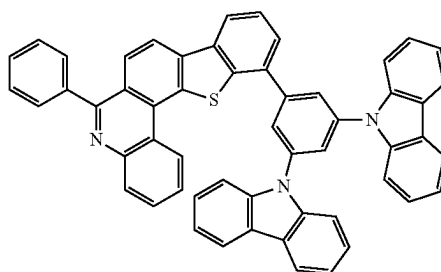


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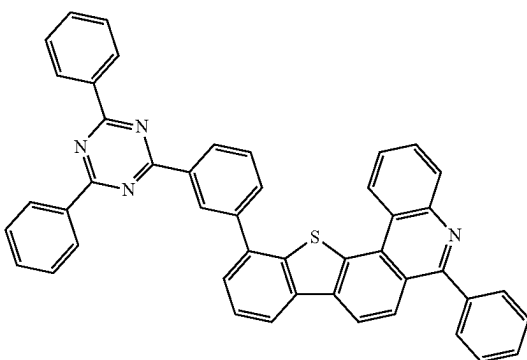
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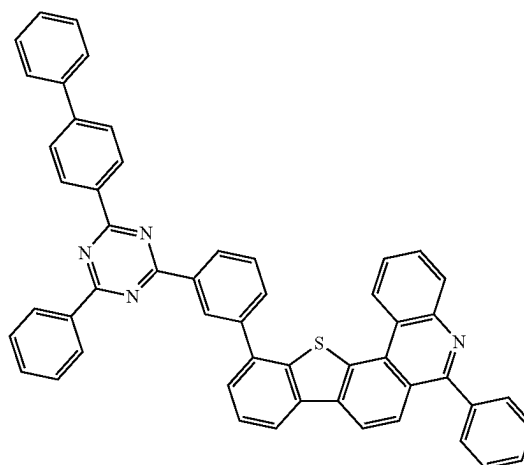
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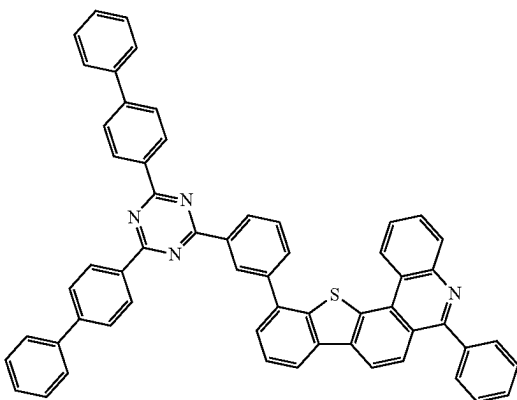


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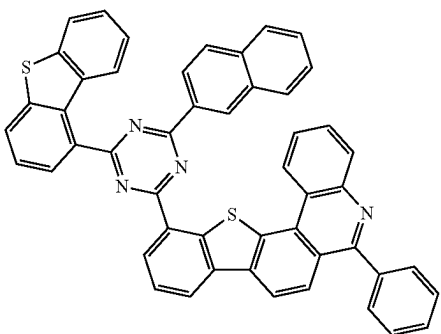
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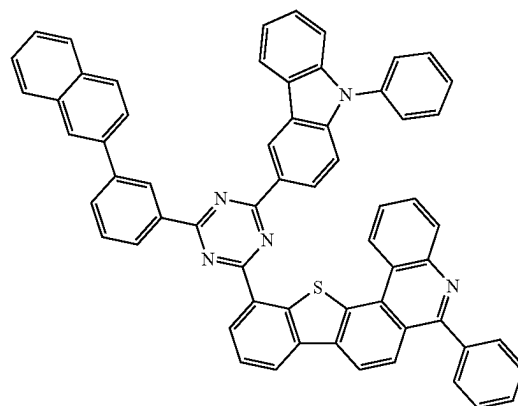
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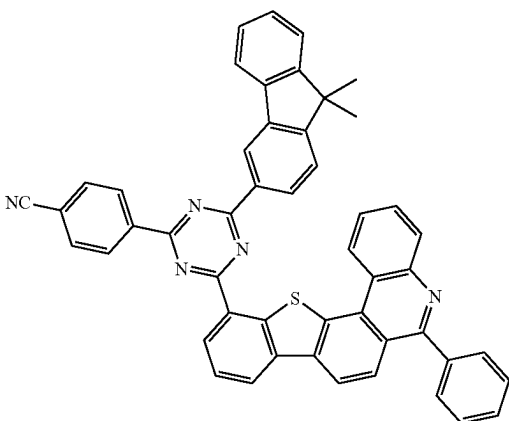
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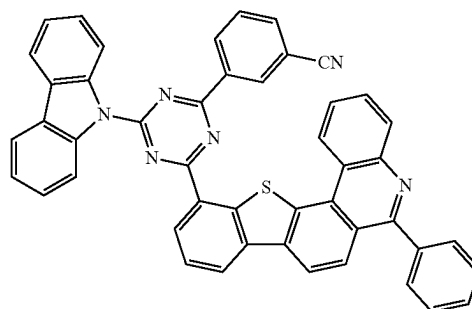


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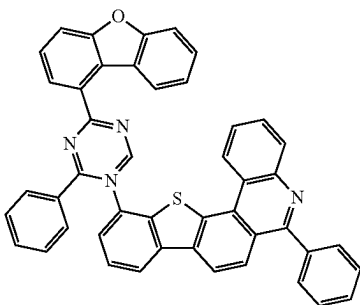
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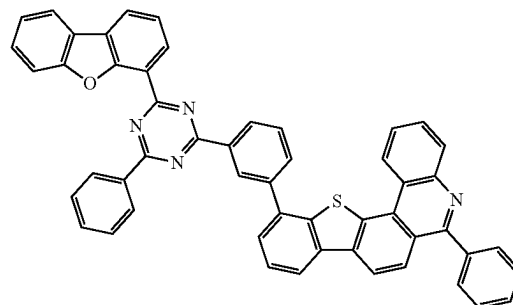
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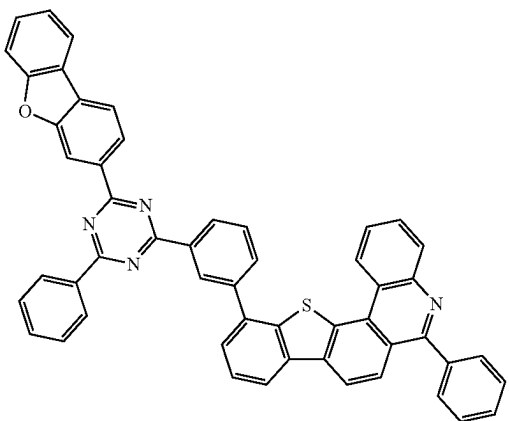
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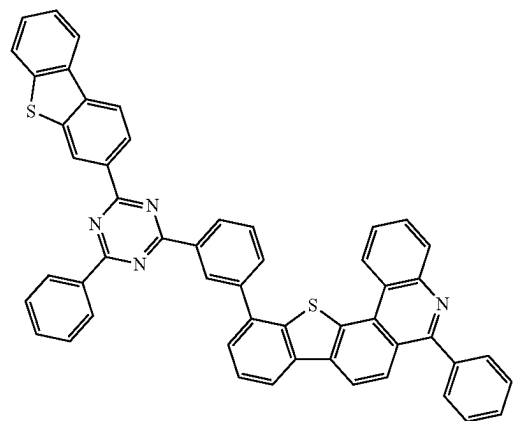
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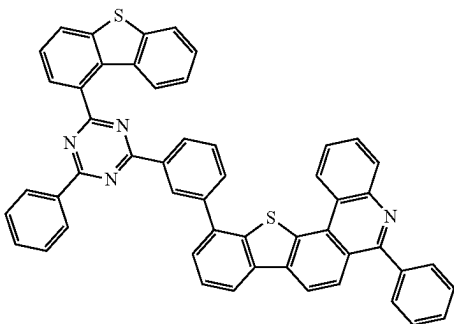
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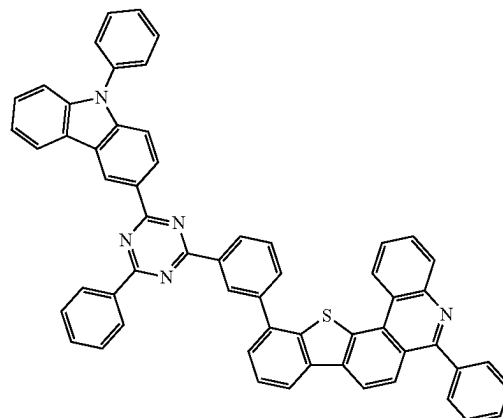
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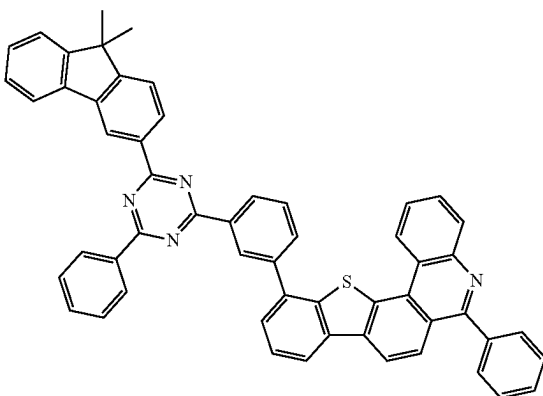
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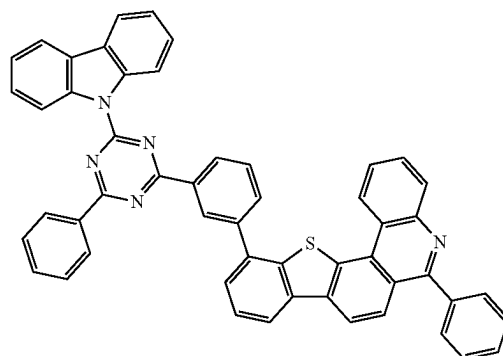


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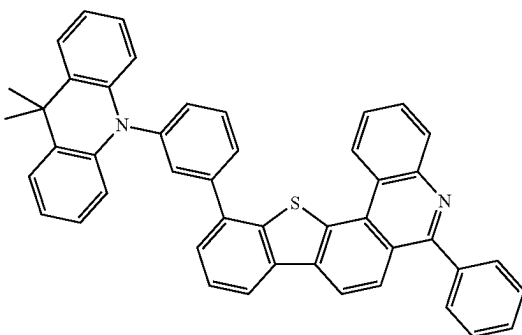
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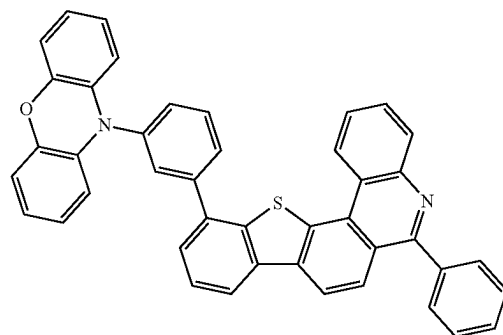
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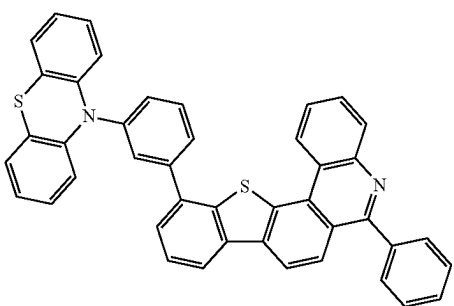
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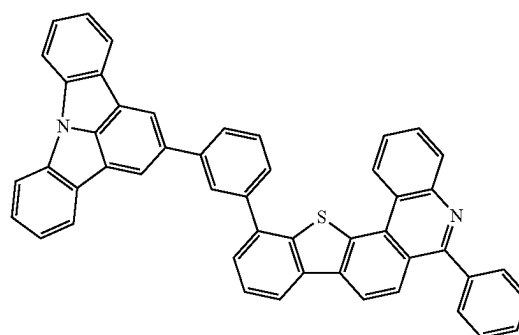
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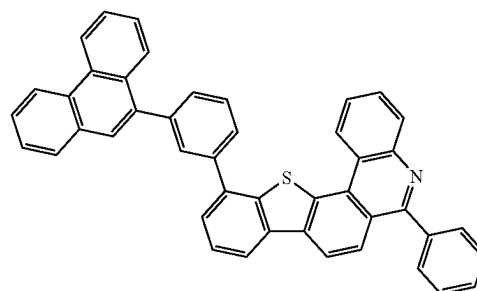
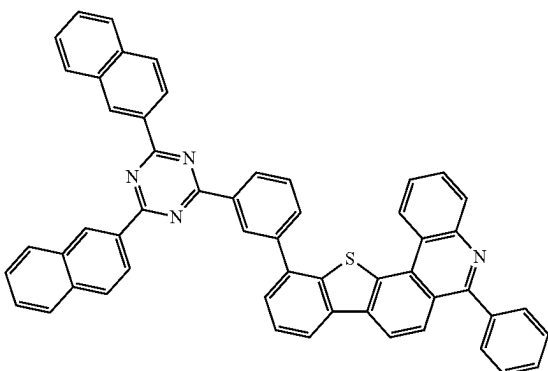
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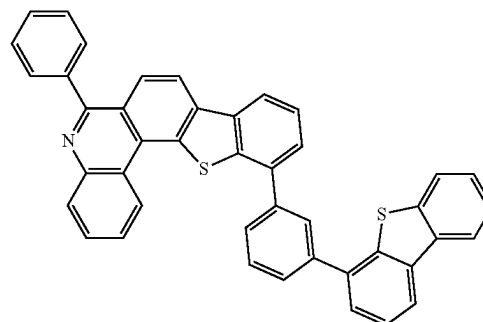
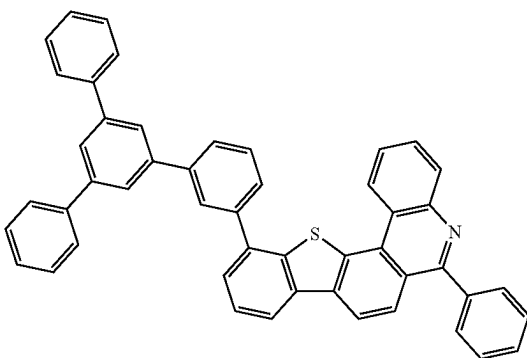
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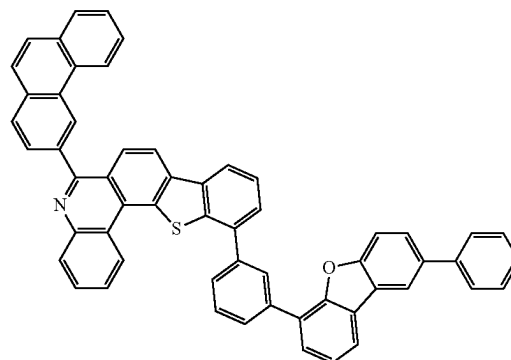
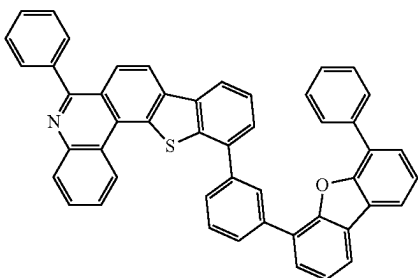
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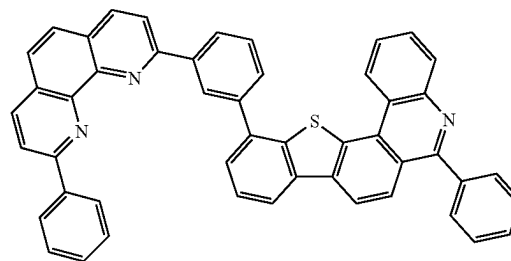
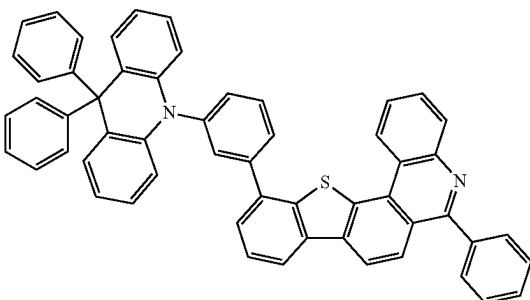
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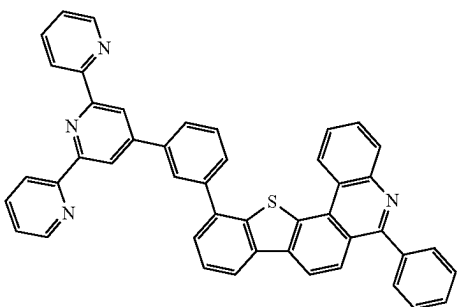


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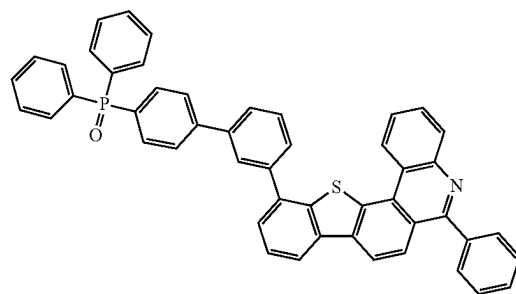
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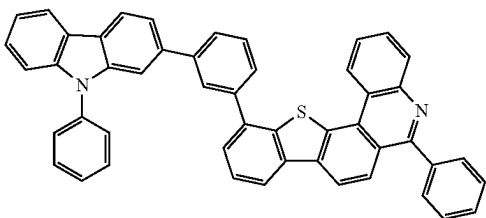
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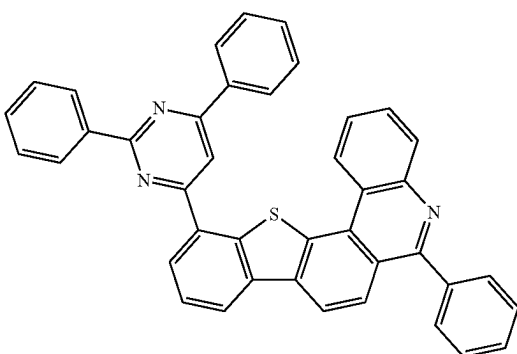
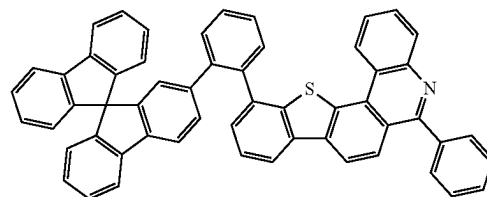


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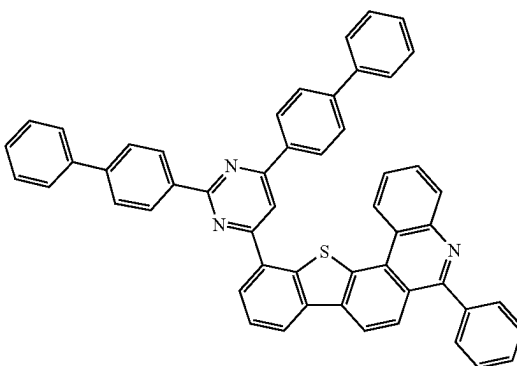
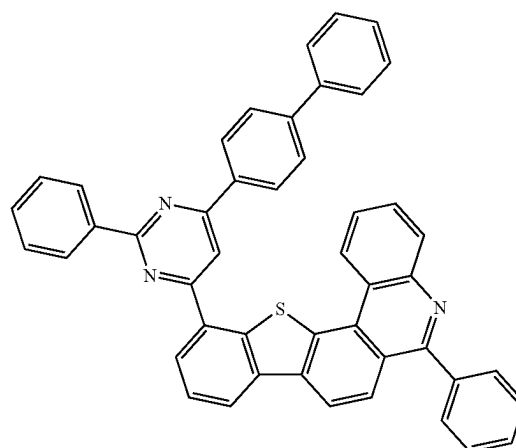
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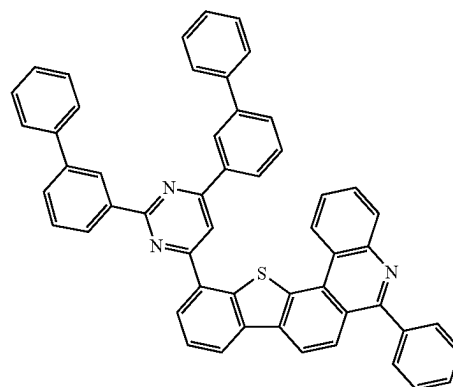
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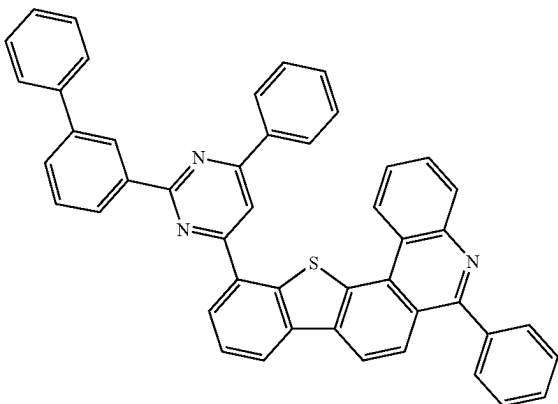
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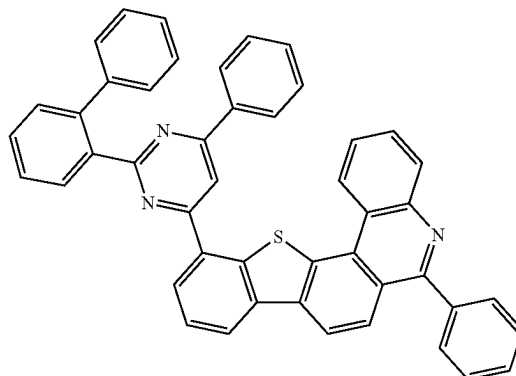
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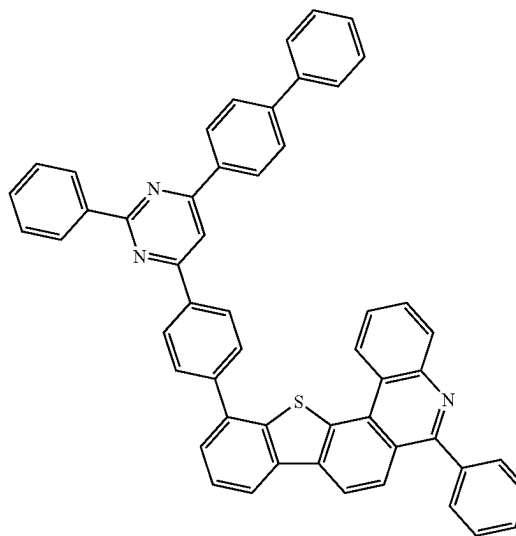
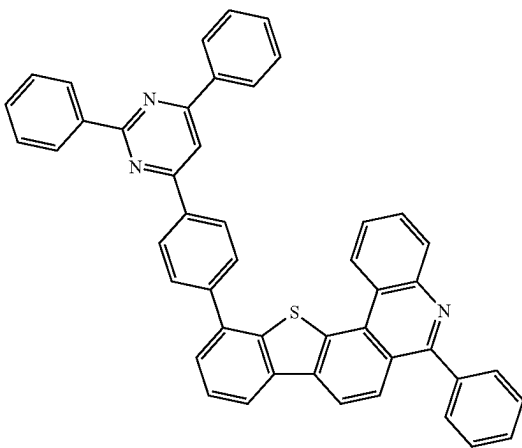
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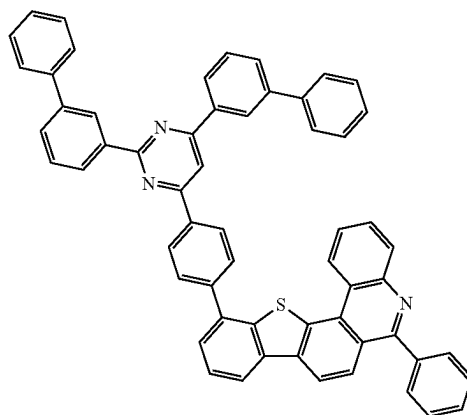
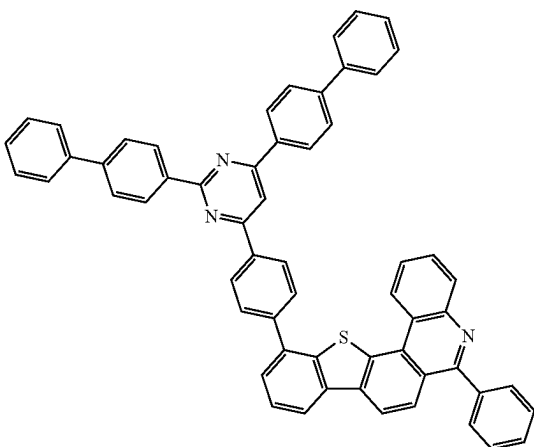
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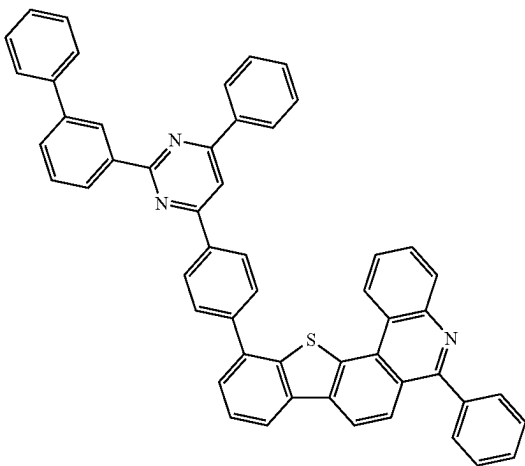


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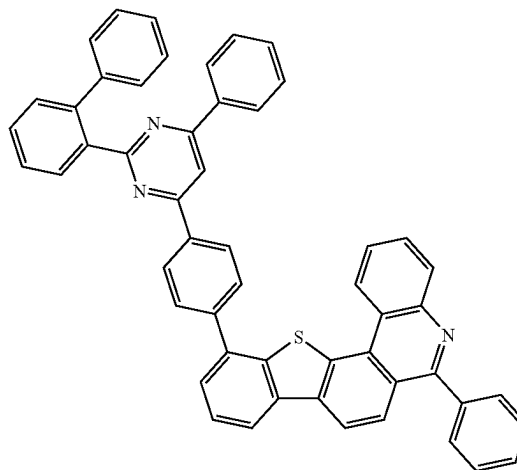
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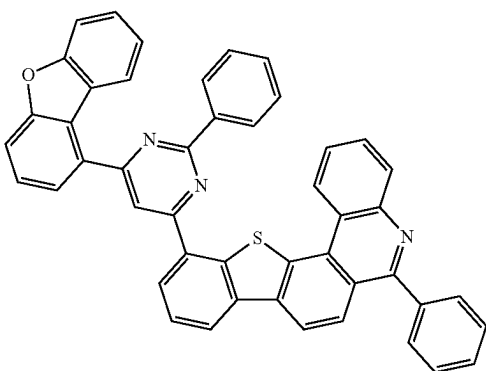
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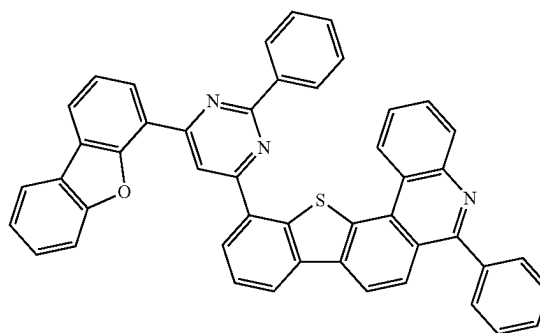


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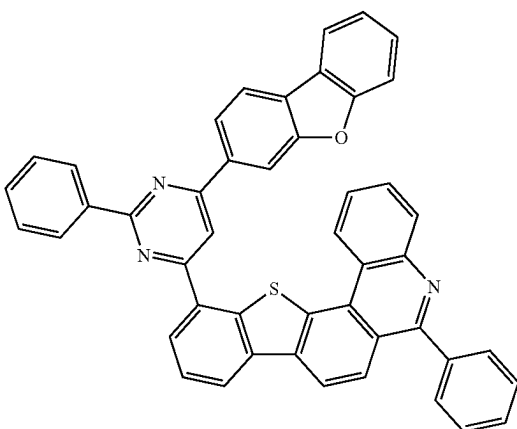
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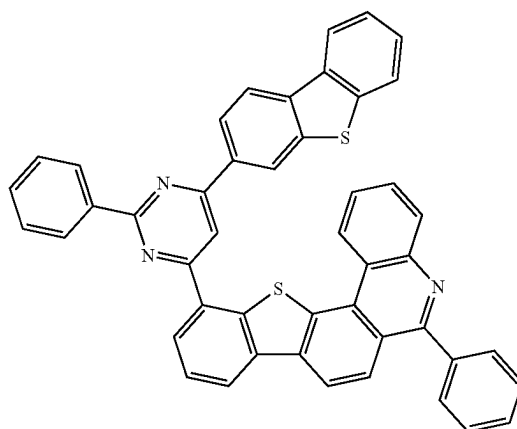
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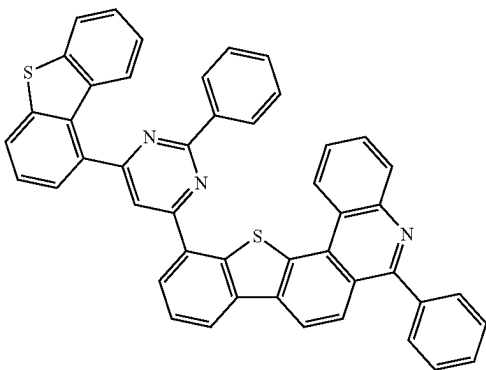
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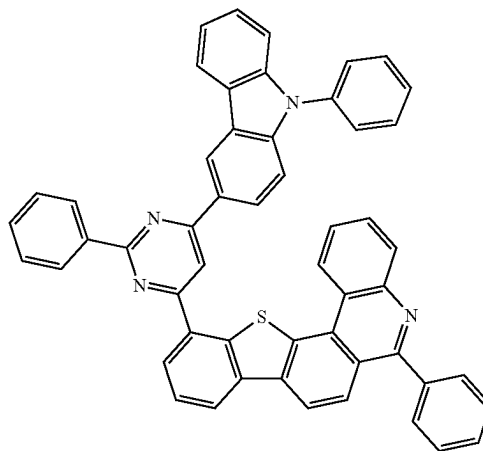


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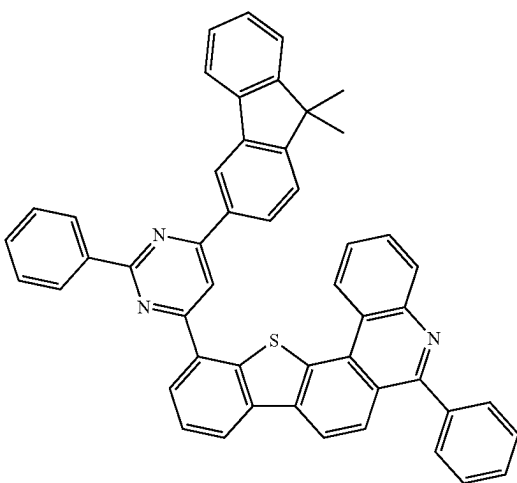
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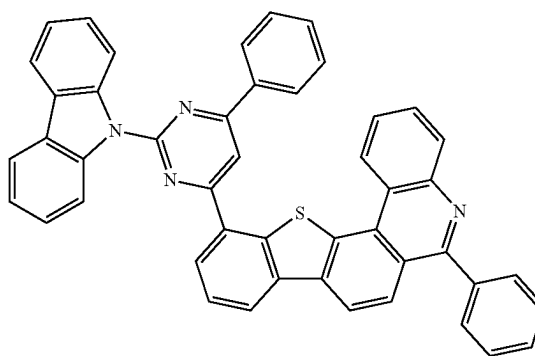


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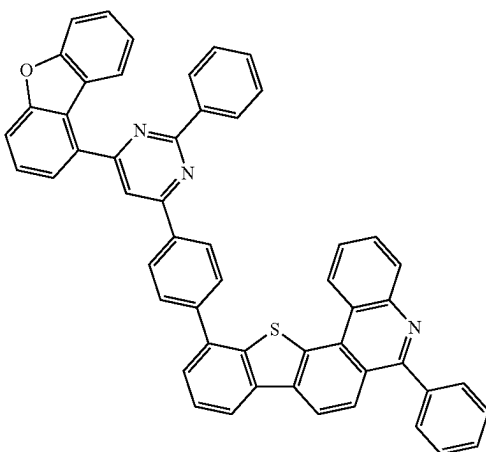
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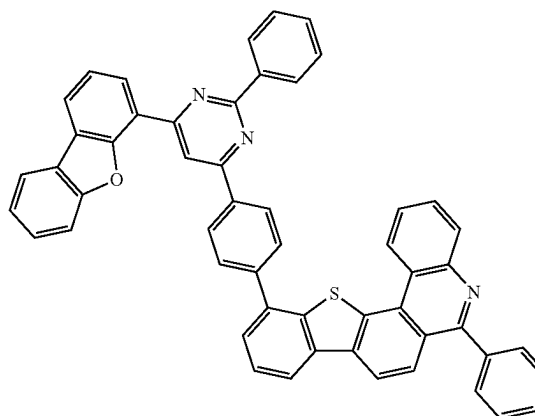
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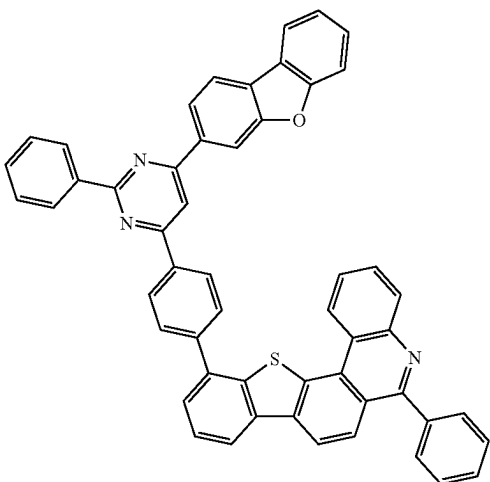


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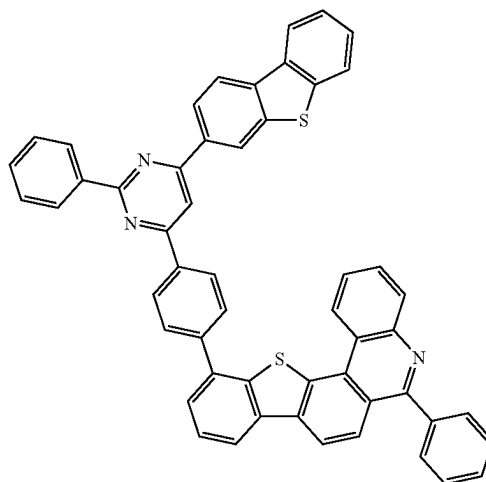


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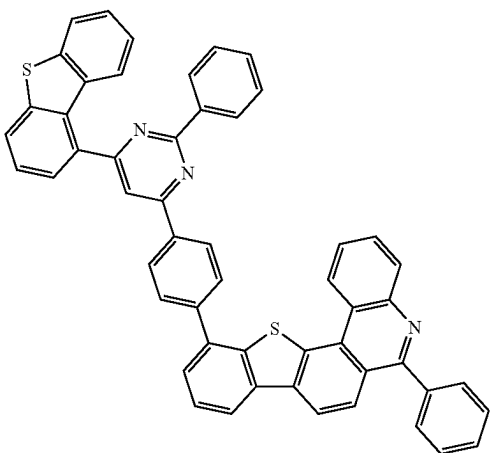


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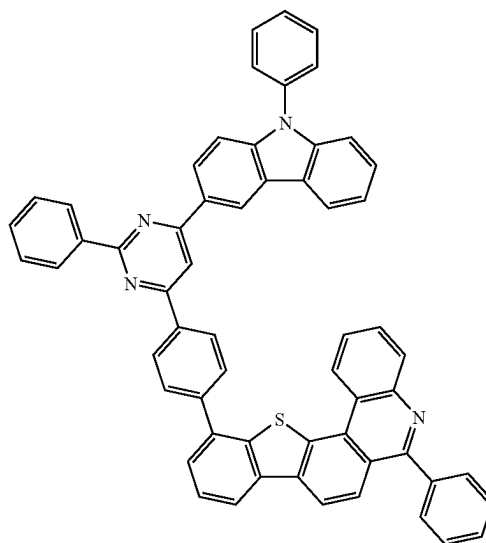
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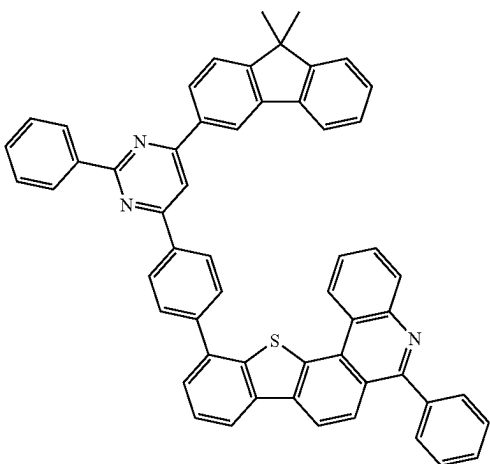
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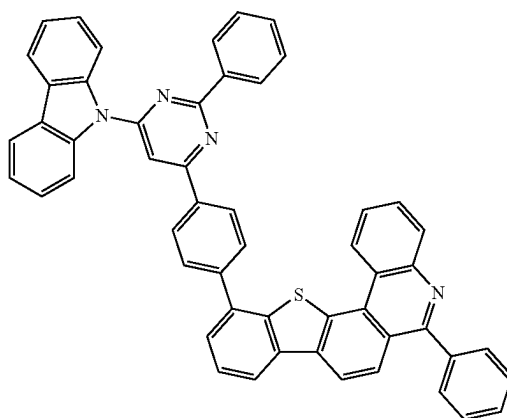
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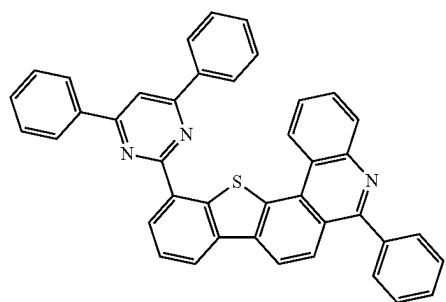


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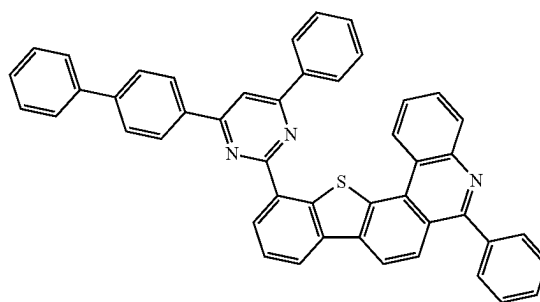
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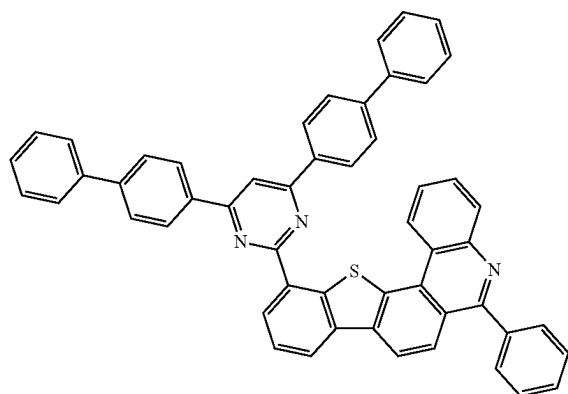
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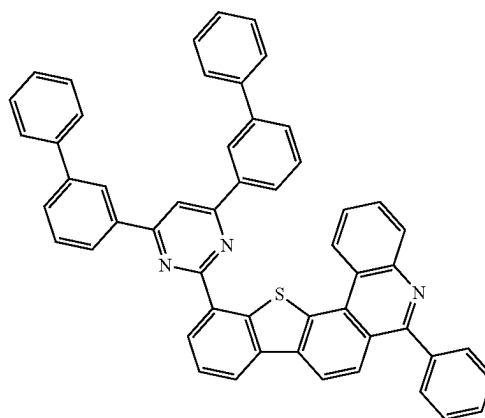


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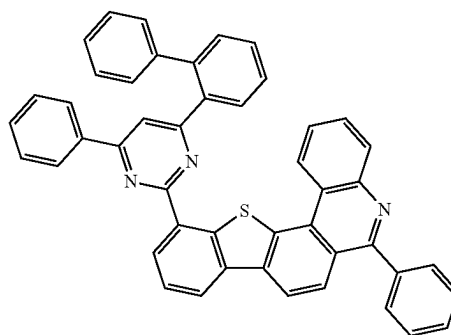
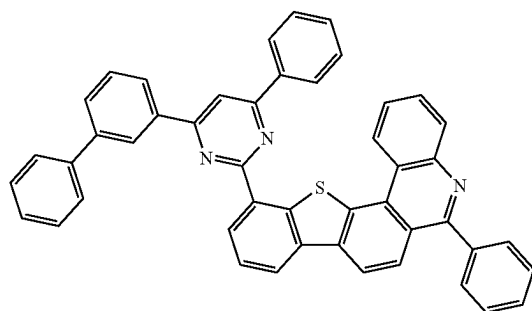


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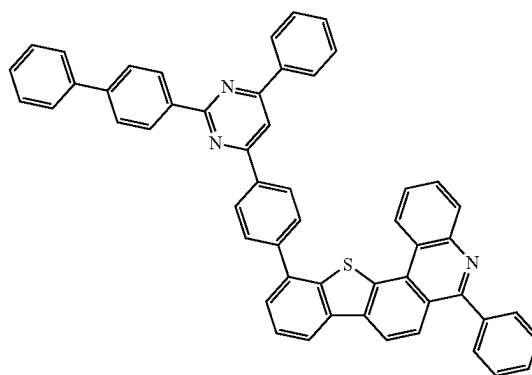
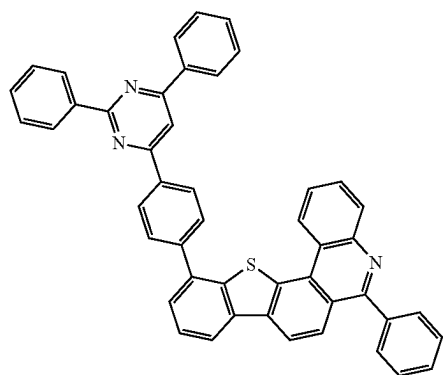
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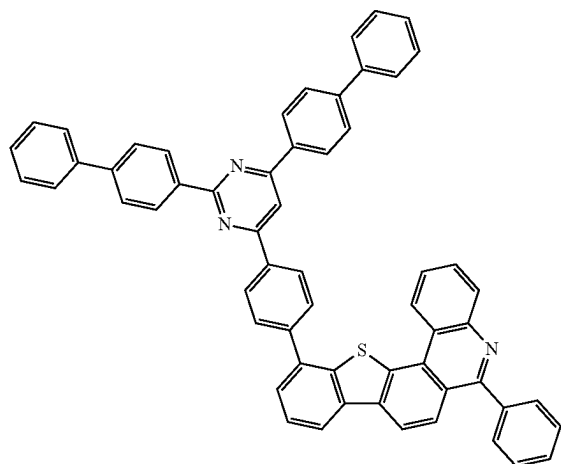


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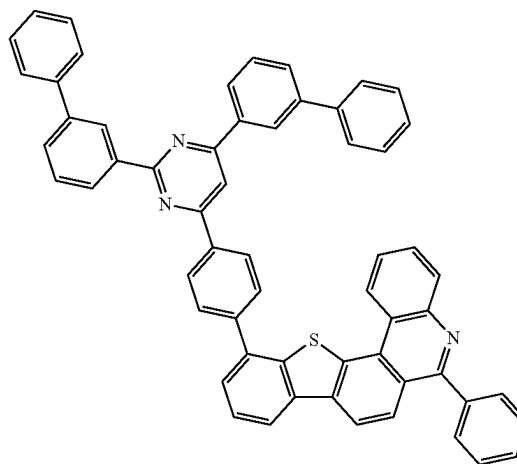


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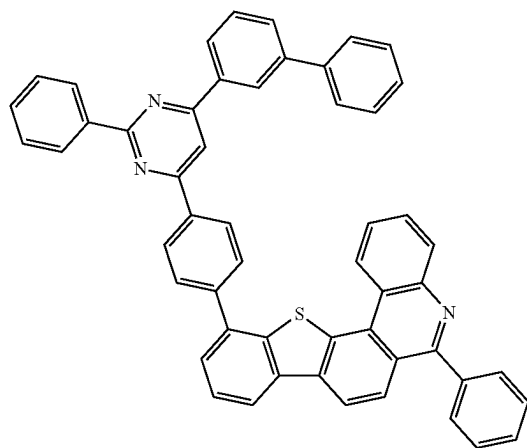


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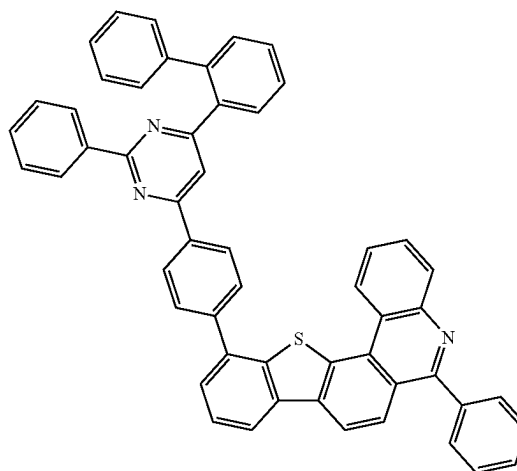
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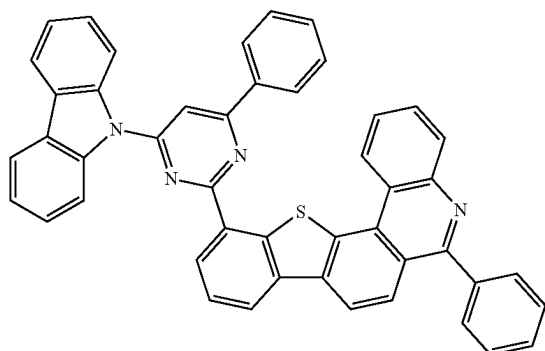
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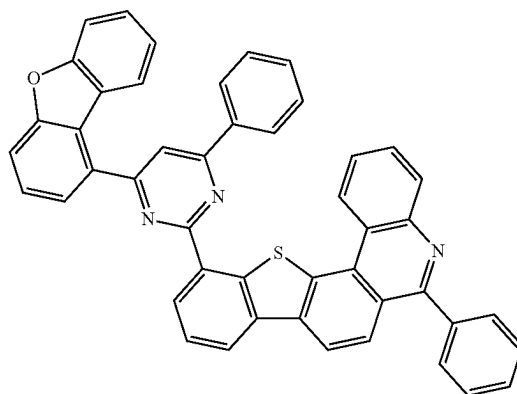
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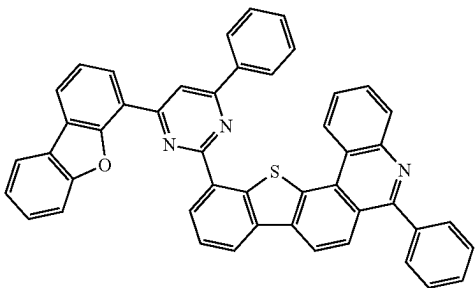


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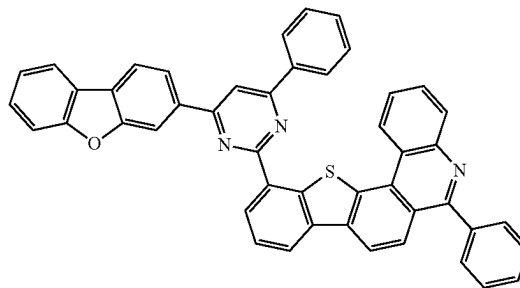


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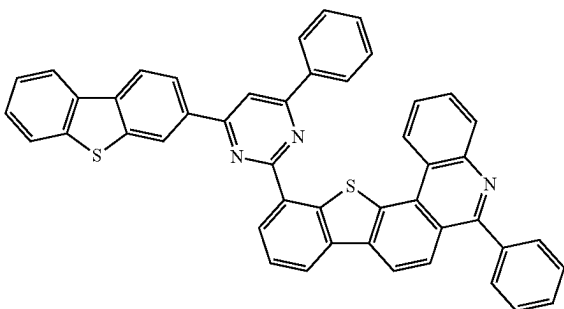
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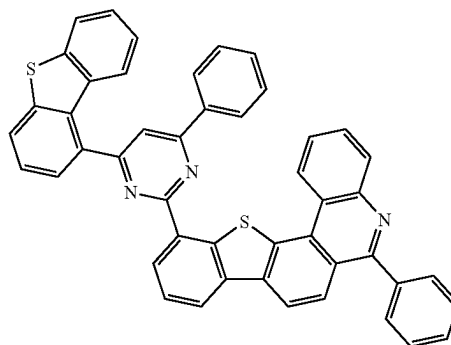


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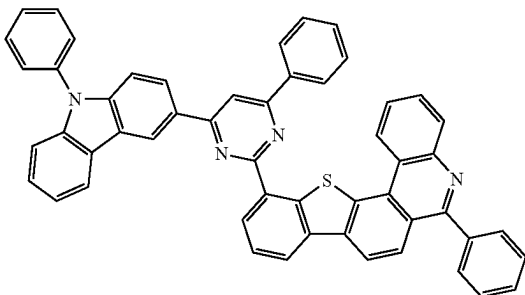
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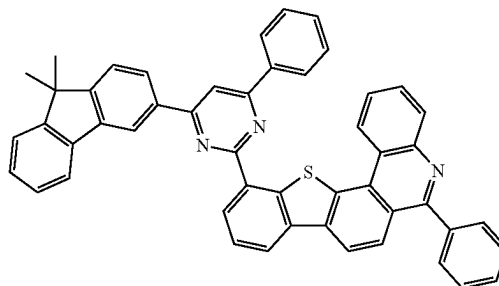
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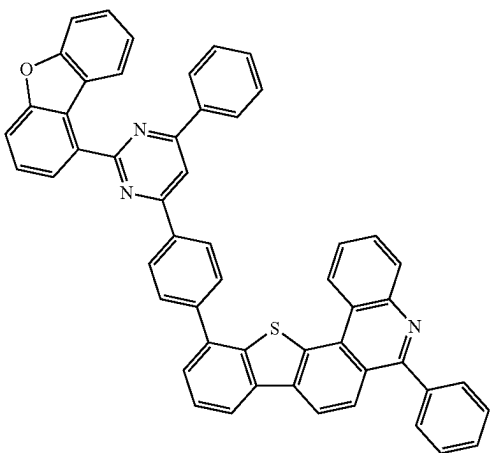
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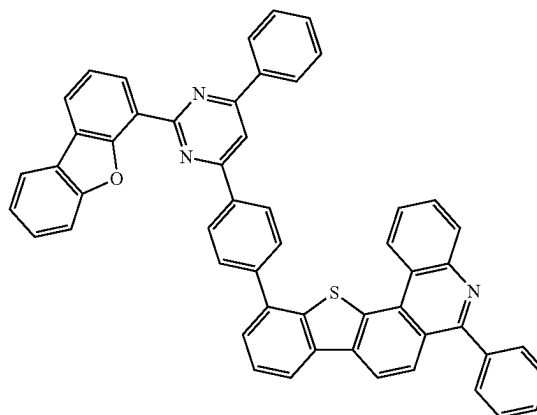
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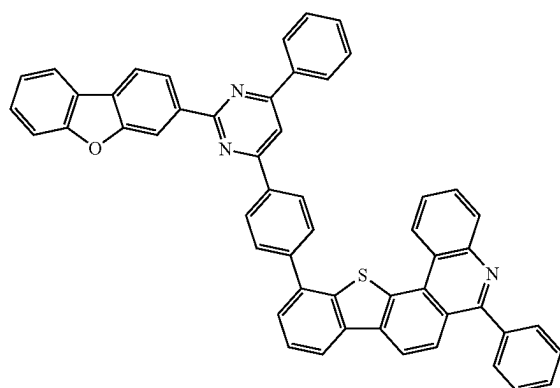
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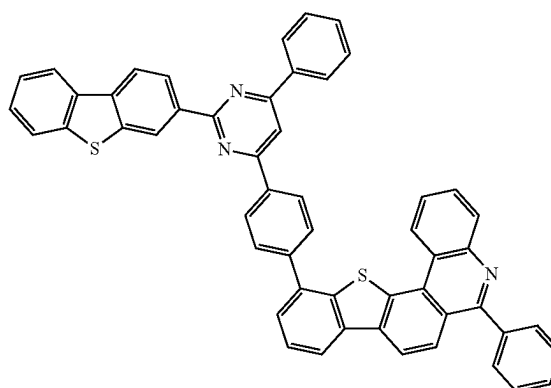


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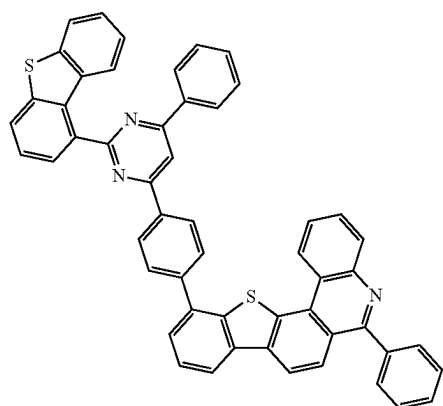
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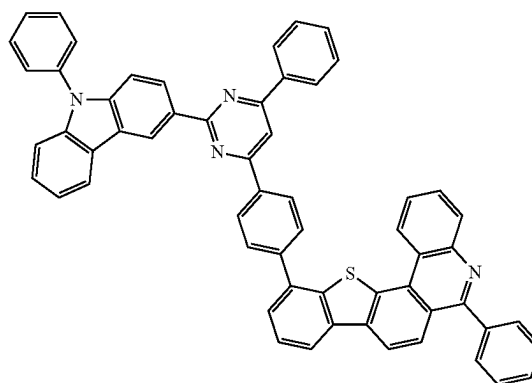


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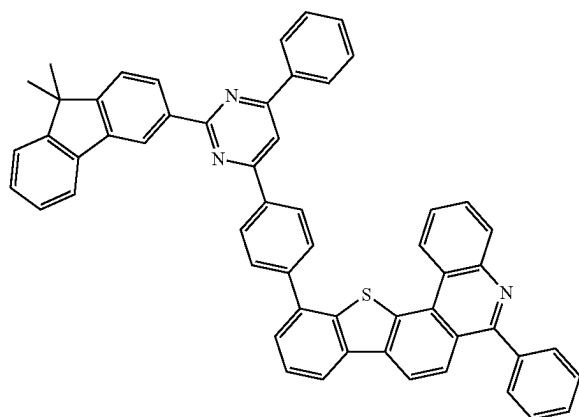
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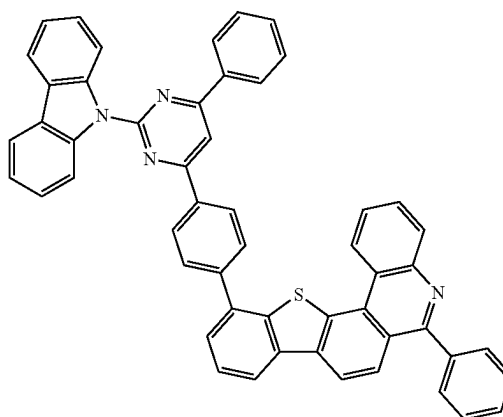
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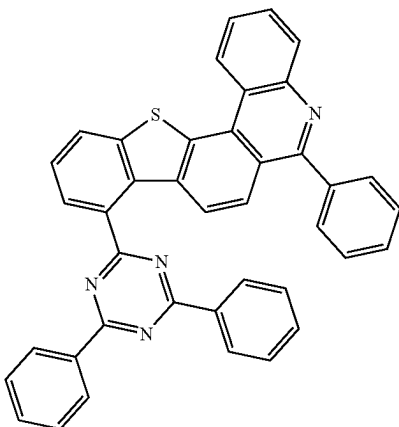
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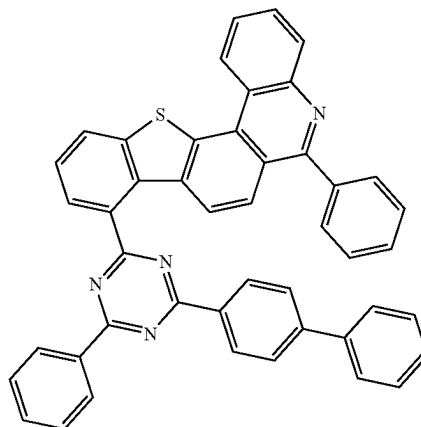


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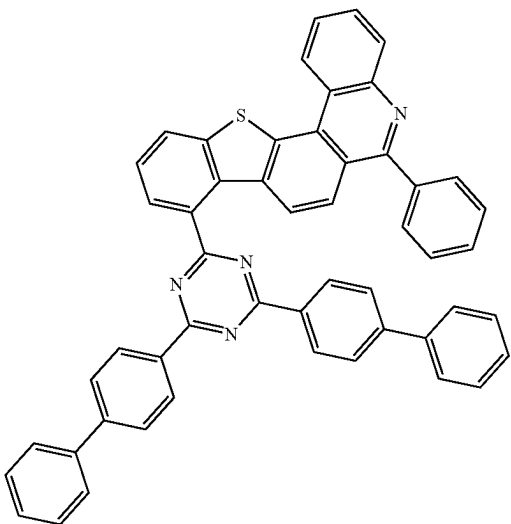
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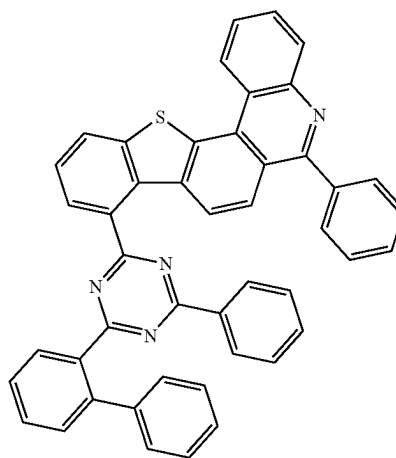
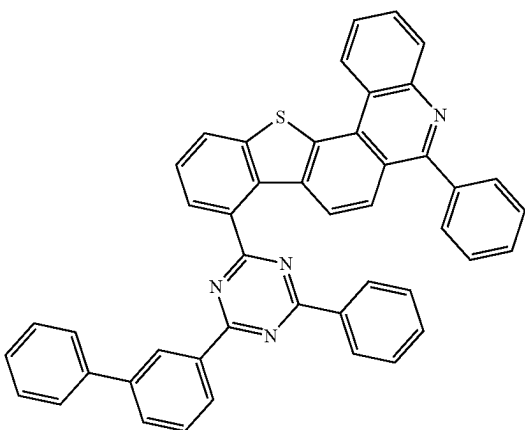
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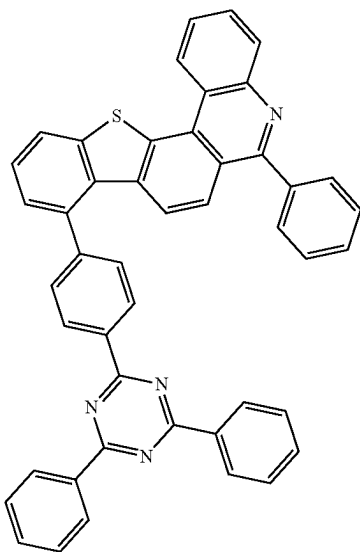


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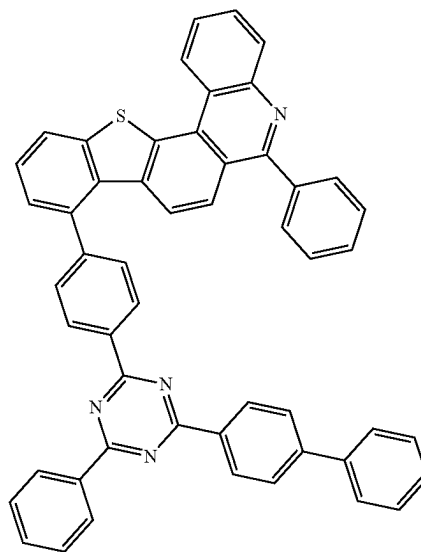


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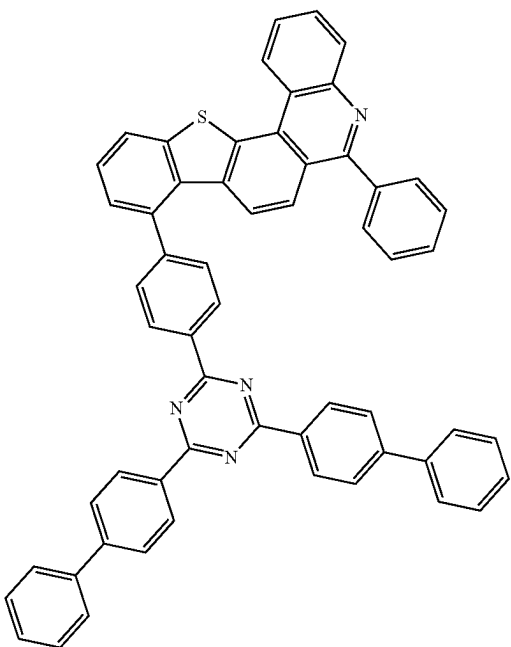
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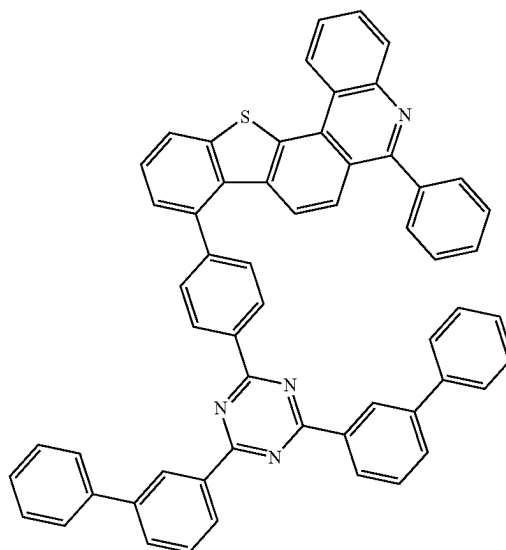


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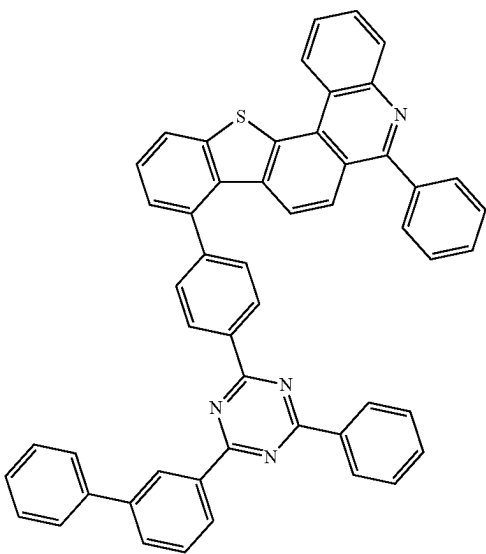
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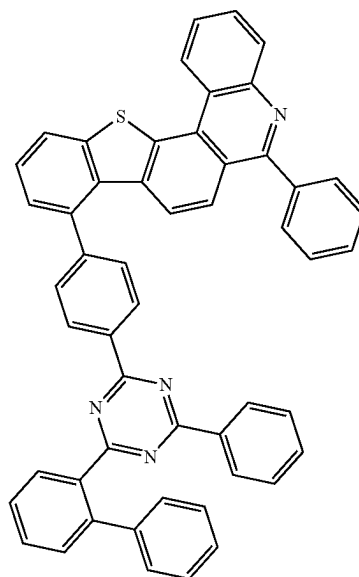
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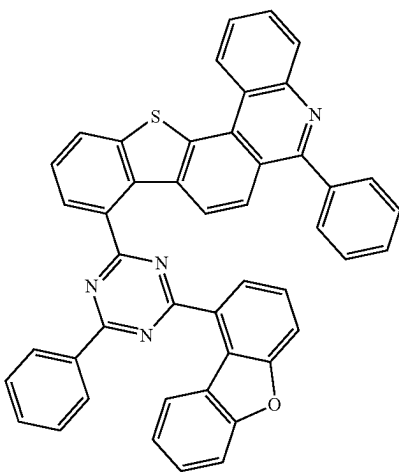
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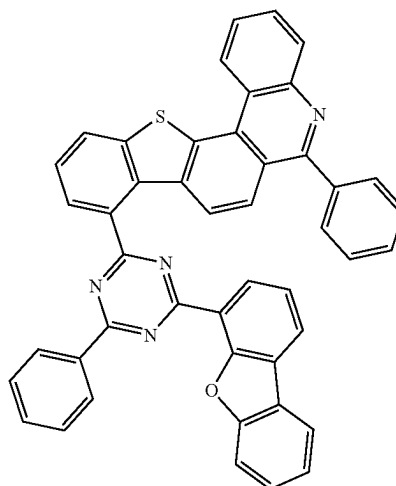


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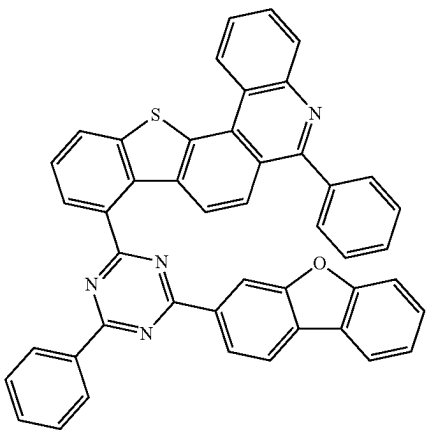
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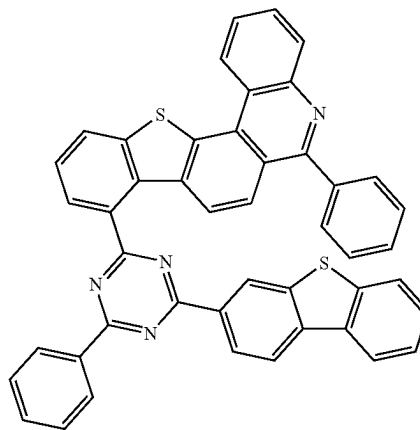
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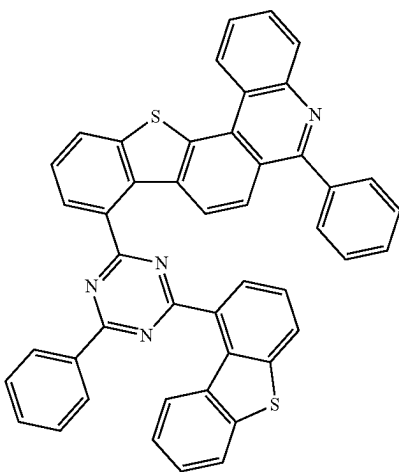
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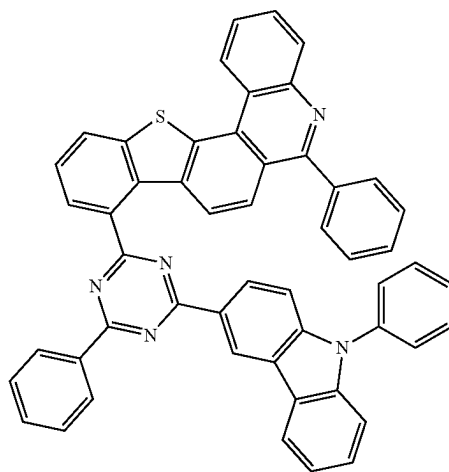


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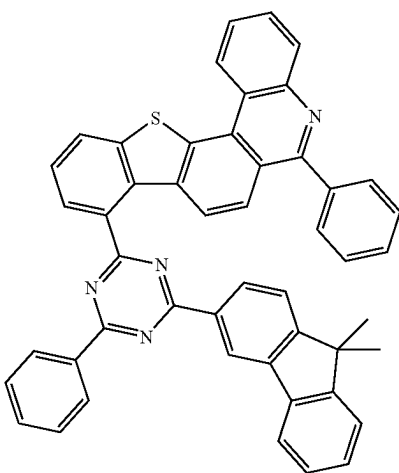
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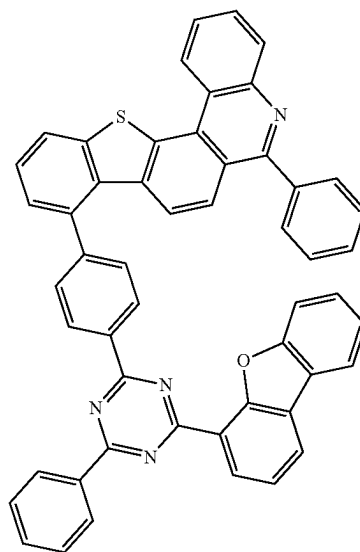
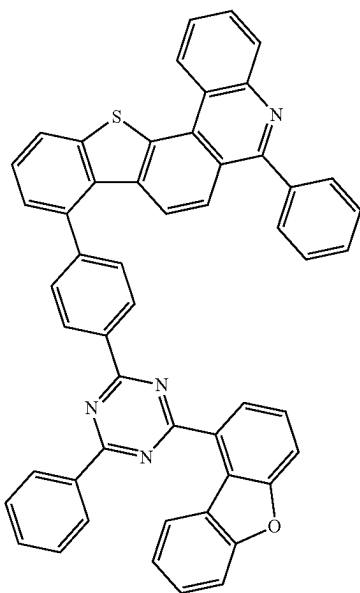
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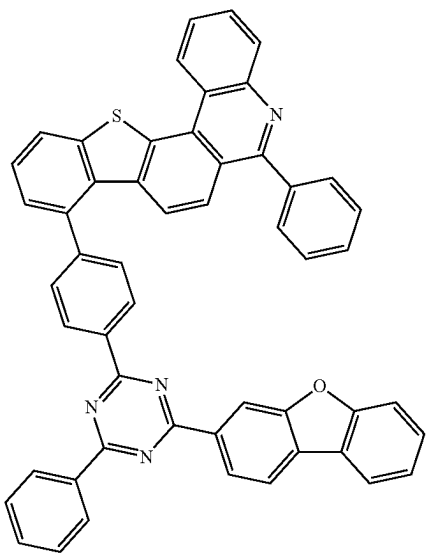


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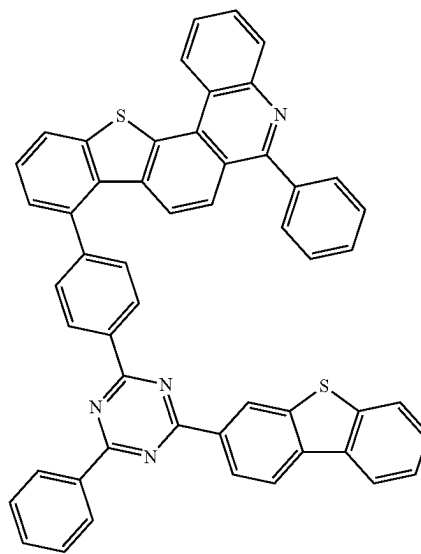
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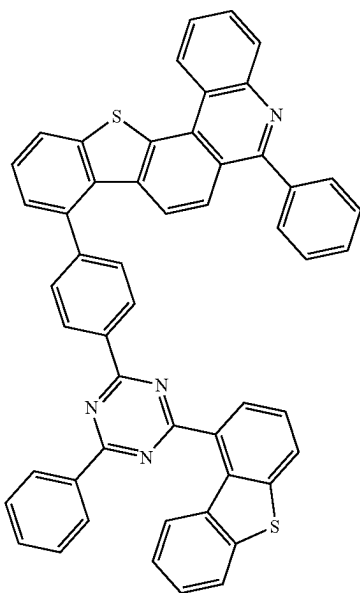
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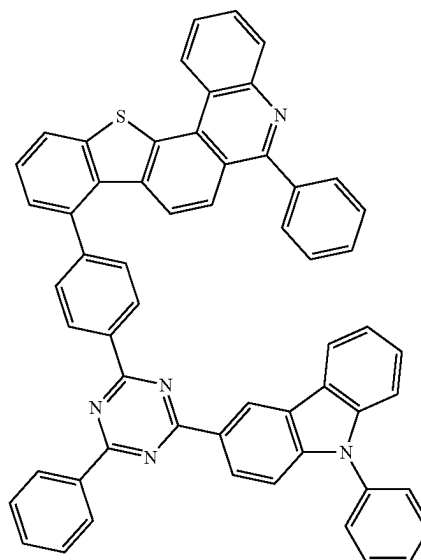
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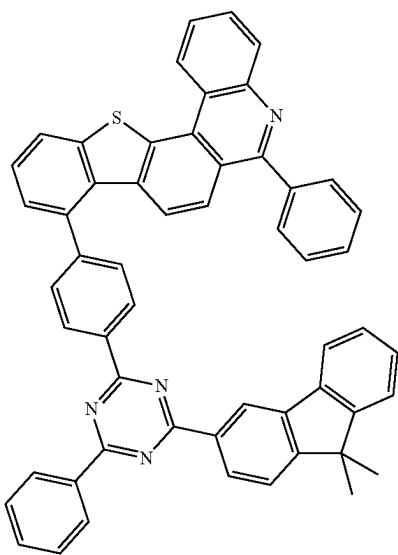


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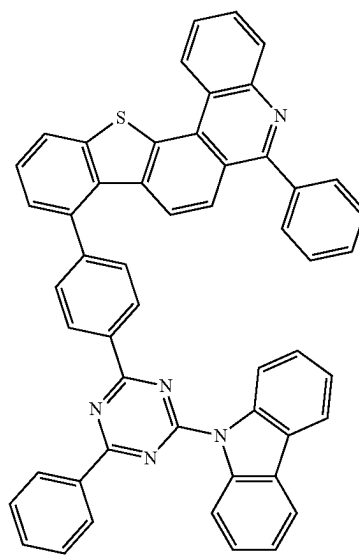
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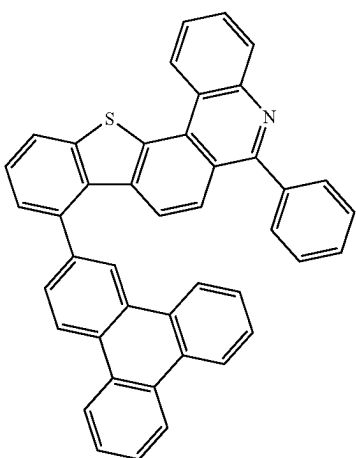
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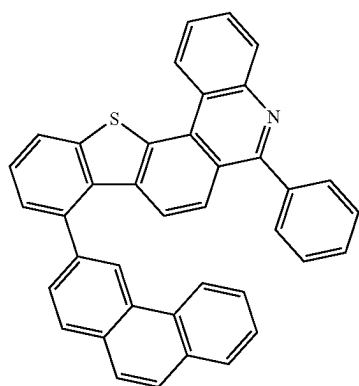


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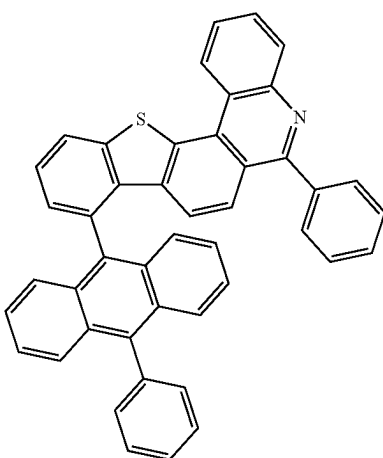
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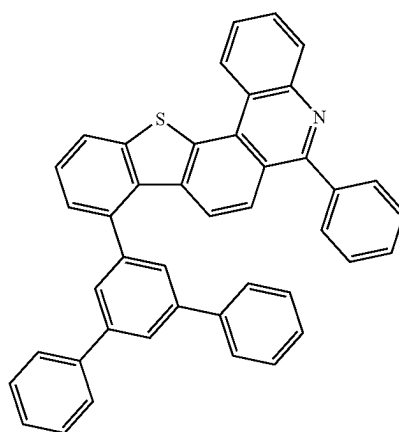
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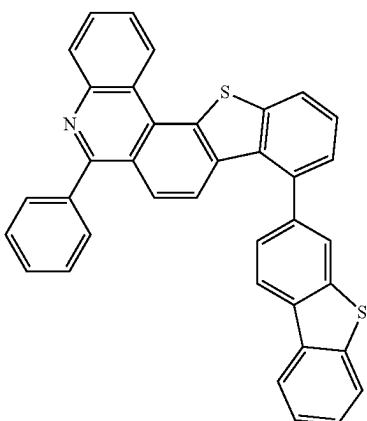
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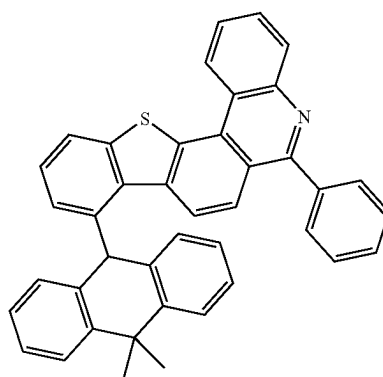


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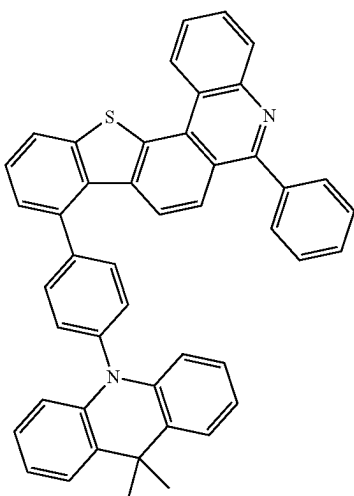


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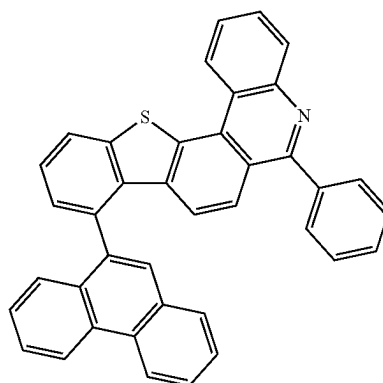
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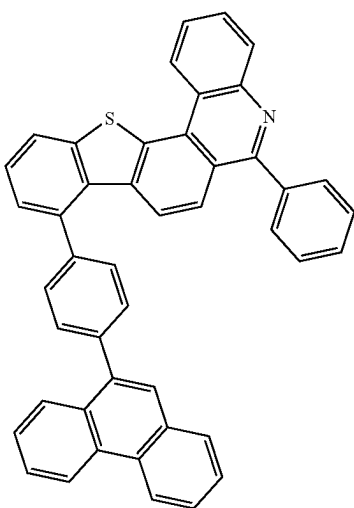
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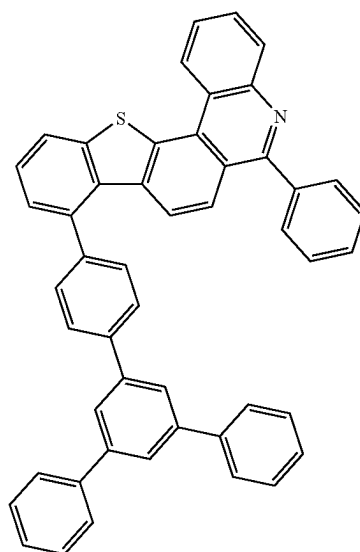
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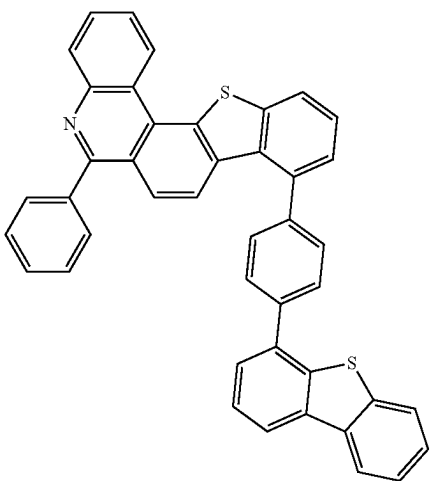


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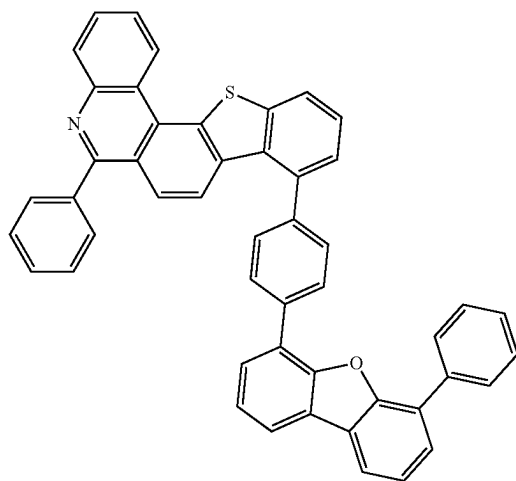
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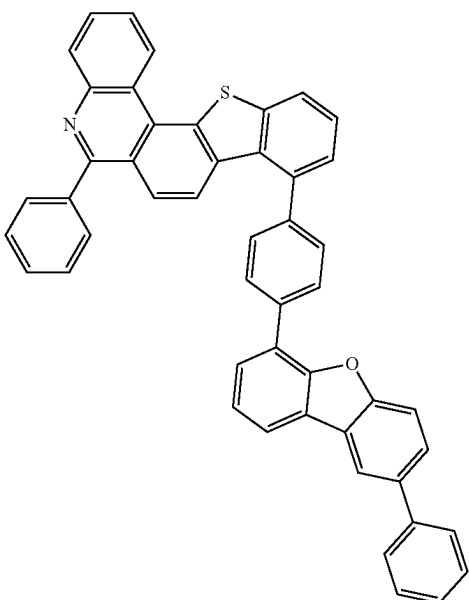
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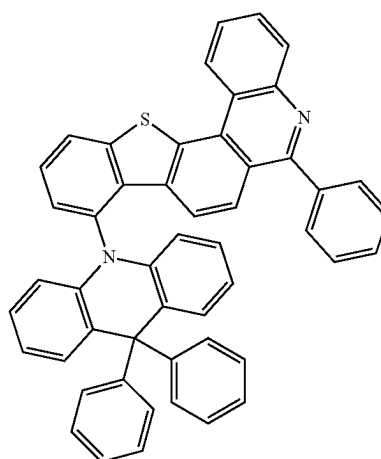


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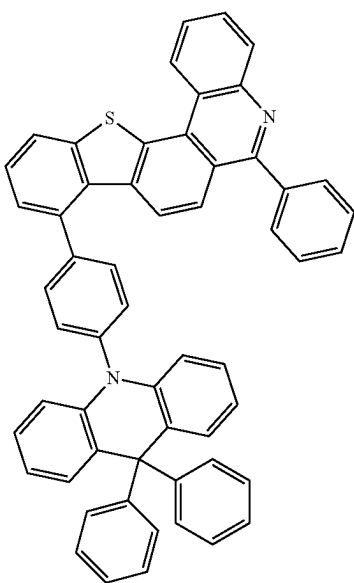
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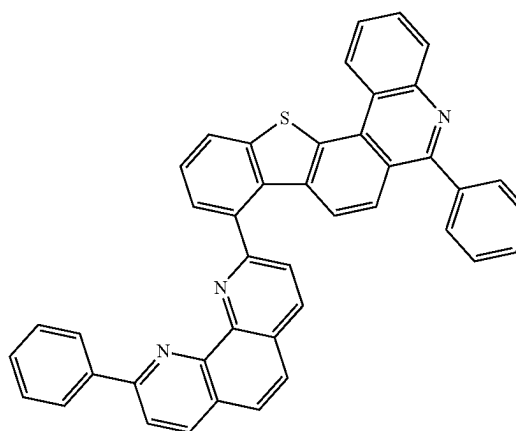
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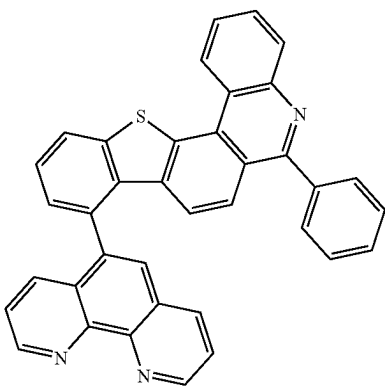
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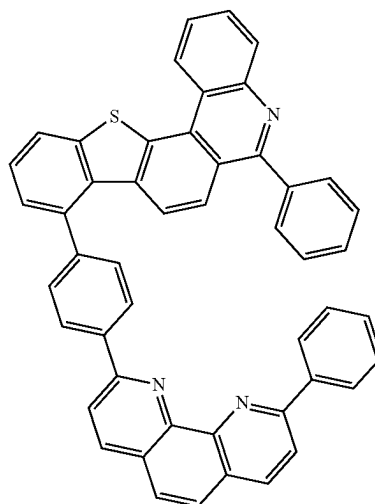


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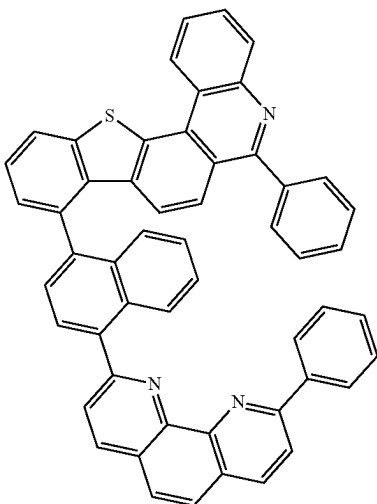


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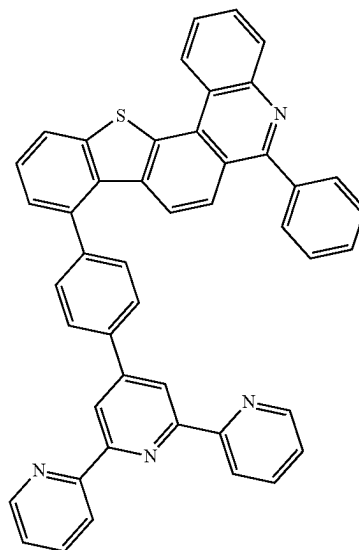


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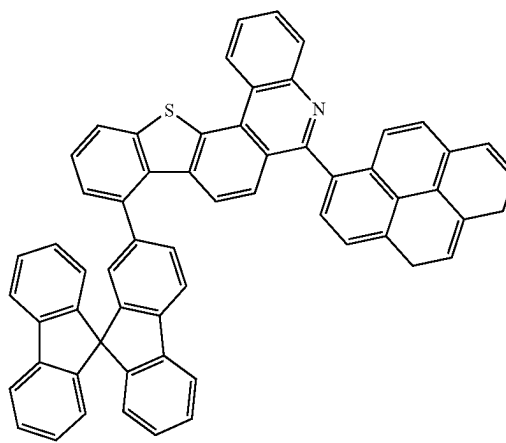
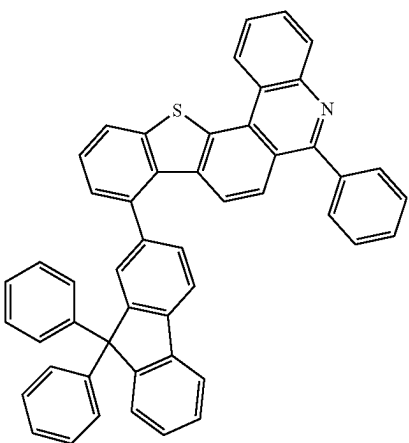
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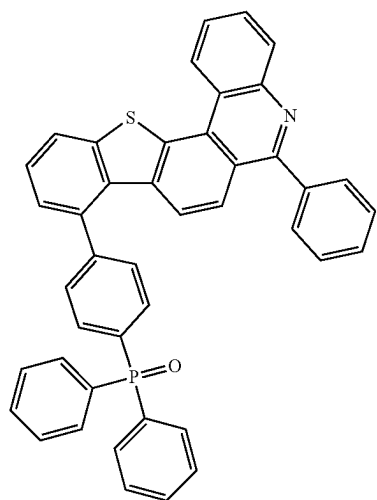


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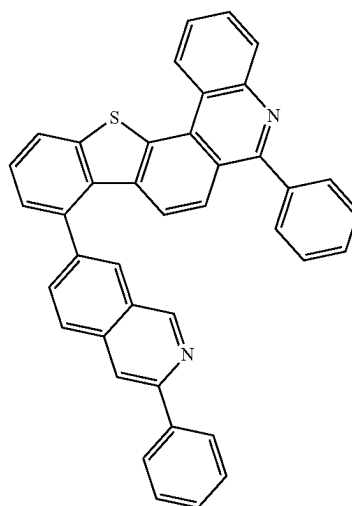


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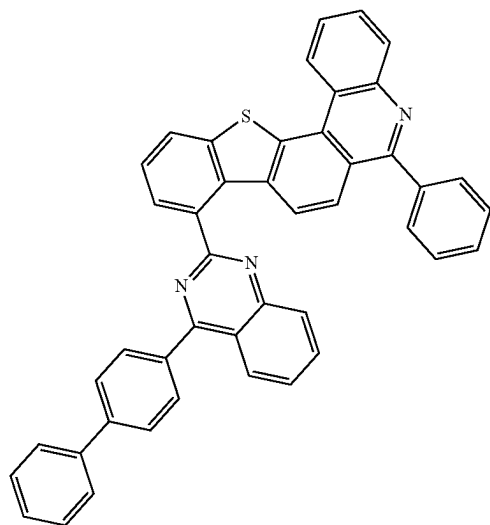
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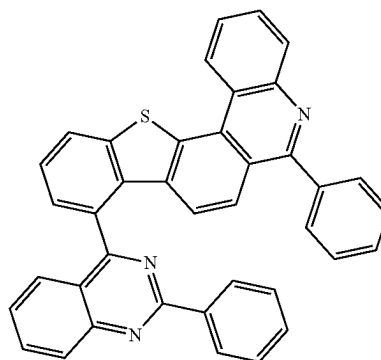


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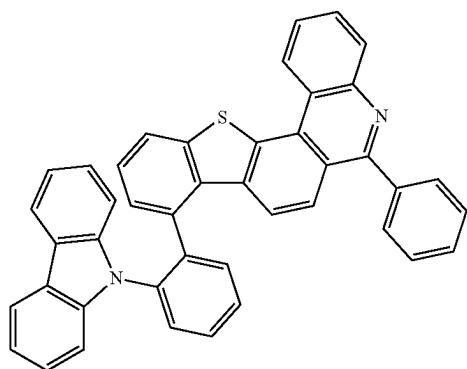
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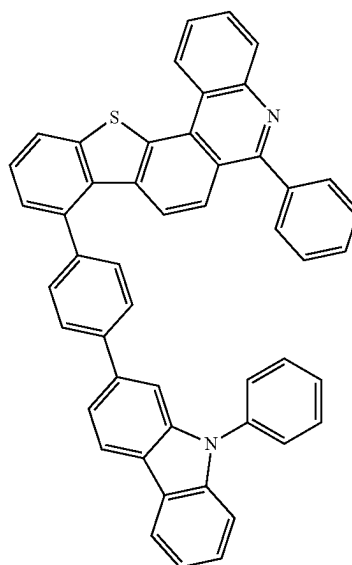
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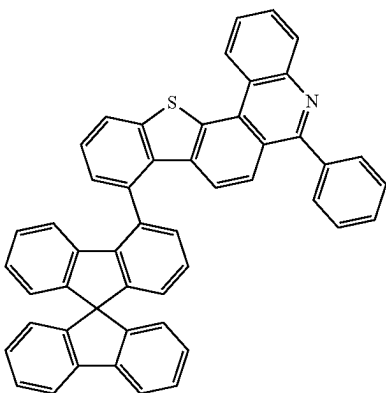
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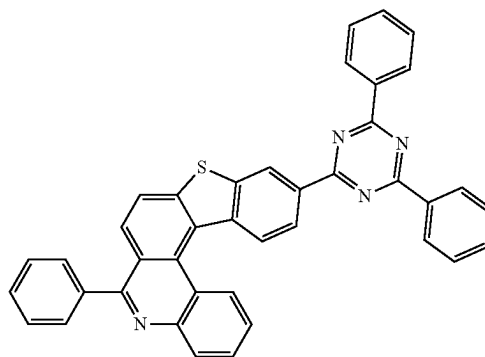


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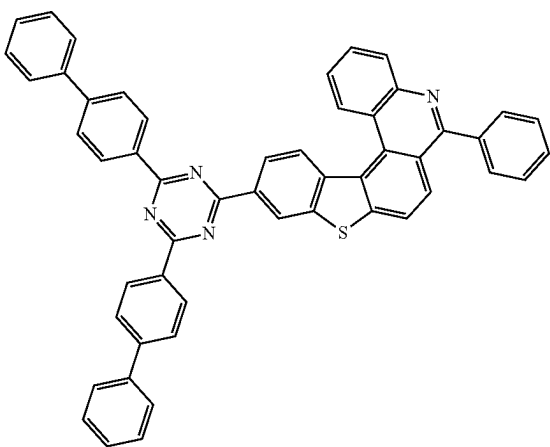
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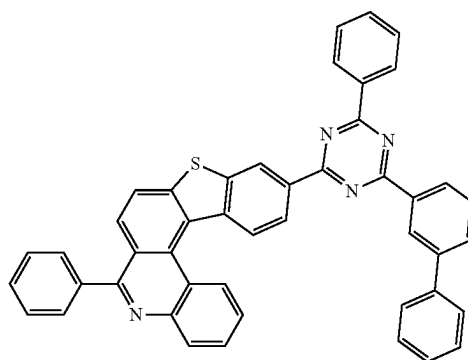


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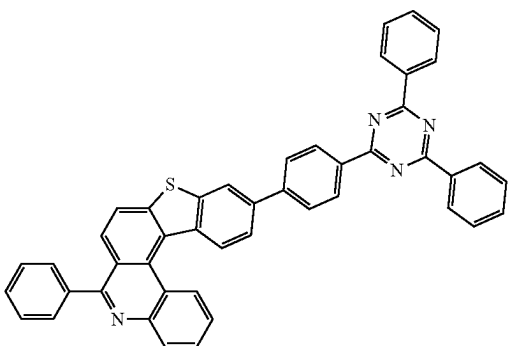
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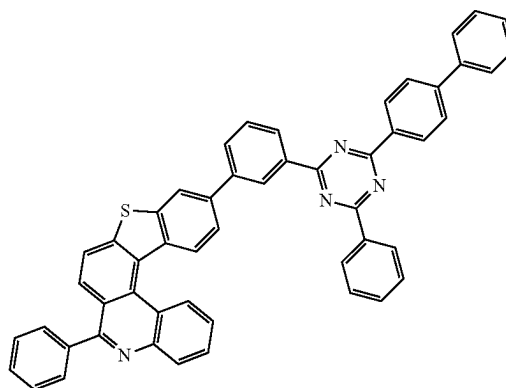
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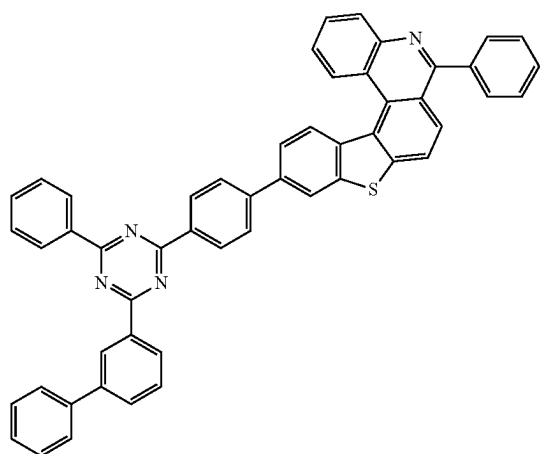
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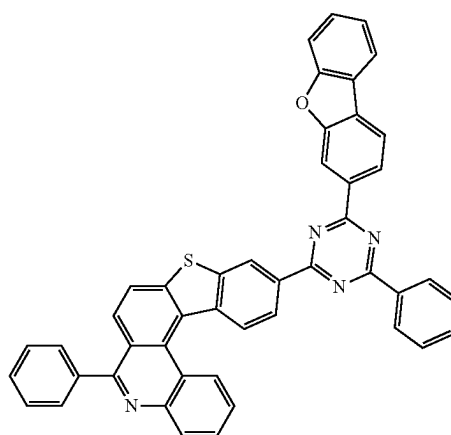


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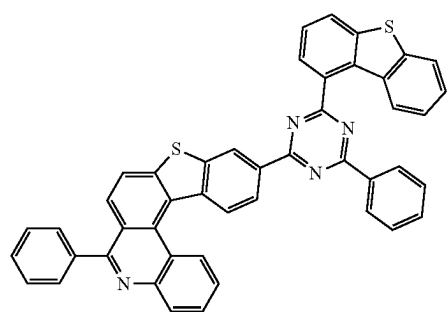


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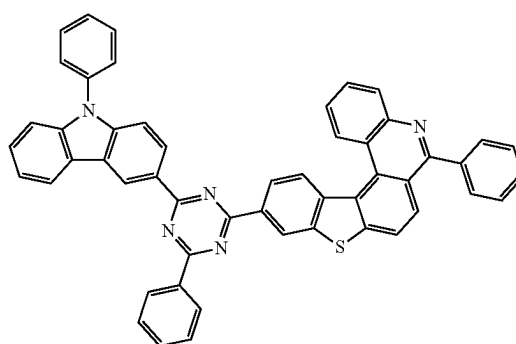
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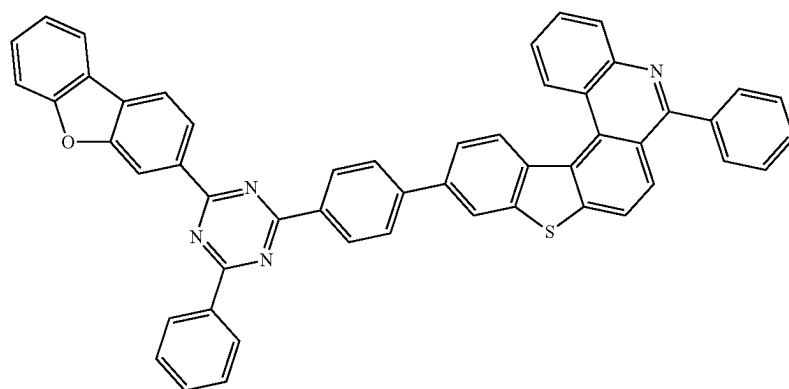
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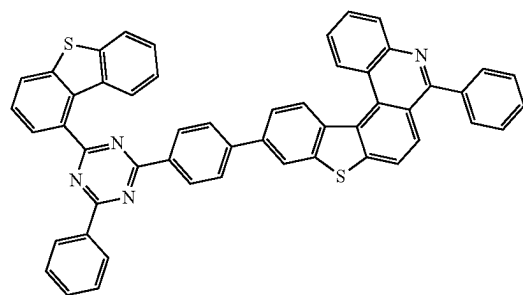
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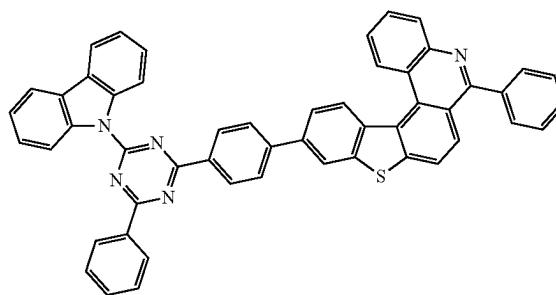
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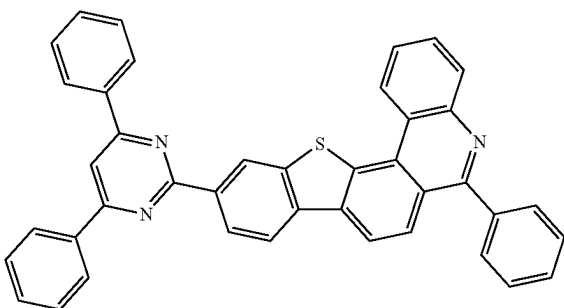
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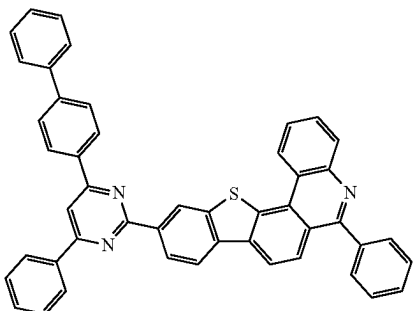
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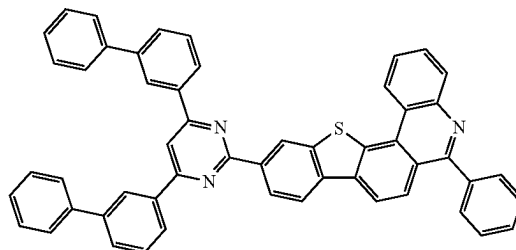
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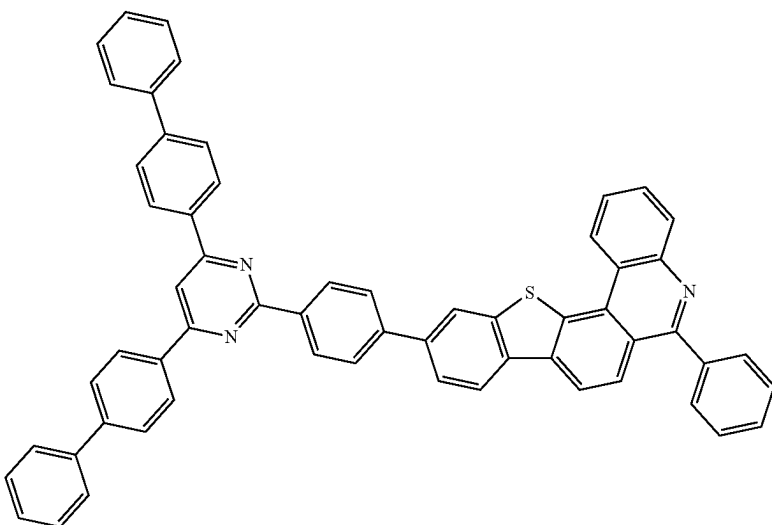
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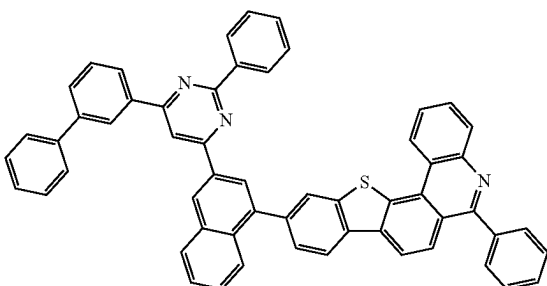
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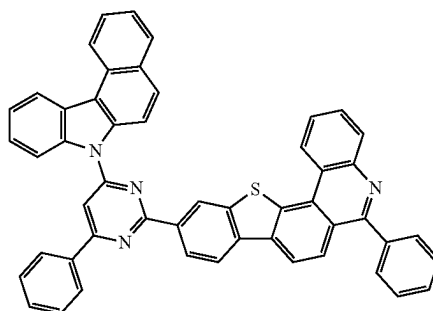
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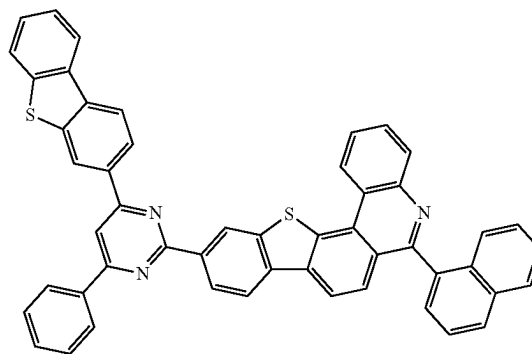
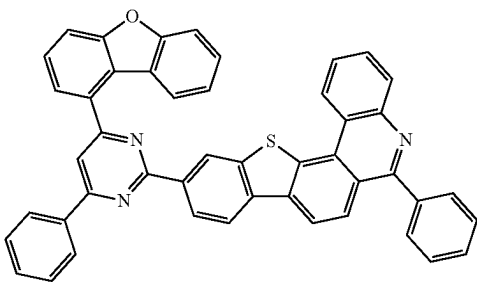


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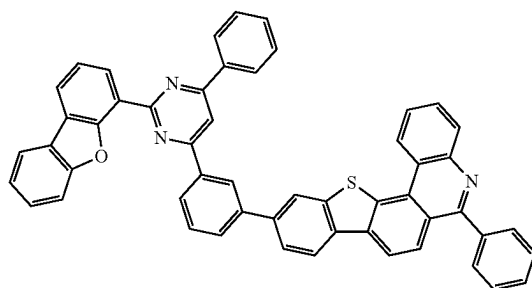
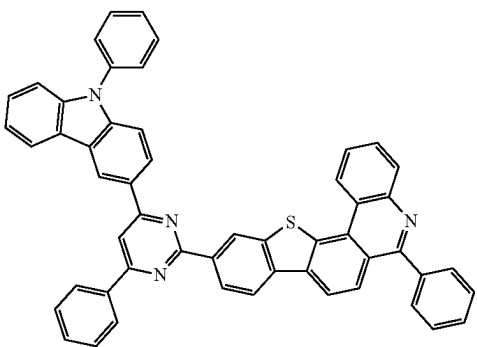
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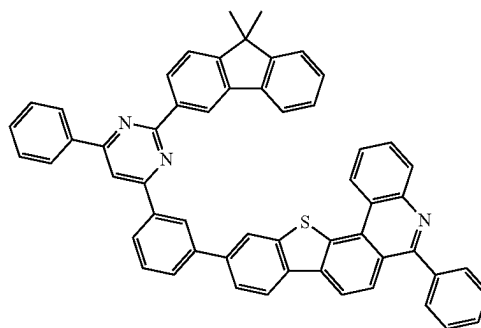
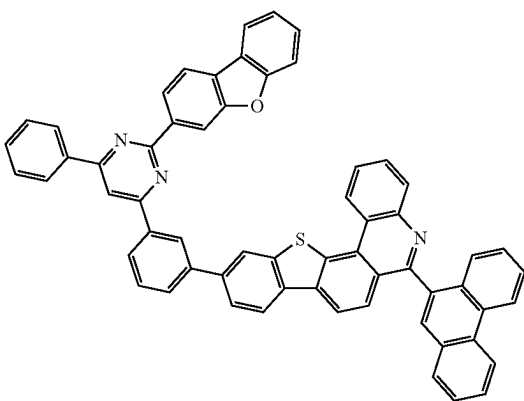
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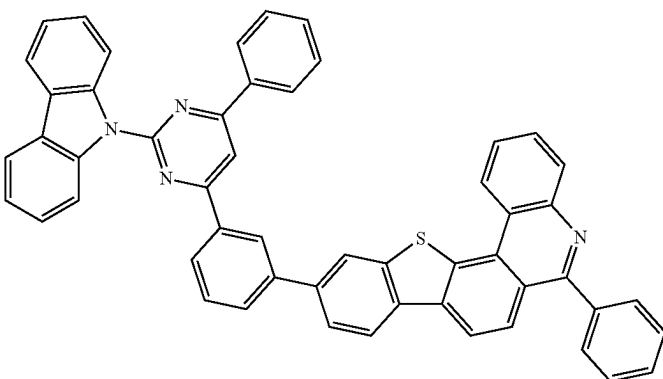


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In addition, by introducing various substituents to the structure of Chemical Formula 1, compounds having unique properties of the introduced substituents may be synthesized. For example, by introducing substituents normally used as hole injection layer materials, hole transfer layer materials, light emitting layer materials, electron transfer

layer materials and charge generation layer materials used for manufacturing an organic light emitting device to the core structure, materials satisfying conditions required for each organic material layer may be synthesized.

In addition, by introducing various substituents to the structure of Chemical Formula 1, the energy band gap may be finely adjusted, and by the high T1 value obtained therefrom, an organic light emitting device with superior efficiency may be provided.

One embodiment of the present specification provides an organic light emitting device including a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes the heterocyclic compound represented by Chemical Formula 1.

In one embodiment of the present specification, the first electrode may be an anode, and the second electrode may be a cathode.

In another embodiment of the present specification, the first electrode may be a cathode, and the second electrode may be an anode.

In one embodiment of the present specification, the organic light emitting device may be a blue organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the blue organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a host material of a light emitting layer of the blue organic light emitting device.

In another embodiment of the present specification, the organic light emitting device may be a green organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the green organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a host material of a light emitting layer of the green organic light emitting device.

In another embodiment of the present specification, the organic light emitting device may be a red organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the red organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a host material of a light emitting layer of the red organic light emitting device.

Specific descriptions on the heterocyclic compound represented by Chemical Formula 1 are the same as the descriptions provided above.

The organic light emitting device of the present specification may be manufactured using common organic light emitting device manufacturing methods and materials except that one or more of the organic material layers are formed using the heterocyclic compound described above.

The heterocyclic compound may be formed into an organic material layer through a solution coating method as well as a vacuum deposition method when manufacturing the organic light emitting device. Herein, the solution coating method means spin coating, dip coating, inkjet printing, screen printing, a spray method, roll coating and the like, but is not limited thereto.

The organic material layer of the organic light emitting device of the present specification may be formed in a single layer structure, but may be formed in a multilayer structure in which two or more organic material layers are laminated. For example, the organic light emitting device of the present disclosure may have a structure including a hole injection layer, a hole transfer layer, a light emitting layer, an electron

transfer layer, an electron injection layer and the like as the organic material layer. However, the structure of the organic light emitting device is not limited thereto, and may include a smaller number of organic material layers.

In the organic light emitting device of the present specification, the organic material layer includes an electron transfer layer, and the electron transfer layer may include the heterocyclic compound of Chemical Formula 1. When using the heterocyclic compound as an electron transfer material, HOMO and LUMO may be adjusted depending on the substituent position, and excellent electron transfer efficiency is obtained. In addition, compared to existing electron transfer materials, migration of electrons in a reverse direction is prevented due to a difference in the T1 value, and as a result, superior device efficiency is obtained.

The organic light emitting device of the present disclosure may further include one, two or more layers selected from the group consisting of a light emitting layer, a hole injection layer, a hole transfer layer, an electron injection layer, an electron transfer layer, an electron blocking layer and a hole blocking layer.

FIG. 1 to FIG. 3 illustrate a lamination order of electrodes and organic material layers of an organic light emitting device according to one embodiment of the present specification. However, the scope of the present application is not limited to these diagrams, and structures of organic light emitting devices known in the art may also be used in the present application.

FIG. 1 illustrates an organic light emitting device in which an anode (200), an organic material layer (300) and a cathode (400) are consecutively laminated on a substrate (100). However, the structure is not limited to such a structure, and as illustrated in FIG. 2, an organic light emitting device in which a cathode, an organic material layer and an anode are consecutively laminated on a substrate may also be obtained.

FIG. 3 illustrates a case of the organic material layer being a multilayer. The organic light emitting device according to FIG. 3 includes a hole injection layer (301), a hole transfer layer (302), a light emitting layer (303), an electron transfer layer (304) and an electron injection layer (305). However, the scope of the present application is not limited to such a lamination structure, and as necessary, layers other than the light emitting layer may not be included, and other necessary functional layers may be further included.

The organic material layer including the heterocyclic compound represented by Chemical Formula 1 may further include other materials as necessary.

The organic light emitting device according to one embodiment of the present specification includes a first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and a second electrode provided on the second stack, wherein the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

The organic light emitting device according to one embodiment of the present specification includes a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes two or more stacks, and the two or more stacks each independently include a light emitting layer, a charge generation layer is included between the two or more stacks, and the charge

generation layer may include the heterocyclic compound represented by Chemical Formula 1.

The organic light emitting device according to one embodiment of the present specification includes a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes a first stack including a first light emitting layer; a charge generation layer provided on the first stack; and a second stack including a second light emitting layer, and the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

In addition, the organic light emitting device according to one embodiment of the present specification includes an anode, a cathode, and two or more stacks provided between the anode and the cathode, the two or more stacks each independently include a light emitting layer, a charge generation layer is included between the two or more stacks, and the charge generation layer includes the heterocyclic compound represented by Chemical Formula 1.

In addition, the organic light emitting device according to one embodiment of the present specification includes an anode, a first stack provided on the anode and including a first light emitting layer, a charge generation layer provided on the first stack, a second stack provided on the charge generation layer and including a second light emitting layer, and a cathode provided on the second stack. Herein, the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1. When the heterocyclic compound is included in the charge generation layer, superior efficiency as a charge generation layer material is obtained by an electron-friendly substituent structure and a hole migration-friendly quinoline-dibenzofuran or quinoline-dibenzothiophene fused structure.

The organic light emitting device according to one embodiment of the present specification includes a first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and a second electrode provided on the second stack, wherein the charge generation layer is an N-type charge generation layer, and the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

In addition, the first stack and the second stack may each independently further include one or more types of the hole injection layer, the hole transfer layer, the hole blocking layer, the electron transfer layer, the electron injection layer and the like described above.

The charge generation layer may be an N-type charge generation layer or a P-type charge generation layer, and the charge generation layer may further include a dopant known in the art in addition to the heterocyclic compound represented by Chemical Formula 1.

As the organic light emitting device according to one embodiment of the present specification, an organic light emitting device having a 2-stack tandem structure is illustrated in FIG. 4.

Herein, the first electron blocking layer, the first hole blocking layer, the second hole blocking layer and the like described in FIG. 4 may not be included in some cases.

In the organic light emitting device according to one embodiment of the present specification, materials other than the heterocyclic compound represented by Chemical Formula 1 are illustrated below, however, these are for

illustrative purposes only and not for limiting the scope of the present application, and the materials may be replaced by materials known in the art.

As the anode material, materials having relatively large work function may be used, and transparent conductive oxides, metals, conductive polymers or the like may be used. Specific examples of the anode material include metals such as vanadium, chromium, copper, zinc and gold, or alloys thereof; metal oxides such as zinc oxide, indium oxide, indium tin oxide (ITO) and indium zinc oxide (IZO); combinations of metals and oxides such as ZnO:Al or SnO₂:Sb; conductive polymers such as poly(3-methylthiophene), poly[3,4-(ethylene-1,2-dioxy)thiophene](PEDOT), polypyrrole and polyaniline, and the like, but are not limited thereto.

As the cathode material, materials having relatively small work function may be used, and metals, metal oxides, conductive polymers or the like may be used. Specific examples of the cathode material include metals such as magnesium, calcium, sodium, potassium, titanium, indium, yttrium, lithium, gadolinium, aluminum, silver, tin and lead, or alloys thereof; multilayer structure materials such as LiF/Al or LiO₂/Al, and the like, but are not limited thereto.

As the hole injection material, known hole injection materials may be used, and for example, phthalocyanine compounds such as copper phthalocyanine disclosed in U.S. Pat. No. 4,356,429, or starburst-type amine derivatives such as tris(4-carbazoyl-9-ylphenyl)amine (TCTA), 4,4',4"-tri[phenyl(m-tolyl)amino]triphenylamine (m-MTDATA) or 1,3,5-tris[4-(3-methylphenylphenylamino)phenyl]benzene (m-MTDAPB) described in the literature [Advanced Material, 6, p. 677 (1994)], polyaniline/dodecylbenzene sulfonic acid, poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate), polyaniline/camphor sulfonic acid or polyaniline/poly(4-styrenesulfonate) that are conductive polymers having solubility, and the like, may be used.

As the hole transfer material, pyrazoline derivatives, arylamine-based derivatives, stilbene derivatives, triphenyl-diamine derivatives and the like may be used, and low molecular or high molecular materials may also be used.

As the electron transfer material, metal complexes of oxadiazole derivatives, anthraquinodimethane and derivatives thereof, benzoquinone and derivatives thereof, naphthoquinone and derivatives thereof, anthraquinone and derivatives thereof, tetracyanoanthraquinodimethane and derivatives thereof, fluorenone derivatives, diphenyldicyanoethylene and derivatives thereof, diphenoquinone derivatives, 8-hydroxyquinoline and derivatives thereof, and the like, may be used in addition to the heterocyclic compound, and high molecular materials may also be used as well as low molecular materials.

As examples of the electron injection material, LiF is typically used in the art, however, the present application is not limited thereto.

As the light emitting material, red, green or blue light emitting materials may be used, and as necessary, two or more light emitting materials may be mixed and used. Herein, two or more light emitting materials may be used by being deposited as individual sources of supply or by being premixed and deposited as one source of supply. In addition, fluorescent materials may also be used as the light emitting material, however, phosphorescent materials may also be used. As the light emitting material, materials emitting light by bonding electrons and holes injected from an anode and a cathode, respectively, may be used alone, however, materials having a host material and a dopant material involving in light emission together may also be used.

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When mixing light emitting material hosts, same series hosts may be mixed, or different series hosts may be mixed. For example, any two or more types of materials among n-type host materials or p-type host materials may be selected and used as a host material of a light emitting layer.

The organic light emitting device according to one embodiment of the present specification may be a top-emission type, a bottom-emission type or a dual-emission type depending on the materials used.

The heterocyclic compound according to one embodiment of the present specification may also be used in an organic electronic device including an organic solar cell, an

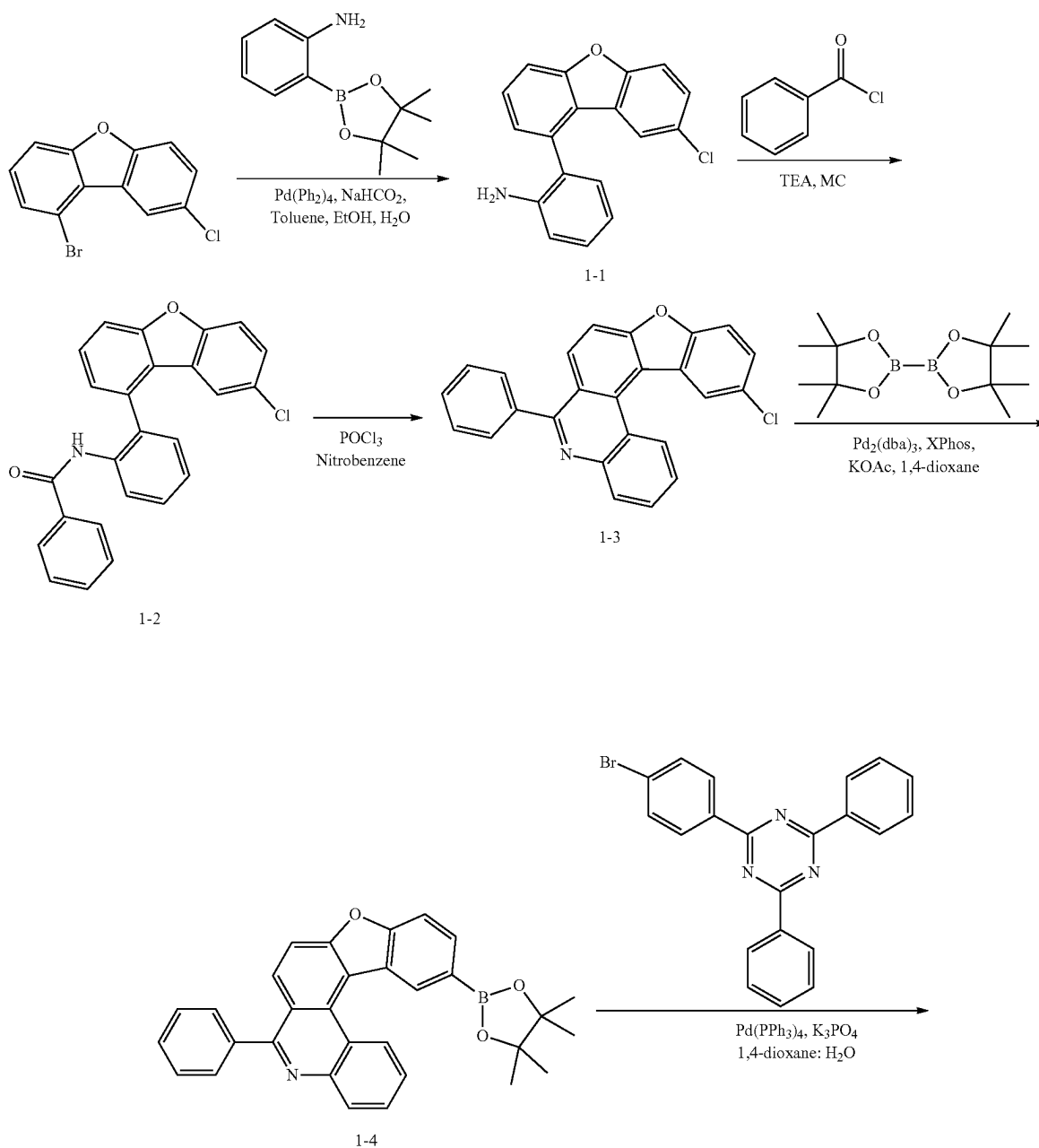
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organic photo conductor, an organic transistor and the like under a similar principle used in the organic light emitting device.

Hereinafter, the present specification will be described in more detail with reference to examples, however, these are for illustrative purposes only, and the scope of the present application is not limited thereto.

[Preparation Example 1] Preparation of Compound

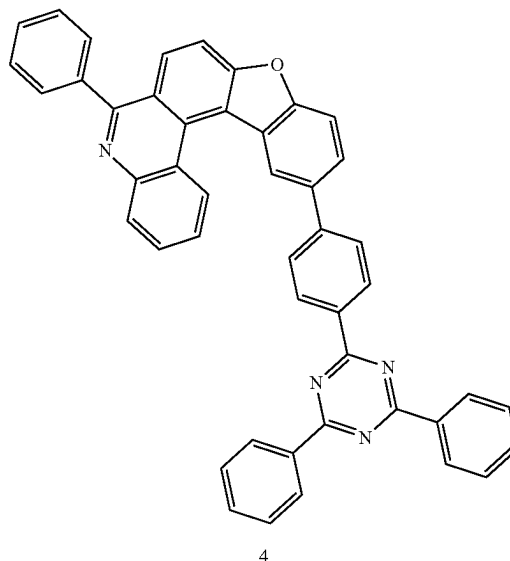
4



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-continued



1) Preparation of Compound 1-1

After dissolving 1-bromo-8-chlorodibenzo[b,d]furan (100 g, 0.3552 mol) and 2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline (77.8 g, 0.3552 mol) in toluene, ethanol and H₂O (2000 mL:200 mL:200 mL), Pd(PPh₃)₄ (tetrakis (triphenylphosphine)palladium(0)) (20.5 g, 0.017 mol) and NaHCO₃ (89 g, 1.06 mol) were introduced thereto, and the result was stirred for 3 hours at 100° C. After the reaction was completed, methylene chloride (MC) and distilled water were introduced to the reaction solution, and the result was extracted, then dried with anhydrous MgSO₄, and the solvent was removed using a rotary evaporator to obtain Compound 1-1 (77 g, 74%) in a liquid form.

2) Preparation of Compound 1-2

Compound 1-1 (77 g, 0.262 mol) and triethylamine (86 mL, 0.393 mol) were introduced to methylene chloride (1200 mL) and dissolved therein. To the mixture, benzoyl chloride (55 g, 0.393 mol) dissolved in methylene chloride (300 mL) was slowly added dropwise at 0° C. After the reaction was completed, white solids in the reaction solution were filtered and washed with hexane. These were dried to obtain Compound 1-2 (86 g, 83%) in a solid form.

3) Preparation of Compound 1-3

After dissolving Compound 1-2 (86 g, 0.216 mol) in nitrobenzene (1500 mL), POCl₃ (30 mL, 0.324 mol) was slowly added dropwise thereto, and the result was reacted for 15 hours at 140° C. After the reaction was completed, a solution obtained by dissolving NaHCO₃ in distilled water was slowly introduced to the reaction solution, and the result

was stirred. Solids produced after that were filtered and collected, and the collected solids were recrystallized in MC and MeOH to obtain Compound 1-3 (41 g, 51%) in a solid form.

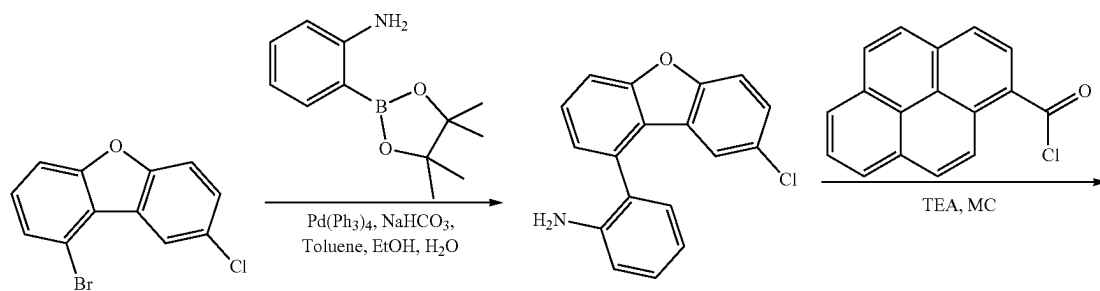
4) Preparation of Compound 1-4

After dissolving Compound 1-3 (41 g, 0.1079 mol) in 1,4-dioxane (800 mL), Pd₂(dba)₃ (tris(dibenzylideneacetone)dipalladium(0)) (9.8 g, 0.0107 mol), XPhos (2-dicyclohexylphosphino-2',4',6'-triisopropylbiphenyl) (5.1 g, 0.0107 mol) and KOAc (potassium acetate) (31 g, 0.3237 mol) were dissolved therein, and the result was reacted for 4 hours at 90° C. After the reaction was completed, water and methylene chloride were introduced to the reaction solution, and the result was extracted. Solids obtained after that were dried, and recrystallized with methylene chloride and methanol to obtain Compound 1-4 (30 g, 60%) in a solid form.

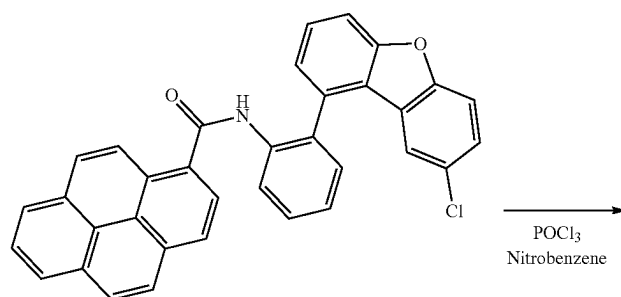
5) Preparation of Compound 4

After dissolving Compound 1-4 (6.4 g, 0.017 mol) and 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine (7.4 g, 0.017 mol) in 1,4-dioxane:H₂O=100 mL:20 mL, Pd(PPh₃)₄ (1.96 g, 0.0017 mol) and K₃PO₄ (20.8 g, 0.051 mol) were introduced thereto, and the result was stirred for 5 hours at 100° C. After the reaction was completed, solids produced in the reaction solution were washed with 1,4-dioxane and H₂O. After that, only the solids were purified using a recrystallization method in dichlorobenzene (DCB) to obtain Compound 4 (6.6 g, 60%).

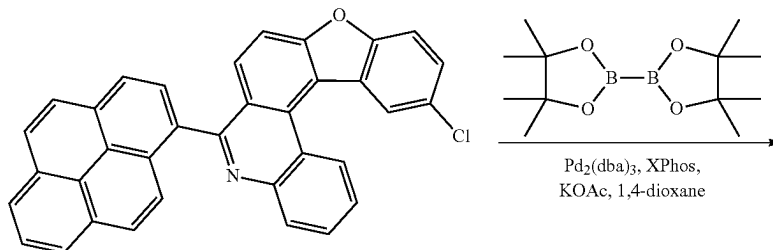
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[Preparation Example 2] Preparation of Compound
88

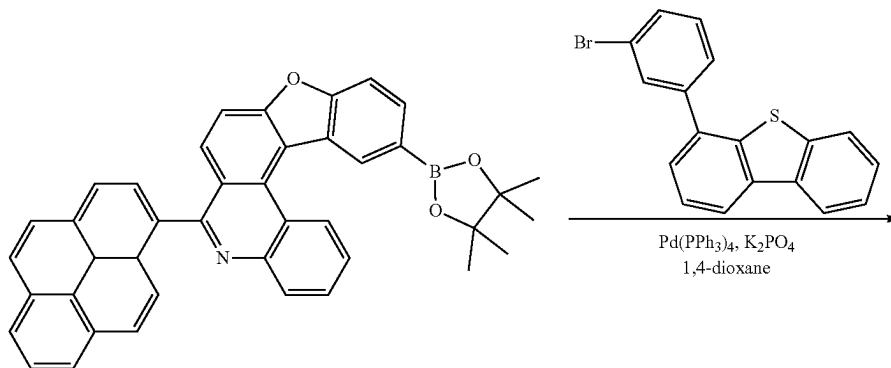
1-1



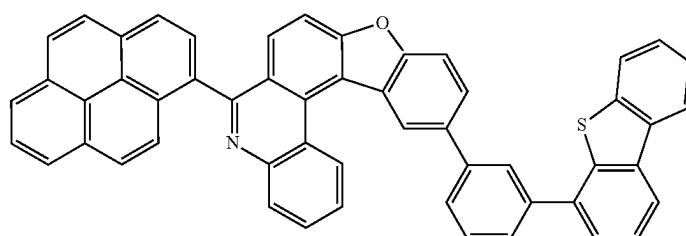
2.2



2.3



2.4



8.8

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1) Preparation of Compound 1-1

Compound 1-1 was the same as Compound 1-1 of Preparation Example 1.

2) Preparation of Compound 2-2

Compound 1-1 (77 g, 0.262 mol) and triethylamine (86 mL, 0.393 mol) were introduced to MC (1200 mL) and dissolved therein. To the mixture, 1-pyrenecarbonyl chloride (104 g, 0.393 mol) dissolved in methylene chloride (300 mL) was slowly added dropwise at 0° C. After the reaction was completed, white solids in the reaction solution were filtered and washed with hexane. These were dried to obtain Compound 2-2 (75 g, 67%) in a solid form.

3) Preparation of Compound 2-3

After dissolving Compound 2-2 (86 g, 0.143 mol) in nitrobenzene (1500 mL), POCl₃ (20 mL, 0.215 mol) was slowly added dropwise thereto, and the result was reacted for 15 hours at 140° C. After the reaction was completed, a solution obtained by dissolving NaHCO₃ in distilled water was slowly introduced to the reaction solution, and the result was stirred. Solids produced after that were filtered and collected, and the collected solids were recrystallized in methylene chloride and methanol to obtain Compound 2-3 (29 g, 41%) in a solid form.

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4) Preparation of Compound 2-4

After dissolving Compound 2-3 (29 g, 0.057 mol) in 1,4-dioxane (800 mL), Pd₂(dba)₃ (5.2 g, 0.0057 mol), XPhos (2.7 g, 0.0057 mol) and KOAc (16.7 g, 0.171 mol) were dissolved therein, and the result was reacted for 4 hours at 90° C. After the reaction was completed, water and MC were introduced to the reaction solution, and the result was extracted. After that, solids obtained by drying the result were recrystallized with methylene chloride and methanol to obtain Compound 2-4 (20 g, 60%) in a solid form.

5) Preparation of Compound 88

After dissolving Compound 2-4 (10 g, 0.017 mol) and 4-(3-bromophenyl)dibenzo[b,d]thiophene (5.7 g, 0.017 mol) in 1,4-dioxane:H₂O=100 mL:20 mL, Pd(PPh₃)₄ (1.96 g, 0.0017 mol) and K₃PO₄ (20.8 g, 0.051 mol) were introduced thereto, and the result was stirred for 5 hours at 100° C. After the reaction was completed, solids produced in the reaction solution were washed with 1,4-dioxane and H₂O. After that, only the solids were purified using a recrystallization method in dichlorobenzene (DCB) to obtain Compound 88 (4.9 g, 40%).

Target compounds were synthesized in the same manner as in Preparation Example 1 except that Intermediate A of the following Table 1 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

TABLE 1

Compound	Intermediate A	Target Compound	Yield
1			61%
5			62%

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TABLE 1-continued

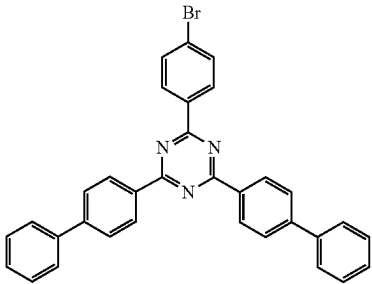
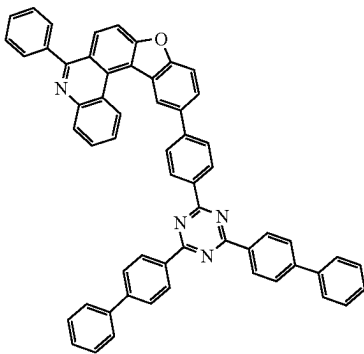
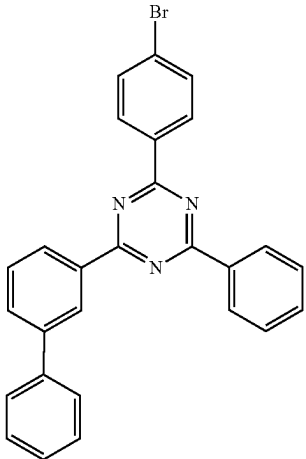
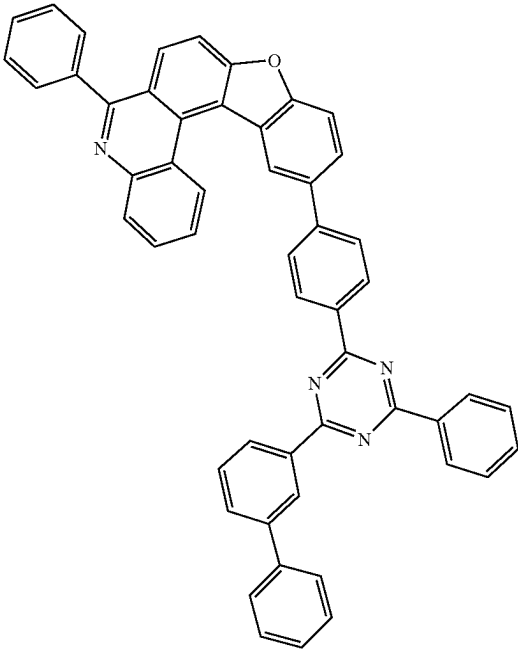
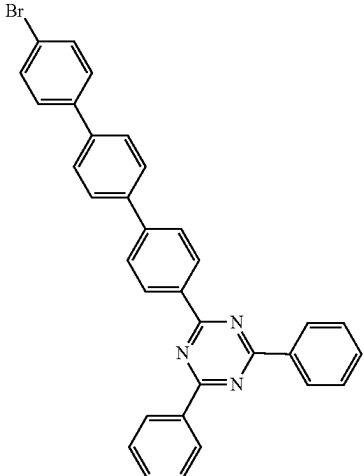
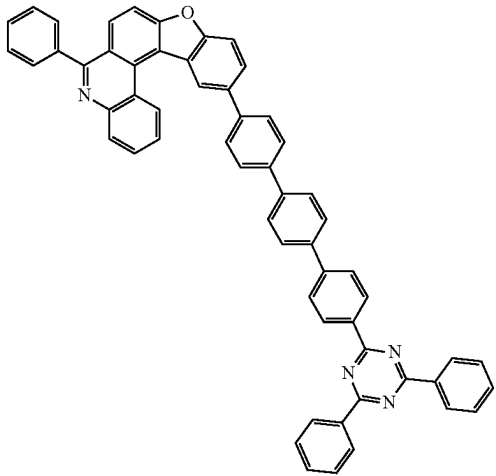
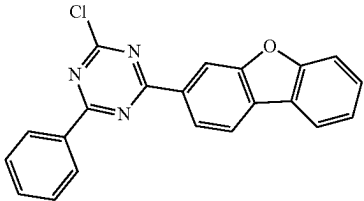
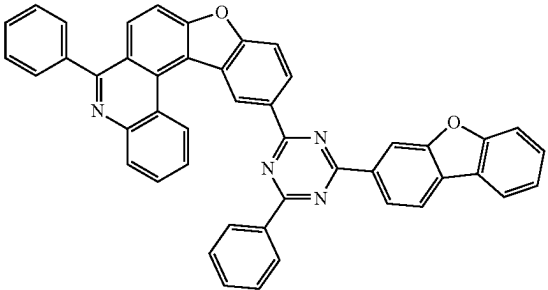
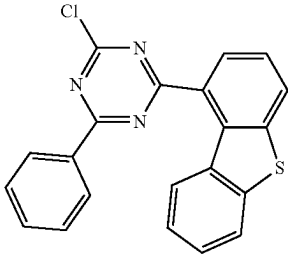
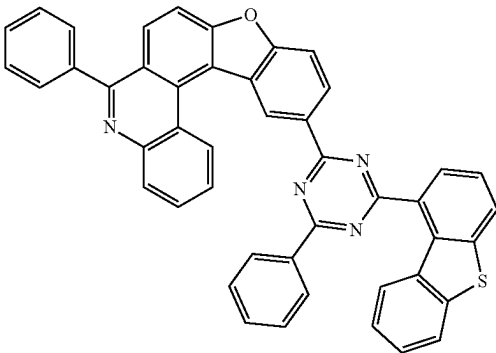
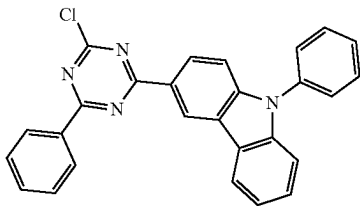
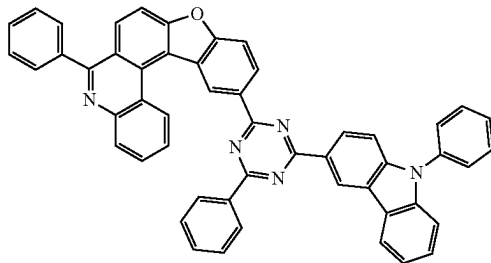
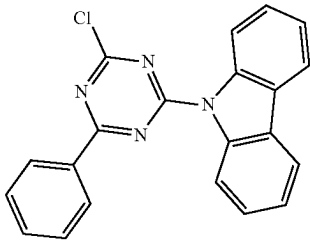
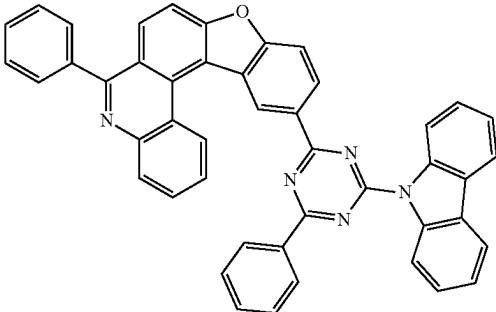
Compound	Intermediate A	Target Compound	Yield
7			51%
8			60%
9			61%

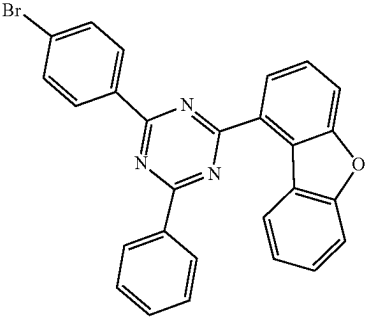
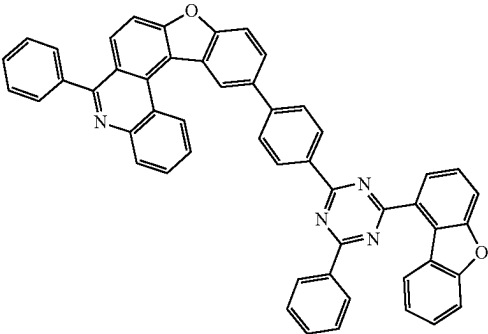
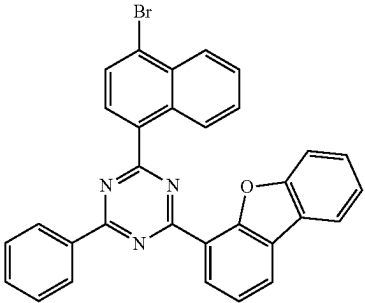
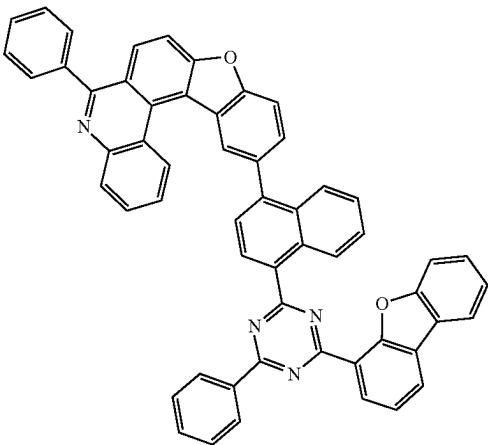
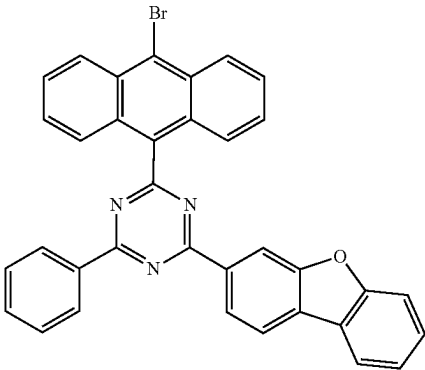
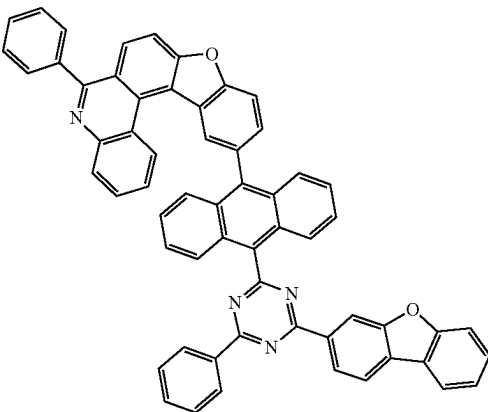
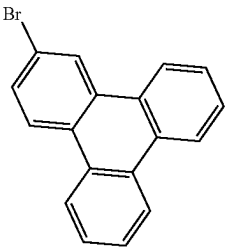
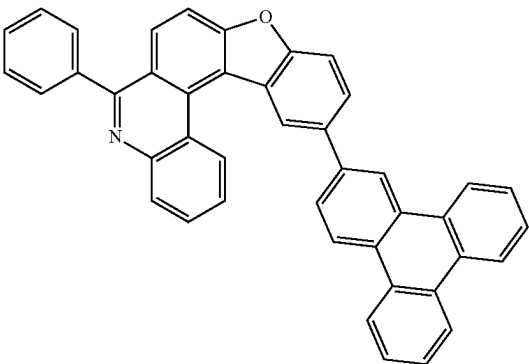
TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
11			62%
12			51%
13			60%
15			61%

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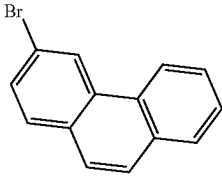
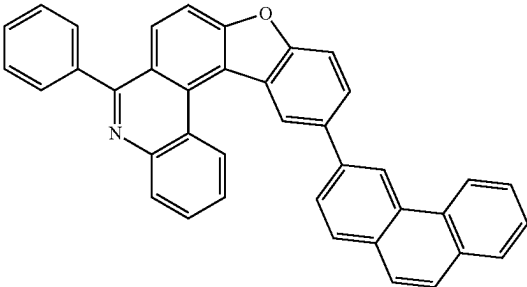
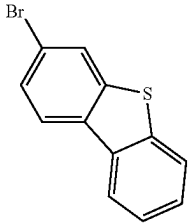
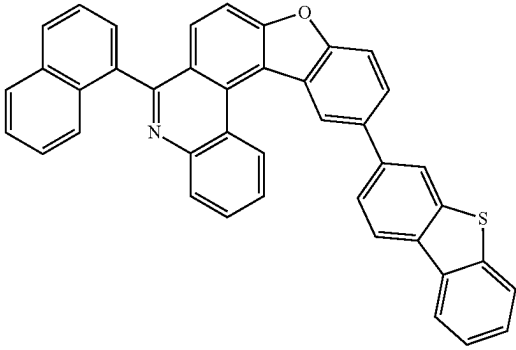
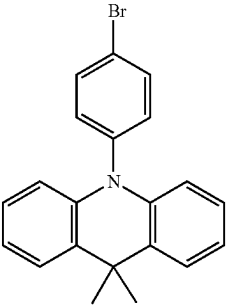
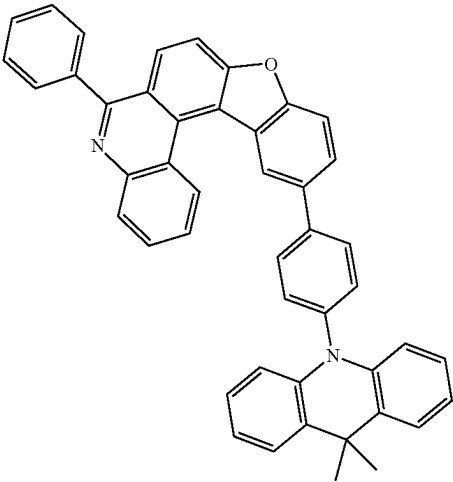
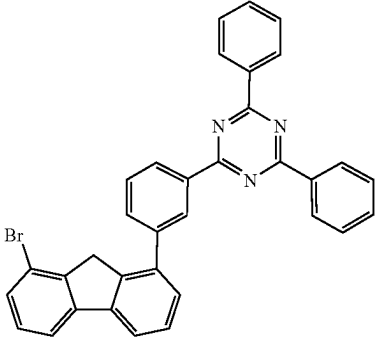
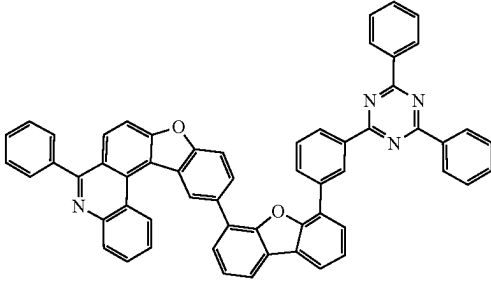
TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
16			62%
17			51%
18			60%
24			61%

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TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
25			62%
28			51%
30			60%
35			61%

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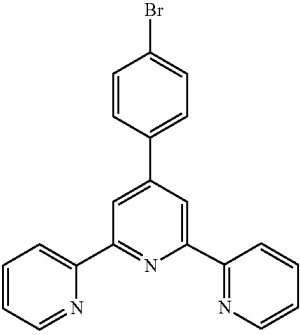
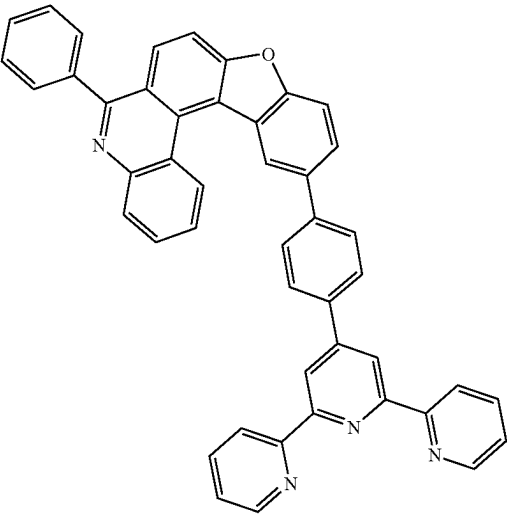
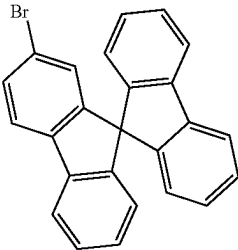
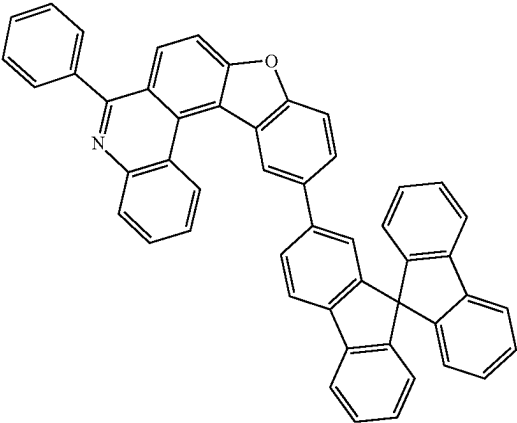
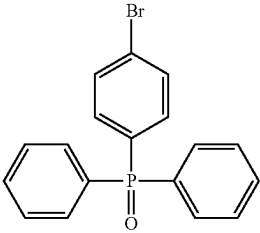
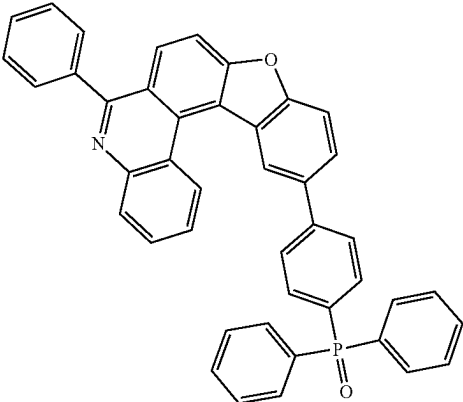
TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
36			62%
37			51%
38			60%
41			61%

461

462

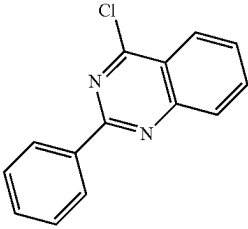
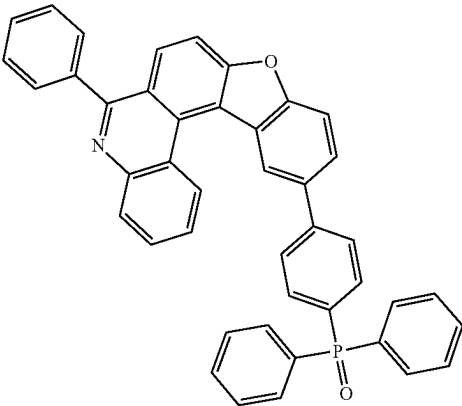
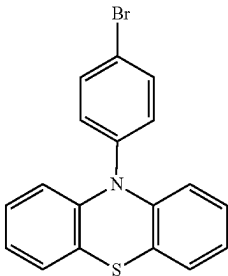
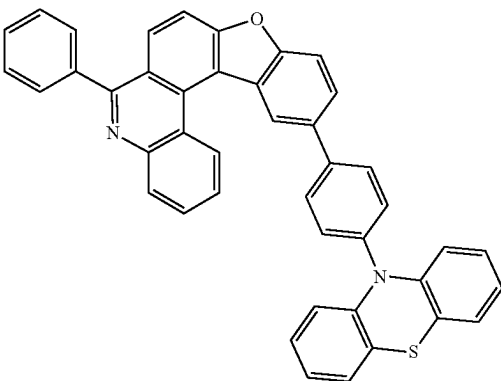
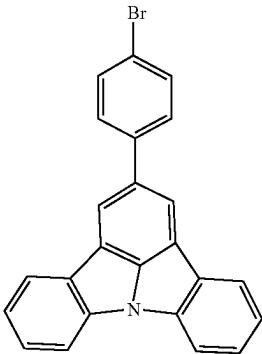
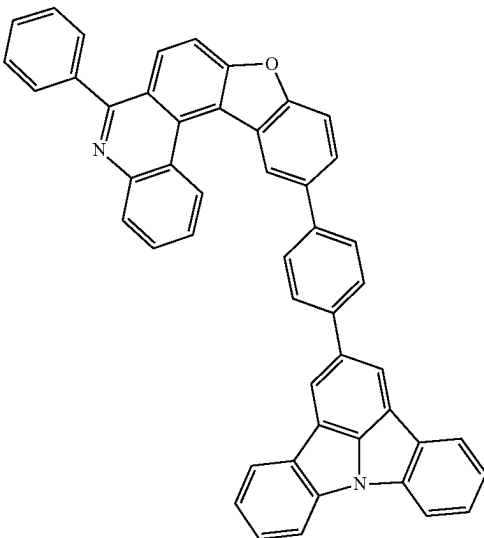
TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
42			62%
44			51%
45			60%

463

464

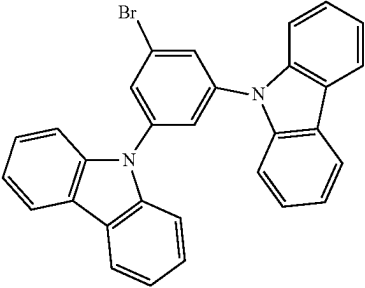
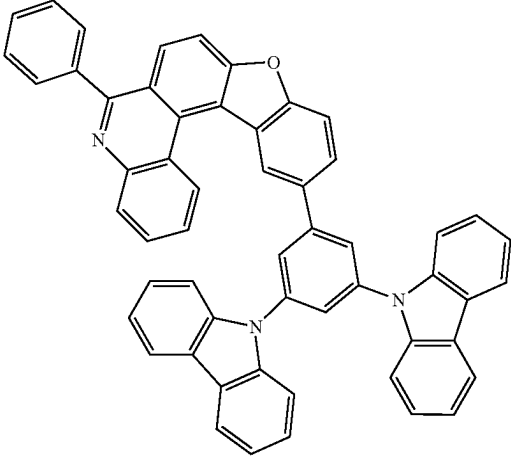
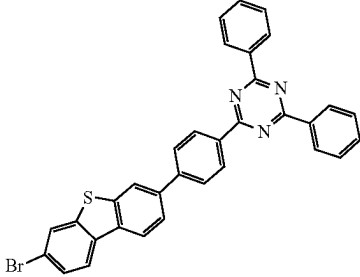
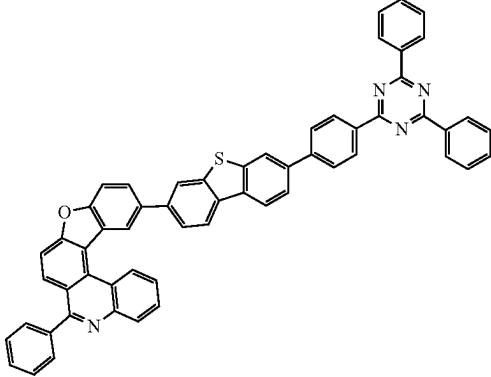
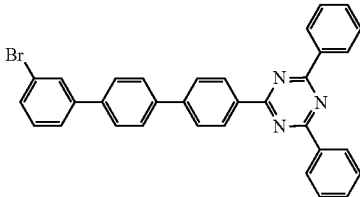
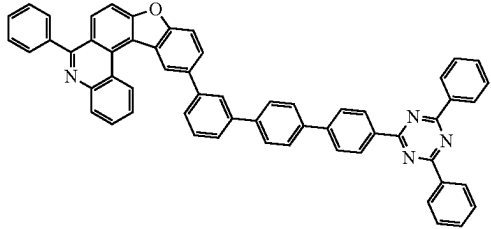
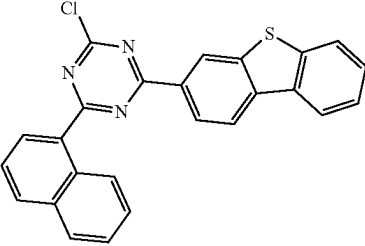
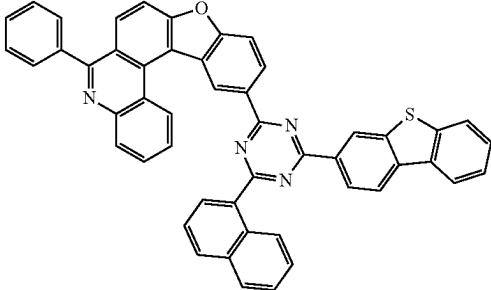
TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
47			61%
52			62%
53			51%

465

466

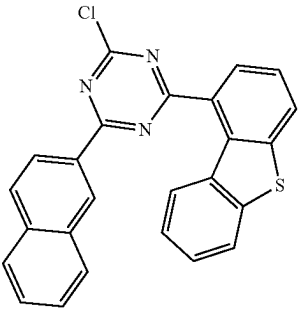
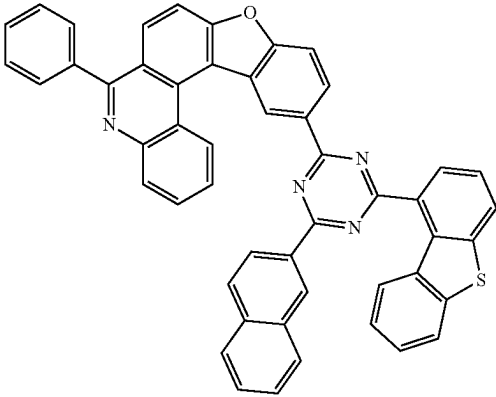
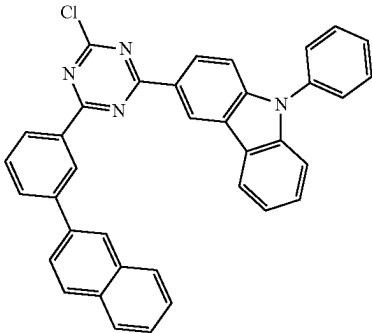
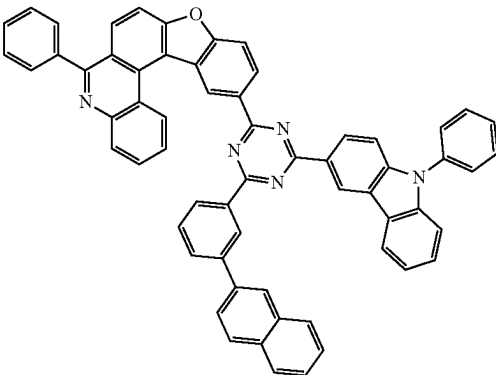
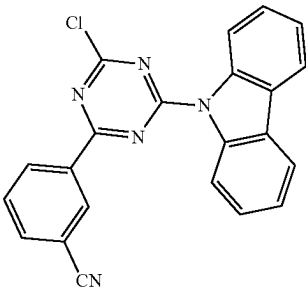
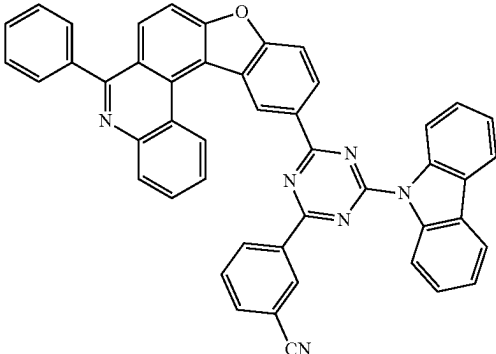
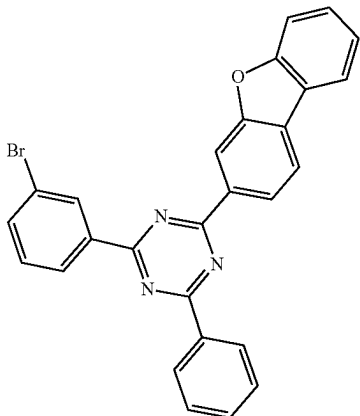
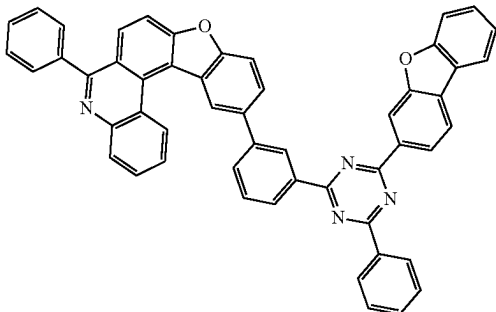
TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
56			60%
57			61%
64			62%
68			51%

467

468

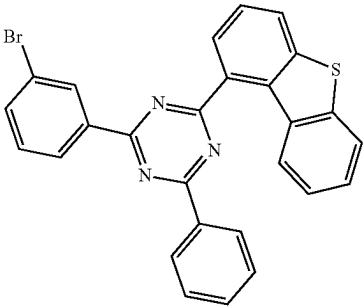
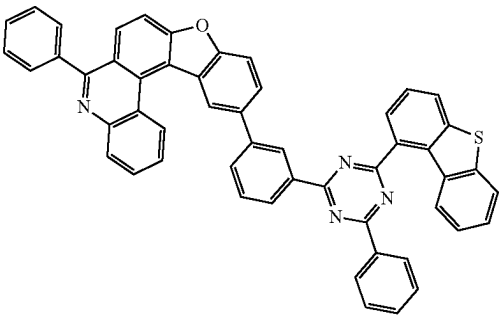
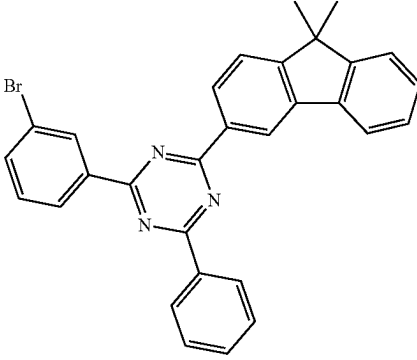
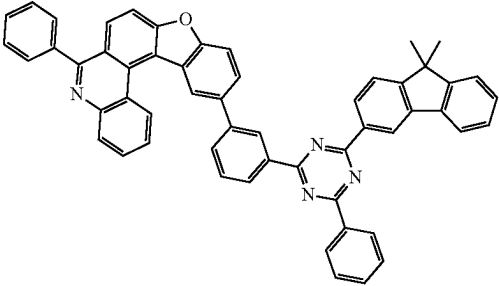
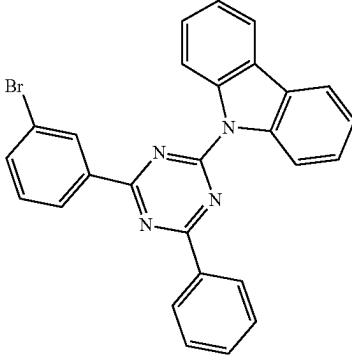
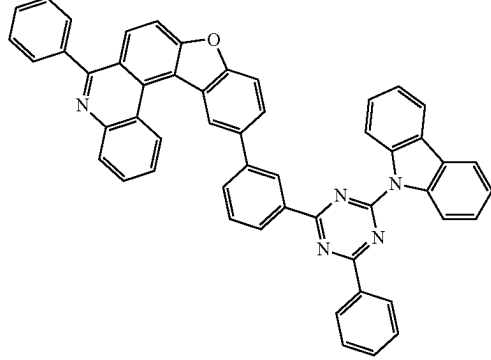
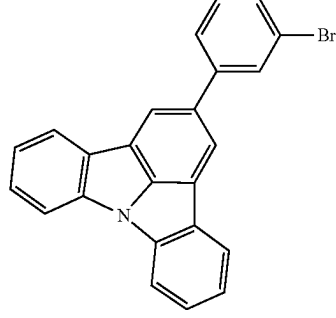
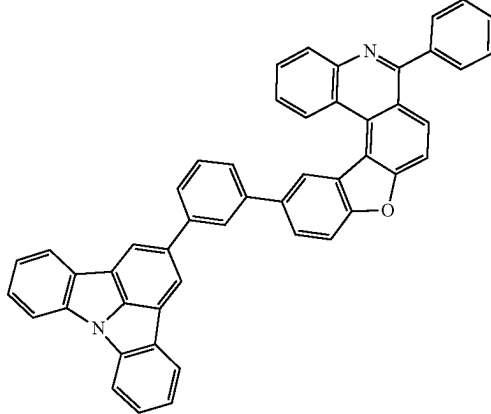
TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
69			60%
70			61%
72			62%
75			51%

469

470

TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
77			60%
79			61%
80			62%
84			51%

471

472

TABLE 1-continued

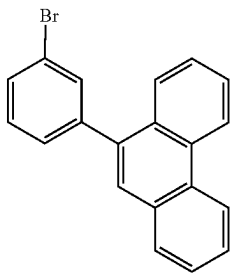
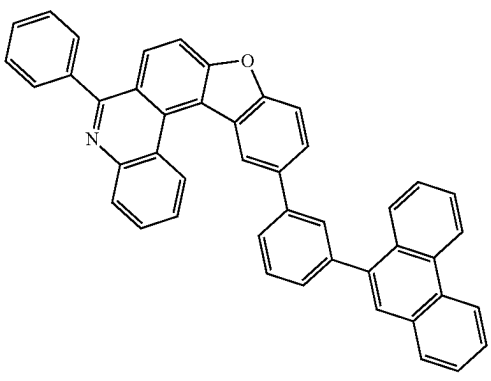
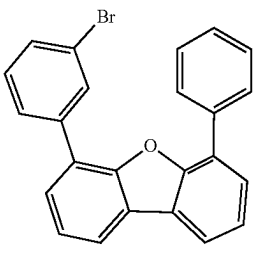
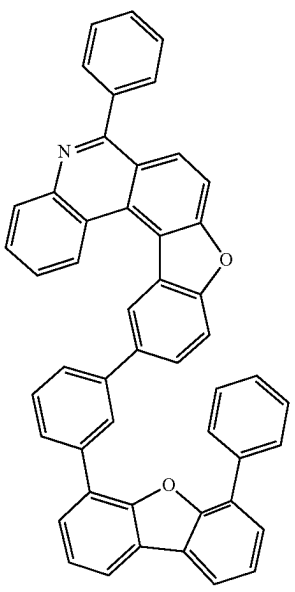
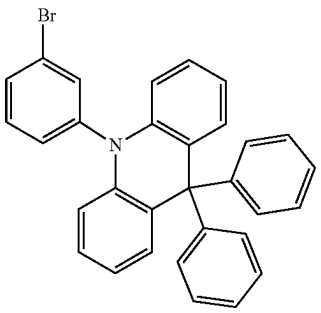
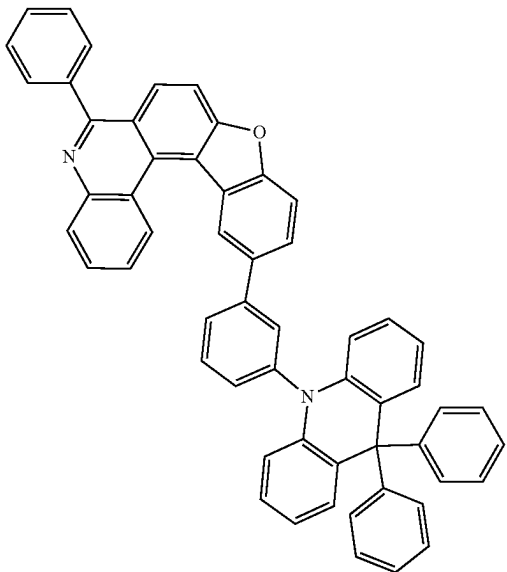
Compound	Intermediate A	Target Compound	Yield
86			60%
89			62%
91			51%

TABLE 1-continued

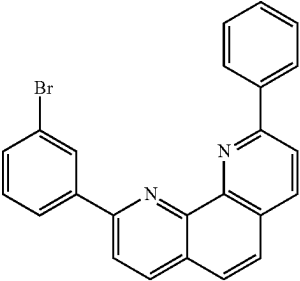
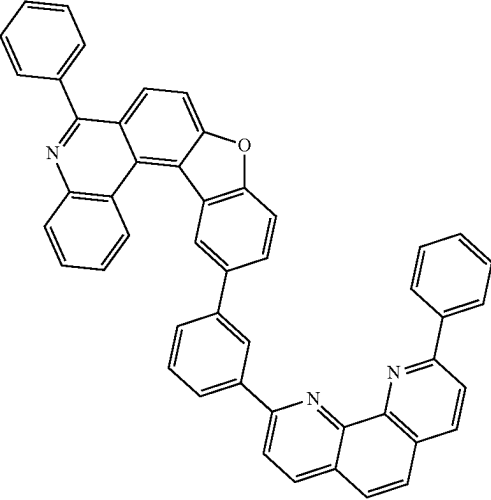
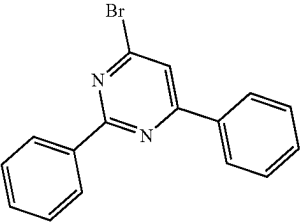
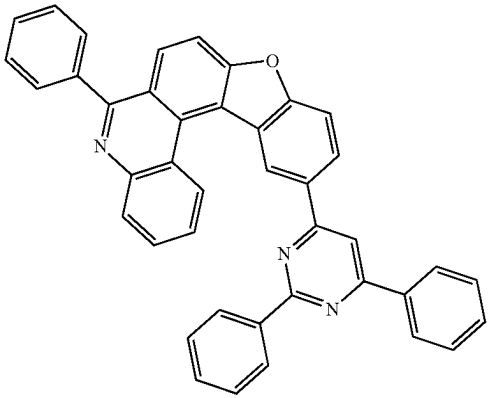
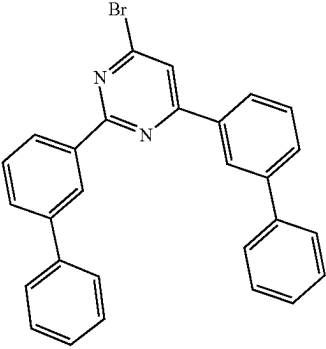
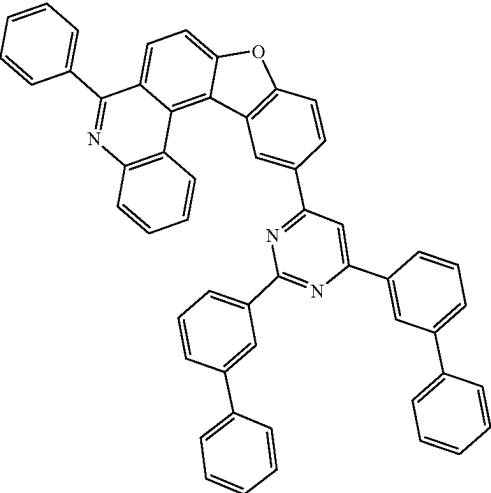
Compound	Intermediate A	Target Compound	Yield
92			60%
97			61%
100			62%

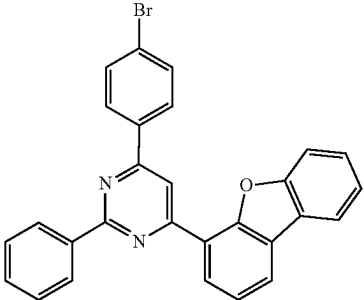
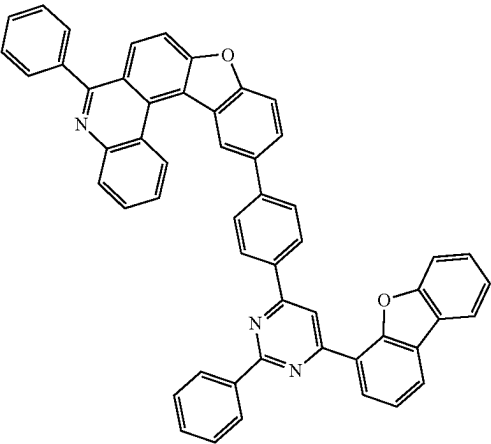
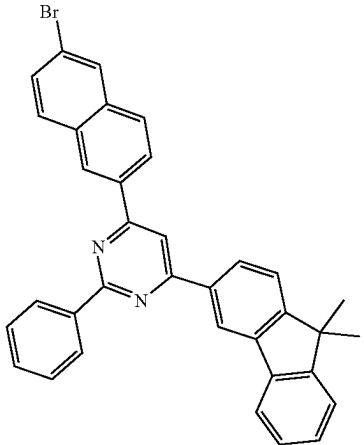
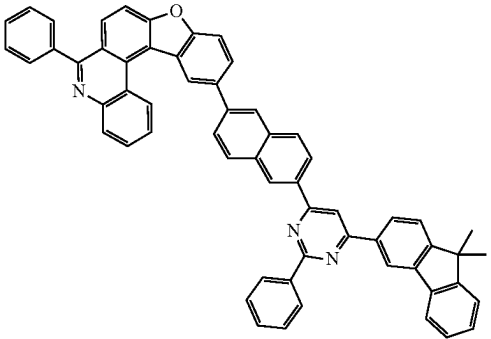
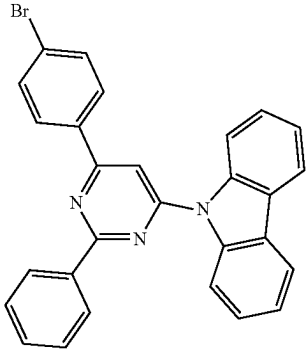
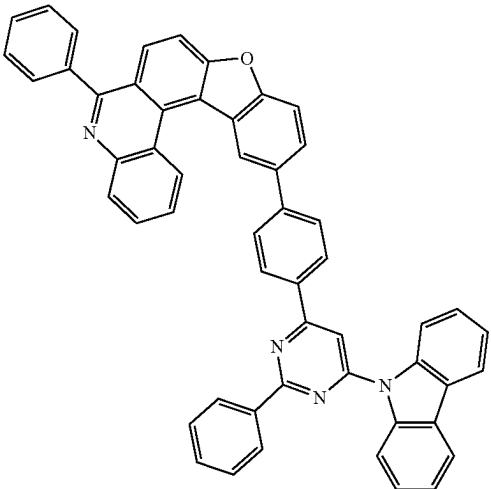
TABLE 1-continued

61%

TABLE 1-continued

60%

TABLE 1-continued

Compound	Intermediate A	Target Compound	Yield
116			61%
119			62%
120			51%

481

482

TABLE 1-continued

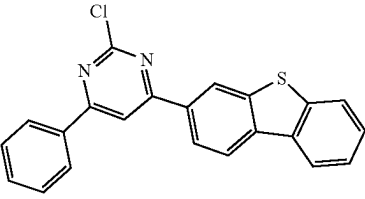
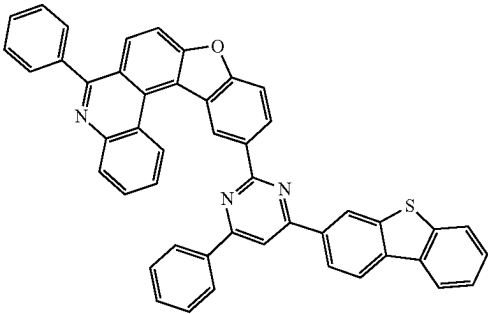
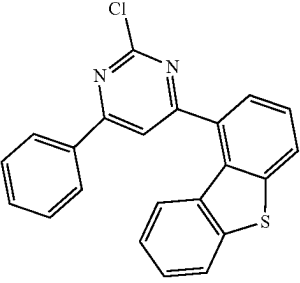
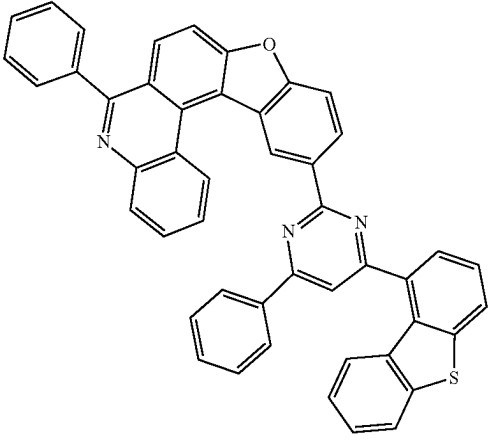
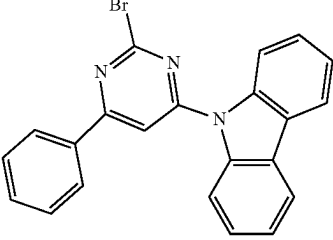
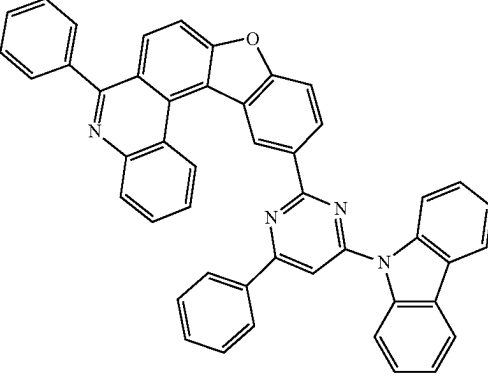
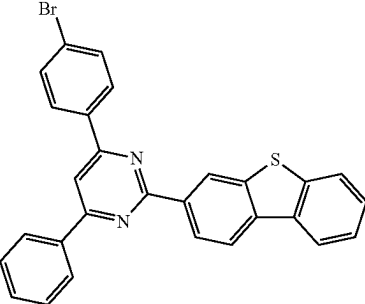
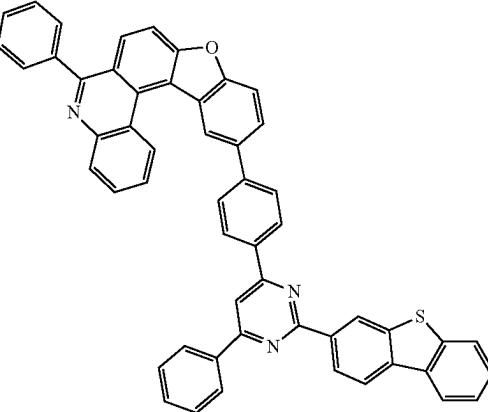
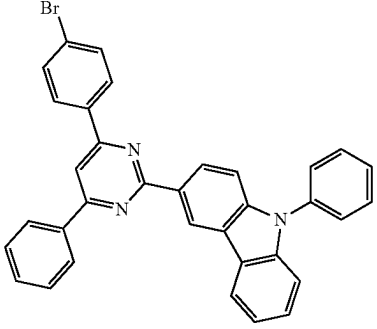
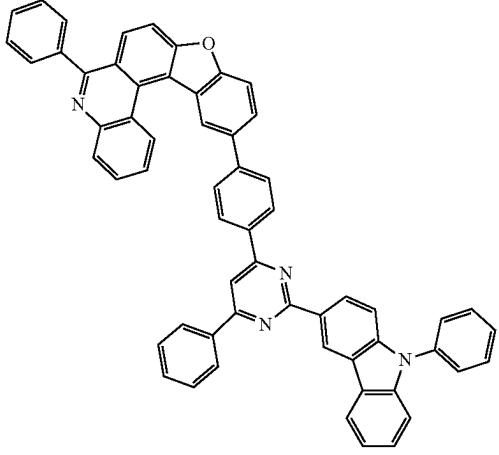
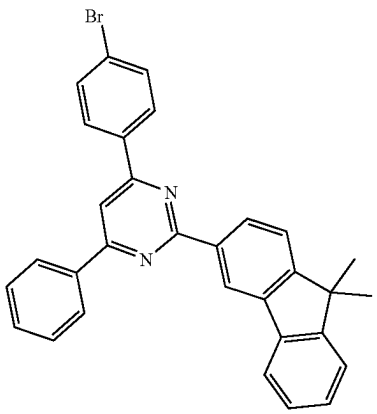
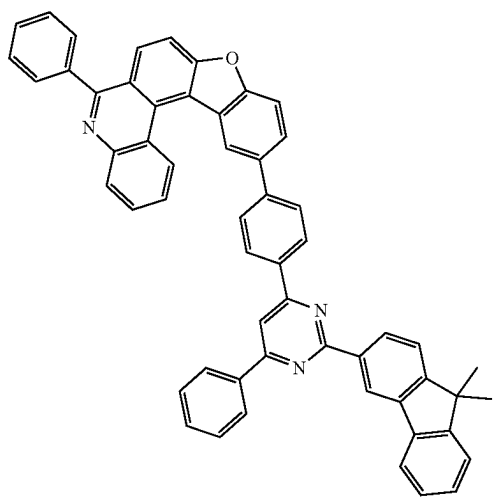
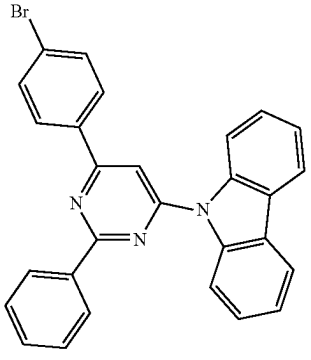
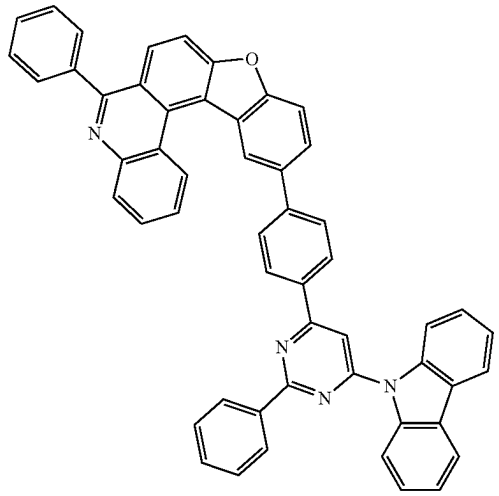
Compound	Intermediate A	Target Compound	Yield
132			62%
133			51%
135			60%
139			61%

TABLE 1-continued

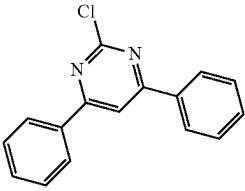
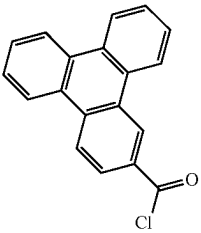
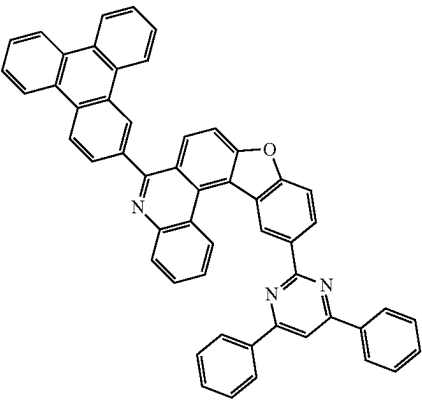
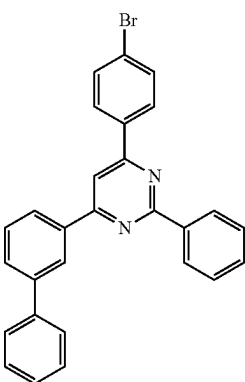
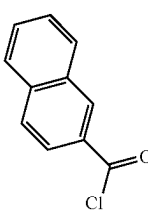
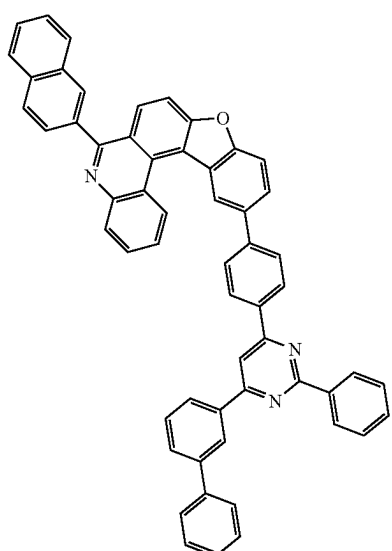
Compound	Intermediate A	Target Compound	Yield
140			62%
141			51%
142			60%

485

486

Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 1 of the following Table 2 was used instead of 1-pyrenecarbonyl chloride, and Intermediate B of the following Table 2 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

TABLE 2

	Intermediate B	Intermediate 1	Target Compound	
121				60%
128				61%

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 6-bromo-2-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate C of the following Table 3 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

487

488

TABLE 3

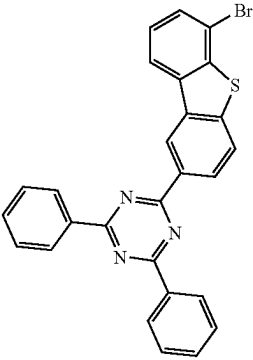
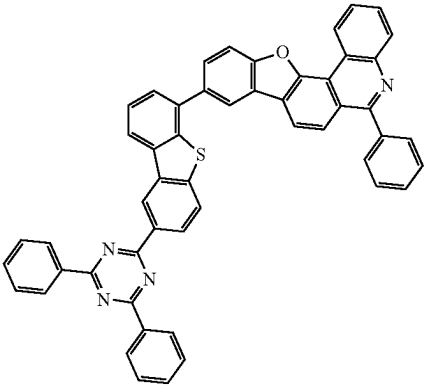
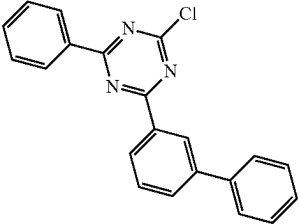
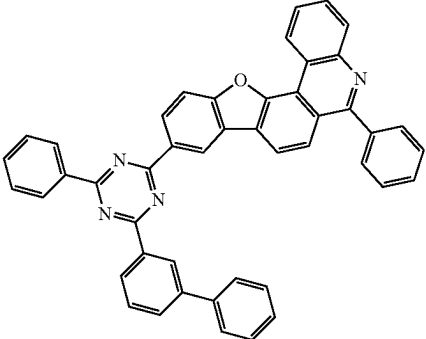
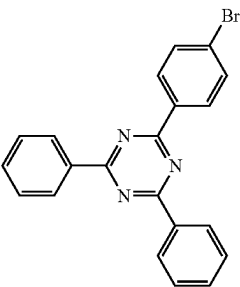
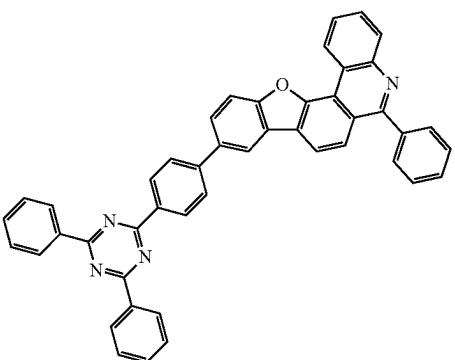
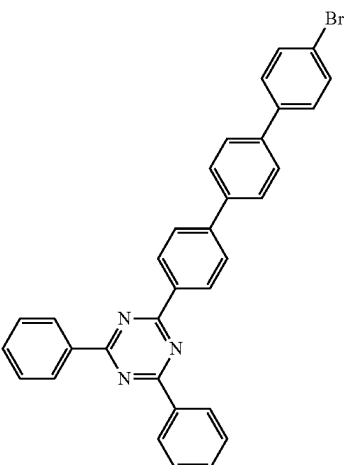
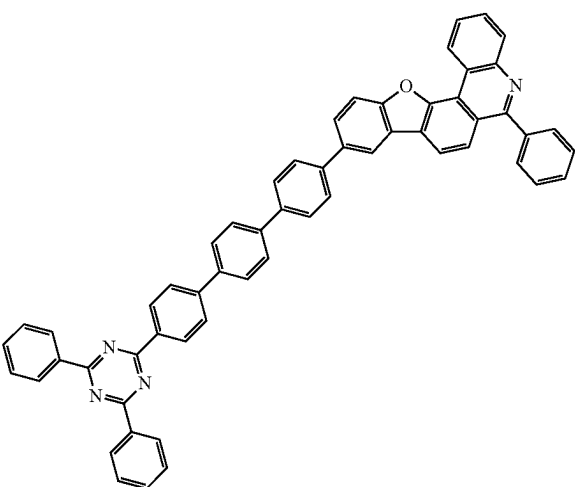
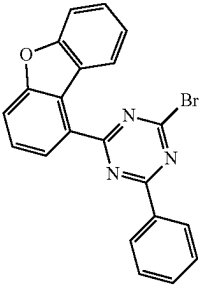
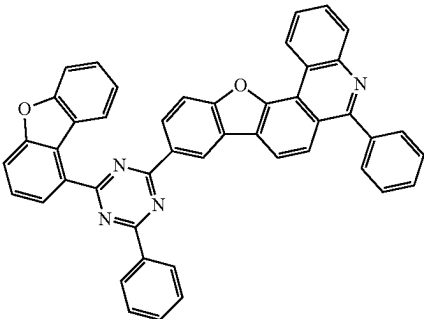
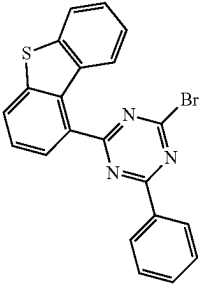
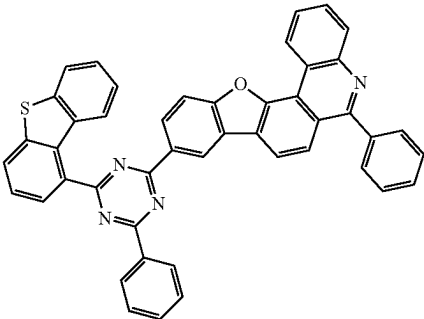
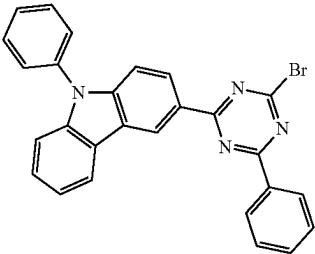
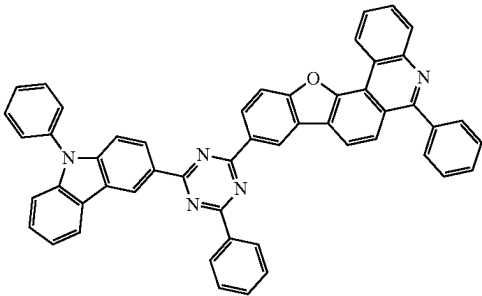
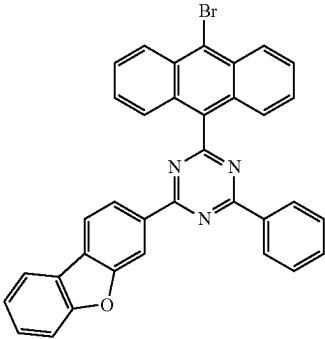
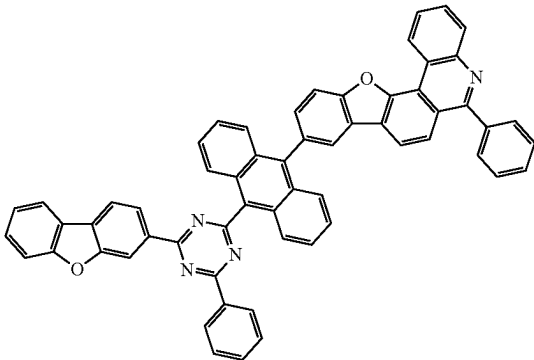
Compound	Intermediate C	Target Compound	Yield
58			62%
156			65%
157			65%
159			62%

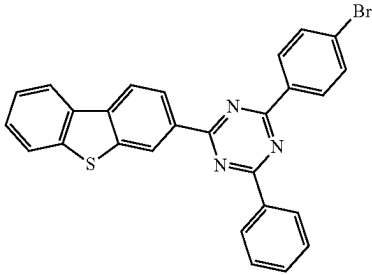
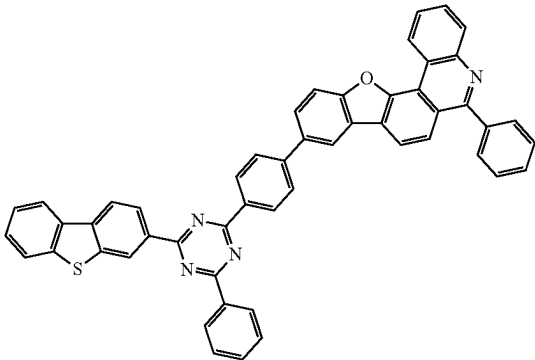
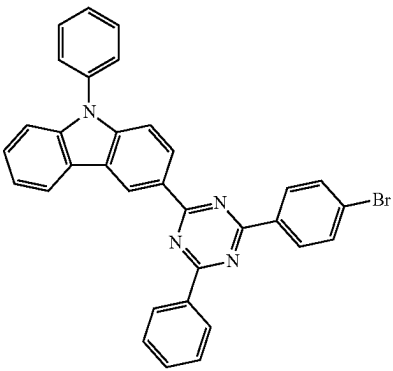
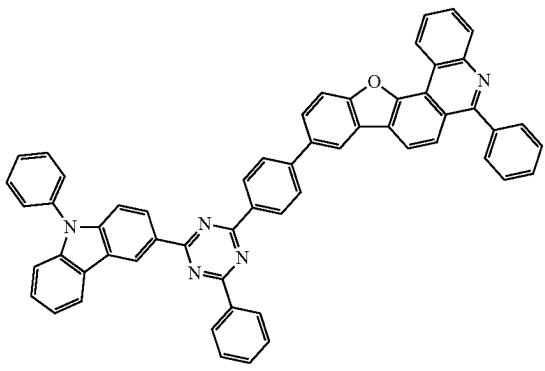
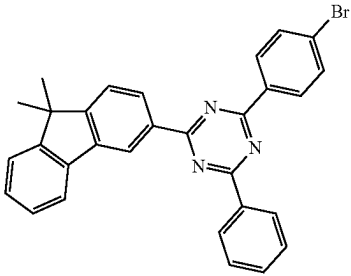
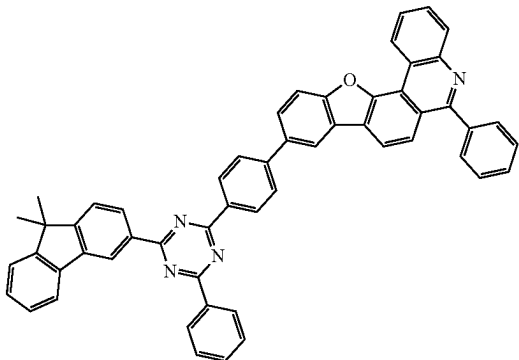
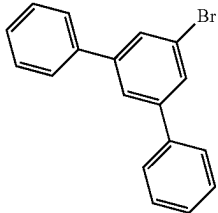
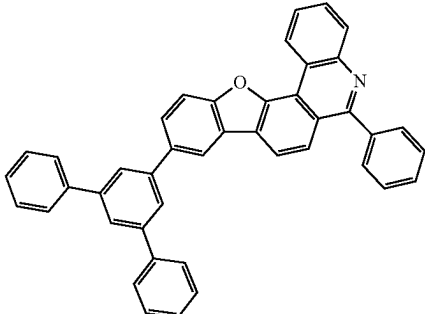
TABLE 3-continued

Compound	Intermediate C	Target Compound	Yield
161			51%
165			60%
166			61%
170			62%

491

492

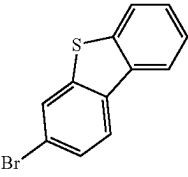
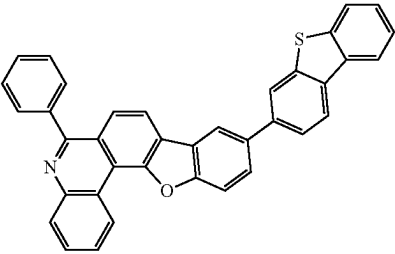
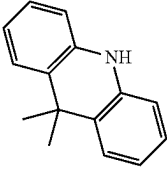
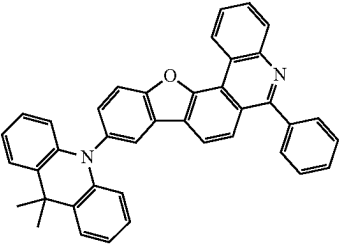
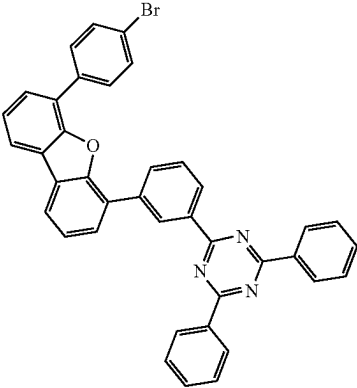
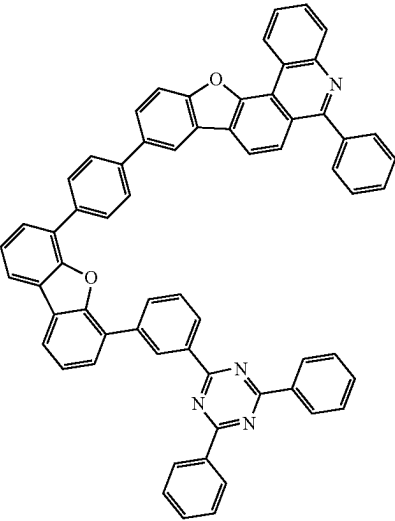
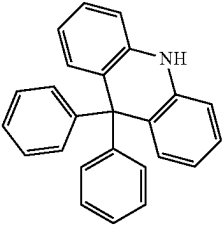
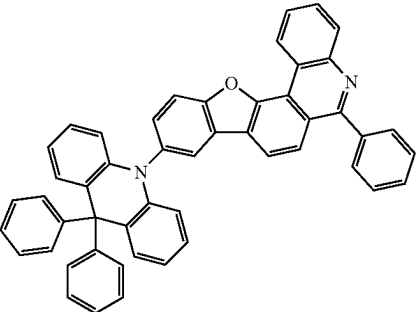
TABLE 3-continued

Compound	Intermediate C	Target Compound	Yield
171			65%
172			65%
173			62%
178			60%

493

494

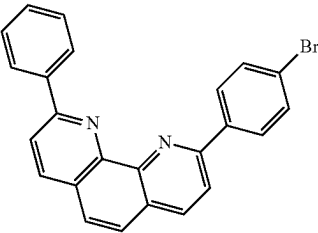
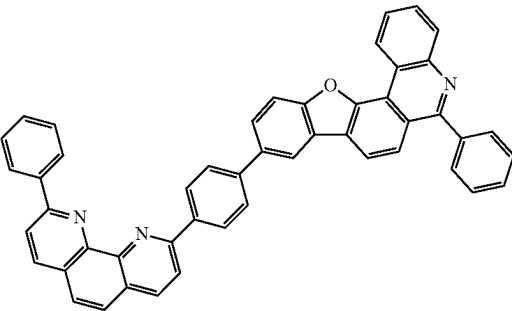
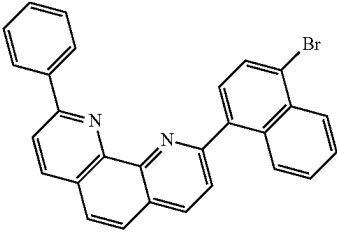
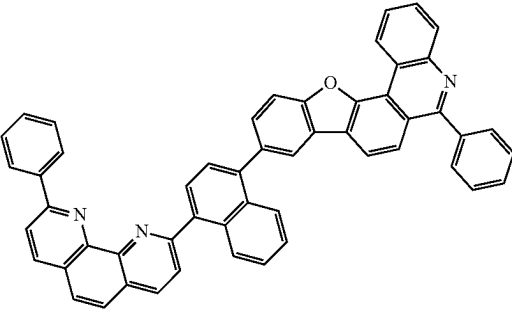
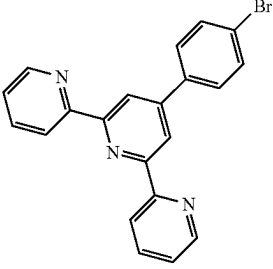
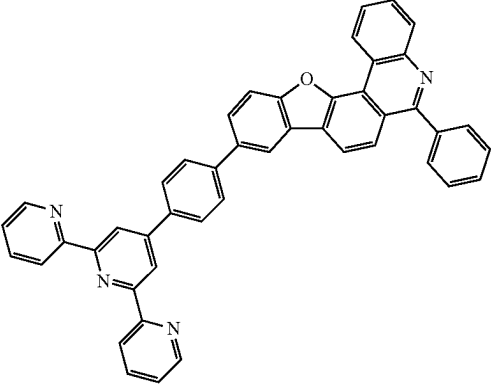
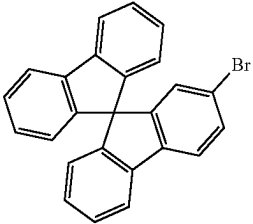
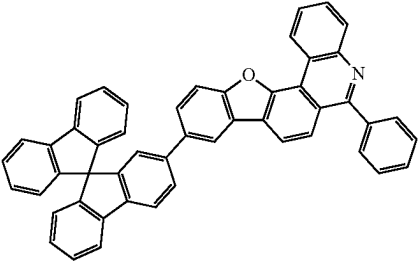
TABLE 3-continued

Compound	Intermediate C	Target Compound	Yield
179			61%
180			60%
185			62%
187			65%

495

496

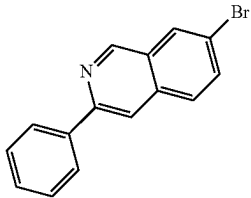
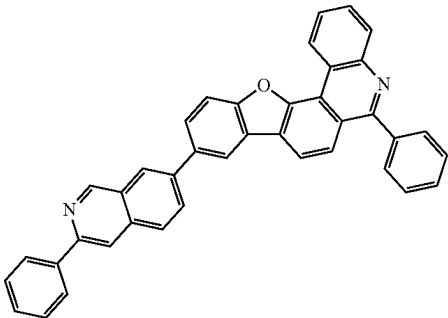
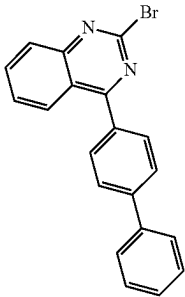
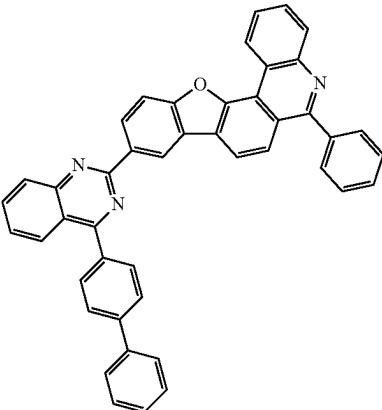
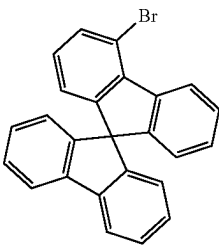
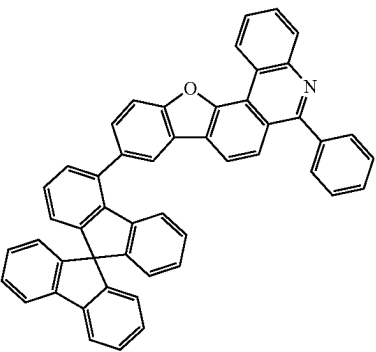
TABLE 3-continued

Compound	Intermediate C	Target Compound	Yield
190			65%
191			62%
192			51%
194			60%

497

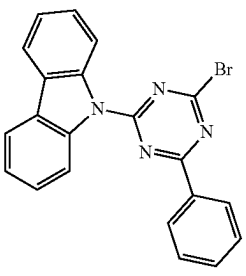
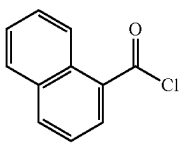
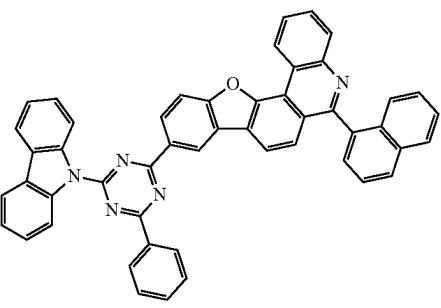
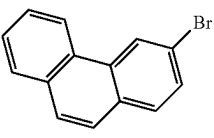
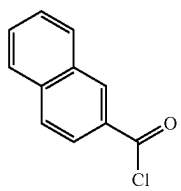
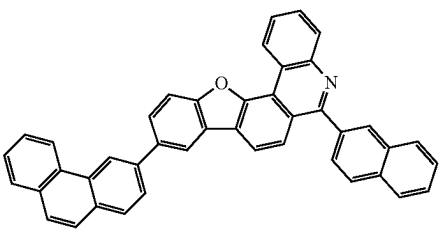
498

TABLE 3-continued

Compound	Intermediate C	Target Compound	Yield
195			61%
196			60%
200			55%

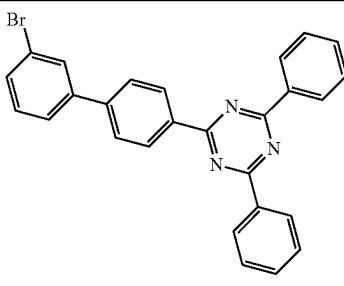
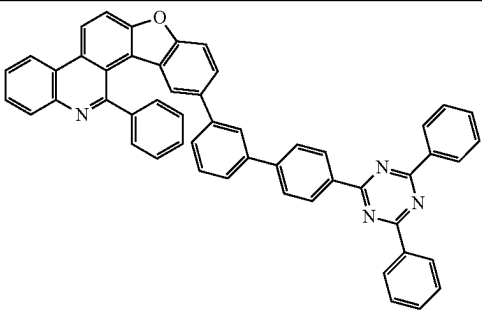
Target compounds were synthesized in the same manner
as in Preparation Example 2 except that Intermediate 2 of the
following Table 4 was used instead of 1-pyrenecarbonyl
chloride, 6-bromo-2-chlorodibenzo[b,d]furan was used
instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Inter-
mediate D of the following Table 4 was used instead of 4-3
bromophenyl)dibenzo[b,d]thiophene.

TABLE 4

	Intermediate D	Intermediate 2	Target Compound	
168				60%
176				61%

A target compound was synthesized in the same manner as in Preparation Example 1 except that 2-bromo-8-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate E of the following Table 5 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

TABLE 5

Compound	Intermediate E	Target Compound	Yield
149			52%

A target compound was synthesized in the same manner as in Preparation Example 2 except that Intermediate 3 of the following Table 6 was used instead of 1-pyrenecarbonyl chloride, 2-bromo-8-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate F of the following Table 6 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

501

502

TABLE 6

Compound	Intermediate F	Intermediate 3	Target Compound	Yield
145				53%

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 1-bromo-6-chloro-rodibenzo[b,d]furan was used instead of 1-bromo-8-chloro-rodibenzo[b,d]furan, and Intermediate G of the following Table 7 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

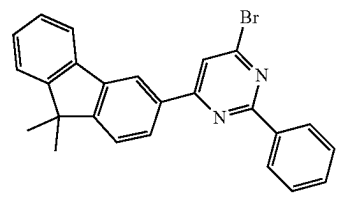
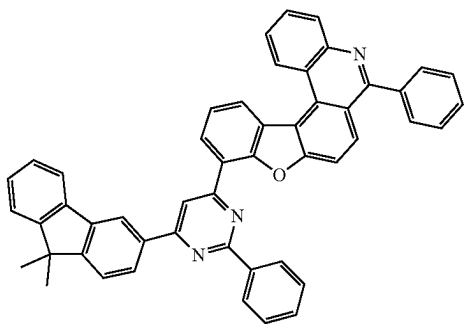
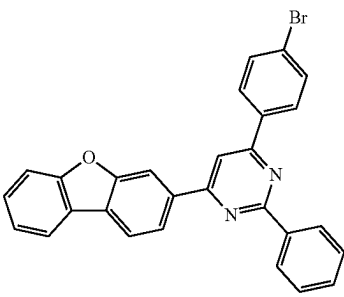
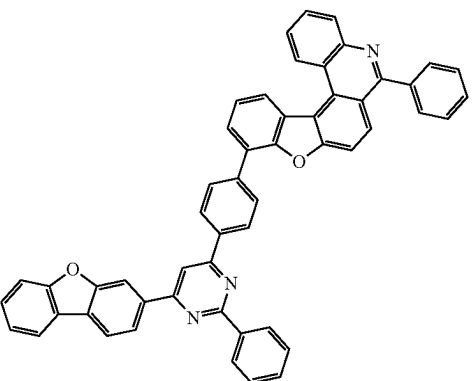
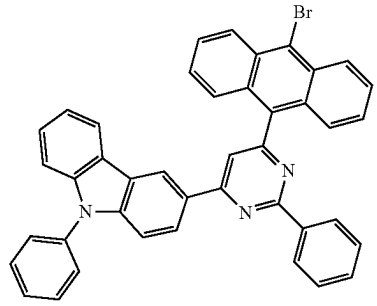
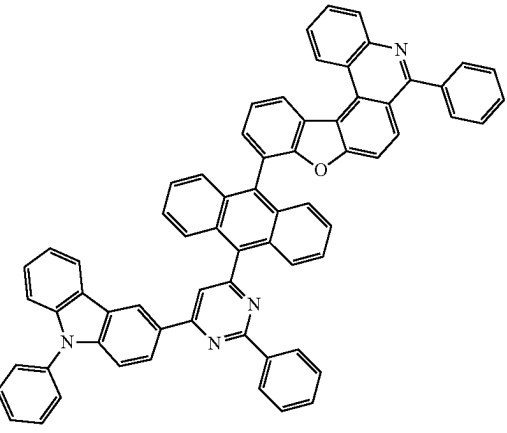
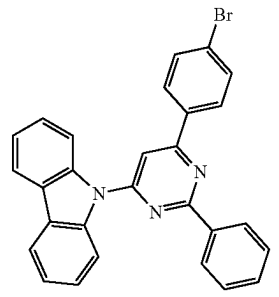
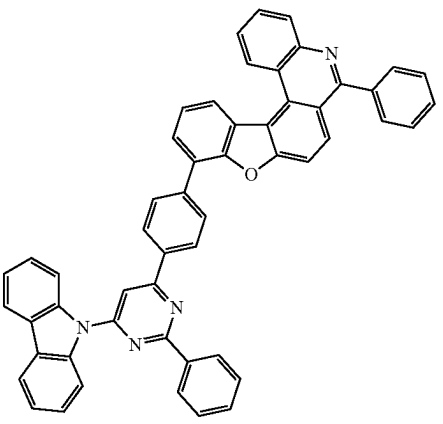
TABLE 7

Compound	Intermediate G	Target Compound	Yield
384			53%
386			62%
387			65%

503

504

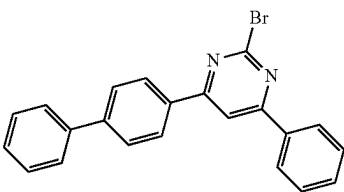
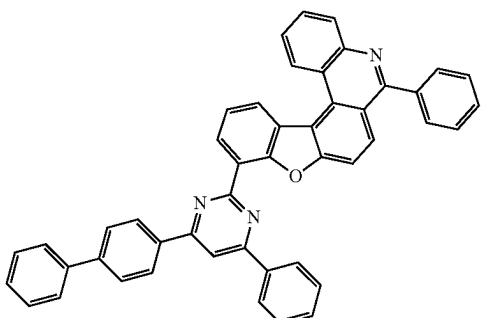
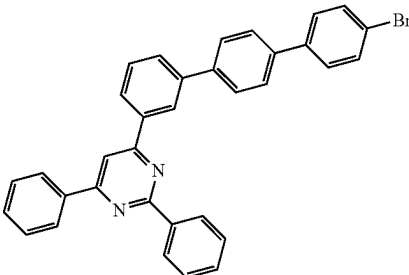
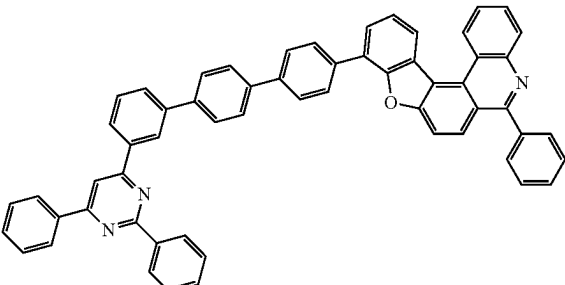
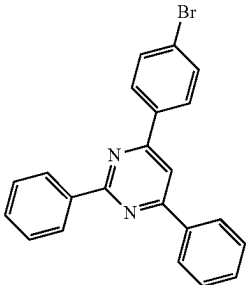
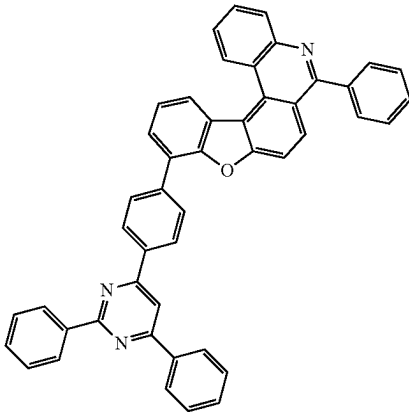
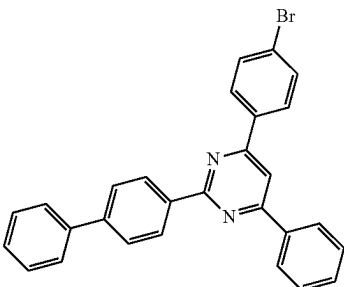
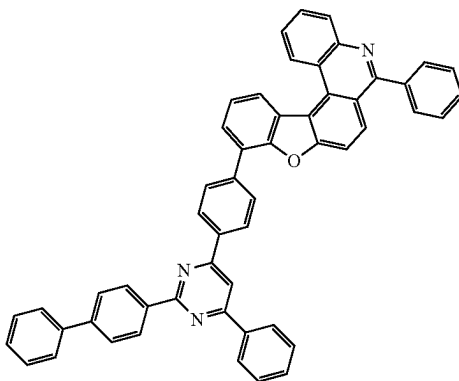
TABLE 7-continued

Compound	Intermediate G	Target Compound	Yield
388			65%
392			62%
395			51%
396			60%

505

506

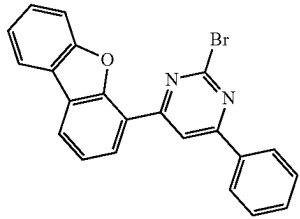
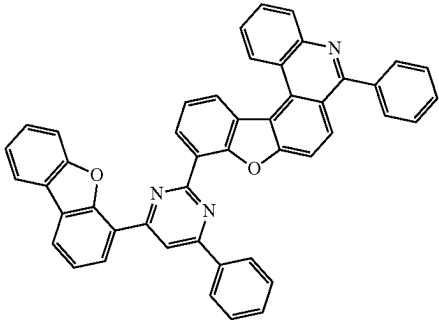
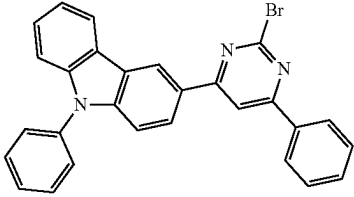
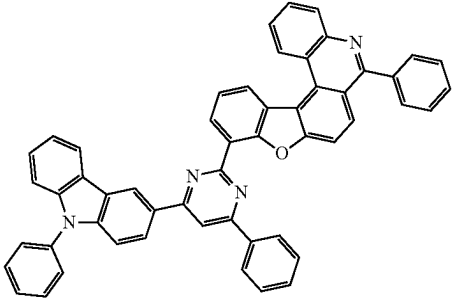
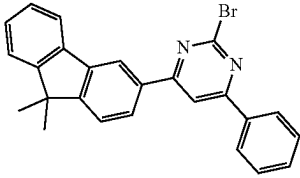
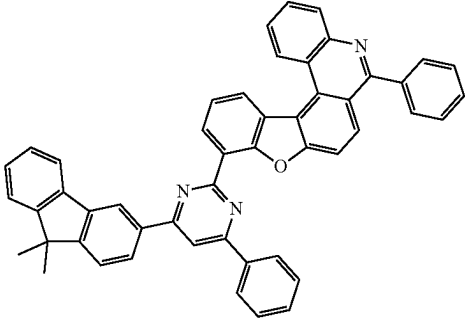
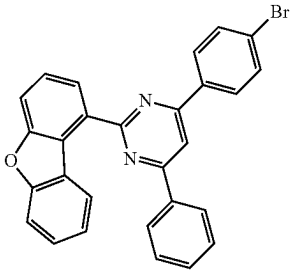
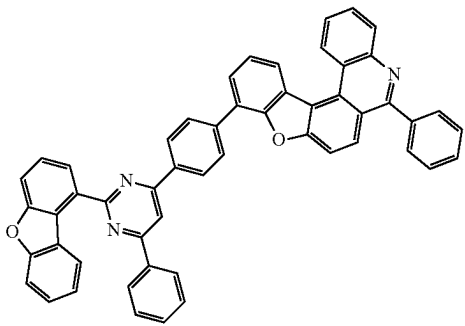
TABLE 7-continued

Compound	Intermediate G	Target Compound	Yield
397			61%
399			60%
400			62%
401			65%

507

508

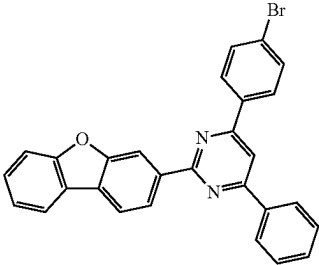
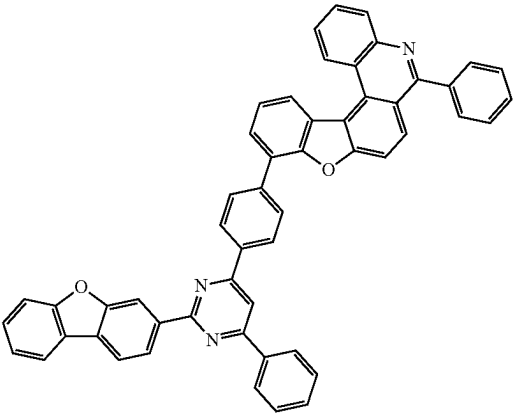
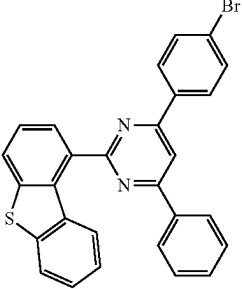
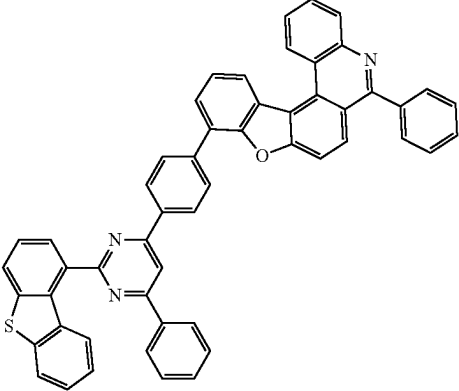
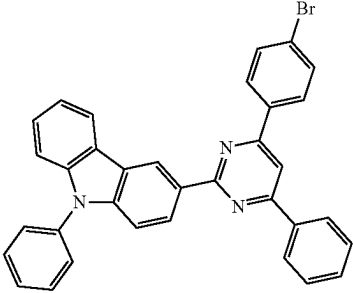
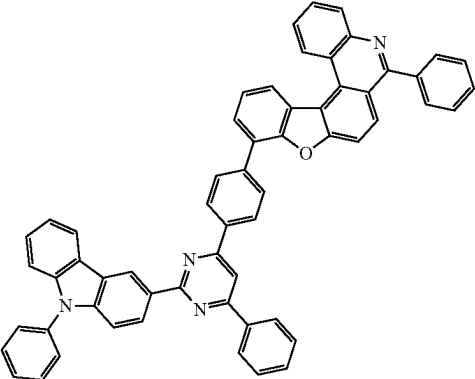
TABLE 7-continued

Compound	Intermediate G	Target Compound	Yield
404			65%
408			51%
409			60%
411			61%

509

510

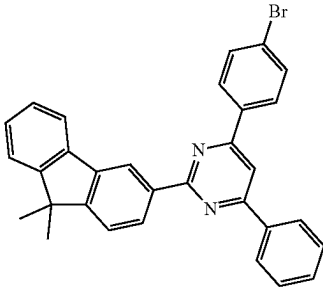
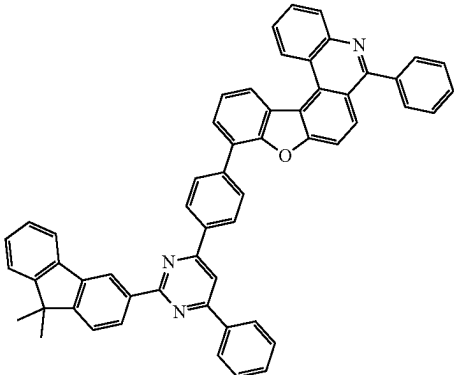
TABLE 7-continued

Compound	Intermediate G	Target Compound	Yield
413			60%
415			62%
416			65%

511

512

TABLE 7-continued

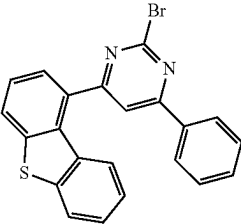
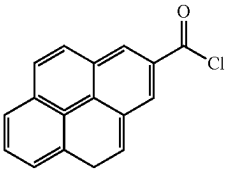
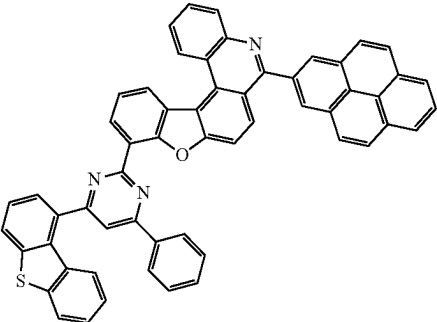
Compound	Intermediate G	Target Compound	Yield
417			65%

20

A target compound was synthesized in the same manner as in Preparation Example 2 except that Intermediate 4 of the following Table 8 was used instead of 1-pyrenecarbonyl chloride, 1-bromo-6-chlorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Inter-
 mediate H of the following Table 8 was used instead of 4-3
 bromophenyl)dibenzo[b,d]thiophene.

25

TABLE 8

Compound	Intermediate H	Intermediate 4	Target Compound	Yield
407				53%

50

55

60

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 4,6-dibromodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate I of the following Table 9 was
 used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triaz-
 ine.

65

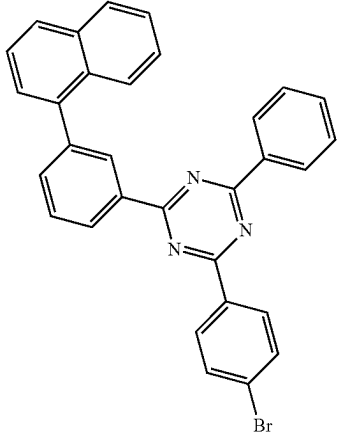
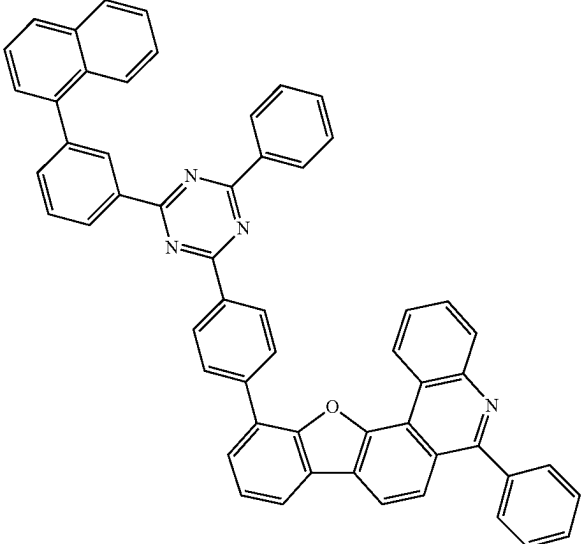
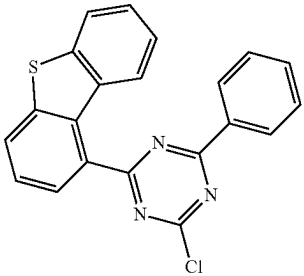
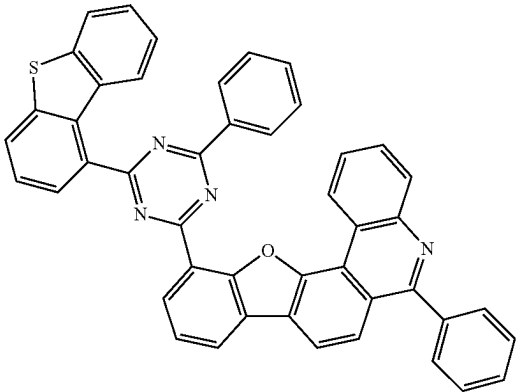
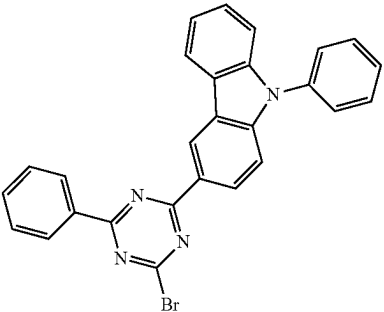
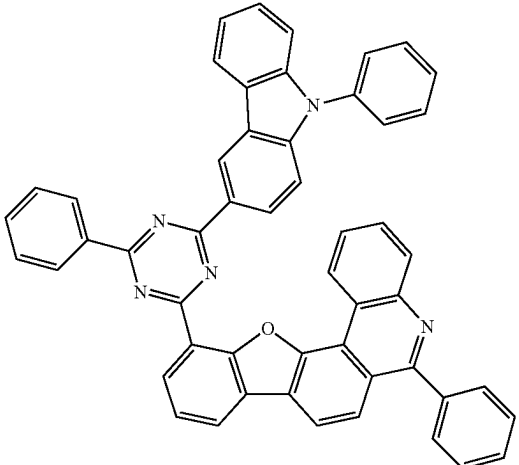
TABLE 9

Compound	Intermediate I	Target Compound	Yield
421			53%
423			62%
424			65%

515

516

TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
426			65%
431			62%
432			51%

517

518

TABLE 9-continued

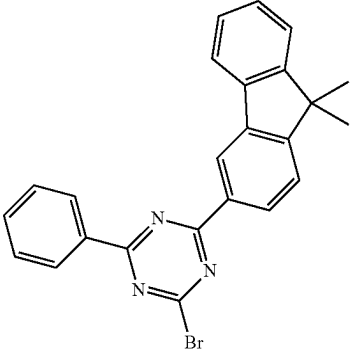
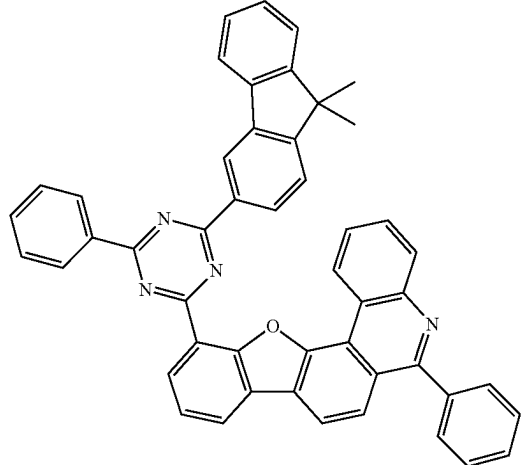
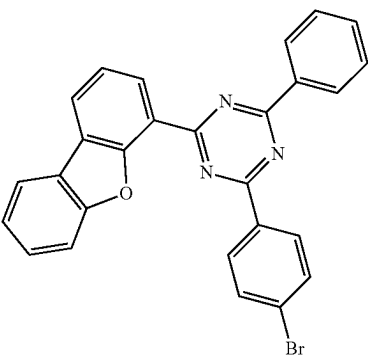
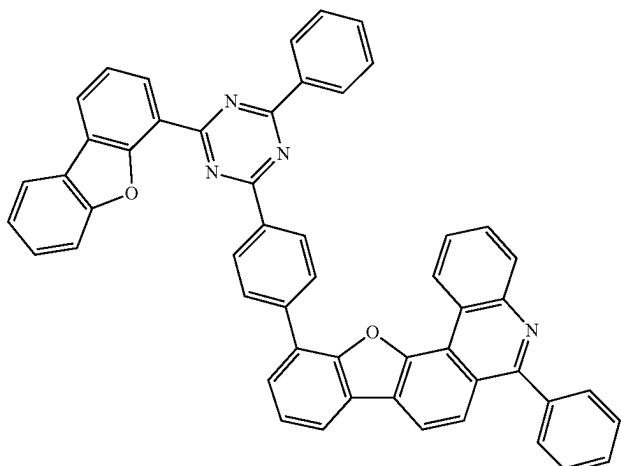
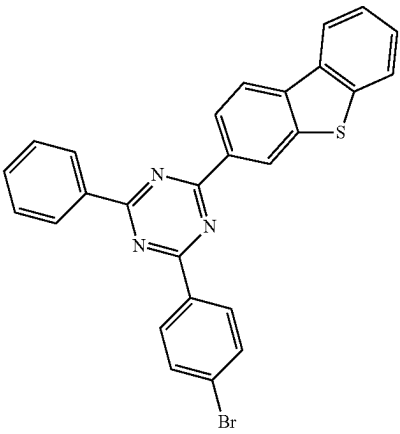
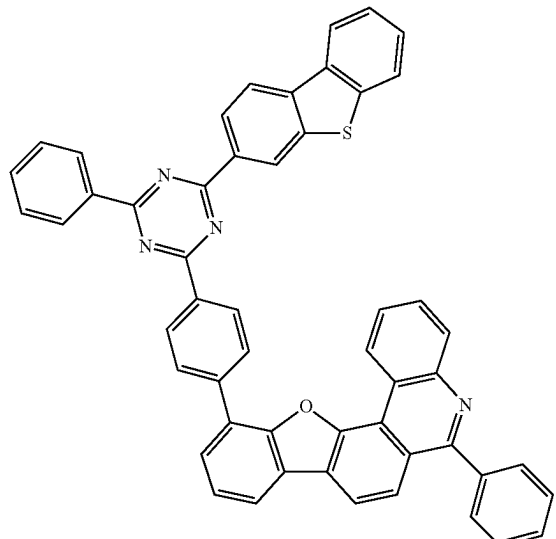
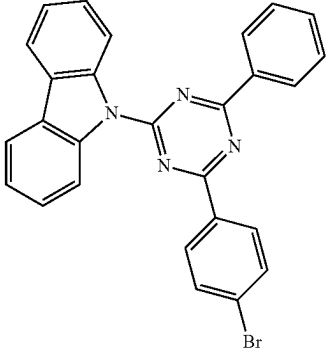
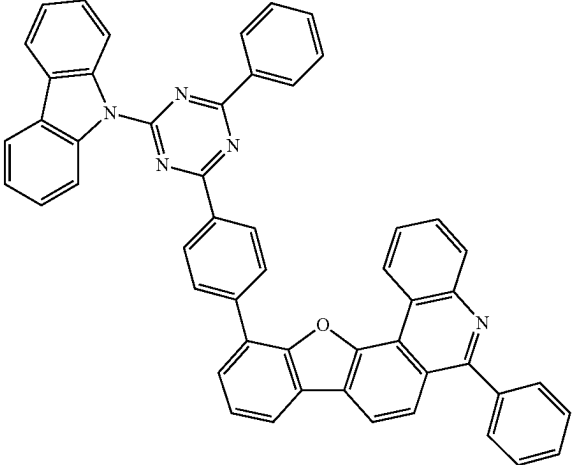
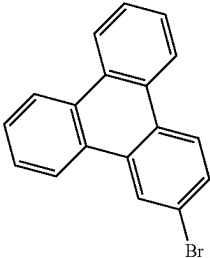
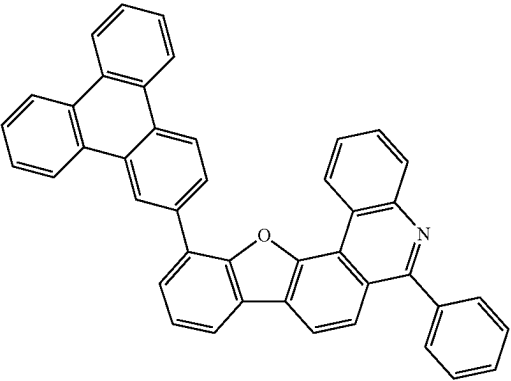
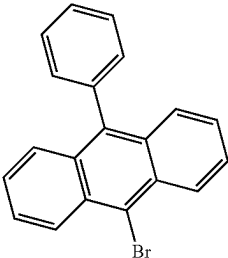
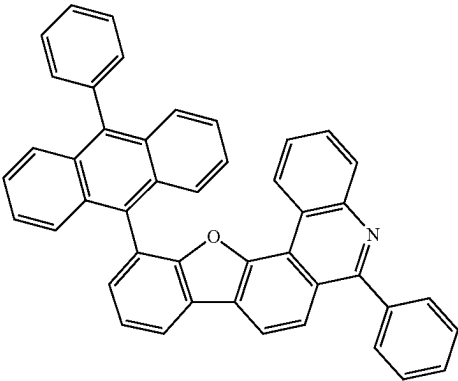
Compound	Intermediate I	Target Compound	Yield
433			60%
436			61%
438			60%

TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
440			62%
441			65%
443			65%

521

522

TABLE 9-continued

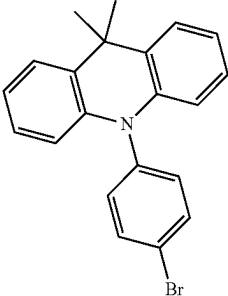
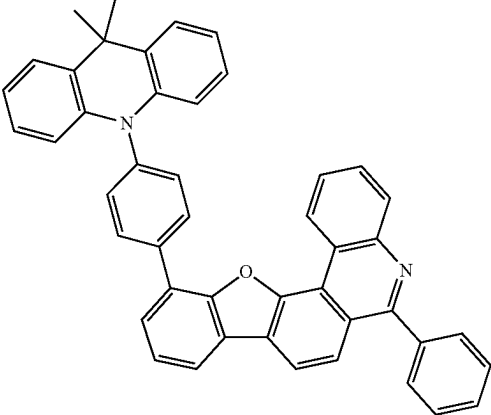
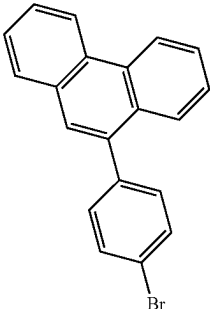
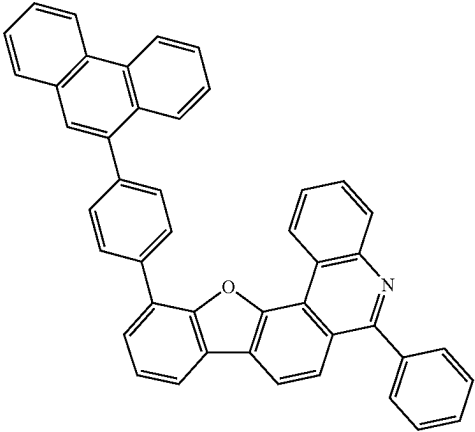
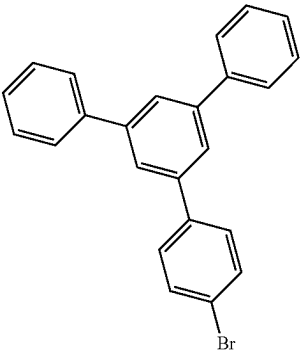
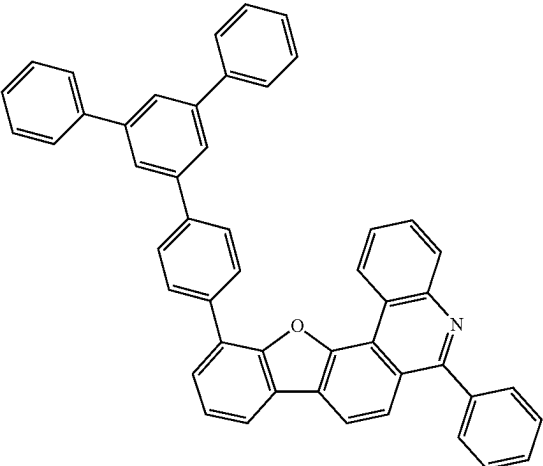
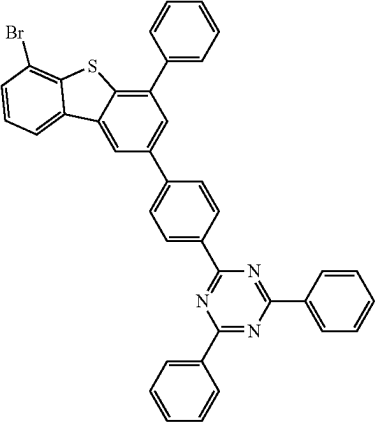
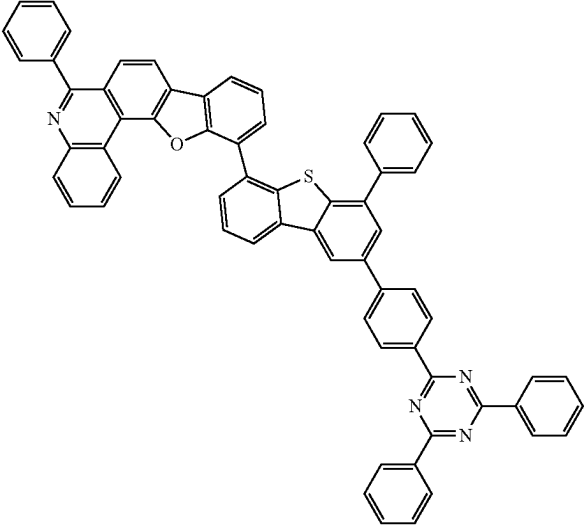
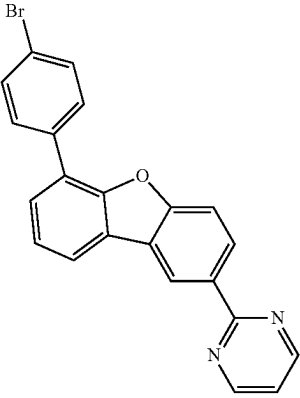
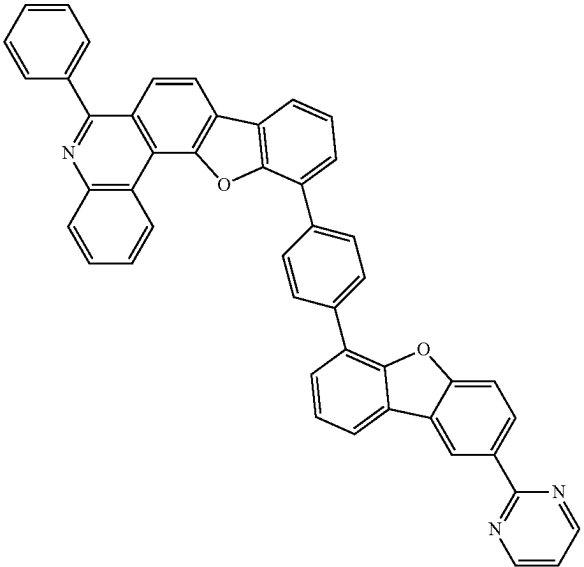
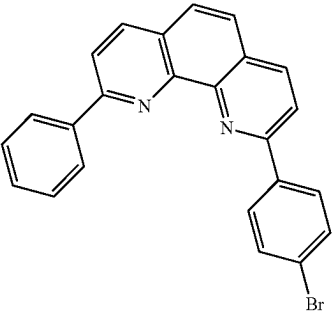
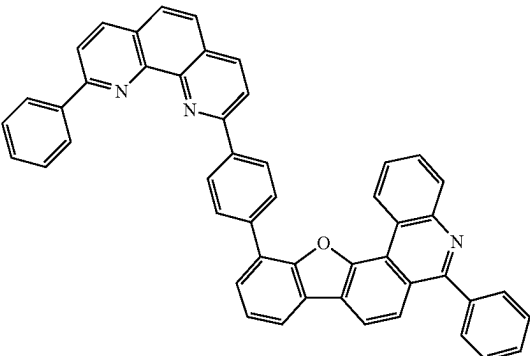
Compound	Intermediate I	Target Compound	Yield
447			62%
448			51%
449			60%

TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
450			61%
452			60%
456			62%

525

526

TABLE 9-continued

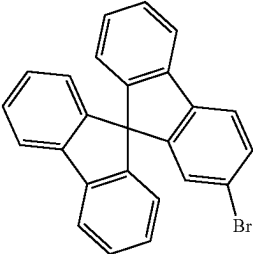
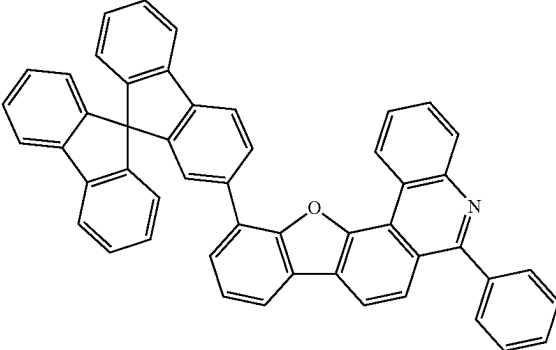
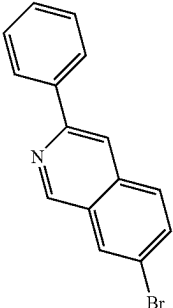
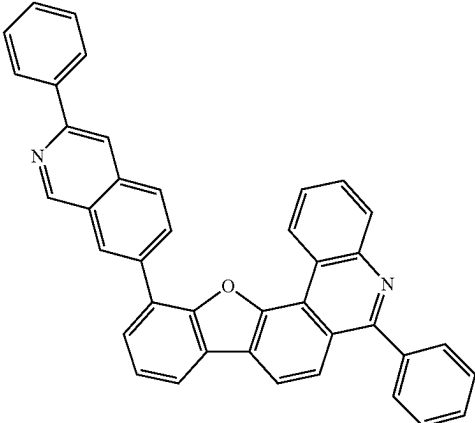
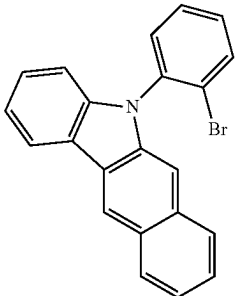
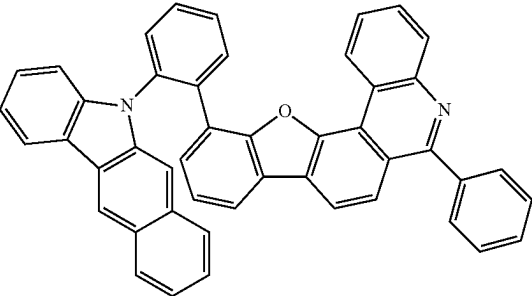
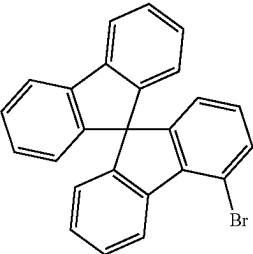
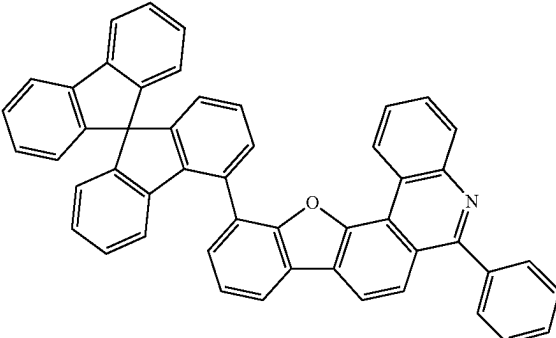
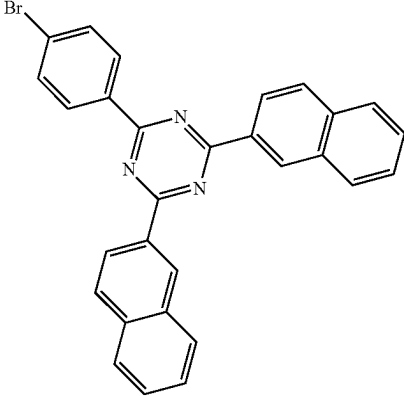
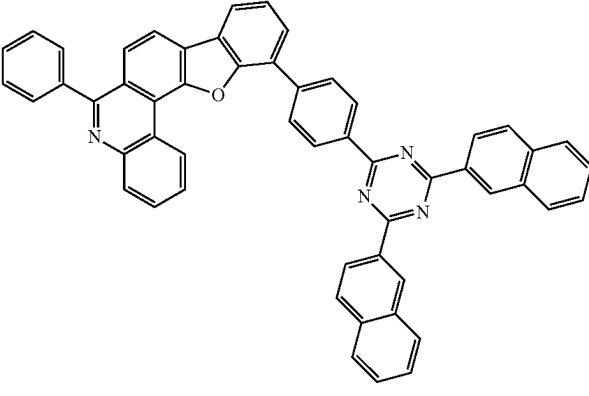
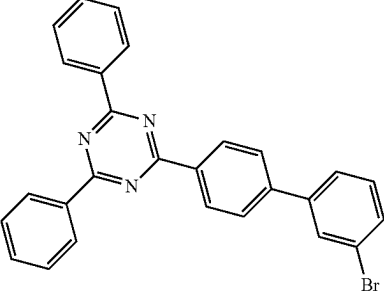
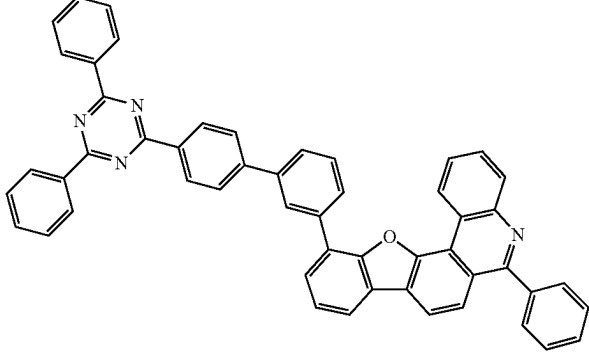
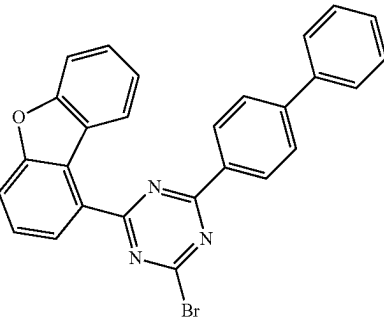
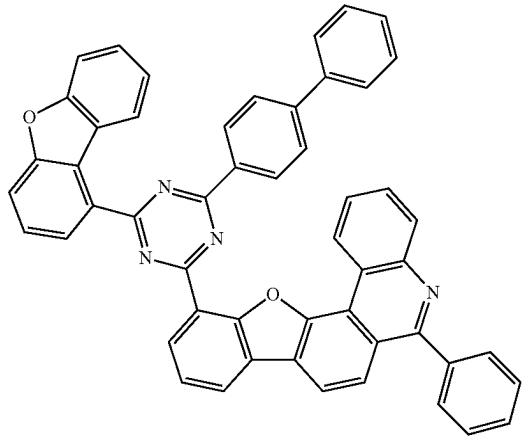
Compound	Intermediate I	Target Compound	Yield
460			65%
462			65%
465			62%
467			51%

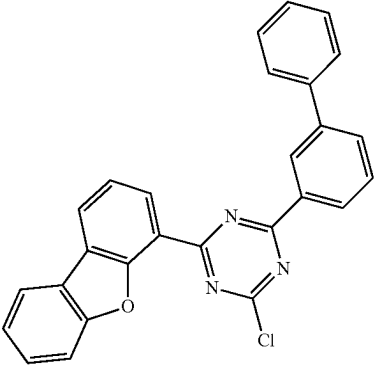
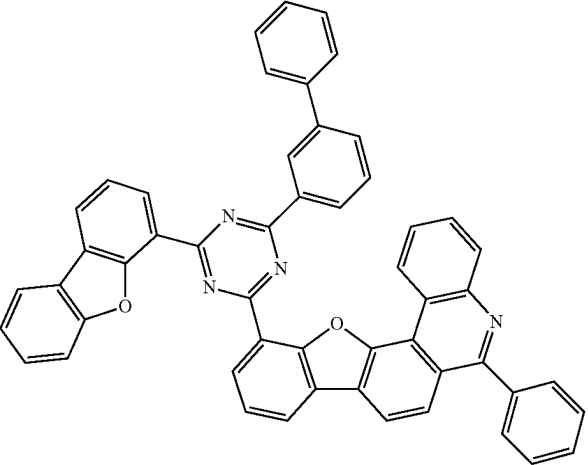
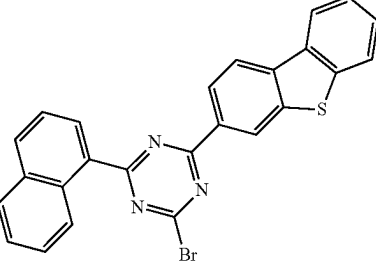
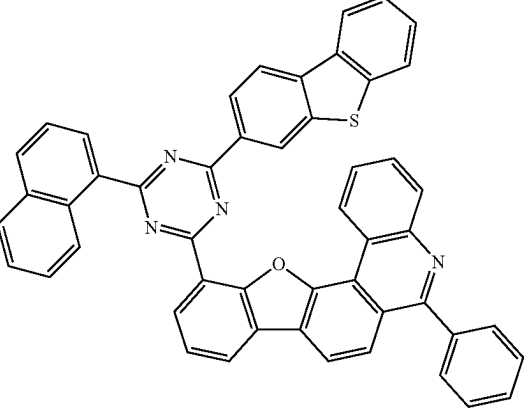
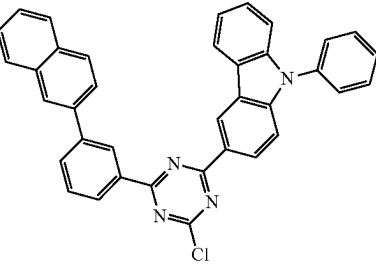
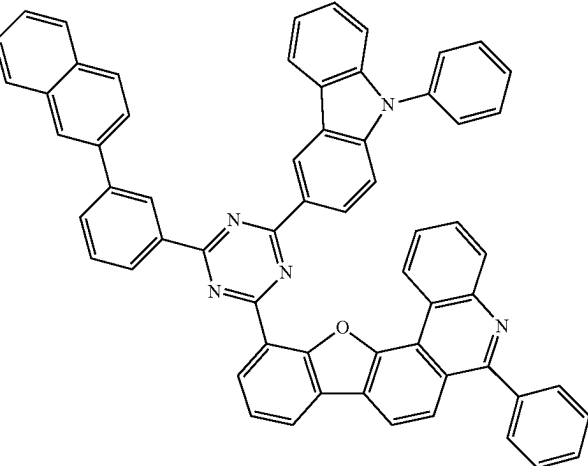
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
470			61%
475			60%
476			62%

529

530

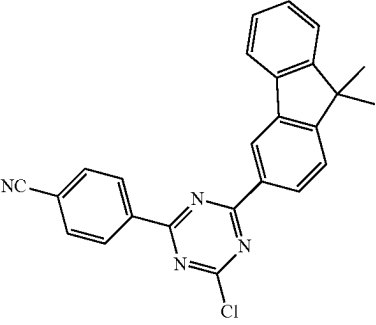
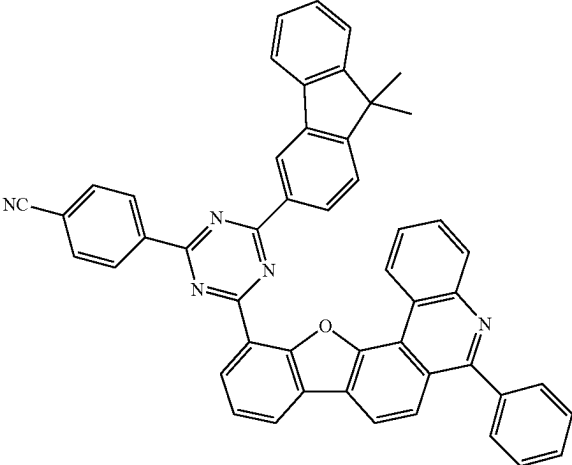
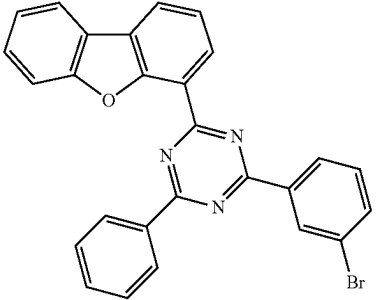
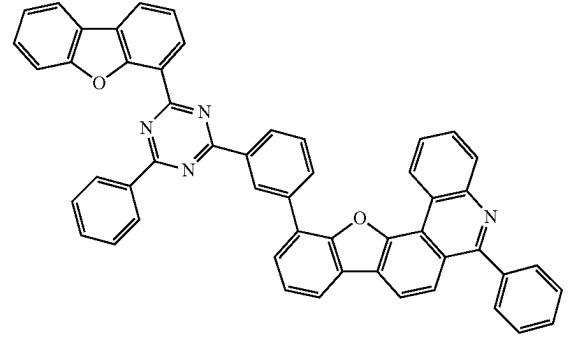
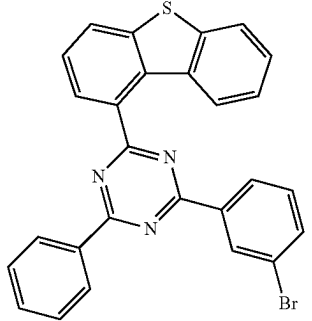
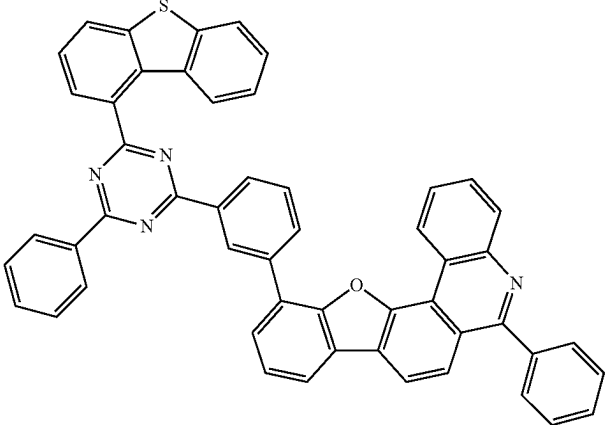
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
477			65%
479			65%
481			62%

531

532

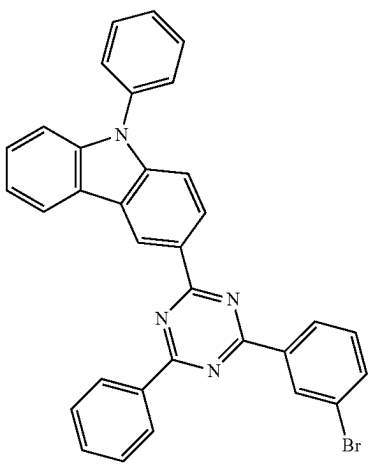
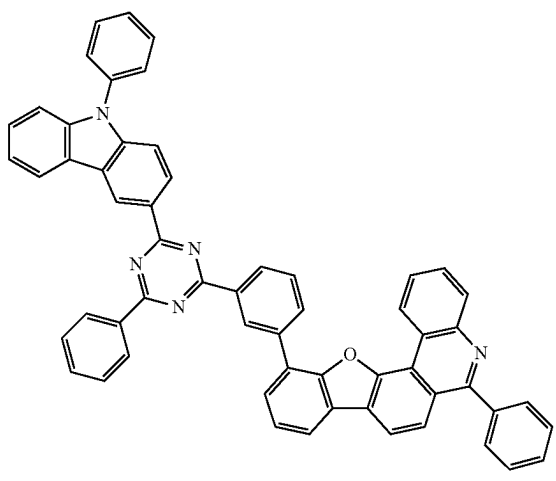
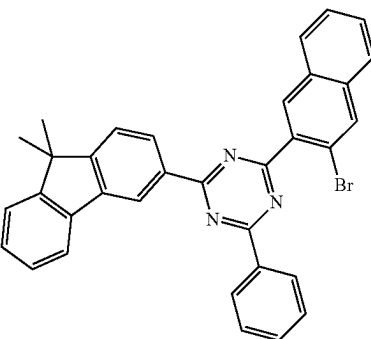
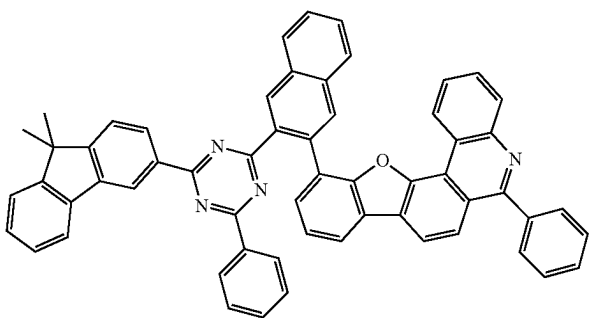
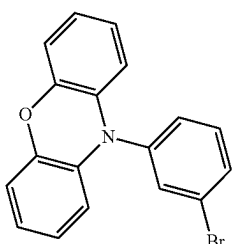
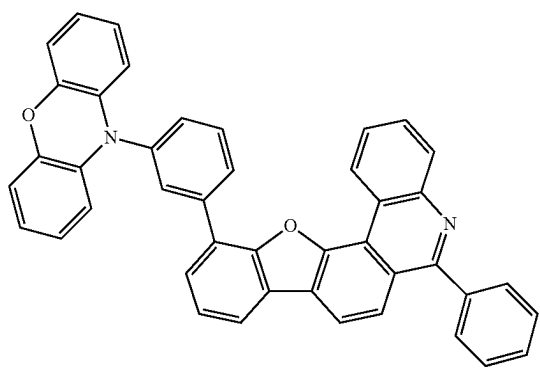
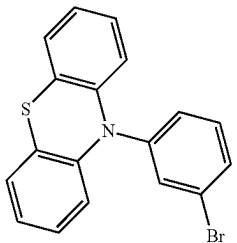
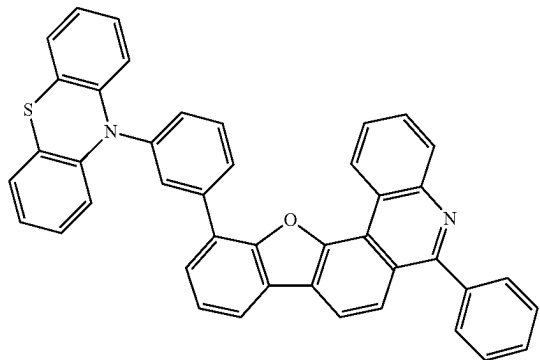
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
482			51%
485			60%
488			61%

533

534

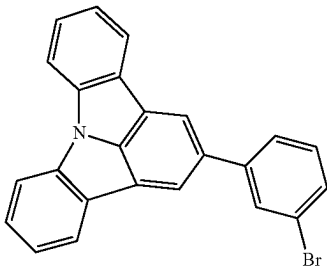
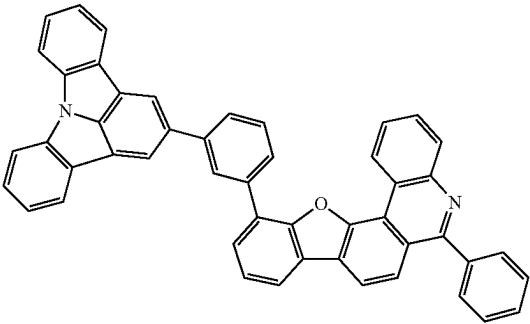
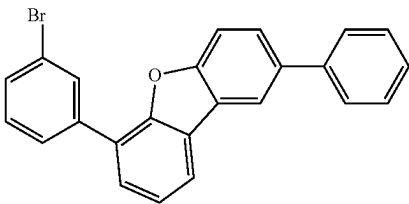
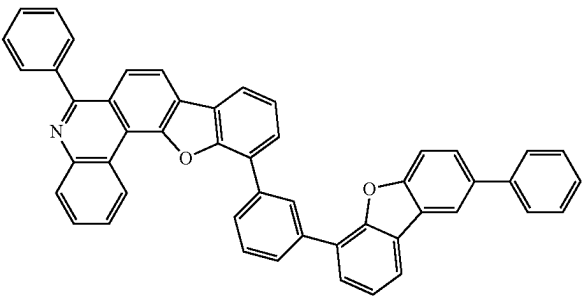
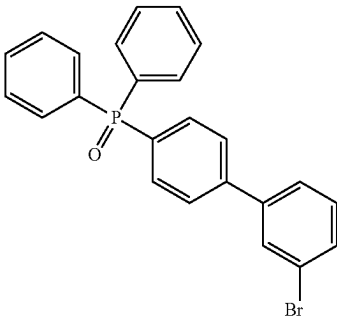
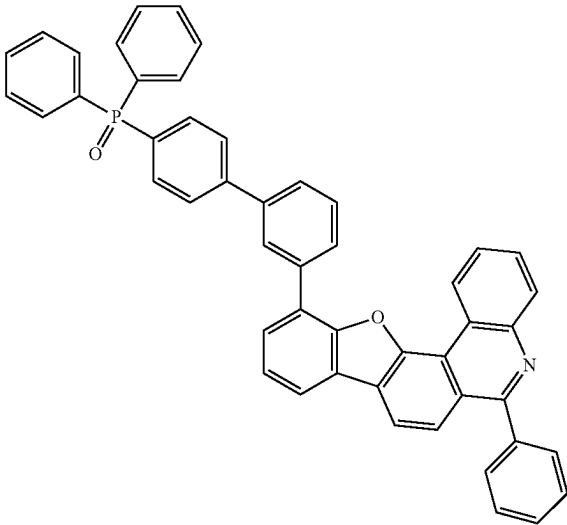
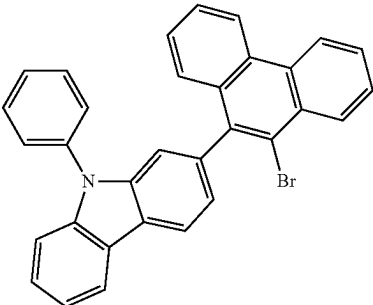
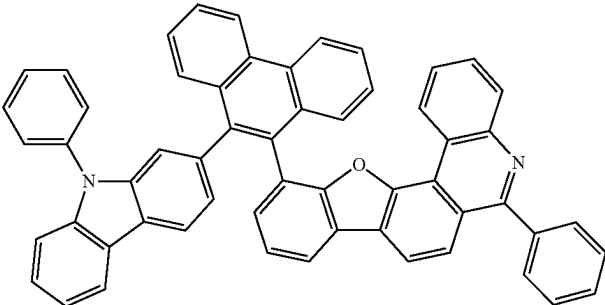
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
489			60%
490			62%
493			65%
494			65%

535

536

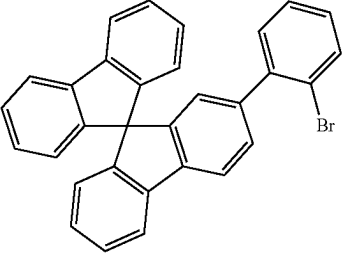
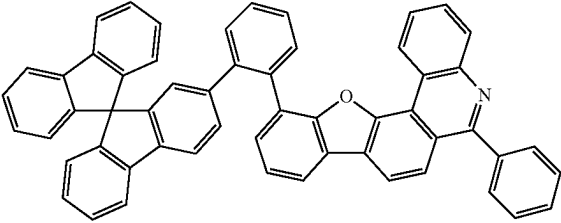
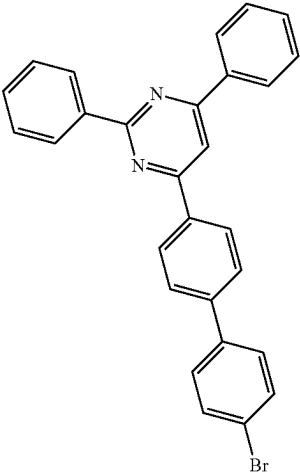
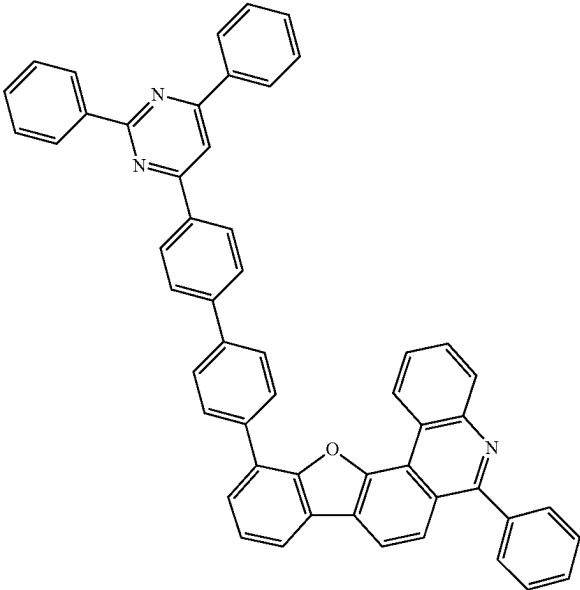
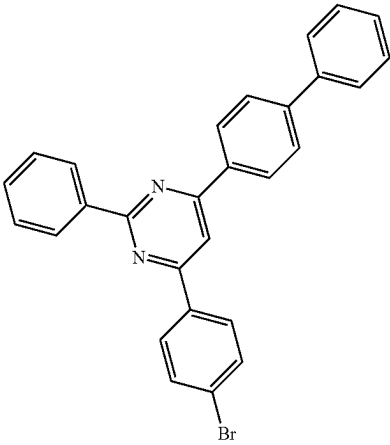
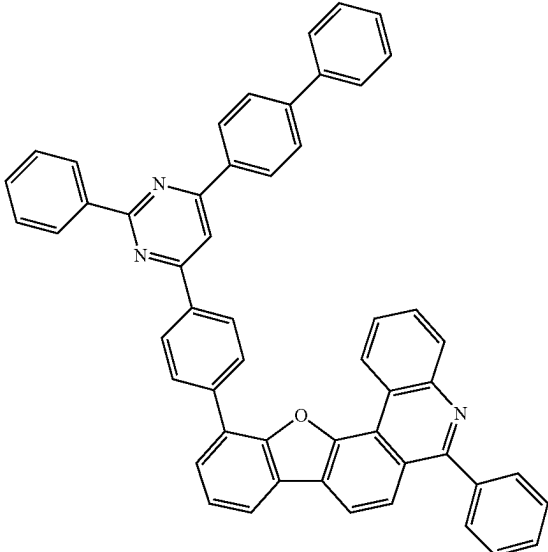
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
495			62%
500			51%
504			60%
505			61%

537

538

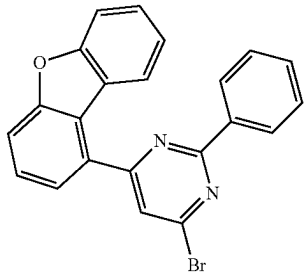
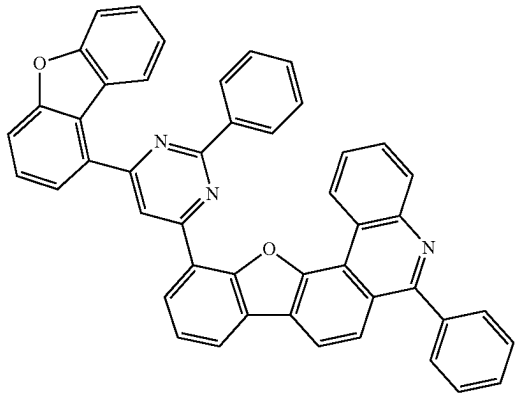
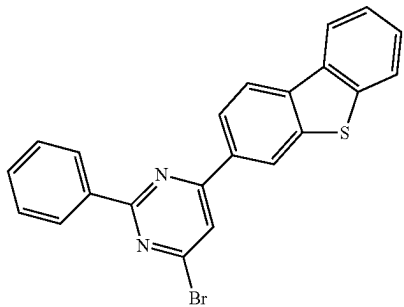
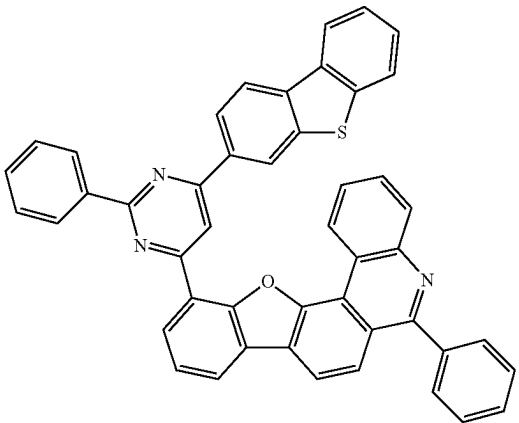
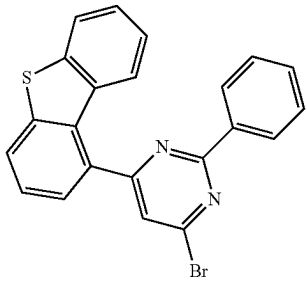
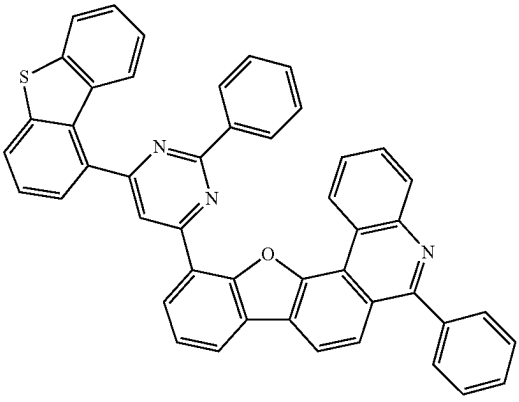
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
506			60%
507			62%
512			65%

539

540

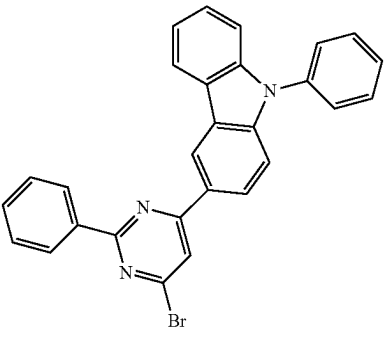
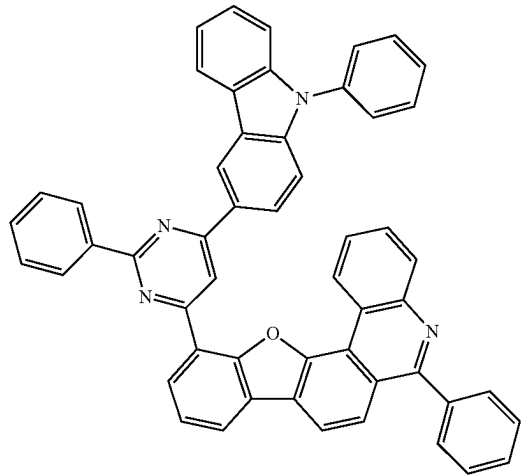
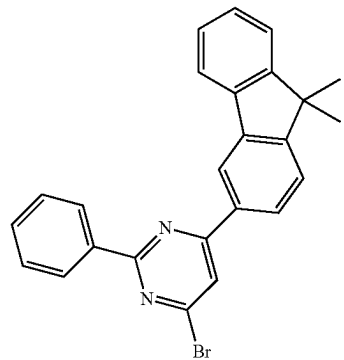
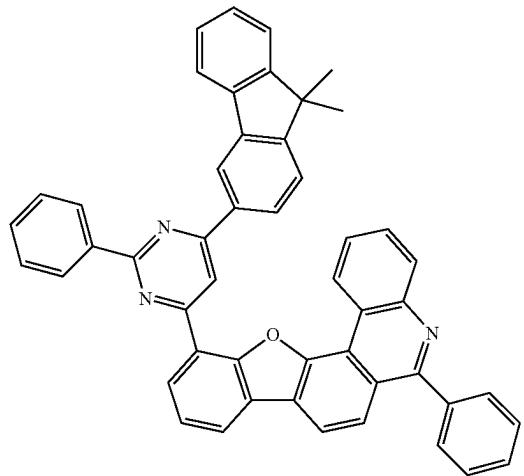
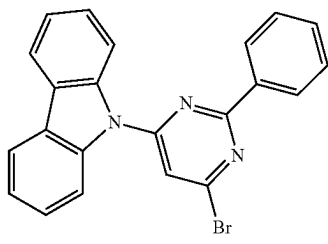
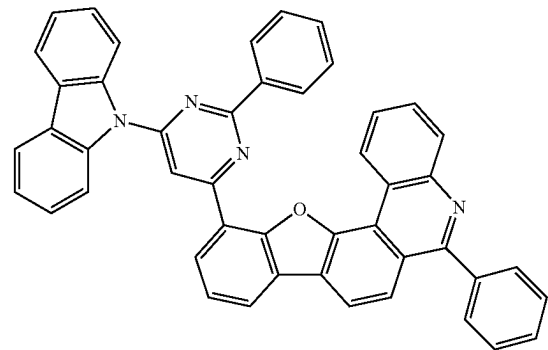
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
513			65%
516			62%
517			51%

541

542

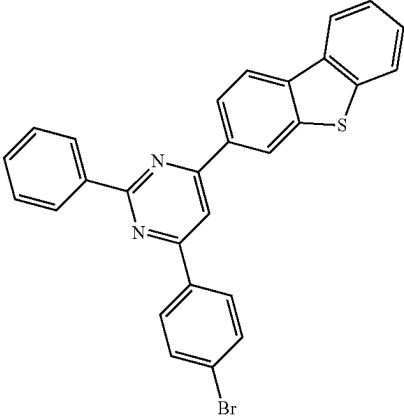
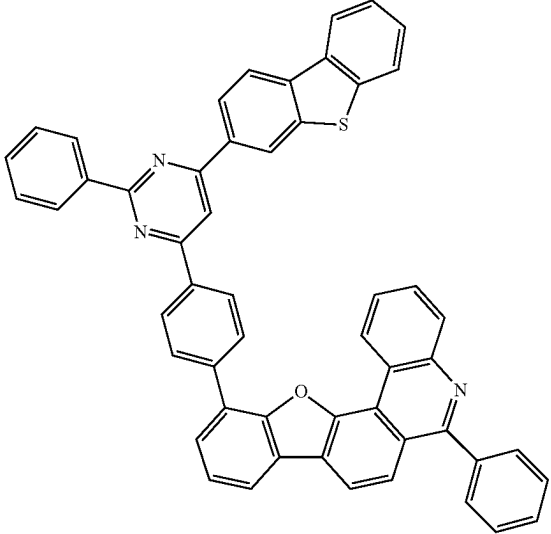
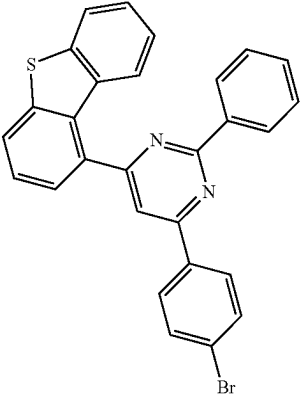
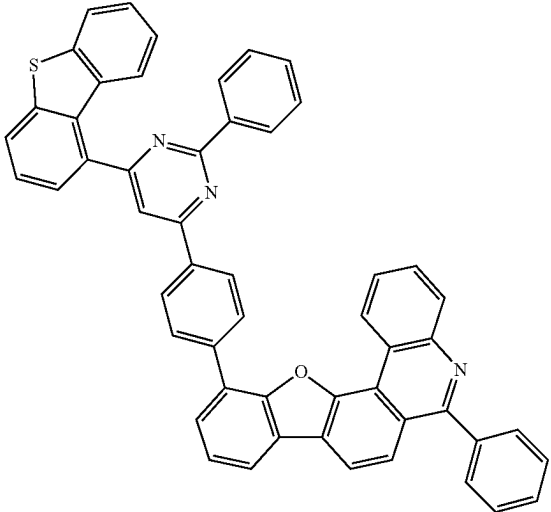
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
518			60%
519			61%
520			60%

543

544

TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
524			65%
525			65%

545

546

TABLE 9-continued

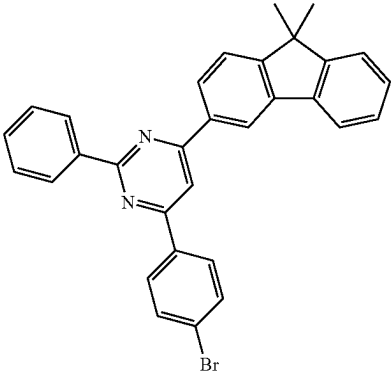
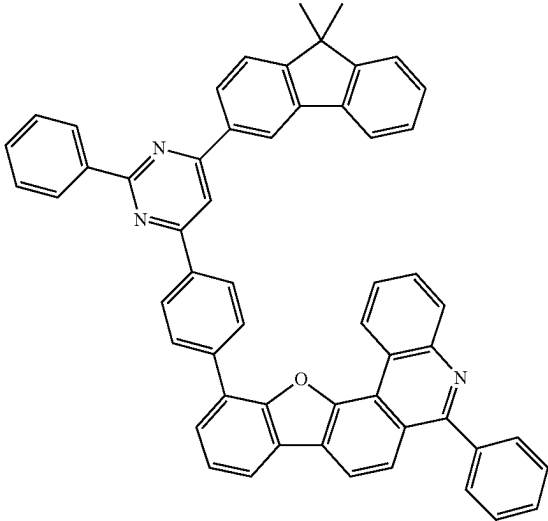
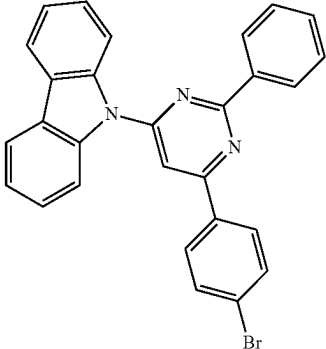
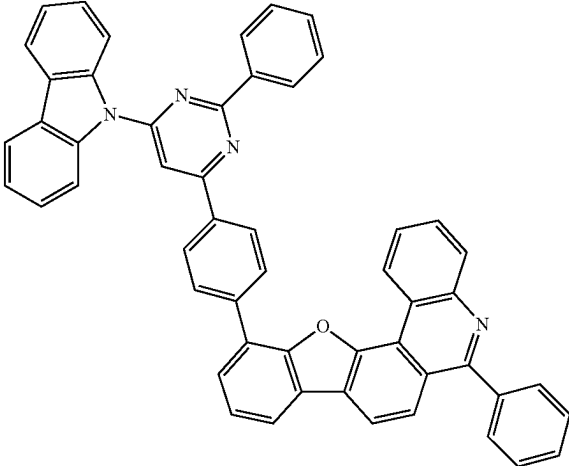
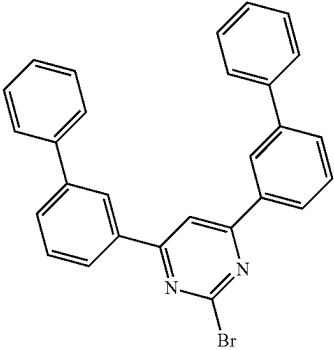
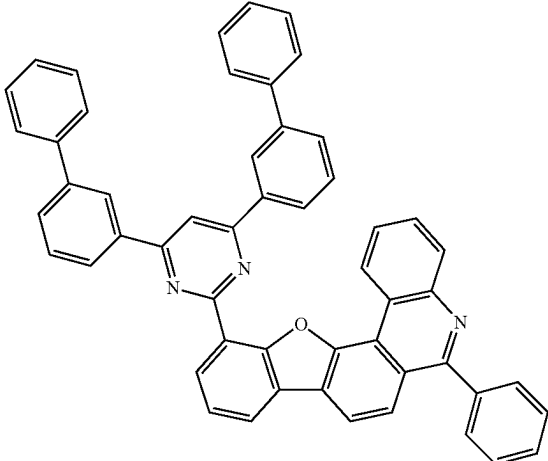
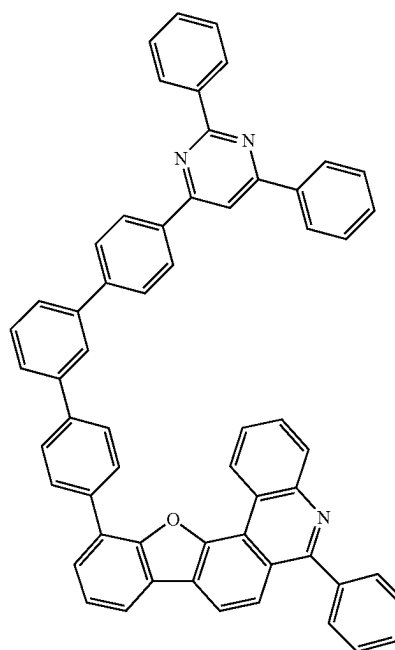
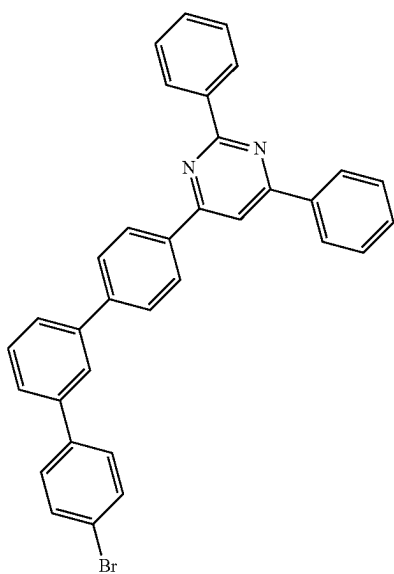
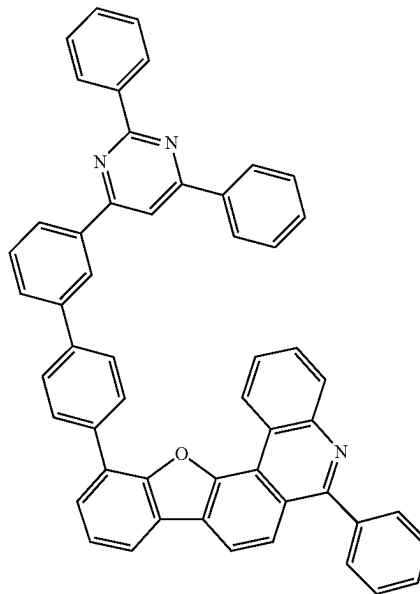
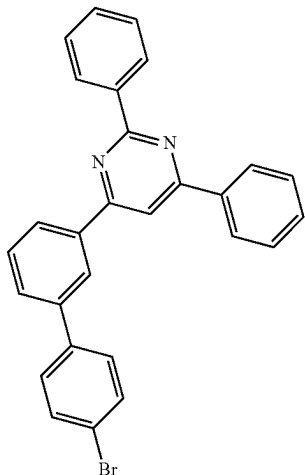
Compound	Intermediate I	Target Compound	Yield
527			62%
528			51%
531			60%

TABLE 9-continued

Com- pound	Intermediate I	Target Compound	Yield
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549

550

TABLE 9-continued

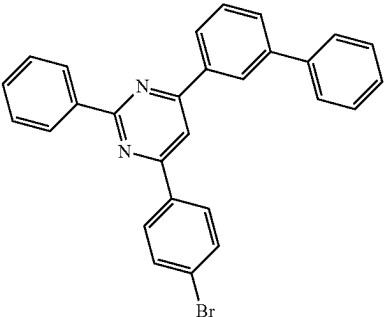
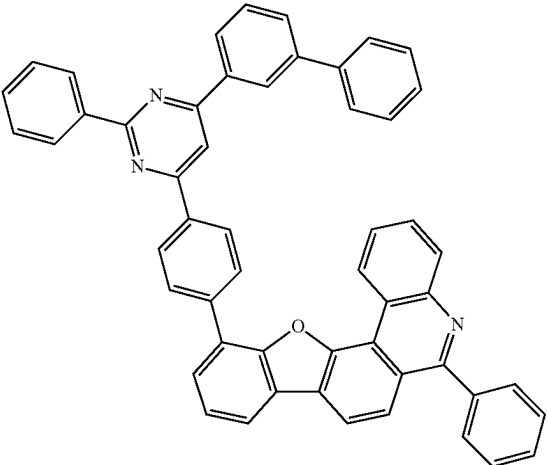
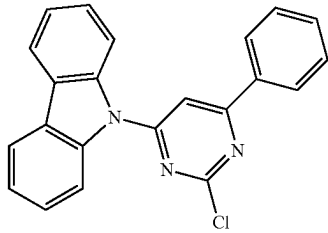
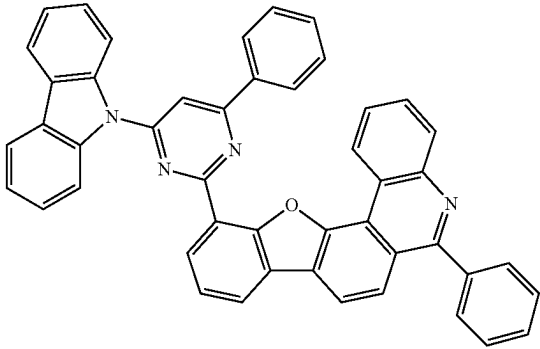
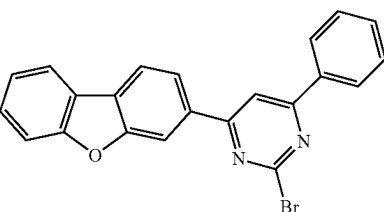
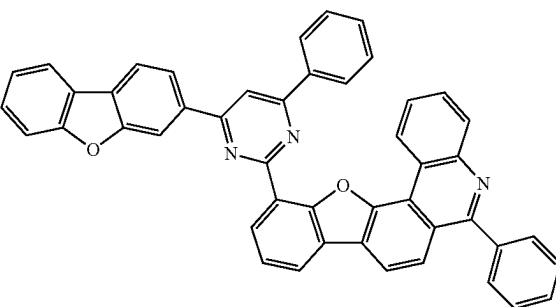
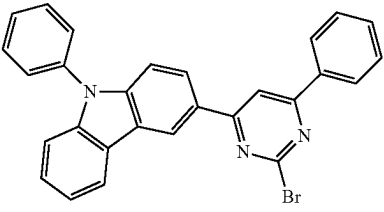
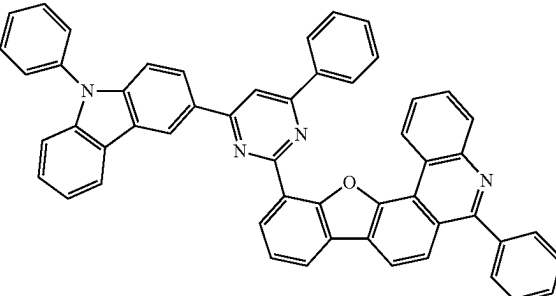
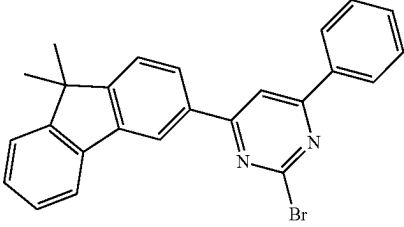
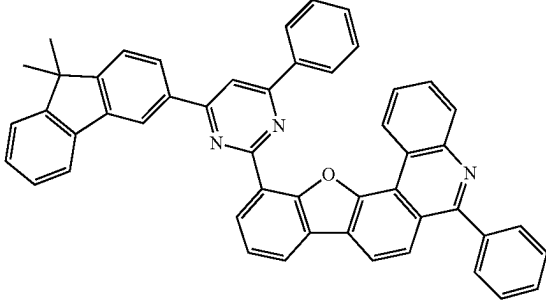
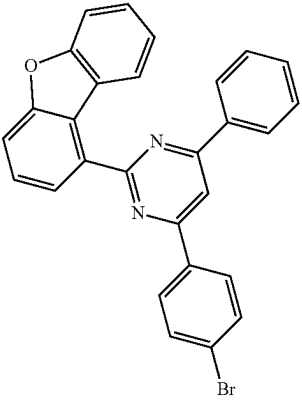
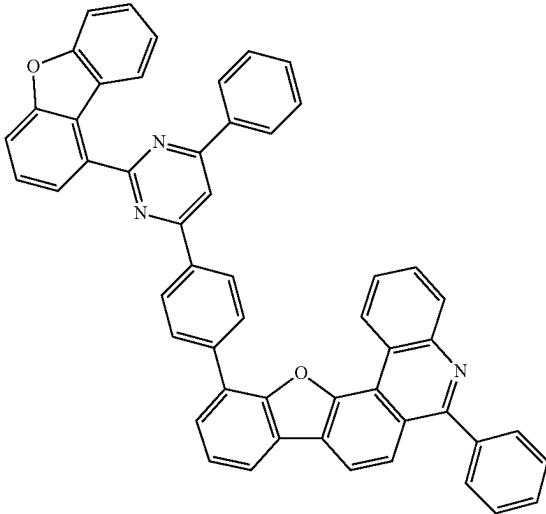
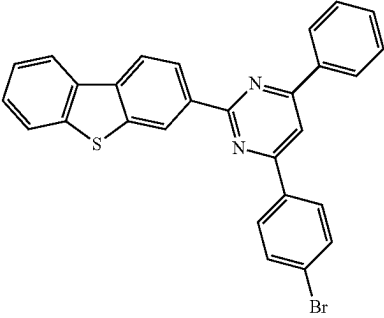
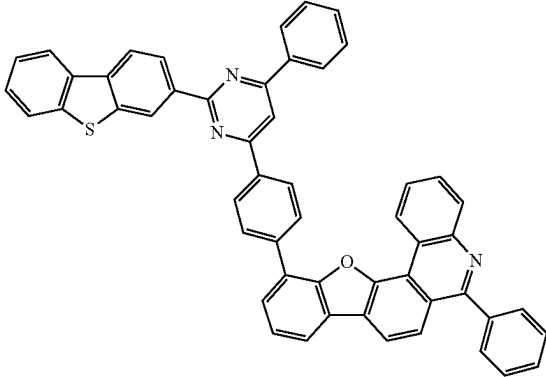
Compound	Intermediate I	Target Compound	Yield
535			62%
536			65%
539			65%
542			62%

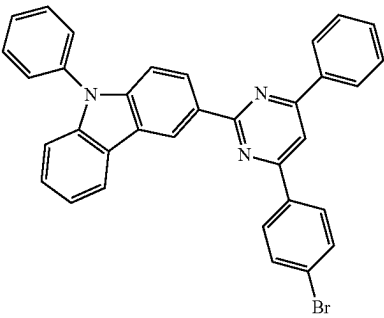
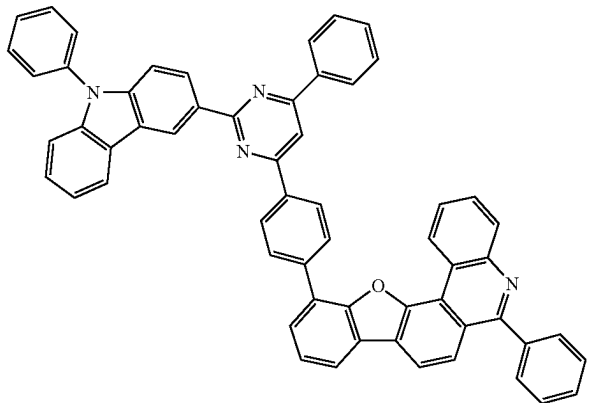
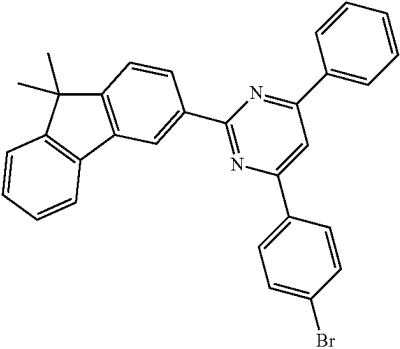
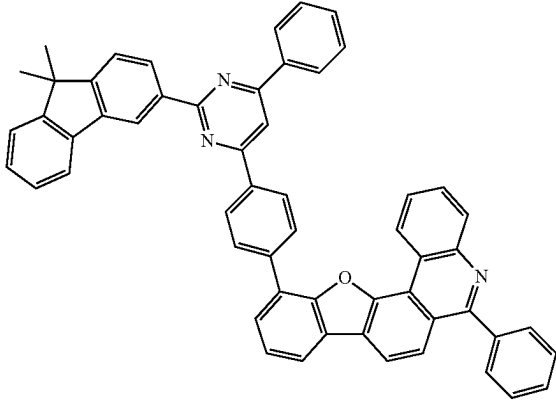
TABLE 9-continued

Compound	Intermediate I	Target Compound	Yield
543			51%
544			60%
547			61%

553

554

TABLE 9-continued

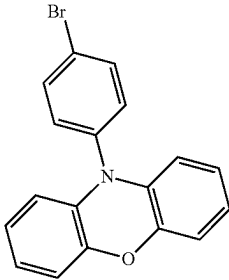
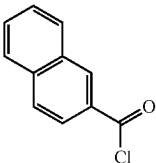
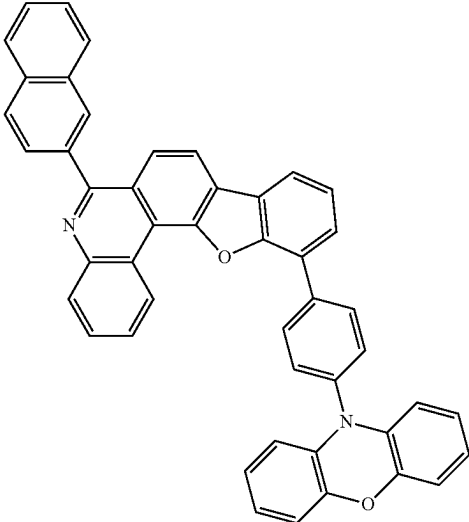
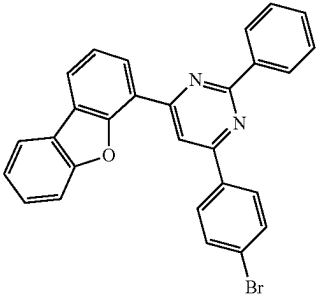
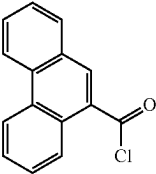
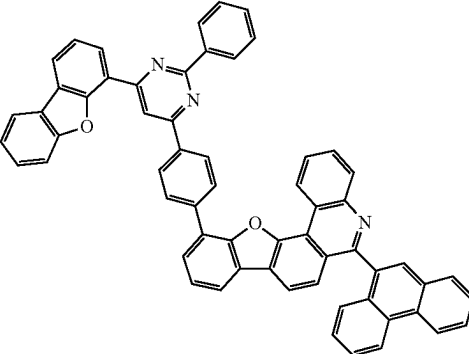
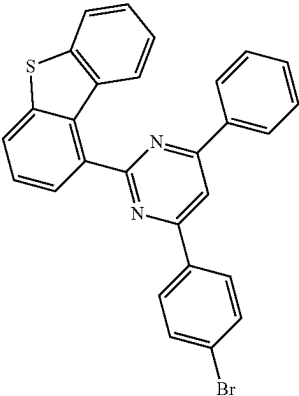
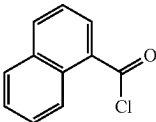
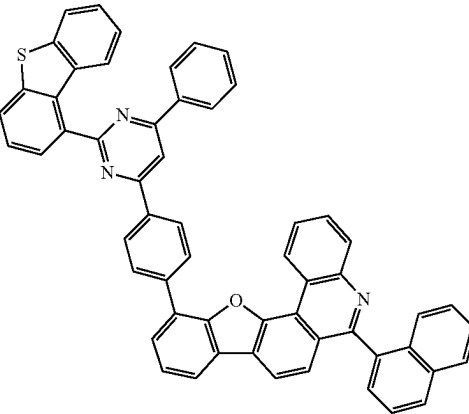
Compound	Intermediate I	Target Compound	Yield
549			62%
550			65%

Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 5 of the following Table 10 was used instead of 1-pyrenecarbonyl chloride, 4,6-dibromodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate I of the following Table 10 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

555

556

TABLE 10

Compound	Intermediate I	Intermediate 5	Target Compound	Yield
468				53%
522				52%
548				50%

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 1-chloro-6-fluorobenzo[b,d]furan was used instead of 1-bromo-8-chlorobenzo[b,d]furan, and Intermediate J of the following Table 11 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

557

558

TABLE 11

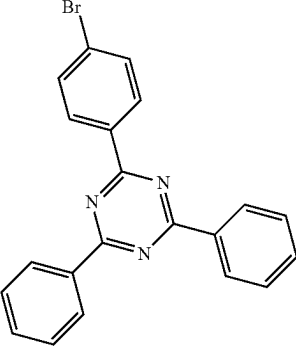
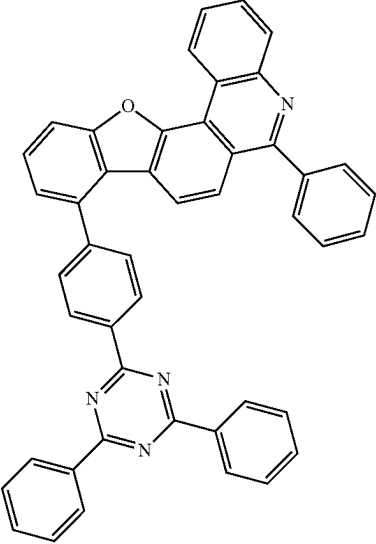
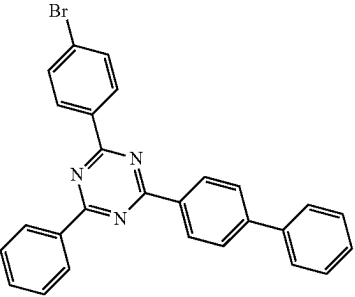
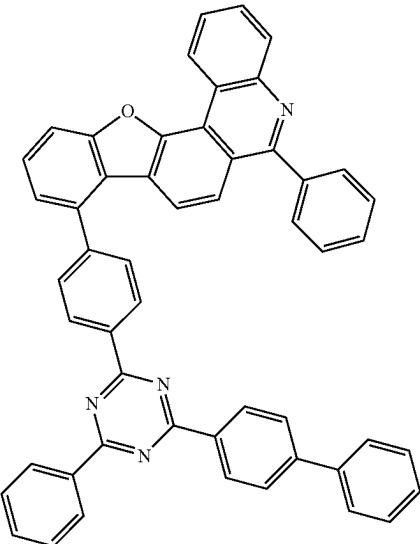
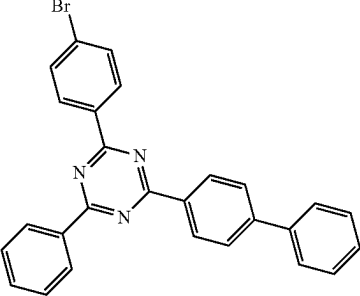
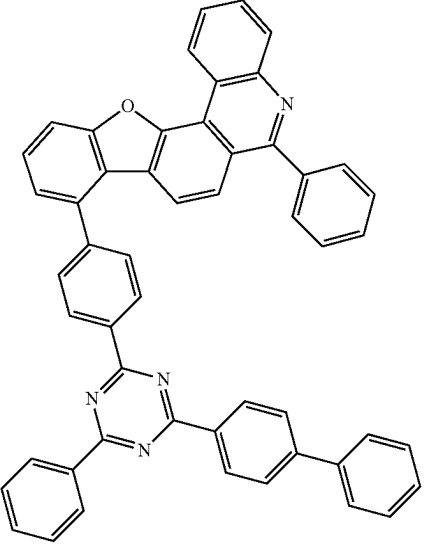
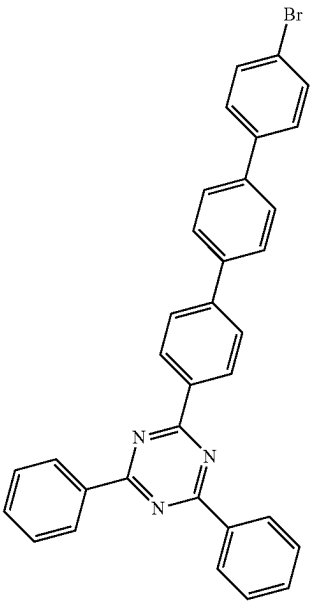
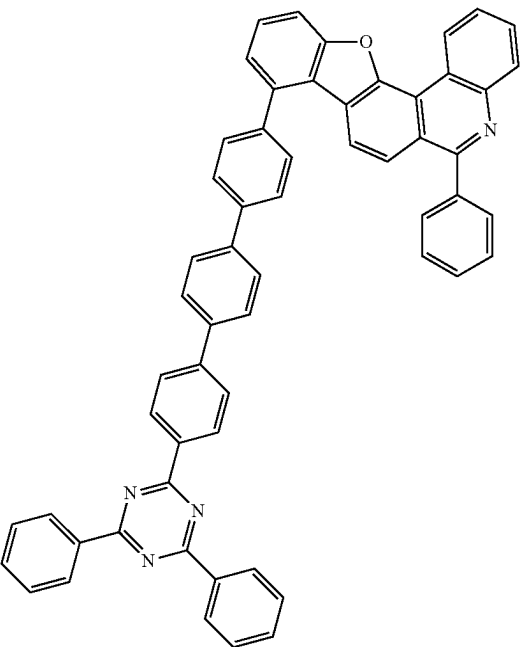
Compound	Intermediate J	Target Compound	Yield
554			53%
555			62%

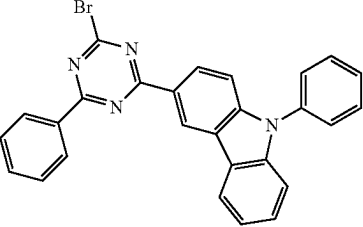
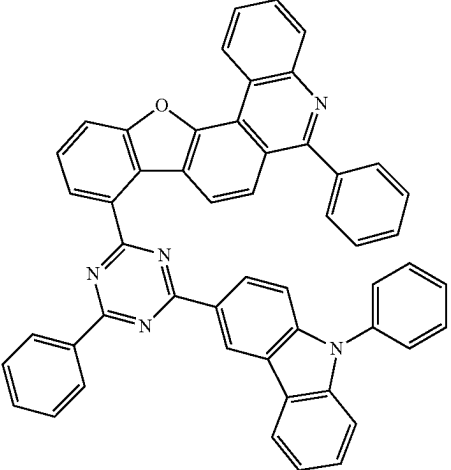
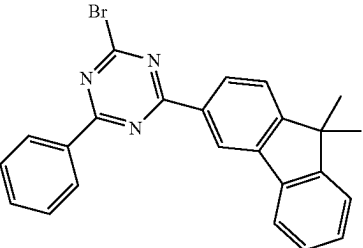
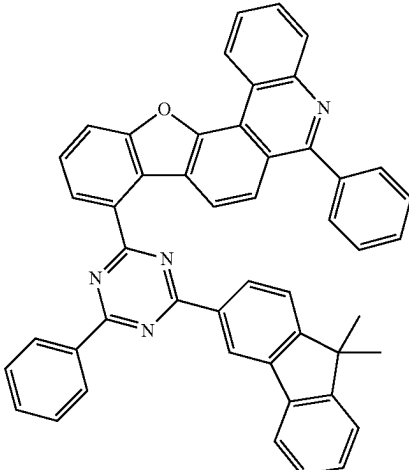
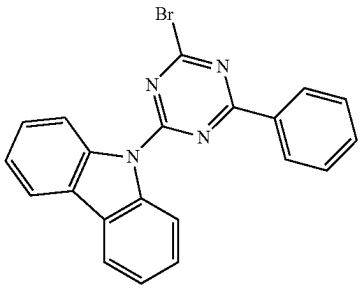
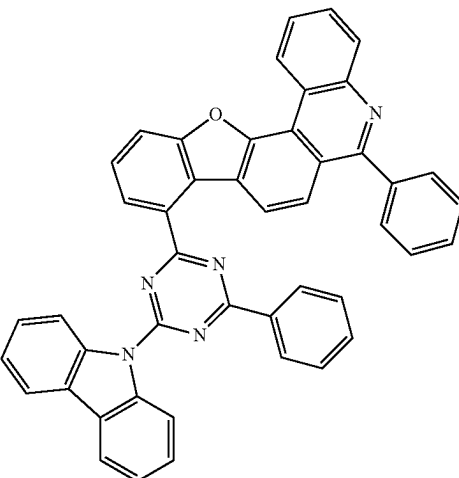
TABLE 11-continued

Compound	Intermediate J	Target Compound	Yield
557			65%
558			65%

561

562

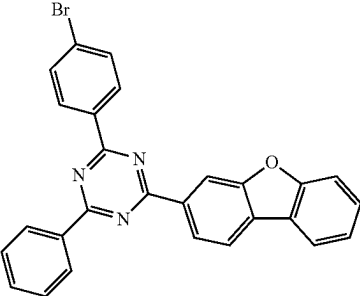
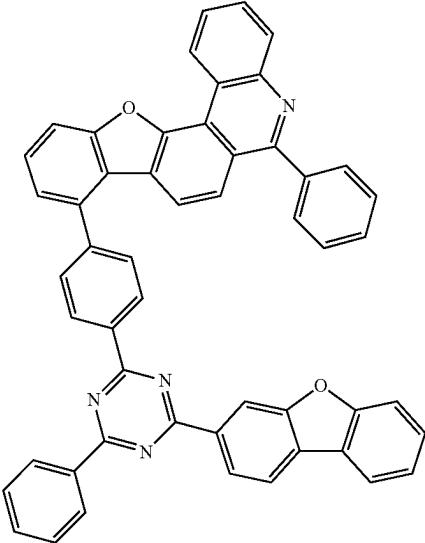
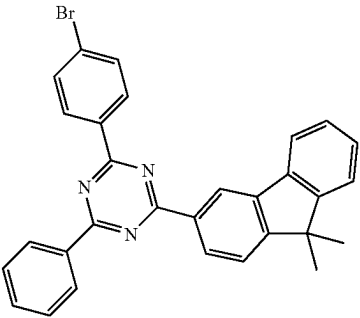
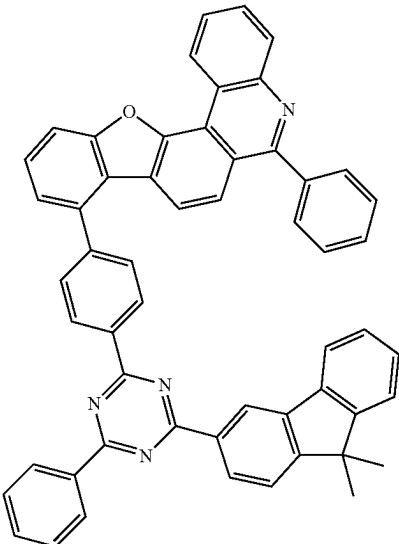
TABLE 11-continued

Compound	Intermediate J	Target Compound	Yield
564			62%
565			51%
566			60%

563

564

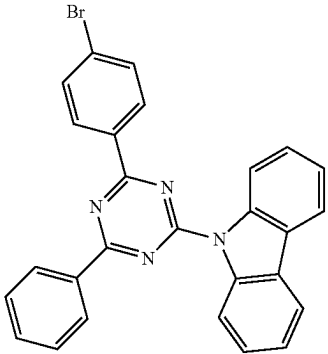
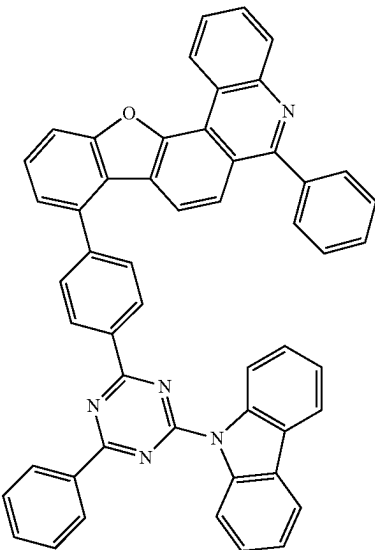
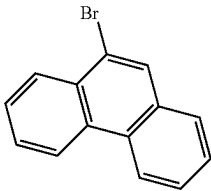
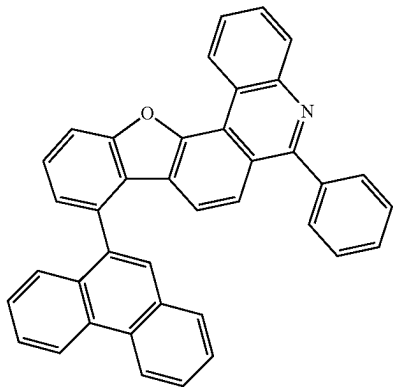
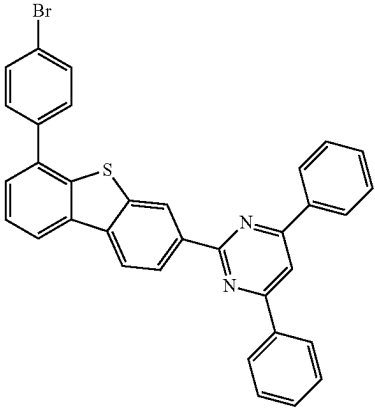
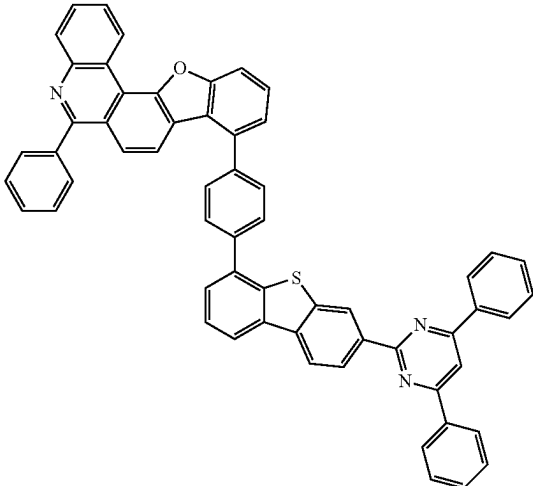
TABLE 11-continued

Compound	Intermediate J	Target Compound	Yield
569			61%
573			60%

565

566

TABLE 11-continued

Compound	Intermediate J	Target Compound	Yield
574			62%
580			65%
583			65%

567

568

TABLE 11-continued

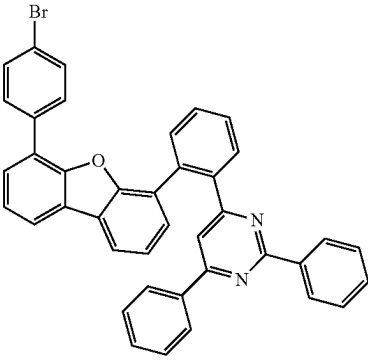
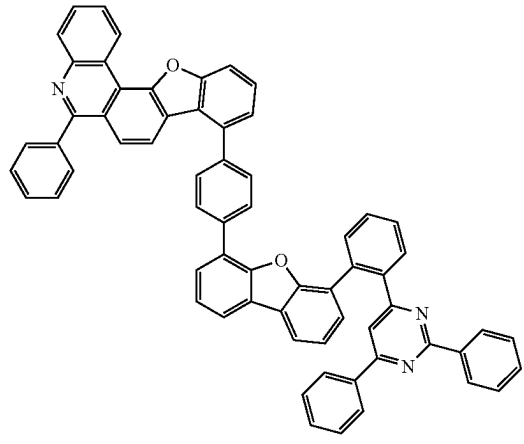
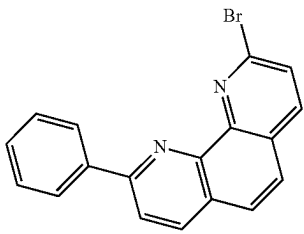
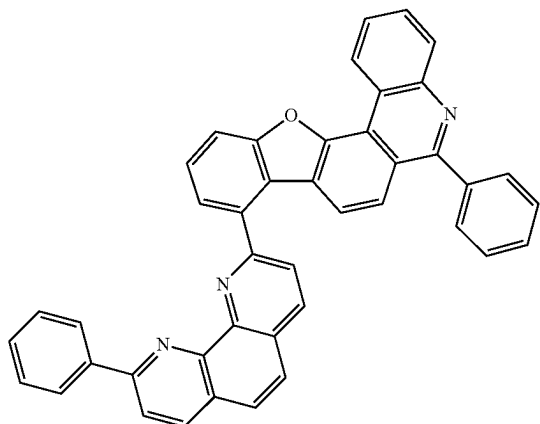
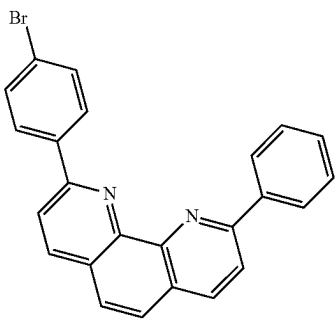
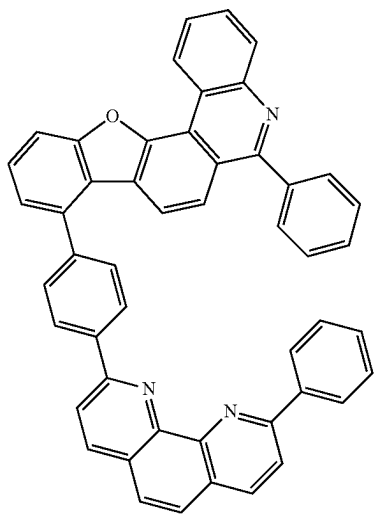
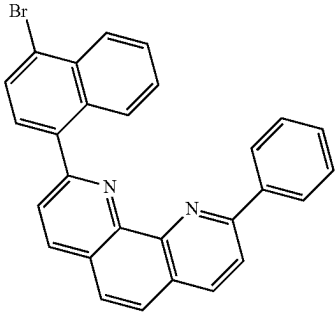
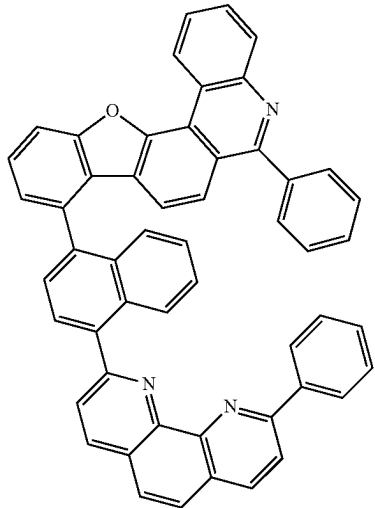
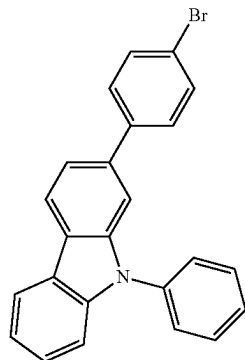
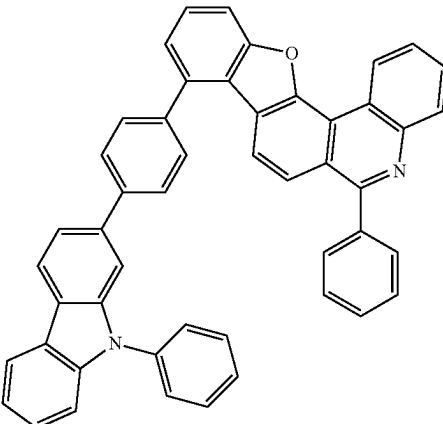
Compound	Intermediate J	Target Compound	Yield
584			62%
585			51%
587			60%

TABLE 11-continued

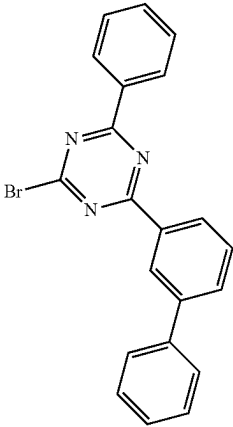
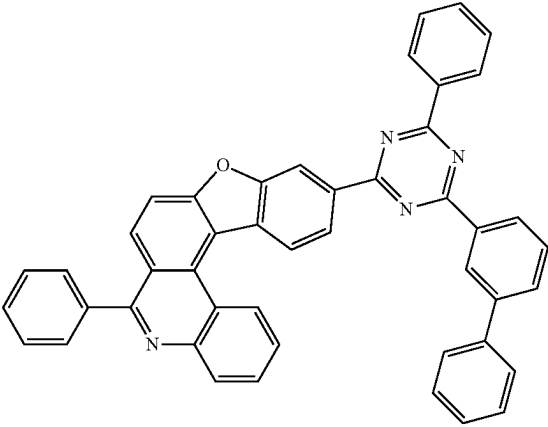
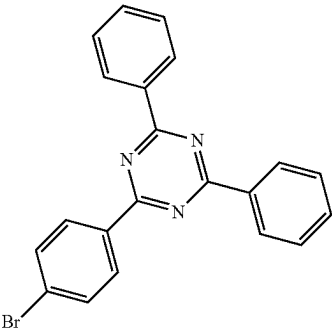
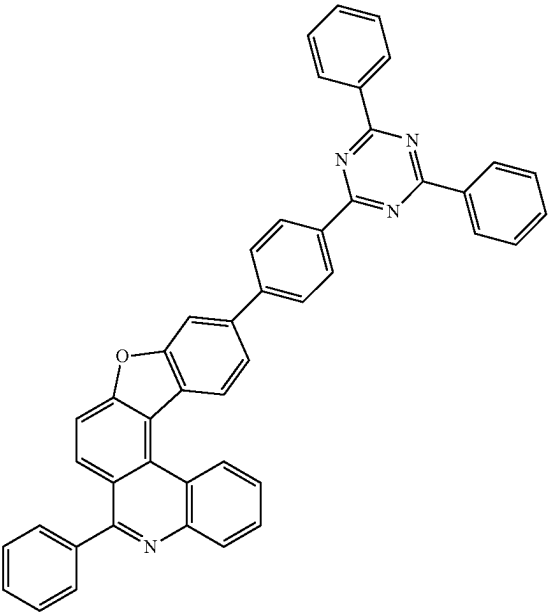
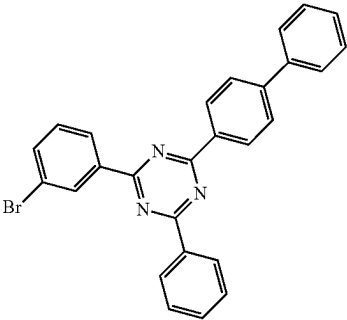
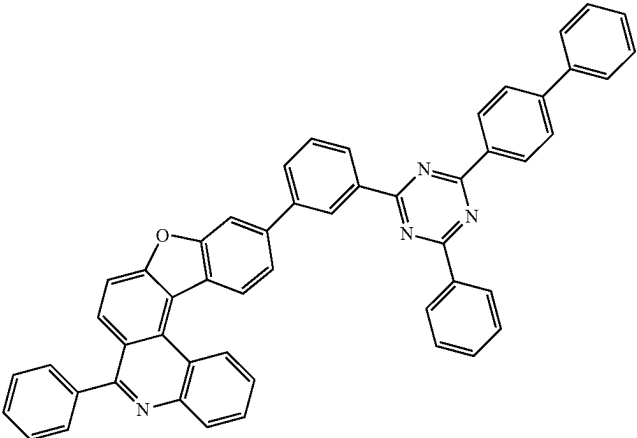
Compound	Intermediate J	Target Compound	Yield
588			61%
594			60%

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 7-chloro-1-fluorodibenzo[b,d]furan was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate K of the following Table 65 12 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

571

572

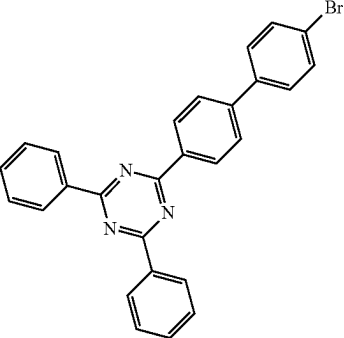
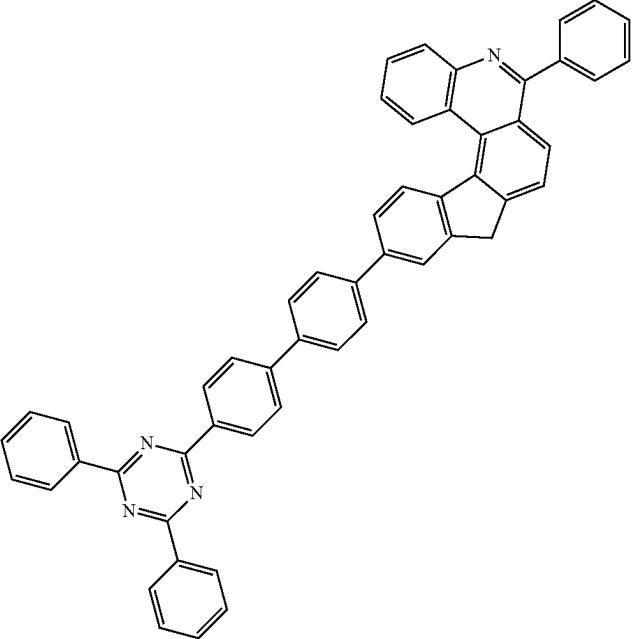
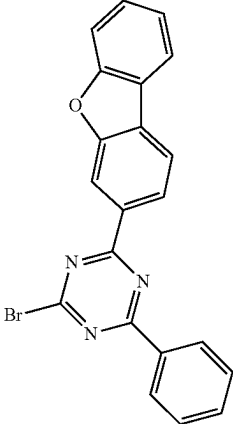
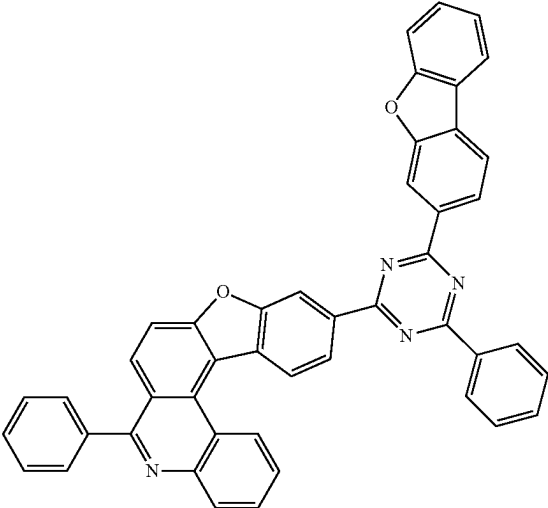
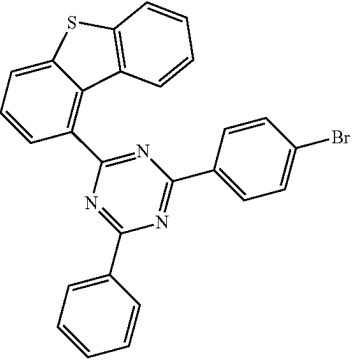
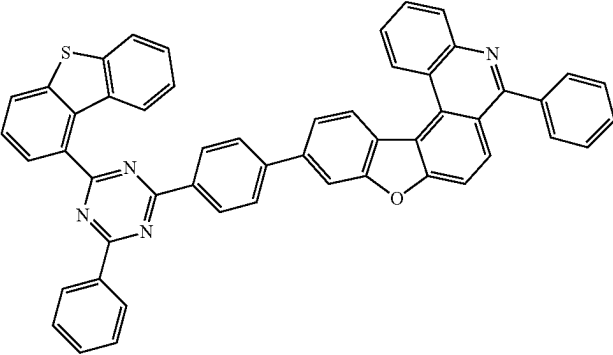
TABLE 12

Compound	Intermediate K	Target Compound	Yield
597			53%
598			62%
599			65%

573

574

TABLE 12-continued

Compound	Intermediate K	Target Compound	Yield
600			65%
601			62%
605			51%

575

576

TABLE 12-continued

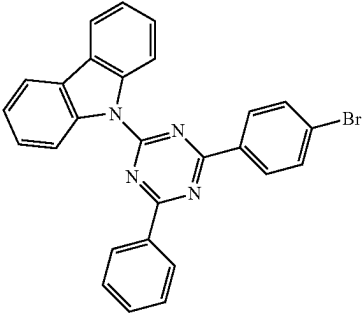
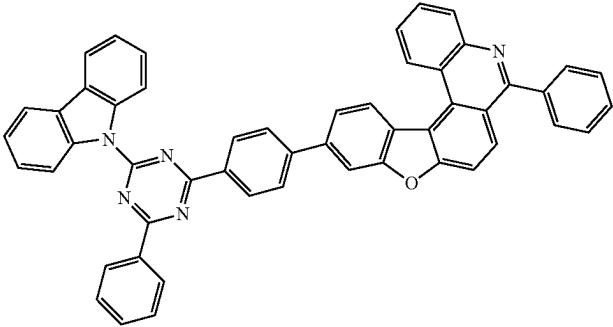
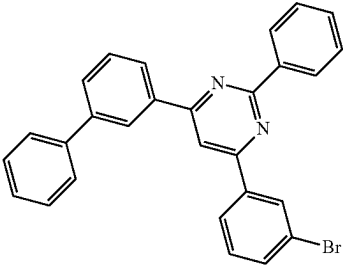
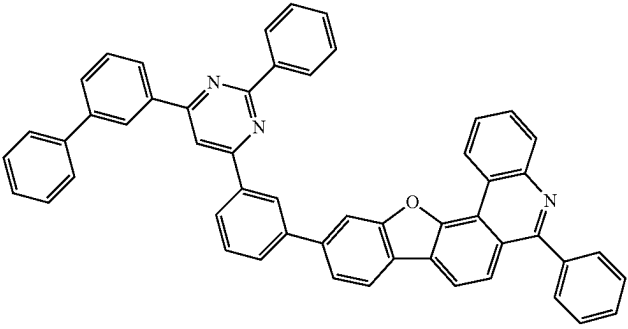
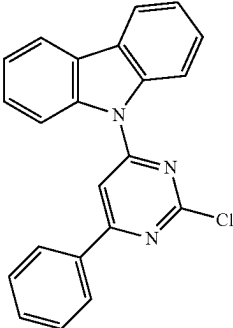
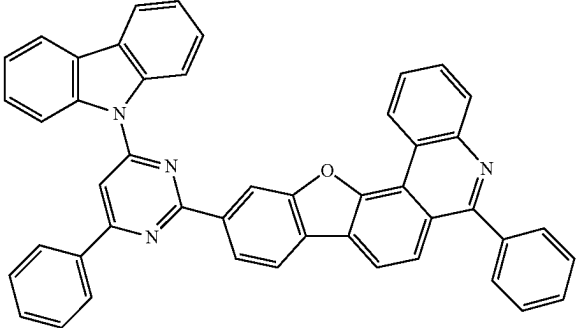
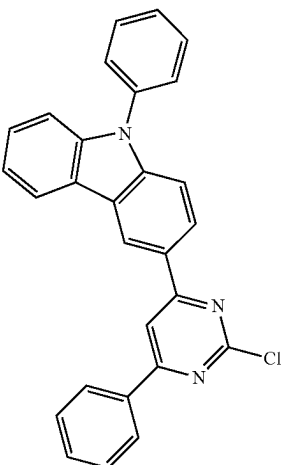
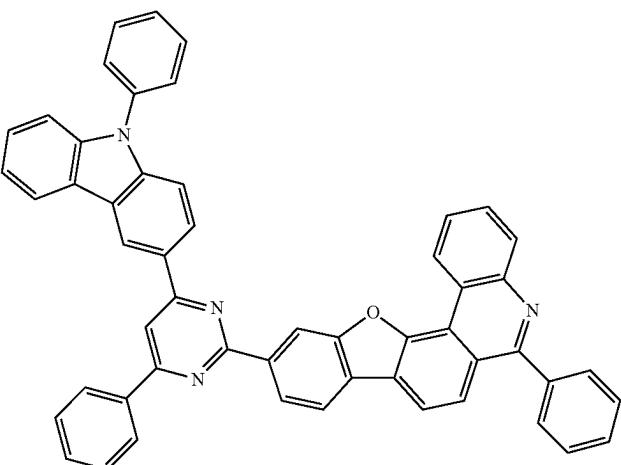
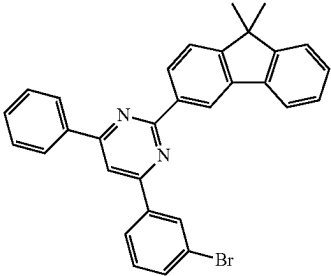
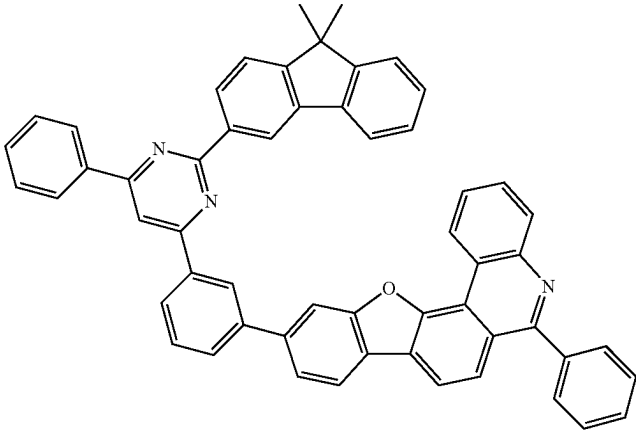
Compound	Intermediate K	Target Compound	Yield
606			60%
611			61%
612			60%
615			51%

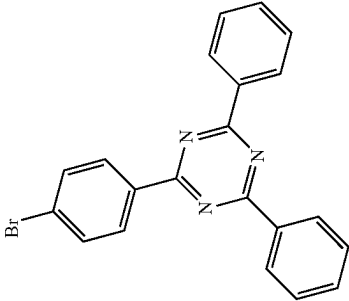
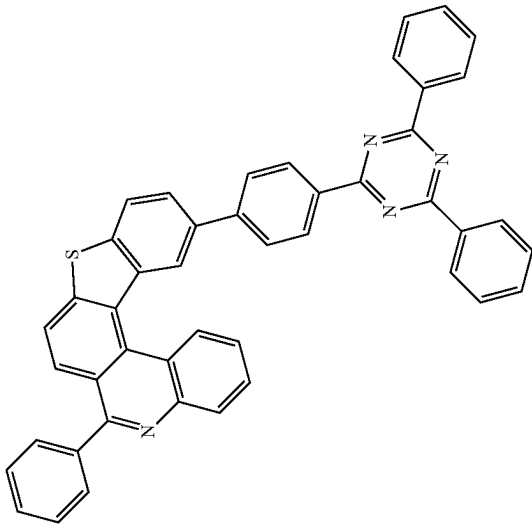
TABLE 12-continued

Compound	Intermediate K	Target Compound	Yield
618			50%

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 1-bromo-8-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-

chlorodibenzo[b,d]furan, and Intermediate L of the following Table 13 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

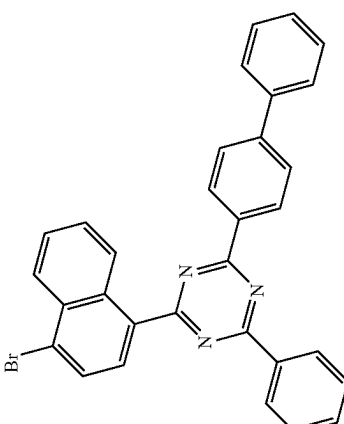
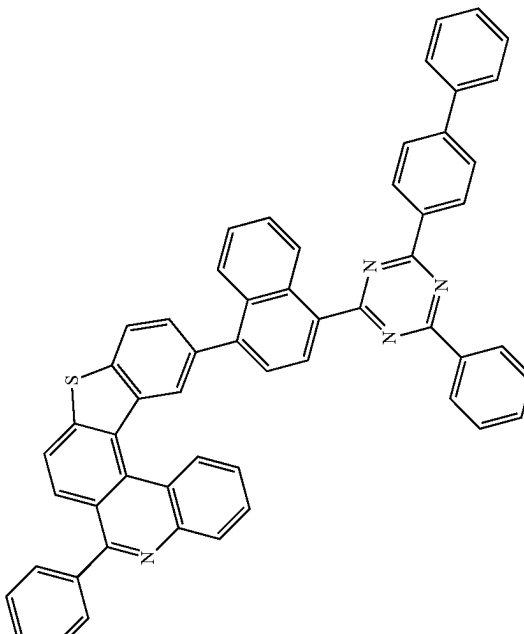
TABLE 13

Compound	Intermediate L	Target Compound	Yield
623			62%

581

582

TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
624			65%

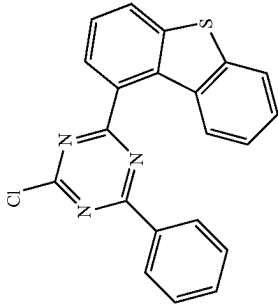
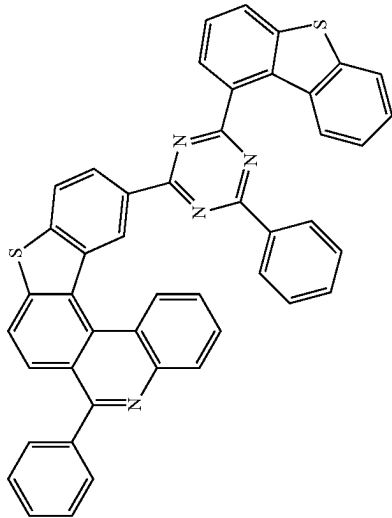
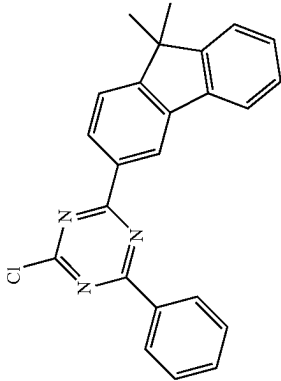
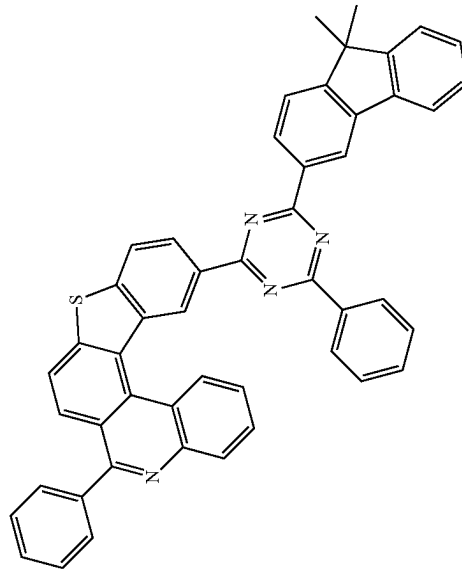
[illegible]

629

585

586

TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
632			51%
634			60%

587

588

TABLE 13-continued

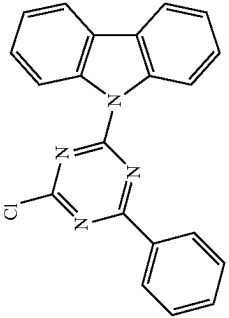
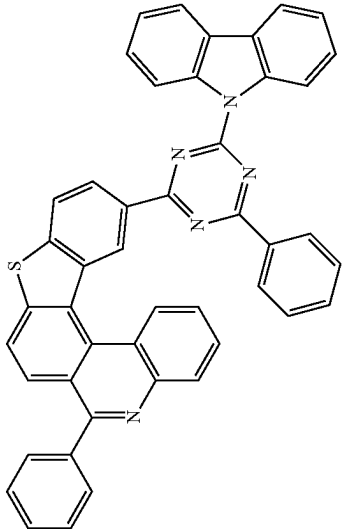
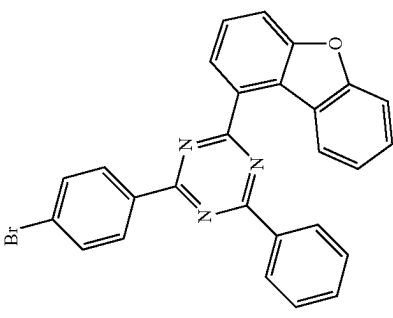
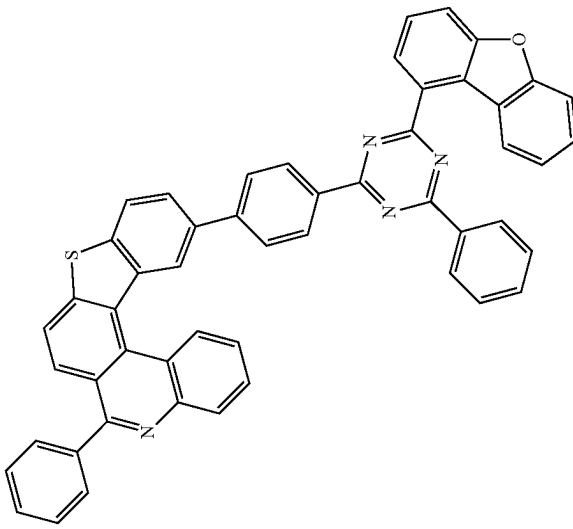
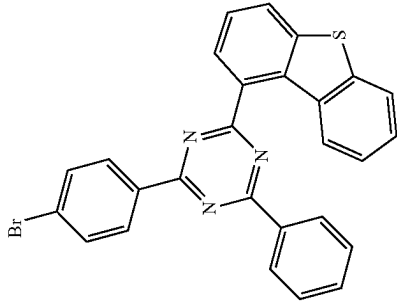
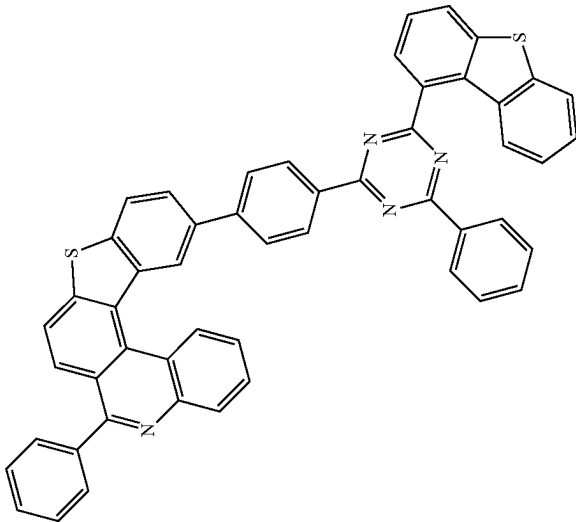
Compound	Intermediate L	Target Compound	Yield
635			61%
636			60%

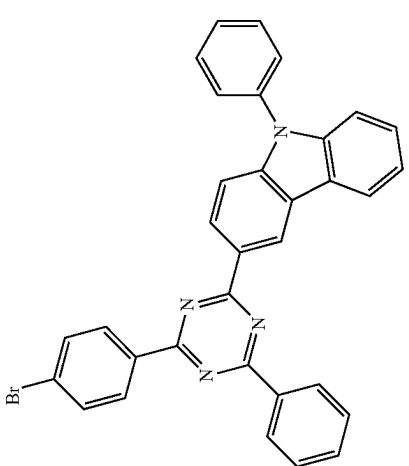
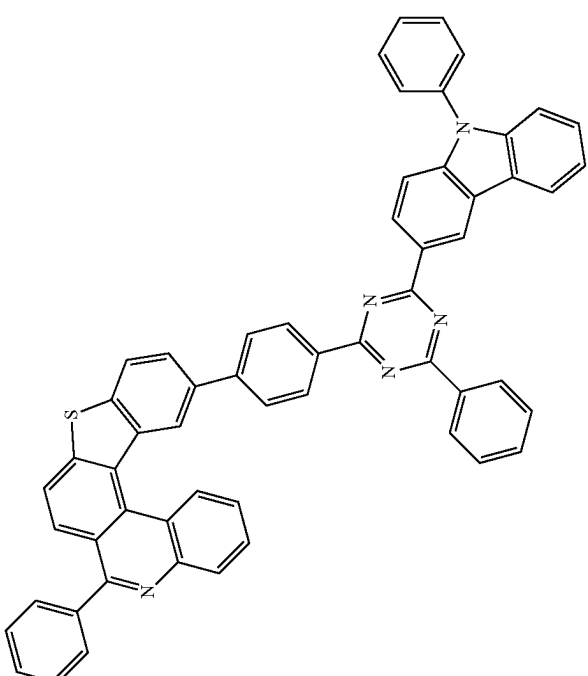
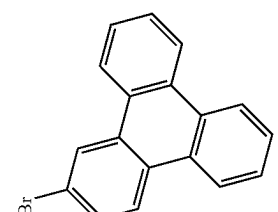
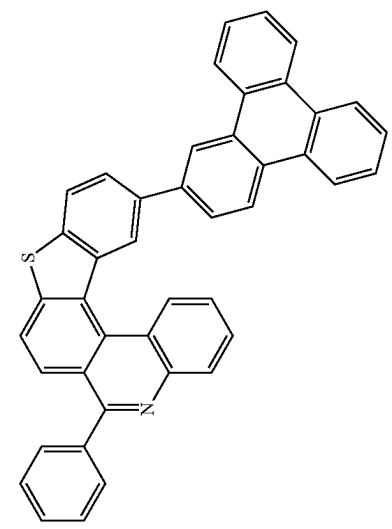
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
640			62%

591

592

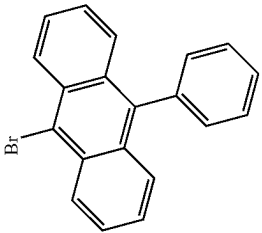
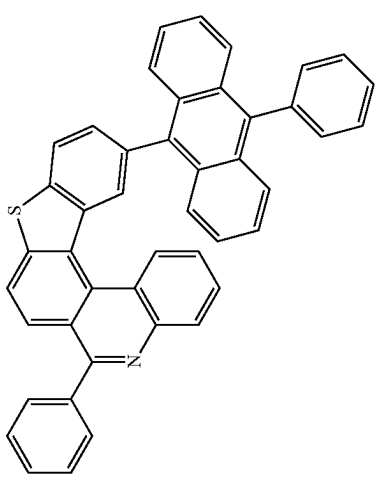
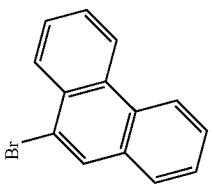
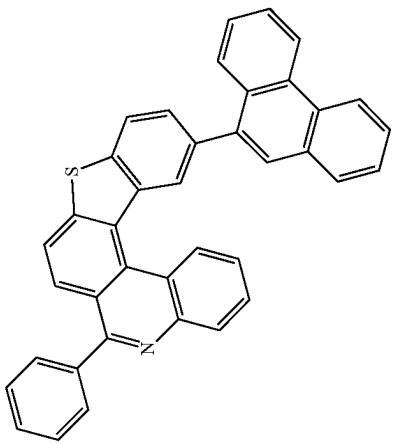
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
641			65%
644			65%

593

594

TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
646			62%
649			51%

595

596

TABLE 13-continued

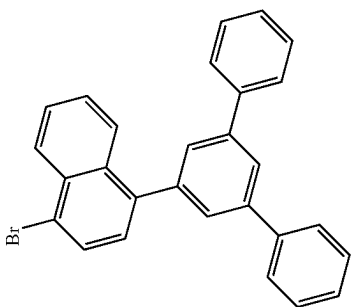
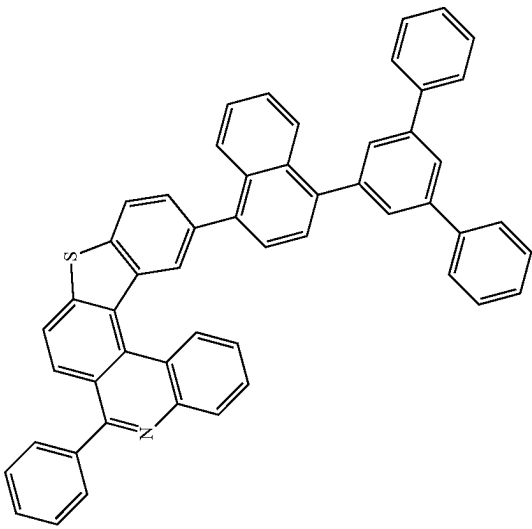
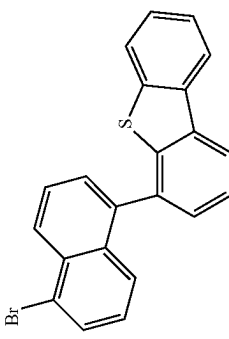
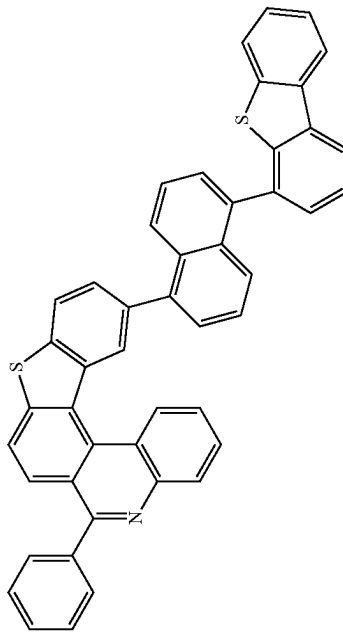
Compound	Intermediate L	Target Compound	Yield
651			60%
652			61%

TABLE 13-continued

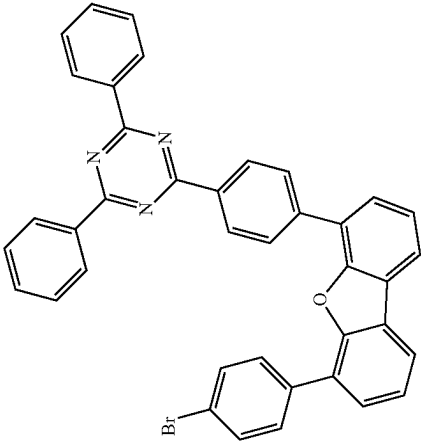
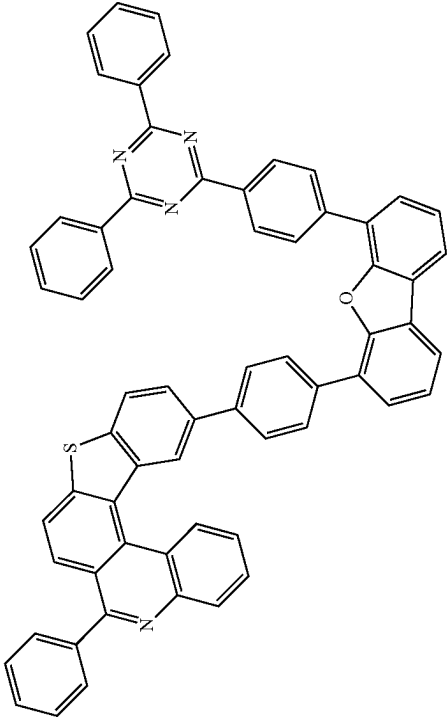
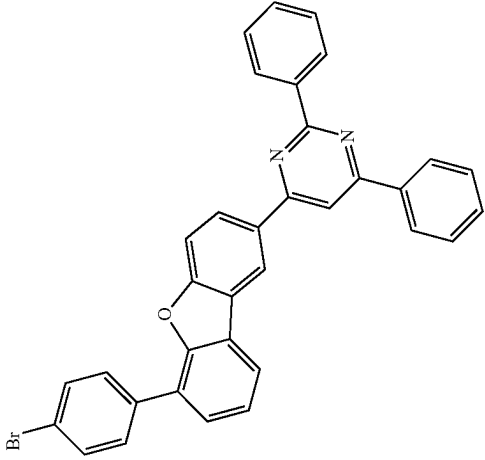
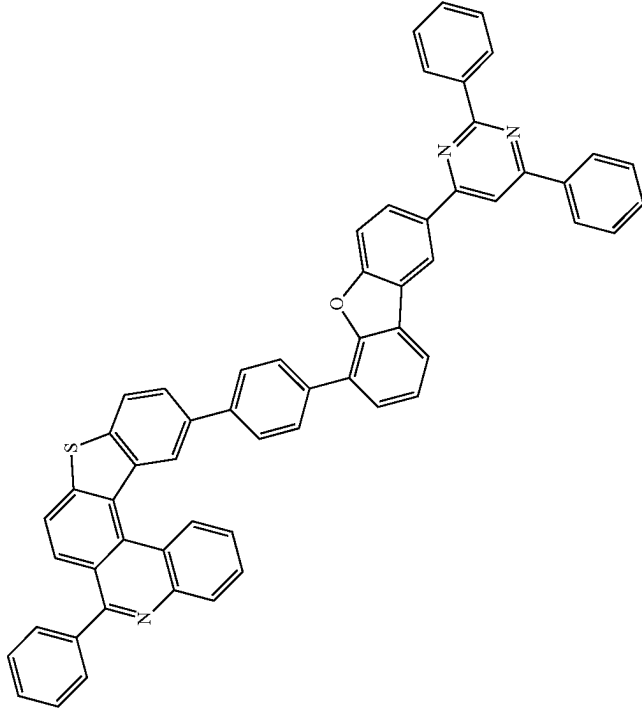
Compound	Intermediate L	Target Compound	Yield
653			60%

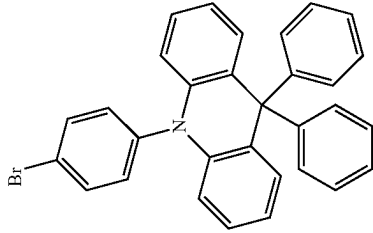
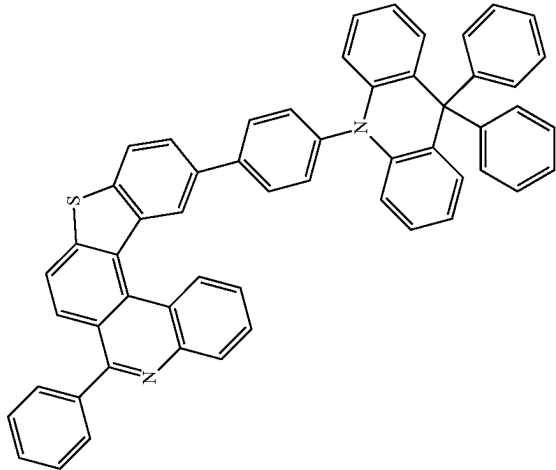
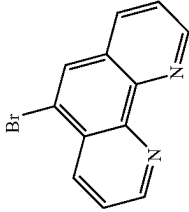
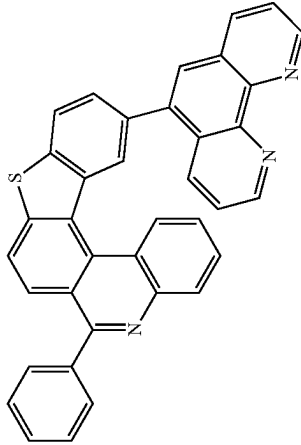
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
654			62%

601

602

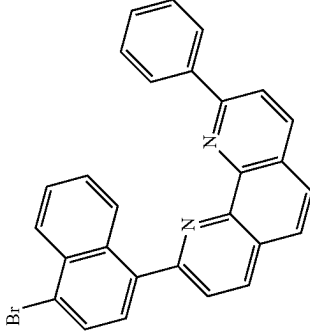
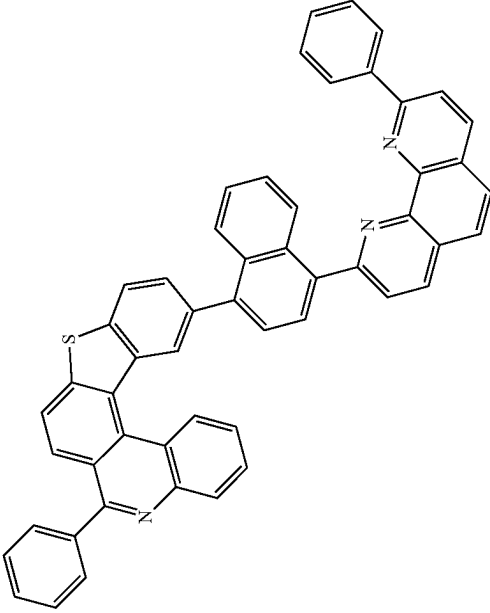
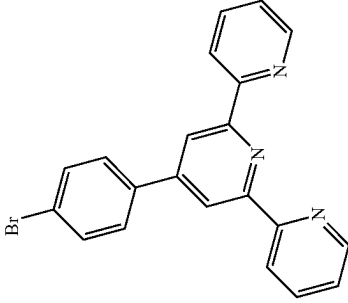
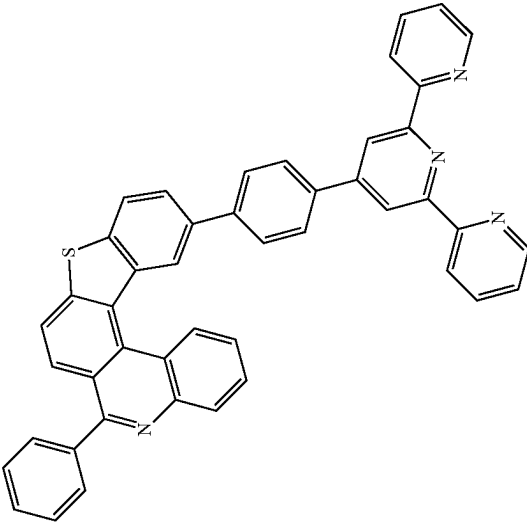
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
655			65%
657			65%

603

604

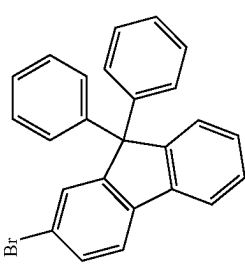
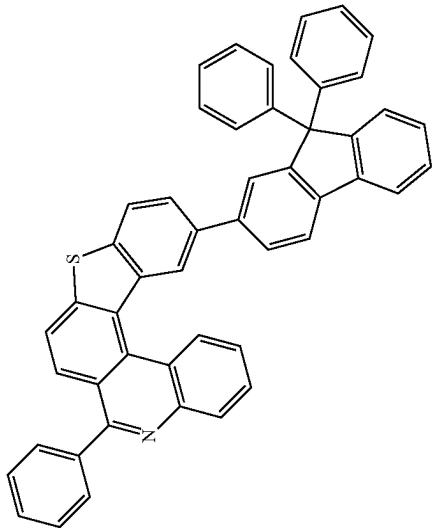
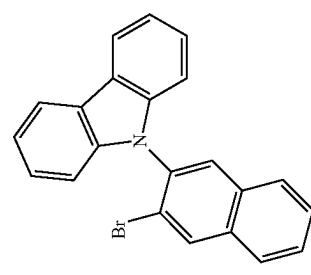
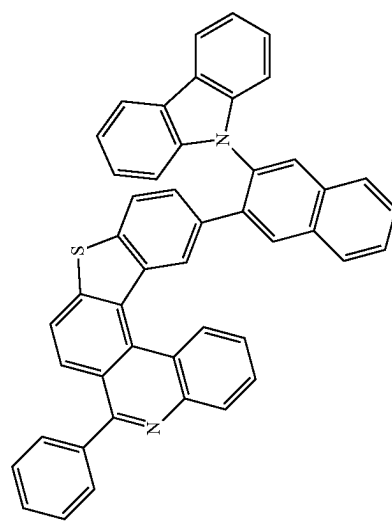
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
659			62%
660			51%

605

606

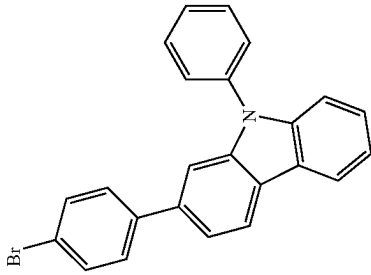
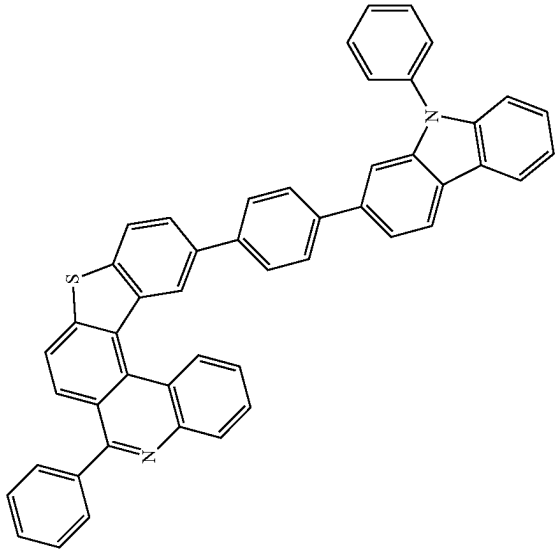
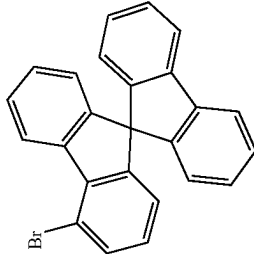
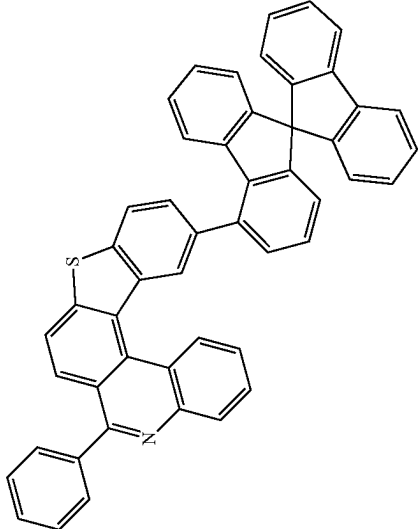
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
661			60%
664			61%

607

608

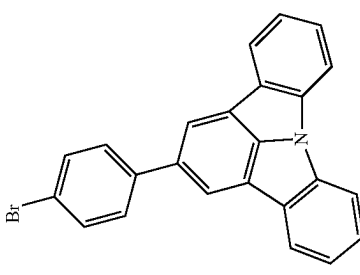
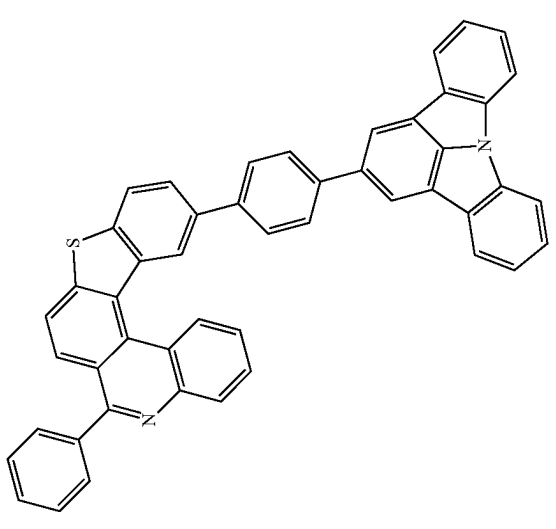
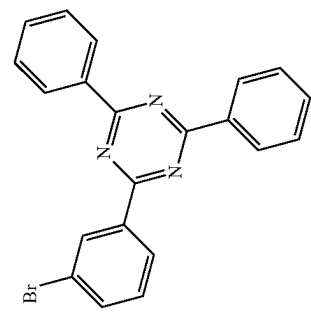
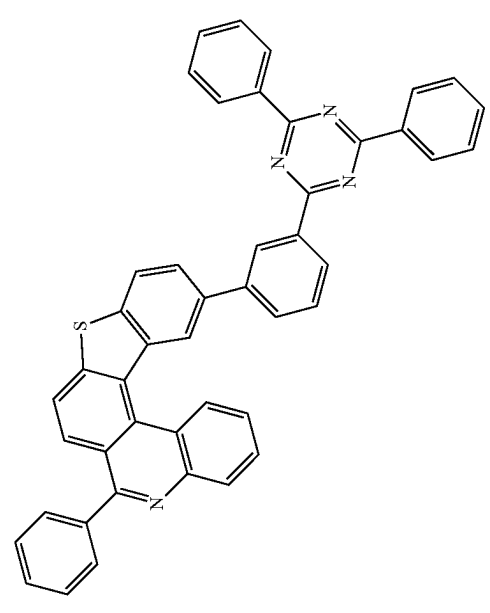
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
665			60%
666			62%

609

610

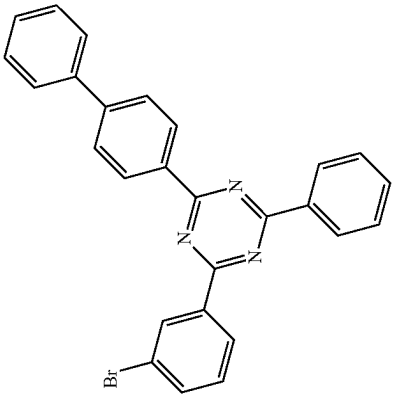
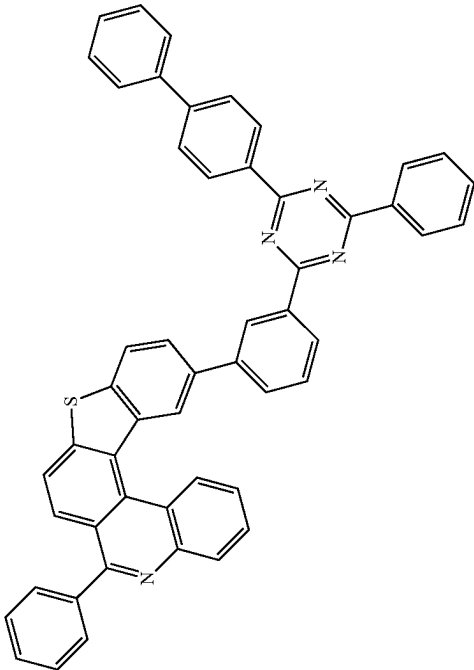
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
670			65%
674			62%

611

612

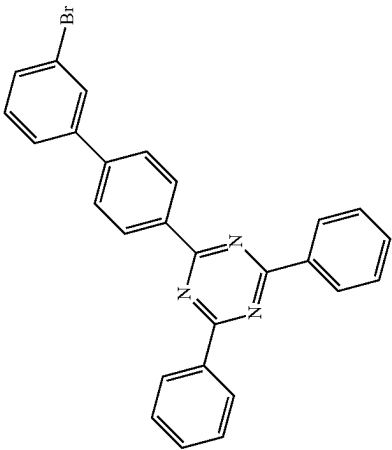
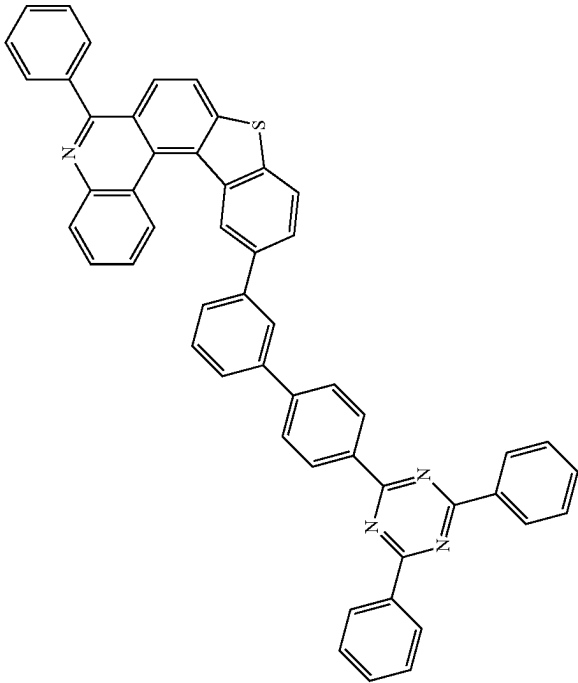
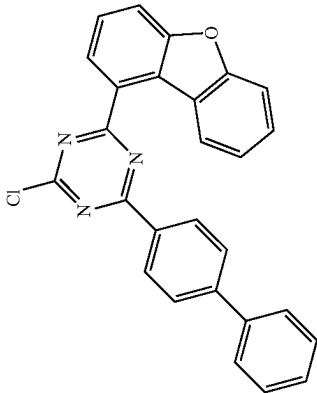
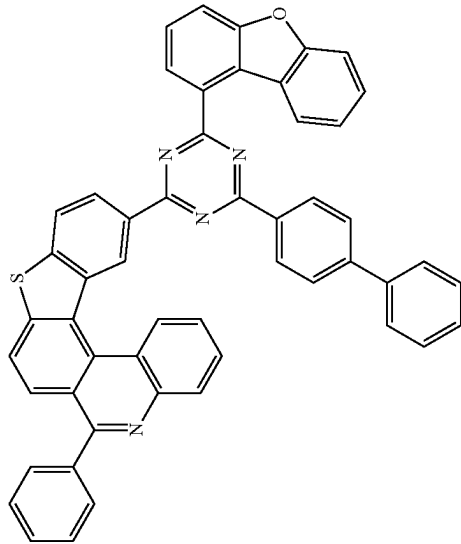
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
675			51%

613

614

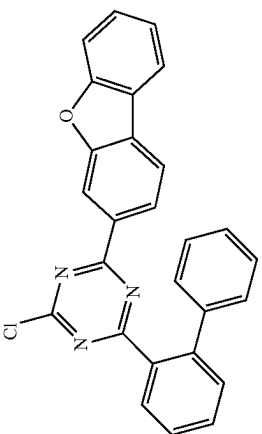
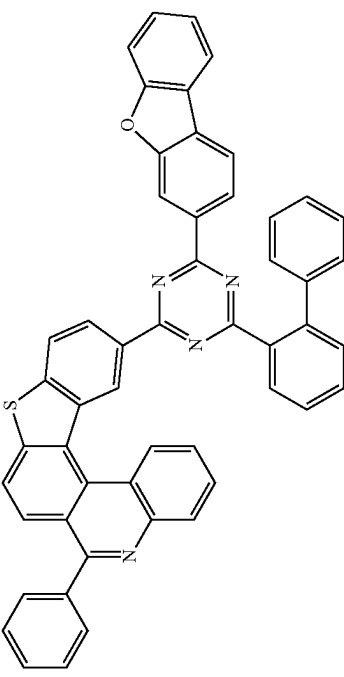
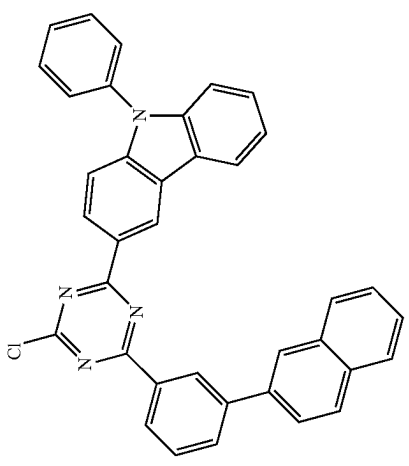
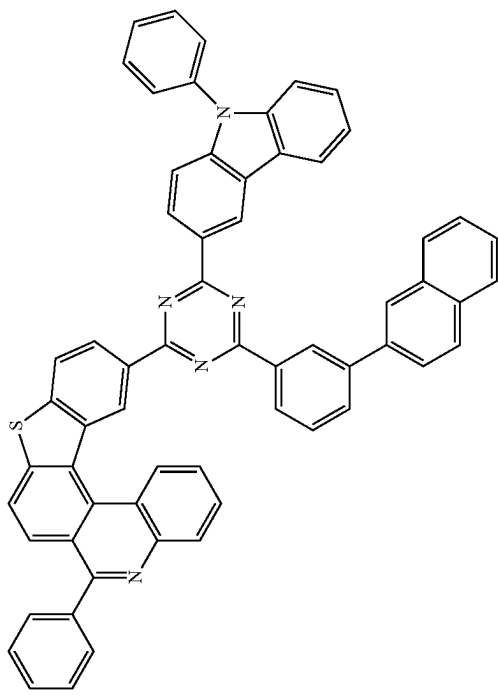
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
677			60%
680			61%

615

616

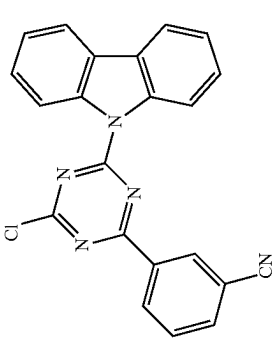
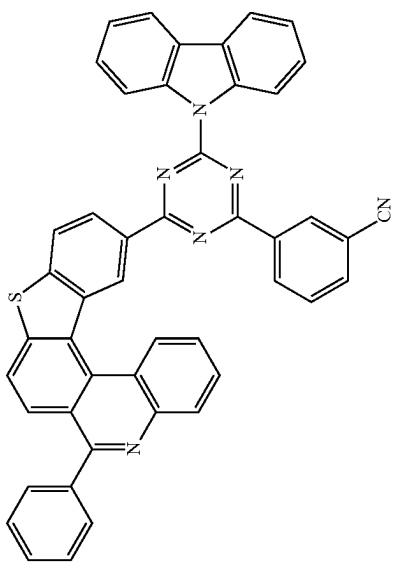
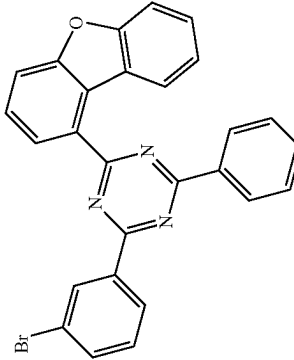
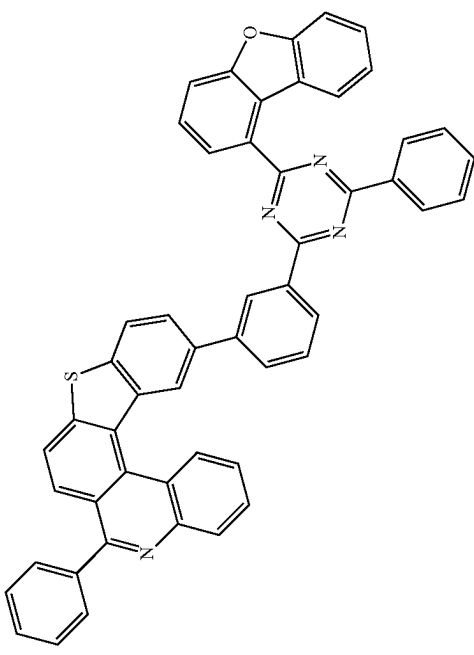
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
682			60%
685			65%

617

618

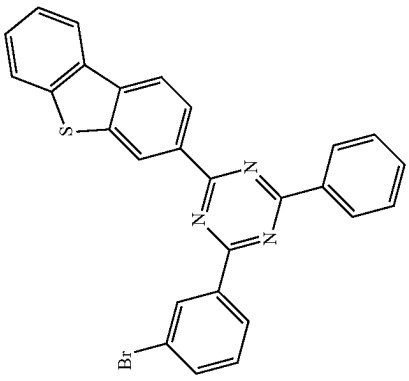
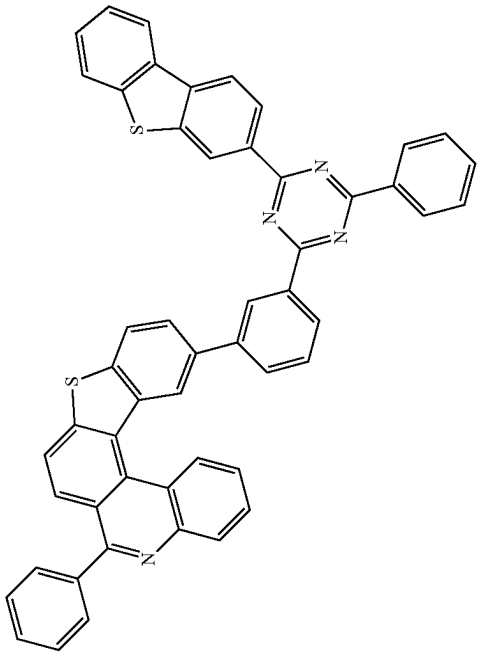
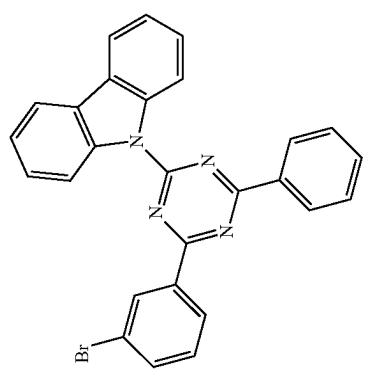
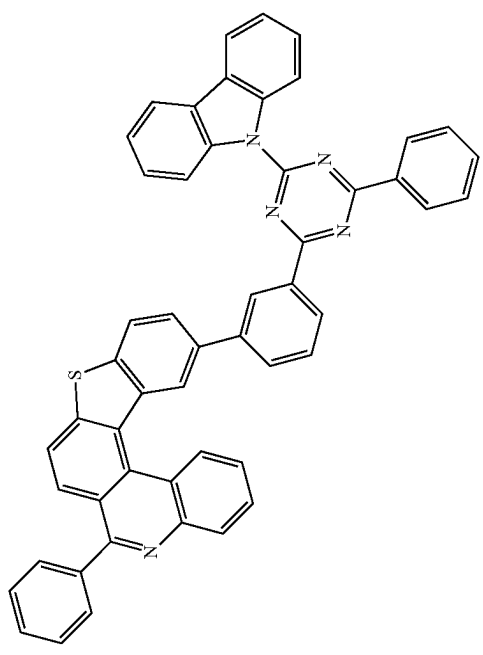
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
687			65%
688			62%

619

620

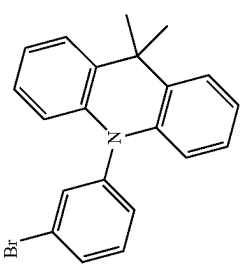
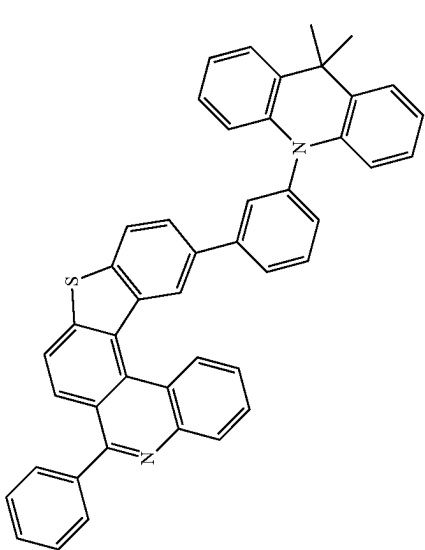
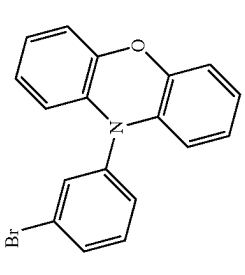
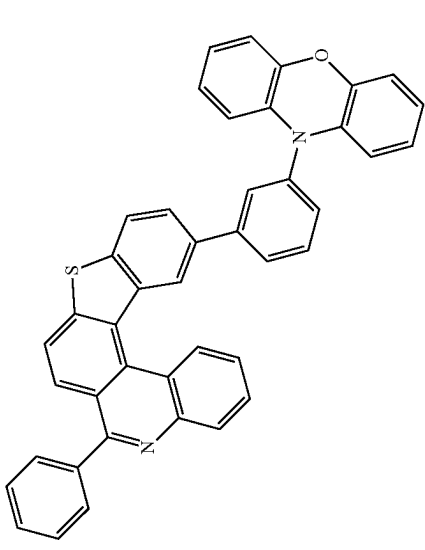
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
691			51%
695			61%

621

622

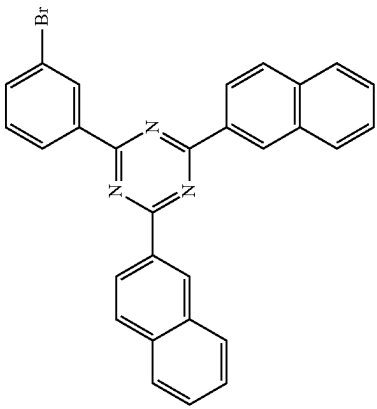
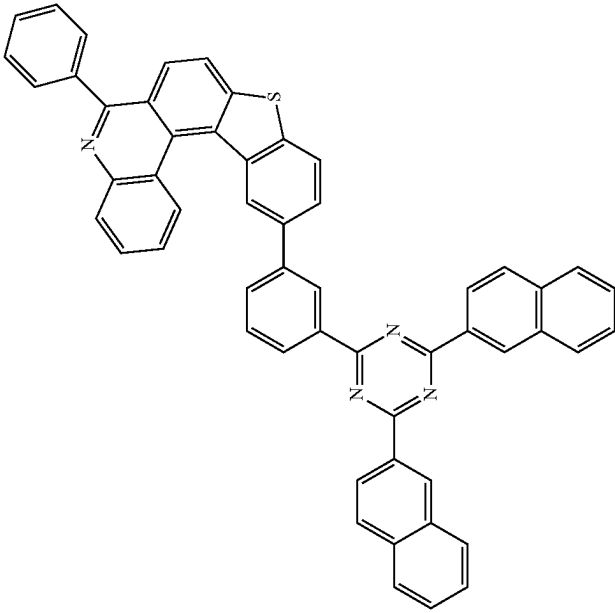
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
696			62%
697			65%

623

624

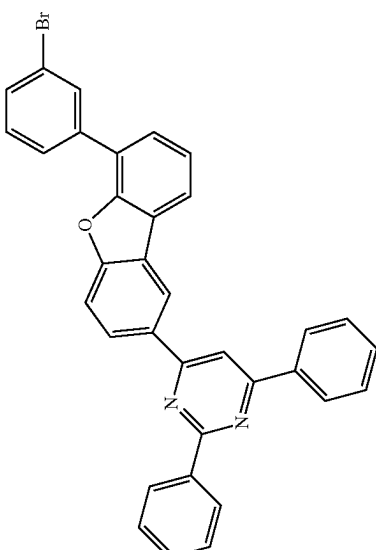
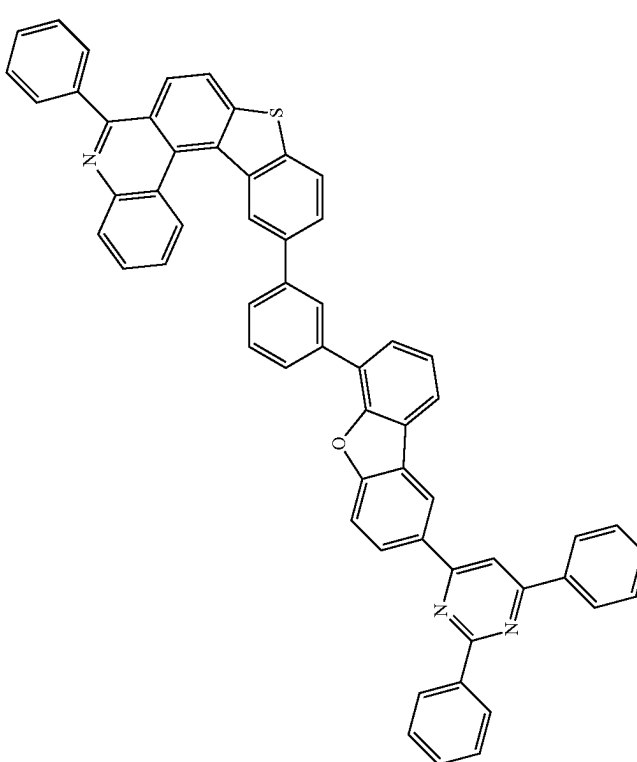
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
700			65%

625

626

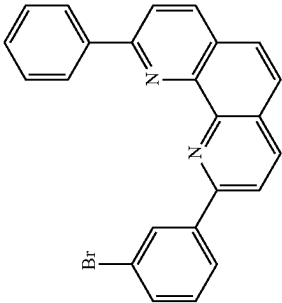
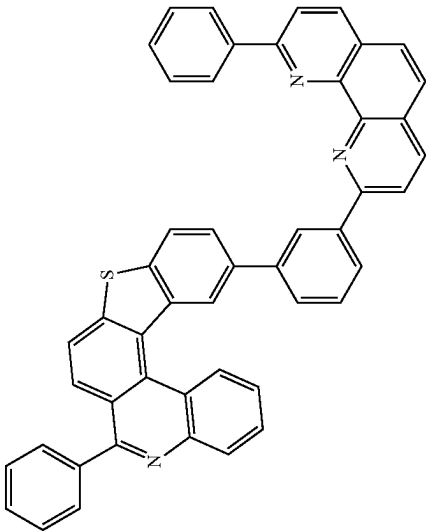
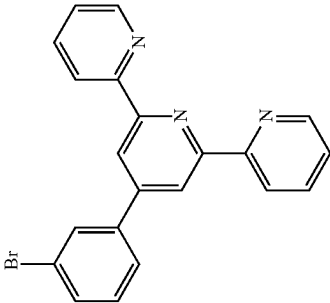
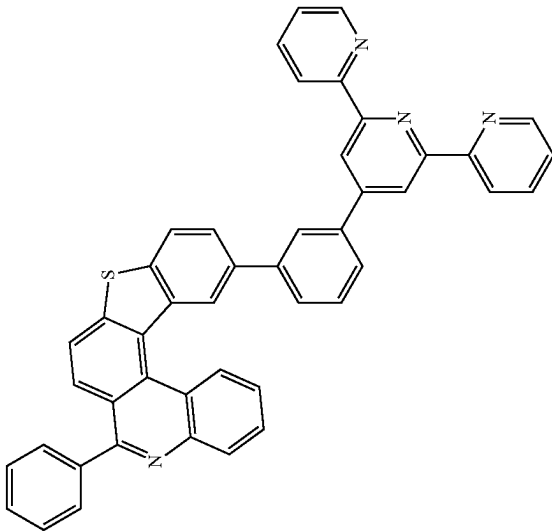
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
704			51%

627

628

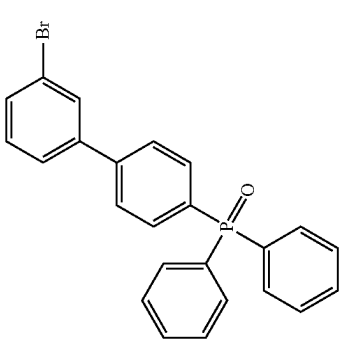
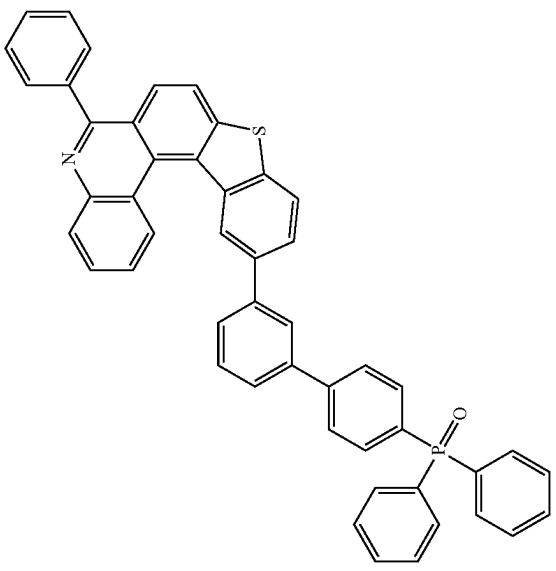
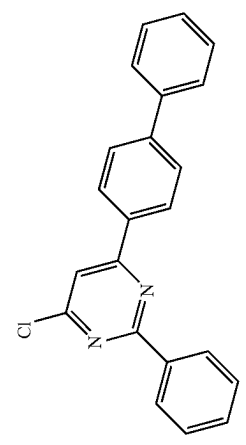
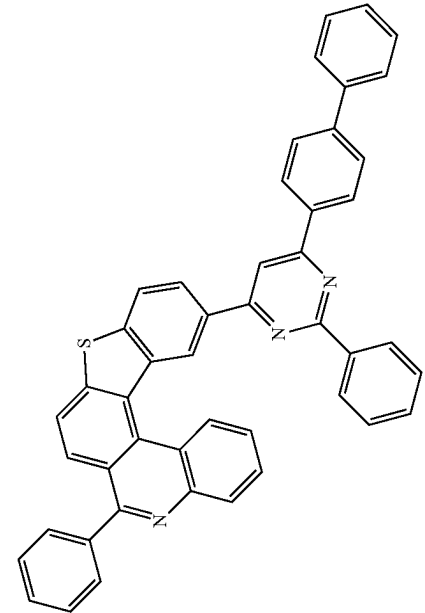
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
707			60%
708			62%

629

630

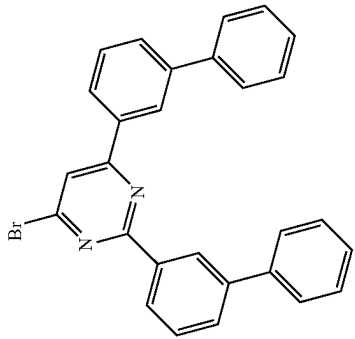
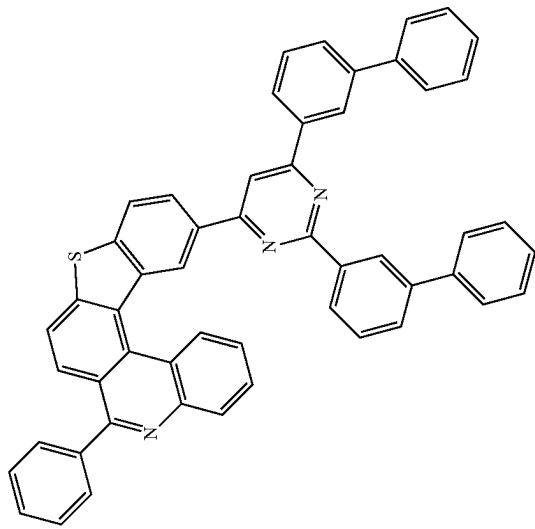
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
709			65%
713			65%

631

632

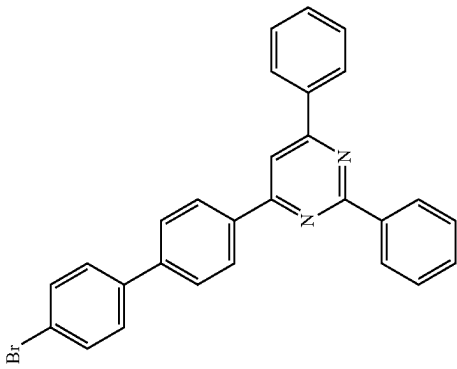
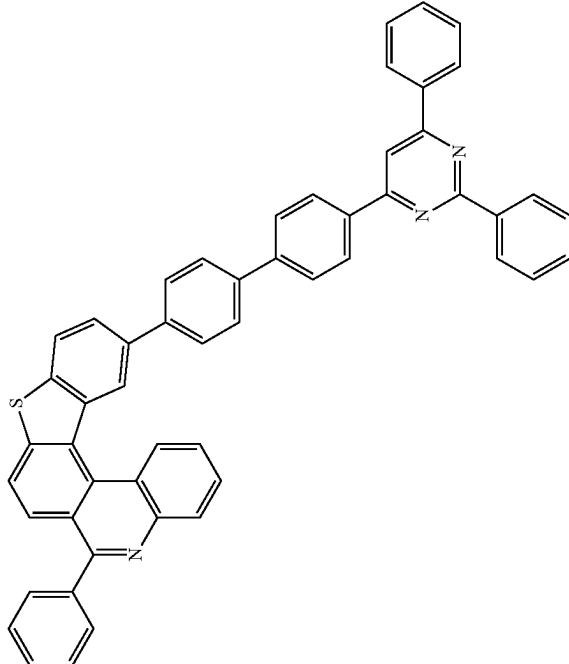
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
714			62%

633

634

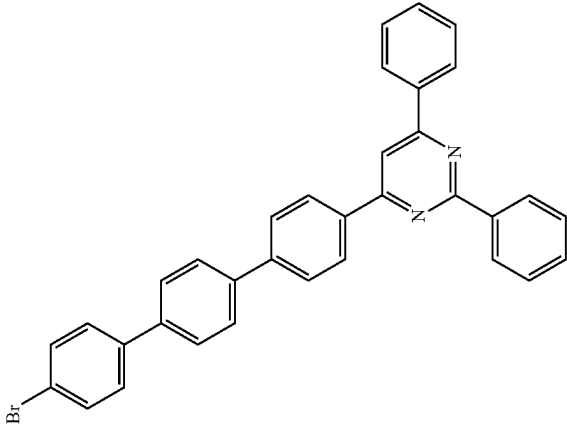
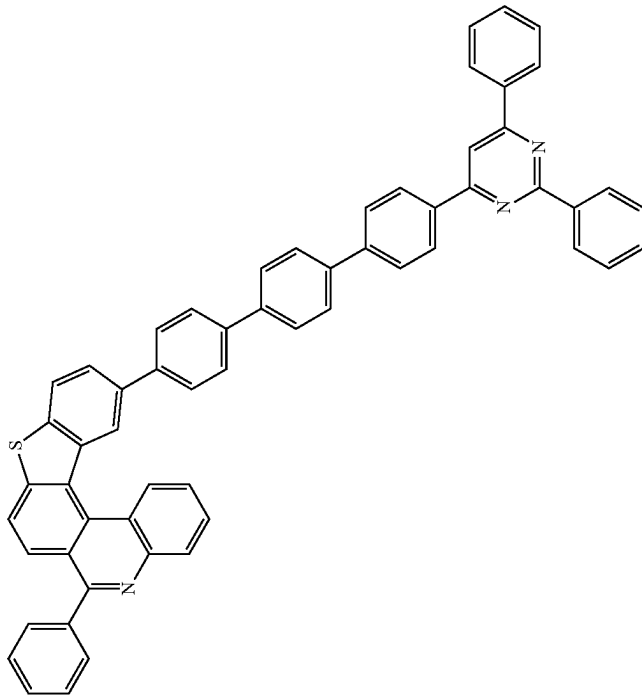
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
716			51%

635

636

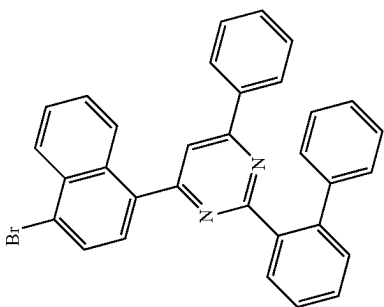
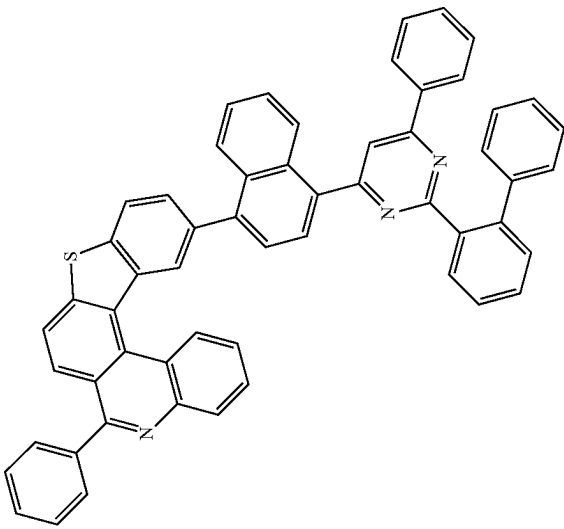
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
717			60%

637

638

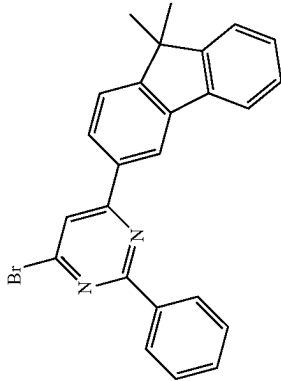
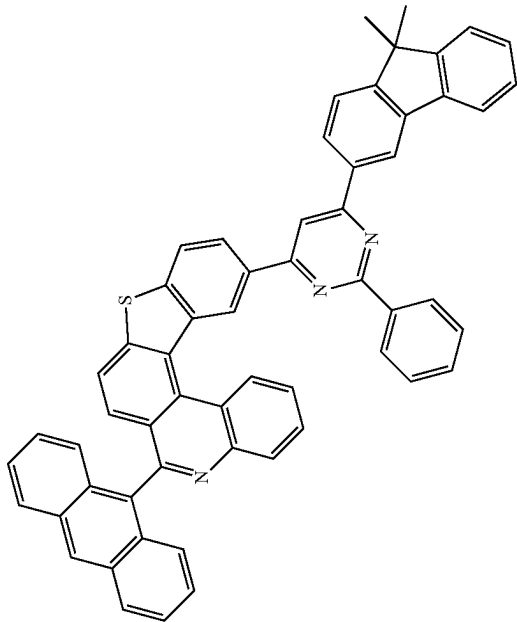
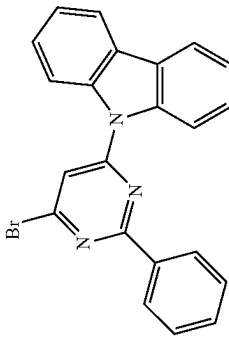
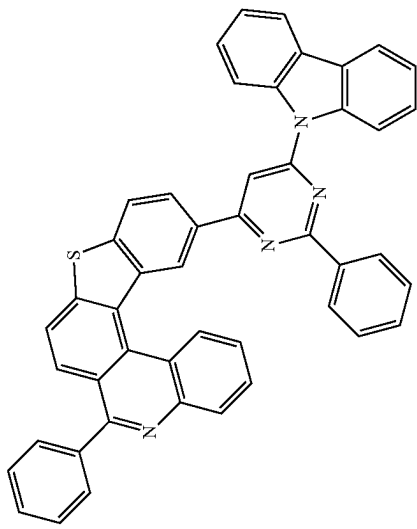
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
721			61%

639

640

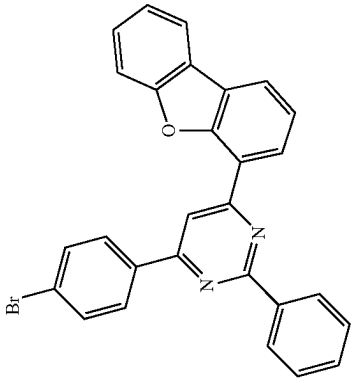
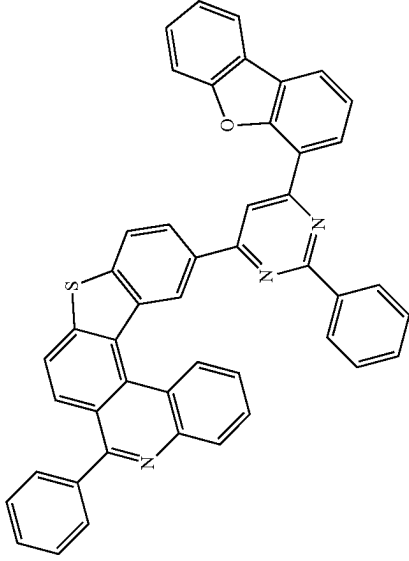
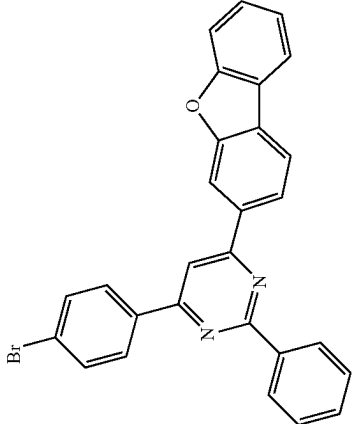
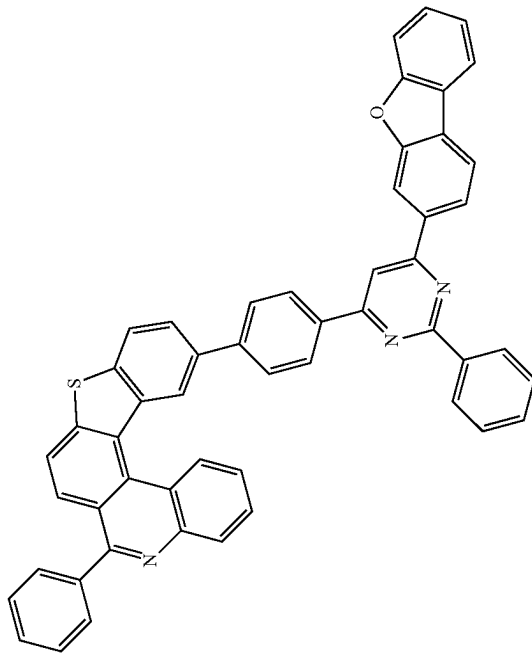
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
728			60%
729			62%

641

642

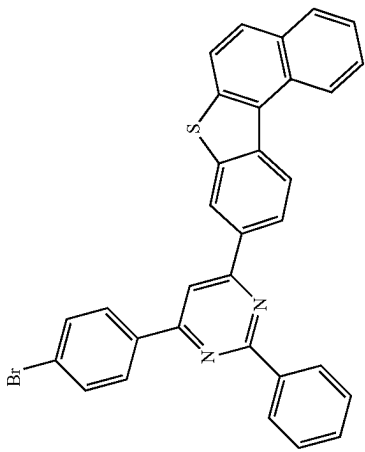
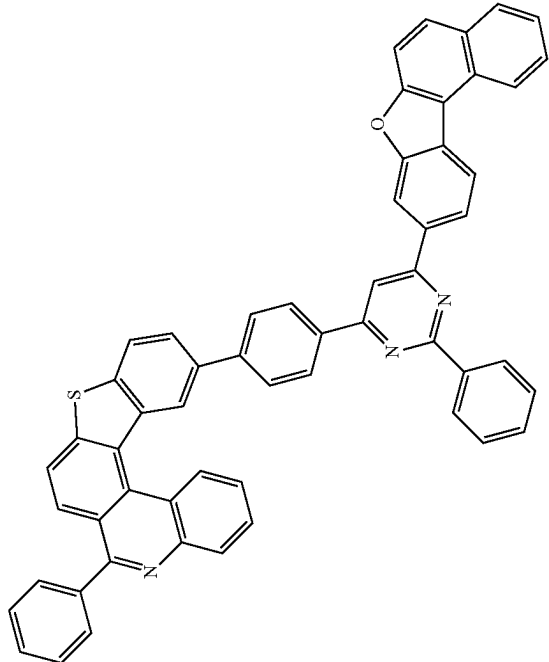
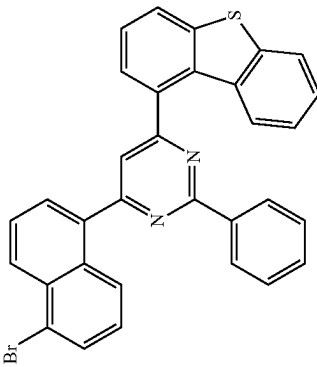
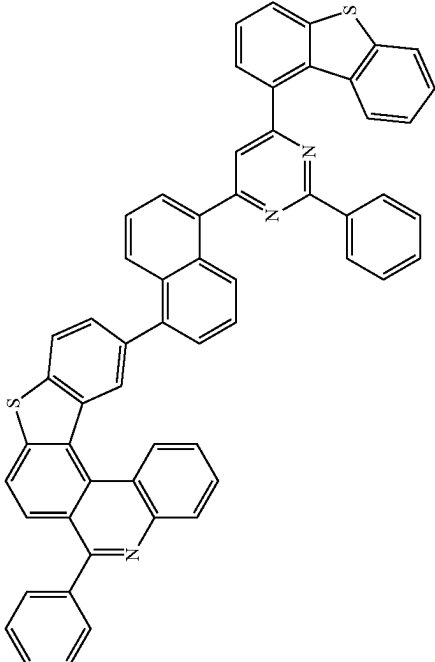
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
731			65%
732			65%

643

644

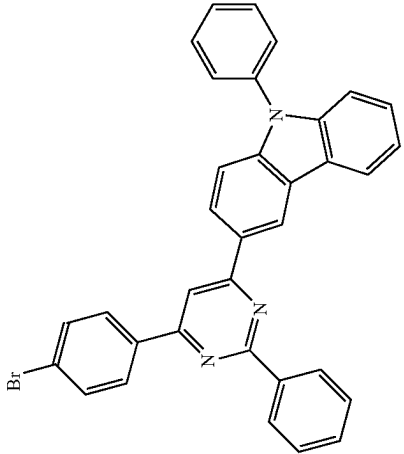
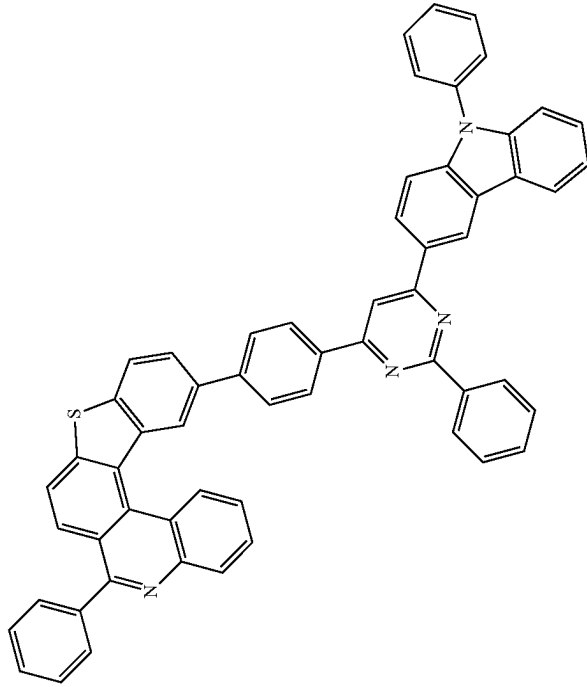
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
733			62%
734			51%

645

646

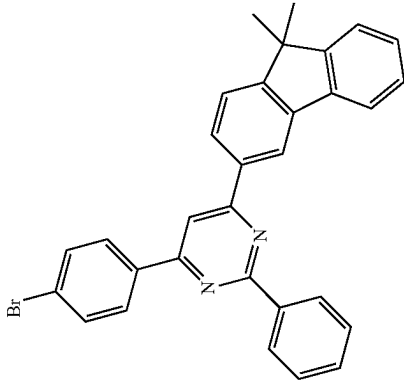
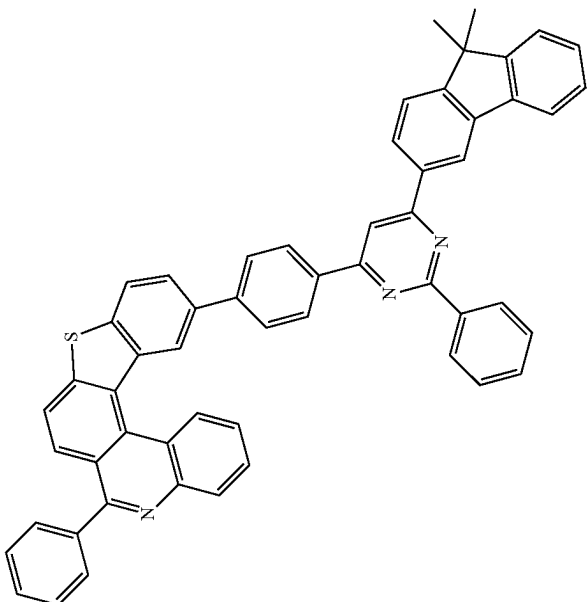
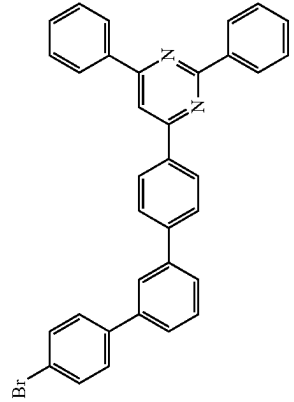
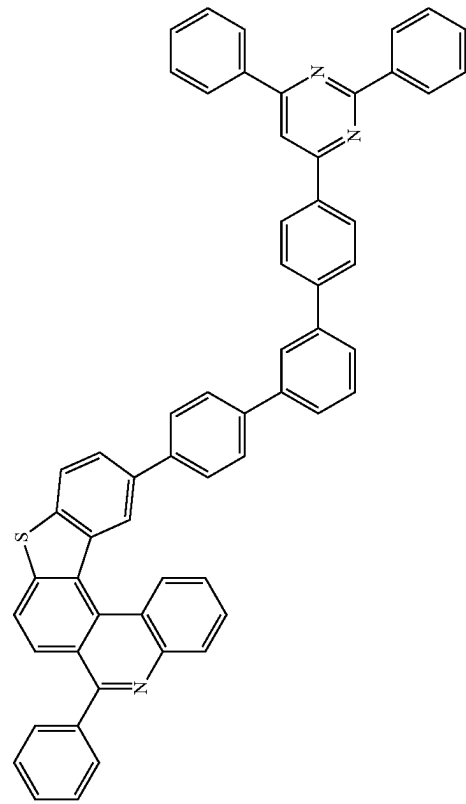
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
735			60%

647

648

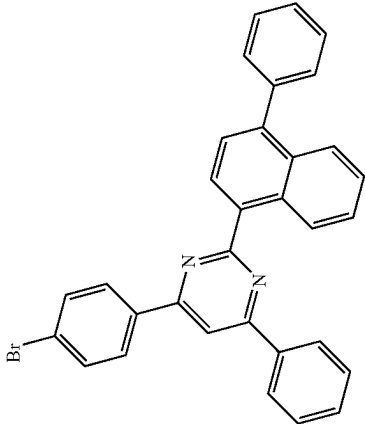
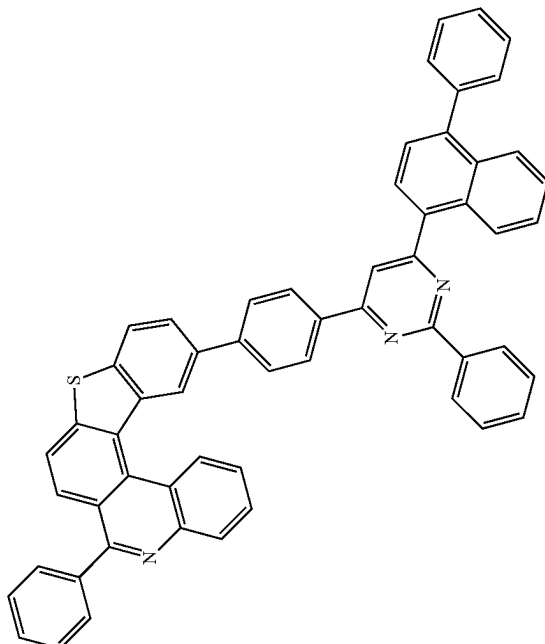
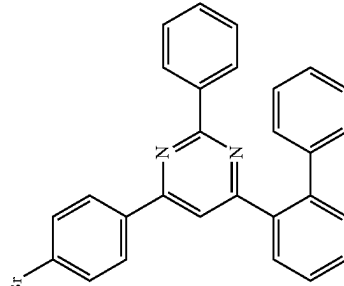
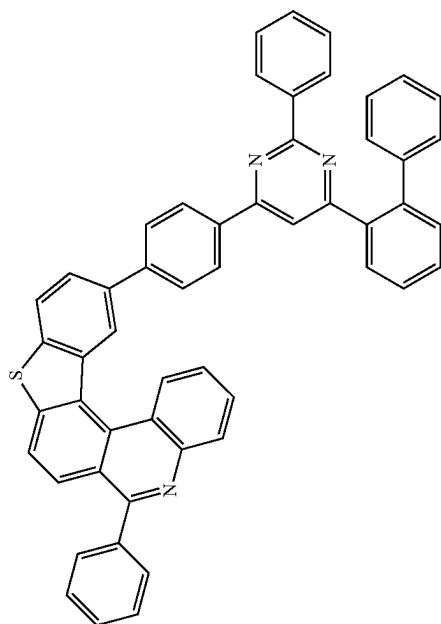
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
736			61%
741			60%

649

650

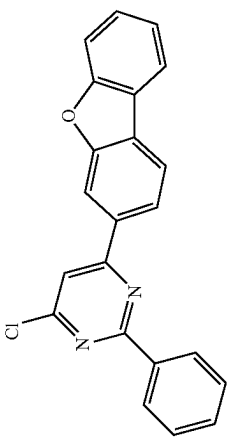
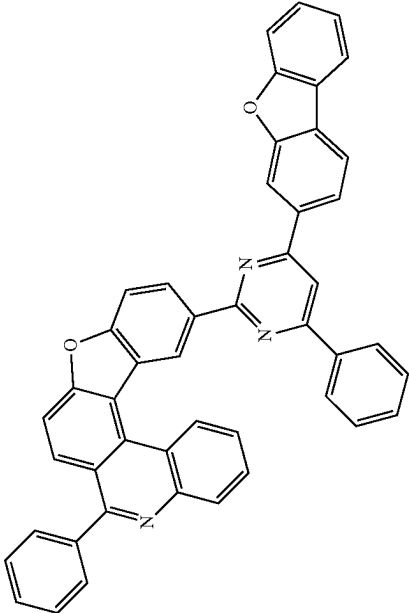
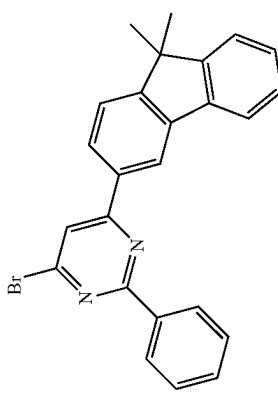
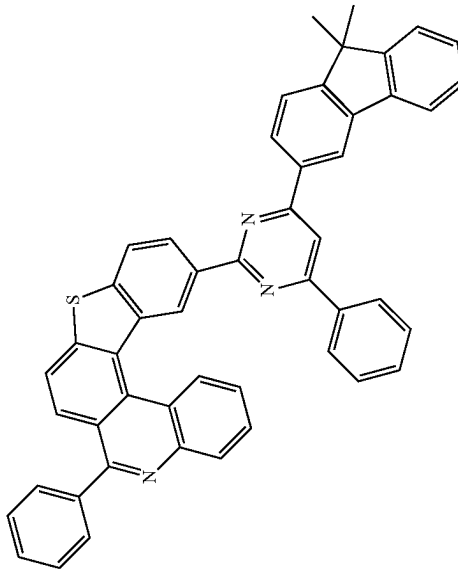
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
743			62%
745			65%

651

652

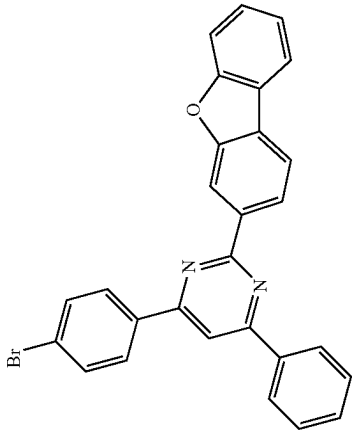
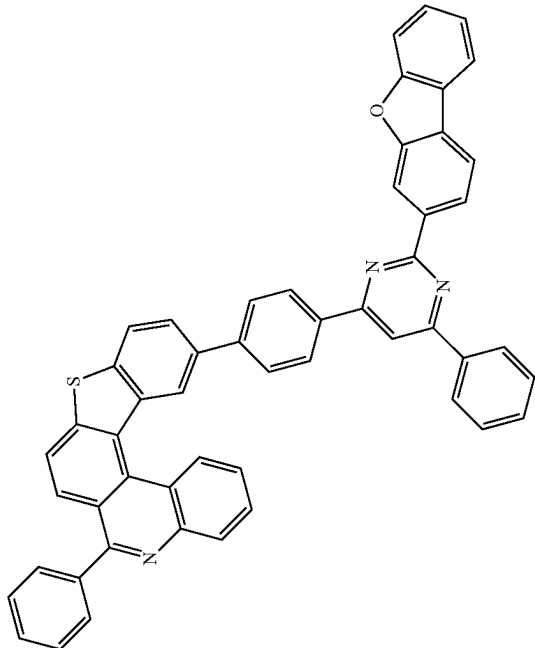
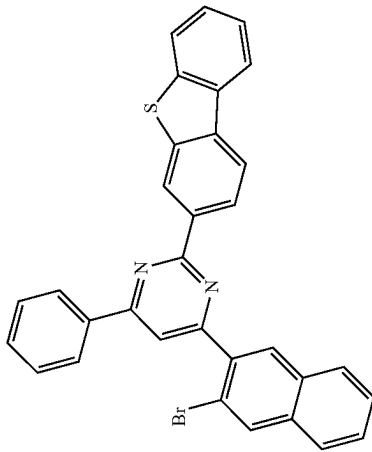
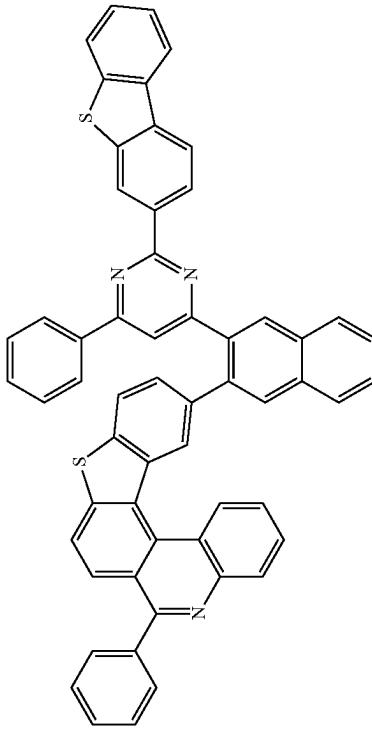
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
748			65%
752			51%

653

654

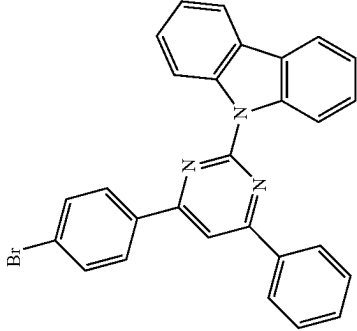
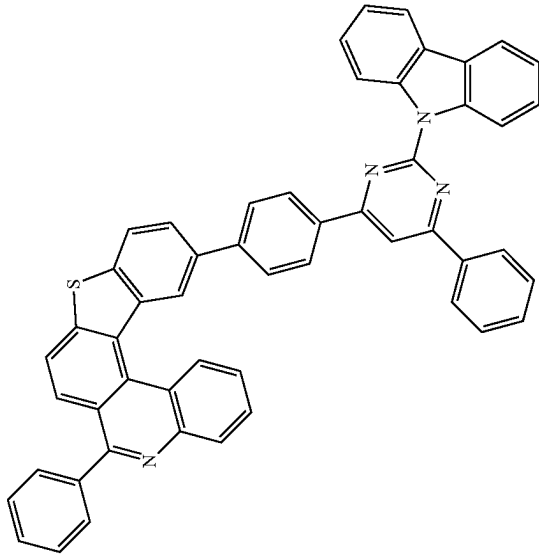
TABLE 13-continued

Compound	Intermediate L	Target Compound	Yield
756			60%
757			61%

655

656

TABLE 13-continued

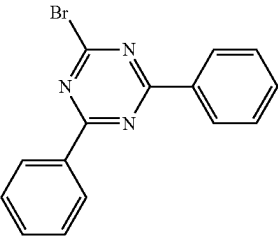
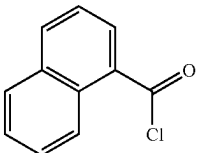
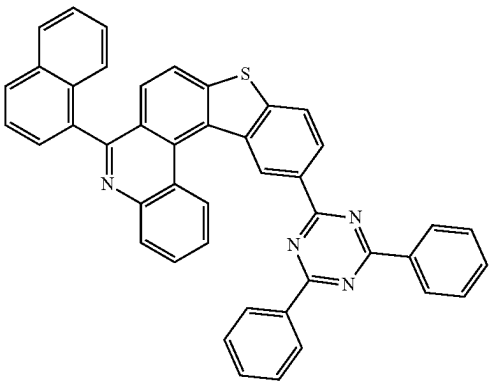
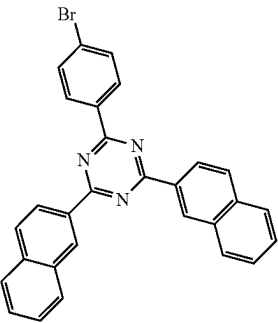
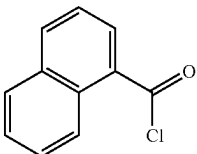
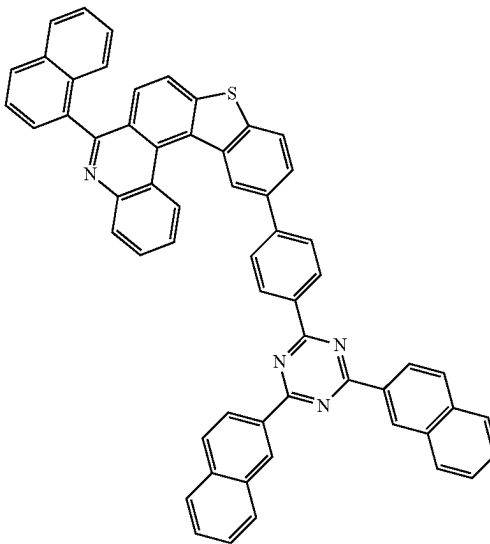
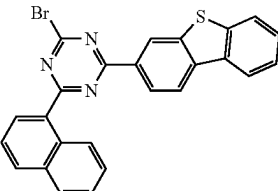
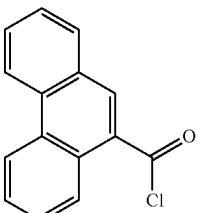
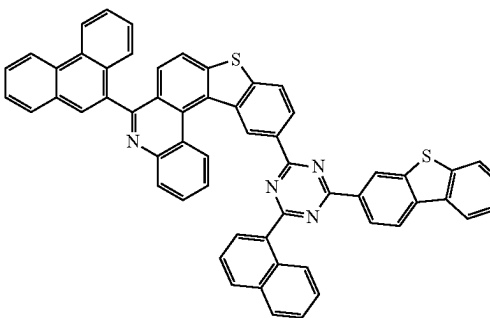
Compound	Intermediate L	Target Compound	Yield
761			60%

657

658

Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 6 of the following Table 14 was used instead of 1-pyrenecarbonyl chloride, 1-bromo-8-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate M of the following Table 14 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

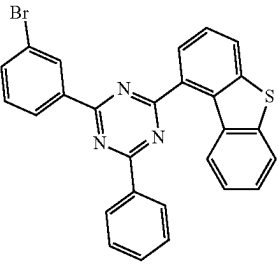
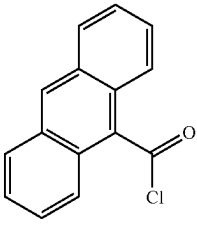
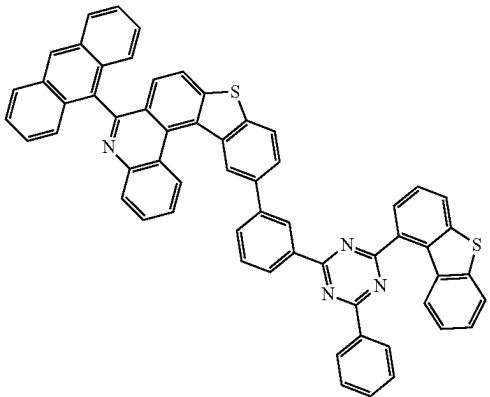
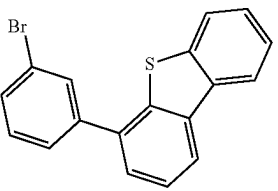
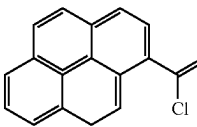
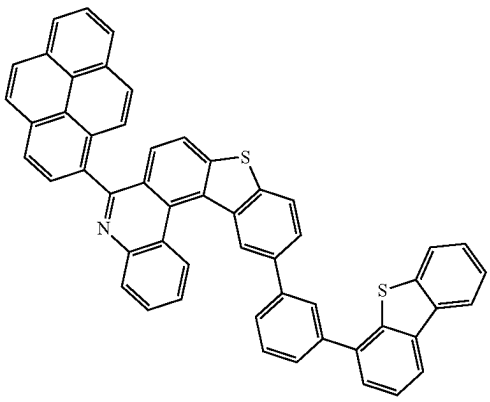
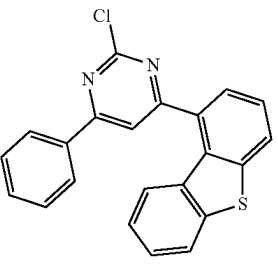
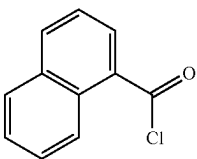
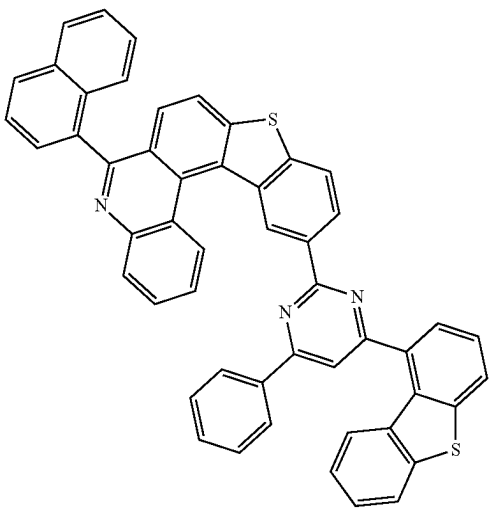
TABLE 14

Compound	Intermediate M	Intermediate 6	Target Compound	Yield
620				53%
672				62%
683				60%

659

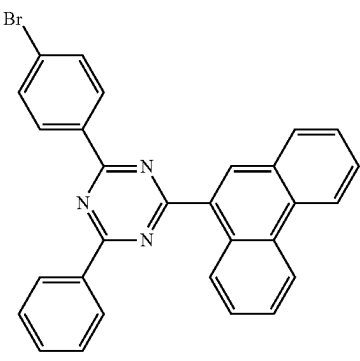
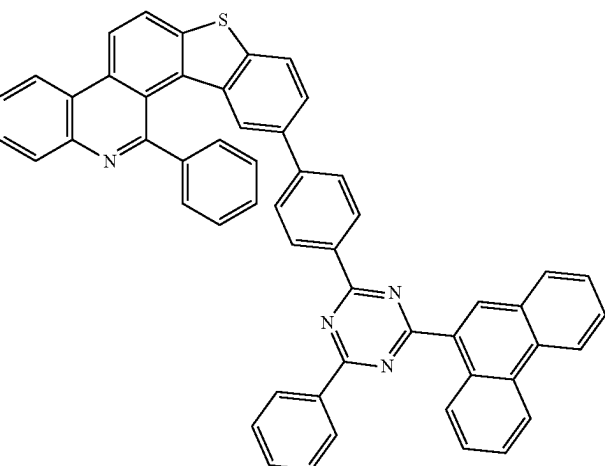
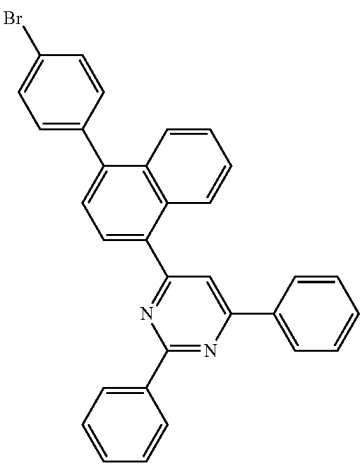
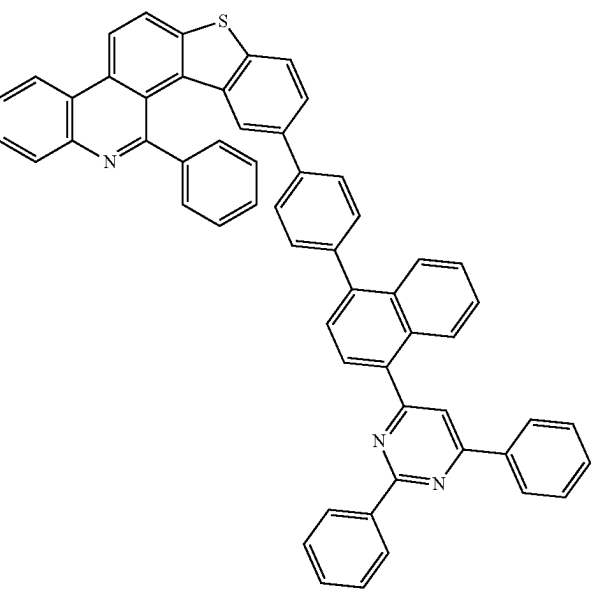
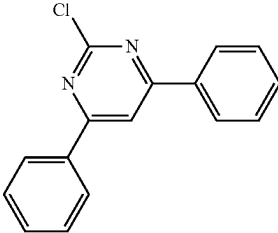
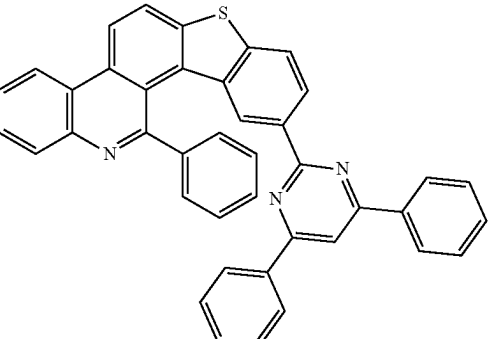
660

TABLE 14-continued

Compound	Intermediate M	Intermediate 6	Target Compound	Yield
692				60%
703				62%
750				52%

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 2-bromo-8-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate N of the following Table 15 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

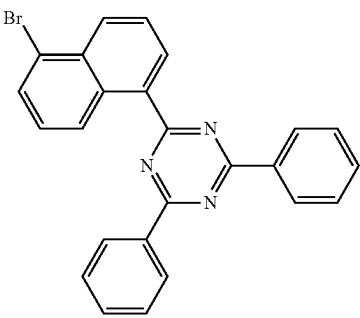
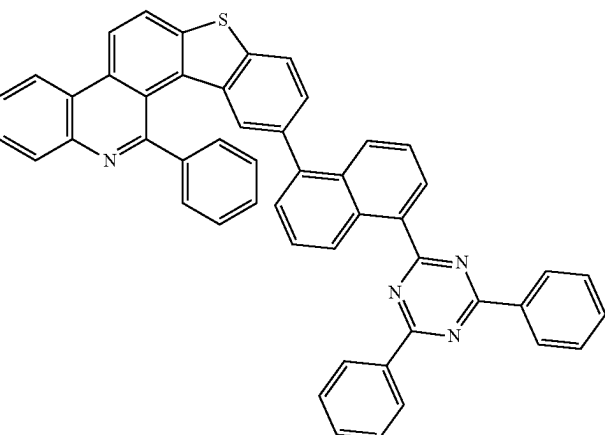
TABLE 15

Compound	Intermediate N	Target Compound	Yield
764			53%
766			62%
767			65%

663

664

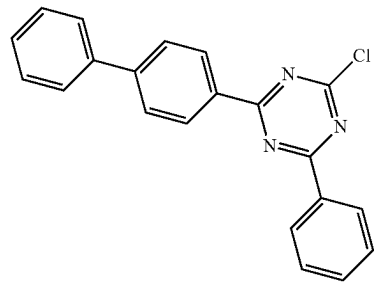
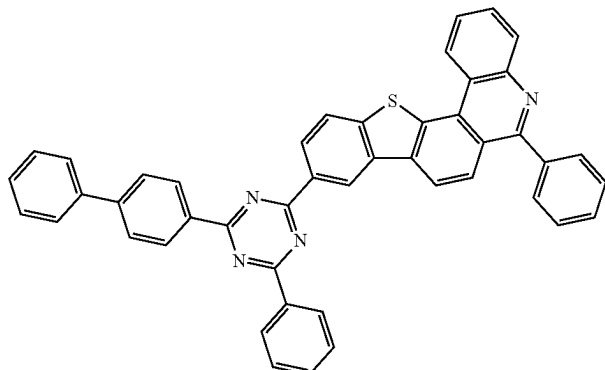
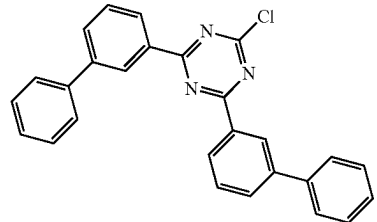
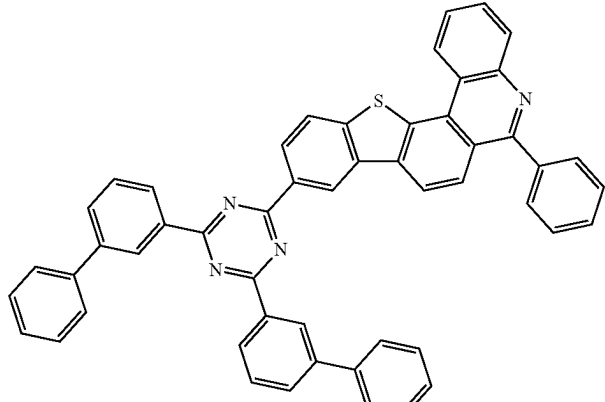
TABLE 15-continued

Compound	Intermediate N	Target Compound	Yield
769			65%

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 6-bromo-2-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-

chlorodibenzo[b,d]furan, and Intermediate O of the following Table 16 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

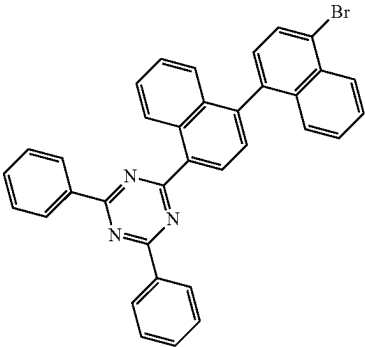
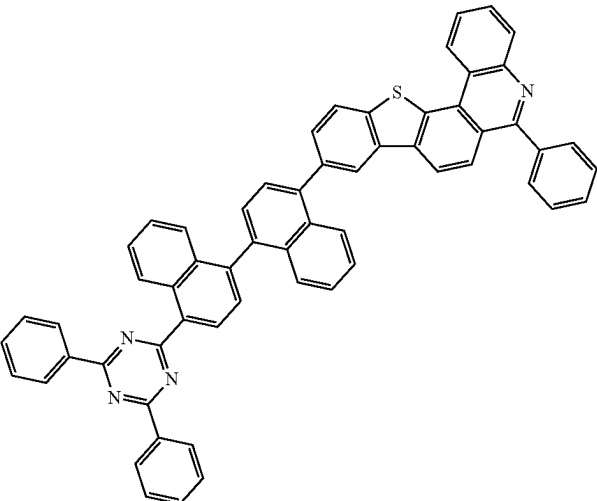
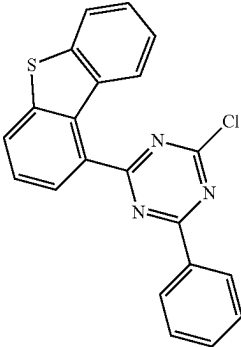
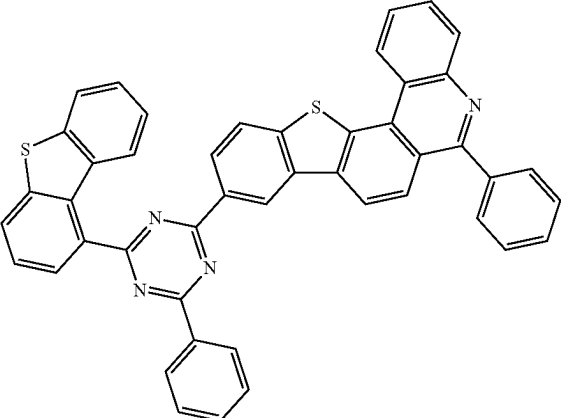
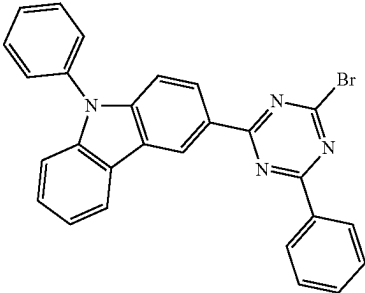
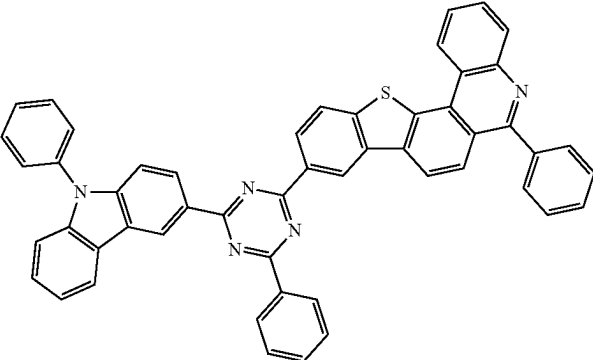
TABLE 16

Compound	Intermediate O	Target Compound	Yield
771			62%
773			62%

665

666

TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
775			65%
783			62%
784			51%

667

668

TABLE 16-continued

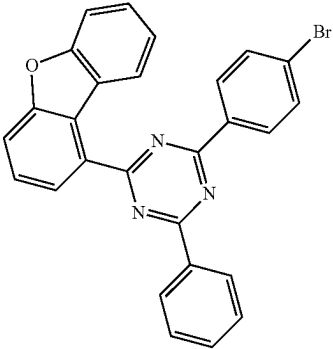
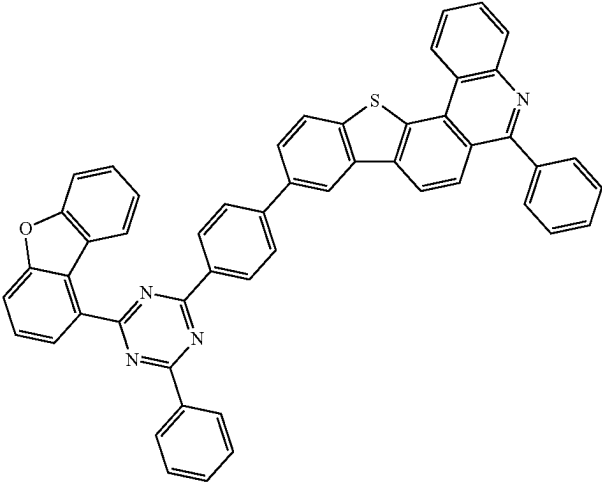
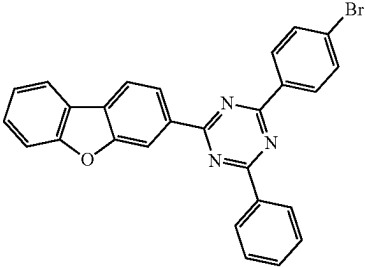
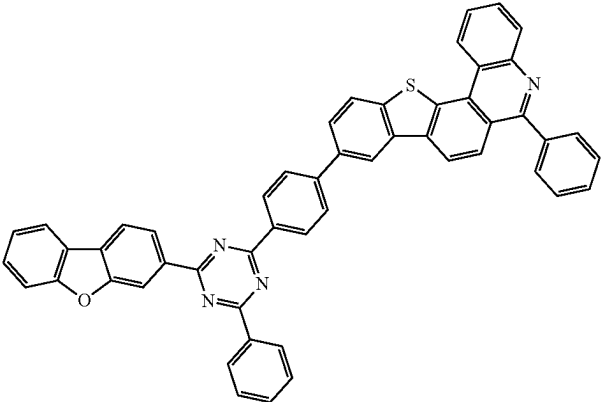
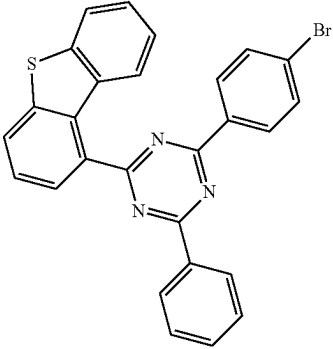
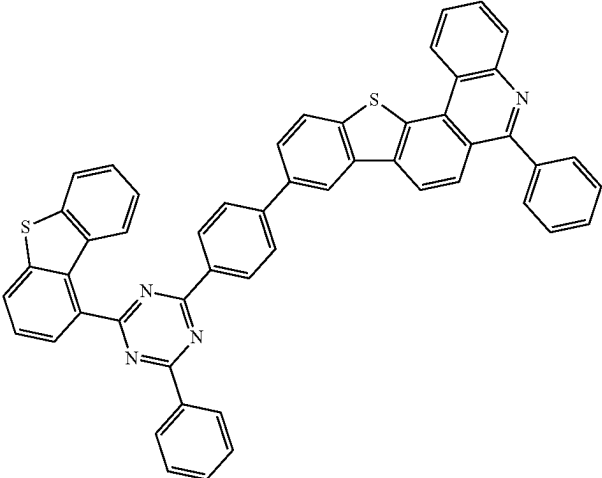
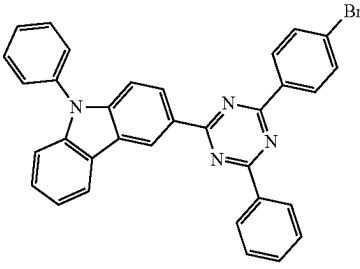
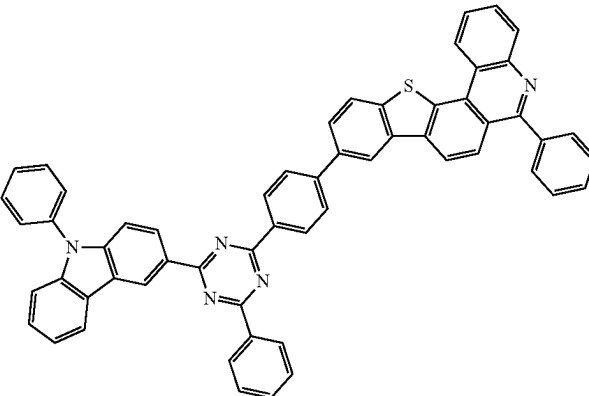
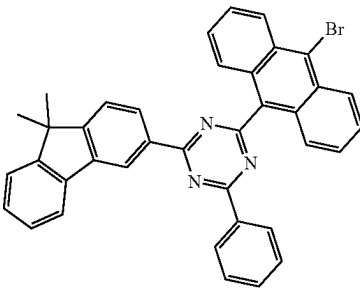
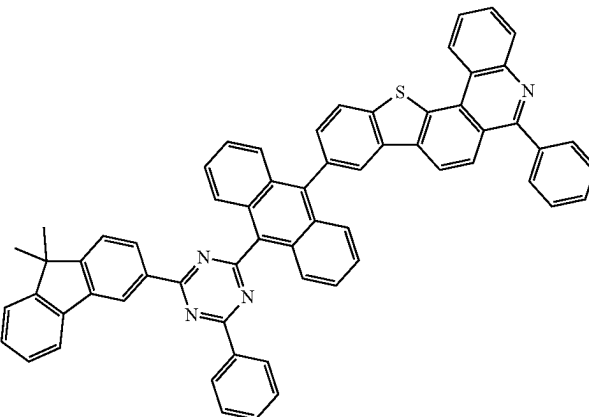
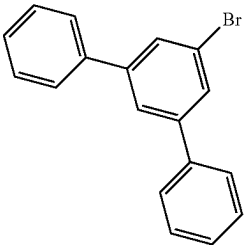
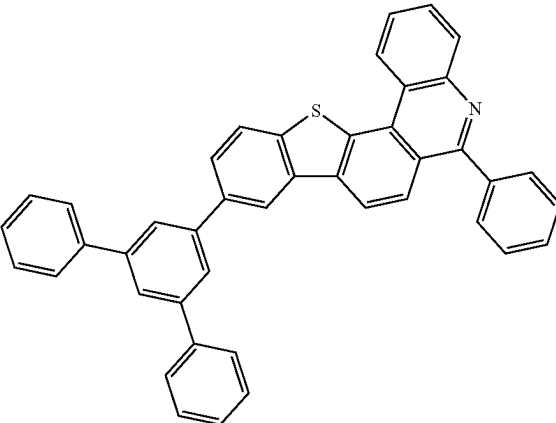
Compound	Intermediate O	Target Compound	Yield
787			61%
789			60%
791			62%

TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
792			65%
793			65%
798			62%

671

672

TABLE 16-continued

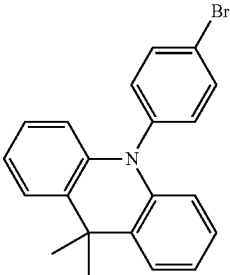
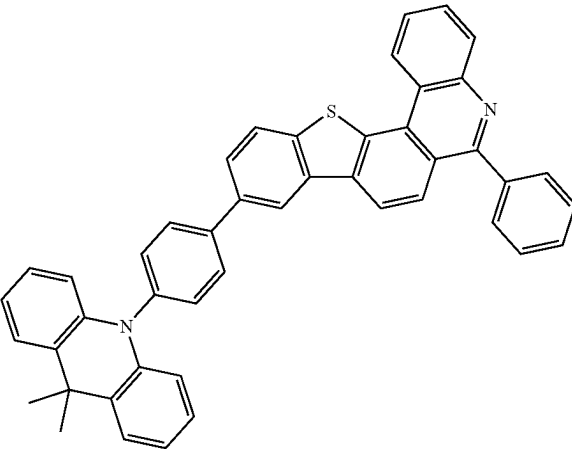
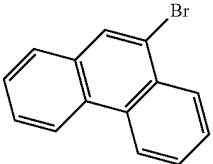
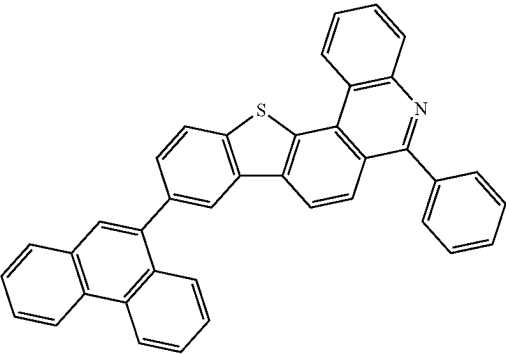
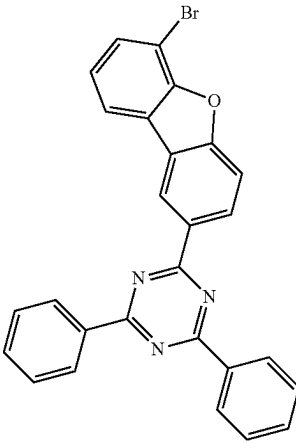
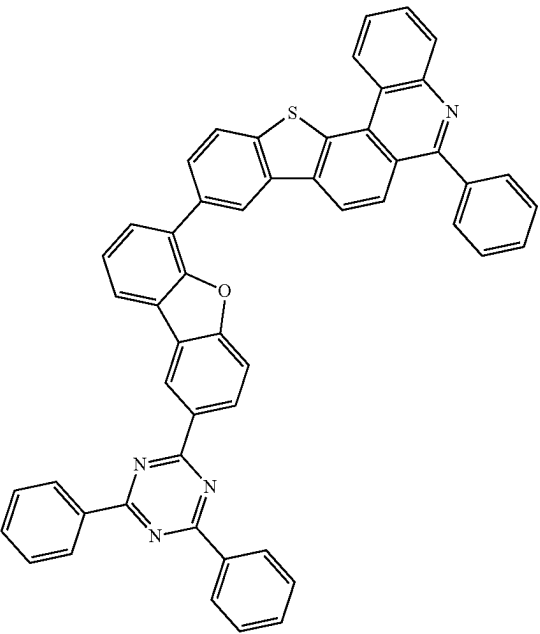
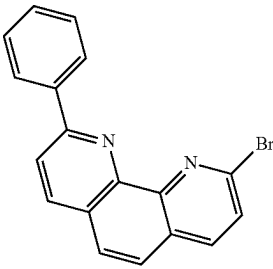
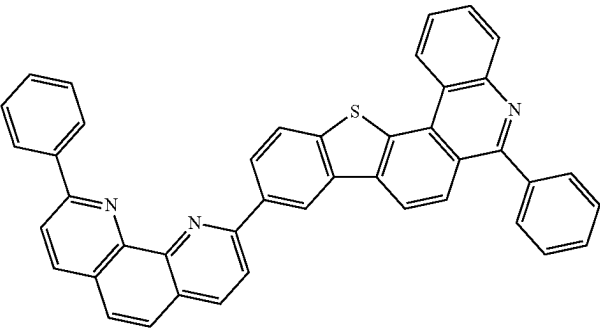
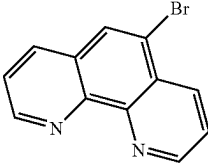
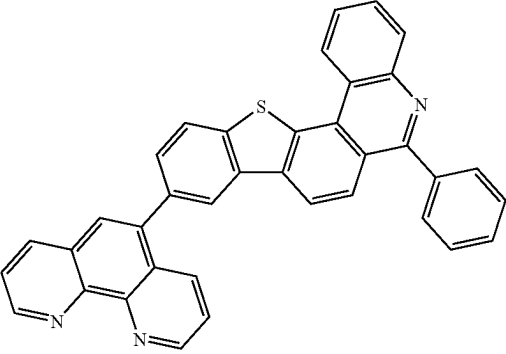
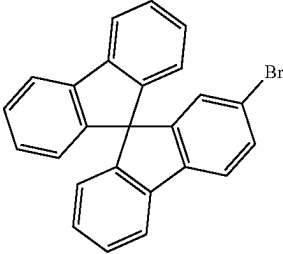
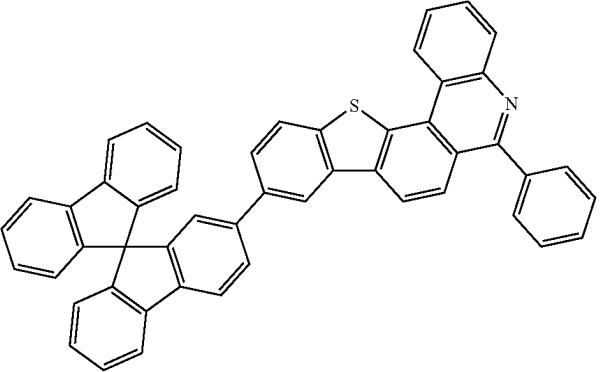
Compound	Intermediate O	Target Compound	Yield
801			51%
802			60%
807			61%

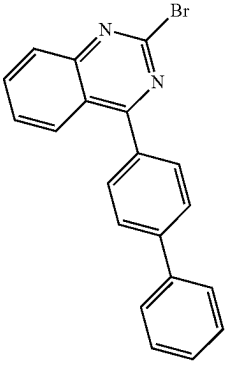
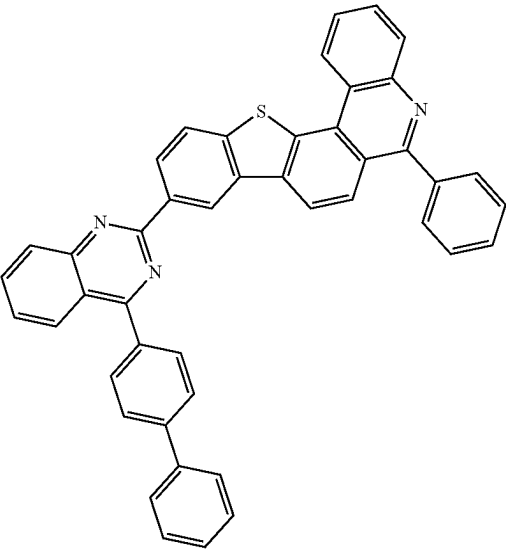
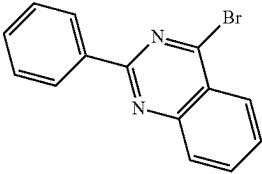
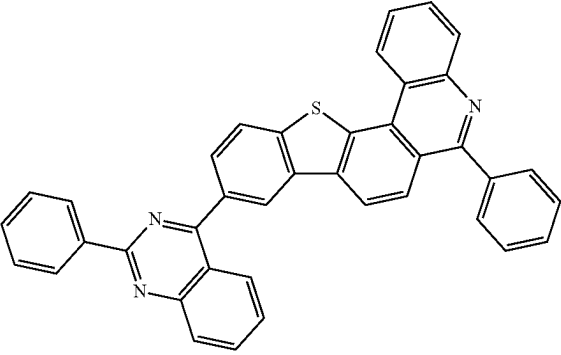
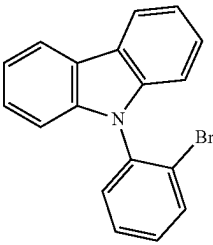
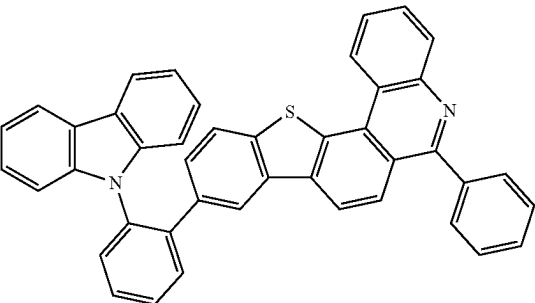
TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
809			60%
810			62%
814			65%

675

676

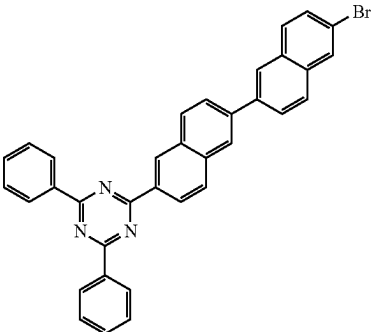
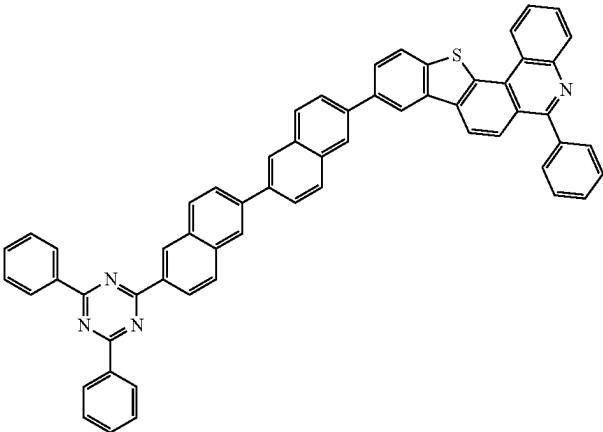
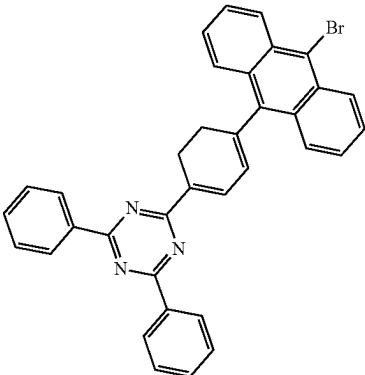
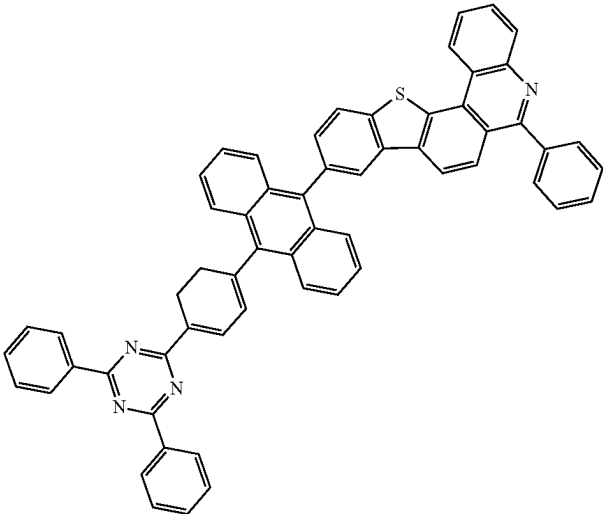
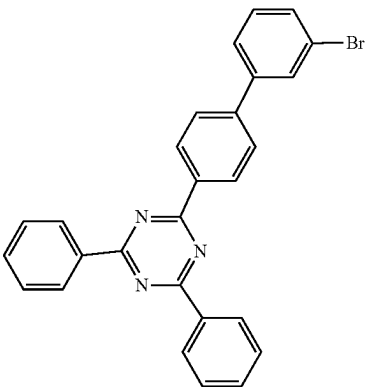
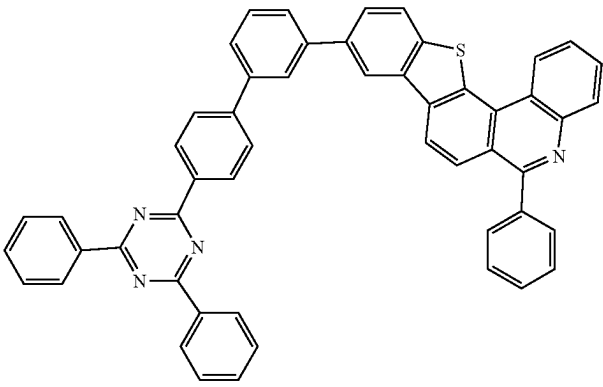
TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
816			65%
817			62%
818			51%

677

678

TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
822			60%
823			61%
829			62%

679

680

TABLE 16-continued

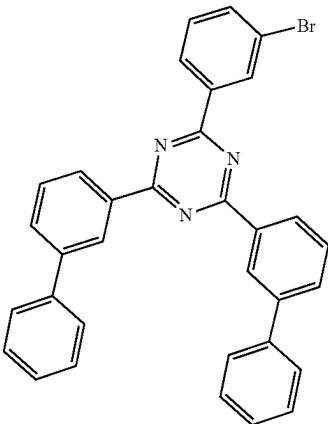
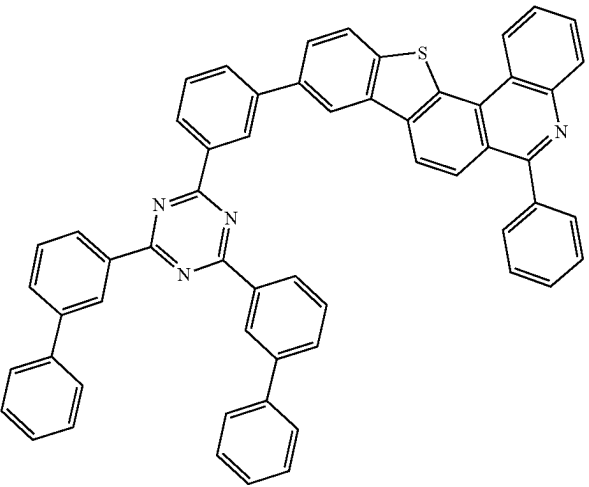
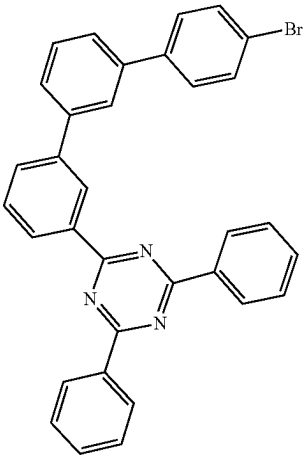
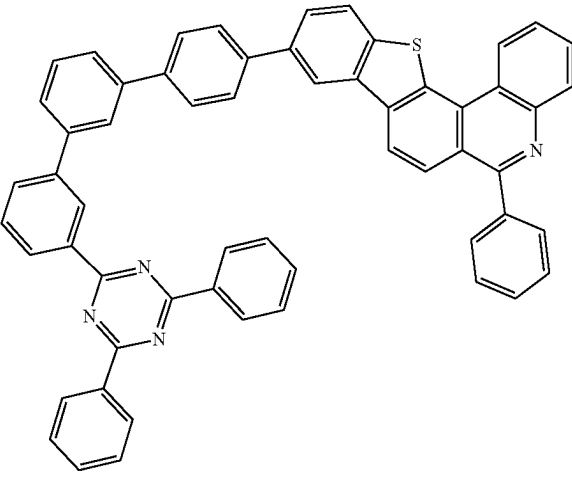
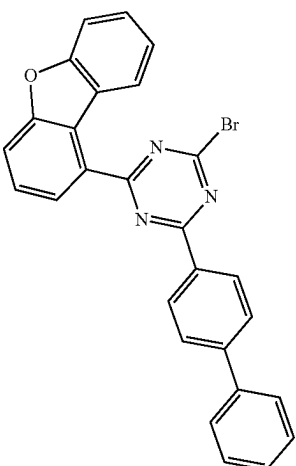
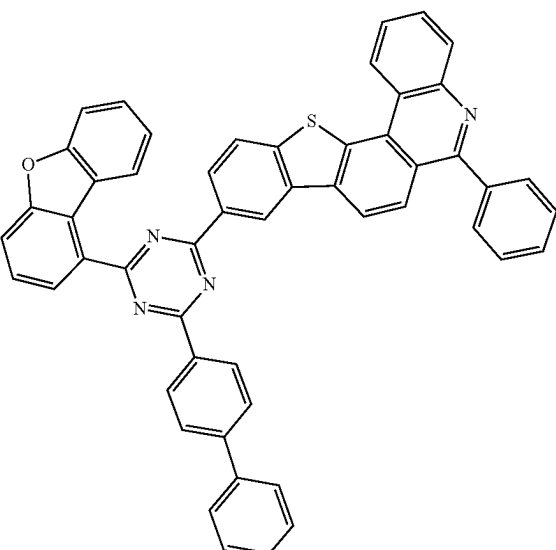
Compound	Intermediate O	Target Compound	Yield
830			65%
831			65%
833			62%

TABLE 16-continued

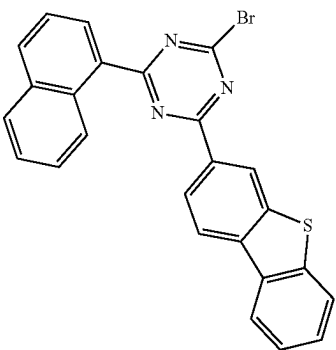
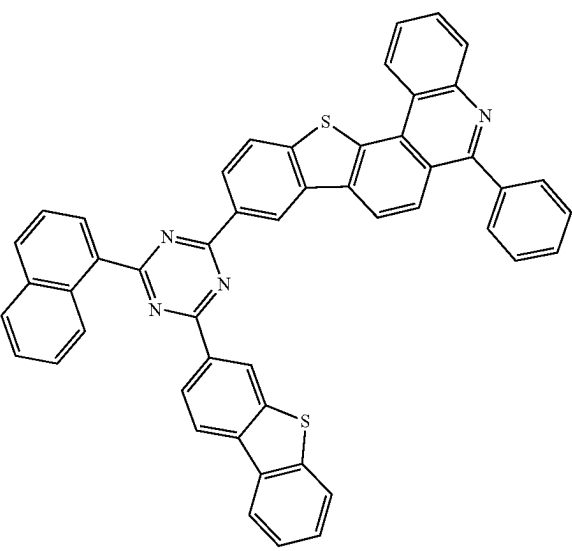
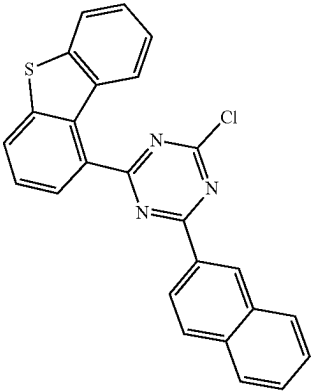
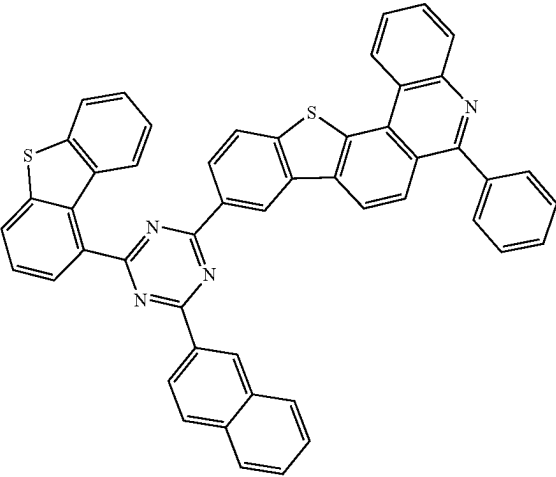
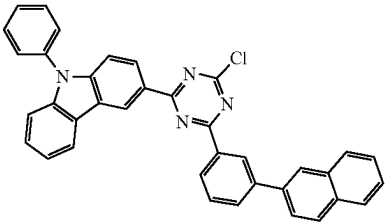
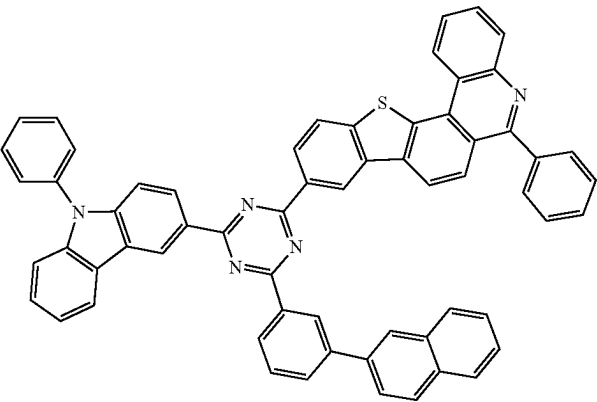
Compound	Intermediate O	Target Compound	Yield
836			51%
837			60%
838			61%

TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
842			60%
844			62%
847			65%
848			65%

685

686

TABLE 16-continued

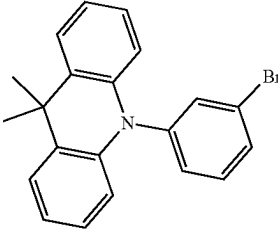
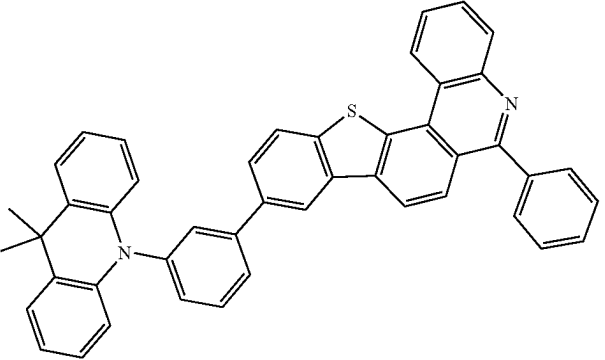
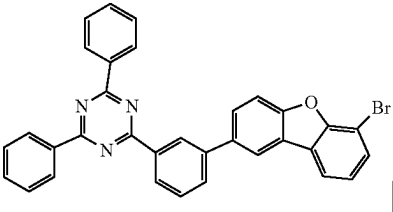
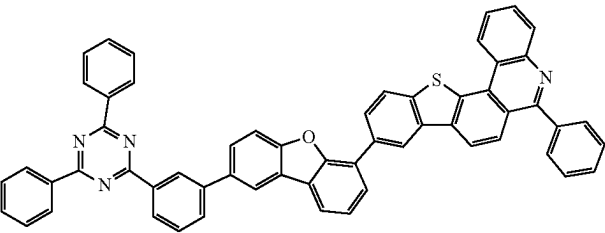
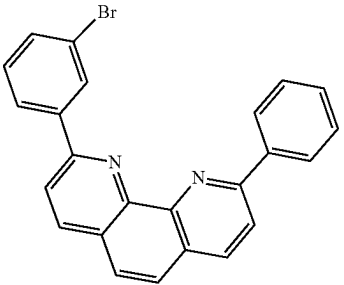
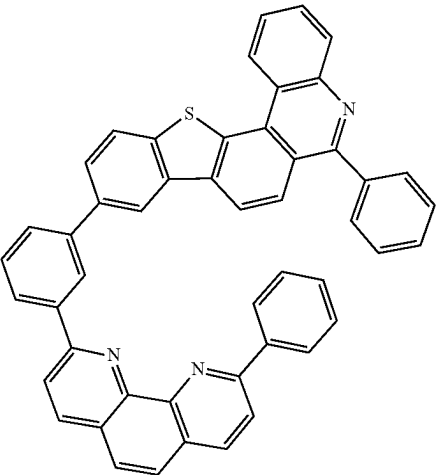
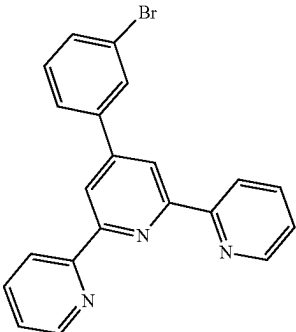
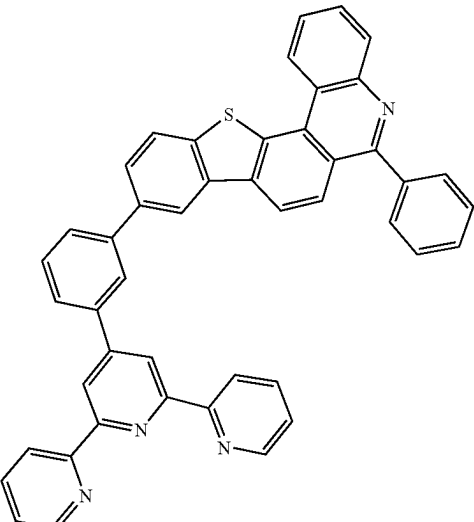
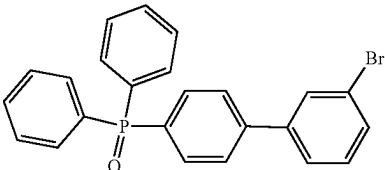
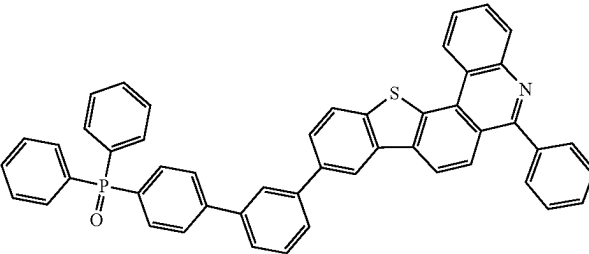
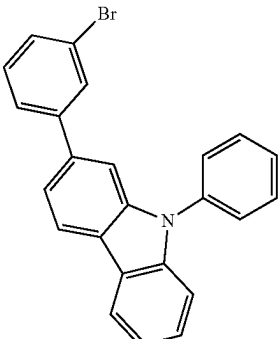
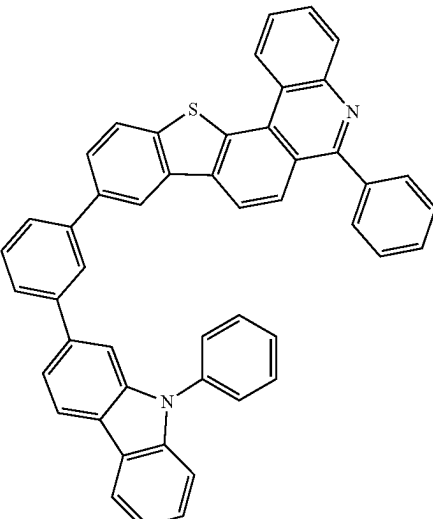
Compound	Intermediate O	Target Compound	Yield
849			62%
858			51%
860			60%

TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
861			61%
862			60%
863			62%

689

690

TABLE 16-continued

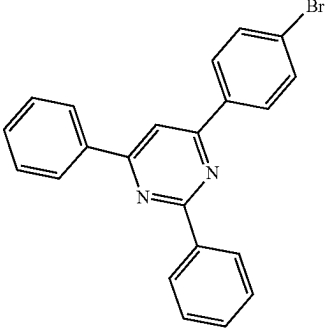
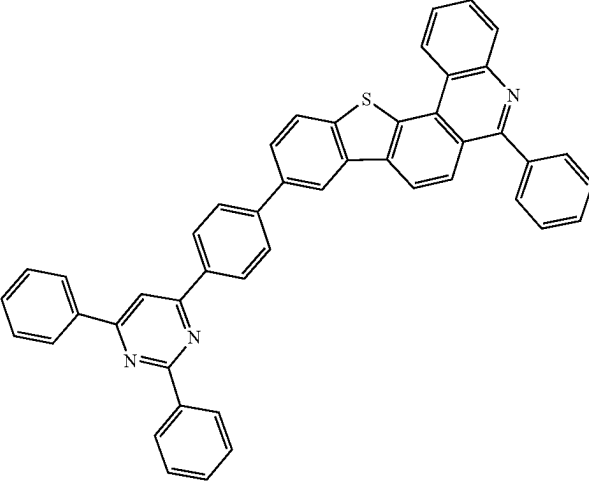
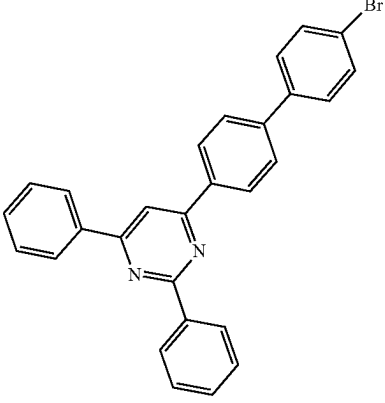
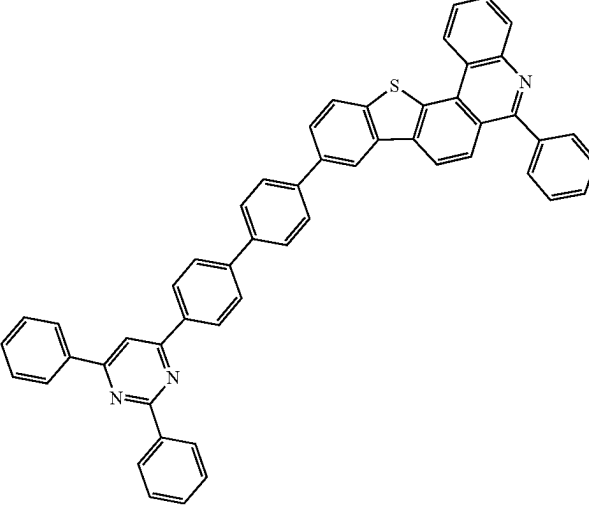
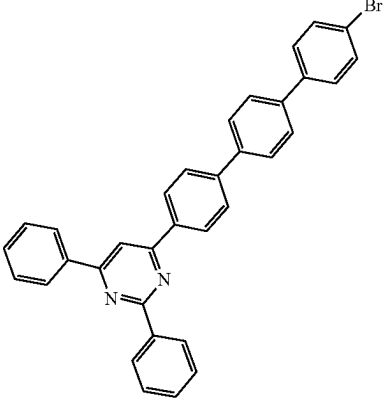
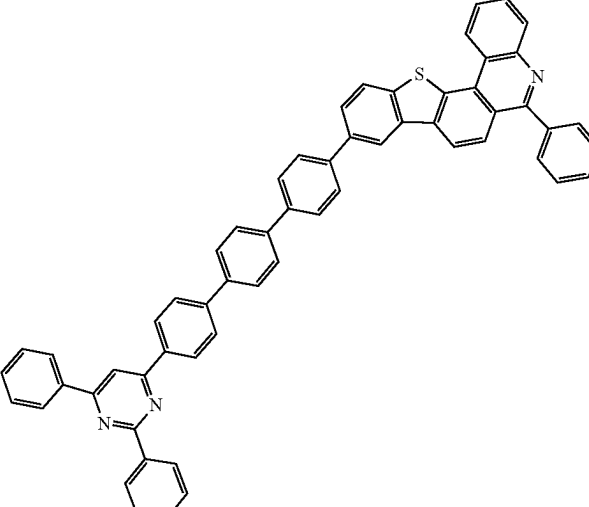
Compound	Intermediate O	Target Compound	Yield
870			65%
871			65%
872			62%

TABLE 16-continued

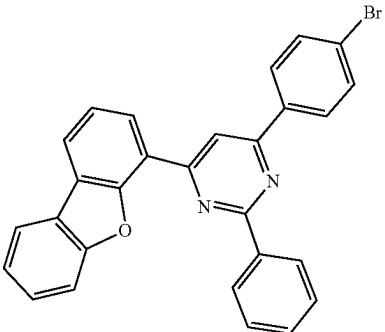
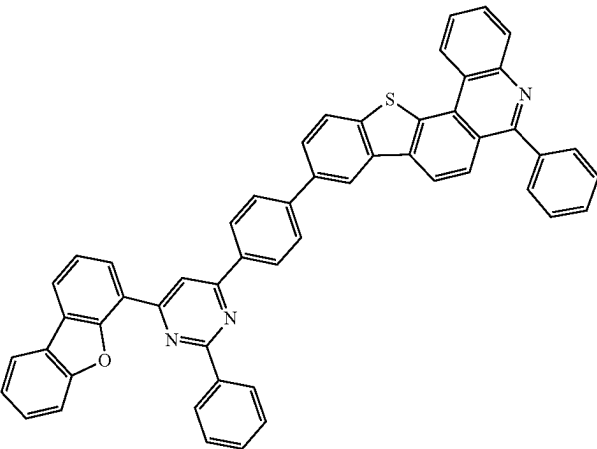
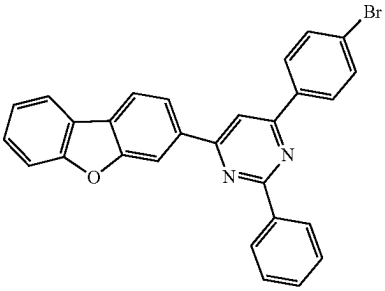
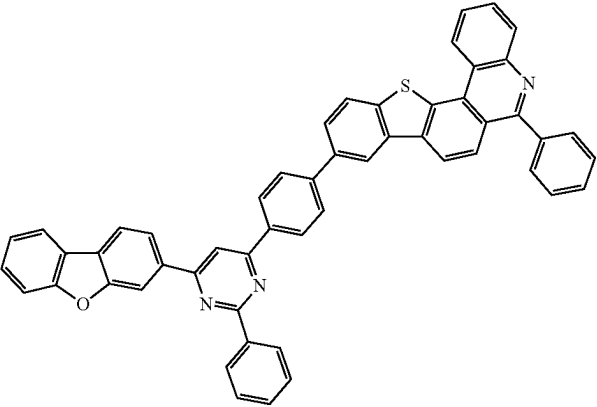
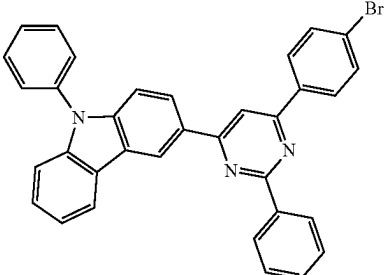
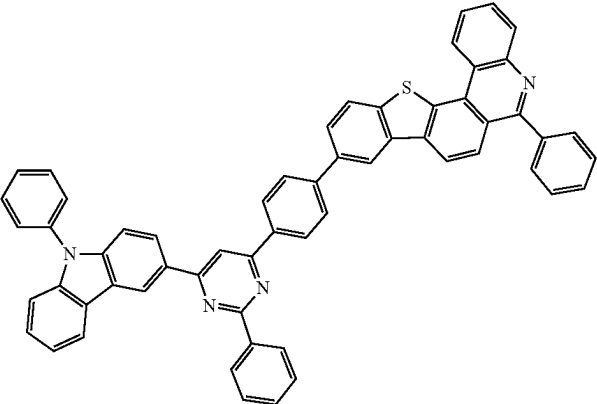
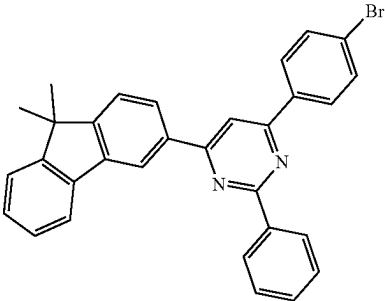
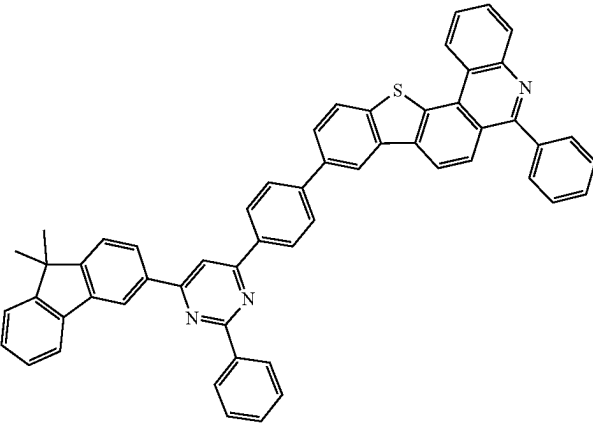
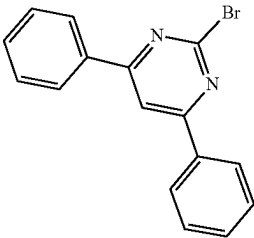
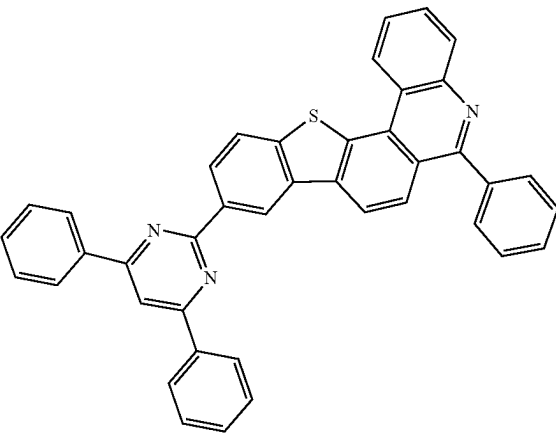
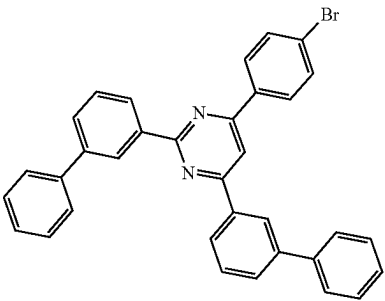
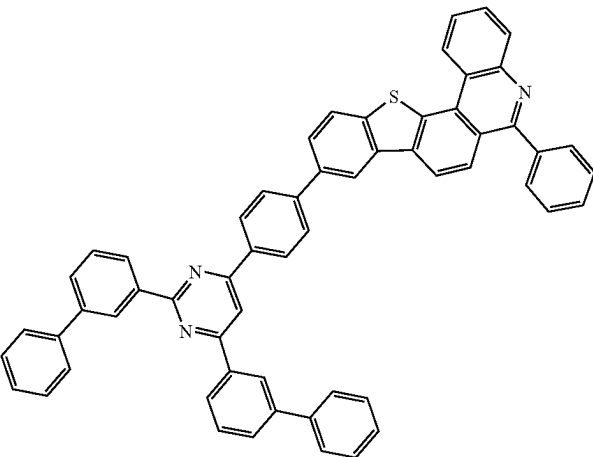
Compound	Intermediate O	Target Compound	Yield
883			51%
884			60%
887			61%

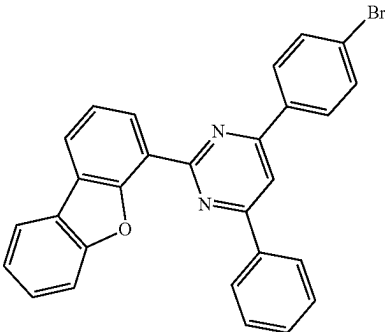
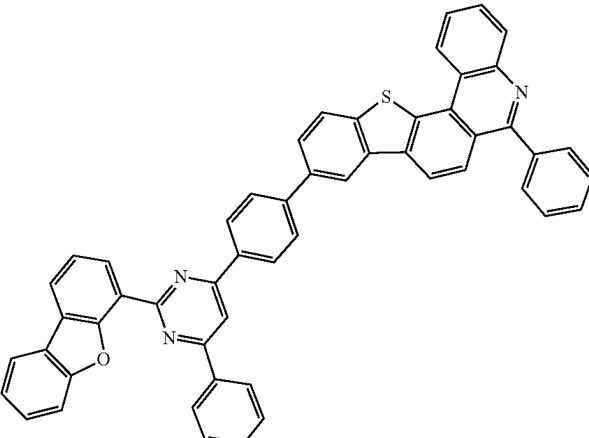
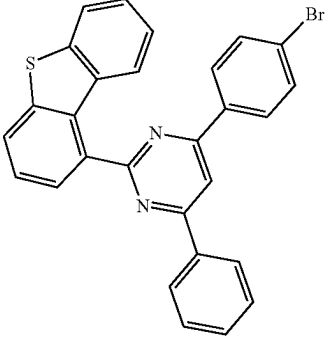
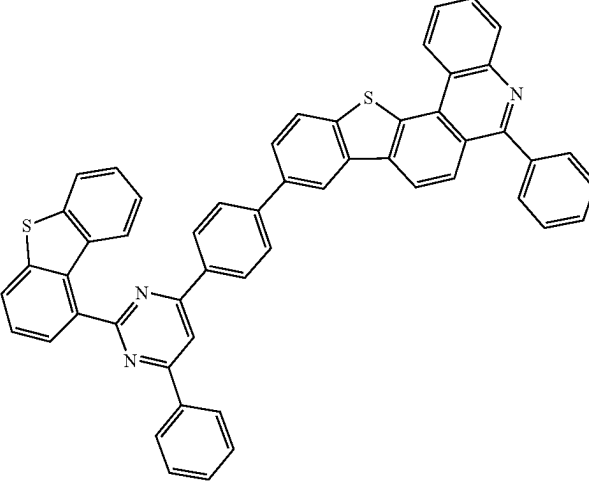
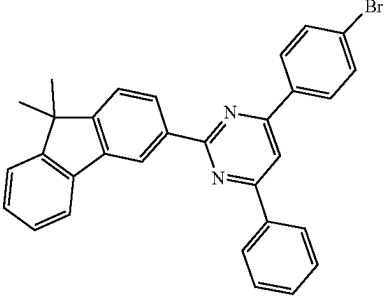
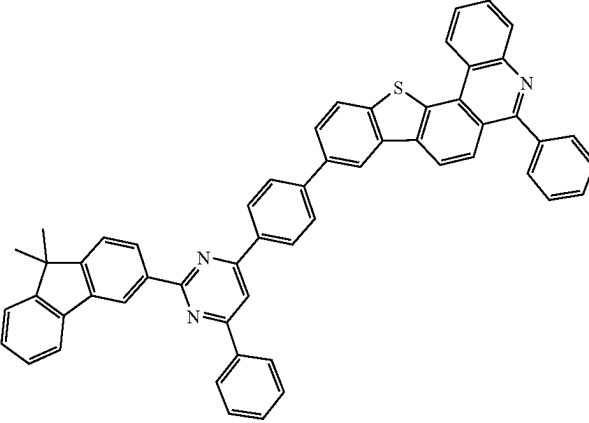
TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
888			60%
890			62%
893			65%

695

696

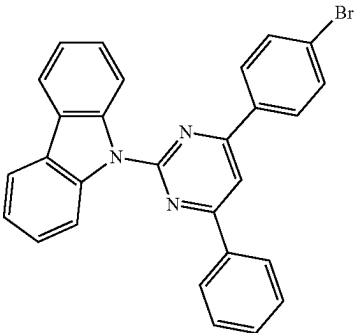
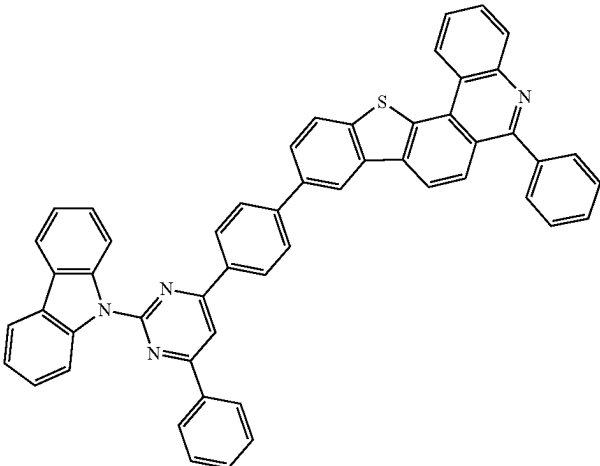
TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
905			65%
908			62%
910			51%

697

698

TABLE 16-continued

Compound	Intermediate O	Target Compound	Yield
911			60%

Target compounds were synthesized in the same manner as in Preparation Example 2 except that Intermediate 7 of the following Table 17 was used instead of 1-pyrenecarbonyl chloride, 6-bromo-2-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate P of the following Table 17 was used instead of 4-(3-bromophenyl)dibenzo[b,d]thiophene.

TABLE 17

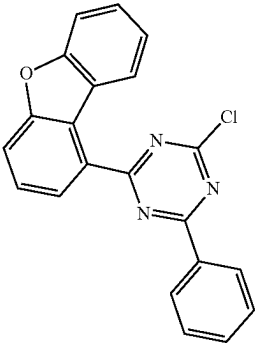
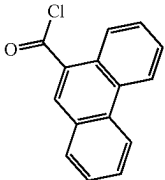
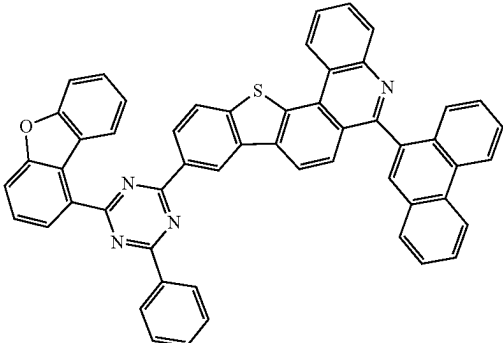
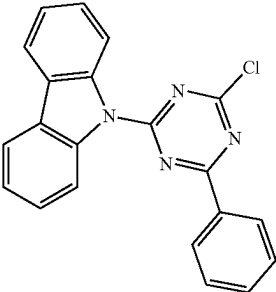
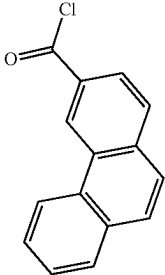
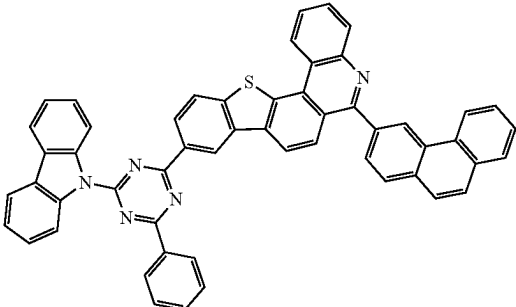
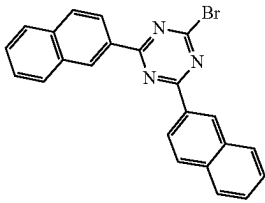
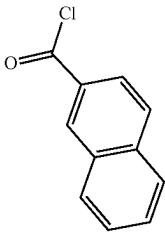
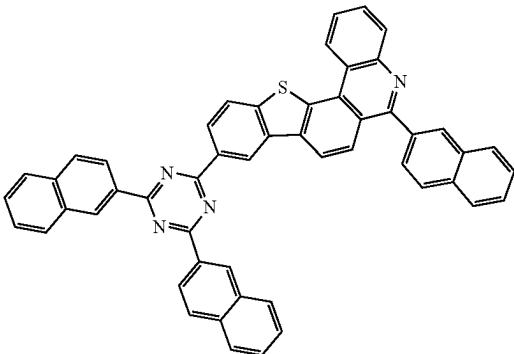
Compound	Intermediate P	Intermediate 7	Target Compound	Yield
779				56%
786				56%

TABLE 17-continued

Compound	Intermediate P	Intermediate 7	Target Compound	Yield
825				60%

Target compounds were synthesized in the same manner as in Preparation Example 1 except that 1-bromo-6-chlorodibenzo[b,d]thiophene was used instead of 1-bromo-8-chlorodibenzo[b,d]furan, and Intermediate Q of the following Table 18 was used instead of 2-(4-bromophenyl)-4,6-diphenyl-1,3,5-triazine.

TABLE 18

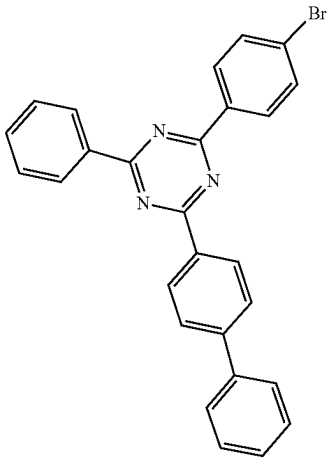
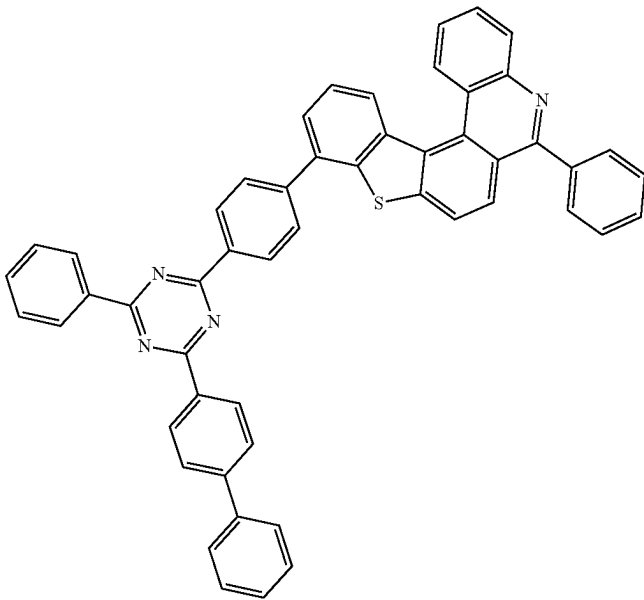
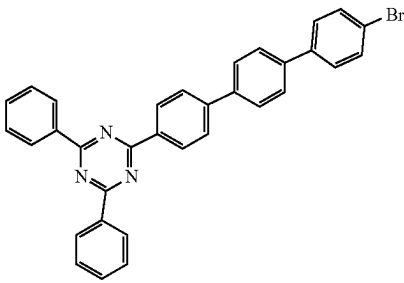
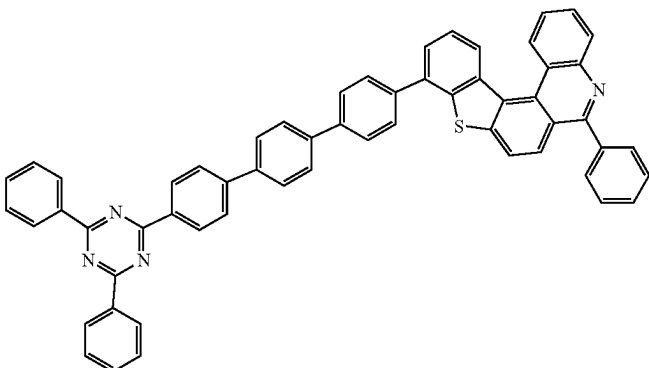
Compound	Intermediate Q	Target Compound	Yield
916			53%
918			62%

TABLE 18-continued

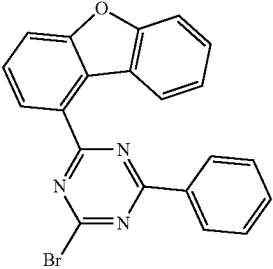
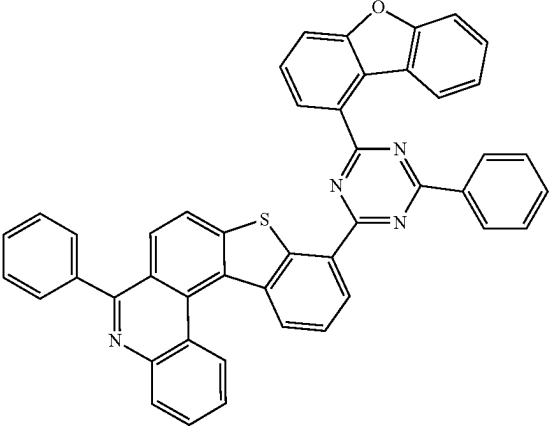
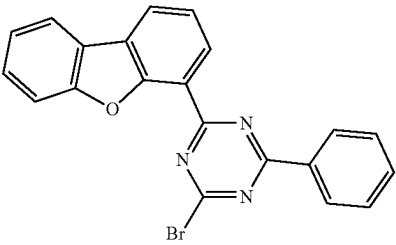
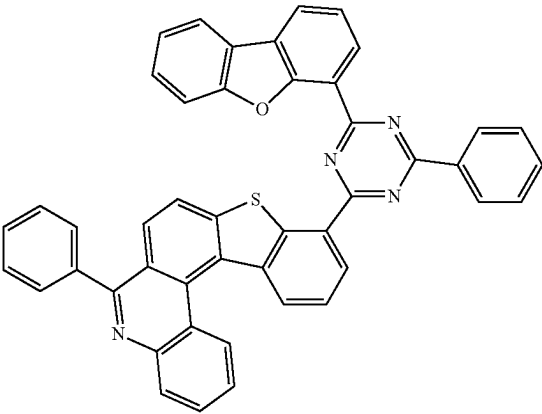
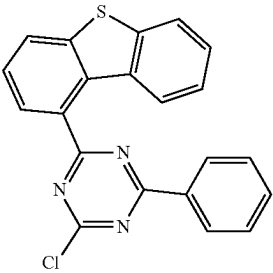
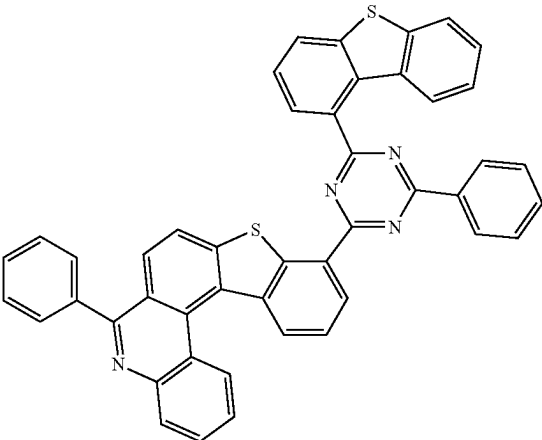
Compound	Intermediate Q	Target Compound	Yield
919			65%
920			65%
923			62%

TABLE 18-continued

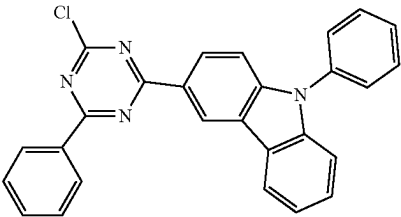
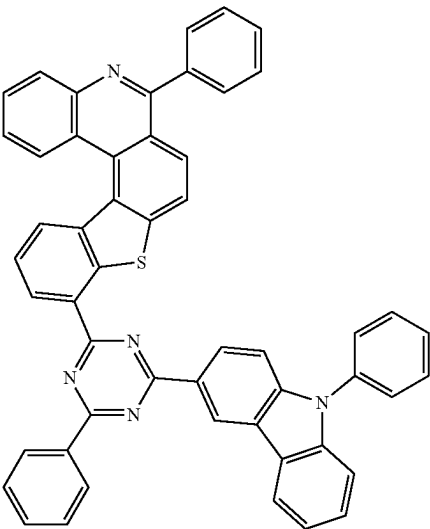
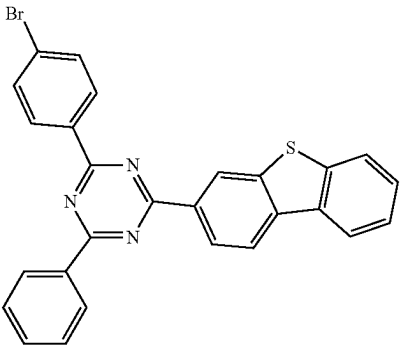
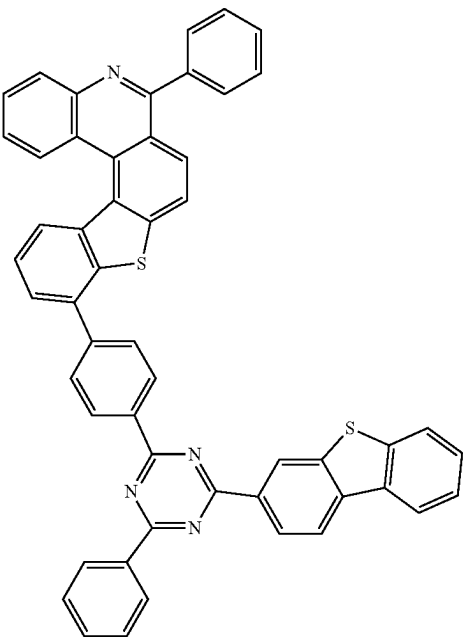
Compound	Intermediate Q	Target Compound	Yield
924			51%
930			60%

TABLE 18-continued

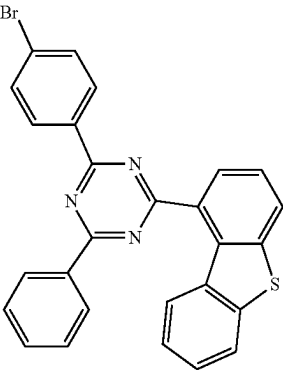
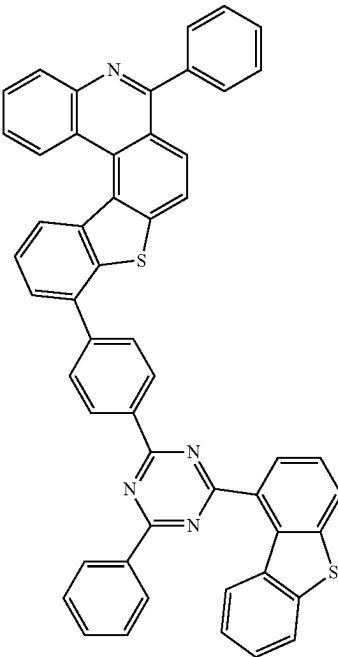
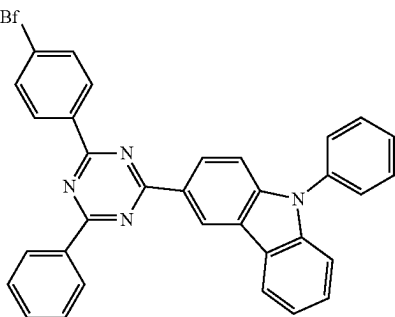
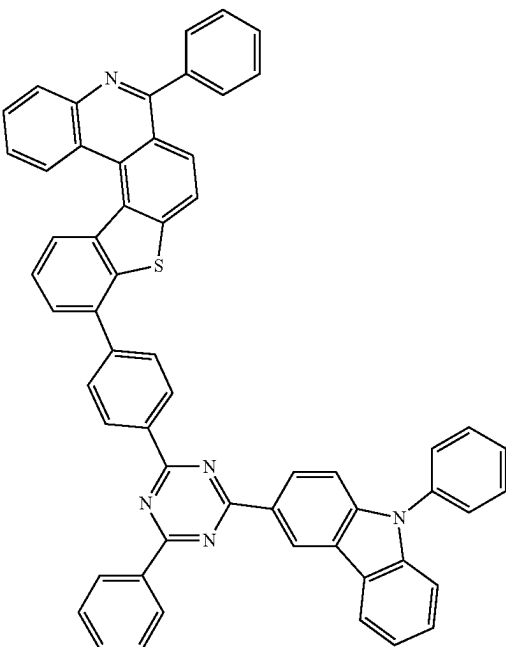
Compound	Intermediate Q	Target Compound	Yield
931			61%
932			62%

TABLE 18-continued

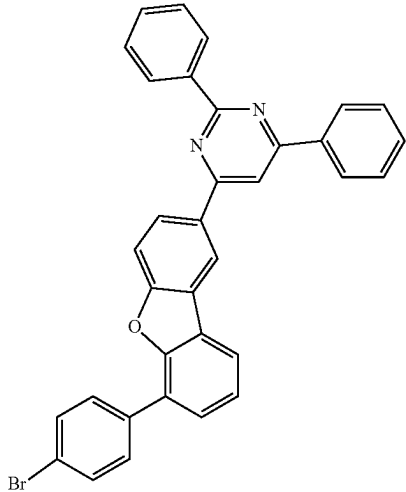
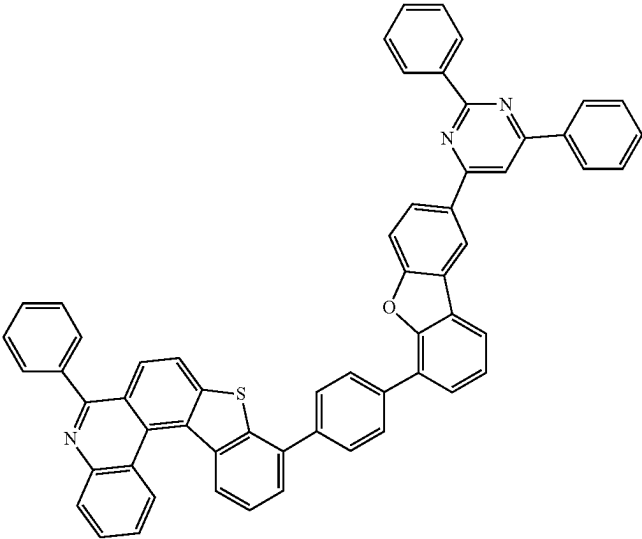
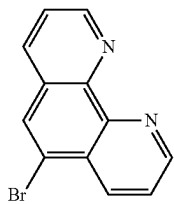
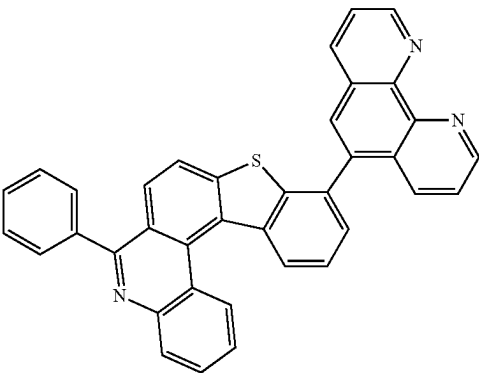
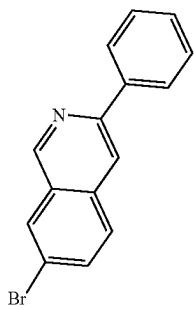
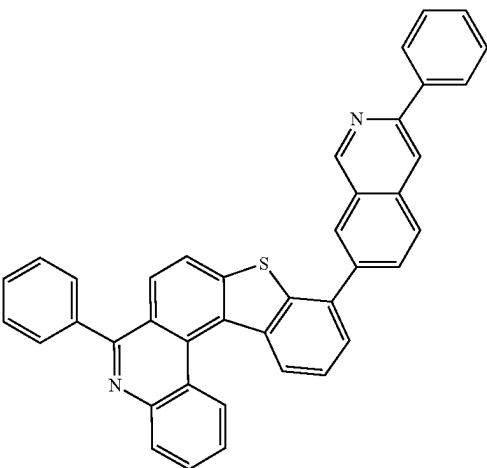
Compound	Intermediate Q	Target Compound	Yield
943			65%
945			65%
952			62%

TABLE 18-continued

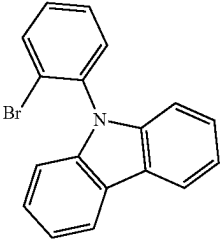
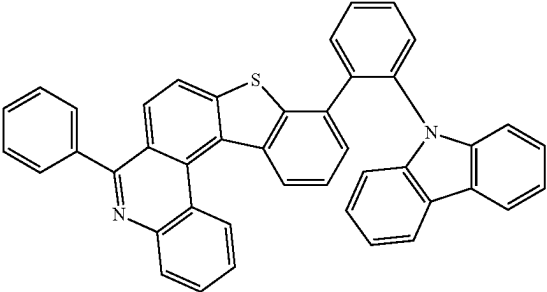
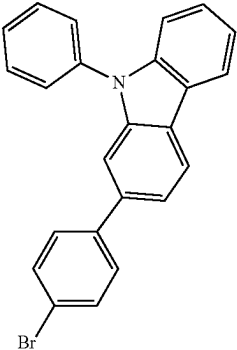
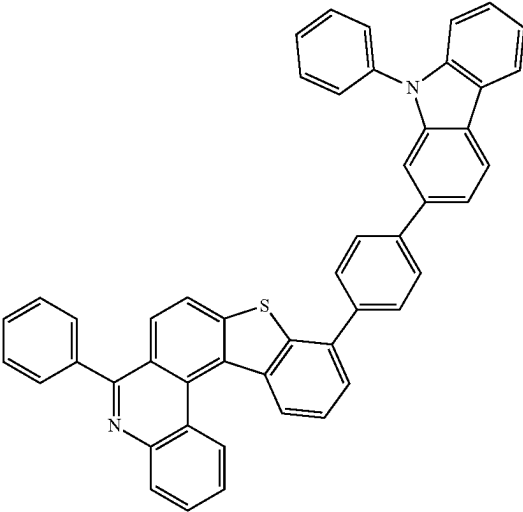
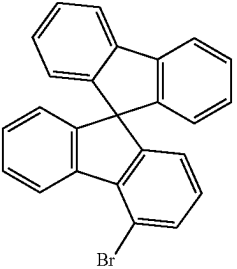
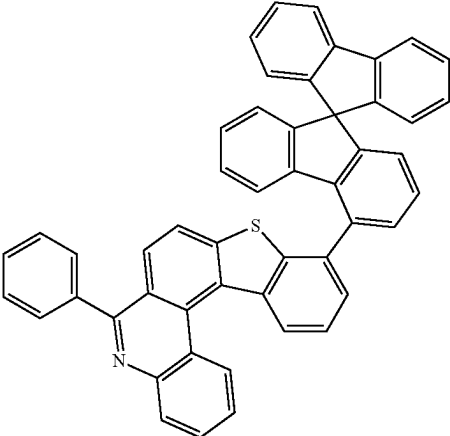
Compound	Intermediate Q	Target Compound	Yield
955			51%
956			60%
957			62%

TABLE 18-continued

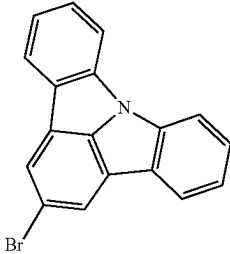
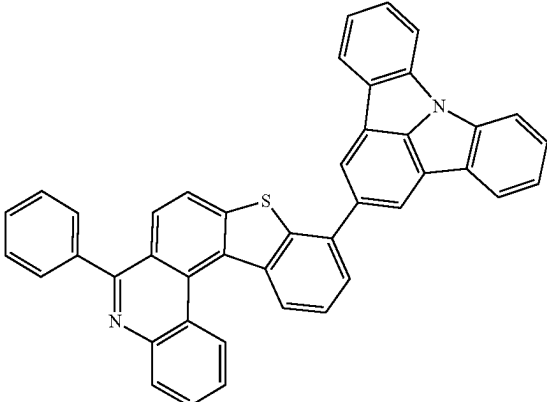
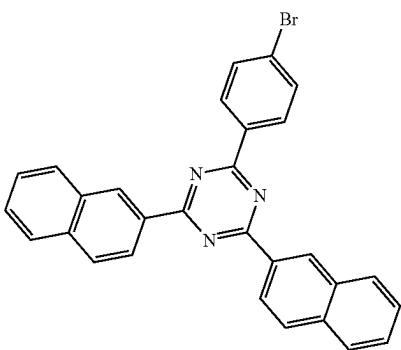
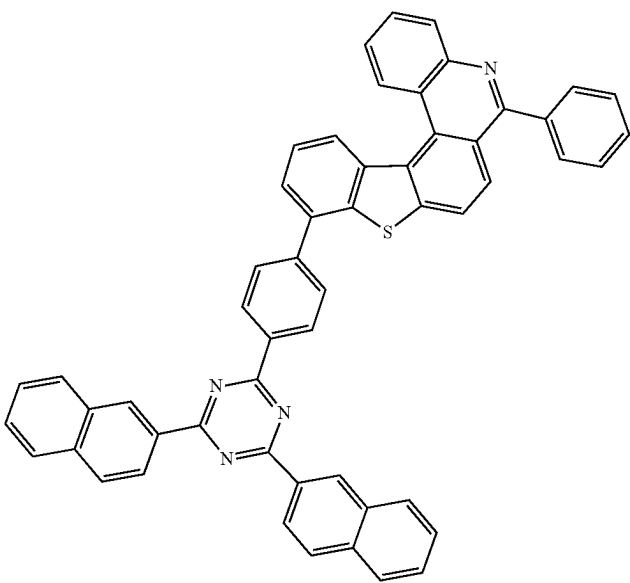
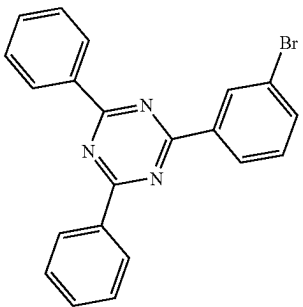
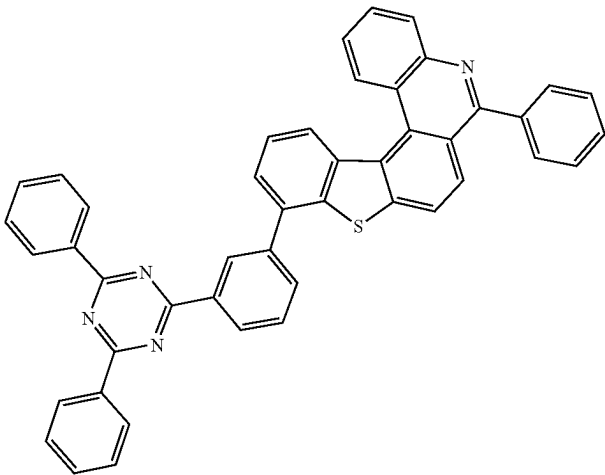
Compound	Intermediate Q	Target Compound	Yield
960			65%
963			65%
965			62%

TABLE 18-continued

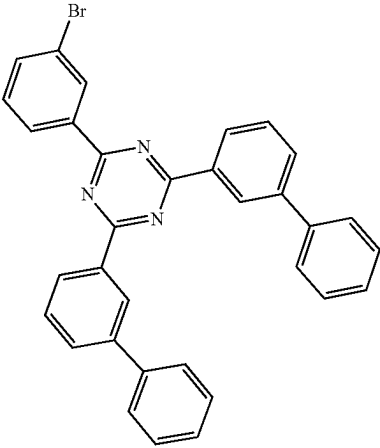
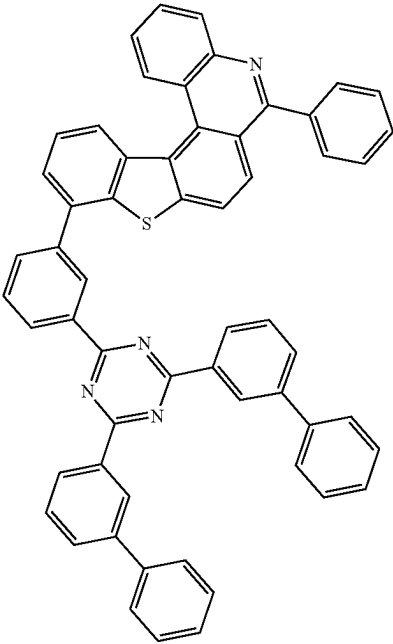
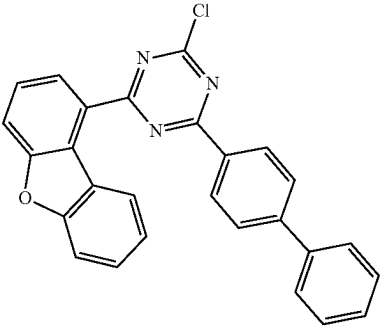
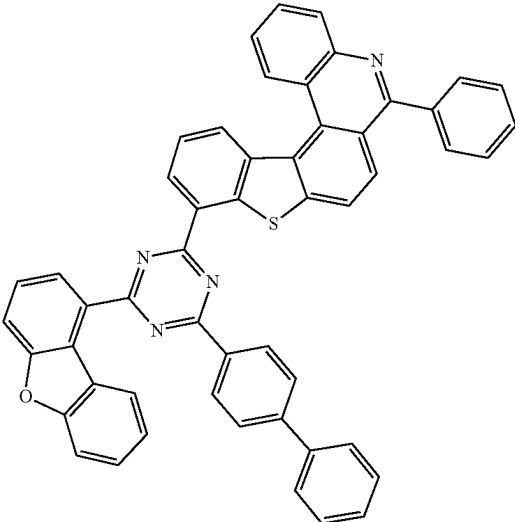
Compound	Intermediate Q	Target Compound	Yield
968			51%
971			62%

TABLE 18-continued

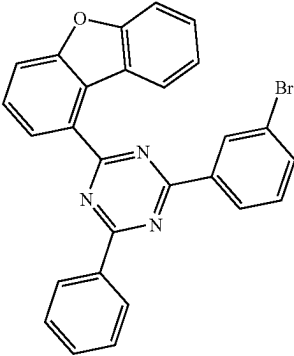
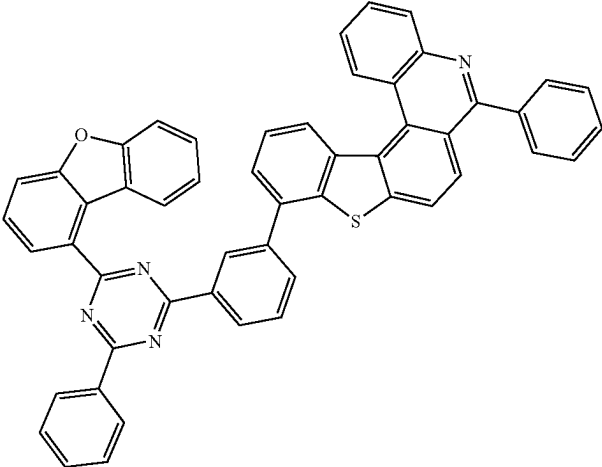
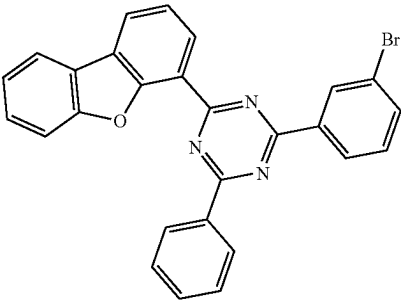
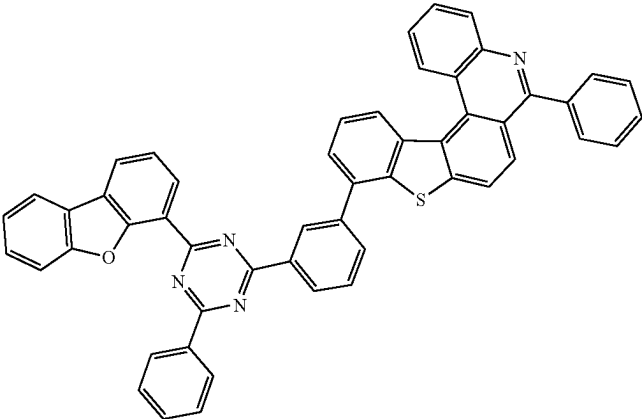
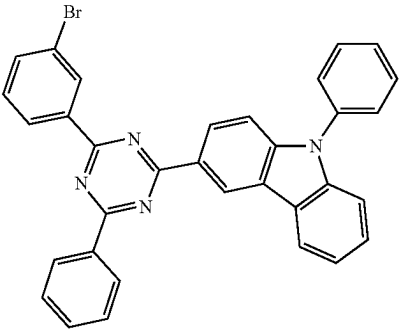
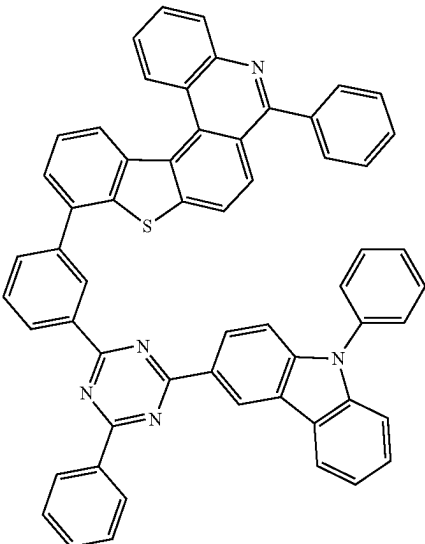
Compound	Intermediate Q	Target Compound	Yield
979			65%
980			65%
984			62%

TABLE 18-continued

Compound	Intermediate Q	Target Compound	Yield
986			65%
998			65%
991			62%

TABLE 18-continued

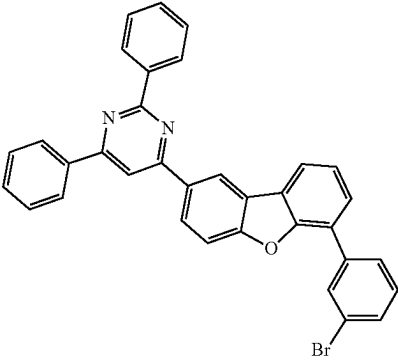
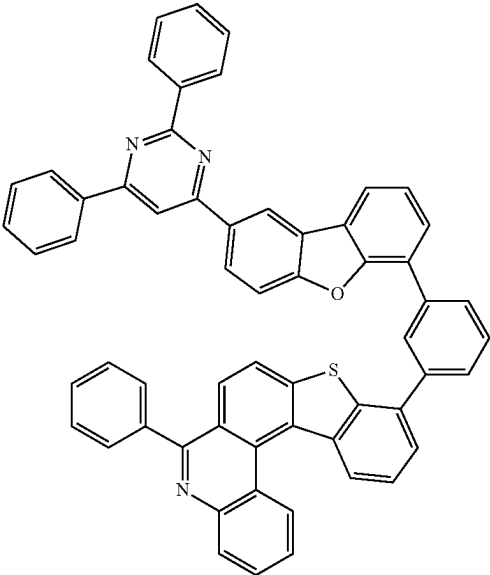
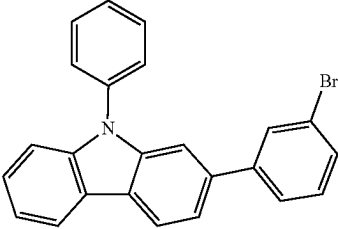
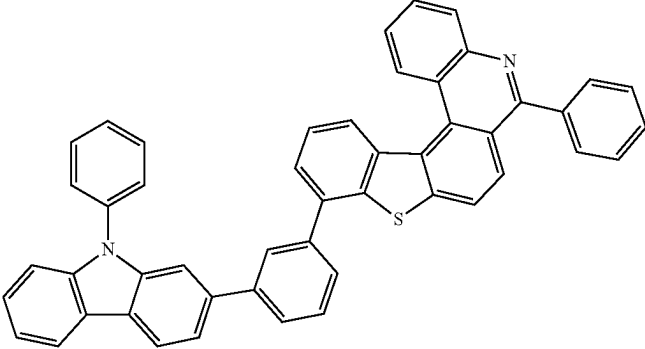
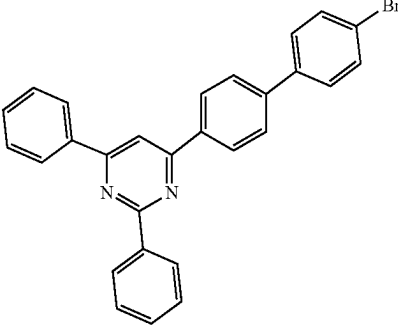
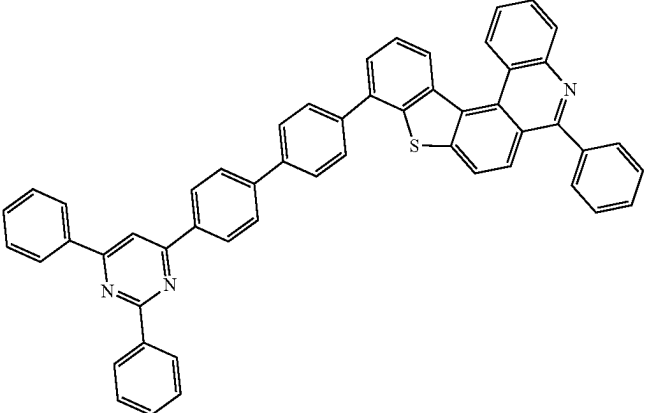
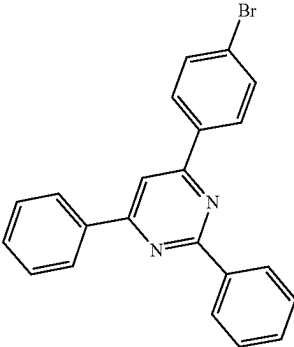
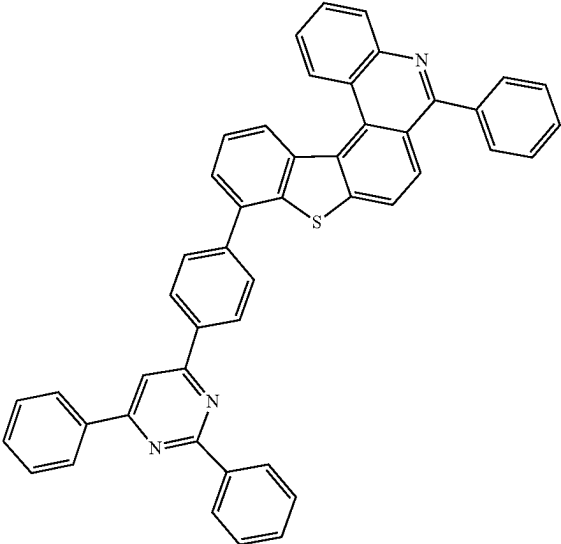
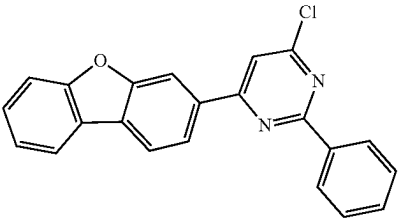
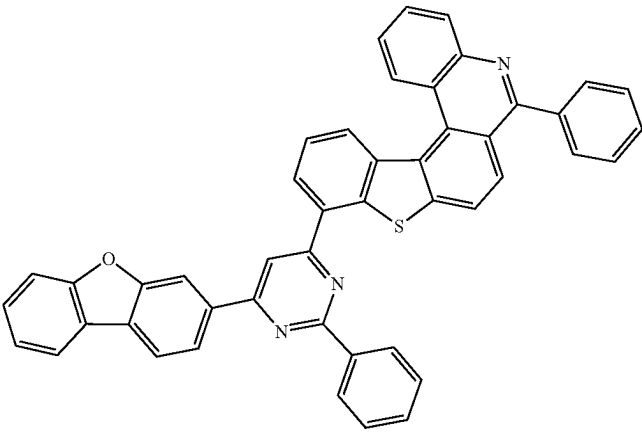
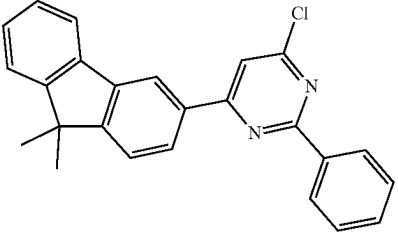
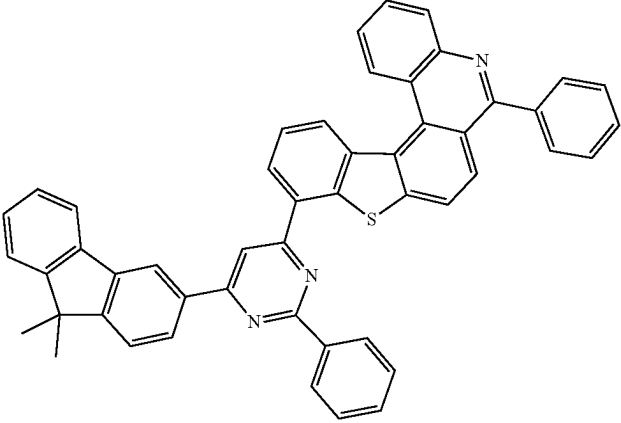
Compound	Intermediate Q	Target Compound	Yield
995			51%
1000			60%
1004			61%

TABLE 18-continued

Compound	Intermediate Q	Target Compound	Yield
1006			60%
1011			62%
1015			65%

Synthesis identification results for the compounds prepared as above are shown in the following Table 19.

TABLE 19

Compound	FD-MS	Compound	FD-MS
1	m/z = 576.65 (C40H24N4O = 576.195)	4	m/z = 652.75 (C46H28N4O = 652.226)
5	m/z = 728.85 (C52H32N4O = 728.25)	7	m/z = 804.95 (C58H36N4O = 804.28)
8	m/z = 728.85 (C52H32N4O = 728.25)	9	m/z = 804.95 (C58H36N4O = 804.28)
11	m/z = 666.74 (C46H26N4O2 = 666.20)	12	m/z = 682.80 (C46H26N4OS = 682.18)
13	m/z = 741.85 (C52H31N5O = 741.25)	15	m/z = 665.75 (C46H27N5O = 665.22)
16	m/z = 742.83 (C52H30N4O = 742.23)	17	m/z = 792.80 (C56H32N4O2 = 792.25)
18	m/z = 842.95 (C60H34N4O2 = 842.26)	24	m/z = 571.67 (C43H25NO = 571.19)
25	m/z = 521.61 (C39H23NO = 521.17)	28	m/z = 577.70 (C41H23NOS = 577.15)
30	m/z = 628.77 (C46H32N2O = 628.25)	35	m/z = 818.93 (C58H34N4O2 = 818.26)
36	m/z = 818.93 (C58H34N4O2 = 818.26)	37	m/z = 752.92 (C56H36N2O = 752.28)
38	m/z = 599.69 (C43H25N3O = 599.2)	41	m/z = 725.85 (C53H1N3O = 725.24)
42	m/z = 652.75 (C46H28N4O = 652.22)	44	m/z = 659.78 (C50H29NO = 659.22)
45	m/z = 621.67 (C43H28NO2P = 621.18)	47	m/z = 549.63 (C39H23N3O = 549.18)
52	m/z = 618.75 (C43H26N2OS = 618.17)	53	m/z = 660.77 (C49H28N2O = 660.22)
56	m/z = 751.88 (C55H33N3O = 751.26)	57	m/z = 834.99 (C58H34N4OS = 834.24)
58	m/z = 758.89 (C52H30N4OS = 758.21)	64	m/z = 804.95 (C58H36N4O = 804.28)
68	m/z = 732.86 (C50H28N4OS = 732.19)	69	m/z = 732.86 (C50H28N4OS = 732.19)
70	m/z = 868.01 (C62H37N5O = 867.30)	72	m/z = 690.76 (C47H26N6O = 690.21)
75	m/z = 742.83 (C52H30N4O2 = 742.23)	77	m/z = 758.89 (C52H30N4OS = 758.21)
79	m/z = 768.92 (C55H36N4O = 768.28)	80	m/z = 741.85 (C5231N5O = 741.25)
84	m/z = 660.77 (C49H28N2O = 660.22)	86	m/z = 597.71 (C45H27NO = 597.20)
88	m/z = 727.88 (C53H29NOS = 727.19)	89	m/z = 663.77 (C49H29NO2 = 663.22)
91	m/z = 752.91 (C56H36N2O = 752.28)	92	m/z = 675.79 (C49H29N3O = 675.23)
97	m/z = 575.67 (C41H25N3O = 575.20)	100	m/z = 727.86 (C53H33N3O = 727.26)
102	m/z = 701.82 (C51H31N3O = 701.24)	103	m/z = 727.86 (C53H33N3O = 727.26)
107	m/z = 727.86 (C53H33N3O = 727.26)	112	m/z = 681.81 (C47H27N3OS = 681.18)
113	m/z = 740.86 (C53H32N4O = 740.25)	114	m/z = 691.83 (C50H33N3O = 691.26)
116	m/z = 741.85 (C53H31N3O2 = 741.24)	119	m/z = 817.99 (C60H39N3O = 817.30)
120	m/z = 740.86 (C53H32N4O = 740.25)	121	m/z = 725.85 (C53H31N3O = 725.24)
128	m/z = 777.92 (C57H35N3O = 777.27)	132	m/z = 681.81 (C47H27N3OS = 681.18)
133	m/z = 681.81 (C47H27N3OS = 681.18)	135	m/z = 664.76 (C47H28N4O = 664.22)
139	m/z = 757.91 (C53H31N3OS = 757.21)	140	m/z = 816.96 (C59H36N4O = 816.28)
141	m/z = 767.93 (C56H37N3O = 767.29)	142	m/z = 740.86 (C53H32N4O = 740.25)
145	m/z = 702.81 (C50H30N4O = 702.24)	149	m/z = 728.85 (C52H32N4O = 728.25)
156	m/z = 652.75 (C46H26N4O = 652.22)	157	m/z = 652.75 (C46H28N4O = 652.22)

TABLE 19-continued

Compound	FD-MS	Compound	FD-MS
159	m/z = 804.95 (C58H36N4O = 804.28)	161	m/z = 666.74 (C46H26N4O2 = 666.20)
165	m/z = 682.80 (C46H26N4OS = 682.18)	166	m/z = 741.85 (C52H31N5O = 741.25)
168	m/z = 715.81 (C50H29N4O = 715.23)	170	m/z = 842.95 (C60H34N4O2 = 842.26)
171	m/z = 758.89 (C52H30N4OS = 758.21)	172	m/z = 817.95 (C58H35N5O = 817.28)
173	m/z = 768.92 (C55H36N4O = 768.28)	176	m/z = 571.67 (C43H25NO = 571.19)
178	m/z = 573.69 (C43H27NO = 573.20)	179	m/z = 527.64 (C37H21NOS = 527.13)
180	m/z = 552.67 (C40H28N2O = 552.22)	185	m/z = 895.03 (C64H38N4O2 = 894.29)
187	m/z = 676.81 (C50H32N2O = 676.25)	190	m/z = 675.79 (C49H29N3O = 675.23)
191	m/z = 725.85 (C53H31N3O = 725.24)	192	m/z = 652.75 (C46H28N4O = 652.22)
194	m/z = 659.78 (C50H29NO = 659.22)	195	m/z = 548.64 (C40H24N2O = 548.18)
196	m/z = 625.73 (C45H27N3O = 625.21)	200	m/z = 659.78 (C50H29NO = 659.22)
202	m/z = 660.77 (C49H28N2O = 660.22)	204	m/z = 802.93 (C58H34N4O = 802.27)
207	m/z = 728.85 (C52H32N4O = 728.25)	210	m/z = 742.83 (C52H30N4O2 = 742.23)
213	m/z = 732.86 (C50H28N4OS = 732.19)	214	m/z = 868.01 (C62H37N5O = 867.30)
216	m/z = 717.83 (C50H31N5O = 717.25)	217	m/z = 690.76 (C47H26N6O = 690.21)
220	m/z = 742.83 (C52H30N4O2 = 742.23)	222	m/z = 758.89 (C52H30N4OS = 758.21)
223	m/z = 817.95 (C58H35N5O = 817.28)	224	m/z = 768.92 (C55H36N4O = 768.28)
225	m/z = 741.85 (C52H31N5O = 741.25)	227	m/z = 602.69 (C43H26N2O2 = 602.19)
228	m/z = 618.75 (C43H26N2OS = 618.17)	229	m/z = 660.77 (C49H26N2O = 660.22)
233	m/z = 911.09 (C64H38N4OS = 910.27)	237	m/z = 675.79 (C49H29N3O = 675.23)
240	m/z = 662.79 (C49H30N2O = 662.23)	243	m/z = 651.76 (C47H29N3O = 651.23)
245	m/z = 651.76 (C47H29N3O = 651.23)	247	m/z = 651.76 (C47H29N3O = 651.23)
249	m/z = 803.96 (C59H37N3O = 803.29)	251	m/z = 715.81 (C51H29N3O2 = 715.22)
256	m/z = 740.86 (C53H32N4O = 740.25)	257	m/z = 691.83 (C50H33N3O = 691.26)
259	m/z = 791.91 (C57H33N3O2 = 791.25)	260	m/z = 741.85 (C53H31N3O2 = 741.24)
262	m/z = 757.91 (C53H31N3OS = 757.21)	265	m/z = 767.93 (C56H37N3O = 767.29)
266	m/z = 740.86 (C53H32N4O = 740.25)	267	m/z = 575.57 (C41H25N3O = 575.20)
269	m/z = 727.86 (C53H33N3O = 727.26)	271	m/z = 727.86 (C53H33N3O = 727.26)
272	m/z = 803.96 (C59H37N3O = 803.29)	273	m/z = 727.86 (C53H33N3O = 727.26)
274	m/z = 841.97 (C61H35N3O2 = 841.27)	276	m/z = 665.75 (C47H27N3O2 = 665.21)
277	m/z = 681.81 (C47H27N3OS = 681.18)	279	m/z = 740.86 (C53H32N4O = 740.25)
281	m/z = 664.76 (C47H28N4O = 662.22)	284	m/z = 741.85 (C53H31N3O2 = 741.24)
285	m/z = 757.91 (C53H31N3OS = 752.21)	287	m/z = 816.96 (C59H36N4O = 816.28)
288	m/z = 767.93 (C56H37N3O = 767.29)	292	m/z = 728.85 (C52H32N4O = 728.25)
294	m/z = 855.01 (C62H38N4O = 854.30)	296	m/z = 652.75 (C46H28N4O = 652.22)
299	m/z = 682.80 (C46H26N4OS = 682.18)	302	m/z = 665.75 (C46H27CN5O = 665.22)
304	m/z = 742.83 (C52H30N4O2 = 742.23)	305	m/z = 758.89 (C52H30N4OS = 758.21)
307	m/z = 817.95 (C58H35N4O = 817.28)	308	m/z = 768.92 (C55H36N4O = 768.28)
309	m/z = 741.85 (C52H31N5O = 741.25)	310	m/z = 571.67 (C43H25NO = 571.19)

TABLE 19-continued

Compound	FD-MS	Compound	FD-MS
312	m/z = 597.71 (C45H27NO = 597.20)	313	m/z = 753.69 (C43H27NO = 573.20)
315	m/z = 521.61 (C39H23NO = 521.17)	316	m/z = 597.71 (C45H27NO = 597.20)
318	m/z = 603.73 (C43H25NOS = 603.16)	319	m/z = 742.83 (C52H30N4O2 = 742.23)
322	m/z = 752.91 (C56H36N2O = 752.28)	325	m/z = 675.79 (C49H29N3O = 675.23)
326	m/z = 725.85 (C53H31N3O = 725.24)	327	m/z = 652.75 (C46H28N4O = 652.22)
329	m/z = 548.64 (C40H24N2O = 548.18)	331	m/z = 636.75 (C47H28N2O = 636.22)
332	m/z = 662.79 (C49H30N2O = 662.23)	336	m/z = 752.87 (C54H32N4O = 752.25)
337	m/z = 751.88 (C55H33N3O = 751.26)	338	m/z = 693.86 (C49H31N3S = 693.22)
339	m/z = 691.85 (C49H29N3S = 691.20)	342	m/z = 728.85 (C52H32N4O = 728.25)
343	m/z = 778.91 (C56H34N4O = 778.27)	345	m/z = 868.01 (C62H37N5O = 867.30)
348	m/z = 782.92 (C54H30N4OS = 782.21)	350	m/z = 717.83 (C50H31N5O = 717.25)
353	m/z = 742.83 (C52H30N4O2 = 742.23)	355	m/z = 758.89 (C52H30N4OS = 758.21)
358	m/z = 768.92 (C55H36N4O = 768.28)	359	m/z = 741.85 (C52H31N4O = 741.25)
360	m/z = 628.77 (C46H32N2O = 628.25)	364	m/z = 597.71 (C45H27NO = 597.20)
365	m/z = 652.75 (C46H28N4O = 652.22)	367	m/z = 945.09 (C68H40N4O2 = 944.31)
368	m/z = 713.83 (C53H31NO2 = 713.23)	373	m/z = 712.85 (C53H32N2O = 712.25)
374	m/z = 735.88 (C56H33NO = 735.25)	377	m/z = 727.86 (C53H33N3O = 727.26)
378	m/z = 803.95 (C59H27N3O = 803.29)	379	m/z = 751.88 (C55H33N3O = 751.26)
380	m/z = 727.86 (C53H33N3O = 727.26)	384	m/z = 665.75 (C47H27N3O2 = 665.21)
386	m/z = 731.87 (C51H29N3OS = 731.20)	387	m/z = 740.86 (C53H32N4O = 740.25)
388	m/z = 691.83 (C50H33N3O = 691.26)	392	m/z = 741.85 (C53H31N3O2 = 741.24)
395	m/z = 917.08 (C67H40N4O = 916.32)	396	m/z = 740.86 (C53H32N4O = 740.25)
397	m/z = 651.76 (C47H2N3O = 651.23)	399	m/z = 803.96 (C59H37N3O = 803.29)
400	m/z = 651.76 (C47H29N3O = 651.23)	401	m/z = 727.86 (C53H33N3O = 727.26)
404	m/z = 665.75 (C47H27N3O2 = 665.21)	407	m/z = 805.95 (C57H31N3OS = 805.21)
408	m/z = 740.86 (C53H32N4O = 740.25)	409	m/z = 691.83 (C50H33N3O = 691.25)
411	m/z = 741.85 (C53H31N3O2 = 741.24)	413	m/z = 741.85 (C53H31N3O2 = 741.24)
415	m/z = 757.91 (C53H31N3OS = 757.21)	416	m/z = 816.96 (C59H36N4O = 816.28)
417	m/z = 767.93 (C56H37N3O = 767.29)	421	m/z = 804.95 (C58H36N4O = 804.28)
423	m/z = 702.81 (C50H30N4O = 702.24)	424	m/z = 652.75 (C46H26N4O = 652.22)
426	m/z = 778.91 (C56H34N4O = 778.27)	431	m/z = 682.80 (C46H26N4OS = 682.18)
432	m/z = 741.85 (C52H31N5O = 741.25)	433	m/z = 692.82 (C49H32N4O = 692.25)
436	m/z = 742.83 (C52H30N4O2 = 742.23)	438	m/z = 758.89 (C52H30N4OS = 758.21)
440	m/z = 741.85 (C52H31N5O = 741.25)	441	m/z = 571.67 (C43H25NO = 571.19)
443	m/z = 597.71 (C54H27NO = 597.20)	447	m/z = 628.77 (C46H32N2O = 628.25)
448	m/z = 597.71 (C45H27NO = 597.20)	449	m/z = 649.79 (C49H31NO = 649.24)
450	m/z = 911.09 (C64H38N4OS = 910.27)	452	m/z = 665.75 (C47H27N3O2 = 665.21)
456	m/z = 675.79 (C49H29N3O = 675.23)	460	m/z = 659.78 (C50H29NO = 659.22)
462	m/z = 548.64 (C40H24N2O = 548.18)	465	m/z = 636.75 (C47H28N2O = 636.22)

TABLE 19-continued

Compound	FD-MS	Compound	FD-MS
467	m/z = 659.78 (C50H29NO = 659.22)	468	m/z = 652.75 (C47H28N2O = 652.21)
470	m/z = 752.87 (C54H32N4O = 752.25)	475	m/z = 728.85 (C52H32N4O = 728.25)
476	m/z = 742.83 (C52H30N4O2 = 742.23)	477	m/z = 742.83 (C52H30N4O2 = 742.23)
479	m/z = 732.86 (C50H28N4OS = 732.19)	481	m/z = 868.01 (C62H37N5O = 867.30)
482	m/z = 717.83 (C50H31N5O = 717.25)	485	m/z = 742.83 (C52H30N4O2 = 742.23)
488	m/z = 758.89 (C52H30N4OS = 758.21)	489	m/z = 817.95 (C58H35N5O = 817.28)
490	m/z = 818.98 (C59H38N4O = 818.30)	493	m/z = 602.69 (C43H26N2O2 = 602.19)
494	m/z = 618.75 (C43H26N2OS = 618.17)	495	m/z = 660.77 (C49H28N2O = 660.22)
500	m/z = 663.77 (C49H29NO2 = 663.22)	504	m/z = 697.77 (C49H32NO2P = 697.21)
505	m/z = 762.91 (C57H34N2O = 762.26)	506	m/z = 735.88 (C56H33NO = 735.25)
509	m/z = 727.86 (C53H33N3O = 727.26)	512	m/z = 727.86 (C53H33N3O = 727.26)
513	m/z = 665.75 (C47H27N3O2 = 556.21)	516	m/z = 681.81 (C47H27N3OS = 681.18)
517	m/z = 681.81 (C47H27N3OS = 681.18)	518	m/z = 740.86 (C53H32N4O = 740.25)
519	m/z = 691.83 (C50H33N3O = 691.26)	520	m/z = 664.76 (C47H26N4O = 664.22)
522	m/z = 841.97 (C61H35N3O2 = 841.27)	524	m/z = 757.91 (C53H31N3OS = 757.21)
525	m/z = 757.91 (C53H31N3OS = 757.21)	527	m/z = 767.93 (C56H37N3O = 767.29)
528	m/z = 740.86 (C53H32N4O = 740.25)	531	m/z = 727.86 (C53H33N3O = 727.26)
532	m/z = 727.86 (C53H33N3O = 727.26)	533	m/z = 803.96 (C59H37N3O = 803.29)
535	m/z = 727.86 (C53H33N3O = 727.26)	536	m/z = 664.76 (C47H28N4O = 664.22)
539	m/z = 665.75 (C47H27N3O2 = 665.21)	542	m/z = 640.86 (C53H32N4O = 740.25)
543	m/z = 691.83 (C50H33N3O = 691.26)	544	m/z = 741.85 (C53H31N3O2 = 741.24)
547	m/z = 757.91 (C53H31N3OS = 757.21)	548	m/z = 807.97 (C57H33N3OS = 807.23)
549	m/z = 816.96 (C59H36N4O = 816.28)	550	m/z = 767.93 (C56H37N3O = 767.29)
554	m/z = 665.75 (C46H26N4O = 652.22)	555	m/z = 728.85 (C52H32N4O = 728.25)
557	m/z = 728.85 (C52H32N4O = 728.25)	558	m/z = 804.95 (C58H36N4O = 804.28)
564	m/z = 741.85 (C52H31N5O = 741.25)	565	m/z = 692.82 (C49H32N4O = 692.26)
566	m/z = 665.75 (C46H27N5O = 665.22)	569	m/z = 742.84 (C52H30N4O2 = 742.24)
573	m/z = 768.92 (C55H36N4O = 768.29)	574	m/z = 741.85 (C52H31N5O = 741.25)
580	m/z = 521.62 (C39H23NO = 521.18)	564	m/z = 741.85 (C52H31N5O = 741.25)
565	m/z = 692.82 (C49H32N4O=692.26)	566	m/z = 665.76 (C46H27N5O=665.22)
569	m/z = 742.84 (C52H30N4O2 = 742.24)	573	m/z = 768.92 (C55H36N4O = 768.29)
574	m/z = 741.85 (C52H31N5O = 741.25)	580	m/z = 521.62 (C39H23NO = 521.18)
583	m/z = 834.01 (C59H35N3OS = 833.25)	584	m/z = 894.05 (C65H39N3O2 = 893.30)
585	m/z = 599.69 (C43H25N3O = 599.20)	587	m/z = 675.79 (C49H29N3O = 675.23)
588	m/z = 725.85 (C53H31N3O = 725.25)	594	m/z = 662.79 (C49H30N2O = 662.24)
597	m/z = 652.76 (C46H28N4O = 652.23)	598	m/z = 652.76 (C46H28N4O = 652.23)
599	m/z = 728.86 (C52H32N4O = 728.26)	600	m/z = 728.86 (C52H32N4O = 728.26)
601	m/z = 666.74 (C46H26H4O2 = 666.21)	605	m/z = 758.90 (C52H30N4OS = 758.21)
606	m/z = 741.85 (C52H31N5O = 741.25)	611	m/z = 727.87 (C53H33N3O = 727.26)

TABLE 19-continued

Compound	FD-MS	Compound	FD-MS
612	m/z = 664.77 (C47H28N4O = 664.23)	615	m/z = 740.87 (C53H32N4O = 740.26)
618	m/z = 767.93 (C56H37N3O = 767.29)	620	m/z = 642.78 (C44H26N4S = 642.19)
623	m/z = 668.82 (C46H28N4S = 668.20)	624	m/z = 794.98 (C56H34N4S = 794.25)
626	m/z = 821.01 (C56H36N4S = 820.27)	629	m/z = 682.80 (C46H26N4OS = 682.18)
632	m/z = 698.86 (C46H26N4S2 = 698.16)	634	m/z = 708.88 (C49H32N4S = 708.23)
635	m/z = 681.82 (C46H27N5S = 681.20)	636	m/z = 758.90 (C52H30N4OS = 758.21)
640	m/z = 774.96 (C52H30N4S2 = 774.19)	641	m/z = 834.01 (C58H35N5S = 833.26)
644	m/z = 587.74 (C43H25NS = 587.17)	646	m/z = 613.78 (C45H27NS = 613.19)
649	m/z = 537.68 (C39H23NS = 537.16)	651	m/z = 715.91 (C53H33NS = 715.23)
652	m/z = 669.86 (C47H27NS2 = 669.16)	653	m/z = 911.10 (C64H38N4OS = 910.28)
654	m/z = 834.01 (C59H35N3OS = 833.25)	655	m/z = 768.98 (C56H36N2S = 768.26)
657	m/z = 539.66 (C37H21N3S = 539.15)	659	m/z = 741.91 (C53H31N3S = 741.22)
660	m/z = 668.82 (C46H28N4S = 668.20)	661	m/z = 677.87 (C50H31NS = 677.22)
664	m/z = 652.82 (C47H28N2S = 652.20)	665	m/z = 678.85 (C49H30N2S = 678.21)
666	m/z = 675.85 (C50H29NS = 675.2)	670	m/z = 676.84 (C49H28N2S = 676.20)
672	m/z = 891.00 (C58H34N4S = 818.25)	674	m/z = 668.82 (C46H28N4S = 668.2)
675	m/z = 744.92 (C52H32N4S = 744.23)	677	m/z = 744.92 (C52H32N4S = 744.23)
680	m/z = 758.90 (C52H30N4OS = 758.21)	682	m/z = 758.90 (C52H30N4OS = 758.21)
683	m/z = 849.04 (C58H32N4S2 = 848.21)	685	m/z = 884.07 (C62H37N5S = 883.28)
687	m/z = 706.83 (C47H26N6S = 706.19)	688	m/z = 758.90 (C52H30N4OS = 758.21)
691	m/z = 774.96 (C52H30N4S2 = 774.19)	692	m/z = 875.08 (C60H34N4S2 = 874.22)
695	m/z = 757.91 (C52H31N5S = 757.23)	696	m/z = 644.84 (C46H32H2S = 644.23)
697	m/z = 618.75 (C43H26N2OS = 618.18)	700	m/z = 768.94 (C54H32N4S = 768.23)
703	m/z = 743.94 (C53H29NS2 = 743.17)	704	m/z = 834.01 (C59H35N3OS = 833.25)
707	m/z = 691.85 (C49H29N3S = 691.21)	708	m/z = 668.82 (C46H28N4S = 668.20)
709	m/z = 713.83 (C49H32NOPS = 713.19)	713	m/z = 667.83 (C47H29N3S = 667.21)
714	m/z = 743.93 (C53H33N3S = 743.24)	716	m/z = 743.93 (C53H33N3S = 743.24)
717	m/z = 820.03 (C59H37N3S = 819.27)	721	m/z = 793.99 (C57H35N3S = 793.26)
728	m/z = 808.01 (C58H37N3S = 807.27)	729	m/z = 680.83 (C47H28N4S = 680.20)
731	m/z = 757.91 (C53H31N3OS = 757.22)	732	m/z = 757.91 (C53H31N3OS = 757.22)
733	m/z = 824.03 (C57H33N3S2 = 823.21)	734	m/z = 824.03 (C57H33N3S2 = 823.21)
735	m/z = 833.03 (C59H36N4S = 832.27)	736	m/z = 783.99 (C56H37N3S = 783.27)
741	m/z = 820.03 (C59H37N3S = 819.27)	743	m/z = 793.99 (C57H35N3S = 793.26)
745	m/z = 743.93 (C53H33N3S = 743.24)	748	m/z = 665.75 (C47H27N3O2 = 664.21)
750	m/z = 747.93 (C51H29N3S2 = 747.18)	752	m/z = 707.89 (C50H33N3S = 707.24)
756	m/z = 757.91 (C53H31N3OS = 757.22)	757	m/z = 824.03 (C57H33N3S2 = 823.21)
761	m/z = 756.93 (C53H32N4S = 756.23)	764	m/z = 768.94 (C54H32N4S = 768.23)
766	m/z = 793.99 (C57H35N3S = 793.26)	767	m/z = 591.73 (C41H25N3S = 591.18)
769	m/z = 718.88 (C50H30N4S = 718.22)	771	m/z = 668.82 (C46H28N4S = 668.20)

TABLE 19-continued

Compound	FD-MS	Compound	FD-MS
773	m/z = 744.92 (C52H32N4S = 744.23)	775	m/z = 845.04 (C60H36N4S = 844.27)
779	m/z = 782.92 (C54H30N4OS = 782.21)	783	m/z = 698.86 (C46H26N4S2 = 698.16)
784	m/z = 757.91 (C52H31N5S = 757.23)	786	m/z = 781.94 (C54H31N5S = 781.23)
787	m/z = 758.90 (C52H30N4OS = 758.21)	789	m/z = 758.90 (C52H30N4OS = 758.21)
791	m/z = 774.96 (C52H30N4S2 = 774.19)	792	m/z = 834.01 (C58H35N5S = 833.26)
793	m/z = 885.10 (C63H40N4S = 884.30)	798	m/z = 589.76 (C43H27NS = 589.19)
801	m/z = 644.84 (C46H32N2S = 644.23)	802	m/z = 537.68 (C39H23NS = 537.16)
807	m/z = 758.90 (C52H30N4OS = 758.21)	809	m/z = 615.75 (C43H25N3S = 615.18)
810	m/z = 539.66 (C37H21N3S = 539.15)	814	m/z = 675.85 (C50H29NS = 675.20)
816	m/z = 641.79 (C4527N3S = 641.19)	817	m/z = 565.69 (C39H23N3S = 565.16)
818	m/z = 602.75 (C43H26N2S = 602.18)	822	m/z = 845.04 (C60H36N4S = 844.27)
823	m/z = 847.05 (C60H38N4S = 846.28)	825	m/z = 742.90 (C52H30N4S = 742.22)
829	m/z = 744.92 (C52H32N4S = 744.23)	831	m/z = 821.01 (C58H36N4S = 820.27)
833	m/z = 758.90 (C52H30N4OS = 758.21)	836	m/z = 748.92 (C50H28N4S2 = 748.18)
838	m/z = 884.07 (C62H37N5S = 883.28)	842	m/z = 758.90 (C52H30N4OS = 758.21)
844	m/z = 774.96 (C52H304S2 = 774.19)	847	m/z = 784.98 (C55H36N4S = 784.27)
848	m/z = 757.91 (C52H31N5S = 757.23)	849	m/z = 644.84 (C46H32N2S = 644.23)
858	m/z = 835.00 (C58H34N4OS = 834.25)	860	m/z = 691.85 (C49H29N3S = 691.21)
861	m/z = 668.82 (C46H28N4S = 668.20)	862	m/z = 713.83 (C49H32OPS = 713.19)
863	m/z = 678.85 (C49H30N2S = 679.21)	870	m/z = 667.83 (C47H29N3S = 667.21)
871	m/z = 743.93 (C53H33N3S = 743.24)	872	m/z = 820.03 (C59H37N3S = 819.27)
883	m/z = 757.91 (C53H31N3OS = 757.22)	884	m/z = 757.91 (C53H31N3OS = 757.22)
887	m/z = 833.03 (C59H36N4S = 832.27)	888	m/z = 783.99 (C56H37N3S = 783.27)
890	m/z = 591.73 (C41H25N3S = 591.18)	893	m/z = 820.03 (C59H37N3S = 819.27)
905	m/z = 757.91 (C53H31N3OS = 757.22)	908	m/z = 773.97 (C53H31N3S2 = 773.20)
910	m/z = 783.99 (C56H37N3S = 783.27)	911	m/z = 756.93 (C53H32N4S = 756.23)
916	m/z = 744.92 (C52H32N4S = 744.23)	918	m/z = 821.01 (C58H36N4S = 820.27)
919	m/z = 682.80 (C46H26N4OS = 682.18)	920	m/z = 682.80 (C46H26N4OS = 682.18)
923	m/z = 698.86 (C46H26N4S2 = 698.16)	924	m/z = 757.91 (C52H31N5S = 757.23)
930	m/z = 774.96 (C52H30N4S2 = 774.19)	931	m/z = 774.96 (C52H30N4S2 = 774.19)
932	m/z = 834.01 (C58H35N5S = 833.26)	943	m/z = 834.01 (C59H35N3OS = 833.25)
945	m/z = 539.66 (C37H21N3S = 539.15)	952	m/z = 564.71 (C40H24N2S = 563.17)
956	m/z = 678.85 (C49H30N2S = 678.21)	957	m/z = 675.85 (C50H29NS = 675.20)
960	m/z = 600.74 (C43H24N2S = 600.17)	963	m/z = 768.94 (C54H32N4S = 768.23)
965	m/z = 668.82 (C46H28N4S = 668.20)	968	m/z = 821.01 (C58H36N4S = 820.27)
971	m/z = 758.90 (C52H30N4OS = 758.21)	979	m/z = 758.90 (C52H30N4OS = 758.21)
980	m/z = 758.90 (C52H30N4OS = 758.21)	984	m/z = 834.01 (C58H35H5S = 833.26)
986	m/z = 757.91 (C52H31N5S = 757.23)	991	m/z = 613.78 (C45H27NS = 613.19)
998	m/z = 668.82 (C46H28N4S = 668.20)	995	m/z = 834.01 (C59H35N3OS = 833.25)

TABLE 19-continued

Compound	FD-MS	Compound	FD-MS
1000	m/z = 678.85 (C ₄₉ H ₃₀ N ₂ S = 678.21)	1004	m/z = 743.93 (C ₅₃ H ₃₃ N ₃ S = 743.24)
1006	m/z = 667.83 (C ₄₇ H ₂₉ N ₃ S = 667.21)	1011	m/z = 681.81 (C ₄₇ H ₂₇ N ₃ SO = 681.19)
1015	m/z = 707.89 (C ₅₀ H ₃₃ N ₃ S = 707.24)	1080	m/z = 835.00 (C ₅₈ H ₃₄ N ₄ OS = 834.25)
1086	m/z = 691.85 (C ₄₉ H ₂₉ N ₃ S = 691.21)	1098	m/z = 676.64 (C ₄₉ H ₂₈ N ₂ S = 676.20)
1099	m/z = 692.84 (C ₄₈ H ₂₆ N ₄ S = 692.20)		

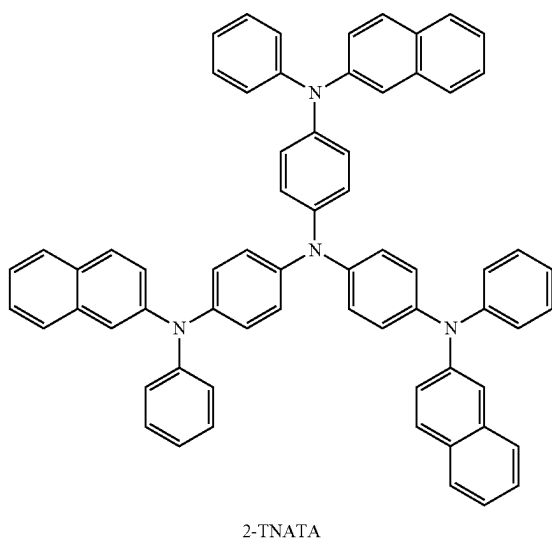
<Experimental Example 1> Manufacture of Organic Light Emitting Device

Comparative Example 1

1) Manufacture of Organic Light Emitting Device

A transparent indium tin oxide (ITO) electrode thin film obtained from glass for an OLED (manufactured by Samsung-Corning Co., Ltd.) was ultrasonic cleaned using trichloroethylene, acetone, ethanol and distilled water consecutively for 5 minutes each, stored in isopropanol, and used.

Next, an ITO substrate was installed in a substrate folder of a vacuum deposition apparatus, and the following 4,4', 4''-tris(N,N-(2-naphthyl)-phenylamino)triphenylamine (2-TNATA) was introduced to a cell in the vacuum deposition apparatus.



Subsequently, the chamber was evacuated until the degree of vacuum therein reached 10^{-6} torr, and then 2-TNATA was evaporated by applying a current to the cell to deposit a hole injection layer having a thickness of 600 Å on the ITO substrate.

To another cell of the vacuum deposition apparatus, the following N,N'-bis(α -naphthyl)-N,N'-diphenyl-4,4'-diamine (NPB) was introduced, and evaporated by applying a current

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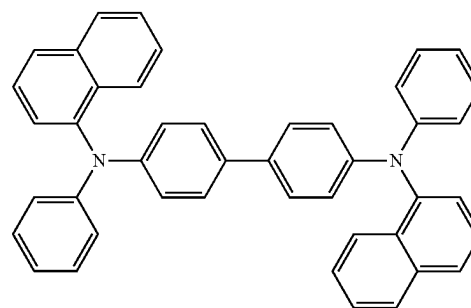
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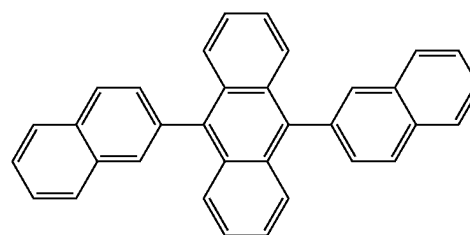
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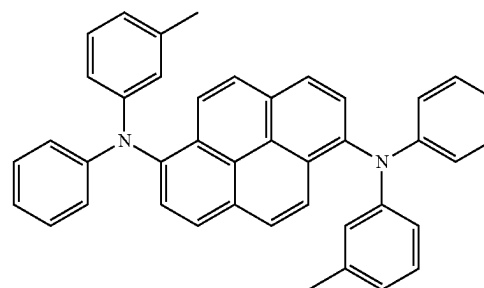


NPB

After forming the hole injection layer and the hole transfer layer as above, a blue light emitting material having a structure as below was deposited thereon as a light emitting layer. Specifically, in one side cell in the vacuum deposition apparatus, H1, a blue light emitting host material, was vacuum deposited to a thickness of 200 Å, and D1, a blue light emitting dopant material, was vacuum deposited thereon by 5% with respect to the host material.



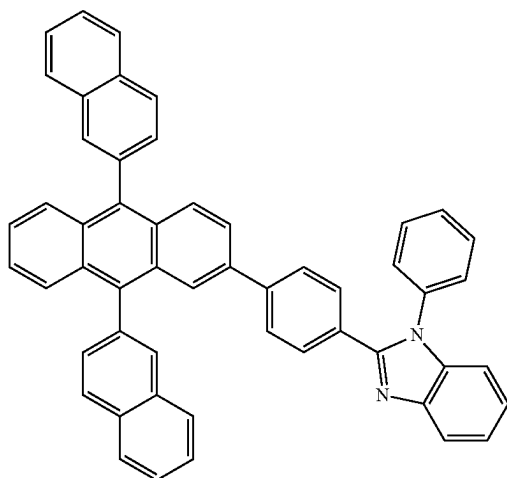
H1



D1

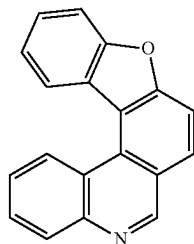
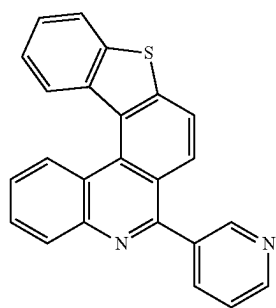
Subsequently, a compound of the following Structural Formula E1 was deposited to a thickness of 300 Å as an electron transfer layer.

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As an electron injection layer, lithium fluoride (LiF) was deposited to a thickness of 10 Å, and an Al cathode was employed to a thickness of 1,000 Å, and as a result, an OLED was manufactured.

Meanwhile, all the organic compounds required to manufacture the OLED were vacuum sublimation purified under 10^{-8} torr to 10^{-6} torr by each material to be used in the OLED manufacture.



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-continued

E1

5

10

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A

B

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C

D

Comparative Example 2 to Comparative Example 5

Organic electroluminescent devices were manufactured in the same manner as in Comparative Example 1 except that Compound A, Compound B, Compound C and Compound D were used instead of E1 used when forming the electron transfer layer.

Example 1 to Example 469

Organic electroluminescent devices were manufactured in the same manner as in Comparative Example 1 except that compounds listed in the following Table 20 were used instead of E1 used when forming the electron transfer layer.

Results of measuring driving voltage, light emission efficiency, color coordinate (CIE) and lifetime of each of the blue organic light emitting devices manufactured according to the present disclosure are as shown in the following Table 20.

TABLE 20

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)
Comparative Example 1	E1	5.21	4.11	(0.134, 0.100)	24
Comparative Example 2	A	4.13	3.20	(0.134, 0.104)	13

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Comparative Example 3	B	4.11	3.12	(0.134, 0.104)	10	5
Comparative Example 4	C	5.67	5.11	(0.134, 0.104)	21	
Comparative Example 5	D	5.67	5.12	(0.134, 0.104)	20	
Example 1	1	5.44	6.47	(0.134, 0.102)	36	10
Example 2	4	4.47	6.87	(0.134, 0.100)	40	
Example 3	5	4.67	6.64	(0.129, 0.100)	38	
Example 4	7	4.82	6.55	(0.130, 0.099)	36	15
Example 5	8	4.44	6.97	(0.134, 0.101)	40	
Example 6	9	4.61	6.89	(0.134, 0.103)	40	
Example 7	11	5.35	6.30	(0.134, 0.102)	33	20
Example 8	12	5.62	5.95	(0.134, 0.103)	42	
Example 9	13	5.40	6.12	(0.134, 0.101)	39	
Example 10	15	5.50	5.89	(0.134, 0.100)	41	25
Example 11	16	5.44	6.01	(0.134, 0.101)	36	
Example 12	17	5.34	6.58	(0.134, 0.100)	45	
Example 13	18	5.38	6.93	(0.134, 0.100)	43	30
Example 14	24	4.91	6.32	(0.134, 0.100)	41	
Example 15	25	4.98	6.44	(0.134, 0.100)	40	
Example 16	28	5.62	6.38	(0.134, 0.100)	35	35
Example 17	30	5.66	6.04	(0.131, 0.102)	30	
Example 18	35	5.40	6.49	(0.134, 0.101)	31	
Example 19	36	5.60	6.22	(0.129, 0.100)	29	40
Example 20	37	4.70	6.01	(0.134, 0.101)	36	
Example 21	38	5.40	6.12	(0.134, 0.103)	44	
Example 22	41	5.60	6.21	(0.134, 0.102)	43	45
Example 23	42	5.45	6.22	(0.134, 0.101)	37	
Example 24	44	5.39	5.95	(0.134, 0.102)	33	
Example 25	45	4.96	5.95	(0.134, 0.101)	42	50
Example 26	47	4.91	6.13	(0.134, 0.101)	39	
Example 27	52	4.91	5.85	(0.134, 0.100)	41	
Example 28	53	4.98	6.38	(0.134, 0.101)	42	55
Example 29	56	4.91	5.85	(0.134, 0.100)	41	
Example 30	57	4.98	6.38	(0.134, 0.101)	42	
Example 31	58	5.44	6.47	(0.134, 0.102)	36	60
Example 32	64	4.47	6.87	(0.134, 0.100)	40	
Example 33	68	4.67	6.64	(0.129, 0.100)	38	
Example 34	69	4.82	6.55	(0.130, 0.099)	36	65

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 35	70	4.44	6.97	(0.134, 0.101)	40	
Example 36	72	4.61	6.89	(0.134, 0.103)	40	
Example 37	75	5.35	6.30	(0.134, 0.102)	33	
Example 38	77	5.62	5.95	(0.134, 0.103)	42	
Example 39	79	5.40	6.12	(0.134, 0.101)	39	
Example 40	80	5.50	5.89	(0.134, 0.100)	41	
Example 41	84	5.44	6.01	(0.134, 0.101)	36	
Example 42	86	5.34	6.58	(0.134, 0.100)	45	
Example 43	88	5.38	6.93	(0.134, 0.100)	43	
Example 44	89	4.91	6.32	(0.134, 0.100)	41	
Example 45	91	4.98	6.44	(0.134, 0.100)	40	
Example 46	92	5.62	6.38	(0.134, 0.100)	35	
Example 47	97	5.66	6.04	(0.131, 0.102)	30	
Example 48	100	5.40	6.49	(0.134, 0.101)	31	
Example 49	102	5.60	6.22	(0.129, 0.100)	29	
Example 50	103	4.70	6.01	(0.134, 0.101)	36	
Example 51	107	5.40	6.12	(0.134, 0.103)	44	
Example 52	112	5.60	6.21	(0.134, 0.102)	43	
Example 53	113	5.45	6.22	(0.134, 0.101)	37	
Example 54	114	5.39	5.95	(0.134, 0.102)	33	
Example 55	116	4.96	5.95	(0.134, 0.101)	42	
Example 56	119	5.44	6.47	(0.134, 0.102)	36	
Example 57	120	4.47	6.87	(0.134, 0.100)	40	
Example 58	121	4.67	6.64	(0.129, 0.100)	38	
Example 59	128	4.82	6.55	(0.130, 0.099)	36	
Example 60	132	4.44	6.97	(0.134, 0.101)	40	
Example 61	133	4.61	6.89	(0.134, 0.103)	40	
Example 62	135	5.35	6.30	(0.134, 0.102)	33	
Example 63	139	5.62	5.95	(0.134, 0.103)	42	
Example 64	140	5.40	6.12	(0.134, 0.101)	39	
Example 65	141	5.50	5.89	(0.134, 0.100)	41	
Example 66	142	5.44	6.01	(0.134, 0.101)	36	
Example 67	145	5.34	6.58	(0.134, 0.100)	45	
Example 68	149	5.44	6.47	(0.134, 0.102)	36	
Example 69	156	4.47	6.87	(0.134, 0.100)	40	
Example 70	157	4.67	6.64	(0.129, 0.100)	38	

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 71	159	4.82	6.55	(0.130, 0.099)	36	5
Example 72	161	4.44	6.97	(0.134, 0.101)	40	
Example 73	165	4.61	6.89	(0.134, 0.103)	40	
Example 74	166	5.35	6.30	(0.134, 0.102)	33	
Example 75	168	5.62	5.95	(0.134, 0.103)	42	
Example 76	170	5.40	6.12	(0.134, 0.101)	39	15
Example 77	171	5.50	5.89	(0.134, 0.100)	41	
Example 78	172	5.44	6.01	(0.134, 0.101)	36	
Example 79	173	5.34	6.58	(0.134, 0.100)	45	
Example 80	176	5.38	6.93	(0.134, 0.100)	43	
Example 81	178	4.91	6.32	(0.134, 0.100)	41	20
Example 82	179	4.98	6.44	(0.134, 0.100)	40	
Example 83	180	5.62	6.38	(0.134, 0.100)	35	
Example 84	185	5.66	6.04	(0.131, 0.102)	30	
Example 85	187	5.40	6.49	(0.134, 0.101)	31	
Example 86	190	5.60	6.22	(0.129, 0.100)	29	30
Example 87	191	4.70	6.01	(0.134, 0.101)	36	
Example 88	192	5.40	6.12	(0.134, 0.103)	44	
Example 89	194	5.60	6.21	(0.134, 0.102)	43	
Example 90	195	5.45	6.22	(0.134, 0.101)	37	
Example 91	196	5.39	5.95	(0.134, 0.102)	33	40
Example 92	200	4.96	5.95	(0.134, 0.101)	42	
Example 93	202	4.91	6.13	(0.134, 0.101)	39	
Example 94	204	4.91	5.85	(0.134, 0.100)	41	
Example 95	207	4.98	6.38	(0.134, 0.101)	42	
Example 96	210	4.74	6.75	(0.134, 0.101)	51	45
Example 97	213	4.81	6.82	(0.134, 0.102)	53	
Example 98	214	5.16	6.20	(0.134, 0.101)	38	
Example 99	216	5.15	6.42	(0.134, 0.102)	39	
Example 100	217	5.31	6.30	(0.134, 0.103)	37	
Example 101	220	4.82	6.35	(0.134, 0.100)	50	55
Example 102	222	4.91	6.12	(0.134, 0.101)	42	
Example 103	223	4.98	6.51	(0.134, 0.101)	39	
Example 104	224	5.62	6.21	(0.134, 0.100)	41	
Example 105	225	5.39	5.95	(0.134, 0.101)	34	
Example 106	227	4.96	6.88	(0.134, 0.100)	45	60
Example 107	228	4.91	6.93	(0.134, 0.102)	43	

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 108	229	4.76	6.95	(0.134, 0.102)	50	5
Example 109	233	4.77	6.90	(0.134, 0.102)	51	
Example 110	237	4.98	6.05	(0.134, 0.101)	34	
Example 111	240	5.22	6.03	(0.134, 0.101)	43	
Example 112	243	4.82	6.84	(0.134, 0.101)	52	
Example 113	245	4.84	6.97	(0.134, 0.102)	51	15
Example 114	247	5.38	6.88	(0.134, 0.100)	41	
Example 115	249	5.60	6.93	(0.134, 0.101)	32	
Example 116	251	5.45	6.95	(0.134, 0.100)	45	
Example 117	256	4.91	6.93	(0.134, 0.100)	43	
Example 118	257	4.91	6.95	(0.134, 0.100)	41	20
Example 119	259	4.98	6.23	(0.134, 0.100)	40	
Example 120	260	5.62	5.98	(0.134, 0.100)	36	
Example 121	262	4.72	6.51	(0.134, 0.102)	48	
Example 122	265	4.91	6.93	(0.134, 0.100)	43	
Example 123	266	4.91	6.95	(0.134, 0.100)	41	30
Example 124	267	4.98	6.26	(0.134, 0.100)	40	
Example 125	269	5.62	5.98	(0.134, 0.100)	33	
Example 126	271	4.91	6.32	(0.134, 0.100)	41	
Example 127	272	4.98	6.44	(0.134, 0.100)	40	
Example 128	273	5.62	6.38	(0.134, 0.100)	35	40
Example 129	274	5.44	6.34	(0.134, 0.102)	36	
Example 130	276	5.62	6.20	(0.134, 0.101)	39	
Example 131	277	5.62	6.22	(0.134, 0.100)	47	
Example 132	279	5.62	5.98	(0.134, 0.100)	33	
Example 133	281	4.72	6.55	(0.134, 0.102)	48	50
Example 134	284	4.72	6.20	(0.134, 0.102)	43	
Example 135	285	5.40	6.12	(0.134, 0.101)	39	
Example 136	287	5.44	6.21	(0.134, 0.100)	41	
Example 137	288	5.39	6.20	(0.134, 0.101)	36	
Example 138	292	4.96	6.88	(0.134, 0.100)	45	60
Example 139	294	4.91	6.93	(0.134, 0.100)	43	
Example 140	296	4.96	6.88	(0.134, 0.100)	45	
Example 141	299	4.91	6.93	(0.134, 0.102)	43	
Example 142	302	4.98	6.22	(0.134, 0.100)	40	
Example 143	304	4.72	6.53	(0.134, 0.102)	48	65

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 144	305	4.74	6.75	(0.134, 0.101)	51	5
Example 145	307	4.81	6.82	(0.134, 0.102)	53	
Example 146	308	5.16	6.20	(0.134, 0.101)	38	
Example 147	309	5.15	6.42	(0.134, 0.102)	39	
Example 148	310	4.74	6.75	(0.134, 0.101)	51	
Example 149	312	4.81	6.82	(0.134, 0.102)	53	15
Example 150	313	5.16	6.20	(0.134, 0.101)	38	
Example 151	315	5.15	6.42	(0.134, 0.102)	39	
Example 152	316	5.31	6.30	(0.134, 0.103)	37	
Example 153	318	4.82	6.35	(0.134, 0.100)	50	
Example 154	319	4.91	6.12	(0.134, 0.101)	42	25
Example 155	322	4.98	6.51	(0.134, 0.101)	39	
Example 156	325	5.62	6.21	(0.134, 0.100)	41	
Example 157	326	5.39	5.95	(0.134, 0.101)	34	
Example 158	327	4.96	6.88	(0.134, 0.100)	45	
Example 159	329	4.91	6.93	(0.134, 0.102)	43	30
Example 160	331	4.76	6.95	(0.134, 0.102)	50	
Example 161	332	4.77	6.90	(0.134, 0.102)	51	
Example 162	336	4.98	6.05	(0.134, 0.101)	34	
Example 163	337	5.22	6.03	(0.134, 0.101)	43	
Example 164	338	4.82	6.84	(0.134, 0.101)	52	40
Example 165	339	4.84	6.97	(0.134, 0.102)	51	
Example 166	342	5.38	6.88	(0.134, 0.100)	41	
Example 167	343	5.60	6.93	(0.134, 0.101)	32	
Example 168	345	5.45	6.95	(0.134, 0.100)	45	
Example 169	348	4.91	6.93	(0.134, 0.100)	43	45
Example 170	350	4.91	6.95	(0.134, 0.100)	41	
Example 171	353	4.98	6.23	(0.134, 0.100)	40	
Example 172	355	5.62	5.98	(0.134, 0.100)	36	
Example 173	358	4.72	6.51	(0.134, 0.102)	48	
Example 174	359	4.91	6.93	(0.134, 0.100)	43	55
Example 175	360	4.91	6.95	(0.134, 0.100)	41	
Example 176	364	4.98	6.26	(0.134, 0.100)	40	
Example 177	365	5.62	5.98	(0.134, 0.100)	33	
Example 178	367	4.91	6.32	(0.134, 0.100)	41	
Example 179	368	4.98	6.44	(0.134, 0.100)	40	65
Example 180	373	5.62	6.38	(0.134, 0.100)	35	

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 181	374	5.44	6.34	(0.134, 0.102)	36	5
Example 182	377	5.62	6.20	(0.134, 0.101)	39	
Example 183	378	5.62	6.22	(0.134, 0.100)	47	
Example 184	379	5.62	5.98	(0.134, 0.100)	33	
Example 185	380	4.72	6.55	(0.134, 0.102)	48	
Example 186	384	4.72	6.20	(0.134, 0.102)	43	15
Example 187	386	5.40	6.12	(0.134, 0.101)	39	
Example 188	387	5.44	6.21	(0.134, 0.100)	41	
Example 189	388	5.39	6.20	(0.134, 0.101)	36	
Example 190	392	4.96	6.88	(0.134, 0.100)	45	
Example 191	395	4.91	6.93	(0.134, 0.100)	43	25
Example 192	396	4.96	6.88	(0.134, 0.100)	45	
Example 193	397	4.91	6.93	(0.134, 0.102)	43	
Example 194	399	4.98	6.22	(0.134, 0.100)	40	
Example 195	400	4.72	6.53	(0.134, 0.102)	48	
Example 196	401	4.72	6.53	(0.134, 0.102)	48	30
Example 197	404	5.45	6.95	(0.134, 0.100)	45	
Example 198	407	4.91	6.93	(0.134, 0.100)	43	
Example 199	408	4.91	6.95	(0.134, 0.100)	41	
Example 200	409	4.98	6.23	(0.134, 0.100)	40	
Example 201	411	5.62	5.98	(0.134, 0.100)	36	40
Example 202	413	4.72	6.51	(0.134, 0.102)	48	
Example 203	415	4.91	6.93	(0.134, 0.100)	43	
Example 204	416	4.91	6.95	(0.134, 0.100)	41	
Example 205	417	4.98	6.26	(0.134, 0.100)	40	
Example 206	421	5.62	5.98	(0.134, 0.100)	33	45
Example 207	423	4.91	6.32	(0.134, 0.100)	41	
Example 208	424	4.98	6.44	(0.134, 0.100)	40	
Example 209	426	4.91	6.95	(0.134, 0.100)	41	
Example 210	431	4.98	6.23	(0.134, 0.100)	40	
Example 211	432	5.62	5.98	(0.134, 0.100)	36	55
Example 212	433	4.72	6.51	(0.134, 0.102)	48	
Example 213	436	4.91	6.93	(0.134, 0.100)	43	
Example 214	438	4.91	6.95	(0.134, 0.100)	41	
Example 215	440	4.98	6.26	(0.134, 0.100)	40	
Example 216	441	5.62	5.98	(0.134, 0.100)	33	65

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 217	443	4.91	6.32	(0.134, 0.100)	41	5
Example 218	447	4.98	6.44	(0.134, 0.100)	40	
Example 219	448	5.62	6.38	(0.134, 0.100)	35	
Example 220	449	5.44	6.34	(0.134, 0.102)	36	
Example 221	450	5.62	6.20	(0.134, 0.101)	39	
Example 222	452	5.62	6.22	(0.134, 0.100)	47	15
Example 223	456	5.62	5.98	(0.134, 0.100)	33	
Example 224	460	4.72	6.55	(0.134, 0.102)	48	
Example 225	462	4.72	6.20	(0.134, 0.102)	43	
Example 226	465	5.40	6.12	(0.134, 0.101)	39	
Example 227	467	5.44	6.21	(0.134, 0.100)	41	25
Example 228	468	5.39	6.20	(0.134, 0.101)	36	
Example 229	470	4.96	6.88	(0.134, 0.100)	45	
Example 230	475	4.91	6.93	(0.134, 0.100)	43	
Example 231	476	4.96	6.88	(0.134, 0.100)	45	
Example 232	477	4.91	6.93	(0.134, 0.102)	43	30
Example 233	479	4.98	6.22	(0.134, 0.100)	40	
Example 234	481	4.72	6.53	(0.134, 0.102)	48	
Example 235	482	4.72	6.53	(0.134, 0.102)	48	
Example 236	485	5.45	6.95	(0.134, 0.100)	45	
Example 237	488	4.91	6.93	(0.134, 0.100)	43	40
Example 238	489	4.91	6.95	(0.134, 0.100)	41	
Example 239	490	4.98	6.23	(0.134, 0.100)	40	
Example 240	493	4.74	6.75	(0.134, 0.101)	51	
Example 241	494	4.81	6.82	(0.134, 0.102)	53	
Example 242	495	5.16	6.20	(0.134, 0.101)	38	45
Example 243	500	5.15	6.42	(0.134, 0.102)	39	
Example 244	504	5.31	6.30	(0.134, 0.103)	37	
Example 245	505	4.82	6.35	(0.134, 0.100)	50	
Example 246	506	4.91	6.12	(0.134, 0.101)	42	
Example 247	509	4.98	6.51	(0.134, 0.101)	39	55
Example 248	512	5.62	6.21	(0.134, 0.100)	41	
Example 249	513	5.39	5.95	(0.134, 0.101)	34	
Example 250	516	4.96	6.88	(0.134, 0.100)	45	
Example 251	517	4.91	6.93	(0.134, 0.102)	43	
Example 252	518	4.76	6.95	(0.134, 0.102)	50	65
Example 253	519	4.77	6.90	(0.134, 0.102)	51	

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)
Example 254	520	4.98	6.05	(0.134, 0.101)	34
Example 255	522	5.22	6.03	(0.134, 0.101)	43
Example 256	524	4.82	6.84	(0.134, 0.101)	52
Example 257	525	4.84	6.97	(0.134, 0.102)	51
Example 258	527	5.38	6.88	(0.134, 0.100)	41
Example 259	528	5.60	6.93	(0.134, 0.101)	32
Example 260	531	5.45	6.95	(0.134, 0.100)	45
Example 261	532	4.91	6.93	(0.134, 0.100)	43
Example 262	533	4.91	6.95	(0.134, 0.100)	41
Example 263	535	4.98	6.23	(0.134, 0.100)	40
Example 264	536	5.62	5.98	(0.134, 0.100)	36
Example 265	539	4.72	6.51	(0.134, 0.102)	48
Example 266	542	4.91	6.93	(0.134, 0.100)	43
Example 267	543	4.91	6.95	(0.134, 0.100)	41
Example 268	544	4.98	6.26	(0.134, 0.100)	40
Example 269	547	5.62	5.98	(0.134, 0.100)	33
Example 270	548	4.91	6.32	(0.134, 0.100)	41
Example 271	549	4.98	6.44	(0.134, 0.100)	40
Example 272	550	5.62	6.38	(0.134, 0.100)	35
Example 273	554	5.44	6.34	(0.134, 0.102)	36
Example 274	555	5.62	6.20	(0.134, 0.101)	39
Example 275	557	5.62	6.22	(0.134, 0.100)	47
Example 276	558	5.62	5.98	(0.134, 0.100)	33
Example 277	564	4.72	6.55	(0.134, 0.102)	48
Example 278	564	4.72	6.20	(0.134, 0.102)	43
Example 279	565	5.40	6.12	(0.134, 0.101)	39
Example 280	565	5.44	6.21	(0.134, 0.100)	41
Example 281	566	5.39	6.20	(0.134, 0.101)	36
Example 282	566	4.96	6.88	(0.134, 0.100)	45
Example 283	569	4.91	6.93	(0.134, 0.100)	43
Example 284	569	4.96	6.88	(0.134, 0.100)	45
Example 285	573	4.91	6.93	(0.134, 0.102)	43
Example 286	573	4.98	6.22	(0.134, 0.100)	40
Example 287	574	4.72	6.53	(0.134, 0.102)	48
Example 288	574	4.74	6.75	(0.134, 0.101)	51
Example 289	580	4.81	6.82	(0.134, 0.102)	53

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 290	580	5.16	6.20	(0.134, 0.101)	38	5
Example 291	583	5.15	6.42	(0.134, 0.102)	39	
Example 292	584	4.74	6.75	(0.134, 0.101)	51	
Example 293	585	4.81	6.82	(0.134, 0.102)	53	
Example 294	587	5.16	6.20	(0.134, 0.101)	38	
Example 295	588	5.15	6.42	(0.134, 0.102)	39	15
Example 296	594	5.31	6.30	(0.134, 0.103)	37	
Example 297	597	4.82	6.35	(0.134, 0.100)	50	
Example 298	598	4.91	6.12	(0.134, 0.101)	42	
Example 299	599	4.98	6.51	(0.134, 0.101)	39	
Example 300	600	5.62	6.21	(0.134, 0.100)	41	25
Example 301	601	5.39	5.95	(0.134, 0.101)	34	
Example 302	605	4.96	6.88	(0.134, 0.100)	45	
Example 303	606	4.91	6.93	(0.134, 0.102)	43	
Example 304	611	4.76	6.95	(0.134, 0.102)	50	
Example 305	612	4.77	6.90	(0.134, 0.102)	51	30
Example 306	615	4.98	6.05	(0.134, 0.101)	34	
Example 307	618	5.22	6.03	(0.134, 0.101)	43	
Example 308	620	4.82	6.84	(0.134, 0.101)	52	
Example 309	623	4.84	6.97	(0.134, 0.102)	51	
Example 310	624	5.38	6.88	(0.134, 0.100)	41	40
Example 311	626	5.60	6.93	(0.134, 0.101)	32	
Example 312	629	5.45	6.95	(0.134, 0.100)	45	
Example 313	632	4.91	6.93	(0.134, 0.100)	43	
Example 314	634	4.91	6.95	(0.134, 0.100)	41	
Example 315	635	4.98	6.23	(0.134, 0.100)	40	45
Example 316	636	5.62	5.98	(0.134, 0.100)	36	
Example 317	640	4.72	6.51	(0.134, 0.102)	48	
Example 318	641	4.91	6.93	(0.134, 0.100)	43	
Example 319	644	4.91	6.95	(0.134, 0.100)	41	
Example 320	646	4.98	6.26	(0.134, 0.100)	40	55
Example 321	649	5.62	5.98	(0.134, 0.100)	33	
Example 322	651	4.91	6.32	(0.134, 0.100)	41	
Example 323	652	4.98	6.44	(0.134, 0.100)	40	
Example 324	653	5.62	6.38	(0.134, 0.100)	35	
Example 325	654	5.44	6.34	(0.134, 0.102)	36	65
Example 326	655	5.62	6.20	(0.134, 0.101)	39	

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 327	657	5.62	6.22	(0.134, 0.100)	47	
Example 328	659	5.62	5.98	(0.134, 0.100)	33	
Example 329	660	4.72	6.55	(0.134, 0.102)	48	
Example 330	661	4.72	6.20	(0.134, 0.102)	43	
Example 331	664	5.40	6.12	(0.134, 0.101)	39	
Example 332	665	5.44	6.21	(0.134, 0.100)	41	
Example 333	666	5.39	6.20	(0.134, 0.101)	36	
Example 334	670	4.96	6.88	(0.134, 0.100)	45	
Example 335	672	4.91	6.93	(0.134, 0.100)	43	
Example 336	674	4.96	6.88	(0.134, 0.100)	45	
Example 337	675	4.91	6.93	(0.134, 0.102)	43	
Example 338	677	4.98	6.22	(0.134, 0.100)	40	
Example 339	680	4.72	6.53	(0.134, 0.102)	48	
Example 340	682	4.72	6.53	(0.134, 0.102)	48	
Example 341	683	5.45	6.95	(0.134, 0.100)	45	
Example 342	685	4.91	6.93	(0.134, 0.100)	43	
Example 343	687	4.91	6.95	(0.134, 0.100)	41	
Example 344	688	4.98	6.23	(0.134, 0.100)	40	
Example 345	691	5.62	5.98	(0.134, 0.100)	36	
Example 346	692	4.72	6.51	(0.134, 0.102)	48	
Example 347	695	4.91	6.93	(0.134, 0.100)	43	
Example 348	696	4.91	6.95	(0.134, 0.100)	41	
Example 349	697	4.98	6.26	(0.134, 0.100)	40	
Example 350	700	5.62	5.98	(0.134, 0.100)	33	
Example 351	703	4.91	6.32	(0.134, 0.100)	41	
Example 352	704	4.98	6.44	(0.134, 0.100)	40	
Example 353	707	4.91	6.95	(0.134, 0.100)	41	
Example 354	708	4.98	6.23	(0.134, 0.100)	40	
Example 355	709	5.62	5.98	(0.134, 0.100)	36	
Example 356	713	4.72	6.51	(0.134, 0.102)	48	
Example 357	714	4.91	6.93	(0.134, 0.100)	43	
Example 358	716	4.91	6.95	(0.134, 0.100)	41	
Example 359	717	4.98	6.26	(0.134, 0.100)	40	
Example 360	721	5.62	5.98	(0.134, 0.100)	33	
Example 361	728	4.91	6.32	(0.134, 0.100)	41	
Example 362	729	4.98	6.44	(0.134, 0.100)	40	

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 363	731	5.62	6.38	(0.134, 0.100)	35	5
Example 364	732	5.44	6.34	(0.134, 0.102)	36	
Example 365	733	5.62	6.20	(0.134, 0.101)	39	
Example 366	734	5.62	6.22	(0.134, 0.100)	47	
Example 367	735	5.62	5.98	(0.134, 0.100)	33	
Example 368	736	4.72	6.55	(0.134, 0.102)	48	15
Example 369	741	4.72	6.20	(0.134, 0.102)	43	
Example 370	743	4.74	6.75	(0.134, 0.101)	51	
Example 371	745	4.81	6.82	(0.134, 0.102)	53	
Example 372	748	5.16	6.20	(0.134, 0.101)	38	
Example 373	750	5.15	6.42	(0.134, 0.102)	39	25
Example 374	752	5.31	6.30	(0.134, 0.103)	37	
Example 375	756	4.82	6.35	(0.134, 0.100)	50	
Example 376	757	4.91	6.12	(0.134, 0.101)	42	
Example 377	761	4.98	6.51	(0.134, 0.101)	39	
Example 378	764	5.62	6.21	(0.134, 0.100)	41	30
Example 379	766	5.39	5.95	(0.134, 0.101)	34	
Example 380	767	4.96	6.88	(0.134, 0.100)	45	
Example 381	769	4.91	6.93	(0.134, 0.102)	43	
Example 382	771	4.76	6.95	(0.134, 0.102)	50	
Example 383	773	4.77	6.90	(0.134, 0.102)	51	40
Example 384	775	4.98	6.05	(0.134, 0.101)	34	
Example 385	779	5.22	6.03	(0.134, 0.101)	43	
Example 386	783	4.82	6.84	(0.134, 0.101)	52	
Example 387	784	4.84	6.97	(0.134, 0.102)	51	
Example 388	786	5.38	6.88	(0.134, 0.100)	41	45
Example 389	787	5.60	6.93	(0.134, 0.101)	32	
Example 390	789	5.45	6.95	(0.134, 0.100)	45	
Example 391	791	4.91	6.93	(0.134, 0.100)	43	
Example 392	792	4.91	6.95	(0.134, 0.100)	41	
Example 393	793	4.98	6.23	(0.134, 0.100)	40	55
Example 394	798	5.62	5.98	(0.134, 0.100)	36	
Example 395	801	4.72	6.51	(0.134, 0.102)	48	
Example 396	802	4.91	6.93	(0.134, 0.100)	43	
Example 397	807	4.91	6.95	(0.134, 0.100)	41	
Example 398	809	4.98	6.26	(0.134, 0.100)	40	65
Example 399	810	5.62	5.98	(0.134, 0.100)	33	

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)	
Example 400	814	4.91	6.32	(0.134, 0.100)	41	5
Example 401	816	4.98	6.44	(0.134, 0.100)	40	
Example 402	817	5.62	6.38	(0.134, 0.100)	35	
Example 403	818	5.44	6.34	(0.134, 0.102)	36	
Example 404	822	5.62	6.20	(0.134, 0.101)	39	
Example 405	823	5.62	6.22	(0.134, 0.100)	47	15
Example 406	825	5.62	5.98	(0.134, 0.100)	33	
Example 407	829	4.72	6.55	(0.134, 0.102)	48	
Example 408	831	4.72	6.20	(0.134, 0.102)	43	
Example 409	833	5.40	6.12	(0.134, 0.101)	39	
Example 410	836	5.44	6.21	(0.134, 0.100)	41	25
Example 411	838	5.39	6.20	(0.134, 0.101)	36	
Example 412	842	4.96	6.88	(0.134, 0.100)	45	
Example 413	844	4.91	6.93	(0.134, 0.100)	43	
Example 414	847	4.96	6.88	(0.134, 0.100)	45	
Example 415	848	4.91	6.93	(0.134, 0.102)	43	30
Example 416	849	4.98	6.22	(0.134, 0.100)	40	
Example 417	858	4.72	6.53	(0.134, 0.102)	48	
Example 418	860	4.74	6.75	(0.134, 0.101)	51	
Example 419	861	4.81	6.82	(0.134, 0.102)	53	
Example 420	862	5.16	6.20	(0.134, 0.101)	38	40
Example 421	863	5.15	6.42	(0.134, 0.102)	39	
Example 422	870	4.74	6.75	(0.134, 0.101)	51	
Example 423	871	4.81	6.82	(0.134, 0.102)	53	
Example 424	872	5.16	6.20	(0.134, 0.101)	38	
Example 425	883	5.15	6.42	(0.134, 0.102)	39	50
Example 426	884	5.31	6.30	(0.134, 0.103)	37	
Example 427	887	4.82	6.35	(0.134, 0.100)	50	
Example 428	888	4.91	6.12	(0.134, 0.101)	42	
Example 429	890	4.98	6.51	(0.134, 0.101)	39	
Example 430	893	5.62	6.21	(0.134, 0.100)	41	60
Example 431	905	5.39	5.95	(0.134, 0.101)	34	
Example 432	908	4.96	6.88	(0.134, 0.100)	45	
Example 433	910	4.91	6.93	(0.134, 0.102)	43	
Example 434	911	4.76	6.95	(0.134, 0.102)	50	
Example 435	916	4.77	6.90	(0.134, 0.102)	51	65

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TABLE 20-continued

	Compound	Driving Voltage (V)	Light Emission Efficiency (cd/A)	CIE (x, y)	Lifetime (T95)
Example 436	918	4.98	6.05	(0.134, 0.101)	34
Example 437	919	5.22	6.03	(0.134, 0.101)	43
Example 438	920	4.82	6.84	(0.134, 0.101)	52
Example 439	923	4.84	6.97	(0.134, 0.102)	51
Example 440	924	5.38	6.88	(0.134, 0.100)	41
Example 441	930	5.60	6.93	(0.134, 0.101)	32
Example 442	931	5.45	6.95	(0.134, 0.100)	45
Example 443	932	4.91	6.93	(0.134, 0.100)	43
Example 444	943	4.91	6.95	(0.134, 0.100)	41
Example 445	945	4.98	6.23	(0.134, 0.100)	40
Example 446	952	5.62	5.98	(0.134, 0.100)	36
Example 447	956	4.72	6.51	(0.134, 0.102)	48
Example 448	957	4.91	6.93	(0.134, 0.100)	43
Example 449	960	4.91	6.95	(0.134, 0.100)	41
Example 450	963	4.98	6.26	(0.134, 0.100)	40
Example 451	965	5.62	5.98	(0.134, 0.100)	33
Example 452	968	4.91	6.32	(0.134, 0.100)	41
Example 453	971	4.98	6.44	(0.134, 0.100)	40
Example 454	979	5.62	6.38	(0.134, 0.100)	35
Example 455	980	5.44	6.34	(0.134, 0.102)	36
Example 456	984	5.62	6.20	(0.134, 0.101)	39
Example 457	986	5.62	6.22	(0.134, 0.100)	47
Example 458	991	5.62	5.98	(0.134, 0.100)	33
Example 459	995	4.72	6.55	(0.134, 0.102)	48
Example 460	998	4.72	6.20	(0.134, 0.102)	43
Example 461	1000	5.40	6.12	(0.134, 0.101)	39
Example 462	1004	5.44	6.21	(0.134, 0.100)	41
Example 463	1006	5.39	6.20	(0.134, 0.101)	36
Example 464	1011	4.96	6.88	(0.134, 0.100)	45
Example 465	1015	4.91	6.93	(0.134, 0.100)	43
Example 466	1080	4.96	6.88	(0.134, 0.100)	45
Example 467	1086	4.91	6.93	(0.134, 0.102)	43
Example 468	1098	4.98	6.22	(0.134, 0.100)	40
Example 469	1099	4.72	6.53	(0.134, 0.102)	48

As seen from Table 20, the organic light emitting device using the compound of the present disclosure as an electron transfer layer material of a blue organic light emitting device had lower driving voltage and improved light emission efficiency and lifetime compared to Comparative Example

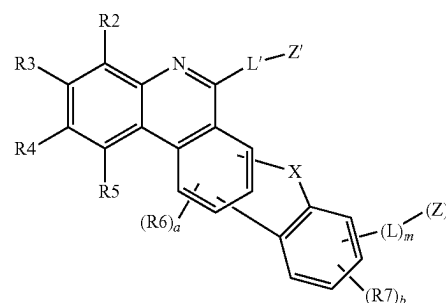
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1, Comparative Example 2, Comparative Example 3 and Comparative Example 4. Such results are due to the fact that the compound is a bipolar type having both a p-type and an n-type, and therefore, hole leakage is prevented and electrons are effectively injected to a light emitting layer.

The invention claimed is:

1. A heterocyclic compound represented by the following Chemical Formula 1:

[Chemical Formula 1-1]



wherein, in Chemical Formula 1,

X is O or S,

L is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group,

Z is a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group; or a substituted or unsubstituted phosphine oxide group,

R1 is a substituted or unsubstituted C6 to C30 aryl group,

R2 to R7 are each independently hydrogen; or deuterium, a is 1 or 2,

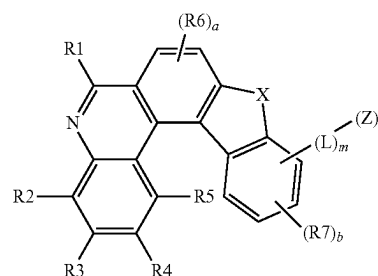
b is an integer of 1 to 3,

m and n are each an integer of 1 to 5, and

when a, b, m and n are each 2 or greater, substituents in the parentheses are the same as or different from each other.

2. The heterocyclic compound of claim 1, wherein Chemical Formula 1 is represented by any one of the following Chemical Formula 2 to Chemical Formula 4:

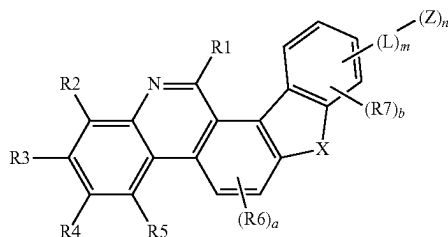
[Chemical Formula 2]



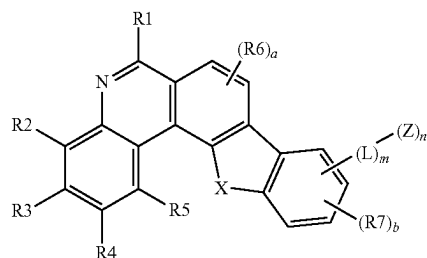
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[Chemical Formula 3]



[Chemical Formula 4]



in Chemical Formulae 2 to 4,

X, L, Z, R1 to R7, a, b, m and n have the same definitions as in Chemical Formula 1.

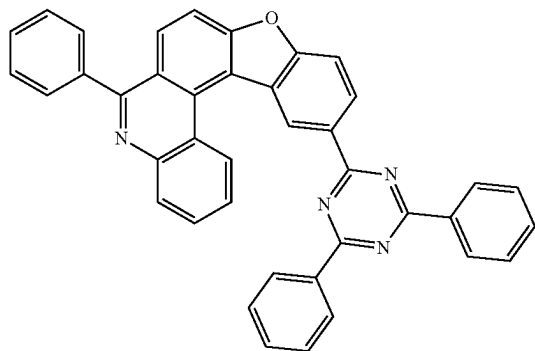
3. The heterocyclic compound of claim 1, wherein L is a direct bond; a substituted or unsubstituted phenylene group;

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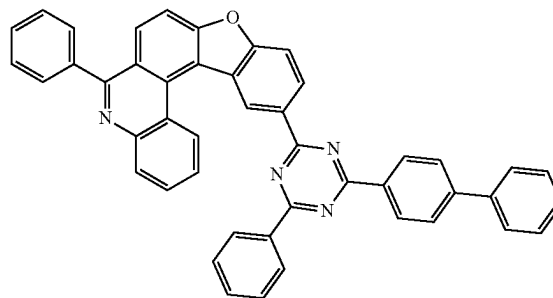
a substituted or unsubstituted biphenylene group; a substituted or unsubstituted terphenylene group; a substituted or unsubstituted naphthylene group; a substituted or unsubstituted anthracenylene group; a substituted or unsubstituted triphenylenylene group; a substituted or unsubstituted phenanthrenylene group; a substituted or unsubstituted divalent pyridine group; a substituted or unsubstituted divalent pyrimidine group; or a substituted or unsubstituted divalent triazine group.

4. The heterocyclic compound of claim 1, wherein Z is a substituted or unsubstituted phenyl group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a substituted or unsubstituted pyridine group; a substituted or unsubstituted pyrimidine group; a substituted or unsubstituted triazine group; a substituted or unsubstituted carbazole group; a substituted or unsubstituted benzocarbazole group; a substituted or unsubstituted phenanthrolinyl group; a substituted or unsubstituted dibenzofuran group; a substituted or unsubstituted dibenzothiophene group; a substituted or unsubstituted dihydroacridine group; or a substituted or unsubstituted phosphine oxide group.

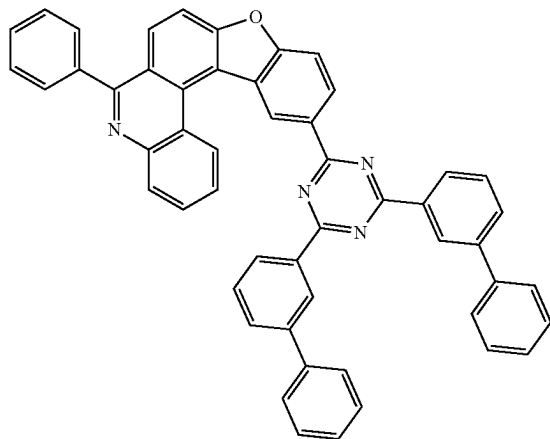
5. The heterocyclic compound of claim 1, wherein Chemical Formula 1 is represented by any one of the following compounds:



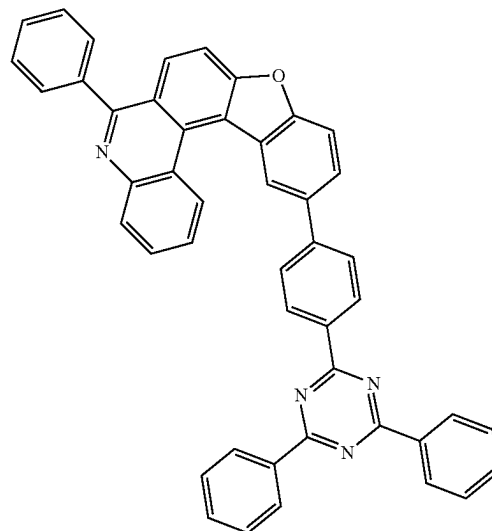
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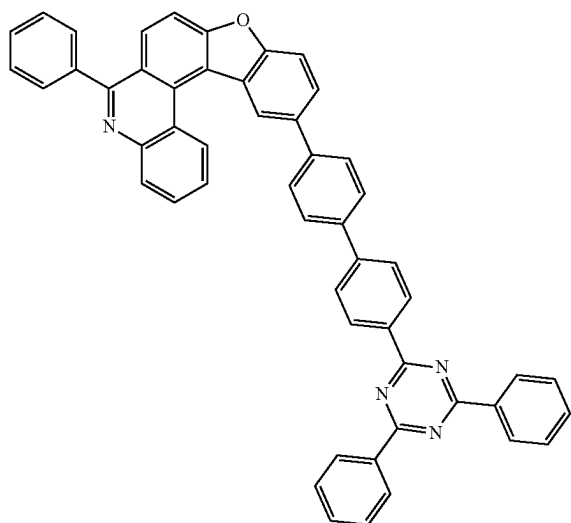


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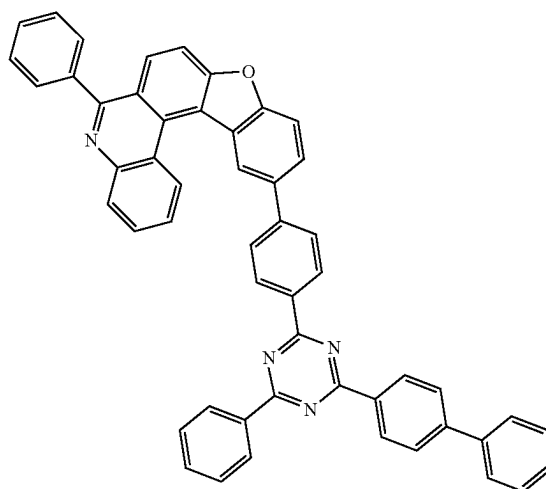
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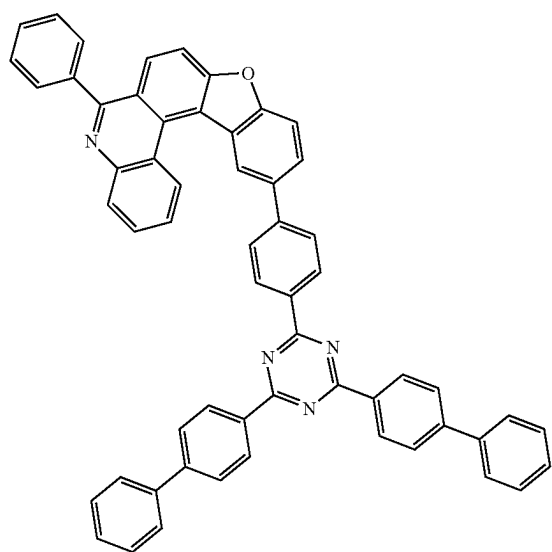
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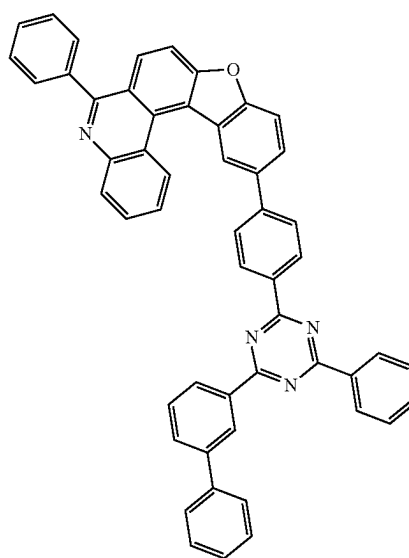
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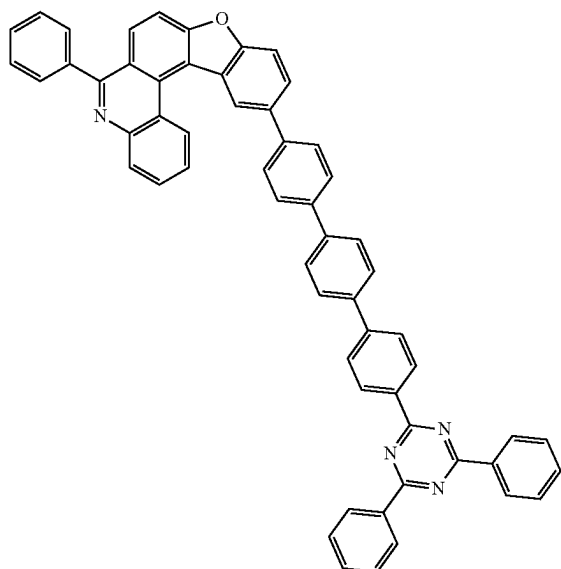
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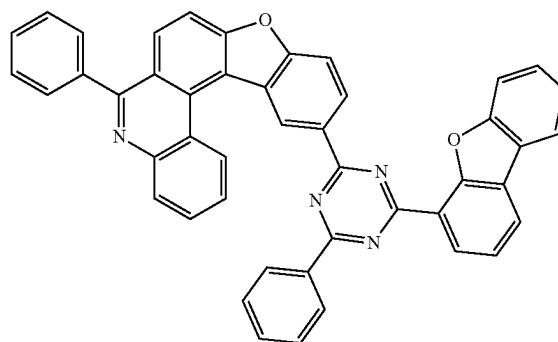
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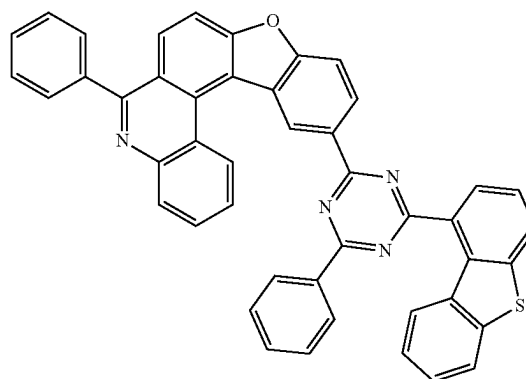
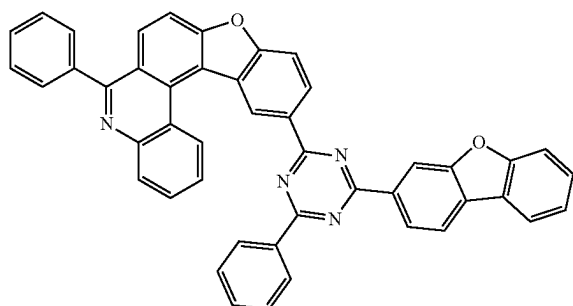
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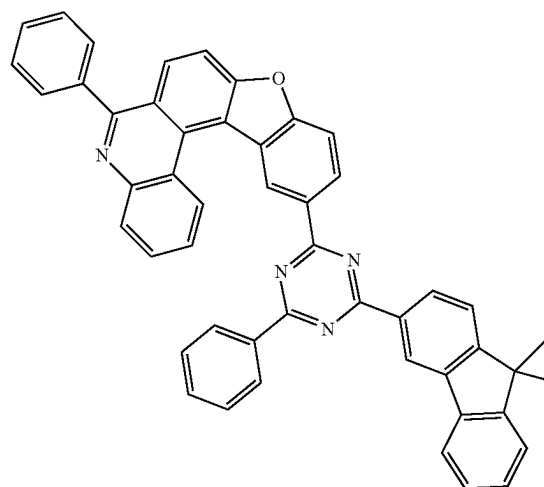
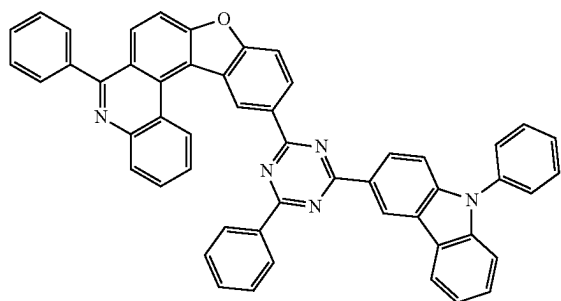


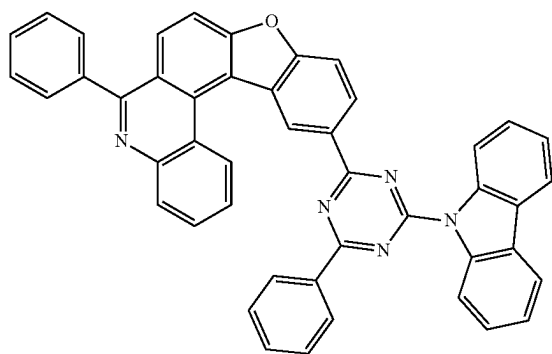
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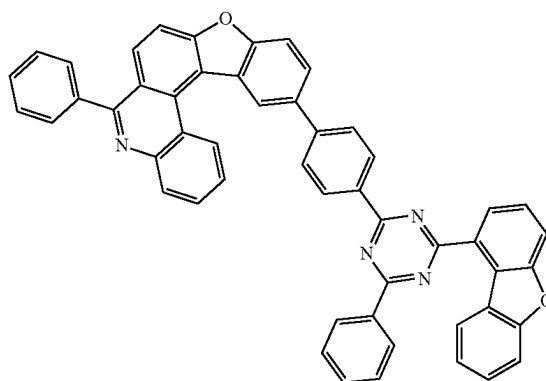
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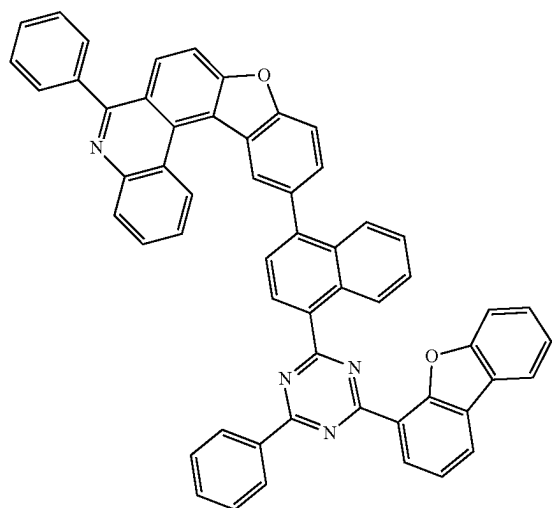
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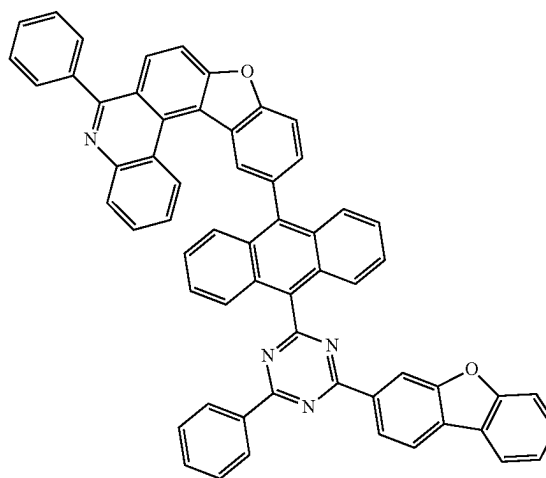
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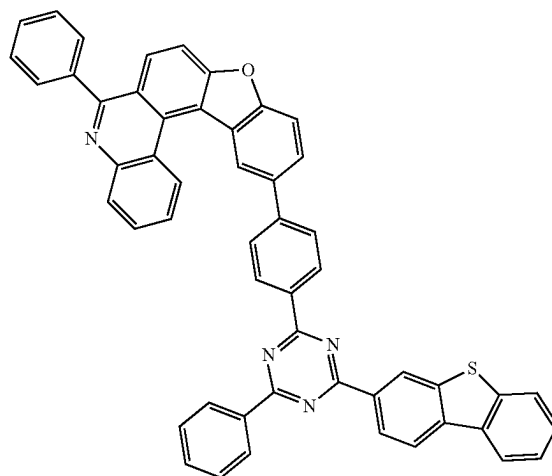
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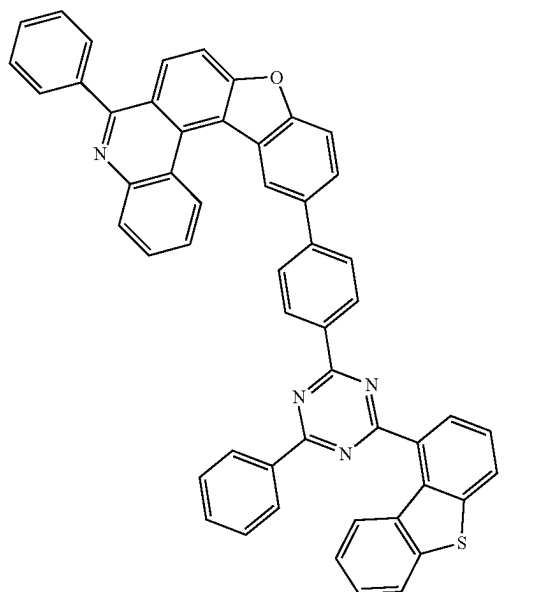
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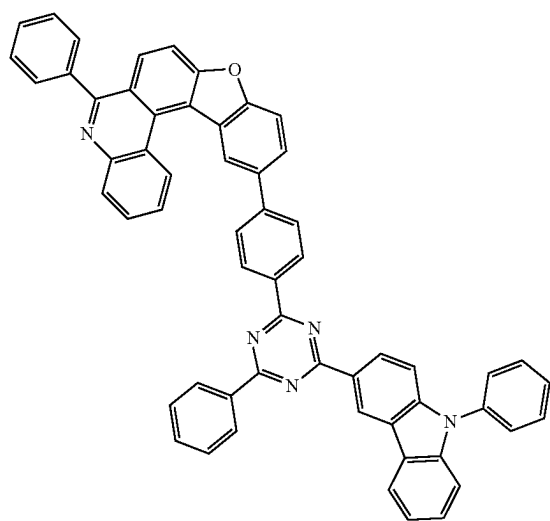
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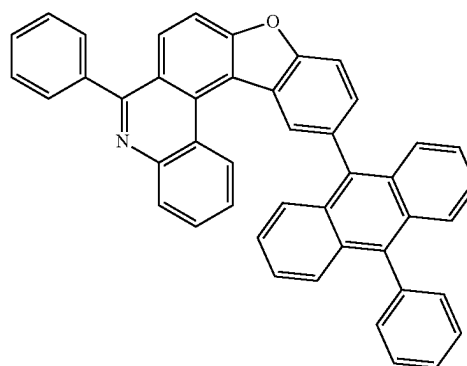
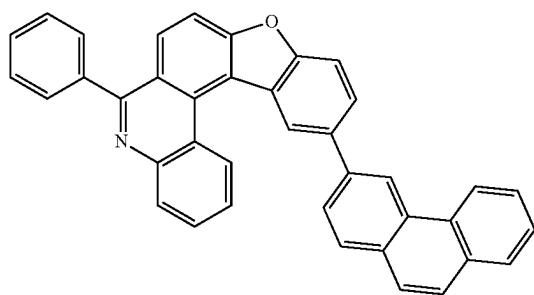
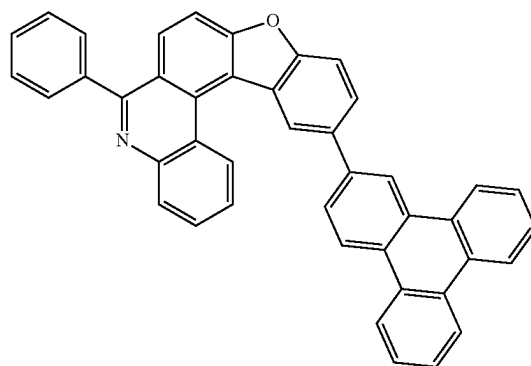
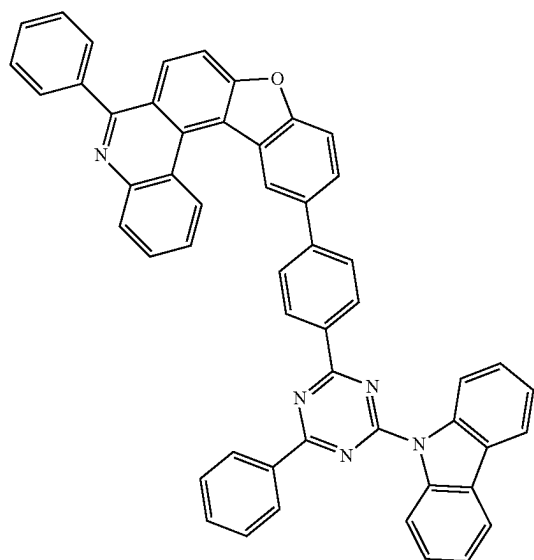
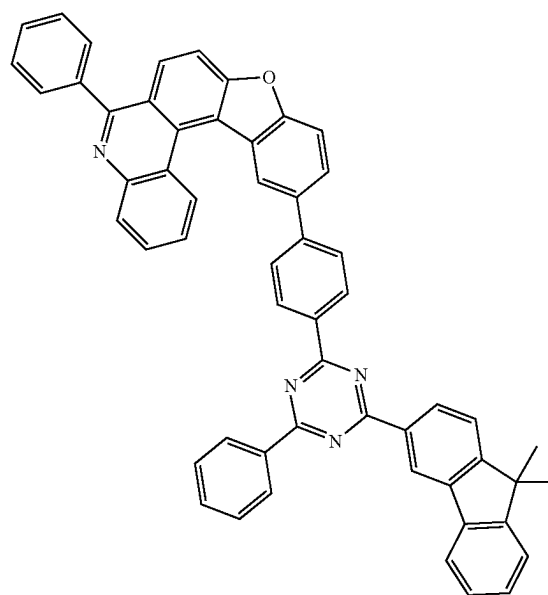


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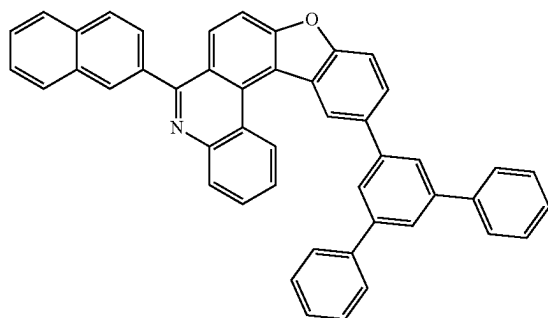


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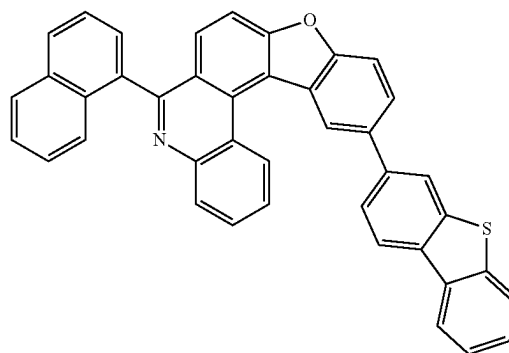
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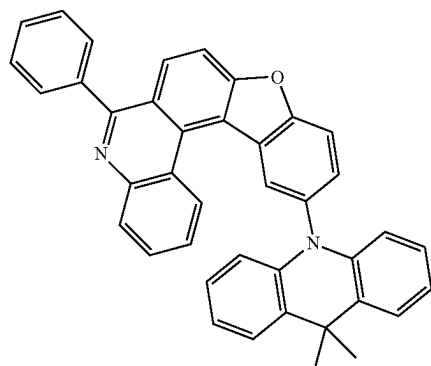
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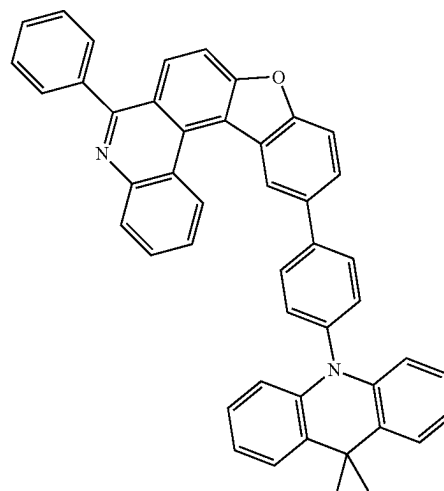


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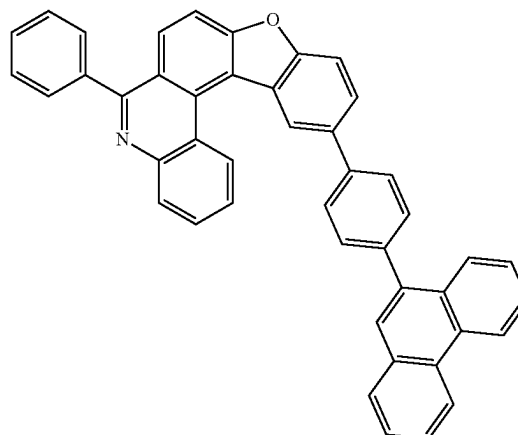
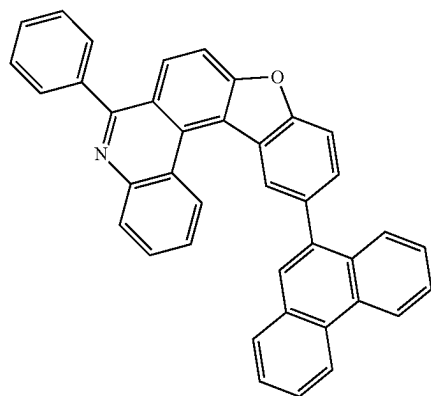
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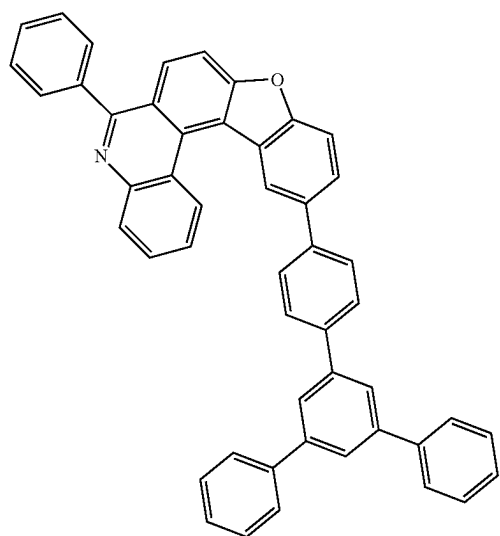


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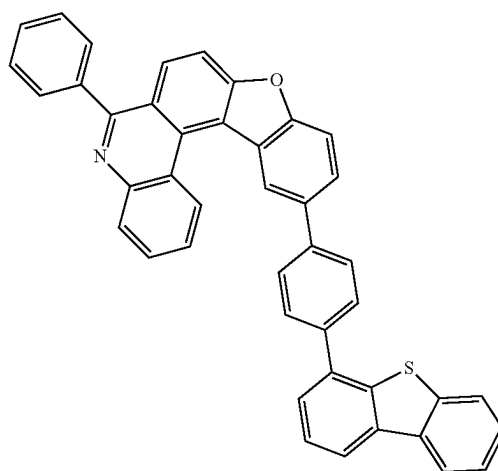
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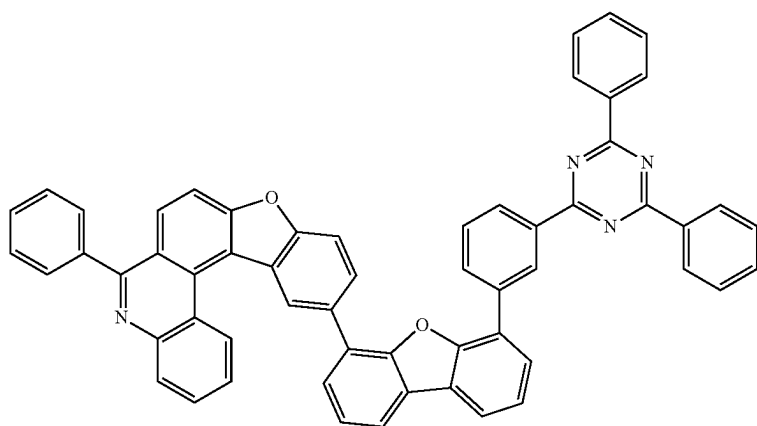
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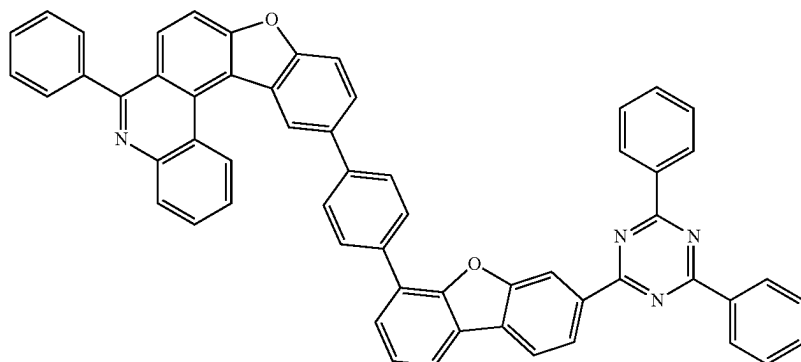
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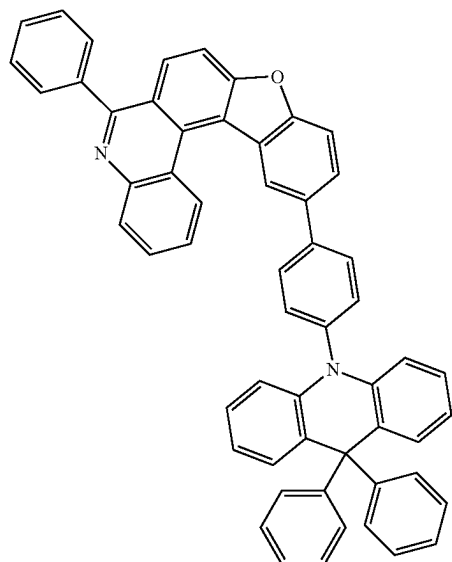


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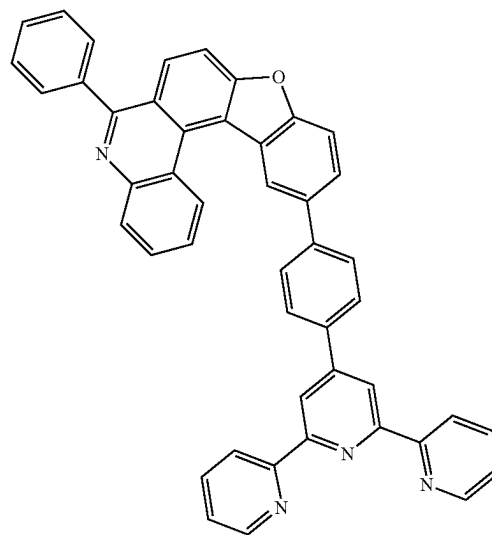
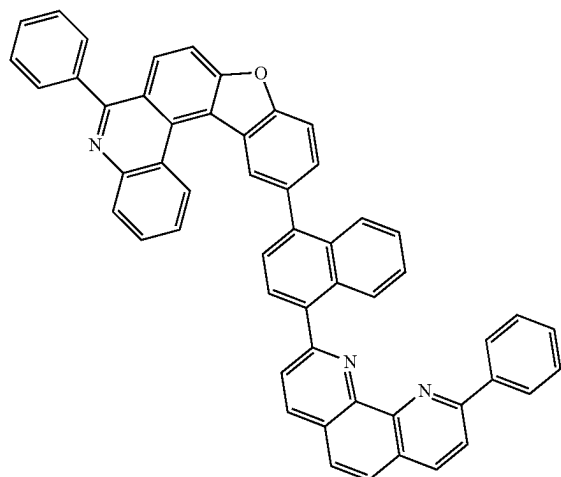
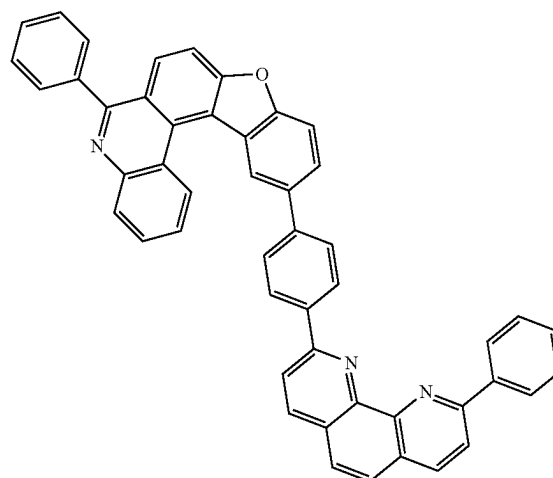
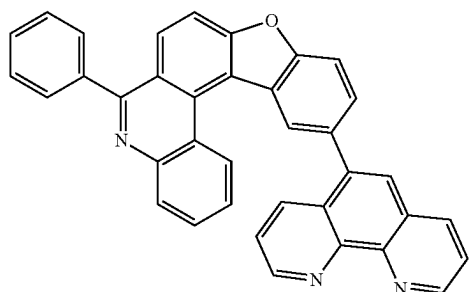
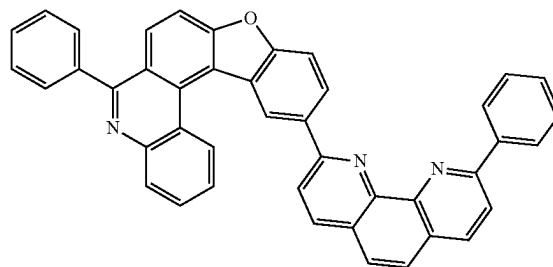
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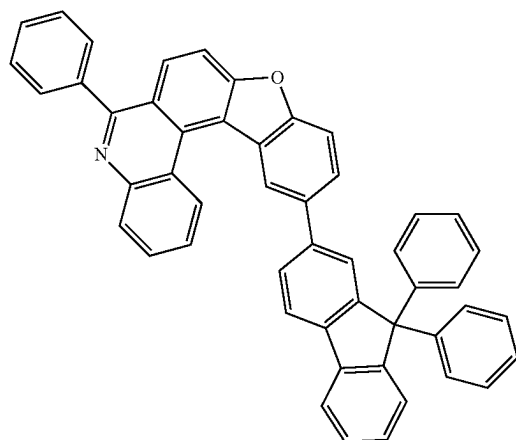


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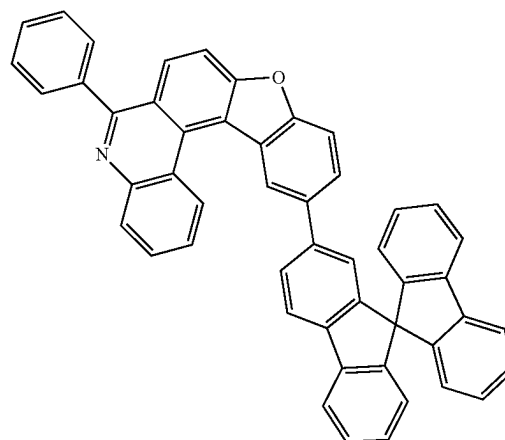
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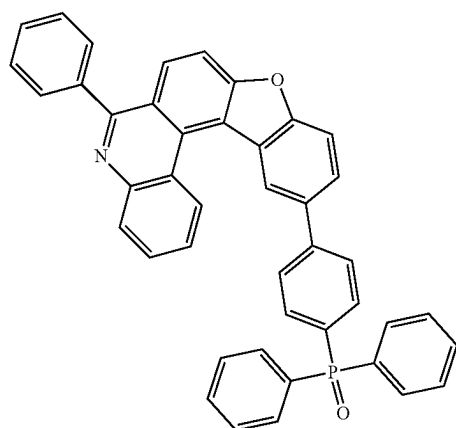
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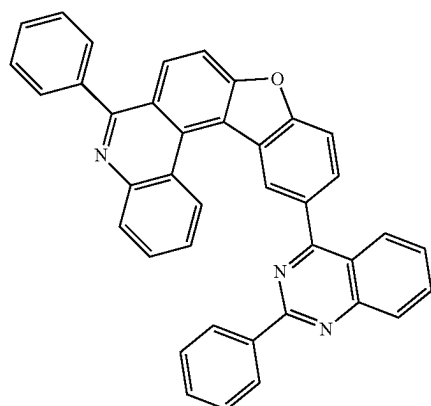
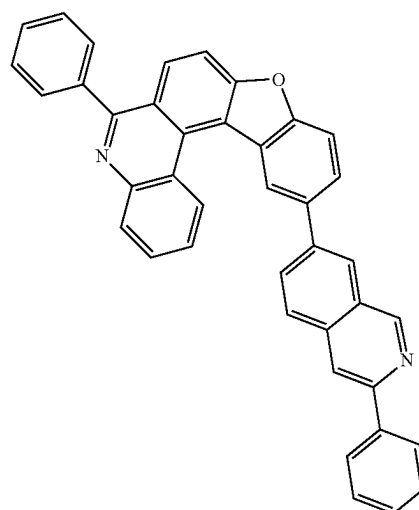


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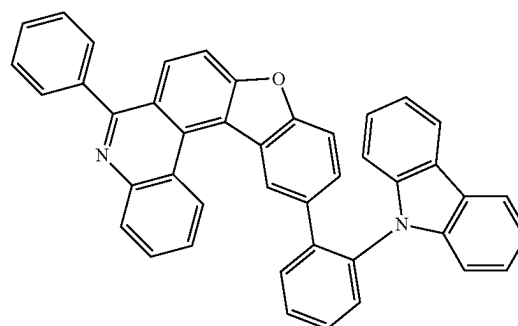
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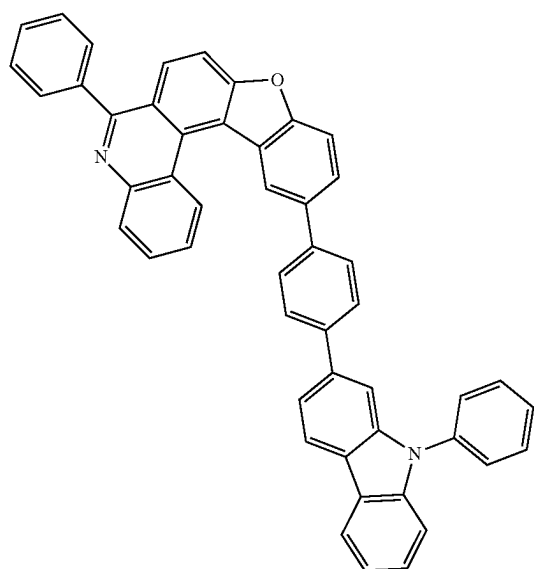


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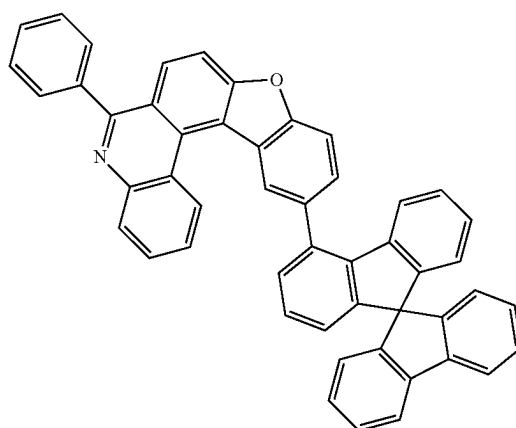
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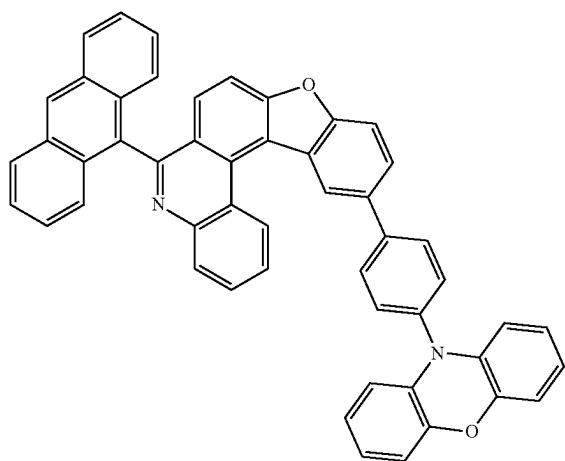
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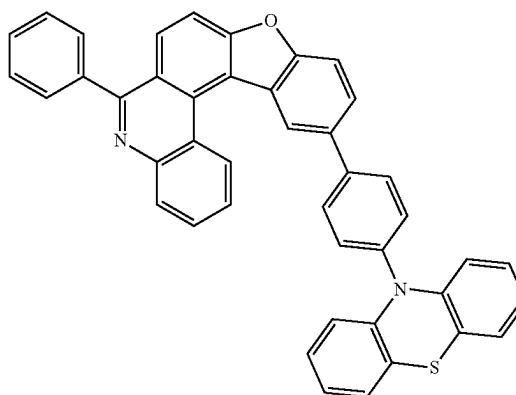


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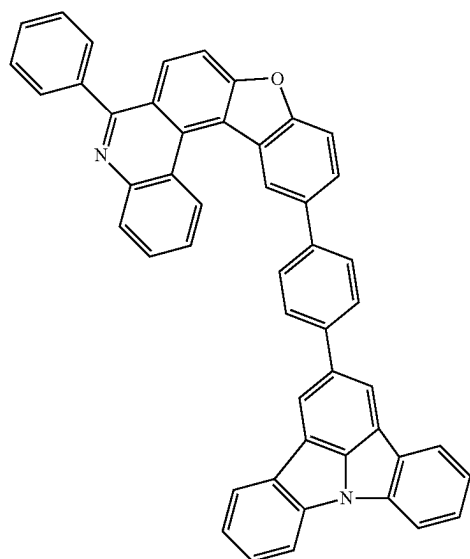
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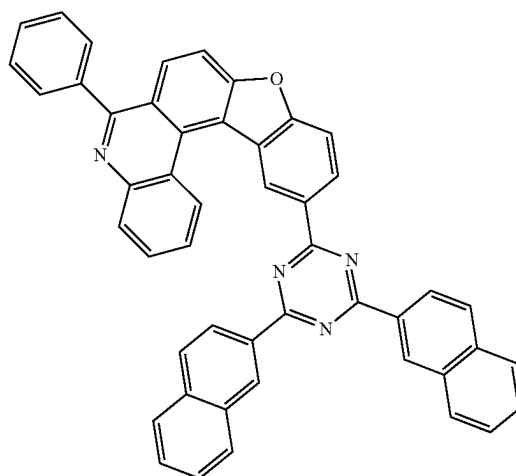
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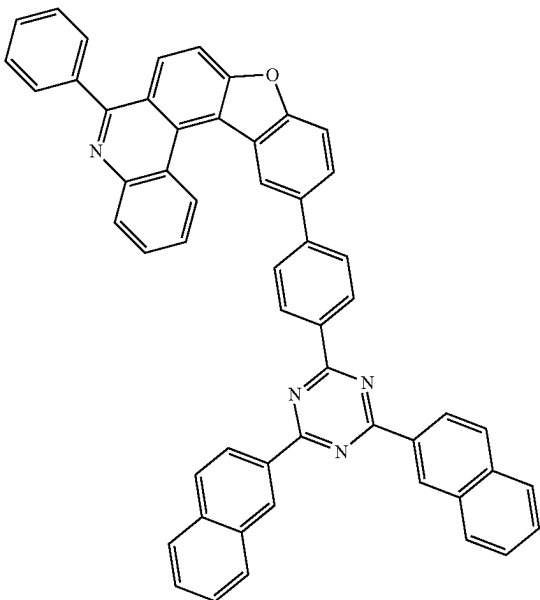
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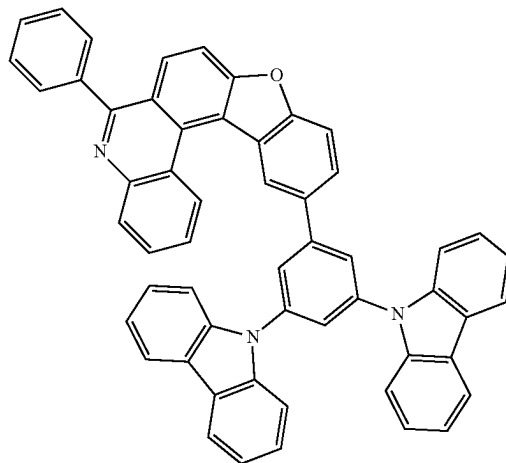
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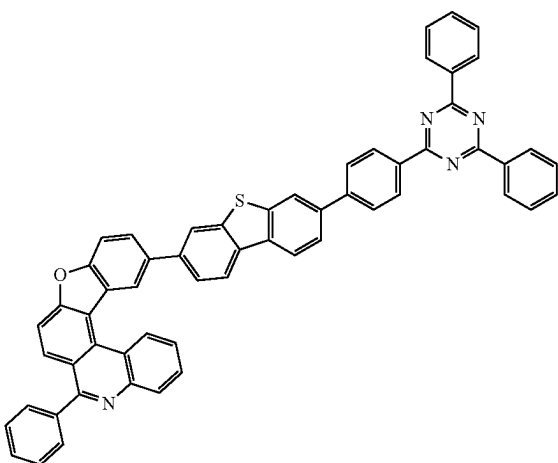
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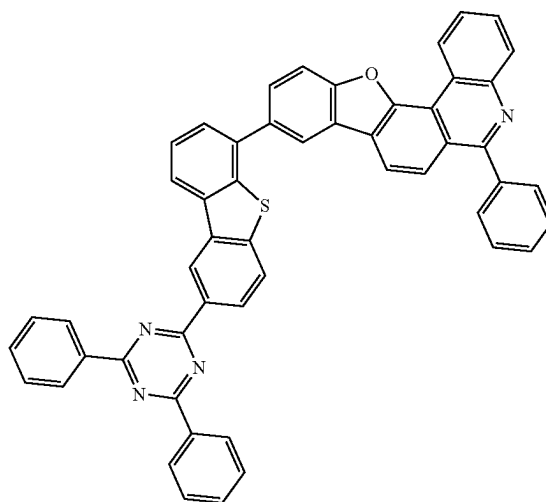


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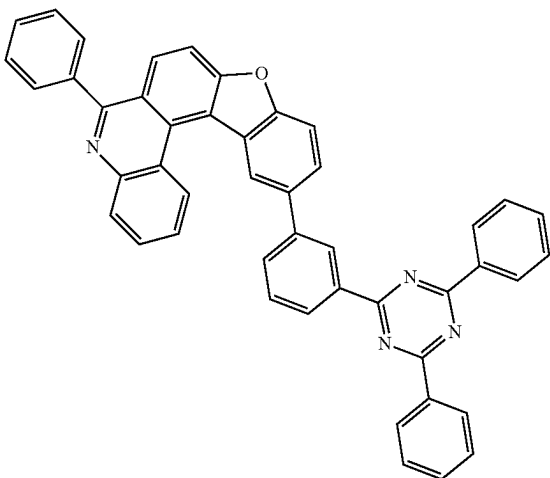
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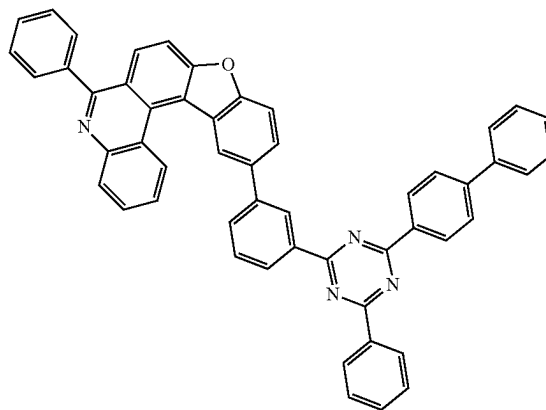
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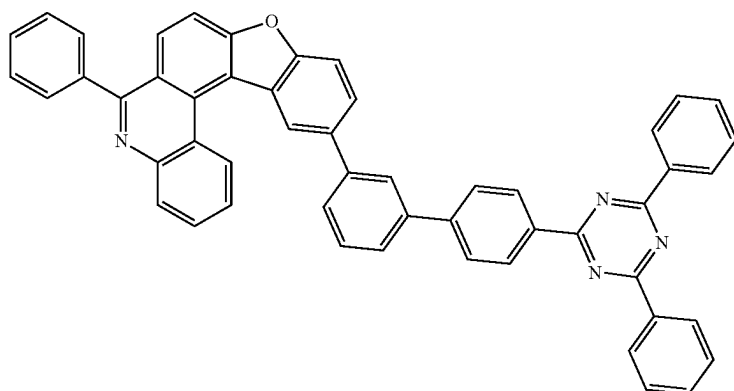


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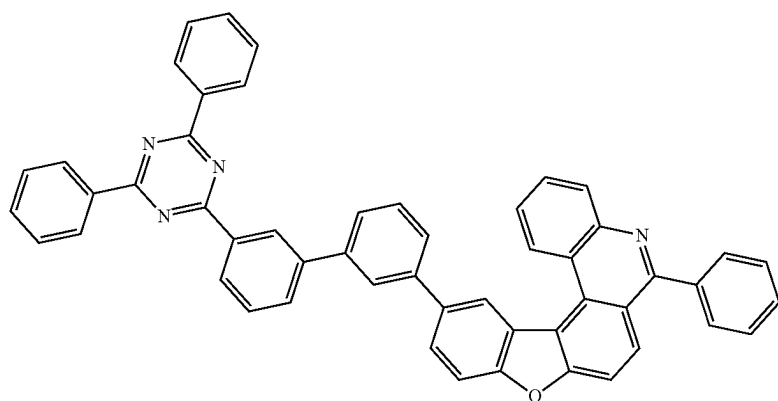
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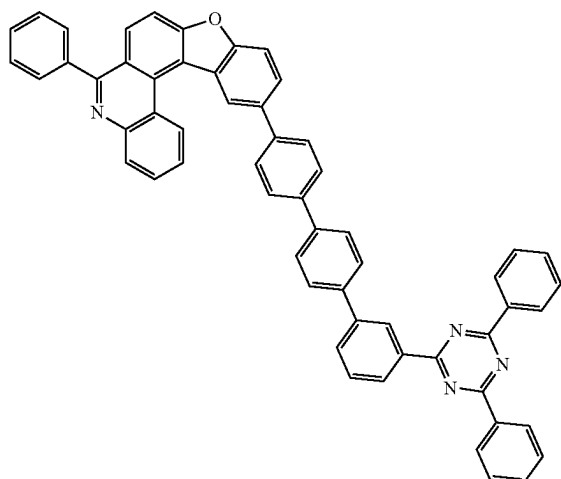
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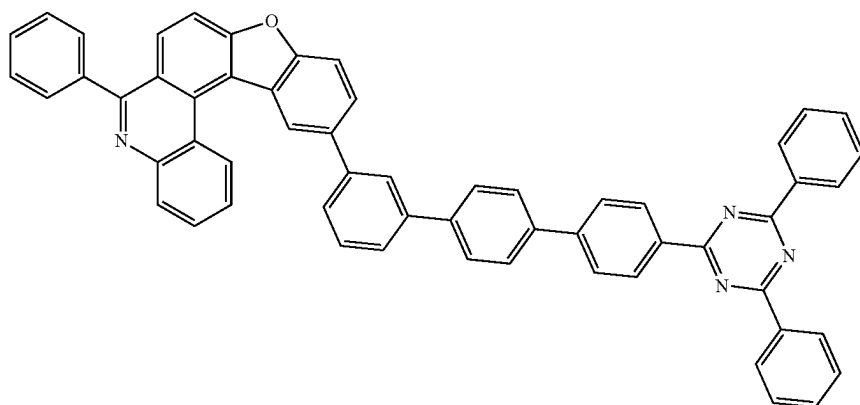


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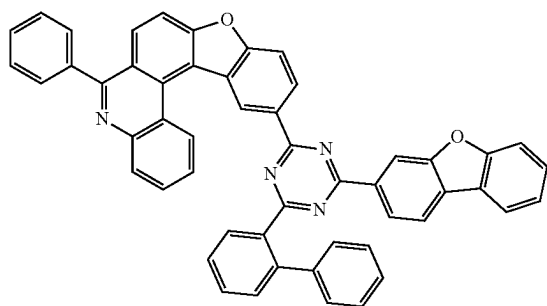
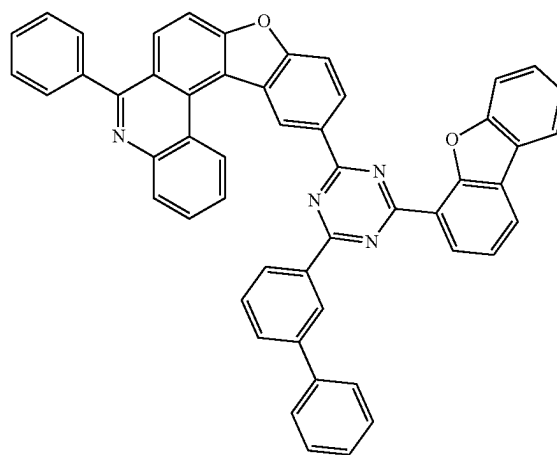
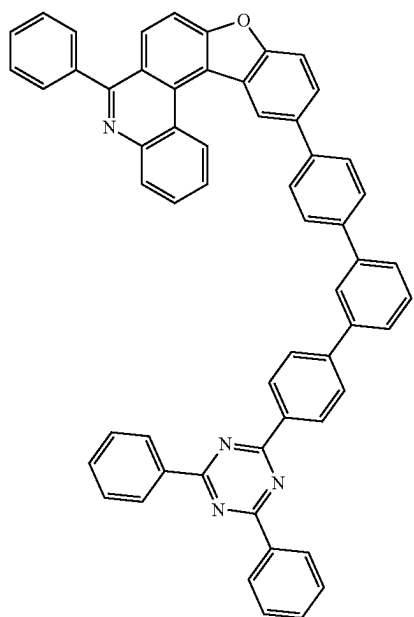
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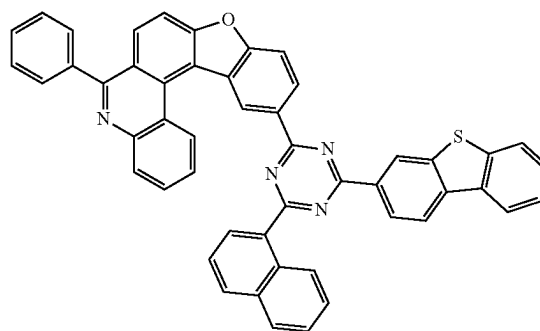
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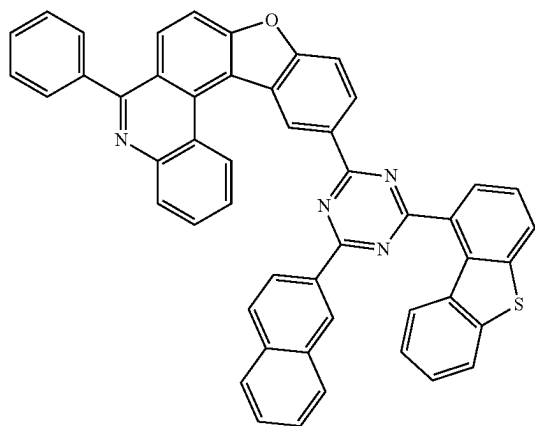


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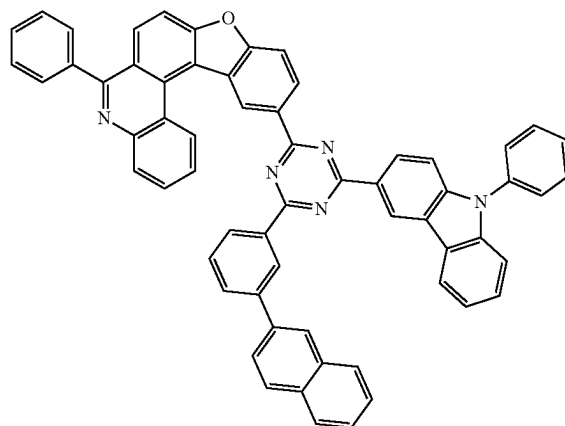
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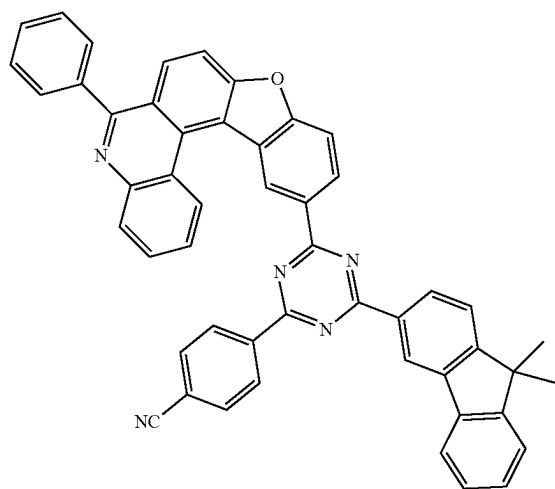
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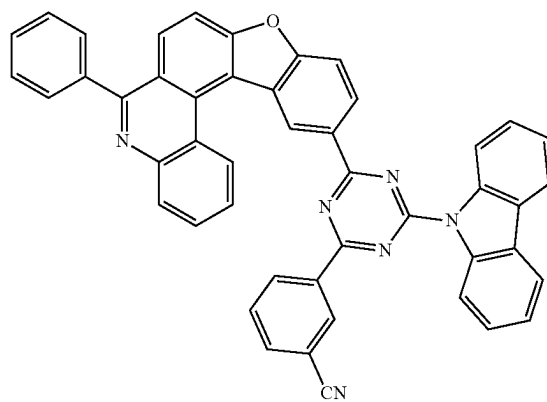


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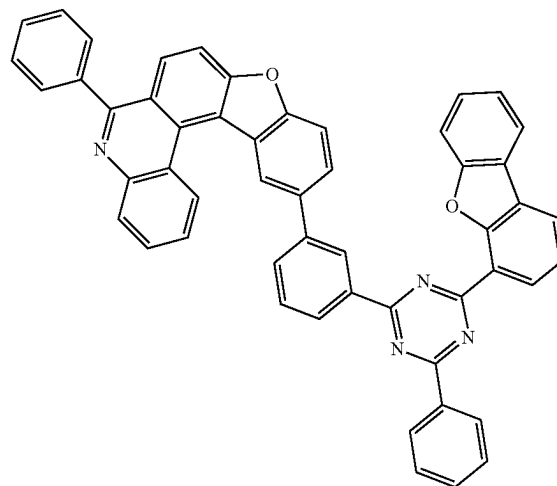
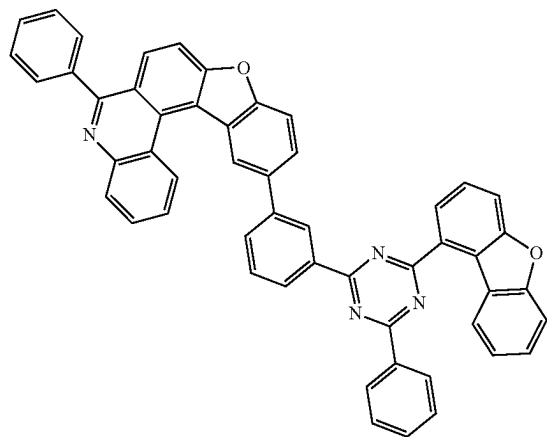
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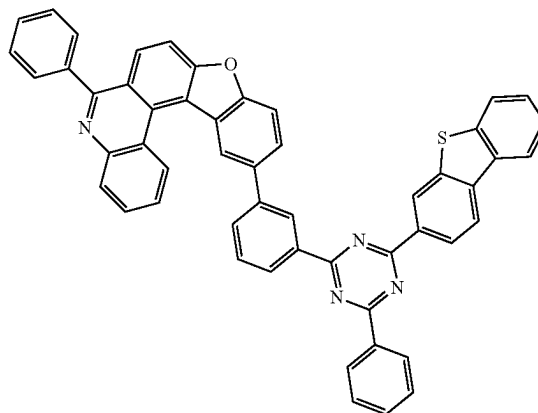
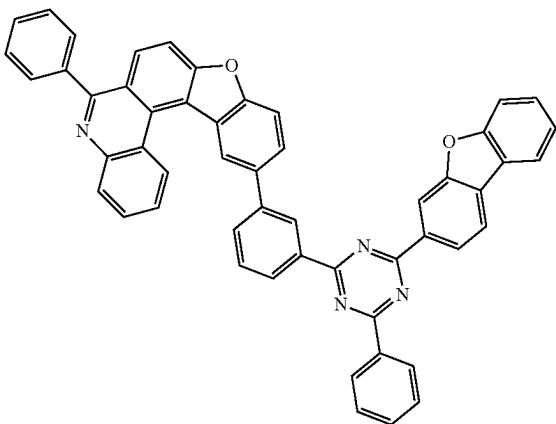
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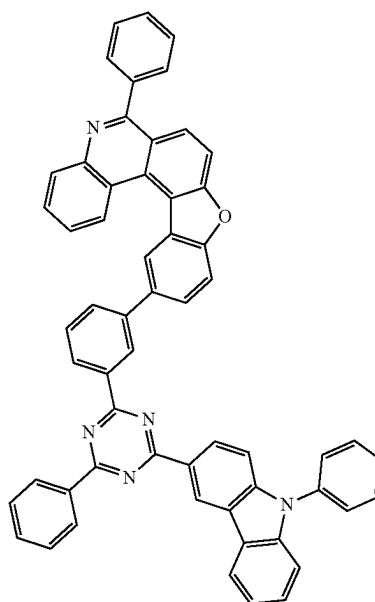
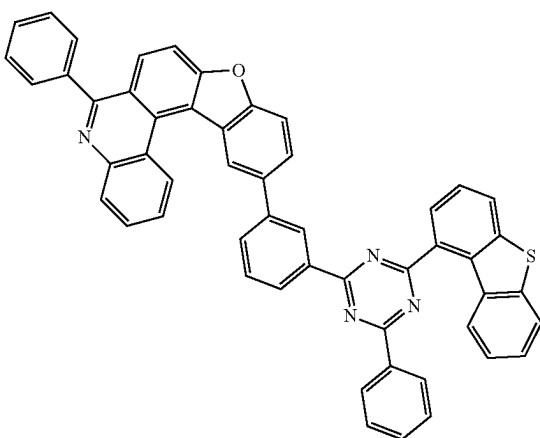
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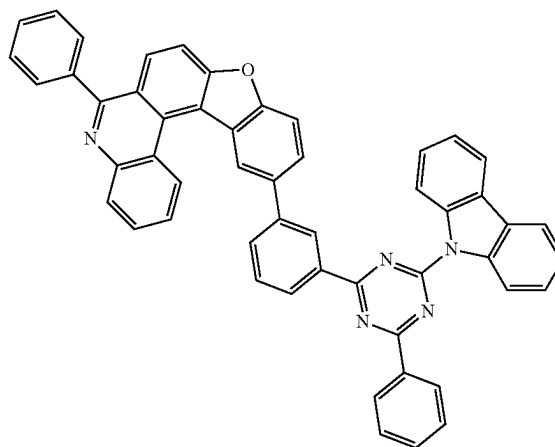
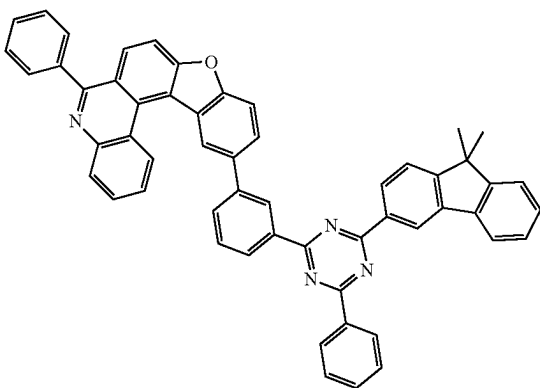
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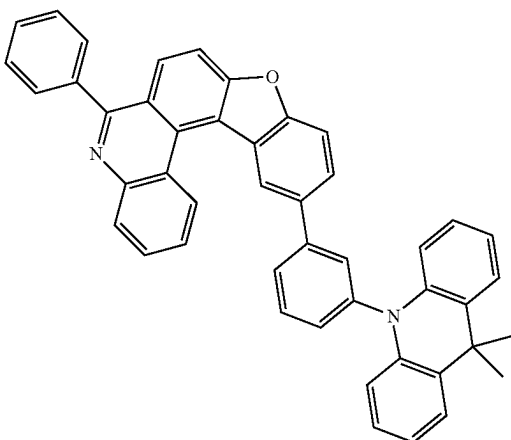


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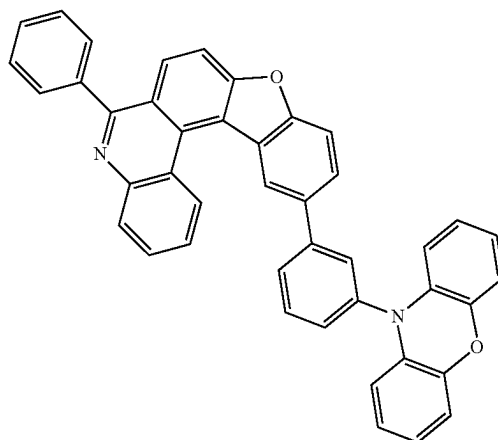
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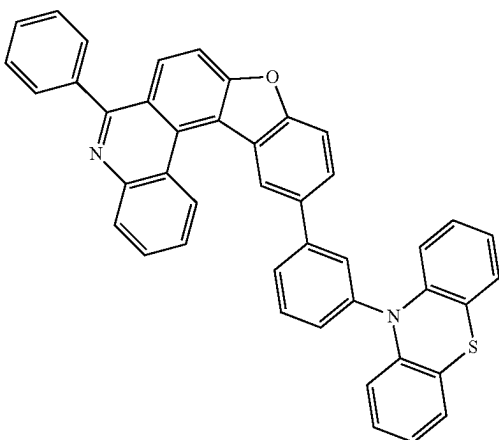
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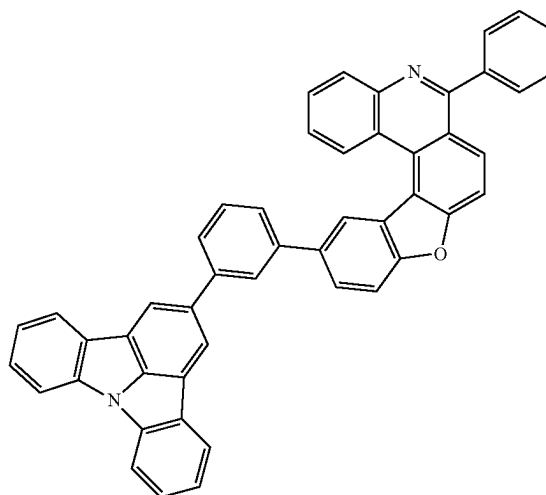


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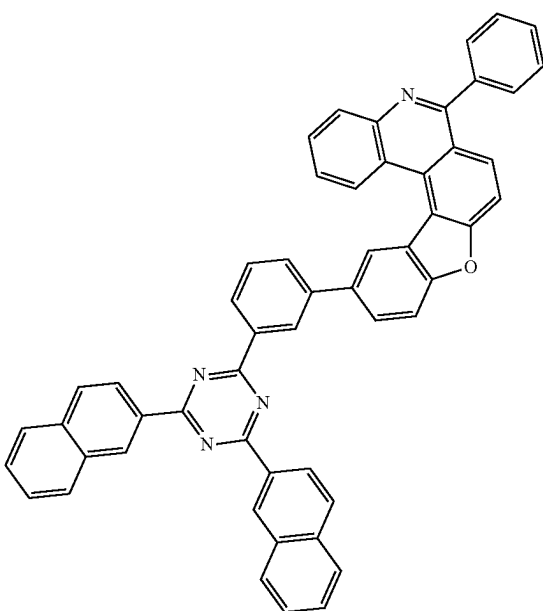
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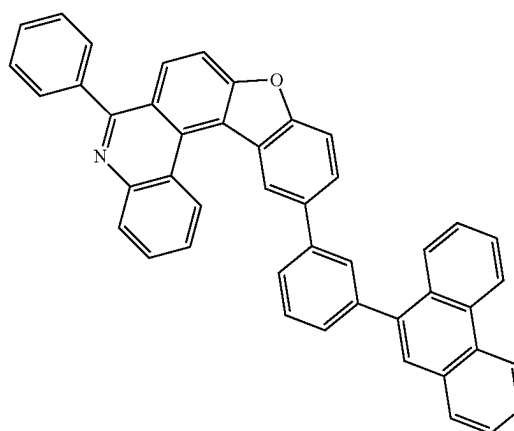
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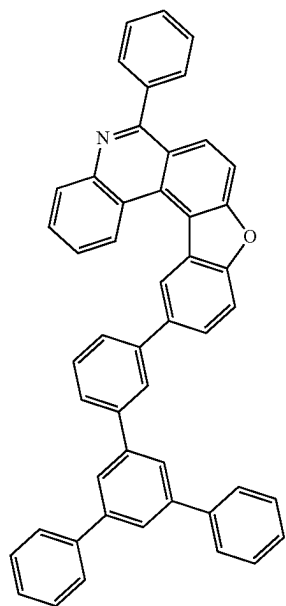
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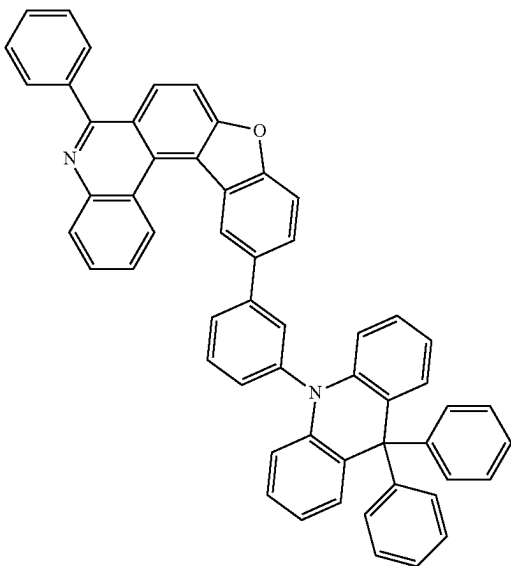
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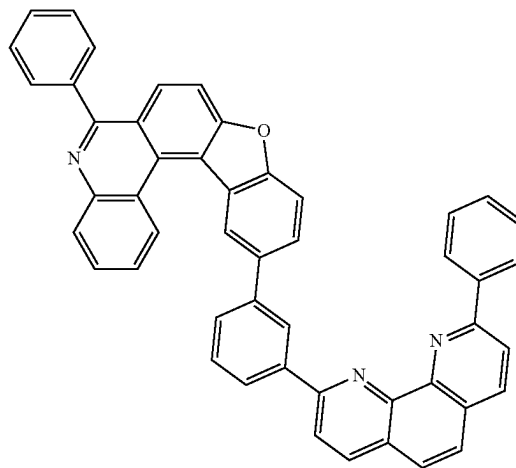
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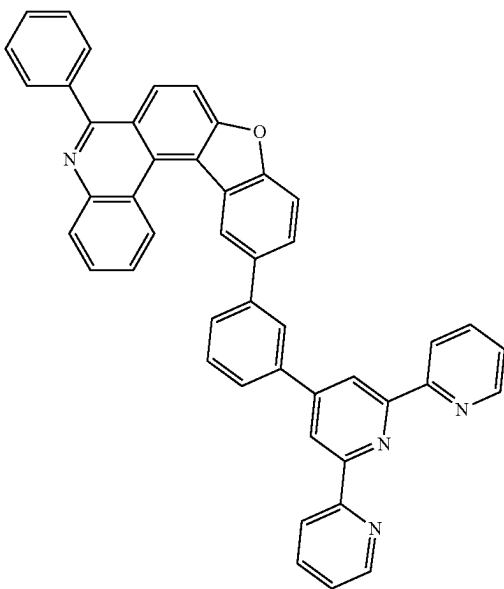
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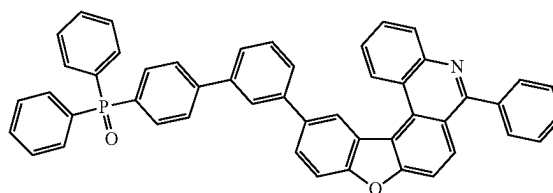


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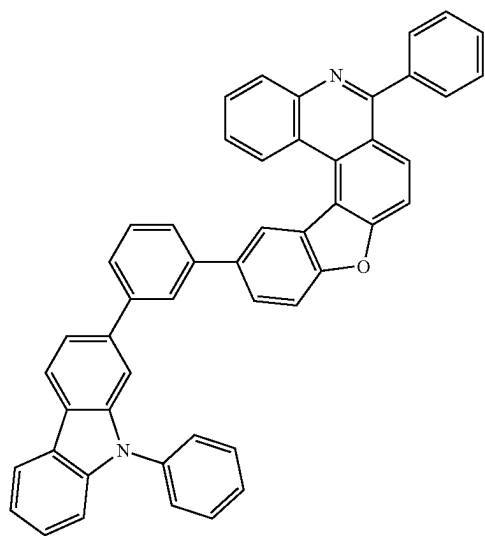
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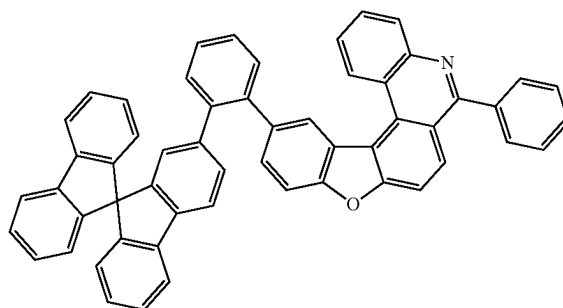
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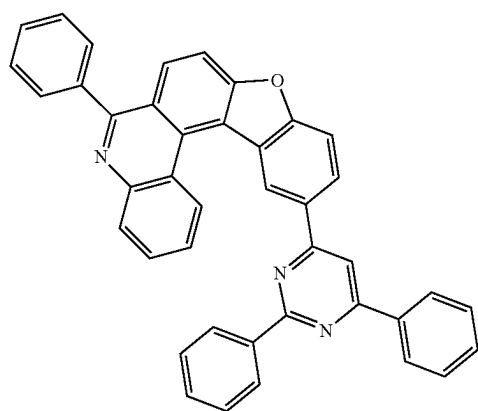
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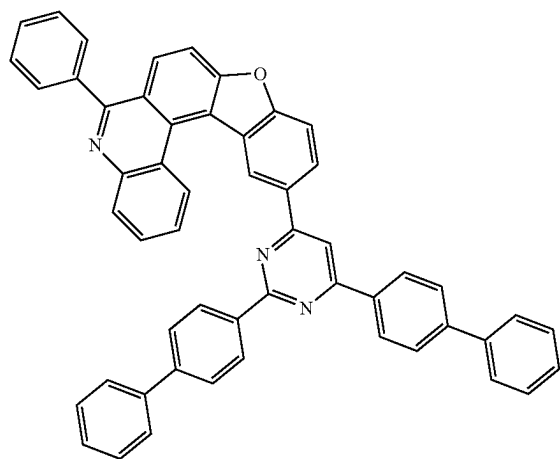
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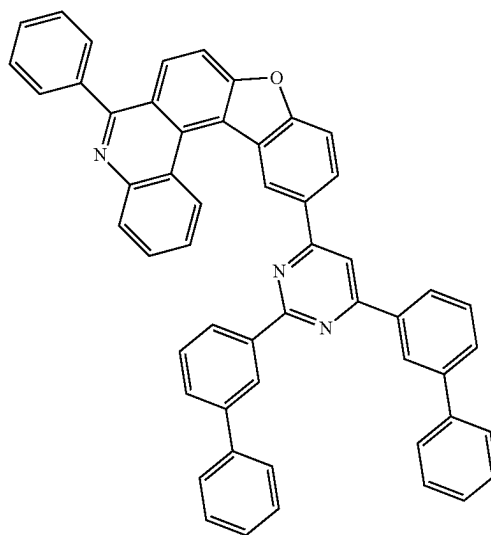


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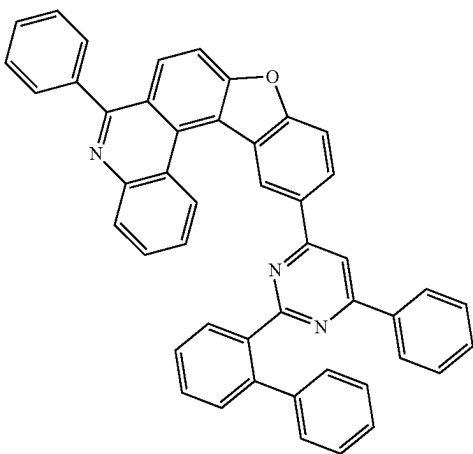
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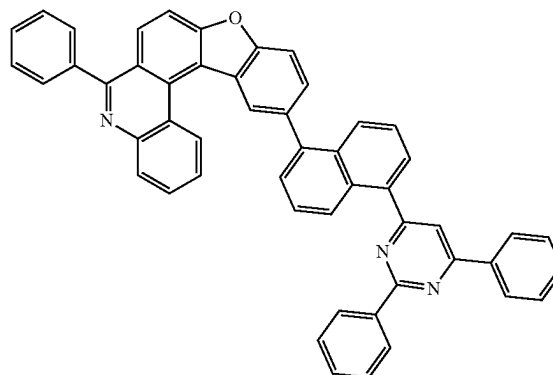


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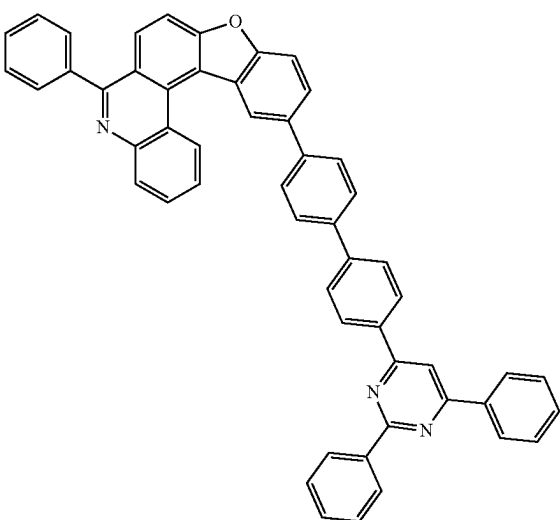
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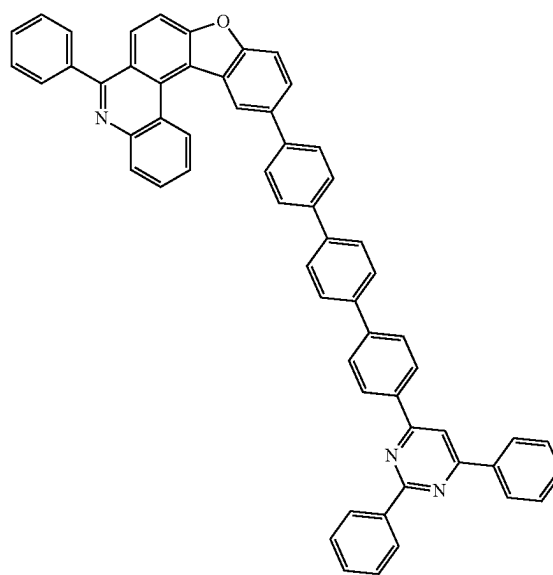


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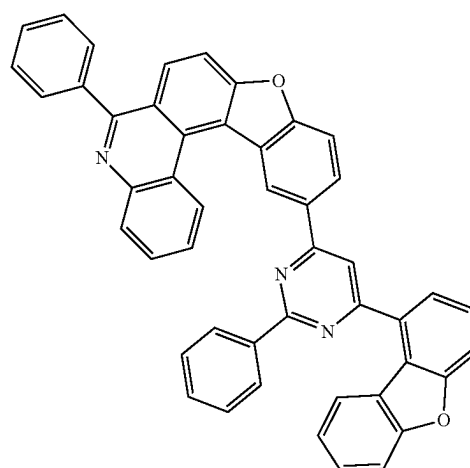
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c1ccc(cc1)C(=N2C3=CC=CC=C3C4=CC=CC=C4O5C6=CC=CC=C6C(=C5)C7=CC=CC=C7C=C8C(=C(C=C8)C9=CC=CC=C9C(=N10C11=CC=CC=C11C(=N10)C12=CC=CC=C12)C13=CC=CC=C13)C2c1ccc(cc1)-c2nc3c(c4ccccc4n2)c5ccccc53-c6cc7ccccc7c8ccccc68

Chemical structure of compound 10, a bis-benzoxazole derivative. The structure consists of two benzoxazole units linked by a biphenyl group. Each benzoxazole unit is substituted with a phenyl group and a 2-phenyl-1,3,4-benzoxazin-5-yl group.

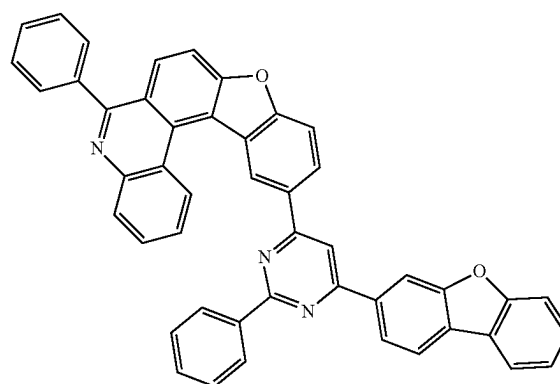
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108



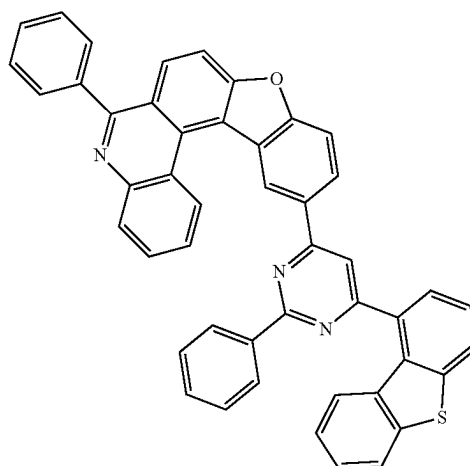
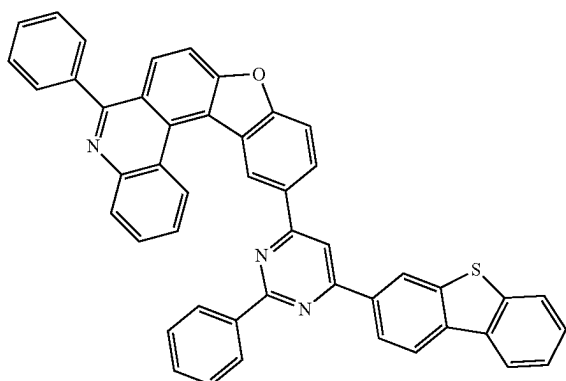
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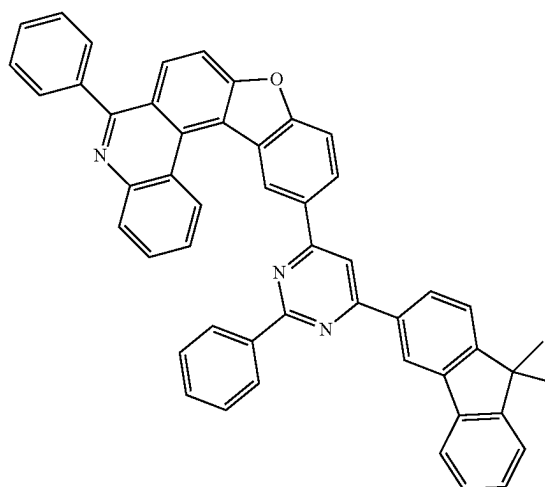
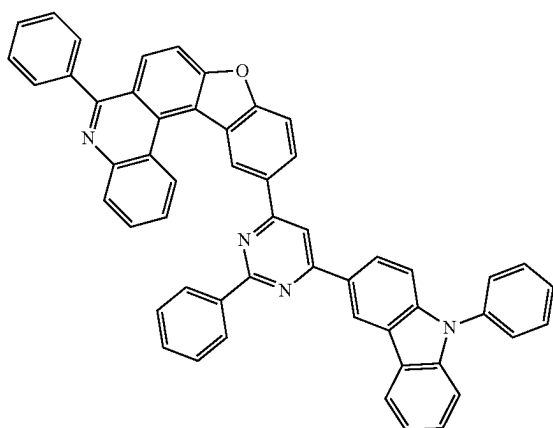
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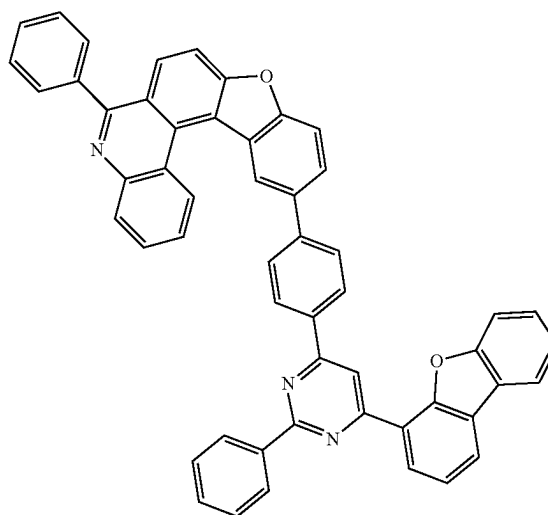
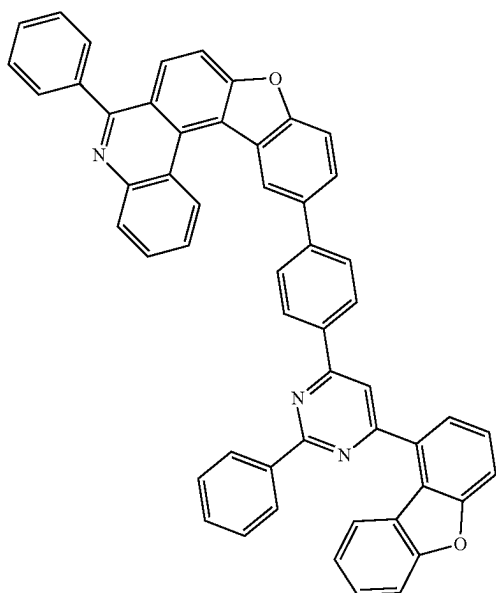
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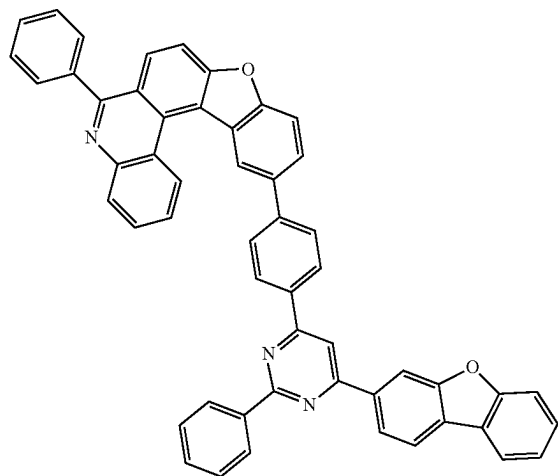


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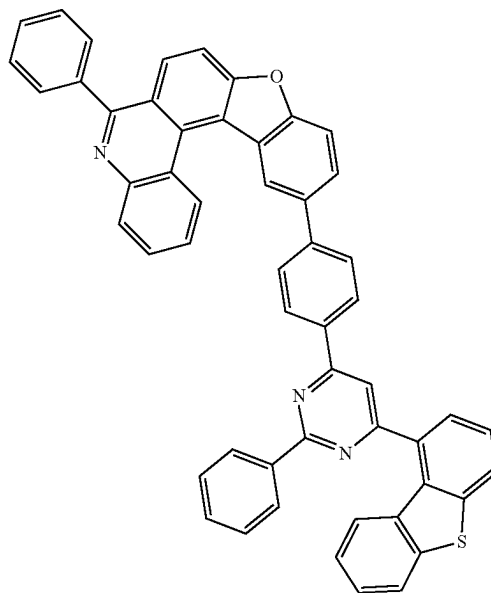


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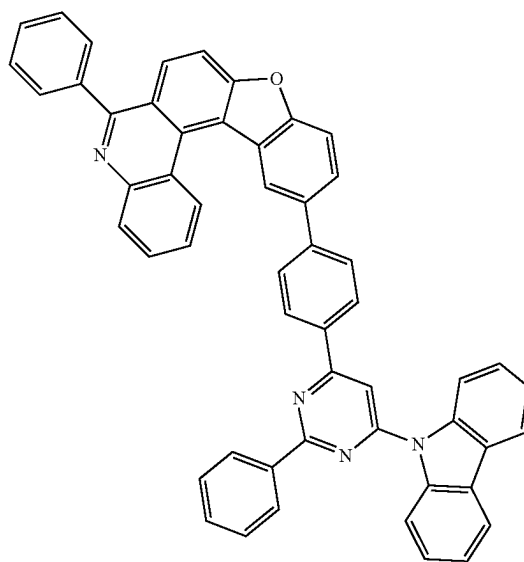
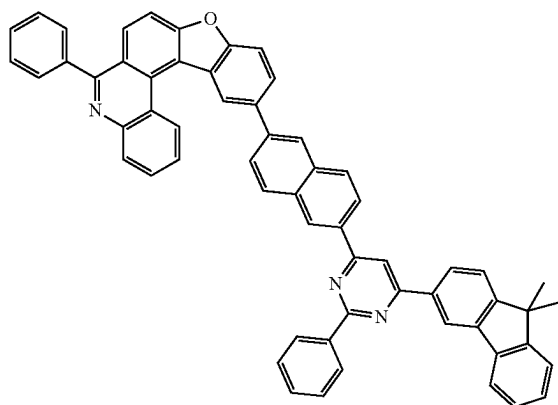
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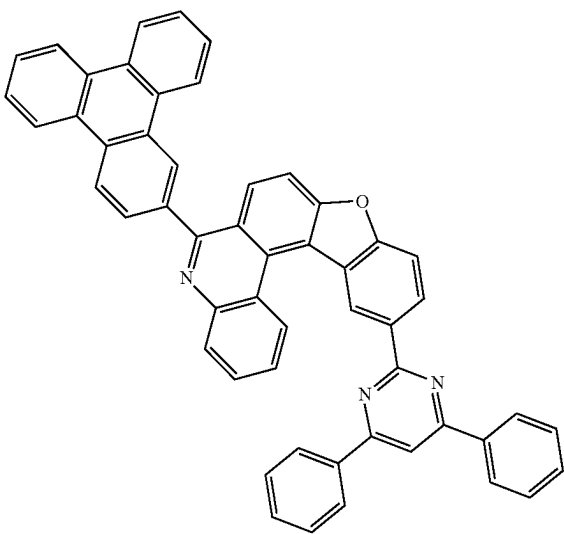
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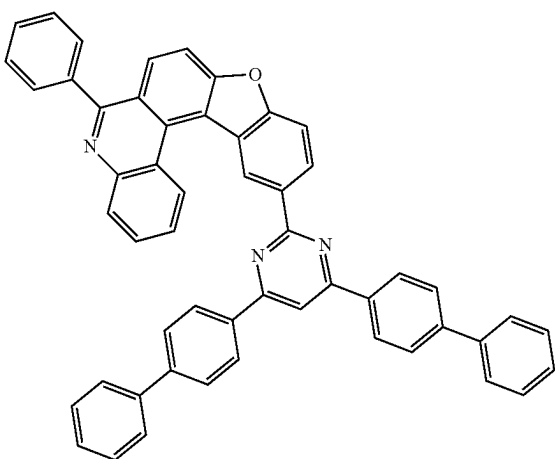
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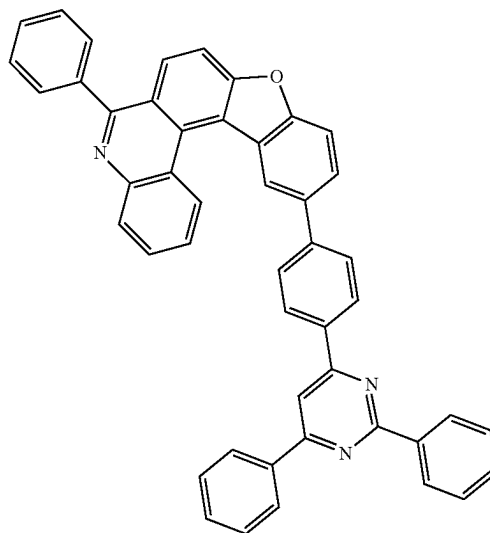
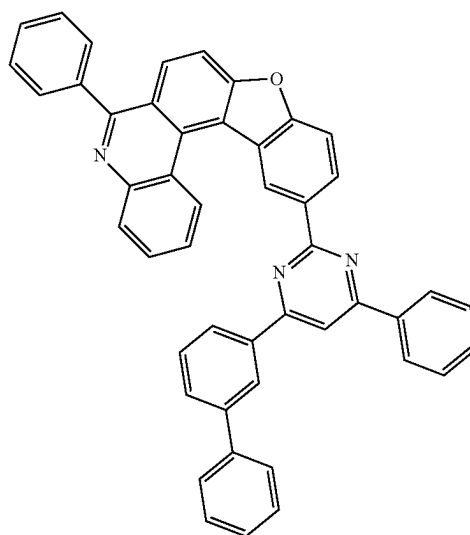
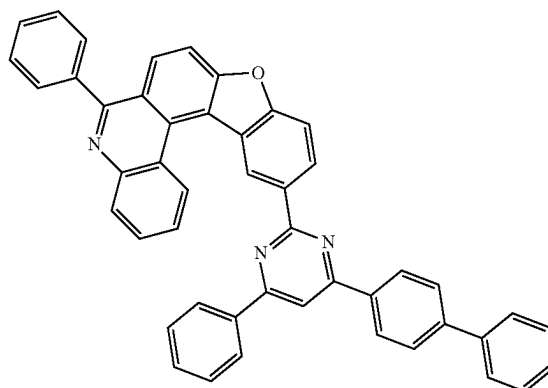
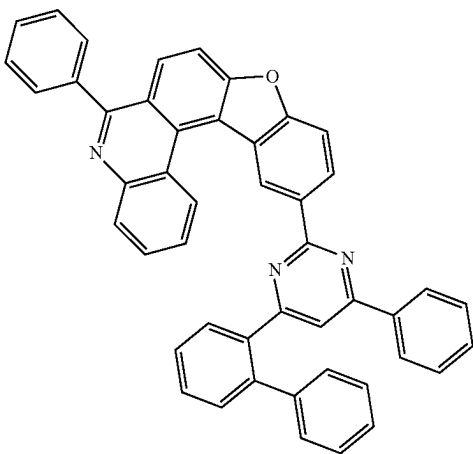
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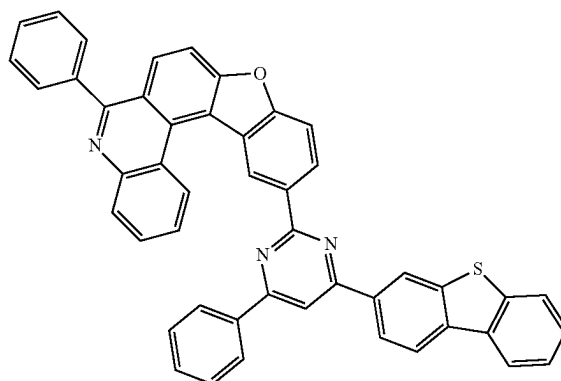
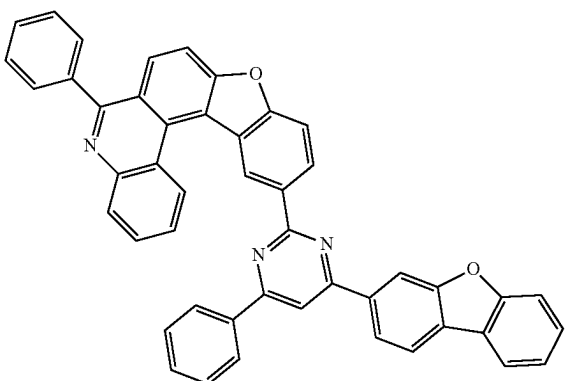
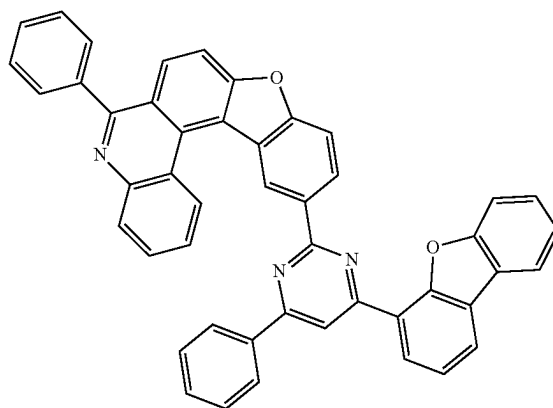
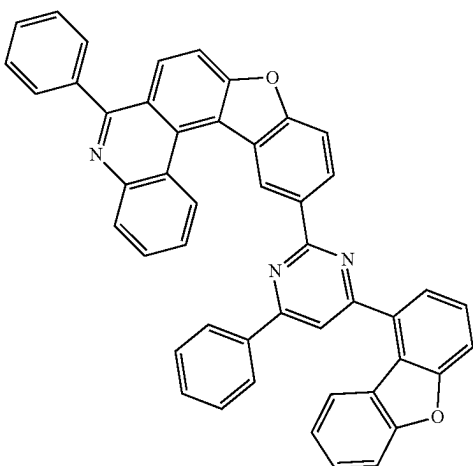
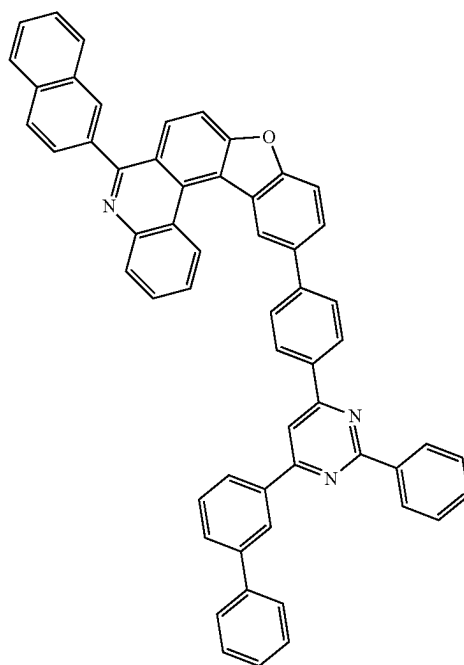
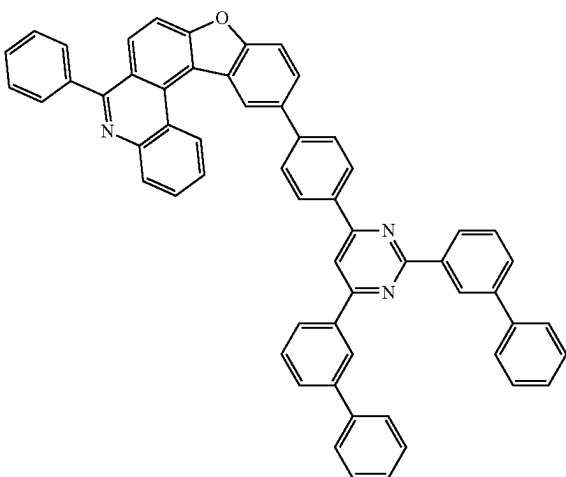
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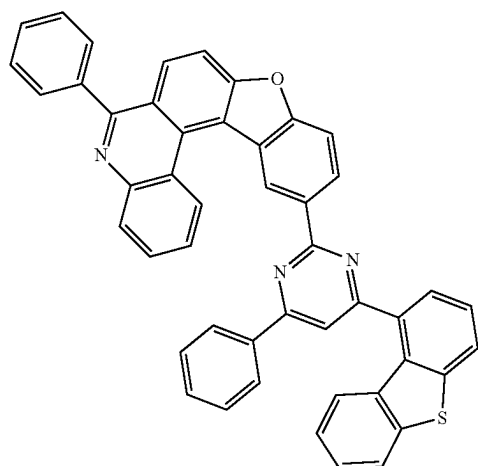


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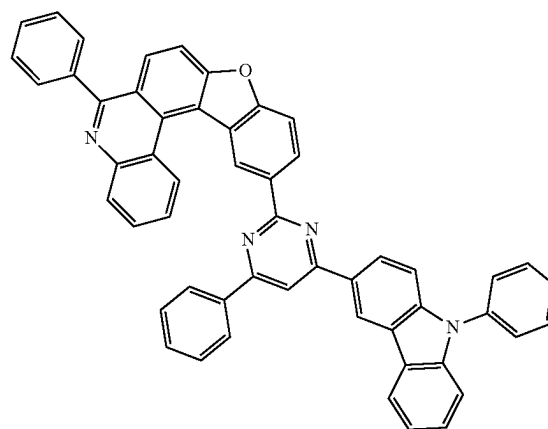


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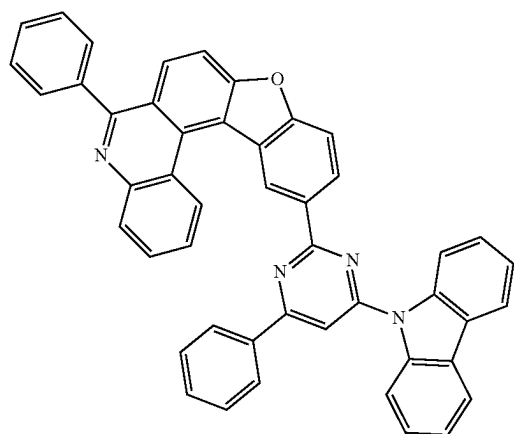


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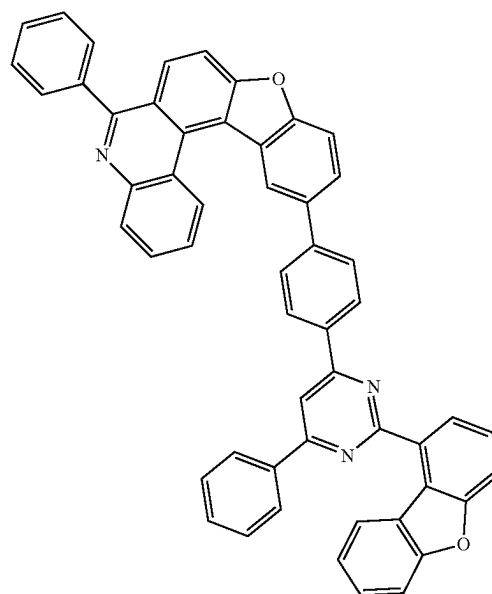


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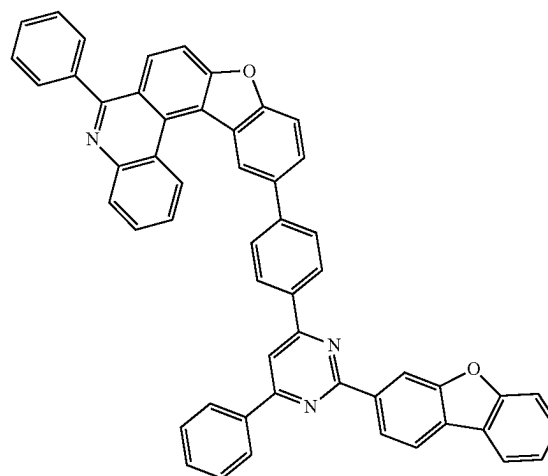
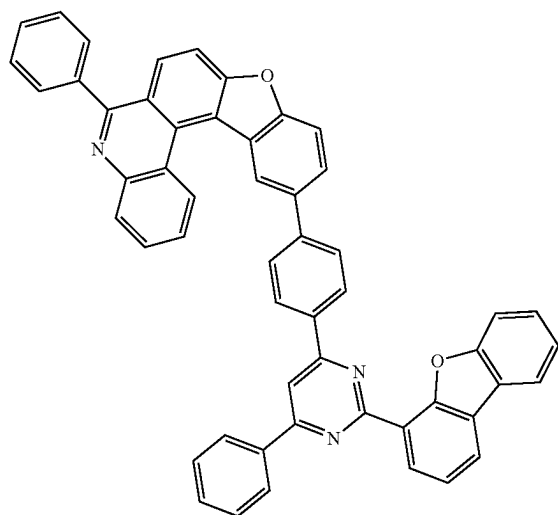
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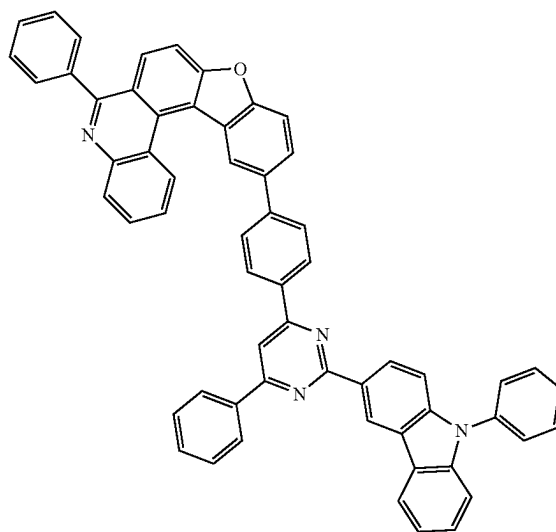
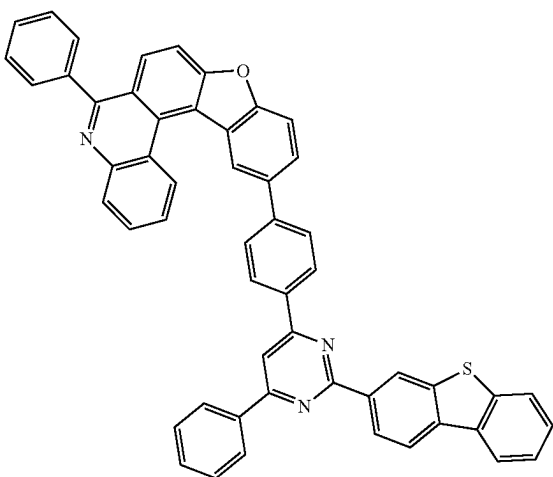
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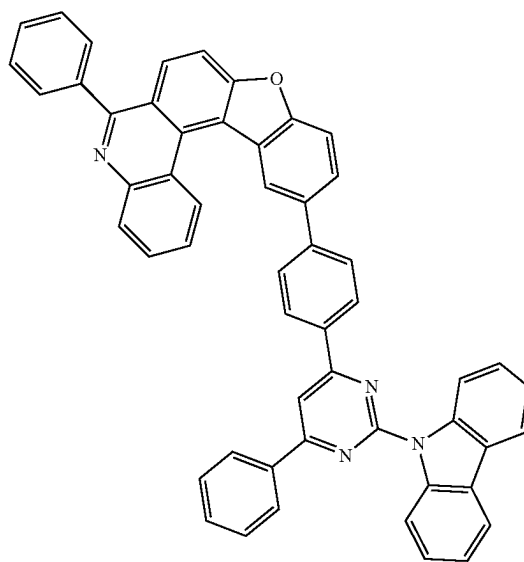
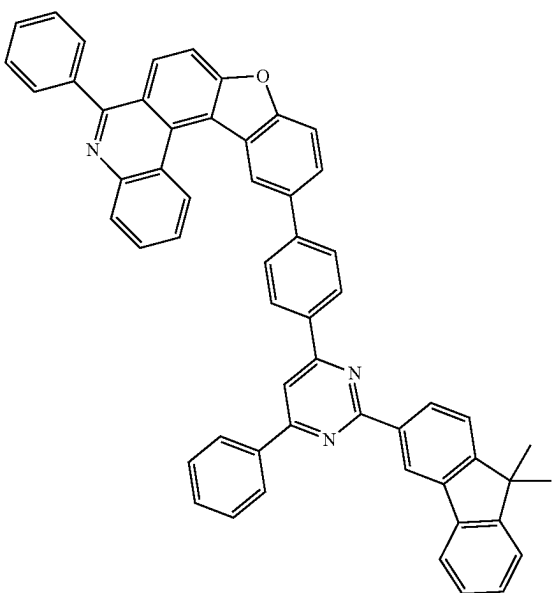
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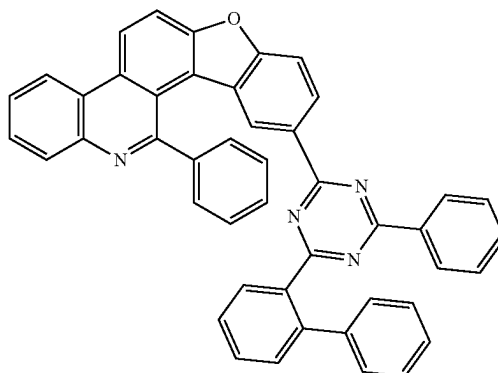
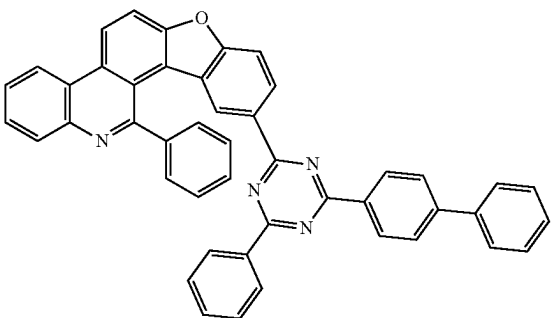
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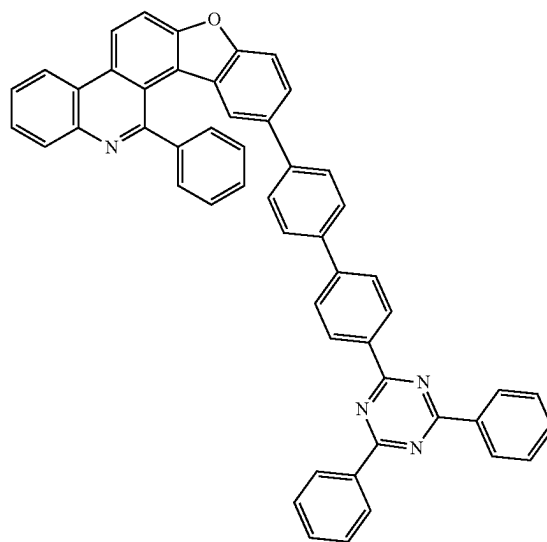
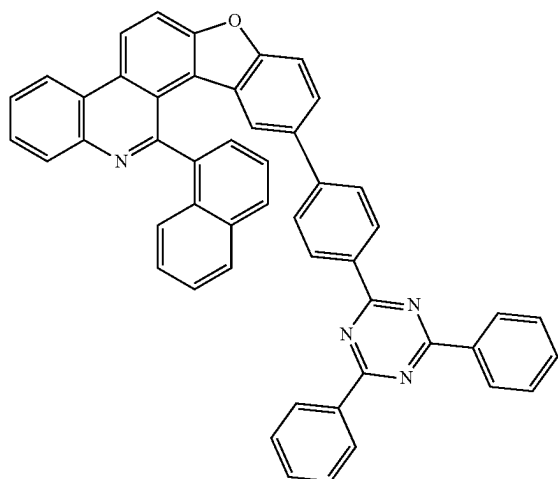
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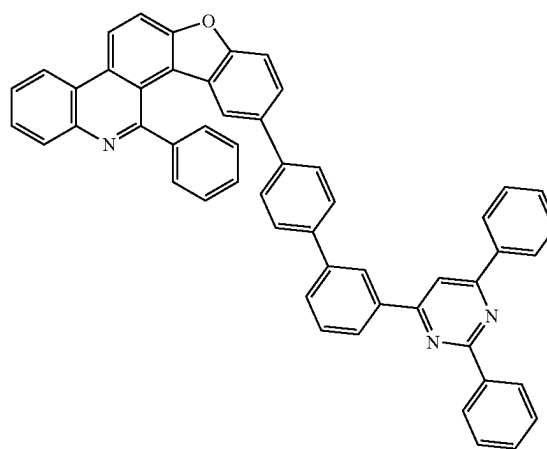
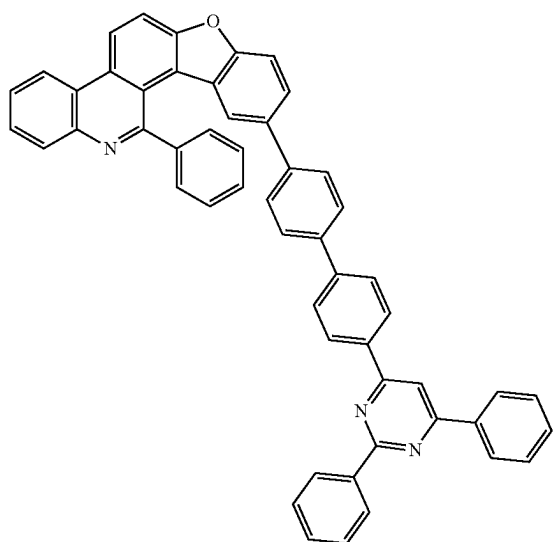
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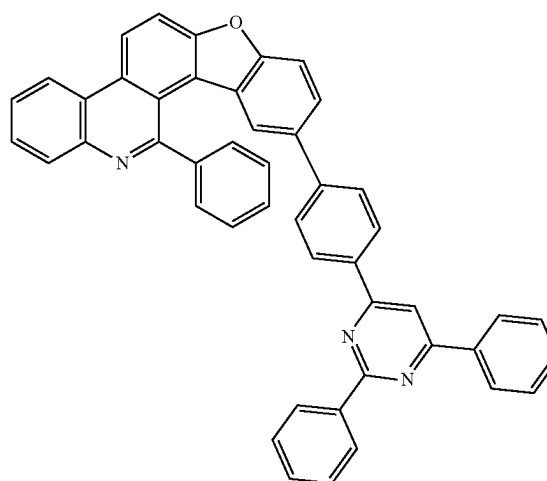
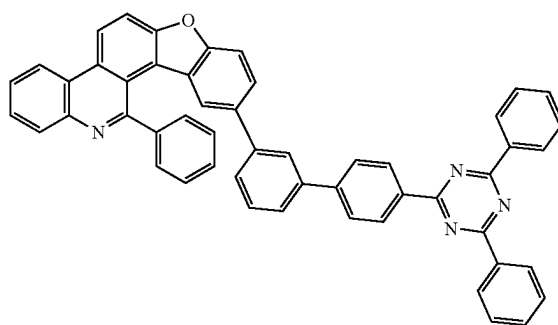
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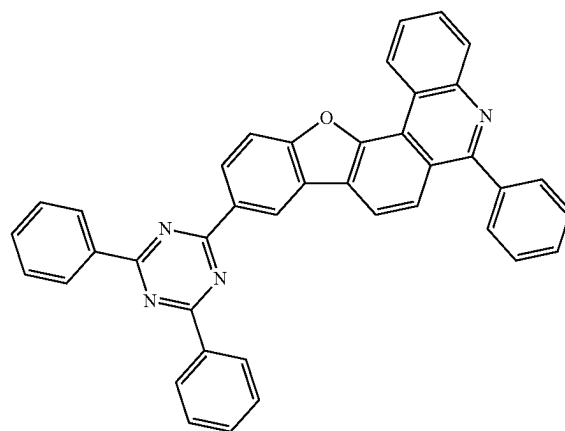
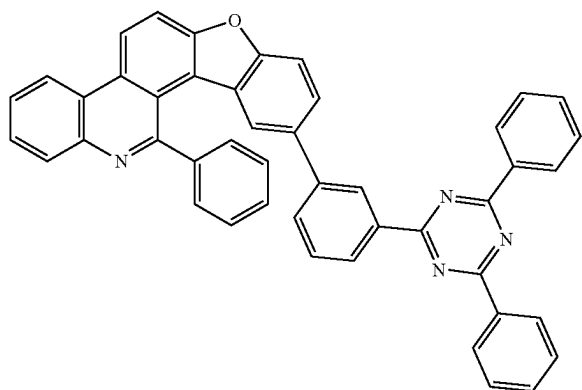
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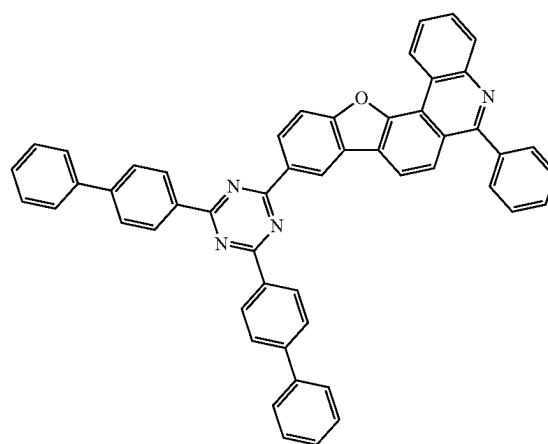
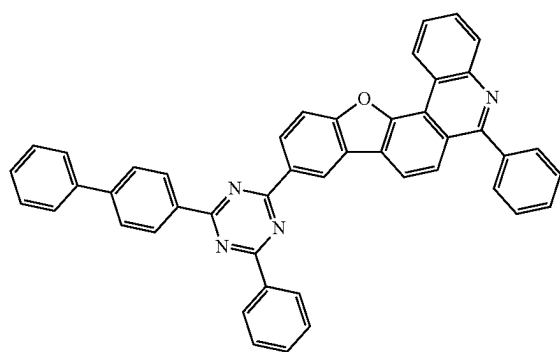
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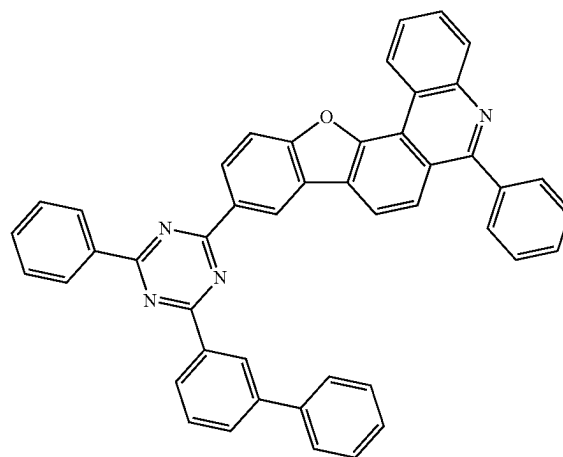
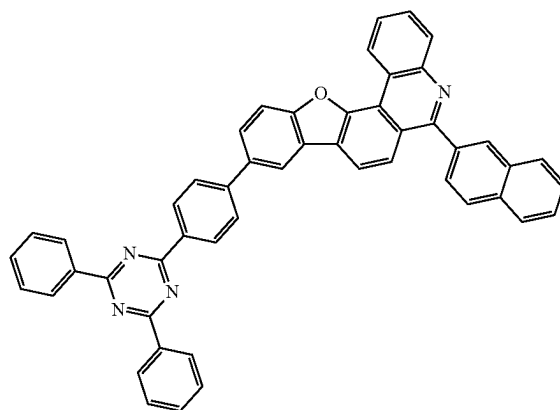
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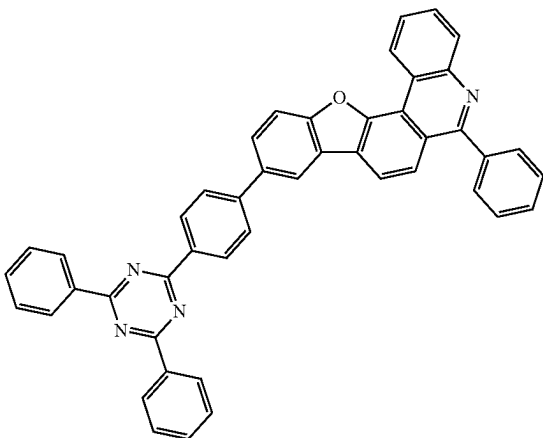
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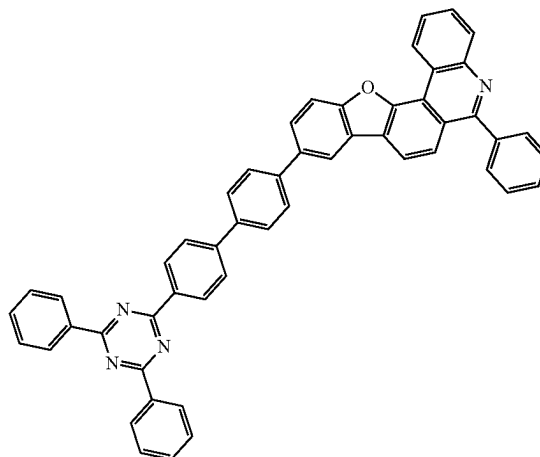
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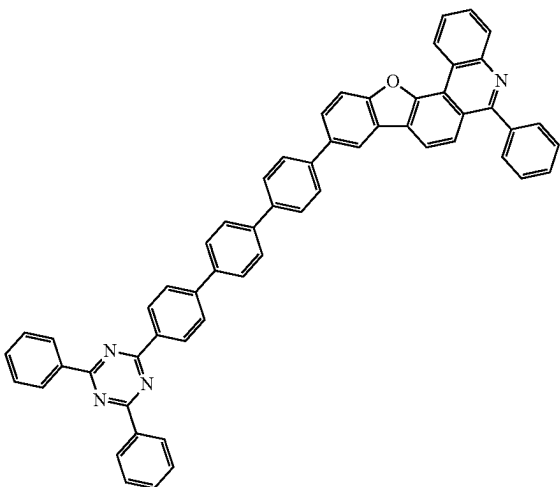


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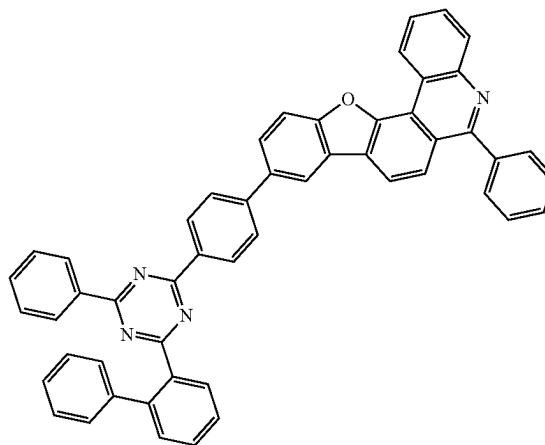
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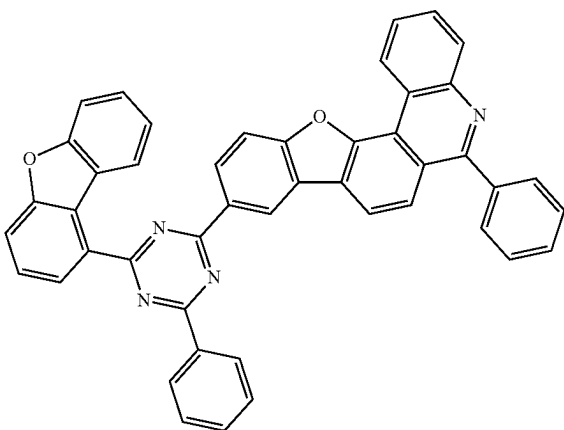
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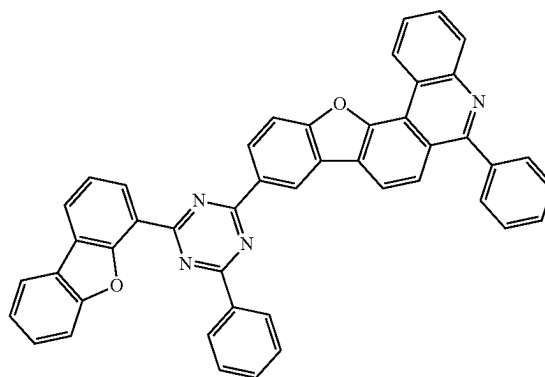
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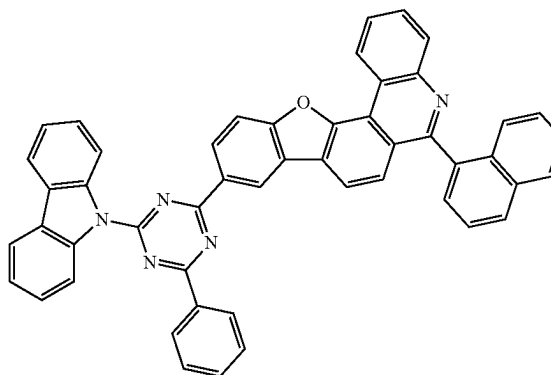
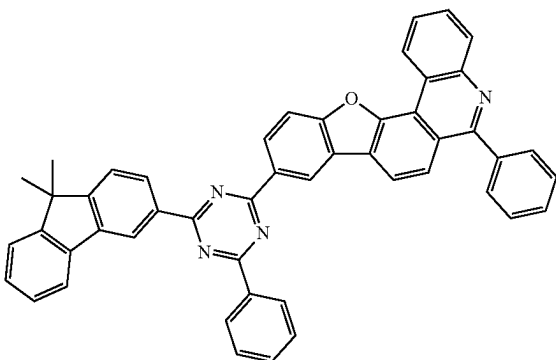
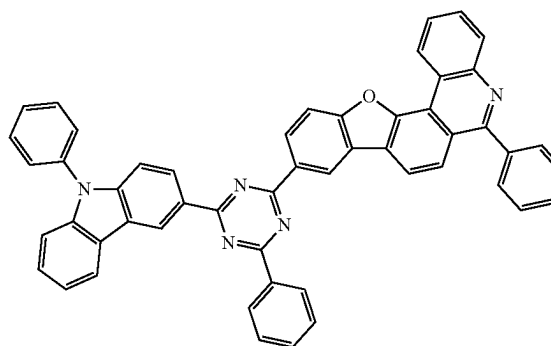
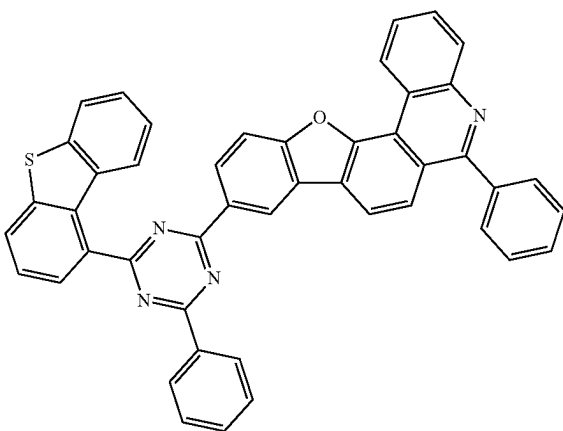
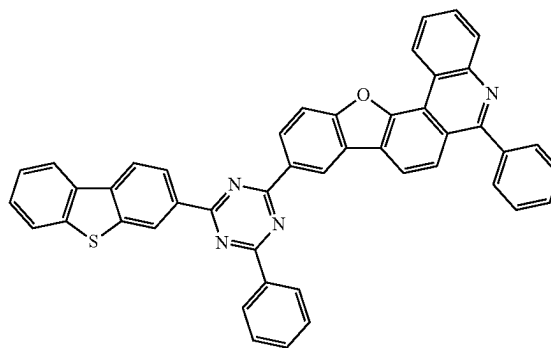
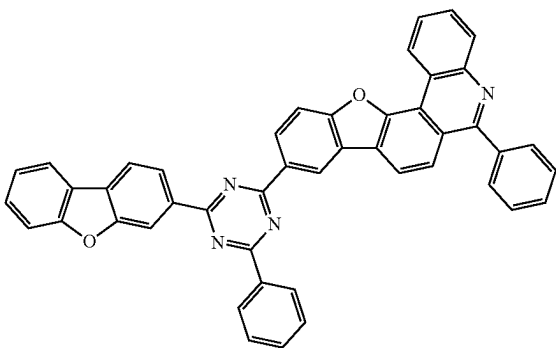


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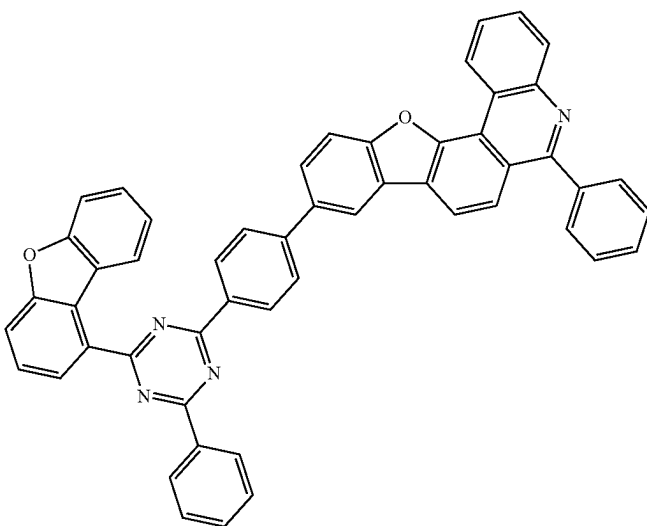


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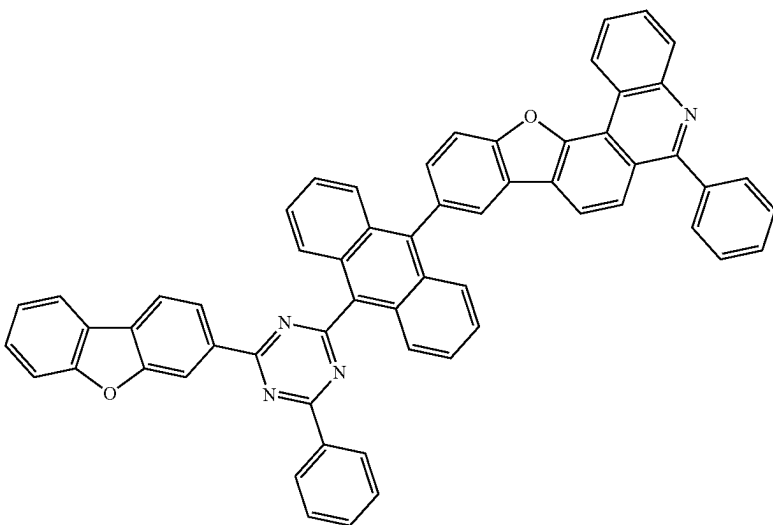
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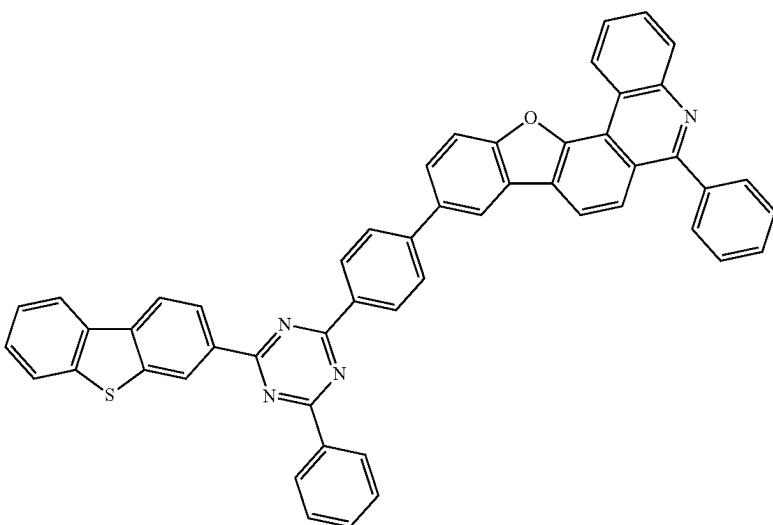
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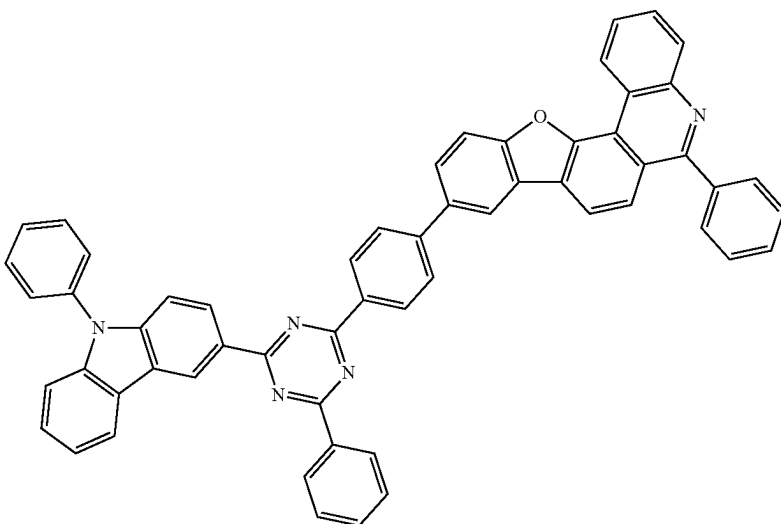


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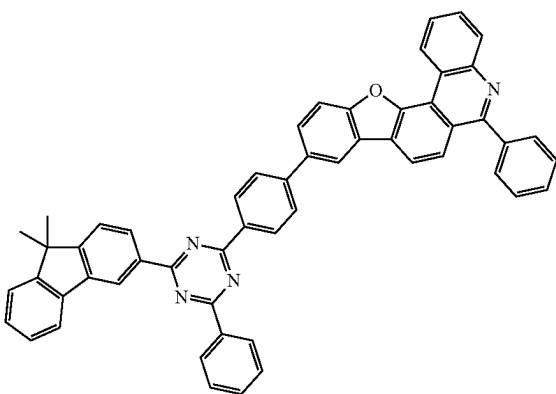
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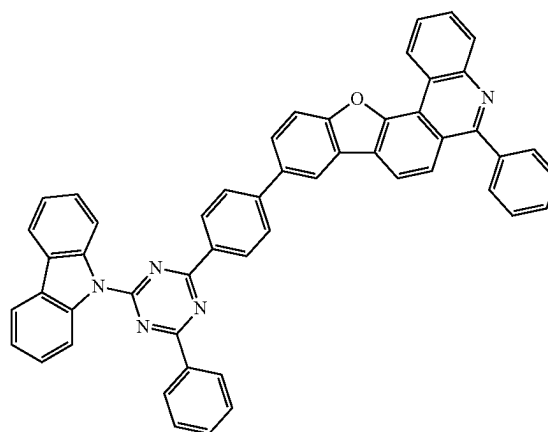
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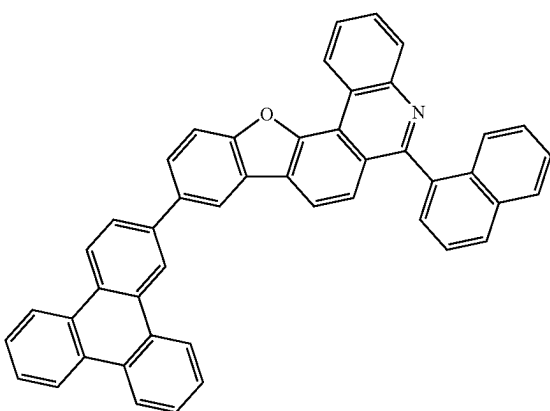
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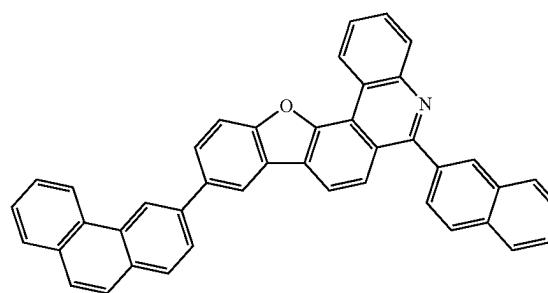
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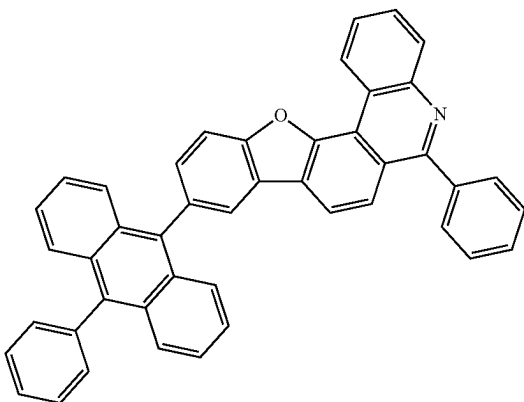
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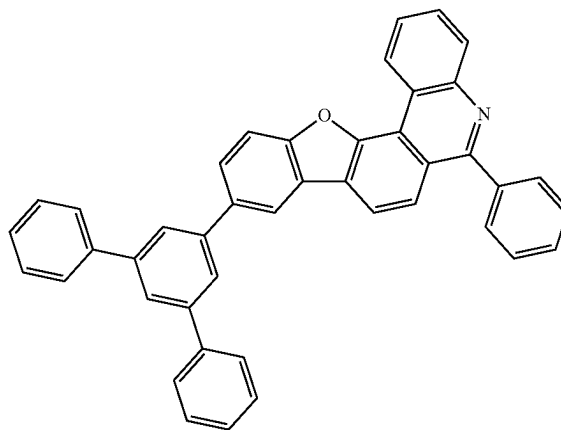
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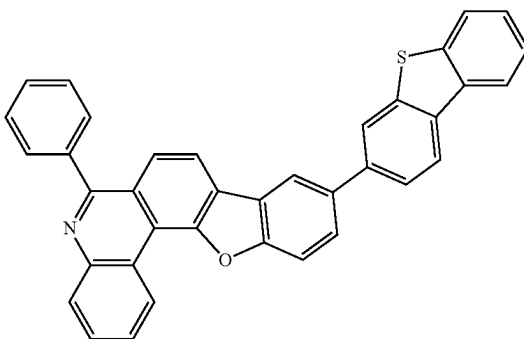


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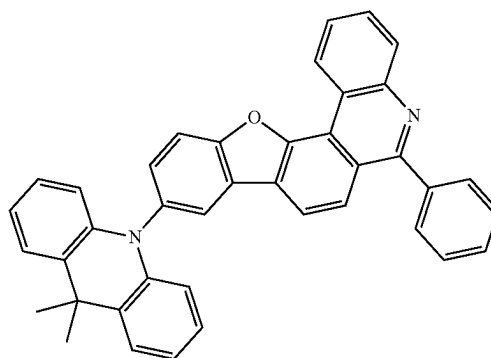
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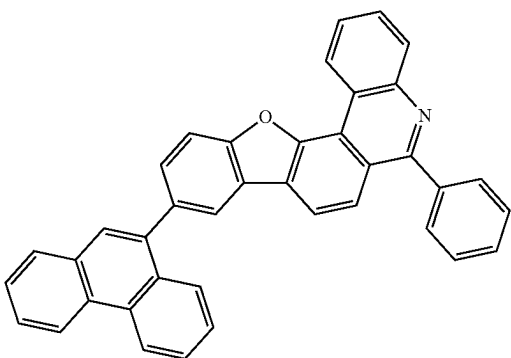
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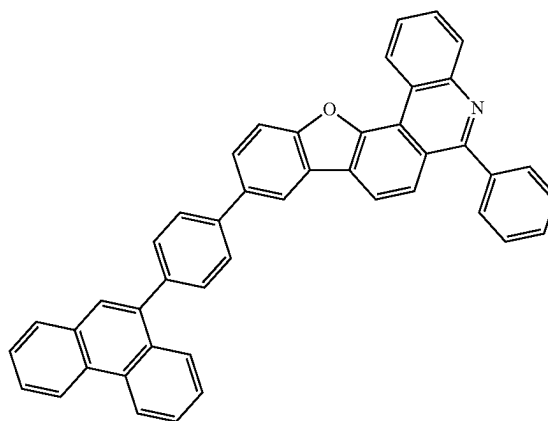
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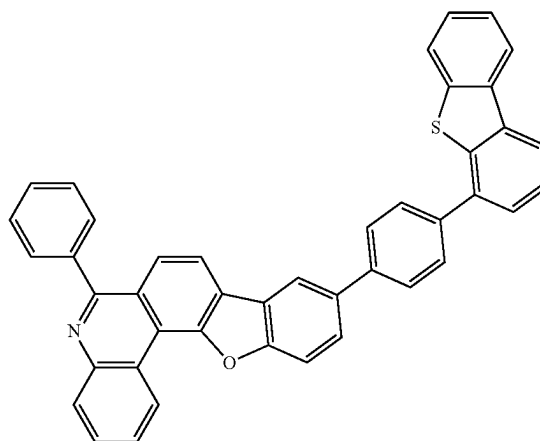
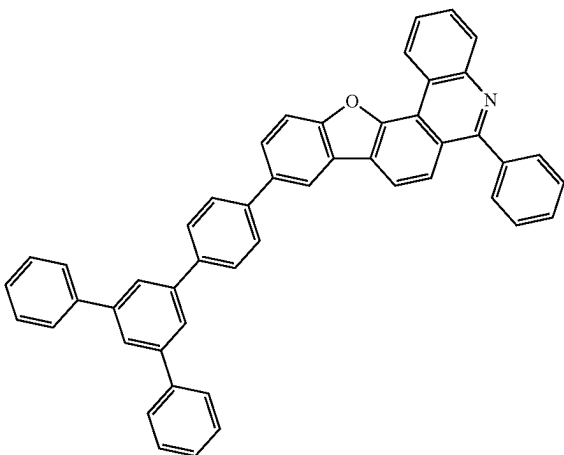
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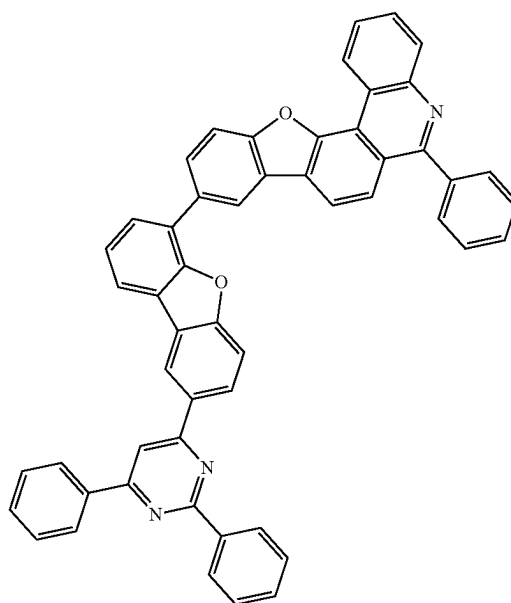
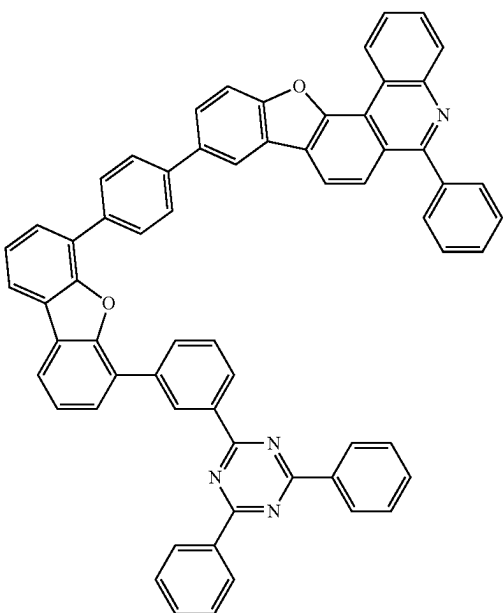
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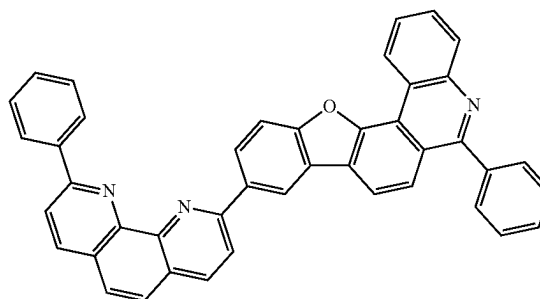
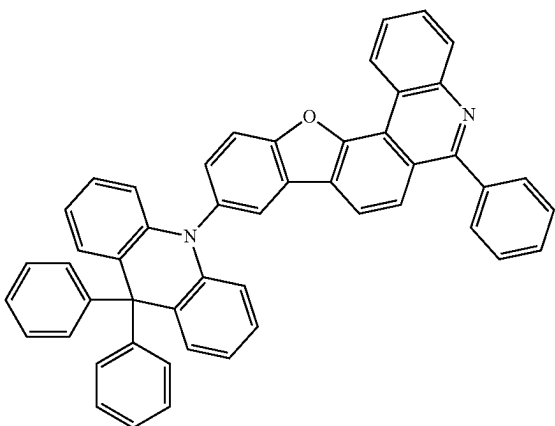
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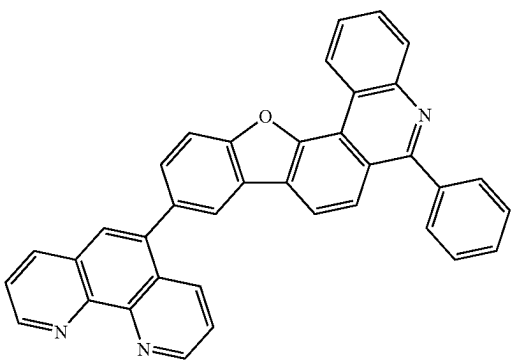


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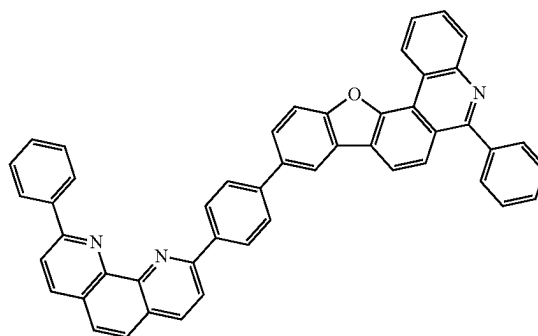
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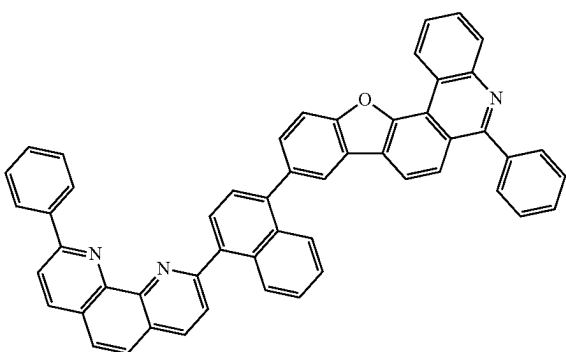
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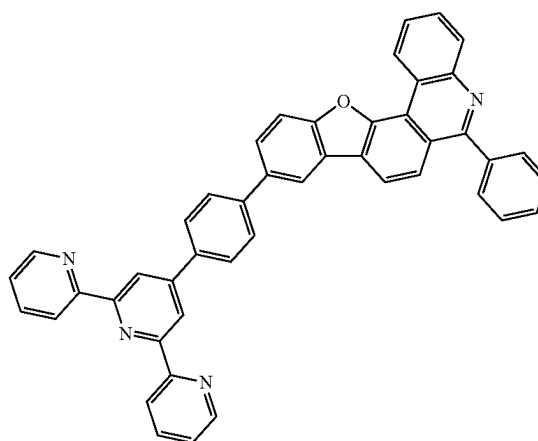


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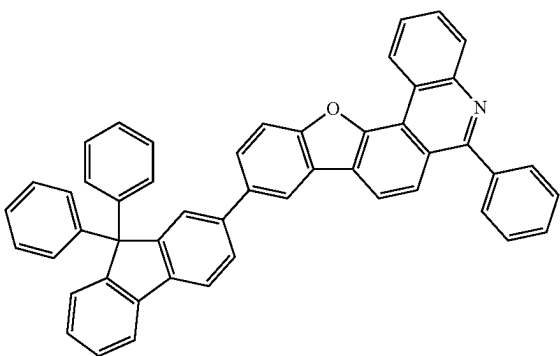
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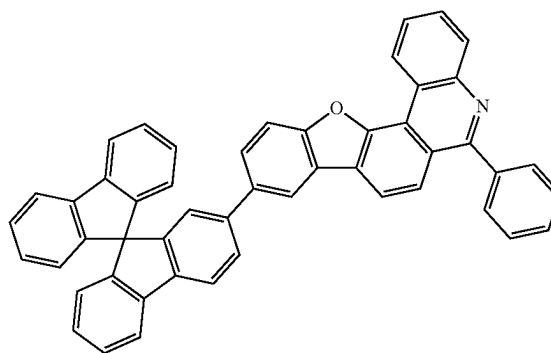
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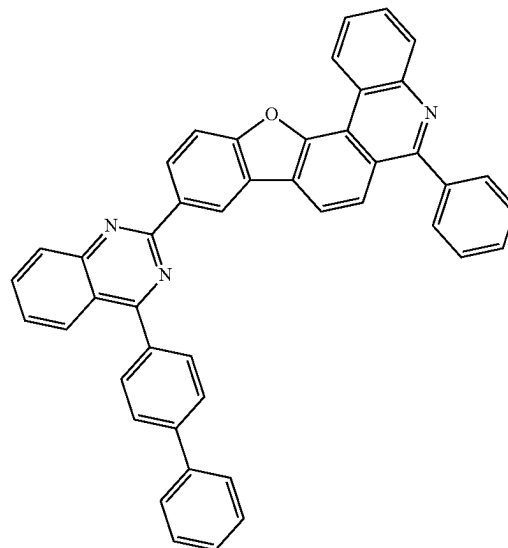
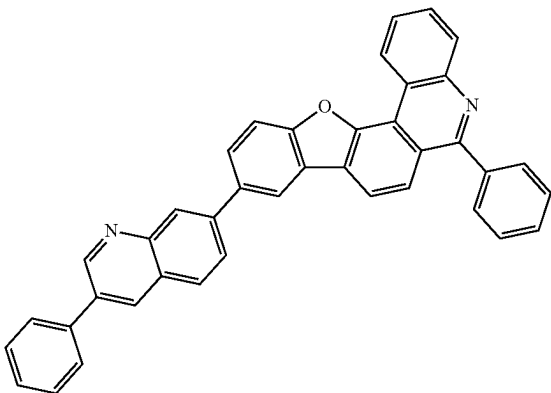
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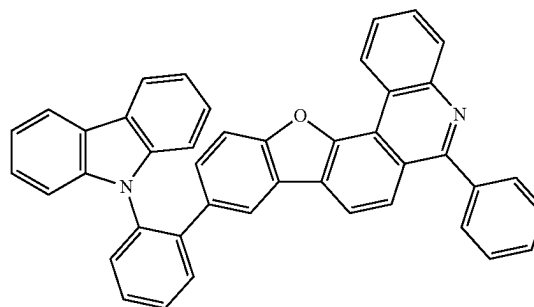
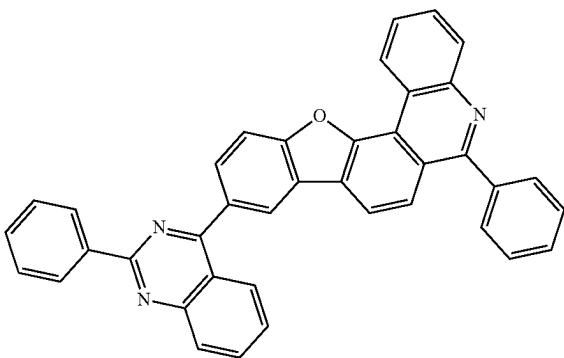
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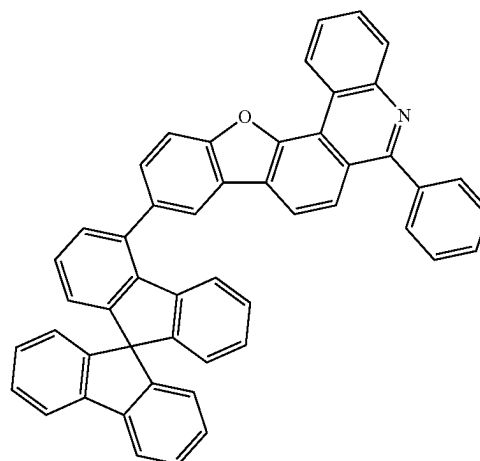
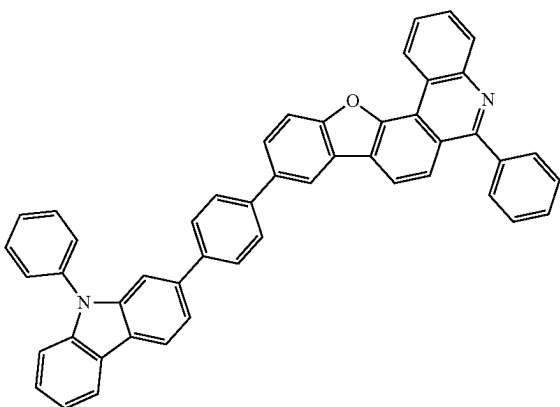
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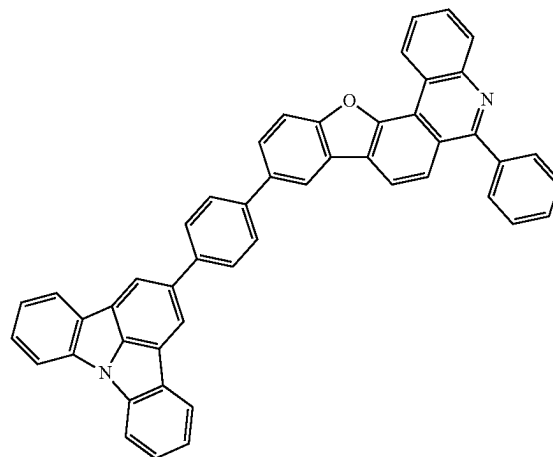
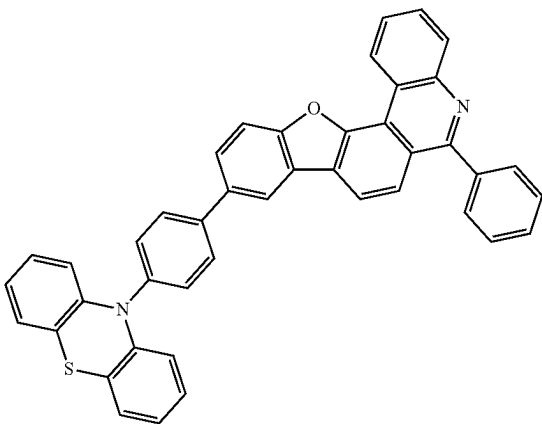
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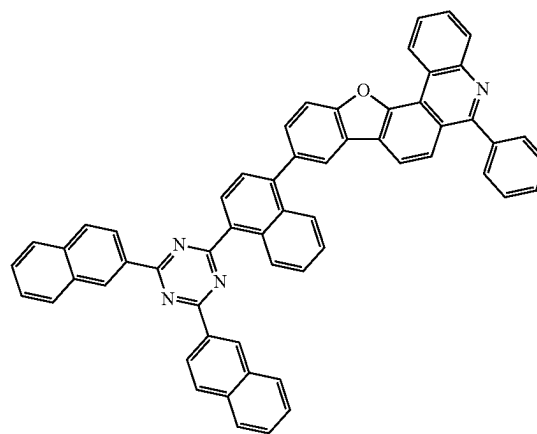
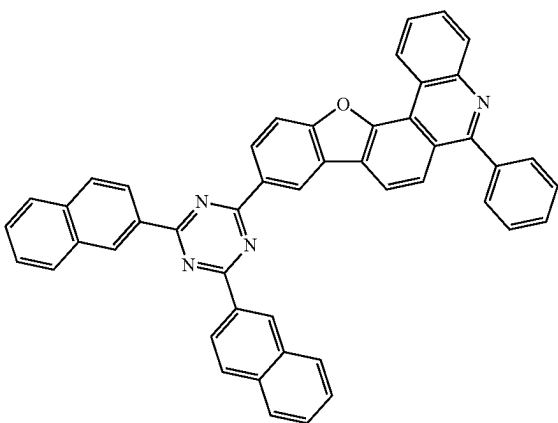
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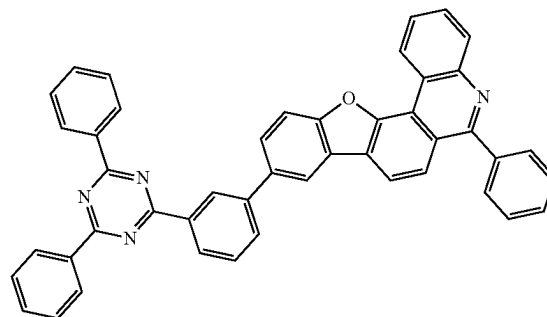
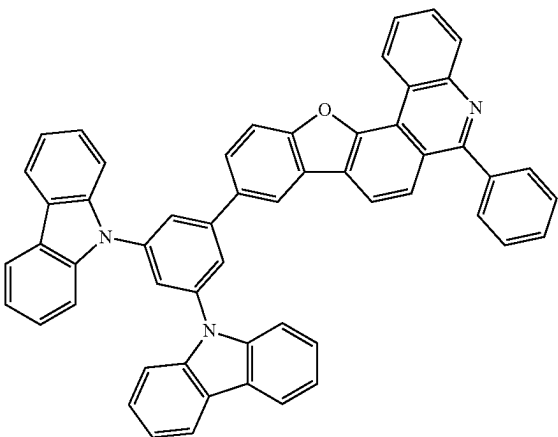
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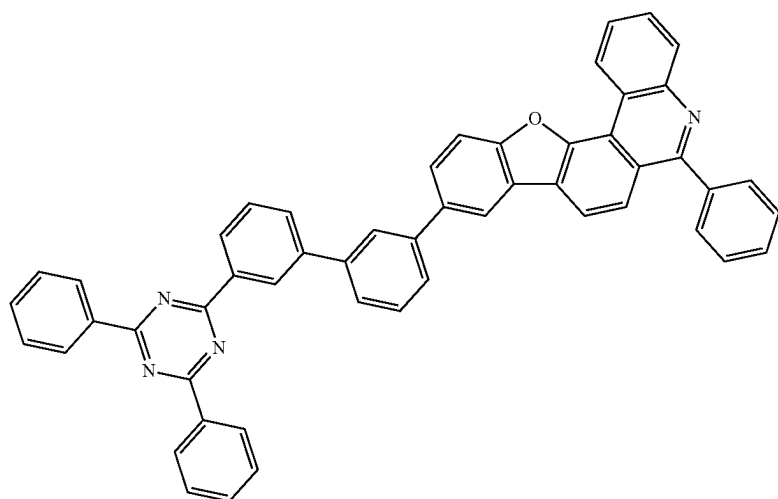


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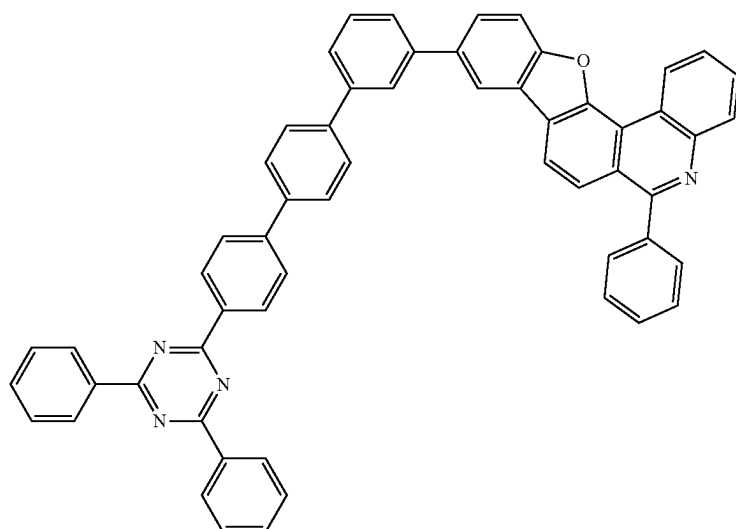
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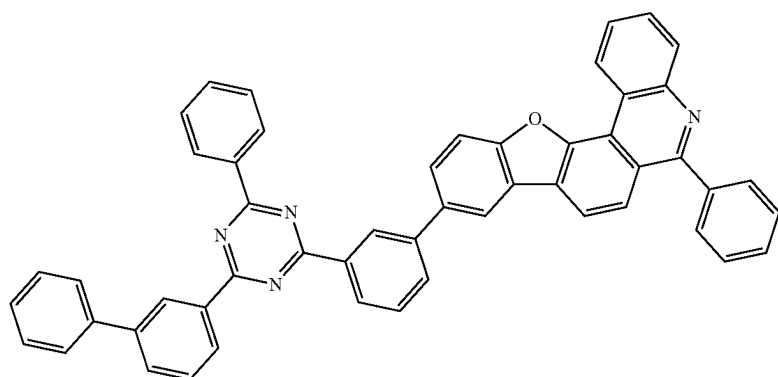
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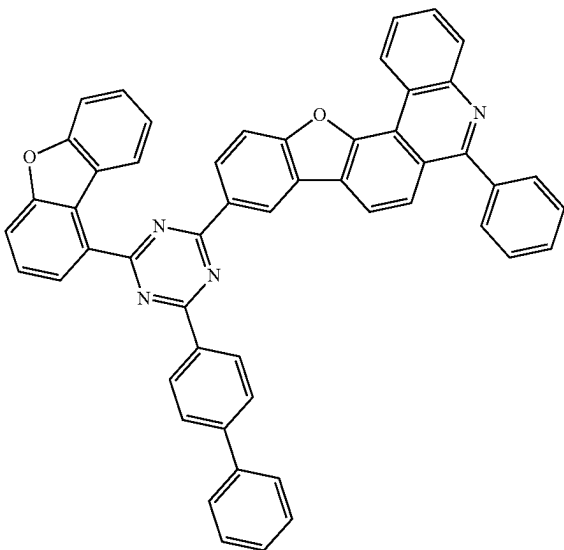
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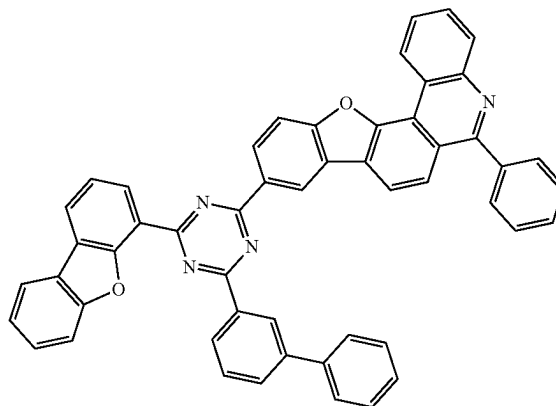
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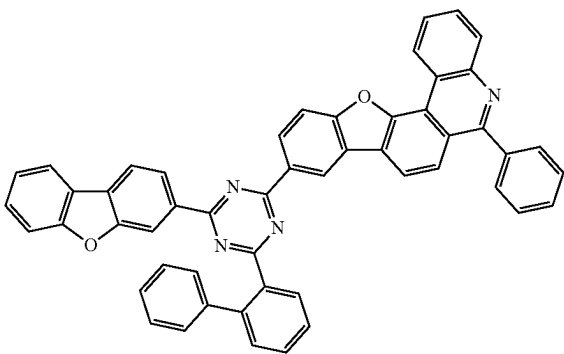


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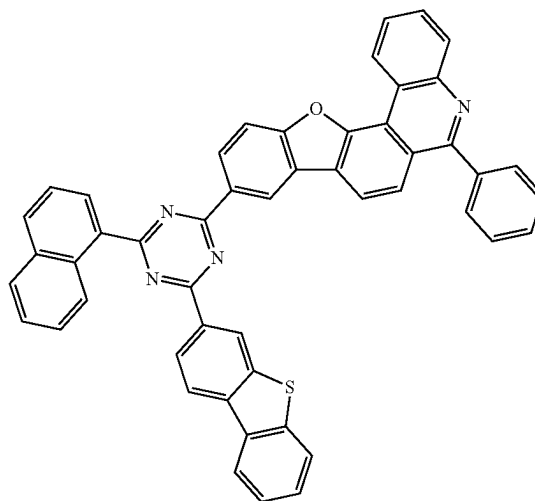
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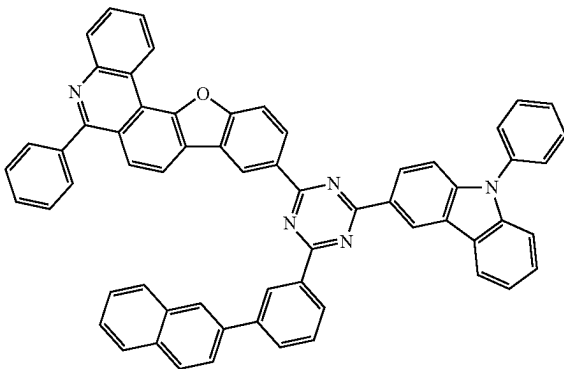
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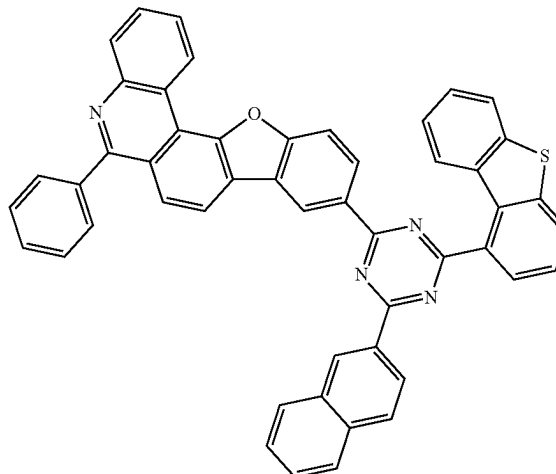
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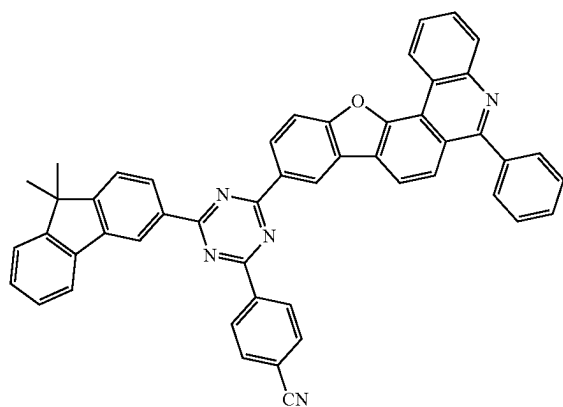
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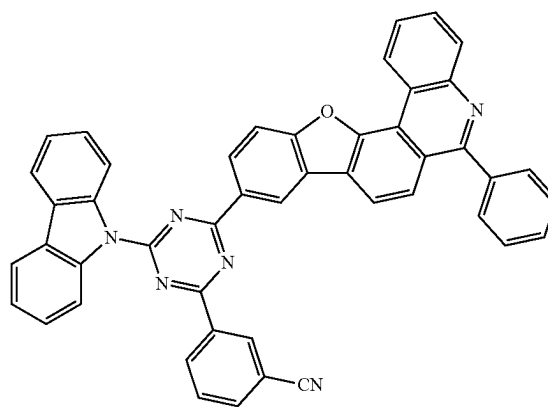


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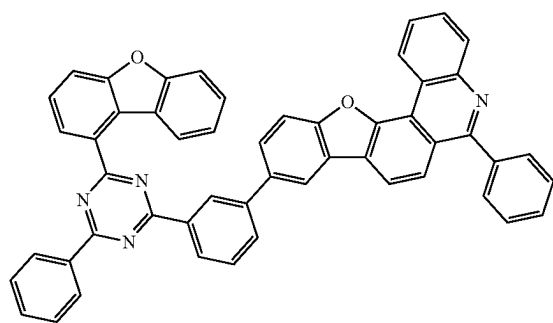
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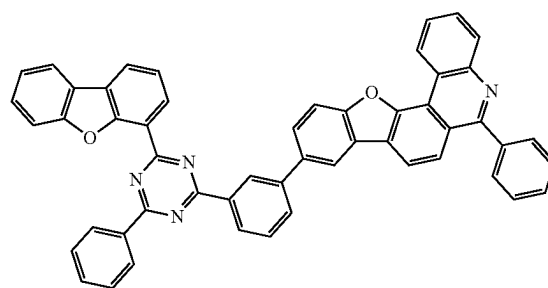
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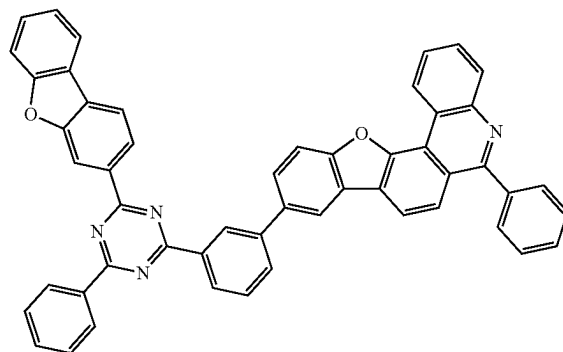
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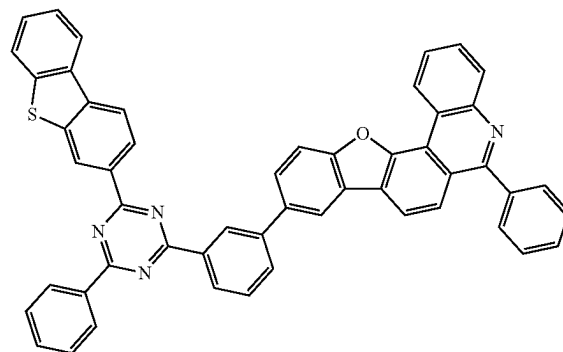
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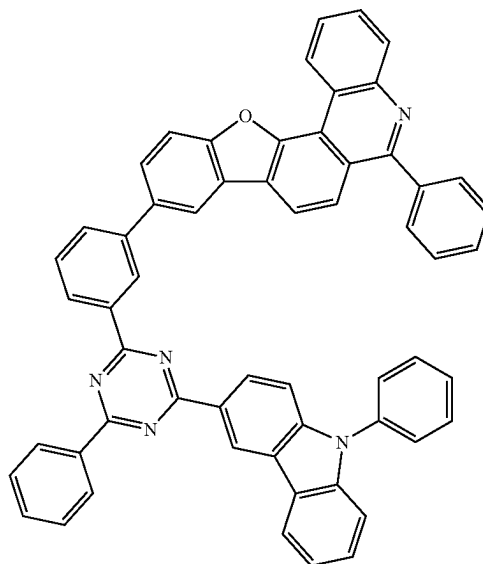
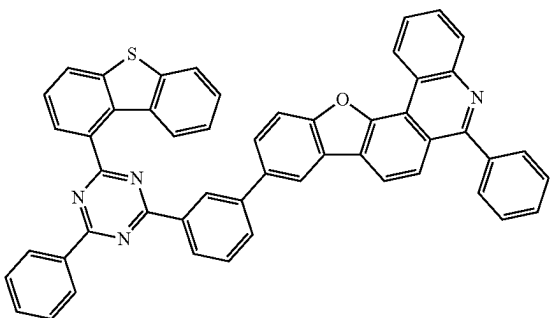


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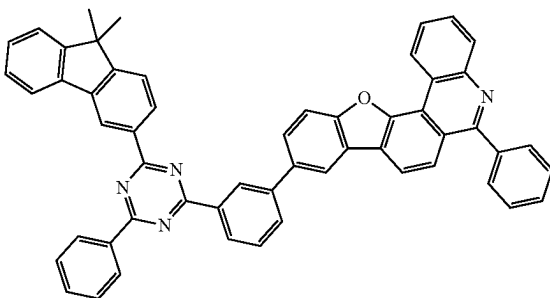
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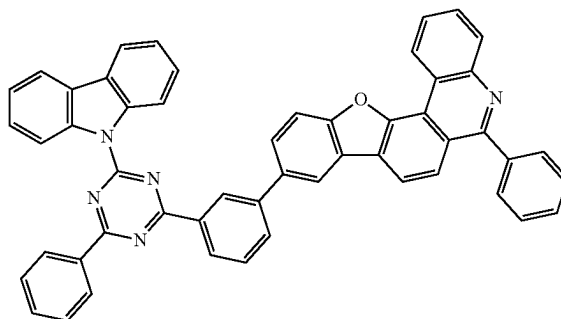
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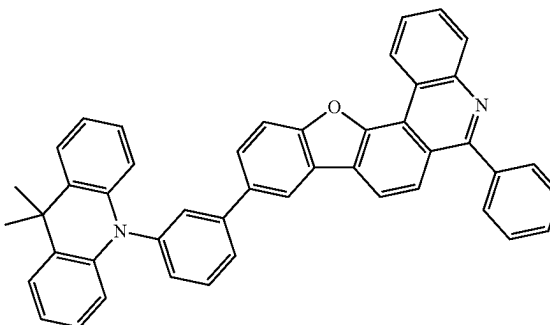
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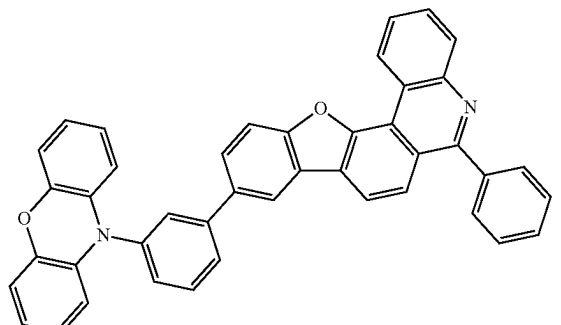
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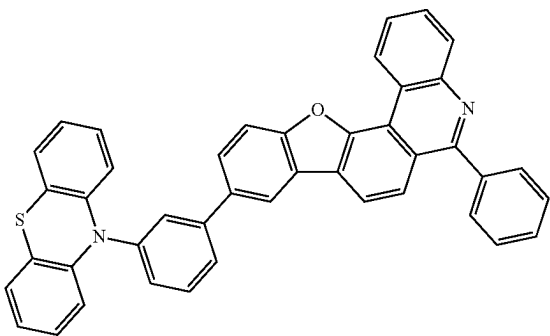
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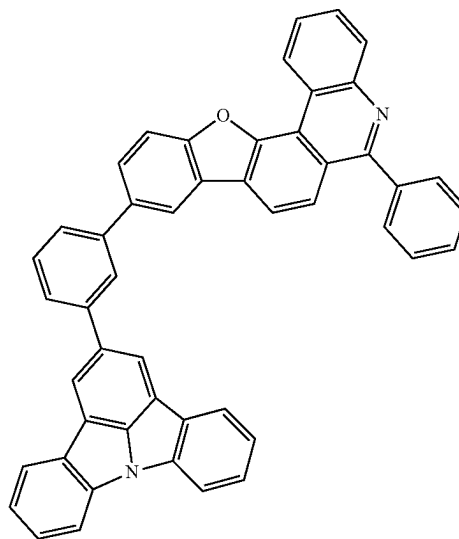


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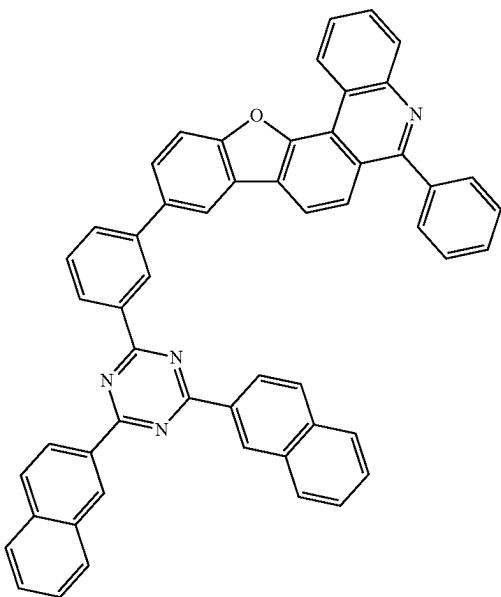
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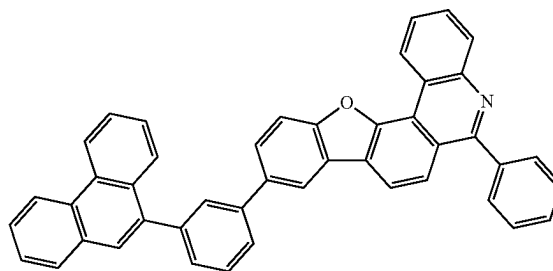


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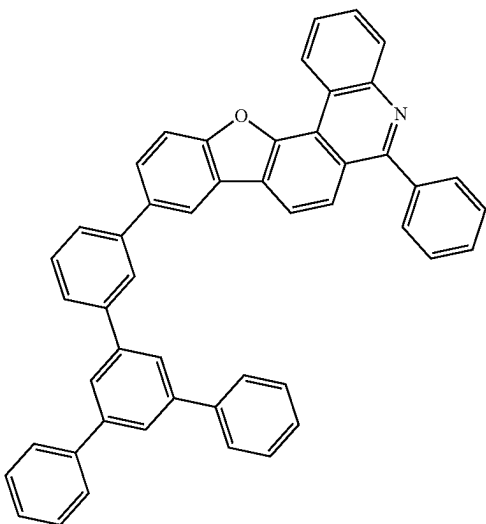
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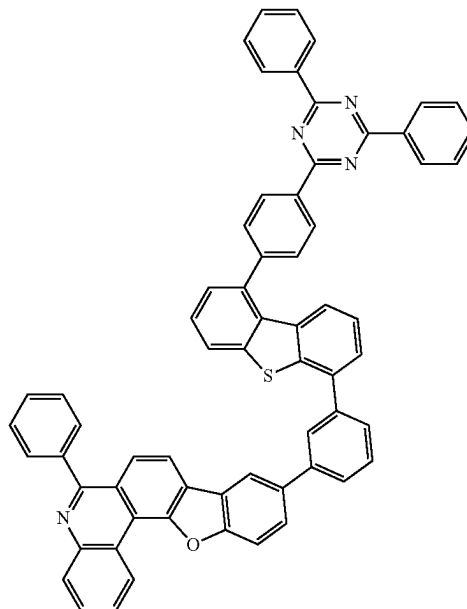


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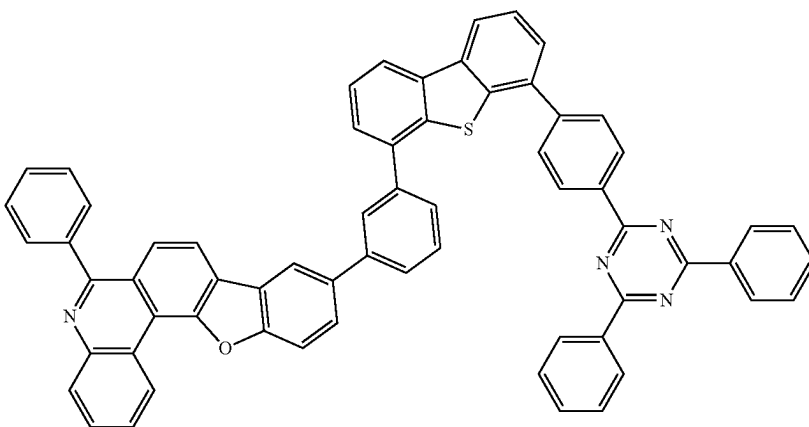


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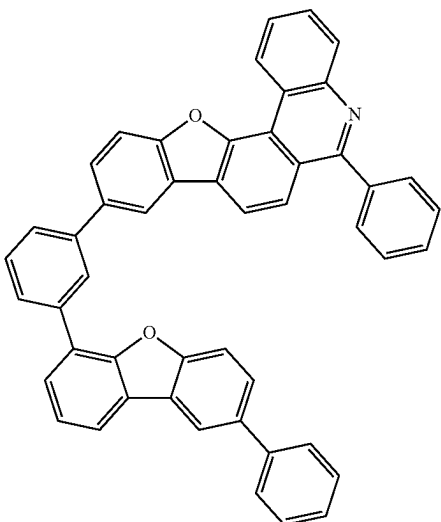


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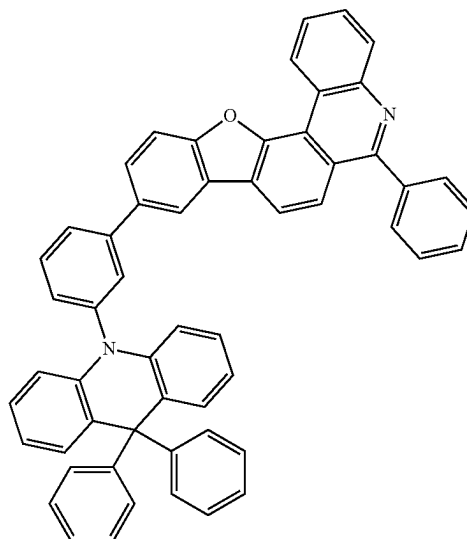


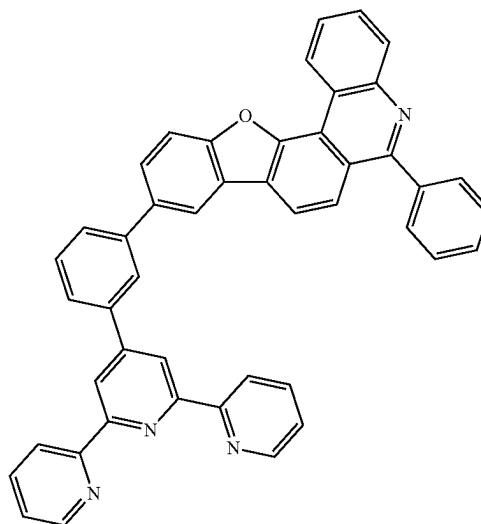
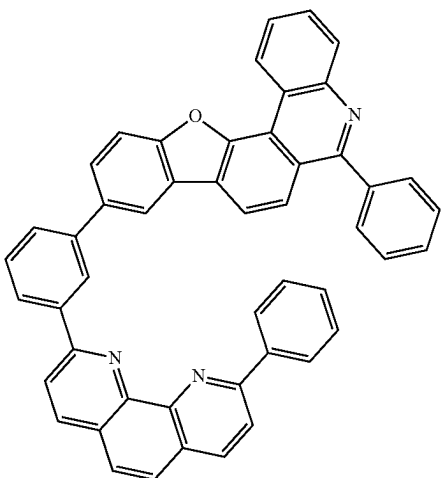
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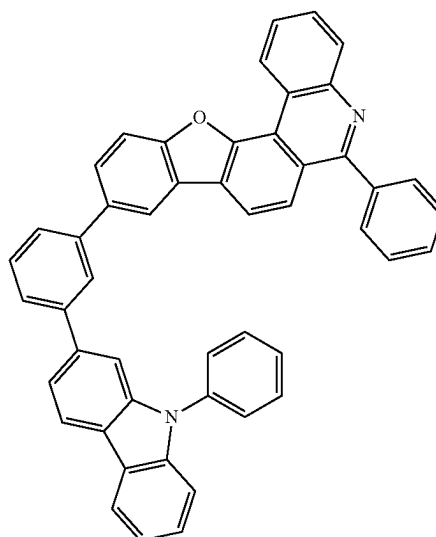
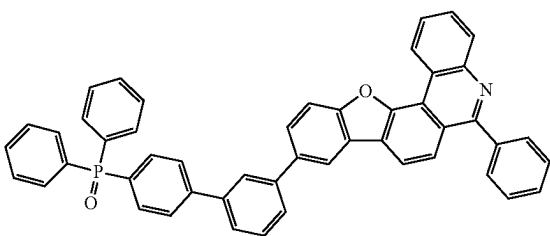
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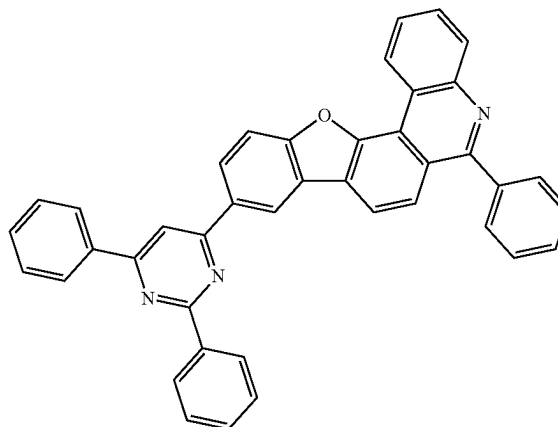
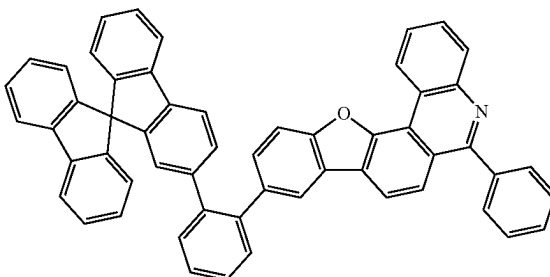
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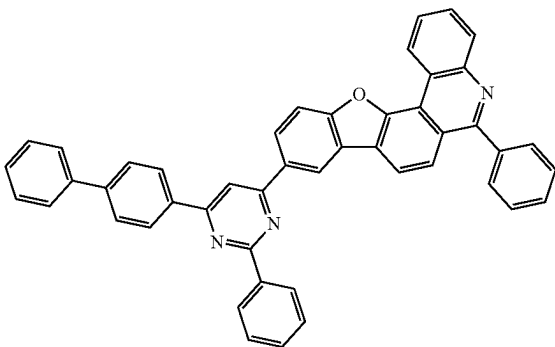


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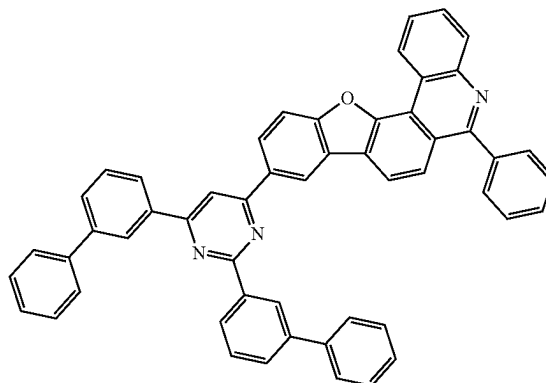


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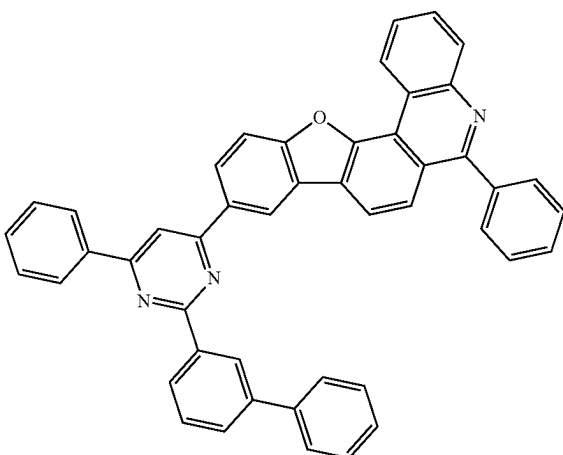
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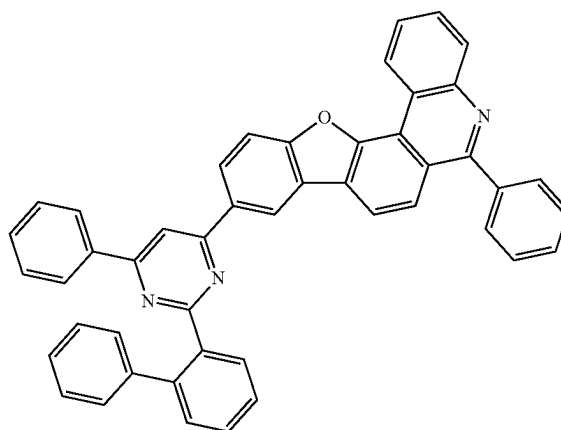
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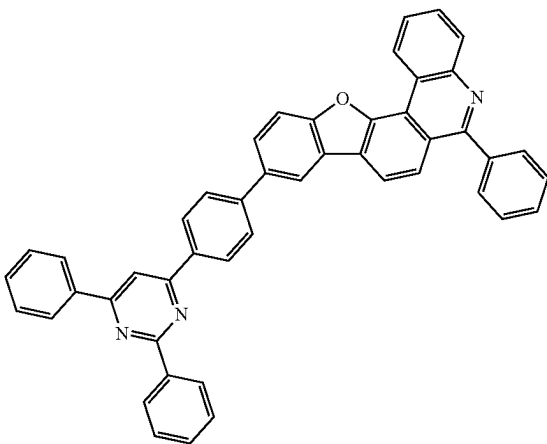
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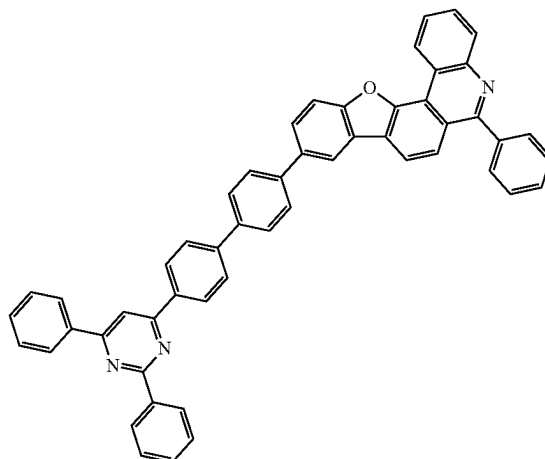
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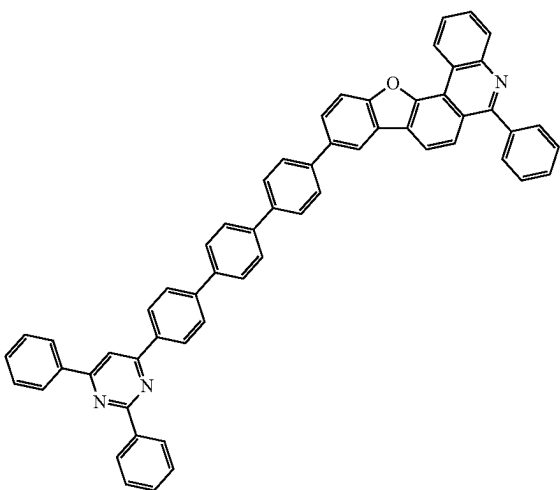
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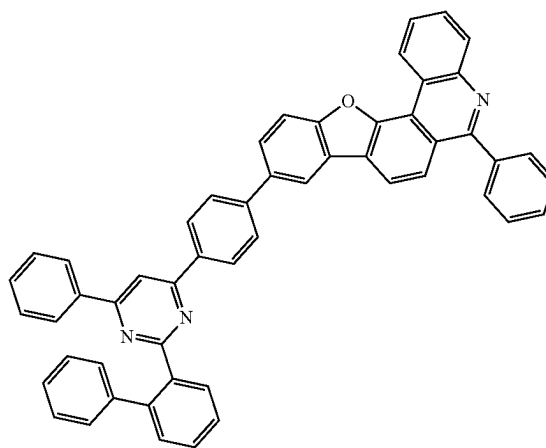


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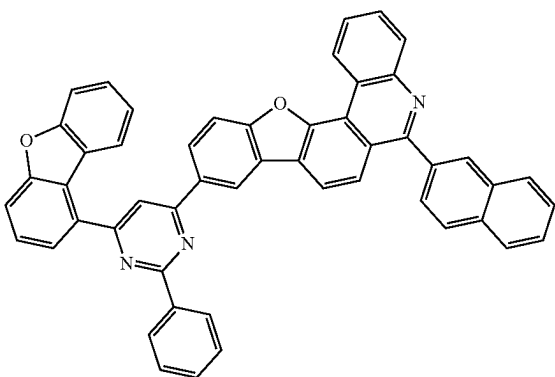
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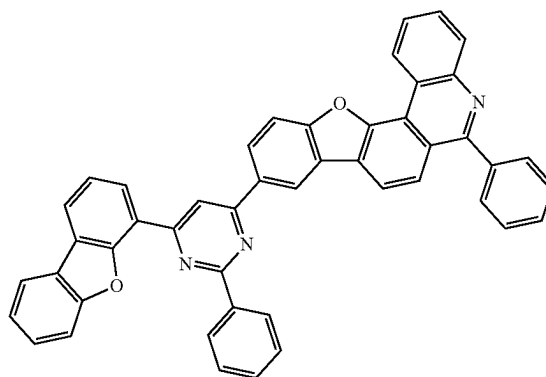
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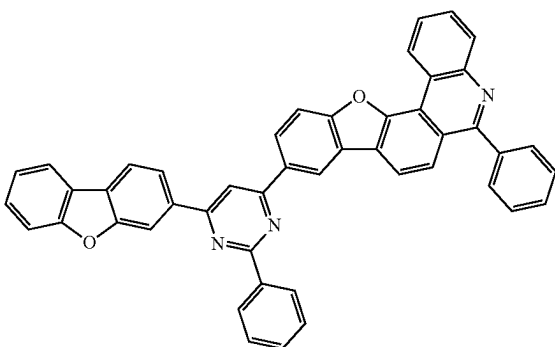
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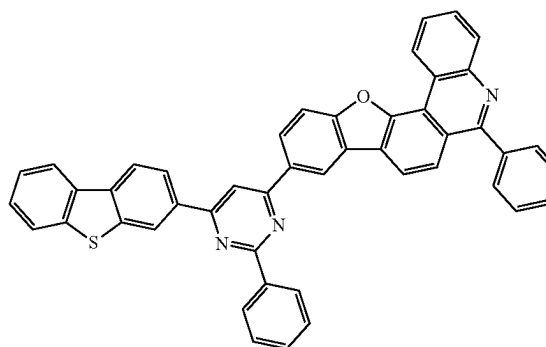
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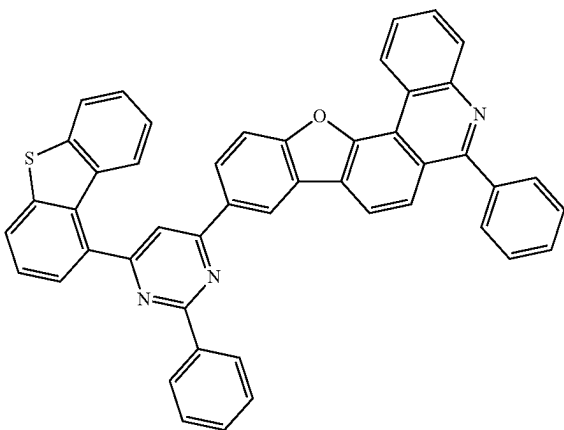
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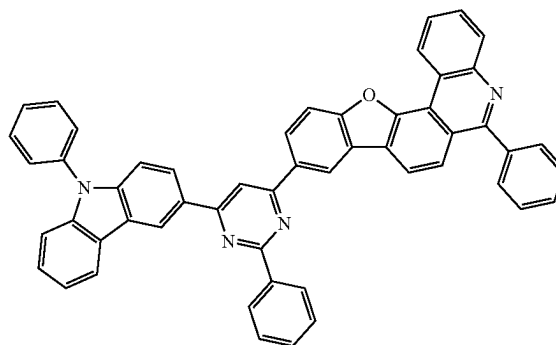


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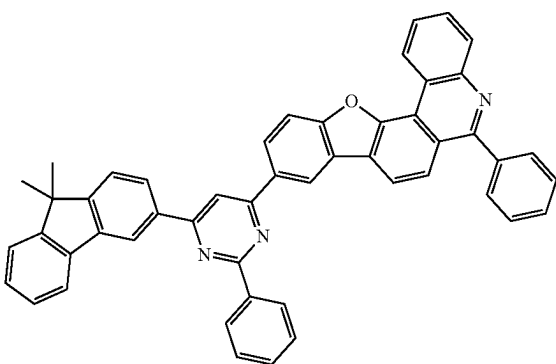
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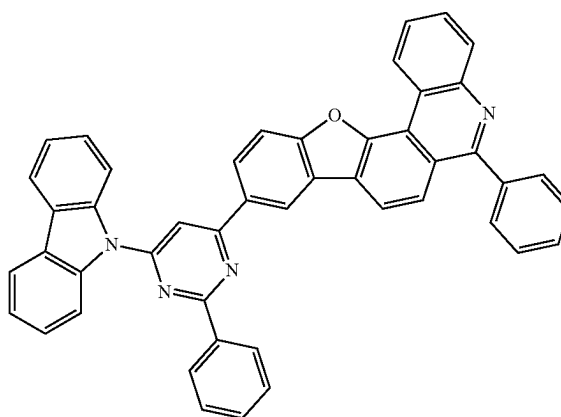
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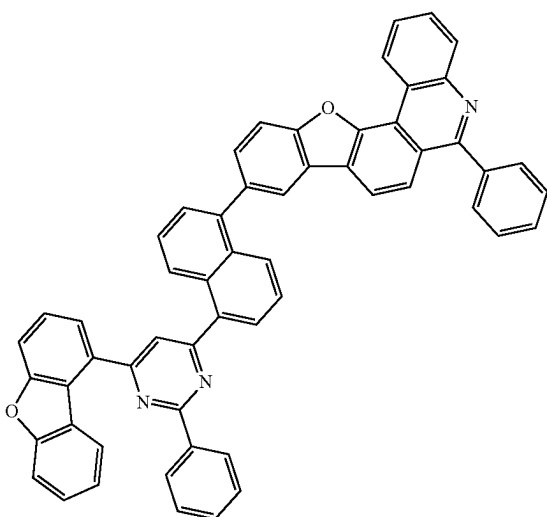
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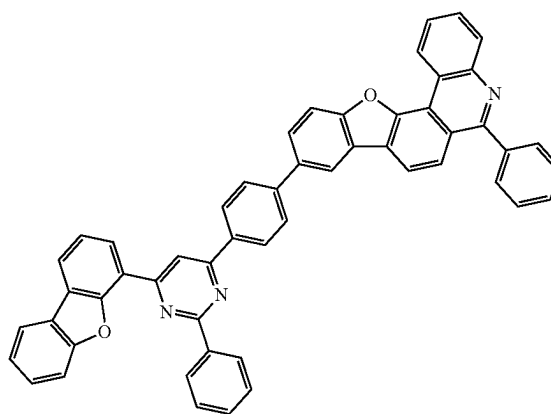
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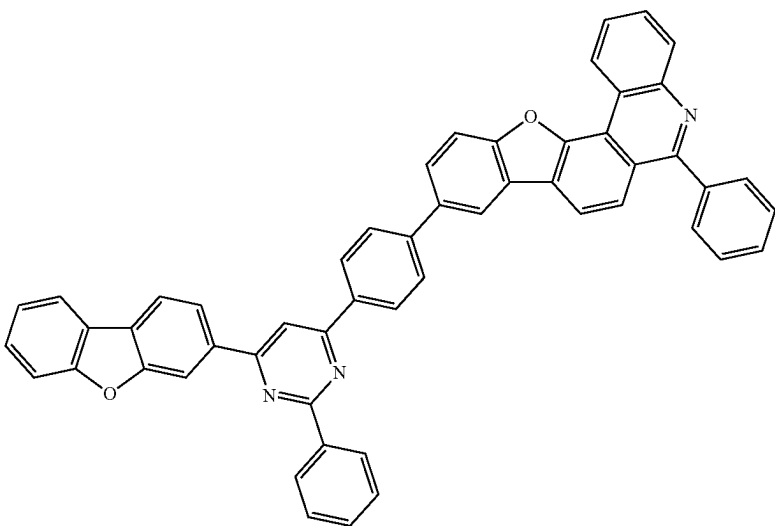


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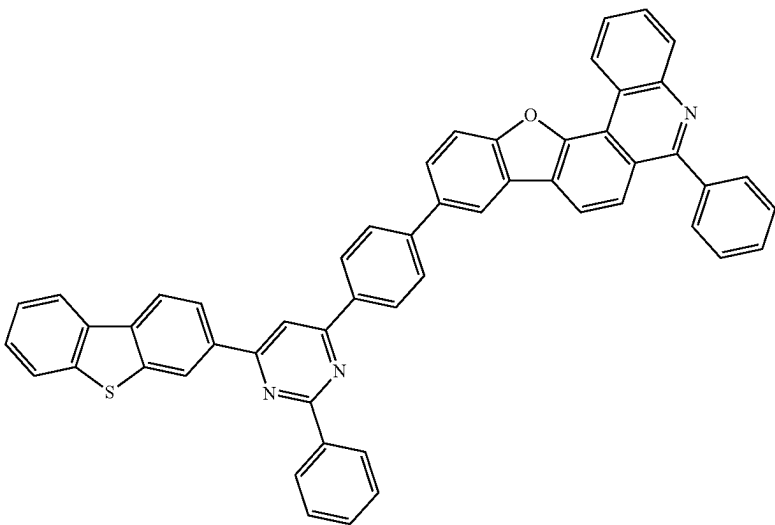
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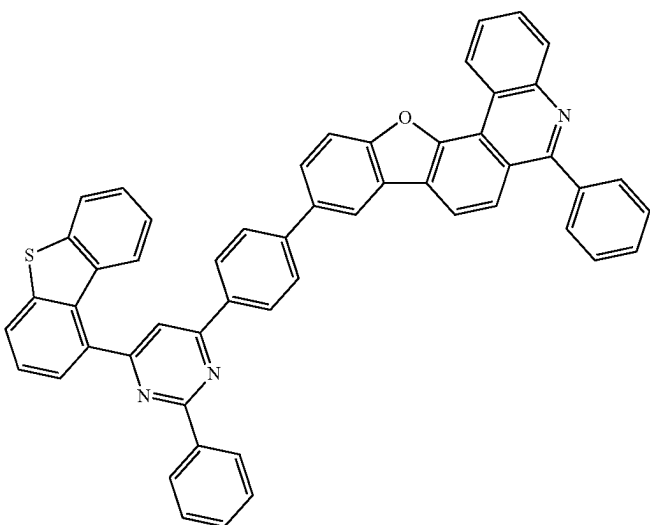
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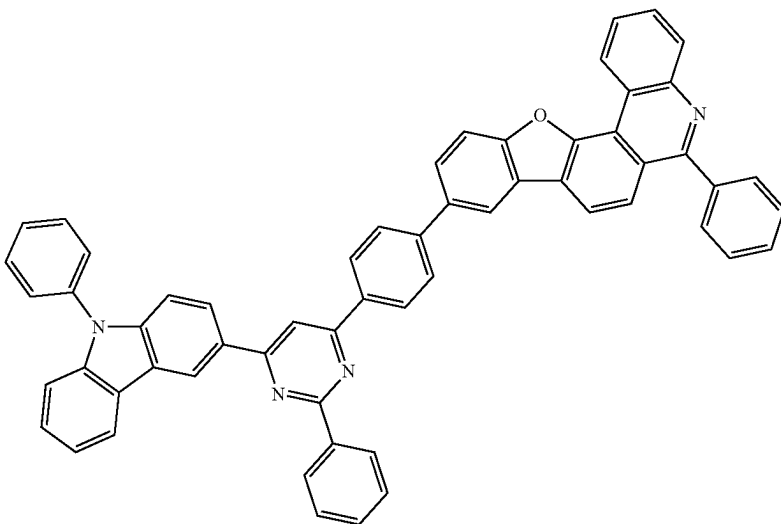


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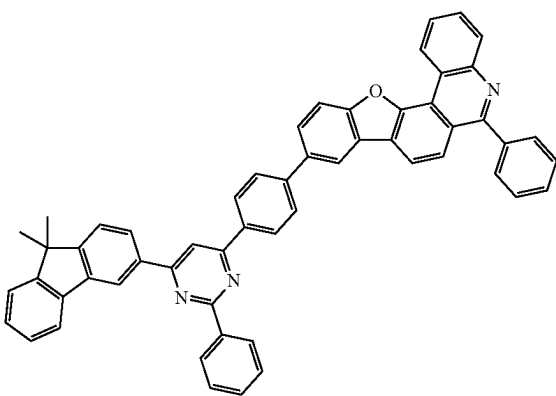
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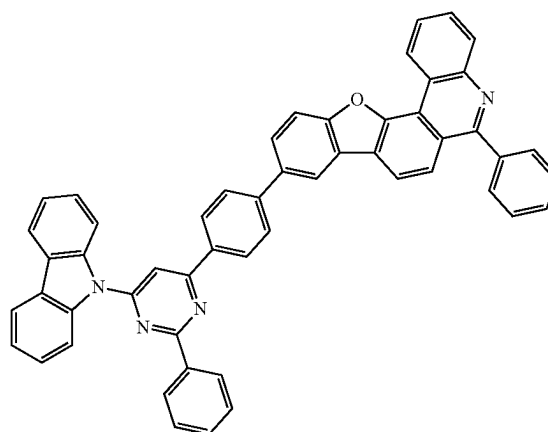
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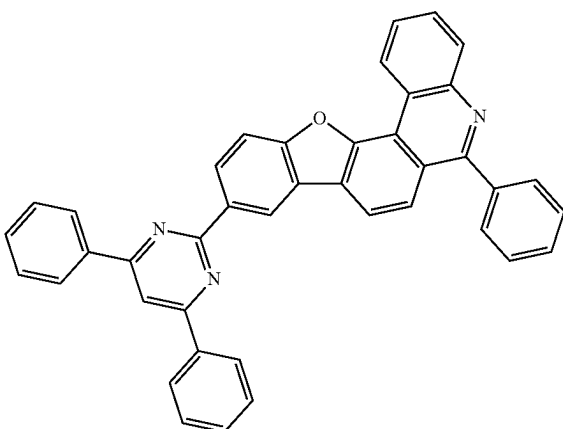
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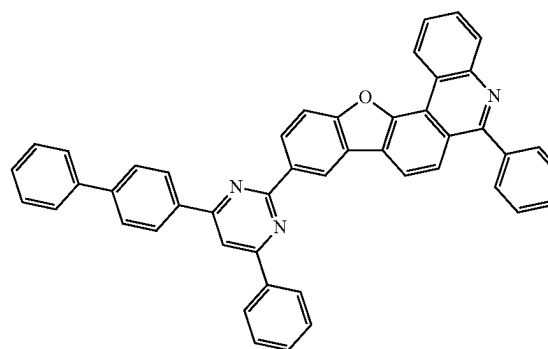
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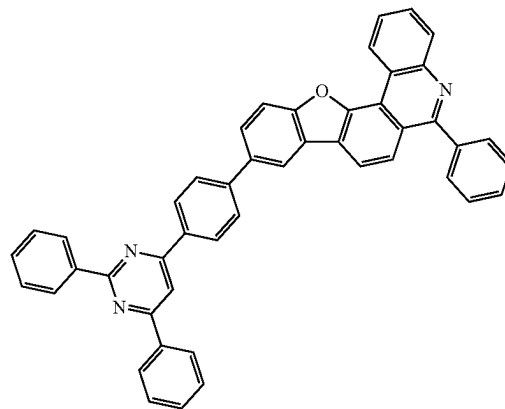
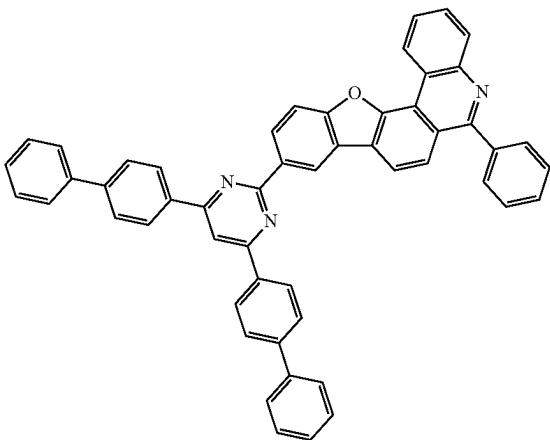
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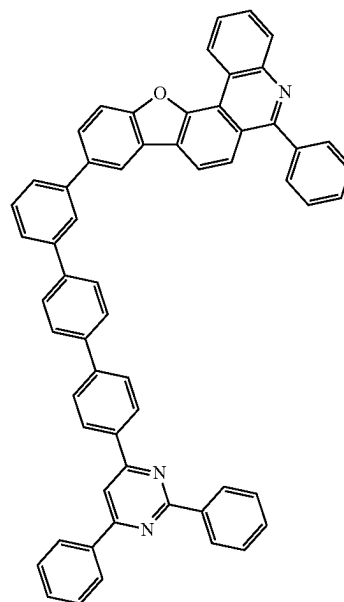
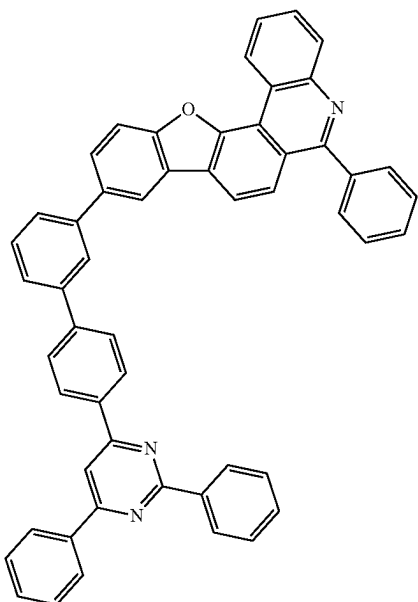
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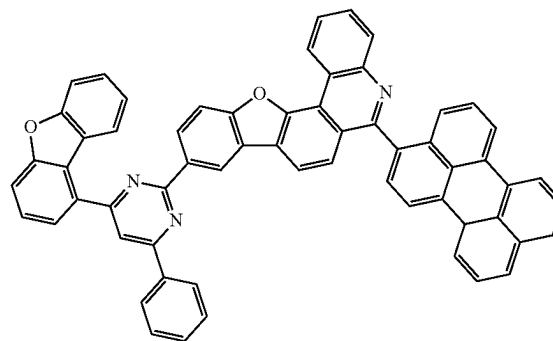
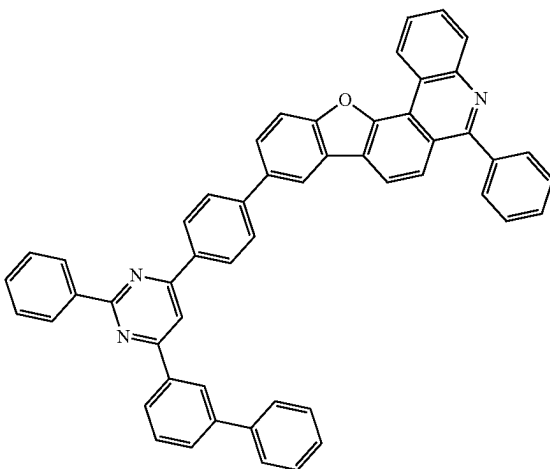
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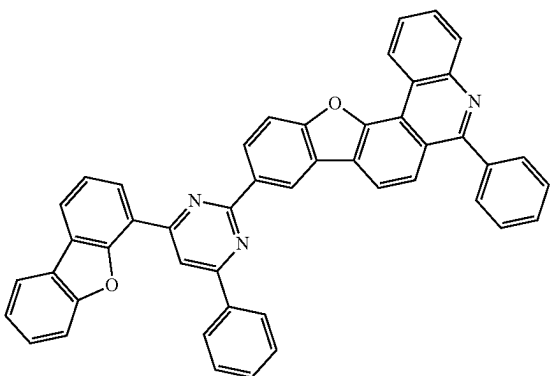
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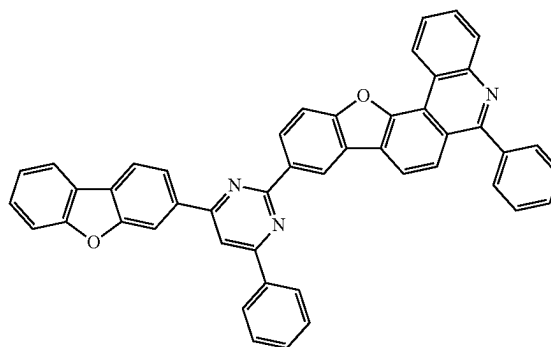
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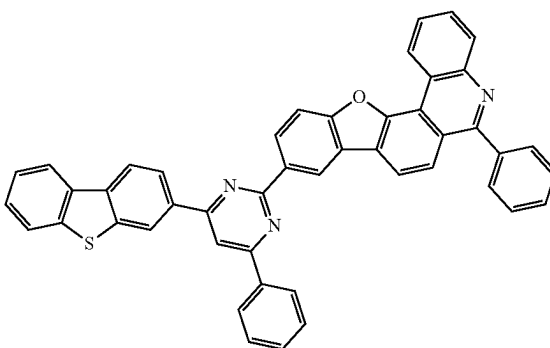


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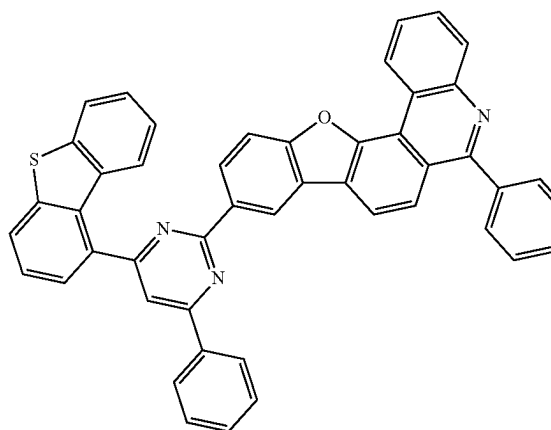
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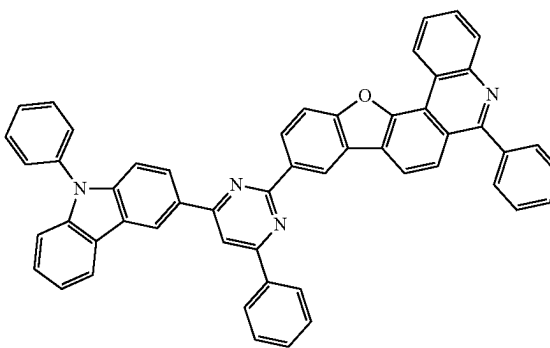
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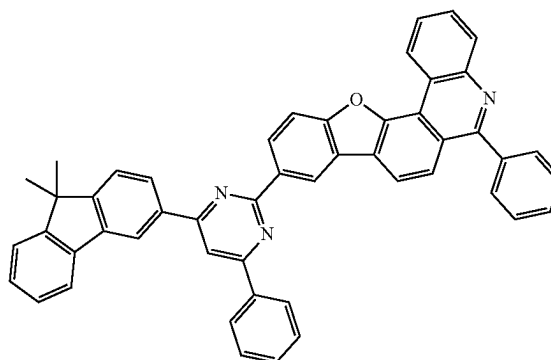
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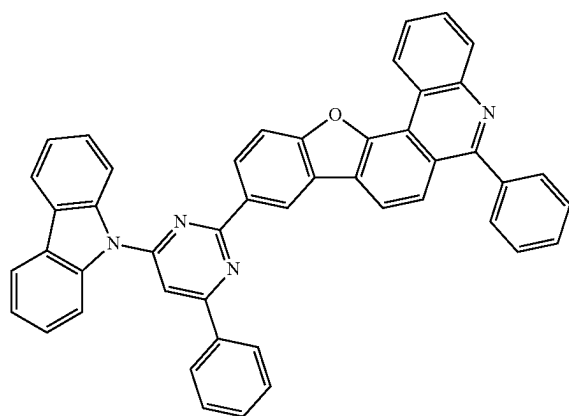
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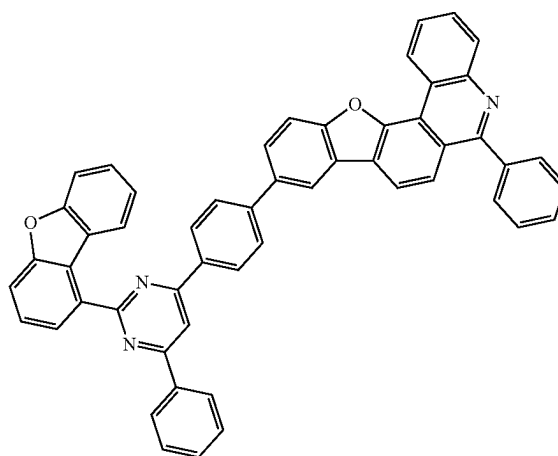
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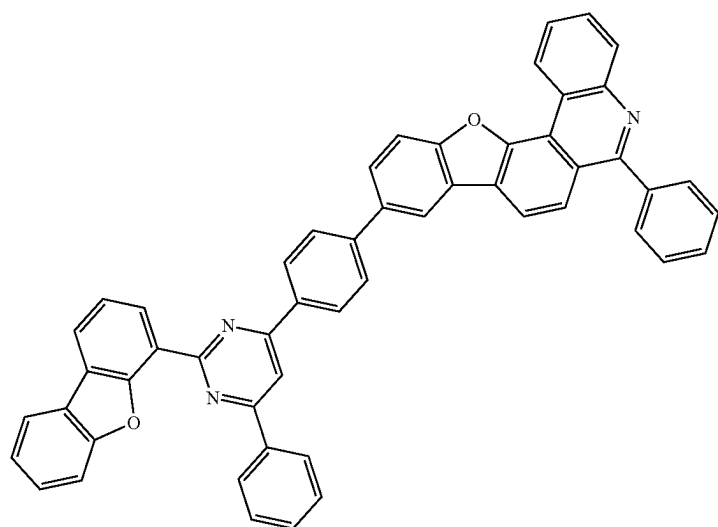


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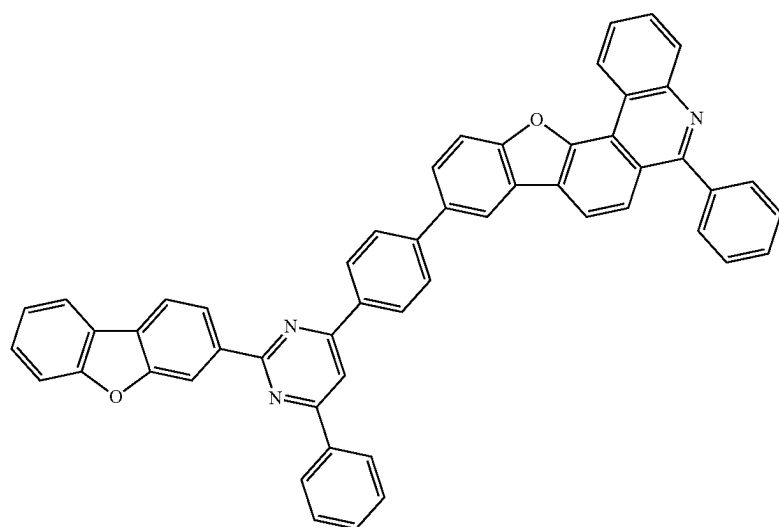
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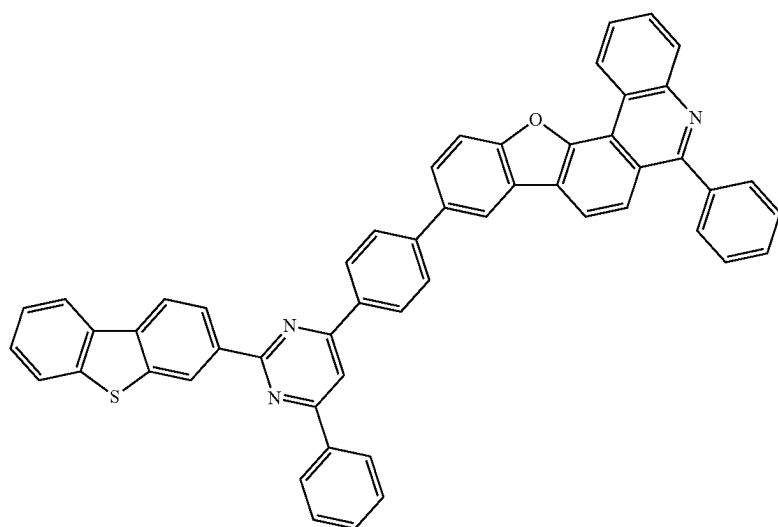


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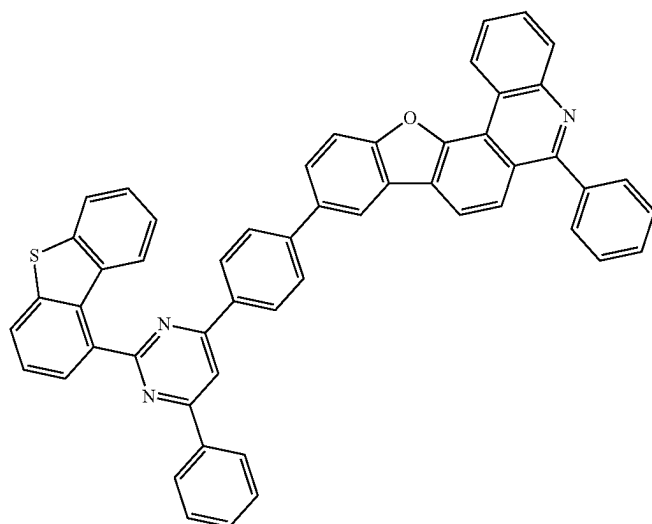
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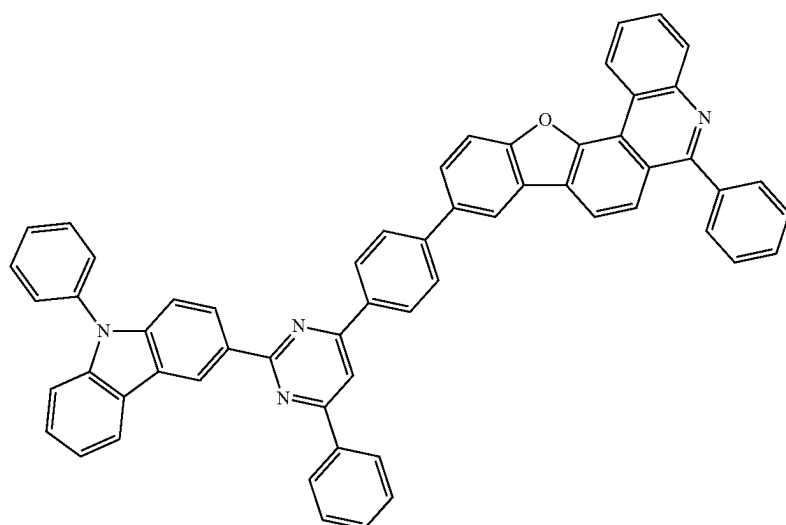
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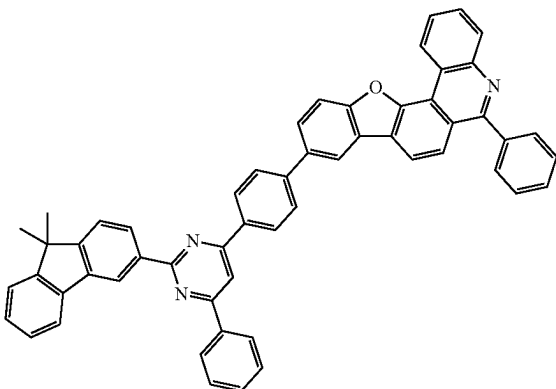
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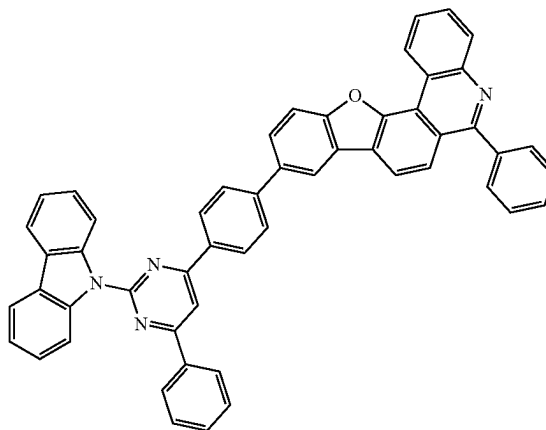


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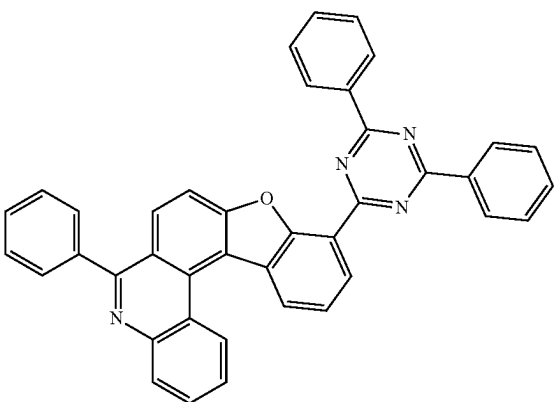
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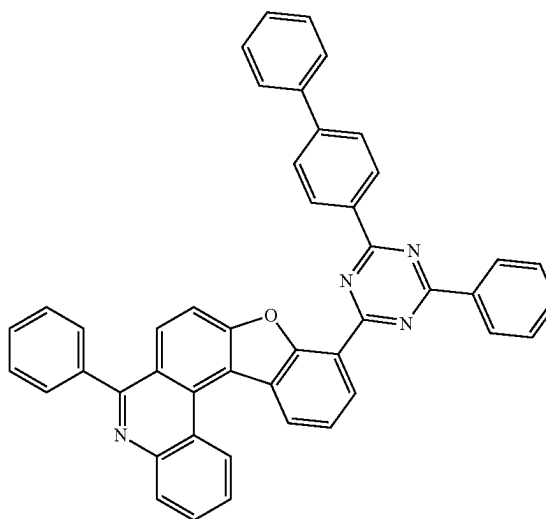
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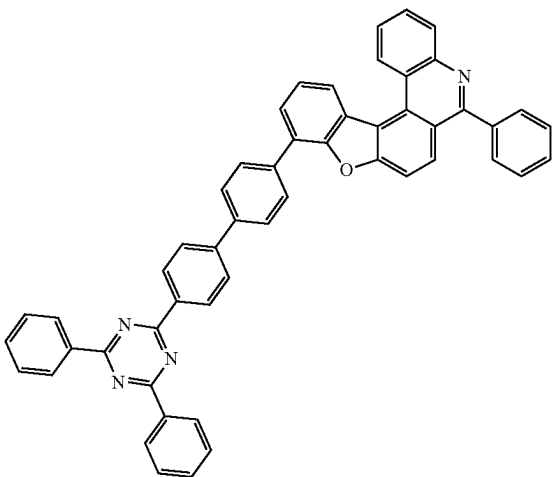
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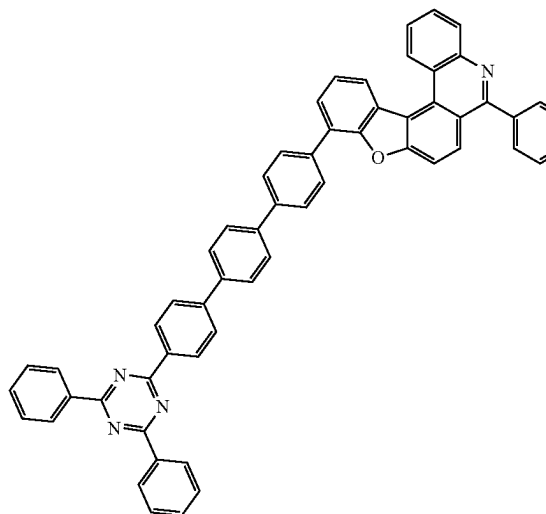
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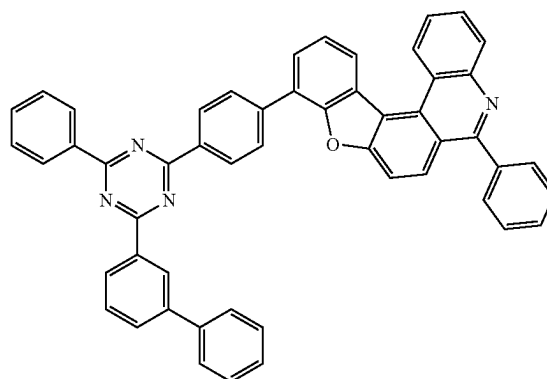
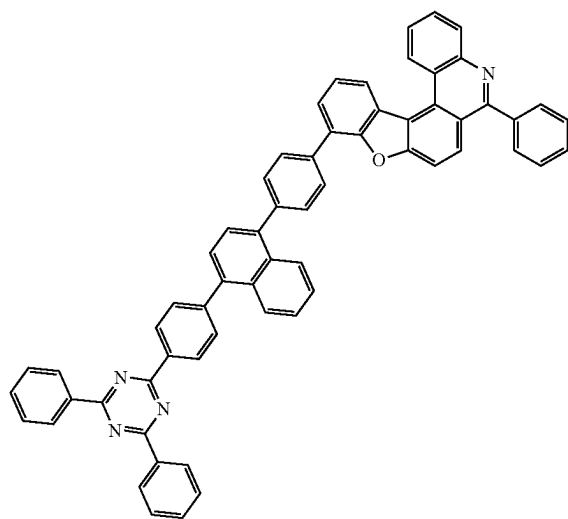
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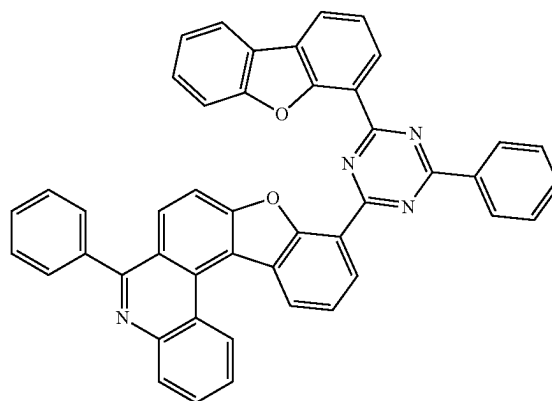
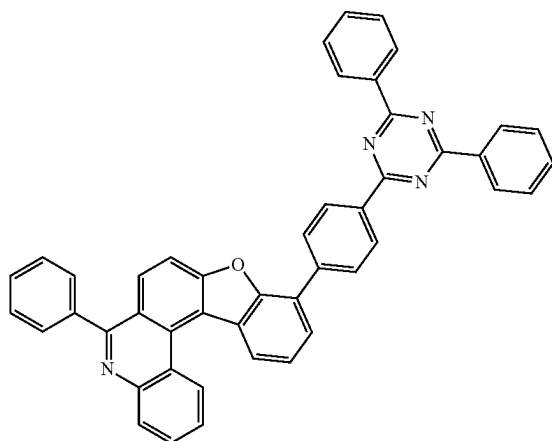
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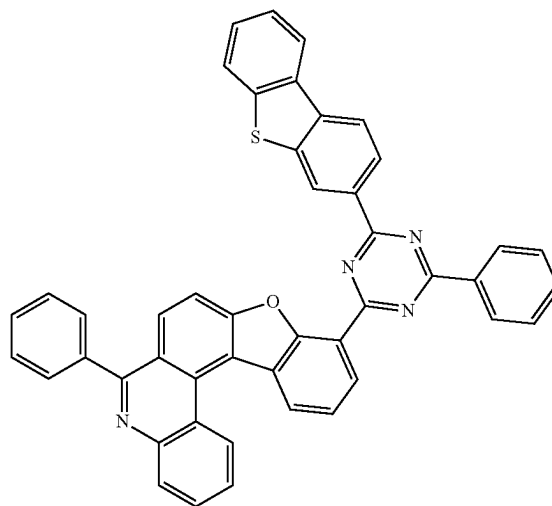
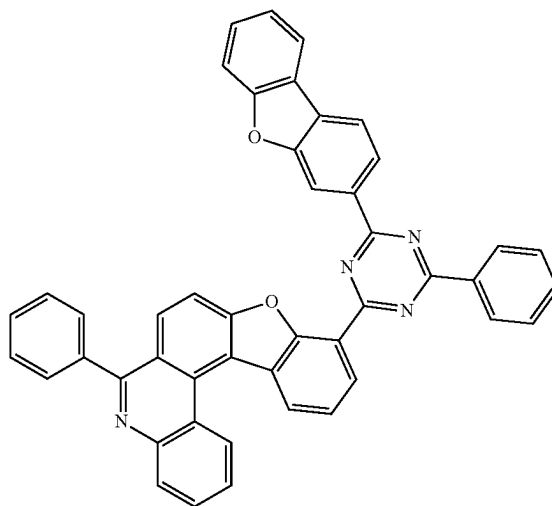
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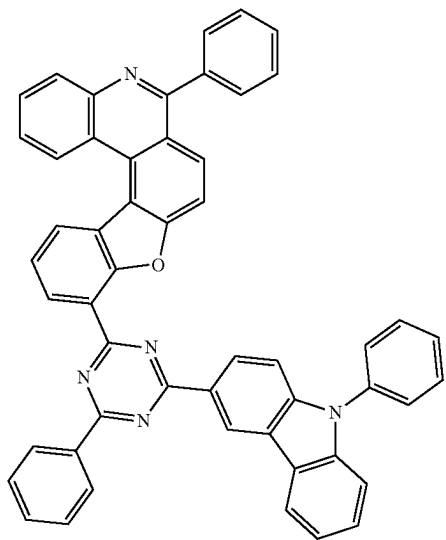


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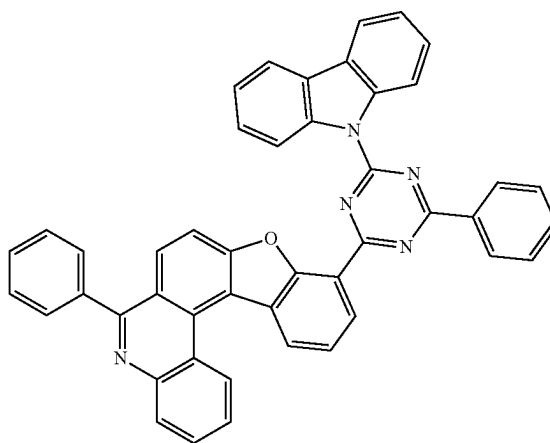
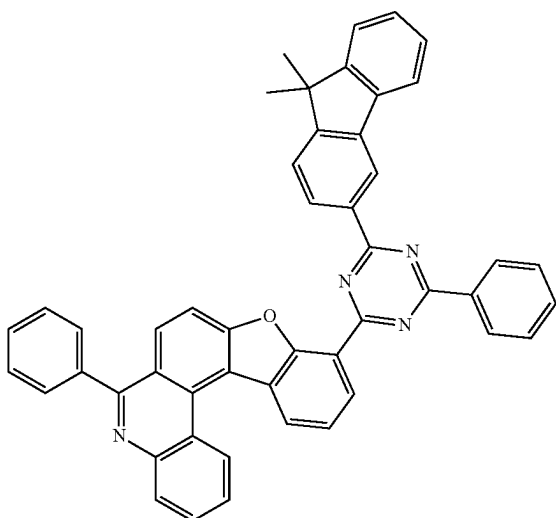
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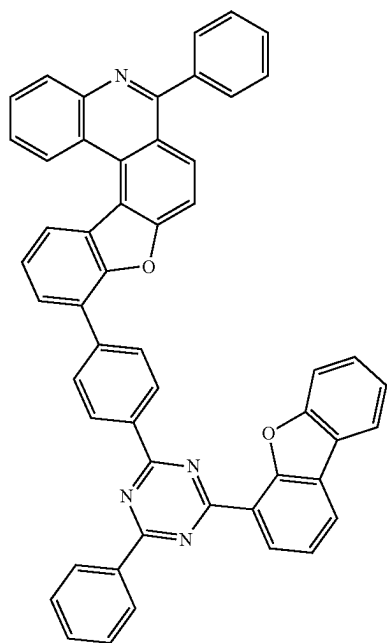
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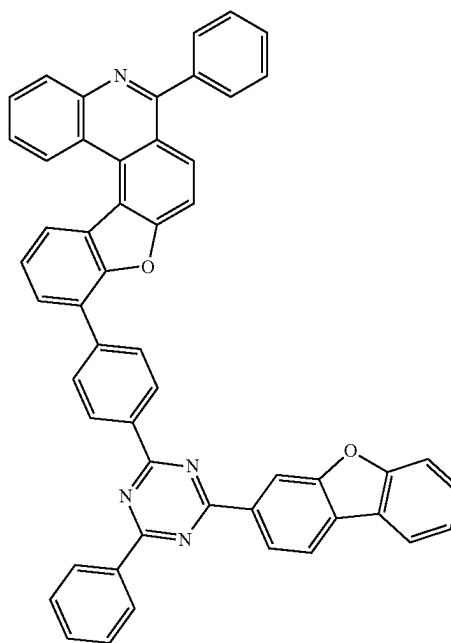


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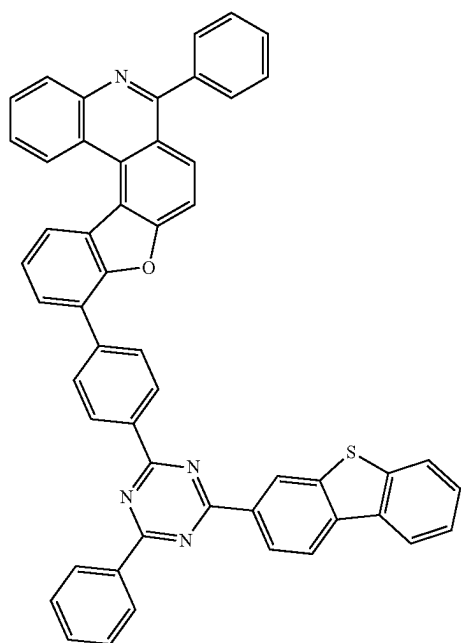
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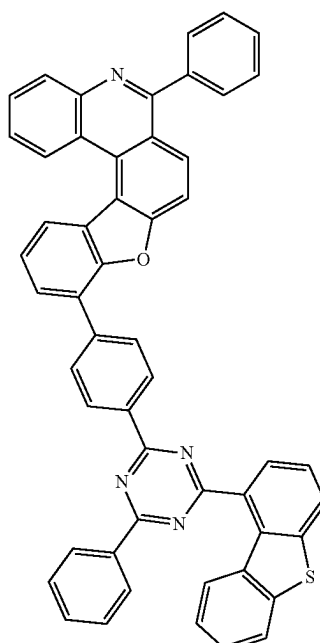


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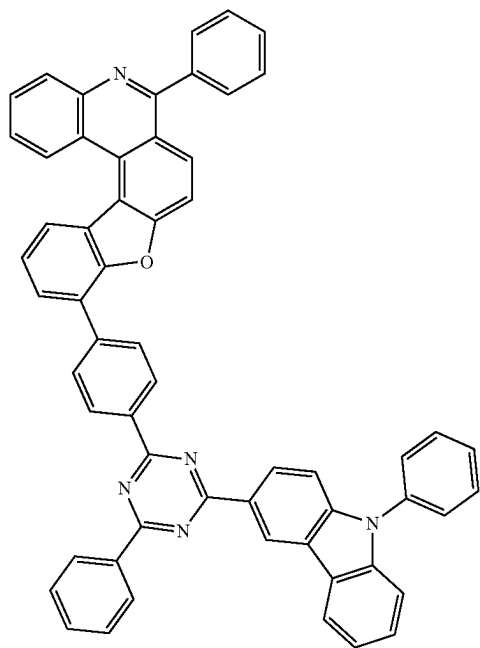
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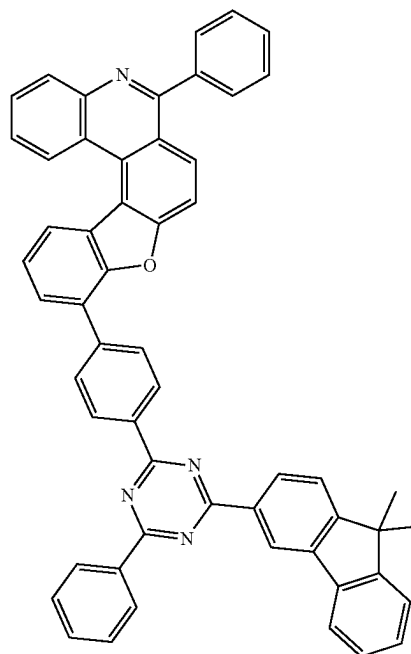
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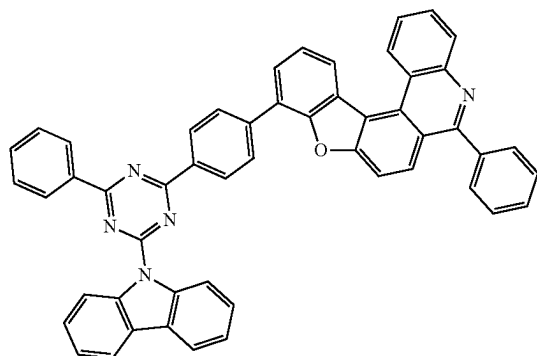
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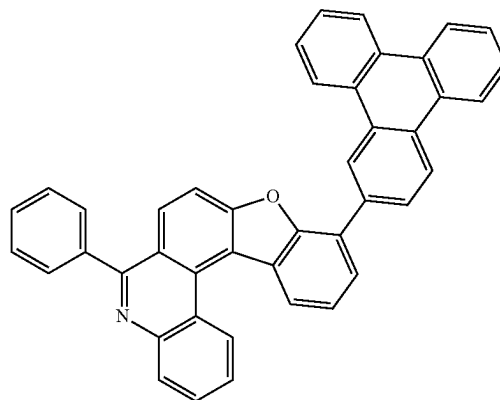


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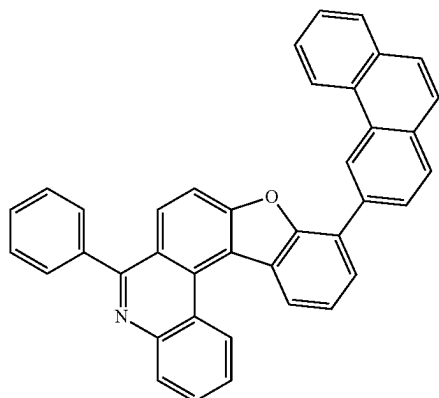
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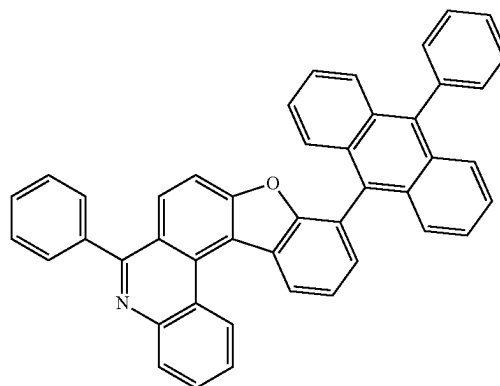
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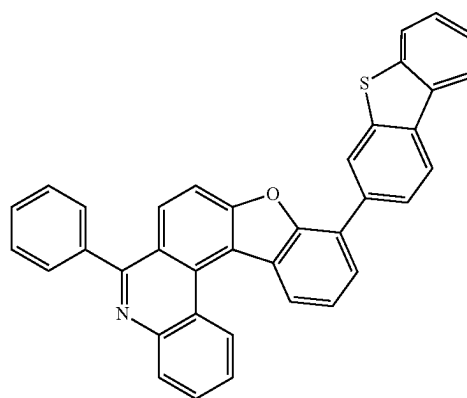
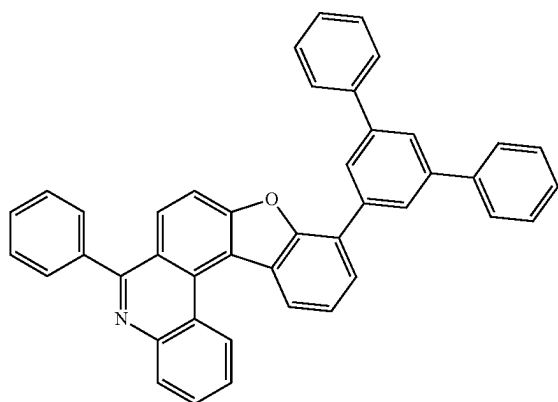
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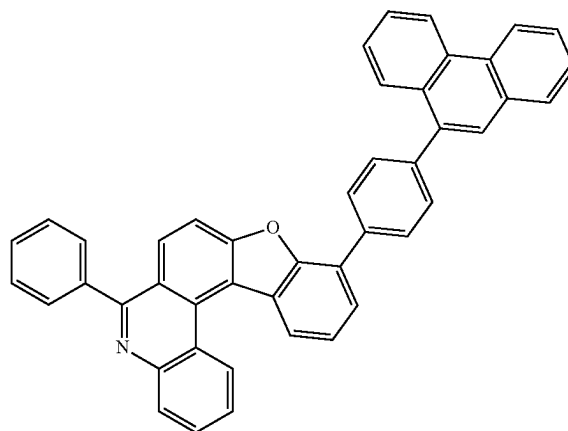
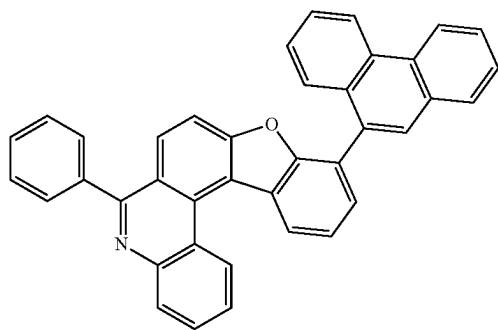
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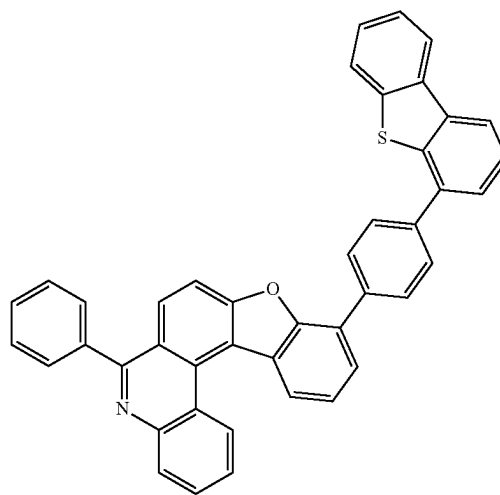
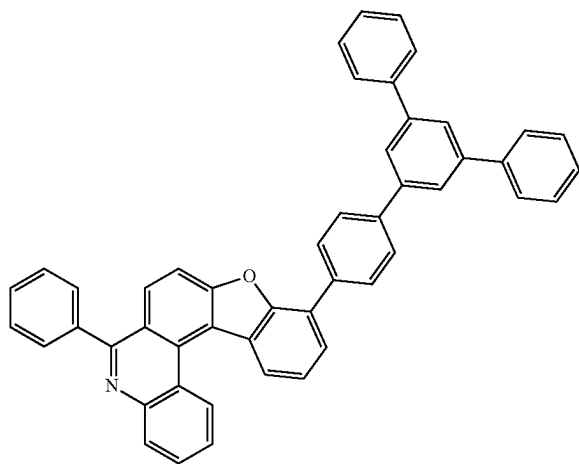
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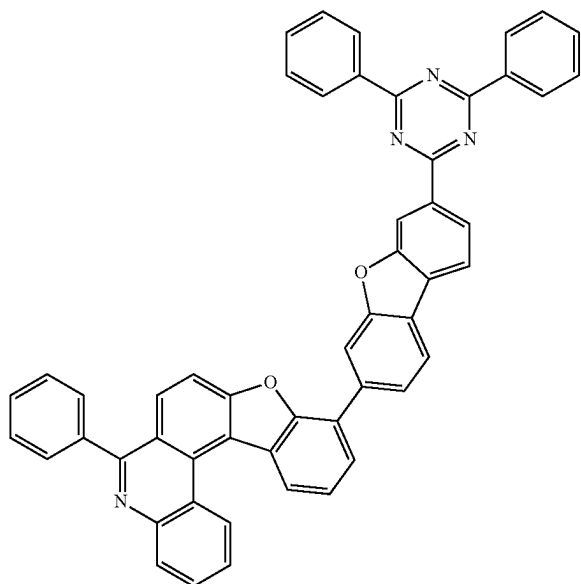


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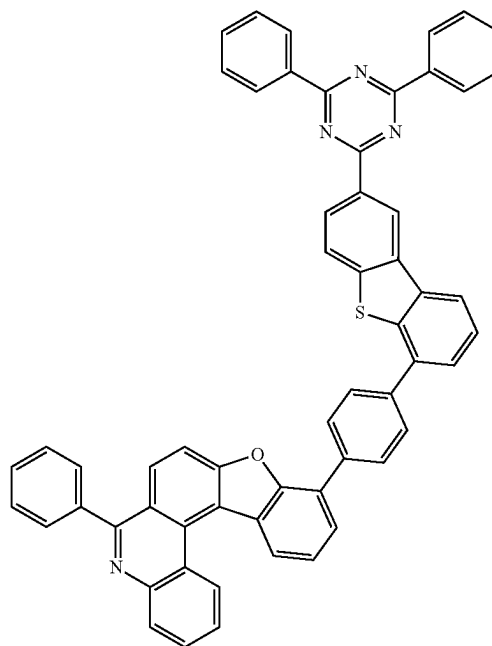


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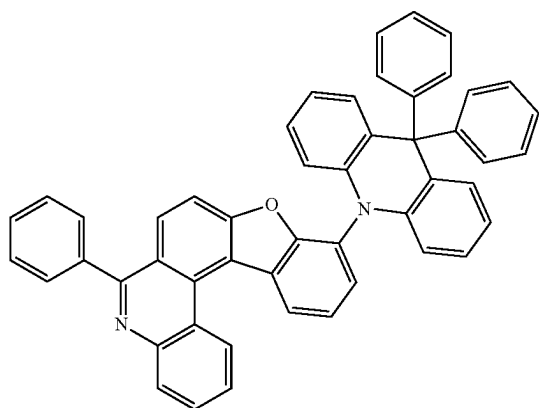
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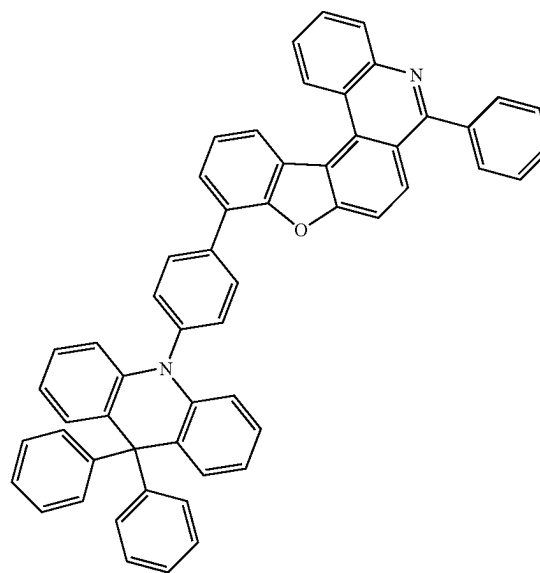
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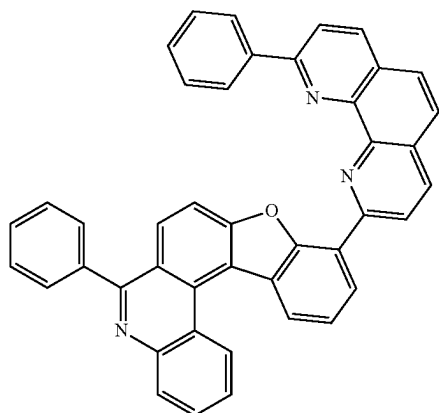
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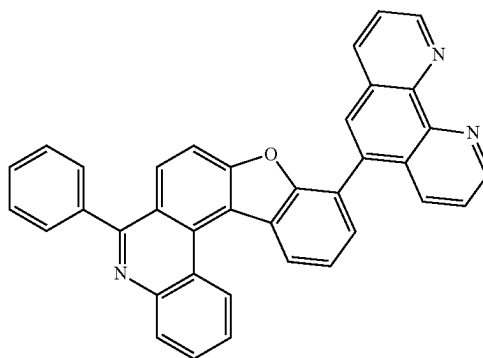
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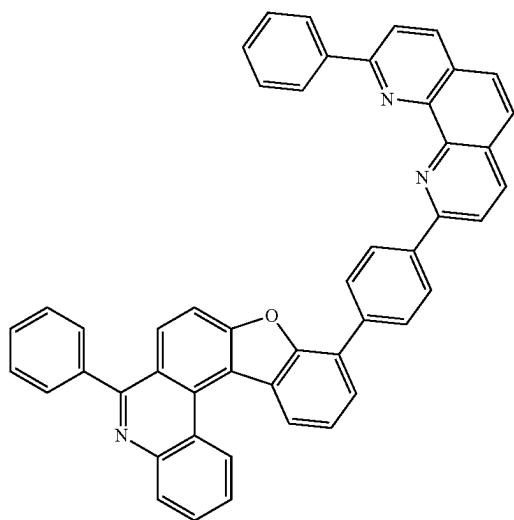
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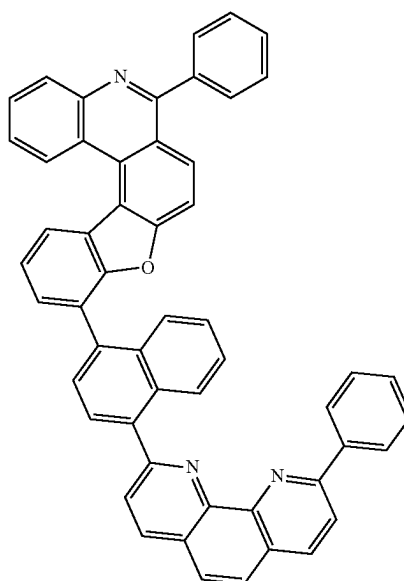


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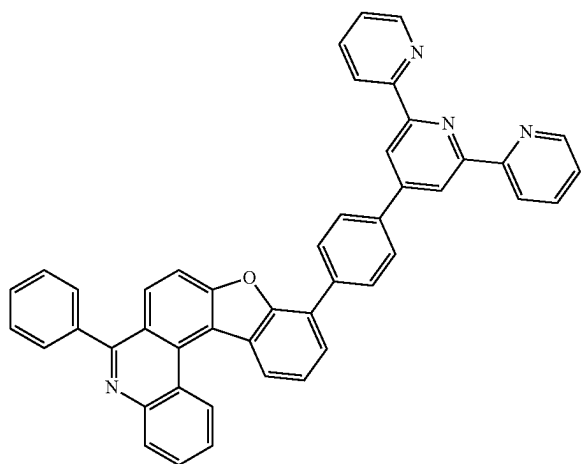
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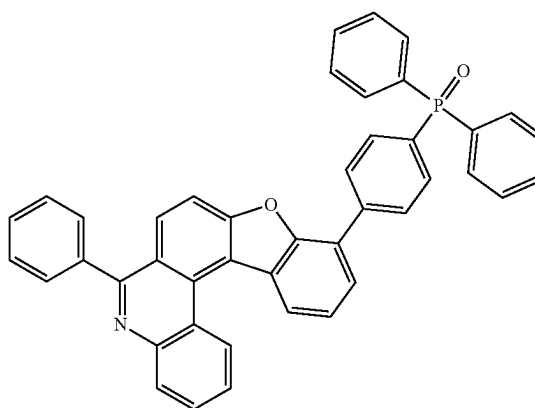
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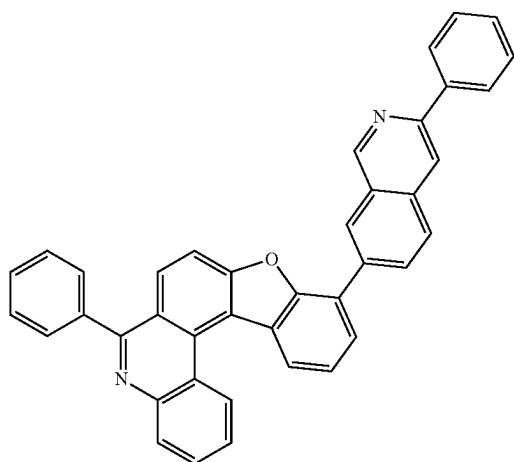
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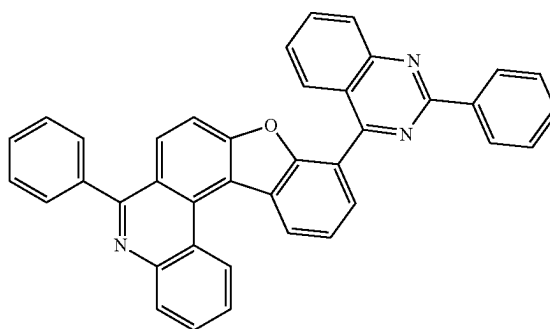


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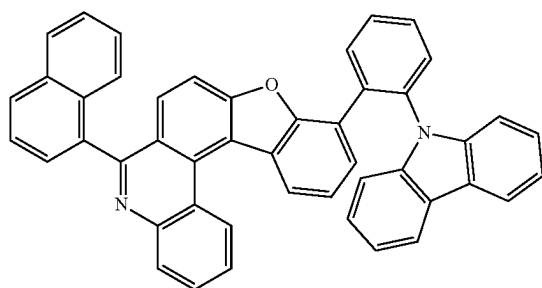
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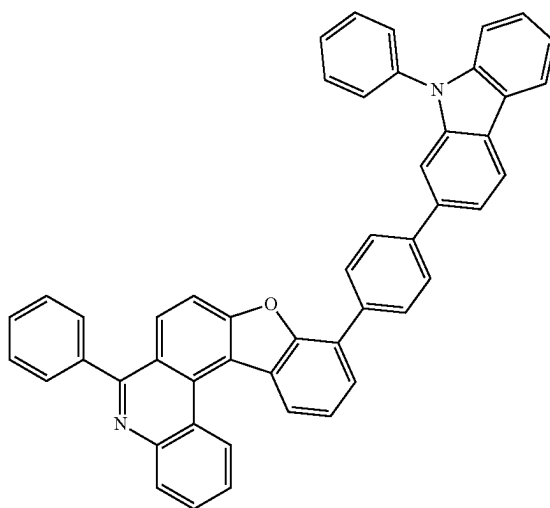
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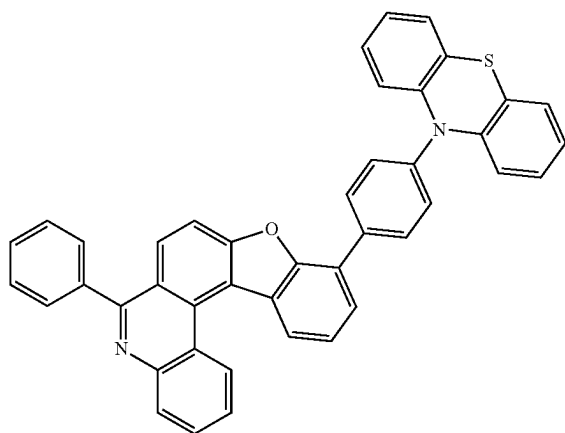
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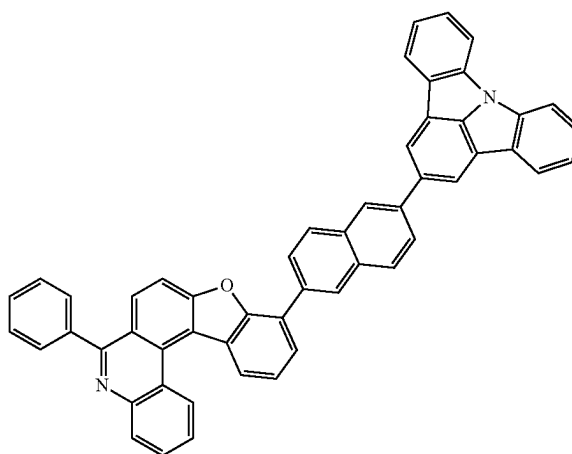
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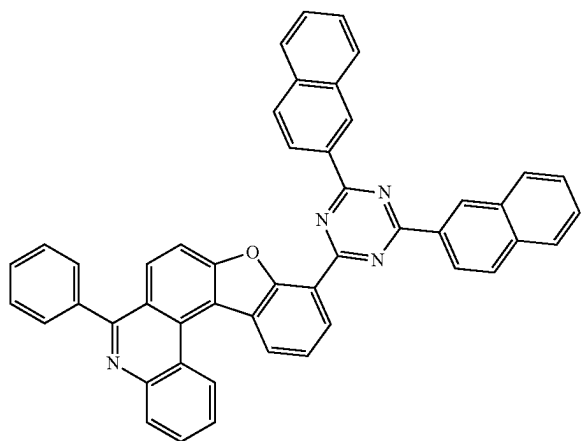
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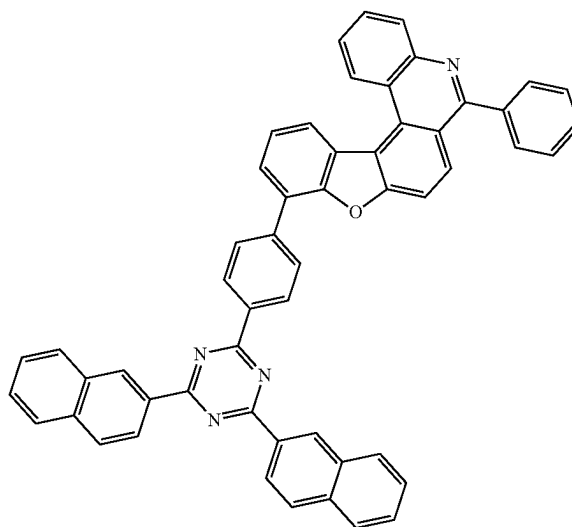


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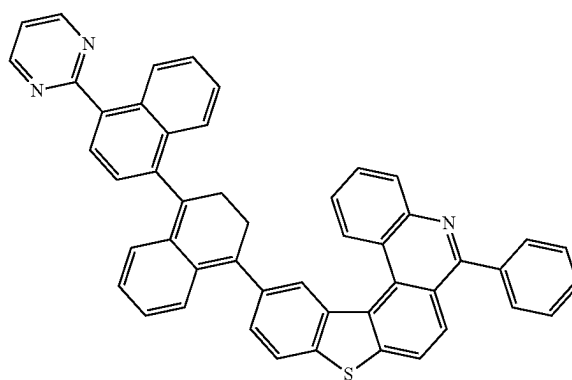
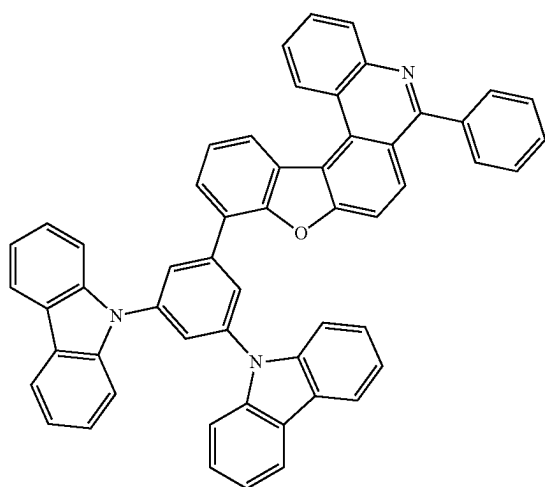
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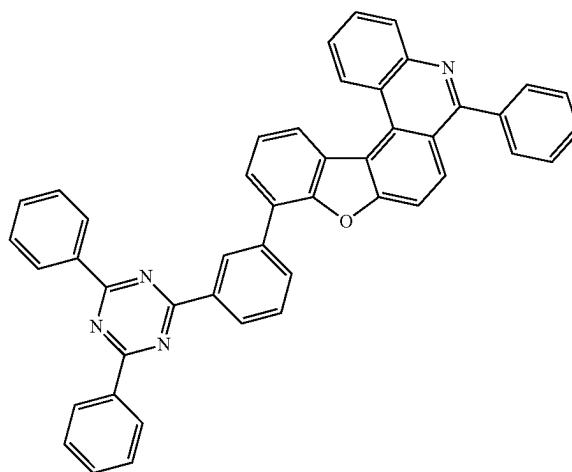
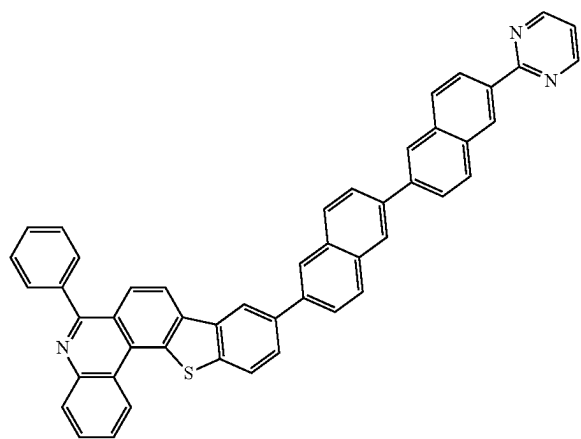
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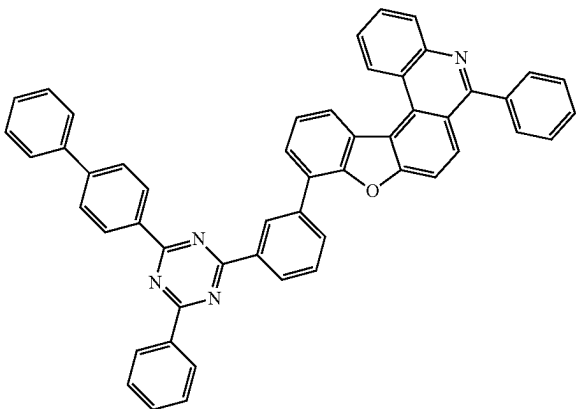
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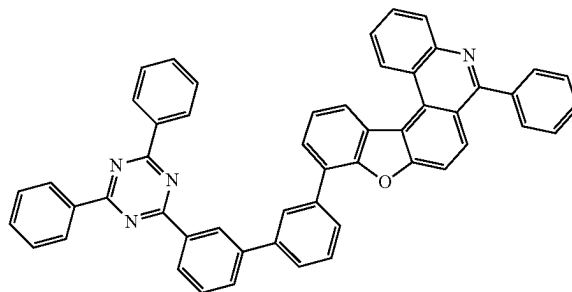
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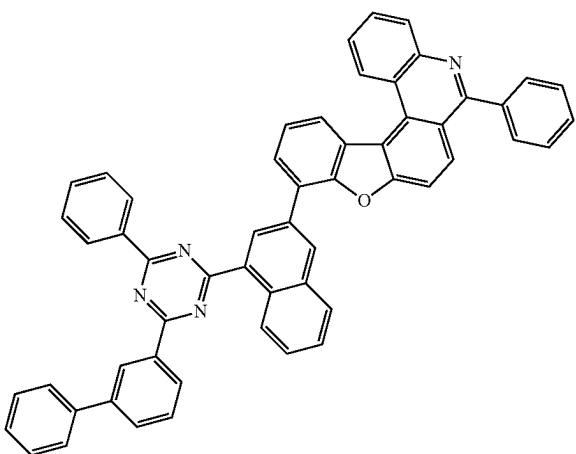


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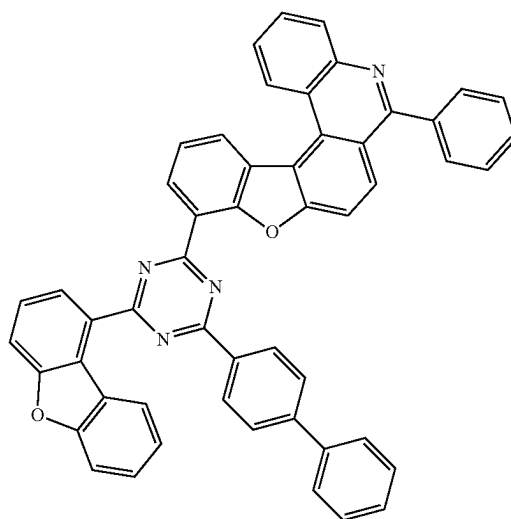
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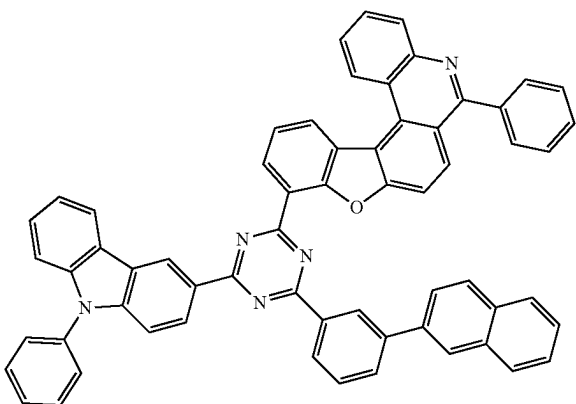
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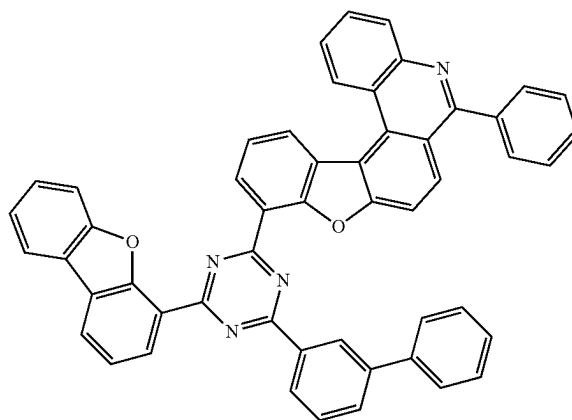
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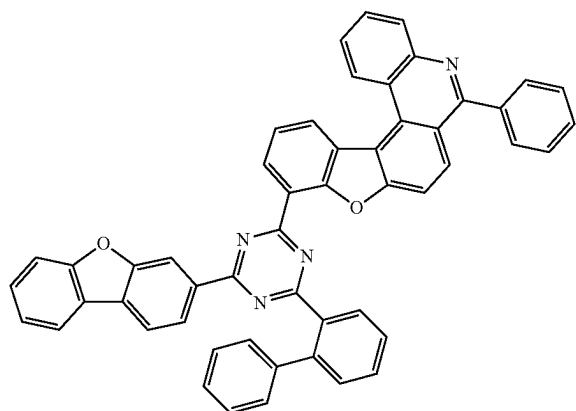
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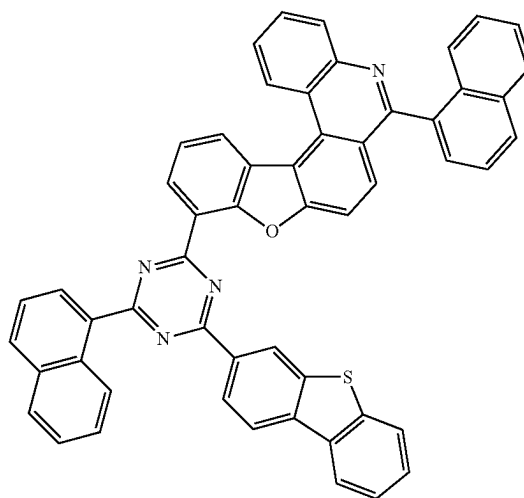


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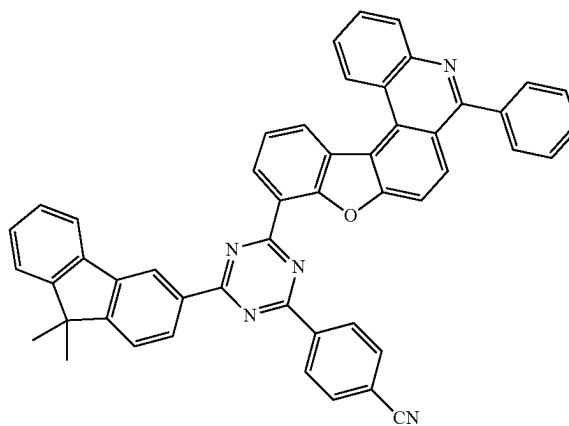
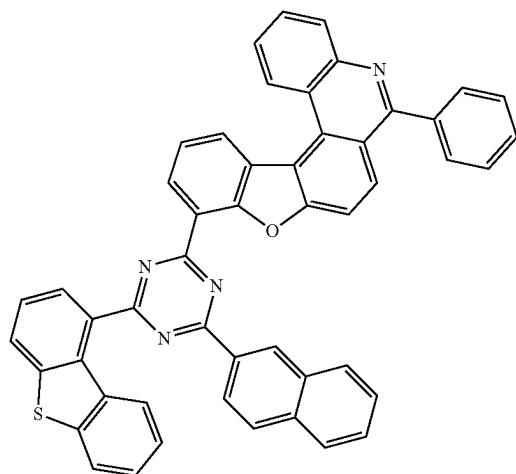
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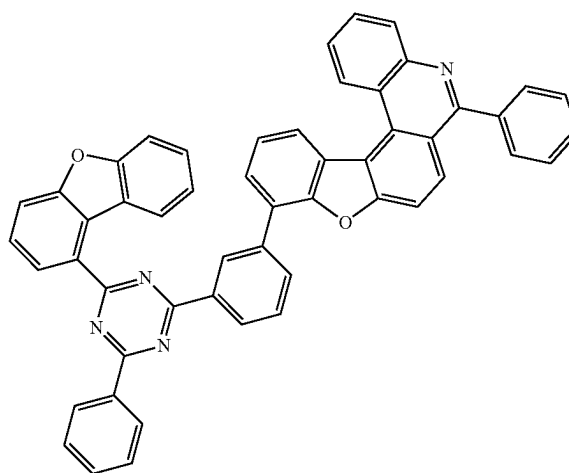
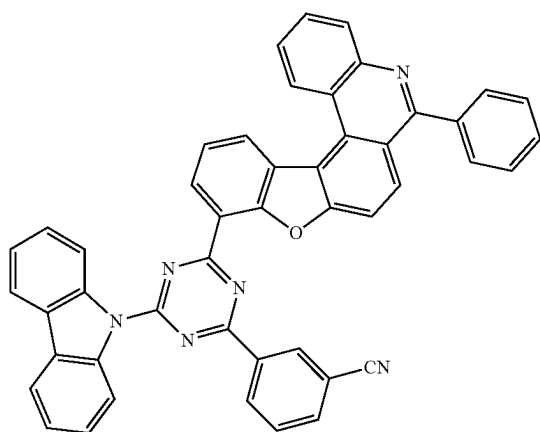
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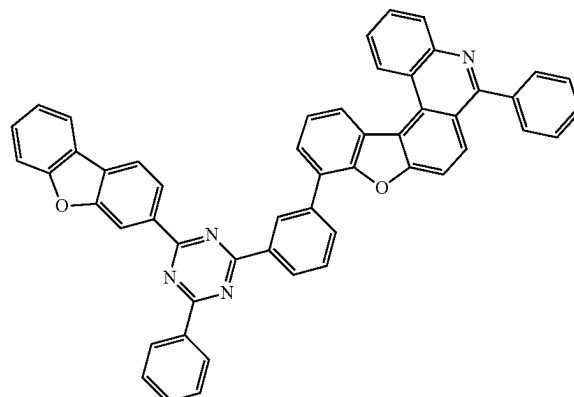
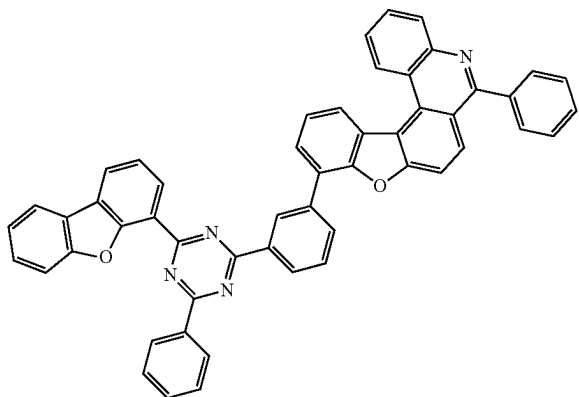
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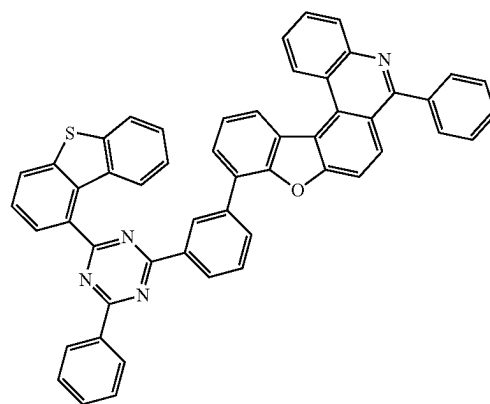
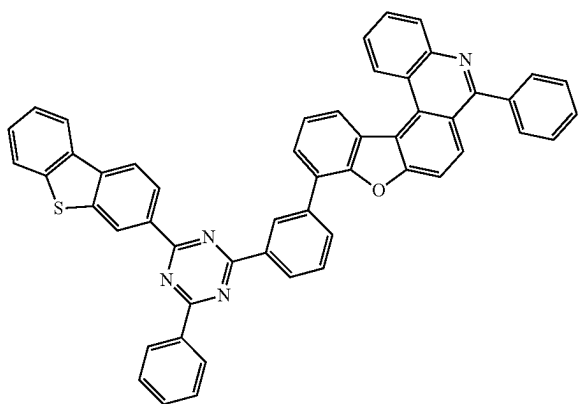
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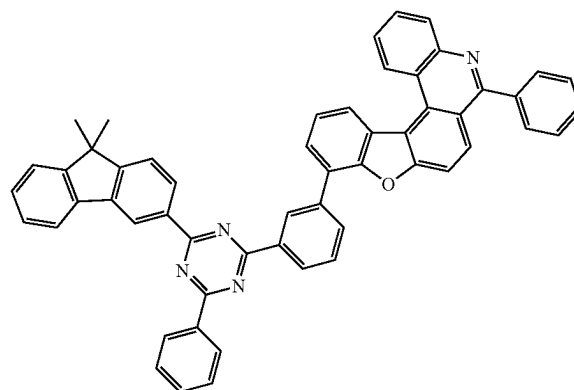
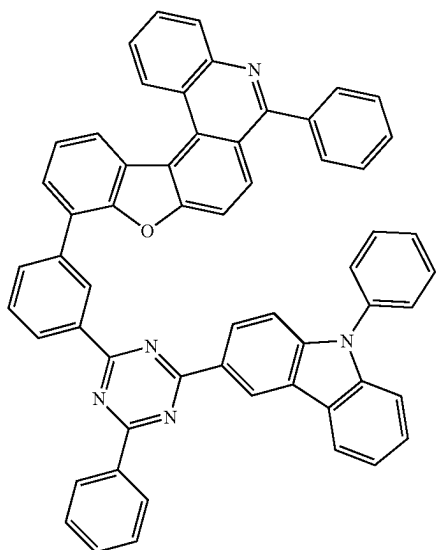
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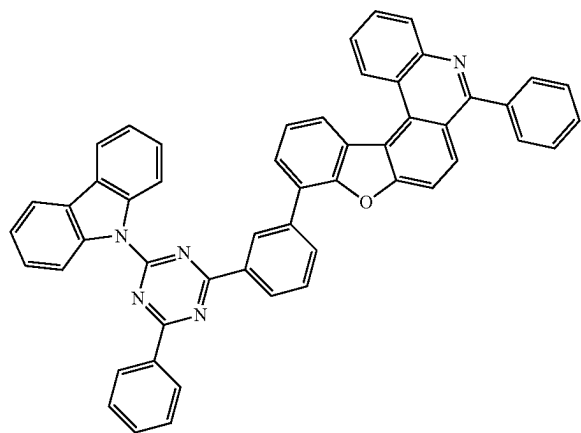


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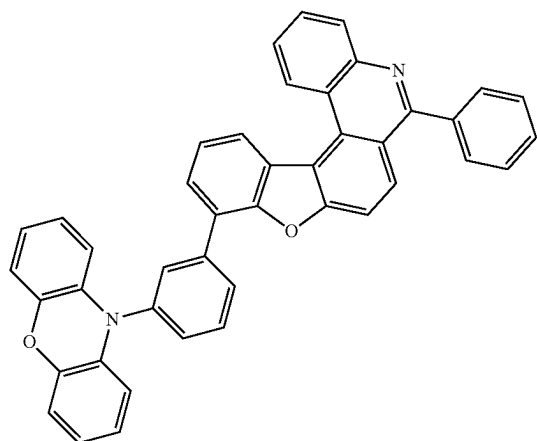
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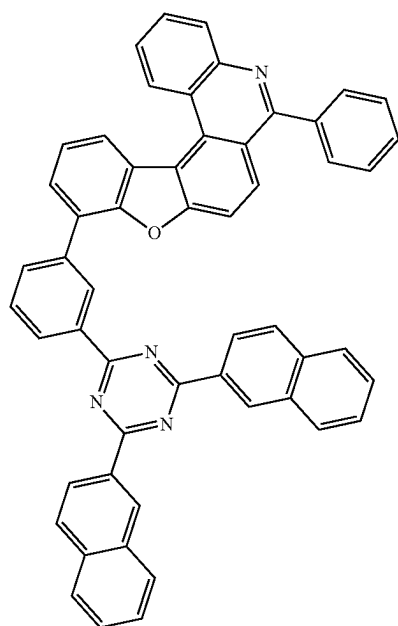
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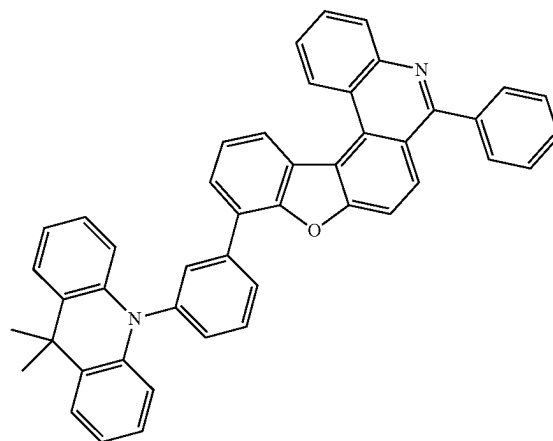


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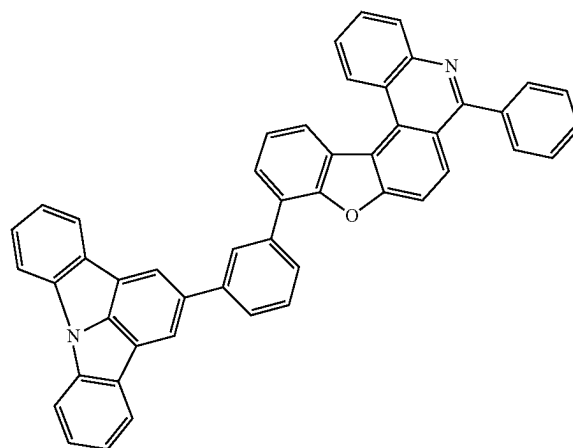


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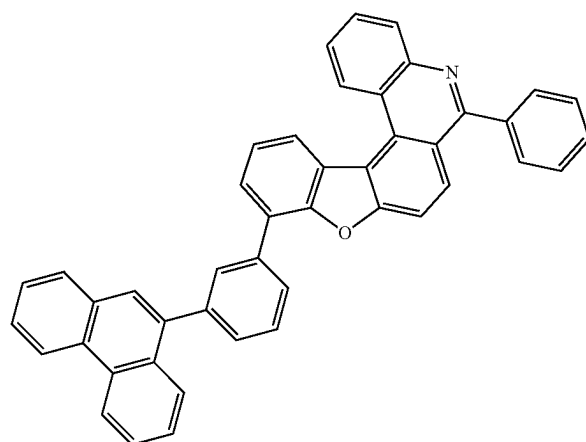
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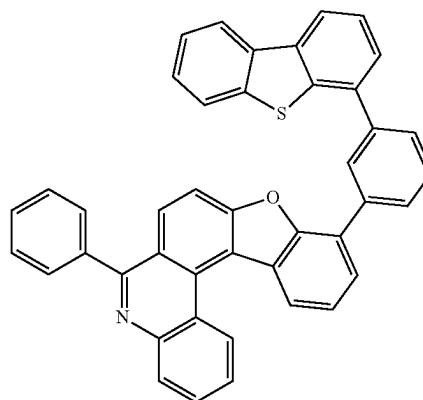
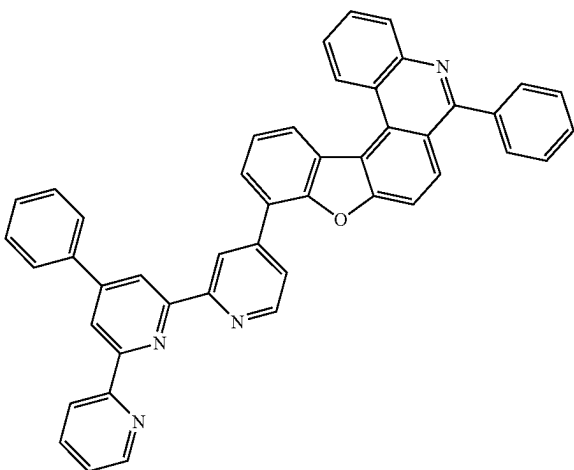


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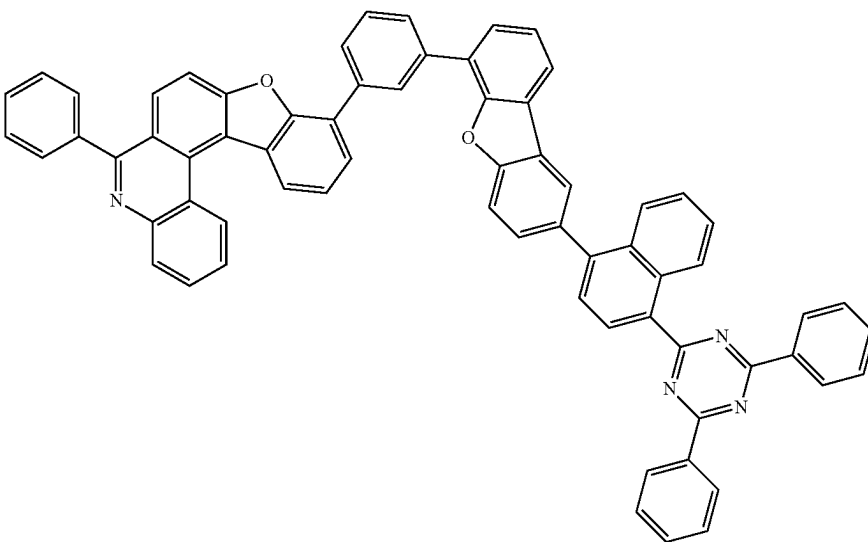
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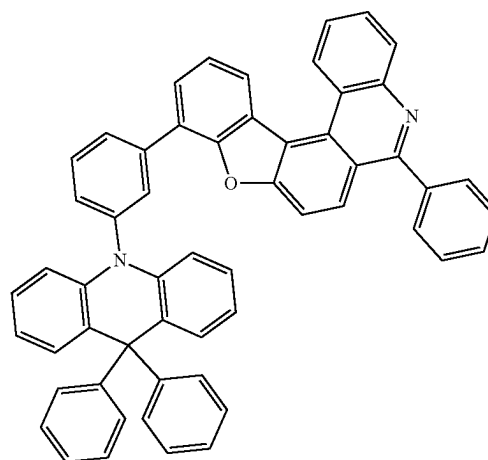
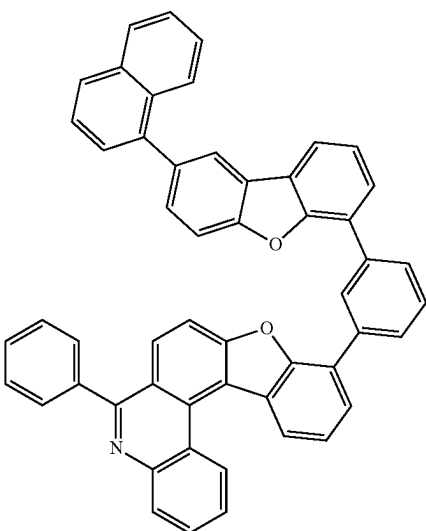


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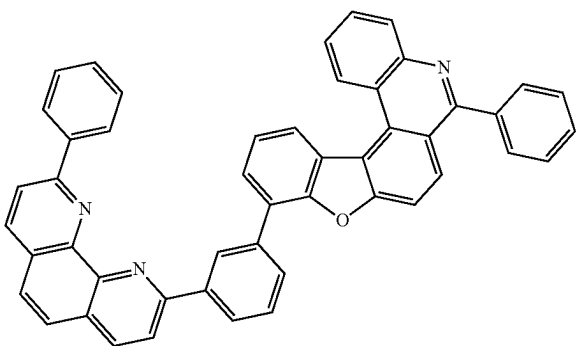


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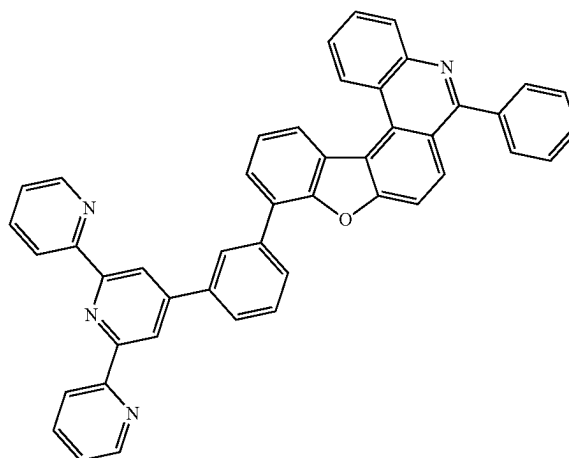


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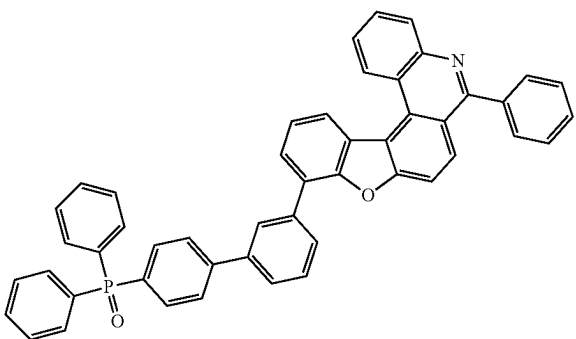
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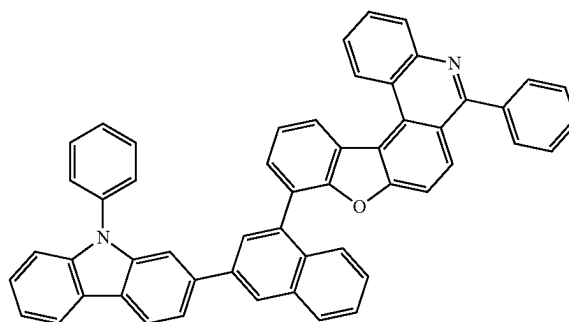
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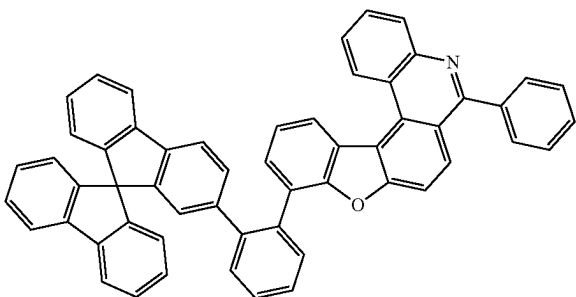
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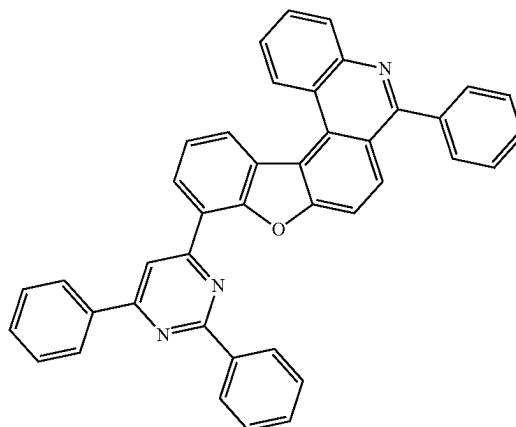
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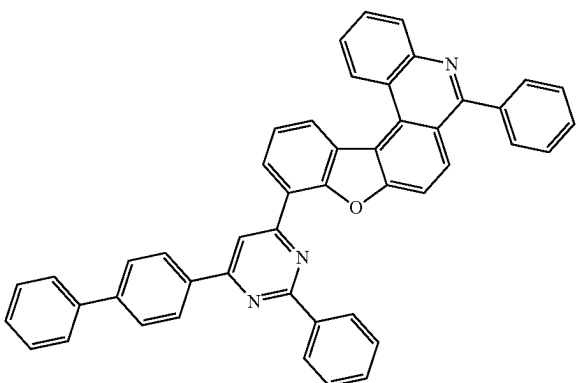
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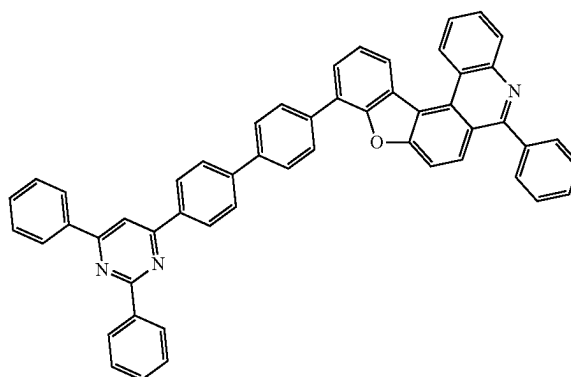


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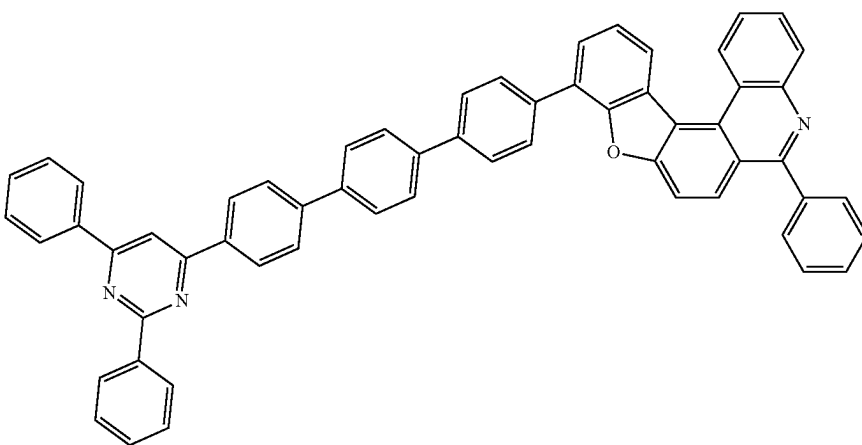
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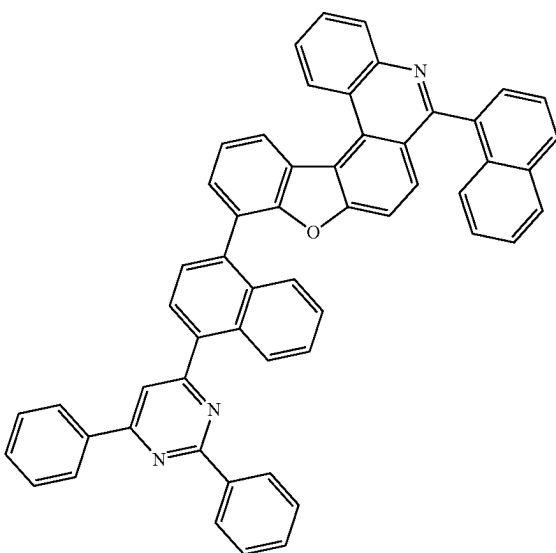
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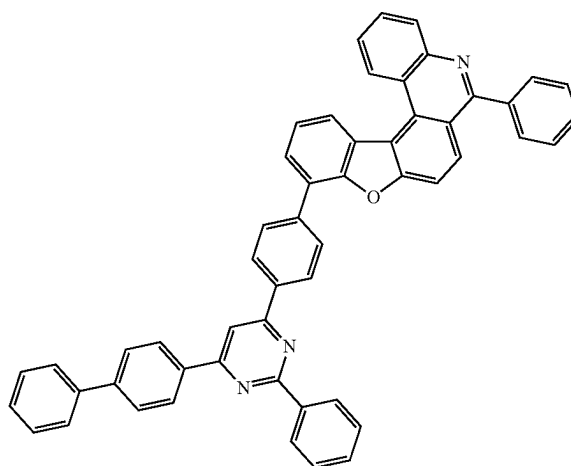
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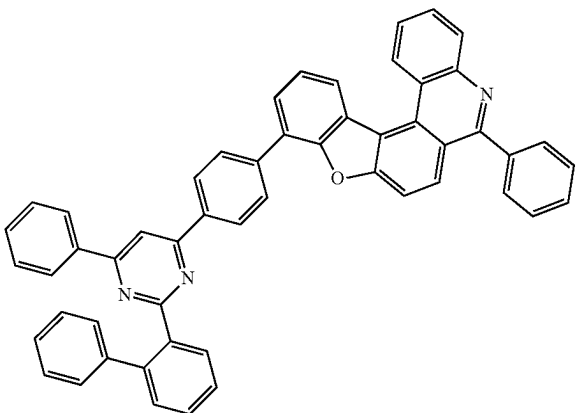
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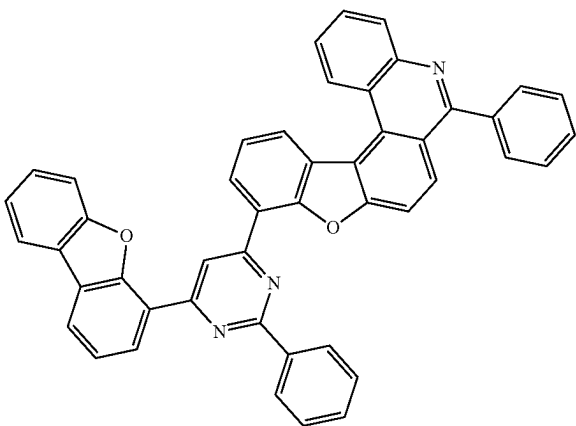
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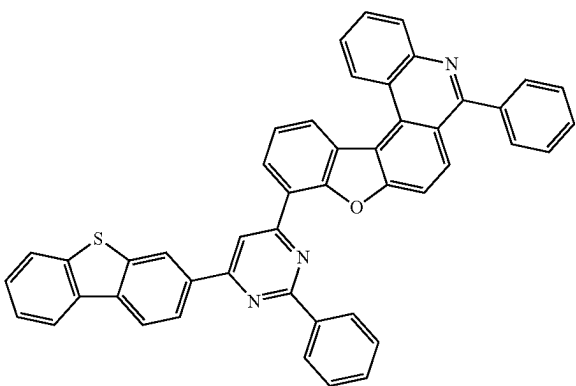
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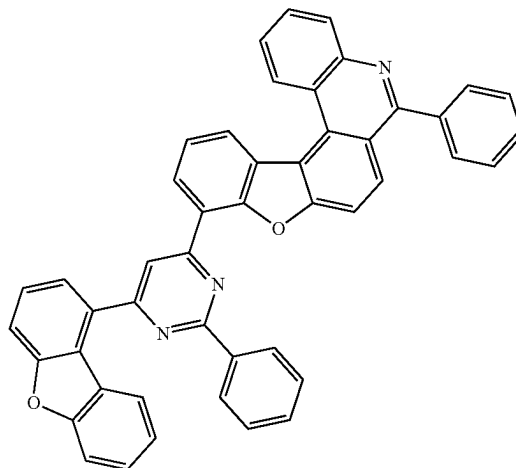


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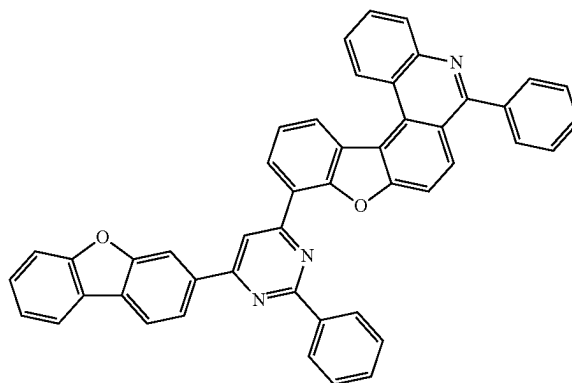


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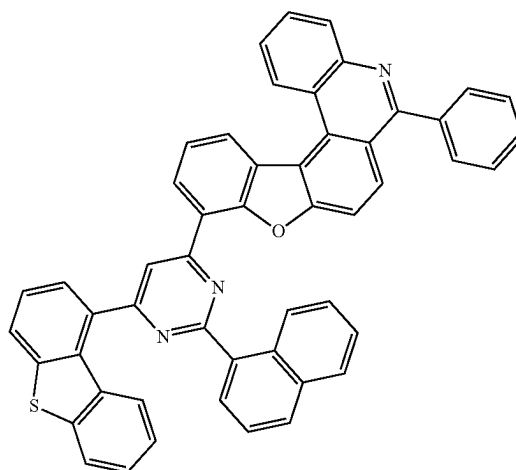
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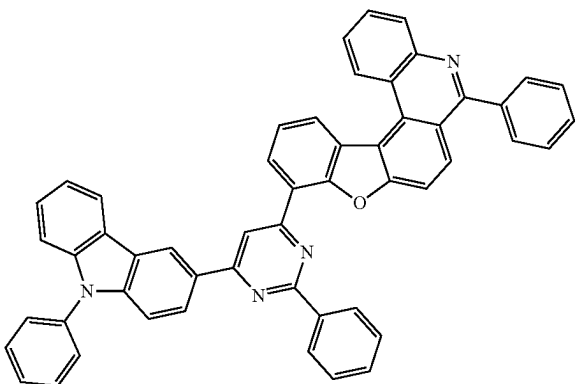
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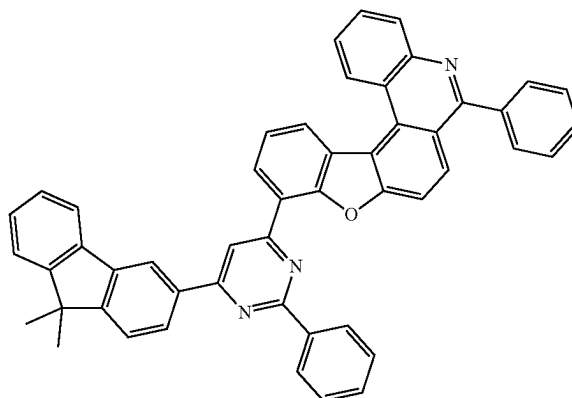


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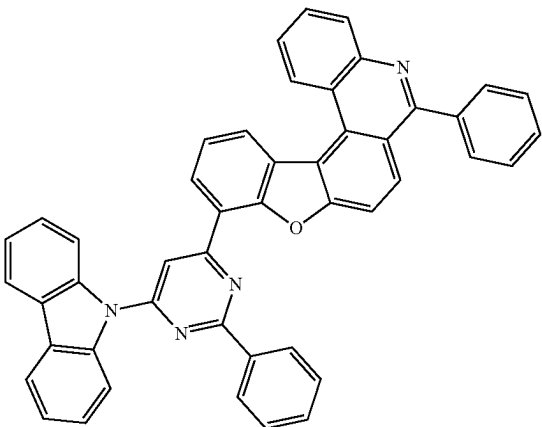
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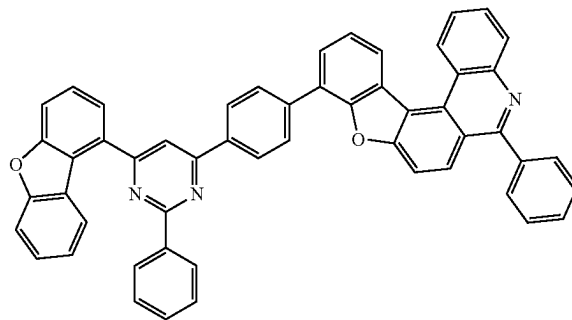
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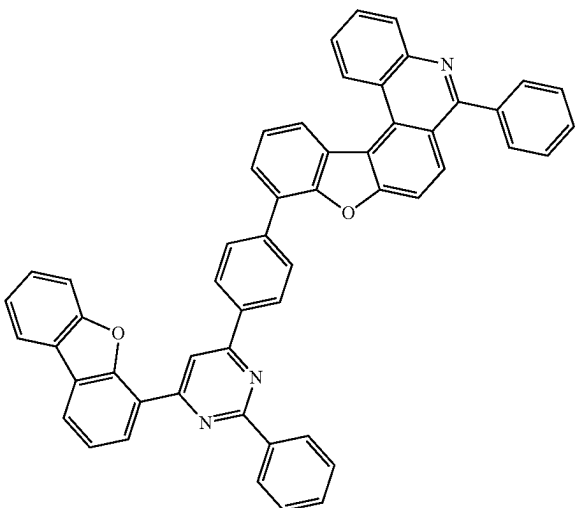
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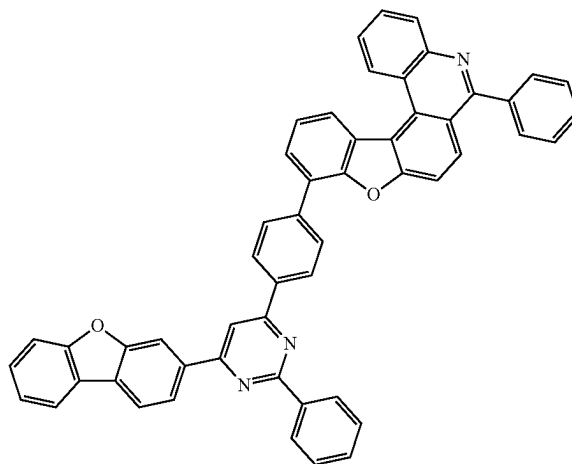
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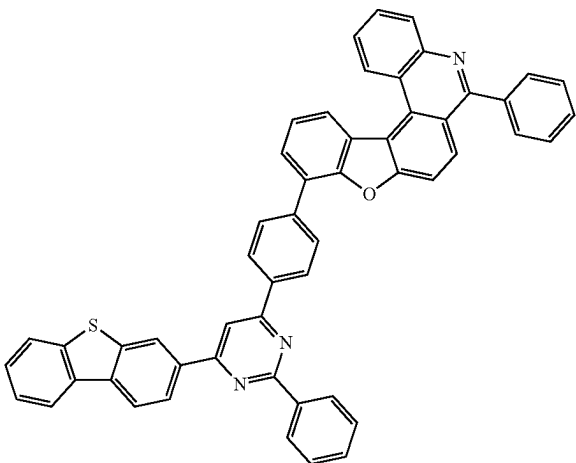
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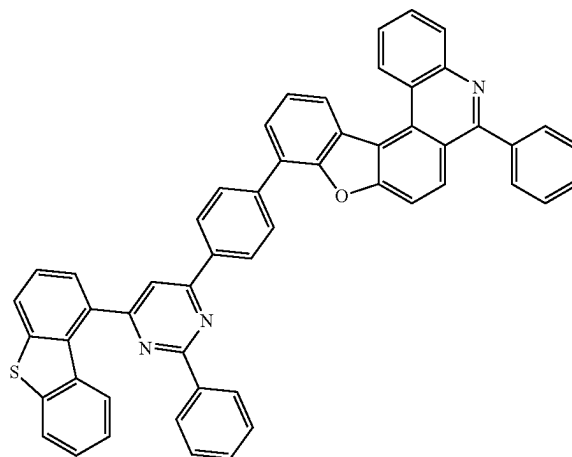
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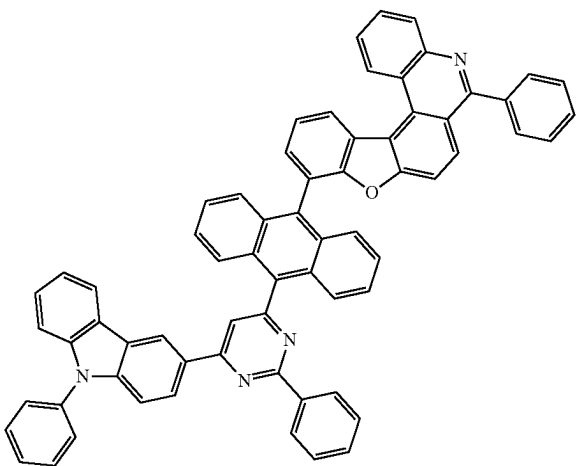


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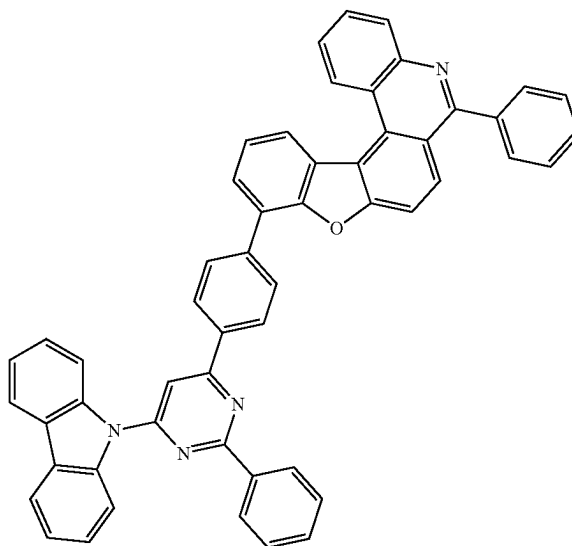
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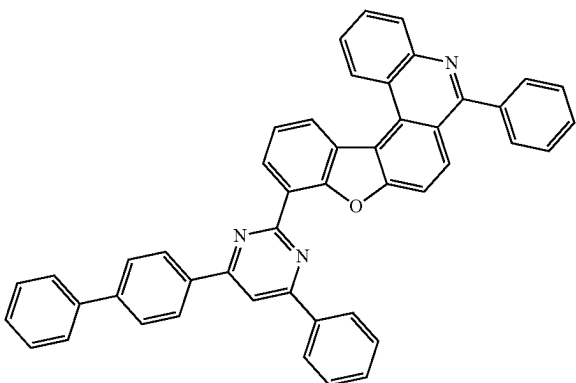
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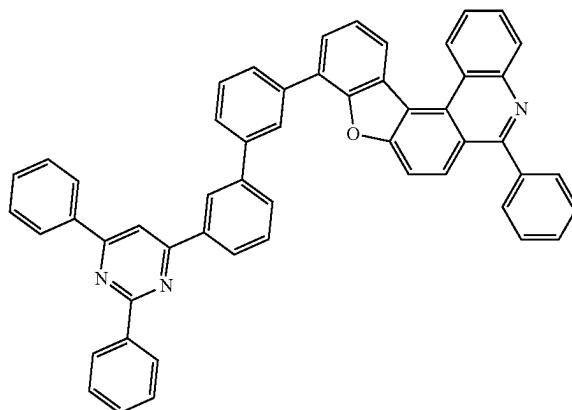
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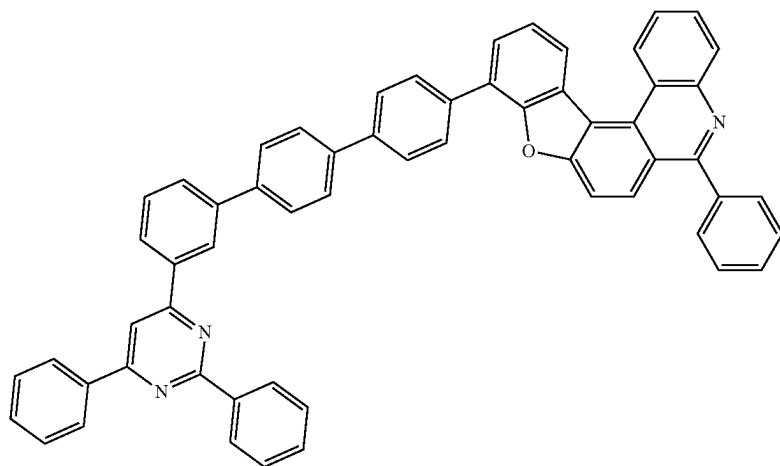


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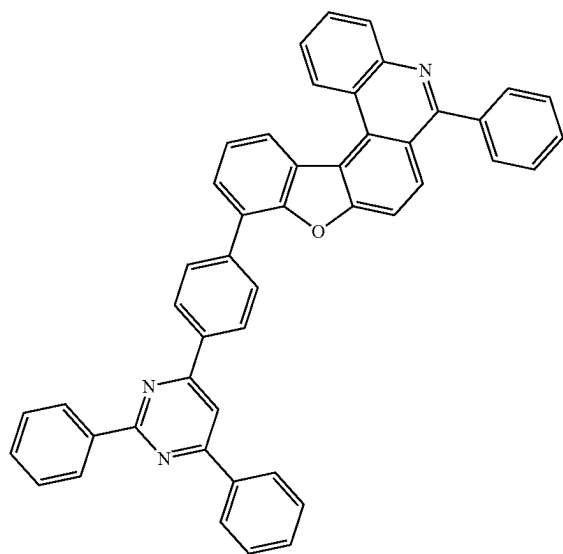
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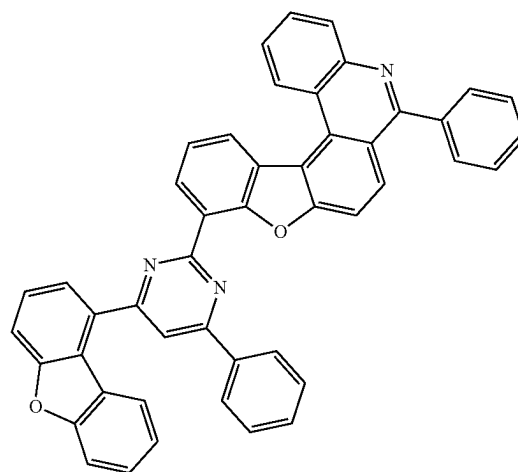
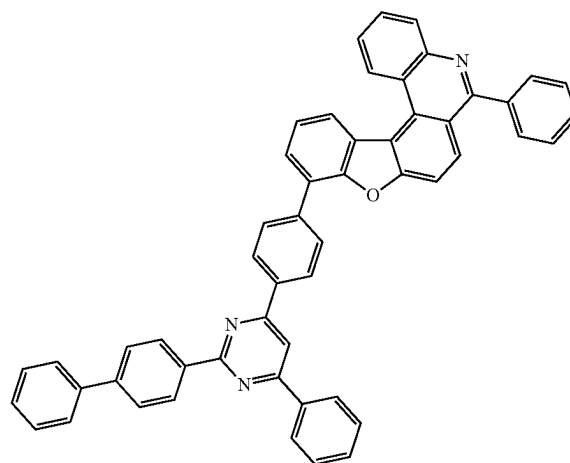
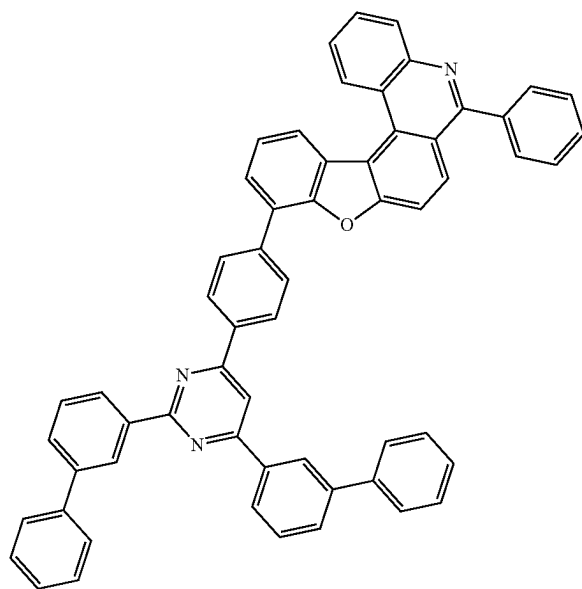
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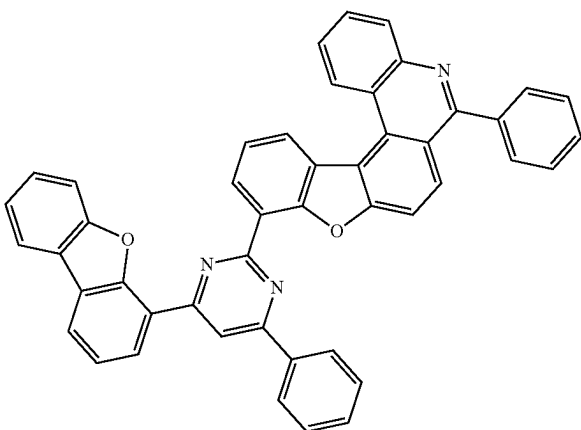


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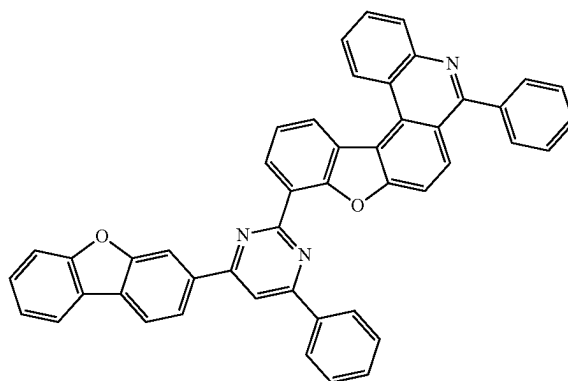


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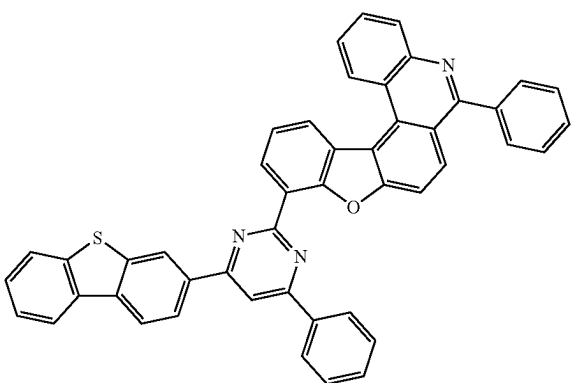
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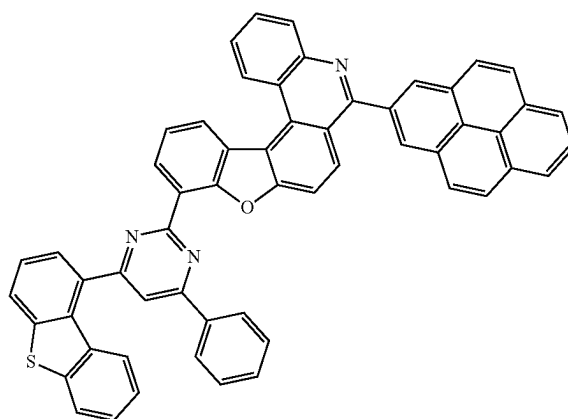
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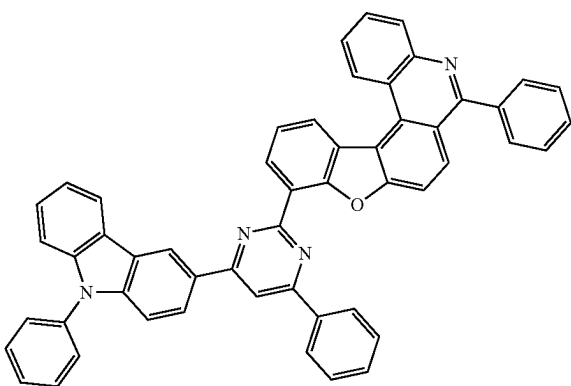
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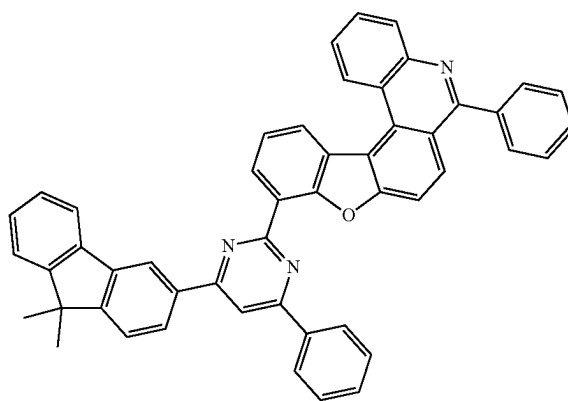
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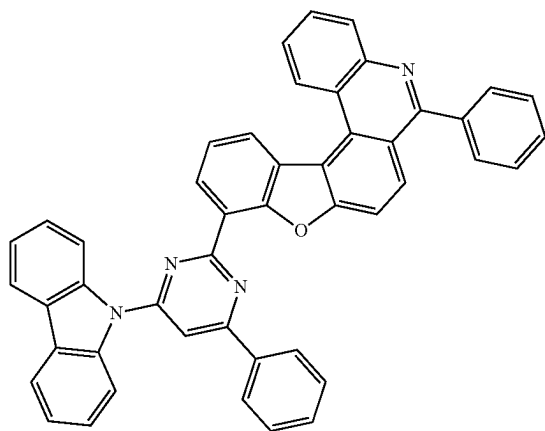
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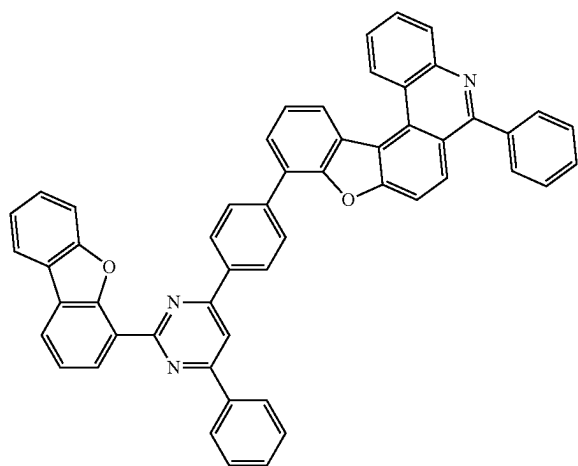
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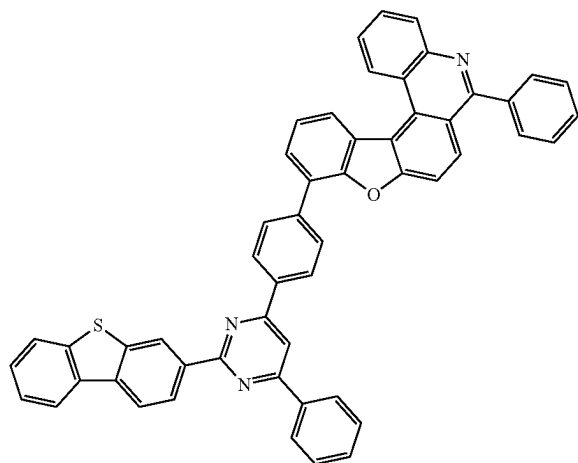
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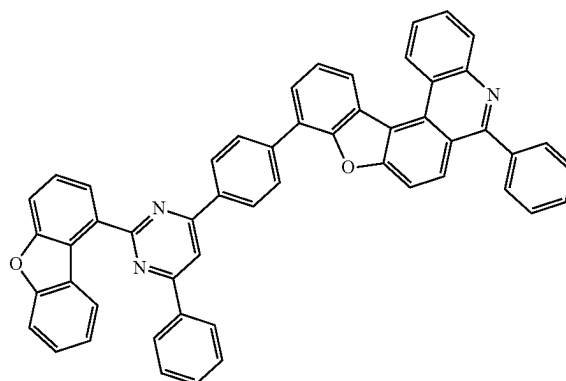


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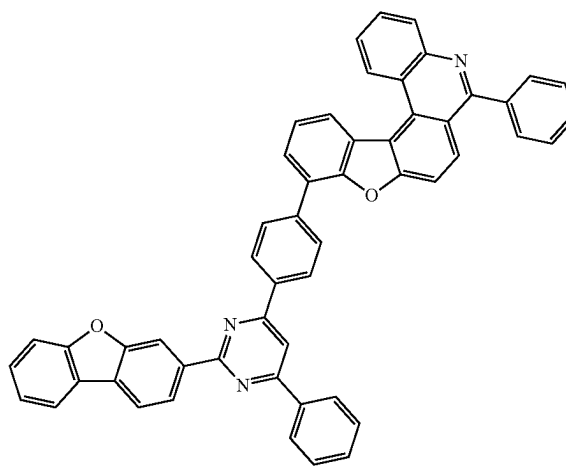


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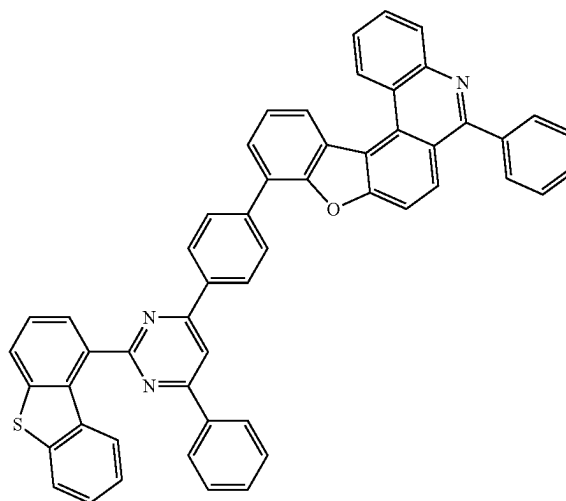
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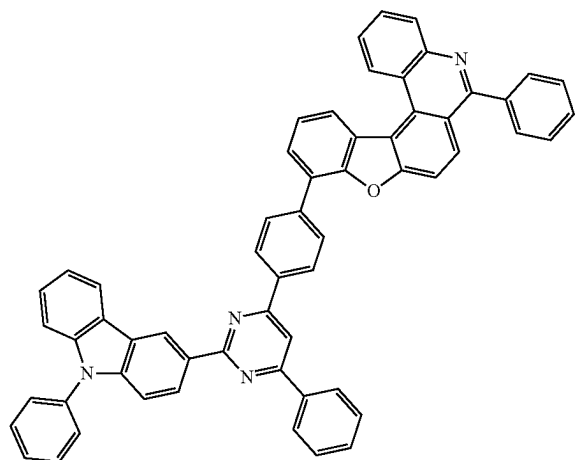
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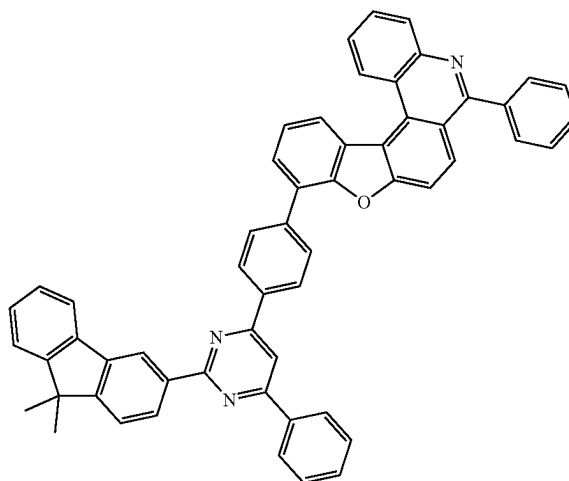


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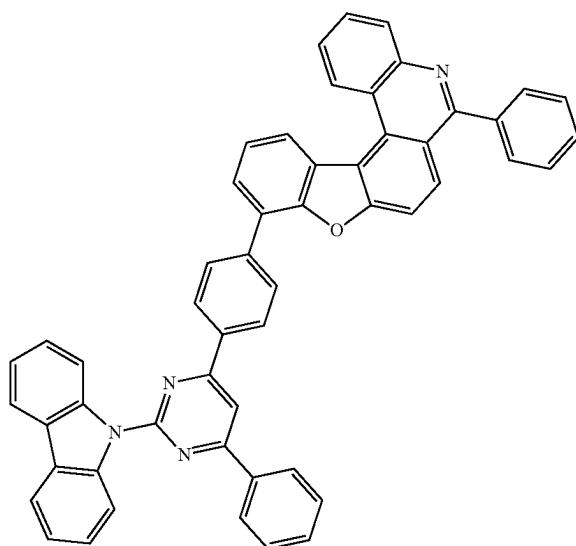
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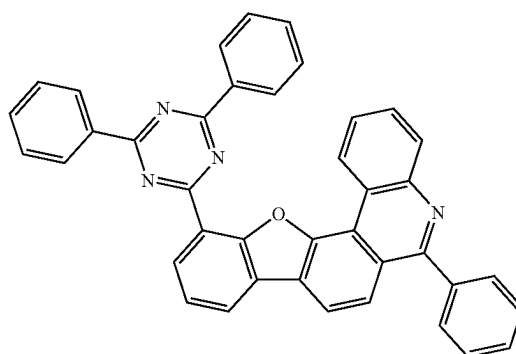
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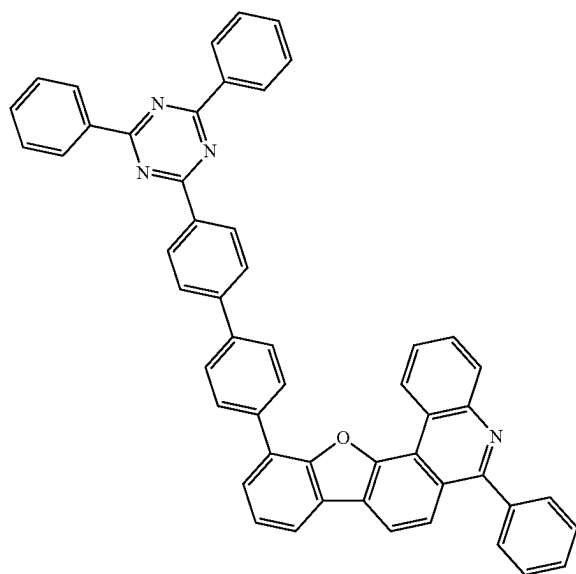
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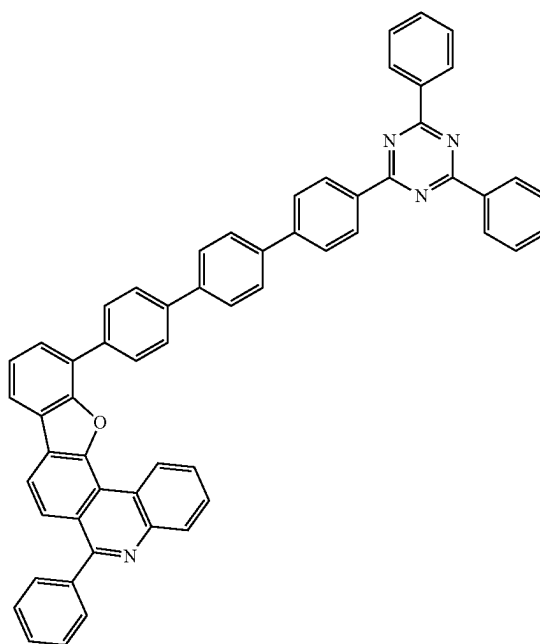


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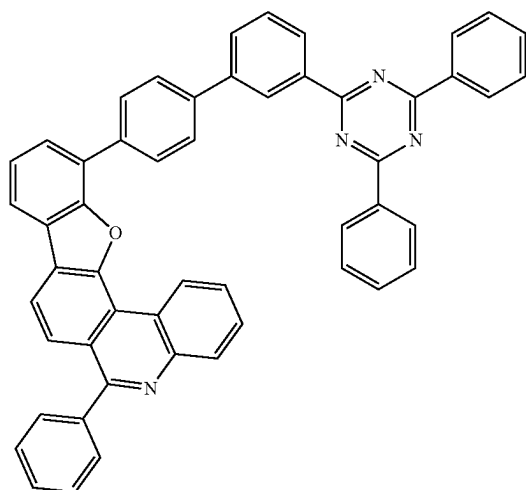
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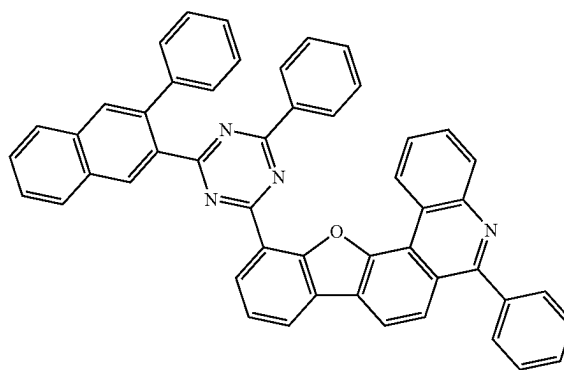
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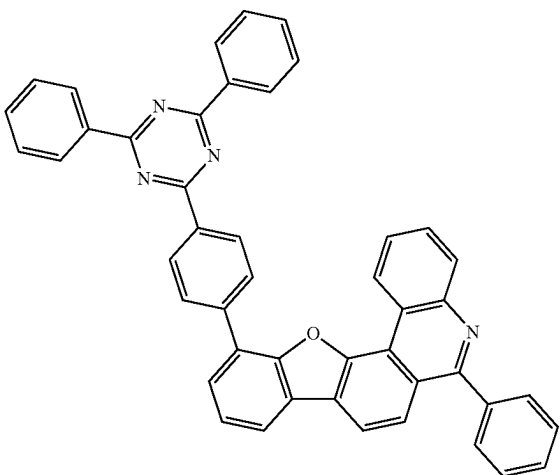
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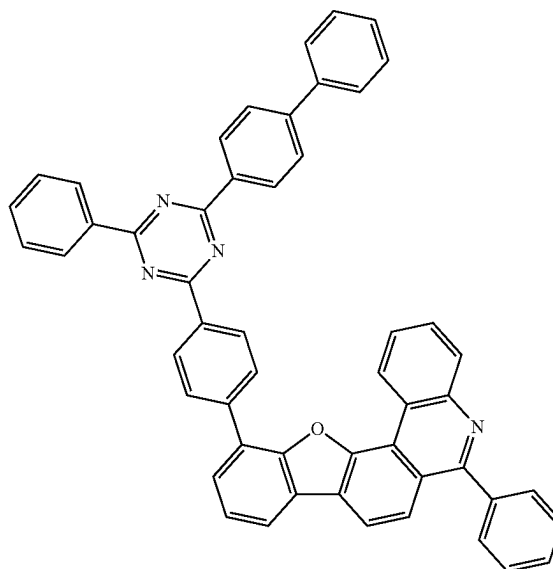
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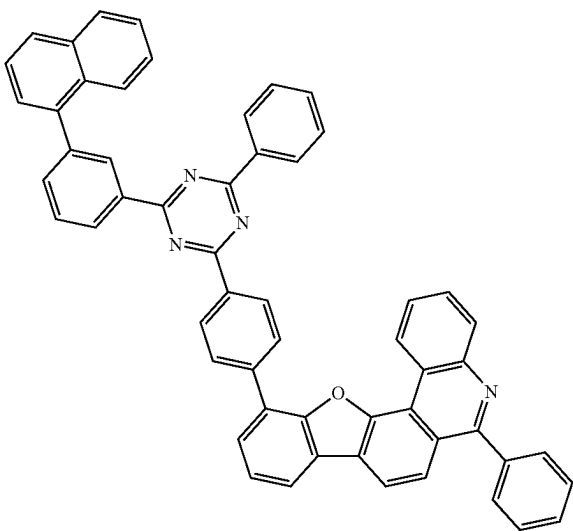
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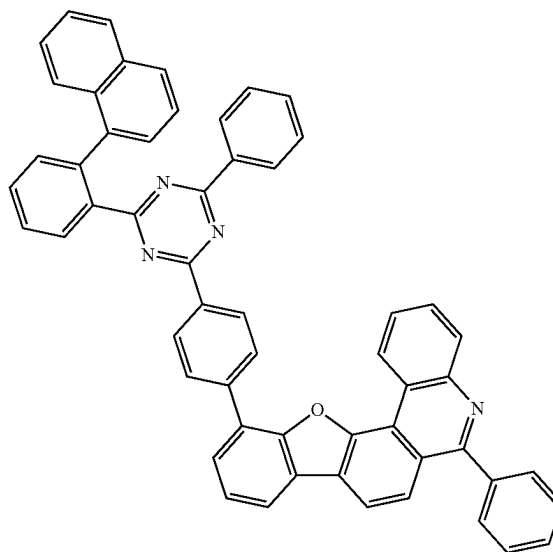
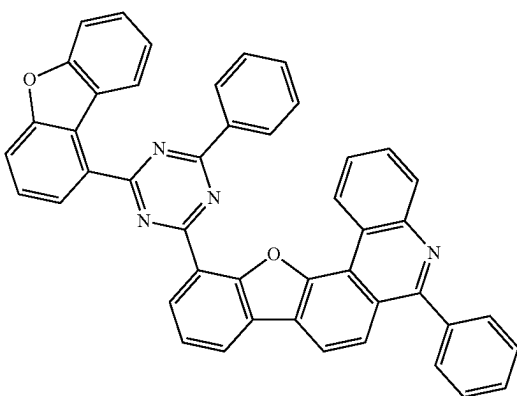


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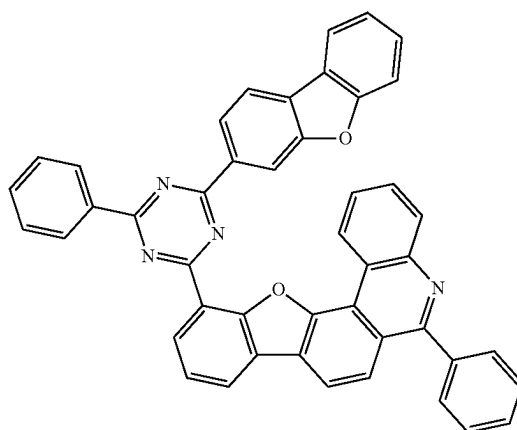


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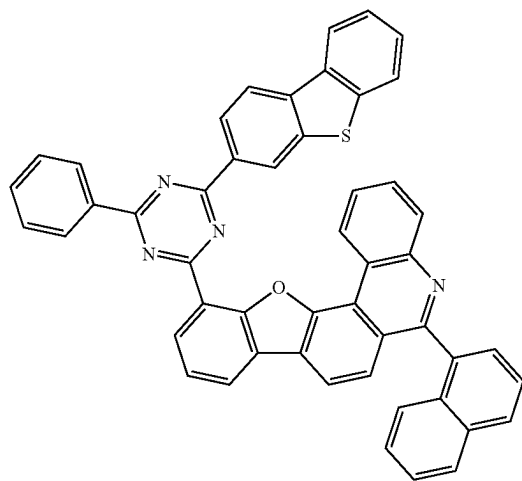


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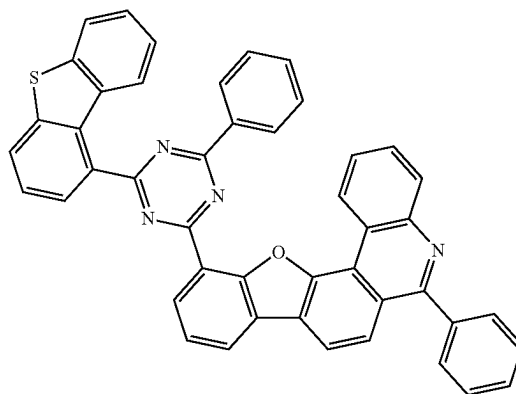
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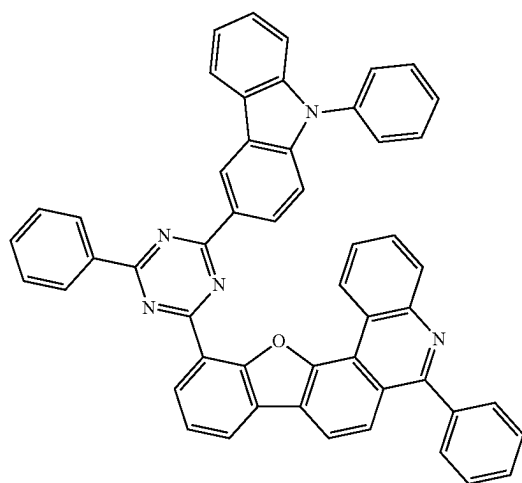
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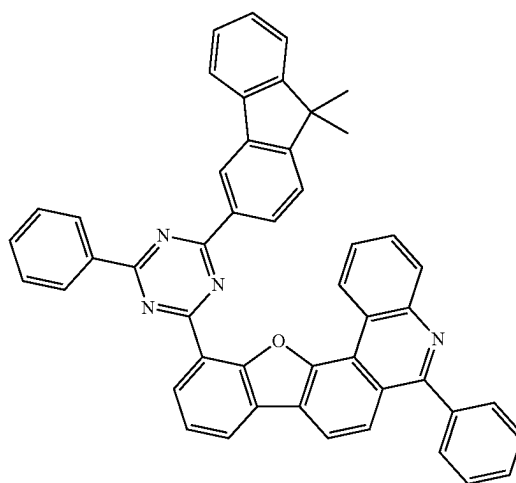


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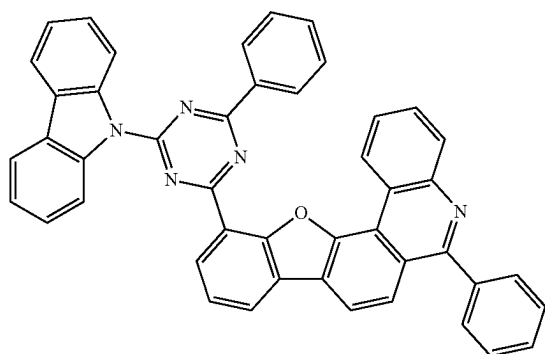
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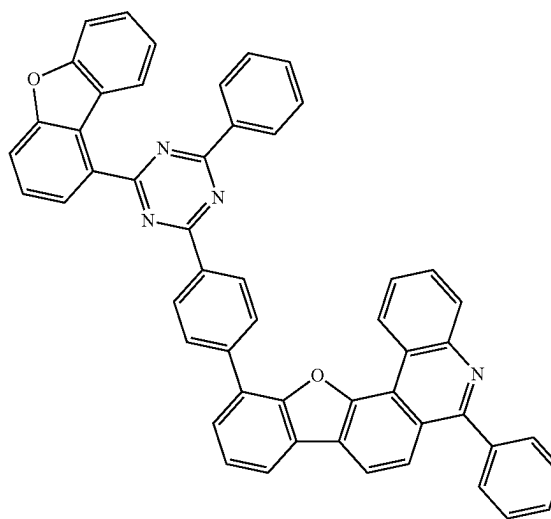
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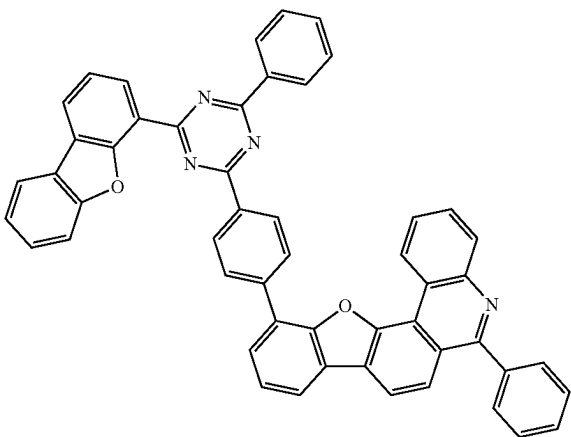
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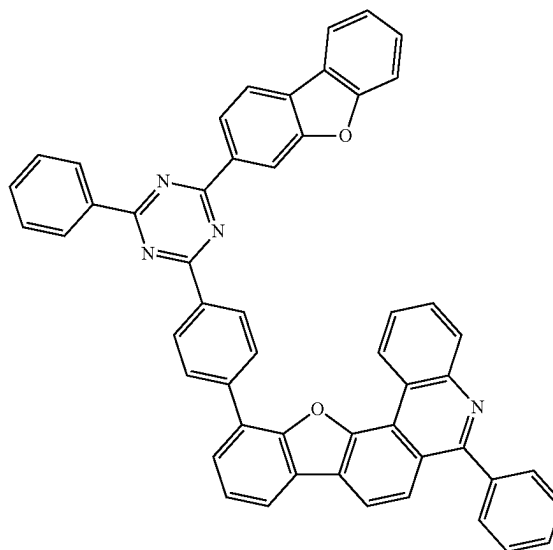


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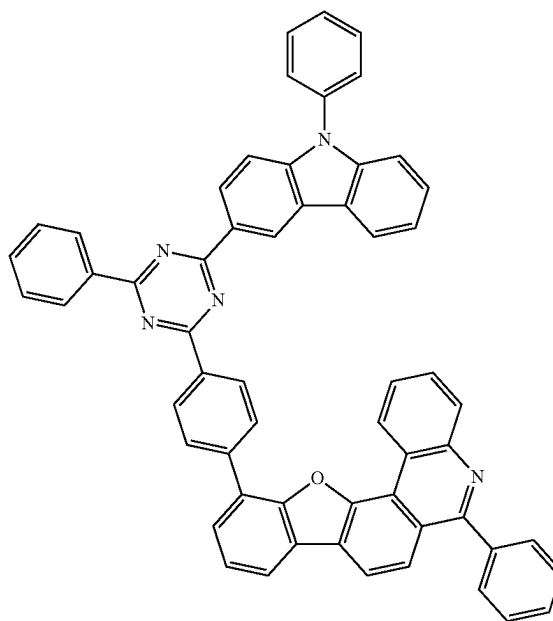
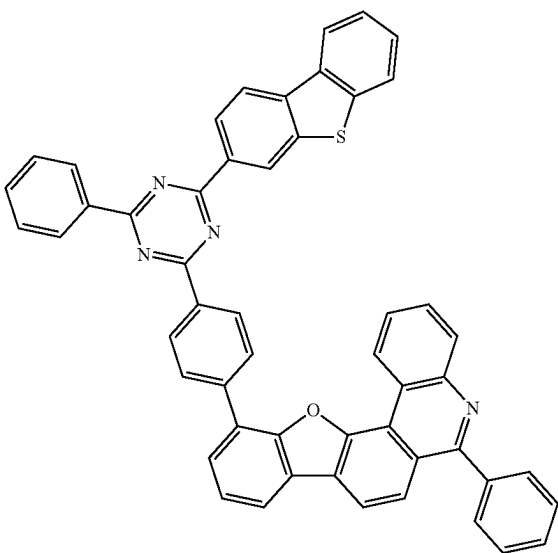
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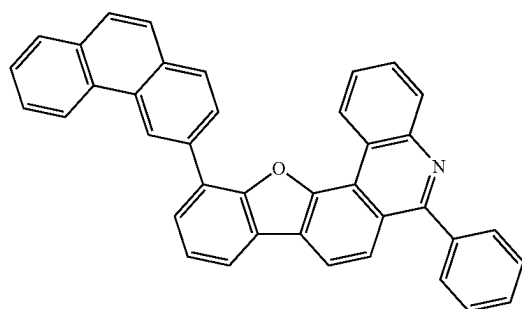
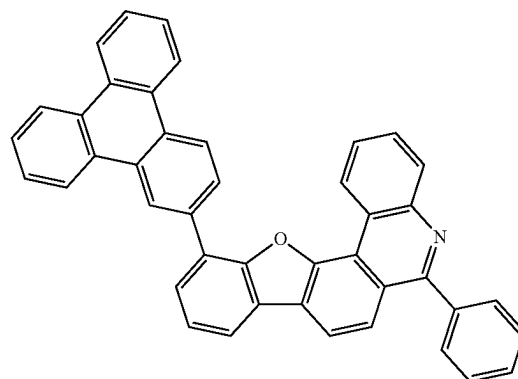
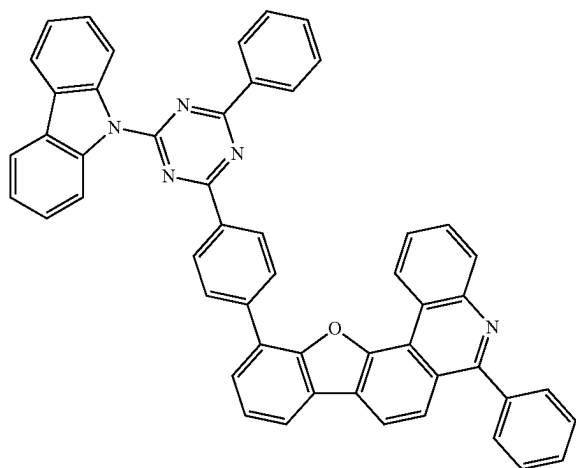


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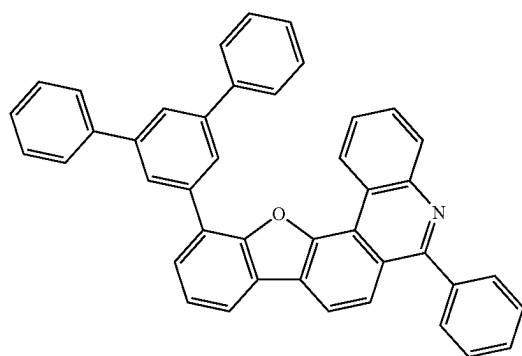
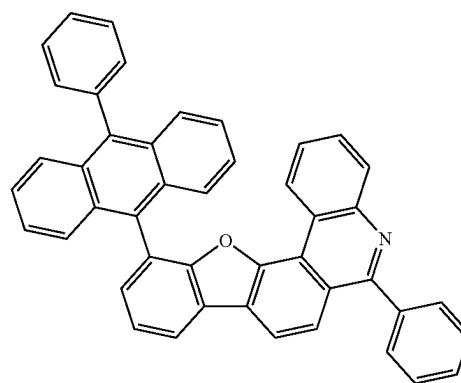
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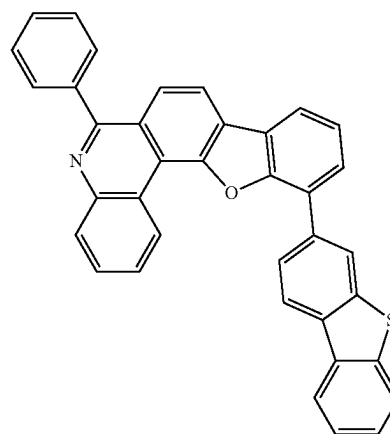
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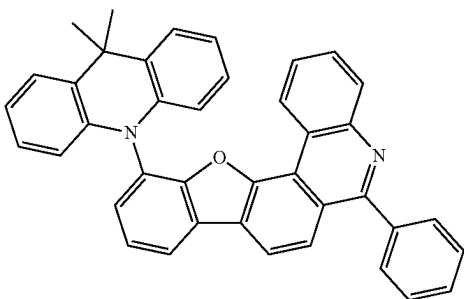


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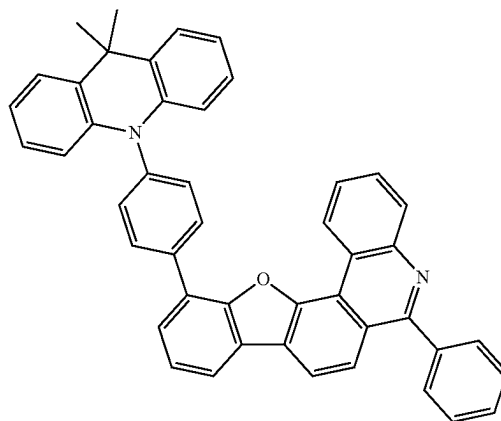
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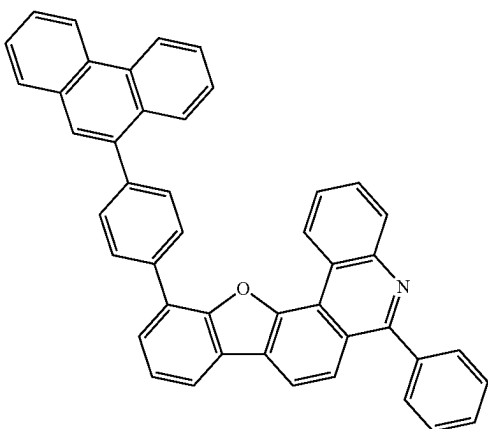
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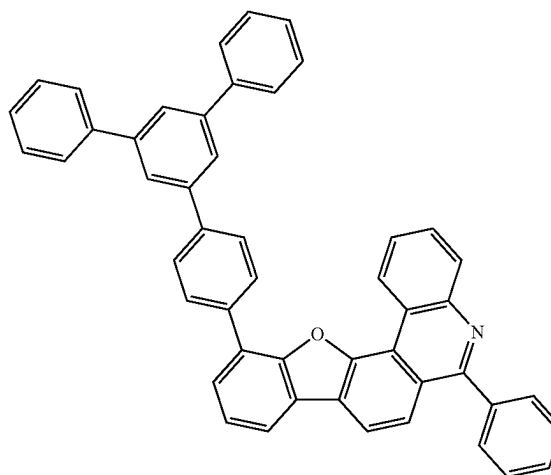


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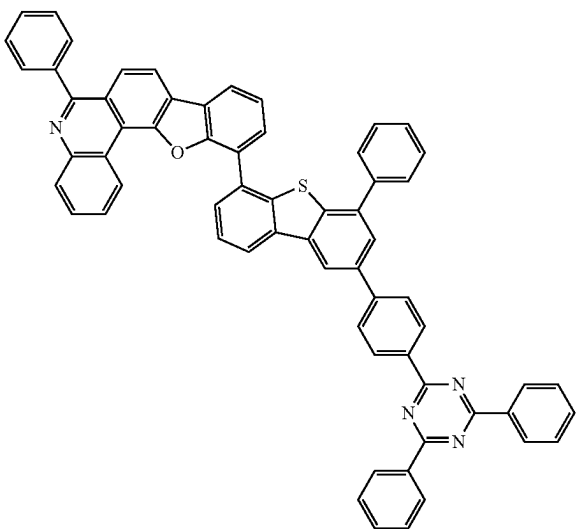
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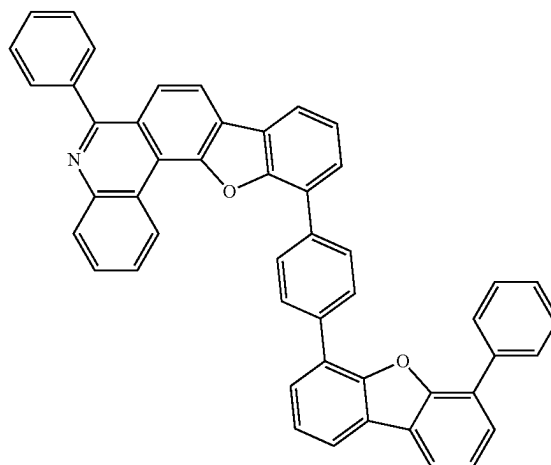
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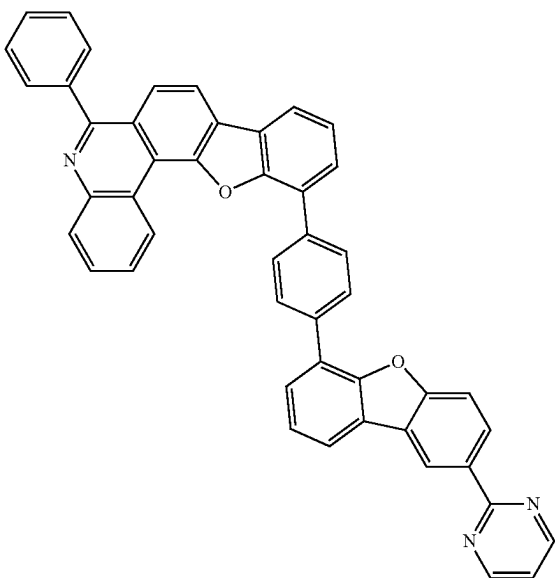
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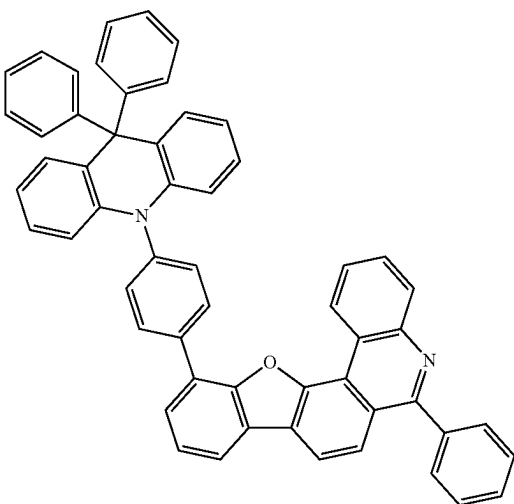
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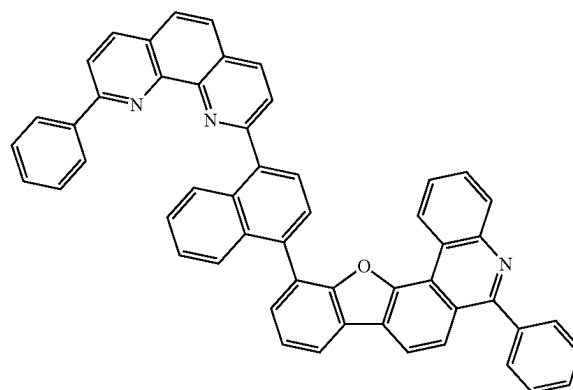
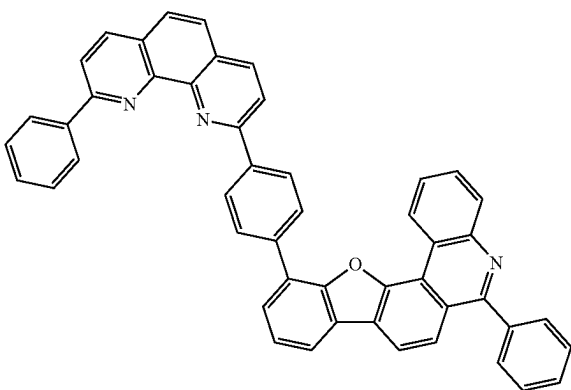
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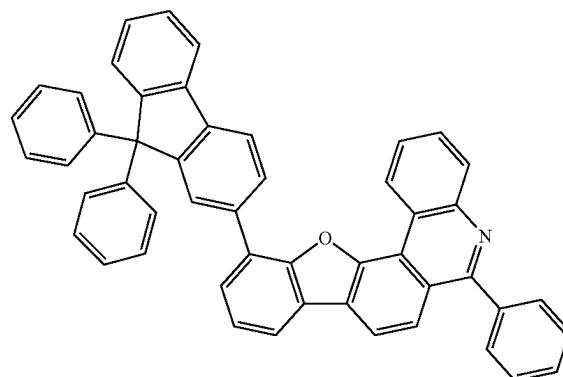
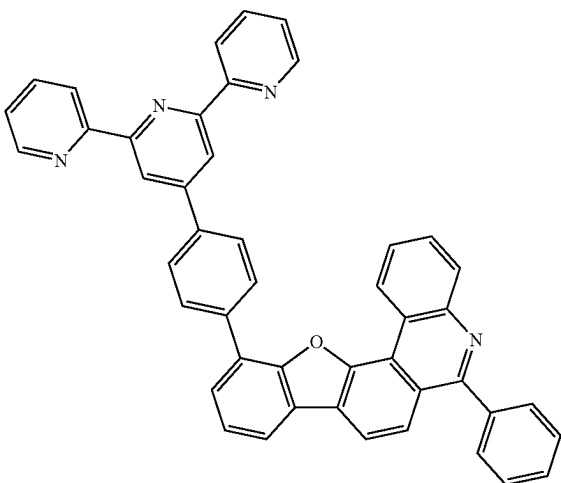


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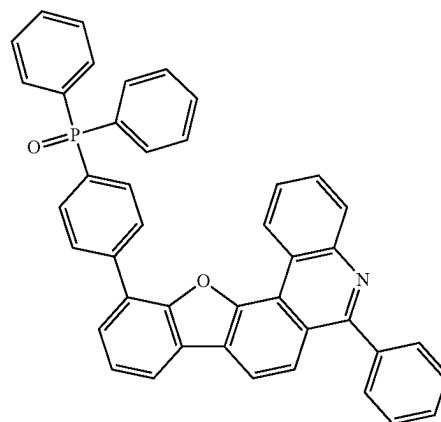
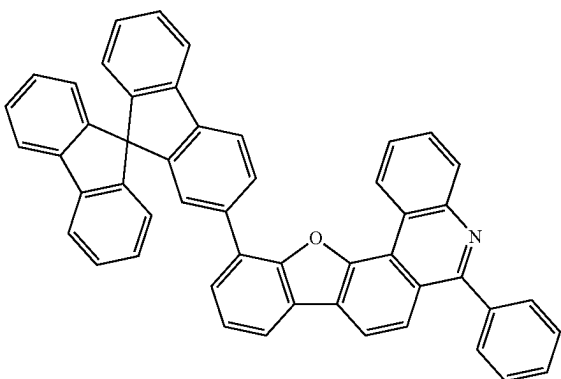
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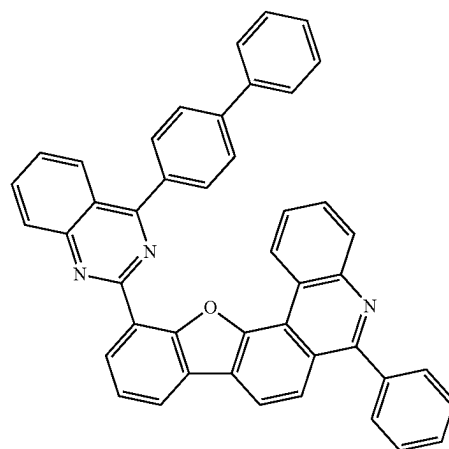
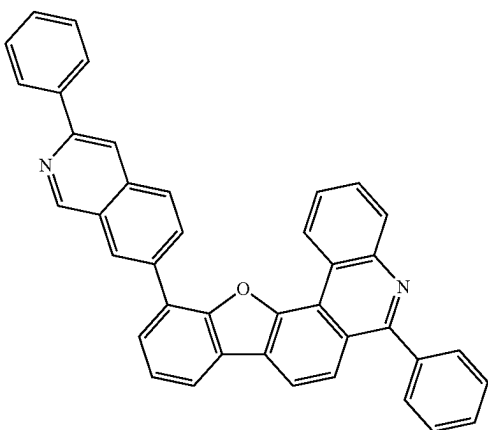
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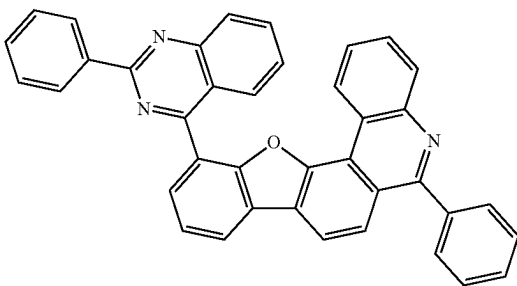


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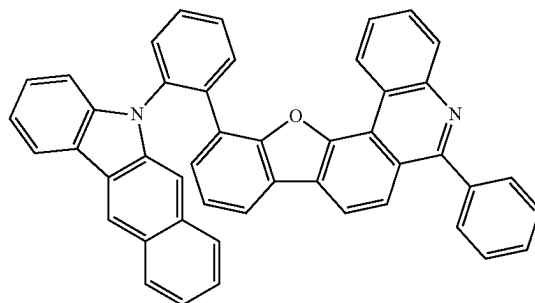
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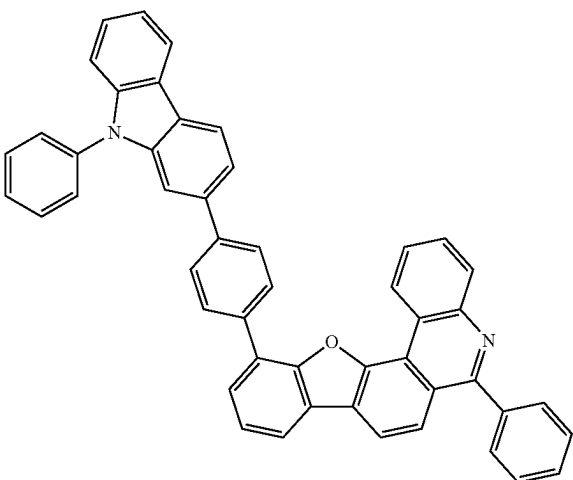


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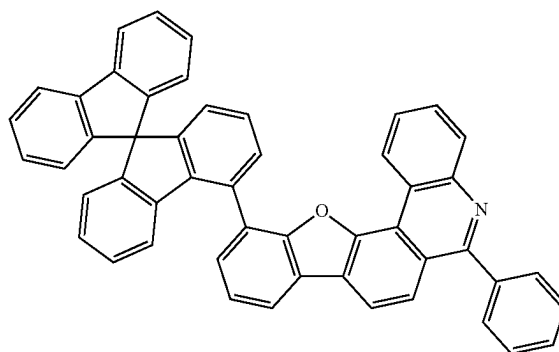


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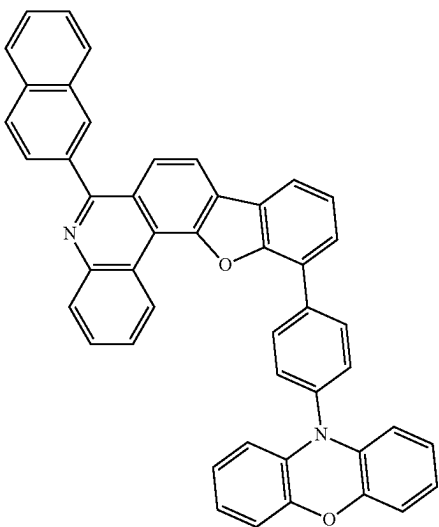
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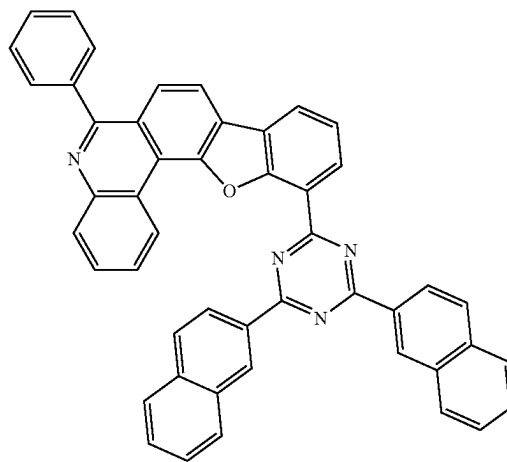
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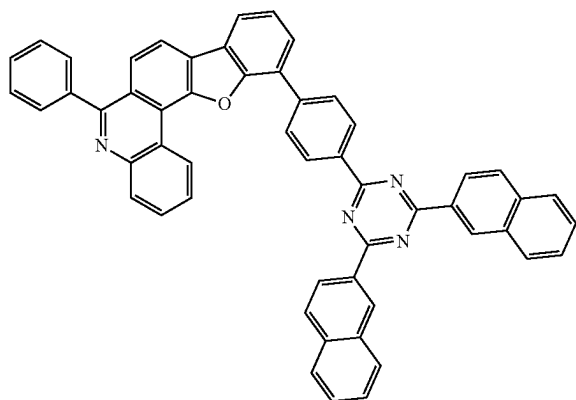
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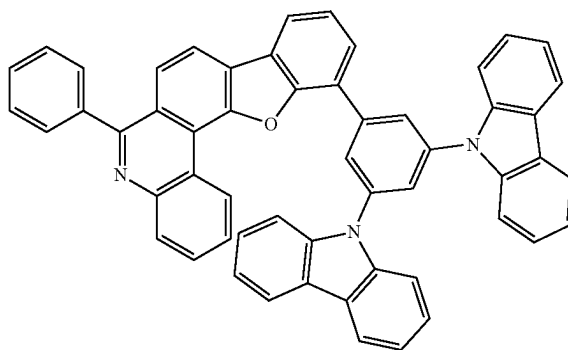
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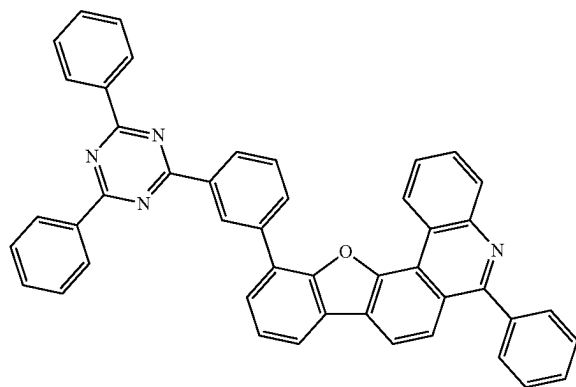
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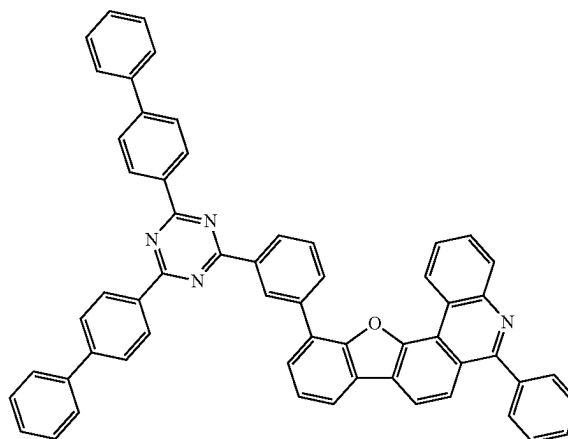


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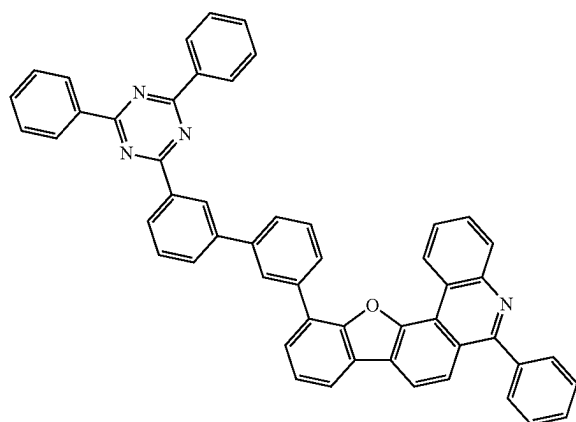
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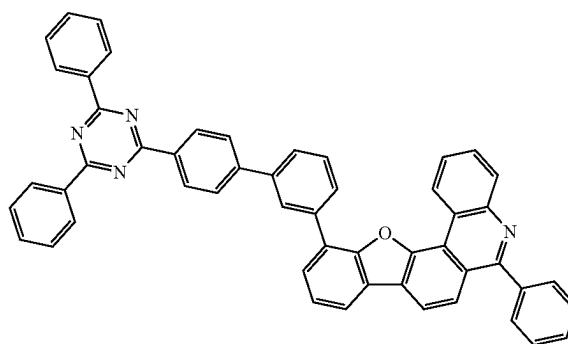
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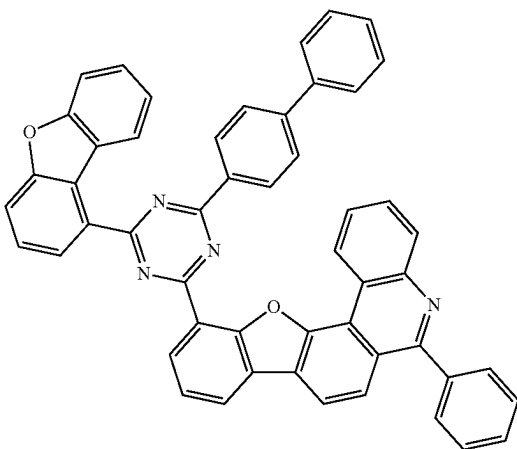
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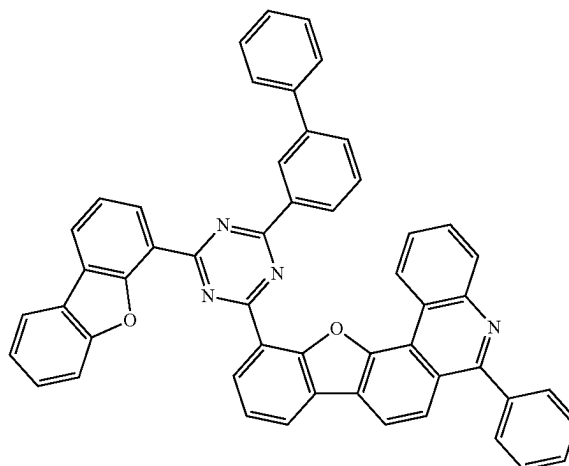
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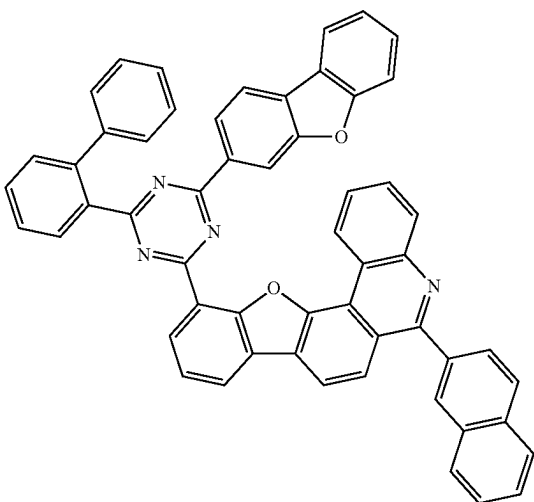
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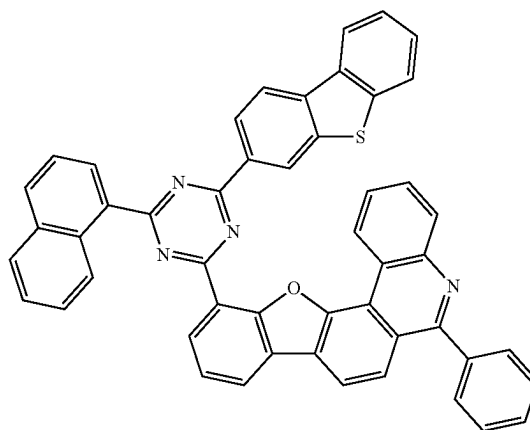


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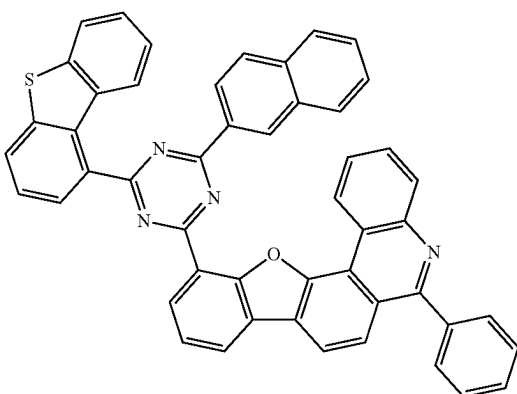
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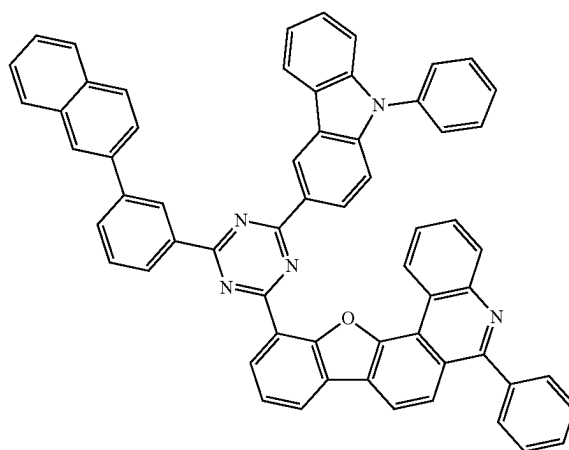
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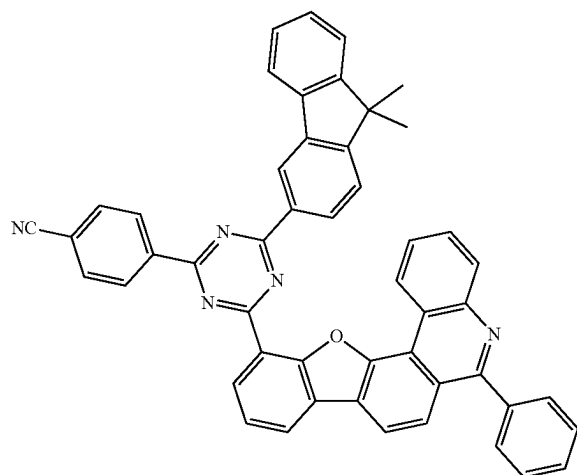
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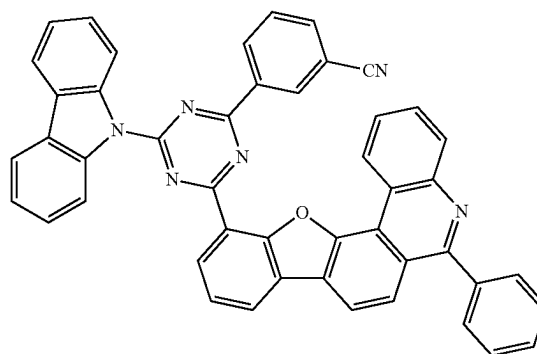
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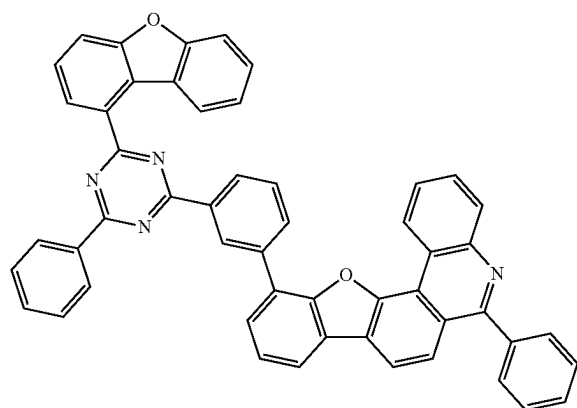
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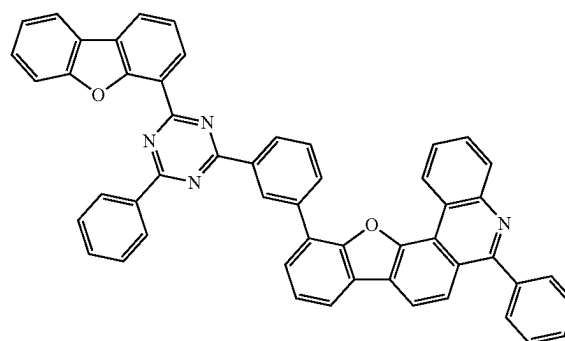
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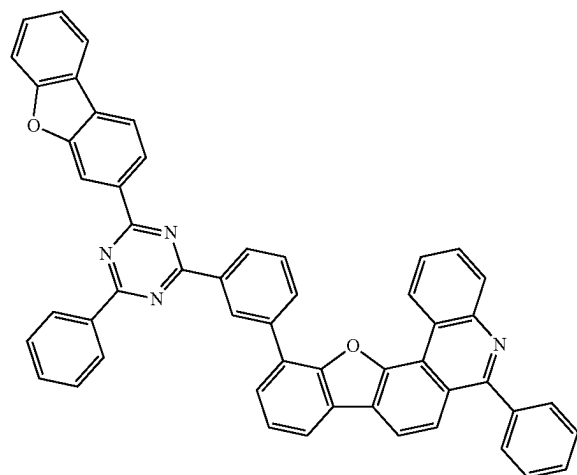
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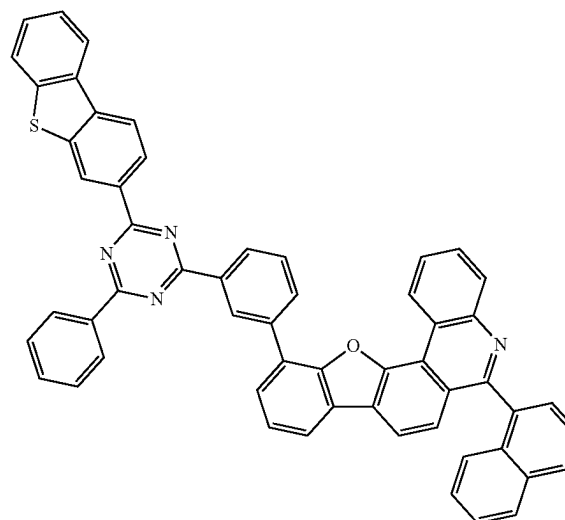
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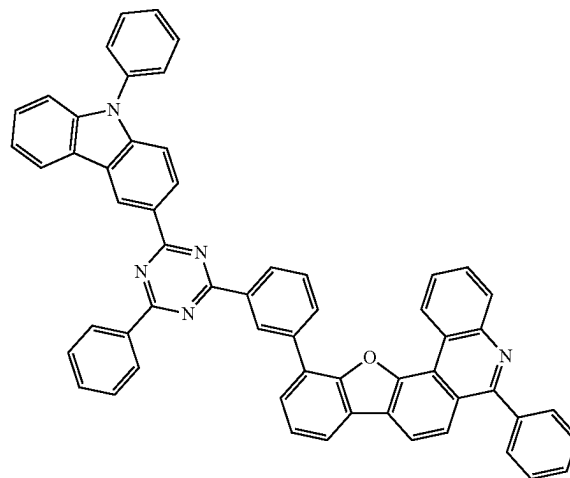
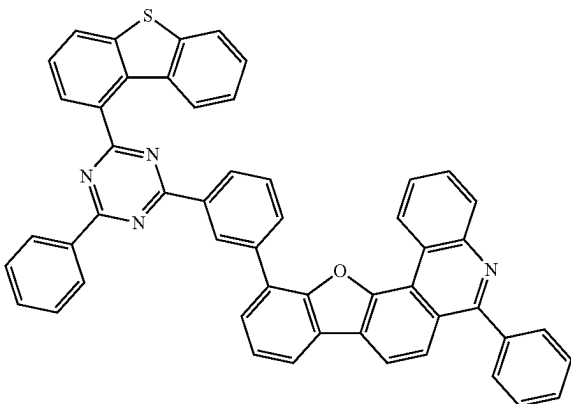
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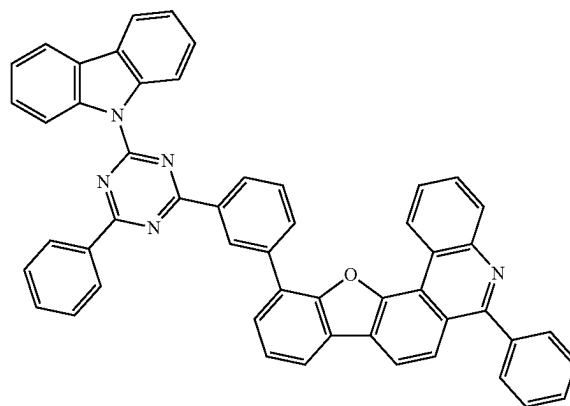
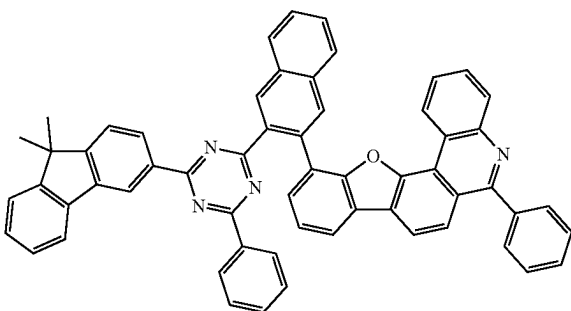
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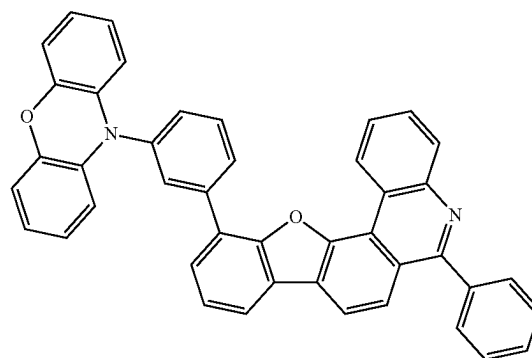
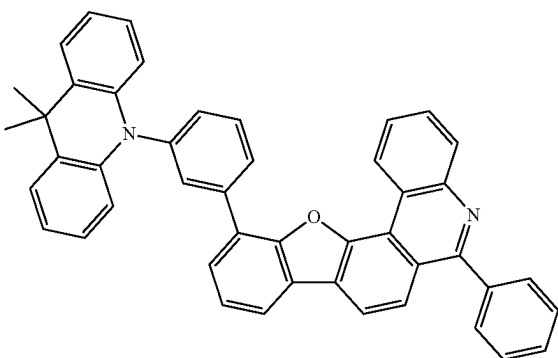
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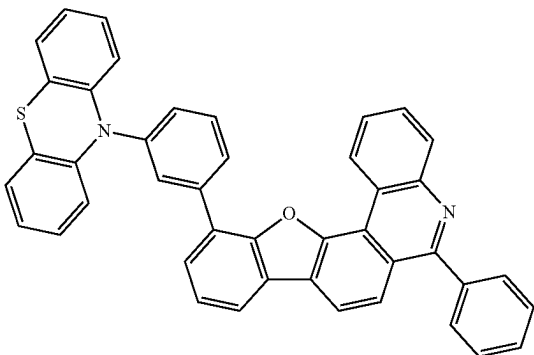
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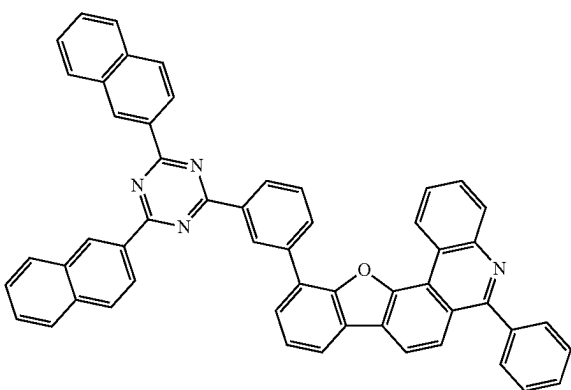
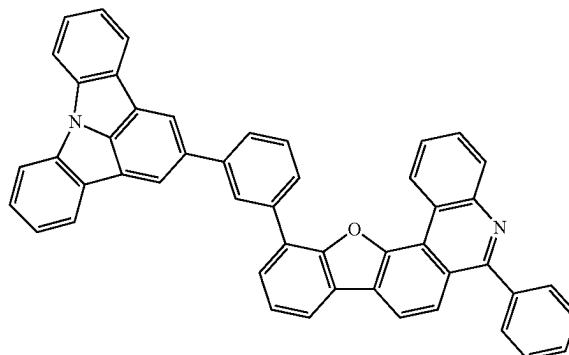
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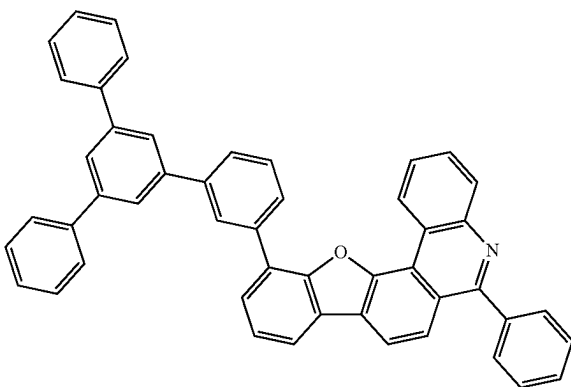
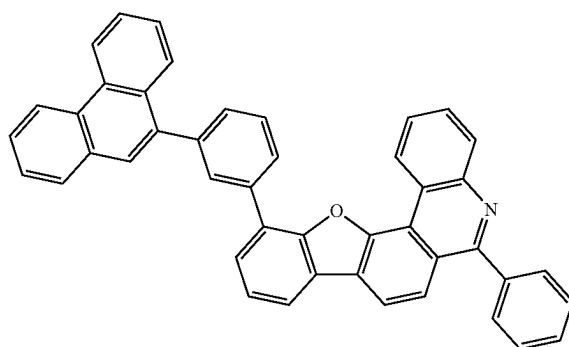
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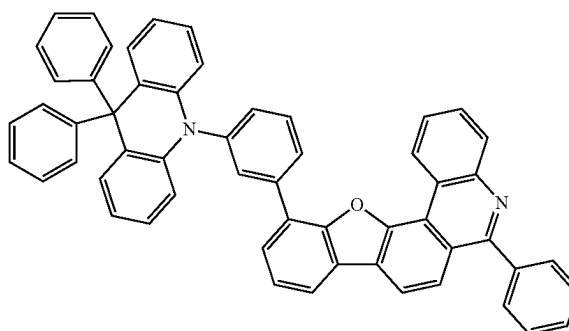
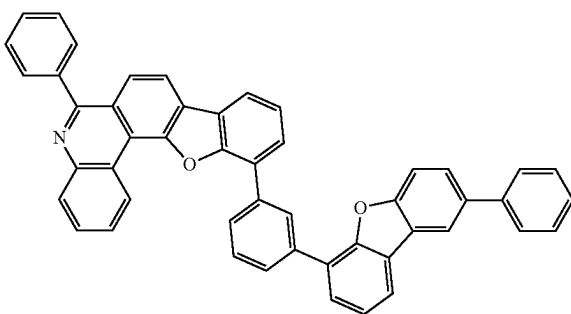
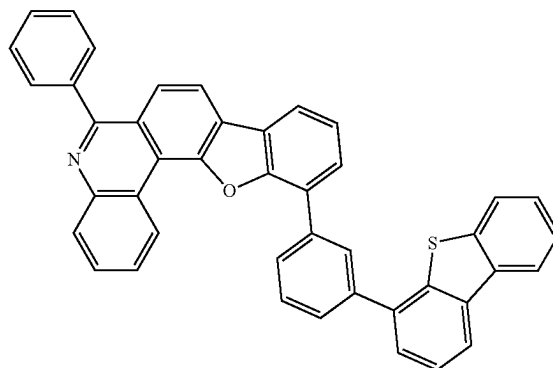
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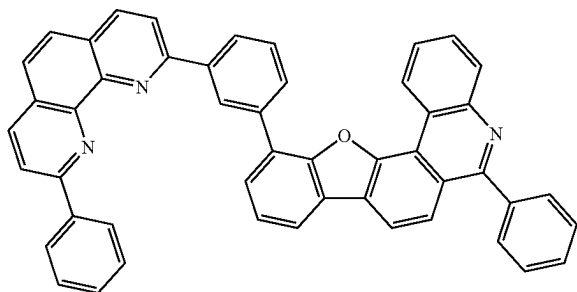


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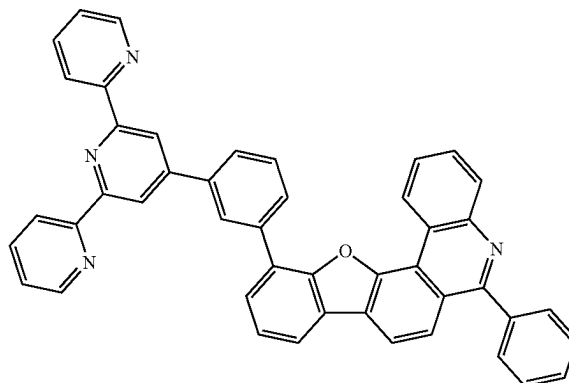


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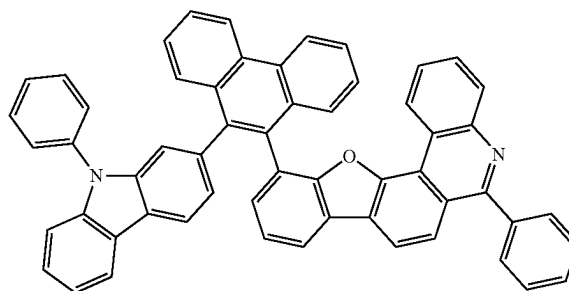
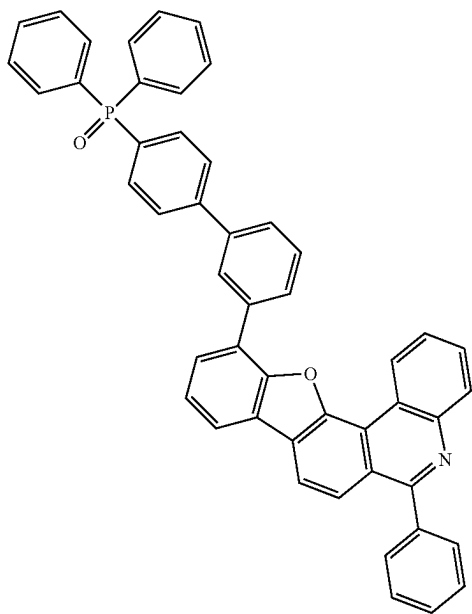
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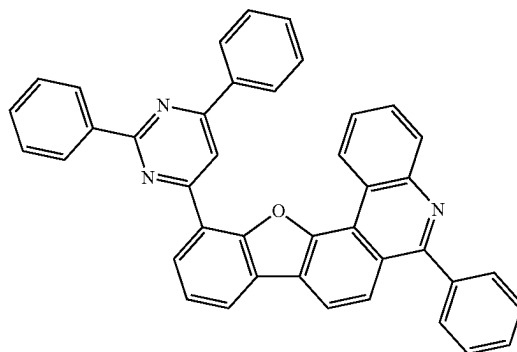
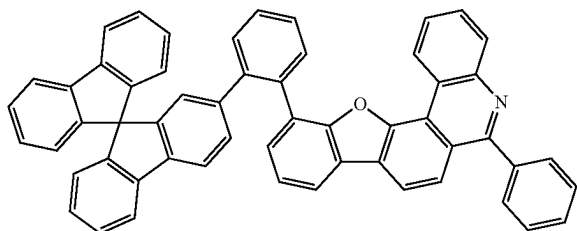
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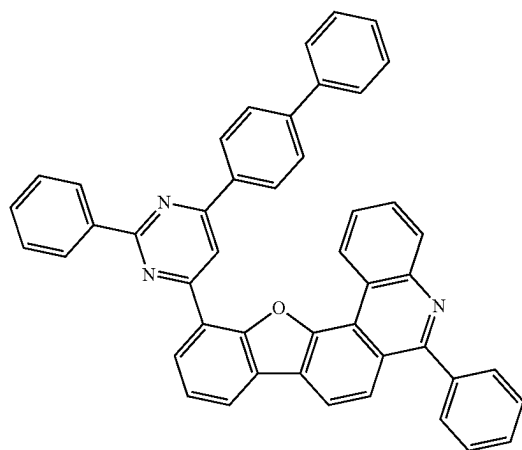


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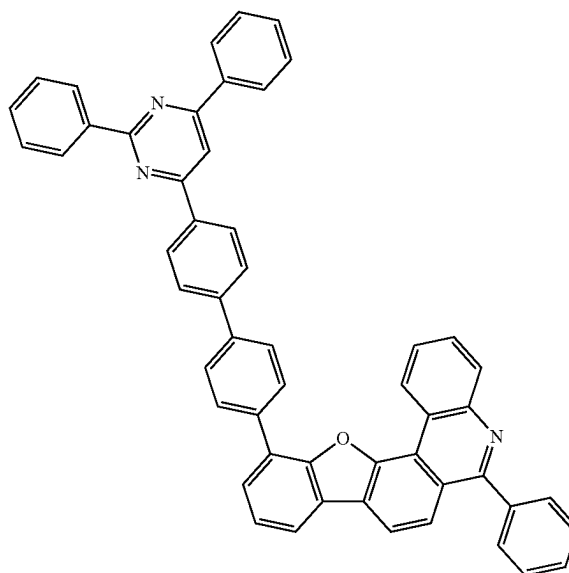


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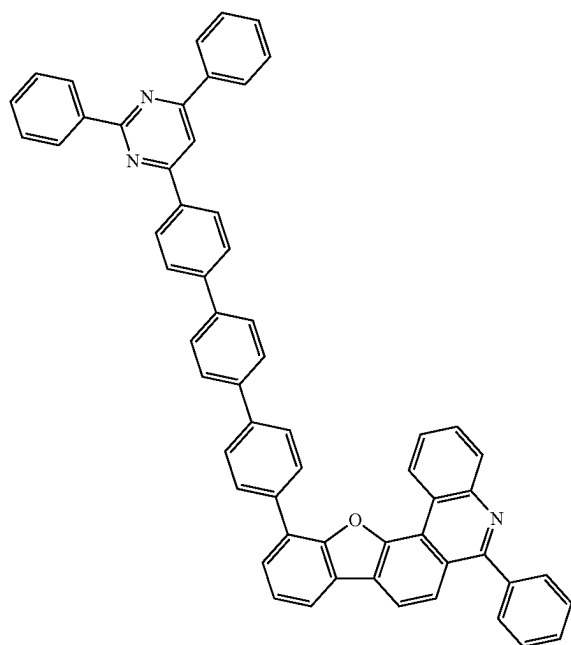
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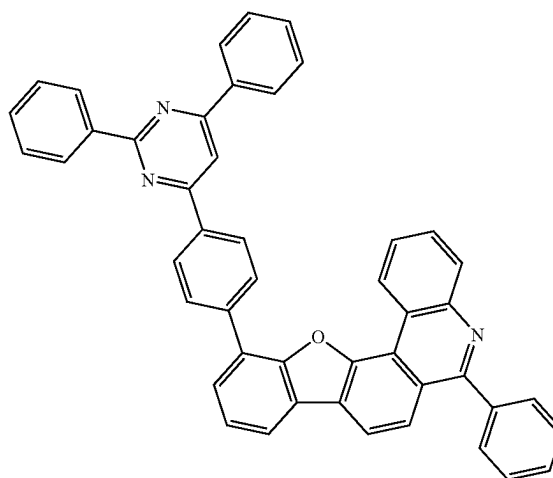


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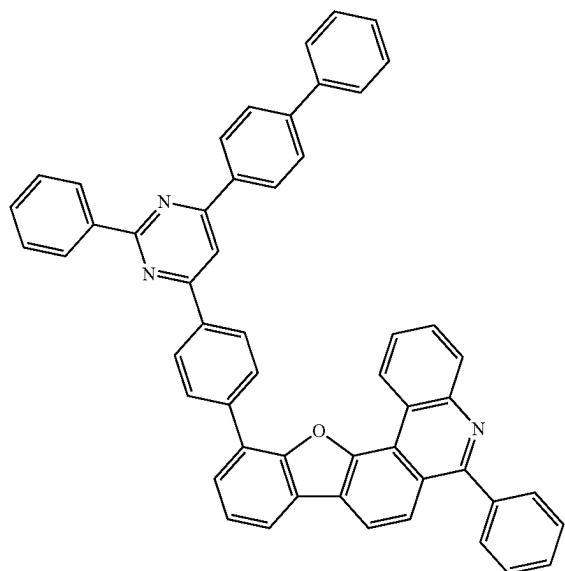
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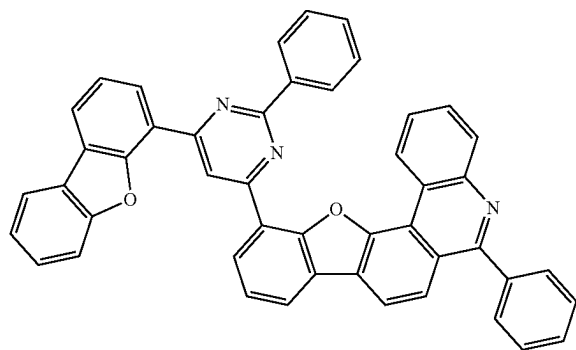
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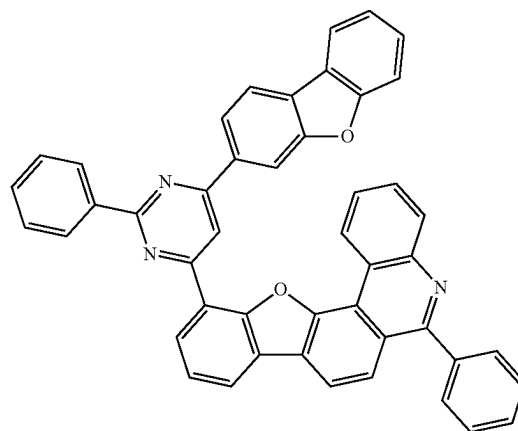
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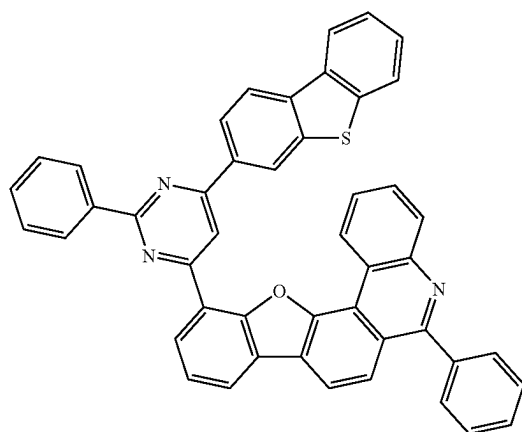
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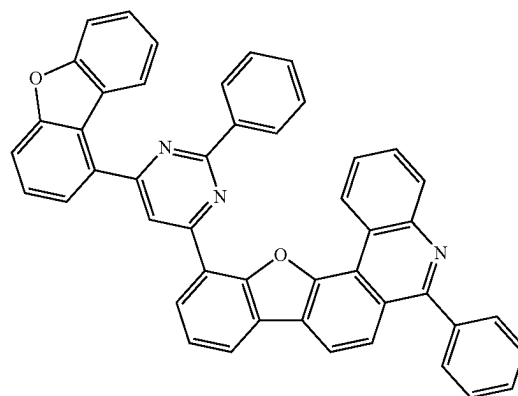
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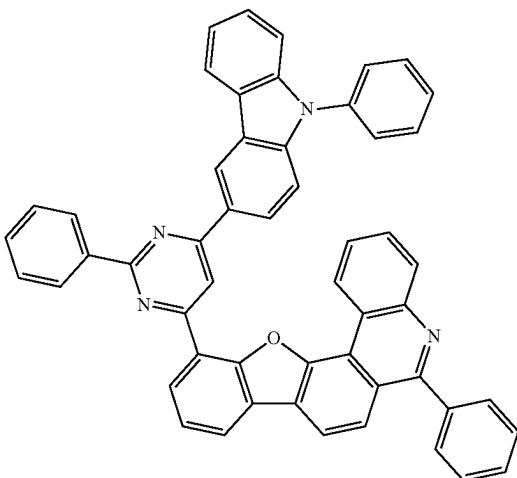
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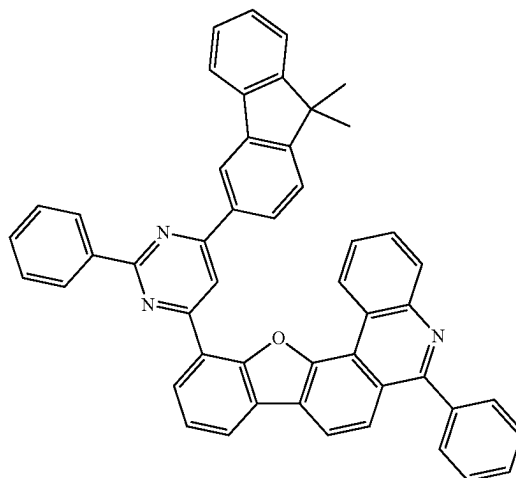
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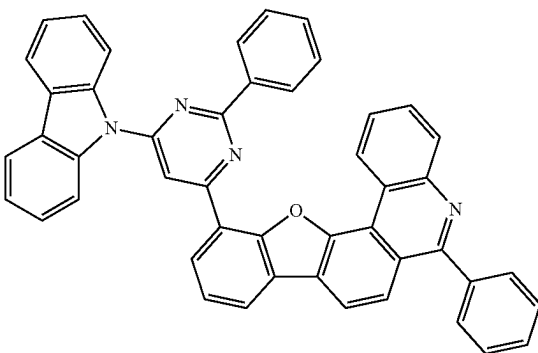
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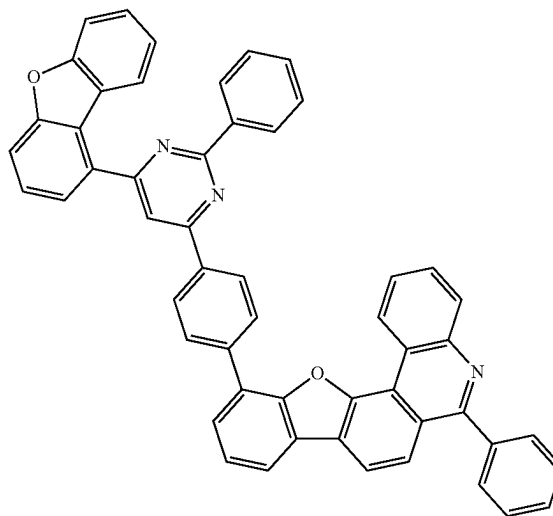
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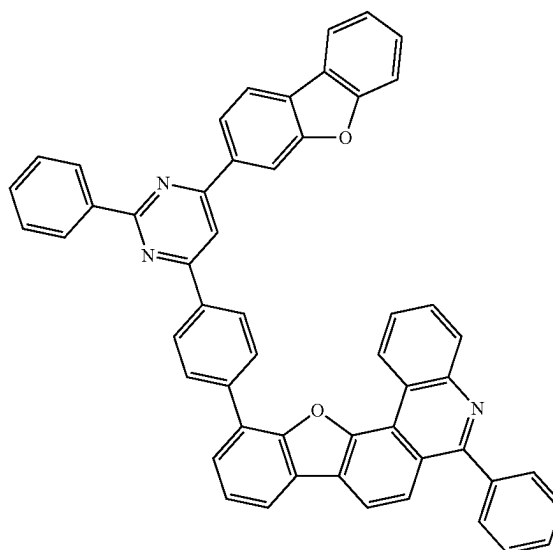
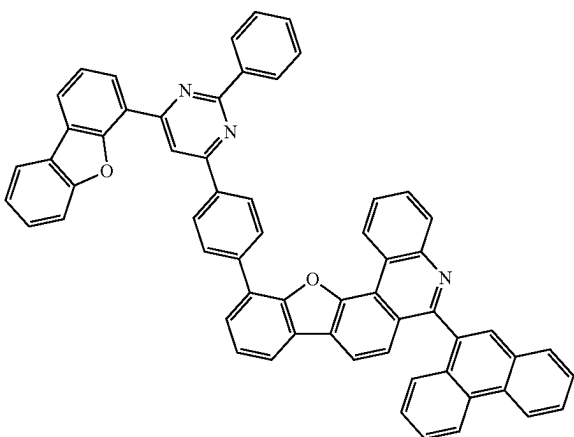
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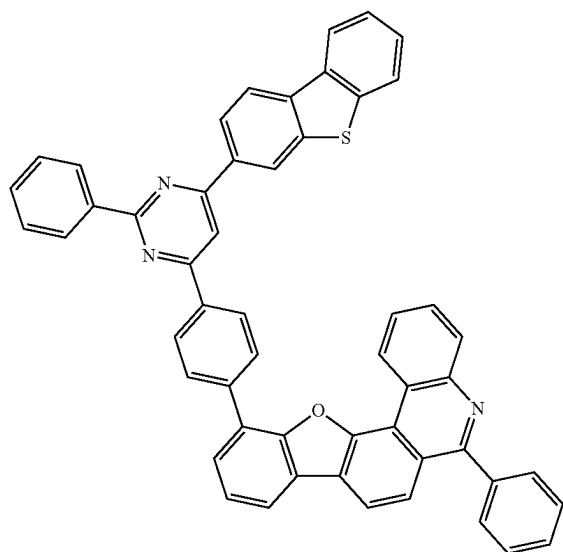
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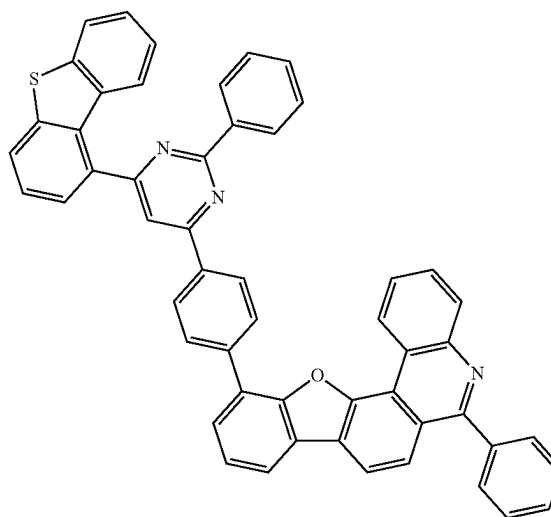


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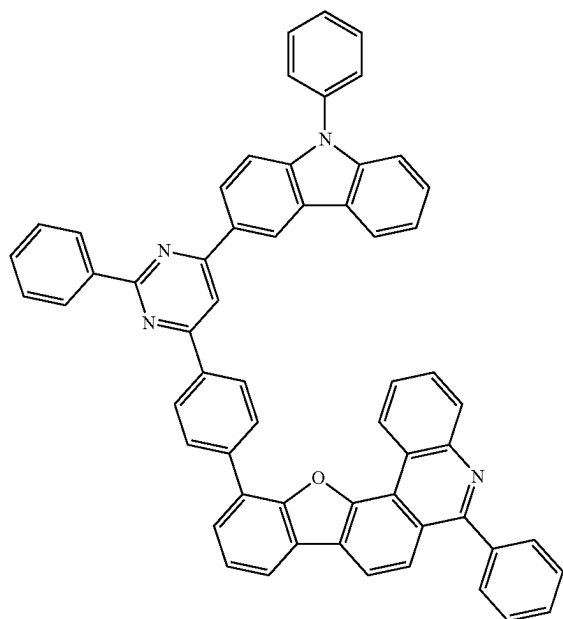
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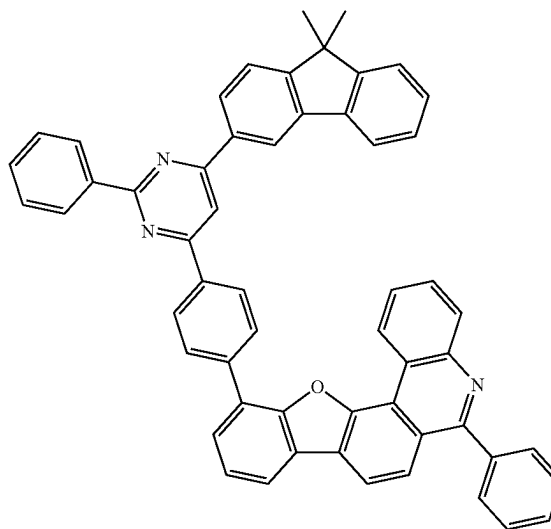


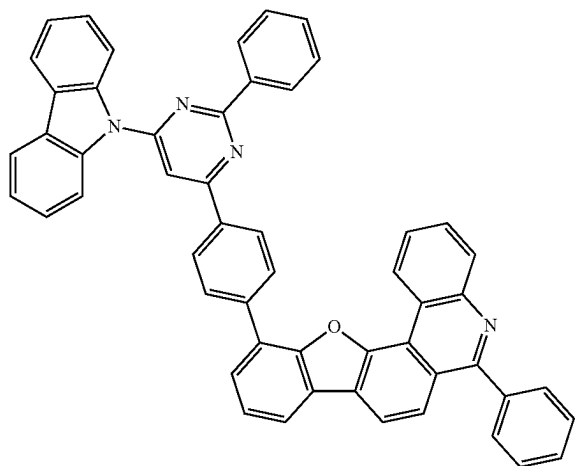
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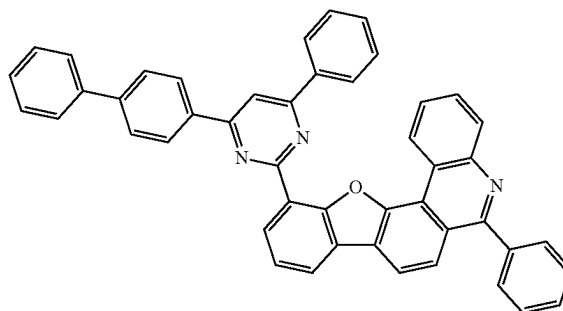


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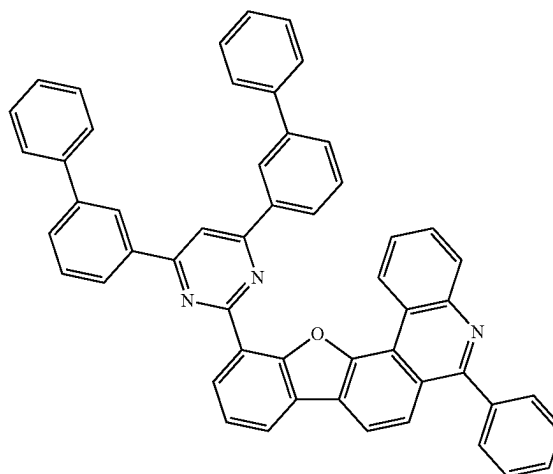
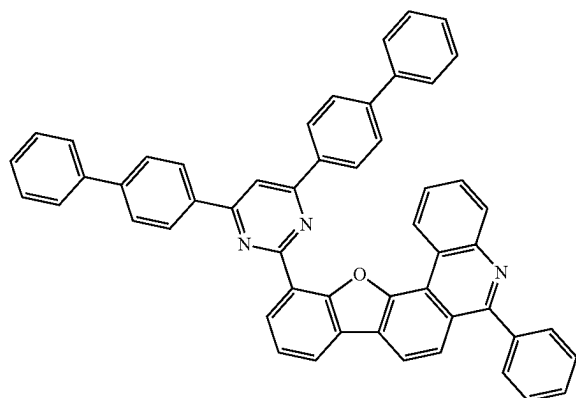




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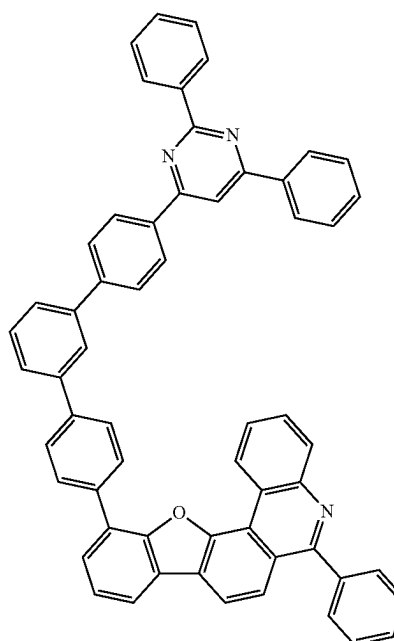
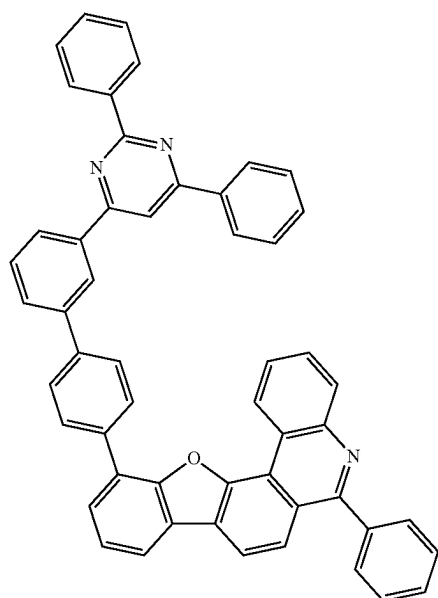


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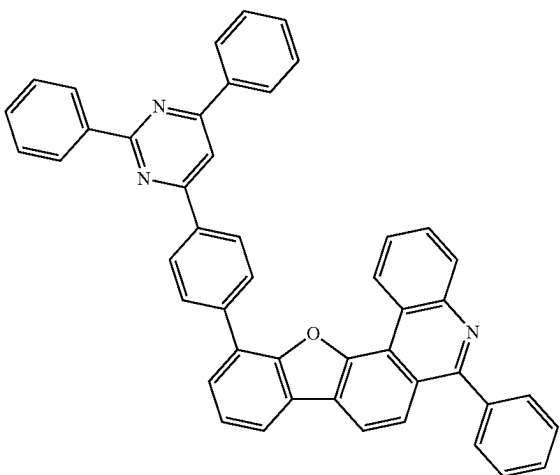


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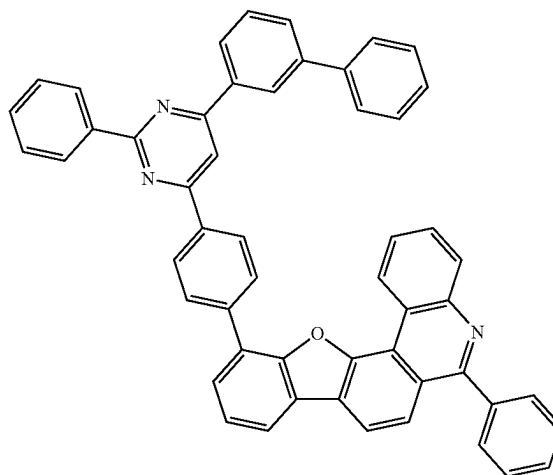
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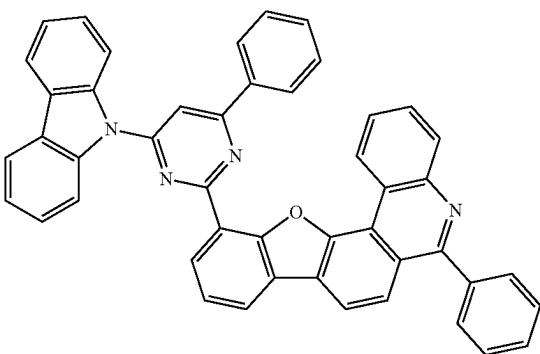
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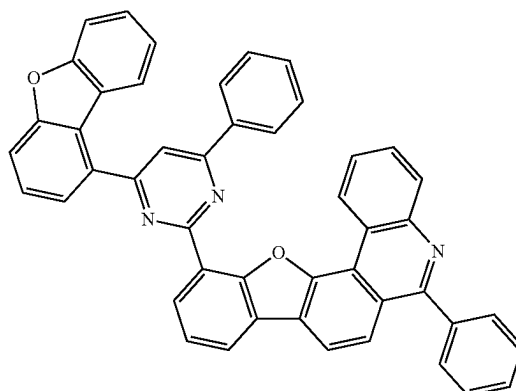
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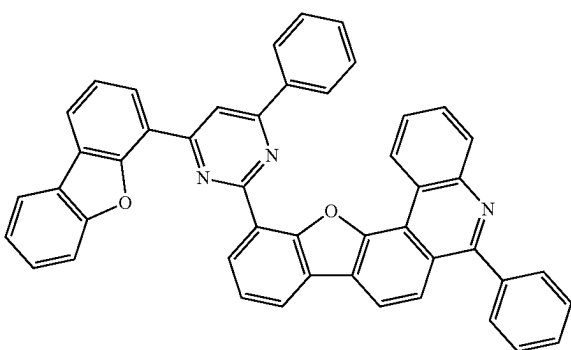
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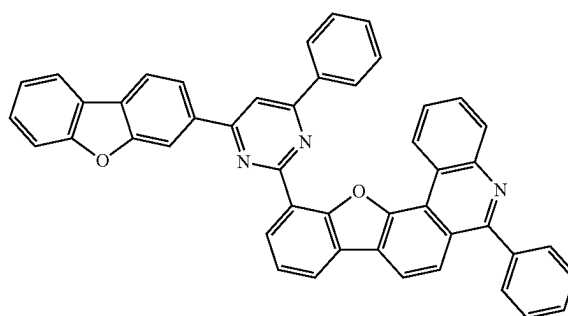
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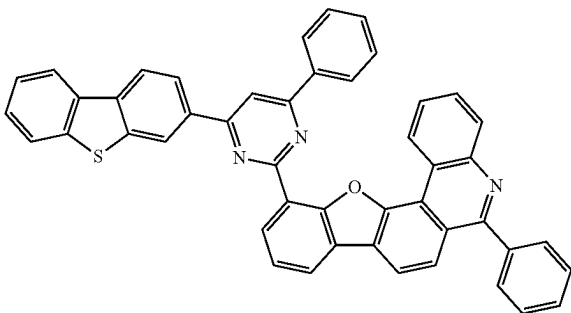


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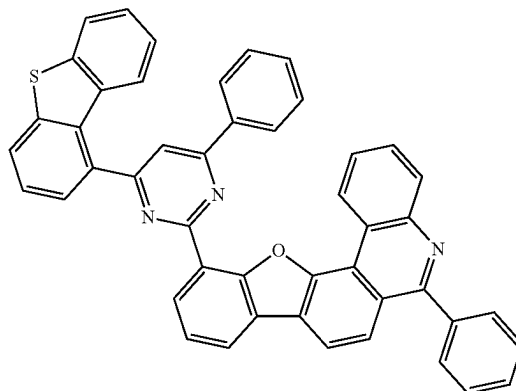
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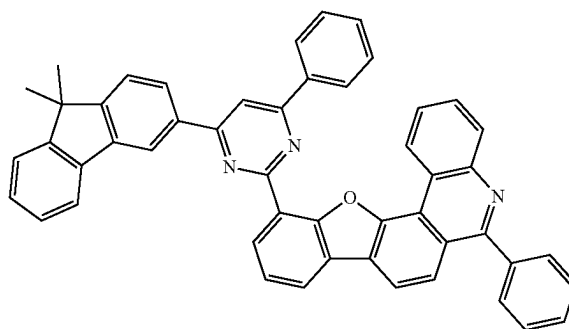
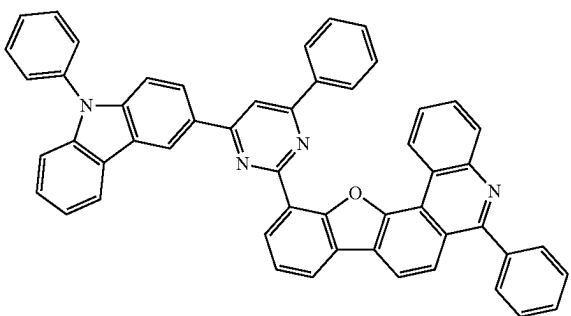
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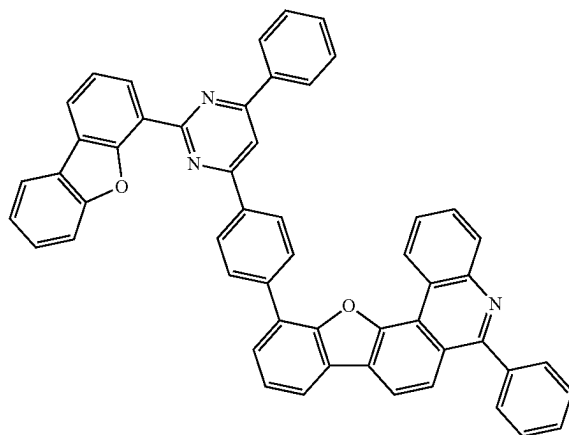
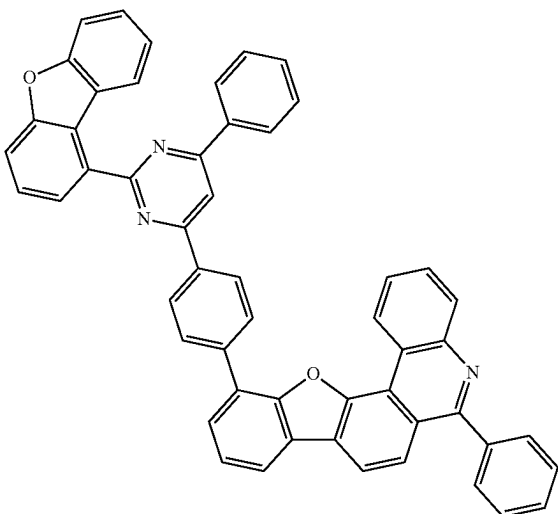
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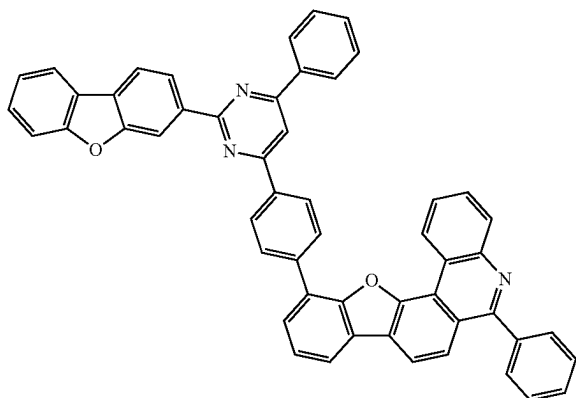


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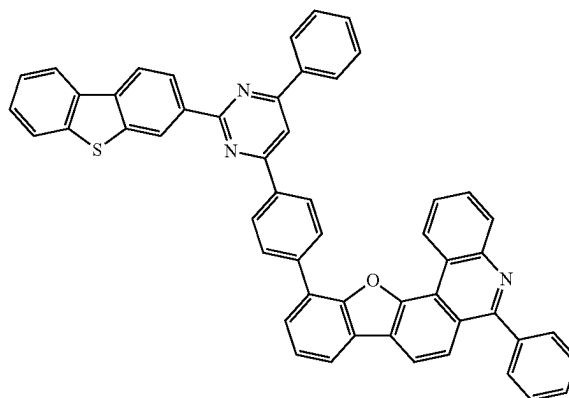


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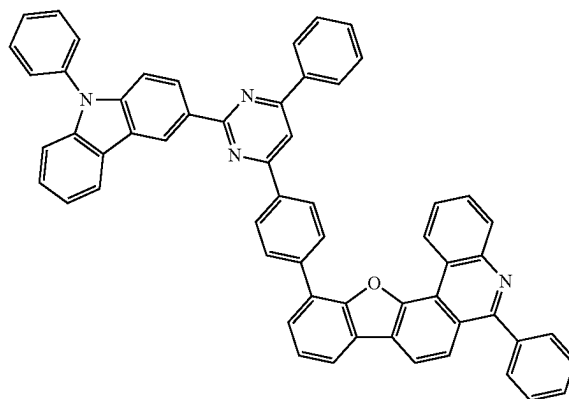
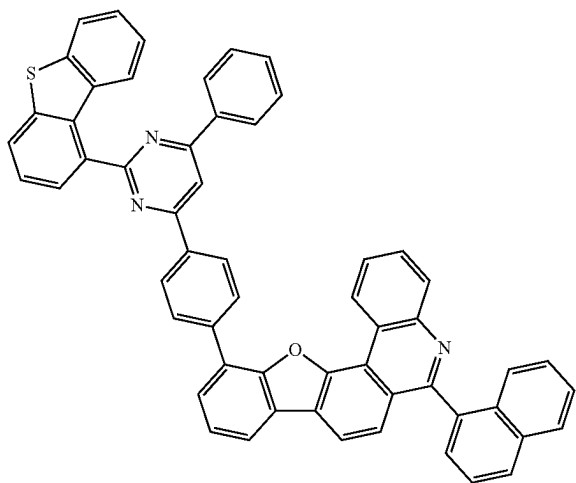
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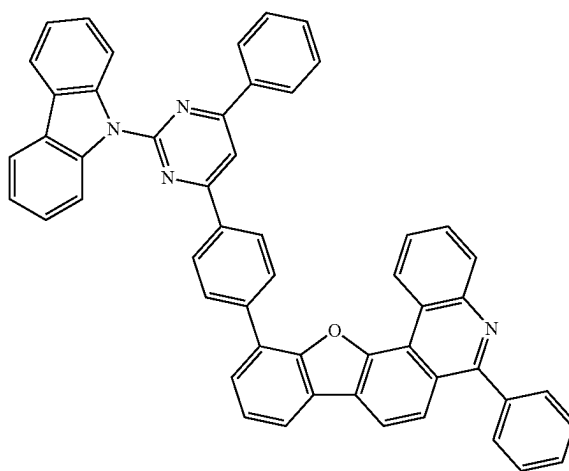
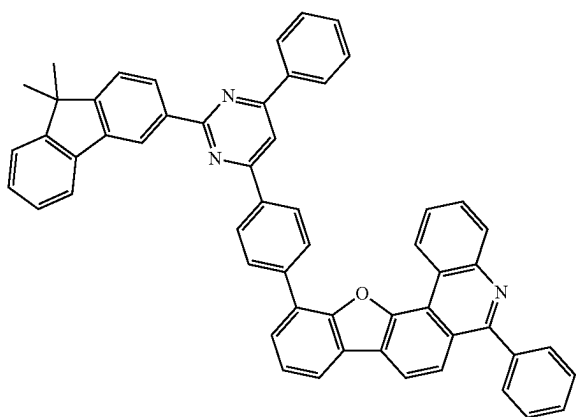
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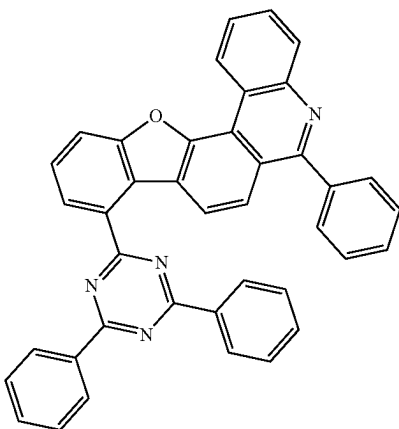


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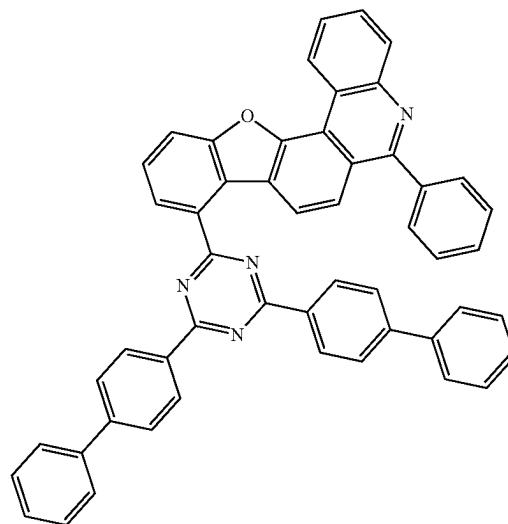
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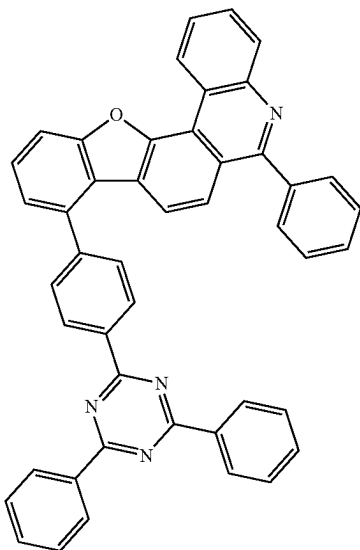
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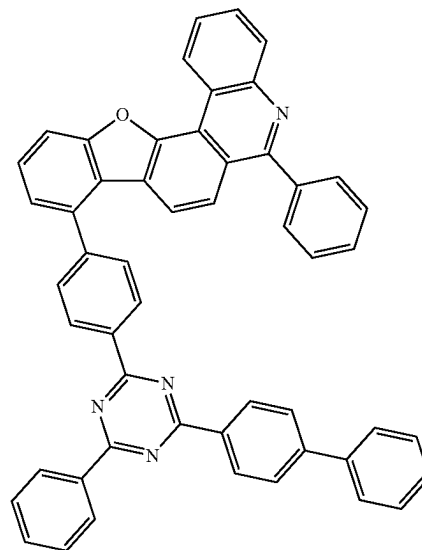


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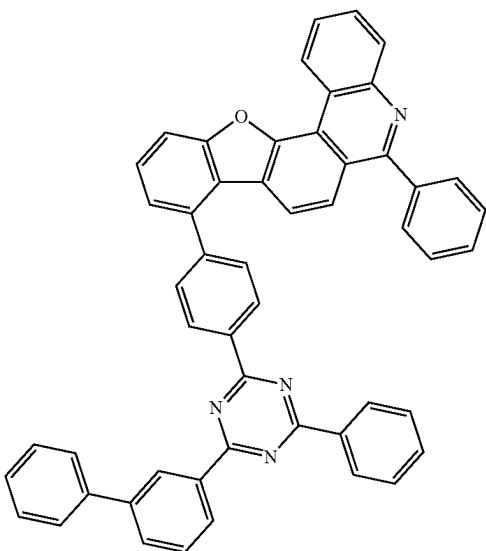
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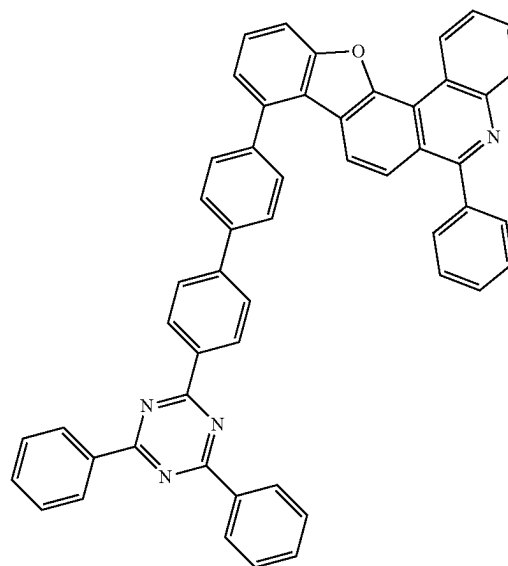
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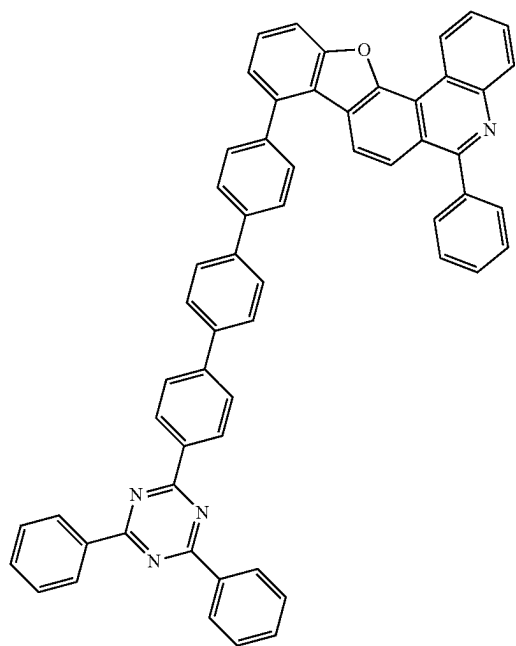
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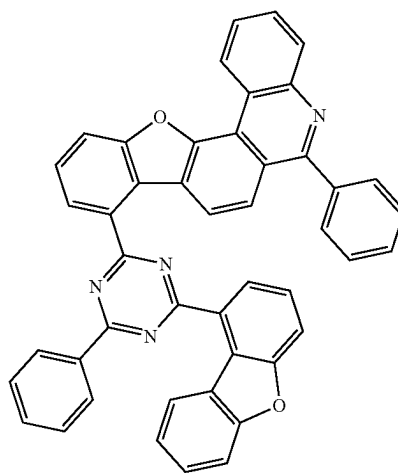
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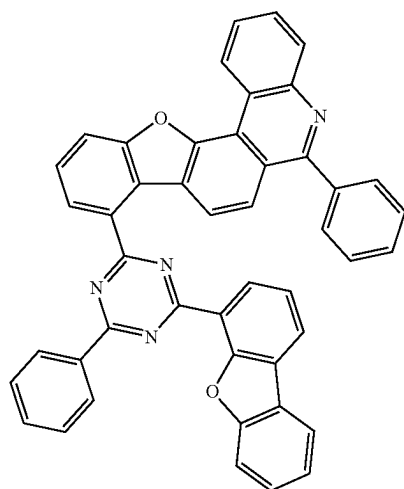
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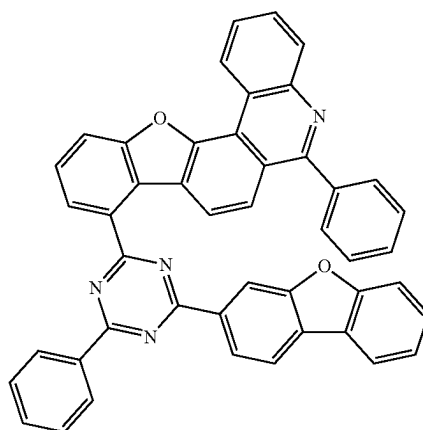


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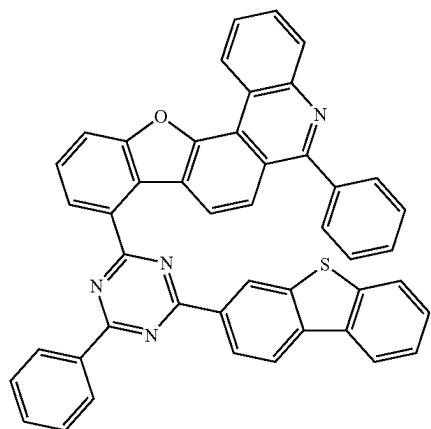
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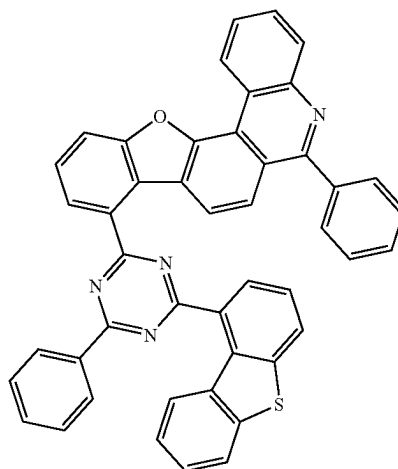
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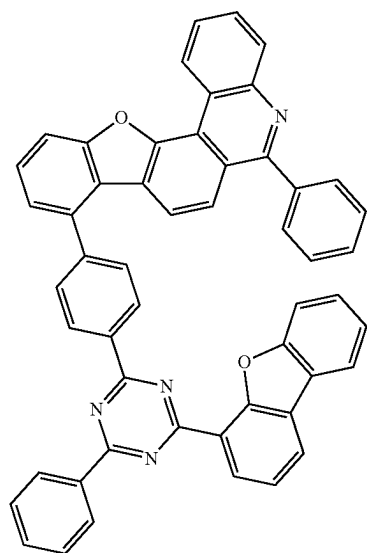
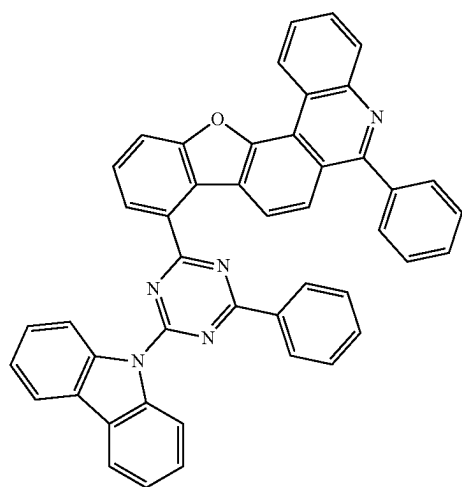
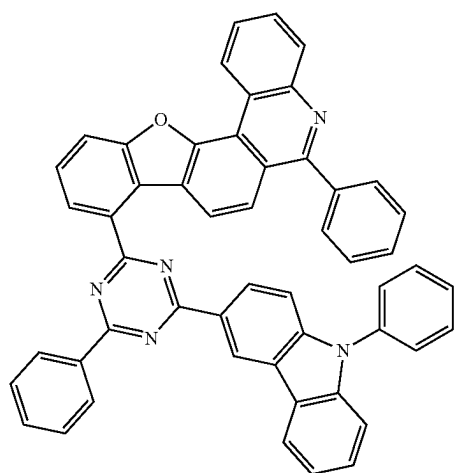
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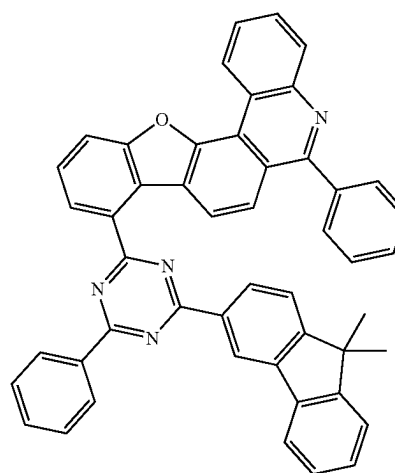
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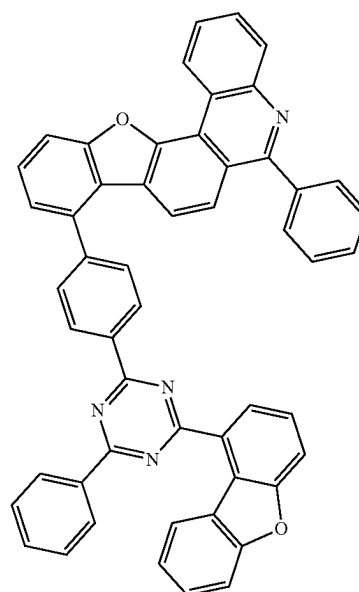
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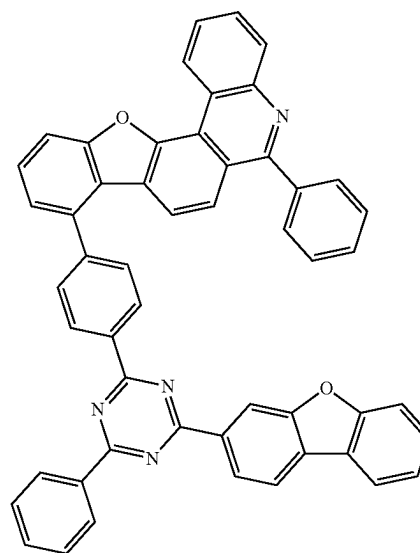
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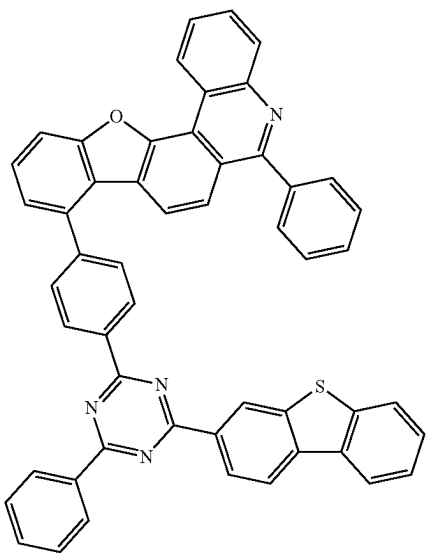


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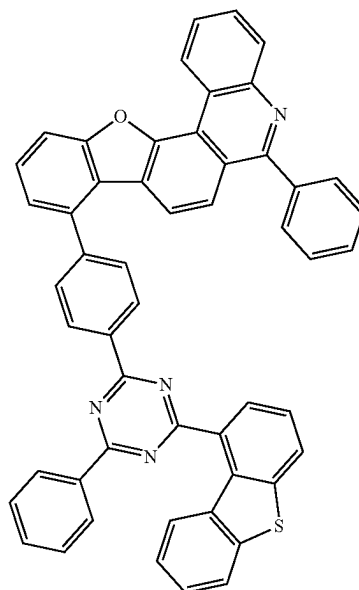
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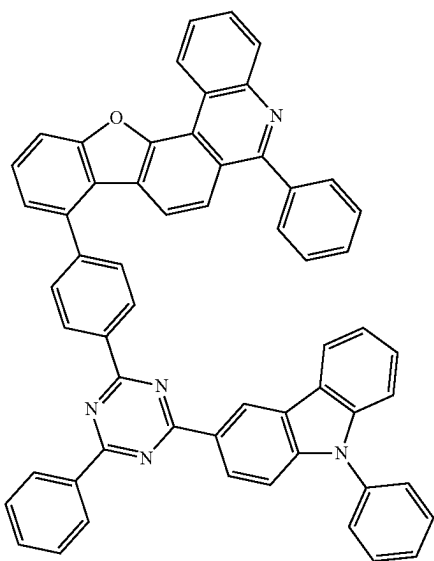


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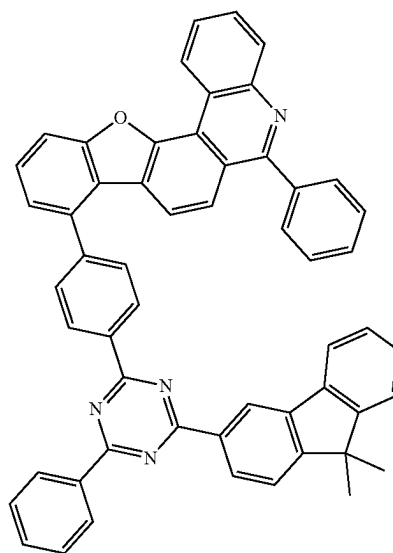
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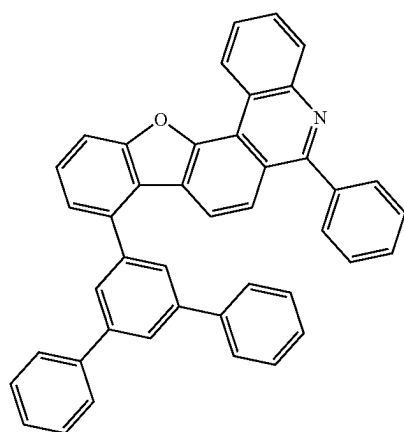
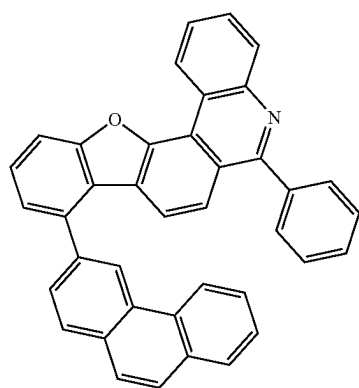
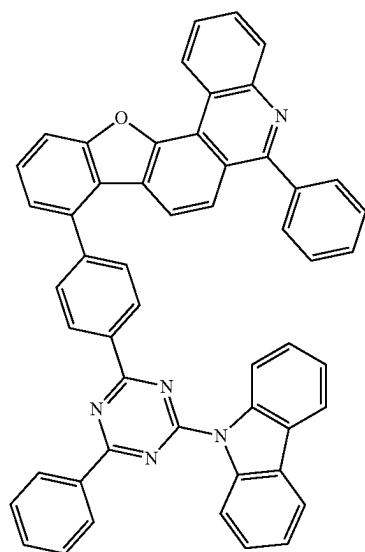


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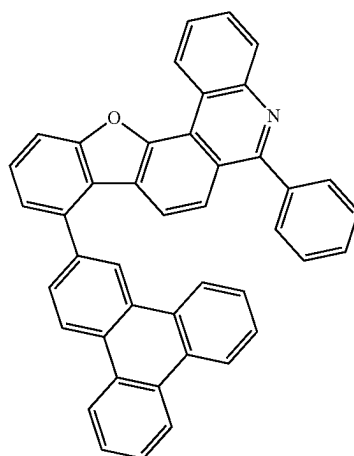
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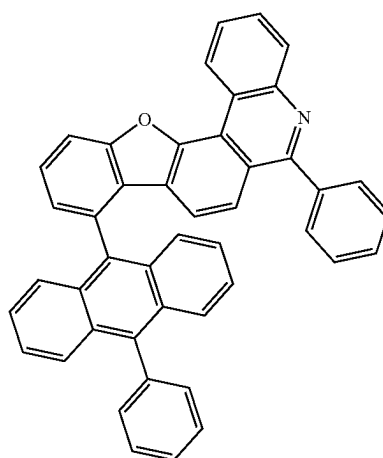


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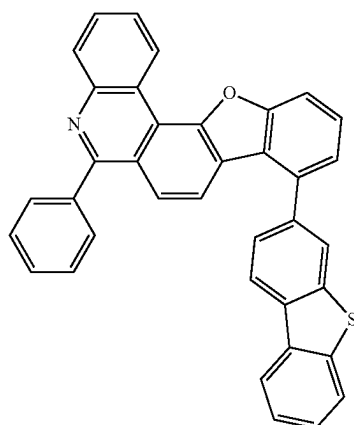
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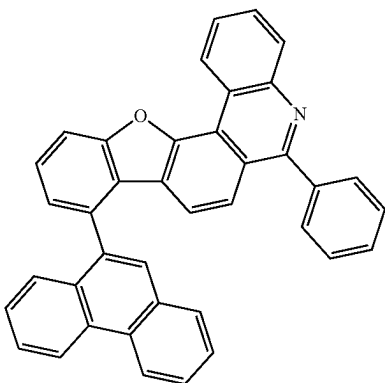
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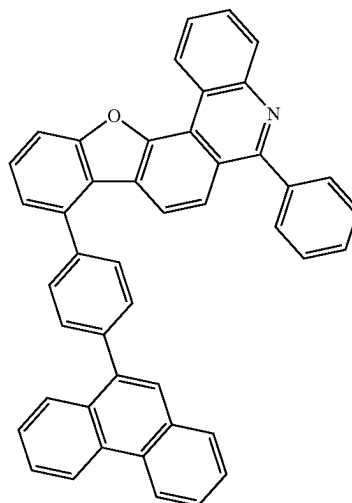
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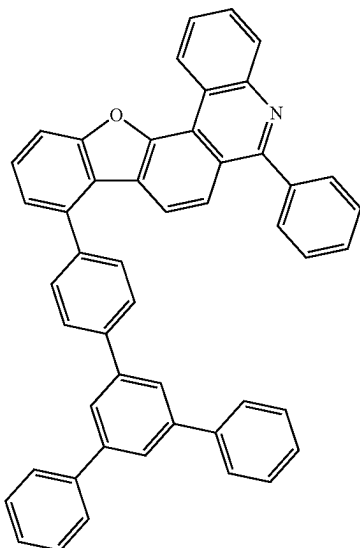
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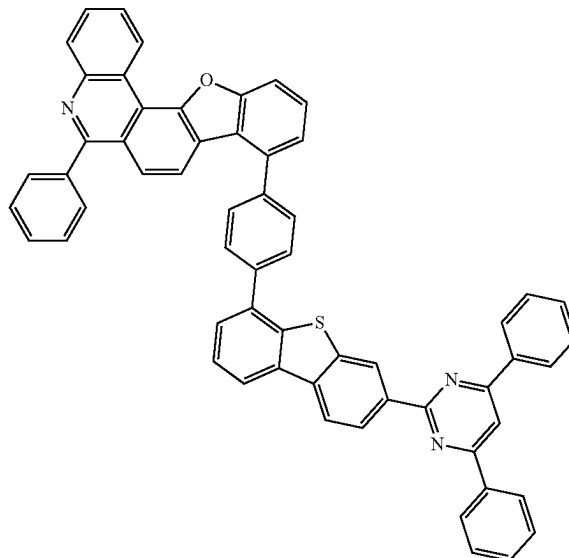


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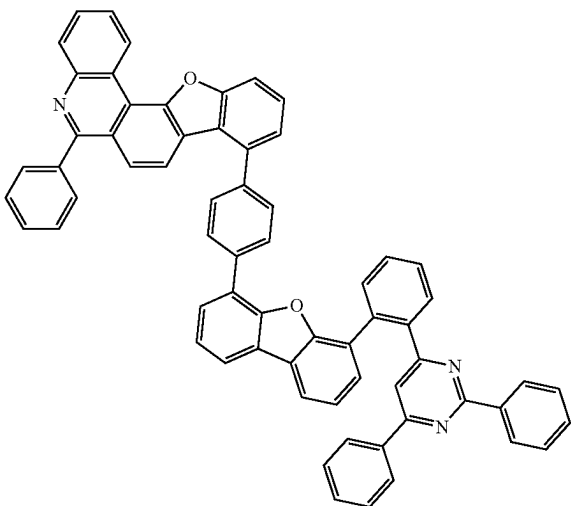
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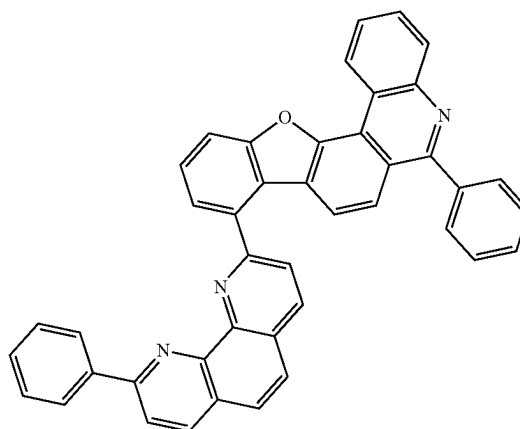
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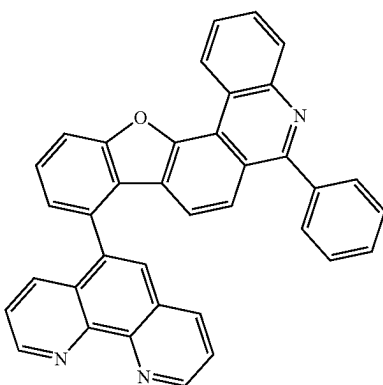
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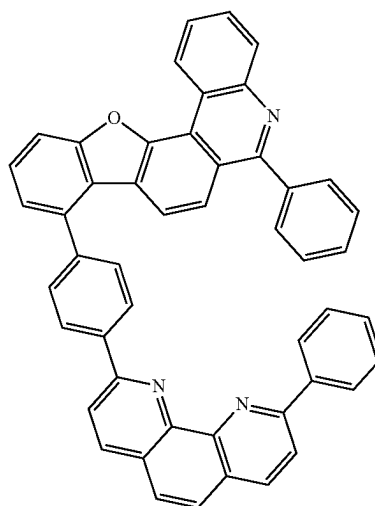


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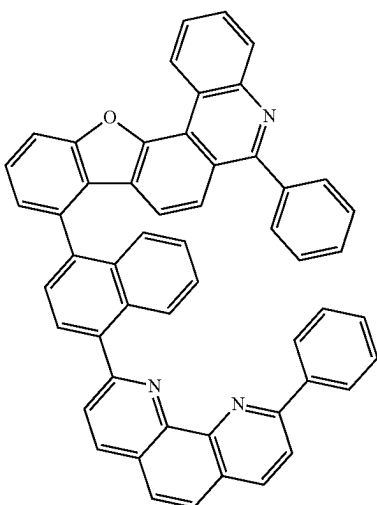
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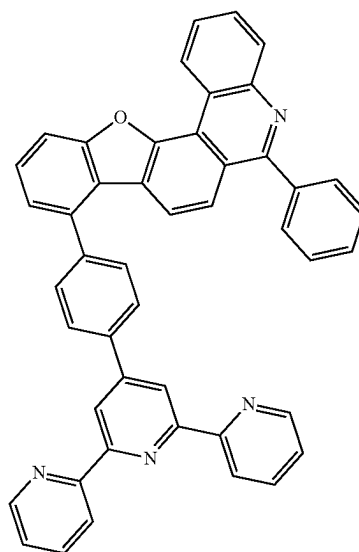


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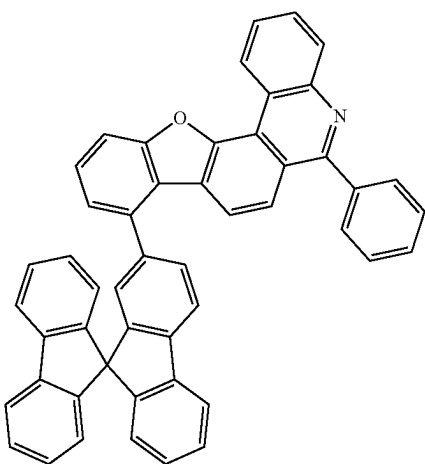
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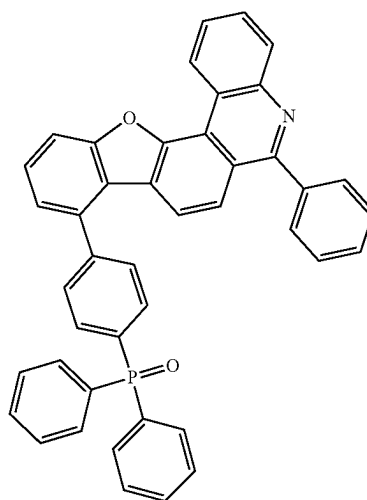
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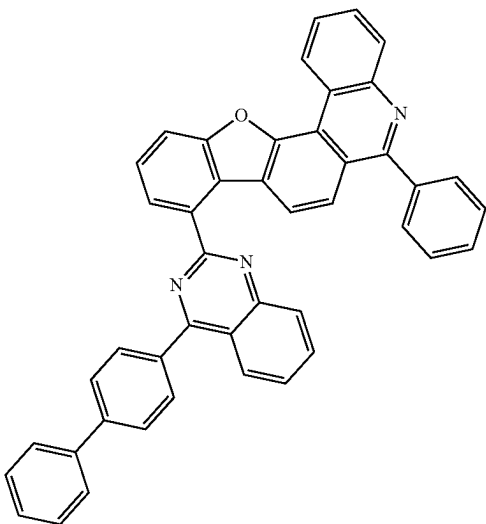
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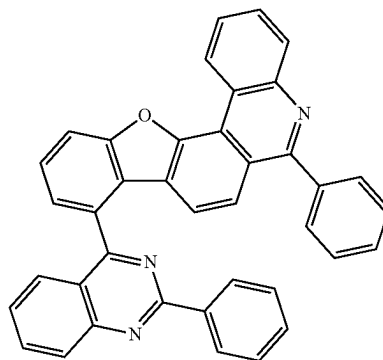


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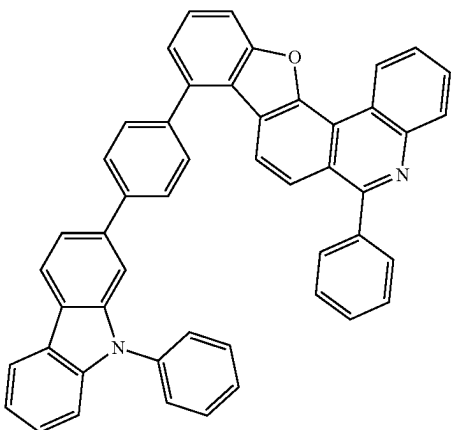
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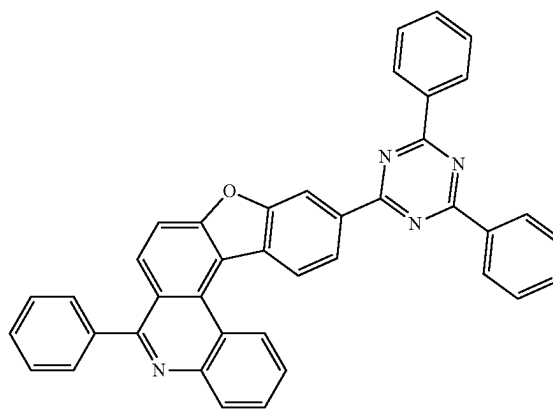


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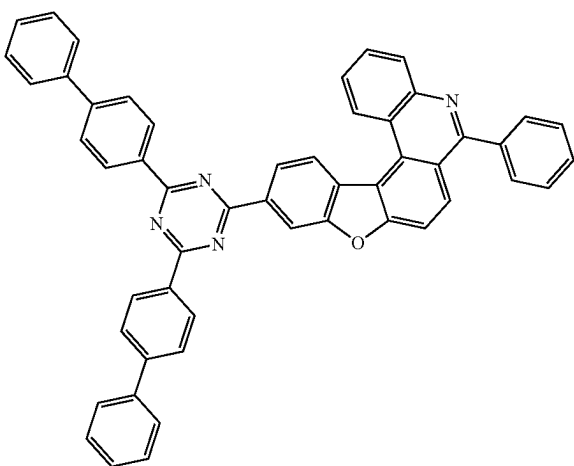
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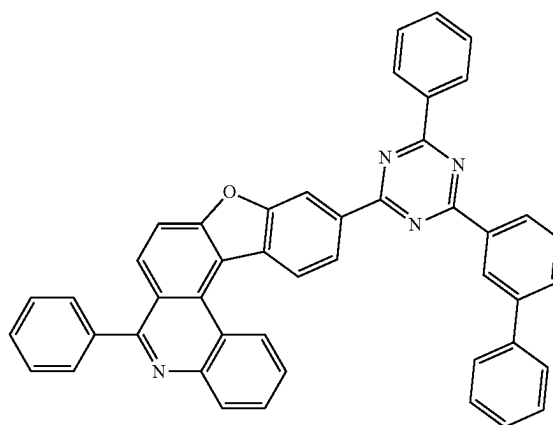
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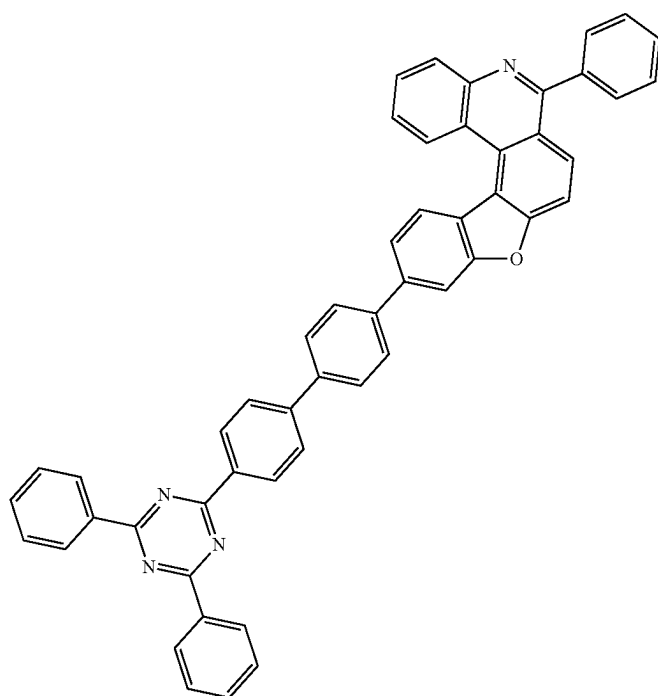
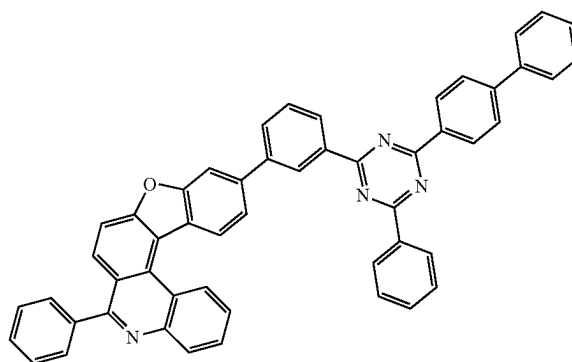
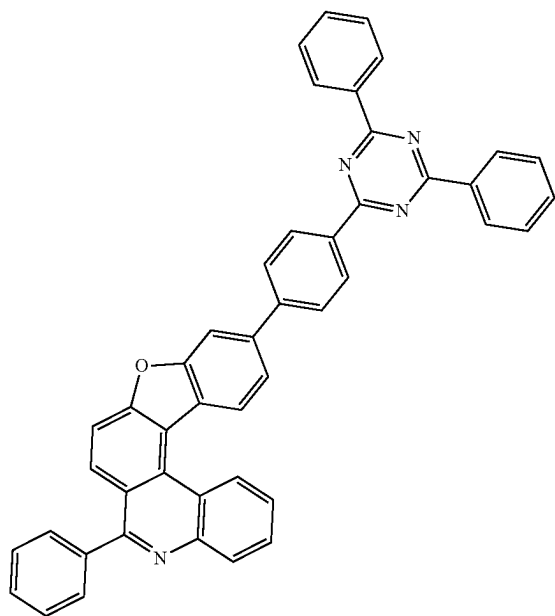
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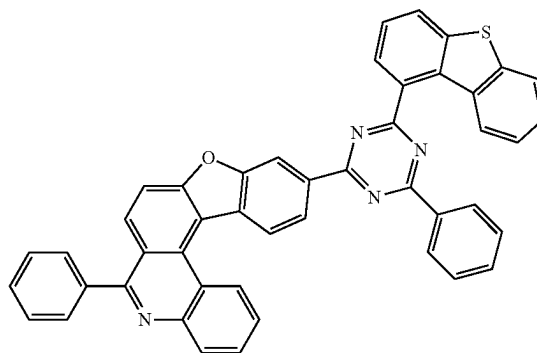
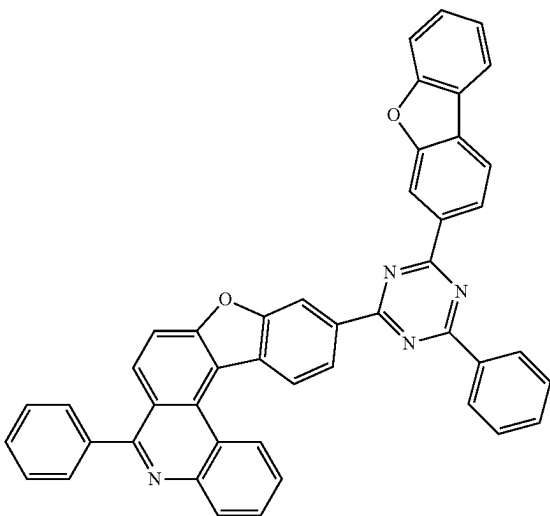


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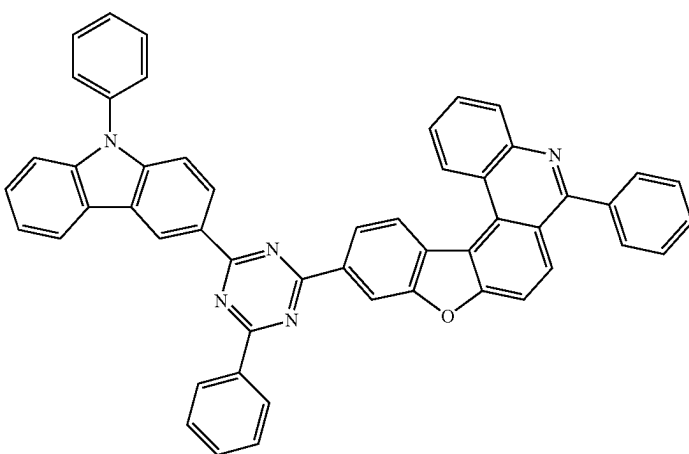
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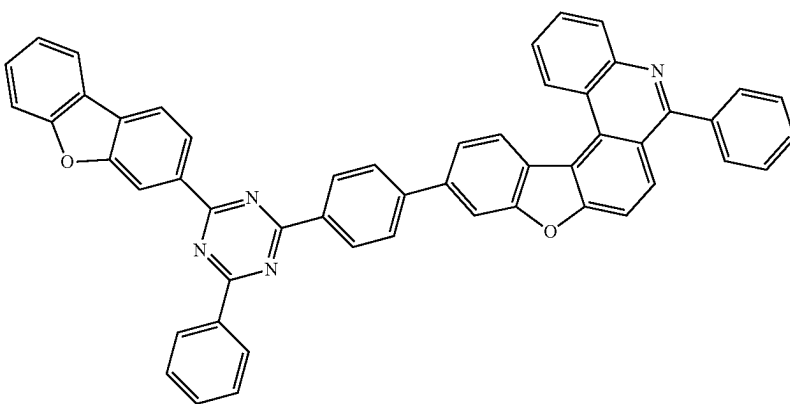
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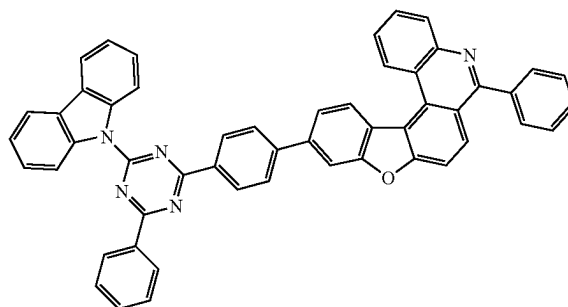
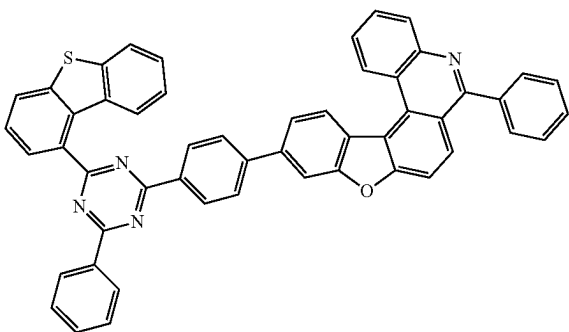
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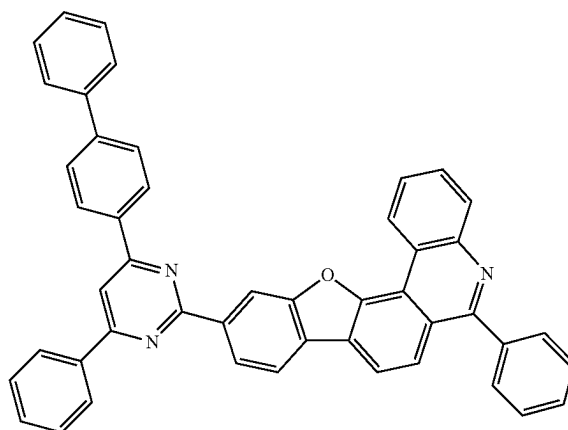
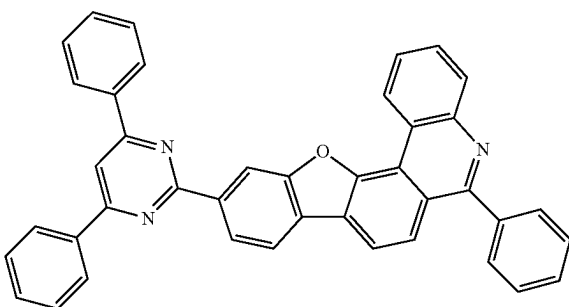
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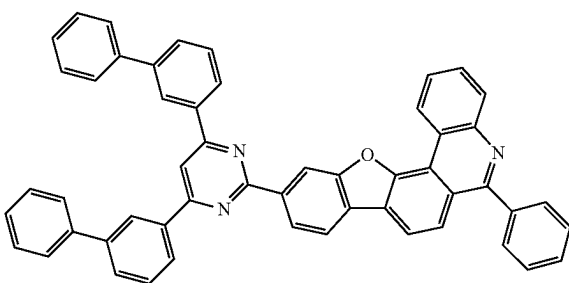


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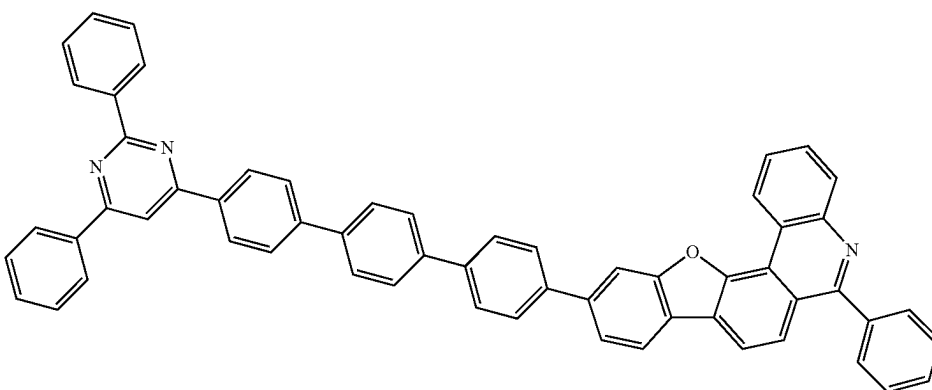
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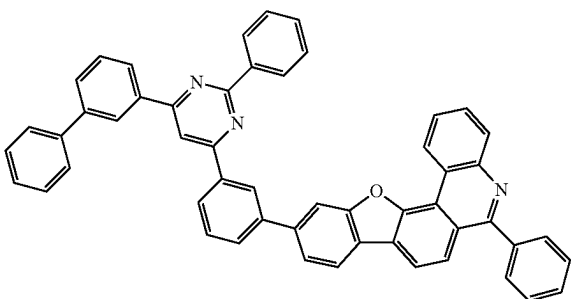


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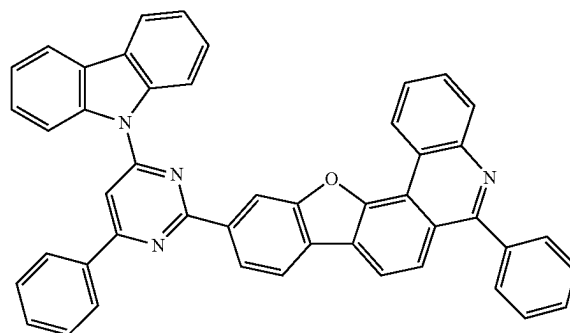
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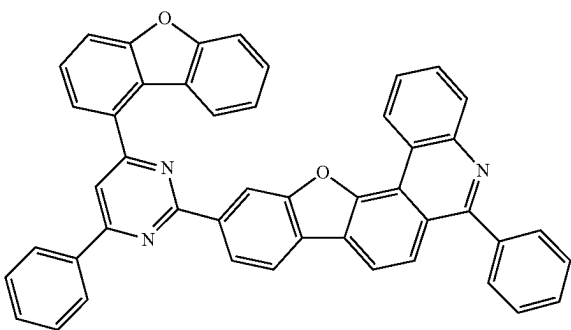
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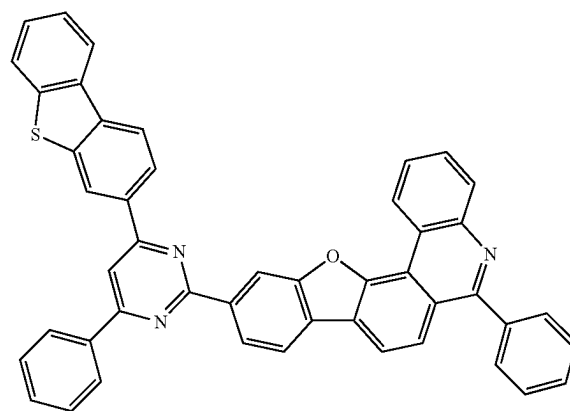
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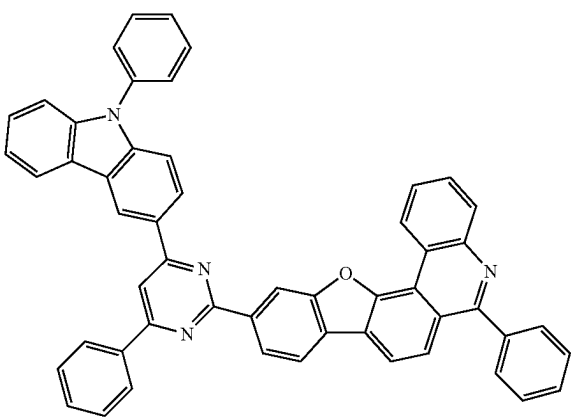
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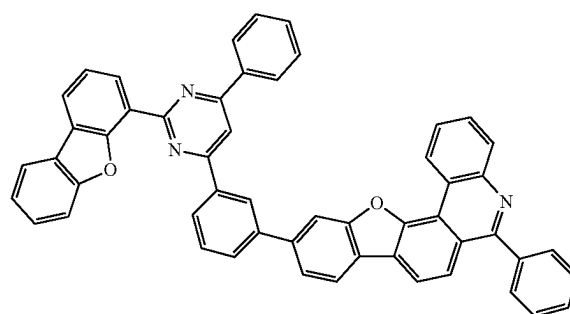
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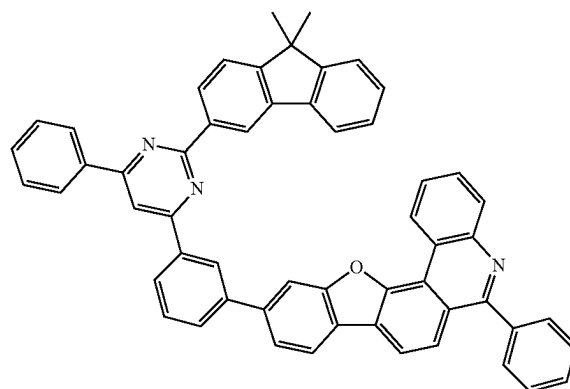
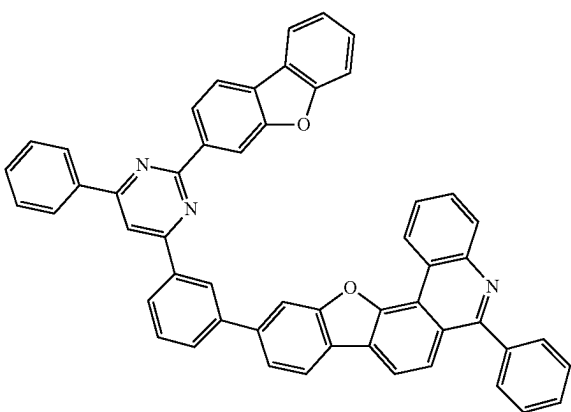
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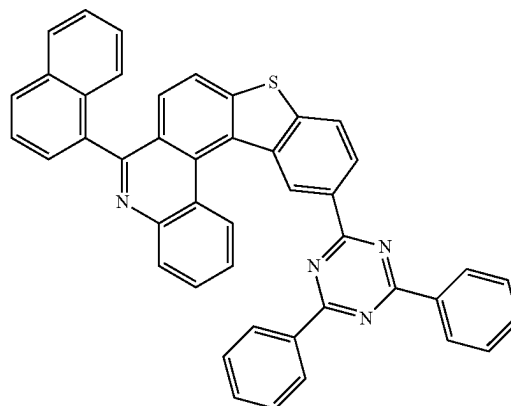
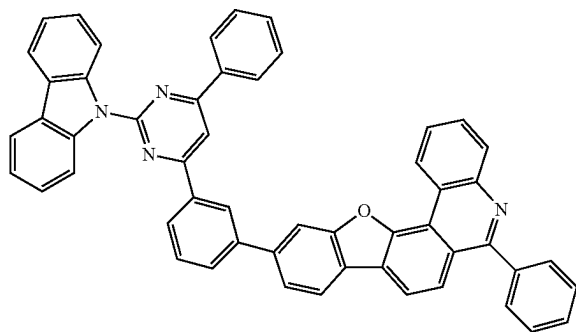
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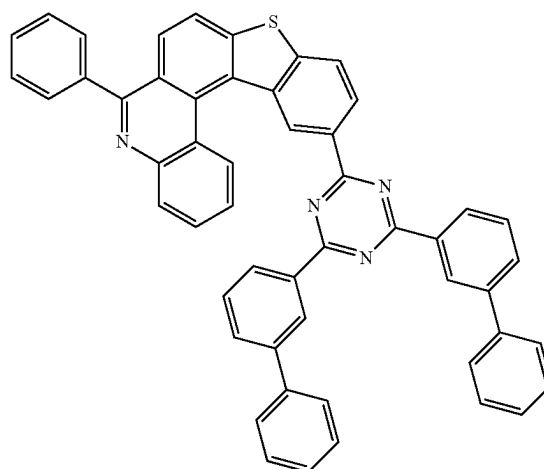
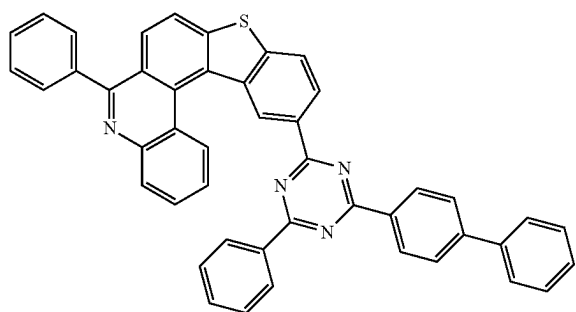
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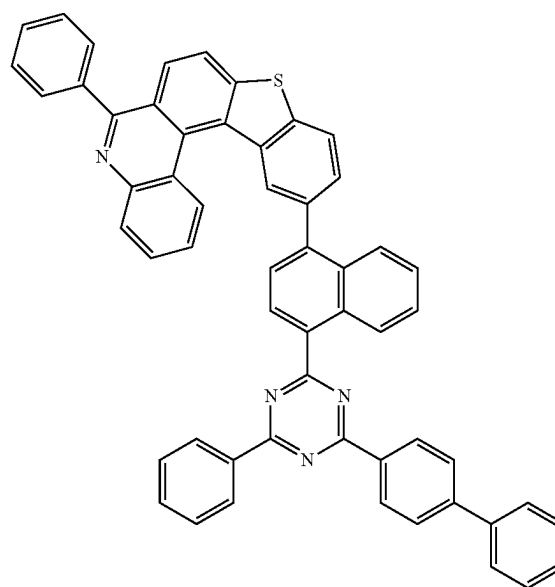
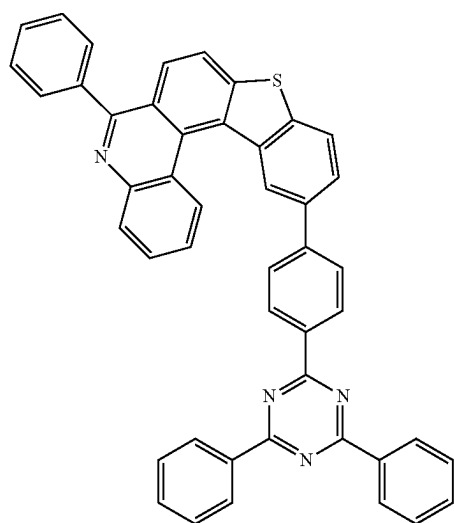
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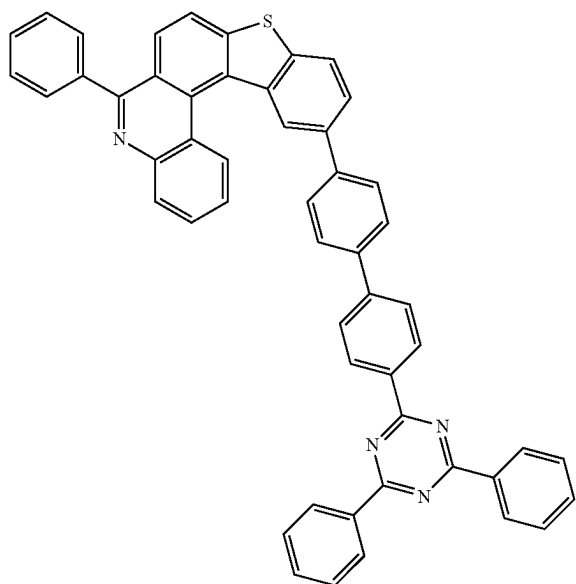


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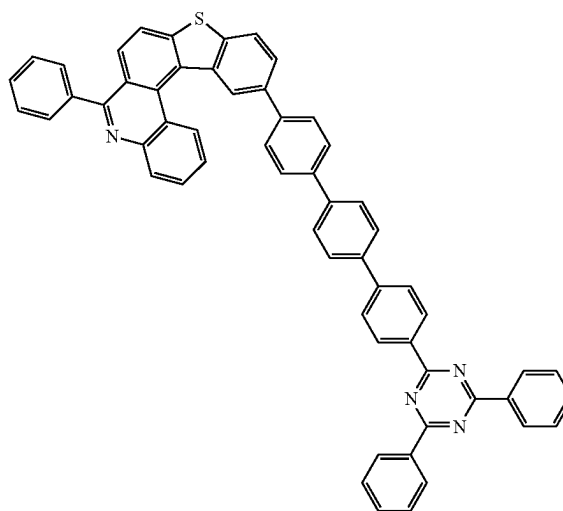


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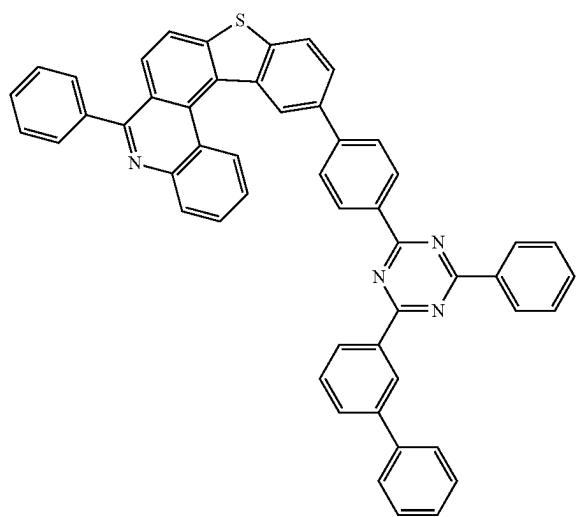
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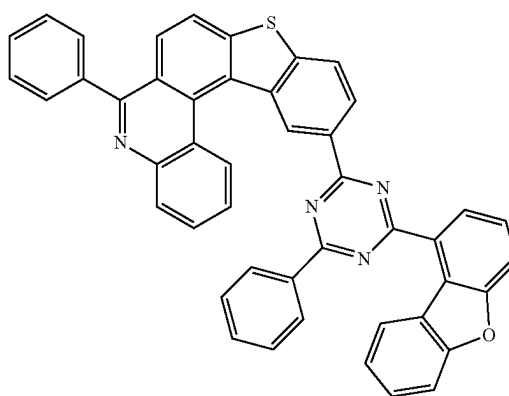
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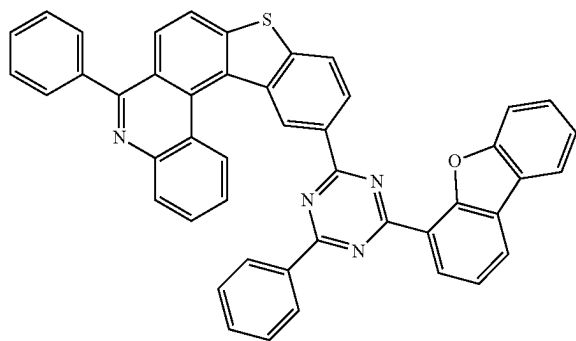
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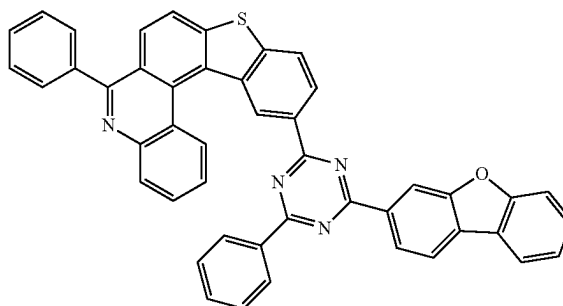
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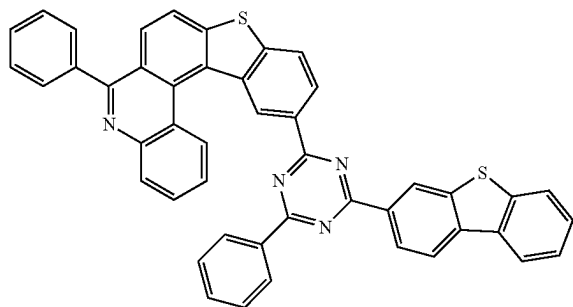
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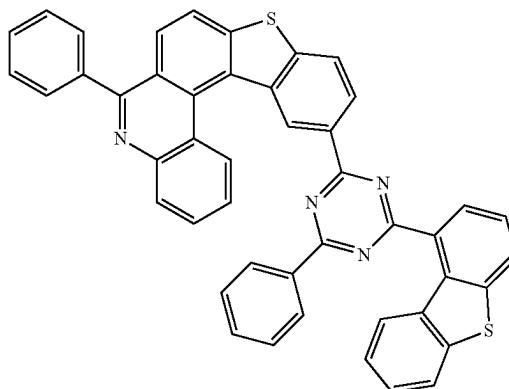
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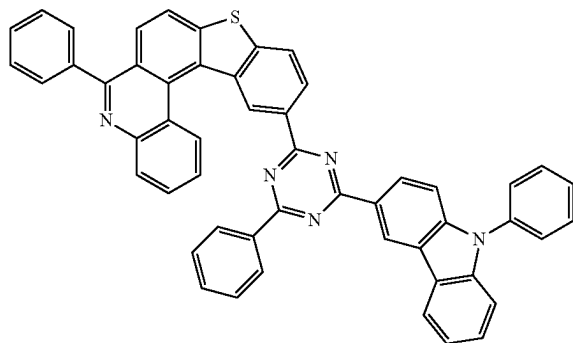


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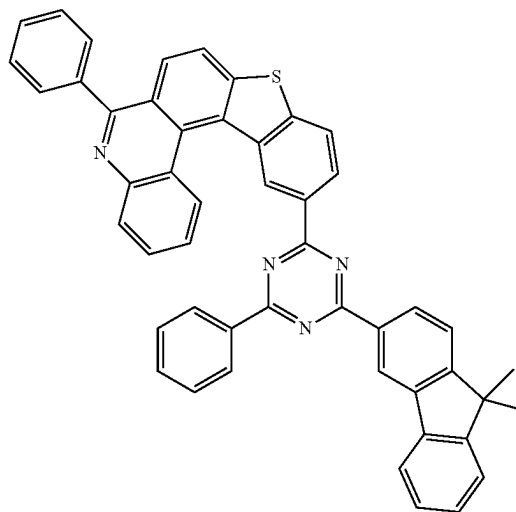
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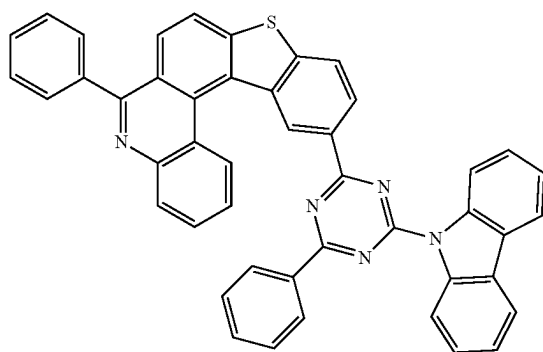
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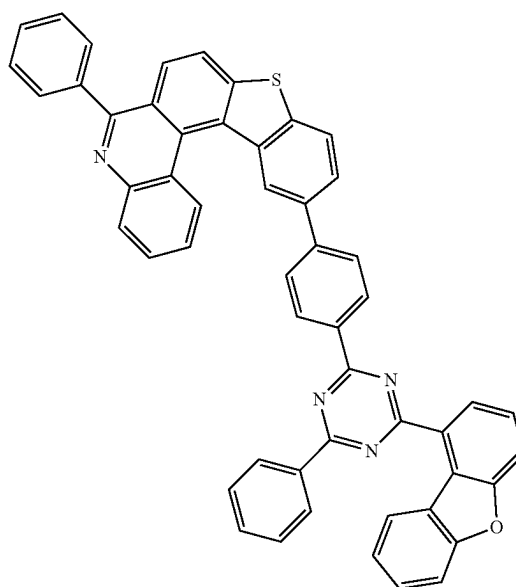
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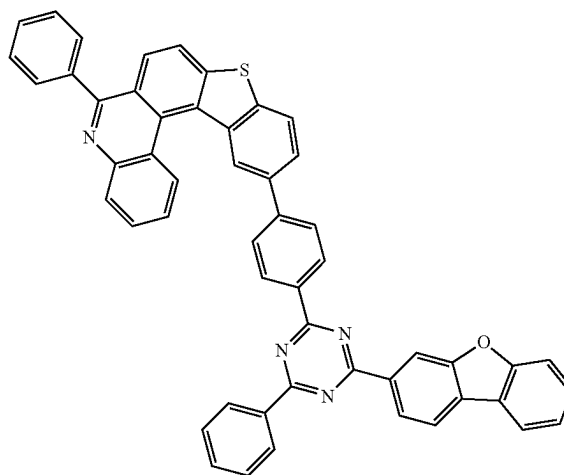
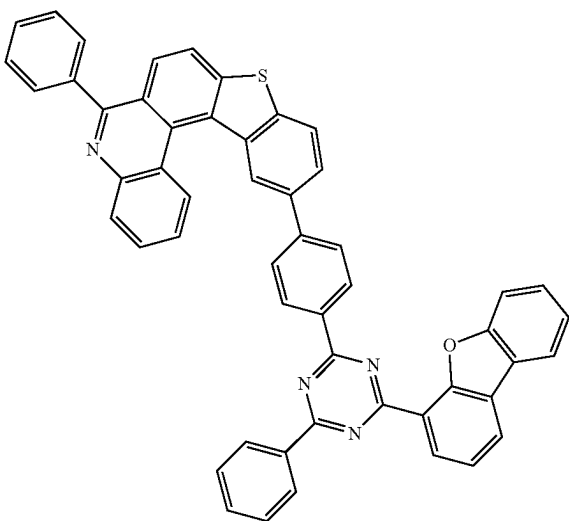


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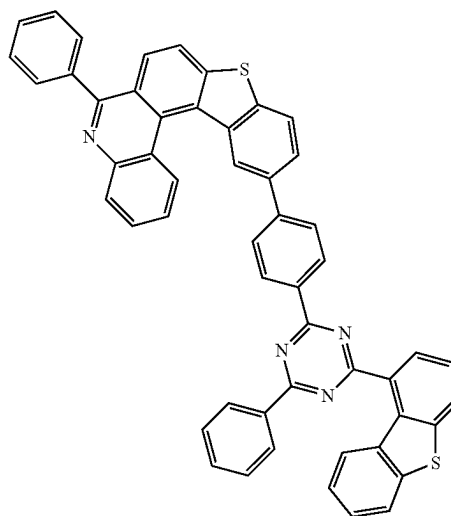
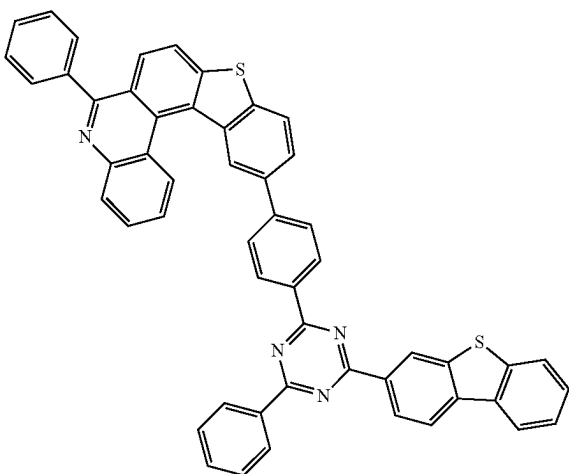
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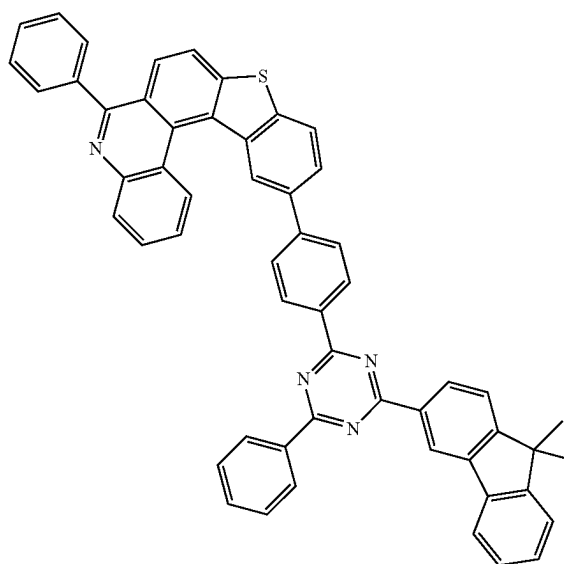
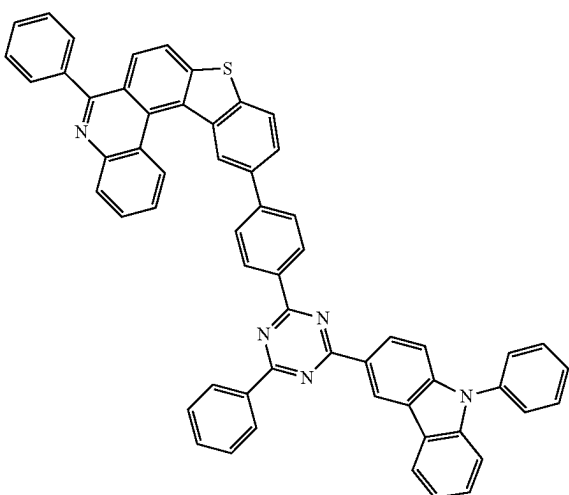
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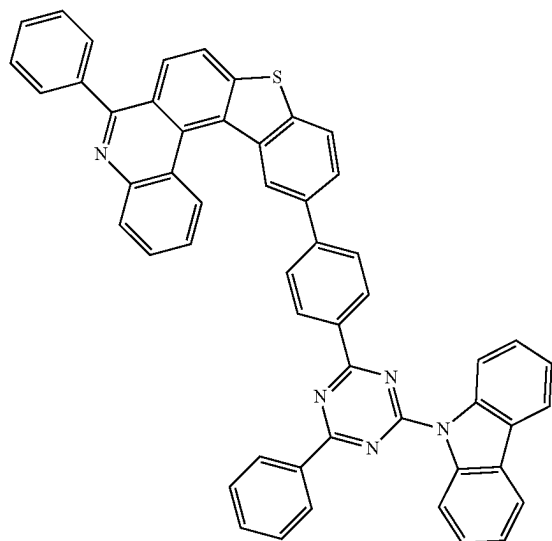


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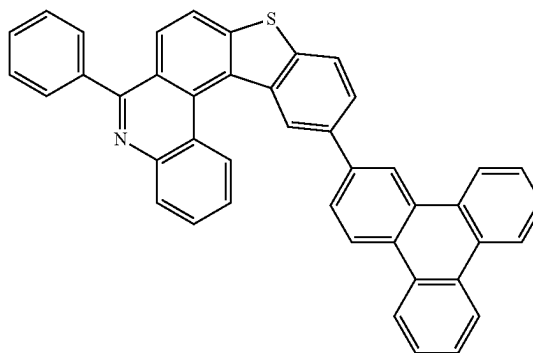


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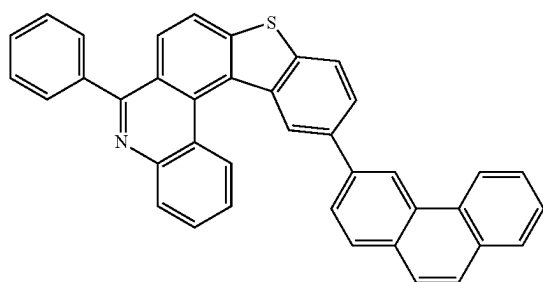


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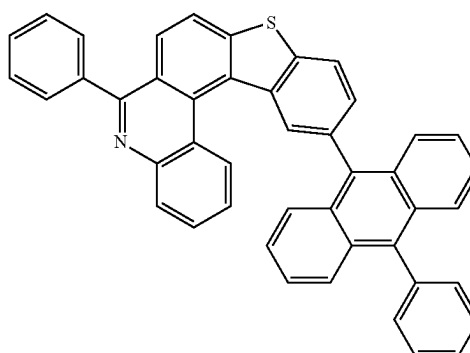
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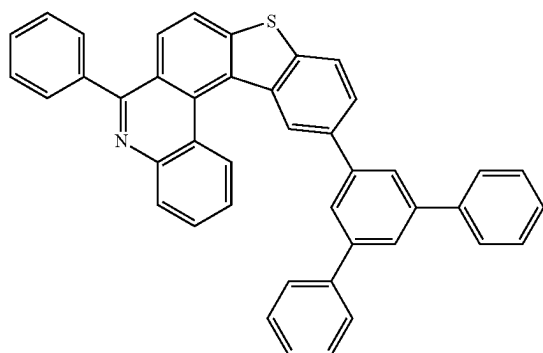
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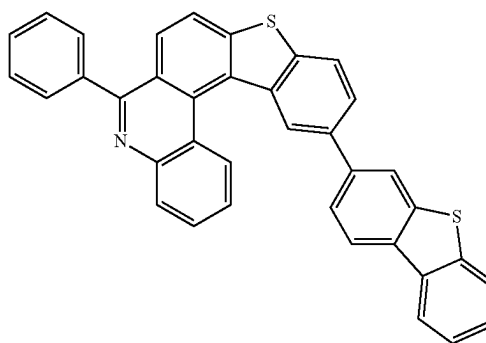
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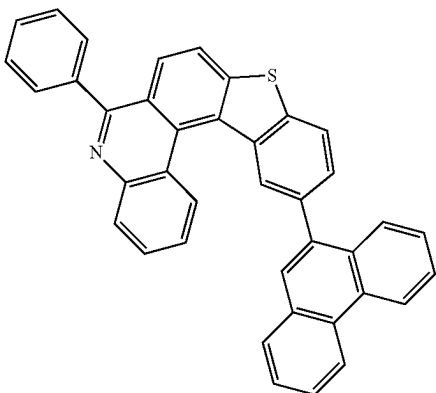


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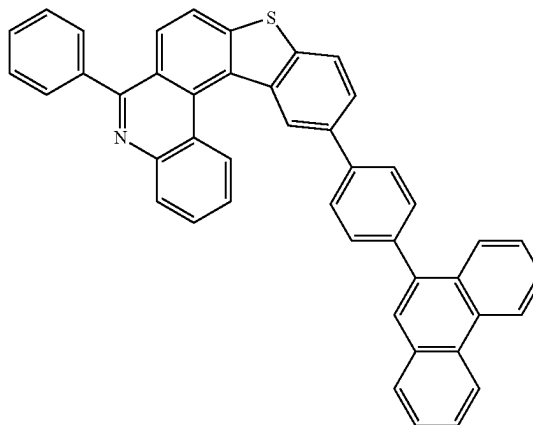


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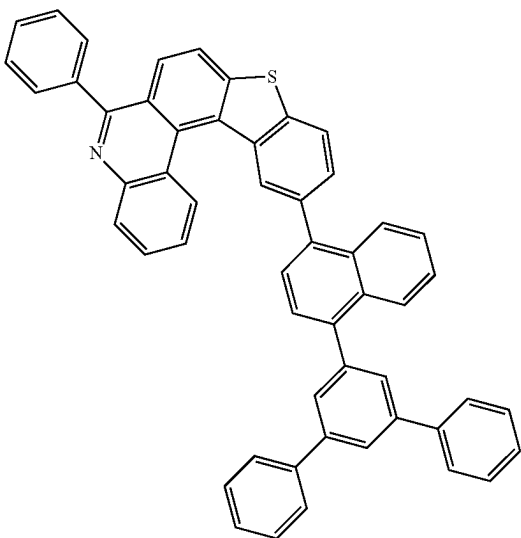
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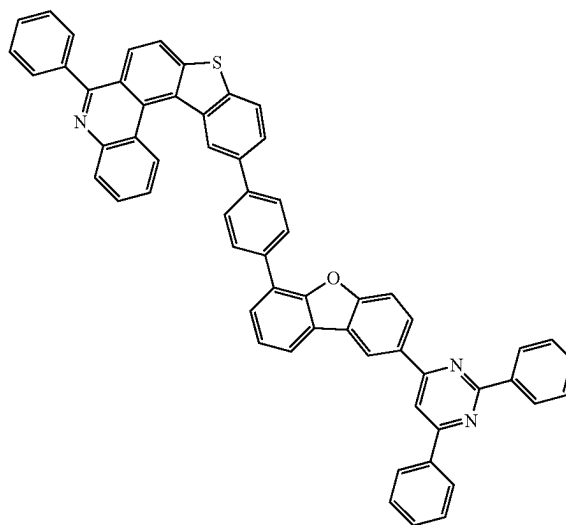
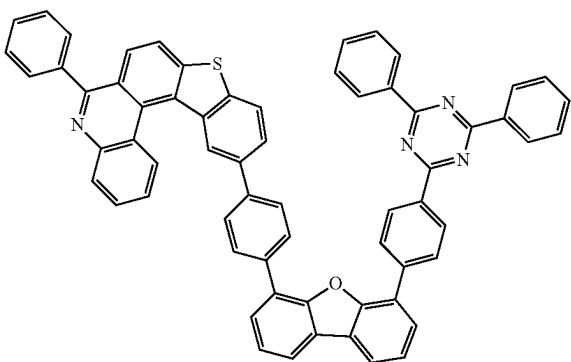
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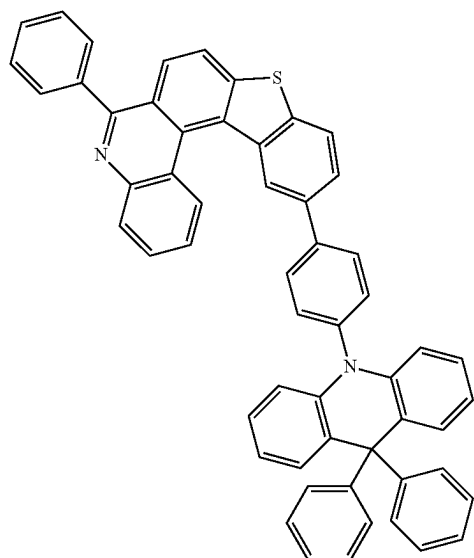


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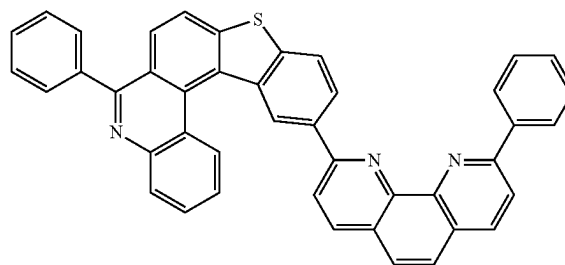
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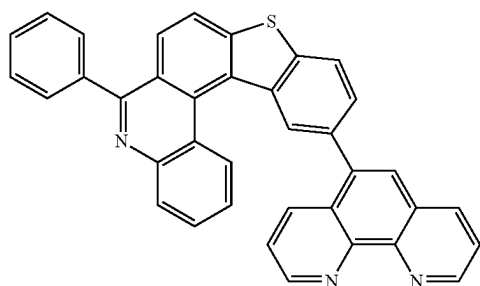
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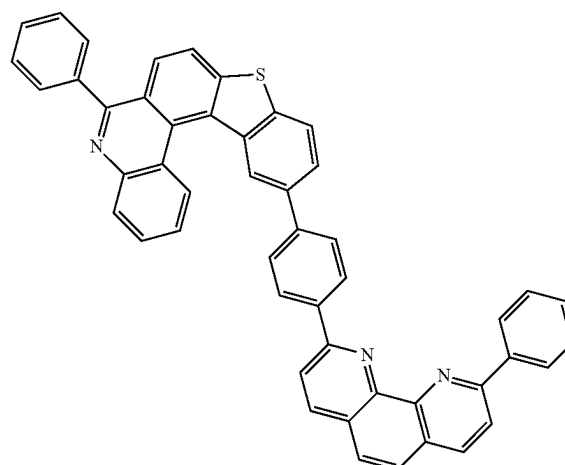


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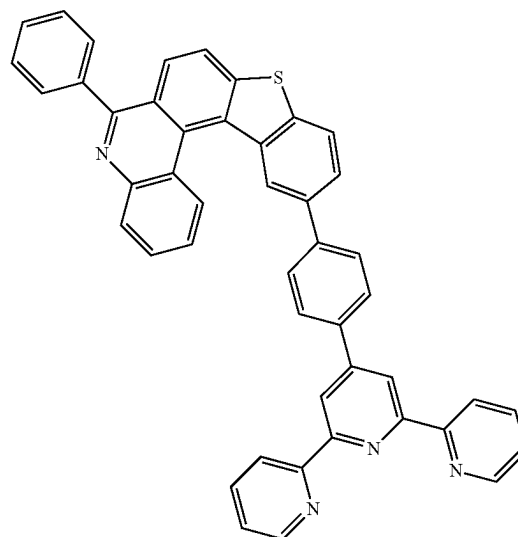
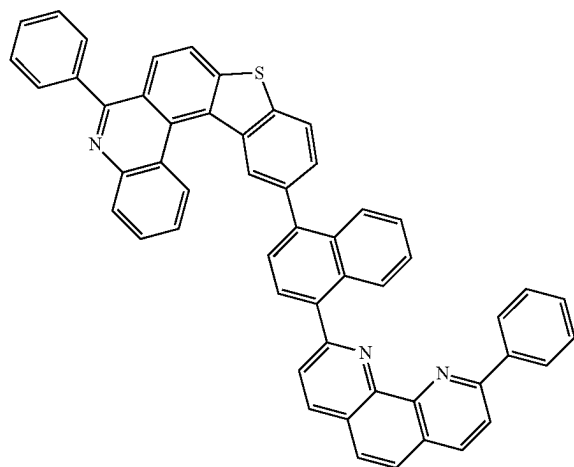
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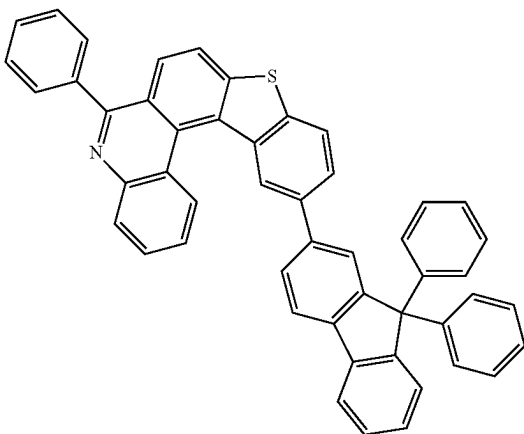


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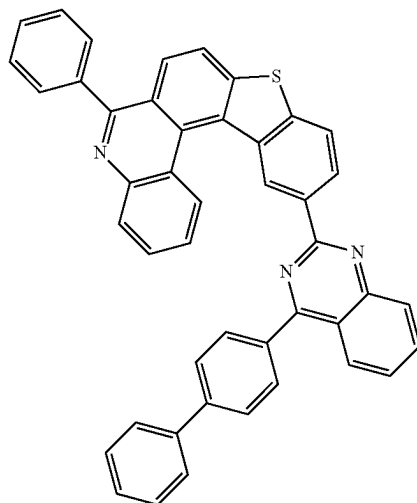
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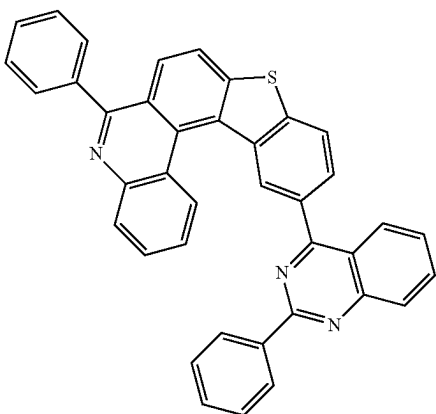
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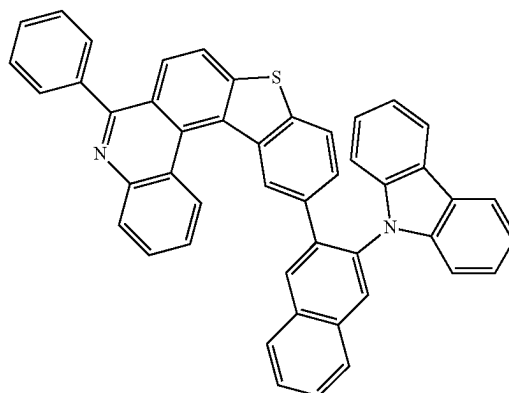


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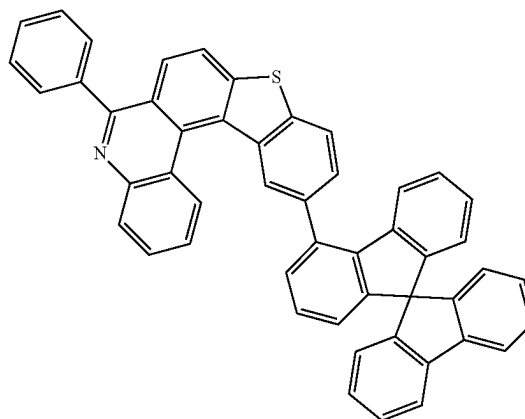
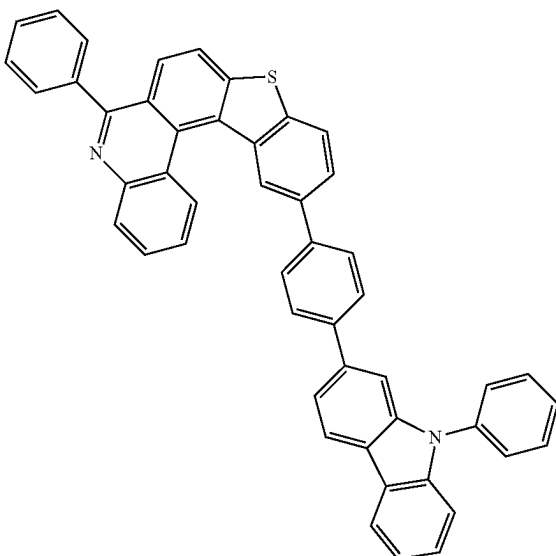
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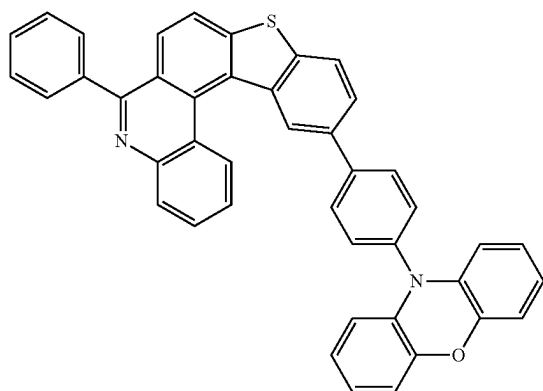


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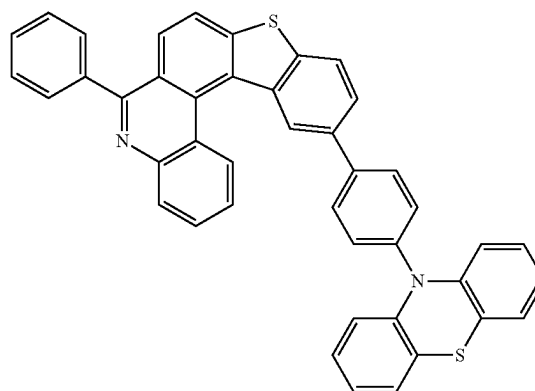
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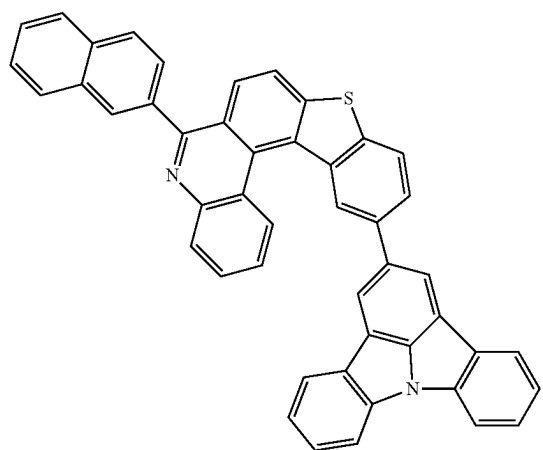
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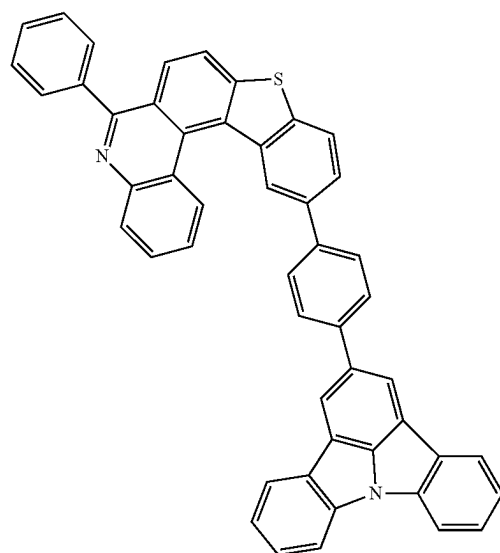


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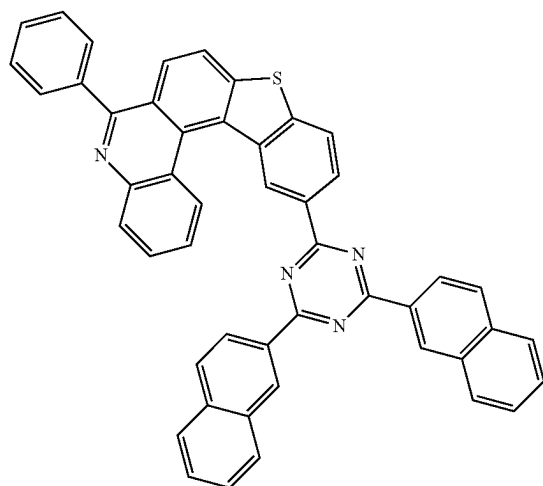
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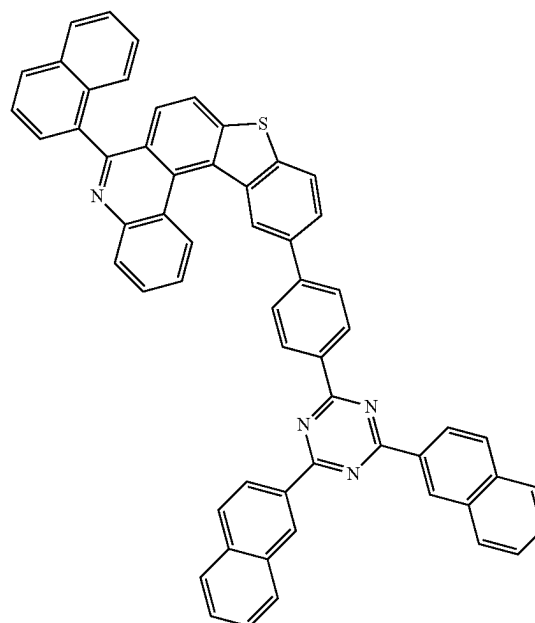
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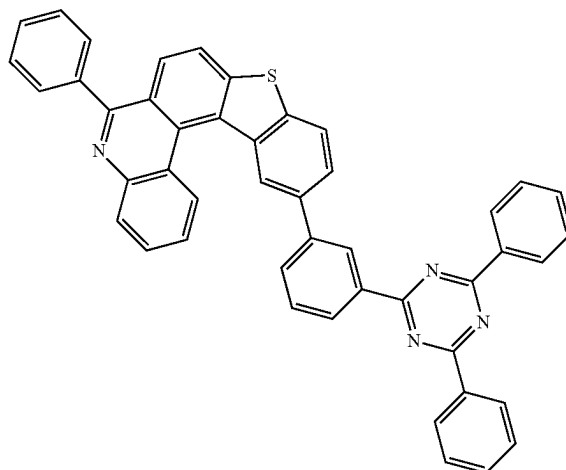
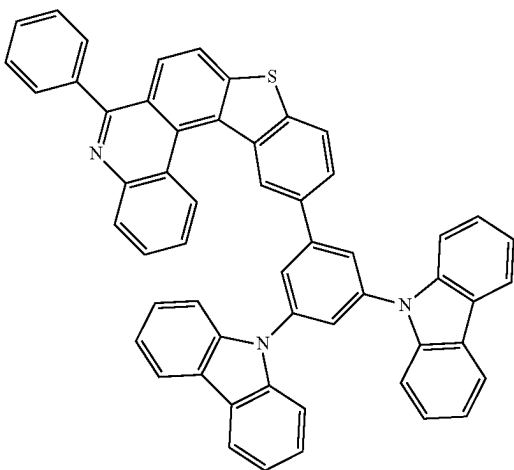
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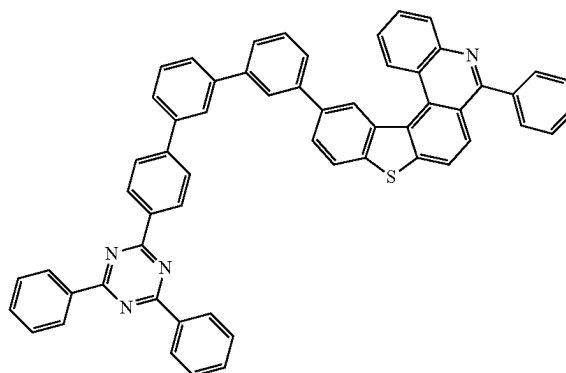
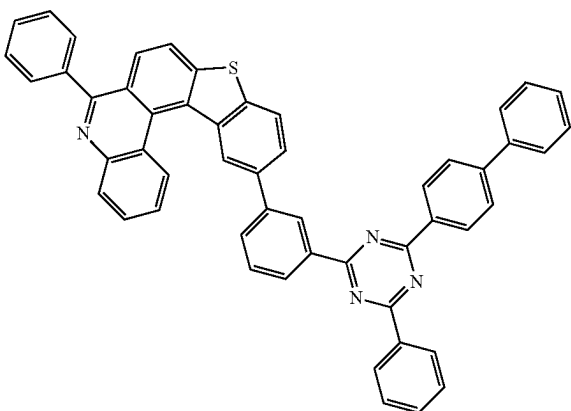
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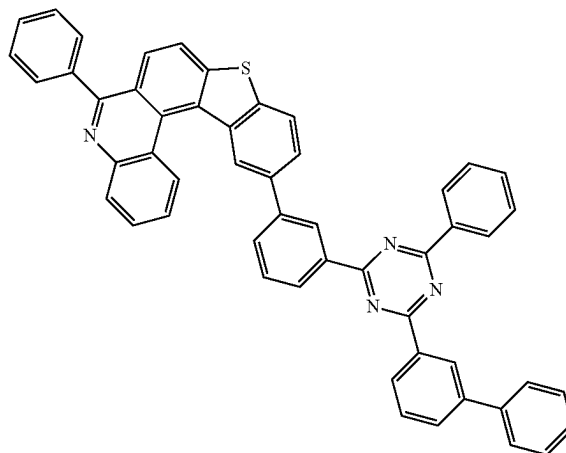
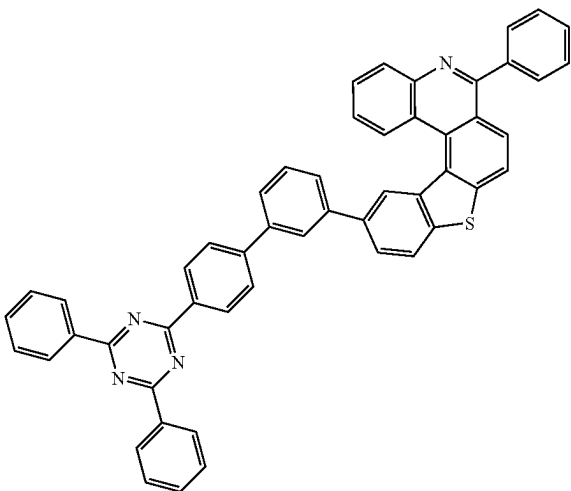
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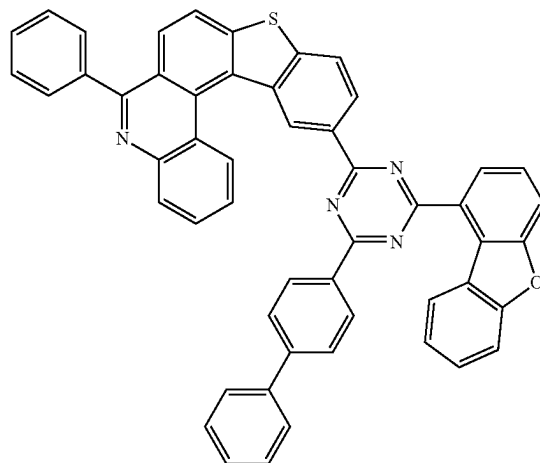
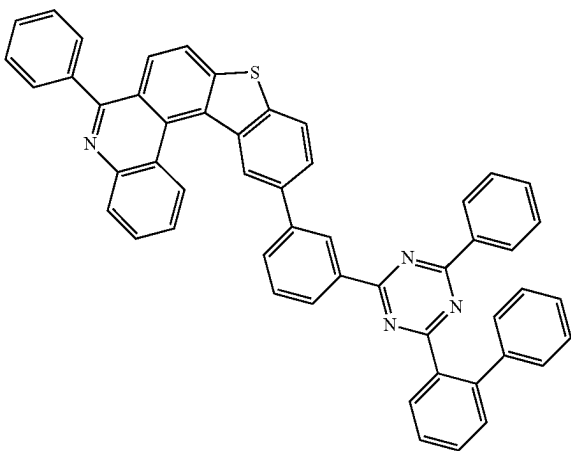
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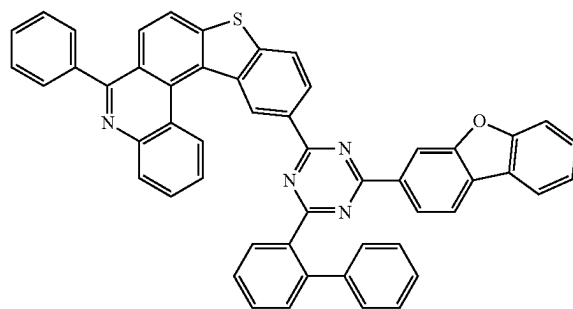
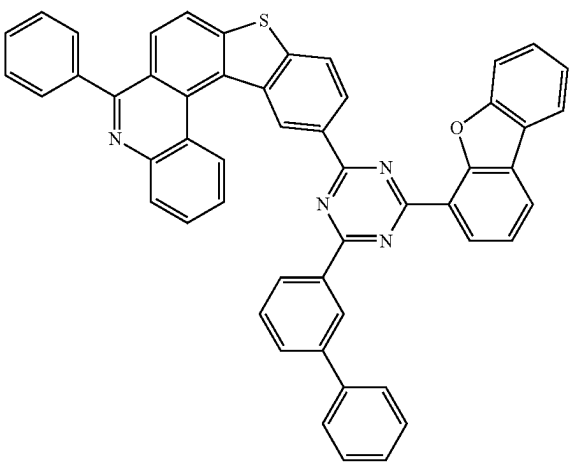
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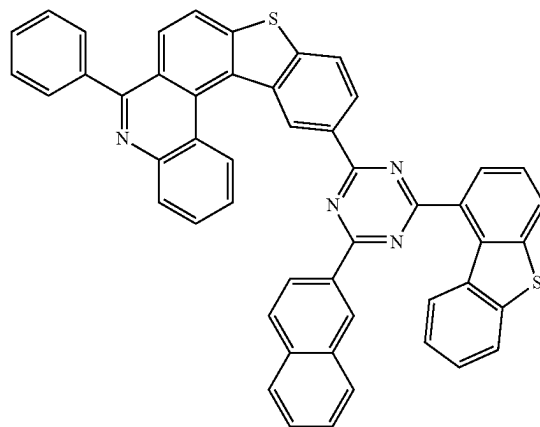
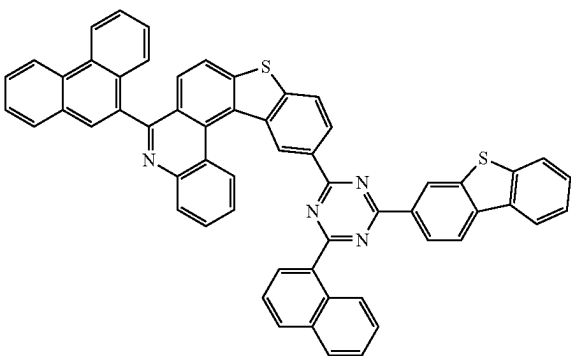
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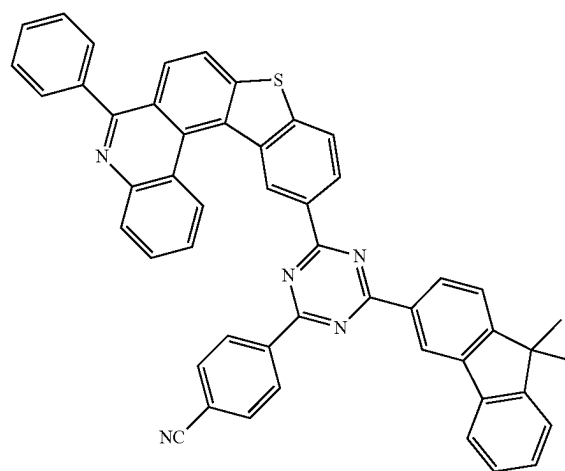
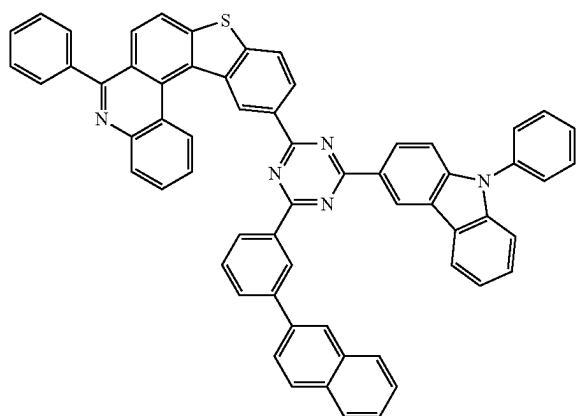
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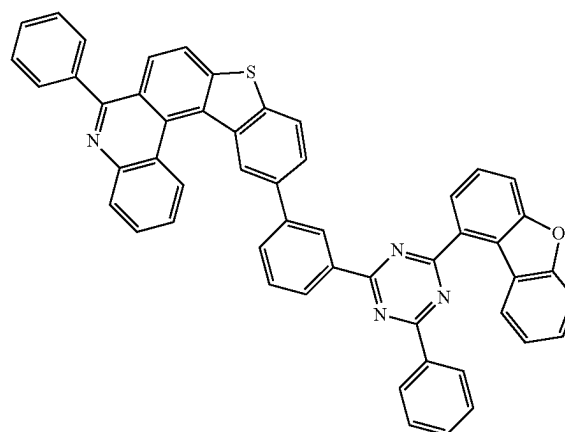
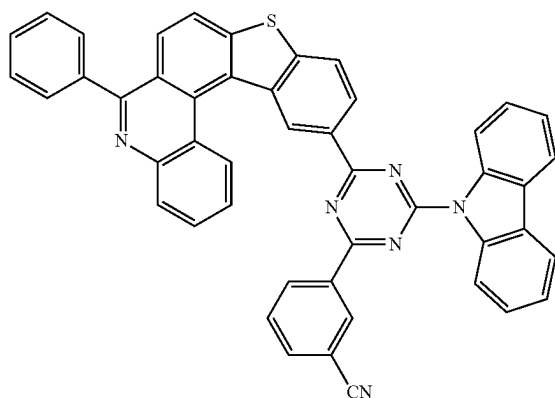
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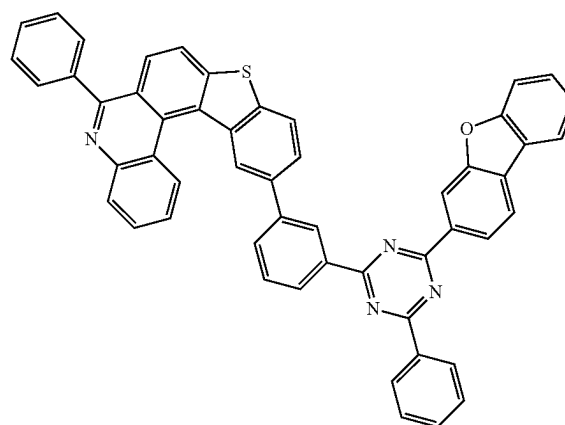
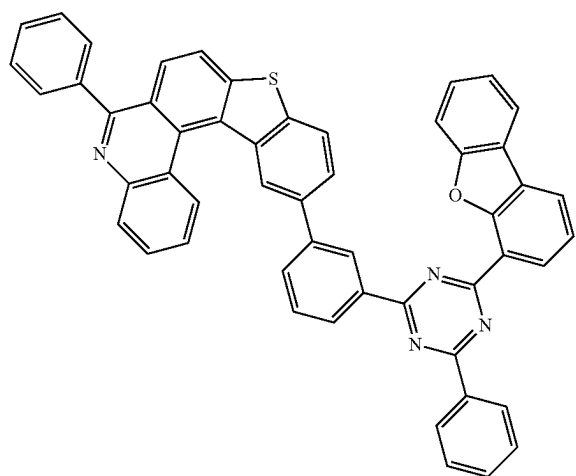
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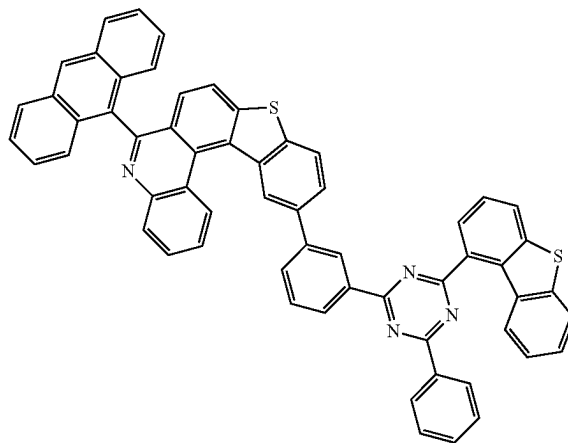
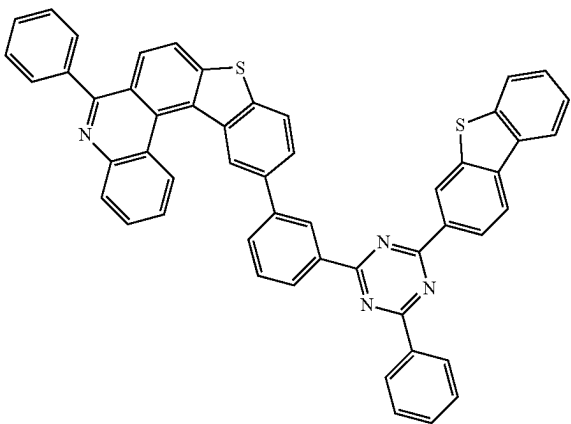
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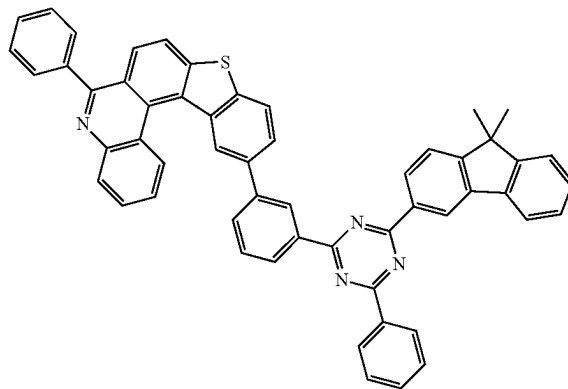
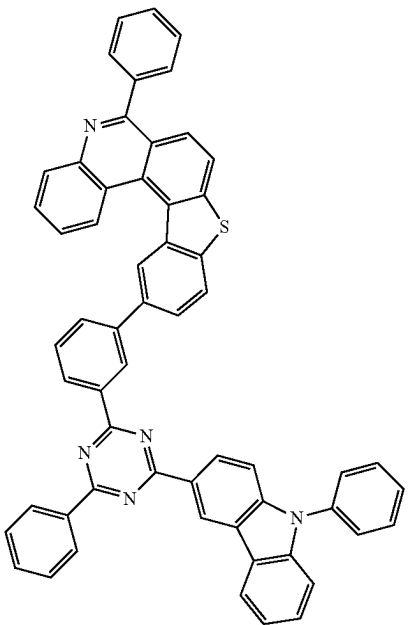
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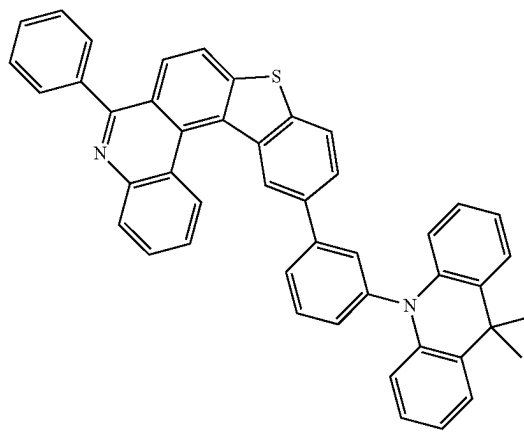
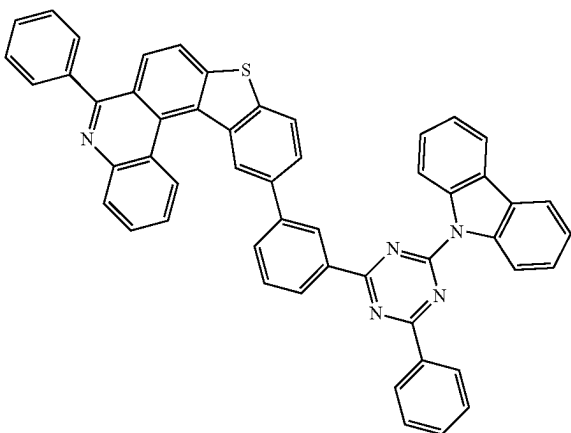
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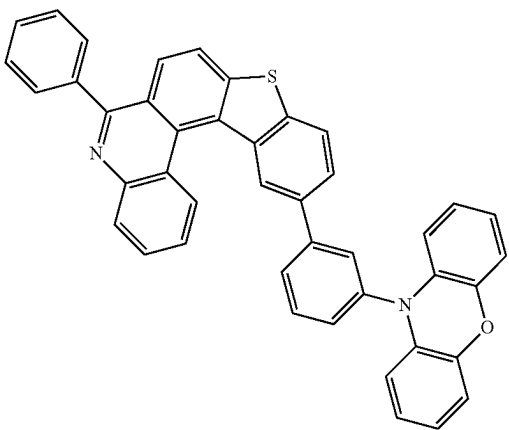


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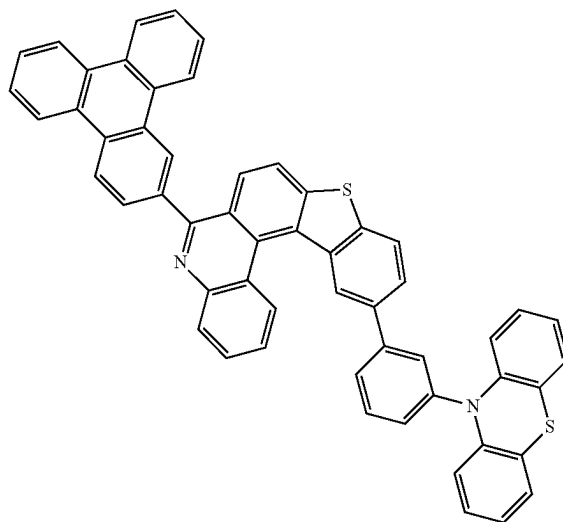
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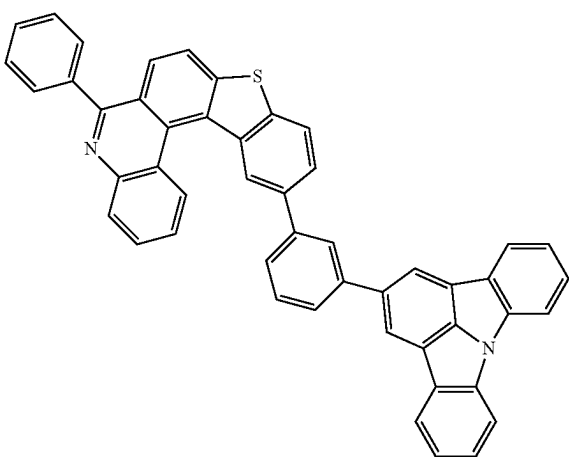
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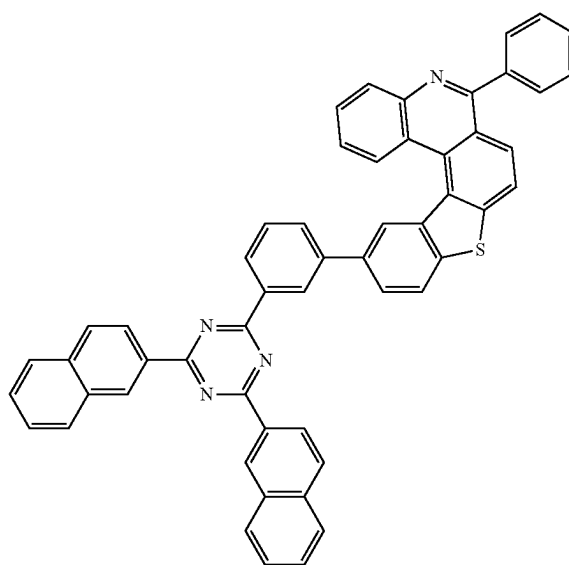


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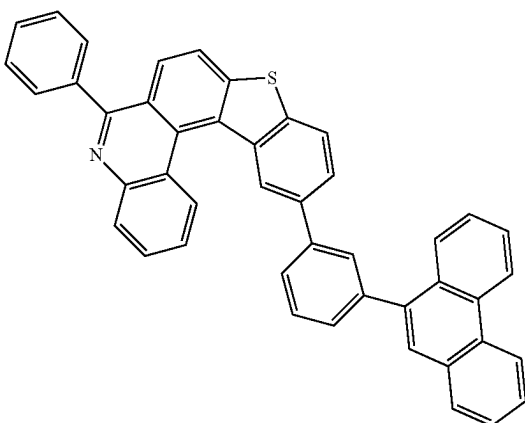
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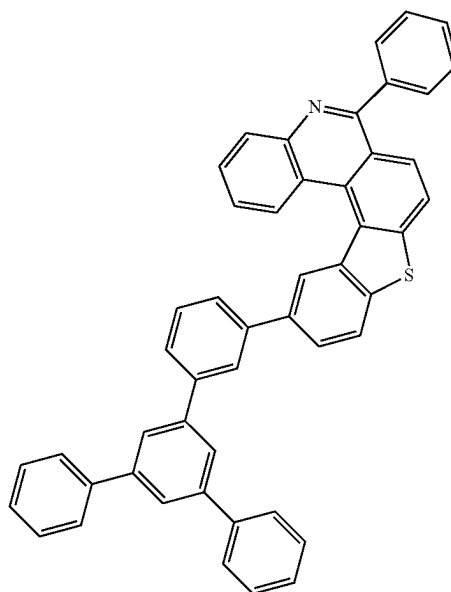


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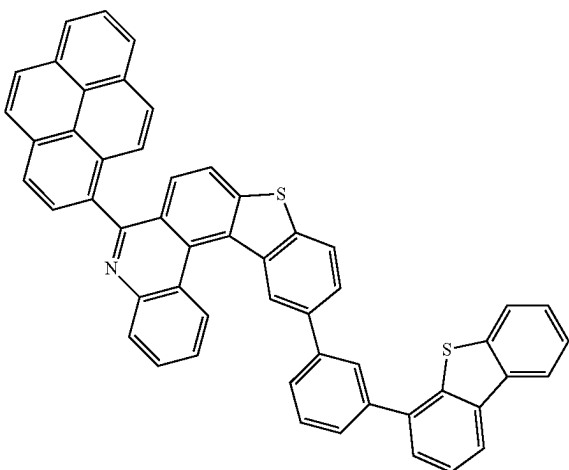
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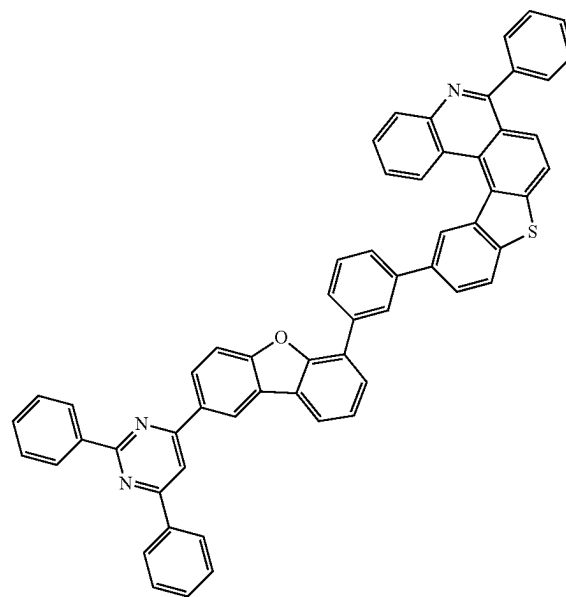


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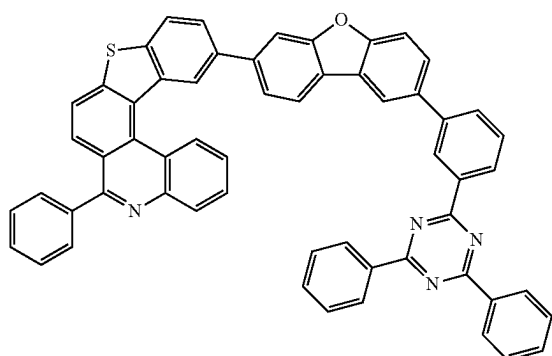
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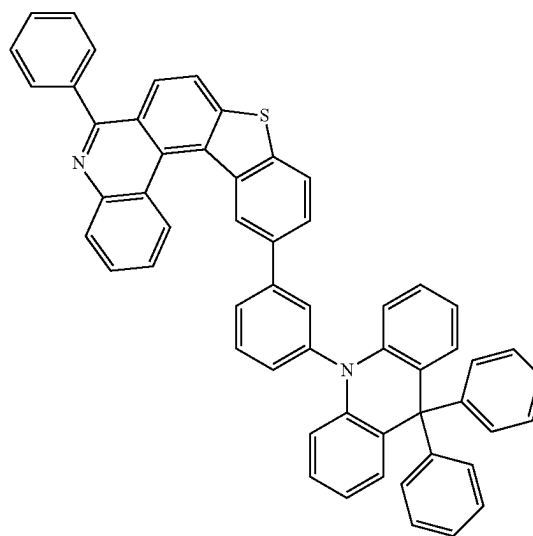
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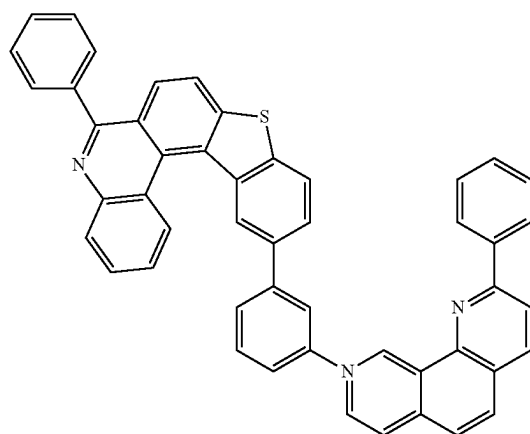
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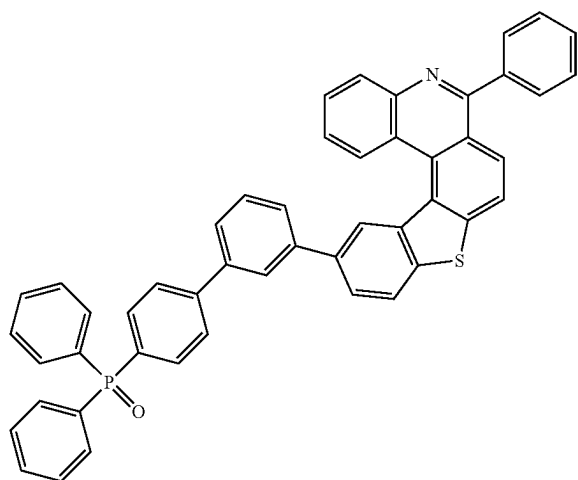
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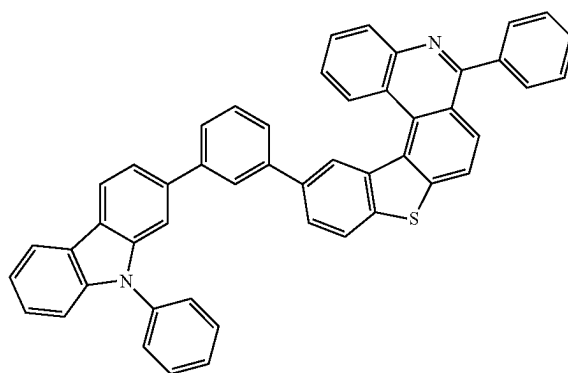


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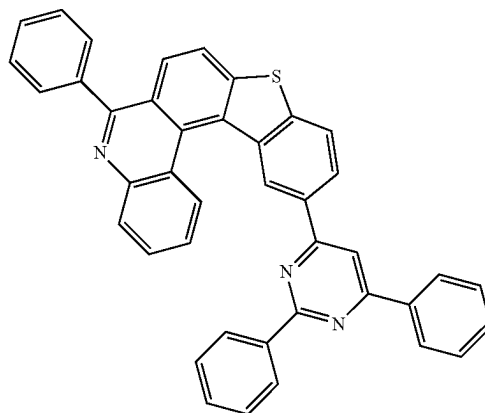
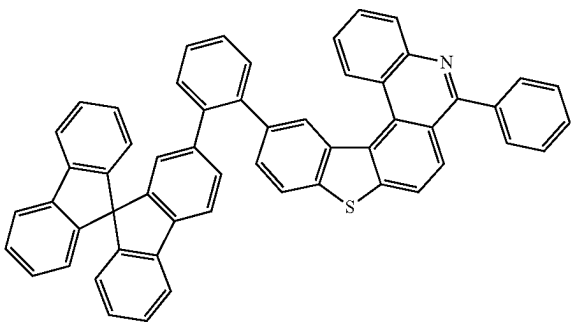
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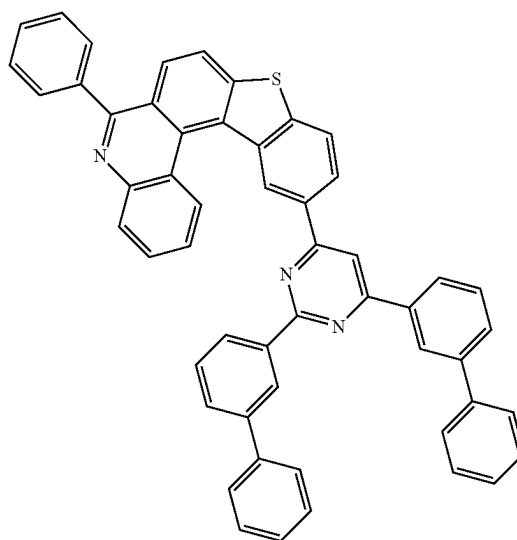
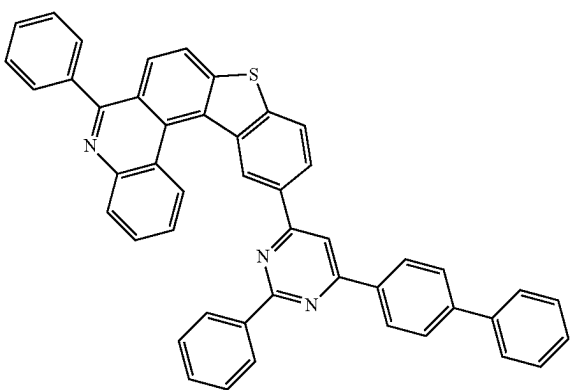
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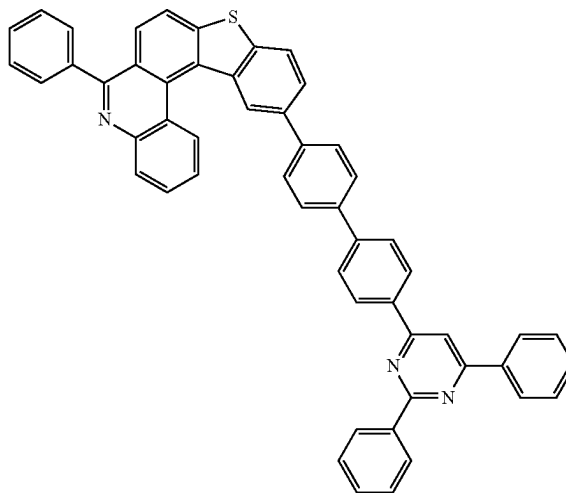
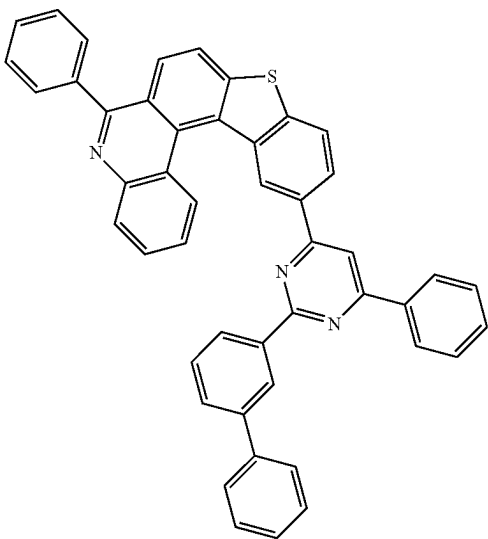
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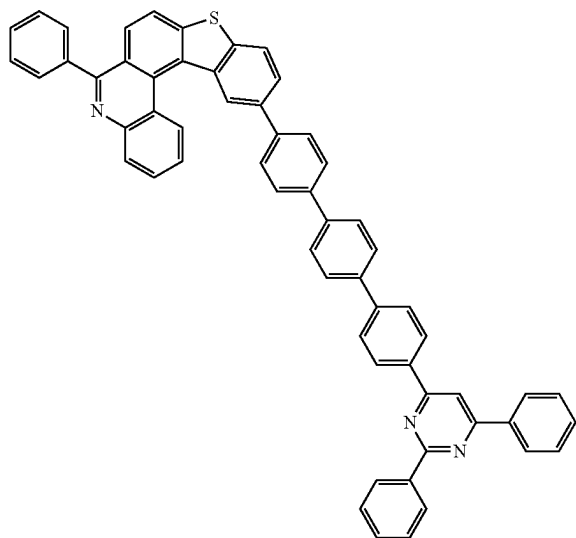


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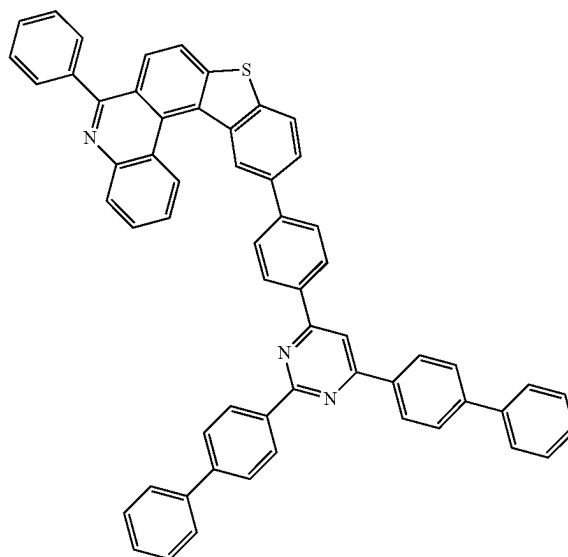


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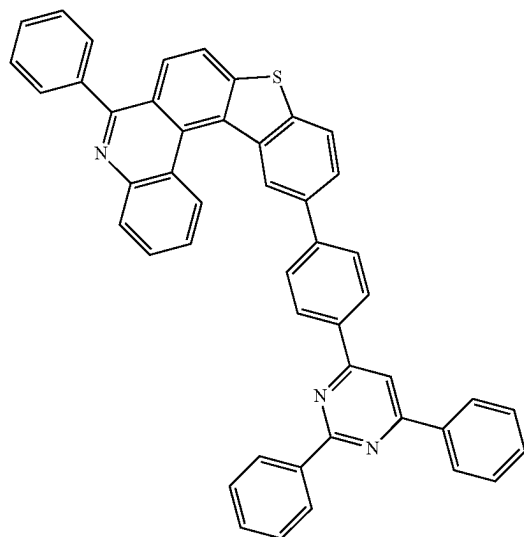


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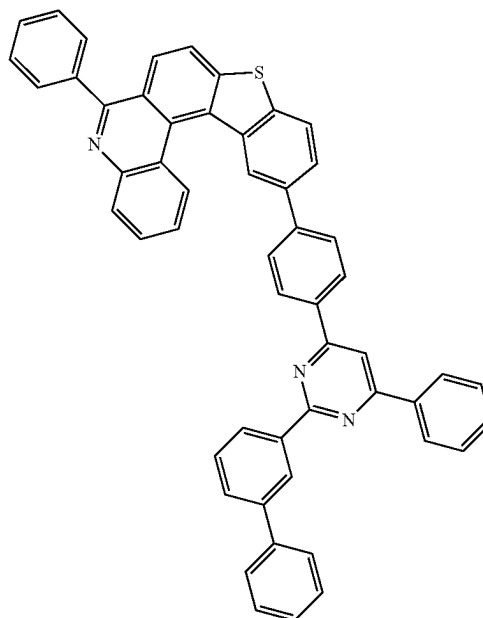
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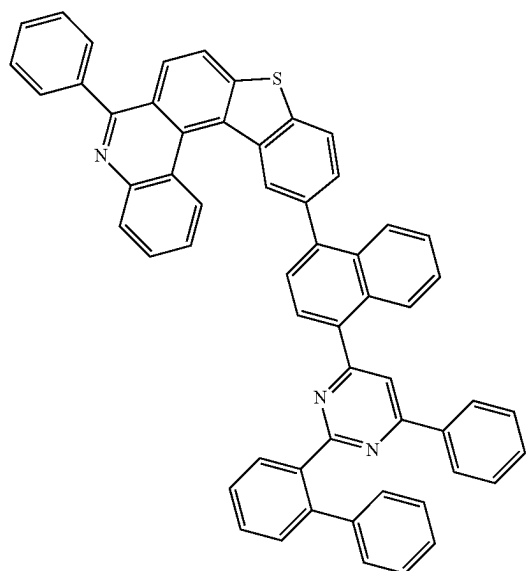


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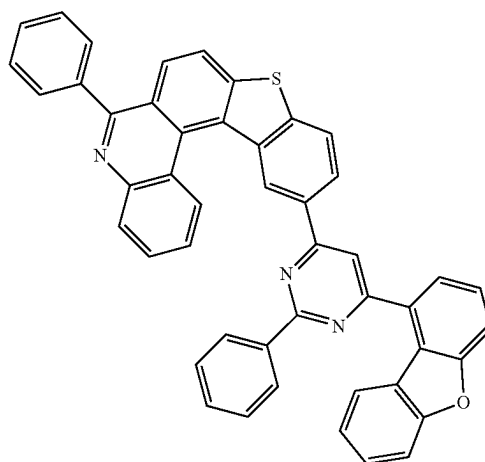
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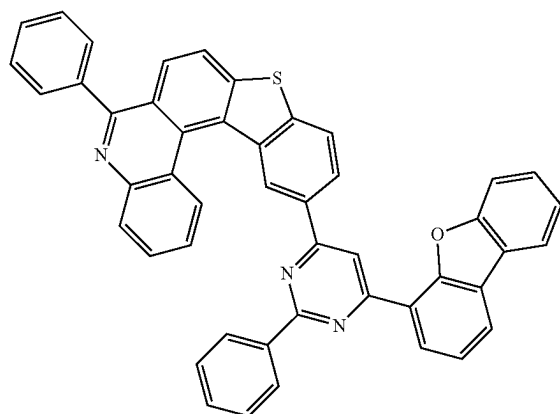
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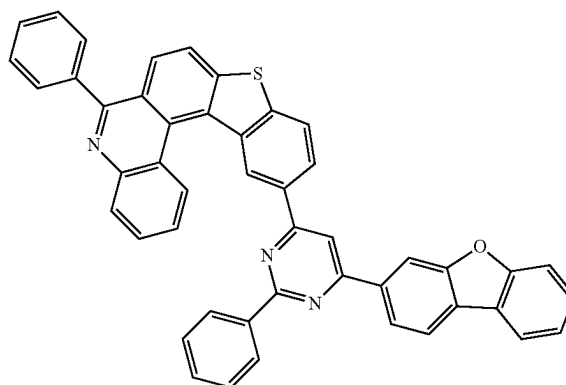


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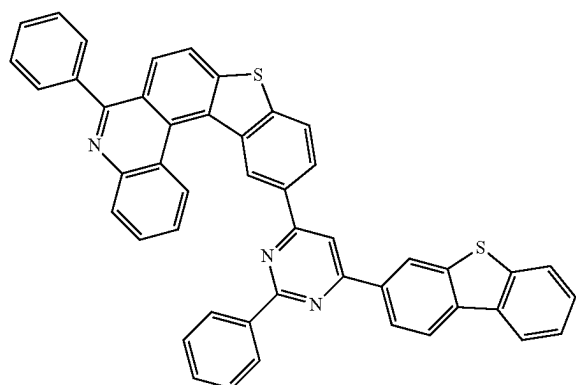
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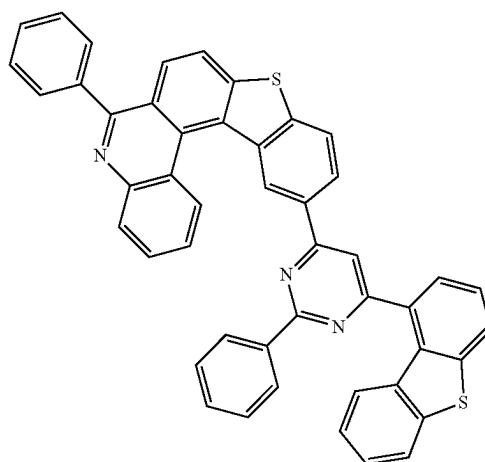
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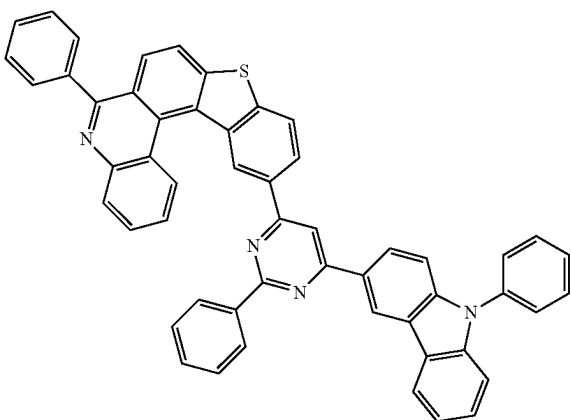
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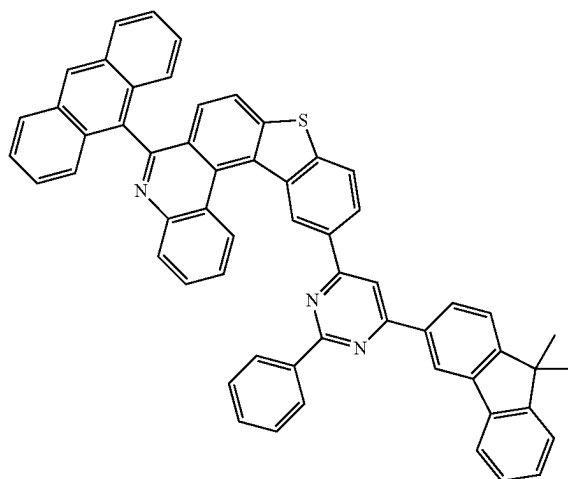
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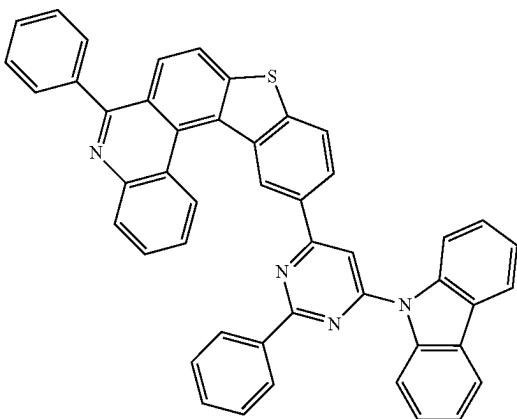


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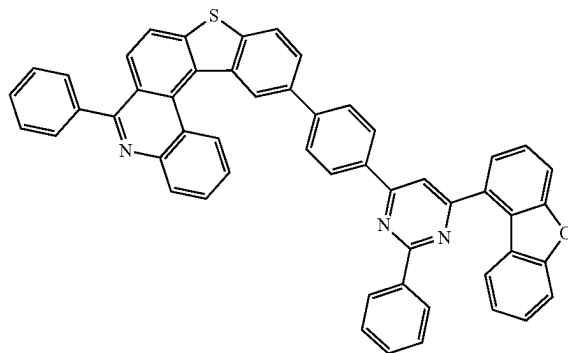
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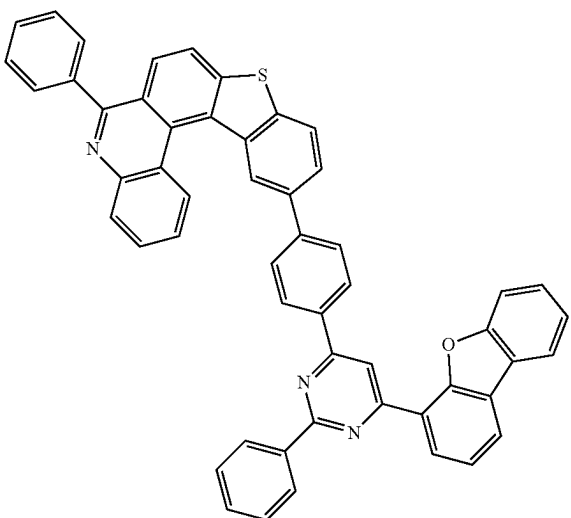
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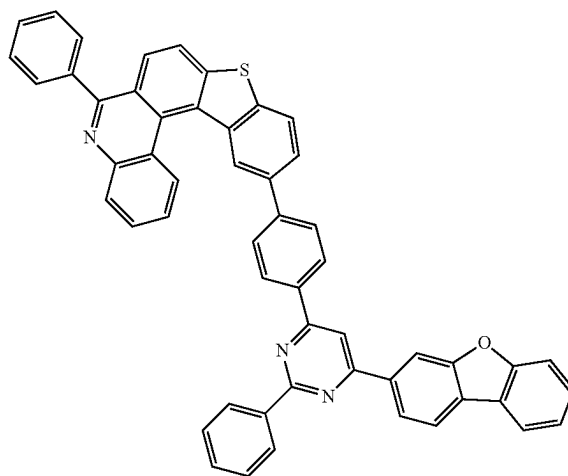
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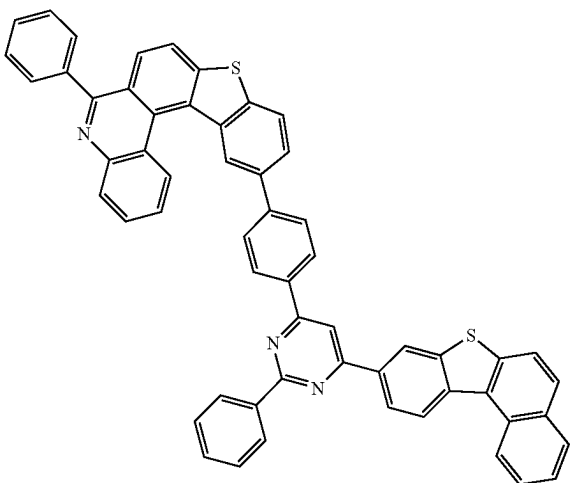
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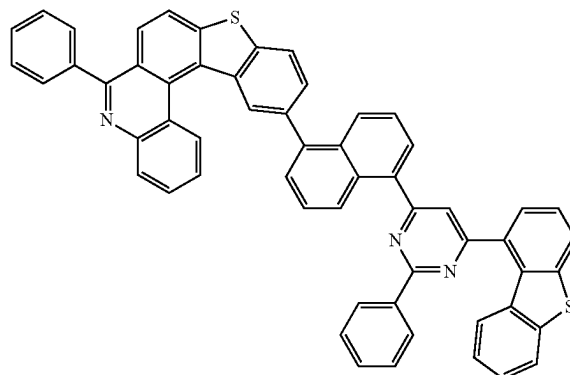
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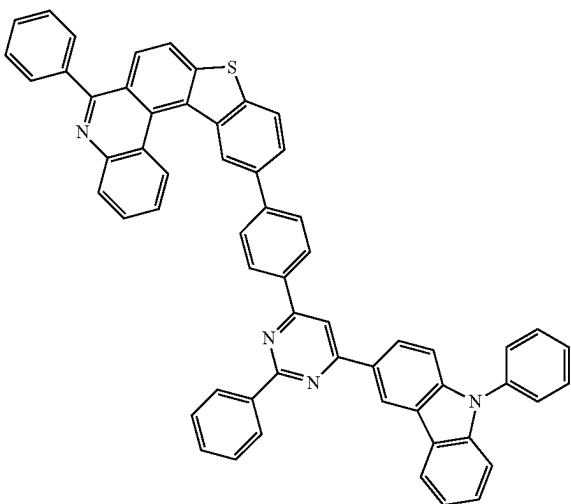
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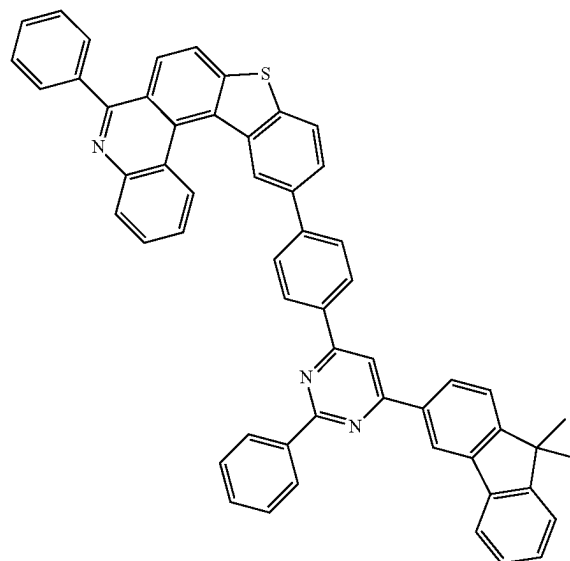
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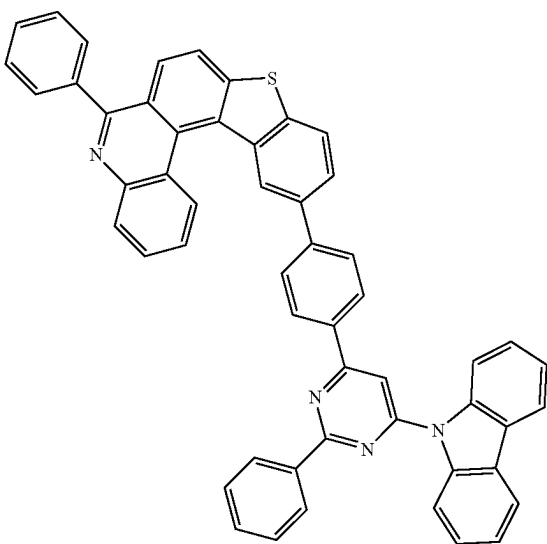
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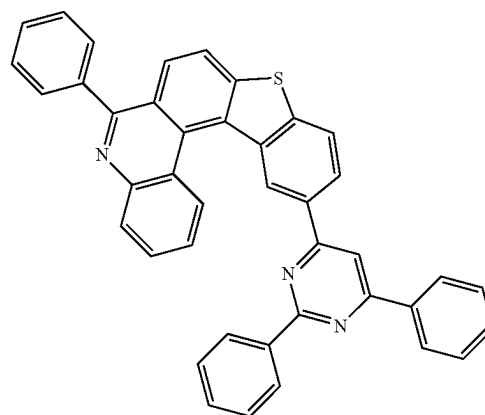
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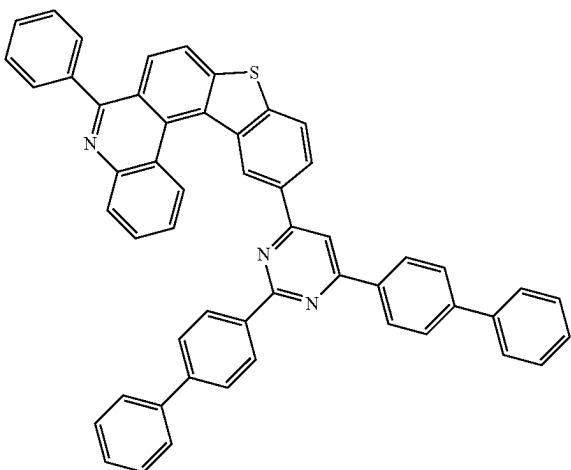


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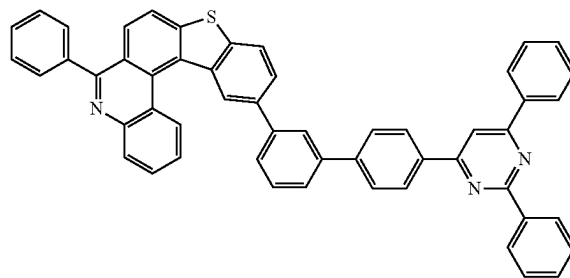
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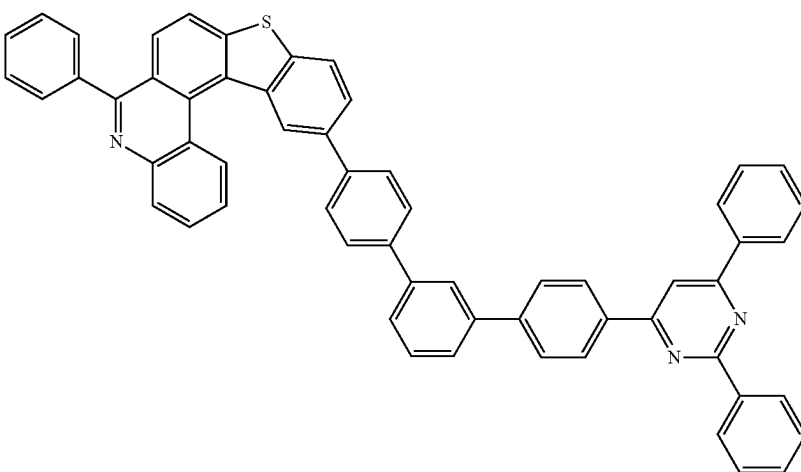
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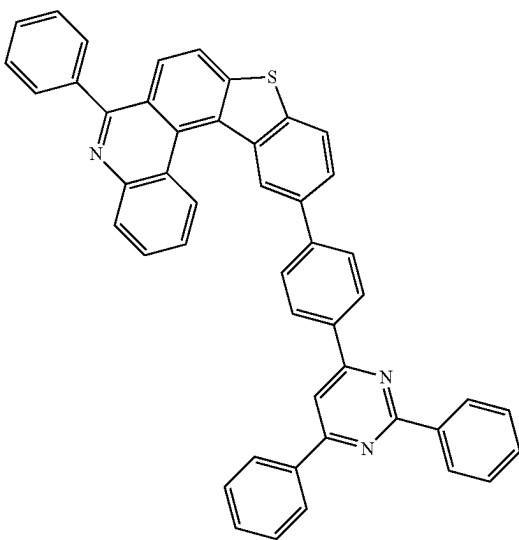
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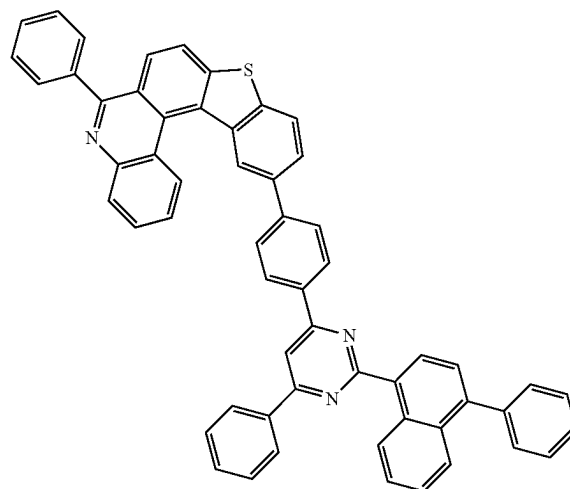
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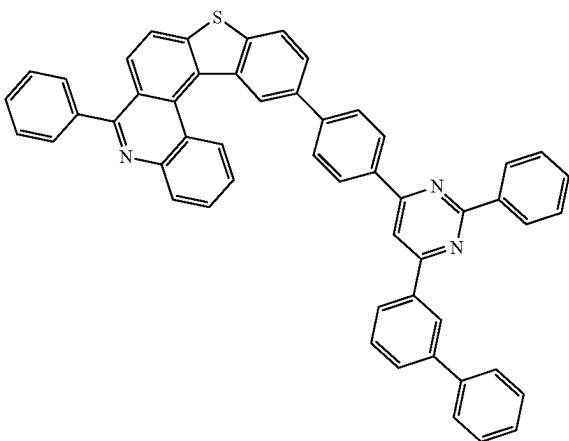
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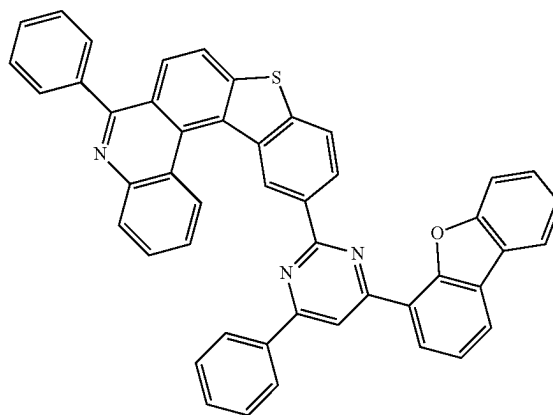
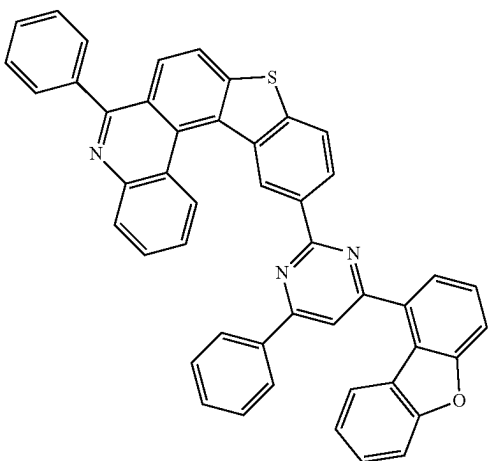
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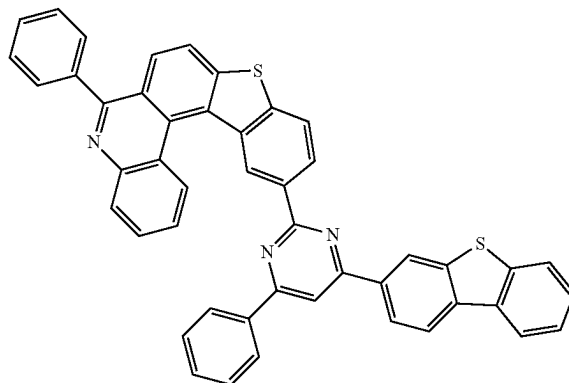
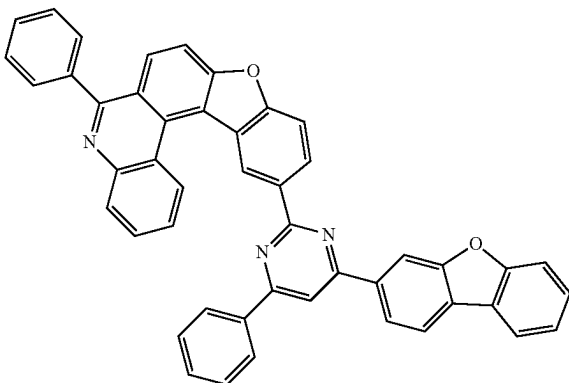
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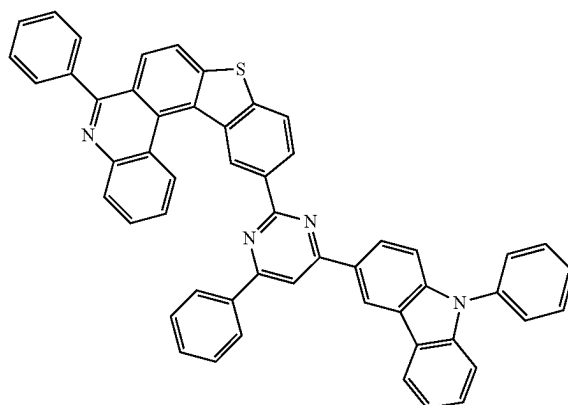
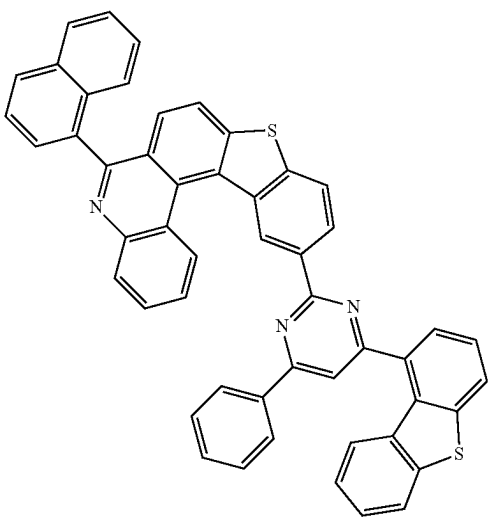
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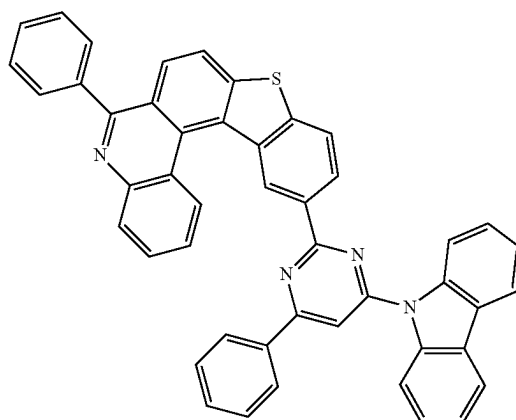
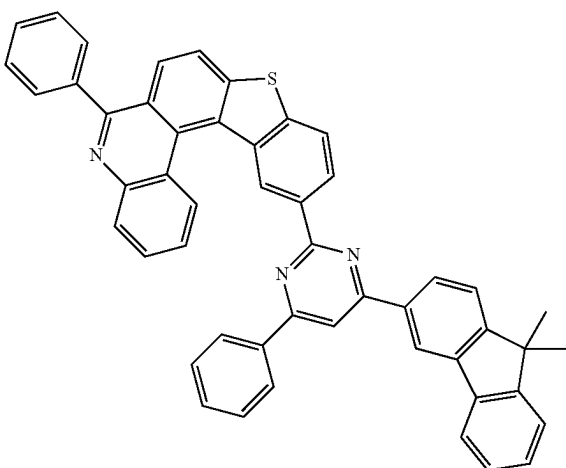
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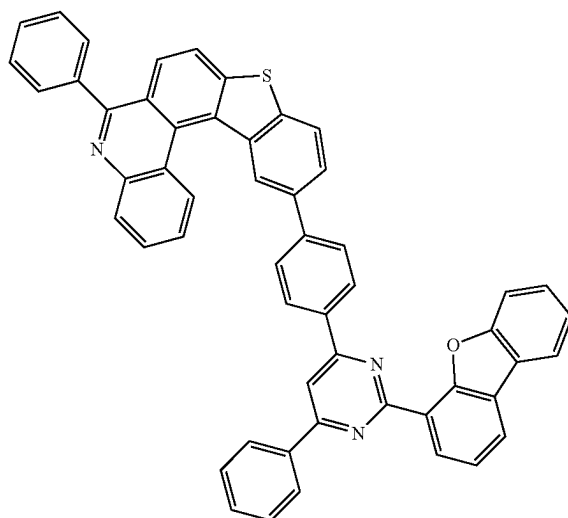
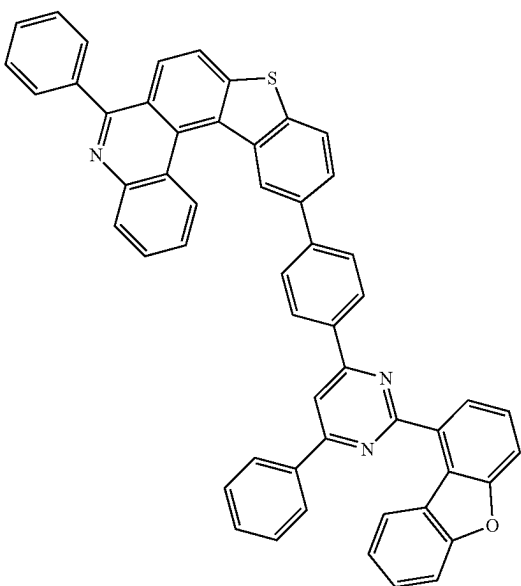
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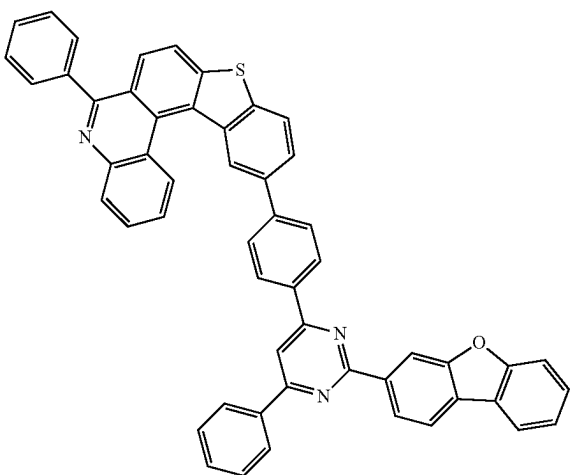


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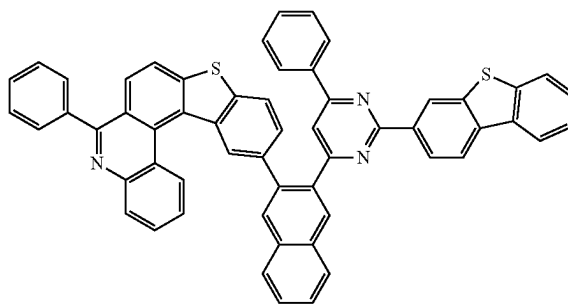


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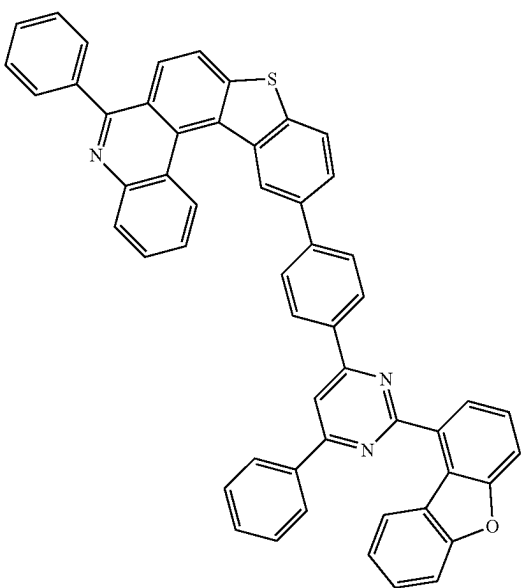
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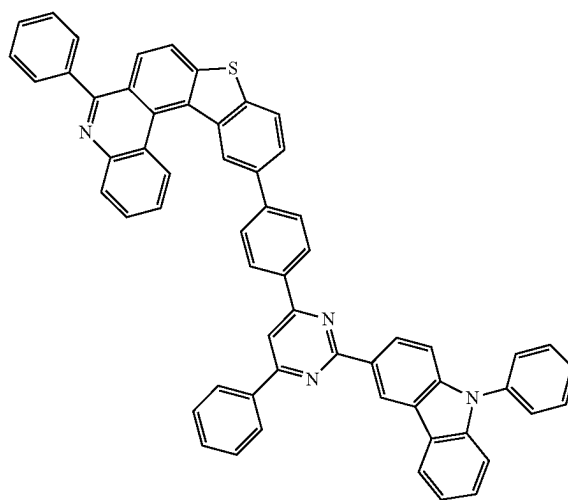


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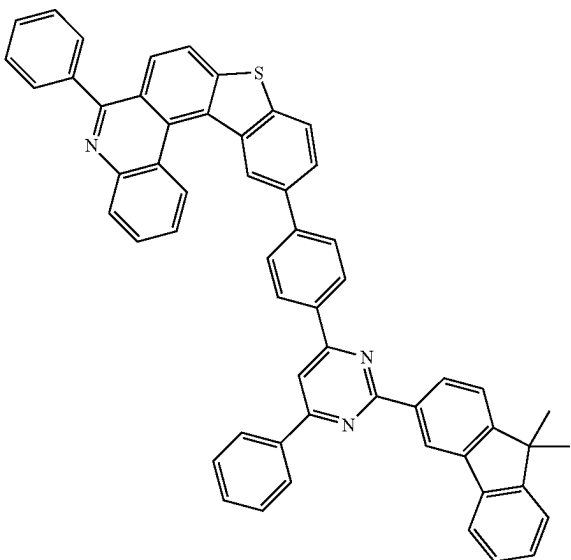
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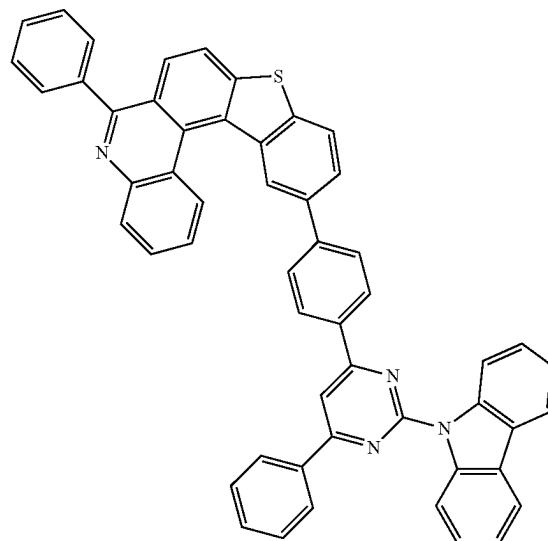
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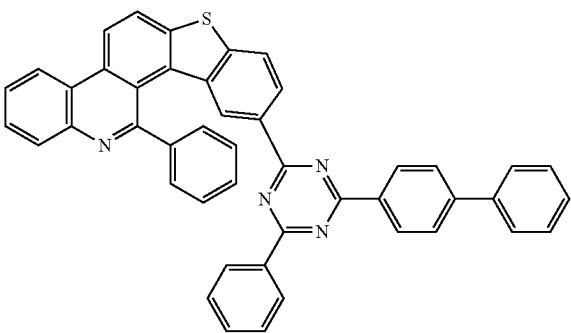
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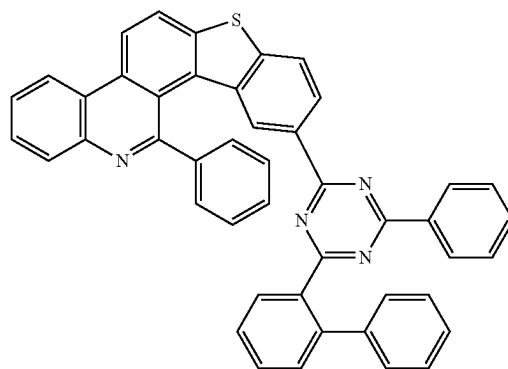
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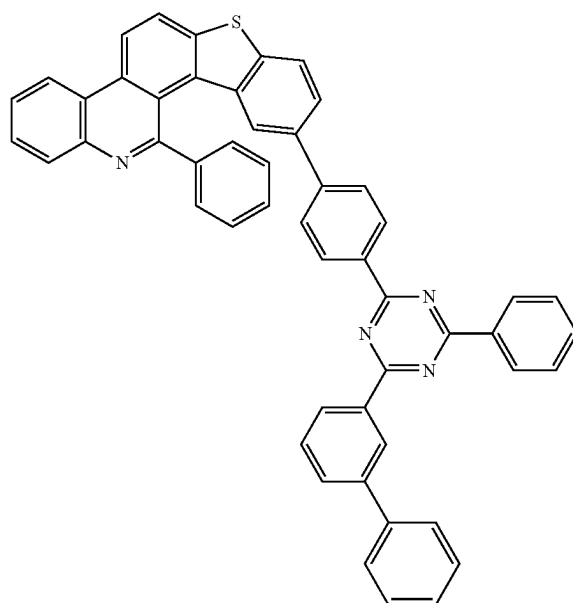
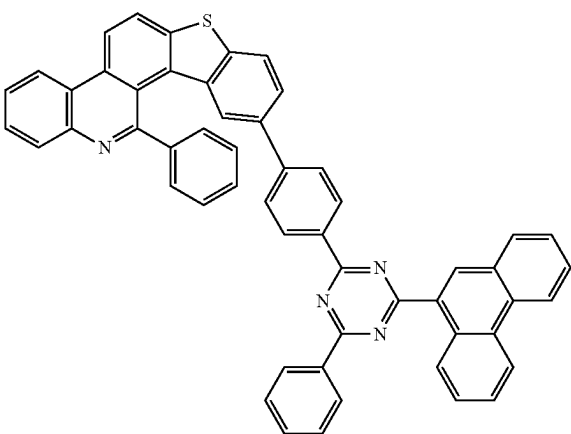
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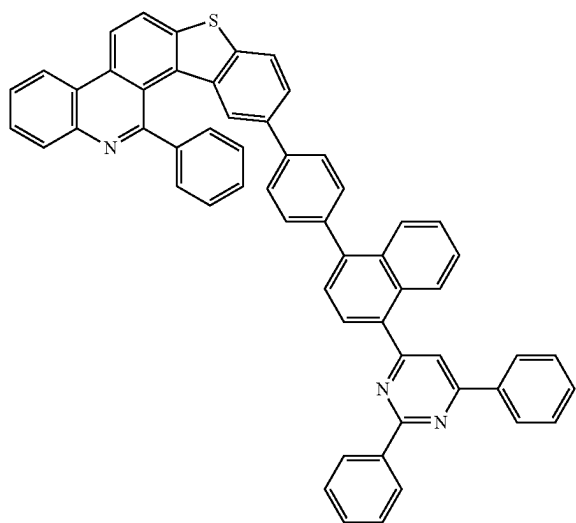
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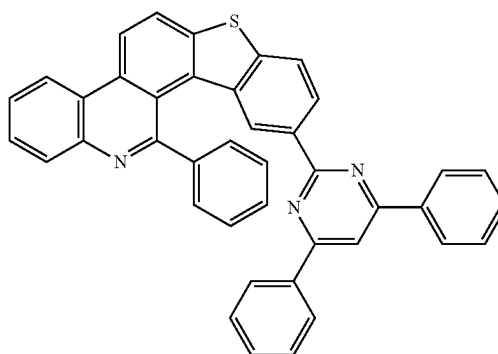
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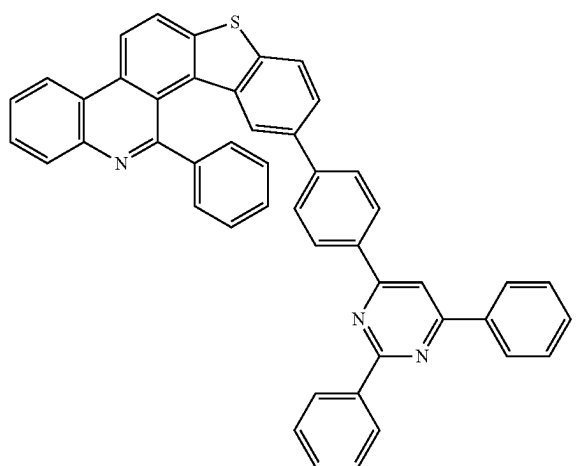
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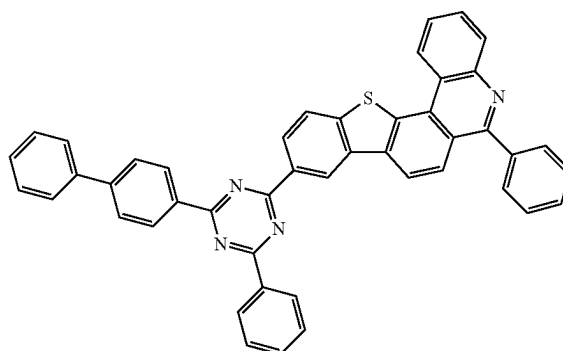
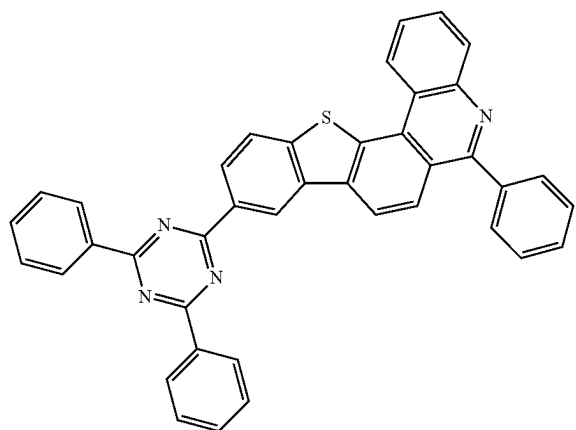
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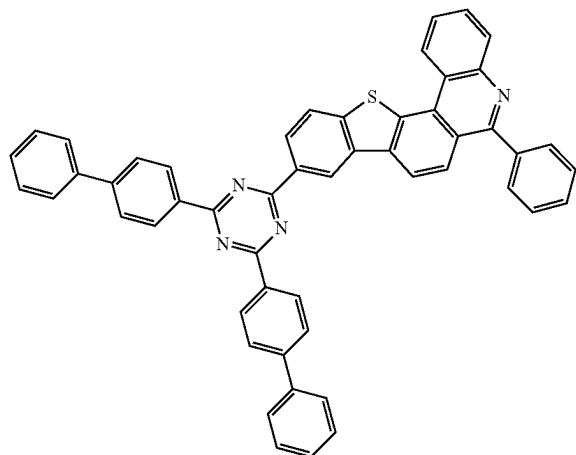
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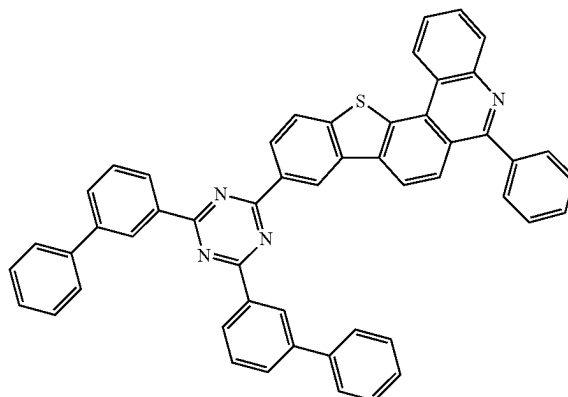
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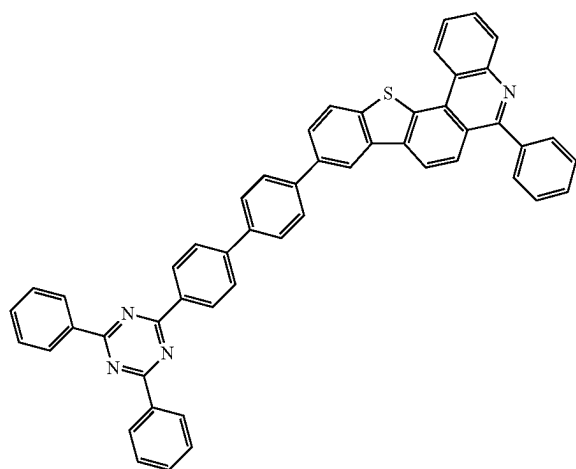


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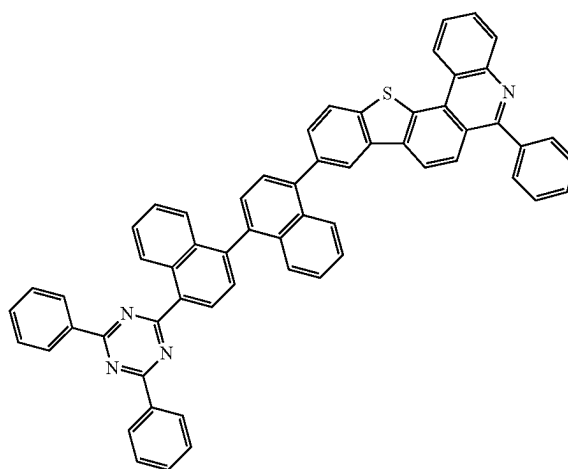
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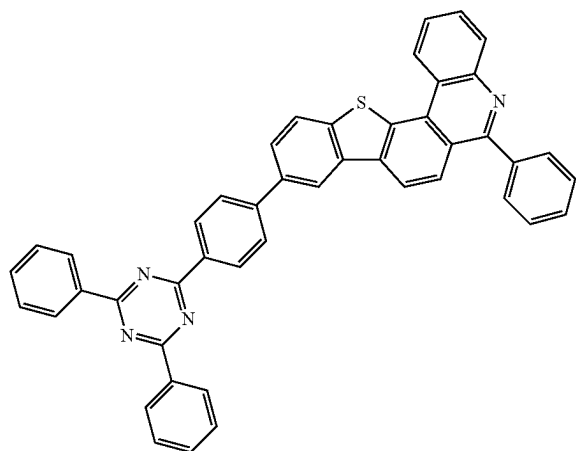
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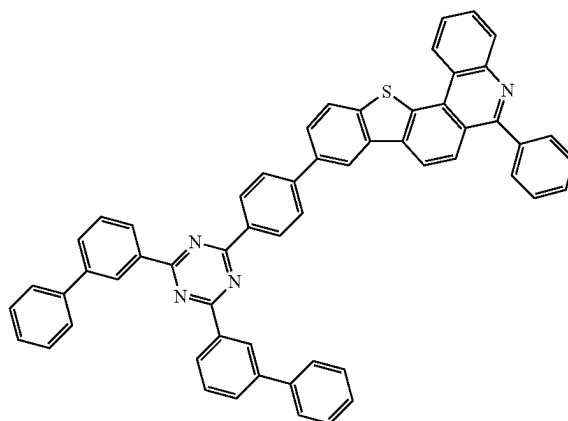
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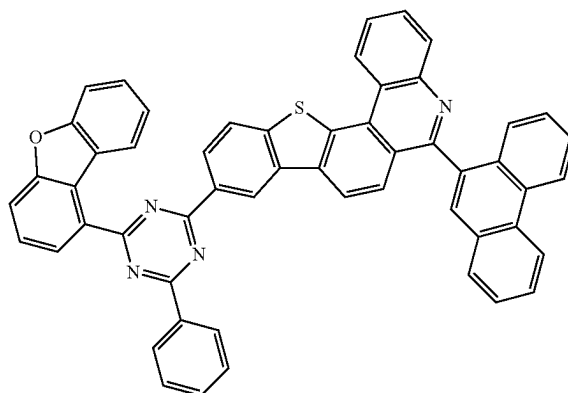
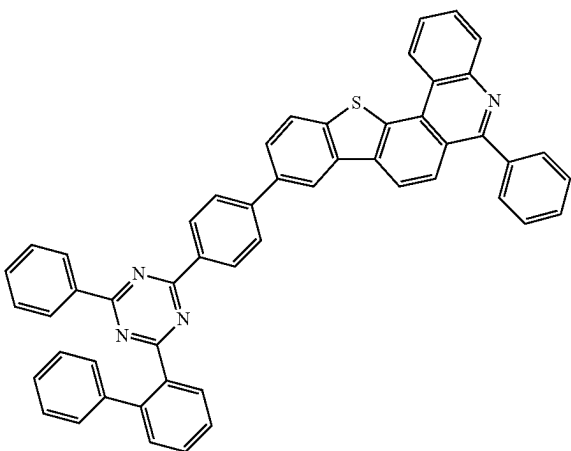
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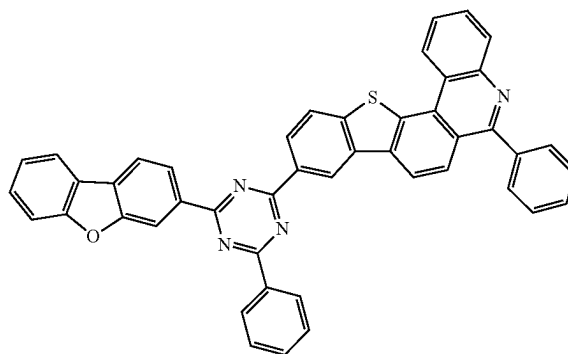
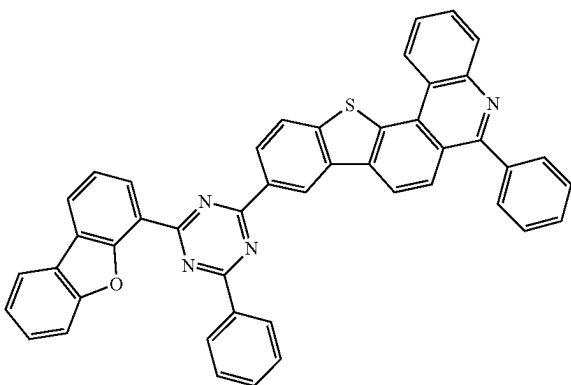
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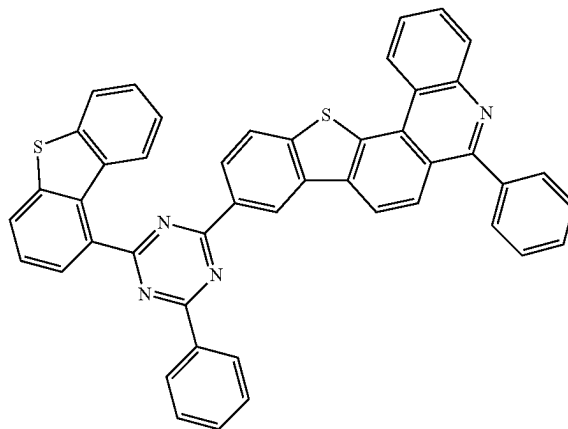
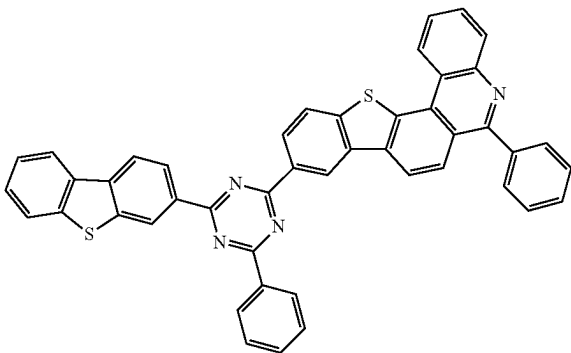
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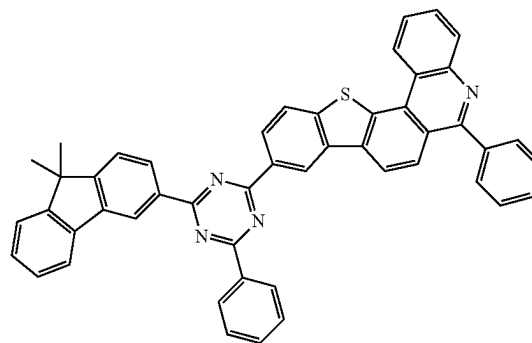
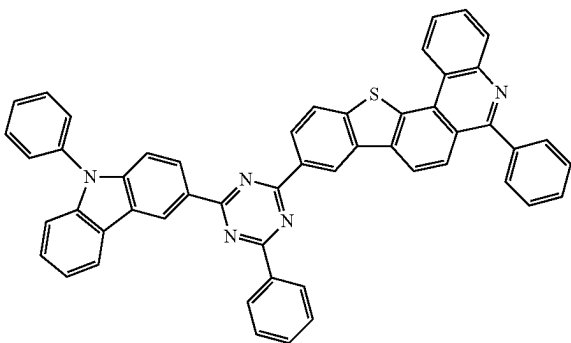
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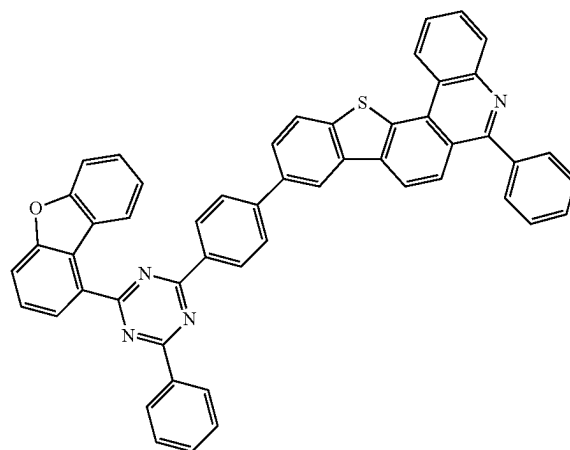
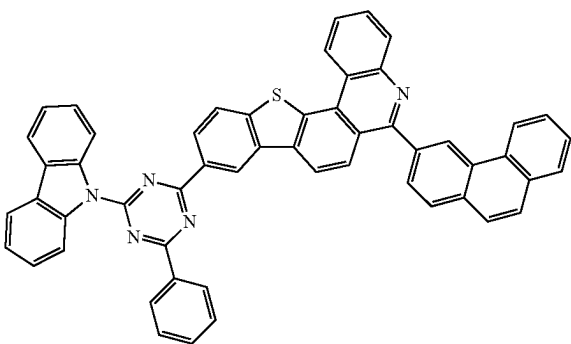
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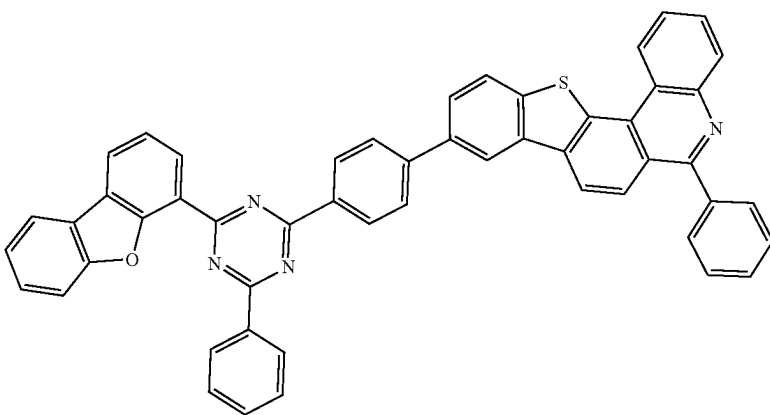


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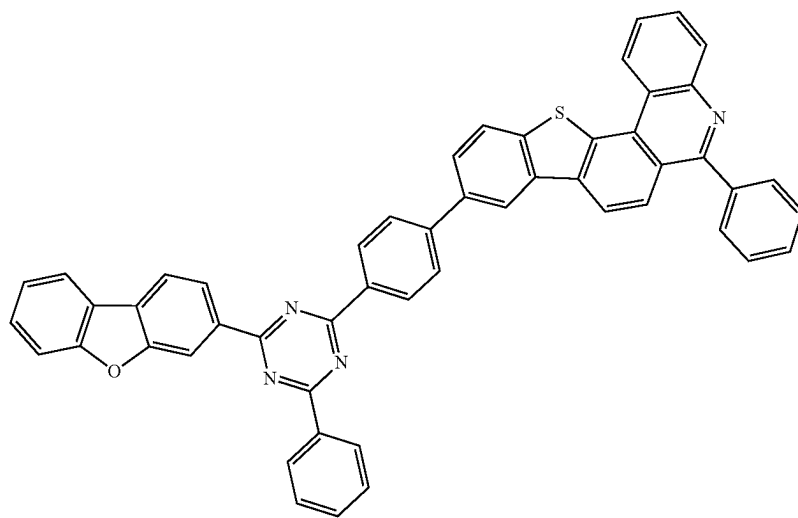


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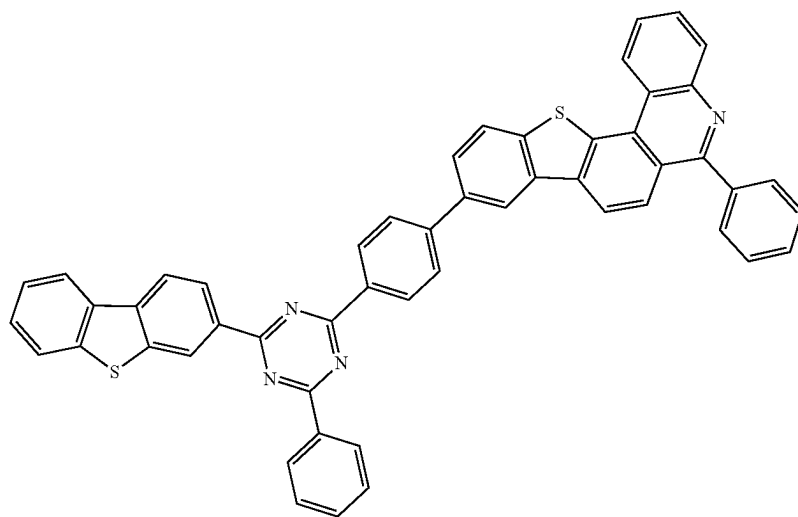
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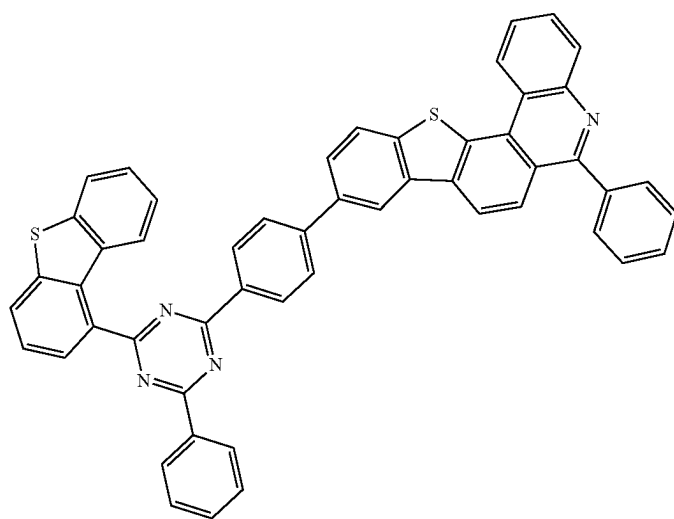
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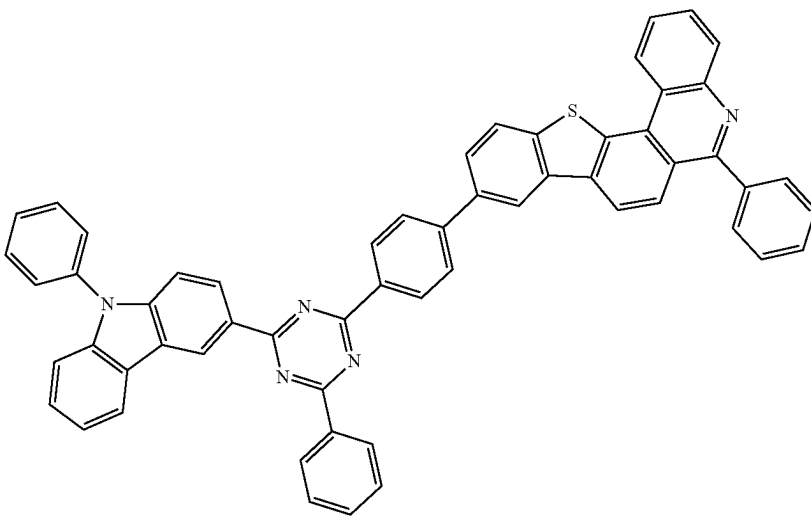


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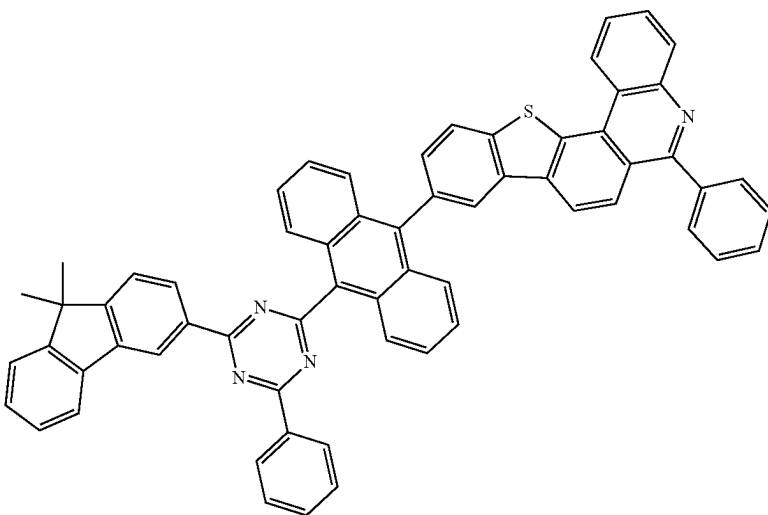
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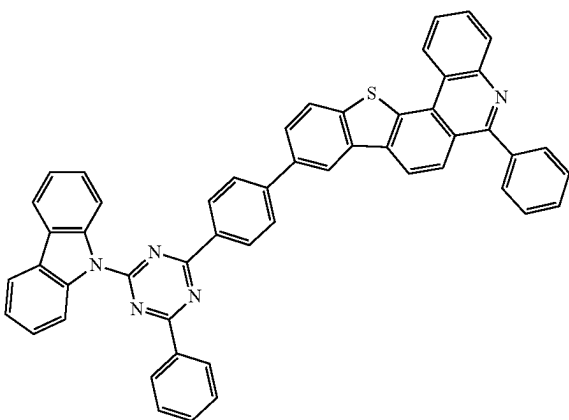
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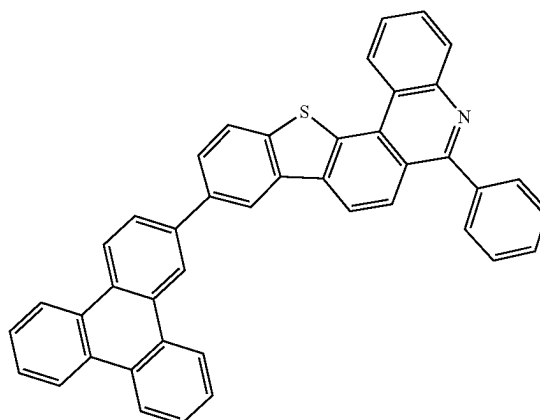
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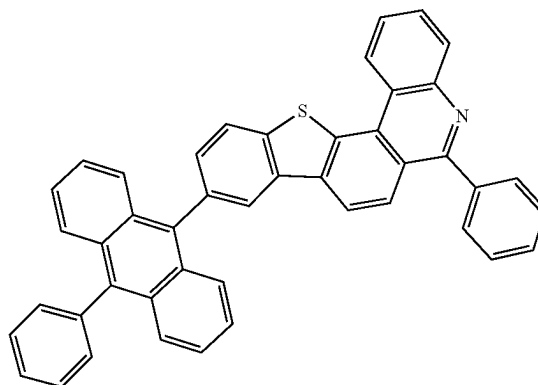
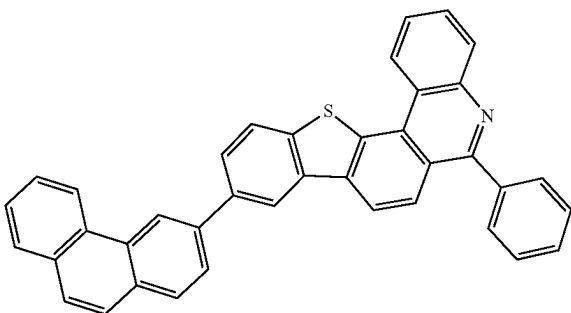
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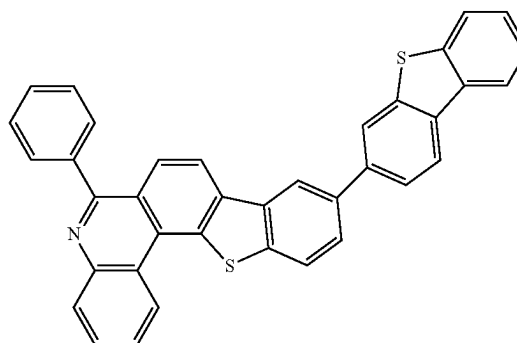
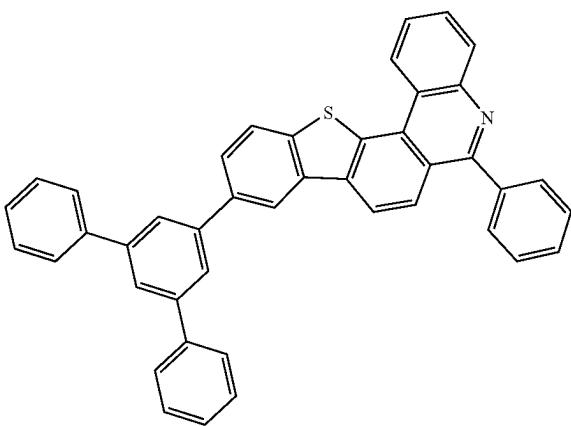
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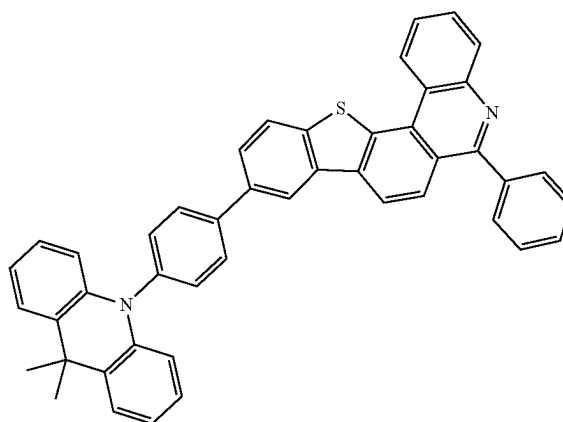
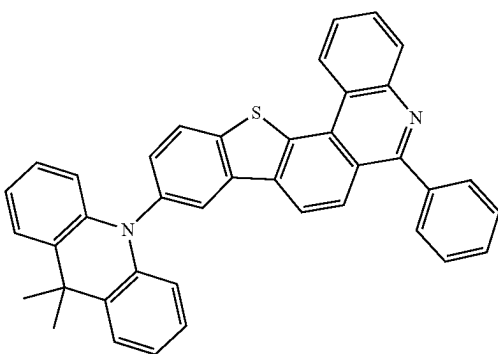
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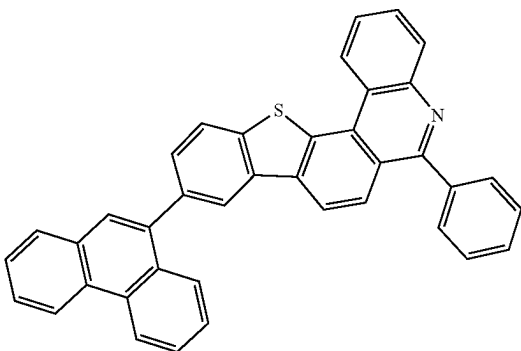


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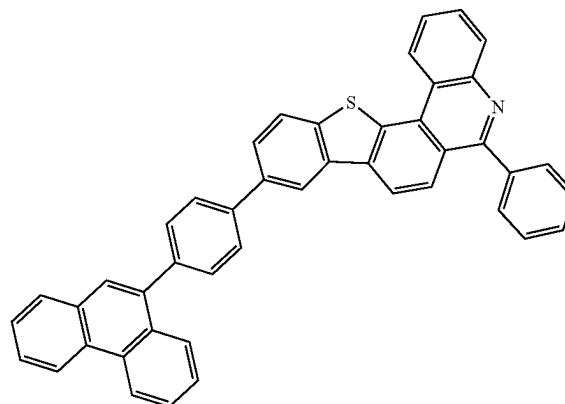
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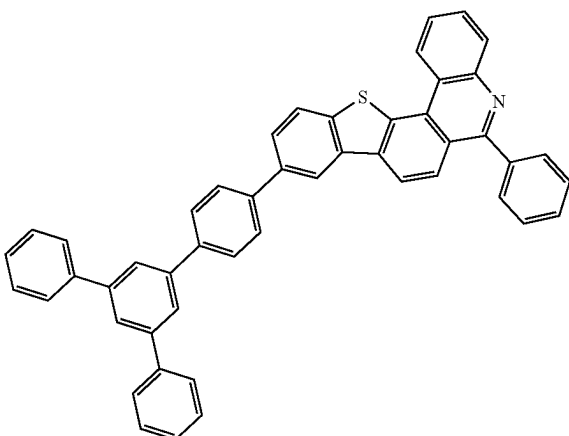
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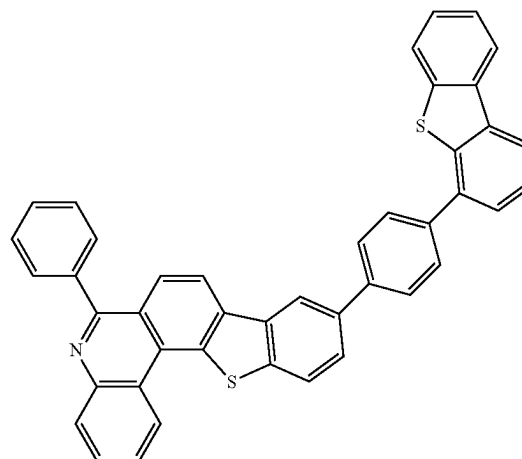


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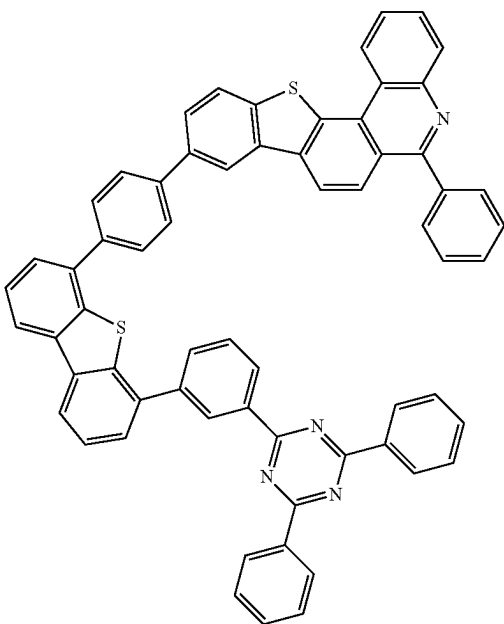
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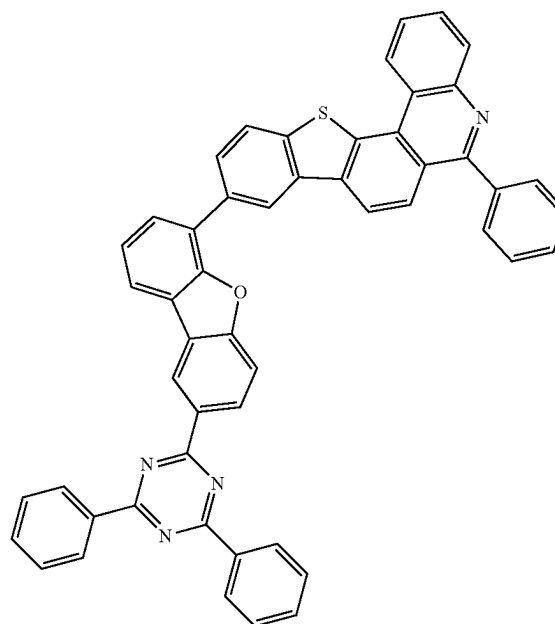
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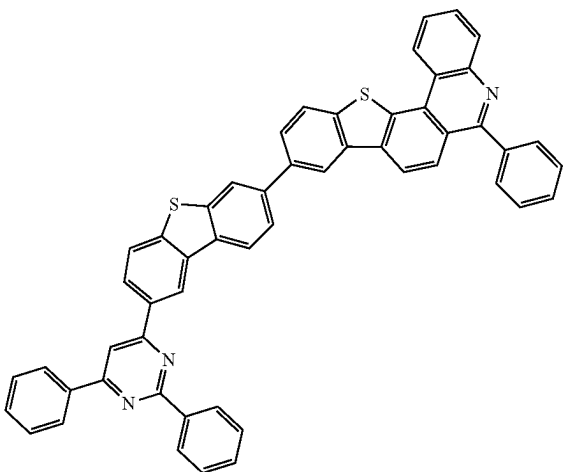
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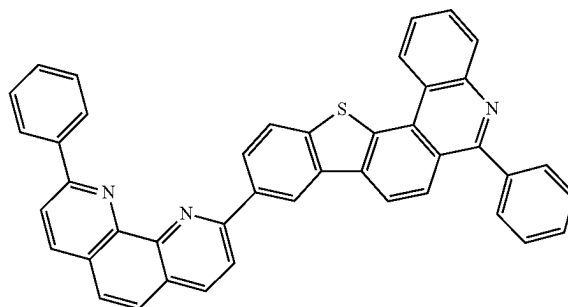


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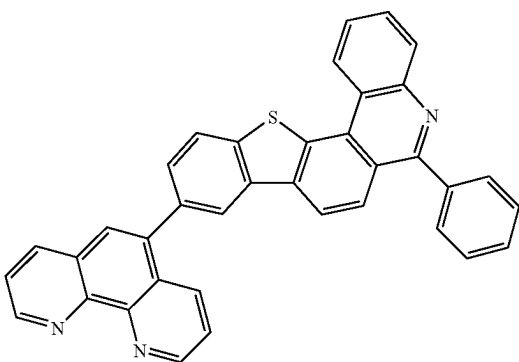
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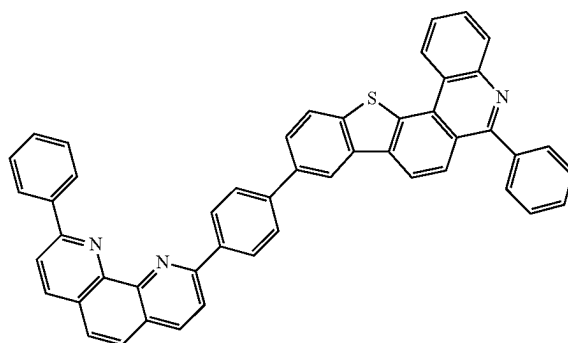
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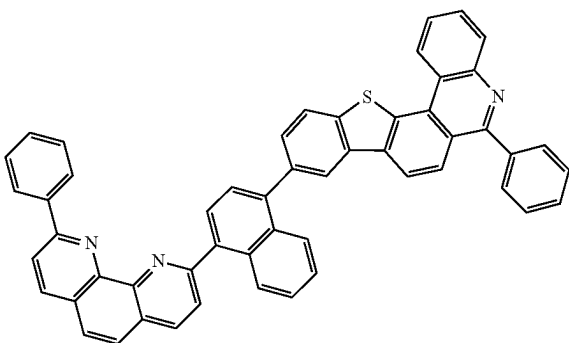
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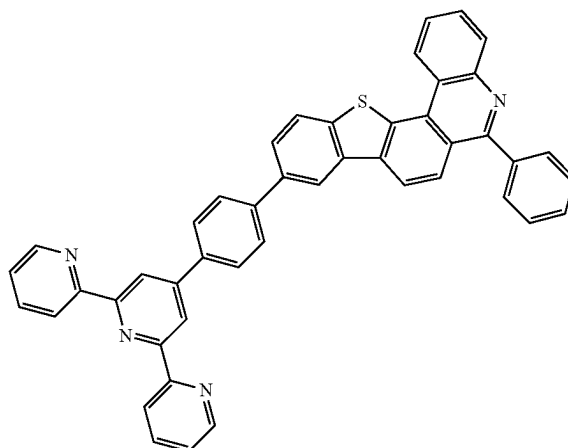
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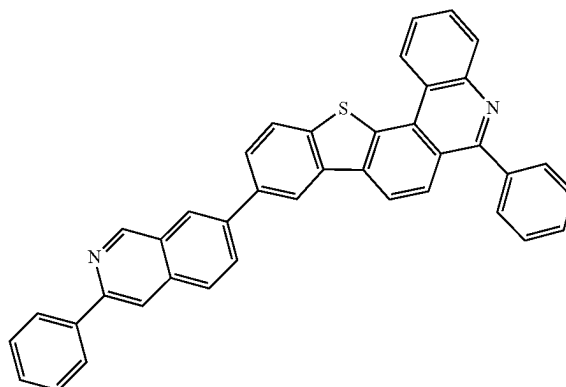
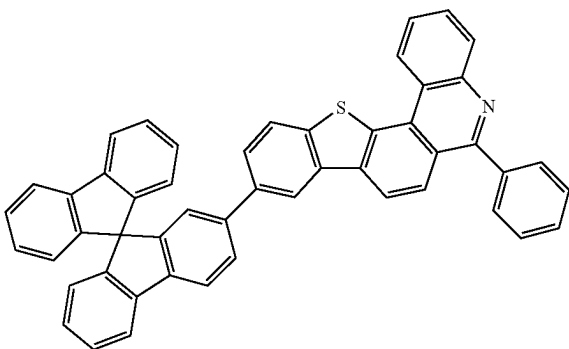


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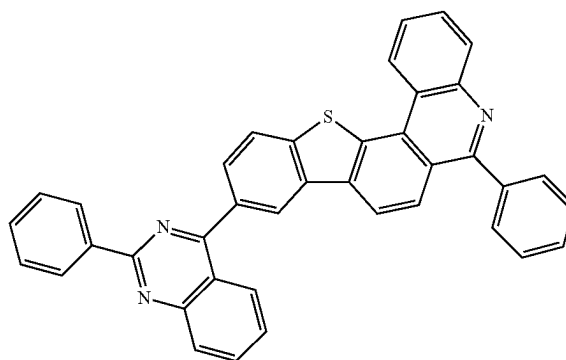
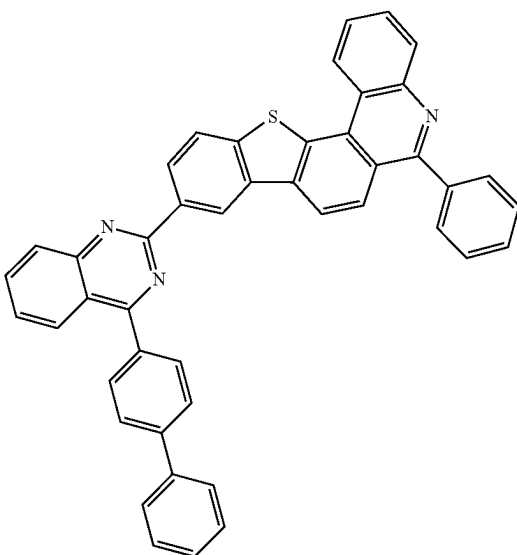
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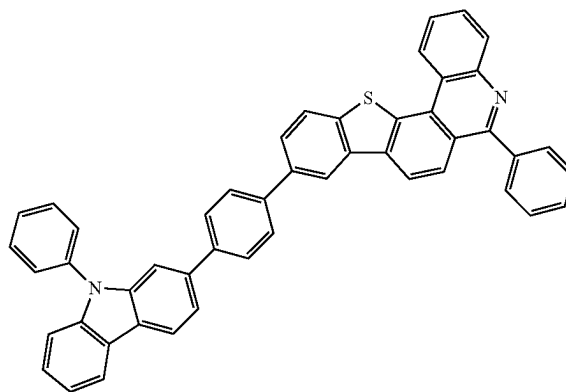
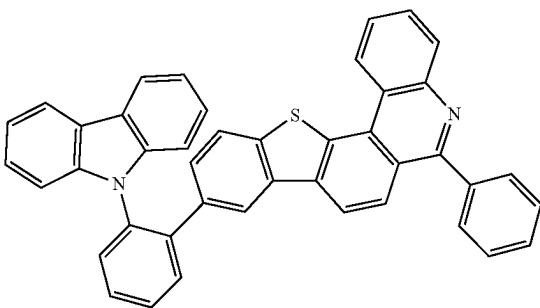
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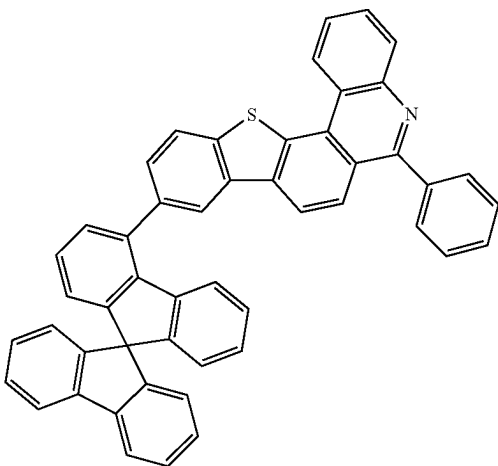


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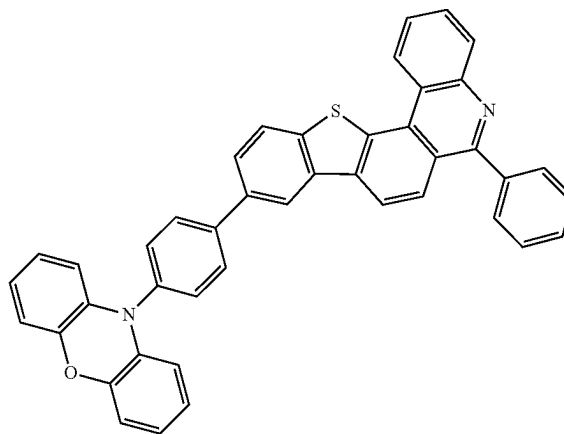


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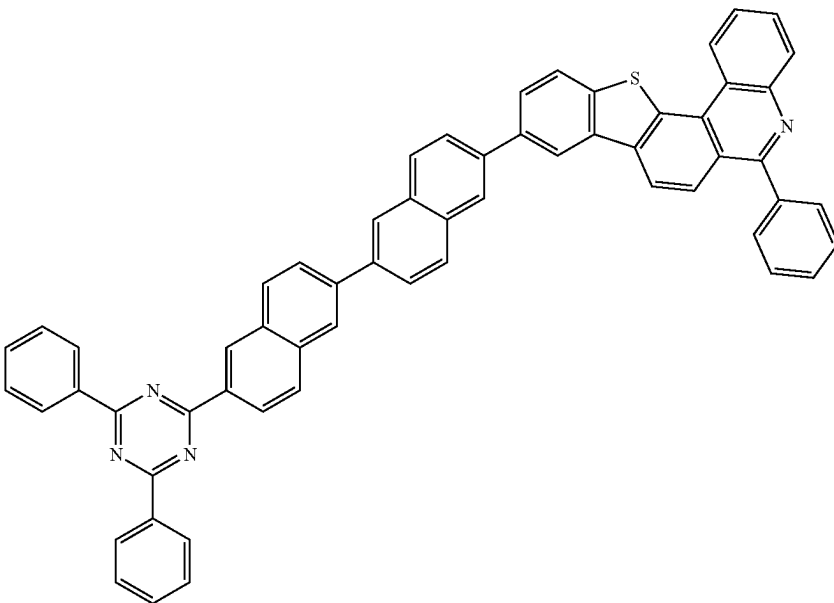


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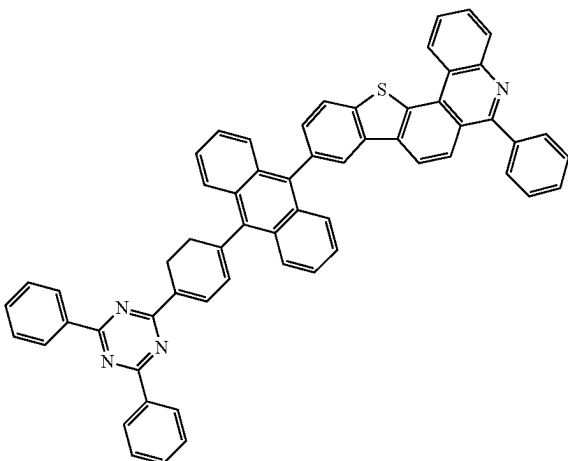


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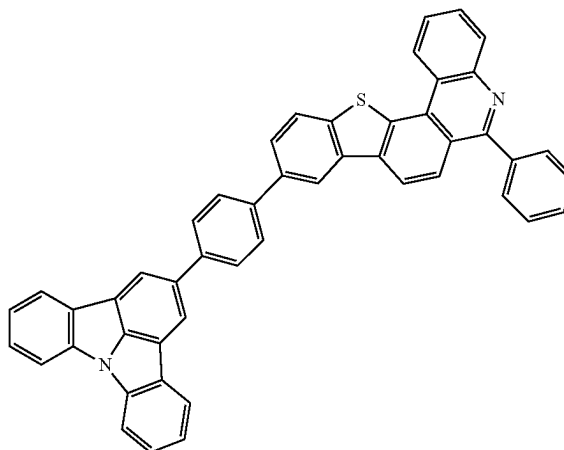


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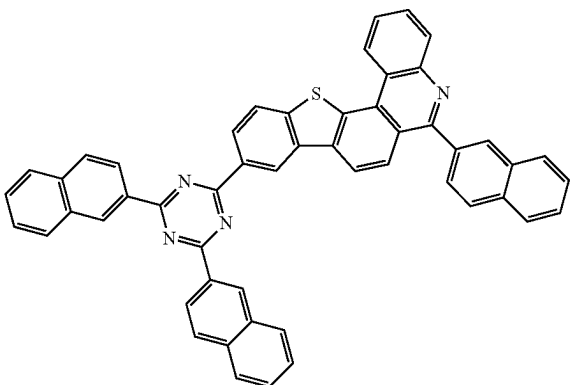
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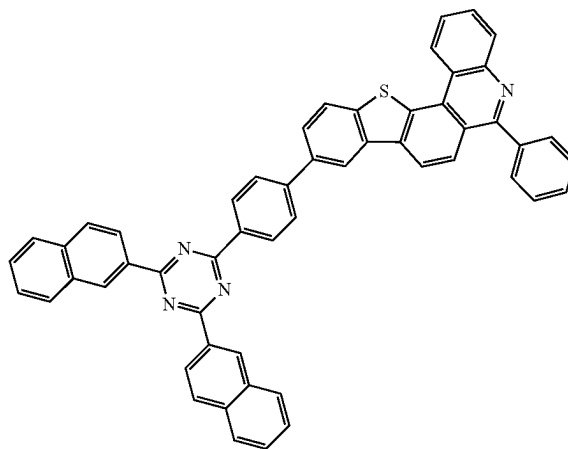


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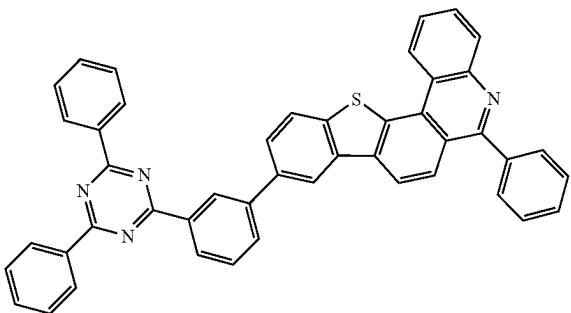
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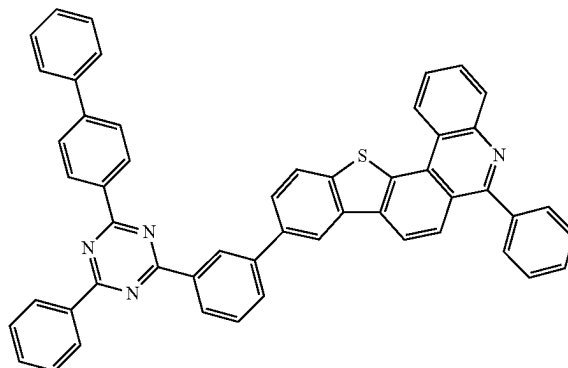
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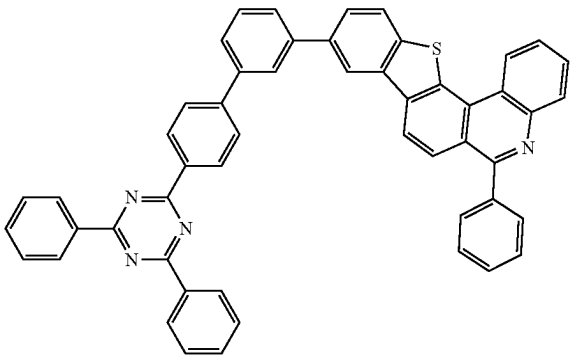
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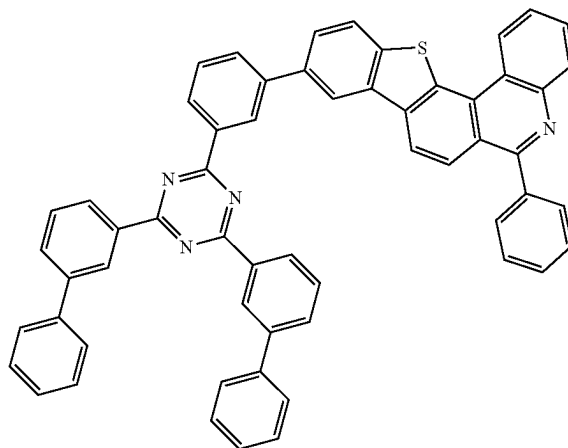
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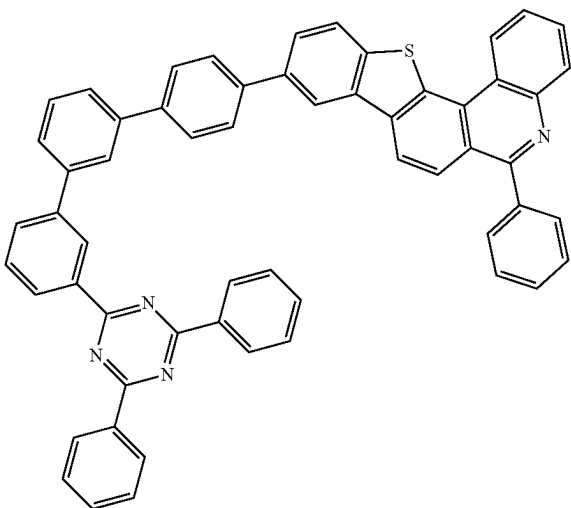
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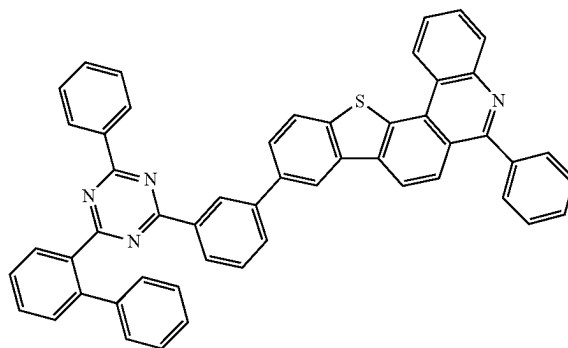


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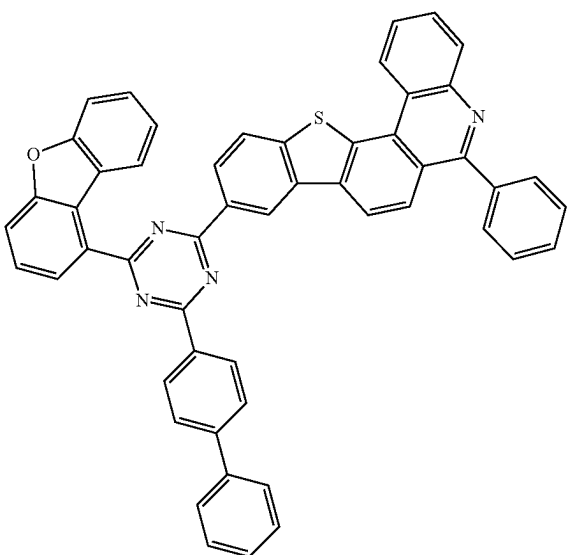
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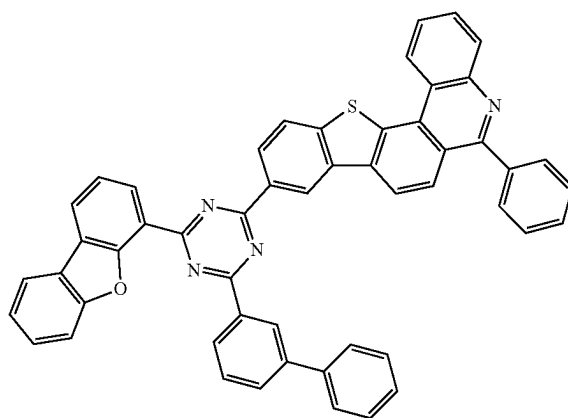
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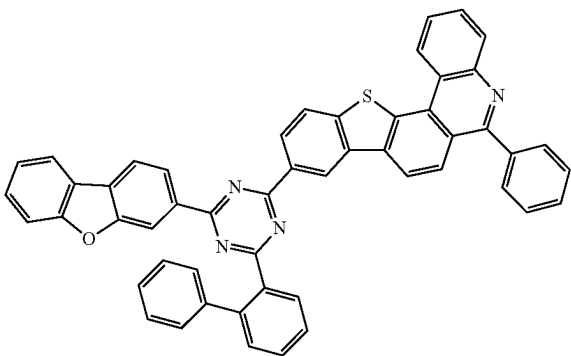
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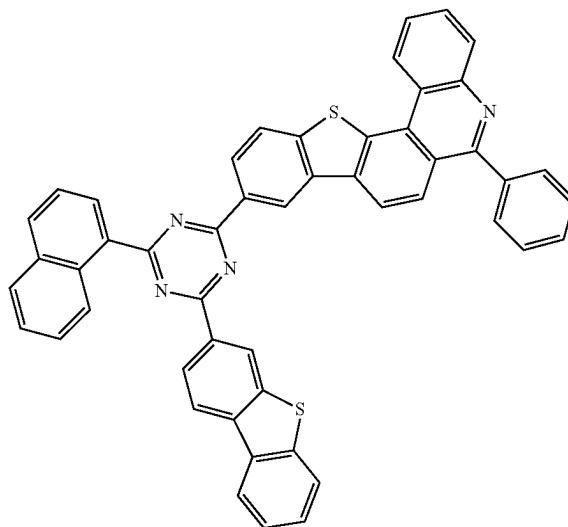
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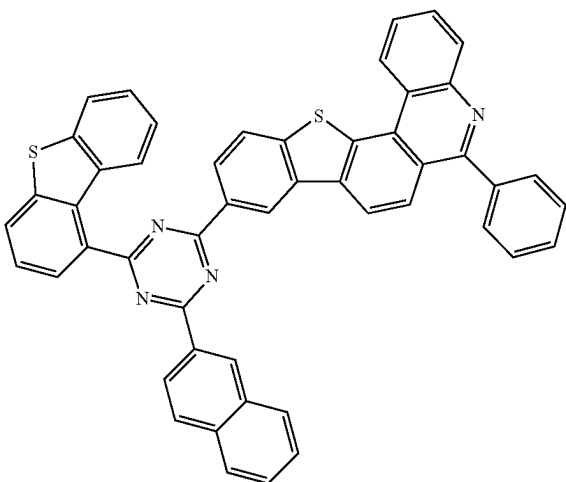
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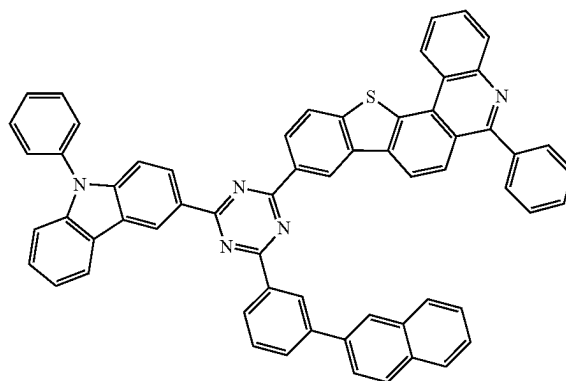
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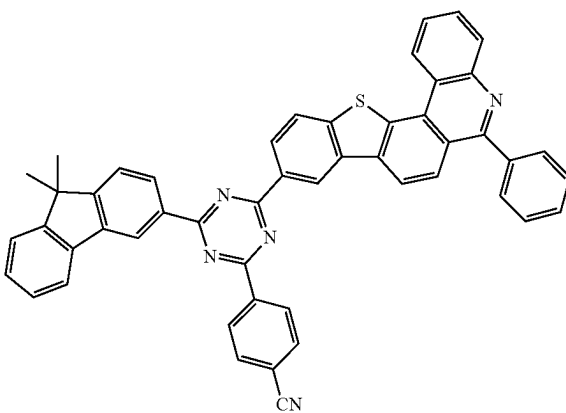


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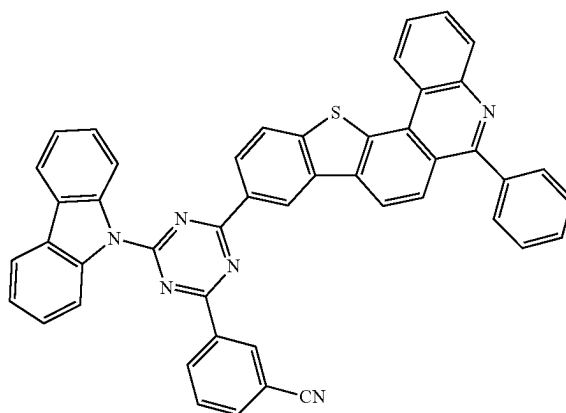
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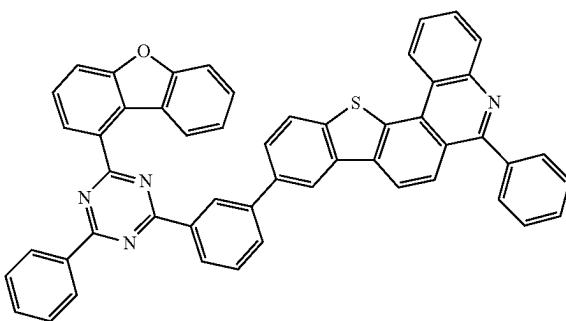
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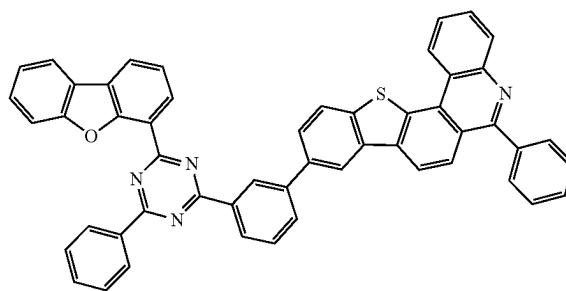
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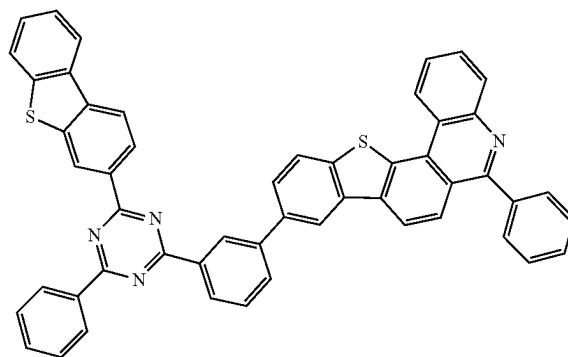
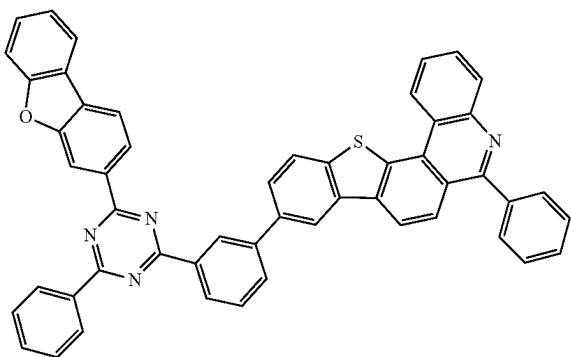
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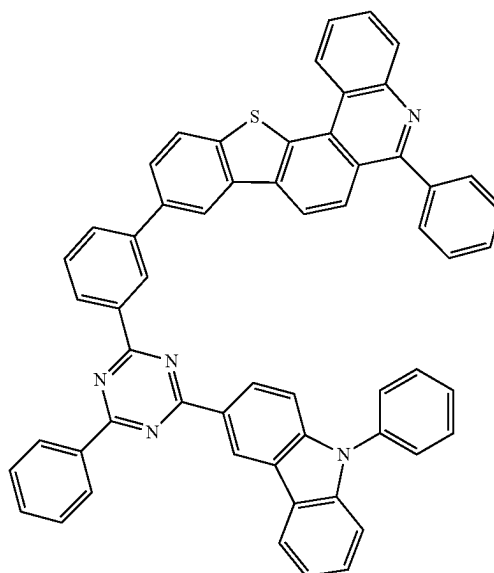
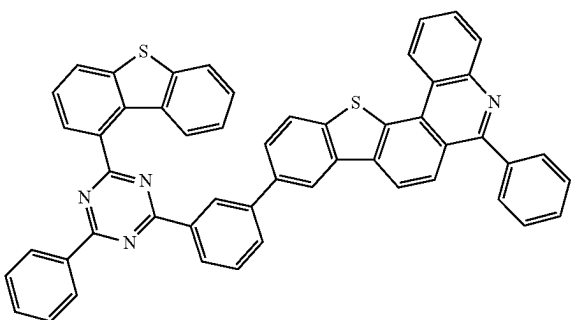
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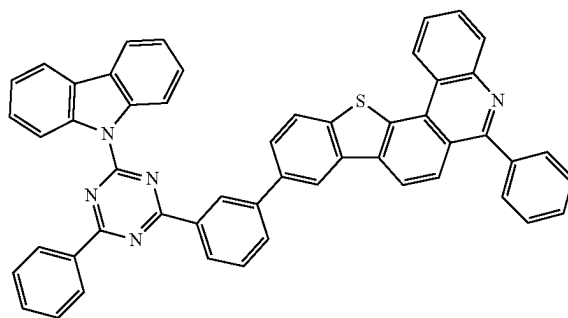
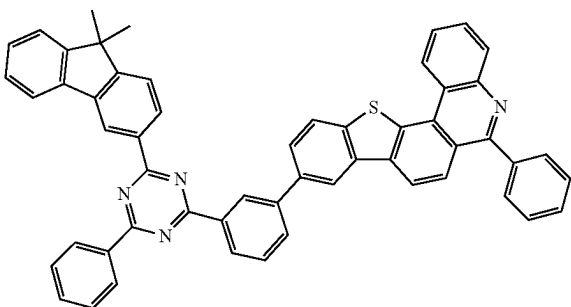
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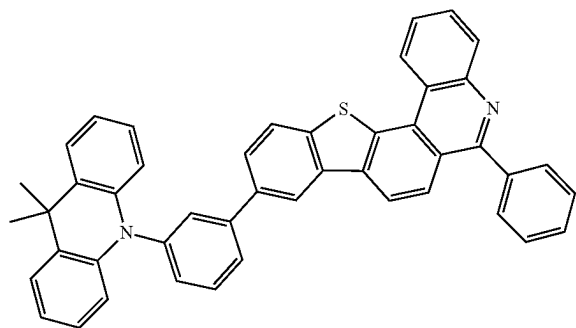
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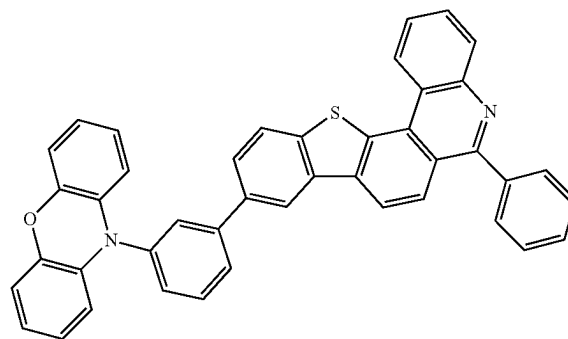
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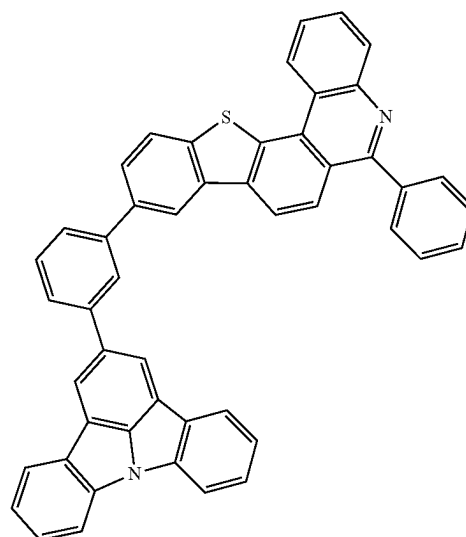
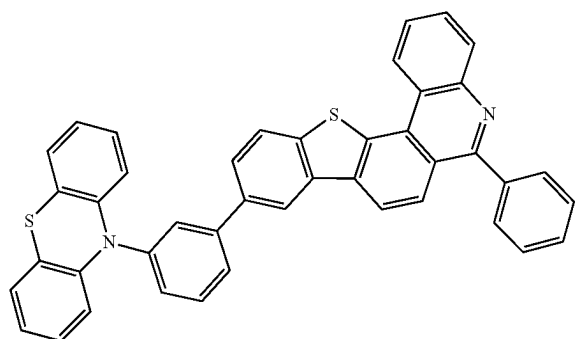
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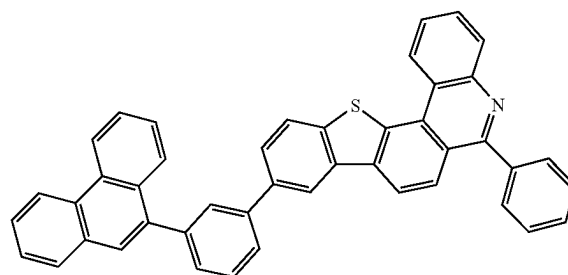
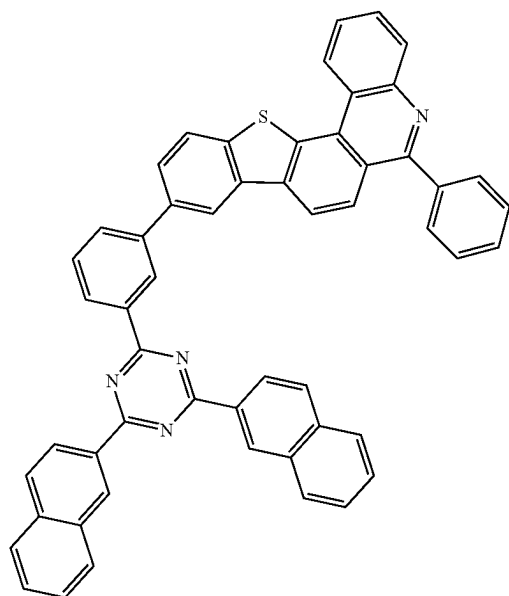


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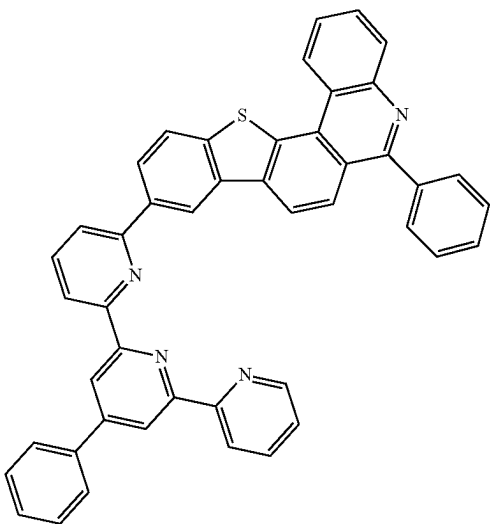


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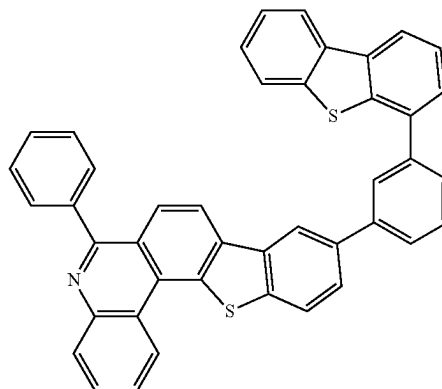


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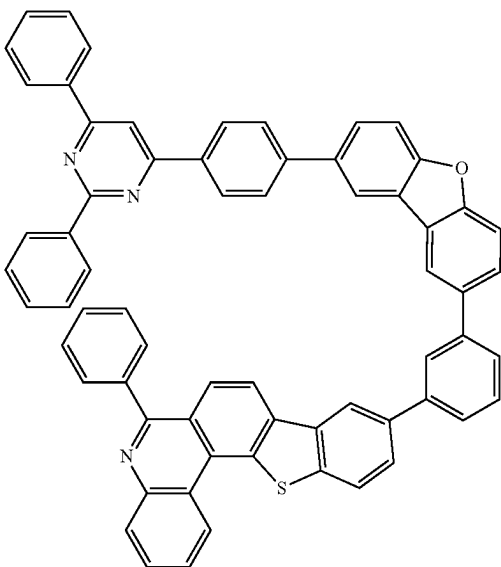


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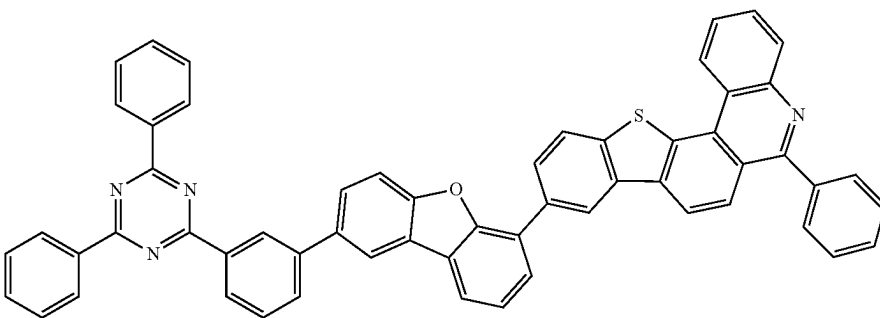
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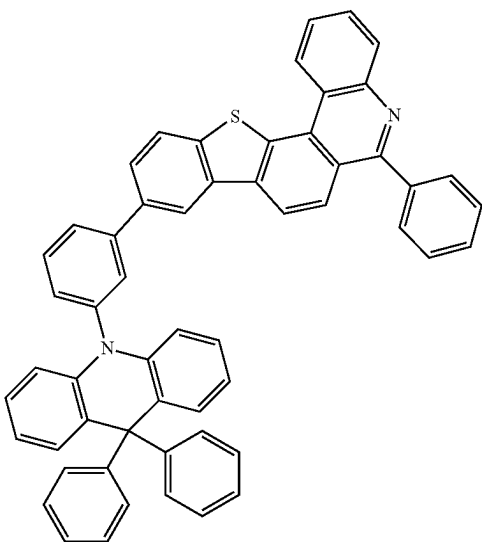
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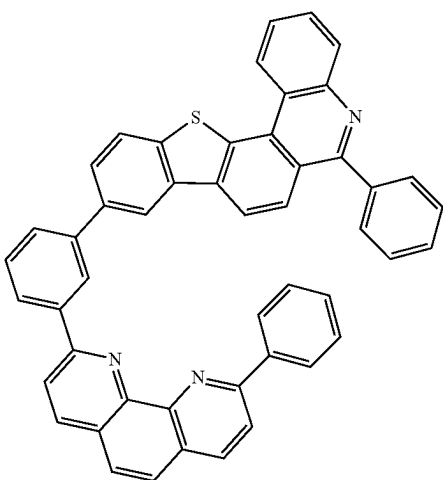
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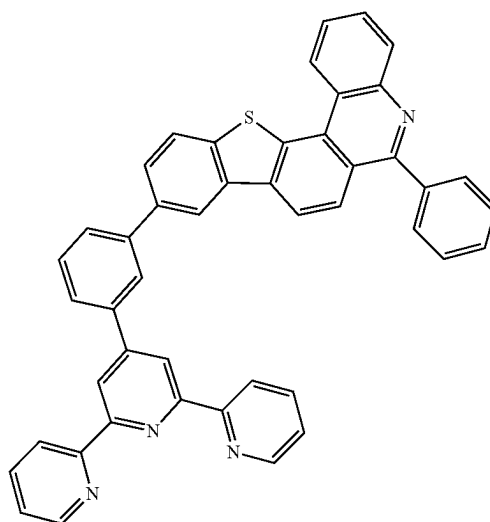
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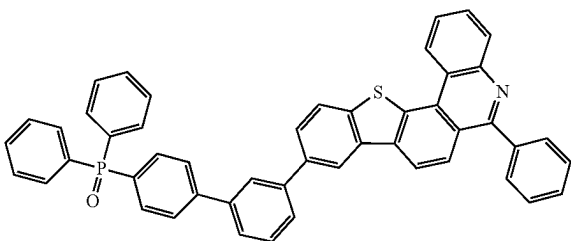
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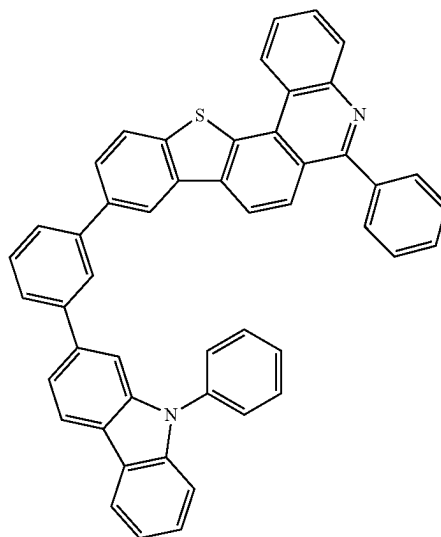
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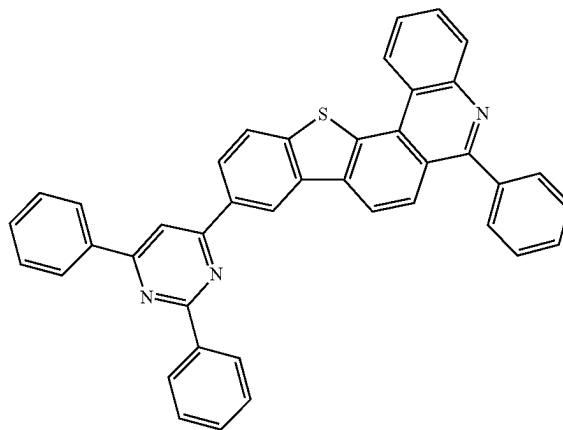
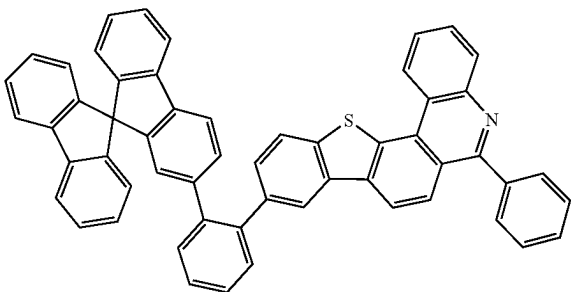
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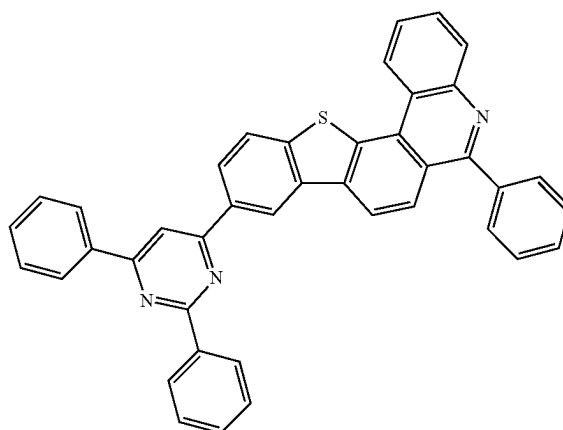
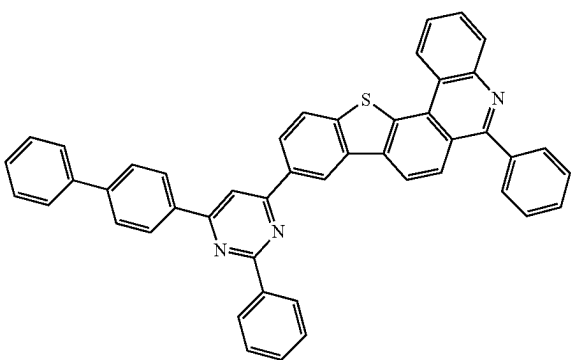
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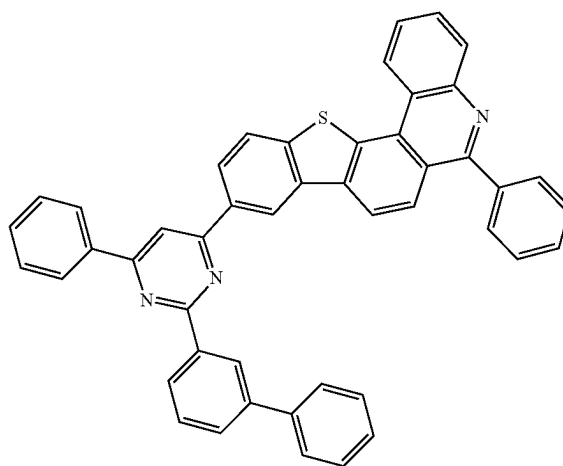
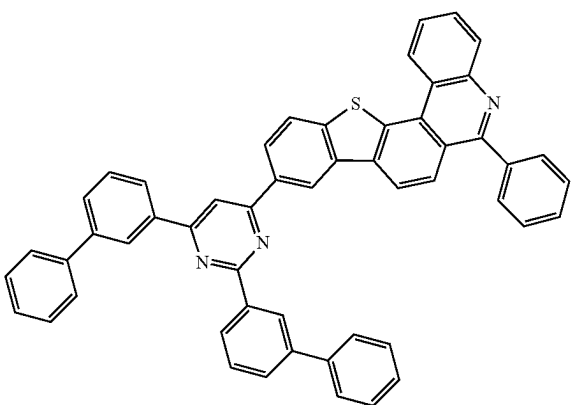
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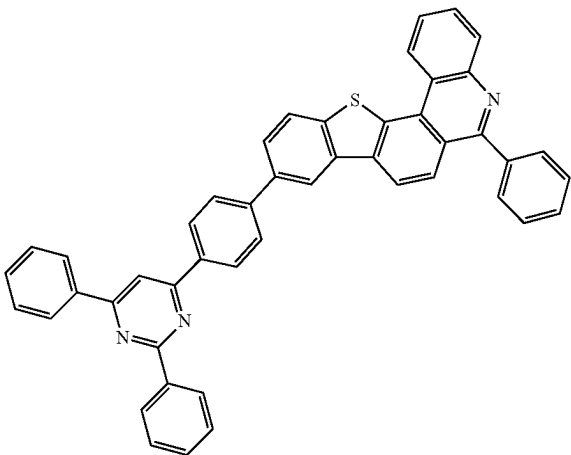


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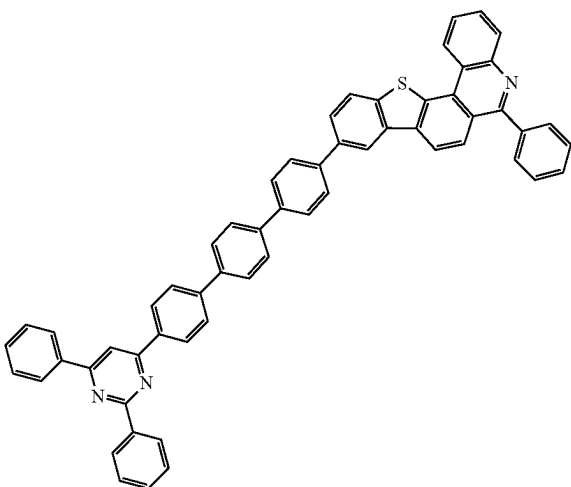
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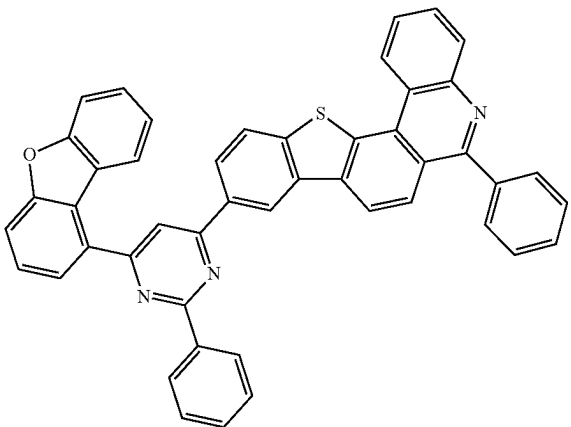
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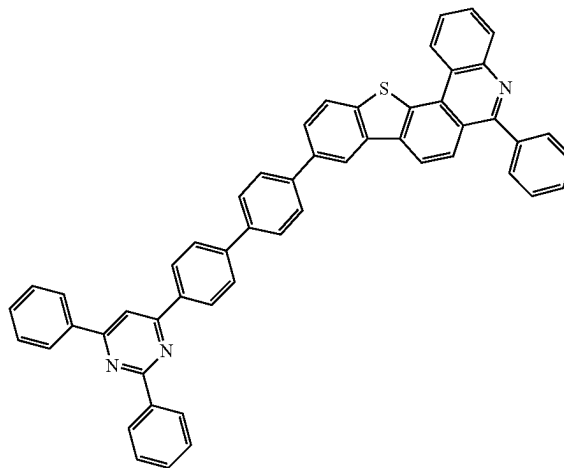


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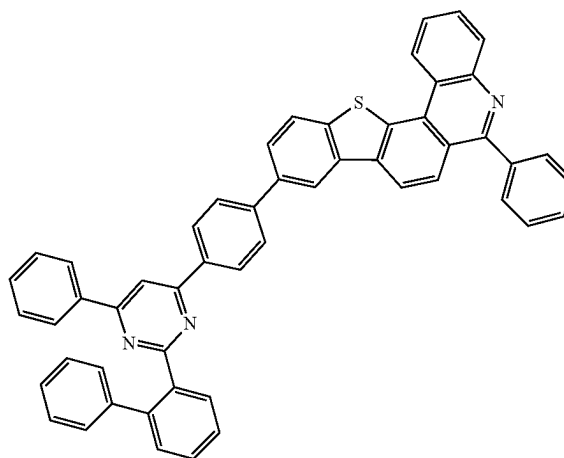


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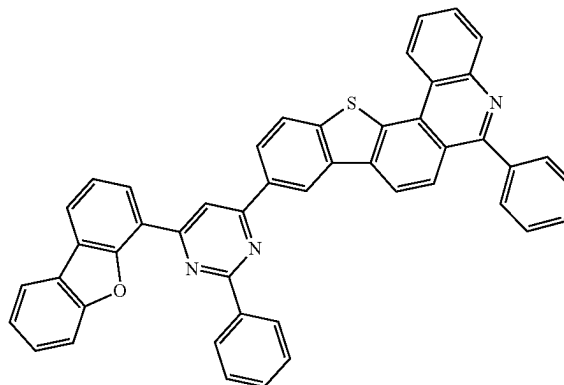
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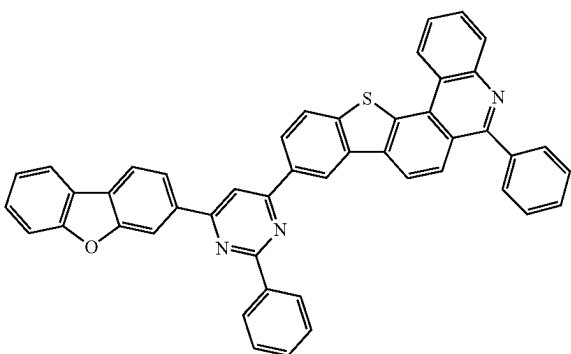
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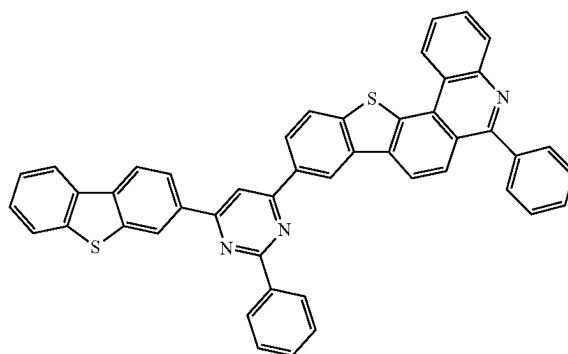


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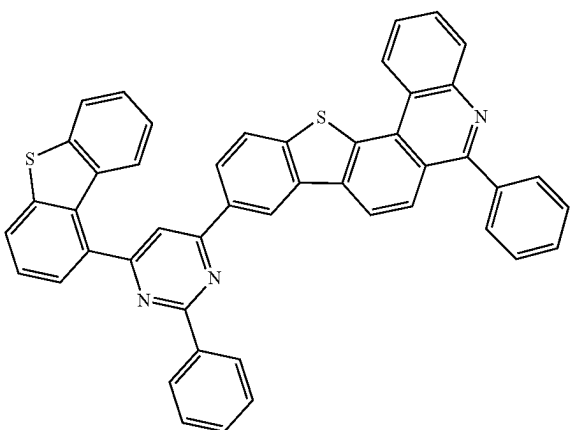
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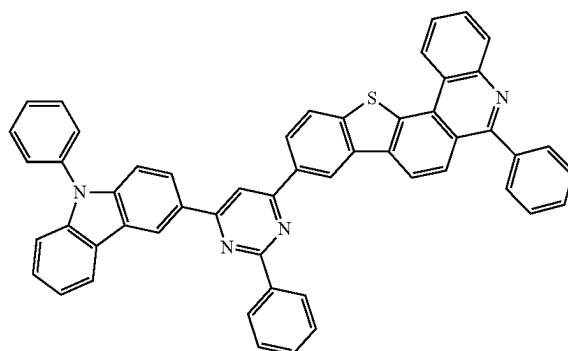
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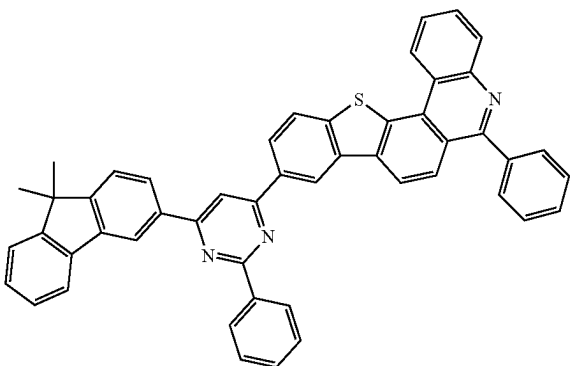
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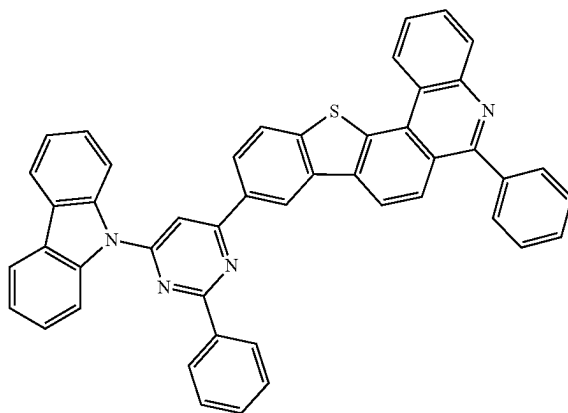
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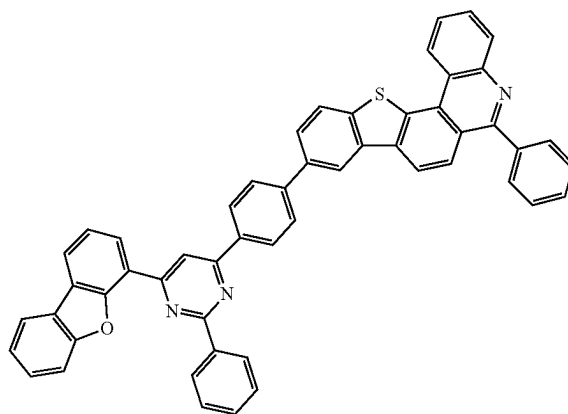
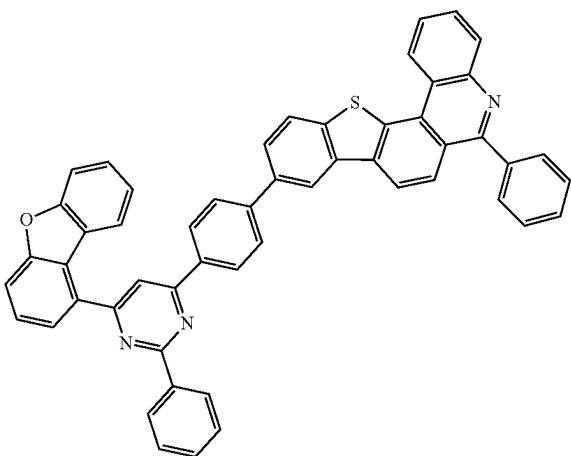
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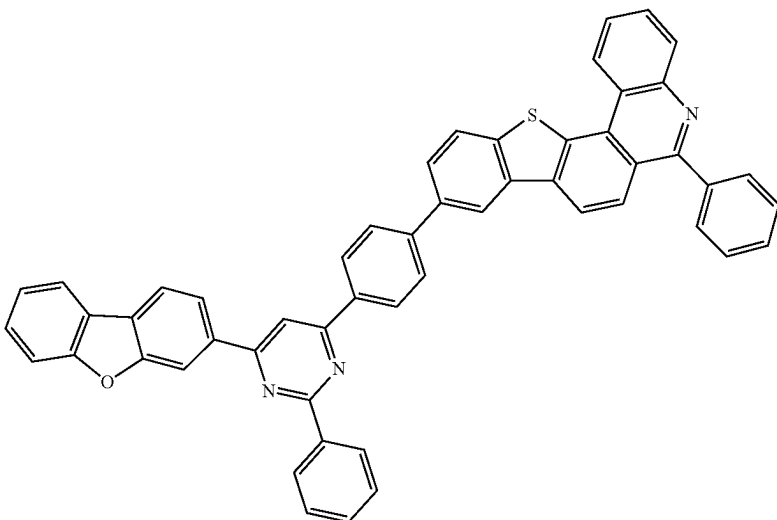
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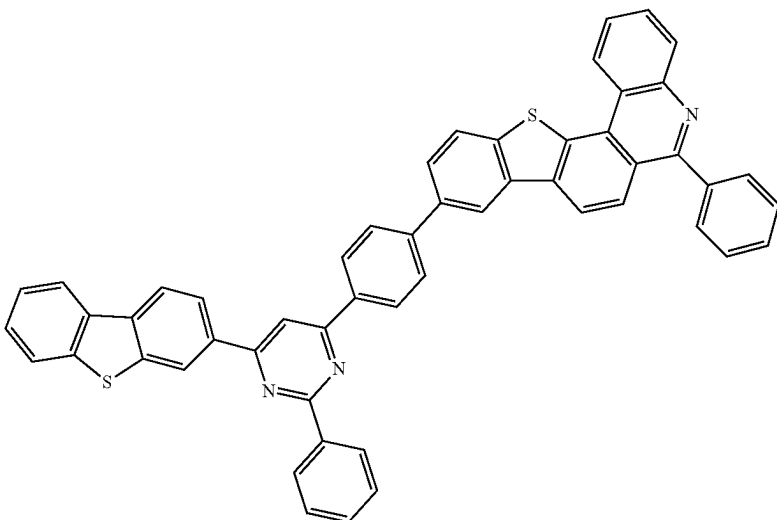
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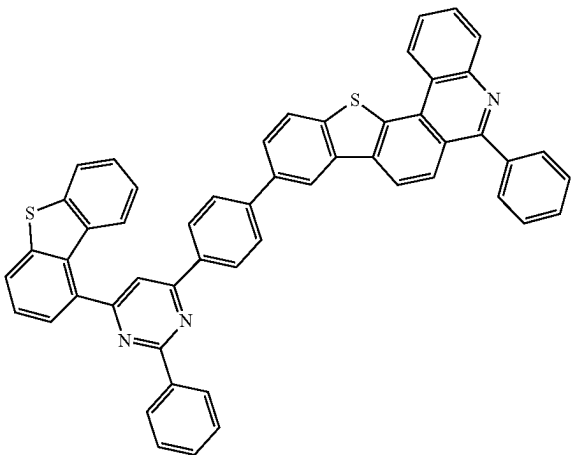


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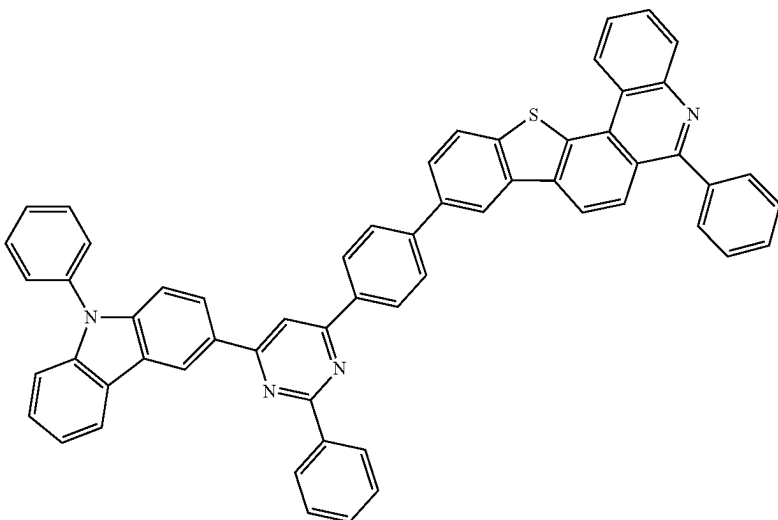
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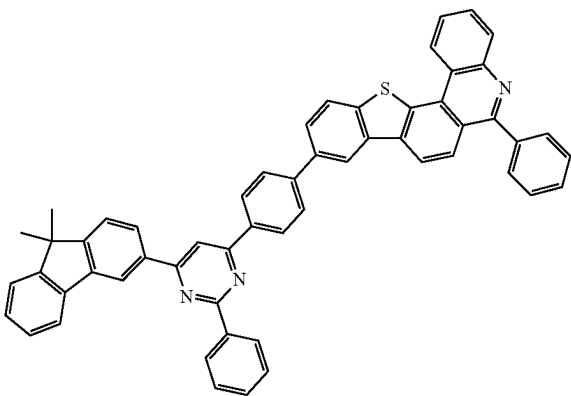
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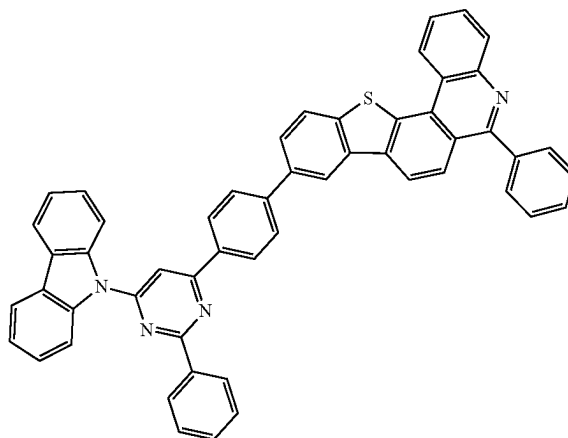
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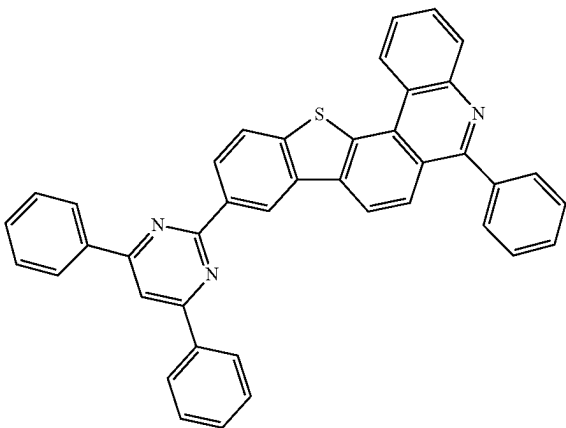
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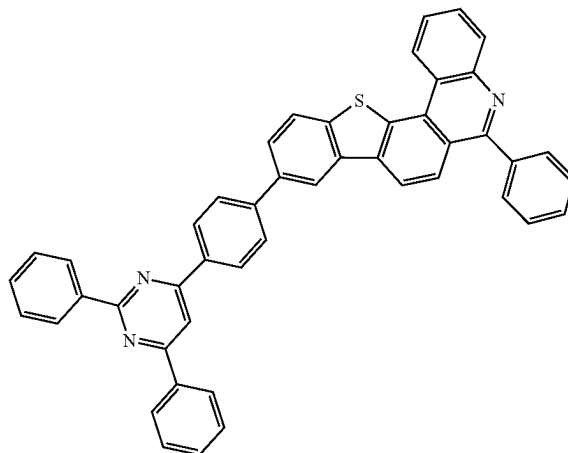


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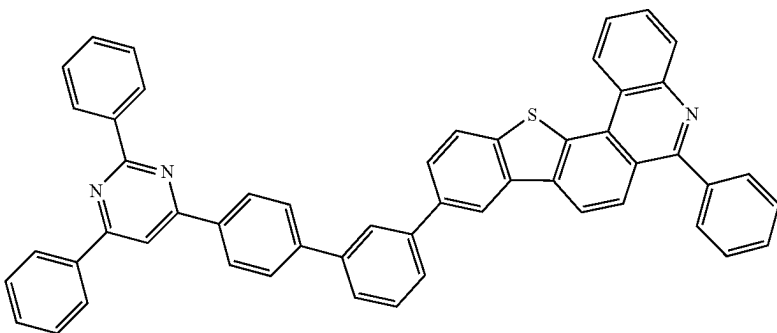
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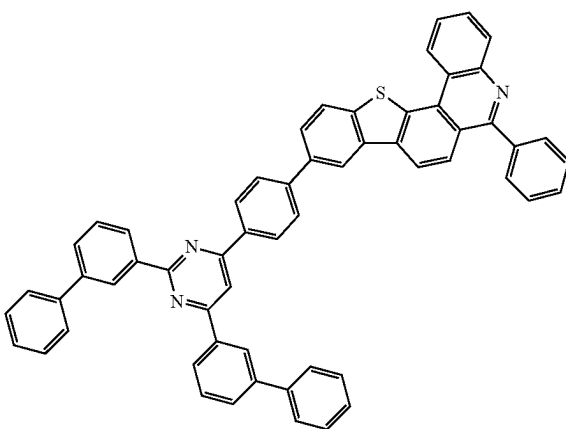
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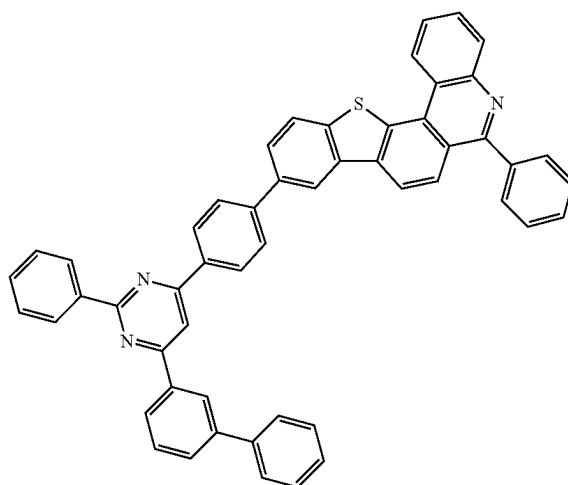
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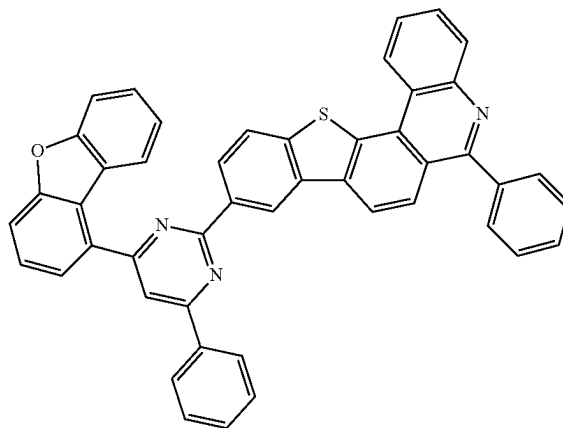
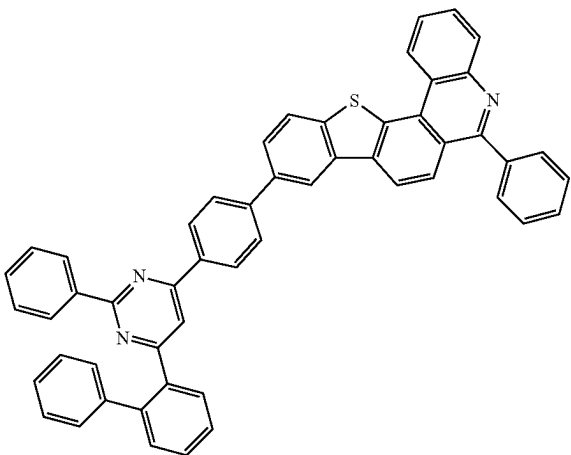
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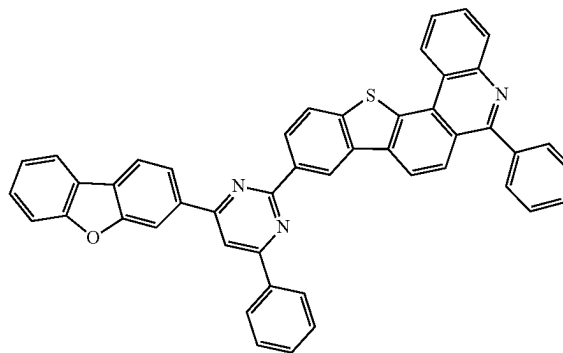
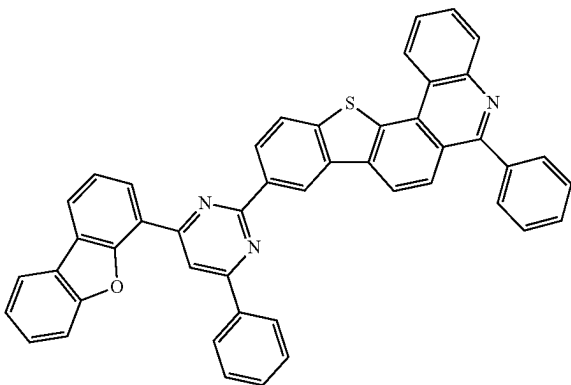
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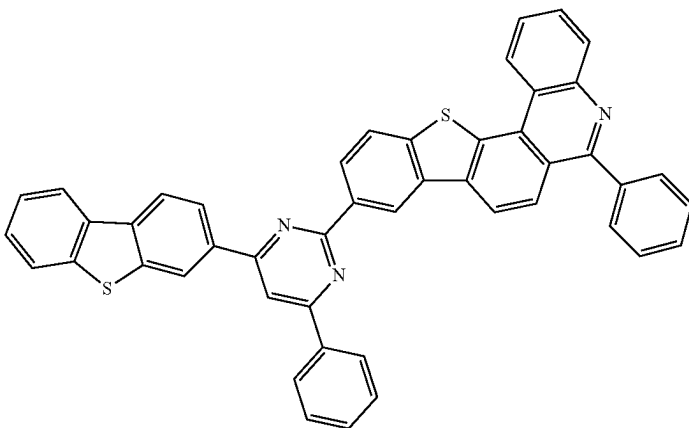


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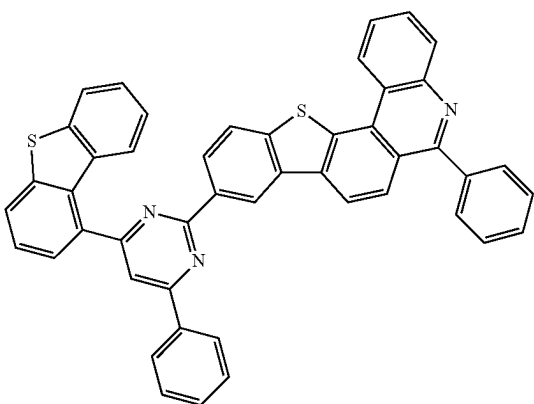
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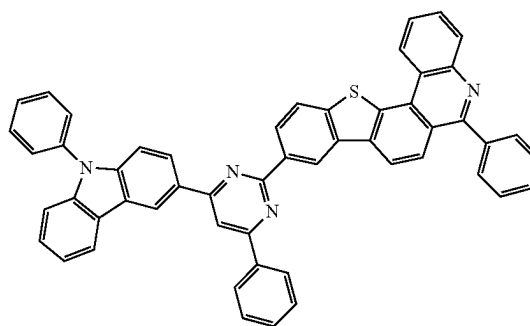


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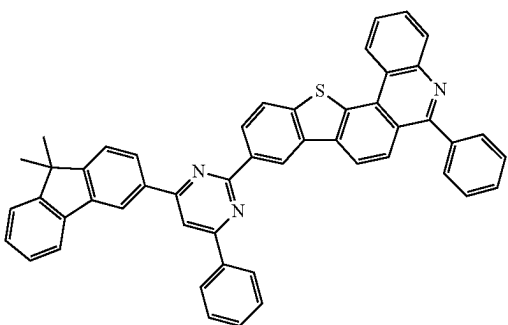


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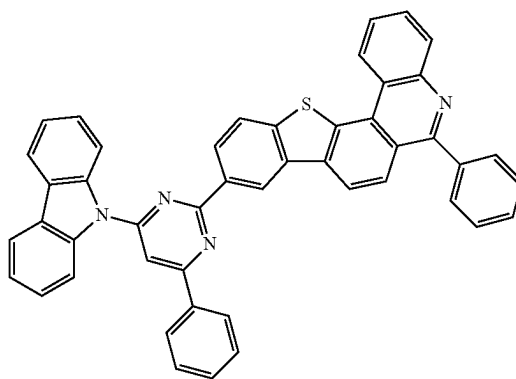
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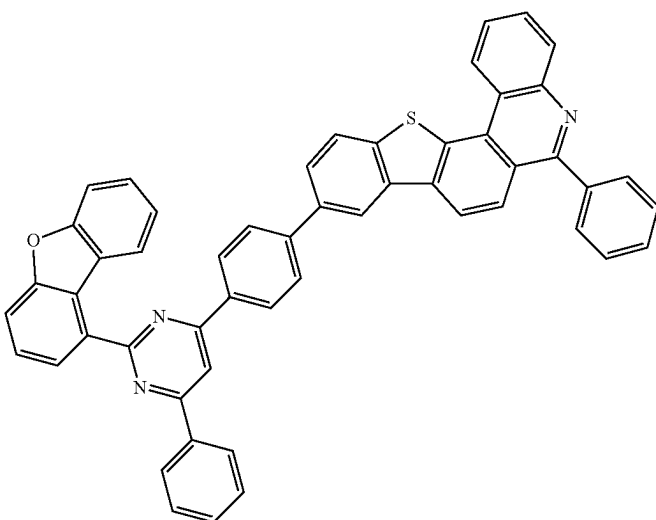
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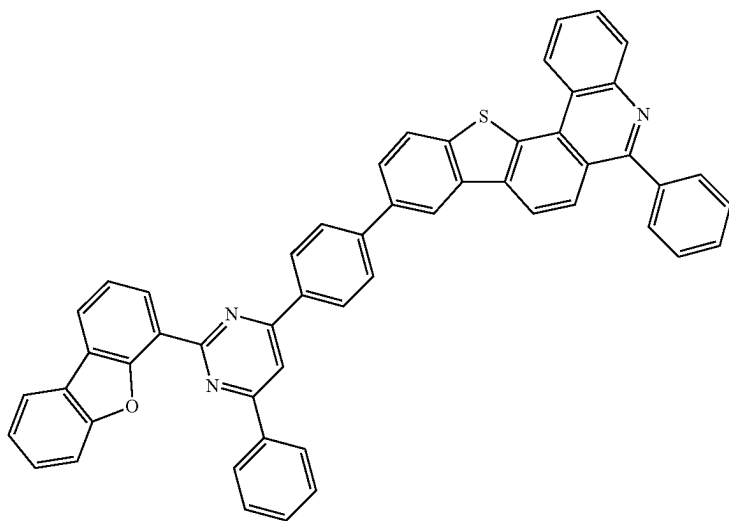
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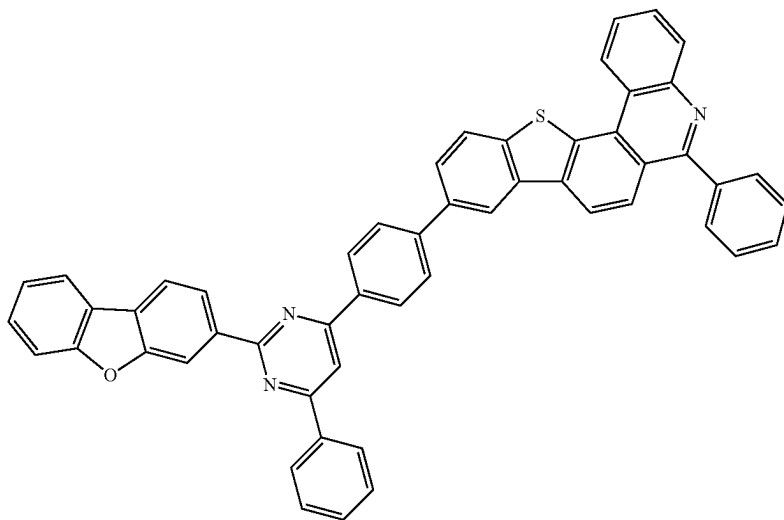
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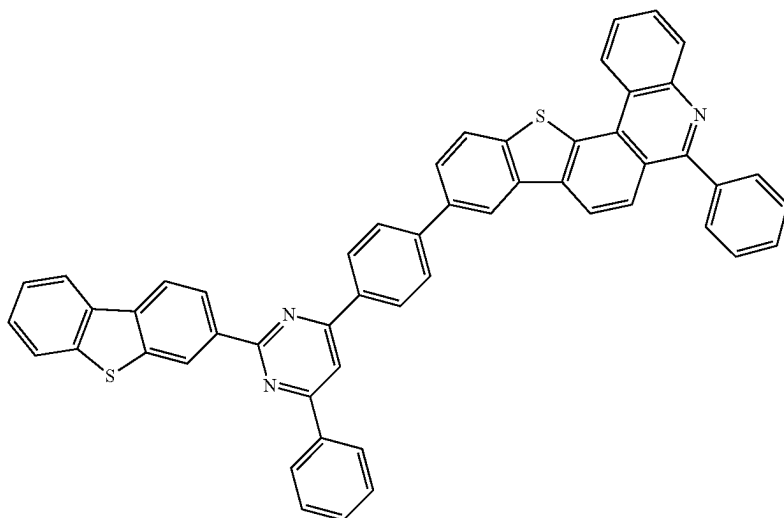
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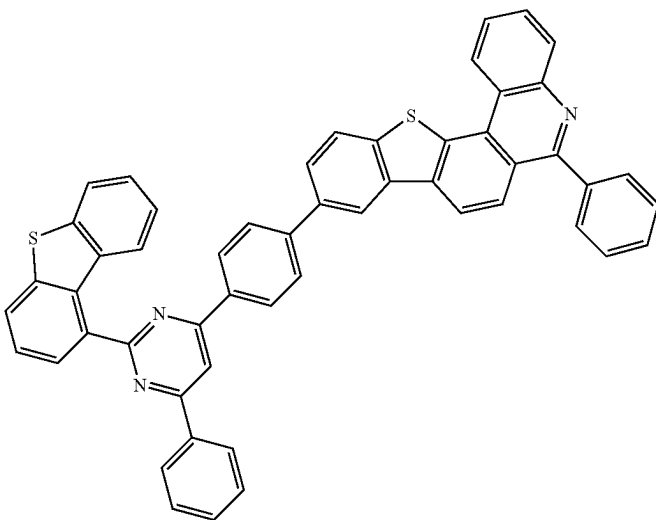


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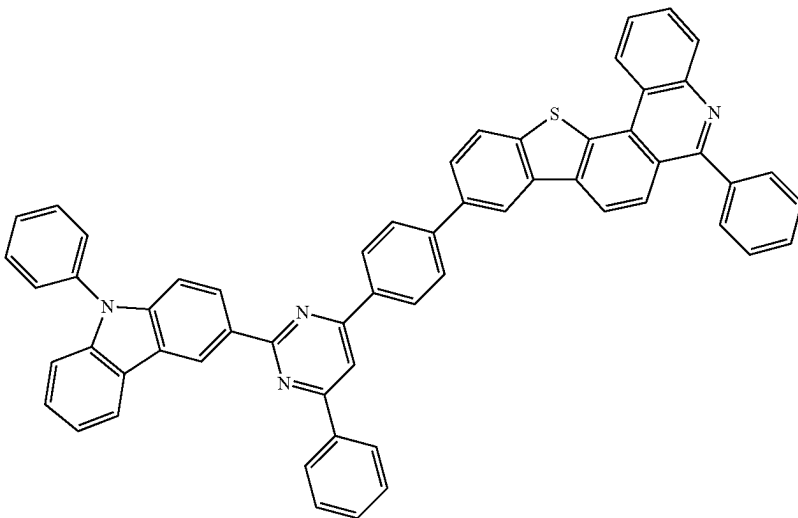
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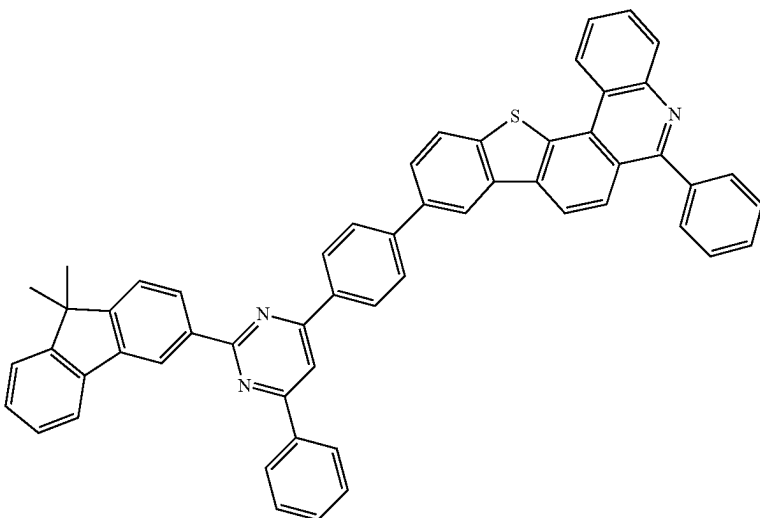
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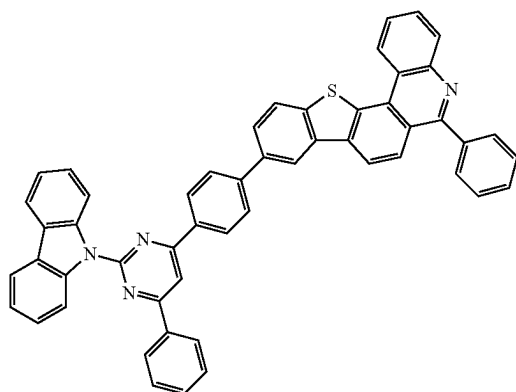
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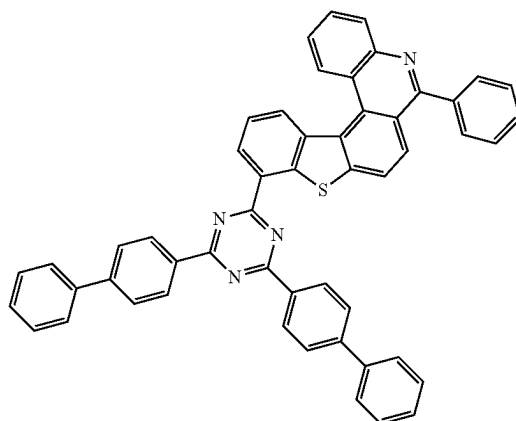
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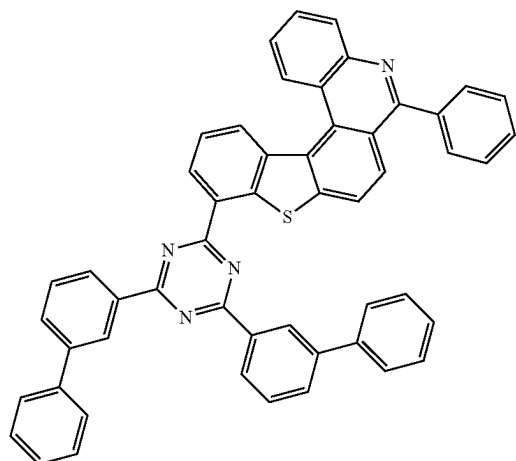
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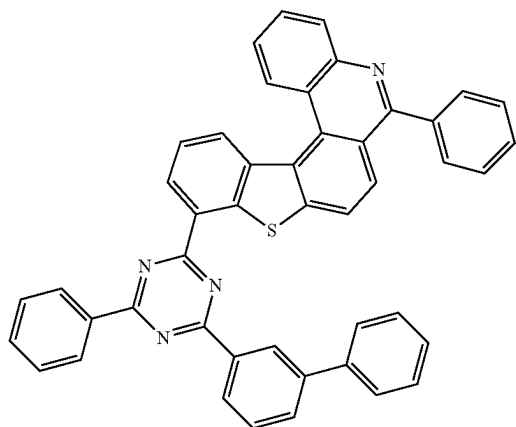


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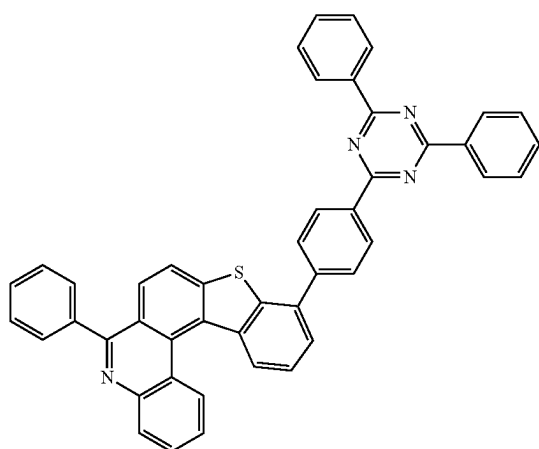
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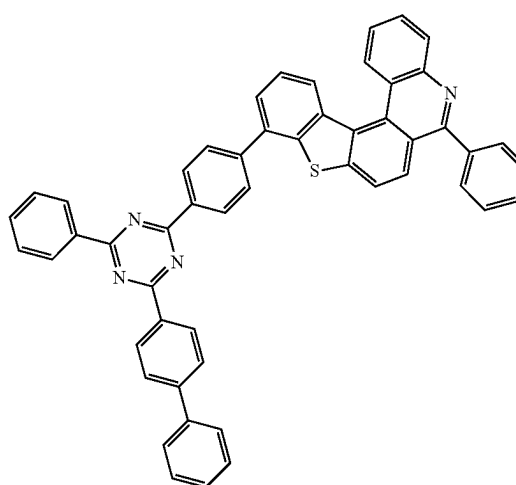
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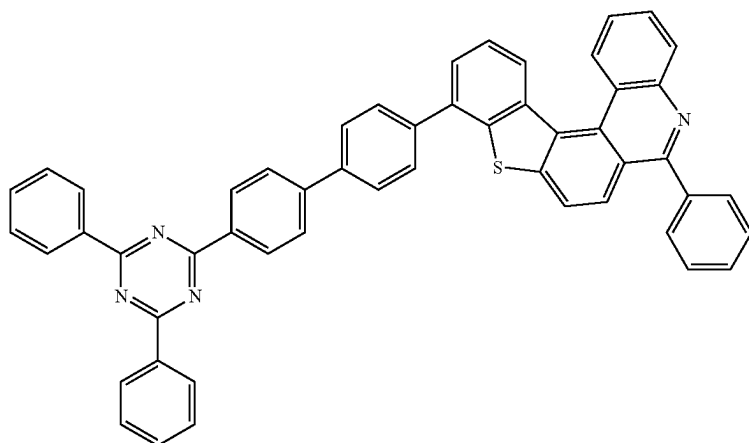


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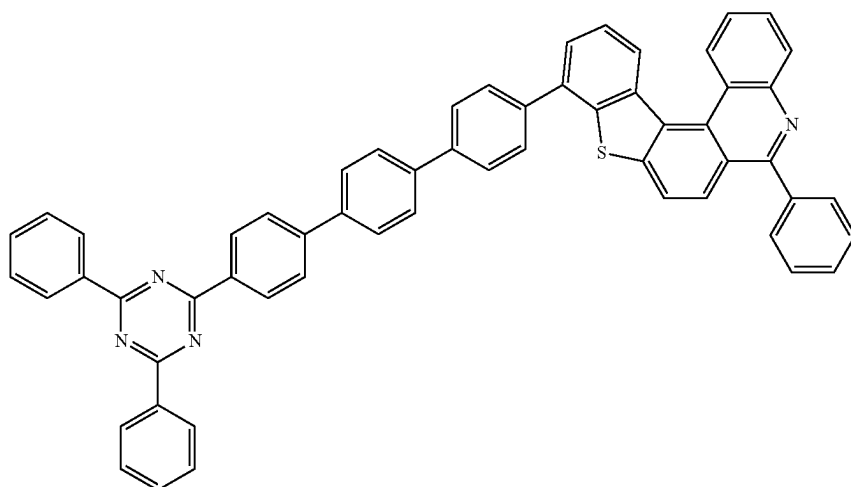
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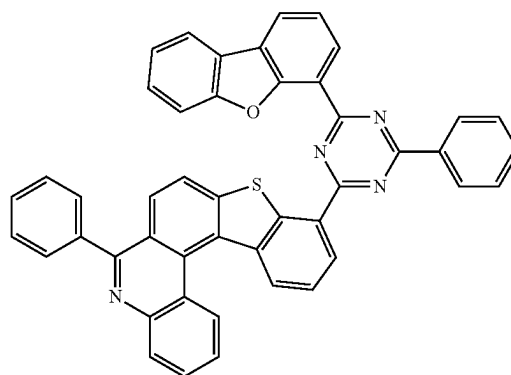
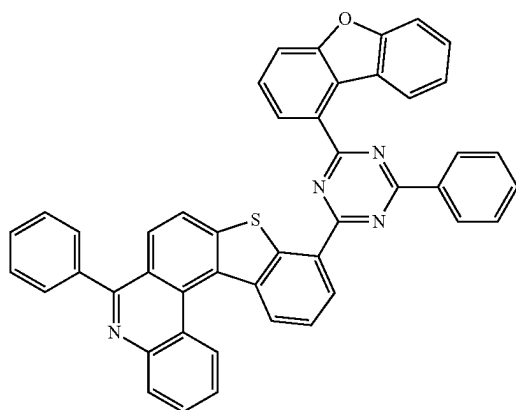


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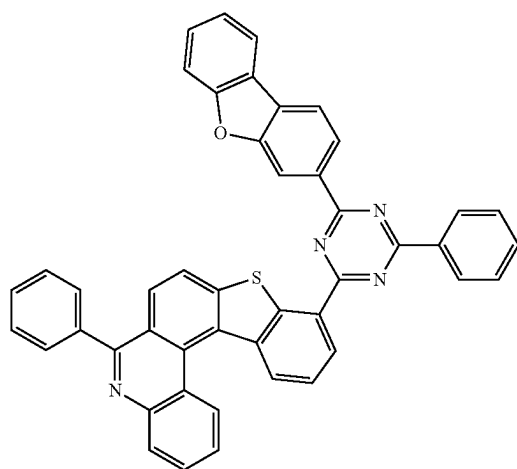


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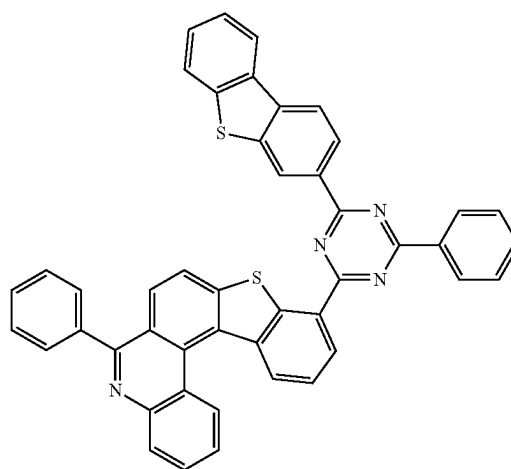
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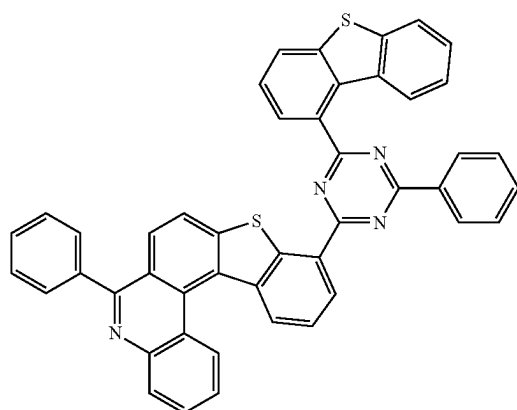
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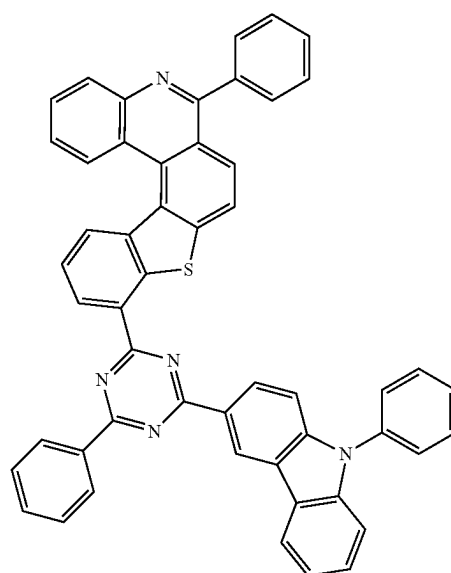
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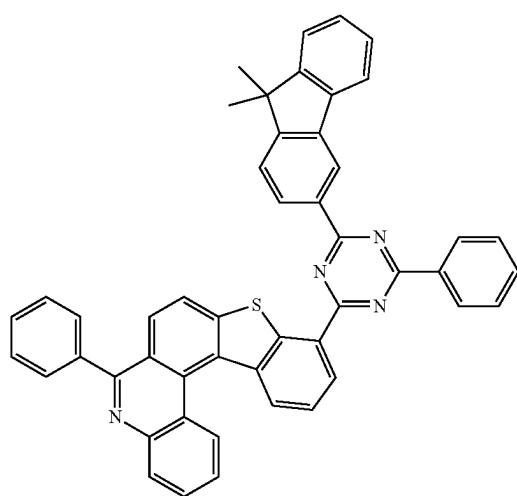
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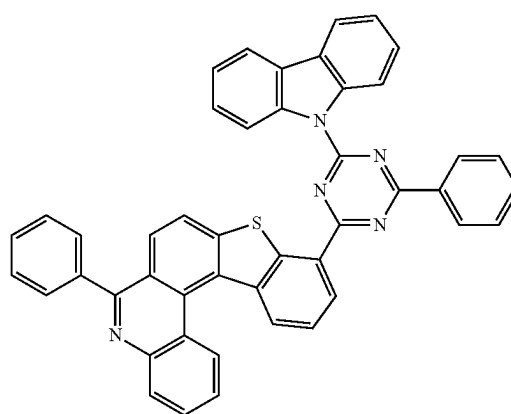
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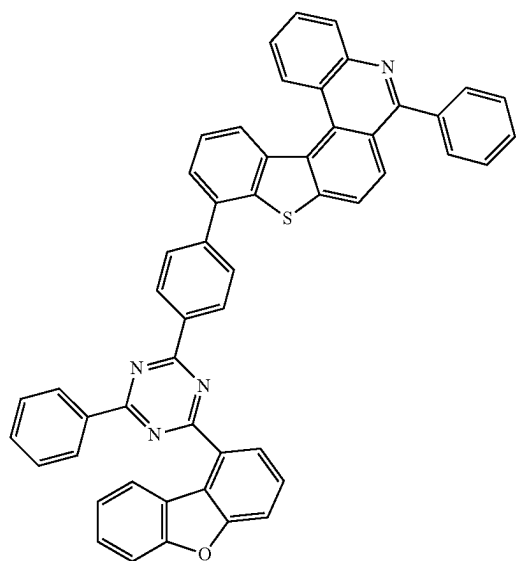


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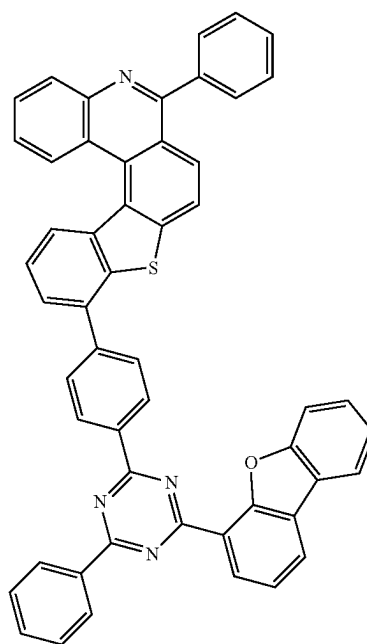


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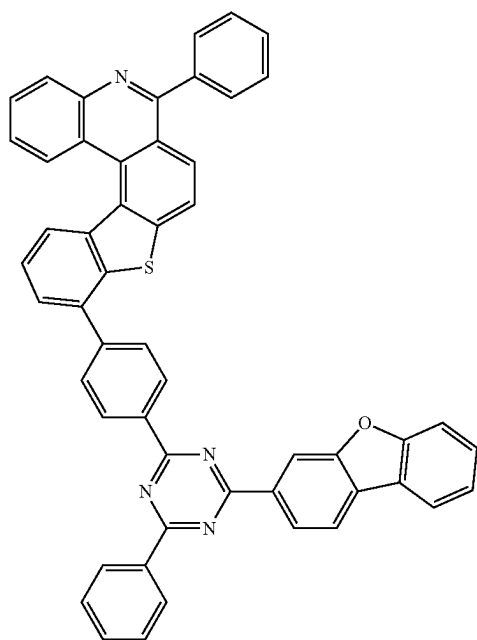
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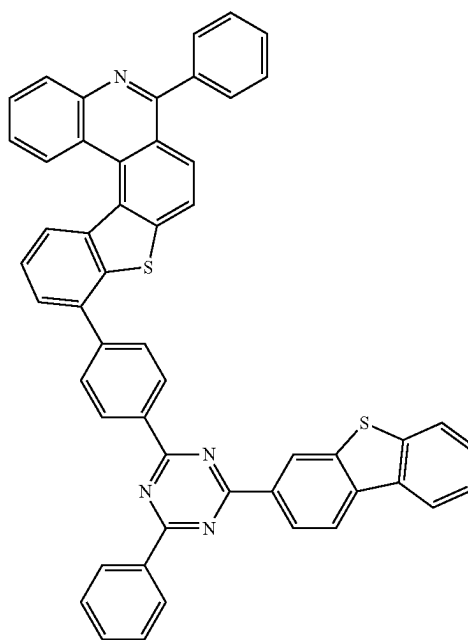


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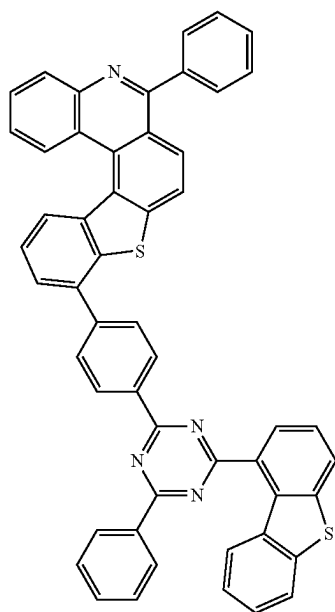
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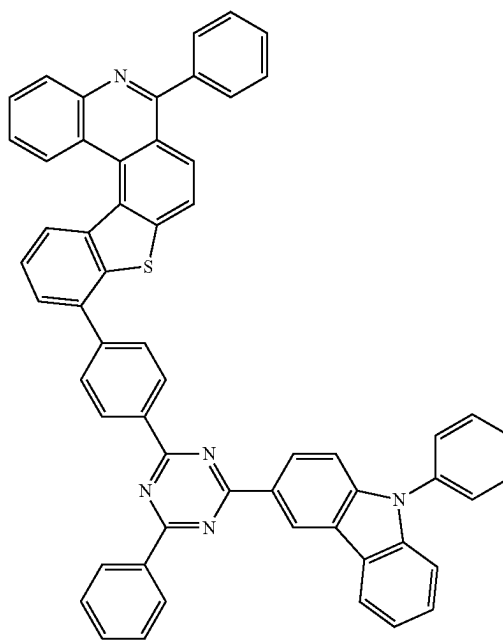
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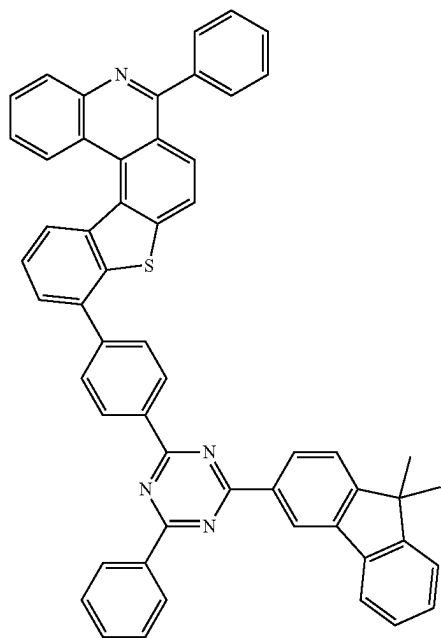
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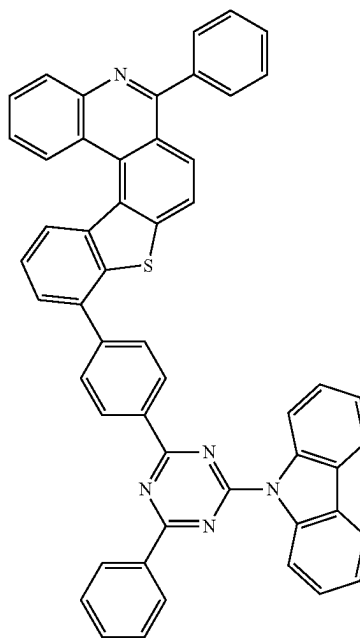


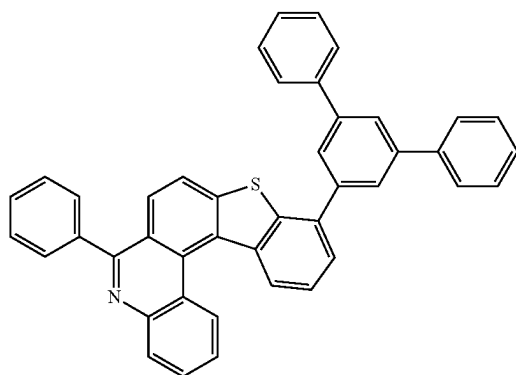
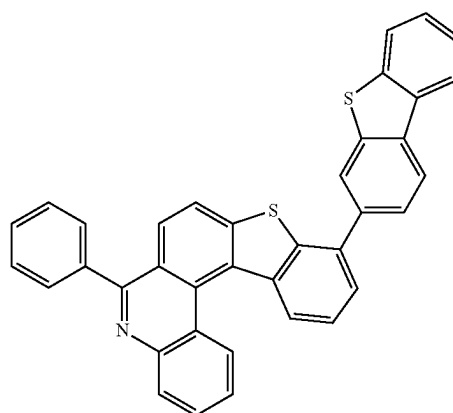
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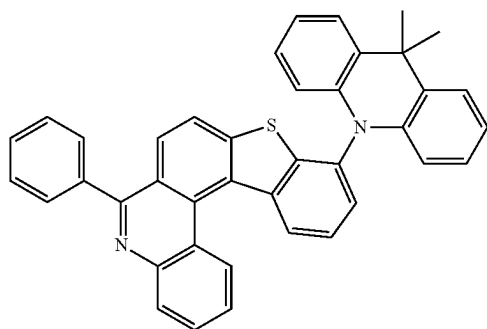


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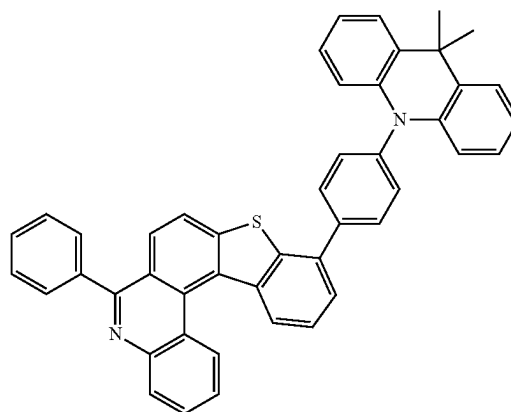


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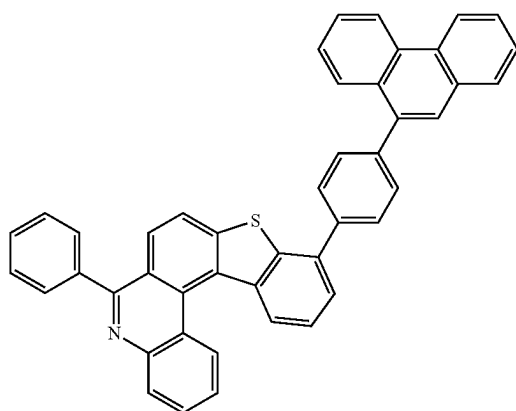
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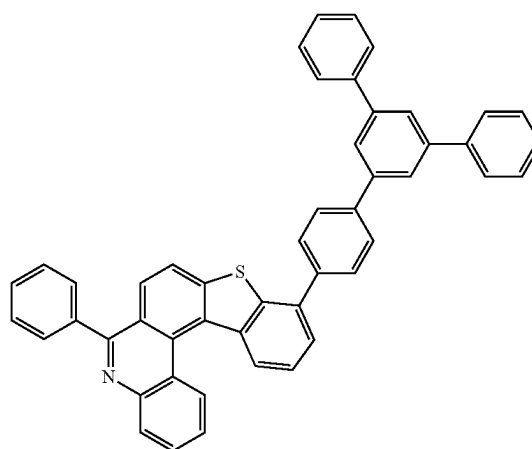
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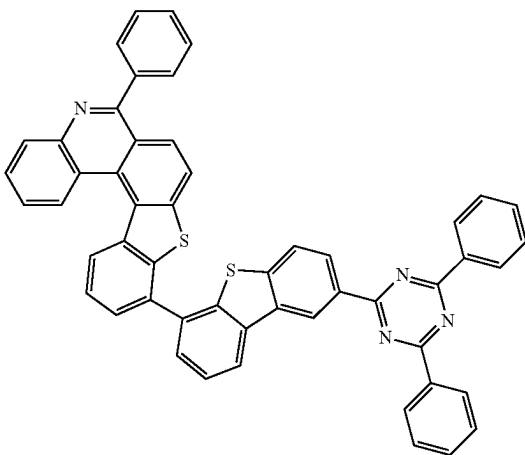
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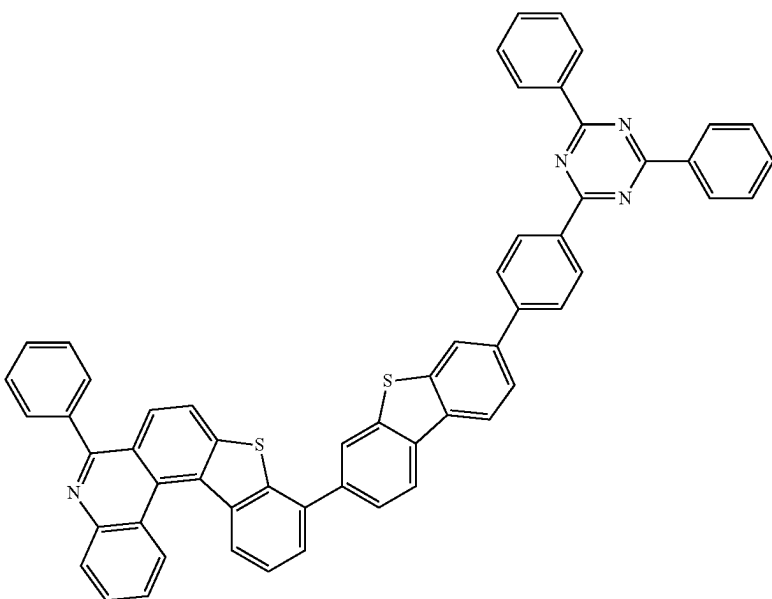
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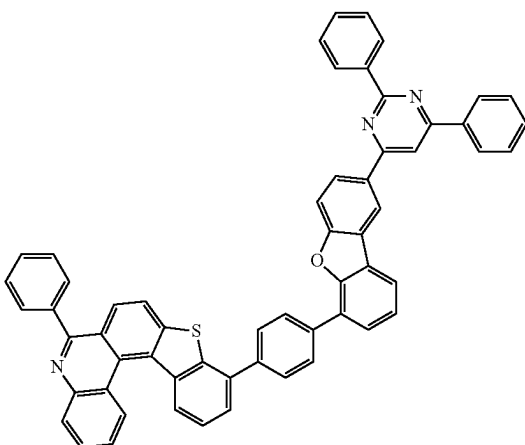
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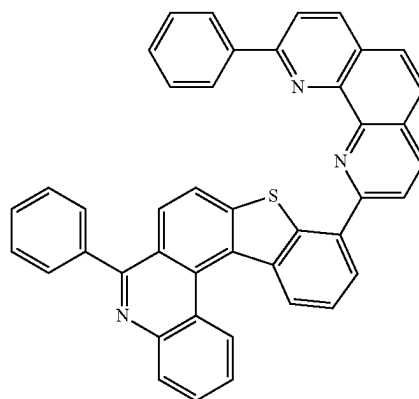
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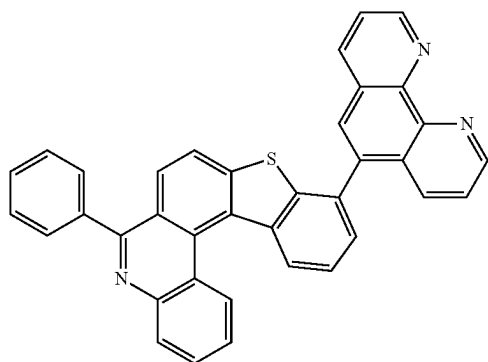


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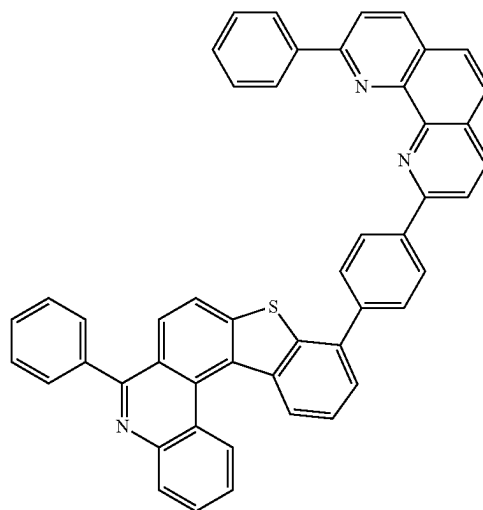
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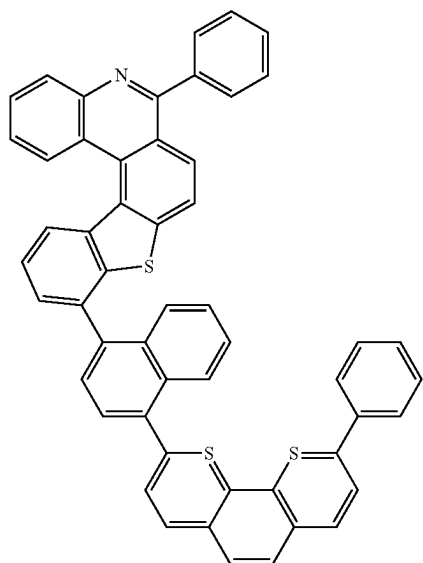
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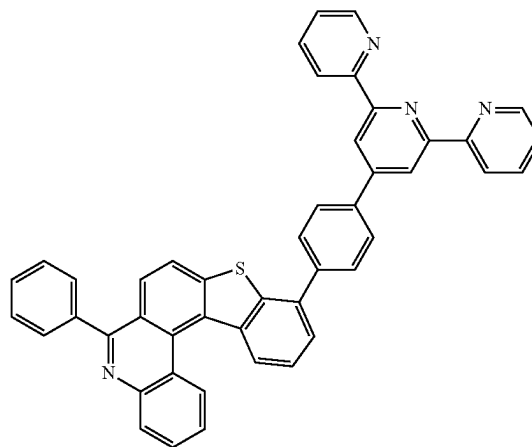
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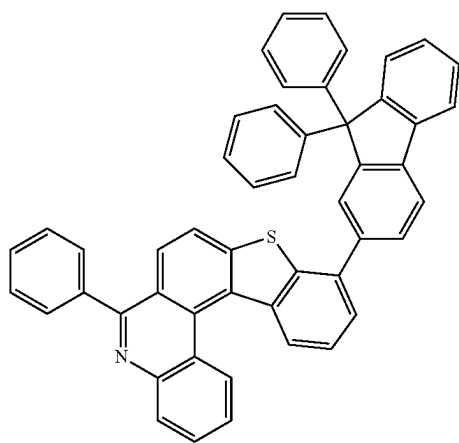
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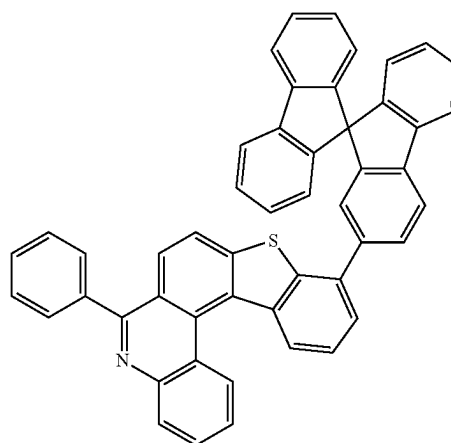
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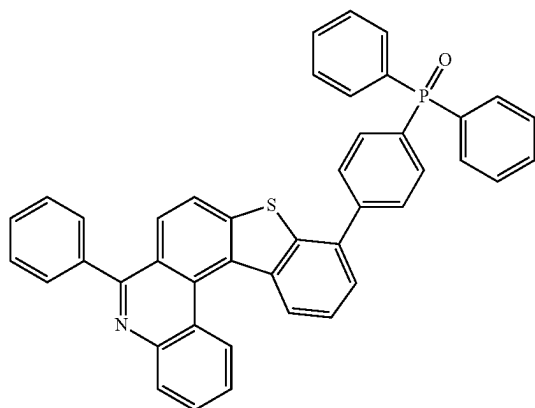


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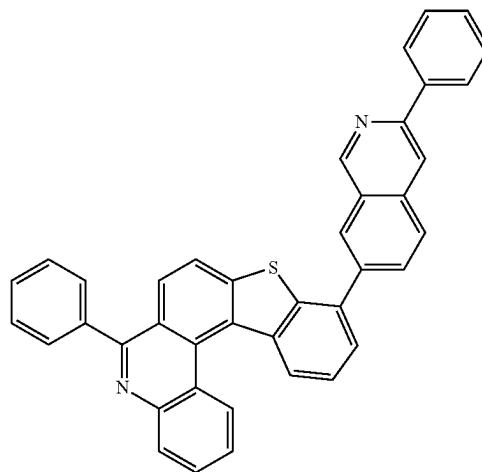
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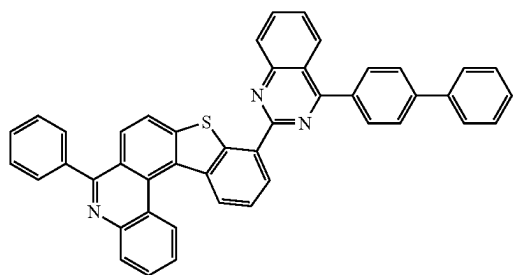


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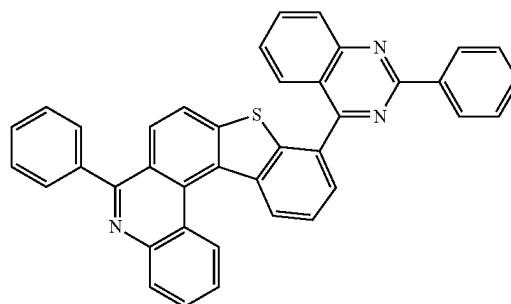
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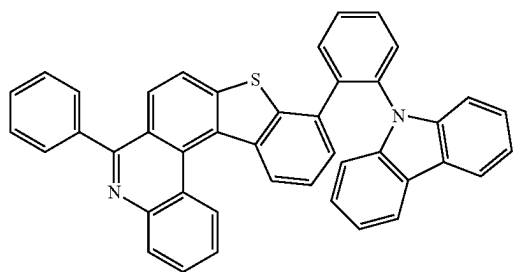
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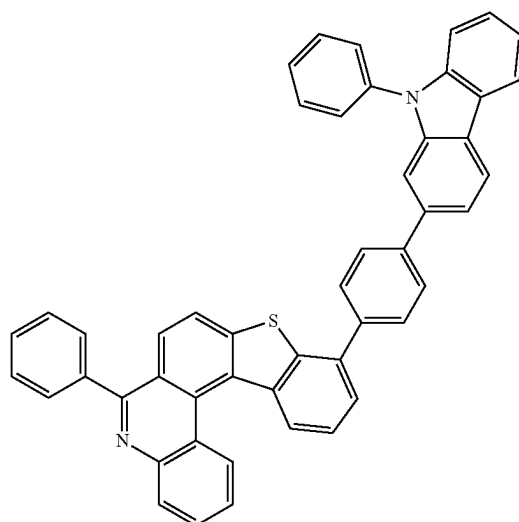
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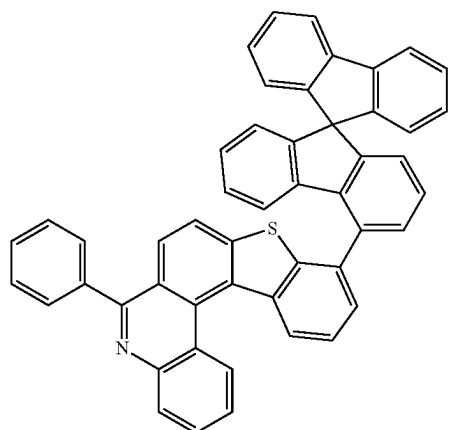


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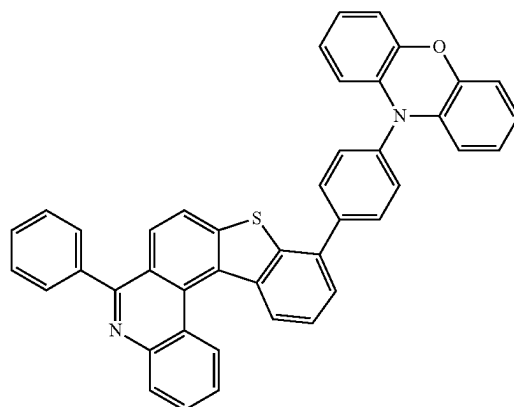
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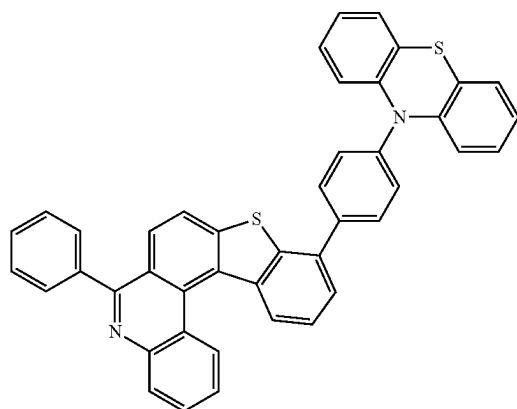


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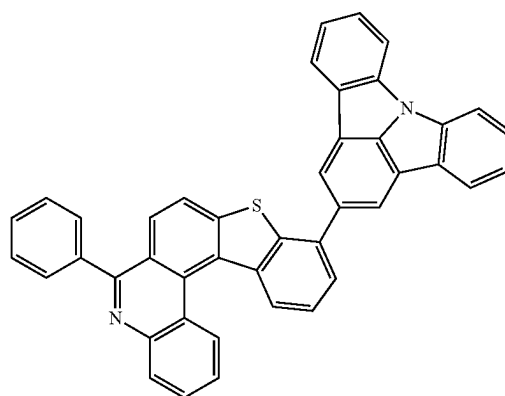
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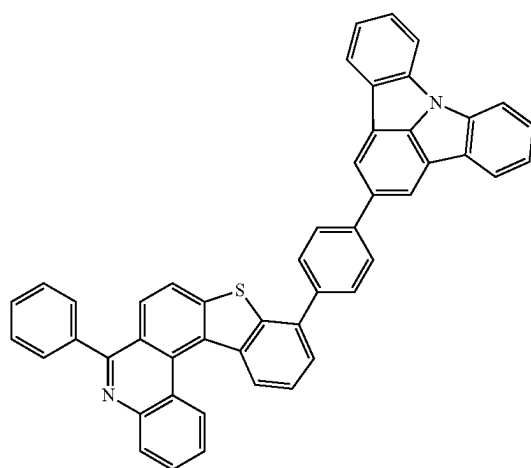
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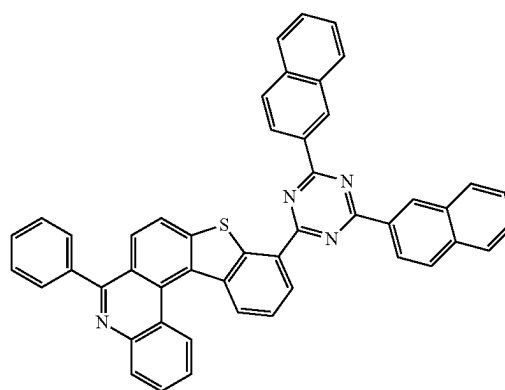
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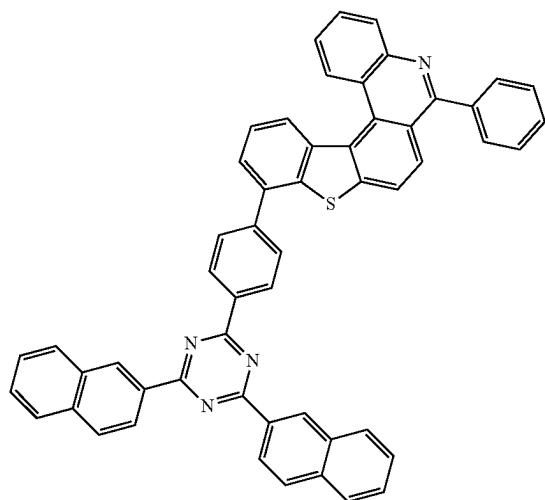


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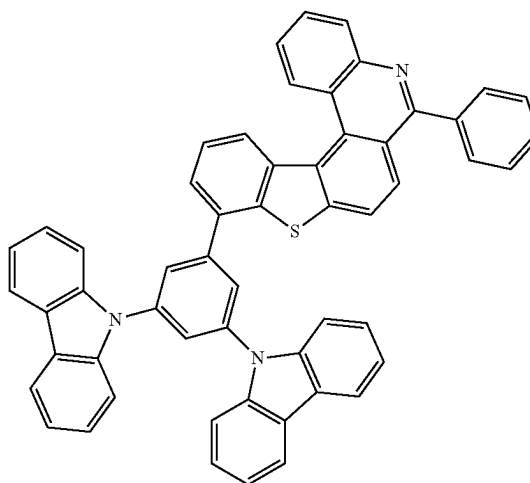


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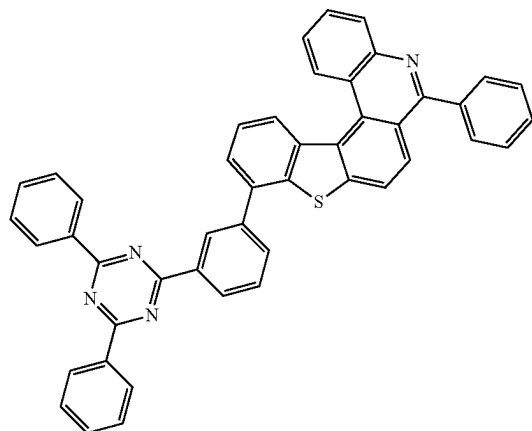
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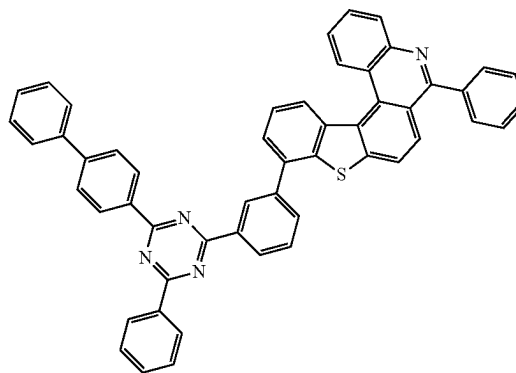


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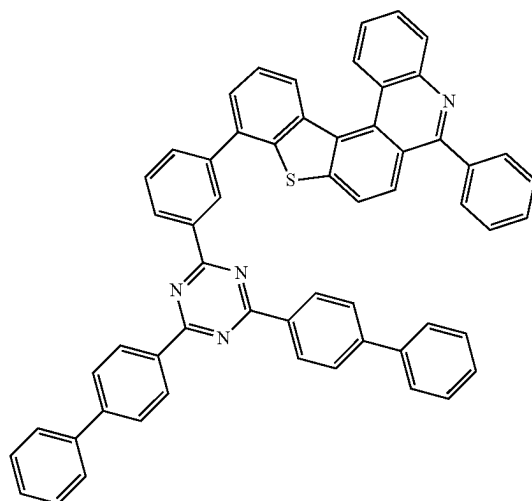
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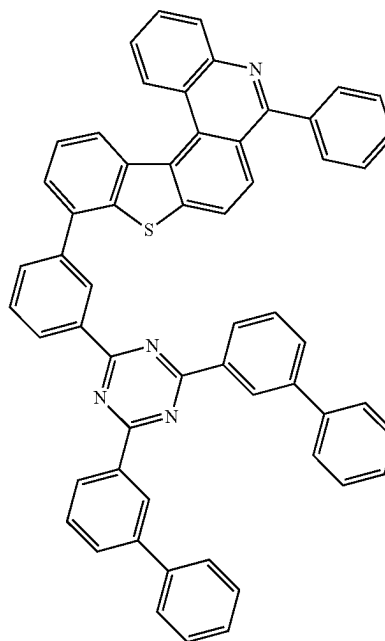
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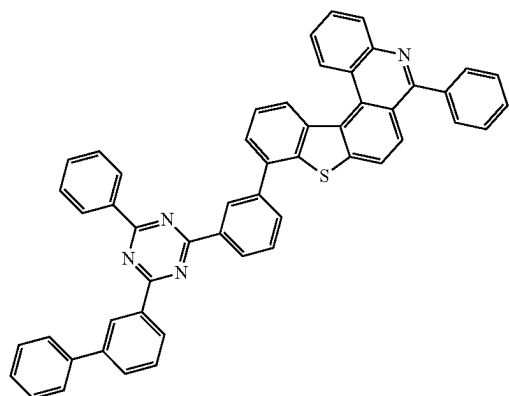
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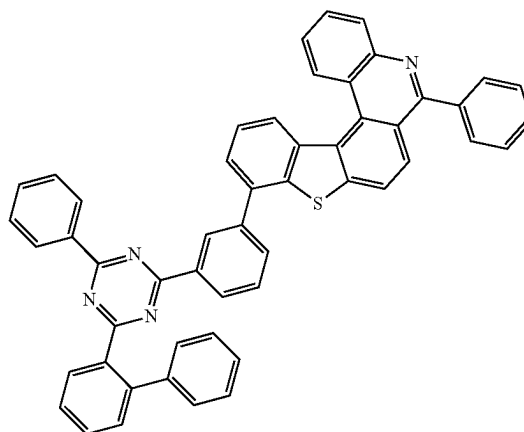
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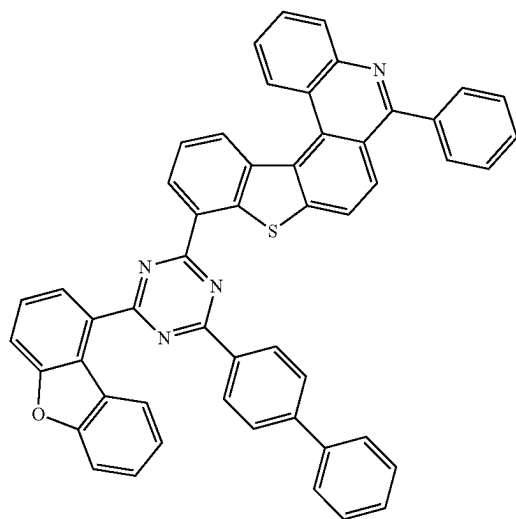
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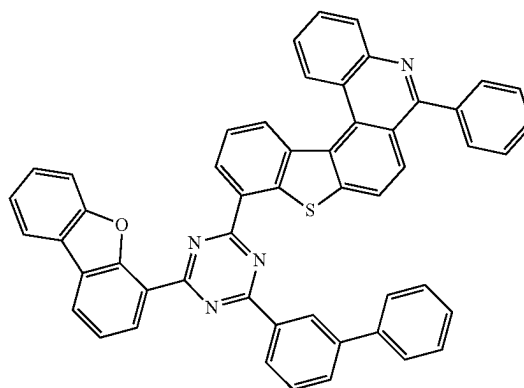


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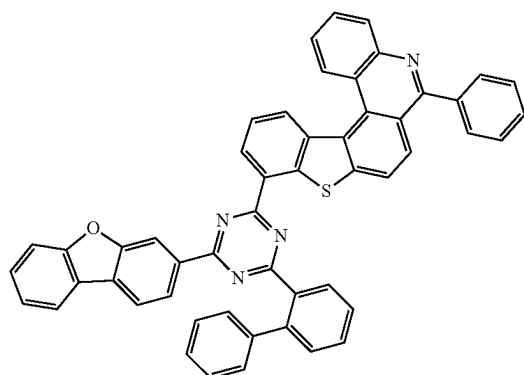
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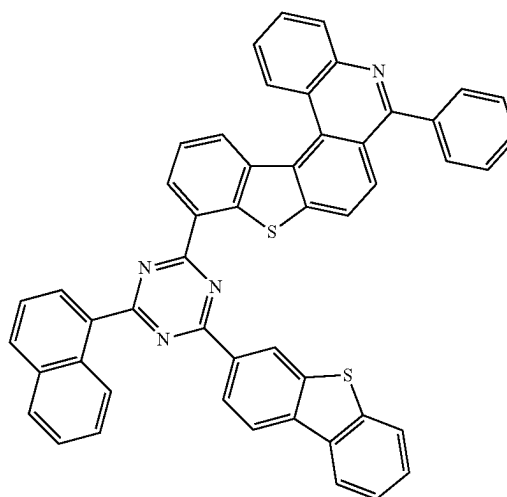
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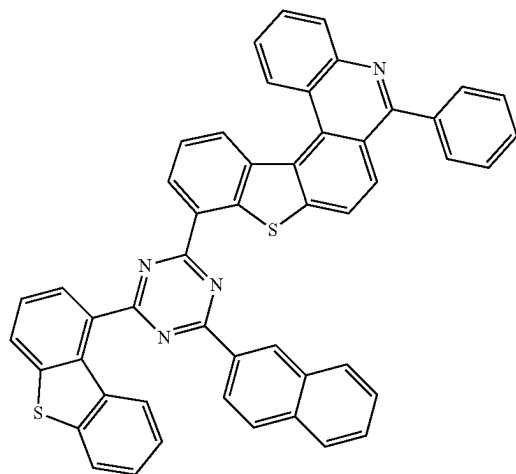
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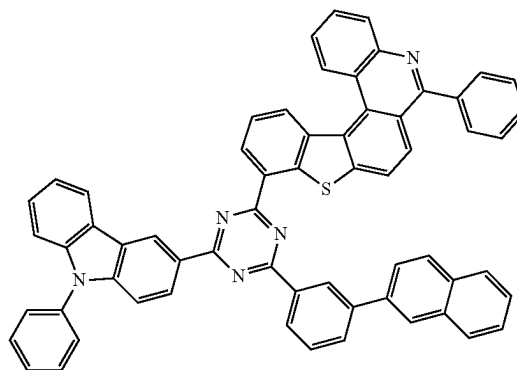


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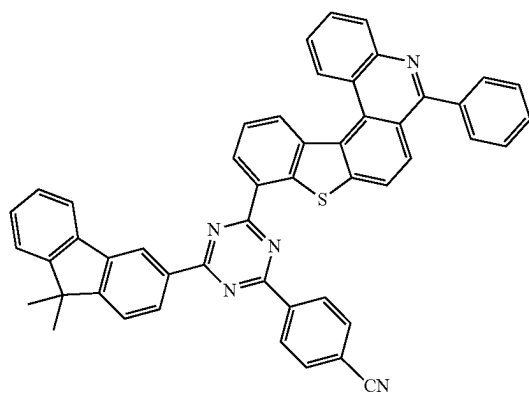
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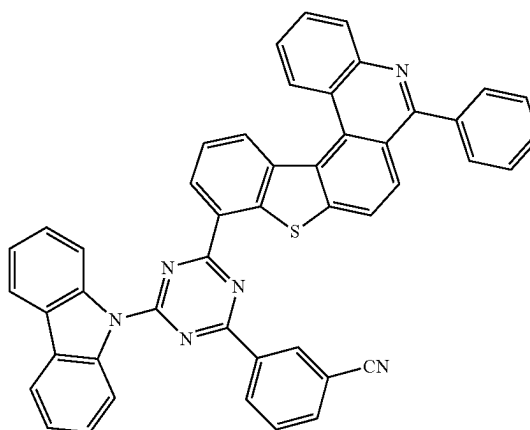


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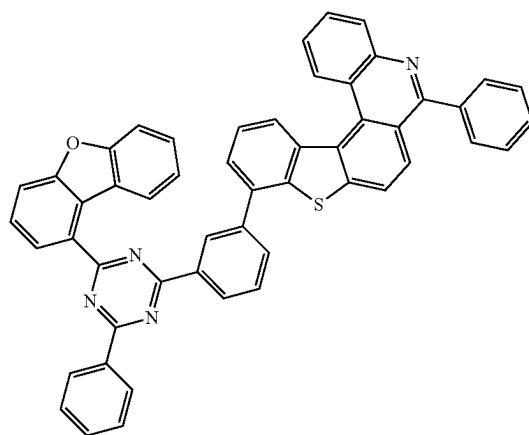
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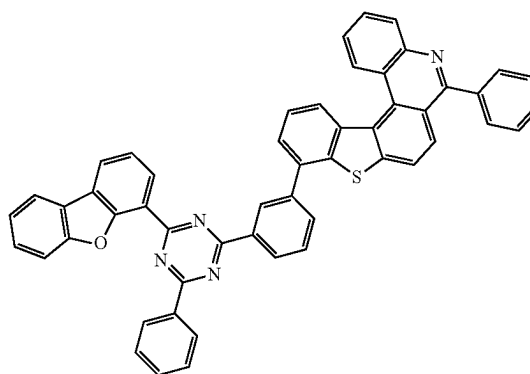
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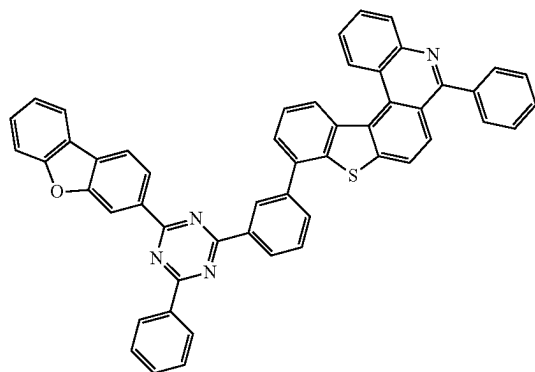
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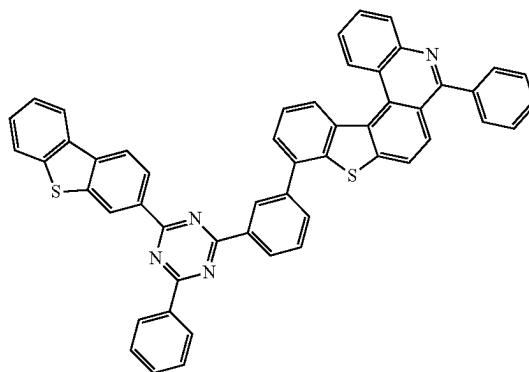


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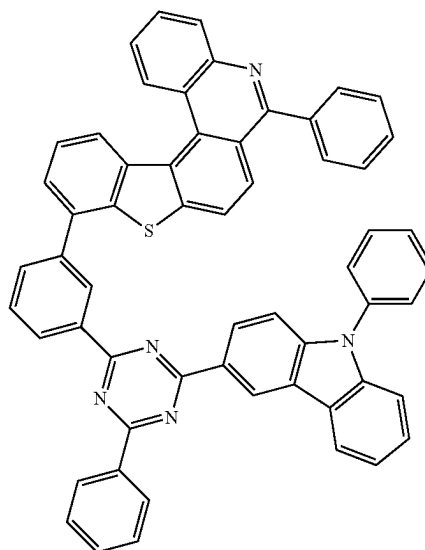
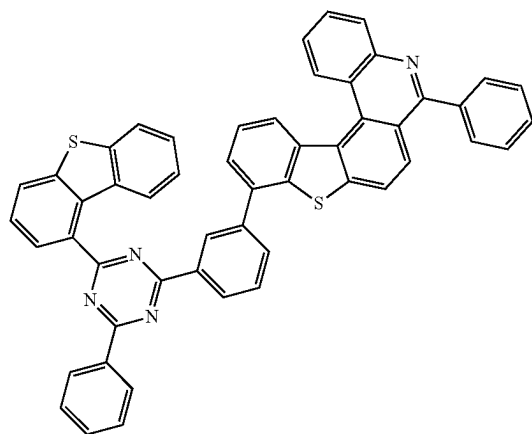
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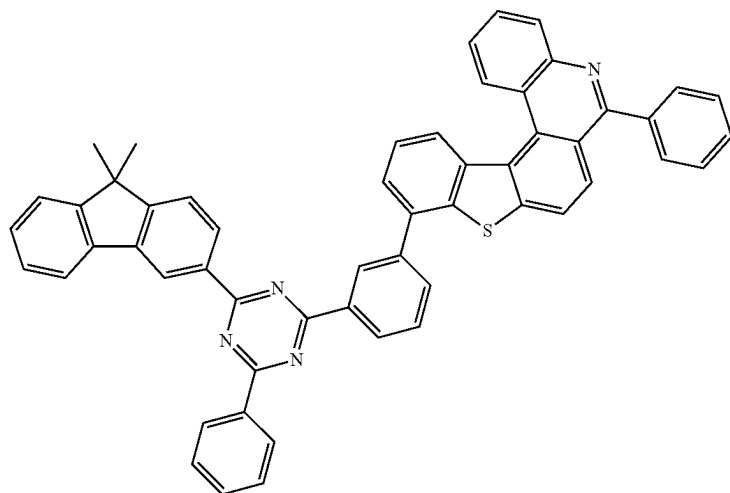


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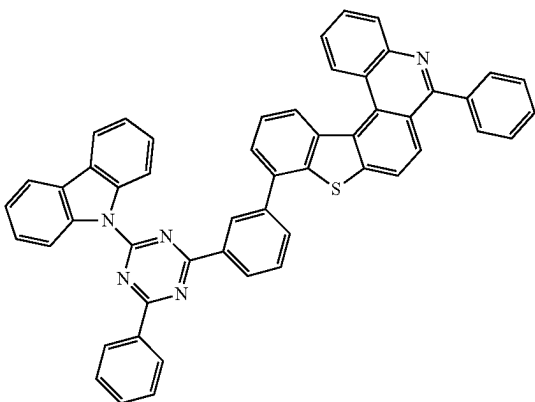
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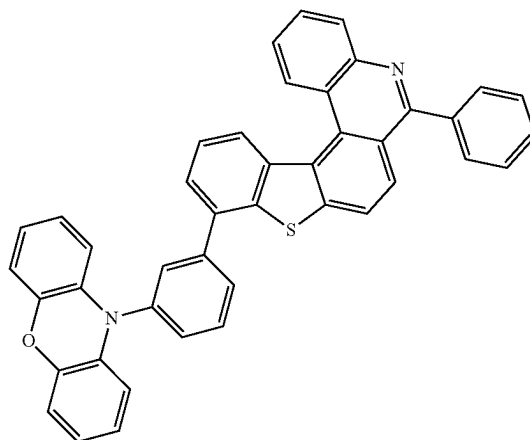
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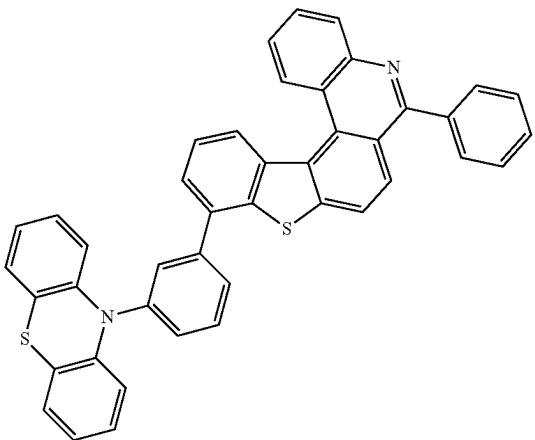
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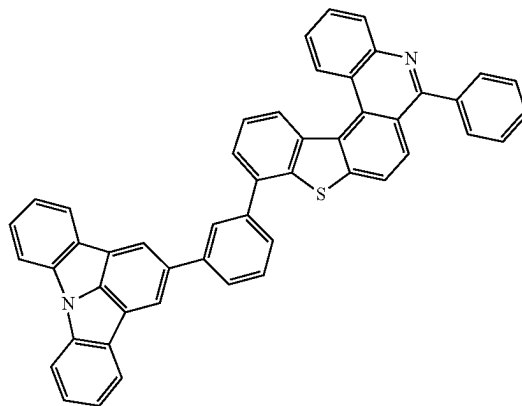
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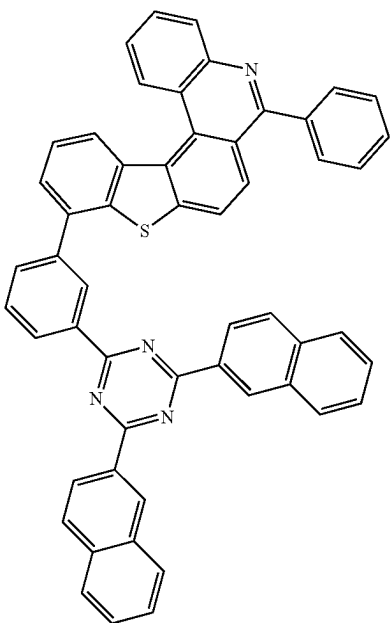
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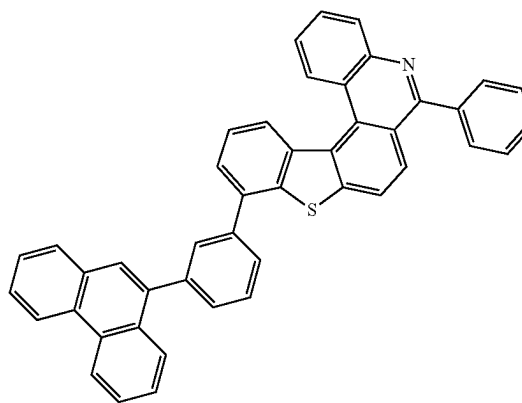
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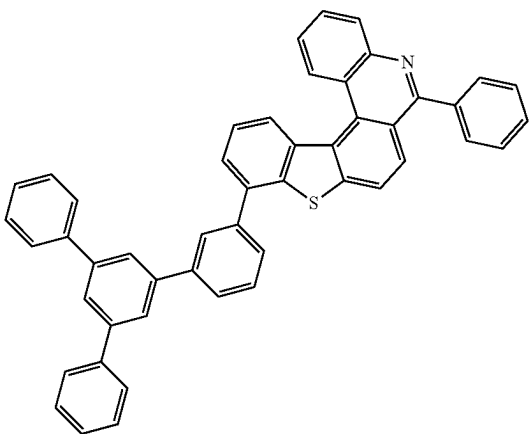


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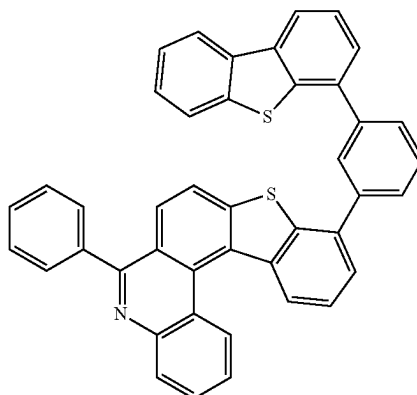
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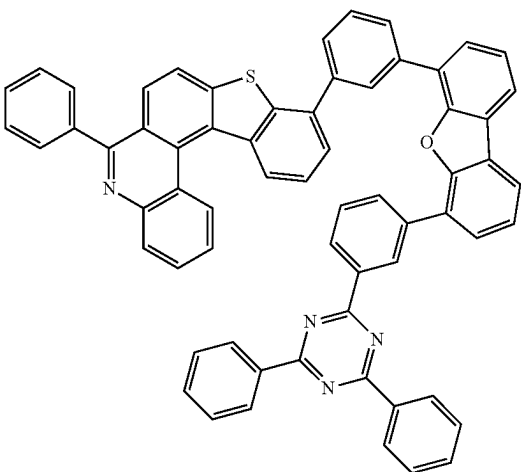


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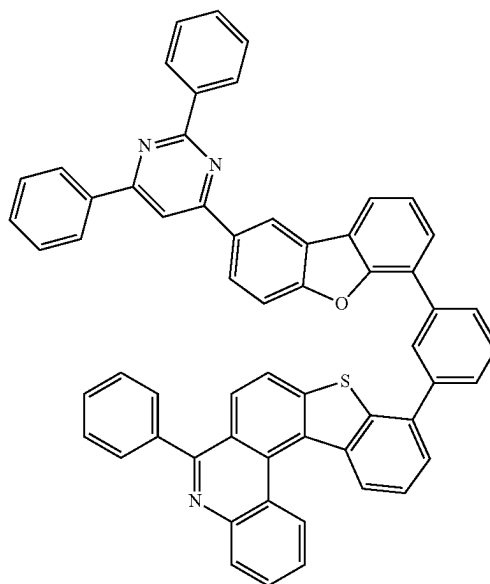


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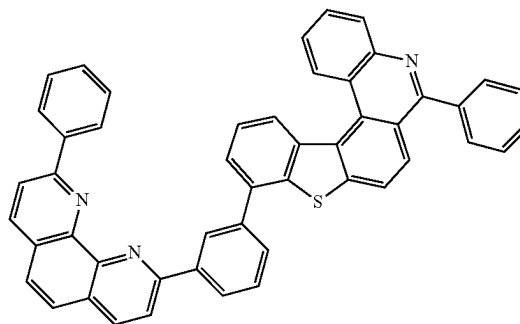
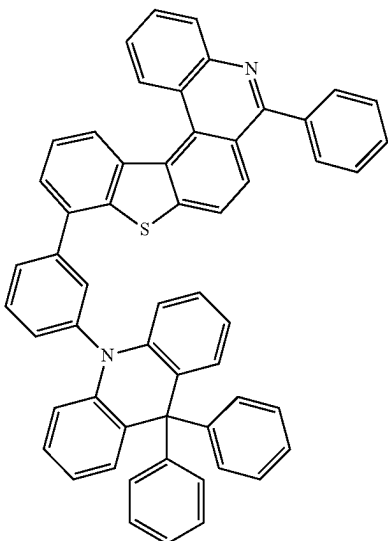
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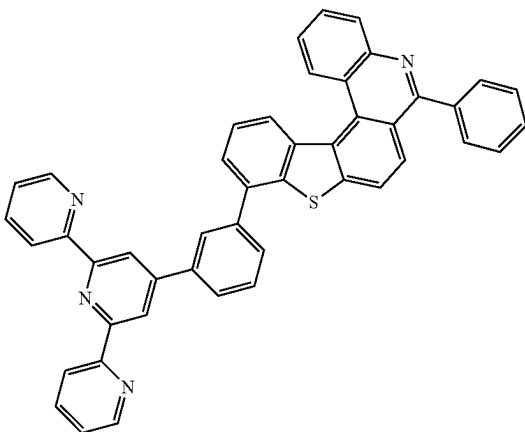


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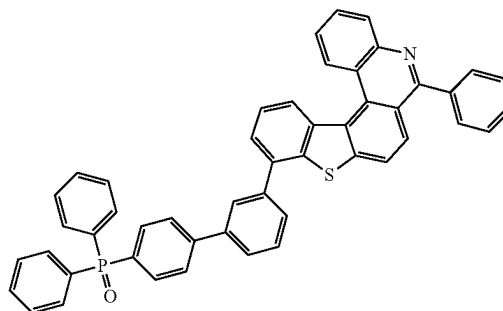
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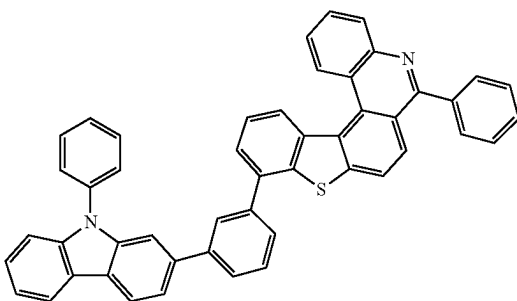
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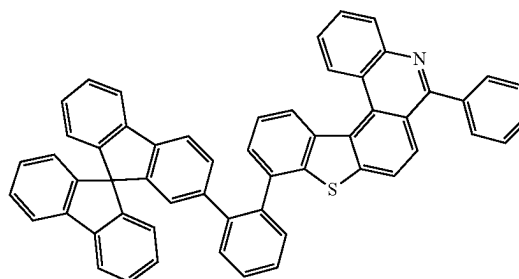


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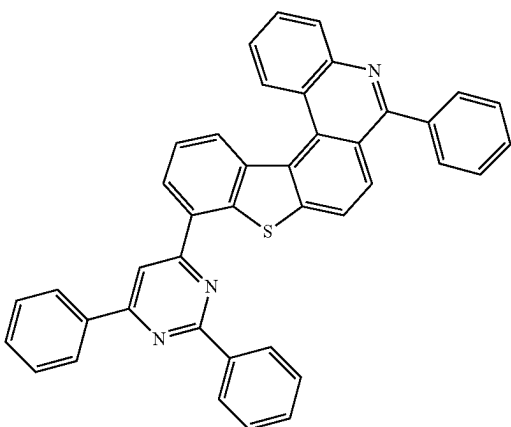
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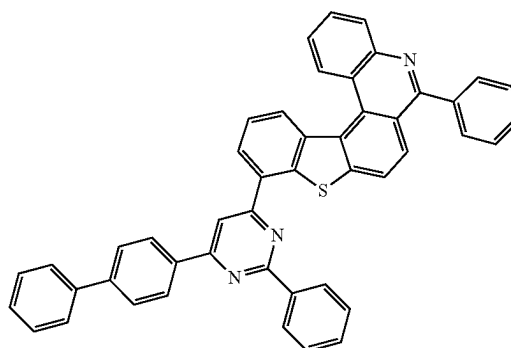
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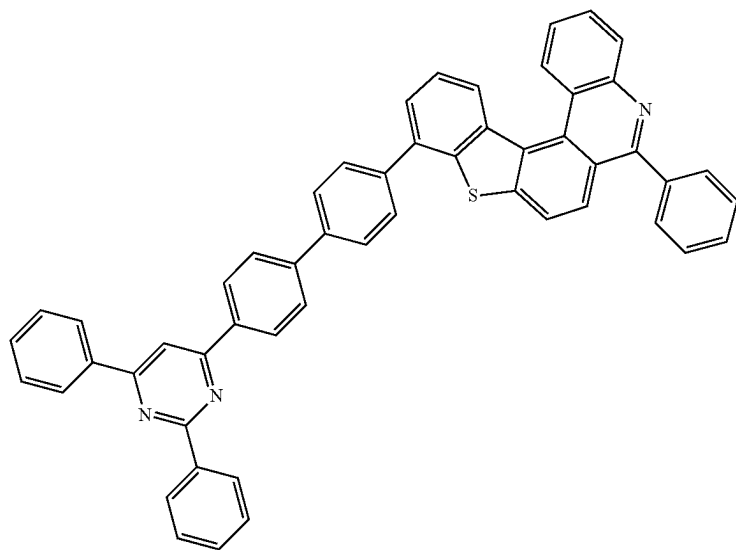
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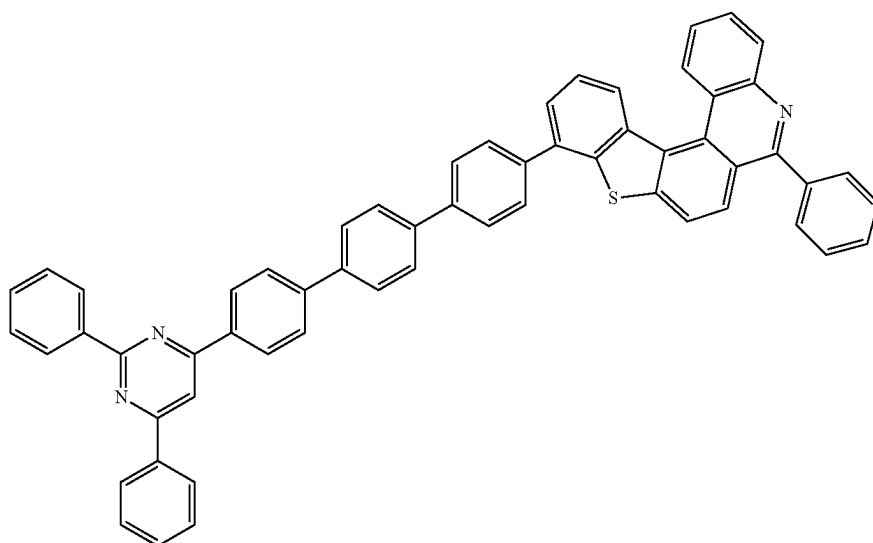
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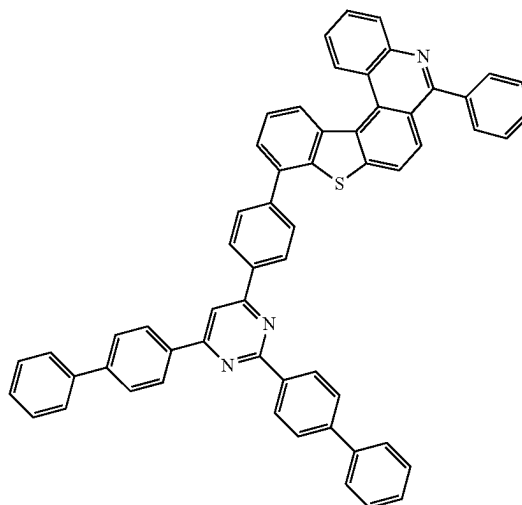
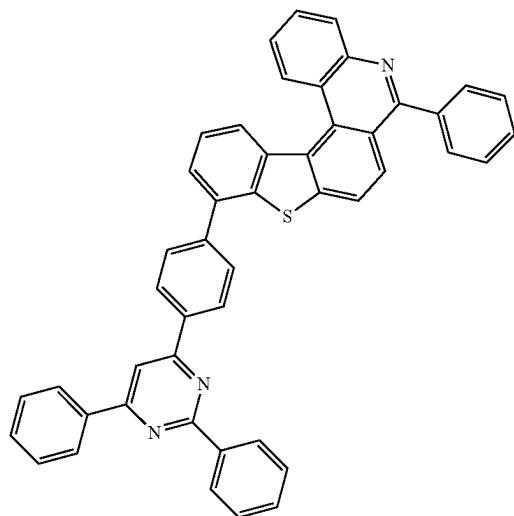


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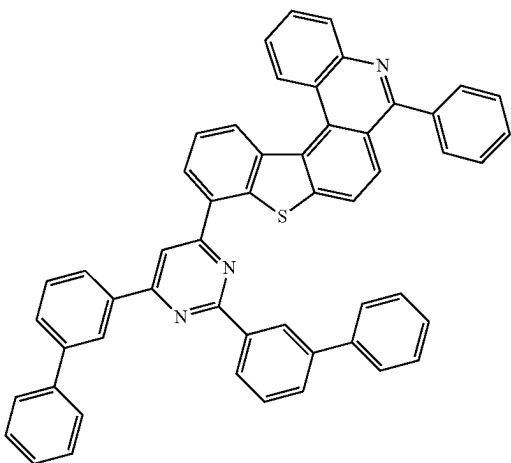


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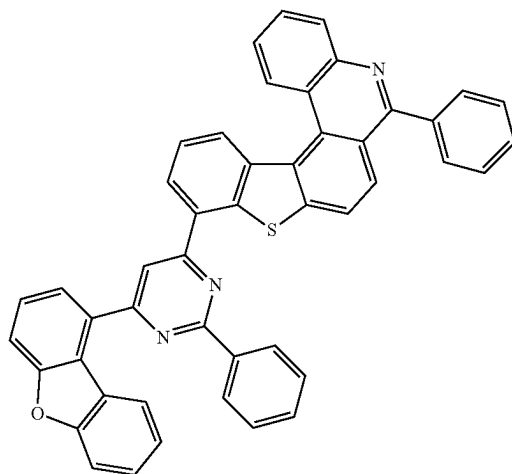


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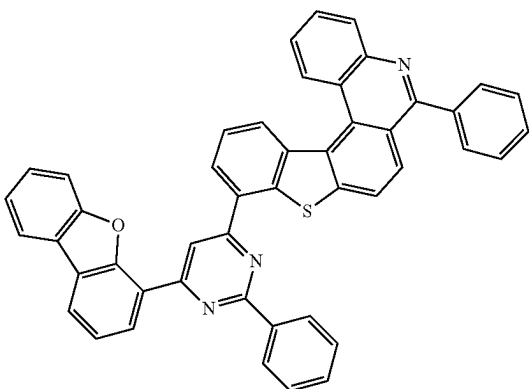
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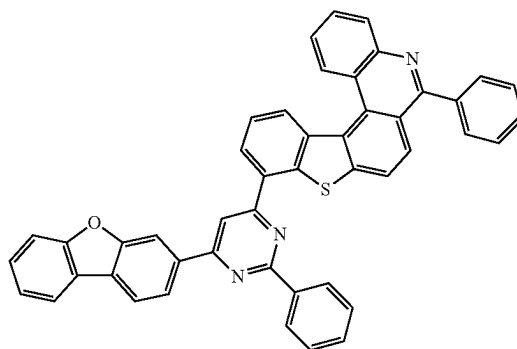


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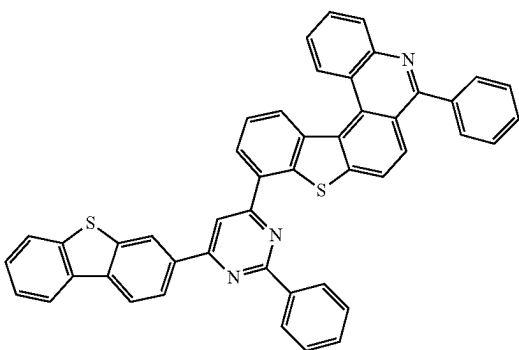
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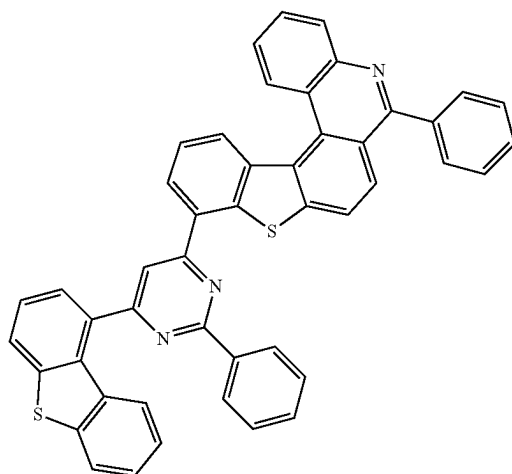
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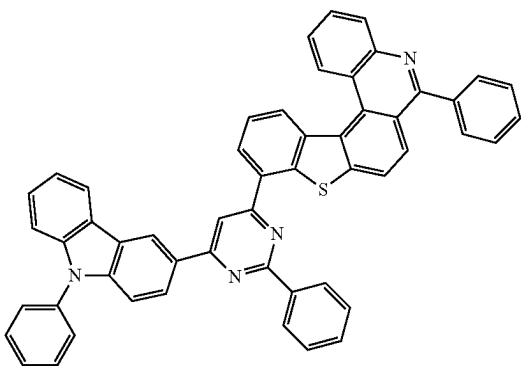
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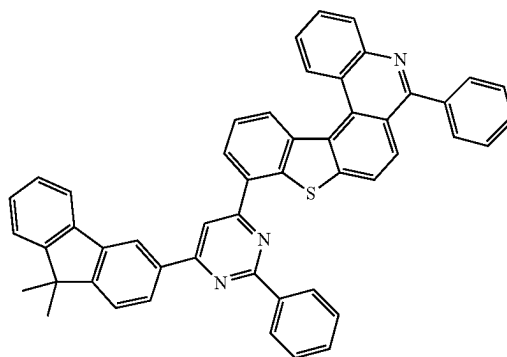
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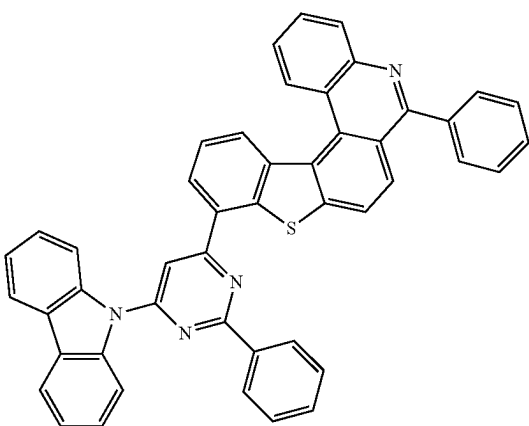
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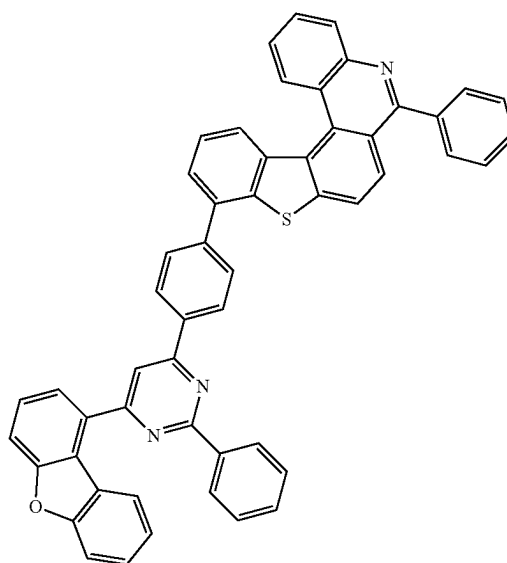


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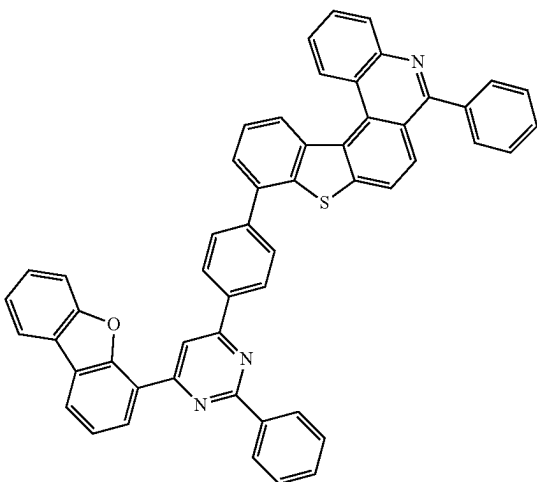
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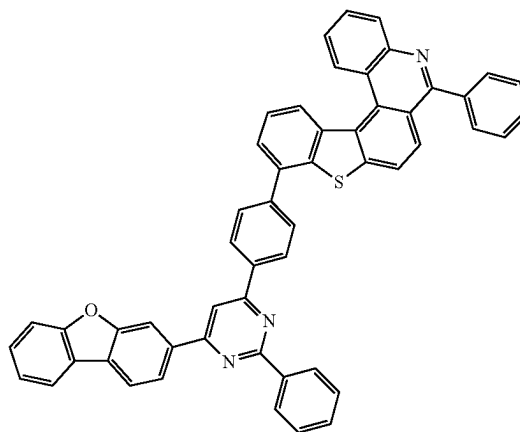
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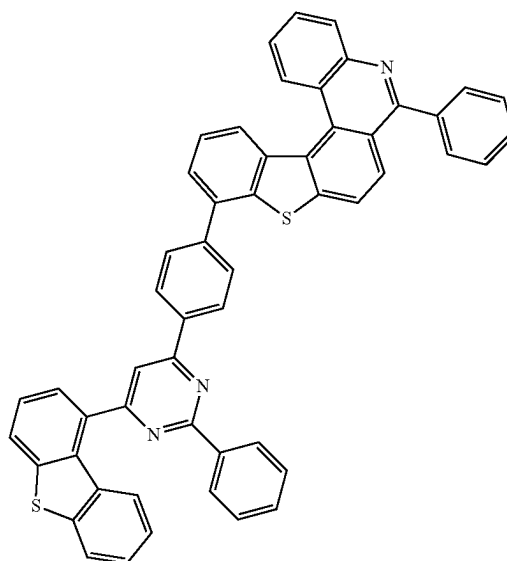
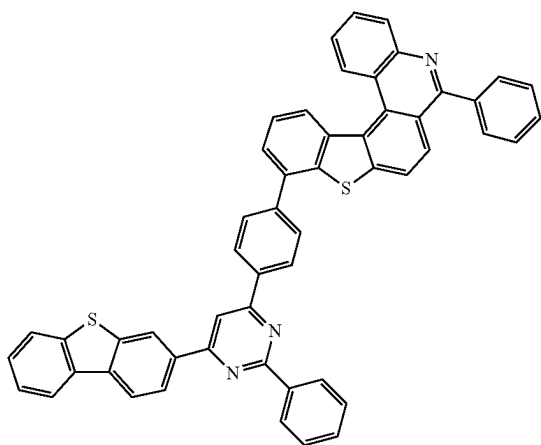
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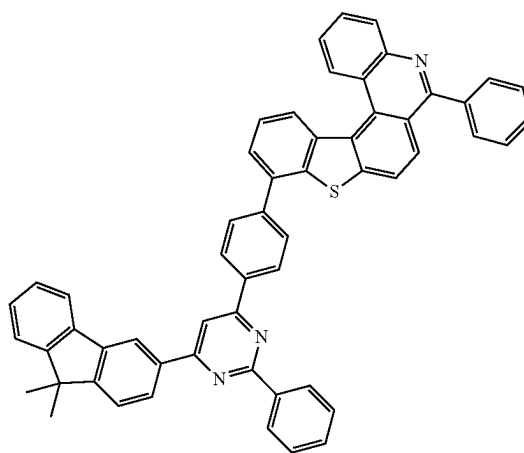
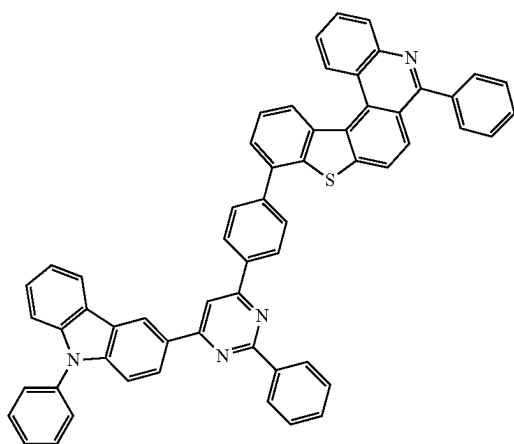
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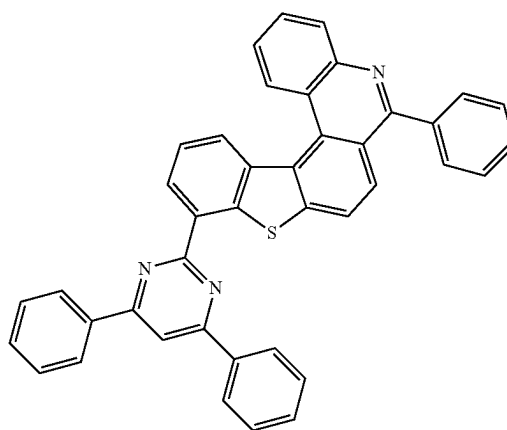
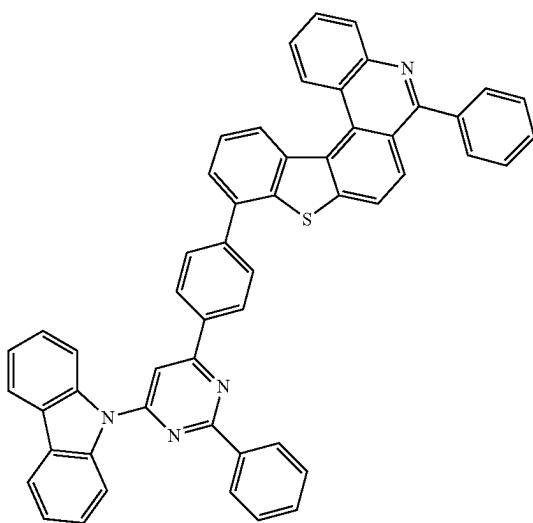
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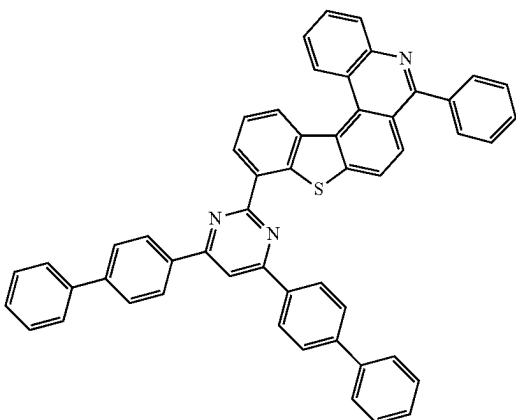


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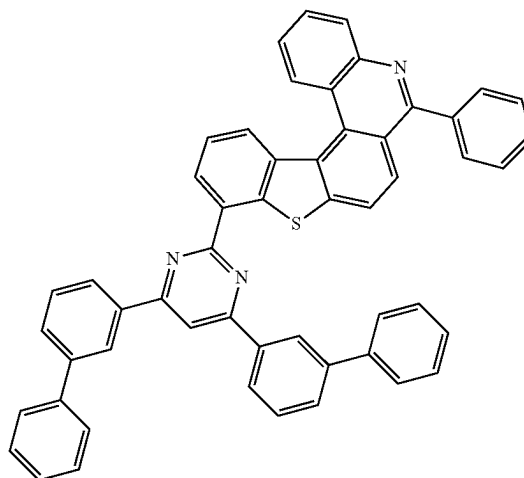
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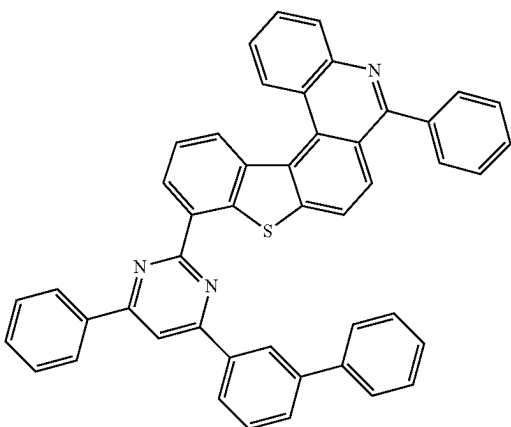
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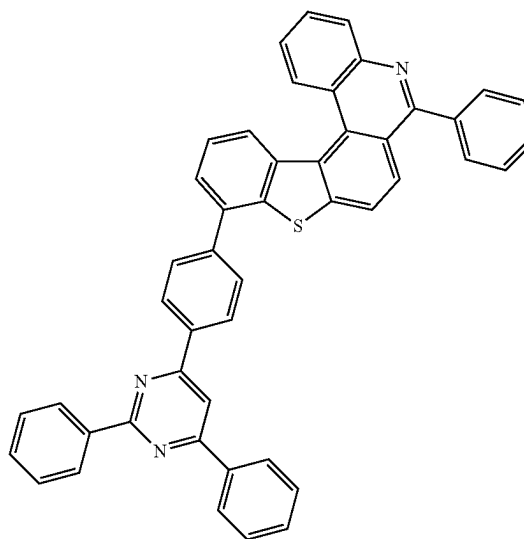
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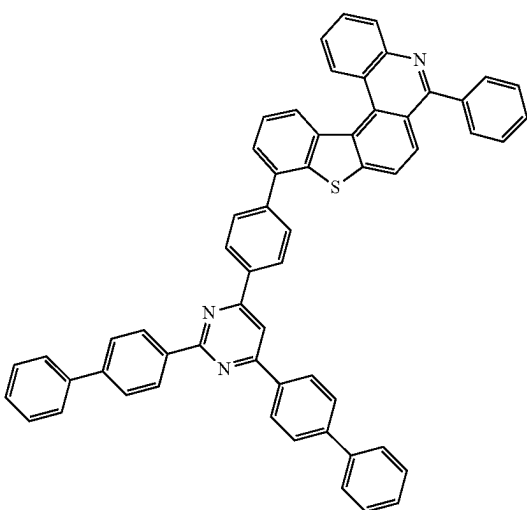
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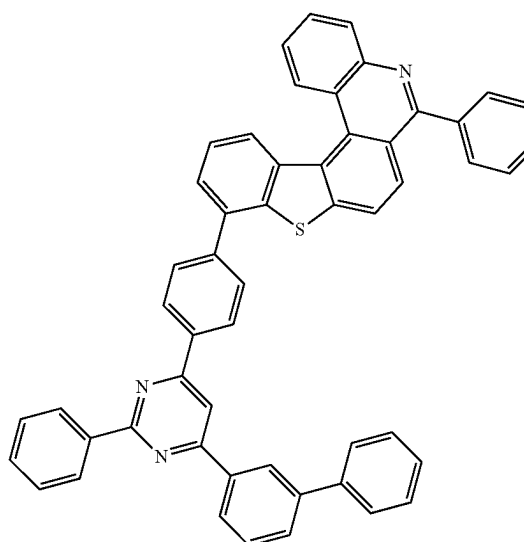
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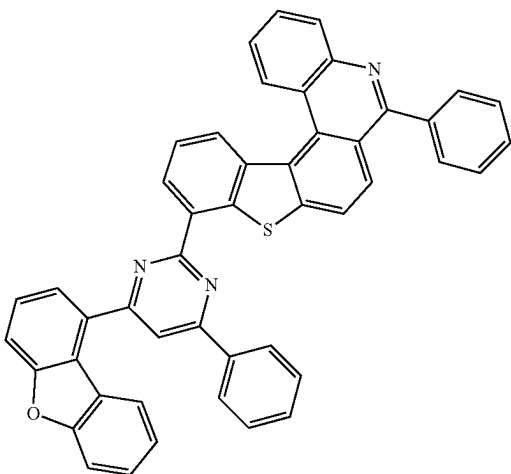


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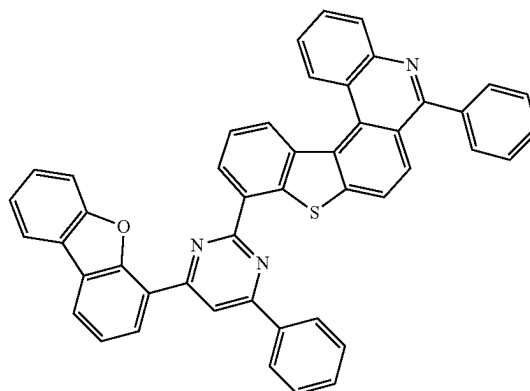
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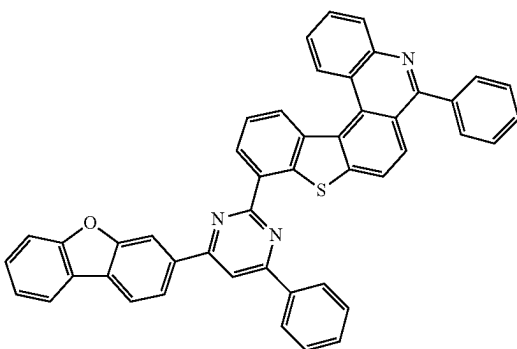


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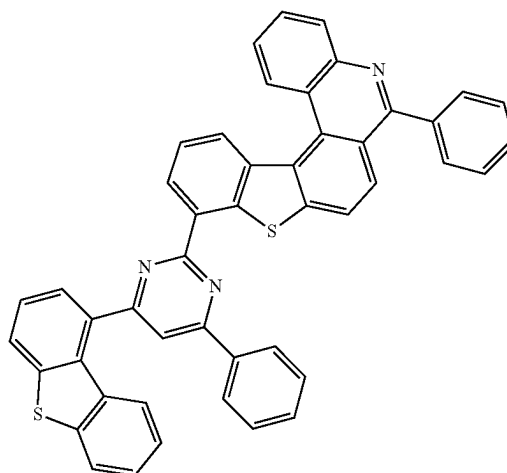
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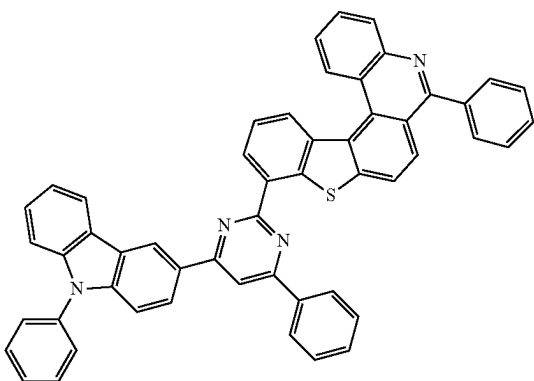
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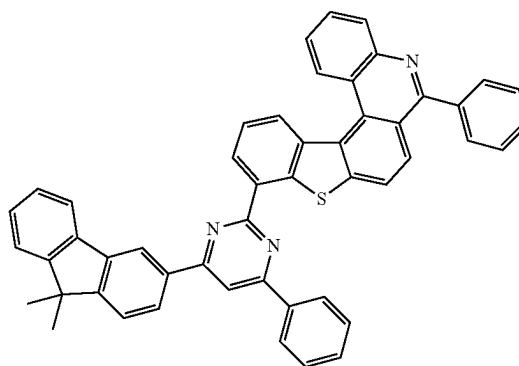
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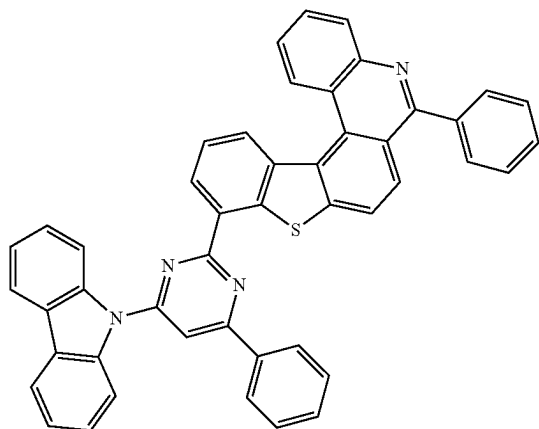


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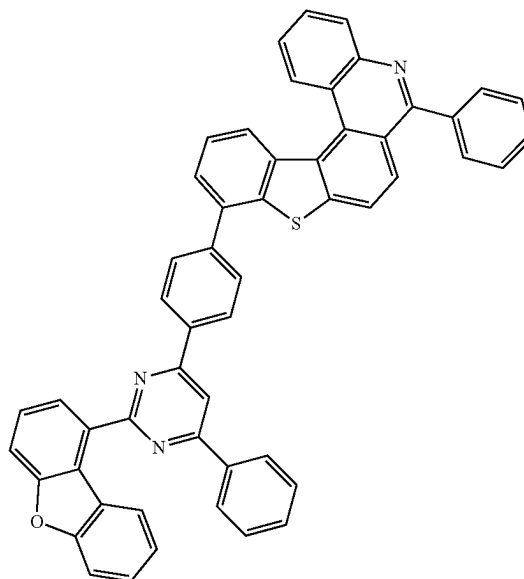
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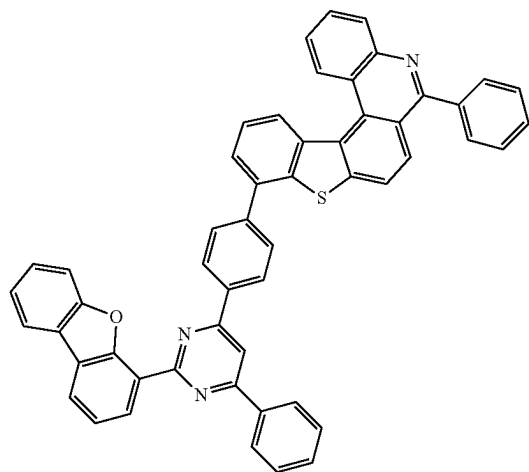


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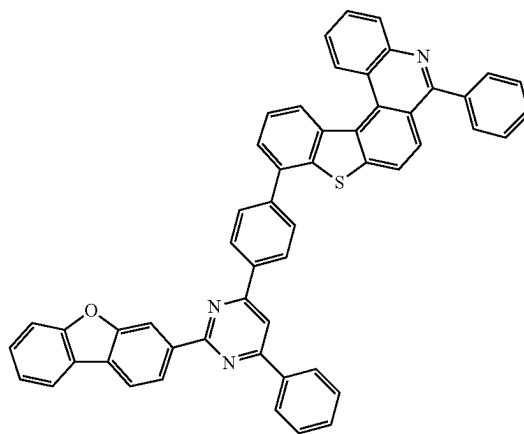
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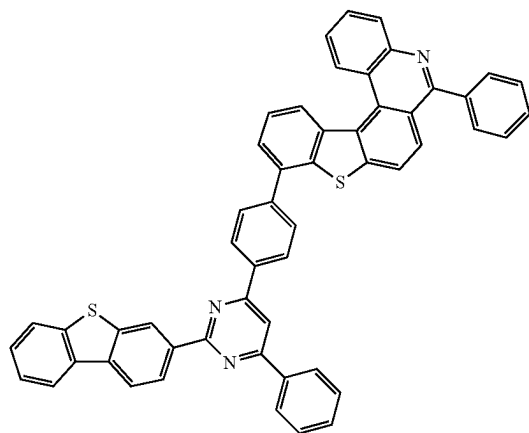
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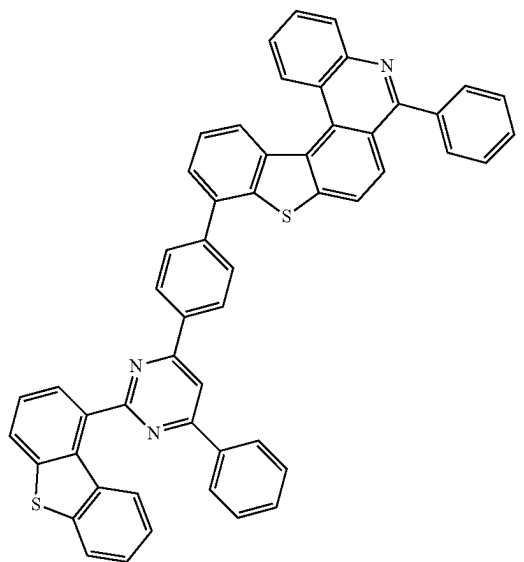
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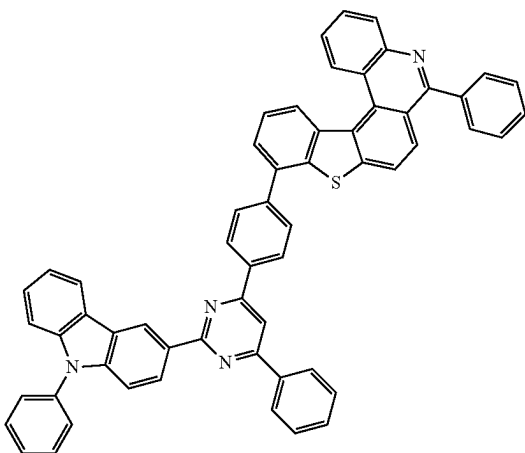


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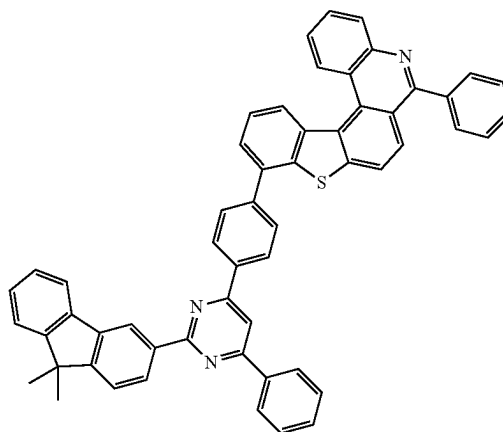
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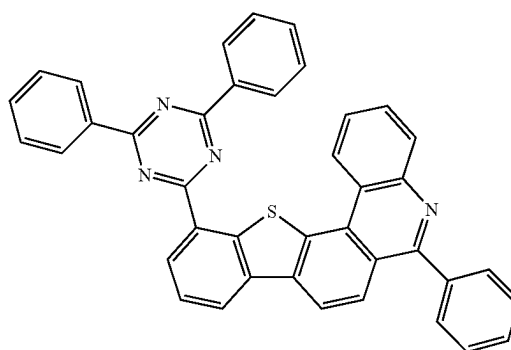
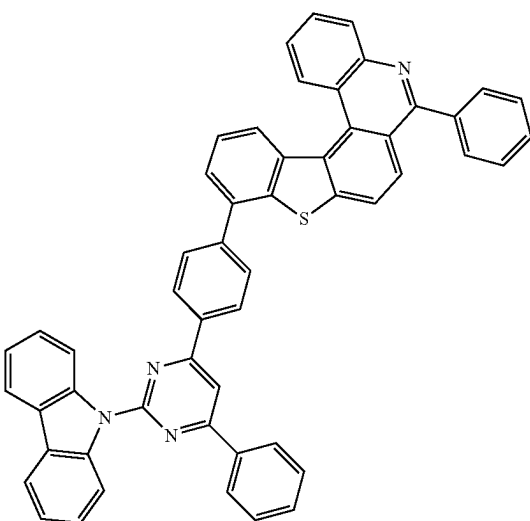
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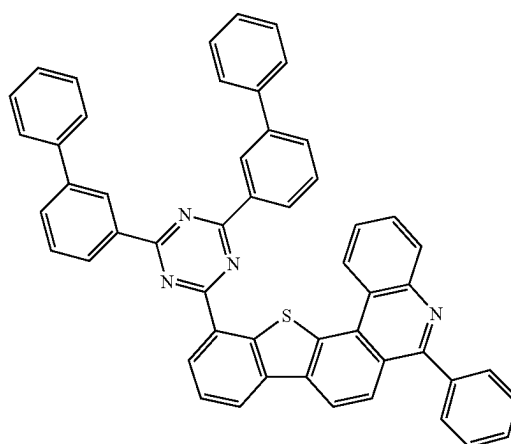
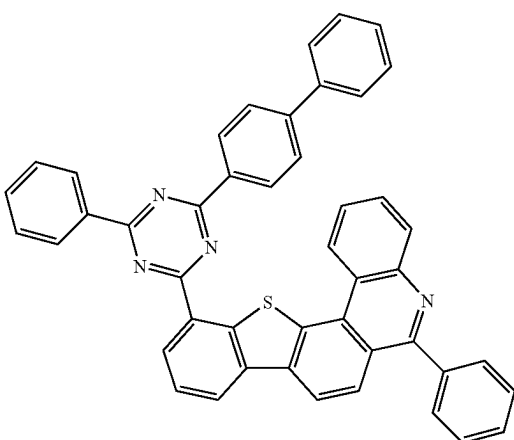
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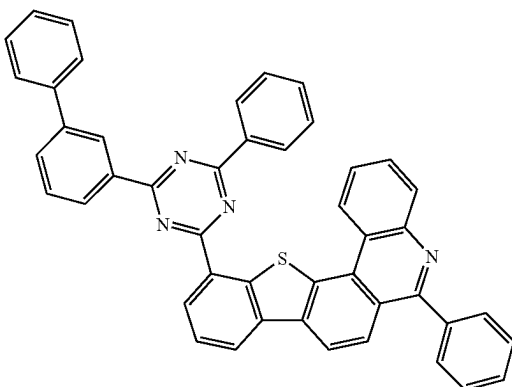


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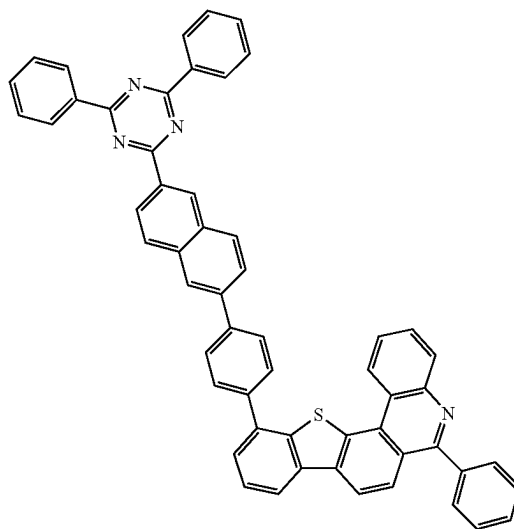


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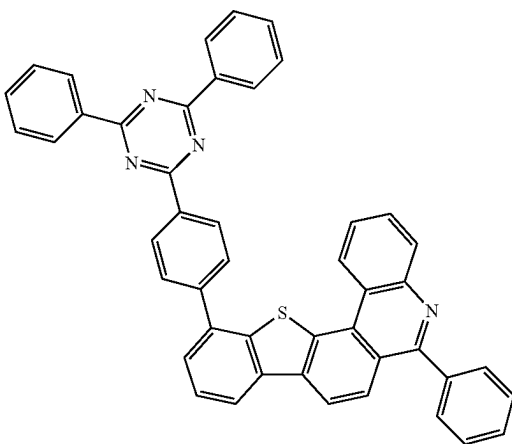


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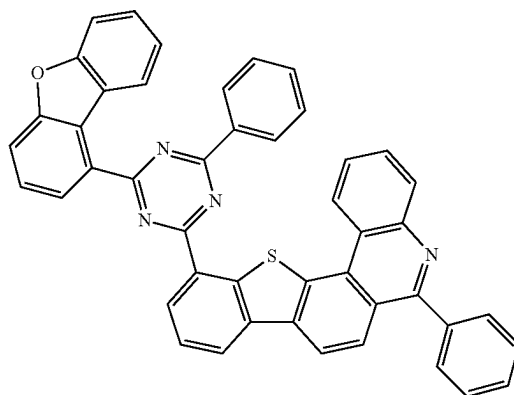
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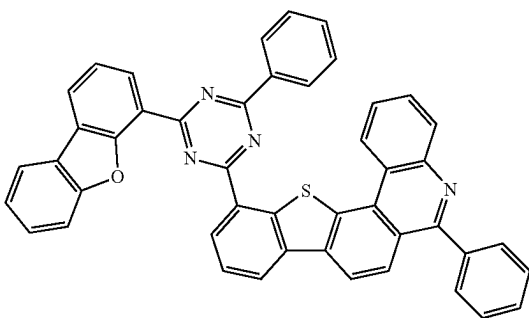
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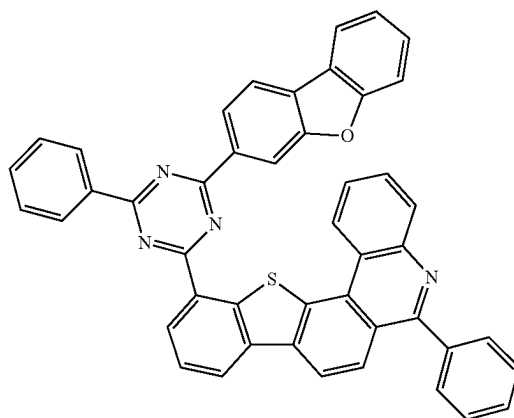
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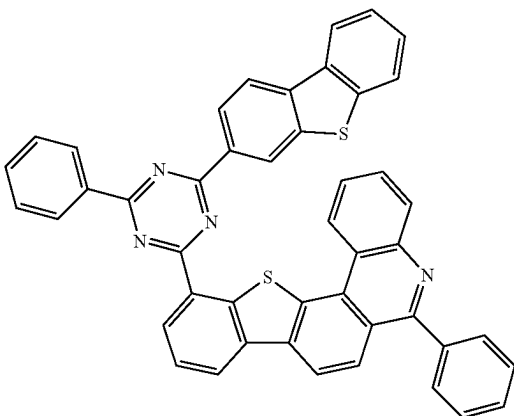


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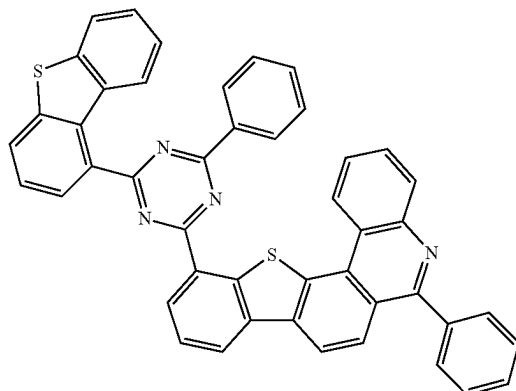
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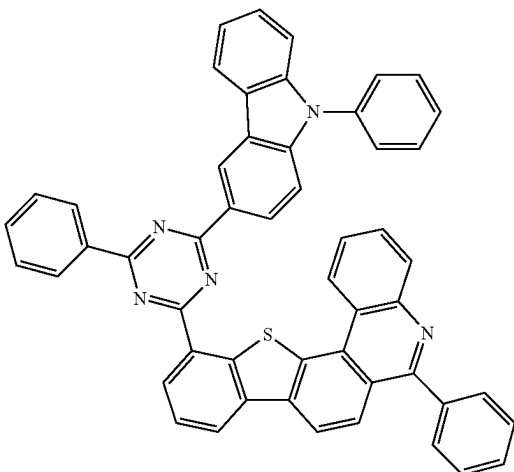
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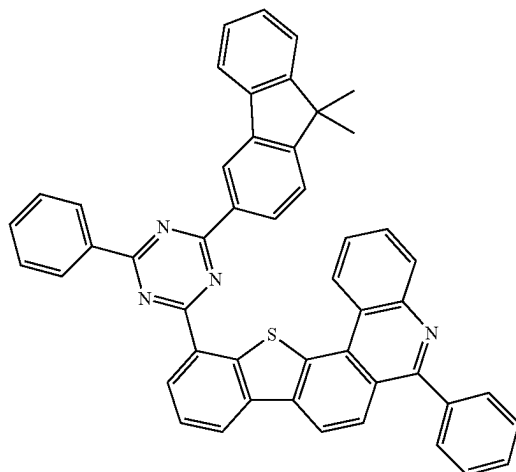


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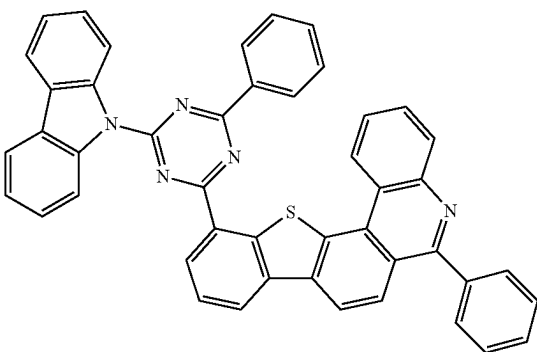
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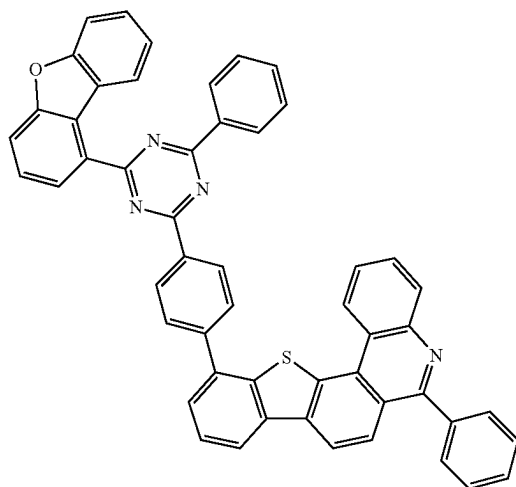
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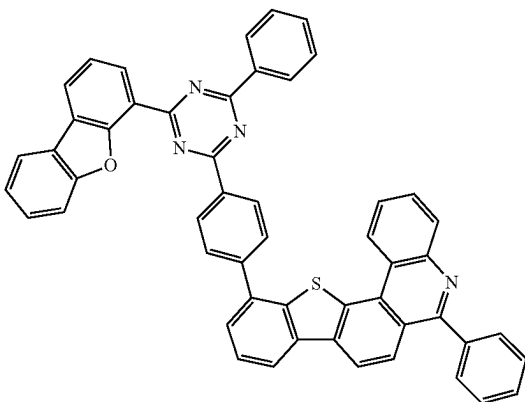
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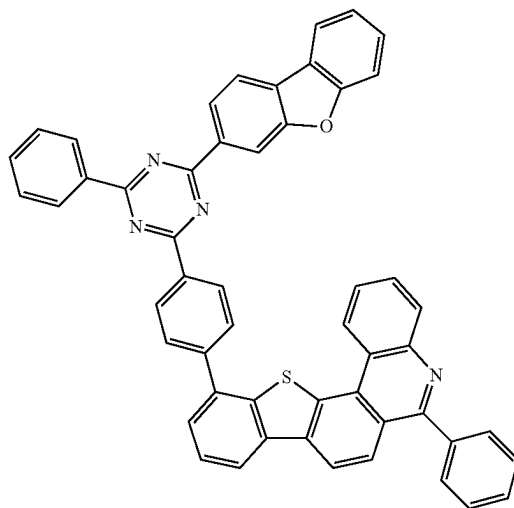


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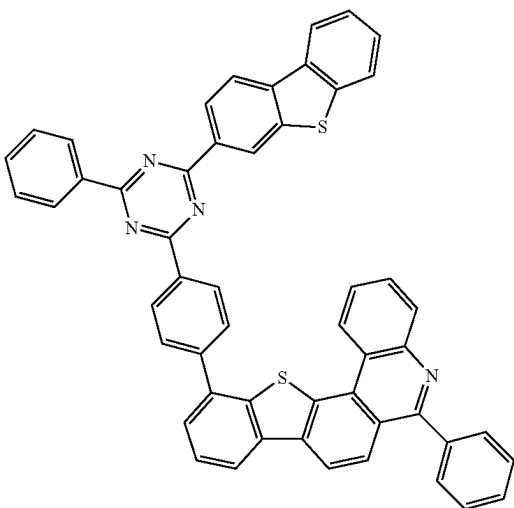
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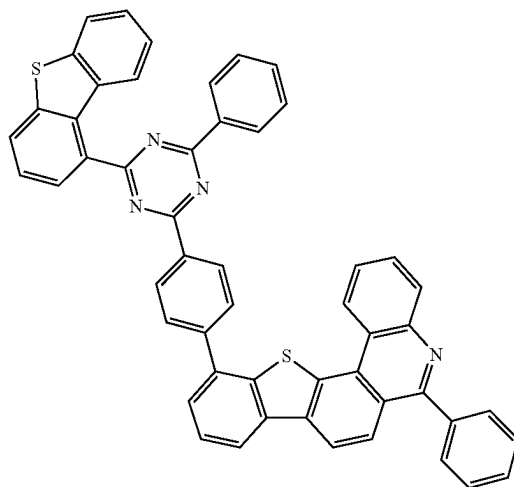


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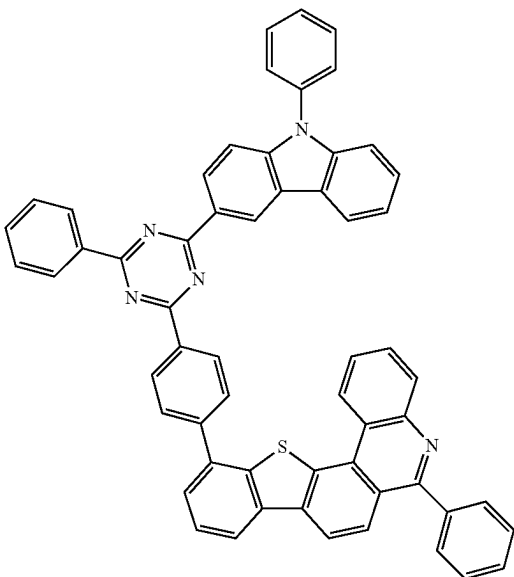
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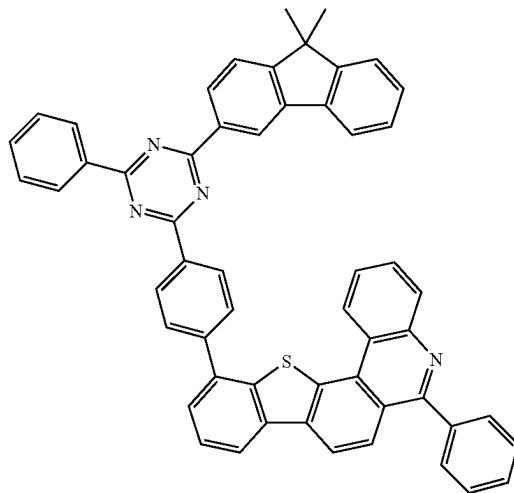
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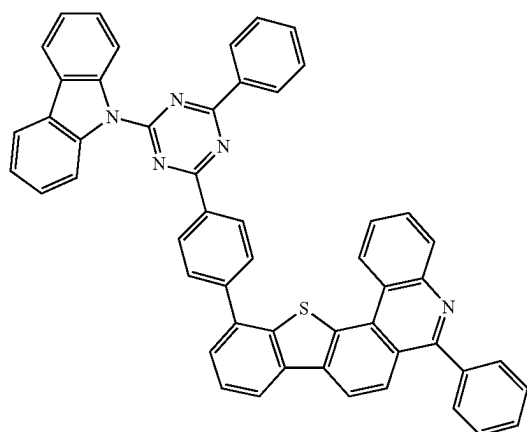
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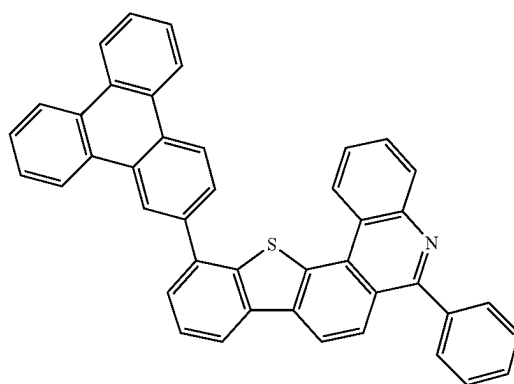


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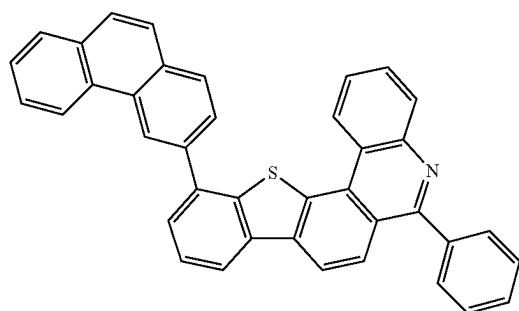
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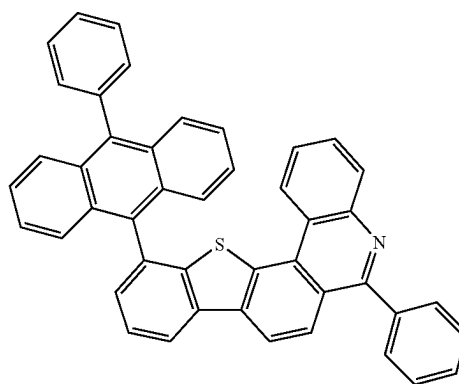


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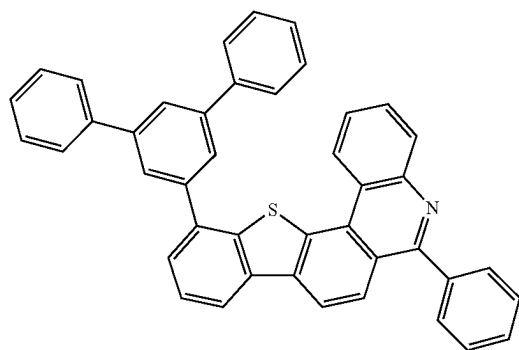
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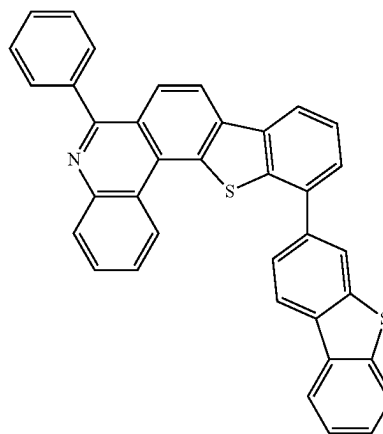
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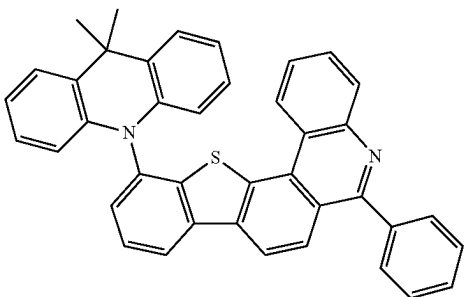
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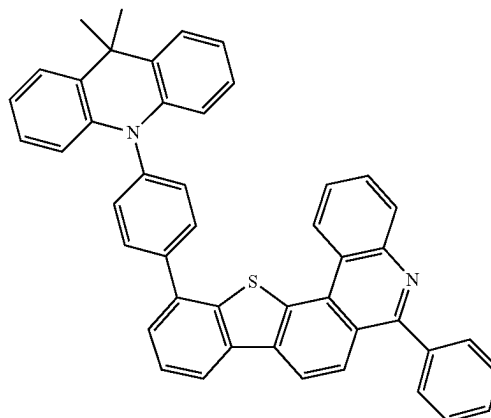


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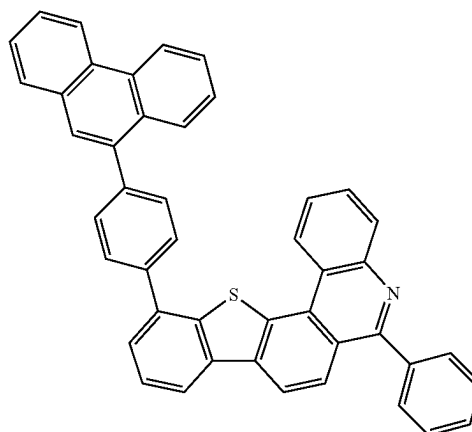
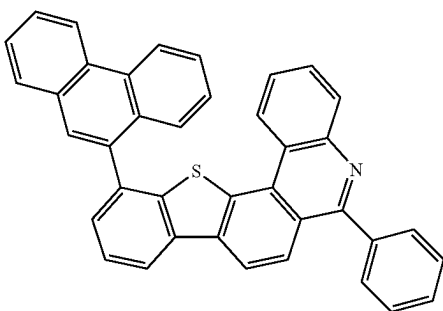
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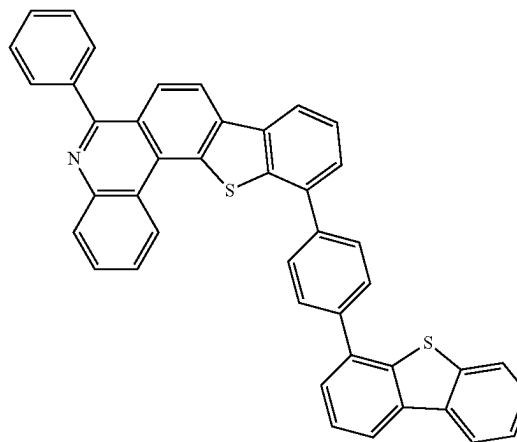
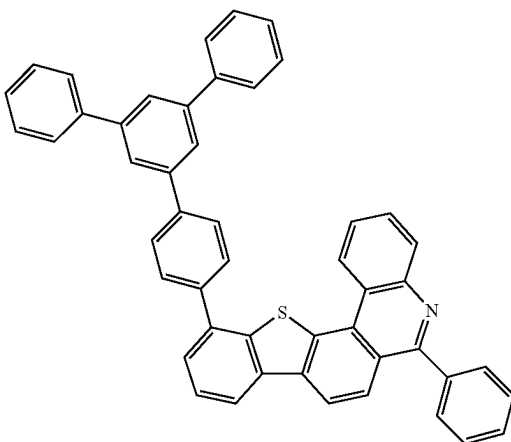
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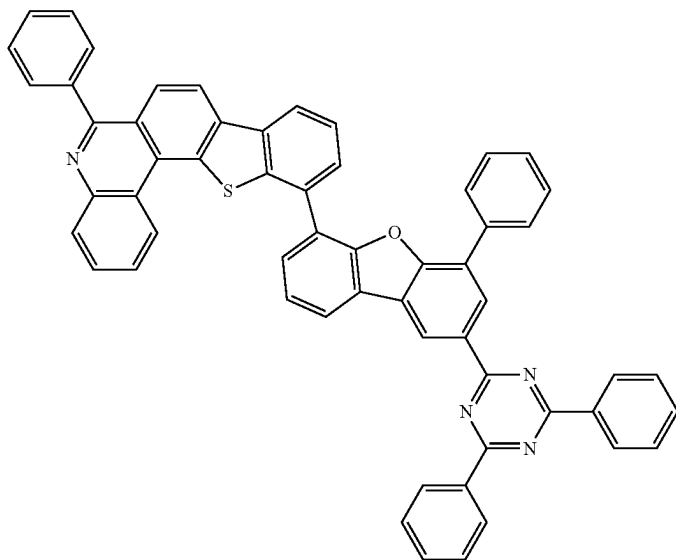


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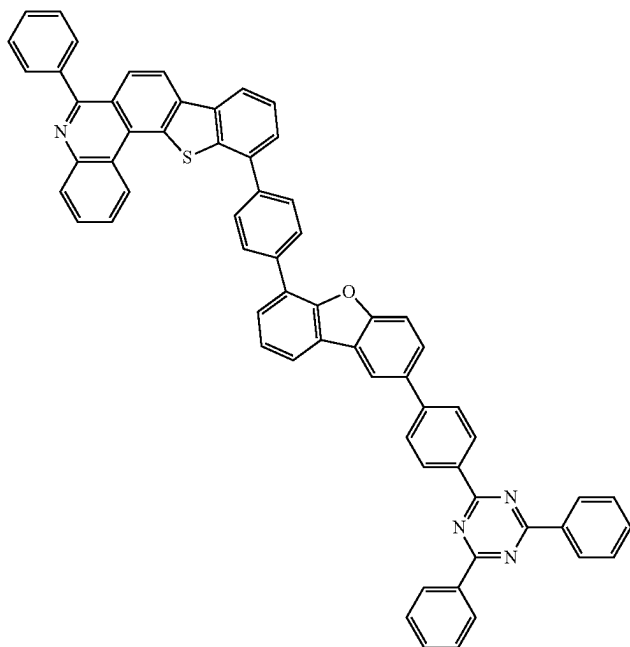
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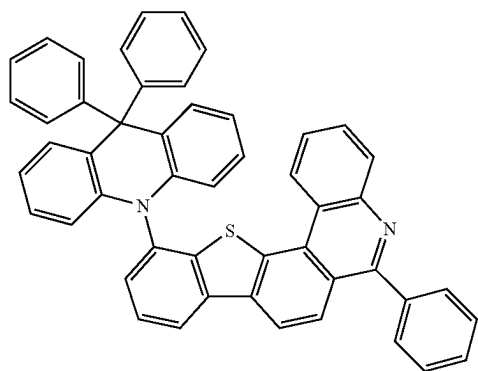
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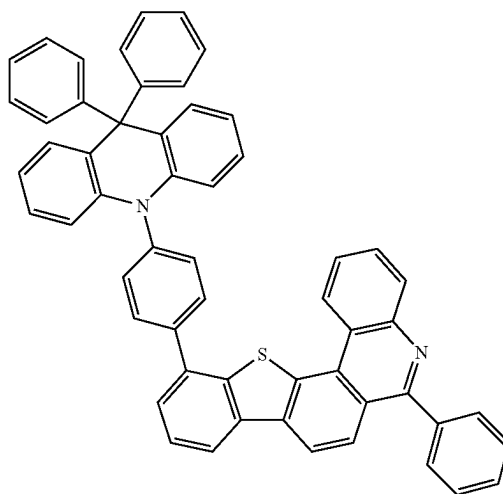
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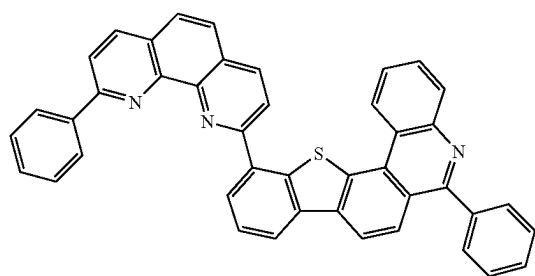
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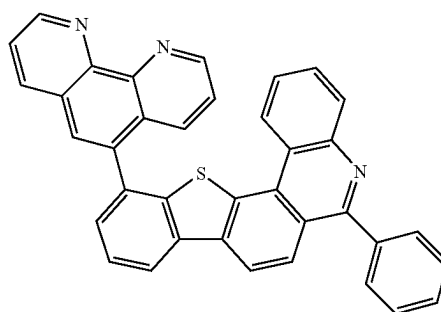
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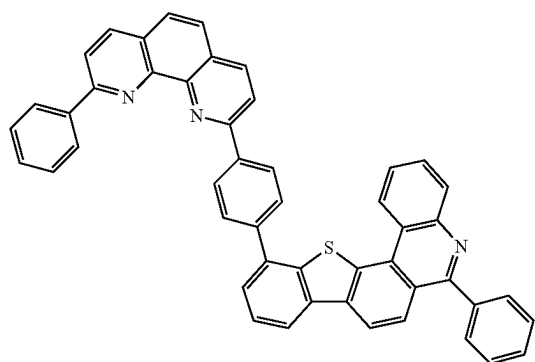
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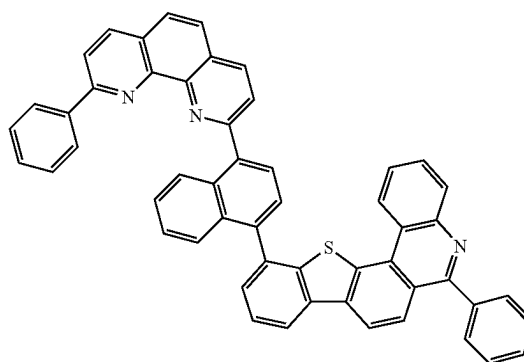
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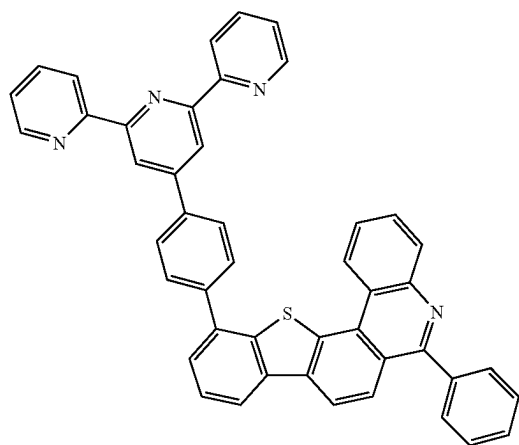
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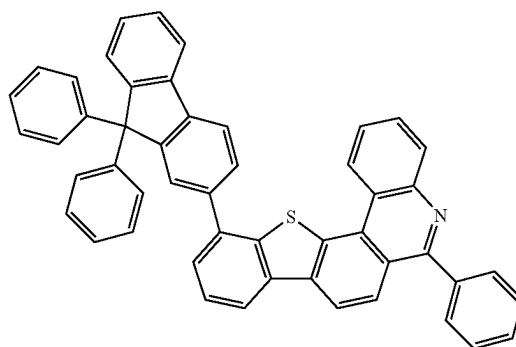
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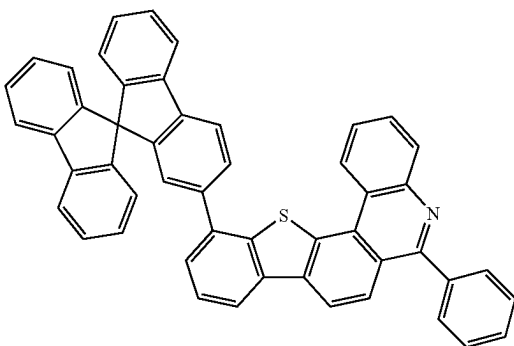
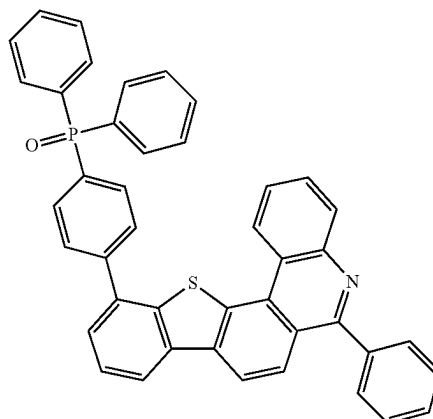
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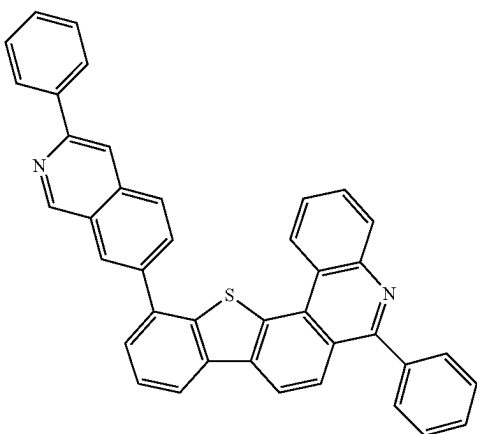
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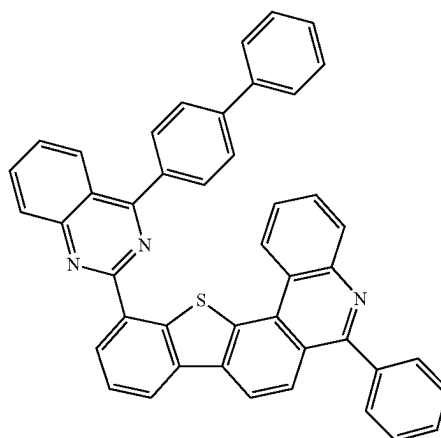
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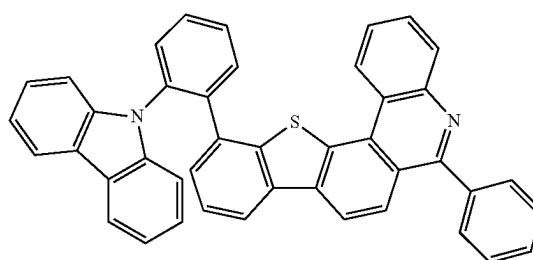
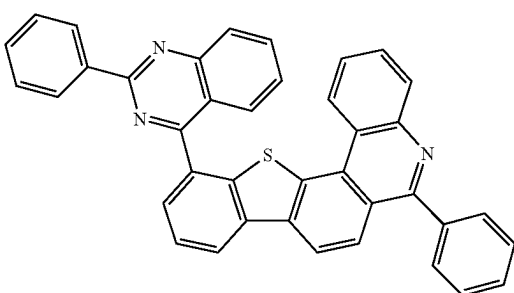
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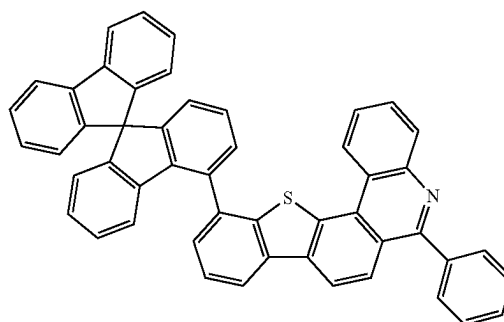
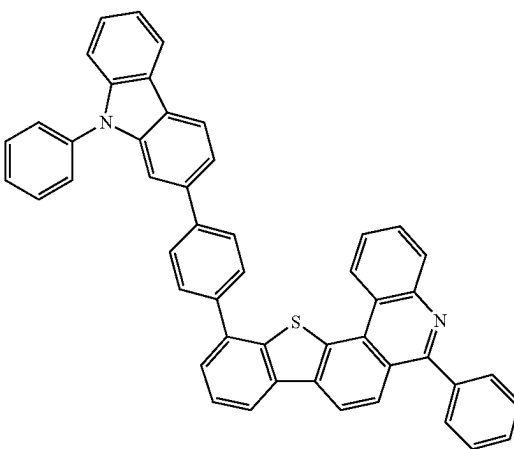
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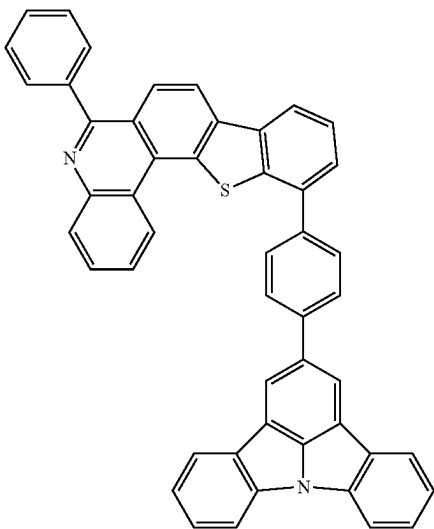


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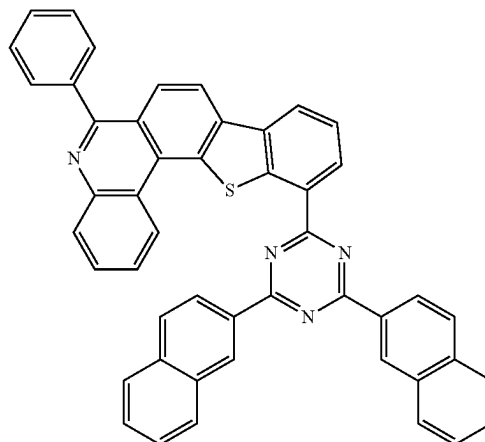


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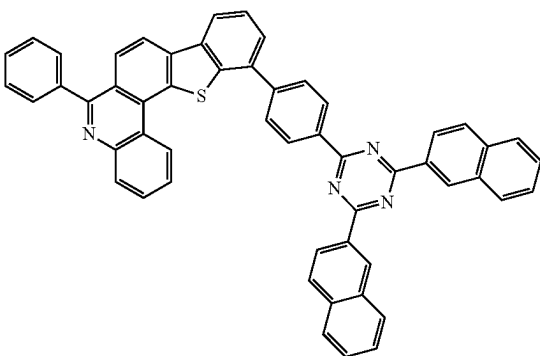
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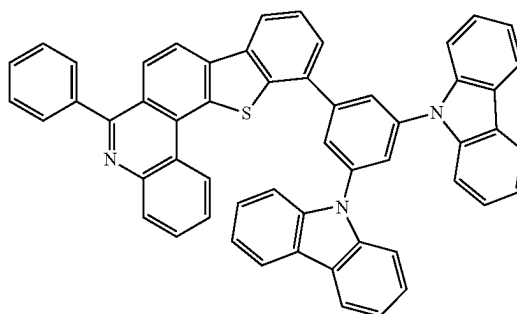


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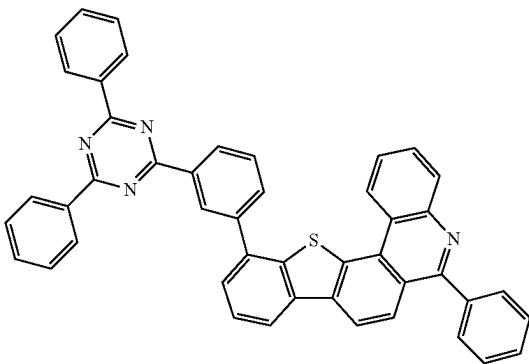
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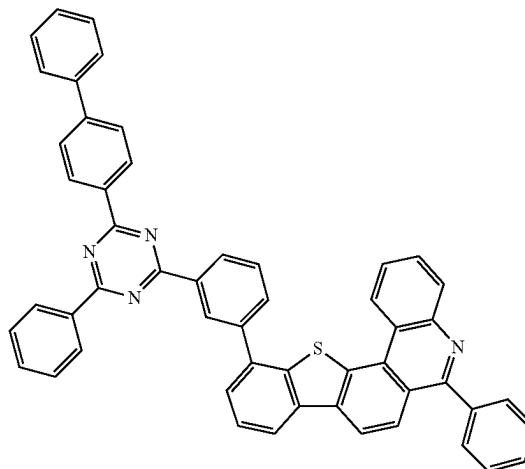
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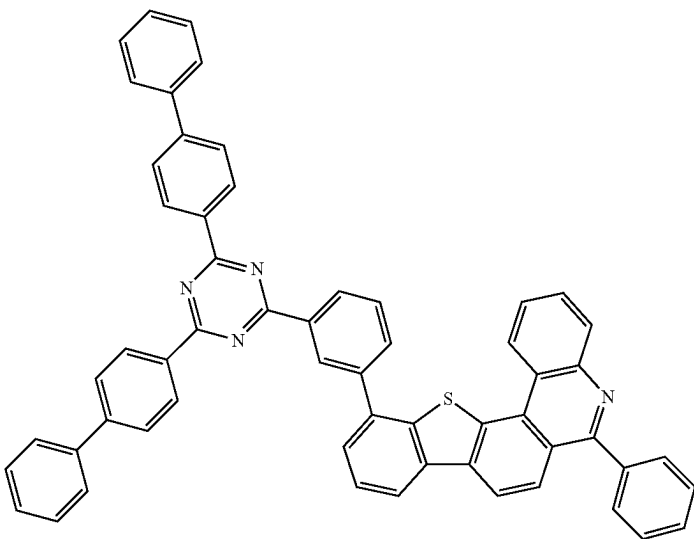
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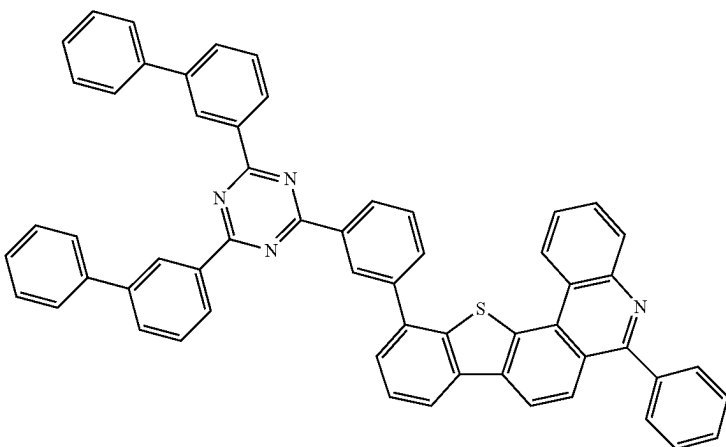
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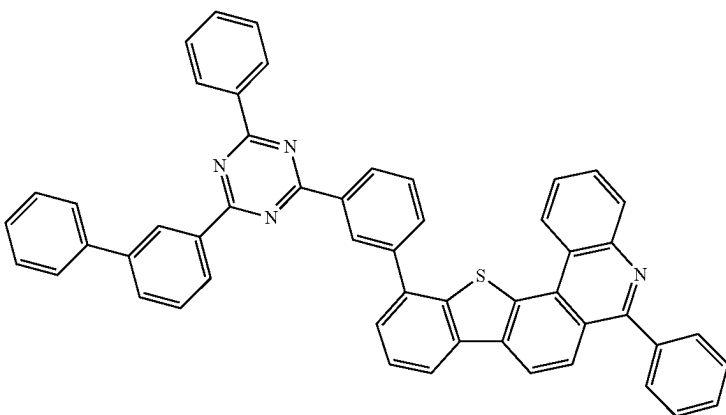
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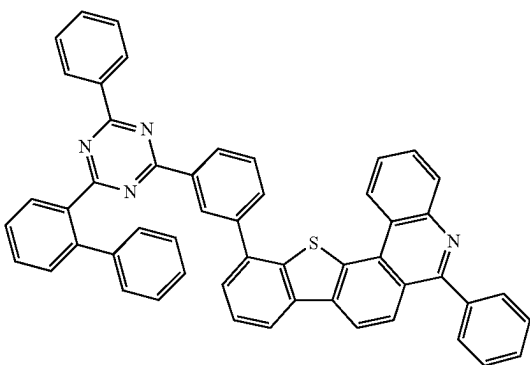
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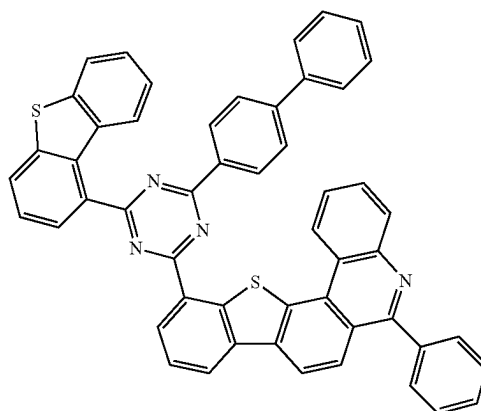


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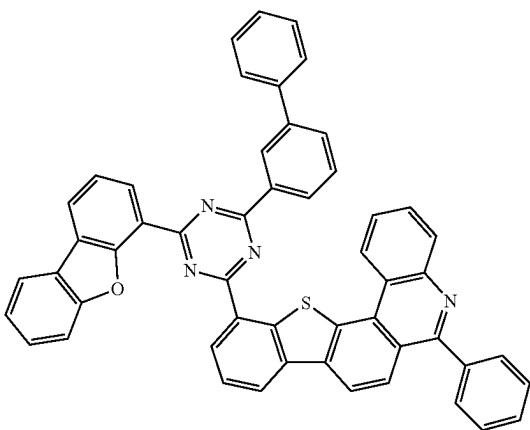
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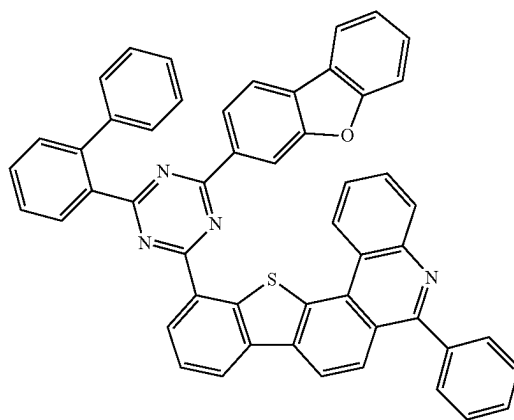


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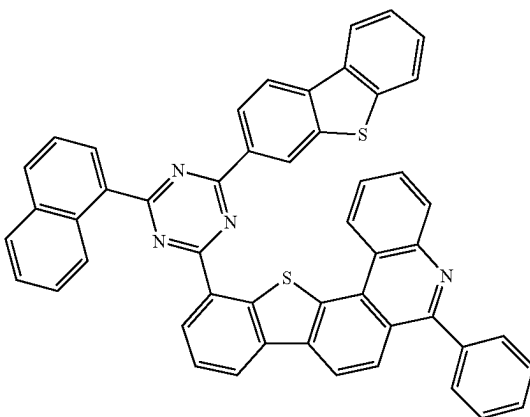
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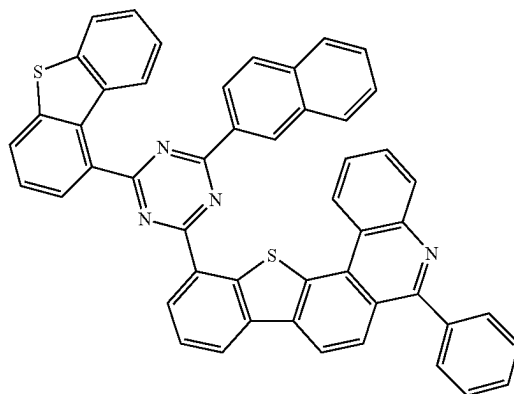
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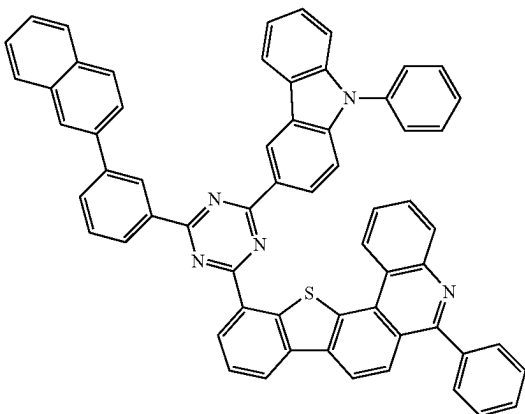
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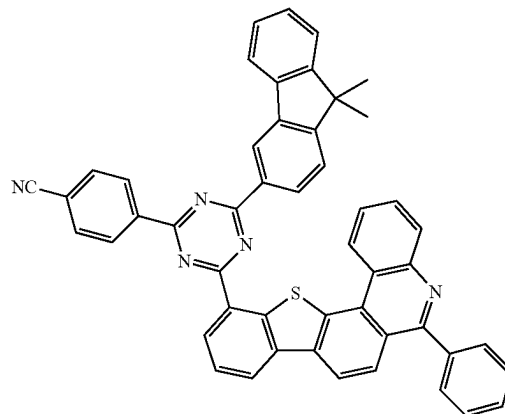
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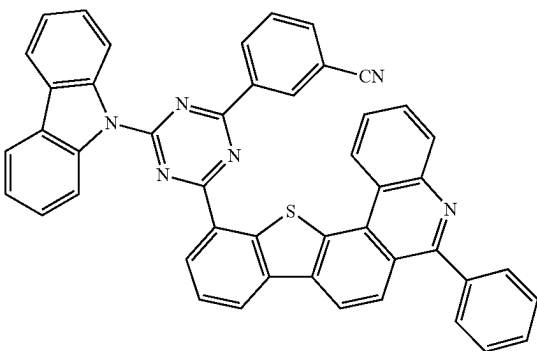
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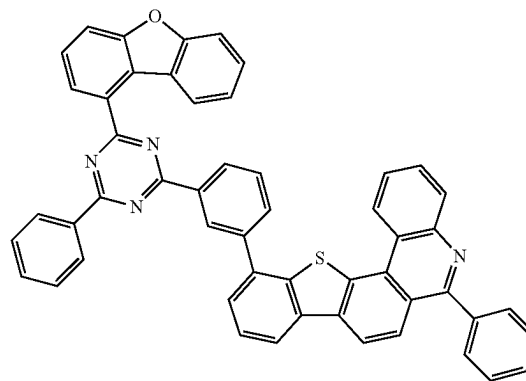


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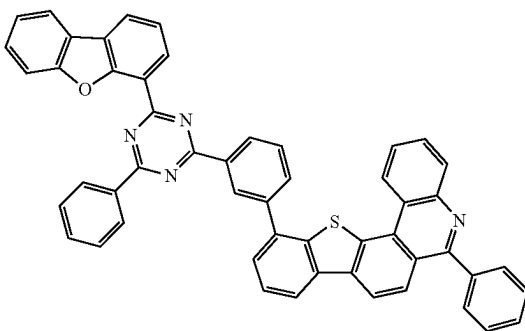
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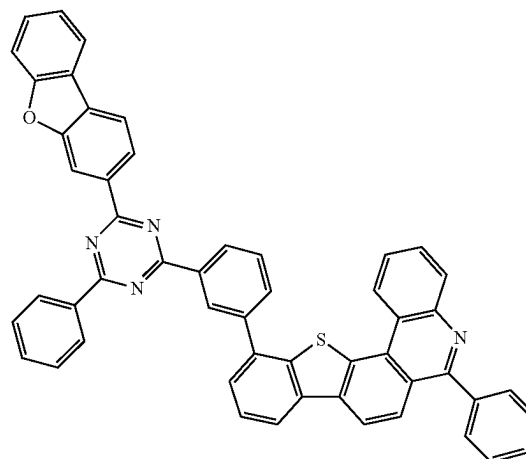
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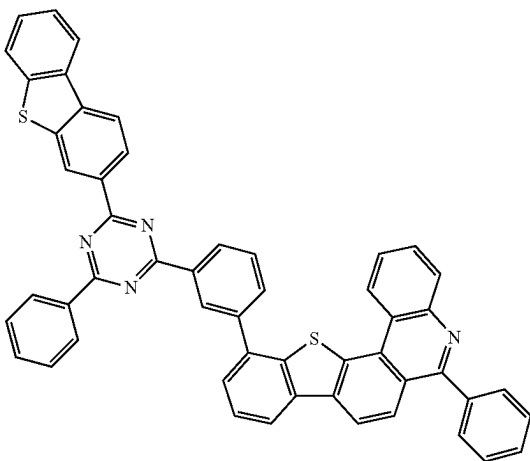
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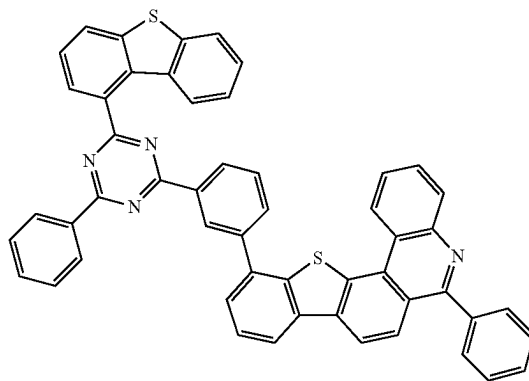


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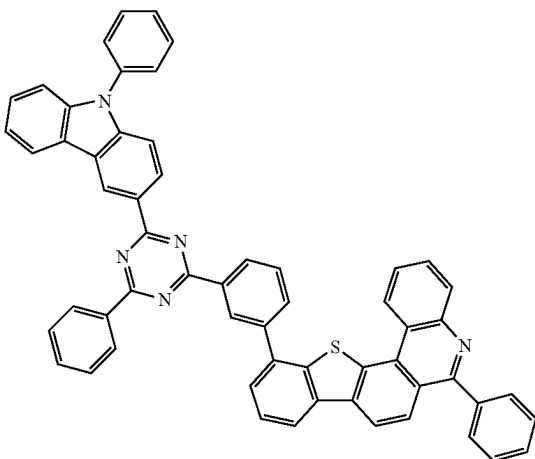
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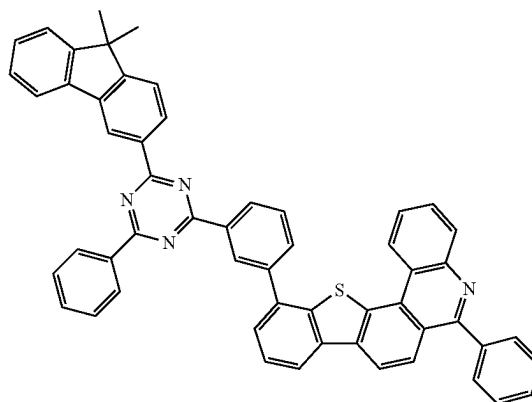


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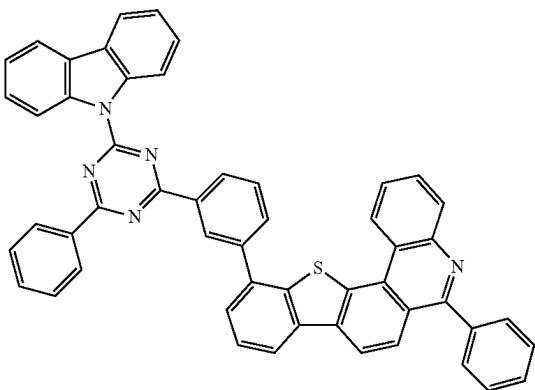
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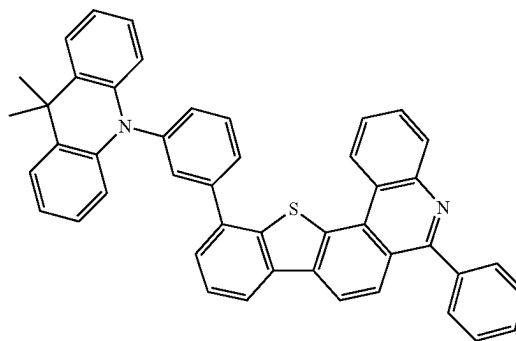
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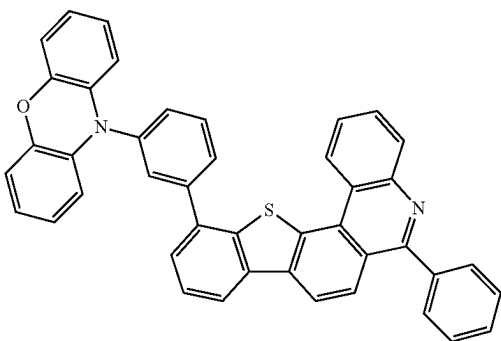
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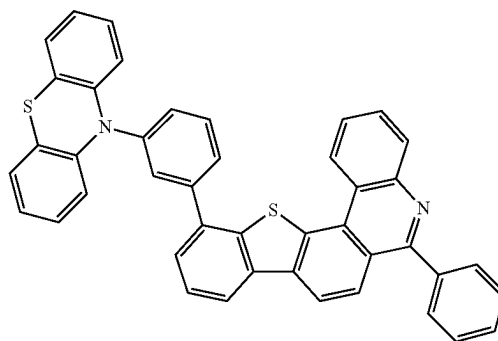
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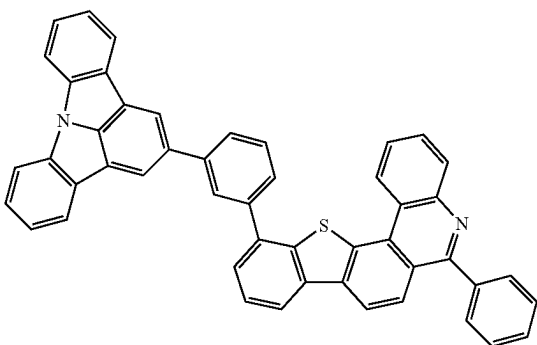
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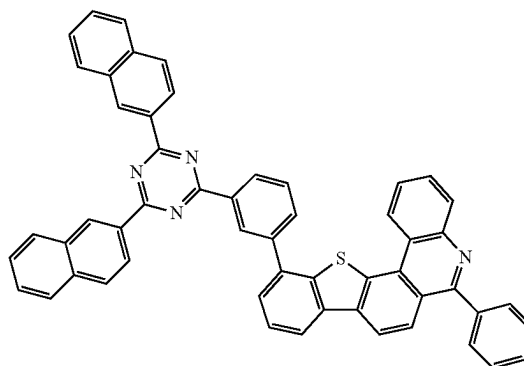


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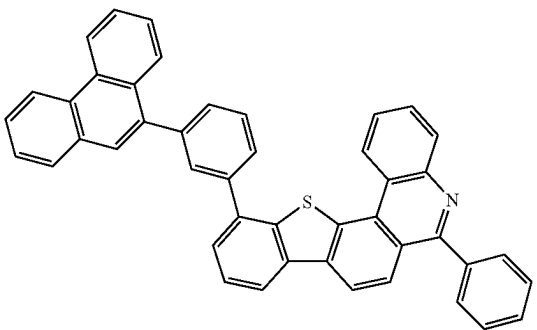
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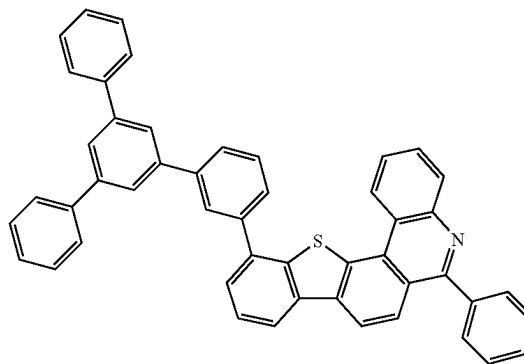
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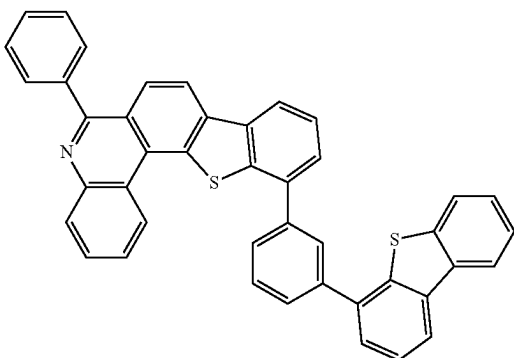
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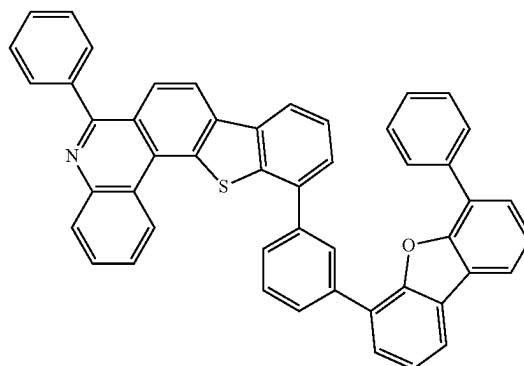
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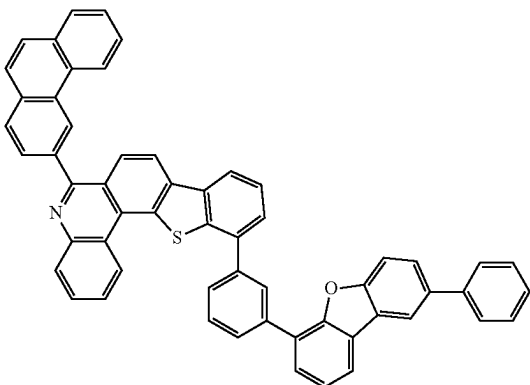
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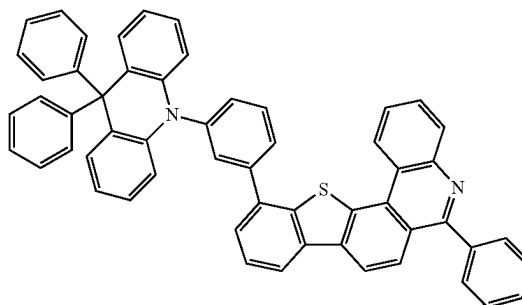


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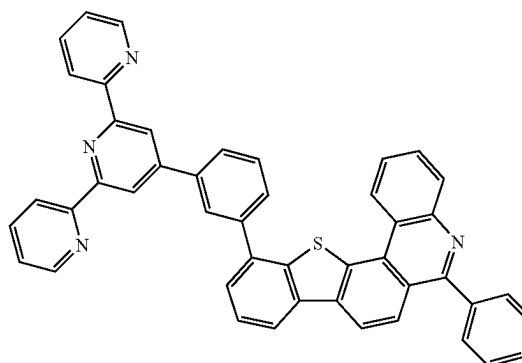
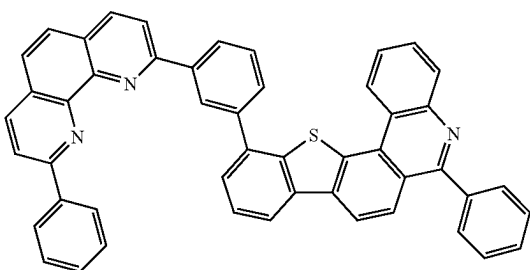
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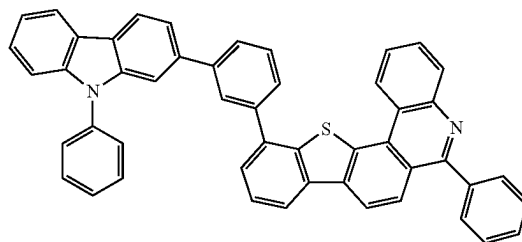
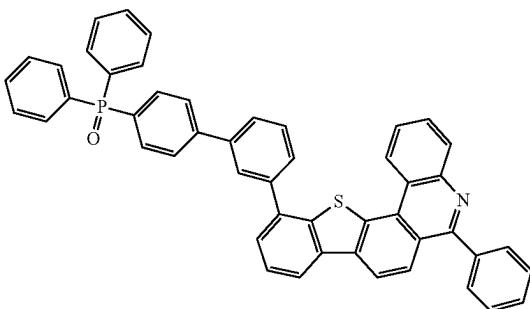
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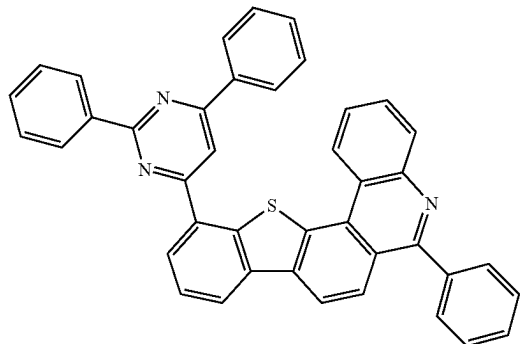
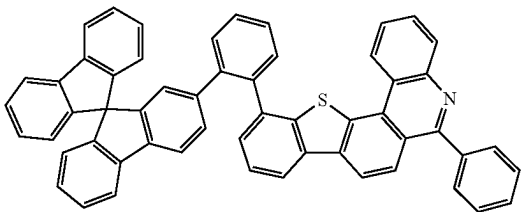
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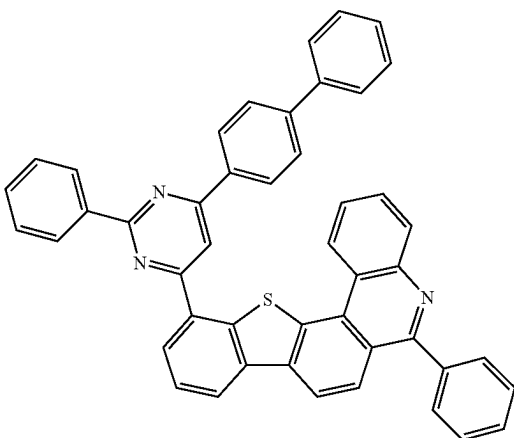
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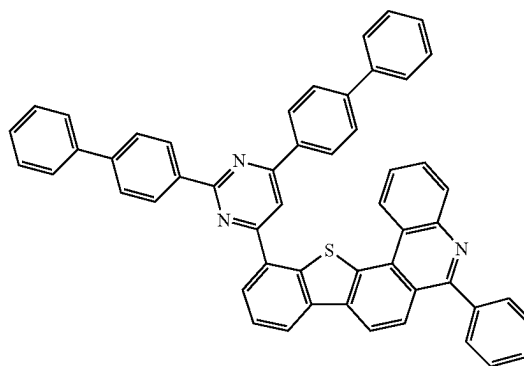
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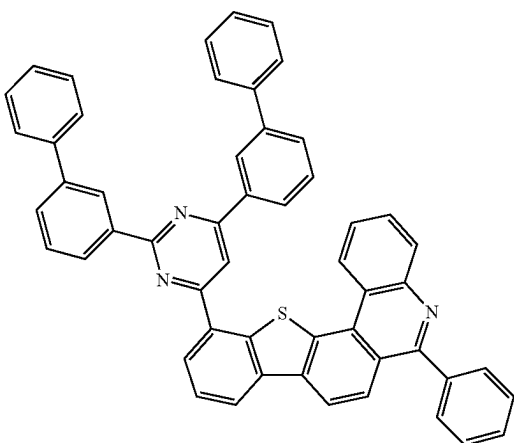
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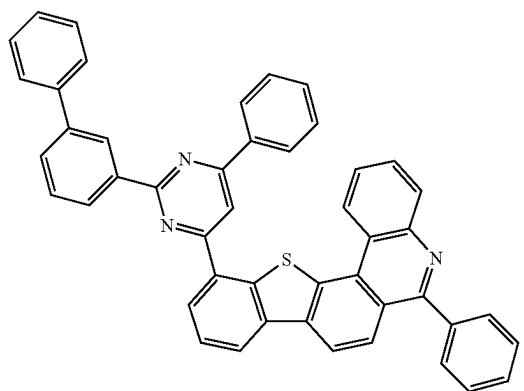
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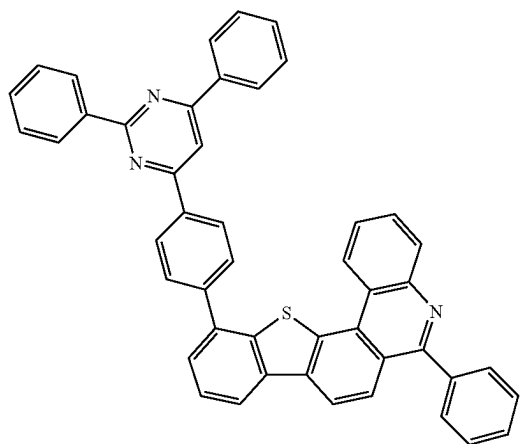
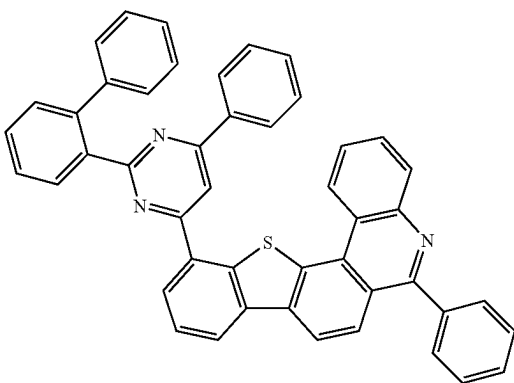
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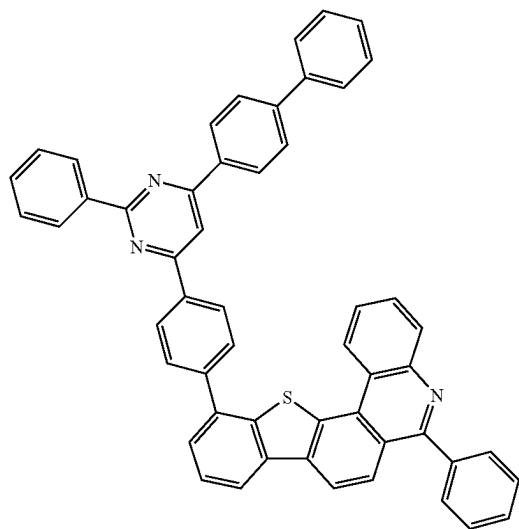


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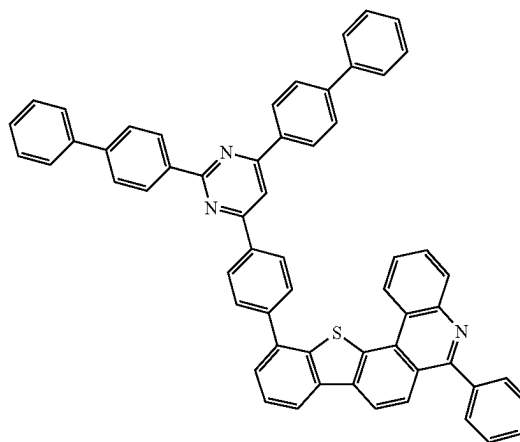
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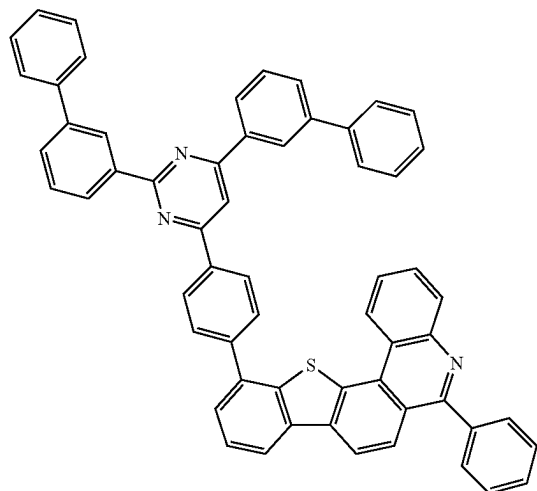
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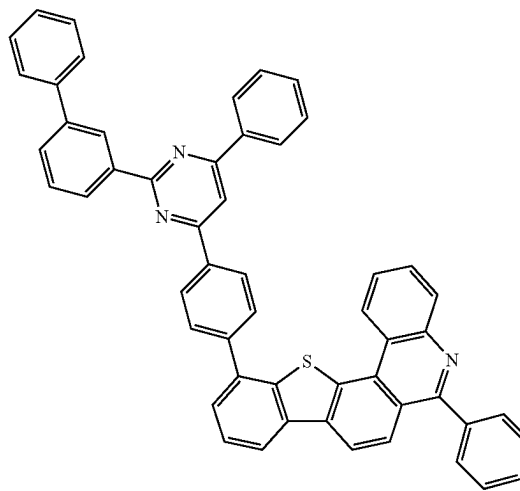


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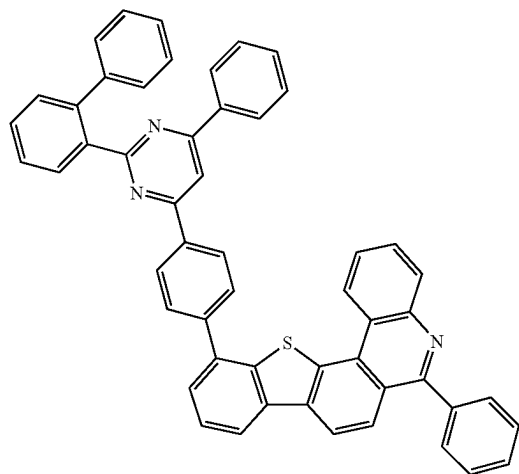
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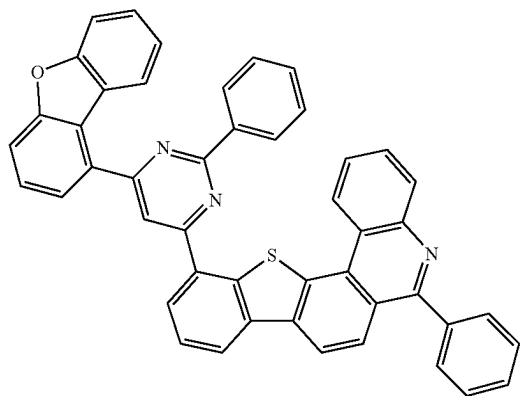
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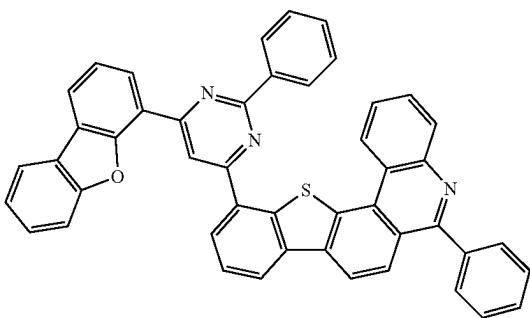
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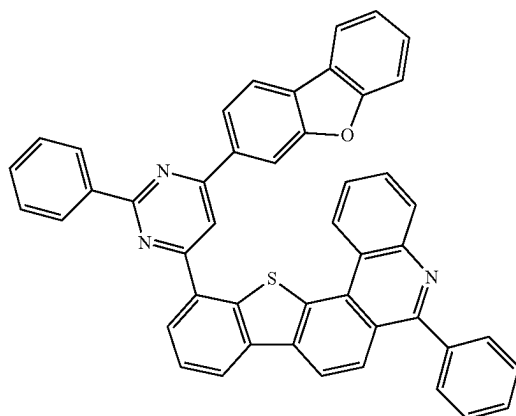


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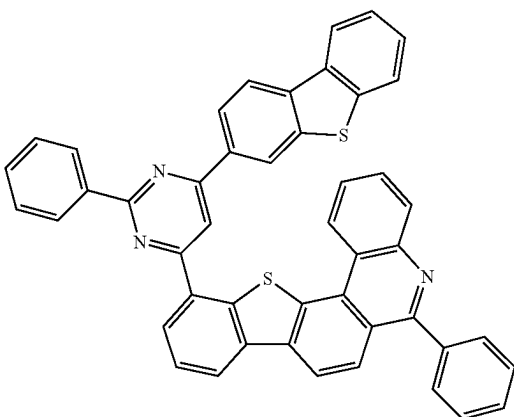
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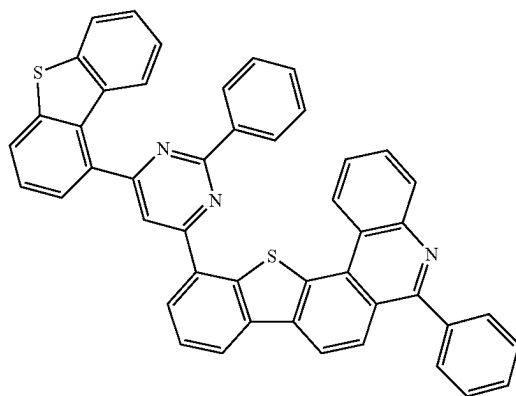


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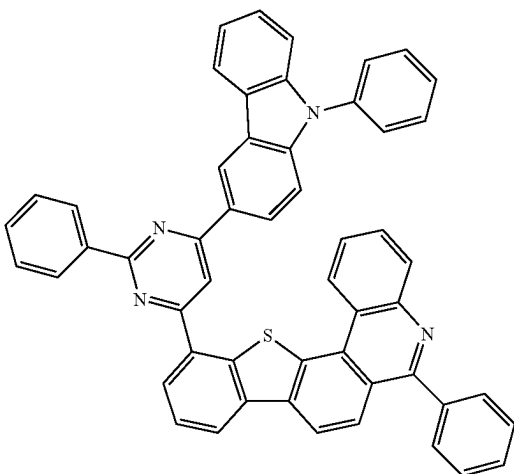
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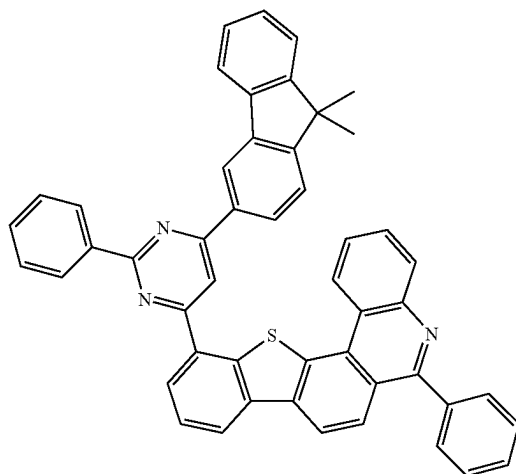
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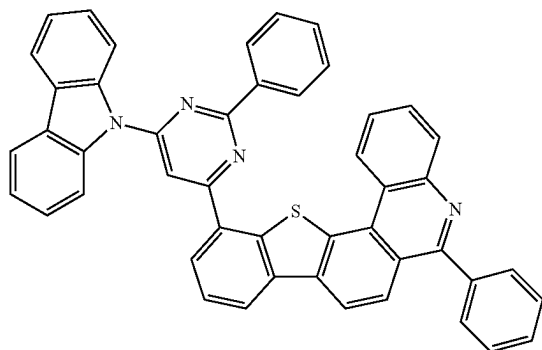
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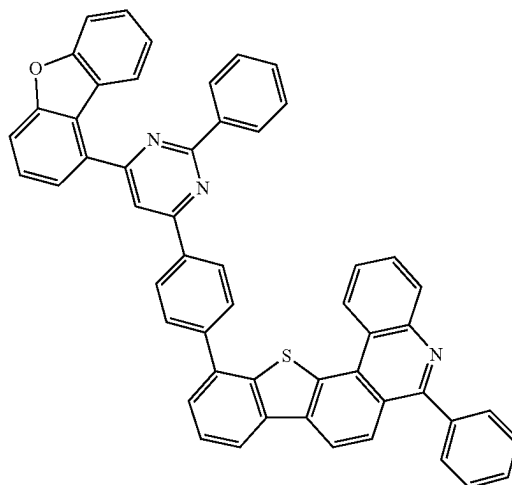


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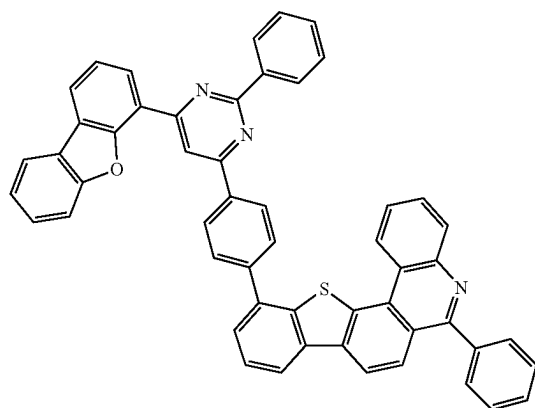
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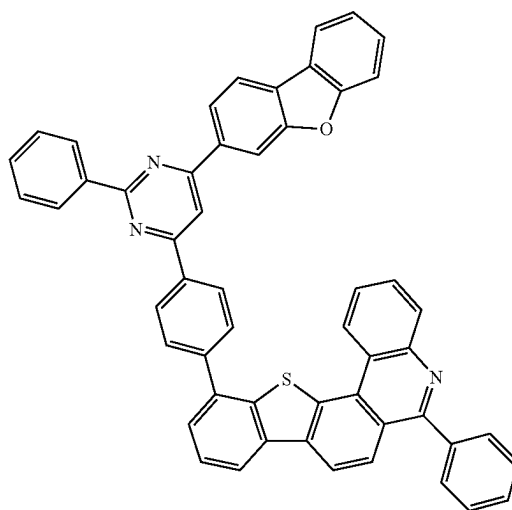


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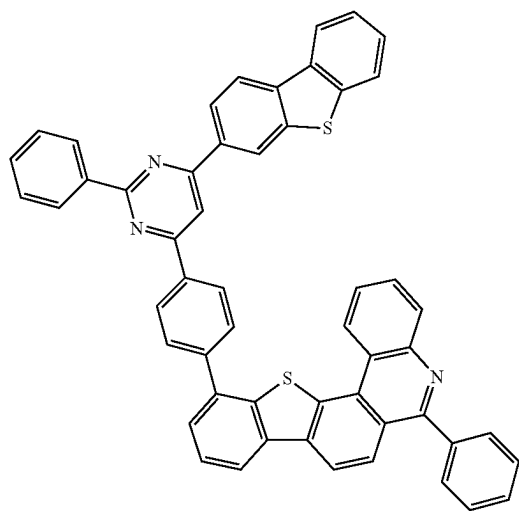
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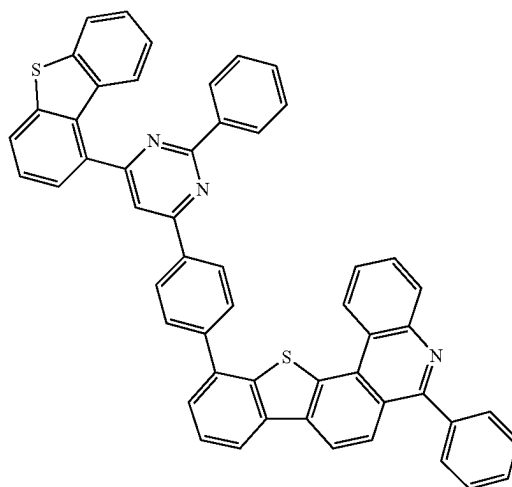
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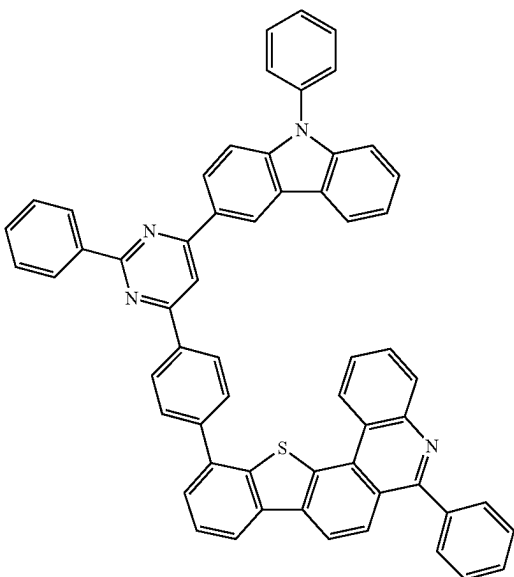
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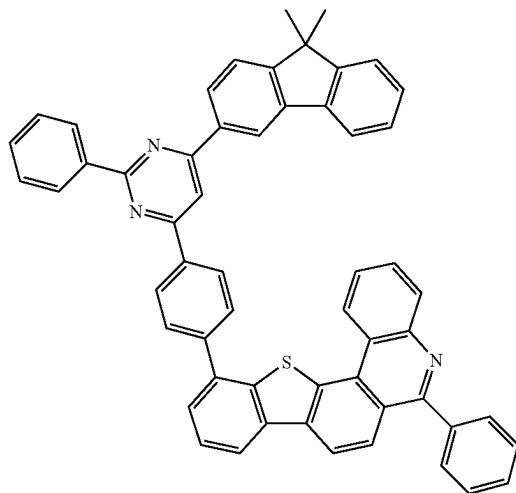


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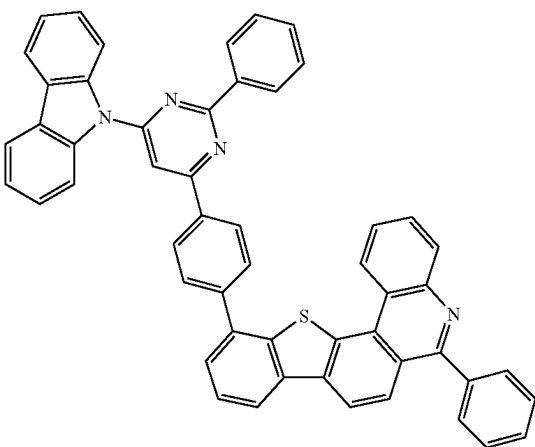
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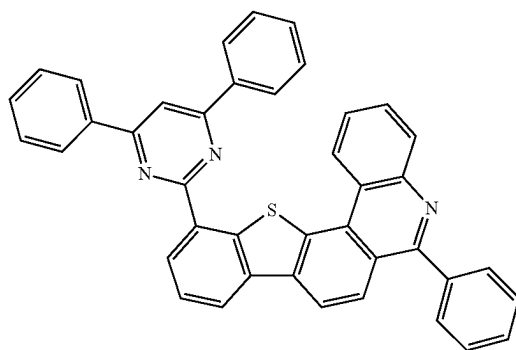


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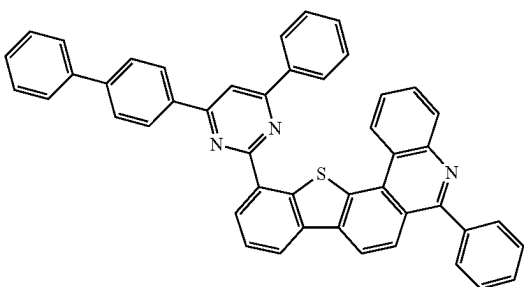
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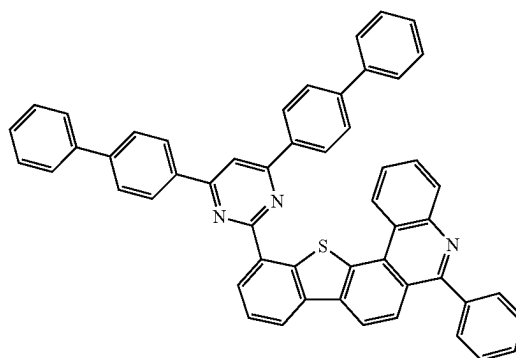
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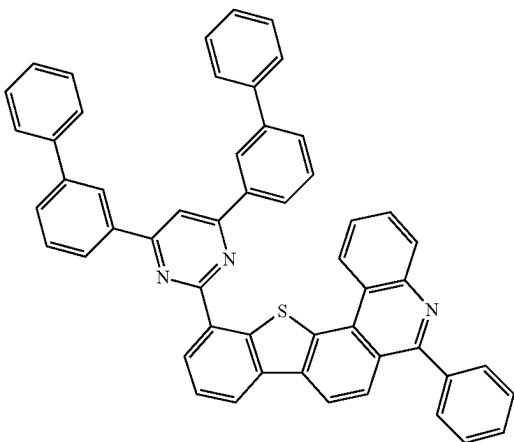
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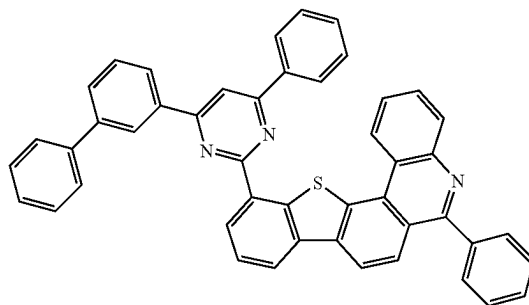
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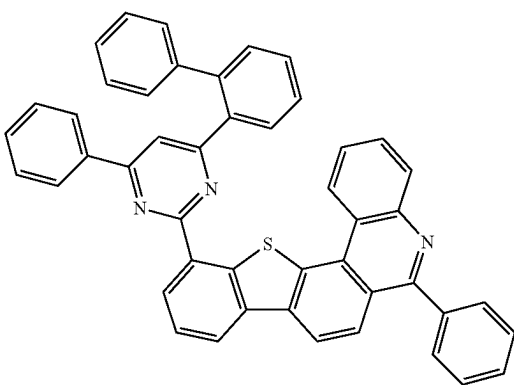
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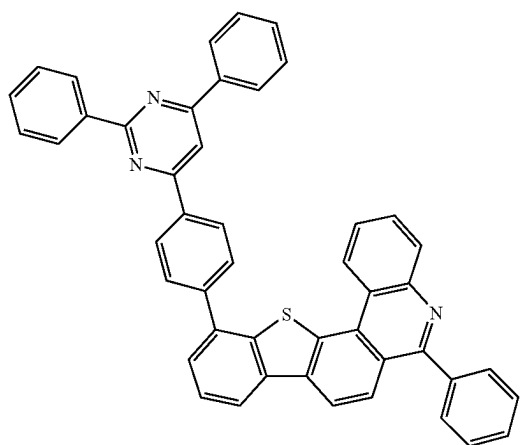


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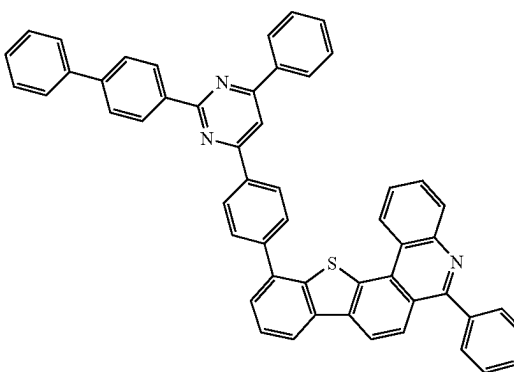
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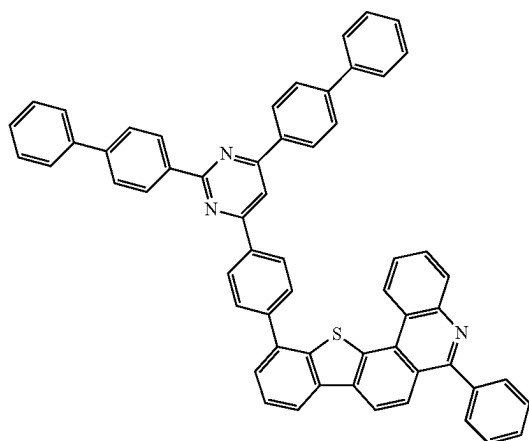
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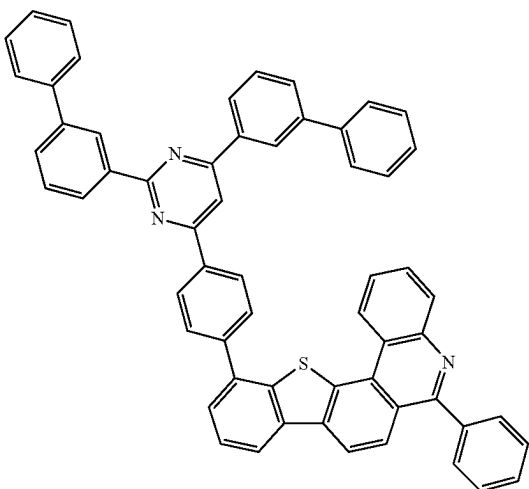
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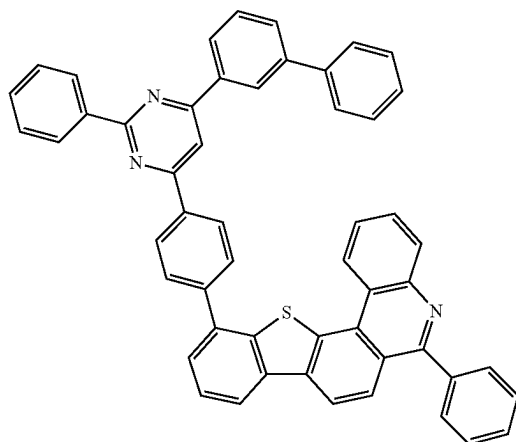


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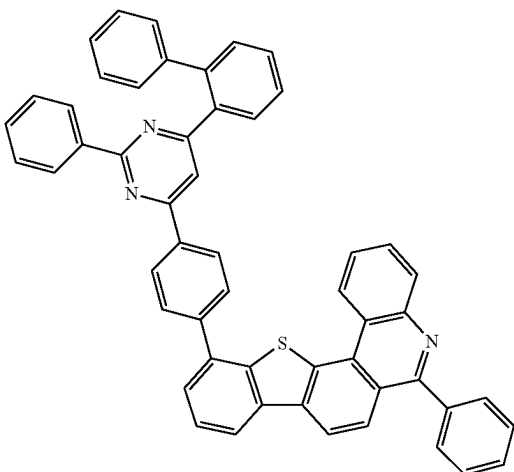
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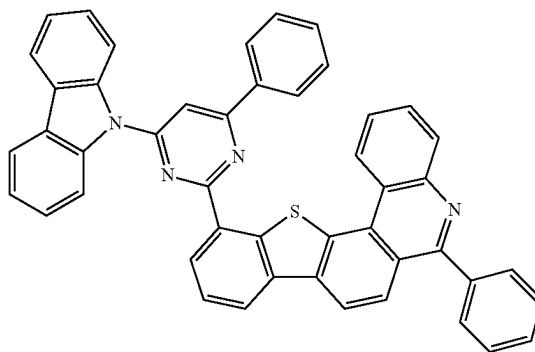


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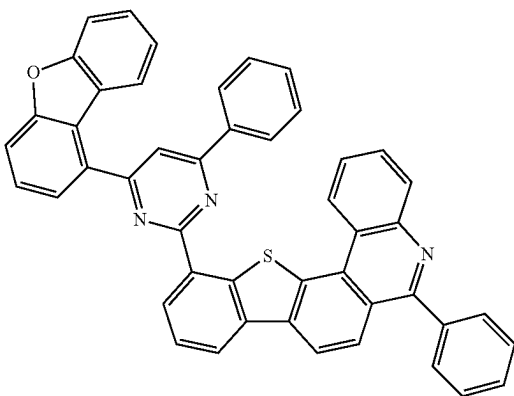
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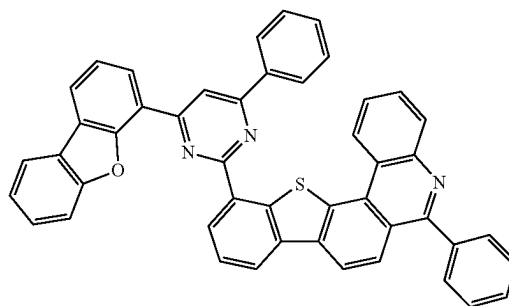
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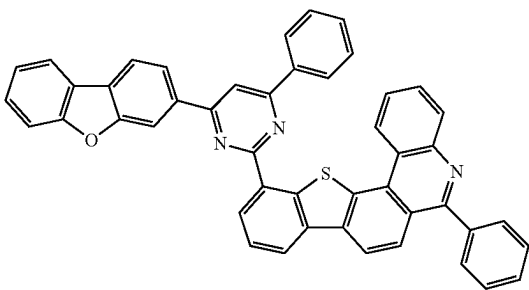
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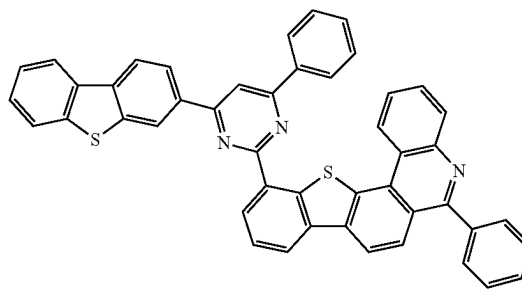
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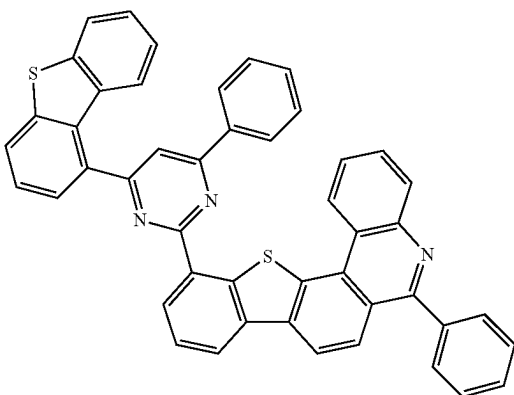
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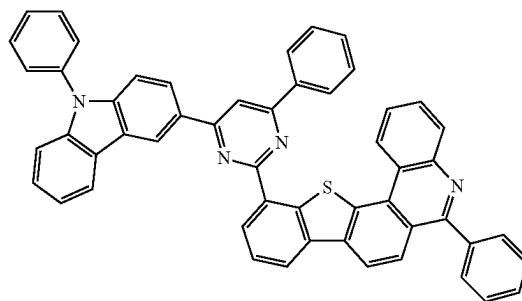


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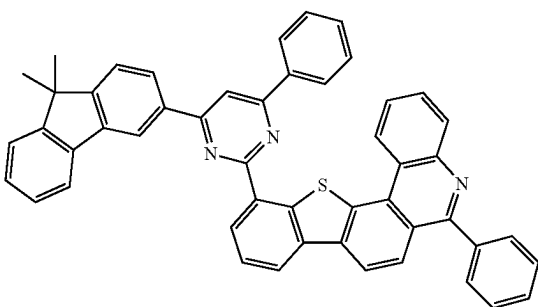
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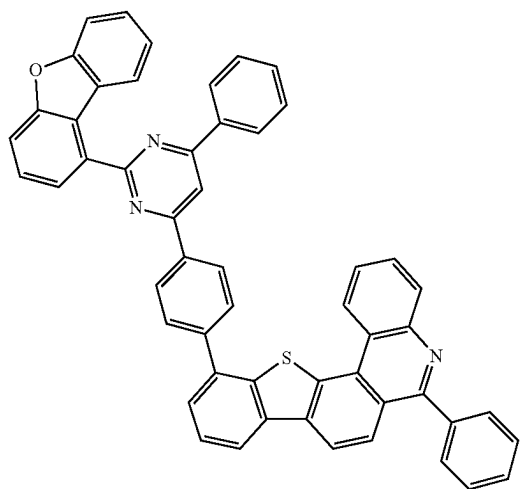
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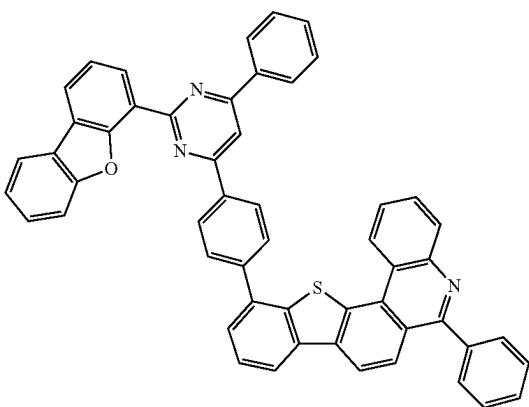
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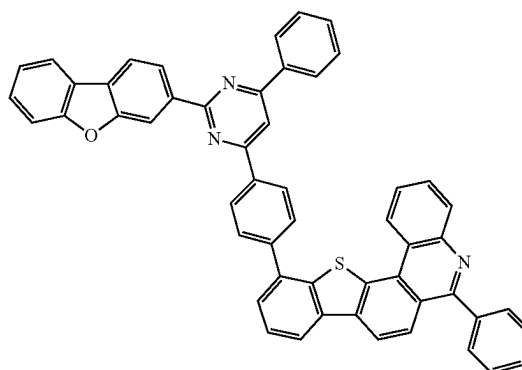
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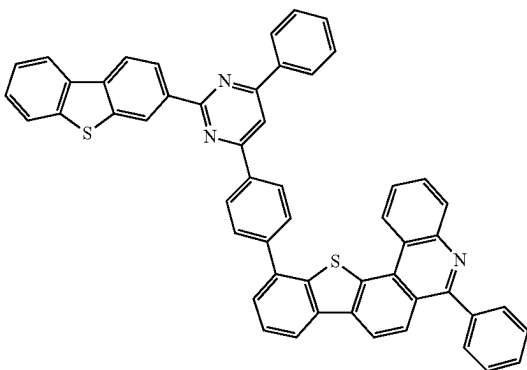
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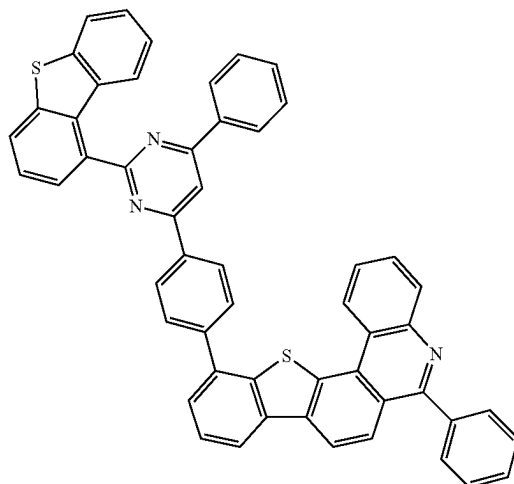


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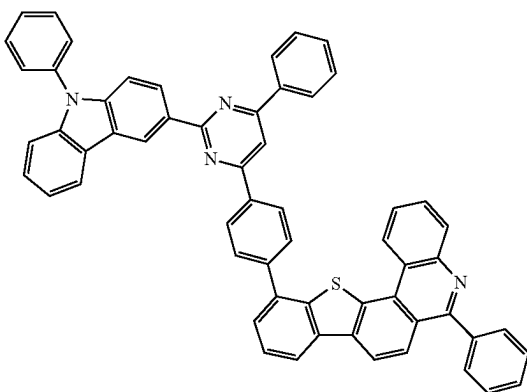


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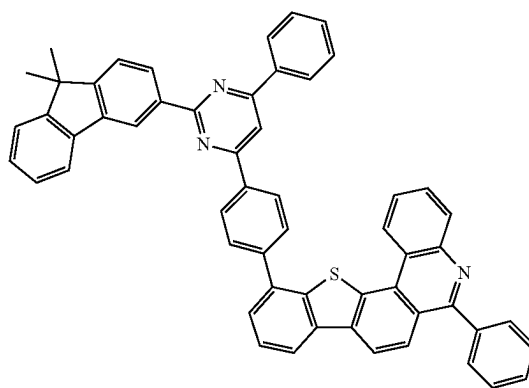
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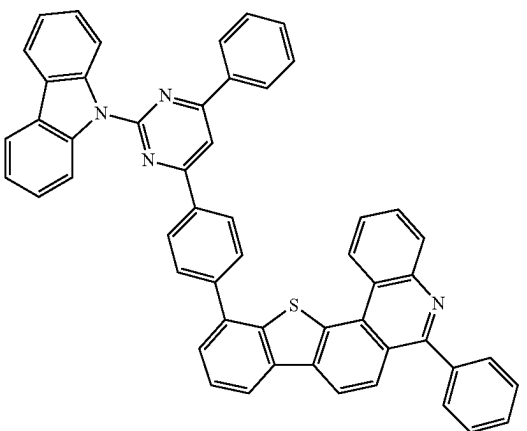
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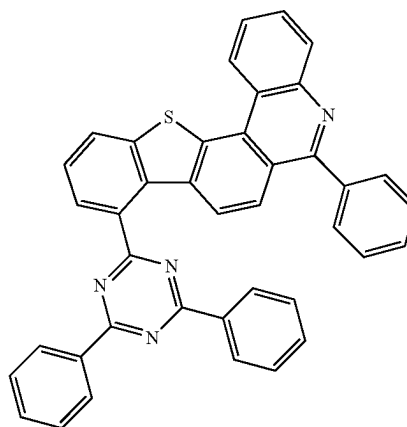
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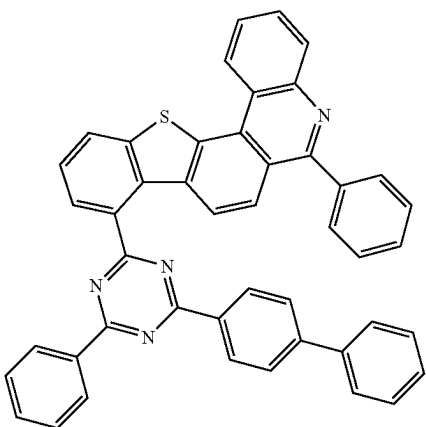


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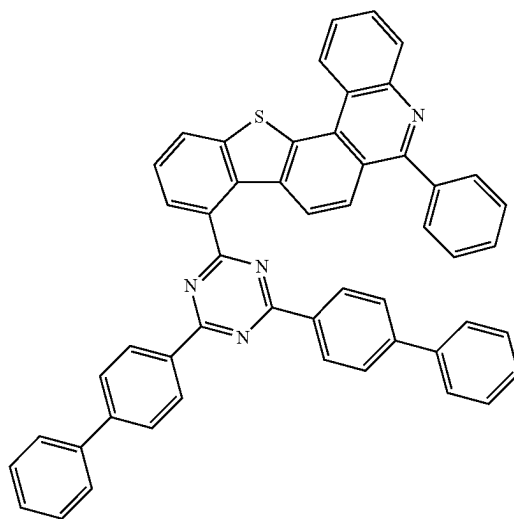


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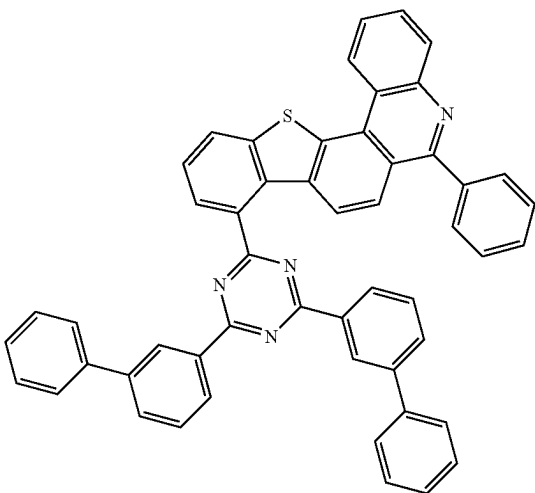
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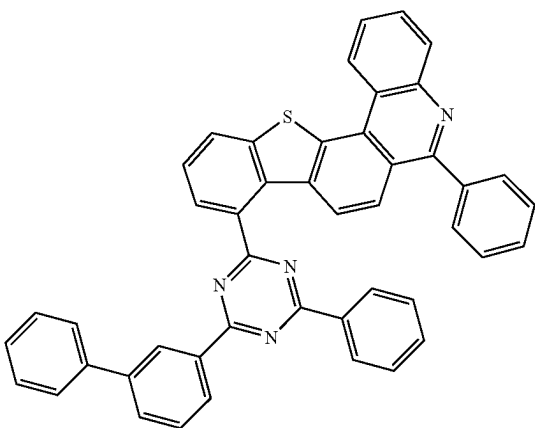


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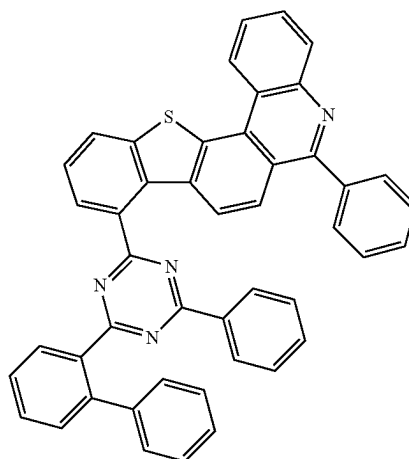


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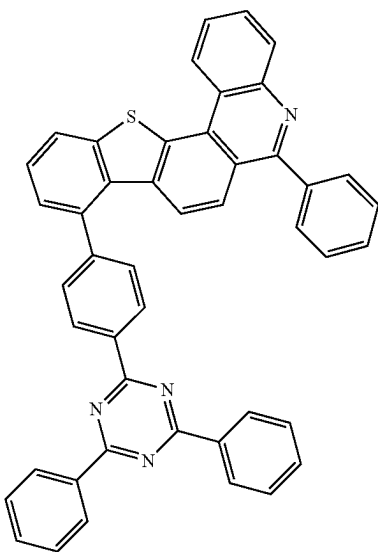
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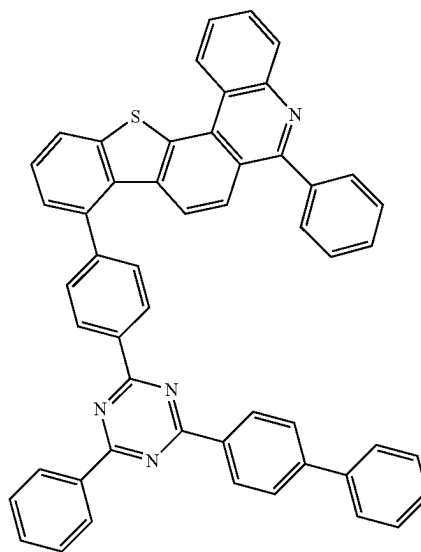


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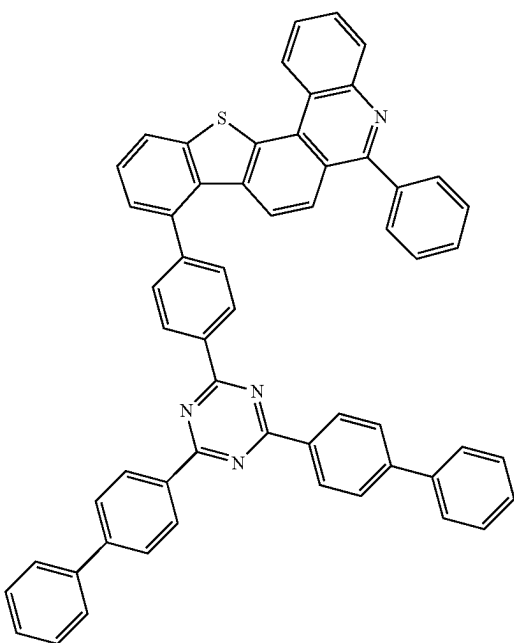
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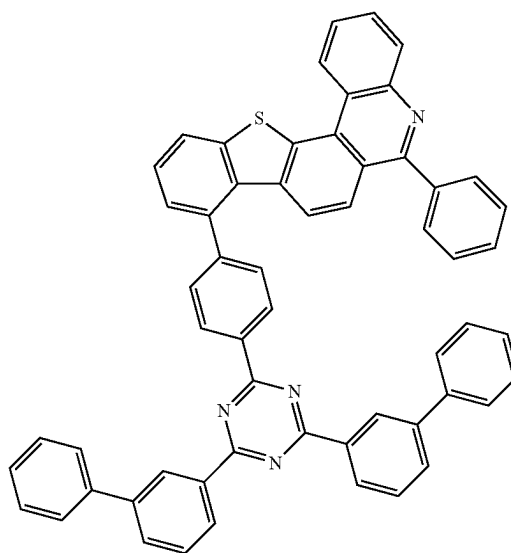


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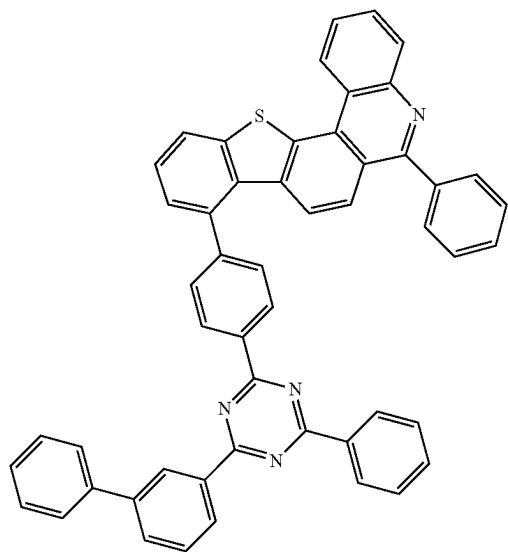
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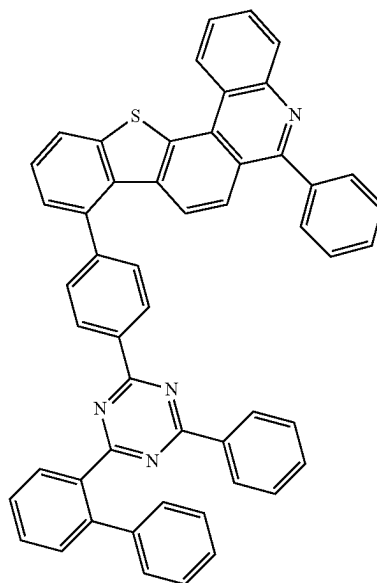
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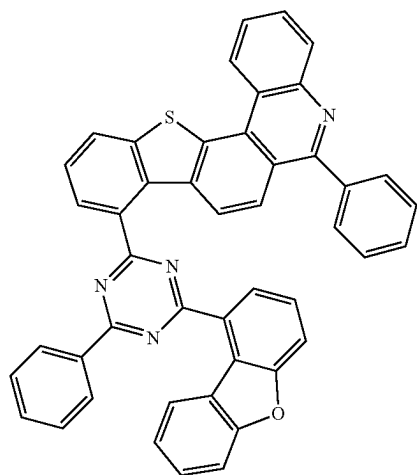
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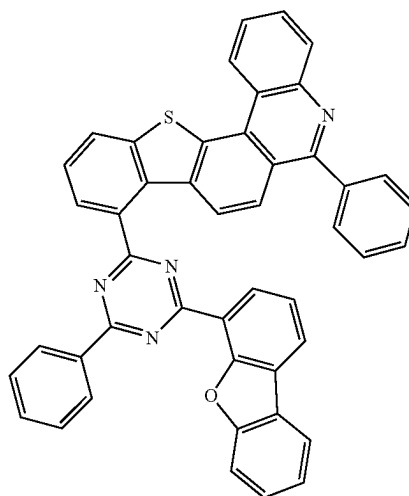


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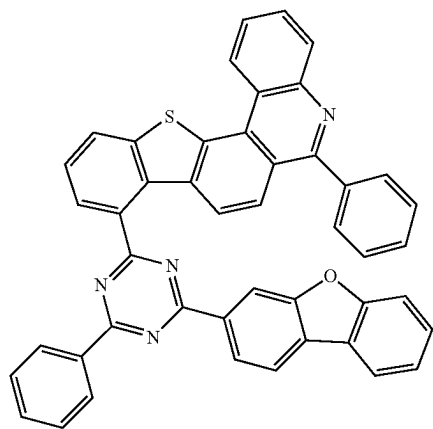
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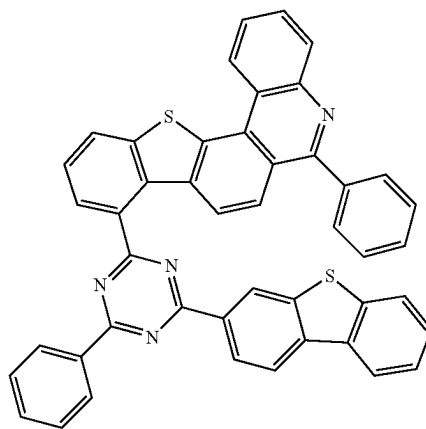
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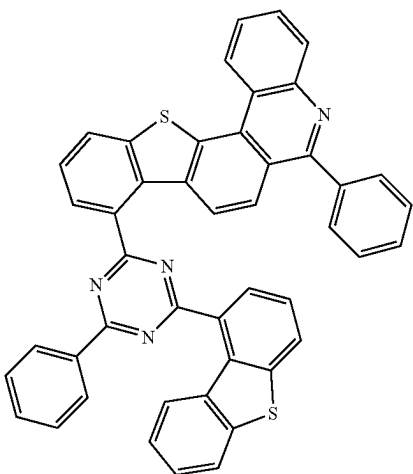
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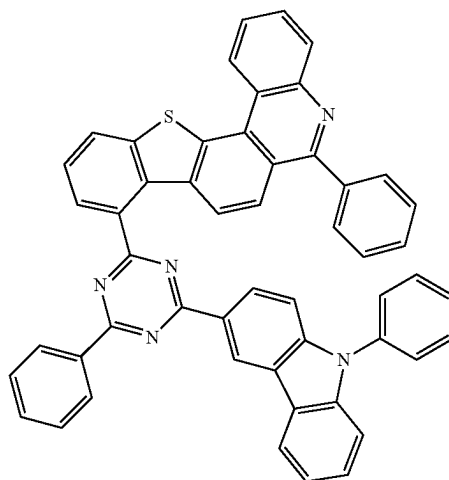
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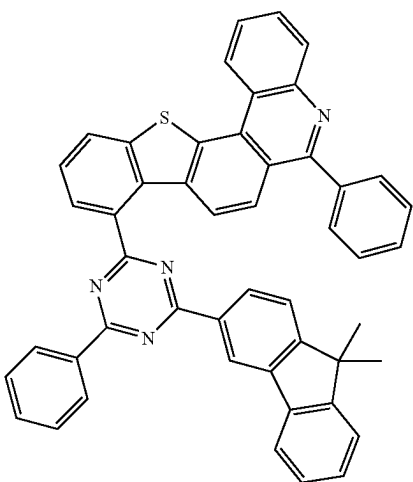
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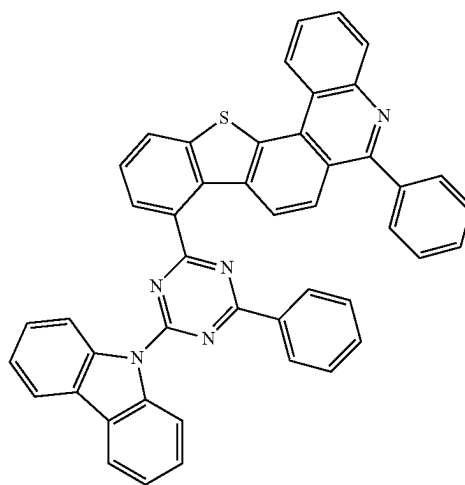


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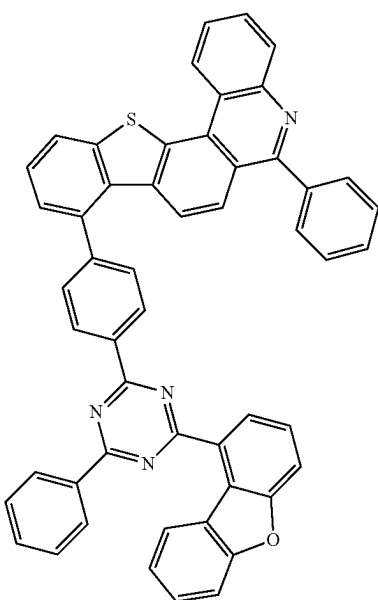
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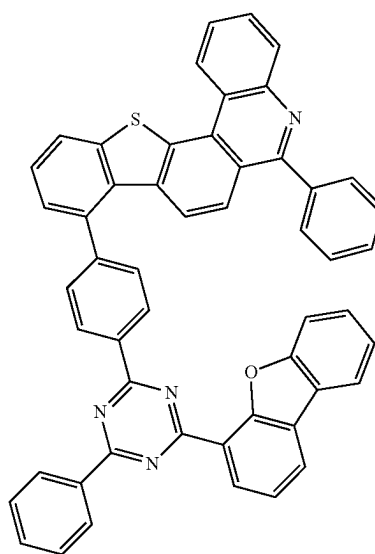
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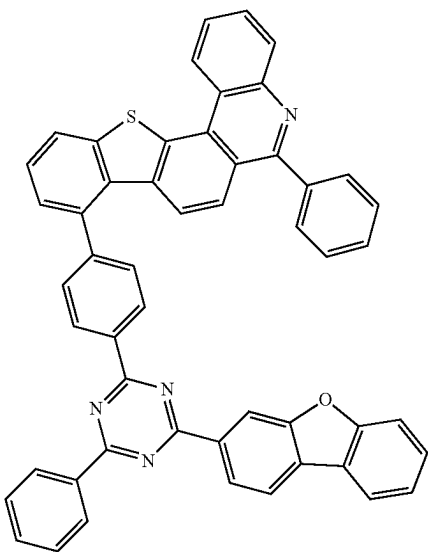
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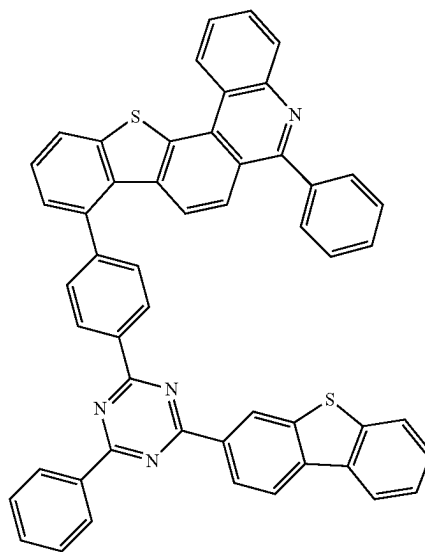


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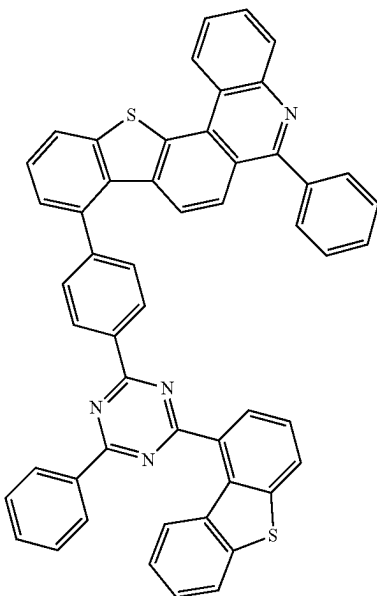
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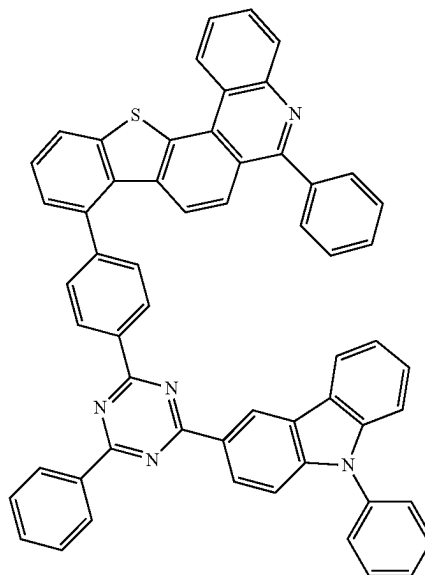


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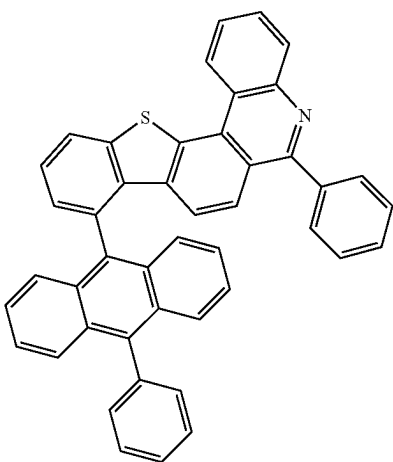
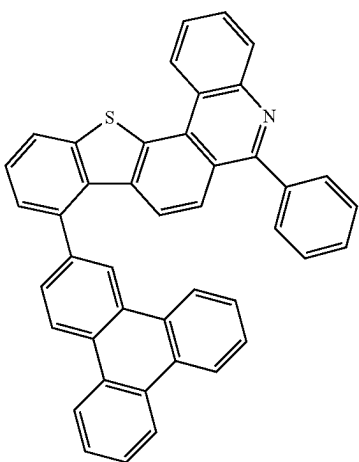
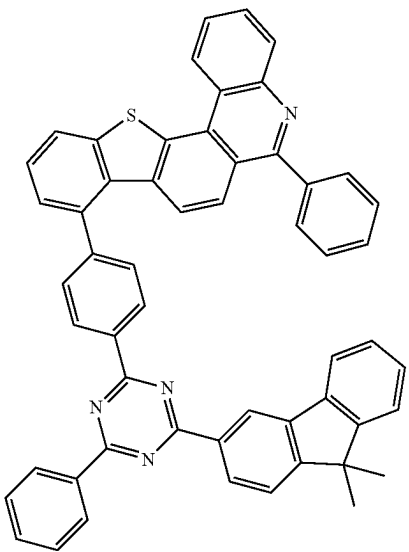
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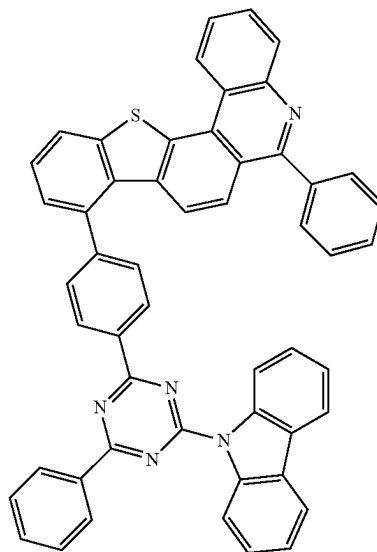


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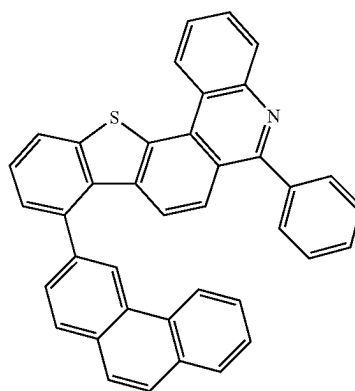


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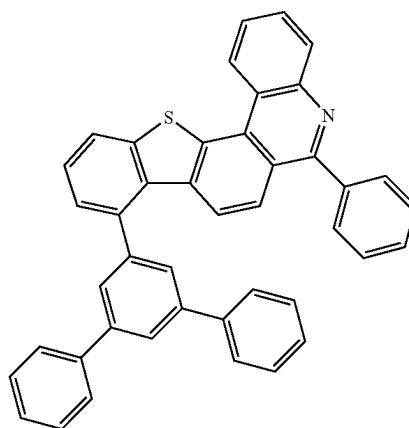
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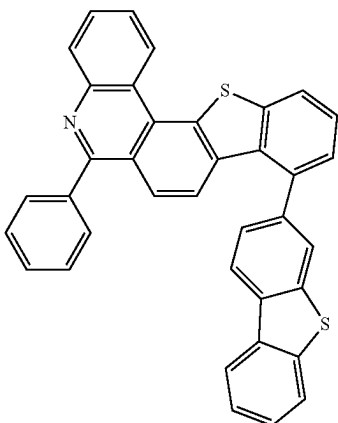
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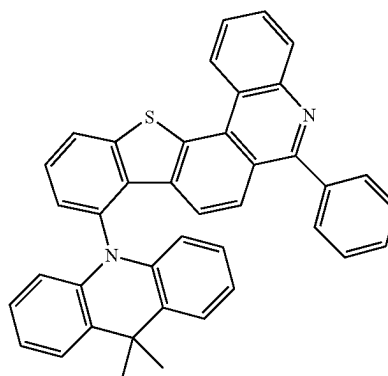
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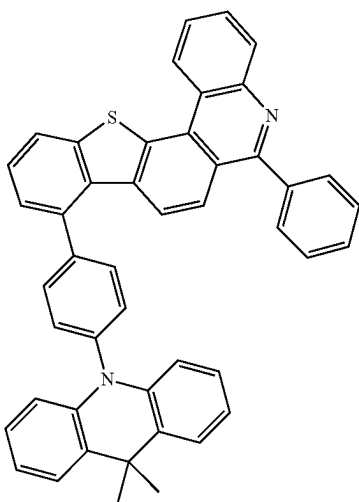
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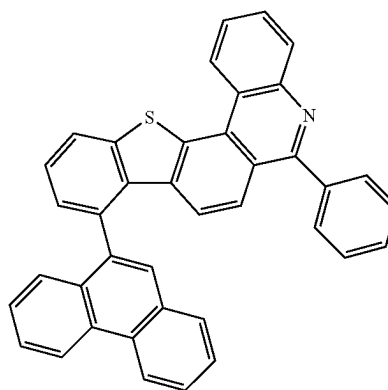


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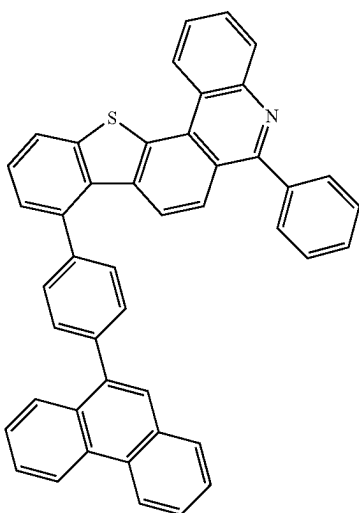
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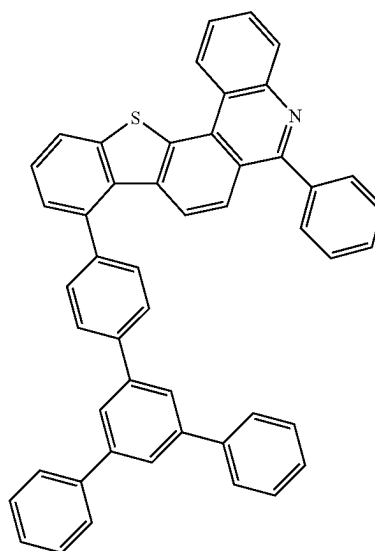
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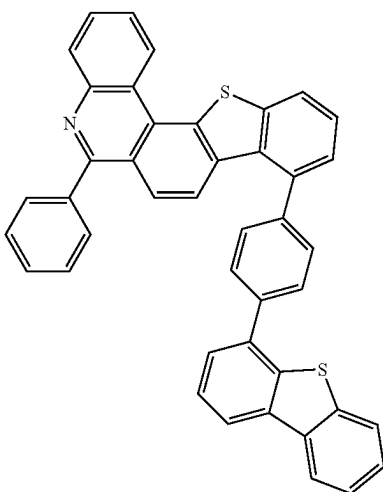
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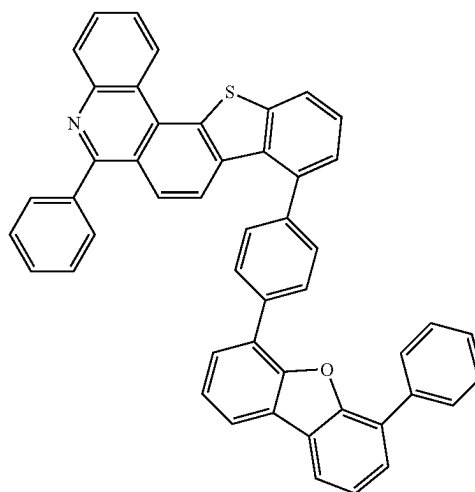


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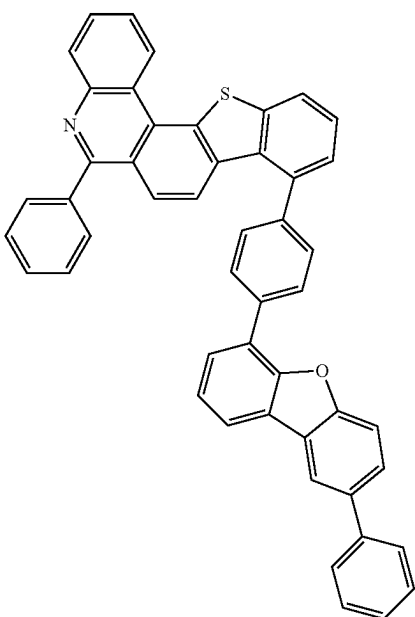


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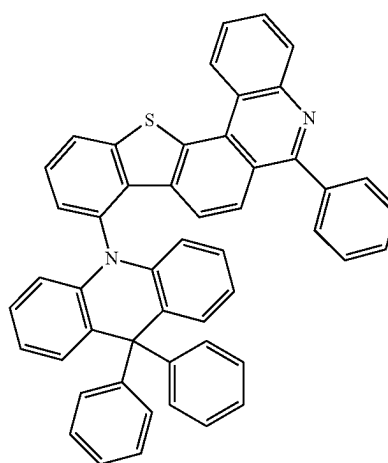
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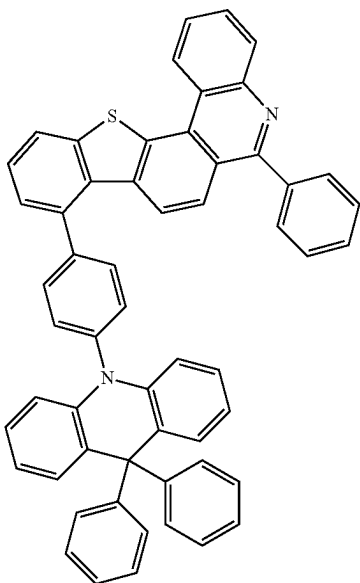


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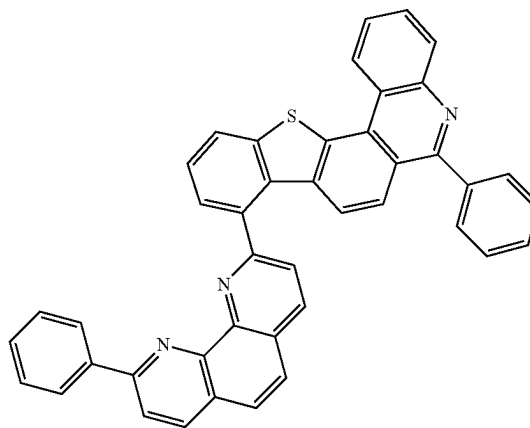
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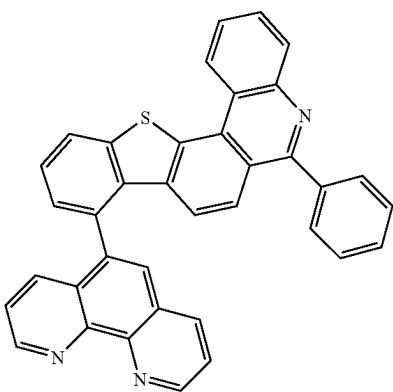
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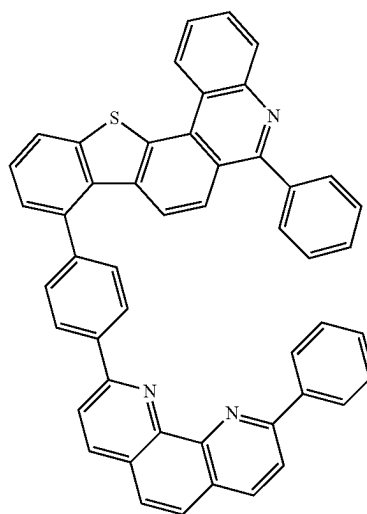


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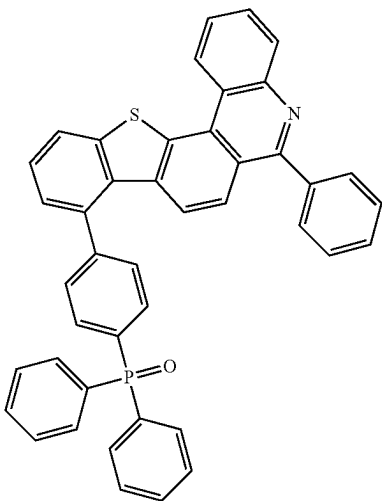
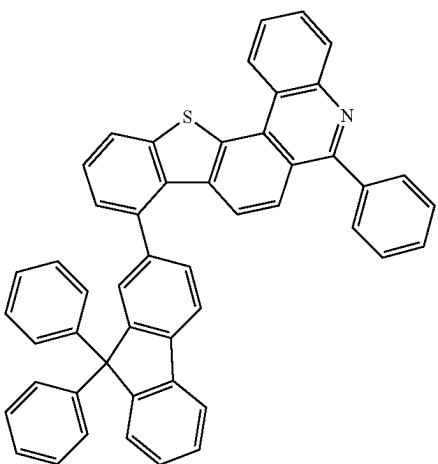
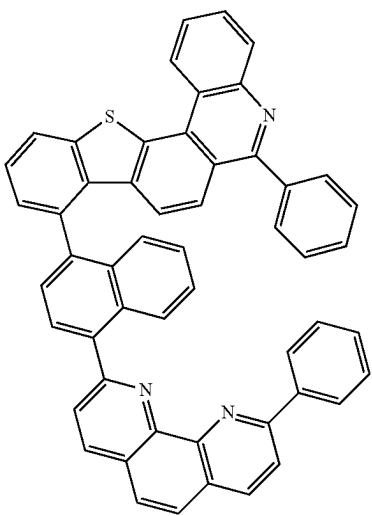
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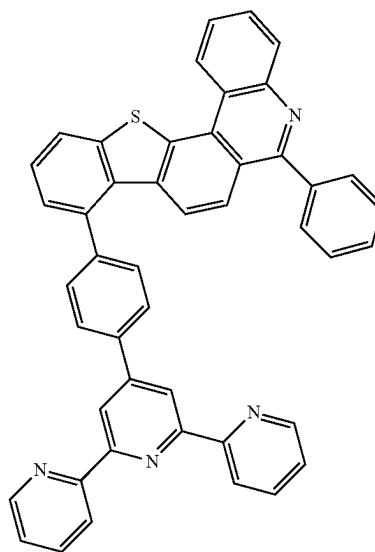


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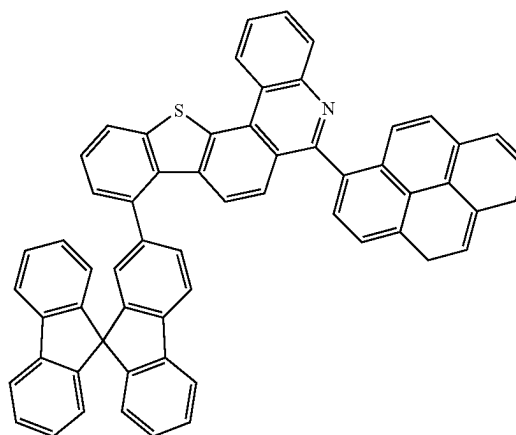
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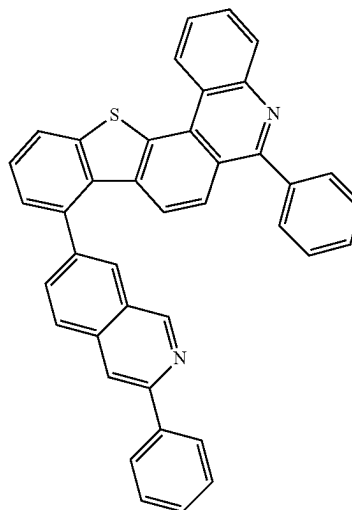
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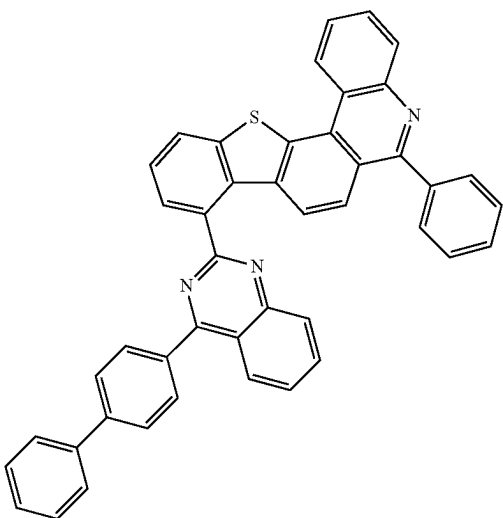
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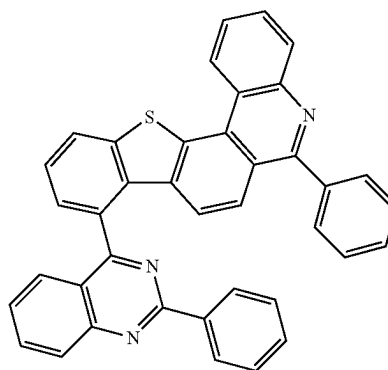
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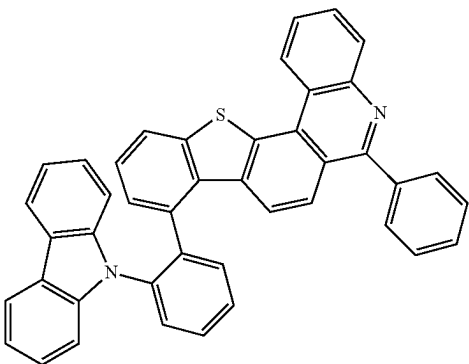
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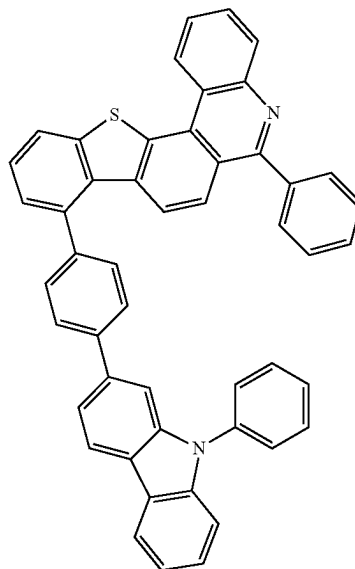
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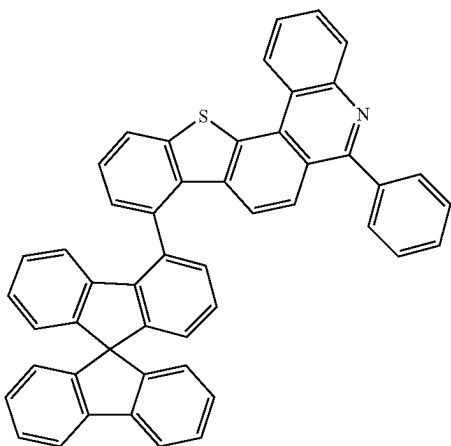
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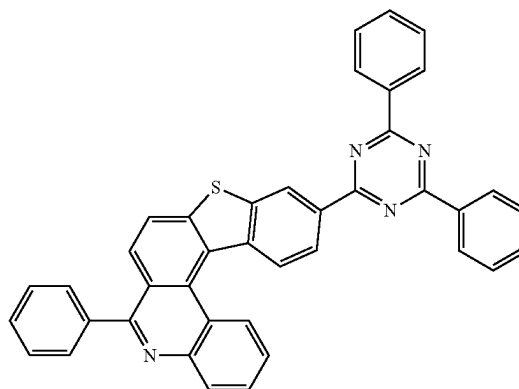
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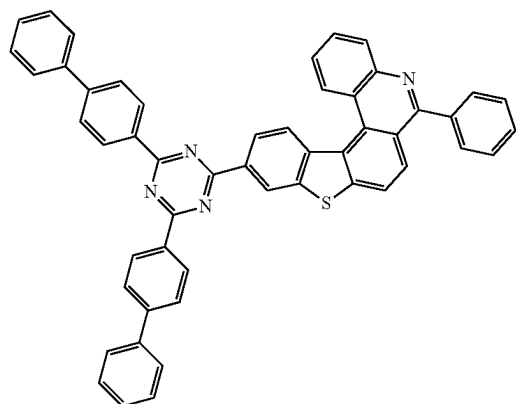
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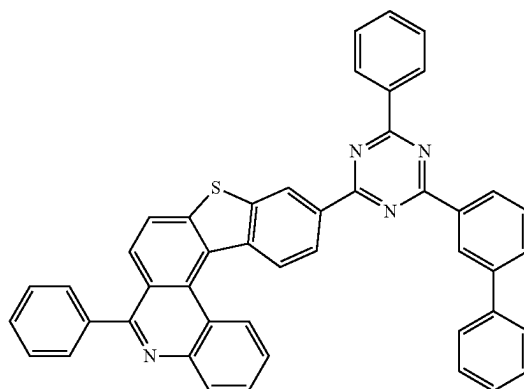


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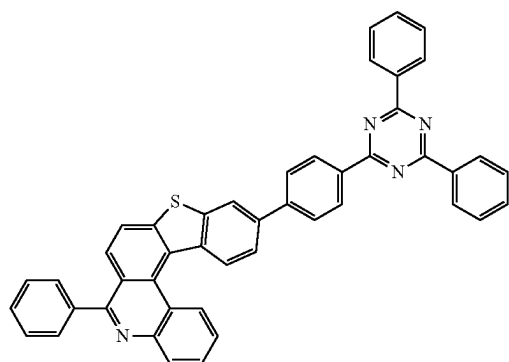
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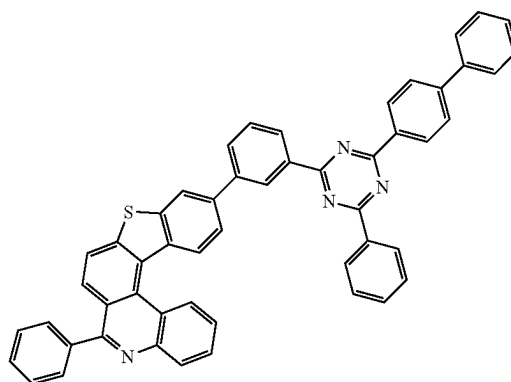


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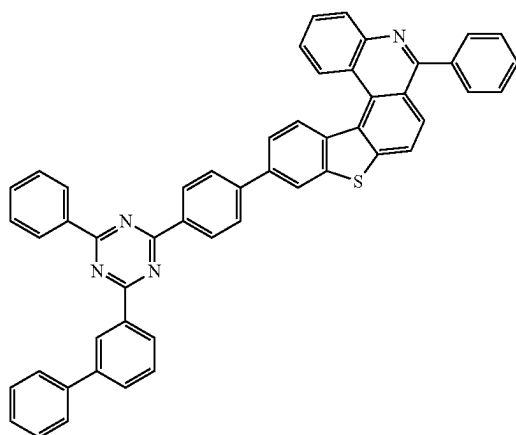
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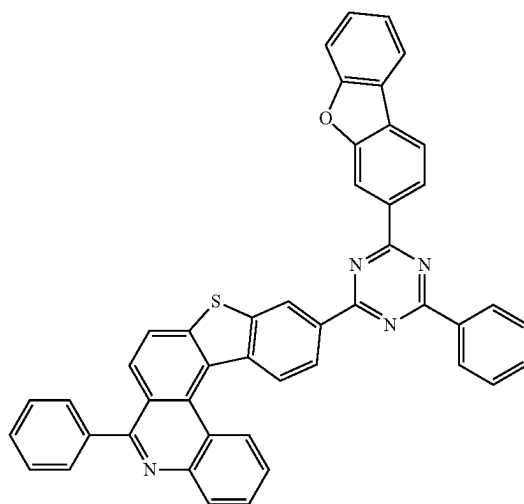
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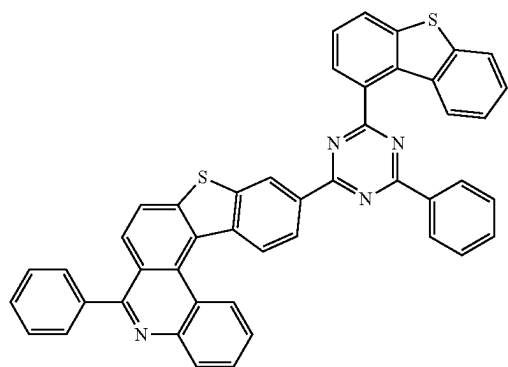
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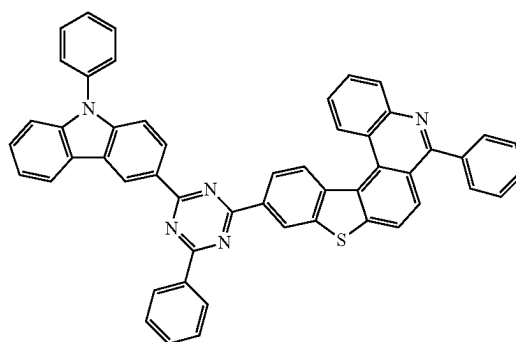


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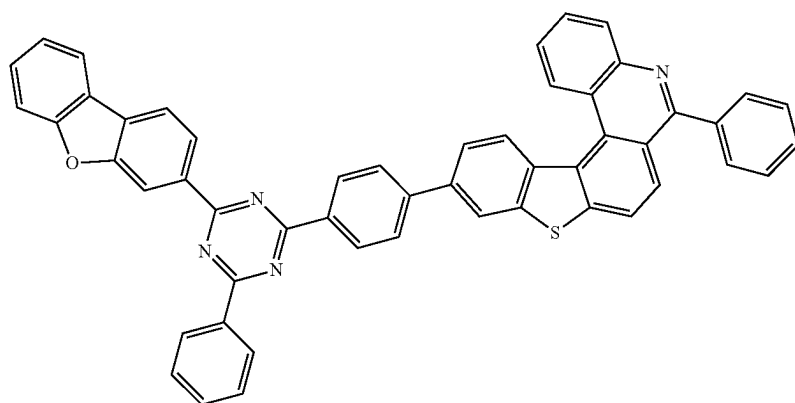


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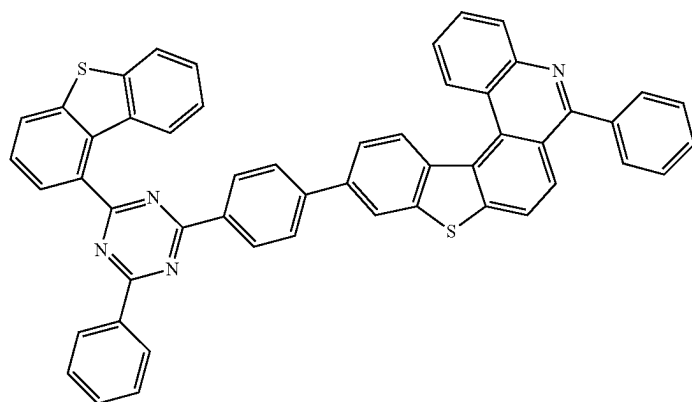
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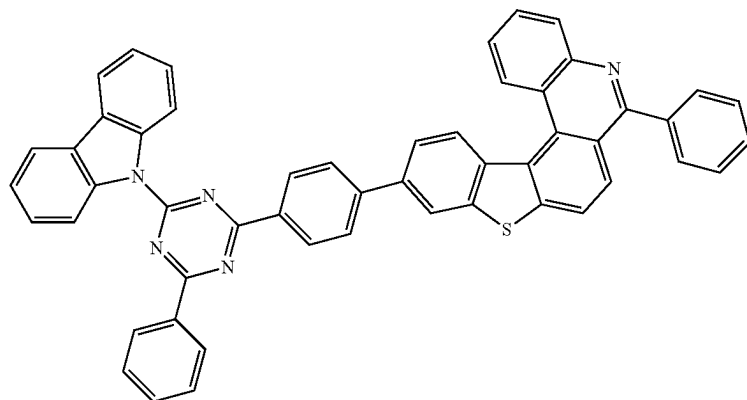
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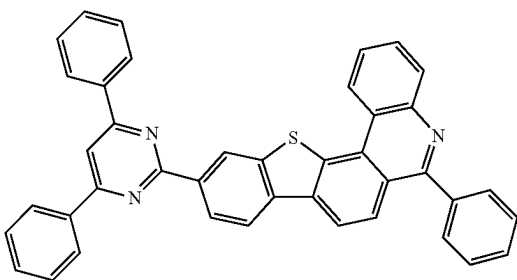


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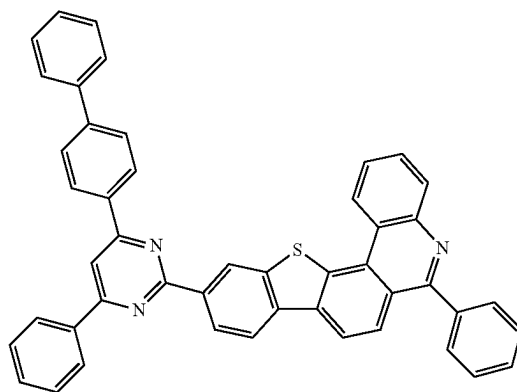


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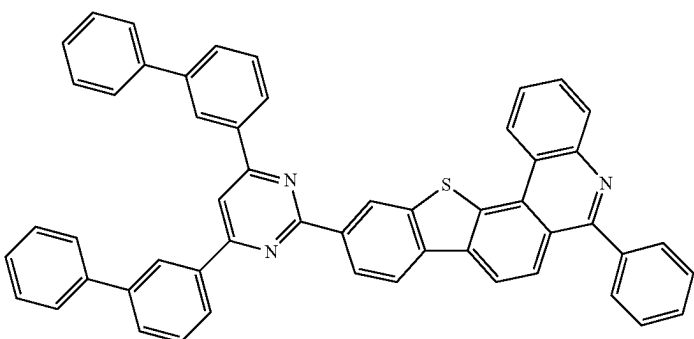
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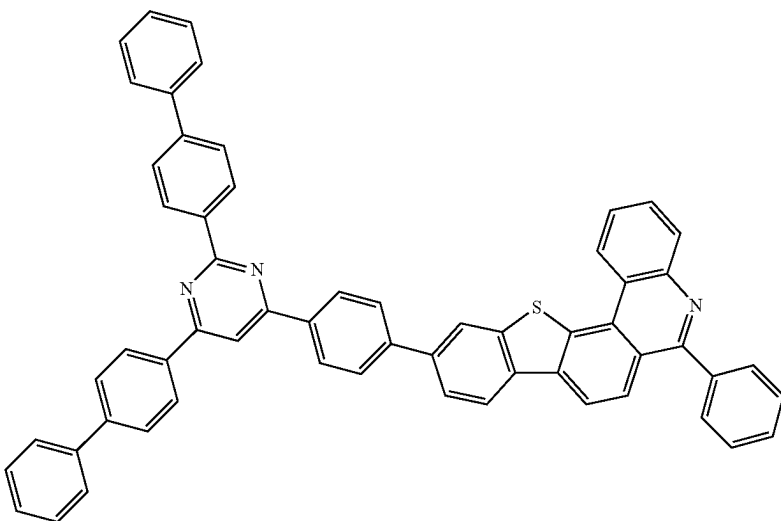
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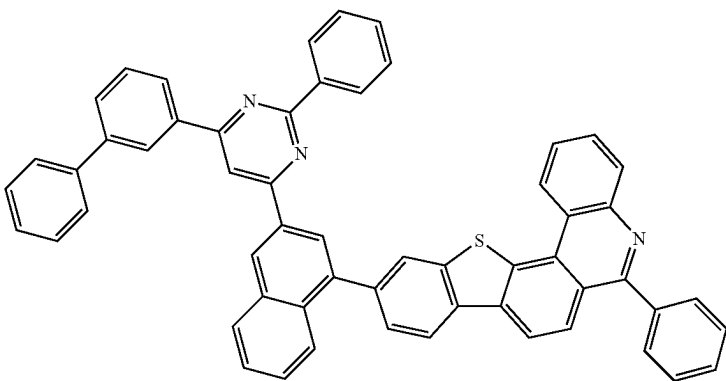
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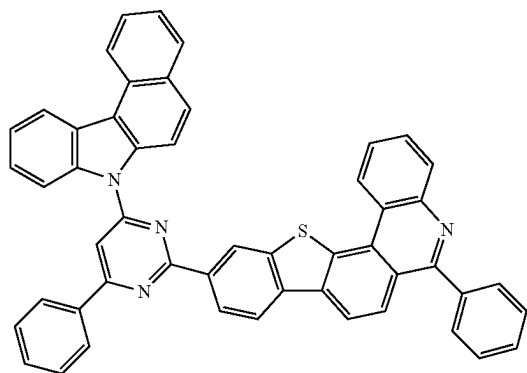


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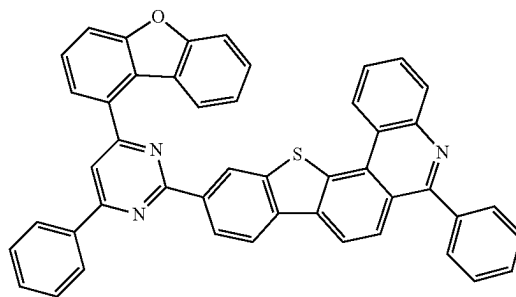
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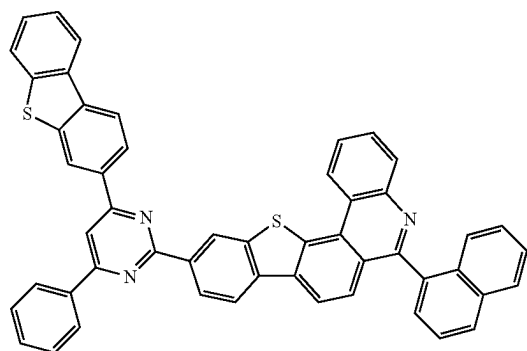
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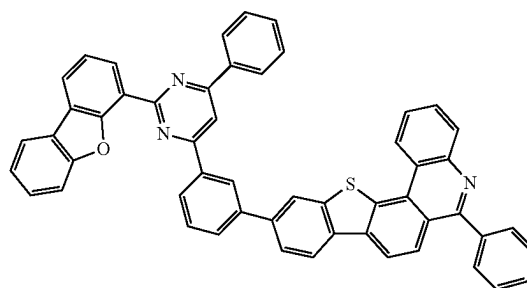
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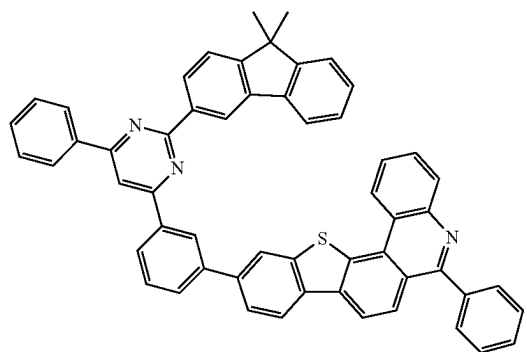
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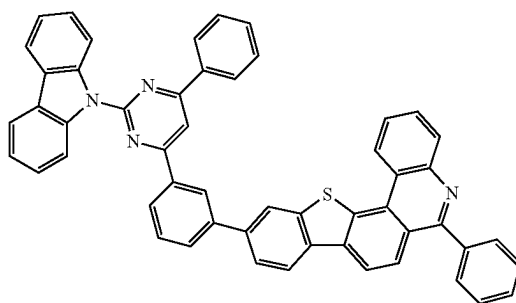


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6. An organic light emitting device comprising:
 a first electrode;
 a second electrode; and
 an organic material layer provided between the first
 electrode and the second electrode,
 wherein the organic material layer includes the heterocyclic
 compound of claim 1.
7. The organic light emitting device of claim 6, wherein
 the organic material layer includes an electron transfer layer,
 and the electron transfer layer includes the heterocyclic
 compound.
8. The organic light emitting device of claim 6, further
 comprising one, two or more layers selected from the group
 consisting of a light emitting layer, a hole injection layer, a
 hole transfer layer, an electron injection layer, an electron
 transfer layer, an electron blocking layer and a hole blocking
 layer.

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9. The organic light emitting device of claim 6 comprising:
 the first electrode;
 a first stack provided on the first electrode and including
 a first light emitting layer;
 a charge generation layer provided on the first stack;
 a second stack provided on the charge generation layer
 and including a second light emitting layer; and
 the second electrode provided on the second stack.
10. The organic light emitting device of claim 9, wherein
 the charge generation layer includes the heterocyclic com-
 pound.
11. The organic light emitting device of claim 9, wherein
 the charge generation layer is an N-type charge generation
 layer, and the charge generation layer includes the hetero-
 cyclic compound.

* * * * *